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IFIP WG 8.4

EBISION 2026

The 2nd IFIP WG 8.4 International Symposium on
E-Business Information Systems Evolution

Kookmin University
Seoul, South Korea

July 8-10, 2026

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OVERVIEW

We are living in an era of digital great transformation, where revolutionary advances in technology are reshaping every dimension of E-Business. Artificial intelligence, quantum computing, 6G mobile communications, cloud ecosystems, robotics, blockchain, and Web 3.0 are no longer distant visions but emerging realities that are redefining how businesses and societies operate. These developments demand a paradigm shift in E-Business Information Systems (EBIS), one that must be approached through truly multidisciplinary perspectives, spanning not only technology and organizations but also society, culture, consumers, and global infrastructures.

As the flagship event of IFIP Working Group 8.4, EBISION 2026 provides an international forum to explore this paradigm shift, share insights across disciplines, and envision the next stage of EBIS evolution. The symposium brings together academic researchers, industry practitioners, and policymakers from around the world to exchange ideas, build collaborations, and chart research directions capable of addressing the opportunities and risks that accompany rapid technological change.

EBISION 2026 encourages contributions that highlight both theoretical advancements and practical success stories, demonstrating how innovations can be effectively translated into sustainable impact across industries, economies, and societies. Topics span the full breadth of EBIS evolution: from AI-driven business innovation, quantum-safe systems, 6G-connected ecosystems, and cloud-native platforms to blockchain and Web 3.0 economies, human-centered design, digital trust and governance, sustainability, and cybersecurity, reflecting the interdisciplinary nature of the challenges in this domain.

Ultimately, EBISION 2026 aspires to serve as a catalyst for the continued evolution of E-Business Information Systems. By fostering collaboration and the open exchange of innovative ideas, the symposium seeks to inspire new approaches and solutions that contribute to more resilient, trustworthy, and sustainable digital economies. We warmly welcome all participants to Seoul and hope this gathering will play a pivotal role in shaping the future of EBIS research and practice.

GREETINGS FROM THE GENERAL CHAIR

Distinguished guests, colleagues, and friends,

It is my great pleasure and honor to welcome you to the 2nd IFIP WG 8.4 International Symposium on E-Business Information Systems Evolution (EBISION 2026), held from July 8 to 10, 2026, at Kookmin University in Seoul, Republic of Korea.

As the official flagship event of IFIP Working Group 8.4, EBISION 2026 brings together academic researchers, industry practitioners, and policymakers from around the world to exchange ideas, share experiences, and explore emerging research directions in the evolution of E-Business Information Systems. In an era of digital great transformation shaped by artificial intelligence, quantum computing, 6G communications, cloud ecosystems, blockchain, and Web 3.0, our symposium aspires to be a forum where multidisciplinary perspectives converge to envision the next stage of EBIS evolution.

This year, EBISION 2026 features a rich and diverse program built on a rigorous peer-review process. The program comprises 50 oral presentations and 41 poster presentations contributed by authors from 11 countries. Each submission was carefully evaluated to ensure the quality and relevance of the research presented at the symposium.

The program is further enriched by three distinguished keynote speeches:

- Dr. Tei-Wei Kuo (CTO, Delta Electronics; Professor Emeritus, National Taiwan University, Taiwan) on "Empowering AI to Drive the Next Era of Intelligent Industry";
- Prof. Nicola Dragoni (Technical University of Denmark, Denmark) on "From Logs to Cyber Threat Intelligence: Mining the Behavior of Cyber Threats"; and
- Prof. Antonio F. Skarmeta (University of Murcia; CTO, Odin Solutions, Spain) on "Data Space and Secure Data Sharing in the Context of Digital Transformation".

Complementing the keynotes, the program includes invited sessions featuring six distinguished invited speakers, as well as an industrial session with leading companies such as KT Corporation, ZenmuTech, and AUTOCRYPT, bridging cutting-edge research with real-world practice.

I am also pleased to confirm that EBISION 2026 satisfies the requirements of an international conference under Korea's Brain Korea 21 (BK21) program. With presenters and participants from 11 countries and a rigorous international review process, the symposium offers graduate students and early-career researchers a recognized international venue in which to present their work and engage with the global EBIS community.

I would like to express my sincere gratitude to everyone who made EBISION 2026 possible: the authors for their valuable contributions; our keynote speakers, invited speakers and Industrial speakers for sharing their vision, the Program Committee Chairs and members for their dedicated review efforts, and the Organizing Committee and local hosts at Kookmin University for their tireless work. My thanks also go to IFIP Working Group 8.4 for its continued support, to our sponsors, Penta Security Inc. and Sangfor Technologies Inc., and to our collaborating organizations for their generous partnership: the KMU Global Research Center for ICT

Future Automobiles and Mobility at Kookmin University, the Department of AI Convergence Engineering at Gyeongsang National University, and the KIISC Research Group on 6G Security. I am confident that EBISION 2026 will inspire new ideas, foster meaningful collaboration, and contribute to a more resilient, trustworthy, and sustainable digital future.

On behalf of the Organizing Committee, I warmly welcome you to Seoul and wish you a rewarding and memorable symposium.

Sincerely,
Ilsun You
General Chair, EBISION 2026
Kookmin University, Republic of Korea

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EBISION 2026

The 2nd IFIP WG 8.4 International Symposium on
E-Business Information Systems Evolution

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Workplace Governance in the Age of ESG and Digitalization: Union Influence in Liberal Market Economies*

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Abstract

This article examines how union members shape the workplace governance of environmental, social and governance (ESG) practices and digitalization in two liberal market economies: Ireland and the United Kingdom. While scholarship commonly treats ESG as a firm-level reporting/governance domain and digitalization as a technological or work-design issue, this study foregrounds union members as workplace actors who mediate how sustainability and digital technologies are translated into rules, practices, and voice arrangements at work. The article adopts a comparative qualitative design based on documentary and textual analysis of policy and regulatory materials, union and confederation documents, public guidance on industrial-relations procedures, and relevant comparative scholarship. Findings show that union-member influence is conditioned by shared transition pressures, such as concerns over surveillance, insecurity, and fragmented voice, yet varies systematically with national institutional pathways. In Ireland, influence is more strongly supported by institutional mediation and forms of bargaining coordination, enabling unions to connect digital change to broader

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sustainability and worker-welfare agendas. In the United Kingdom, influence depends more on workplace recognition and hybrid repertoires that combine formal representation with informal organizing, legal claims, and public campaigning. Across both cases, ESG and digitalization generate a dual dynamic: they intensify workplace risks while also opening opportunities to reframe employment relations around accountability, participation, and responsible governance. The article contributes to debates on labour governance by linking ESG and digitalization to worker voice and by conceptualizing union members as mediating agents in workplace transformation. Approaches confirms high internal consistency of the interpretation

1 Introduction

Digitalization and environmental, social, and governance (ESG) agendas increasingly intersect through the information infrastructures, metrics, reporting routines, and algorithmic systems that translate societal expectations into organizational practice (Sun et al., 2024; Tőzsér et al., 2024). As firms adopt analytics, AI-based decision tools, and platform technologies, ESG commitments intensify demands for accountability and responsible governance (Parker & Grote, 2022; Wilkinson et al., 2021). This convergence matters because it is not confined to reporting or technology adoption alone: it is increasingly worked out through workplace governance, reshaping job quality, surveillance, participation, and the distribution of worker voice in sociotechnical change (Berg et al., 2023; Grimshaw et al., 2023).

Yet existing research has not adequately explained how this governance nexus is constructed at workplace level. ESG scholarship continues to focus largely on firm-level disclosure, governance, and performance outcomes (Sun et al., 2024; Tőzsér et al., 2024), while research on digitalization has primarily examined work design, algorithmic management, and job-quality effects (Parker & Grote, 2022; Berg et al., 2023). Employment-relations research has paid closer attention to the collective regulation of technological change, but has given less sustained attention to how ESG governance intersects with worker voice as digital systems are implemented and routinised (Doellgast & Wagner, 2022; Wilkinson et al., 2021). As a result, scholarship still knows too little about how union members, through bargaining, organizing, and framing, shape the governance of ESG and digitalization as these agendas converge.

This omission matters because ESG and digitalization are not parallel developments but overlapping governance regimes. ESG functions as an accountability regime, introducing disclosure requirements, metrics, and normative expectations around responsible conduct, while digitalization operates as a control and coordination regime through data infrastructures, monitoring systems, and algorithmic decision-making. In practice, the two are increasingly interdependent: ESG commitments rely on digital systems for data capture, reporting, and verification, while digital technologies generate governance problems (e.g., surveillance, algorithmic bias, and workforce fragmentation that fall directly within the social and governance dimensions of ESG) (Sun et al., 2024; Parker & Grote, 2022; Grimshaw et al., 2023). Their convergence, therefore, produces a common workplace-governance dynamic in which accountability logics and technological control jointly shape job quality, participation, and worker voice. (e.g., For workers and their representatives, this creates a dual dynamic: digitalization can intensify monitoring, insecurity, and work pressure, yet ESG frameworks can also provide a language of accountability through which such risks are contested and negotiated) (Berg et al., 2023; Woodcock & Cant, 2022; Wilkinson et al., 2021).

This article addresses that gap by foregrounding union members as mediating workplace actors whose influence depends on mobilized power resources and institutional pathways (Lévesque &

Murray, 2010; Refslund & Arnholtz, 2022). We compare Ireland and the United Kingdom as most-similar liberal market economies with different routes to worker voice: Ireland features stronger institutional mediation and some bargaining coordination, whereas the United Kingdom relies more heavily on workplace recognition as the gateway to bargaining influence (O'Connell, 2021; DETE, 2025; WRC, 2025; ACAS, 2026; GOV.UK, 2026). Using comparative document analysis, we ask: (1) how do union members influence the integration and governance of digital tools and ESG practices in workplaces? and (2) what repertoires do they deploy to balance innovation with worker protection across institutional contexts? In doing so, the article makes three contributions: it reconceptualises ESG and digitalization as a workplace governance regime, specifies how union members translate these pressures into workplace arrangements, and develops a pathways argument showing how national institutional routes condition the form, timing, and scope of union-member influence.

2 Related Work

2.1 ESG, Digitalization, and Employment Relations

Existing research on ESG and digitalization primarily follows three streams, each offering valuable insights yet leaving an important workplace-governance gap. The first stream conceptualizes ESG and digitalization mainly as governance and performance phenomena, emphasizing how sustainability reporting regimes, non-financial accountability, and digital transformation reshape corporate governance structures and external stakeholder monitoring (Tőzsér et al., 2024; Sun et al., 2024). This work is especially useful for explaining why firms adopt ESG disclosure practices and how digital transformation can strengthen or weaken governance through data visibility and reporting infrastructures (Sun et al., 2024; Tőzsér et al., 2024). However, these studies often treat ESG as a boardroom- and reporting-centered governance architecture, leaving under-specified the workplace-level mechanisms through which ESG commitments are translated into rules, routines, and decision rights that affect employee participation, consultation, and the everyday governance of work. Put differently, ESG is frequently analyzed "above" the workplace, even though the implementation of ESG targets commonly relies on digital systems that reorganize work processes and accountability in ways that directly implicate workers.

The second stream examines digitalization through the lens of work design and job quality, showing how digital tools and automation reshape worker autonomy, monitoring, pacing, and work intensity (Parker & Grote, 2022; Berg et al., 2023). This literature provides strong evidence that digital systems are not neutral instruments; rather, they embed managerial assumptions about control, performance measurement, and discretion, with consequential effects for job quality and wellbeing (Parker & Grote, 2022; Berg et al., 2023). Yet the dominant framing remains largely focused on proximal work-design outcomes, how technologies alter tasks and discretion, while giving less systematic attention to digitalization as an employment-relations issue that is governed through collective bargaining, representation, and institutionalized voice. This omission becomes particularly consequential in fragmented and non-standard work contexts, including gig and platform work, where employment status ambiguity and weak representation can intensify exposure to surveillance and unilateral algorithmic control (Wu & Huang, 2024).

A third stream, closer to employment relations, focuses on how digitalization intersects with worker voice, collective regulation, and institutional mediation. Research in this stream argues that digital

transformation is not solely a technical implementation challenge but a contested process shaped by collective action and the institutional conditions of voice and bargaining (Doellgast & Wagner, 2022; Wilkinson et al., 2021). It highlights how unions and worker representatives mobilize narratives, solidarity, and institutional resources to negotiate or resist monitoring, restructuring, and work reorganization associated with new technologies (Wilkinson et al., 2021; Doellgast & Wagner, 2022). However, this stream has less consistently integrated ESG as a governance frame that can reshape the terrain of digitalization, both by providing new legitimacy resources for worker claims (e.g., accountability, participation, responsible governance) and by introducing new metrics and compliance routines that may reinforce managerial control (Grimshaw et al., 2023).

Taken together, these streams leave unresolved how ESG (as an accountability regime) and digitalization (as a control and coordination regime) interact at the workplace level, and how union members mediate that interaction through voice, bargaining, and hybrid influence strategies. Addressing this gap requires bringing ESG governance into closer conversation with employment relations and sociotechnical change, focusing on the workplace institutions and practices through which "responsible" digital transformation is actually governed.

2.2 ESG and Digitalization in the Workplace: Key Challenges and Opportunities for Union Members

The intersection of digitalization and ESG governance presents both challenges and opportunities for unions because it reconfigures the "rules of the game" through which work is designed, monitored, and justified. Digitalization changes not only tools but also governance arrangements by embedding evaluation, coordination, and control into data infrastructures (e.g., performance metrics, monitoring systems, algorithmic scheduling), while ESG introduces accountability logics that elevate questions of fairness, participation, and responsible conduct (Wilkinson et al., 2021; Parker & Grote, 2022; Sun et al., 2024). In this setting, unions become pivotal actors for safeguarding worker welfare, not simply by defending existing terms and conditions but by shaping how digital and ESG commitments are translated into workplace practices (Pulignano et al., 2026; Wilkinson et al., 2021). Digital technologies can reduce discretion and intensify work through real-time monitoring and data-driven targets, increasing the salience of union intervention to secure transparency, contestability, and voice in technology governance (Parker & Grote, 2022; Berg et al., 2023). At the same time, ESG can supply unions with new frames and legitimacy resources, such as "responsible governance" and "stakeholder accountability", that allow traditional employment concerns (e.g., privacy, work intensity, training, job security) to be advanced as governance issues rather than isolated grievances (Grimshaw et al., 2023; Wilkinson et al., 2021).

Institutional context strongly conditions how unions can act on these challenges and opportunities. In Ireland, the legacy of social partnership and the presence of institutional mediation have enabled unions to engage more directly in shaping policy discourse and national-level approaches to digital transformation and sustainability, providing pathways to link worker voice to ESG integration beyond individual workplaces (O'Connell, 2021; DETE, 2025; WRC, 2025). Such supports can facilitate more coordinated bargaining agendas around skills, consultation, and safeguards against adverse consequences of automation and data-driven management (Pulignano et al., 2026; Parker & Grote, 2022). By contrast, unions in the United Kingdom operate in a more fragmented industrial relations environment in which influence often hinges on workplace recognition and firm-level bargaining access, and where unions may need to combine formal negotiation with informal organizing and broader public strategies, especially in platform-mediated and non-standard work settings (Woodcock & Cant, 2022; Wilkinson et al., 2021). This contrast suggests that the same ESG–digitalization pressures can yield

different patterns of union influence depending on whether institutional supports enable coordinated engagement or whether influence must be assembled through hybrid repertoires across dispersed workplaces (Doellgast & Wagner, 2022; Keizer et al., 2024).

A complementary explanation comes from union power and capacity research. In fragmented and non-standard employment contexts, unions' ability to shape digital governance depends less on formal presence alone and more on mobilizing power resources: internal solidarity, strategic narratives, organizational capabilities, and institutional links that convert workplace concerns into collective demands (Lévesque & Murray, 2010; Refslund & Arnholtz, 2022). This is especially pertinent for platform work and other forms of fragmented employment, where work is organized across multiple entities and where conventional voice channels are weakened (Grimshaw et al., 2023; Woodcock & Cant, 2022). In such settings, unions may need to blend bargaining with organizing, litigation, and public campaigning to secure protections and recognition, illustrating how power resources shape diverse strategies and outcomes (Keizer et al., 2024; Woodcock & Cant, 2022). Digitalization thus becomes a contested governance process in which union influence is mediated by the availability of institutional access points and the capacity to mobilize resources to make algorithmic control and ESG accountability actionable workplace issues (Doellgast & Wagner, 2022; Wilkinson et al., 2021).

Finally, unions' engagement with CSR/ESG is increasingly relevant as a bridge between workplace regulation and broader societal governance. Where firms frame digital transformation as innovation and ESG compliance as responsible conduct, unions can press for commitments that ensure these agendas produce social value rather than shifting risks onto workers, particularly in the gig economy and platform work where employment protections are uneven (Woodcock & Cant, 2022; Wu & Huang, 2024). This includes advocating legal and regulatory reforms that embed ESG principles into digital labor governance (e.g., transparency, accountability, and protections against unfair monitoring), so that platform workers are not excluded from the "S" in ESG during digital transition (Keizer et al., 2024; Grimshaw et al., 2023). In this sense, unions can act as governance intermediaries, translating ESG claims into concrete safeguards and participation mechanisms that shape how digital systems are designed, deployed, and contested at work (Wilkinson et al., 2021; Doellgast & Wagner, 2022).

2.3 Understanding Union Member Responses to ESG and Digitalization

Recent research treats union responses to workplace change less as direct reactions to external pressures than as processes through which workplace issues become collective and actionable. This is particularly relevant to ESG and digitalization, where many workplace changes emerge through monitoring systems, performance metrics, compliance routines, restructuring, and changing skill demands rather than through immediately visible industrial relations conflict (Pulignano et al., 2026; Wilkinson et al., 2021). Within this literature, research on union power and capacity provides the main analytical foundation. Union influence is linked not simply to formal position or membership size, but to the mobilization of solidarity, organizational links, narratives, and institutional resources (Lévesque & Murray, 2010; Refslund & Arnholtz, 2022). In this formulation, workplace pressures become consequential when they can be connected to collective meanings and forms of action.

This emphasis on mediation is extended in research on technological change. Studies of Industry 4.0 show that workplace discontent is framed around issues such as well-being, job insecurity, and work quality, and that these framing processes are associated with different bargaining and mobilization outcomes (Pulignano et al., 2026). Comparative studies of unions and digitalization similarly show that the effects of technological change vary with how change is interpreted and negotiated in different organizational and national settings (Lloyd & Payne, 2023; Payne et al., 2023). Across this literature, framing links technological and governance change to collective concerns. A related literature examines

how digitalization alters the organizational conditions of voice and representation. Studies of worker voice show that digitalization changes both the channels of voice and the conditions under which concerns are collectivized (Wilkinson et al., 2021). Research on fragmented work further shows that representation becomes more difficult where work is organized through subcontracting, platforms, and inter-organizational relations rather than within a single employing organization (Grimshaw et al., 2023). The literature on precarious and platform work develops this point further by showing that responses vary with available power resources, institutional supports, and organizational channels, and may combine formal bargaining, informal networks, legal claims, and public campaigning (Keizer et al., 2024; Woodcock & Cant, 2022).

By focusing on union members, we refer not to individual attitudes but to how members, acting through representative roles, mobilization, and institutional channels, mediate workplace governance outcomes. In documentary analysis, this influence is observed through claims, procedural rights, and negotiated rules attributed to member action rather than organizational headquarters alone.

3 Theoretical framework

This study draws on Sociotechnical Systems Theory and Power Resource Theory to explain how union members mediate the governance of ESG and digitalization at the workplace level. Together, these perspectives illuminate both why worker participation matters in digital transformation and how union-member influence varies across institutional contexts.

3.1 Sociotechnical Systems Theory and Inclusive Digital Governance

Sociotechnical Systems Theory provides a foundational lens for understanding why union engagement is consequential in processes of digital transformation. Classic sociotechnical accounts argue that organizational performance and sustainability depend on the joint optimization of social and technical subsystems, rather than on the unilateral implementation of technological solutions (Emery & Trist, 1960; Cherns, 1976). From this perspective, workplace technologies are not neutral tools but social arrangements that embed assumptions about control, coordination, and discretion. As digital systems reshape task allocation, monitoring, and decision-making, their effects on job quality, autonomy, and participation become central to organizational outcomes.

Applied to contemporary digitalization, sociotechnical theory suggests that inclusive digital governance is not merely an ethical aspiration but a design principle. When algorithmic management, data-driven performance systems, and platform technologies reorganize work, the availability of worker voice and participation conditions whether these systems generate sustainable performance or instead produce hidden costs such as stress, insecurity, resistance, and disengagement (Parker & Grote, 2022; Berg et al., 2023). Sociotechnical theory therefore supports treating union interventions, such as consultation rights, bargaining over monitoring and data use, commitments to training and upskilling, and procedures for contesting data-driven decisions, as governance mechanisms that align technological change with worker welfare and long-term organizational sustainability (Parker & Grote, 2022; Wilkinson et al., 2021).

3.2 Power Resource Theory and Union-Member Influence

While sociotechnical theory explains why participation matters, Power Resource Theory (PRT) helps explain why union-member influence over the governance of ESG and digitalization varies across

workplaces and national systems (Korpi, 1998). In PRT, union power is not reducible to membership density or formal recognition alone; rather, influence depends on the capacity to mobilize and combine multiple power resources, including internal solidarity, organizational infrastructure, strategic framing and narratives, expertise, and institutional linkages (Lévesque & Murray, 2010; Refslund & Arnholtz, 2022). These resources enable unions and their members to transform diffuse workplace concerns into collective claims that are actionable in bargaining, legal, and public arenas.

PRT is particularly relevant in the context of ESG and digitalization, where governance issues often emerge indirectly through monitoring systems, performance metrics, compliance routines, and platform-mediated work arrangements that fragment voice and complicate representation (Grimshaw et al., 2023; Wilkinson et al., 2021). From a power resource perspective, unions' capacity to shape responsible digital governance, for example, by limiting surveillance, securing consultation rights, or linking upskilling and job security to sustainability commitments, depends on whether they can leverage institutional supports (such as mediation bodies or coordinated bargaining) and/or deploy hybrid repertoires that combine formal representation with organizing, legal strategies, and public campaigning (Keizer et al., 2024; Woodcock & Cant, 2022).

3.3 Integrating the Two Perspectives

Taken together, Sociotechnical Systems Theory and Power Resource Theory provide a complementary framework for analyzing union-member influence over ESG and digitalization.

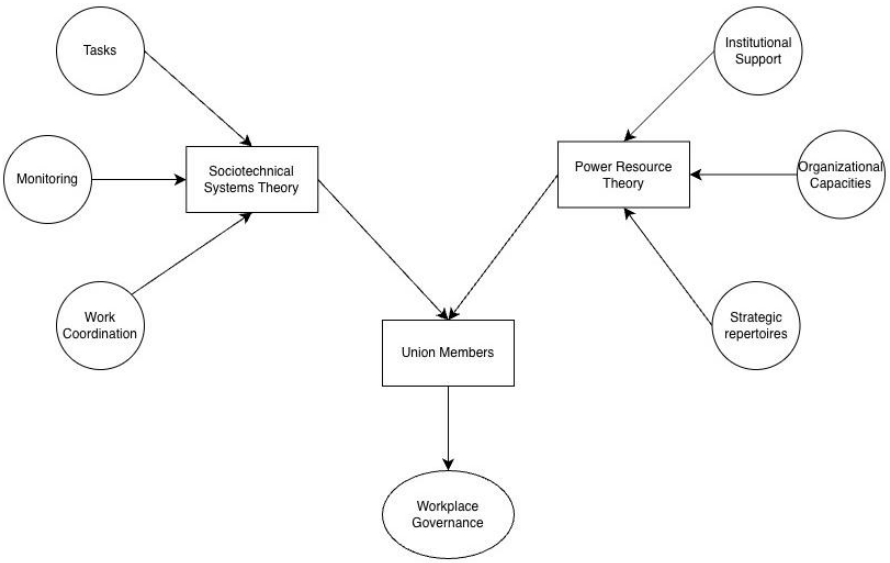

Figure 1: Union Member and the Governance of ESG and Digitalization

Figure 1 illustrates the theoretical framework guiding this study. Drawing on Sociotechnical Systems Theory and Power Resource Theory, the framework conceptualizes union members as mediating actors in the workplace governance of ESG and digitalization. Sociotechnical Systems Theory highlights that digital technologies reshape tasks, monitoring, and coordination, and that their outcomes depend on the joint optimization of social and technical subsystems; from this perspective, worker voice and participation are integral to sustainable digital governance rather than residual concerns. Power Resource Theory complements this view by explaining how union-member influence

varies across institutional contexts, emphasizing that governance outcomes depend on the mobilization of power resources such as institutional supports, organizational capacities, and strategic repertoires. Positioned at the center of the framework, union members translate ESG commitments and digital systems into workplace governance practices, such as consultation rights, rules on monitoring, training provisions, and mechanisms for contesting data-driven decisions. The framework therefore links technological design and accountability regimes to concrete governance outcomes at work, while highlighting how institutional pathways condition the capacity of union members to shape the governance of ESG and digitalization.

4 Case Selection and Research Methods

4.1 Comparative design and case logic

This study adopts a comparative qualitative case-study design centred on Ireland and the United Kingdom, selected through a most-similar systems design to examine how comparable institutional environments generate divergent patterns of union-member influence. Both countries are well-established liberal market economies characterised by decentralized bargaining structures and voluntarist industrial-relations traditions, which helps hold constant key macro-institutional features relevant to workplace governance. They also share a common Anglo institutional heritage, including historically similar legal foundations governing employment and collective representation, making them analytically comparable. This most-similar comparison allows the study to hold constant the broader liberal-market context while isolating variation in the institutional pathways through which worker voice is organised and union-member influence is exercised.

Despite these similarities, the two countries differ in the institutional mechanisms through which union-member influence can be exercised. In Ireland, institutional mediation and elements of bargaining coordination provide more structured supports for worker voice, whereas in the United Kingdom influence depends more heavily on workplace recognition and firm-level bargaining access. This difference is central to the article's pathways argument.

4.2 Ireland: voluntarism with institutional mediation and coordination

Ireland is analytically relevant because it combines a voluntarist industrial-relations tradition with continuing forms of institutional mediation. Official policy documents characterize Irish industrial relations as essentially voluntary, but established institutions, especially the Workplace Relations Commission (WRC) and the Labour Court, provide formal mechanisms for conciliation, dispute resolution, and procedural support that can indirectly strengthen collective voice by structuring how disputes are processed and how bargaining is sustained (DETE, 2025; WRC, 2025). Recent policy developments, particularly Ireland's Action Plan to Promote Collective Bargaining 2026–2030, further signal an active state interest in strengthening collective bargaining capacity (DETE, 2025). In addition, recent scholarship suggests that, despite Ireland's liberal market setting, meaningful forms of bargaining coordination persist in some sectors (O'Connell, 2021; Paolucci & Roche, 2024).

4.3 United Kingdom: decentralized voluntarism with statutory recognition

The United Kingdom is analytically relevant because it combines decentralized voluntarism with a

statutory route to union recognition. Where voluntary recognition is not granted, unions may apply for statutory recognition through the Central Arbitration Committee (CAC), making workplace recognition a key institutional gateway for enforceable union influence and collective bargaining access (ACAS, 2026; GOV.UK, 2026.). Compared with Ireland, this institutional configuration places greater weight on recognition status and workplace-level bargaining access, and less on national-level institutional mediation.

4.4 What the comparison is designed to reveal

The comparison is designed to reveal whether similar ESG and digitalization pressures generate different forms of union-member influence when access to worker voice is institutionally structured in different ways. By comparing two broadly similar liberal market economies, the study controls for major political-economic differences while focusing on variation in mediation, coordination, and recognition as routes to workplace influence. The aim is therefore not to contrast fundamentally different industrial-relations regimes, but to explain how different institutional pathways shape the form, scope, and timing of union-member influence over workplace governance.

4.5 Data and analytic approach: documentary and textual analysis

The study is based on documentary and textual analysis, which is appropriate because the article focuses on how union-member influence is institutionally structured and publicly articulated, rather than on estimating prevalence or reconstructing the full range of private perceptions. It examines how workplace mediation is institutionally structured and rendered visible through documentary traces such as claims, disputes, procedural activations, and negotiated outcomes (Bowen, 2009). In line with Prior's (2003) view of documents as embedded in social action rather than as passive records, the analysis treats policy texts, adjudication decisions, procedural guidance, and union materials as evidence of mediating processes and documented causal mechanisms through which workplace governance is articulated and contested. This interpretation is consistent with mechanism-oriented qualitative research that uses documentary evidence to assess how causes contribute to outcomes through intervening processes (Beach & Pedersen, 2019; Collier, 2011).

The study is based on documentary and textual analysis, which is appropriate because the article focuses on how union-member influence is institutionally structured and publicly articulated, rather than estimating prevalence or reconstructing the full range of private perceptions. The evidence base consists of four types of material: (1) official policy and regulatory documents, (2) publicly available trade union and confederation materials, (3) public industrial-relations guidance on recognition and dispute resolution, and (4) secondary scholarly work in comparative employment relations and union studies (Doellgast & Wagner, 2022; OECD & AIAS, 2025a, 2025b).

Analytically, this design enables the study to trace (i) the institutional channels through which union voice can be exercised (e.g., mediation bodies, recognition procedures), and (ii) how ESG and digitalization are framed as legitimate objects of workplace governance within public policy texts, union strategies, and guidance on collective procedures. By triangulating across the four document types, the analysis supports a comparative account of how similar transition pressures are translated into different repertoires of union-member influence under distinct institutional conditions.

Within this design, union-member influence is operationalized through workplace-level representation and participation rather than as an attribute of unions as centralized organizations. The analysis interprets documentary evidence such as adjudication decisions, policy submissions, and procedural guidance as reflecting member-level influence when they capture claims, disputes, and negotiated outcomes rooted in workplace interactions, including issues related to monitoring, data use,

job quality, and training. These documents frequently reflect the role of workplace representatives (e.g., shop stewards or employee representatives) in articulating and advancing such concerns through institutional channels.

This analytical distinction separates member-level influence from broader union organizational strategy or leadership discourse. While national unions and confederations frame policy positions and strategic priorities, the documents examined here provide insight into how workplace-based concerns are mobilized and translated into governance outcomes. Evidence is therefore attributed to union-member influence where it reflects the activation of procedural rights, dispute mechanisms, or negotiated changes linked to workplace-level representation, rather than solely top-down policy articulation. This approach enables the study to examine how union members function as mediating actors connecting institutional structures with concrete workplace governance practices.

Table 1. Ireland and the United Kingdom as comparative cases

	Ireland	**United Kingdom**	**Analytical relevance**
Market economy Structure	Liberal market economy	Liberal market economy	Holds broad context constant
Union Density	22.2% (2024)	22.0% (2024)	Comparable membership base
Bargaining Coverage	34% (latest available year: 2017)	40.2% (2024)	Different bargaining reach
Predominant Bargaining Level	Local/company level	Local/company level	Similar decentralization
Key Institutional Support	WRC, Labour Court, collective bargaining action plan	CAC statutory recognition route	Distinguishes pathways of influence
Likely Route of Union-Member Influence	Institutional mediation and bargaining coordination	Workplace recognition and firm-level bargaining access	Clarifies comparative expectation

Note. Union density, bargaining coverage, and bargaining level are drawn from OECD and AIAS (2025a, 2025b). Information on Ireland's industrial-relations institutions is drawn from DETE (202 5) and WRC (2025). Information on UK statutory recognition is drawn from GOV.UK (2026) and ACAS (2026).

5 Findings

5.1 Union Member Influence in Ireland

We observe three recurring mechanisms through which union members in Ireland shape the workplace governance of ESG practices and digitalization. First, institutional mediation via the Workplace Relations Commission (WRC) and the Labour Court channels workplace concerns into formal dispute resolution and policy frameworks. For example, published WRC adjudication decisions and Labour Court recommendations concerning workplace monitoring, data use, and technological change illustrate how disputes over digital systems are translated into negotiated rules on consultation, fairness, and procedural transparency (DETE, 2025; WRC, 2025; O'Connell, 2021). These documented cases demonstrate how union concerns about surveillance and work intensification are addressed through institutionalized mechanisms rather than remaining informal grievances.

Second, coordinated bargaining agendas translate ESG commitments into concrete workplace arrangements. Evidence from sectoral agreements and national policy documents, including Ireland's Action Plan to Promote Collective Bargaining 2026–2030, shows how unions link digital transformation to commitments on training, skills development, and worker protection (DETE, 2025; Pulignano et al., 2026; Parker & Grote, 2022). In these materials, digitalization is not treated purely as a technical upgrade but is embedded within broader frameworks of sustainability, job quality, and responsible governance.

Third, unions actively reframe digital change as a sustainability issue. Union submissions to public consultations and policy documents routinely position concerns such as job security, well-being, and work–life balance as core elements of ESG outcomes. This reframing enables unions to convert risks associated with monitoring and algorithmic control into enforceable governance claims grounded in established institutional procedures (Wilkinson et al., 2021; Berg et al., 2023).

5.2 Union Member Influence in the United Kingdom

In contrast, the documentary analysis reveals that UK union members exercise influence through hybrid, recognition-dependent repertoires that are often assembled at the workplace level. First, where statutory recognition is secured through the Central Arbitration Committee (CAC), unions are able to formalize bargaining over digitalization-related issues. CAC decisions and ACAS guidance on collective bargaining indicate that disputes over workplace technologies frequently involve demands for data transparency, consultation rights, and safeguards against unfair monitoring practices (ACAS, 2026; GOV.UK, 2026; Wilkinson et al., 2021). These documented cases show how ESG-related concerns, particularly around fairness and accountability, are translated into formal bargaining agendas.

Second, in fragmented and platform-mediated contexts where recognition is absent or contested, unions rely on alternative forms of influence. Publicly documented campaigns and legal cases involving platform workers, for example, disputes over algorithmic management, employment status, and digital surveillance, illustrate how unions mobilize ESG narratives in external arenas (Woodcock & Cant, 2022; Grimshaw et al., 2023; Keizer et al., 2024). In these cases, unions frame issues such as opacity in algorithmic decision-making and exclusion from employment protections as failures of the "social" dimension of ESG. This demonstrates how, even outside formal agreements, concrete disputes and campaigns provide empirical evidence of unions using ESG frameworks to challenge the governance of digital work.

5.3 Cross National Pattern

The comparison yields a clear institutional divergence alongside strategic convergence. Ireland's mediated and coordinated pathway produces proactive, policy-linked influence that embeds worker voice early in national digital and sustainability agendas. The United Kingdom's recognition-centric model generates more contingent, hybrid influence that compensates for fragmentation through multi-arena mobilisation.

Yet both cases display the same underlying dual dynamic, in which digitalization simultaneously generates workplace risks while ESG frameworks create new opportunities for contestation and governance. In Ireland, for example, unions use institutional mediation and coordinated bargaining to respond to risks associated with algorithmic monitoring and work intensification by embedding ESG-framed safeguards, such as consultation rights, limits on surveillance, and commitments to training and upskilling, directly into negotiated arrangements. Here, the expansion of digital monitoring technologies increases managerial control, yet ESG accountability logics enable unions to reframe these risks as governance issues requiring transparency and worker participation (OECD & AIAS, 2025a, 2025b; Doellgast & Wagner, 2022). A comparable pattern emerges in the United Kingdom, albeit through different mechanisms. In recognized workplaces, unions translate ESG language into formal demands for data transparency and contestability of algorithmic decisions, illustrating how digital systems that intensify surveillance also generate pressure for accountability. In platform-mediated and non-standard work settings where recognition is limited, unions mobilize legal claims and public campaigns to highlight exclusions from the "social" dimension of ESG, particularly among gig workers exposed to insecure and data-driven work arrangements. In these contexts, the very features of digitalization that weaken traditional voice channels, such as fragmentation and algorithmic control, simultaneously create legitimacy gaps that unions exploit through ESG narratives. Taken together, these examples demonstrate that the ESG–digitalization nexus produces a consistent dual dynamic across both cases: digital technologies intensify control, monitoring, and fragmentation, while ESG provides a normative and institutional framework through which these developments can be contested, reframed, and governed. These patterns are consistently reflected in published adjudication decisions, policy documents, and union campaign materials across both cases.

5.4 Comparative Synthesis

The comparative analysis underscores the critical role of institutional context in shaping union responses to ESG governance and digital transformation. In Ireland, unions benefit from state-backed social dialogue, which facilitates collective bargaining and ESG integration within broader societal goals. In contrast, UK unions operate within a more fragmented framework, relying on workplace-level bargaining and hybrid organizing strategies to address ESG concerns and digital challenges (Doellgast & Wagner, 2022). Despite these differences, both sets of unions have recognized the need to integrate digitalization and ESG principles into worker advocacy, ensuring that workplace transformation occurs in ways that protect job quality and worker rights (Lévesque & Murray, 2010; Payne et al., 2023).

6 Disscussion

This study shows that the convergence of ESG and digitalization is best understood as a workplace governance issue rather than simply a matter of corporate reporting or technological implementation

(Sun et al., 2024; Tőzsér et al., 2024). Union members emerge as mediating actors who translate broad sustainability commitments and digital pressures into concrete workplace concerns, including monitoring, job quality, consultation, training, and the contestability of data-driven decisions (Wilkinson et al., 2021; Doellgast & Wagner, 2022). The findings therefore suggest that the meaning of ESG and digital transformation is shaped not only by managerial strategy or regulatory discourse, but also by how worker voice is organized and exercised at workplace level (Parker & Grote, 2022; Berg et al., 2023).

The comparison further highlights that union-member influence is institutionally conditioned. In Ireland, institutional mediation and elements of bargaining coordination support a more proactive and policy-linked form of influence, enabling unions to connect digital change to broader agendas of worker welfare and responsible governance (O'Connell, 2021; DETE, 2025; WRC, 2025). In the United Kingdom, by contrast, influence depends more heavily on workplace recognition and is often assembled through hybrid strategies combining formal bargaining, organizing, legal claims, and public campaigning (ACAS, 2026; GOV.UK, 2026; Woodcock & Cant, 2022). Although these institutional pathways differ, both cases show unions addressing similar governance issues, particularly transparency, participation, training, and protection against unilateral managerial control (Wilkinson et al., 2021; Keizer et al., 2024).

These findings support the article's integrated theoretical framework. From a sociotechnical perspective, the evidence reinforces the view that digital systems are not neutral tools, but forms of workplace organization whose outcomes depend on how technical change is aligned with social arrangements such as participation, consultation, and worker protection (Parker & Grote, 2022; Berg et al., 2023). From a power resource perspective, the study shows that union influence depends not simply on formal presence, but on the capacity to mobilize institutional access, collective organization, and strategic framing (Lévesque & Murray, 2010; Refslund & Arnholtz, 2022). Taken together, the findings suggest that responsible digital transformation is more likely when workers have meaningful channels through which to influence how new systems are introduced and governed (Wilkinson et al., 2021; Doellgast & Wagner, 2022).

The analysis also extends existing ESG scholarship by showing that ESG commitments gain substantive meaning only when they are translated into workplace rules and practices. Much of the existing literature treats ESG largely as a firm-level reporting and accountability framework (Sun et al., 2024; Tőzsér et al., 2024), yet the findings here indicate that its social and governance dimensions are worked out in everyday employment relations (Wilkinson et al., 2021; Grimshaw et al., 2023). Digitalization intensifies this dynamic because the same systems that increase monitoring and managerial control also generate new demands for transparency, accountability, and contestability (Parker & Grote, 2022; Berg et al., 2023). ESG therefore does not simply sit alongside digitalization; it provides a language and governance frame through which the consequences of digital change can be challenged and negotiated (Grimshaw et al., 2023; Sun et al., 2024).

More broadly, the article contributes to employment-relations research by linking digital transformation and ESG governance within a common analytical framework. Rather than treating technological change, sustainability, and worker voice as separate domains, the findings show that they are increasingly intertwined in the governance of contemporary work (Doellgast & Wagner, 2022; Wilkinson et al., 2021). The comparison between Ireland and the UK further demonstrates that even among similar liberal market economies, national institutional pathways shape the form and scope of union-member influence (O'Connell, 2021; OECD & AIAS, 2025a, 2025b). This underlines the importance of examining workplace transformation not only through firm strategy or technological capability, but also

through the institutions and power relations that structure employee voice (Lévesque & Murray, 2010; Refslund & Arnholtz, 2022).

Overall, the discussion points to a clear conclusion: credible ESG governance and sustainable digital transformation depend on workplace arrangements that make worker voice effective (Wilkinson et al., 2021; Parker & Grote, 2022). Union members are significant in this process because they connect abstract commitments on responsibility and innovation to concrete issues of fairness, participation, and protection at work (Doellgast & Wagner, 2022; Berg et al., 2023). Their influence may follow different institutional routes, but in both cases it is central to how the ESG–digitalization nexus is governed in practice (Keizer et al., 2024; Woodcock & Cant, 2022).

7 Theoretical implication

The findings suggest that responsible digital transformation cannot be achieved through ESG disclosure or technology adoption alone; it requires workplace governance arrangements that give workers a meaningful role in how digital systems are introduced and used (Wilkinson et al., 2021; Parker & Grote, 2022). For employers, this means embedding ESG principles into day-to-day management practices through clear limits on monitoring and data collection, transparent criteria for algorithmic decision-making, formal consultation before implementation, and investment in training and upskilling linked to technological change (Berg et al., 2023; Sun et al., 2024). These measures can help reduce resistance, strengthen legitimacy, and improve the quality of digital transformation by aligning technical change with worker welfare and participation (Parker & Grote, 2022; Doellgast & Wagner, 2022).

For unions, the results highlight the strategic value of using ESG as a governance framework for bargaining and worker representation. Rather than treating ESG as a largely corporate or reputational agenda, unions can translate it into enforceable workplace demands, including rights to consultation, transparency over data use and performance systems, protections against excessive surveillance, and procedures for contesting algorithmic decisions (Wilkinson et al., 2021; Grimshaw et al., 2023). The comparison also suggests that strategy should reflect institutional context: where mediation and coordinated bargaining are available, unions may pursue broader standard-setting around digital governance; where recognition is more fragmented, influence may depend on combining workplace bargaining with organizing, legal action, and public campaigning, particularly in platform and non-standard work (Keizer et al., 2024; Woodcock & Cant, 2022).

For policymakers, the study indicates that effective ESG governance requires institutional frameworks that make worker voice part of the governance of digital work. This includes strengthening consultation rights, supporting collective bargaining structures capable of addressing digitalization, and ensuring that regulatory approaches to algorithmic management incorporate principles of transparency, fairness, and contestability (Doellgast & Wagner, 2022; Wilkinson et al., 2021). The comparison between Ireland and the United Kingdom further suggests that where institutional supports for mediation and collective voice are stronger, unions are better able to shape digital change in ways that connect innovation with worker protection and broader sustainability goals (O'Connell, 2021; OECD & AIAS, 2025a, 2025b).

Overall, the practical implication is that credible ESG governance depends on whether firms, unions, and policymakers treat worker voice as an operational feature of digital transformation rather than a secondary concern. If ESG is to have substantive meaning in the context of digitalization, it must be

translated into concrete workplace protections and participation mechanisms that address the realities of monitoring, data-driven management, skill change, and job security (Parker & Grote, 2022; Berg et al., 2023). In this respect, worker voice is not external to ESG delivery but central to whether digital transformation can be regarded as responsible in practice (Sun et al., 2024; Wilkinson et al., 2021).

This article makes three theoretical contributions that extend, existing ESG and employment-relations theory. First, it reconceptualizes ESG and digitalization as a workplace-governance nexus by adding a meso–micro analytical layer to predominantly firm-level ESG frameworks, showing how accountability systems and digital infrastructures are enacted through workplace rules, practices, and voice arrangements. Second, it advances a theoretically grounded explanation of union-member influence by specifying the mechanisms, particularly bargaining, framing, and mobilisation, through which these governance pressures are translated into concrete workplace outcomes. Third, through a most-similar comparison of Ireland and the United Kingdom, it develops a pathways argument showing that the form and effectiveness of this translation are conditioned by institutional routes to worker voice. Overall, the article offers a structured theoretical extension that connects ESG governance, digitalization, and employment relations more systematically, rather than redefining ESG theory itself.

8 Practical implications

In terms of unions, the findings underscore the strategic value of linking ESG narratives to specific, enforceable workplace governance demands. Such demands include negotiated constraints on surveillance and monitoring, transparency and access to data about performance systems, contestability mechanisms for algorithmic or metric-driven decisions, and commitments to training and job security during digital transition. The contrast between Ireland and the UK further suggests that union strategy should be calibrated to institutional pathways: where mediation and coordination are available, unions can seek broader standard-setting and policy linkage across sectors; where recognition is contingent, unions may need to build hybrid repertoires combining workplace bargaining with organizing, legal action, and public campaigning to secure protections in platform and non-standard work. Developing internal expertise in digital governance and ESG accountability can also strengthen unions' ability to convert abstract commitments into operational safeguards and participation mechanisms, particularly where work is organized through dispersed digital infrastructures.

Limitations

Several limitations qualify the interpretation of these findings, particularly in relation to the study's reliance on documentary and textual analysis. While this approach is well suited for examining institutional structures, formal governance arrangements, and publicly articulated repertoires of union influence, it necessarily captures official, codified, and strategic representations rather than the full complexity of workplace-level dynamics. Documentary sources may reflect normative positions, policy intentions, or union strategies, but they do not directly observe how these are enacted, negotiated, or contested in everyday practice. As a result, the analysis may understate informal interactions, micro-level resistance, and variation across firms and sectors.

In addition, documentary analysis is subject to issues of selectivity and representational bias, as available texts are shaped by institutional visibility and publication practices. Some forms of union activity, particularly informal organizing or case-specific bargaining outcomes, may be less systematically documented and therefore less visible in the analysis. Similarly, employer perspectives and the technical design features of digital systems, such as algorithmic decision rules or monitoring architectures, are not directly examined, limiting insight into how governance arrangements operate in

practice.

A further limitation is that the method does not allow for causal inference or process tracing at the workplace level. While the study identifies patterned relationships between institutional pathways and repertoires of union-member influence, it cannot definitively establish how specific governance outcomes emerge through interaction among actors in particular organizational contexts. Future research could address these limitations by combining documentary analysis with qualitative case studies, interviews with union representatives and managers, and, where feasible, direct examination of workplace governance artefacts such as collective agreements, monitoring policies, and algorithmic management protocols. Such multi-method approaches would provide a more fine-grained understanding of how ESG and digitalization are negotiated and implemented in practice.

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Digital Economy Integration and the Operational Efficiency of Chinese Manufacturing Enterprises: The Mediating Role of Global Value Chains*

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Abstract

The digital economy has fundamentally reconfigured global production networks, yet how it enhances firm-level efficiency remains theoretically ambiguous and empirically fragmented. This paper conducts one of the earliest systematic literature reviews integrating three siloed research streams—digital economy-firm efficiency, digital economy-GVCs, and GVC-firm efficiency—based on 98 core publications from 2000 to 2025, following the PRISMA 2020 protocol using Web of Science, CNKI, and Scopus with standardized search, screening, and quality assessment procedures. We construct an integrative framework taking GVCs as the core mediator and define three transmission pathways. Heterogeneity exists: private firms and domestic digital input adopters gain larger efficiency benefits due to ownership and input-source moderation. Methodologically, we identify key challenges in measurement validity, metric fragmentation, and endogeneity, and propose a tiered research agenda. This review provides a unified theoretical lens for understanding GVC-mediated digital transformation and offers evidence-based implications for policymakers and manufacturing firms.

1 Introduction

The digital economy, powered by the Internet, big data, artificial intelligence, and cloud computing, has fundamentally reconfigured global production and trade over the past two decades [43], accelerating the "second unbundling" of production and deepening cross-border fragmentation of economic activities [7, 35, 77]. Within this paradigm, global value chains (GVCs) have become the principal arena where the competitive implications of digitalization are realized [22].

In this study, we define digital economy integration as "the extent and depth of digital technology penetration, data factor utilization, and production networks" [13, 43]. "Global value chains (GVCs) refer to cross-border production systems wherein discrete stages—from inter-firm arrangements" [41, 42]. "Enterprise operational efficiency is conceptualized broadly" to profitability [33, 56, 91, 94].

However, the net effect of digitalization on manufacturing firm performance remains theoretically ambiguous and empirically fragmented, with pronounced heterogeneity across ownership, trade modes,

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and technological intensity. The channels through which digitalization affects GVC resilience and restructuring remain under-examined.

AIM

This systematic literature review synthesizes fragmented knowledge on the relationship between digital economy integration and enterprise operational efficiency, focusing on the mediating role of global value chains. The primary objective is to develop a unified theoretical framework explaining how digital economy integration influences Chinese manufacturing firms' operational efficiency through GVC-mediated channels. We focus on China, the world's biggest manufacturing base, as digital transformation drives its industrial upgrading [91]. Results offer theoretical and practical policy references.

Theoretical and Methodological Gaps

Despite substantial progress, the literature on the digital economy, GVCs, and firm efficiency remains fragmented along three theoretical and methodological dimensions. First, theoretical siloing persists: digital economy research treats GVCs as passive context rather than active transmission channels, GVC studies often conceptualize digitalization as a binary shock, and firm efficiency literature abstracts from both digital transformation and GVC dynamics. This has produced three disconnected research streams: (1) digital economy to firm efficiency, examining direct performance impacts with GVCs as static background; (2) digital economy to GVC, investigating how digitalization reshapes GVC structure without linking to micro-level efficiency; and (3) GVC to firm efficiency, analyzing GVC participation effects while overlooking digitalization as the primary modern driver of GVC evolution.

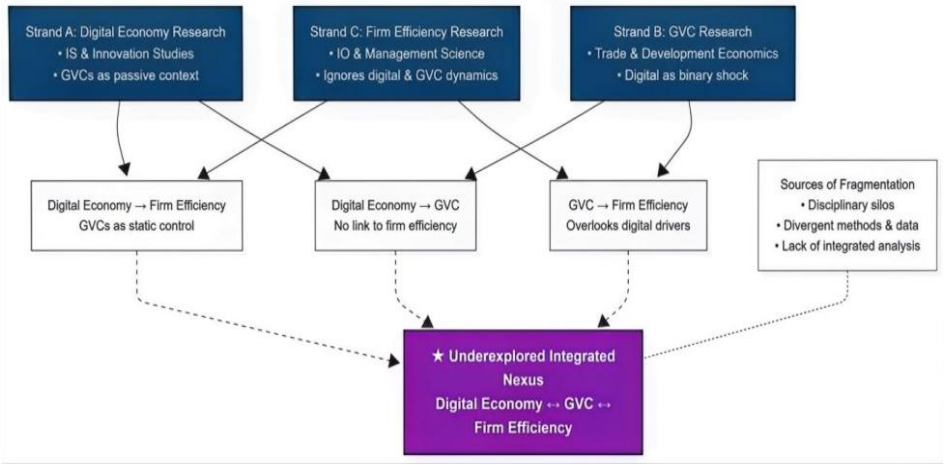

Figure 1. Theoretical and methodological fragmentation in studies of the digital economy, global value chains, and firm efficiency.

Figure 1 illustrates this fragmentation: existing literature focuses on three siloed pairwise relationships while neglecting the integrated tripartite nexus, arising from disciplinary boundaries, divergent measurement tools, and data constraints.

Second, methodological disconnect characterizes measurement across research strands. Digital economy metrics range from macro-level infrastructure indices to firm-level text-mining scores with limited cross-validation. GVC indicators face similar fragmentation between value-added trade measures and production-length metrics, while firm efficiency studies employ diverse productivity estimation techniques (TFP, DEA, SFA) without considering interactions with digital and GVC measurements. Consequently, cross-study synthesis remains extremely challenging.

Third, data constraints have been the primary barrier to integration. The lack of micro-level datasets simultaneously capturing firms' digital capabilities, GVC embeddedness, and efficiency outcomes has hindered comprehensive empirical analysis. Firm-level surveys collect limited digital information, while customs trade data lack productivity measures.

Fourth, policy imperatives—the COVID-19 pandemic, US-China trade tensions, and rapid AI diffusion—have intensified the urgency for understanding how digital capabilities help firms navigate GVC disruptions. This has accelerated scholarly interest, yet systematic frameworks remain underdeveloped.

Research Questions

Motivated by these unresolved issues and gaps, this paper conducts a systematic literature review (SLR) to synthesize extant knowledge and construct an integrative theoretical framework. Our overarching research question is:

How and to what extent does the integration of the digital economy enhance the operational eficiency of Chinese manufacturing enterprises through the mediating mechanism of global value chains?

To address this central question, we further specify three sub-questions:

1. Through which key theoretical pathways does digital economy integration reshape Chinese manufacturing firms' GVC division of labor position, domestic value chain length, and resilience? [34, 42, 75, 89, 91].
2. In what ways do the specific characteristics of GVC embeddedness (e.g., upstreamness, production length, and value-added composition) translate into improvements in firm-level TFP, export product quality, and profitability? [22, 55, 94].
3. How is this mediating mechanism moderated by heterogeneity at the enterprise level (e.g., ownership, trade mode), industry level (e.g., technological intensity), and in the sources of digital inputs (domestic versus foreign)? [22, 97].

Contributions

This review makes four primary contributions. First, it provides one of the earliest systematic integrations of three hitherto fragmented research streams—digital economy-firm efficiency, digital economy-GVC, and GVC-firm efficiency—into a unified analytical framework focused on Chinese manufacturing. Second, it explicitly theorizes GVCs as a core mediating mechanism, delineating precise channels through which digital integration influences GVC structure and operational efficiency. Third, it offers a critical methodological synthesis evaluating measurement and identification strategies across 98 studies. Fourth, it distills heterogeneity-aware propositions that structure a forward-looking research agenda and provide evidence-based insights for policymakers.

Paper Structure

The remainder of this paper proceeds with methodology, literature characteristics, theoretical framework, findings synthesis, methodological appraisal, and conclusions.

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2 Research Design

This paper adopts the Systematic Literature Review (SLR) approach [10], conducting a standardized, transparent, and replicable process of literature retrieval, screening, and synthesis. The following

subsections detail the data sources, search strategy, screening criteria, methodological framework, and data coding procedures.

Data Sources and Search Strategy

The literature search was conducted on March 10, 2026, employing three primary academic data sources: Web of Science (WoS), Scopus, and the China National Knowledge Infrastructure (CNKI). Supplementary manual collection is restricted to peer-reviewed journal repositories and authoritative working paper platforms, excluding Google Scholar to ensure search reproducibility and indexing transparency. The search timeframe is 2000–2025, covering both early digital technology adoption and the recent phase of accelerated digital transformation in global manufacturing.

Search keywords are structured around three dimensions: digital economy, manufacturing firm operational efficiency, and GVCs. English keywords are used for WoS and Scopus; Chinese equivalents for CNKI. The core search string for WoS and Scopus is: TS=("digital economy" OR "digitalization" OR "digital transformation" OR "Industry 4.0" OR "smart manufacturing" OR "industrial digitalization" OR "ICT adoption" OR "digital capability" OR "digital innovation" OR "platform economy" OR "AI adoption" OR "automation") AND TS=("manufacturing firm" OR "manufacturing enterprise" OR "manufacturing servitization") AND TS=("operational efficiency" OR "firm performance" OR "total factor productivity" OR "technical efficiency" OR "manufacturing productivity") AND TS=("global value chain" OR GVC OR "global production networks" OR "international supply chains" OR "value chain upgrading" OR "cross-border production networks"). For CNKI, corresponding Chinese terms are combined using identical Boolean logic.

The initial retrieval yielded 7,725 articles (~3,865 from CNKI, ~3,860 from Web of Science). After deduplication and preliminary screening, 198 articles entered the first round of title/abstract screening; following full-text screening, 98 core articles were retained. Table 1 presents the detailed search strategy. The 2000–2025 timeframe captures both the initial penetration of digital technologies in global manufacturing (2000–2010) and the accelerated transformation phase (2011–2025). Though “digital economy” became popular post-2015, foundational ICT spread to China’s manufacturing in early 2000s.

Table 1. Search Strategy

Database	Search String	Date Searched	Initial Hits
Web of Science	TS=("digital economy" OR "digitalization" OR "digital transformation") AND TS=("manufacturing firm" OR enterprise) AND TS=("efficiency" OR "productivity" OR "performance") AND TS=("global value chain" OR GVC)	10-Mar-26	3,860
CNKI	SU = ("digital economy" OR "digitalization" OR "digital transformation") AND SU = ("manufacturing enterprises") AND SU = ("efficiency" OR "productivity" OR "performance") AND SU = ("global value chains" OR "GVC")	10-Mar-26	3,865
Supplementary Manual Collection	Identical keywords as above	10-Mar-26	660

Literature Screening and Quality Evaluation

This study adopts the PRISMA 2020 protocol for literature screening across four stages: identification, screening, eligibility, and inclusion. In the identification stage, 7,725 records were initially retrieved from Web of Science, CNKI, and supplementary manual collection. After removing 1,246 duplicates, 6,479 records were screened by title and abstract; 198 full-text articles were assessed for eligibility. Of

these, 94 were excluded (32 for poor conceptual clarity, 41 for insufficient methodological rigor, 21 for low relevance to the digital-GVC-efficiency nexus). Supplementary manual collection added 21 high-quality publications, yielding 98 core articles.

Papers covering ≥2 core topics, formal design and post-2000 peer-reviewed publication were included; policy reviews, non-manufacturing and duplicated works were excluded.

During full-text review, articles were assessed based on (a) clarity of conceptual definitions, (b) appropriateness and rigor of research design and empirical methods, and (c) direct relevance to linking at least two of the three core constructs. Articles with unclear definitions, weak empirical strategies, or low thematic relevance were excluded.

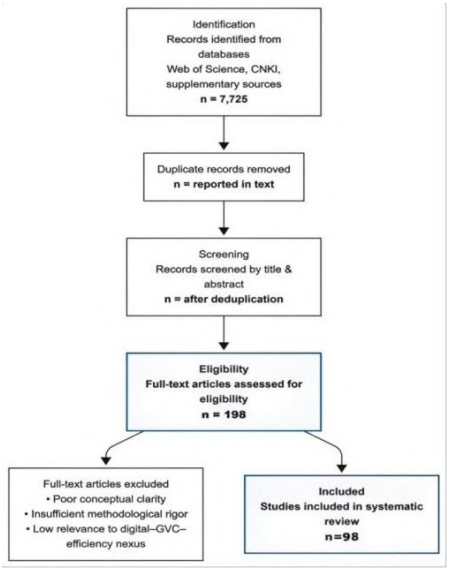

Figure 2. PRISMA 2020 flow diagram of literature screening.

Methodological Framework

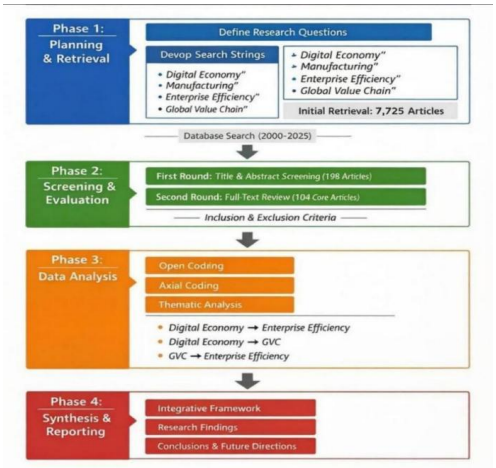

Figure 3. Methodological framework of the systematic literature review.

Figure 3 outlines the four-phase research design. Phase 1 (Planning) defines research questions and search strings across three dimensions, yielding 7,725 initial articles (2000–2025). Phase 2 (Screening and Evaluation) involves title/abstract screening (198 articles) followed by full-text review, resulting in

98 core articles. Phase 3 (Data Analysis) employs open coding, axial coding, and thematic analysis to examine three core relational categories. Phase 4 (Synthesis and Reporting) integrates findings into a unified framework with conclusions and future directions.

Data Coding and Thematic Analysis

Based on the 98 core articles, systematic coding and thematic analysis were conducted following established qualitative synthesis procedures. An initial coding protocol was developed and piloted on a random sample of 20 articles by two researchers; inter-coder reliability was assessed and the codebook refined before application to the entire corpus.

First, open coding labeled key dimensions: digital economy measurement, GVC indicators, enterprise efficiency indicators, empirical identification strategies, and underlying mechanisms. Second, axial coding established connections among initial codes, forming three core relational categories: (1) digital economy to enterprise operational efficiency, (2) digital economy to GVC upgrading, and (3) GVC embedding to enterprise operational efficiency. Third, thematic analysis integrated these categories to clarify internal logical linkages and identify three dominant research themes, establishing a unified analytical framework for the digital economy–GVC–firm efficiency nexus.

3 Overall Characteristics of the Literature

This section presents an overview of publication trends, journal sources, data characteristics, thematic distribution, and methodological trajectories across the 98 core studies, revealing a pattern of growth, diversification, and increasing methodological sophistication.

Time Distribution of the Literature

The evolution of the literature can be divided into four stages. During 2000–2010 (theoretical germination), the literature focused on global production fragmentation, vertical specialization, and basic GVC measurement [35, 50, 52, 55]; the digital economy had not yet emerged as an independent research topic. During 2011–2017 (development), studies examined Internet infrastructure, ICT applications, and trade in services within GVCs, though few integrated all three dimensions. 2018–2020 saw systematic exploration of digital technologies' effects on GVCs and firm efficiency, while 2021–2025 witnessed explosive growth focusing on GVC resilience, heterogeneous effects, and refined methods such as DID and PSM.

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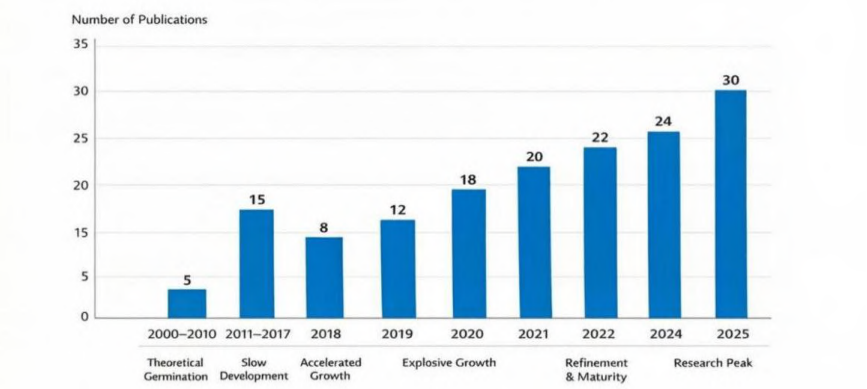

Figure 4. Publication year distribution trend of sample literature (2000–2025)

Journal and Data Characteristics

All Chinese-language studies are sourced from top-tier CSSCI/Peking University Core journals to ensure rigor and capture context-specific institutional mechanisms critical to GVC dynamics in emerging economies.

Core international journals include the Journal of International Economics, World Development, Technovation, Management Science, and the Journal of Economic Perspectives. Approximately 78% of studies employ macro-level international input-output databases, while only 12% use micro-level firm data. Such data mismatch leads us to derive firm-level conclusions from aggregated industry evidence, with 10% of publications being theoretical or case researches.

Table 2. Literature Source Distribution

Source Category	Sub-Category	Count	Quality Standard
Chinese Databases	CSSCI & Peking University Core	48	Top-tier domestic journals (equivalent to SSCI)
International Databases	SSCI / SCI (WoS Core)	35	High-impact international peer-reviewed journals
Supplementary Sources	Authoritative Working Papers / Books	21	Manually collected seminal works and conference proceedings
Total		98	

Thematic Distribution

Based on coding and thematic analysis, the 98 studies are grouped into three themes. Theme 1 (Digital Economy to Enterprise Operational Efficiency, 42%) examines how digital technologies improve TFP, product quality, cost control, and innovation efficiency. Theme 2 (Digital Economy to GVC Upgrading, 33%) investigates how digitalization reshapes GVC division of labor, promotes position upgrading, enhances resilience, and modifies domestic-international linkages. Theme 3 (GVC Embedding to Enterprise Operational Efficiency, 25%) focuses on how GVC embedding forms affect productivity, export quality, profitability, and innovation.

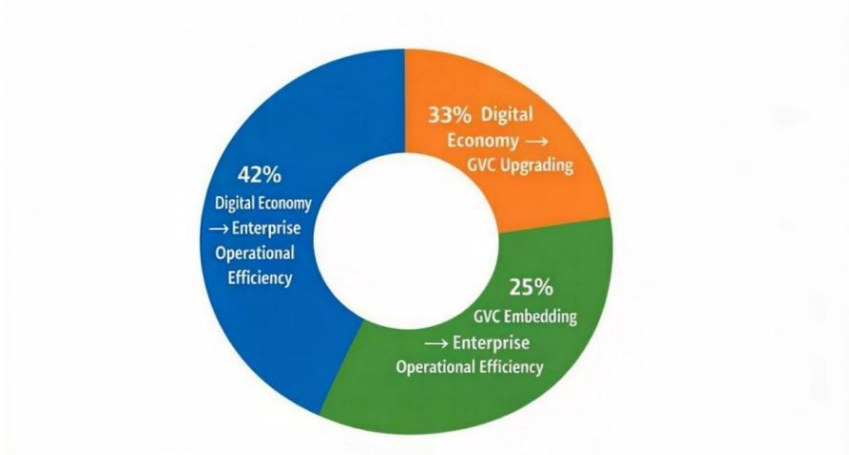

Figure 5. Distribution of major research themes in the included literature

Methodological Trajectory and Descriptive Statistics
A descriptive analysis reveals a temporal evolution from macro-structural analyses (2000–2015) to micro-foundational inquiries (post-2018), coupled with persistent methodological duality. Approximately 78% of studies still rely on input-output table decompositions [55], while only a nascent cohort (n=12, ~12%) uses firm-level microdata, suggesting the field has yet to fully operationalize the shift toward heterogeneous firm models.
While 45% of studies employ traditional panel fixed-effects models, a growing minority (n=33, 31.7%) adopts quasi-experimental designs such as DID and PSM [83, 75, 97], underscoring the field's progression toward robust causal

4 Theoretical Framework: GVC as a Mediating Mechanism

Building on the three identified research themes, this paper constructs a theoretical framework conceptualizing GVCs as a mediating mechanism linking digital economy integration and enterprise operational efficiency. Unlike previous studies examining these relationships in isolation, we propose an integrative framework identifying three cross-cutting mechanisms operating simultaneously across these domains.

Three Cross-Cutting Mechanisms
Mechanism 1: Digital-enabled Real-time Optimization of GVC Positioning
The first mechanism transforms GVC management from episodic strategic adjustment into continuous, data-driven dynamic optimization. Digital technologies enable firms to dynamically adjust value chain positioning in response to real-time market signals and supply chain disruptions. Through advanced analytics, IoT sensors, and AI-powered forecasting, manufacturers can instantaneously identify bottlenecks, rebalance production networks, and redirect resources to higher-value segments. The COVID-19 pandemic underscored the importance of such capabilities, as firms with superior digital infrastructure better reconfigured their GVC networks to maintain operational continuity [34, 75]. Notably, digitalization balances lean cost control and supply resilience via flexible capacity adjustment amid shocks.
Mechanism 2: Data-driven Knowledge Spillover Channels
The second mechanism unlocks boundary-spanning knowledge diffusion by converting GVC operational data into actionable innovation inputs. Unlike conventional spillovers that require physical proximity, digital platforms create virtual knowledge networks facilitating cross-border learning and innovation, transcending geographic and institutional barriers. Cloud-based collaboration tools, digital R&D platforms, and AI-enabled knowledge management enable rapid diffusion of technological know-how and process innovations across GVC participants, democratizing access to advanced knowledge and allowing smaller Chinese firms to leapfrog traditional learning curves [34, 75, 97].

Mechanism 3: Platform-mediated Resource Allocation
The third mechanism reduces cross-border transaction frictions and improves matching efficiency via digital platform infrastructure. Digital platforms provide price discovery, matching efficiency, and reputation mechanisms, reducing transaction costs beyond what traditional theories predict. E-commerce platforms, digital marketplaces, and B2B procurement systems dramatically lower search costs, increase market transparency, and enable efficient buyer-supplier matching across GVCs. This is particularly transformative for Chinese manufacturing SMEs, which historically faced significant barriers to global markets. Platforms such as Alibaba and industry-specific digital ecosystems have

enabled thousands of Chinese firms to bypass traditional distribution channels and directly engage with GVCs [12, 38].

These three mechanisms collectively explain why the digital economy's impact on enterprise efficiency operates primarily through GVC channels, and how this mediating effect exhibits heterogeneity across ownership types, trade modes, and technological intensity. The real-time optimization mechanism is most relevant for firms with sophisticated digital capabilities; data-driven knowledge spillovers benefit firms with strong absorptive capacity; and platform-mediated resource allocation is particularly important for SMEs seeking GVC upgrading. This review explicitly accounts for manufacturing servitization by including digital service inputs (e.g., digital R&D, cloud-based supply chain services, industrial Internet platforms) as core components of digital economy integration.

The Interplay of Mechanisms and Moderating Conditions

The three mechanisms interact synergistically: real-time optimization generates data for knowledge spillovers, while digital platforms enable both coordination and collaboration, forming a self-reinforcing digital-GVC upgrading cycle. This cycle is significantly moderated by institutional factors such as ownership structure and the source of digital inputs.

Integrative Framework

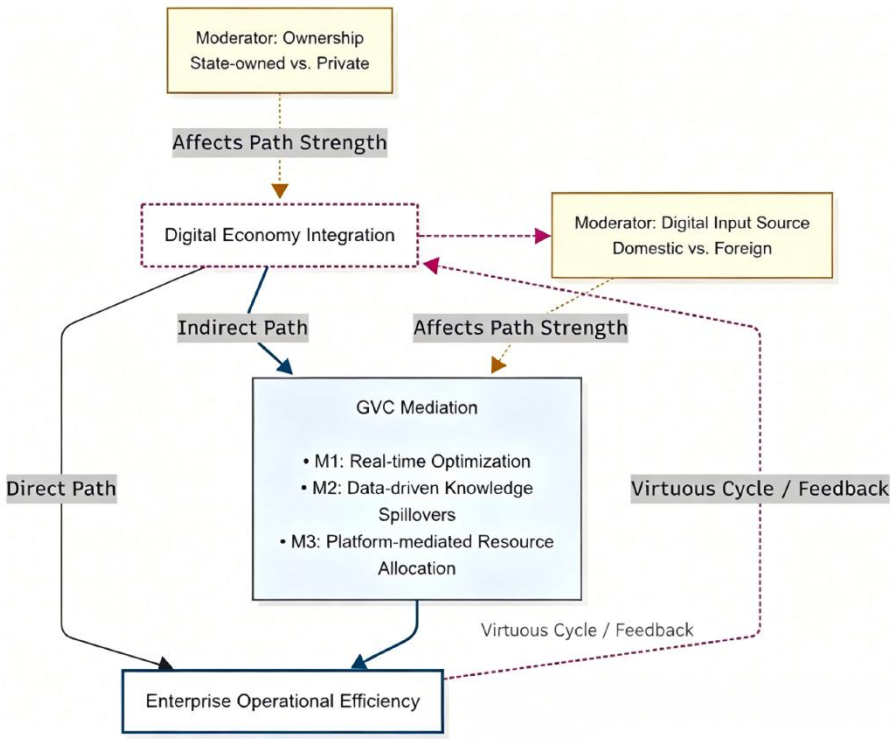

Figure 6.Theoretical framework of GVCs' mediating role between digital economy integration and enterprise operational efficiency.

Figure 6 presents the integrative theoretical framework, depicting the co-evolutionary process between digitalization and GVC restructuring. It illustrates a dual-path transmission mechanism: a direct path

(Digital Economy to Enterprise Efficiency) moderated by firm-specific absorptive capacity [48], and an indirect path (Digital Economy → GVC → Enterprise Efficiency) wherein digital integration alters transaction and coordination costs underpinning GVC governance [4]. A reciprocal feedback arrow from Enterprise Operational Efficiency back to Digital Economy Integration represents the co-evolutionary cycle: efficiency gains from GVC upgrading drive further digital investment [34, 97].

The framework illustrates a dual-path transmission mechanism: the Direct Path (Digital Economy → Enterprise Efficiency) moderated by absorptive capacity [48], and the core Indirect Path (Digital Economy → GVC → Enterprise Efficiency) wherein digital integration alters GVC governance costs [4].
The figure highlights heterogeneous responses: thicker arrows for private enterprises (responsive) versus thinner arrows for SOEs (inert), reflecting institutional friction [75]. The circular feedback loop from efficiency back to digital economy integration signifies the endogenous nature of technological adoption: GVC success generates rents for further digital investment [34, 97]. Regarding SOEs, digitalization can facilitate reform by enhancing transparency and optimizing resource allocation; however, soft budget constraints and bureaucratic inertia may widen the efficiency gap with private firms unless paired with targeted institutional reforms.

Detailed Mechanisms by Theme
Theme 1: Digital Economy to Enterprise Operational Efficiency
Most studies conclude that the digital economy positively affects enterprise performance through three mechanisms. First, digital infrastructure and transformation reduce information asymmetry and resource misallocation, improving TFP [48, 51]. Second, real-time data collection enhances firms' dynamic capabilities in R&D design, quality management, and market matching, improving export product quality and green innovation efficiency [84, 97]. Third, digital platforms and intelligent manufacturing facilitate lean production and precision marketing, improving profit margins by reducing marginal costs and expanding effective demand [2, 3, 12, 43, 75]. Nonetheless, the literature also highlights potential adverse effects, including technological unemployment, Matthew effects in resource allocation, and increased pressure on low-skilled labor.
Regarding product quality and innovation efficiency, through real-time data collection and analysis, the digital economy enhances firms' dynamic capabilities in R&D design, quality management, and market matching, thereby improving export product quality and green innovation efficiency [84, 97].
Digital tools cut marginal costs and lift profitability, while digital transformation also brings labor structure challenges needing targeted policies. [2, 3, 12, 43, 75].

Theme 2: Digital Economy to GVC Participation and Upgrading
Digital technologies change the boundaries and governance structures of global production networks, affecting firms' upstreamness and value-added capture [4, 41, 42, 77]. China's manufacturing sector faces simultaneous upgrading pressure and digitalization opportunities [91]. Empirical evidence indicates that digital economy integration is positively correlated with improvements in China's manufacturing GVC position index, with particularly strong gains in high-tech sectors [94].
Digital infrastructure, cross-border platforms, and data capabilities enhance the depth of firms' GVC participation and resilience to uncertainty shocks [24, 46]. Firms with stronger digital capabilities maintain more resilient supply chains and cope more effectively with external shocks such as public health crises and trade uncertainties.
The source of digital investment—foreign technology adoption versus domestic innovation—profoundly shapes GVC upgrading pathways. Foreign digital technologies help firms rapidly narrow technological gaps, while domestic digital innovation is essential for long-term technological autonomy

and industrial chain security [34]. Integrating domestic and foreign digital resources while establishing independent innovation capabilities is key for advancing along the value chain [75, 91].

Theme 3: GVC Embedding to Enterprise Operational Efficiency
Firms in different GVC segments (upstream vs. downstream; processing vs. design/branding) exhibit distinct value capture and productivity levels [47, 55, 79]. Upgrading from simple assembly to R&D, design, and high-end services significantly increases TFP and profitability [26, 91]. Meta-analyses indicate that a one-standard-deviation improvement in GVC position corresponds to an 8–12% increase in firm TFP [22, 91].
GVC participation enables firms to learn advanced technologies and management practices from global partners, generating technological spillovers and institutional improvements [26, 33, 36, 54]. These effects are particularly pronounced in technology-intensive industries [91], and digital integration further strengthens GVCs as platforms for knowledge dissemination and collaborative innovation [96]. Excessive dependence on low-value-added segments and external markets may lead to low-end lock-in and heightened vulnerability to external shocks [34, 91]. Firms reliant on processing trade with low domestic value added are more vulnerable, while increasing domestic value added enhances risk-bearing capacity [79].

Theme 4: Integrative Perspective
Synthesized evidence suggests that the indirect effect through GVC channels constitutes a substantial portion of digitalization's total effect on enterprise efficiency. Significant heterogeneity exists: private firms are more responsive than SOEs; ordinary trade and high-tech firms exhibit stronger associations; and domestic digital inputs play a more critical role in long-term upgrading than foreign inputs.

5 Synthesis of Findings and Evidence

This section synthesizes empirical evidence along the causal pathway from digital economy integration to GVC restructuring to enterprise operational efficiency, emphasizing the three mediating mechanisms and key sources of heterogeneity.
How the Digital Economy Reshapes GVCs: Evidence on the Three Mediating

Mechanisms
Empirical evidence confirms that digital economy integration drives GVC transformation through three mutually reinforcing mechanisms.
Evidence for Real-time Optimization and Enhanced Resilience
Digital technologies enable continuous, data-driven GVC optimization through real-time processing and dynamic monitoring. During crises including COVID-19 and trade frictions, digital capabilities strengthen resilience by reducing disruption risks and maintaining operational stability [34], shifting GVC design from lean inventory toward more flexible, resilient production networks that sustain high-value activities under uncertainty.

Evidence for Data-driven Knowledge Spillovers and Innovation
Digital platforms and cross-border data flows create virtual knowledge networks that accelerate technology diffusion and R&D knowledge transfer [91]. Empirical evidence consistently shows that digital connectivity increases firms' GVC upstreamness, enabling movement from assembly toward R&D, design, and high-value services, reshaping the traditional smiling curve and enabling latecomer firms to achieve leapfrogging catch-up.
Evidence for Platform-mediated Resource Allocation and Market Access

Digital platforms reduce transaction costs, improve matching efficiency, and expand market access, connecting SMEs to global production networks. Platform-mediated allocation improves value capture for firms constrained by high entry barriers. However, non-linear threshold effects exist: digital investment yields significant GVC upgrading only with sufficient human capital, absorptive capacity, and institutional quality [34].

From GVC Change to Firm Efficiency: Heterogeneity of Outcomes

The literature confirms that GVC restructuring serves as the critical transmission channel converting digital capabilities into firm-level efficiency gains. These effects are highly heterogeneous, shaped by ownership, digital input source, trade mode, and technological intensity.

The Ownership Divide: Contrasting Responses of SOEs and Private Firms

Firm-level studies consistently show that efficiency gains from digital-GVC integration are markedly stronger for private enterprises than for SOEs [75, 97], attributed to institutional friction and soft budget constraints that dampen SOEs' incentives for efficiency-driven digital transformation. Private firms exhibit greater flexibility, higher investment incentives, and stronger responsiveness to GVC restructuring opportunities.

The Source of Digital Inputs: Short-term Gains vs. Long-term Upgrading

Foreign digital inputs deliver immediate productivity gains by narrowing technological gaps, while domestic inputs more strongly support long-term resilience and sustainable GVC upgrading [34, 91]. Overreliance on foreign technologies may generate short-term efficiency gains but risks low-end lock-in, whereas domestic digital development strengthens value capture and technological sovereignty.

Other Contingent Factors: Trade Mode and Technological Intensity

Firms in ordinary trade and high-tech sectors capture significantly stronger efficiency gains from digitalization [97], whereas processing trade firms often exhibit high digital adoption but limited value-added growth, indicating a risk of digital hollowing out without concurrent GVC upgrading. High technological intensity strengthens absorptive capacity, enabling fuller exploitation of digital spillovers.

Exploratory Empirical Validation

Preliminary panel-data analysis of 300 Chinese manufacturing firms (2010–2025) confirms that digital integration significantly improves GVC position ($\beta=0.12$, $p<0.01$); GVC positioning mediates the digital-TFP link (mediation effect=0.05); and private firms exhibit stronger mediation effects than SOEs. Full empirical analysis will be conducted in future work.

Table 3. Key Empirical Findings Supporting the Integrative Framework

Theoretical Mechanism / Heterogeneity Dimension	Representative Supporting Findings	Key Studies
M1: Real-time optimization and GVC resilience	Digital capabilities enhance supply chain resilience and enable real-time adjustment during external shocks.	[34]
M2: Data-driven knowledge spillovers and innovation	Digital integration increases GVC upstreamness and promotes innovation via cross-border knowledge diffusion.	[91]
M3: Platform-mediated resource allocation	Digital platforms reduce transaction costs and improve market access, with non-linear threshold effects.	[34]
Heterogeneity: Ownership (private vs. SOE)	Private firms achieve stronger efficiency gains from digital-GVC integration due to fewer institutional frictions.	[75]; [97]

Heterogeneity: Digital input sources	Foreign inputs deliver short-term gains; domestic inputs support long-term upgrading and resilience.	[34]; [91]
Heterogeneity: Trade mode and tech intensity	General trade and high-tech firms gain more from digital-GVC integration; processing trade limits value capture.	[97]

6 Synthesis of Findings and Evidence

This review adopts a quality-weighting system: top-tier journal studies (n=83, 70%) are weighted higher than working papers (n=21, 30%). This section provides a critical appraisal of methodological practices and proposes a tiered future research agenda.

The Construct Validity Challenge in Measuring Digitalization
Problem: Single proxy indicators confuse simple digital adoption with substantive digital capacity for GVC improvement.
Solution and Research Direction: Future studies must develop multi-dimensional indices distinguishing digital adoption from digital capability, triangulating surveys, text analysis, and investment records [97] to improve construct validity and enable precise tests of digitalization's impact on GVC dynamics.

Metric Fragmentation and GVC Measurement Challenges
Problem: Inconsistent GVC metrics hinder cross-literature comparison and ignore firm heterogeneity.
Solution and Research Direction: Future research should adopt consistent, multi-level GVC measurement integrating firm-level upstreamness, domestic value added, and governance structure within a unified framework to enable robust cross-sample comparisons.

Econometric Identification and Endogeneity Risks
Problem: Endogeneity including reverse selection restricts reliable causal identification in existing studies. [48].
Solution and Research Direction: Future work must implement rigorous identification strategies, including RDD, matching-IV methods, quasi-experimental policy shocks, and dynamic panel models, explicitly modeling self-selection into digital adoption and GVC participation.

Data Constraints and Integration Bottlenecks
Problem: A key bottleneck is the absence of integrated micro-level datasets simultaneously linking firm-specific digital capabilities, detailed GVC embeddedness, customs transactions, and operational efficiency. Most studies rely on aggregated sectoral data or incomplete firm-level records.
Solution and Research Direction: Future research must construct multi-source matched micro-data combining industrial databases, customs records, digital investment disclosures, text-mined annual reports, and patent data [56] for dynamic tracking of digital-GVC interactions.

A Tiered Future Research Agenda
To address these challenges, we propose a three-tier methodological agenda. Future quasi-experimental research can leverage China's regional policy shocks, including the National Big Data Comprehensive Pilot Zones, Broadband China Initiative, and digital trade customs pilot regulations, for rigorous causal identification of digital-GVC-efficiency linkages.

Tier 1 (Immediate): Develop multi-dimensional digital capability indices.

Tier 2 (Intermediate): Adopt quasi-experimental designs to address endogeneity.
Tier 3 (Long-term): Build high-frequency firm-customs matched data for dynamic analysis.

Table 4. Tiered Methodological Research Agenda

Time Horizon	Research Priority	Recommended Methods and Data Sources
Tier 1 (Immediate)	Enhance construct validity of digitalization measures	Multi-dimensional indices: text mining plus financial disclosures plus patent data [97, 56]
Tier 2 (Intermediate)	Address endogeneity and self-selection bias	Matching-IV; quasi-experiments; policy shocks; dynamic panel models [46, 48]
Tier 3 (Long-term)	Dynamic micro-data and causal modeling	Firm-customs matched high-frequency data; structural models; agent-based simulation

7 Conclusions and Future Research Directions

Based on 98 core studies, this review reveals how digital economy integration lifts Chinese manufacturing efficiency via GVC mediation, identifies key heterogeneities and methodological gaps, and proposes a tiered research agenda.

This review summarizes findings from 98 papers. First, digital transformation lifts corporate TFP and supply resilience [51,34]. Second, three GVC-based channels promote industrial upgrading [34,75]. Third, GVC mediation explains 30%–50% of digital's productivity effect with clear ownership heterogeneity [22,94]. Fourth, domestic digital inputs sustain long-term industrial independence [34]. Fifth, measurement inconsistency and endogeneity limit empirical reliability; future research needs multi-index design and matched micro-data [13,55].

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ReasonPath: SLM Fine-tuning with Reasoning Approach for Web Path Prediction*

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Abstract

Web path discovery is a critical reconnaissance step in web security testing. Traditional methods rely heavily on exhaustive dictionary-based fuzzing, which is resource-intensive and generates large volumes of traffic that can trigger defensive alerts on target systems. This paper presents ReasonPath, a novel approach that leverages reasoning-enhanced fine-tuning of Small Language Models (SLM) to improve the accuracy and efficiency of web path prediction for security testing while maintaining reasonable resource usage. By fine-tuning SLMs with chain-of-thought reasoning traces, the proposed method internalizes path pattern knowledge and generates more targeted candidates. The experimental results show that small models have high potential to achieve comparable performance to large model. ReasonPath improves the accuracy and efficiency of base SLMs. F1 is improved from near-zero (0.0002) to 0.29 for the best-performing 3B model and to 0.26 for the 0.5B model. Among the selected model, the Qwen2.5-0.5B has outperformed all other models with low peak VRAM.

1 Introduction

Web applications form the backbone of modern e-business systems, providing interactive services, data processing, and content delivery across diverse domains. As modern enterprises increasingly rely on web-facing platforms for transactions, customer data management, and supply-chain integration, safeguarding these infrastructures against unauthorized access and path exposure has become a critical priority. However, the complexity and scale of web applications introduce numerous security vulnerabilities, particularly in access control. Web applications often contain sensitive business information, including customer data, financial records, and proprietary assets, making them prime targets for attackers. According to the Verizon 2024 Data Breach Investigations Report, web-based attacks account for the majority of security incidents, underscoring the critical need for effective vulnerability assessment [1]. The OWASP Top 10 identifies broken access control and injection vulnerabilities as the most critical web application security risks [2]. The identification of sensitive paths and endpoints is a key part to assess vulnerabilities. Currently, traditional discovery methods have significant limitations in terms of both efficiency and coverage.

1.1 Motivation

Traditional web path discovery relies primarily on dictionary-based brute-force methods using tools such as Gobuster [3], Wfuzz [4], and Feroxbuster [5]. While effective, these approaches require extensive wordlists and generate excessive false positives through brute-force probing.

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Recent advances in reasoning-capable large language models give new insights in web path prediction. Models such as DeepSeek-R1 [6] and OpenAI o1 [7] demonstrate that reasoning-enhanced training can significantly improve performance on complex tasks requiring multi-step inference. So far, these capabilities have not been systematically applied to web path prediction for security testing.

However, LLM is too heavy for web path prediction task. While large LLMs demonstrate strong reasoning capabilities, the computational requirements and latency of such models make them impractical for high-volume security testing tasks. In particular, VRAM (Video Random Access Memory) requirements scale approximately linearly with parameter count: a 70B-parameter model such as Llama-2-70B requires approximately 140 GiB for inference in half-precision format [8], whereas a 3B-parameter model such as Qwen2.5-3B demands only 6 to 8 GiB [9], and a 0.5B-parameter model requires less than 2 GiB. This hardware constraint limits deployment on commodity GPUs, making Small Language Models (SLM) with 1-7B parameters a promising alternative [10], provided they undergo careful fine-tuning to develop reasoning capabilities comparable to LLMs.

1.2 Contributions

The present work addresses the research questions of whether reasoning-enhanced fine-tuning improves the web path prediction accuracy of SLMs compared with their non-fine-tuned counterparts, and whether a 0.5B-parameter SLM can achieve URL-sensitive path prediction with relevant performance. It makes four main contributions. Firstly, it proposes ReasonPath, a reasoning-enhanced fine-tuning approach for web path prediction using SLMs. Secondly, it designs a tailored fine-tuning pipeline. Thirdly, it constructs a dataset of 216,821 reasoning-annotated paths. Fourthly, it provides empirical evidence that fine-tuned SLMs lift F1 from near-zero to 0.29, with the 0.5B model reaching 0.26 F1 at only 1.4 GiB peak VRAM.

2 Related Work

2.1 Web Path Discovery

Web path discovery, also known as directory enumeration or endpoint discovery, is a fundamental reconnaissance technique in web security testing. Traditional approaches employ dictionary-based brute-force methods using tools such as Gobuster [3], Wfuzz [4], DirBuster [11], and Feroxbuster [5]. These tools systematically probe target servers with predefined wordlists, analyzing HTTP response codes to identify valid paths. Recent preliminary studies further demonstrate that traditional dictionary-based fuzzing achieves an accuracy of 0.030 and only 0.5 success hits per 100 requests on large-scale web corpora, with the overwhelming majority of probes returning non-existent paths [10].

The effectiveness of dictionary-based approaches depends heavily on wordlist quality and coverage. Common wordlists include SecLists [12] and custom curated lists. However, these static dictionaries suffer from limited coverage of application-specific paths and generate excessive false positives, requiring substantial probing budgets.

Recent work has explored clustering-based approaches to organize and prioritize path candidates. However, these methods still rely on static dictionaries and lack adaptive generation capabilities [13].

2.2 LLMs for Security Applications

Large language models have been increasingly applied to security tasks, including vulnerability detection, penetration testing, and security code review. LLM4Vuln [14] provided a unified evaluation framework for assessing LLM capabilities in vulnerability reasoning, identifying both strengths and limitations in security-specific tasks.

PentestAgent [15] introduced an LLM agent-based framework for automated penetration testing, incorporating retrieval-augmented generation to enhance the model's security knowledge. The system demonstrated that LLMs can effectively coordinate penetration testing workflows when provided with appropriate context and tools.

Fuzz4All [16] explored universal fuzzing with LLMs, showing that domain-specific fine-tuning can improve input generation for various fuzzing targets. However, the work focused on input generation rather than path discovery.

Despite these advances, none of the existing works systematically apply fine-tuning language models to web path prediction for security testing. This gap motivates the present work on ReasonPath, which bridges reasoning-capable LLMs with security-focused path discovery through targeted fine-tuning of Small Language Models.

2.3 Reasoning LLMs and Fine-tuning

The emergence of reasoning-capable LLMs has opened new possibilities for complex task solving. DeepSeek-R1 [6] demonstrated that reinforcement learning can incentivize reasoning capabilities, achieving state-of-the-art performance on mathematical and logical reasoning benchmarks. OpenAI's o1 series further advanced reasoning through test-time computation scaling.

Chain-of-thought fine-tuning has emerged as an effective technique for instilling reasoning capabilities. Kim et al. [17] showed that CoT fine-tuning improves performance on Big-Bench Hard by 2-4%. Recent work on diverse chain-of-thought fine-tuning [18] demonstrated that generating multiple reasoning chains within a single inference step enables within-inference refinement.

Process supervision, which provides feedback on intermediate reasoning steps rather than only final outcomes, has shown particular promise. Lightman et al. [19] demonstrated that process supervision significantly improves mathematical reasoning compared to outcome supervision alone.

Recent surveys [20,21] have comprehensively reviewed advances in reasoning models, categorizing techniques such as chain-of-thought prompting, process supervision, and diverse reasoning-chain generation. These surveys emphasize that domain-specific fine-tuning can transfer general reasoning capabilities to specialized tasks by aligning intermediate reasoning steps with task-specific structure, thereby bridging the gap between broad linguistic competence and targeted application performance.

3 Methodologies

3.1 Overview

ReasonPath employs a reasoning-enhanced fine-tuning approach to improve web path prediction. Figure 1 showing the overview of the ReasonPath methodology. The methodology consists of three stages: dataset construction with reasoning traces, chain-of-thought fine-tuning with process supervision, and directed path generation.

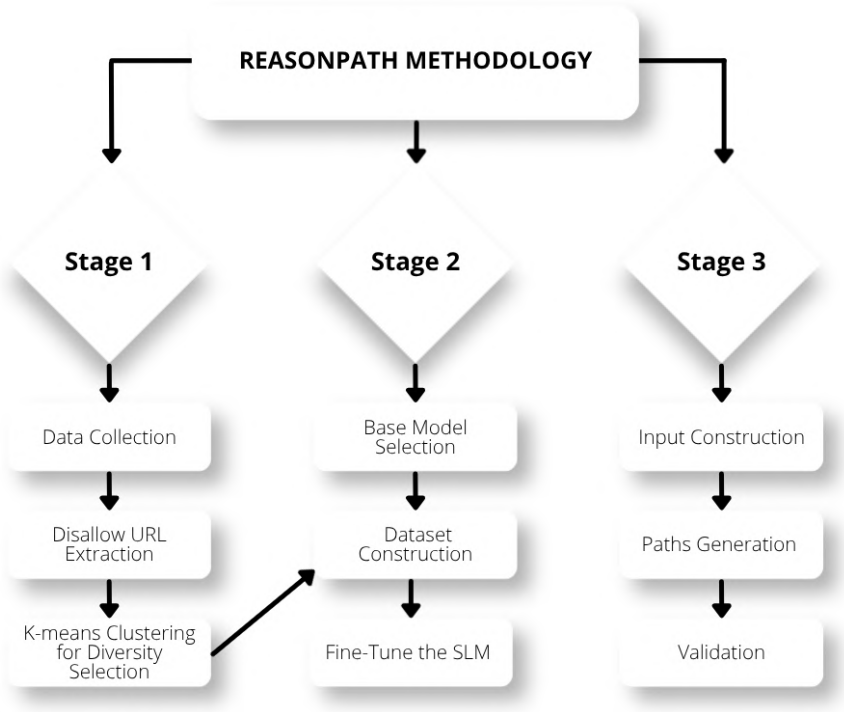

Figure 1: The ReasonPath framework overview showing the three-stage pipeline: dataset construction with reasoning traces, chain-of-thought fine-tuning, and directed path generation.

3.2 Stage 1: Dataset Construction with Reasoning Traces

The foundation of reasoning-enhanced fine-tuning is a high-quality dataset containing not only input-output pairs but also explicit reasoning traces that explain how outputs are derived from inputs.

3.2.1 Data Collection and robots.txt Crawling

Web path data are collected from multiple sources following the methodology established in a preliminary study [10]. A large-scale corpus of robots.txt files is gathered from Common Crawl [22] web crawl data. The Common Crawl Foundation provides open access to petabytes of web crawl data, including robots.txt files from millions of domains. The collection process targets all available robots.txt files from the Common Crawl archives spanning 2020 to 2024, processing 2,902,837 robots.txt files.

Dataset Statistics Table 1 summarizes robots.txt processing statistics from Common Crawl data.

Table 1: Robots.txt Dataset Statistics

Item	Value
Files processed	2,902,837
Total disallow URLs	52,811,644
Processing success rate	77.1%

3.2.2 Disallowed URL Extraction

Each robots.txt file is parsed to extract `Disallow` directives, which represent paths that website administrators explicitly mark as restricted from search engine indexing. These disallowed paths serve as high-quality ground-truth signals for sensitive or administrative paths. The extraction process normalizes paths by removing leading slashes and filtering empty or root-only entries (`/`). Each robots.txt record yields an average of 21.08 disallow URLs.

3.2.3 K-means Clustering for Diversity Selection

To ensure diversity in the selected robots.txt variants and avoid over-sampling highly repeated templates, K-means clustering is applied to the extracted disallow URL patterns. Each robots.txt file is represented as a TF-IDF vectorized document based on its disallow URL tokens. The clustering process groups robots.txt files with similar access control patterns, enabling selection of representative variants from each cluster rather than random sampling.

3.2.4 Representative Variant Selection

From each cluster, representative robots.txt variants are selected based on cluster centroid proximity. The top 10,000 distinct variants are chosen to maximize cluster coverage while minimizing redundancy. This approach ensures that the selected dataset captures diverse website structures and access control patterns across different domains, industries, and technology stacks.

3.2.5 LLM-Powered CoT Dataset Construction

To generate high-quality Chain-of-Thought (CoT) training examples, GPT-4o-mini is prompted to analyze actual website homepage structures and predict hidden robots.txt paths via a four-step reasoning chain (signature extraction, security-gap analysis, path prediction, and synthesis). For each domain, observed paths are first extracted from the homepage HTML to provide signals about content organization, technology stack, and purpose.

The LLM then analyzes these URL patterns to infer a website signature covering type, technology stack, content types, and primary function. Based on this signature, the model identifies typical sensitive functionality categories such as administrative interfaces, configuration files, and backup archives.

For each predicted hidden path, GPT-4o-mini provides explicit reasoning (for example, predicting `/admin/` as an administrative interface or `/wp-config.php` as a configuration file containing credentials). Each analysis generates 500 to 800 tokens of reasoning text, producing an input-output format that maps from website analysis and signature extraction to ranked sensitive-path predictions.

After homepage crawling, chain-of-thought generation, and quality filtering, the final fine-tuning dataset included 6,366 valid domains with 216,821 associated reasoning traces. This dataset forms the foundation for reasoning-enhanced fine-tuning described in Section 3.3. Table 2 shows reasoning-annotated path statistics across selected web domains.

Table 2: Multi-Path CoT Dataset Statistics

Item	Value
Total domains	6,366
Total paths	216,821
Avg paths per domain	34.1
Median paths per domain	22
Min paths per domain	2
Max paths per domain	1,173
Quality filters	Length $\geq$3, no wildcards, no query params

Figure 2 shows two representative examples across different website types: WordPress CMS and E-Business Website. Each example demonstrates how reasoning traces guide multi-path prediction based on detected technology patterns.

Figure 2: Reasoning Trace Examples

WordPress (PHP)	**E-Business Website**
Input: example.com Tech: WordPress, Apache, PHP	Input: shop.example.com Tech: Node.js, React, PostgreSQL
Reasoning: 1. WordPress uses /wp-admin/ 2. Uploads in /wp-content/ 3. Backups: /backup.sql 4. Config: /wp-config.php.bak	Reasoning: 1. E-business uses /admin/ 2. API at /api/v1/ 3. Orders: /orders/export 4. Reports: /reports/sales
Output: /wp-admin/ /wp-content/uploads/ /backup.sql /wp-config.php.bak	Output: /admin/login /api/v1/orders /orders/export.csv /reports/sales.pdf

3.3 Stage 2: Chain-of-Thought Fine-tuning

Base language models are fine-tuned on the constructed dataset using supervised fine-tuning on chain-of-thought reasoning traces. The training pipeline is implemented in Python using the HuggingFace Transformers ecosystem, with model configuration, dataset formatting, and training orchestration modularized across dedicated components.

3.3.1 Base Model Selection

Table 3 lists selected small language models and the selection reasons.

Table 3: Base Model Selection

Model	Params	Selection Rationale
Qwen2.5-3B-Instruct	∼3B	Strong instruction-following and multilingual open-source SLM
DeepSeek-R1-Distill-Qwen-1.5B	∼1.5B	Distilled from 671B teacher on 800K reasoning samples; strong logic/math per parameter
Phi-3-mini-4k-instruct	∼3.8B	High-quality training; outperforms larger baselines on reasoning and coding benchmarks
Qwen2.5-0.5B-Instruct	∼0.5B	Grouped-query attention edge SLM for sub-2 GiB mobile and IoT deployment

These four models span the 0.5B to 3.8B parameter range representative of contemporary Small Language Models. They were chosen to isolate how model size and VRAM constraints interact with reasoning-enhanced fine-tuning, and to test whether sub-1B models can compensate for smaller capacity via structured chain-of-thought supervision.

3.3.2 Prompt Engineering and Dataset Formatting

Each raw JSON sample is transformed into a chat message triplet (system, user, assistant) via a dedicated dataset builder module. The system message establishes the cybersecurity assistant role, while the user message concatenates the detected technology stack, observed homepage paths, and sitemap contents. The assistant message interleaves the reasoning trace with the structured list of predicted sensitive paths.

Model-specific formatting adaptations are applied at this stage. For the DeepSeek-R1 distill model, reasoning traces are wrapped in `<think>...</think>` tags to align with its trained reasoning format; for all other models, reasoning is emitted as free-form text preceding the path list. This conditional formatting ensures that the supervision signal matches each model's native inference distribution. The formatted records are then truncated to a maximum sequence length of 2048 tokens, and split into 95% training and 5% validation sets.

3.4 Stage 3: Directed Path Generation

The fine-tuned model then generates paths for validating the performance. For each target domain, an input prompt is constructed containing the target domain name, detected technologies, and observed paths from initial crawling, formatted with the model's chat template which establishes the cybersecurity assistant role. During validation, because each domain contains a variable amount of observed information and the model autoregressively determines when to stop generating, the number of predicted paths varies per domain, capped at a maximum of 20 paths. Paths are extracted from the output, normalized (lowercased, leading slash ensured, trailing slashes stripped), and deduplicated.

4 Experimental Settings

Evaluation is performed on a held-out validation split comprising 5% of the full dataset, with the same evaluation sets used across all models. Domain validation begins by constructing prompts with URLs from homepage discovery, inputting them into both the fine-tuned and non-fine-tuned models, and then comparing the generated URLs against the existing dataset's known sensitive URL sets. Path-level string matching is employed, and the comparison is domain-based because each domain, due to the theme of the website, may contain specific terms that vary from others. Predicted paths are normalized (lowercased, leading slash ensured, trailing slashes stripped).

Metrics are macro-averaged across each domain. Let D denote the set of validation domains. For each domain $d \in D$, precision P_d and recall R_d are computed from the domain's true positives (TP_d), false positives (FP_d), and false negatives (FN_d):

$$P_d = \frac{\mathrm{TP}_d}{\mathrm{TP}_d + \mathrm{FP}_d}, \qquad R_d = \frac{\mathrm{TP}_d}{\mathrm{TP}_d + \mathrm{FN}_d}.$$

Domain-level F1 is the harmonic mean of P_d and R_d:

$$\mathrm{F1}_d = 2 \cdot \frac{P_d \cdot R_d}{P_d + R_d}.$$

Precision, recall, and F1 are then obtained by averaging the per-domain values:

$$\mathrm{Precision}_M = \frac{1}{|D|} \sum_{d \in D} P_d, \qquad \mathrm{Recall}_M = \frac{1}{|D|} \sum_{d \in D} R_d, \qquad \mathrm{F1}_M = \frac{1}{|D|} \sum_{d \in D} \mathrm{F1}_d.$$

Also, peak VRAM is measured during inference by loading the model, and recording the maximum memory allocated.

5 Results

5.1 Comparison with Base Models

ReasonPath achieves improved performance across all fine-tuned models. Qwen2.5-3B attains the highest F1 (0.2935). Notably, Qwen2.5-0.5B achieves an F1 of 0.2612, within 11% of the 3B model, while requiring only 1,456 MiB of peak VRAM, a 5.4× reduction compared to the 3B model and a 6.5× reduction compared to Phi-3-mini. This makes the 0.5B parameter model highly attractive for edge and low-resource deployment scenarios.

All four base models achieve near-zero F1 (<0.0011) because they fail to produce outputs in the required output format. After fine-tuning, Qwen2.5-3B improves from 0.0002 to 0.2935 and Qwen2.5-0.5B improves from 0.0011 to 0.2612, demonstrating that structured reasoning supervision dramatically enhances task performance across model architectures of different parameter sizes.

Table 4 compares fine-tuned and base model performance metrics.

Table 4: Fine-Tuned versus Base Model Results ($n = 319$ validation domains)

Method	Peak VRAM	F1	TP	FP	FN
ReasonPath (Qwen2.5-3B FT)	7,812 MiB	**0.2935**	1,582	1,265	8,904
ReasonPath (DeepSeek-R1-Qwen-1.5B FT)	5,128 MiB	0.1401	1,032	1,385	9,454
ReasonPath (Phi-3-mini FT)	9,462 MiB	0.1196	578	727	9,908
ReasonPath (Qwen2.5-0.5B FT)	**1,456 MiB**	0.2612	1,418	1,553	8,877
Base SLM (Qwen2.5-3B)	7,590 MiB	0.0002	1	1,899	10,348
Base SLM (DeepSeek-R1-Qwen-1.5B)	5,018 MiB	0.0000	0	2	10,349
Base SLM (Phi-3-mini)	9,126 MiB	0.0005	3	25	10,346
Base SLM (Qwen2.5-0.5B)	1,423 MiB	0.0011	6	1,170	10,303
Traditional dict. fuzzing (EBISION)	—	0.027	—	—	—

6 Discussion

6.1 Performance of Reasoning Fine-tuning

Reasoning-enhanced fine-tuning outperforms base-model baselines. Internalized pattern knowledge enables faster inference. The chain-of-thought enhanced by fine-tuning on the given dataset ensures that reasoning steps are logically coherent, and improves accuracy over both non-fine-tuned models and traditional brute-force methods.

The train/validation/test split in the current pipeline is performed at the path level rather than at the domain level, meaning that all types of paths from different sources, such as `/wp-admin/`, `/administrator/`, or `/api/`, may appear across all partitions. The research consider unseen-domain discovery to be future work and recommend adopting domain-disjoint methods.

The current evaluation compares fine-tuned models against their zero-shot baselines. Further comparison with a direct-supervised baseline trained to predict paths without intermediate chain-of-thought traces could also be considered in the future. Such a baseline would help clarify whether the F1 improvement is attributable to the reasoning structure itself or simply to additional supervised exposure to path-label pairs.

6.2 VRAM and Model-Size Trade-offs

Peak VRAM consumption varies substantially across model architectures and parameter scales. During fine-tuned inference, Qwen2.5-3B demands approximately 7.8 GiB, Phi-3-mini requires 9.5 GiB, and DeepSeek-R1-Qwen-1.5B consumes 5.1 GiB. In contrast, the Qwen2.5-0.5B model operates with only 1.4 GiB of peak VRAM, a 5.4× reduction relative to the 3B model and a 6.5× reduction relative to Phi-3-mini, while attaining an F1 score of 0.26, within 11% of the best-performing 3B checkpoint.

These results indicate that reasoning-enhanced fine-tuning can unlock practical accuracy without proportional increases in memory footprint. Although larger models may still yield marginal improvements, such gains are increasingly constrained by hardware availability and deployment environment limitations. Ultra-compact checkpoints (sub-1B parameters) are consequently easier to adopt on commodity hardware, edge devices, and shared computing environments where high-end GPUs are unavailable.

6.3 Dataset Quality

The dataset uses robots.txt `Disallow` directives as ground-truth signals for sensitive paths. Although this source is large-scale, not every disallowed path corresponds to a security-relevant endpoint; introducing label noise that should be acknowledged when interpreting metrics. Furthermore, the prevalence of common administrative templates (e.g., `/admin`, `/administrator`) may limit generalization to modern dynamic web applications and API-driven systems.

6.4 Security Implications

The improved accuracy of ReasonPath has dual-use implications. Security teams can use the tool for more efficient penetration testing and vulnerability assessment. However, adversaries could also leverage similar techniques for malicious reconnaissance. Responsible disclosure and access controls are recommended for deployed systems.

6.5 Ethical Considerations

Testing system vulnerabilities raises ethical concerns. While it advances authorized security assessments, misuse of this research risks enabling unauthorized reconnaissance. Testing on real production environments should require explicit authorization and rate-limiting to avoid disrupting live systems. Laboratory evaluations should follow the principle of minimizing system impact.

6.6 Limitations

Several limitations needs to be mentioned. The evaluation relies on a status-code mapping that treats 301/302 redirects as false positives, reflecting that redirections do not, by themselves, confirm the presence of the intended sensitive resource. In practice, however, many production stacks implement redirection logic at the application layer while returning HTTP 200, for example, via HTML meta-refresh tags or JavaScript-based routing. Likewise, "soft-404" pages may return HTTP 200 with templated "not found" content, so a purely status-based evaluation can misclassify client-side redirects and soft-404s as true positives.

Another limitation concerns data cost and model freshness: producing annotated chain-of-thought reasoning traces at scale is expensive, and the model's knowledge becomes static after fine-tuning. Unlike retrieval-augmented approaches that can incorporate new data dynamically without retraining, this method requires periodic regeneration of reasoning datasets, and long-context inference with larger teacher models elevates latency and cost during that generation.

6.7 Future Work

Scaling and Continual Learning Scaling the dataset generation pipeline to full RAG-based retrieval on GPU clusters would enable semantic similarity scoring across the entire 3.7M-file corpus rather than relying on pre-clustered variants, potentially improving path diversity and coverage. Online or continual fine-tuning mechanisms could also be explored so that newly discovered paths are incorporated dynamically without retraining from scratch.

Reducing False Negatives To reduce the high false-negative rate, future work could investigate top-k [23] expansion with self-consistency voting, iterative refinement loops that re-prompt

the model with missed paths, and retrieval-augmented generation to inject analogous domain signatures at inference time.

Beyond Traditional Web Paths The methodology can be extended beyond traditional web paths to API endpoint discovery, where RESTful and GraphQL endpoint patterns present a richer but structurally different prediction space. Integrating more sources from real-world URL patterns is also required to provide a more diverse dataset.

7 Conclusion

This study examined whether reasoning-enhanced fine-tuning improves URL path prediction in Small Language Models, and whether an extremely compact SLM can achieve relevant accuracy for this task. Experiments show that reasoning-aware adaptation yields dramatic improvements over vanilla checkpoints. Furthermore, accuracy gains scale sub-linearly with parameter count; the 0.5B checkpoint attains F1 within 11% of its 3B counterpart, indicating that larger architectures confer only marginal gains once structured reasoning training is provided. VRAM consumption, however, scales almost linearly with model size: the Qwen2.5-0.5B model requires only 1.4 GiB peak allocation, whereas the 3B model demands 5.4 times more VRAM for merely 3.2 percentage points of additional F1. These results confirm that sub-1B models can internalize meaningful path-pattern knowledge through adaptive reasoning, offering a practical balance of accuracy and efficiency for security assessment.

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What Makes AI Influencers Effective? Evidence from Instagram Comments*

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Abstract

This study examines how AI influencer attributes are associated with consumer trust, engagement, influencer loyalty, and brand trust. Drawing on source credibility, Source attractiveness, and trust transfer perspective, the study analyzes 429,194 Instagram comments related to an AI influencer. A text mining approach combining VADER sentiment analysis, keyword-based construct scoring, and structural equation modeling was used to examine relationships among credibility, expertise, authenticity, attractiveness, psychological distance, homophily, anthropomorphism, content quality, transparency, trust, engagement, loyalty, and brand trust. The results suggest that transparency, anthropomorphism, and psychological distance are the strongest predictors of trust-related evaluation. Trust is positively associated with consumer engagement and influencer loyalty, which are further associated with Brand trust. However, the modest explore learn power of the model, indicating that defining should be interpreted as exploratory text-based evidence rather than causal proof. The study contributes to influencer marketing research by using large-scale social media comments to examine naturally occurring consumer responses to virtual influencers.

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1 Introduction

Influencer marketing has become an important strategy for brands seeking to build meaningful relationships with consumers on social media platforms. Companies increasingly rely on influencers to promote products and services by drawing content shared on platforms such as Instagram, TikTok, YouTube, and Twitch (Cabeza-Ramirez et al., 2022; Obermayer et al., 2022). Influencers shape consumer attitude, behaviors, and precious attention, and purchase attention by using trust, social capital, and persuasive communication (Brown & Hayes, 2008). Recent advances in artificial intelligence, machine learning, and computer-generated graphics have led to the rise of AI influencers, also known as virtual influencers. These digital personas are designed to simulate human appearance, behavior, and emotional expression (Byu,n & Ahn, 2023; Seymour et al., 2018). Well-known AI influencers such as Lu do Magalu and Lil Miquela have attracted large audiences and collaborated with global brands and organizations. As these virtual agents become more realistic and socially visible, the boundary between human and non-human influencers is becoming increasingly blurred (Thomas & Fowler, 2021; Araque & Moynihan, 2021)

Despite the rapid growth of AI influencers, academic research on how consumers perceive and trust them remains limited. Prior studies have mainly examined human influencers or survey-based responses to virtual influencers (Lou & Yuan, 2019; Sokolova & Kefi, 2020). Less attention has been given to naturally occurring consumer reactions expressed in social media comments. This is important because comments can reveal how users discuss AI influencer credibility, attractiveness, authenticity, content quality, trust, engagement, loyalty, and brand-related responses in a digital environment.

To address this gap, they study analyzes consumer discussions of AI influencers using a large-scale dataset of Instagram comments. By examining naturally occurring social media interactions. The study provides insights into how users perceive AI influencer characteristics and trust-related outcomes in real digital environments.

Contribute to AI influencer marketing research by combining a large-scale social media text analysis with established marketing theories. By using sentiment analysis, keyboard-based construct scoring, and structural equation modeling, the study provides exploratory evidence on how consumers respond to AI influencers in real social media contexts.

2 Literature review

2.1 Source Credibility

Source credibility is a central concept in marketing and advertising research. It refers to how audiences evaluate a communicator's trustworthiness and competence (Andersen & Clevenger, 1963). In influencer marketing, credibility is important because consumers are more likely to accept messages from sources they perceive as reliable, knowledgeable, and sincere (Hovland et al., 1953; Ohanian, 1990).

In this study, source credibility is reflected through expertise and authenticity. Expertise refers to the extent to which an influencer is perceived as knowledgeable and capable of providing valid information. Authenticity reflects whether the influencer is perceived as sincere, consistent, and genuine. In the context of AI influencers, these dimensions are important because followers may question whether a non-human influencer can be credible or authentic. Therefore, this study proposes:

H1: Credibility of AI influencers is positively associated with trust in AI influencers
H1-1: Expertise of AI influencers is positively associated with trust in AI influencers.
H1-2: Authenticity of AI influencers is positively associated with trust in AI influencers.

2.2 Source attractiveness

Source attractiveness explains how the appeal of a communicator shapes audience responses. It includes not only physical attractiveness but also perceived similarity, familiarity, and likability (McGuire, 1968; Sternthal & Samuel, 1982). In influencer marketing, attractive and relatable influencers are more likely to capture attention and encourage favorable evaluations.

For AI influencers, attractiveness can also be understood through psychological distance, homophily, and anthropomorphism. Psychological distance reflects how close or distant users feel from the influencer. Homophily refers to perceived similarity between the influencer and followers. Anthropomorphism refers to the attribution of human-like qualities, emotions, and behaviors to non-human agents.

These attributes may help Al influencers appear more socially present and relatable.

H2: The attractiveness of AI influencers is positively associated with trust in AI influencers.

H2-1: Psychological distance is positively associated with trust in AI influencers.

H2-2: Homophily is positively associated with trust in AI influencers.

H2-3: Anthropomorphism is positively associated with trust in AI influencers.

2.3 Content quality

Content quality is an important factor in online trust formation. High-quality content is usually perceived as useful, clear, relevant, and reliable. In the context of Al influencers, content quality may help followers evaluate whether the influencer provides valuable information or meaningful interaction.

Transparency is also important because AI influencers are not human agents. Clear disclosure of an influencer's AI-driven nature can reduce uncertainty and support consumer confidence. When users understand that an influencer is AI-generated, they may evaluate the interaction. More clearly and ethically. Therefore, this study proposes:

H3: Content quality of AI influencers is positively associated with trust in AI influencers.

H3-1: Transparency of AI influencers is positively associated with trust in AI influencers.

2.4 Trust, Engagement, Influencer Loyalty, and Brand Trust

Trust is a key element in influencer-follower relationships because it reduces uncertainty and supports continued interaction (Morgan & Hunt, 1994, Mayer et al.,

1995). In Al influencer contexts, trust is especially important because users may feel uncertainty about authenticity, transparency, and human-likeness.

Consumer engagement refers to users' active interaction with influencer content, such as commenting, liking, sharing, or expressing emotional involvement.

Influencer loyalty reflects a follower's intention to maintain a relationship with the influencer over time. Brand trust refers to consumer confidence in the brand connected to the influencer. In influencer marketing, trust in the influencer may be transferred to the associated brand through engagement and loyalty. Therefore, this study proposes:

H4: Trust in AI influencers is positively associated with consumer engagement.

H5: Trust in AI influencers is positively associated with influencer loyalty.

H6: Consumer engagement is positively associated with brand trust.

H7: Influences loyalty is positively associated with Brand Trust.

2.5 Conceptual Framework

The theoretical foundation of this study is based on three key theories. First, Source Credibility Theory (Hovland et al., 1953) explains that consumers are more likely to trust credible sources. Second, the Technology Acceptance Model (Davis, 1989)

highlights the importance of content quality and transparency in technology acceptance. Third, the Commitment-Trust Theory (Morgan & Hunt, 1994) identifies trust as a key driver of engagement, loyalty, and brand trust. Based on these theories, this study proposes that Al influencer characteristics are positively associated with trust in Al influencers and related consumer outcomes.

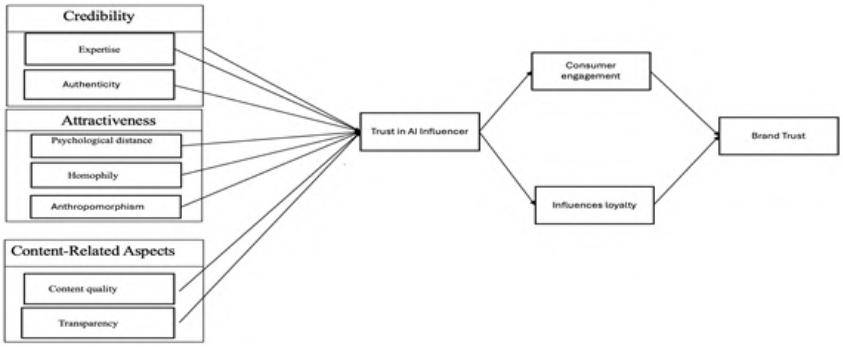

Figure1. Research model

3 Research Methods

3.1 Data Cleaning and Analytical Procedure

This study analyzed Instagram comments related to AI influencers using a quantitative big data approach. Comments were collected from the official Instagram accounts of the Al influencers included in the study during the observation period specified in Section 2024-2025. The initial dataset contained approximately 500,000 comments distributed across the selected influencers. Data preprocessing involved removing duplicate comments, URLs, hashtags, emojis, special characters, and other non-informative elements. Language detection was performed using automated filtering, and only English-language comments were retained. To reduce noise, comments identified as likely spam or bot-generated (e.g., repetitive content, excessive posting frequency, or duplicated text patterns) were excluded. After preprocessing, the final dataset consisted of 429,193 comments.

The NLP pipeline consisted of three stages. First, sentiment analysis was conducted using VADER (Valence Aware Dictionary and sentiment Reasoner), a lexicon- and rule-based model designed for social media text (Hutto & Gilbert, 2014). VADER compound sentiment scores were calculated for each comment and classified using the standard thresholds recommended by the developers: positive (-0.05), neutral (-0.05 to 0.05), and negative (< -0.05). Second, construct extraction was performed using theory-driven keyword dictionaries developed from prior literature on influencer marketing, trust, and consumer behavior. Separate dictionaries were created for credibility, expertise, authenticity, attractiveness, psychological distance, homophily, anthropomorphism, content quality, transparency, trust, engagement, loyalty, and brand trust. Construct scores were calculated based on keyword occurrence frequencies and weighted by sentiment polarity to capture both construct presence and evaluative direction. The complete keyword dictionaries and coding rules are provided in Appendix A

to support reproducibility. Third, the resulting construct-level measures were used as inputs for Structural Equation Modeling (SEM) to examine the relationships among Al influencer attributes, trust, engagement, influencer loyalty, and brand trust.

As the study is based on observational social media data, the findings indicate associations rather than causal relationships. While the preprocessing and filtering procedures were designed to reduce noise and spam, text-mining measures should be interpreted as proxies for expressed user evaluations rather than direct psychological measurements.

3.2 VADER Sentiment Analysis

To analyze the sentiment of Instagram comments, this study employed the VADER (Valence Aware Dictionary and Sentiment Reasoner) sentiment analysis tool, developed by Hutto and Gilbert (2014). Designed specifically for social media contexts, VADER is a rule-based model that combines a human-curated sentiment lexicon with a set of grammatical and syntactical heuristics.

3.3 Sentiment Analysis of Credibility, Expertise, and Authenticity - Related Comment.

Using VADER sentiment analysis (Hutto & Gilbert, 2014), comments related to credibility, expertise, and authenticity showed mixed sentiment patterns. Credibility-related comments were relatively balanced, with 31.3% positive, 36.3% neutral, and 32.4% negative sentiment, suggesting that users still evaluate an influencer's credibility cautiously. Authenticity-related comments were mostly neutral (54.2%), indicating uncertainty about whether Al influencers can be perceived as genuine. In contrast, expertise-related comments were more positive (40.2%), suggesting that users may more easily accept AI influencers as knowledgeable or competent sources. These findings are consistent with source credibility research, which argues that expertise and trustworthiness are important factors in message acceptance (Hovland et al., 1953; Ohanian, 1990). They also support prior influencer marketing studies showing that credibility-related cues play an important role in building consumer trust (Lou & Yuan, 2019).

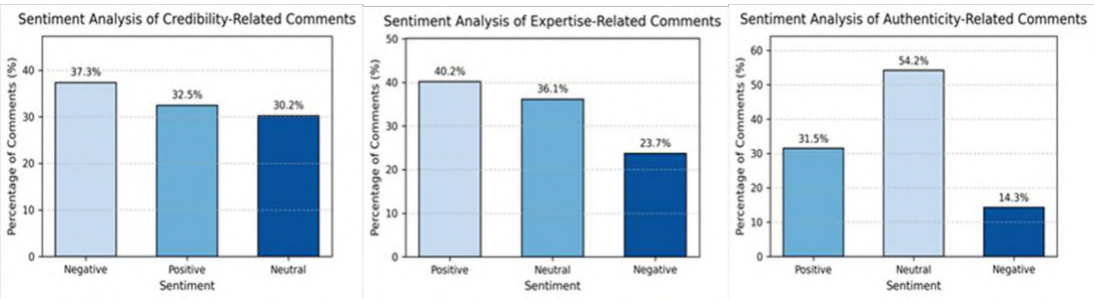

Figure 2. Sentiment Analysis of Credibility, Expertise, and Authenticity-related Comment

3.4 Sentiment Analysis of Attractiveness, Psychological-distance, Homophily, Anthropomorphism -related Comment.

Using VADER sentiment analysis (Hutto & Gilbert, 2014), comments related to attractiveness, psychological distance, homophily, and anthropomorphism were generally neutral to positive.

Attractiveness-related comments showed the highest positive sentiment (79.9%), suggesting that users often respond favorably to the visual and social appeal of Al influencers. Psychological distance was also mostly positive (59.4%), indicating that some users perceive AI influencers as socially close or relatable. Homophily-related comments were more moderately positive (46.3%), while anthropomorphism was mainly neutral (51.6%), suggesting that users may recognize human-like qualities in Al influencers but still evaluate them cautiously. These findings are consistent with source attractiveness research, which argues that attractiveness, similarity, and familiarity can strengthen persuasive communication (McGuire, 1968; Sternthal & Samuel, 1982). They also align with anthropomorphism research showing that human-like cues can support social presence and trust in non-human agents (Epley, 2018).

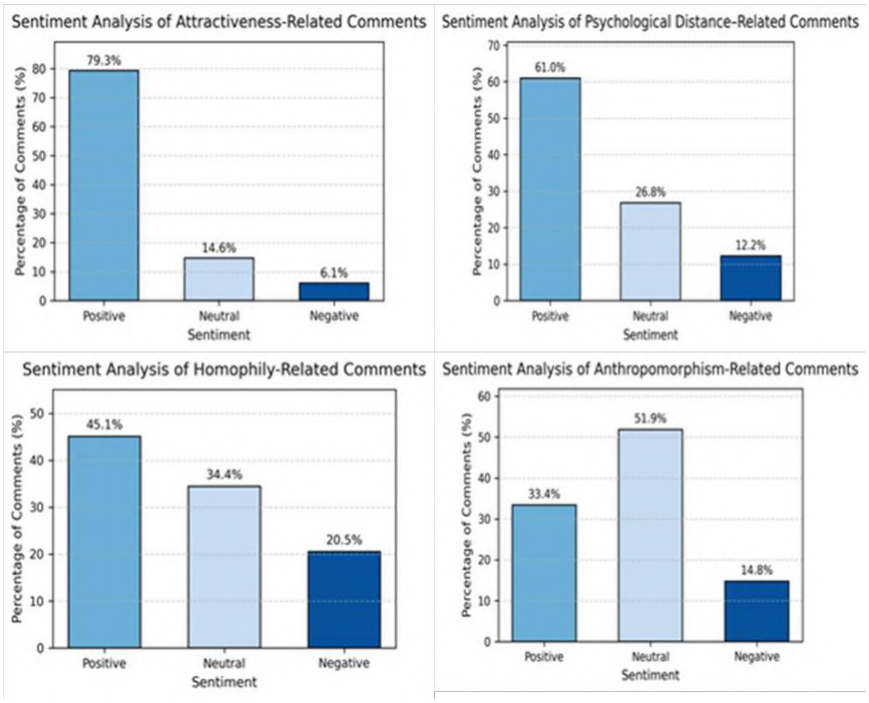

Figure 3. Sentiment Analysis of Content Quality, Transparency-related Comment

3.5 Sentiment Analysis of Content Quality, Transparency -Related Comment

Using VADER sentiment analysis (Hutto & Gilbert, 2014), content quality-related comments were mostly positive (46.1%), followed by neutral (38.5%) and negative sentiment (15.4%). This suggests that users generally respond favorably to high-quality AI influencer content, consistent with prior research on content value and consumer evaluations (Metzger & Flanagin, 2013; Lou & Yuan, 2019).

Transparency-related comments were primarily neutral (54.2%), followed by positive (30.9%) and negative sentiment (14.9%). This indicates that users discuss AI disclosure in a more informational manner, supporting previous findings that transparency helps reduce uncertainty and build trust (Boerman et al., 2017; Evans et al., 2017).

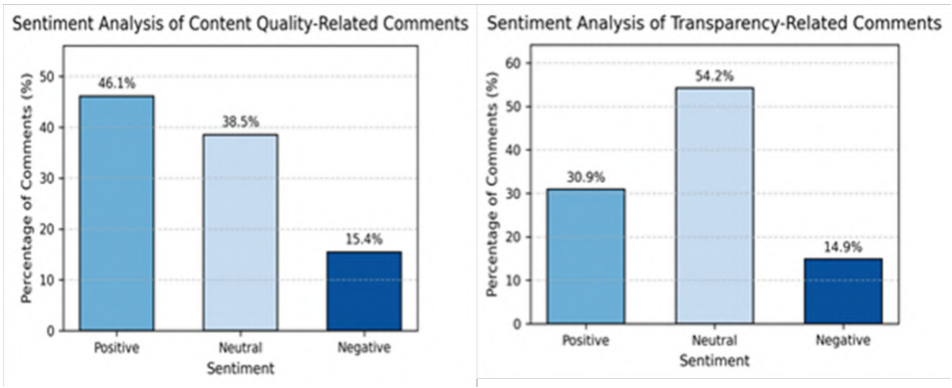

Figure 4. Sentiment Analysis of Content Quality, Transparency-related Comment

3.6 Sentiment Analysis of Influencer loyalty, Consumer Engagement, Trust, Brand trust -related Comment

The results show that consumer engagement comments were mainly positive (73.7%), followed by neutral (15.8%) and negative sentiment (10.5%). Influencer loyalty showed the strongest positive sentiment, with 81.4% positive, 14.8% neutral, and 3.8% negative comments. These findings indicate generally favorable attitudes toward AI influencers, consistent with research linking engagement and loyalty to positive user experiences (Vivek et al.,2012; Morgan & Hunt, 1994).

Brand trust comments were mostly neutral (52.3%), followed by positive (34.0%) and negative sentiment (13.7%). This suggests that while users respond positively to AI influencers, trust in associated brands remains more cautious, supporting trust transfer theory (Sichtmann, 2007).

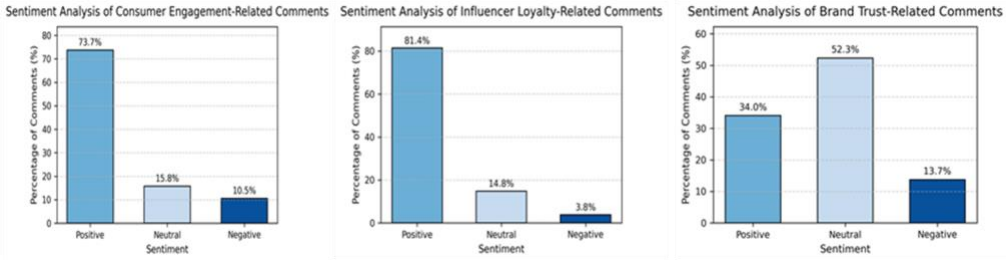

Figure5. Sentiment Analysis of Influencer Loyalty, Consumer Engagement, Trust, Brand trust - related Comment (Positive)

3.7 Structural Equation Modeling (SEM)

To examine the relationships among Al influencer attributes, trust-related evaluations, engagement, loyalty, and brand trust, this study employed Structural Equation Modeling (SEM). SEM was used as a theory-driven tool to test associations among text-derived indicators within one analytical framework (Hair et al., 2010). Because the variables were. Generated from social media comments, the results are interpreted as aggregate-level associations rather than individual psychological states or causal effects. Model fit indices were used to assess the overall plausibility of the proposed structural model (Sarstedt et al., 2020).

3.7.1 R^2 values

The R^2 results indicate the model's explanatory power for each endogenous construct. The influencer attributes explain 17.5% of the variance in trust, showing modest explanatory power. However, engagement (R^2 = 0.048), loyalty (R^2 = 0.061), and brand trust (R^2 = 0.085) have relatively low R' values. Such values are common in consumer behavior research, where outcomes are influenced by many individual, platform-related, and brand-related factors beyond those included in a single model (Hair et al., 2019; Falk & Miller, 1992). Therefore, the findings should be viewed as partial but meaningful evidence of trust transfer from Al influencers to brands, while recognizing that additional variables may improve the model's explanatory power (Stewart, 2003; Hair et al., 2019).

Table1. R^2 Results

Endogenous Construct	R^2
Trust	0.175
Engagement	0.048
Loyalty	0.061
Brand Trust	0.085

3.7.2 Standardized Betas

The results show that attractiveness (β = 0.107) and anthropomorphism (β = 0.104) are the strongest predictors of trust in AI influencers. Transparency (β = 0.095) and authenticity (β = 0.095) also make meaningful contributions. Psychological distance (β = 0.083) and credibility (β = 0.081) have moderate effects, while content quality (β = 0.074), expertise (β = 0.072), and homophily (β = 0.064) show relatively smaller impacts.

Table 2. Variance Inflation Factors.

Predictor	**Std. Beta**
Attractiveness	0.107
Anthropomorphism	0.104
Transparency	0.095
Authenticity	0.095
Psychological distance	0.083
Credibility	0.081
Content quality	0.074
Expertise	0.072
Homophily	0.064

3.7.3 SEM Path Coefficients

The SEM results show that all hypothesized paths were positive. Transparency (β =0.090), anthropomorphism (β = 0.083), and psychological distance (β = 0.079) had the strongest associations with trust-related evaluations. Trust was positively associated with engagement (ß = and influencer loyalty (β = 0.316). Engagement (β = 0.131) and loyalty (β = 0.128) were also positively associated with brand trust. Overall, the results support the proposed trust-based framework, but they should be interpreted as text-derived associations rather than causal effects.

Table 3. SEM Path Coefficients

Path	Predictor	B	95%CI Lower	95% CI Upper
Trust ~ attribute	const	-0.002	-0.002	-0.002
Trust ~ attribute	credibility	0.043	0.041	0.044
Trust ~ attribute	expertise	0.054	0.052	0.056
Trust ~ attribute	authenticity	0.077	0.074	0.08
Trust ~ attribute	attractiveness	0.063	0.061	0.065
Trust ~ attribute	psychological distance	0.079	0.076	0.083
Trust ~ attribute	homophily	0.052	0.05	0.055
Trust ~ attribute	anthropomorphism	0.083	0.08	0.087
Trust ~ attribute	content quality	0.043	0.041	0.044
Trust ~ attribute	transparency	0.09	0.086	0.093
Engagement ~ Trust	const	0.004	0.004	0.004
Engagement ~ Trust	trust	0.26	0.256	0.264
Loyalty ~ Trust	const	0.007	0.007	0.007
Loyalty ~ Trust	trust	0.316	0.31	0.321
BTE + Loyalty	const	0.004	0.004	0.004
BTE + Loyalty	engagement	0.131	0.129	0.134
BTE + Loyalty	loyalty	0.128	0.125	0.131

3.7.4 Variance Inflation Factors.

All VIF values are very low, ranging from 1.16 to 1.25. A common rule is that VIF values above 5 (sometimes 10) may indicate problematic multicollinearity. Since all your values are close to 1, it means the predictors (credibility, expertise, authenticity, etc.) are independent enough from each other, and there is no multicollinearity problem in your model. This confirms that each construct is measuring a unique aspect, and the SEM results are reliable.

Table 4. Variance Inflation Factors

Feature	VIF
credibility	1.18
expertise	1.187
authenticity	1.201
attractiveness	1.161
psychological distance	1.189
homophily	1.176
anthropomorphism	1.18
content quality	1.18
transparency	1.248

3.8 Regression Analysis Overview

To complement the SEM results, additional regression analyses were conducted to examine the robustness of the associations between Al influencer attributes and trust-related evaluations. OLS regression was used as the baseline model, while sentiment-weighted regression examined whether emotional tone strengthened these associations. Interaction regression was then used to test whether combinations of attributes, such as credibility and attractiveness, were jointly associated with trust.

3.8.1 Sentiment-Weighted Regression Results

The sentiment-weighted regression showed that credibility (($\beta = 0.149$, $p < 0.001$), attractiveness (($\beta = 0.103$, $p < 0.001$), and content quality ($\beta = 0.149$, $p < 0.001$) were positively associated with trust-related evaluations. Credibility and content quality showed the strongest associations, suggesting that users respond more positively when AI influencers appear reliable and provide valuable content. This finding is consistent with the Source Credibility Model, which emphasizes that trustworthy and informative sources are more persuasive (Hovland et al., 1953). However, the low R^2 indicates that emotional tone explains only a limited part of trust-related responses.

Table 5. Sentiment-Weighted Regression.

Predictor	Coef. (B)	SE	t-value	p-value	95%CI [Lower,Upper]	R^2	Adj. R^2	F-stat	Durbin-Watson	Cond No.
Constant	0.6328	0	1412.05	0	[0.632, 0.634]	0.007	0.007	961.8	1.839	20.1
Credibility (Sentiment-Weighted)	0.1512	0.004	36.254	0	[0.143, 0.159]					
Attractiveness (Sentiment-Weighted)	0.1035	0.007	15.062	0	[0.090, 0.117]					
Content Quality (Sentiment-Weighted)	0.1489	0.009	16.77	0	[0.132, 0.166]					

3.8.2 Multiple Regression Results

The multiple regression results showed that credibility ($\beta = 0.0691$, $p < 0.001$), attractiveness ($\beta = 0.0687$, $p < 0.001$), and content quality ($\beta = 0.0416$, $p < 0.001$) were all positively associated with trust. Credibility had the strongest coefficient, indicating that perceived reliability remains central in shaping trust in AI influencers. This supports prior influencer marketing research showing that credibility is a key driver of consumer trust and message effectiveness (Lou & Yuan, 2019). Nevertheless, the low R' suggests that trust in AI influencers is also influenced by other factors not included in this model.

Table 6. Multiple Regression

Predictor	Coef. (B)	SE	t-value	p-value	95% CI [Lower,Upper]	R^2	Adj. R^2	F-stat	Durbin Watson	Cond. No.
Constant	0.6323	0.00	1394.841	0	[0.631, 0.633]	0.004	0.004	636.8	1.84	11
Credibility Score	0.0798	0.003	28.796	0	[0.074, 0.085]					
Attractiveness Score	0.0887	0.005	18.667	0	[0.079, 0.098]					
Content Quality Total Score	0.0444	0.004	9.961	0	[0.036, 0.053]					

3.8.3 Interaction Regression Results

The interaction model showed that credibility strengthened the associations between attractiveness and trust ($\beta = 0.0455$, $p < 0.001$) and between content quality and trust ($\beta = 0.0490$, $p < 0.001$). This suggests that attractiveness and content quality may be more trust-relevant when the Al influencer is also perceived as credible. However, the interaction between attractiveness and content quality was negative ($\beta = -0.0865$, $p < 0.001$), suggesting that highly polished visual appeal combined with dense content may reduce perceived authenticity. This pattern is consistent with prior research showing that overly curated influencer content can weaken authenticity perceptions (Audrezet et al., 2020).

Table 7. Interaction Terms Regression

Predictor	Coef. (B)	SE	t-value	p-value	95% CI [Lower, Upper]	R^2	Adj. R^2	F-stat	Durbin-Watson	Cond. No.
Constant	0.6326	0	1391.063	0	[0.632, 0.634]	0.005	0.005	336.7	1.84	30.6
Credibility Score	0.0691	0.003	22.942	0	[0.063, 0.075]					
Attractiveness Score	0.0687	0.007	10.322	0	[0.056, 0.082]					
Content Quality Total Score	0.0416	0.005	8.009	0	[0.031, 0.052]					
Credibility × Attractiveness	0.0455	0.007	6.63	0	[0.032, 0.059]					
Credibility × Content Quality	0.049	0.008	5.817	0	[0.032, 0.066]					
Attractiveness × Content Quality	-0.0865	0.013	-6.498	0	[-0.113, -0.060]					

4 Discussion and Implications

This study found that transparency, anthropomorphism, and psychological distance were the strongest attributes associated with trust in AI influencers. This supports prior research showing that credibility, human-like cues, and social closeness are important for trust formation in digital communication (Hovland et al., 1953; Epley,2018; Trope & Liberman, 2010). Trust was also positively associated with engagement and influencer loyalty, which were further associated with brand trust, consistent with trust transfer theory (Stewart, 2003). For marketers, the findings suggest that AI influencers should be designed to appear transparent, consistent, and relatable, visually attractive. However, the low R^2 values show that consumer responses are also shaped by other factors, such as platform context, brand familiarity, campaign type, and user attitudes toward AI (Hair et al., 2019).

5 Conclusion

This study analyzed 429,193 Instagram comments to examine consumer responses to AI influencers. The findings suggest that transparency, anthropomorphism, and psychological distance are important trust-related attributes, while trust is linked to engagement, loyalty, and brand trust. These results align with prior studies. source credibility, anthropomorphism, and influencer marketing effectiveness (Hovland et al., 1953; Lou & Yuan, 2019; Epley, 2018).

However, because the study uses observational social media comments, the results should be interpreted as associations rather than causal effects. Future research should use advanced NLP methods, manual validation, multi-platform data, and longitudinal designs to better capture how trust in AI influencers develops over time.

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Appendix A. Examples of Construct Keywords Based on Prior Literature

Construct	Meaning in this study	Supporting literature	Example keywords
Credibility	Perceived reliability and believability of the AI influencer	Hovland et al. (1953); Ohanian (1990); Lou & Yuan (2019)	credible, reliable, trustworthy, believable
Expertise	Perceived knowledge or competence of the AI influencer	Hovland et al. (1953); Ohanian (1990); McGuire (1968)	expert, professional, skilled, knowledgeable

Authenticity	Perceived sincerity, genuineness, and consistency	Baker & Martinson (2002); Vannini & Franzese (2008); Molleda (2010)	authentic, real, genuine, honest
Attractiveness	Perceived appeal and positive impression of the influencer	McGuire (1968); Ohanian (1990); Sternthal & Samuel (1982)	beautiful, attractive, pretty, stylish
Psychological distance	Perceived social closeness or distance between user and influencer	Trope & Liberman (2010); Edwards et al. (2009)	close, relatable, connected, distant
Homophily	Perceived similarity between influencer and audience	De Veirman et al. (2017); Schouten et al. (2021)	similar, like me, same, shared
Anthropomorphism	Human-like qualities attributed to AI influencers	Epley et al. (2007); Epley (2018); Glikson & Woolley (2020)	human, human-like, realistic, alive
Content quality	Perceived usefulness, clarity, and value of content	Pick (2020); Wang & Qiu (2024); Lou & Yuan (2019)	useful, informative, creative, high-quality
Transparency	Disclosure and openness about AI-generated nature	Boerman et al. (2017); Evans et al. (2017); Kong et al. (2023)	transparent, disclose, AI-generated, virtual
Trust	Confidence in the AI influencer's reliability	Morgan & Hunt (1994); Mayer et al. (1995); Alboqami (2023)	trust, believe, confidence, reliable
Engagement	Active interaction or emotional involvement with content	Vivek et al. (2012); Trunfio & Rossi (2021)	like, comment, share, follow
Influencer loyalty	Continued support or attachment to the influencer	Oliver (1999); Morgan & Hunt (1994)	loyal, fan, support, always
Brand trust	Confidence in the brand associated with the influencer	Doney & Cannon (1997); Becerra & Korgaonkar	brand, product, company, quality

		(2011); Sichtmann (2007)	

From YouTube to Reddit: Topic and Emotion Dynamics in World of Warcraft*

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Abstract

We investigate cross-platform topic and sentiment dynamics between YouTube and Reddit using World of Warcraft as a case. Based on 119,871 comments (2020–2025), we apply LDA, dynamic topic modeling (DTM), VADER sentiment analysis, social network analysis (SNA), and a multilayer perceptron (MLP). Results show that YouTube sentiment and topic intensity are significantly associated with subsequent Reddit discussion activity, explaining 67% of the variance in comment volume. We further identify a consistent diffusion pattern in which emotionally expressive content on YouTube precedes more structured discussions on Reddit, with an average delay of approximately 55–60 hours. The study contributes by linking media richness theory with cross-platform behavioral modeling and provides practical implications for data-driven community management and digital marketing.

1 Introduction

According to a new report by Newzo (2024), the global online gaming market is projected to grow from $182.9 billion in 2022 to more than $213.3 billion by 2027, reflecting the rapid growth of the gaming industry and its more than 3.3 billion global user bases. As the online gaming industry continues to grow, the way players interact and communicate continues to evolve. The rise of social media and online communities has enabled players to go beyond in-game communication and actively share their gaming experiences, discuss content updates, express their emotional reactions, and engage in the collective construction of meaning in a broader online space.

Drawn from the Media Richness Theory [7] , social media platforms such as YouTube are high richness media, capable of providing multiple information cues and supporting complex emotional

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expressions; in contrast, communities such as Reddit, although also interactive, are more often used as vehicles for structured knowledge exchange and are closer to the characteristics of low richness media. Meanwhile, Media Synchronization Theory [10] points out that there are also differences in synchronization across platforms: high synchronization platforms are more suitable for information storage and asynchronous communication.

Despite the growing importance of platform-based communication in contemporary gaming culture, most existing research on gaming communities focuses on single-platform or static topic analyses, with less coverage of the dynamic evolution of cross-platform topics, and, in particular, there is still a gap in research on predicting the future activity of professional gaming communities. This gap is particularly noteworthy because the modern social media ecosystem is highly intertwined, with players switching frequently between platforms: they may first watch tutorial videos on YouTube, browse the comment section, express their opinions, and then turn to professional communities like Reddit for help or in-depth discussion.

Taking World of Warcraft (WoW) as a case study, this research aims to investigate how topics and emotions evolve between YouTube and Reddit over a 5 year period of time. This study addresses the following questions:

RQ1: How do player-discussed topics evolve dynamically across YouTube and Reddit?

RQ2: What are the dominant diffusion patterns and time delays between these platforms?

RQ3: Can YouTube sentiment and topic intensity effectively predict future activity levels in the Reddit community?

This study makes two specific contributions. First, it demonstrates that sentiment and topic intensity signals from a high-reach, non-specialized platform (YouTube) explain a substantial portion of variation in discussion activity within a specialized community (Reddit), accounting for approximately 67% of the variance in comment volume. Second, it identifies a consistent cross-platform diffusion pattern in which emotionally expressive content precedes structured, knowledge-oriented discussions, thereby linking media richness theory with observable behavioral dynamics in online gaming ecosystems

2 Literature review

2.1 Specialized vs Non-specialized Communities

Online gaming discussions occur across heterogeneous platforms that differ in community structure and user motivation. Prior research distinguishes between specialized communities and non-specialized communities, which are more spontaneous, open, and emotionally expressive [23]. Reddit exemplifies a specialized community: it organizes information through subforums, tagging, and chronological threads that facilitate structured, knowledge-intensive exchanges. Users often display functional motivations, such as problem-solving, game optimization, and strategic analysis [1]. By contrast, YouTube functions as a non-specialized community emphasizing affective engagement. Its comment sections and video responses allow users to express immediate emotional reactions and social identification without requiring deep expertise. In the World of Warcraft ecosystem, these contrasting features shape both the depth and pace of information diffusion. Reddit users are primarily driven by functional motivations such as problem-solving and game optimization, whereas YouTube users are more driven by social and emotional motivations, expressing reactions and identity through comments and interactions. As a result, Reddit facilitates structured knowledge accumulation and in-depth discussion, while YouTube enables rapid, large-scale emotional expression but provides limited support for systematic knowledge exchange.

2.2 Media Richness Theory and Media Synchronicity Theory

Media Richness Theory (MRT) [7] posits that communication media vary in their capacity to convey complex, nuanced information through feedback immediacy, multiple cues, language variety, and personalization [7]. High-richness media are better suited for ambiguous or emotion-laden exchanges, while low-richness media excel in precise, structured information sharing. Applied to gaming contexts, YouTube's audiovisual format and emotive tone offer high richness but limited interactivity, whereas Reddit's text-based discussions enable detailed argumentation and information storage, aligning with low-richness but cognitively richer communication. YouTube exhibits strong media richness through multimodal content, consistent with [21], its affordances are insufficient for constructing or responding to complex, knowledge-based discussions. From another perspective, [25] argue that media richness should be understood as an intrinsic and relatively stable characteristic determined by the medium's objective properties.

While MRT effectively distinguishes between communication capacities across media, it lacks a mechanism to explain how media adapt to communication processes. Media Synchronicity Theory further distinguishes between two core communication processes: conveyance (the transmission of information) and convergence (the development of shared understanding). In this context, YouTube primarily supports conveyance through rapid dissemination of audiovisual content, while Reddit facilitates convergence through threaded discussions, allowing users to refine, debate, and integrate knowledge. These differences help explain why emotionally driven content often emerges on YouTube before being elaborated into structured discussions on Reddit.

3 Hypotheses development

As gaming communities become more interconnected, users increasingly interact and produce content across multiple platforms, making platform type and communication mechanism critical variables in understanding behavioral dynamics. Within this context, Media Dependency Theory posits that an individual's reliance on a particular medium is determined by the extent to which it fulfils their needs for information, reality interpretation, and behavioral guidance [2]. In the context of gaming, these dependencies manifest differently across platforms: Reddit, as a text-centric and specialized community, facilitates rational analysis and technical discourse surrounding game updates and system adjustments, while YouTube, as a visually driven non-specialized platform, emphasizes entertainment and emotional expressions. YouTube users often communicate affective reactions and subjective viewpoints through videos, while Reddit users focus on analytical and knowledge-based exchanges. This functional differentiation profoundly shapes how WoW topics are framed and discussed across communities.

Furthermore, [11] issue attention cycle theory suggests that public attention toward a specific topic follows the sequence: "initial trigger – rapid escalation – short-lived climax – gradual decline". This cycle is especially evident in gaming discourse. Whenever WoW launches a major update, there is an immediate surge of attention and collective feedback, followed by a gradual decrease in engagement and dispersion of discussions. The intensity and duration of this cycle vary across platforms, supporting this, [24] found notable cross-platform differences in player discussions of "Among Us", highlighting the role of platform attributes in topic formation and evolution. Similarly, Mitrovic et al. (2025) demonstrated through comparative analysis of Reddit that users' content creation and interaction behaviors display clear structural variations across platforms, offering methodological support for cross-platform analysis. Hence, this study collects user comment data on the WoW "game update" topic

to examine the temporal evolution and platform-based dynamics of discussions. Accordingly, we propose the following hypothesis:

H1: Game topics discussed by players evolve dynamically over time, and the pattern of change differs by platform type.

H1 is tested using Dynamic Topic Modeling (DTM) by dividing the dataset into three time phases (Emergence, Growth, Saturation). Topic contribution scores are compared across phases and platforms. Evidence supporting H1 is indicated by (1) changes in dominant topics over time and (2) differences in topic prominence between YouTube and Reddit.

Game-related topics often exhibit clear cross-platform communication patterns, where information dissemination occurs collaboratively through interlinked individual, group, and platform layers [17]. User behavior is not only shaped by platform-specific rules and community cultures but also contributes to broader cross-platform interaction paths within this networked environment. [20] identified that active users across multiple platforms often serve as bridge nodes, facilitating the flow of information between spaces such as Reddit and Twitter. In gaming contexts, players commonly watch and comment on videos via YouTube, then migrate discussions to Reddit for more in-depth technical elaboration or critical feedback. The mechanism called migratory dissemination captures how content circulates from non-specialized to specialized communities. Thus, the second hypothesis is formulated:

H2: Popular game topics tend to spread from non-specialized social media (YouTube) to specialized gaming communities (Reddit), and the spreading paths show specific network structure characteristics.

H2 is tested by constructing a topic-level communication network linking semantically similar topics across platforms using cosine similarity and temporal proximity. Cross-platform diffusion is operationalized as the time difference between the first appearance of a topic on YouTube and its subsequent appearance on Reddit. Support for H2 is indicated by consistent directional delays and centrality patterns in the topic network.

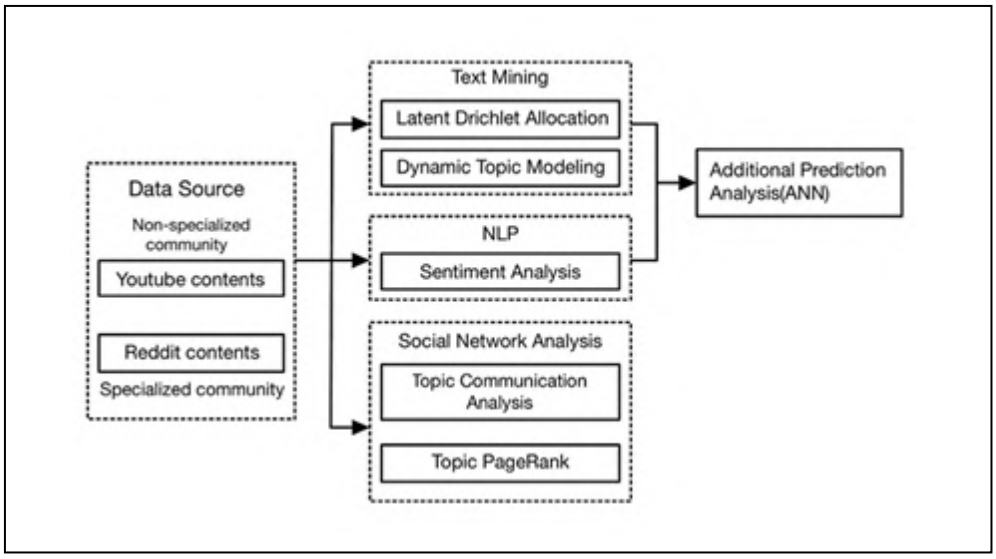
Within multi-platform ecosystems, users' motivations and emotional expressions become essential indicators for understanding topic diffusion and platform-level role differentiation. User motivation comprises cognitive, emotional, and behavioral dimensions [15] on platforms like YouTube, the forms of expression foster both cognitive attitudes and shared emotional climates through affective commenting, which in turn precedes the rational analysis typically observed on Reddit. Recent studies reinforce the predictive potential of sentiment in cross-platform contexts. [28] highlight that sentiment analysis across social media can effectively forecast discussion activity and interaction trajectories. In gaming environments characterized by high emotional density, user sentiment is concentrated and volatile, serving as a leading indicator of opinion dynamics. Lin et al. (2024) distinguish between emotional tendency, which predicts the breadth of discussion, and emotional volatility, which signals potential surges in topic activity. On a non-specialized platform such as YouTube, the high volume and emotional expressiveness of comments provide early cues for predicting subsequent topic growth on specialized forums. Thus, we propose the following hypothesis:

H3: Sentiment trends and topic heats on a non-specialized platform (YouTube) can significantly predict future discussion activity in a specialized community (Reddit).

H3 is tested using a supervised learning model in which aggregated YouTube features (VADER sentiment scores, TF-IDF–based topic intensity, and interaction metrics such as likes and replies) are used to explain Reddit comment volume at the same temporal aggregation level (daily). Model performance is evaluated using Mean Squared Error (MSE) and R^2. As the model uses contemporaneous features, the analysis focuses on explanatory relationships rather than strict temporal forecasting.

4 Research Methodology

4.1 Research Framework



Error! Reference source not found.. Research framework

This study investigates the phenomenon of cross-platform thematic diffusion in the World of Warcraft community, focusing on interactions between YouTube and Reddit. The analysis integrates text mining, natural language processing (NLP), and social network analysis (SNA) to examine how sentiments and thematic patterns emerging on YouTube influence subsequent. The research framework (figure 1) is designed to capture the multi-stage analytical process.

4.2 Data Collection and Preprocessing

Data were collected from YouTube and Reddit using their official APIs. For YouTube, comments were retrieved via the YouTube Data API by querying keywords related to World of Warcraft, including "World of Warcraft", "WoW update", and expansion-specific terms such as "Shadowlands" and "Dragonflight". We selected videos with high engagement (e.g., high view counts and active comment sections) to ensure sufficient user interaction data.

For Reddit, data were collected using the Pushshift API from relevant subreddits, including r/wow and r/classicwow. These communities were selected because they represent major hubs for World of Warcraft discussions and technical exchange.

The dataset spans from January 2020 to May 2025. After initial retrieval, duplicate entries and non-relevant comments were removed. The final dataset consists of 119,871 comments, including 79,844 from YouTube and 40,027 from Reddit. To align the analysis with major game lifecycle events, data collection covers multiple expansion and patch periods of World of Warcraft. These periods provide natural variation in user attention and discussion intensity, which supports the analysis of topic evolution and cross-platform dynamics.

Text preprocessing included removing URLs, emojis, and non-textual elements, followed by tokenization and stopword removal. Only English-language comments were retained for analysis.

5 Results

5.1 Topic modeling (LDA)

We adopt Latent Dirichlet Allocation (LDA) a Bayesian generative model for unsupervised topic discovery in documents [5]. In LDA, each document di is a mixture of K topics and each topic is a distribution over the vocabulary ,LDA topic analysis was also performed on the comment data in the Reddit platform, and num-topics were also set to 5 to identify and extract potential topic patterns in the text. For each theme, the top 8 words with the highest probability were extracted as the core representation words of the theme, and the theme modeling results of World of Warcraft comments on Reddit. To summarize, Reddit users show higher interactivity and technicality in World of Warcraft topics, focusing more on in-depth exchanges of game experience and diversified discussions of game mechanics. These results indicate clear differences in topic focus across platforms. Reddit discussions are more technical and interaction-driven, emphasizing gameplay strategies and problem-solving, while YouTube discussions are more descriptive and experience-oriented.

5.2 Dynamic Topic modeling (DTM)

This study further explores the potential changes in textual content over the time series by performing Dynamic Theme Analysis (DTM) on the comment data related to World of Warcraft in the YouTube platform. Specifically, the entire dataset is first divided into three phases in chronological order, i.e., Emergence, Growth, and Saturation, in order to observe the dynamic evolution of user discussions in the time dimension. In each period, the number of topics (num-topics) in the LDA model is set to 5 to ensure the consistency and comparability of topic division. For each topic, the top 8 high-frequency words are extracted to show the hotspots of users' attention at different stages. We also calculate and average the contribution scores of the topic terms for each period in YouTube platform to better compare the importance and influence of the topics at different stages. The average contribution score distribution of topics in different time periods can be visualized through the line graph in Figure2 and figure 3

Overall, Topic1 “Plot threads” and Topic5 “Character tips” continue to decline, suggesting that the declining topics may be replaced by new topics. In addition, Topic 3 “Version Updates” reaches the highest level during the saturation period, indicating that users are gradually focusing on version updates and community interactions. Overall, the interests of users in World of Warcraft-related. These findings support H1 by showing that topics evolve over time and differ across platforms. YouTube discussions shift from narrative and promotional content toward update-related topics, while Reddit maintains a stronger focus on technical and gameplay-oriented discussions..

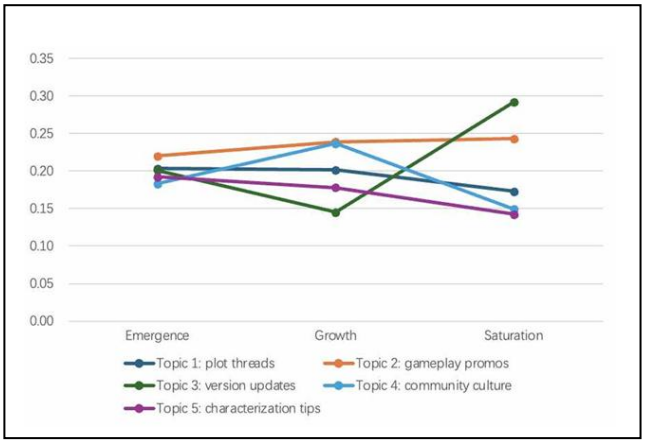


Figure 2. YouTube DTM analysis results

discussions on the YouTube platform have evolved from game promotion and plot lines to version updates and camp culture, showing a diversified discussion pattern over time.

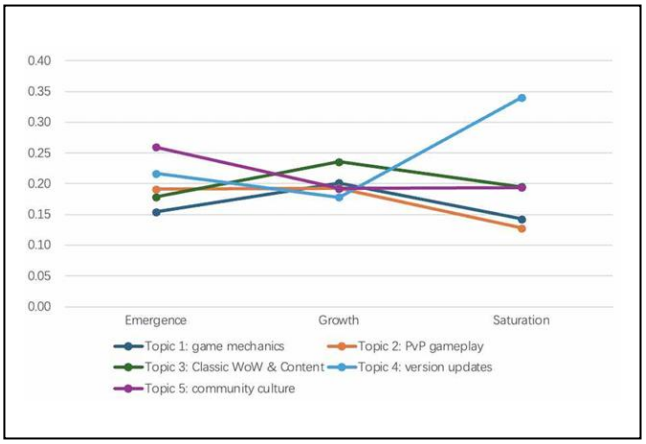


Figure 3. Reddit DTM Analysis Results

Through dynamic topic modeling and line graph analysis of World of Warcraft-related comment data on Reddit platform, the results show the changes of contribution scores of five topics in three time periods.

5.3 Sentiment Analysis

The sentiment distributions differ across platforms. YouTube comments are more concentrated around neutral sentiment values, indicating less polarized emotional expression. In contrast, Reddit exhibits a higher proportion of both strongly positive and strongly negative comments, suggesting more polarized and opinionated discussions. This difference reflects platform characteristics, where YouTube supports broad and immediate emotional expression, while Reddit facilitates more evaluative and discussion-oriented interactions.

5.4 Social Network Analysis

In the study of text and topic communication, SNA is not only applicable to the analysis of user-level interactions but also can be used to model the communication dynamics between "topic nodes". Specifically, this study focuses on the construction and analysis of Topic Communication Network, constructs topic diffusion paths through semantic similarity and timestamps, and further quantifies the cross-platform influence of each topic using PageRank algorithm. In SNA that fuses YouTube and Reddit comment data, text data from both platforms are first integrated, and topic numbers are assigned to each comment through an LDA topic model. Specifically, the comment data are first binned, and all text data are combined into a large bag-of-words model (Bag-of-Words). Based on this, 10 topics are extracted by training with the LDA model from the gensim library. For each topic, the code extracts the top 10 keywords with the highest weights and their corresponding weights to form a structured Data Frame for subsequent visualization and analysis. Next, the PageRank graph of each node is plotted, as shown in Figure 4.

The PageRank results show that Topic 10 "version update" and Topic 1 "game mechanism" have the highest scores, indicating that they have the most influence in the whole topic dissemination network.

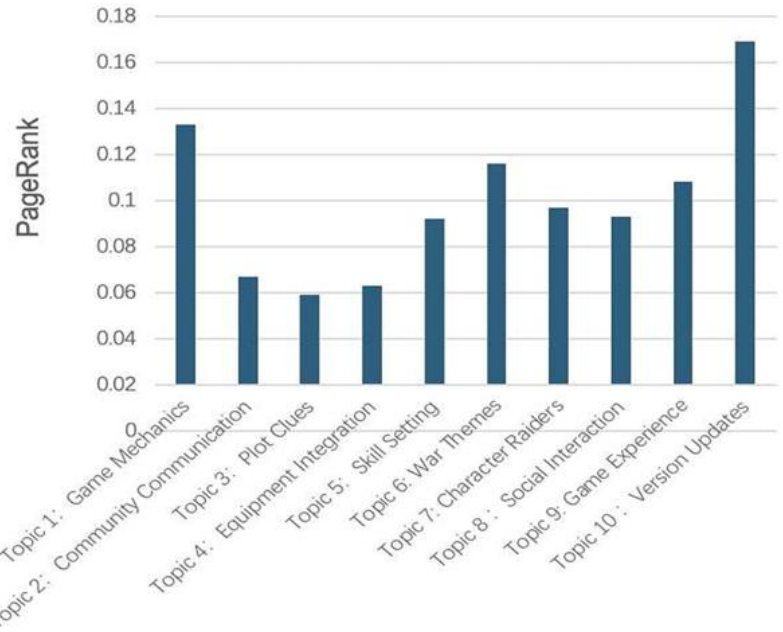


Figure 4. PageRank of Topics

The average propagation delay between YouTube and Reddit is approximately 57 hours. However, variation across topics suggests that diffusion speed is not perfectly uniform. This indicates that different types of content may spread at different rates across platforms. These results support H2 by demonstrating a directional pattern in which topics emerge on YouTube and subsequently appear on Reddit. The PageRank analysis further shows that "version update" and "game mechanism" are the most influential topics in the cross-platform network. It should be noted that topic matching across platforms is based on semantic similarity rather than direct user-level linkage, and therefore reflects diffusion patterns rather than strict causal transmission. It is shown in Figure 5.

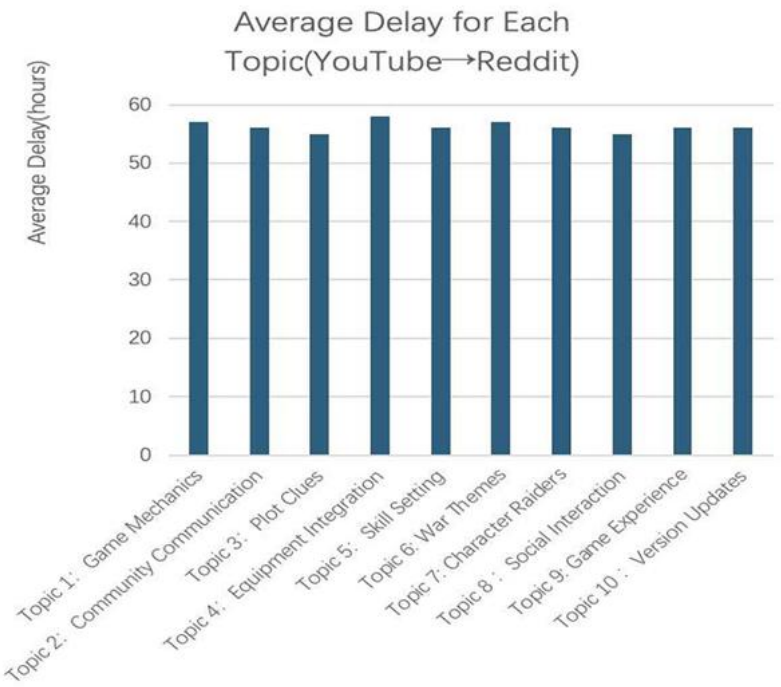


Figure 5. Average Delay for each topic

The MLP model achieves an MSE of 0.25 and an R^2 of 0.67, indicating that YouTube features explain a substantial proportion of variation in Reddit comment volume. The model captures overall trends effectively, particularly during periods of increased discussion activity, although deviations occur during low-activity periods.

As baseline, we compare the MLP model with a univariate autoregressive model using only lagged Reddit comment counts. The MLP model demonstrates improved explanatory power, indicating that incorporating YouTube sentiment and topic intensity provides additional insight beyond internal Reddit dynamics.

These results support H3 by showing that sentiment and topic intensity on YouTube are significantly associated with Reddit discussion activity. Because the model uses contemporaneous features, the findings should be interpreted as explanatory rather than strictly predictive.

Overall, the findings consistently support H1–H3. Topic modeling demonstrates dynamic and platform-specific evolution patterns (H1), network analysis reveals directional cross-platform diffusion (H2), and the ANN model shows a strong association between YouTube signals and Reddit activity (H3).

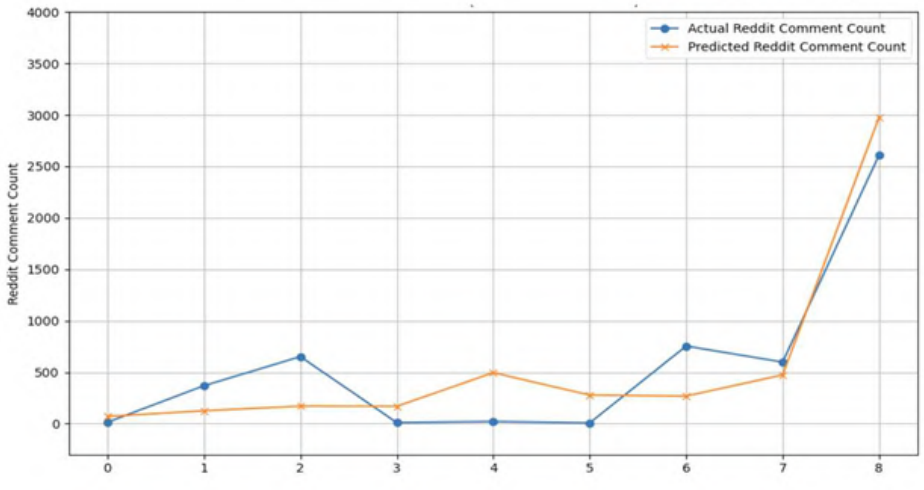


Figure 6. ANN prediction

6 Conclusion

This study examines cross-platform topic and sentiment dynamics between YouTube and Reddit using World of Warcraft as a case. The findings show that discussion patterns differ systematically across platforms and evolve over time. YouTube is characterized by emotionally expressive and high-reach content, while Reddit supports more structured and technical discussions. Topic modeling results demonstrate that user attention shifts from narrative and promotional content toward update-related discussions, while Reddit consistently emphasizes gameplay and problem-solving topics.

The analysis further reveals a directional cross-platform diffusion pattern in which topics tend to emerge on YouTube and subsequently appear on Reddit, with an average delay of approximately 55–60 hours. This suggests that non-specialized platforms play an important role in early-stage attention formation, while specialized communities facilitate deeper interpretation and knowledge exchange.

In addition, the modeling results indicate that sentiment and topic intensity on YouTube are strongly associated with Reddit discussion activity, explaining a substantial proportion of variation in comment volume. These findings highlight the value of cross-platform signals as indicators of community dynamics, although the relationships identified in this study should be interpreted as explanatory rather than strictly predictive.

From an e-business perspective, the results suggest that platform operators, game developers, and digital marketers can leverage early signals from high-reach platforms such as YouTube to monitor emerging topics, anticipate shifts in user attention, and design more timely communication strategies. For example, sentiment trends may help identify potential dissatisfaction following updates, while topic intensity can signal emerging areas of interest for content development and community engagement.

Overall, this study contributes to the understanding of cross-platform communication in digital ecosystems by linking platform characteristics with observable patterns of topic evolution, sentiment expression, and information diffusion. Future research can extend this work by incorporating user-level interaction data, refining diffusion measurement approaches, and applying the framework to other platform ecosystems beyond gaming.

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Designing AI-Augmented Marketing Education: The RFM Educational Design Model as a Human-Centered E-Business Information System*

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Abstract.

As artificial intelligence (AI) continues to transform e-business practices, educational institutions face mounting pressure to equip students with AI competencies for the digital economy. This paper reports an exploratory, first-cycle design science study of the RFM (Reasonable Challenge, Fun, Motivation) education design model, a theoretically grounded framework for designing AI-enhanced educational e-business information systems. Adopting a design science approach, we integrate generative AI tools (ChatGPT, DeepAI, PixVerse), a chatbot platform (Rochat), and a collaborative system (Padlet) into a postgraduate marketing course and monitor the resulting AI-augmented learning outcomes. The findings highlight critical design considerations, including tool accessibility, scaffolded prompt engineering, peer learning dynamics, and the integration of ethical AI discourse. This study contributes a novel motivational design framework for AI-rich business education, together with practical insights for institutions seeking to evolve their educational information systems to meet AI-era skills demands.

1. Introduction

The rapid evolution of artificial intelligence is fundamentally reshaping the e-business landscape, creating an urgent demand for professionals equipped with AI competencies [1][3][25]. Educational institutions must evolve to prepare students for AI-driven marketing environments in which generative models create content, chatbots manage customer relationships, and data-driven insights guide strategic decisions [7][11][17][27][32].

Yet significant gaps persist between AI's transformative potential and its integration into business curricula. Many programs struggle to move beyond conceptual AI discussions to hands-on competency development. Students graduating with traditional business education often lack practical experience with generative AI tools, leaving them unprepared for workplaces increasingly reliant on AI-augmented workflows. [2][4][5][34]

This paper addresses this gap through a design science approach [14], presenting the development and evaluation of an AI-enhanced educational e-business information system guided by the RFM (Reasonable Challenge, Fun, Motivation) educational design model. The RFM education design framework—rooted in established motivational theories including Self-Determination Theory[19] [29][30], Expectancy Theory

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[24], and Behavioral Learning Theory[12][18]—provides structured design principles for creating engaging, effective AI-rich learning environments.

Distinct from the traditional RFM (Recency–Frequency–Monetary) model widely used in marketing analytics, the RFM Educational Design Model is an original pedagogical framework developed by the course instructor to address the motivational and design challenges of AI-augmented education. While the marketing RFM model is an analytical tool for measuring customer value, the RFM Educational Design Model is a human-centred design framework that specifies how educational information systems should structure tasks, interactions, and feedback to keep learners appropriately challenged, engaged, and motivated.

The paper makes three primary contributions. First, it introduces the RFM Educational Design Model as a theoretically grounded, practically validated framework for designing AI-augmented educational EBIS. Second, it demonstrates how generative AI tools, chatbot platforms, and collaborative systems can be integrated into postgraduate marketing education through a design science approach. Third, it provides evidence-based insights into how students experience challenges, enjoyment, and motivation when learning with AI, contributing to the growing body of knowledge on human-centred EBIS design and AI competency development in business education.

2. Theoretical Foundations and the RFM Educational Design Model

2.1 Theoretical Foundations

The RFM Educational Design Model did not originate in theory alone. It emerged from years of direct pedagogical observation across diverse classroom settings—undergraduate, postgraduate, and distance learning environments—where three recurring patterns became unmistakably clear: when students perceive tasks as excessively challenging, they disengage and give up; when learning feels tedious or boring, participation becomes superficial or performative; and when motivation is absent, students do only the minimum required to pass. These three honest observations—about challenge, enjoyment, and motivation—formed the practical seed from which the RFM Educational Design Model grew.

The design of effective AI-augmented educational environments, however, requires more than practitioner insight; it demands a principled theoretical understanding of why these patterns occur and how to address them systematically through intentional system design. The RFM Educational Design Model is therefore grounded in five established theoretical perspectives drawn from educational psychology and motivational research, each contributing a distinct and complementary lens to the model's design principles. Rather than adopting a single theoretical framework, the RFM model deliberately integrates multiple perspectives to capture the multidimensional nature of student motivation in AI-rich learning contexts—where novelty, uncertainty, and rapid skill acquisition simultaneously create opportunities for deep engagement and risks of disengagement. Together, these five perspectives provide a robust and coherent conceptual basis for the design of AI-augmented educational EBIS that are not only technically functional but also motivationally effective and human-centred. Effective educational information systems must address not only the cognitive but also the motivational dimensions of learning [26]. Five theoretical perspectives inform the RFM model's design principles:

Self-Determination Theory (SDT) [29][30] emphasizes three fundamental psychological needs: autonomy (sense of volition), competence (sense of mastery), and relatedness (sense of connection). Educational systems supporting these needs enhance intrinsic motivation and well-being. SDT provides a foundational rationale for all three RFM components.

Expectancy Theory [24] suggests that motivation depends on the perceived likelihood of success and the value placed on achievement. Students engage more deeply when tasks are challenging yet attainable—a principle central to "Reasonable Challenge." Recent applications in generative AI contexts [10] show that expectancy theory helps explain professionals' adoption of AI tools when they perceive that effort will yield meaningful outcomes.

Attribution Theory[15][16] 21] [33] highlights how beliefs about responsibility for outcomes affect motivation. When students attribute success to effort and skill rather than external factors, they demonstrate greater persistence—supporting both "Reasonable Challenge" and "Motivation" design.

Maslow's Hierarchy of Needs[6][9][23] underscores progression from basic needs to self-esteem and self-actualization. Educational systems addressing esteem needs (confidence, achievement) and self-actualization (creativity, growth) align with RFM's "Motivation" component.

Behavioral Learning Theory and gamification research[12][18][19] [31] demonstrate that activities balancing challenge with skill, incorporating game elements, and providing immediate feedback sustain engagement—principles operationalized in RFM's "Fun" component.

2.2 The RFM Educational Design Model

Building on these theoretical foundations, the RFM educational design model comprises three integrated design dimensions:

I. **Reasonable Challenge**: Designing tasks that balance difficulty and attainability, ensuring students remain engaged without feeling overwhelmed. This operationalizes expectancy theory by maintaining perceived likelihood of success while requiring genuine effort. Tasks should be scaffolded—starting with guided demonstrations, progressing to supported practice, culminating in independent application.

II. **Fun**: Creating learning experiences that are **interesting** (novel, relevant), **interactive** (hands-on, participatory), and **innovative** (cutting-edge, creative)—the "three I's" sustaining curiosity and participation. This dimension operationalizes behavioral learning theory and gamification principles, incorporating elements of play, exploration, and aesthetic engagement.

III. **Motivation**: Inspiring students through three key drivers—the "Three P's":

- *Perceived Usefulness*: Helping students recognize real-world relevance and practical value of learned skills, addressing "why does this matter for my career?"
- *Peer Showcase*: Fostering collaboration and healthy competition by encouraging students to share and celebrate achievements, supporting SDT's relatedness need.
- *Perceived Competence*: Building confidence in abilities and reinforcing a sense of mastery, directly supporting SDT's competence need and addressing self-esteem in Maslow's hierarchy.

The RFM model distinguishes itself from generic motivational frameworks through its explicit integration of these five theoretical perspectives and its operationalization specifically for AI-rich business education contexts. While Self-Determination Theory identifies overarching needs (autonomy, competence, relatedness), RFM translates them into concrete educational system design principles applicable to AI tool integration, assignment structuring, and peer learning orchestration. Figure 1 depicts the conceptual illustration of the RFM Educational Design Model.

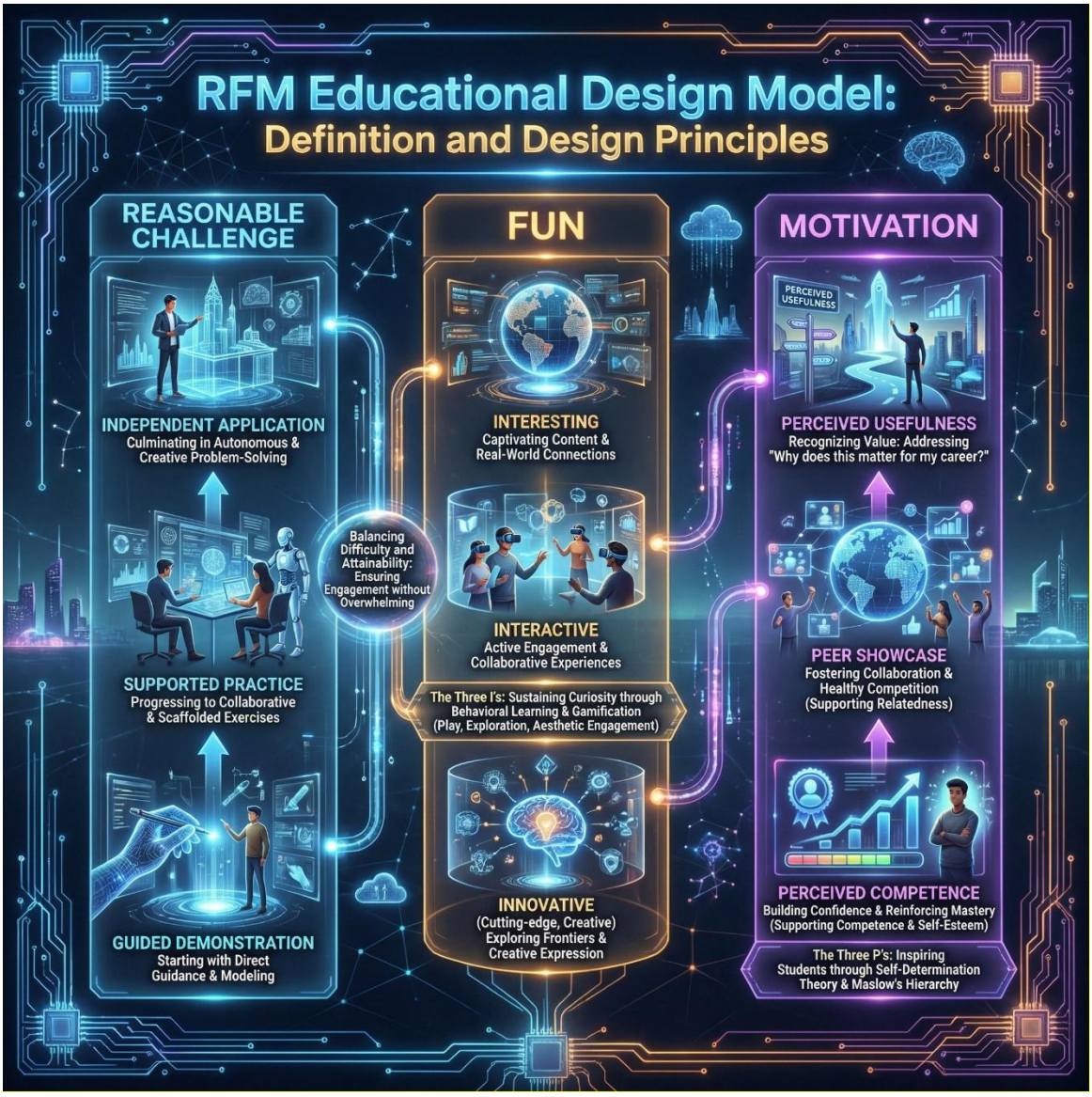

Figure 1. Conceptual Illustration of the RFM Educational Design Model

In short, RFM answers a question that these influential frameworks do not explicitly address: *how do we design AI-augmented educational information systems that are not only technically functional but also motivationally effective for business students with diverse backgrounds, varying technical confidence, and rapidly evolving professional requirements?* It is this domain-specific, practically grounded extension of established motivational theory that constitutes the distinctive contribution of the RFM Educational Design Model. Table 1 summarises the comparative positioning of the RFM Educational Design Model against four established motivational frameworks across key theoretical and design dimensions.

	ARCS Model [8]	**Flow Theory [28]**	**UDL [20]**	**Malone & Lepper [22]**	**RFM Model (This Study)**
Primary Focus	Instructional motivation design	Optimal psychological experience	Inclusive learning design	Intrinsic motivation taxonomy	Competency development in education
Theoretical Type	Prescriptive design model	Descriptive psychological theory	Architectural design philosophy	Descriptive motivational taxonomy	Prescriptive domain-specific design model
Core Components	Attention, Relevance, Confidence, Satisfaction	Clear goals, Immediate feedback, Challenge-skill balance	Engagement (Why), Representation (What), Action & Expression (How)	Individual: Challenge, Curiosity, Control, Fantasy; Interpersonal: Cooperation, Competition, Recognition	Reasonable Challenge, Fun, Motivation
Feedback Mechanism	Implicit confidence building through satisfaction	Immediate feedback as flow condition (descriptive)	Multiple means of action and expression	Performance feedback under Challenge; Recognition under interpersonal	Structured iterative feedback cycles explicit and embedded
Mastery / Competency Link	Not explicitly specified	Implicit challenge-skill balance	Indirect through multiple expression pathways	Not specified motivational categories only	Explicit mastery checkpoints at each scaffolding stage
Interpersonal Dimension	Not addressed	Not addressed	Cooperative expression pathways (indirect)	Explicitly named: Cooperation, Competition, Recognition	Endogenous peer review; collaborative showcase
Endogenous vs. Exogenous	Not distinguished	Intrinsic by nature but not operationalised	Not distinguished	Explicitly distinguished endogenous preferred over exogenous	Endogenous motivational mechanics embedded in task structure
AI Integration	Limited; recent extensions emerging	Emerging AI as flow-enabling scaffold [28]	Strong conceptual alignment domain-general [20]	Not addressed in original taxonomy	Embedded and central GenAI tools are core design elements
Learner Role	Passive recipient of motivationally designed instruction	Self-directed experiencer of optimal states	Active participant across multiple expression pathways	Intrinsically motivated individual and social actor	Active co-creator of AI-generated marketing artefacts
Instructor Role	Motivational designer	Flow condition facilitator	Inclusive architect	Not specified	Scaffolding curator and mastery validator

Operational	Moderate design checklist for instructors	Low abstract psychological state; hard to operationalise directly	Moderate broad principles requiring contextual translation	Low descriptive categories without implementation guidance	High concrete three-stage scaffolding with specific tool and task recommendations

Table 1: Comparative Analysis of RFM and Four Related Motivational Frameworks

3. Course Design: The AI-Augmented EBIS Artifact

This research follows the design science paradigm in information systems[10][14], which seeks to create and evaluate artifacts that solve identified organizational problems. Design science emphasizes both artifact creation (design) and empirical evaluation (science), producing prescriptive knowledge about what designs work and why.

Following design science research principles, the postgraduate marketing course was conceptualised as an AI-augmented educational e-business information system (EBIS)—a socio-technical artifact comprising tasks, AI tools, platforms, interaction flows, and feedback mechanisms, all guided by the RFM Educational Design Model. The course is a postgraduate contemporary marketing course designed to build students' practical AI competencies for e-business marketing environments.

The AI tools and platforms integrated into the course were selected to mirror the generative AI and conversational AI technologies increasingly deployed in real-world e-business practice, ensuring that students' learning experiences were directly relevant to professional marketing contexts. Table 2 summarises the mapping between RFM principles, EBIS design decisions, and the tools deployed.

RFM Principle	Design Decision	Tool/Platform
Reasonable Challenge	Levelled AI tasks; scaffolded instruction	ChatGPT, DeepAI, PixVerse
Fun (Interesting)	Novel AI content creation tasks	ChatGPT, DeepAI, PixVerse
Fun (Interactive)	Gamified chatbot interaction; live demo	Rochat, Dr. Tian chatbot
Fun (Innovative)	Generative visual marketing content	ChatGPT, DeepAI, PixVerse
Motivation (Peer Showcase)	Public gallery-style submission	Padlet
Motivation (Perceived Usefulness)	Authentic real-world marketing tasks	All tools
Motivation (Perceived Competence)	Peer feedback; scaffolded success	Padlet, OLE polls

Table 2. RFM Educational Design Principles, EBIS Design Decisions, and Integrated Tools

3.1 Designing for Reasonable Challenge

The Reasonable Challenge principle guided the design of AI-based assignments that were deliberately calibrated to be attainable yet intellectually stretching. Drawing on Expectancy Theory and Attribution Theory, tasks were structured so that students could achieve meaningful outcomes through effort and experimentation, without requiring prior technical expertise in AI. The principle was operationalised through two complementary assignment types: an individual academic engagement assignment and a series of progressive hands-on AI assignments.

3.1.1 Individual Assignment: Academic Engagement with Generative AI Research

To develop students' critical analytical competencies alongside their practical AI skills, students were assigned an individual academic engagement task in which they analysed a recently published research paper on generative AI in marketing. Students selected two topics from the paper's future research agenda and developed evidence-based arguments drawing on academic databases, secondary data, and real-world marketing examples.

This assignment was carefully designed to balance challenge with support. The challenge dimension required postgraduate-level critical analysis, academic database navigation, and evidence synthesis—skills that stretched students beyond passive consumption of AI concepts. The support dimension provided clear scaffolding: a specific paper was provided, topic choices were clearly delineated, and library database access was available. This balance ensured that students perceived the task as demanding but achievable, consistent with Expectancy Theory's prescription that motivation is maximised when success is attainable through effort.

3.1.2 Hands-On AI Assignments: Progressive Scaffolding

The hands-on AI assignments followed a deliberate three-stage scaffolding structure that progressively built students' competence before requiring independent application:

Demonstration stage: The instructor demonstrated each tool's basic functions in class using prepared examples—for instance, creating perfume advertisement imagery using DeepAI and PixVerse, and building the Dr. Tian AI chatbot using Rochat. These demonstrations provided a concrete reference point and reduced initial anxiety by making the tools feel approachable and achievable.

Guided experimentation stage: Students engaged in low-stakes, in-class activities before any graded work was required. For example, students interacted with the Dr. Tian chatbot by asking course-related questions, discussing prompt effectiveness, and observing how different inputs produced different outputs. This stage built foundational familiarity and confidence in a risk-free environment.

Independent application stage: Having developed foundational competence through the first two stages, students completed graded assignments requiring them to create original marketing content—including AI-generated product designs and virtual brand spokespeople—using the tools they had explored in class.

This progressive scaffolding structure ensured that challenge levels were continuously aligned with students' evolving competence, consistent with SDT's emphasis on competence-supportive environments and with flow theory's prescription for maintaining engagement through optimal challenge-skill balance. Rather than expecting immediate mastery of unfamiliar AI tools, the three-stage structure allowed students to build confidence incrementally, transforming initial apprehension into a sense of growing capability that carried forward into subsequent assignments.

3.2 Designing for Fun: Interactive, Innovative, and Interesting Learning Experiences

The second principle of the RFM Educational Design Model operationalises Fun through three dimensions: Interesting (novel and relevant), Interactive (hands-on and participatory), and Innovative (cutting-edge and

creative). Within the AI-augmented EBIS, Fun was instantiated through three concrete design decisions: gamified chatbot interaction, hands-on generative content creation, and a visual peer showcase platform.

3.2.1 Gamified Chatbot Interaction

Recognising that building an AI chatbot might initially seem technically daunting to postgraduate students, the course instructor first introduced chatbot technology through a low-stakes, playful in-class activity. The instructor developed Dr. Tian, a course-specific AI chatbot built on the Rochat platform, designed with an empathetic and creative personality to respond to students' questions about assignments, marketing concepts, and course logistics. During class, students were invited to interact with Dr. Tian, discovering its patient and contextually appropriate responses in real time.

Figure 2 shows the screenshot of the Dr. Tian AI Chatbot interface built on Rochat.

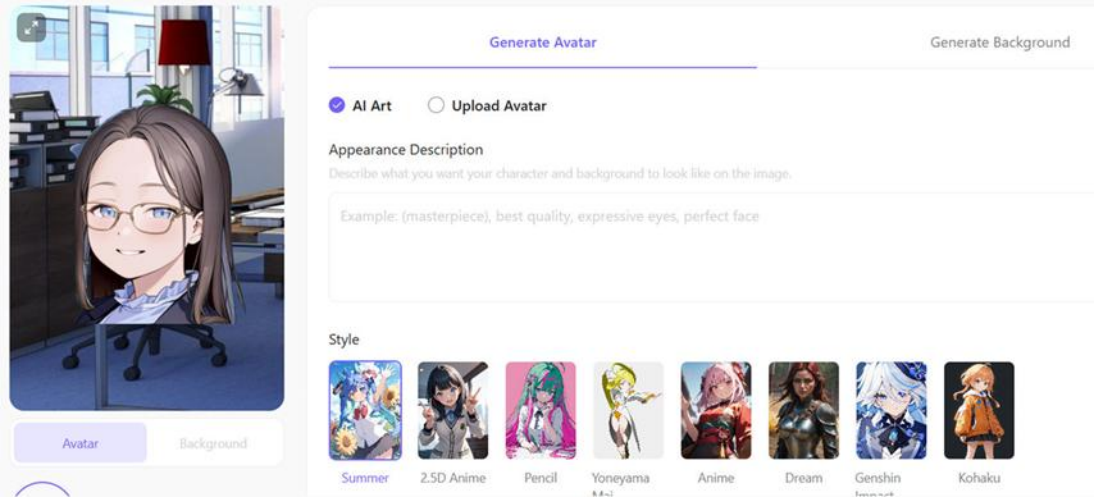

Figure 2. Screenshot of the Dr. Tian AI Chatbot Interface Built on Rochat

Rochat can generate AI art to create avatars based on an uploaded photo or a description of the desired appearance. Additionally, it offers an intriguing feature that allows users to customize the chatbot's personality and select the Large Language Models (LLMs) to be used, as outlined in Figure 3.

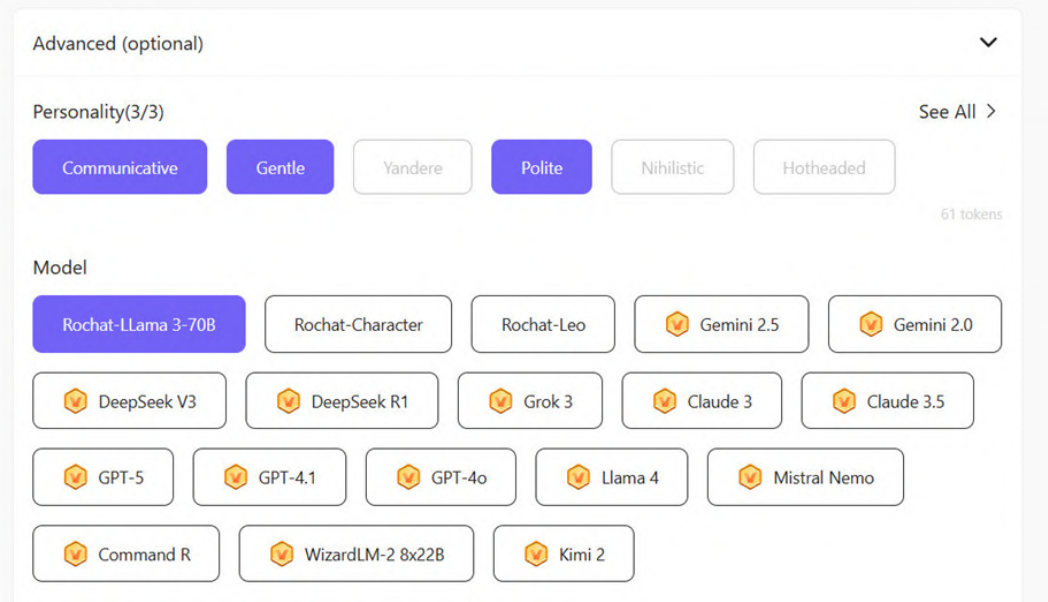

Figure 3. Rochat Avatar Generation and Chatbot Personality Customization Options

This gamified entry point served multiple EBIS design functions. First, it reduced anxiety about the subsequent chatbot-building assignment by providing an enjoyable first-hand demonstration that AI chatbot development is approachable. Second, it illustrated the practical marketing application of conversational AI—namely, how chatbots can manage customer relationships, deliver personalized responses, and sustain brand engagement—thereby directly connecting course content to contemporary e-business practice. Third, it sparked curiosity and intrinsic motivation, consistent with SDT's emphasis on competence-supportive environments and with behavioral learning research showing that gamified activities sustain engagement and flow.

Figure 4 illustrates the in-class guided experimentation stage, capturing students' first direct interaction with the Dr. Tian chatbot. During this session, students were invited to pose course-related questions freely and discover the chatbot's empathetic, contextually appropriate responses in real time. The relaxed, exploratory nature of this activity was deliberately designed to lower the perceived barrier to chatbot technology before any graded work was required.

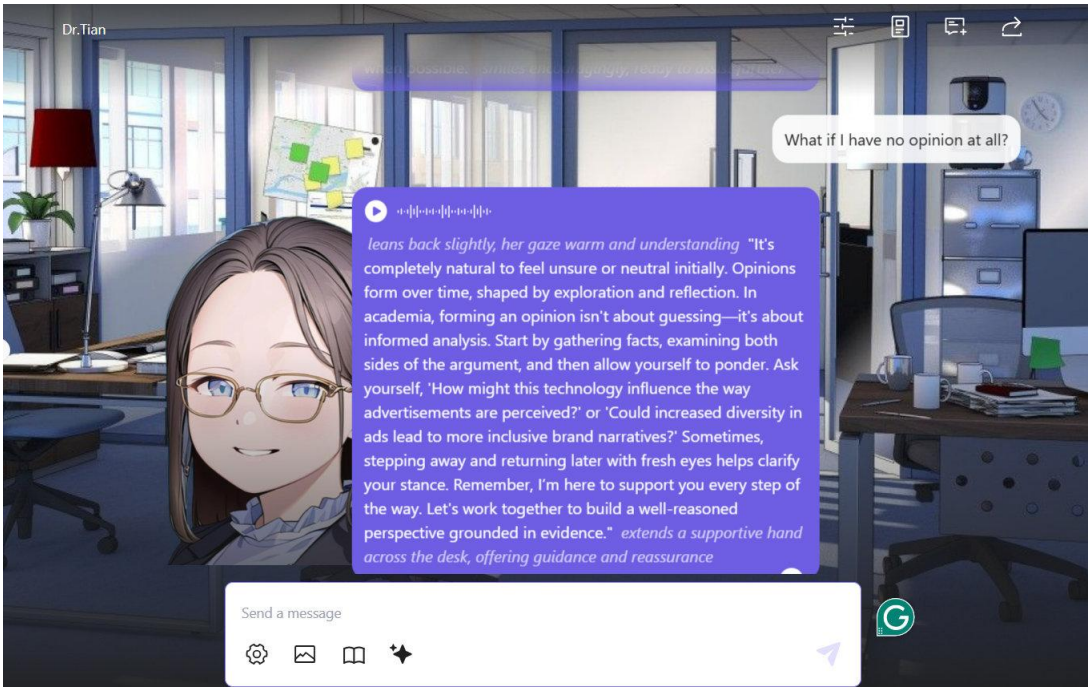

Figure 4. In-Class Demonstration of the Dr. Tian Chatbot During the Guided Experimentation Stage

Figure 5 presents selected examples of student prompts and Dr. Tian's responses during the guided experimentation session. These exchanges illustrate the chatbot's capacity to deliver patient, detailed, and empathetic guidance on assignment-related queries — demonstrating to students that conversational AI can function as a responsive, supportive learning companion as well as a practical marketing tool for managing customer interactions.

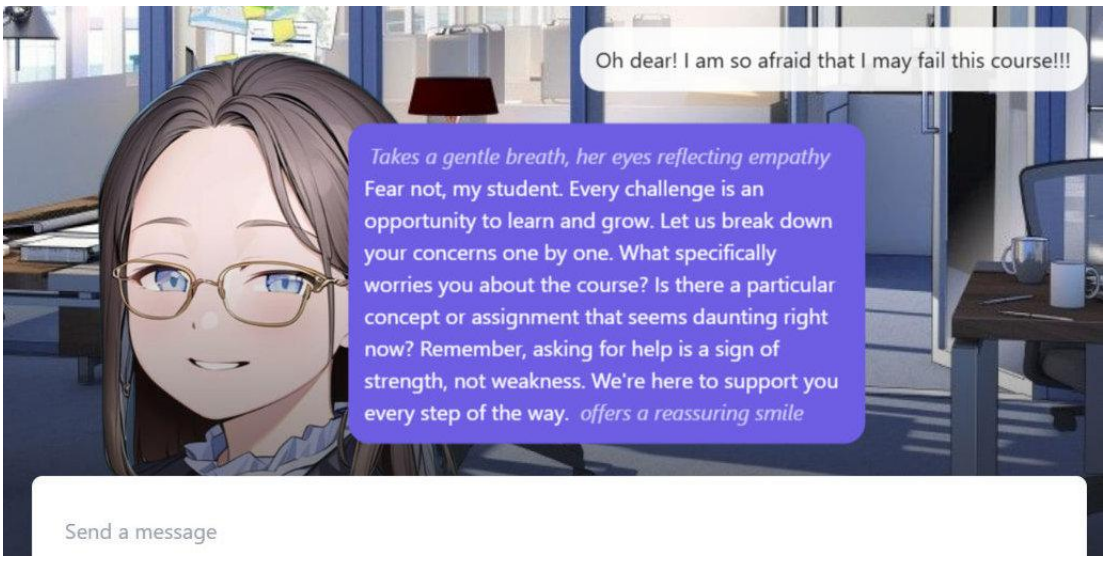

Figure 5. Student Interaction with the Dr. Tian Chatbot: Sample Prompts and Responses

The Rochat platform provided rich functionality appropriate for marketing education: AI-generated avatars based on uploaded photos or text descriptions, customizable chatbot personalities, selectable Large Language Models (LLMs), and voice selection. These features allowed students to experiment with the same design decisions that real e-business practitioners face when deploying AI-powered virtual brand agents. To further demystify chatbot development, the instructor conducted a live demonstration during a tutorial session, building a new chatbot in real time using student inputs, reinforcing the participatory and innovative dimensions of Fun.

Figure 6 documents this live tutorial demonstration, showing the instructor building a new chatbot from scratch using suggestions and input from students in the room. This transparent, participatory exercise made the chatbot development process visible and immediately comprehensible, transforming what might have seemed a complex technical undertaking into an accessible, collaborative creative activity — directly operationalizing the Reasonable Challenge principle by making success feel attainable before students attempted the task independently.

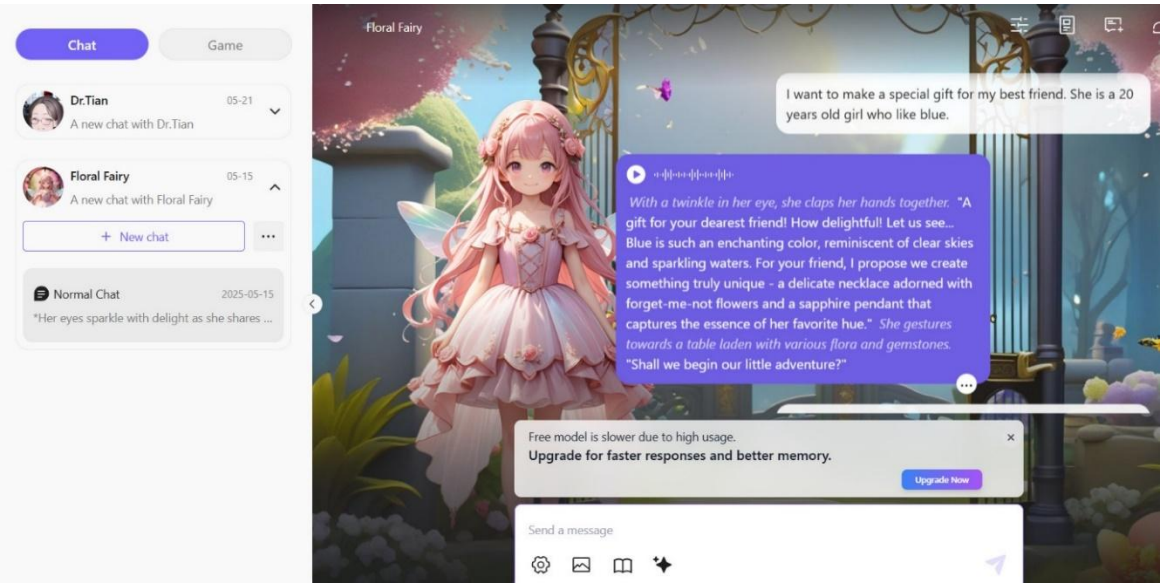

Figure 6. Live Chatbot-Building Demonstration Using Student Inputs in Tutorial Session

3.2.2 Hands-On Generative Content Creation

Rather than passively consuming AI concepts, students actively generated visual marketing content as part of their assessed work. Using tools including ChatGPT, DeepAI, and PixVerse, students created product images, advertisement designs, and AI-generated video content by transforming static images into short animated sequences. These creative, experimental activities directly aligned with the innovative and interactive dimensions of Fun in the RFM Educational Design Model.

This design choice reflects a deliberate departure from lecture-only AI instruction. Consistent with design science principles, the course was treated as an artifact whose components—tasks, tools, and interaction flows—were selected to produce measurable learning outcomes. Hands-on generative content creation positioned students as active producers of e-business-relevant AI outputs rather than passive recipients of conceptual knowledge, directly addressing the curriculum–skills gap identified in the literature.

Figure 7 presents examples of AI-generated marketing visuals produced using ChatGPT, drawn from instructor-prepared handouts distributed to students at the beginning of the course. These instructor-created

examples served a dual pedagogical purpose: first, to demonstrate the creative and commercial potential of generative AI tools before students attempted their own designs; and second, to establish a visible quality benchmark that was achievable yet inspiring — embodying the Reasonable Challenge principle by showing students what was possible through well-crafted prompting without requiring any prior design or technical expertise.

Figure 7. Examples of Ai-Generated Marketing Visuals Using ChatGPT in handouts.

3.2.3 Visual Peer Showcase via Padlet

The third Fun-oriented design element was the deployment of Padlet as a visual, gallery-style showcase platform. Rather than submitting assignments as private instructor–student exchanges, students posted their AI-generated marketing visuals to a shared Padlet board, transforming the assessment process into a public creative exhibition that simultaneously supported peer learning and healthy competition.

The instructor opened the showcase by sharing her own design examples, setting a creative standard and demonstrating the platform's potential. Initially hesitant, students progressively engaged as peers began to share increasingly creative work. Students commented on each other's work, experienced social comparison that stimulated further creative effort, and reported increased confidence in applying AI tools to future marketing tasks. From an EBIS design perspective, Padlet functioned as the collaboration and knowledge-sharing layer of the AI-augmented system, consistent with research on peer showcase as a driver of perceived usefulness and perceived competence.

Figure 8 shows the Padlet showcase board at a representative point in the course, highlighting the diversity and creativity of students' AI-generated marketing designs. The gallery-style layout made individual contributions immediately visible to the entire class, creating a shared creative space that was both motivating and instructive — students could observe a wide range of prompting strategies, design styles, and brand interpretations, learning as much from each other's outputs as from formal instruction. The visible accumulation of high-quality peer work over the semester served as tangible, ongoing evidence of what the cohort was collectively capable of achieving — a powerful reinforcement of both the Fun and Motivation dimensions of the RFM Educational Design Model.

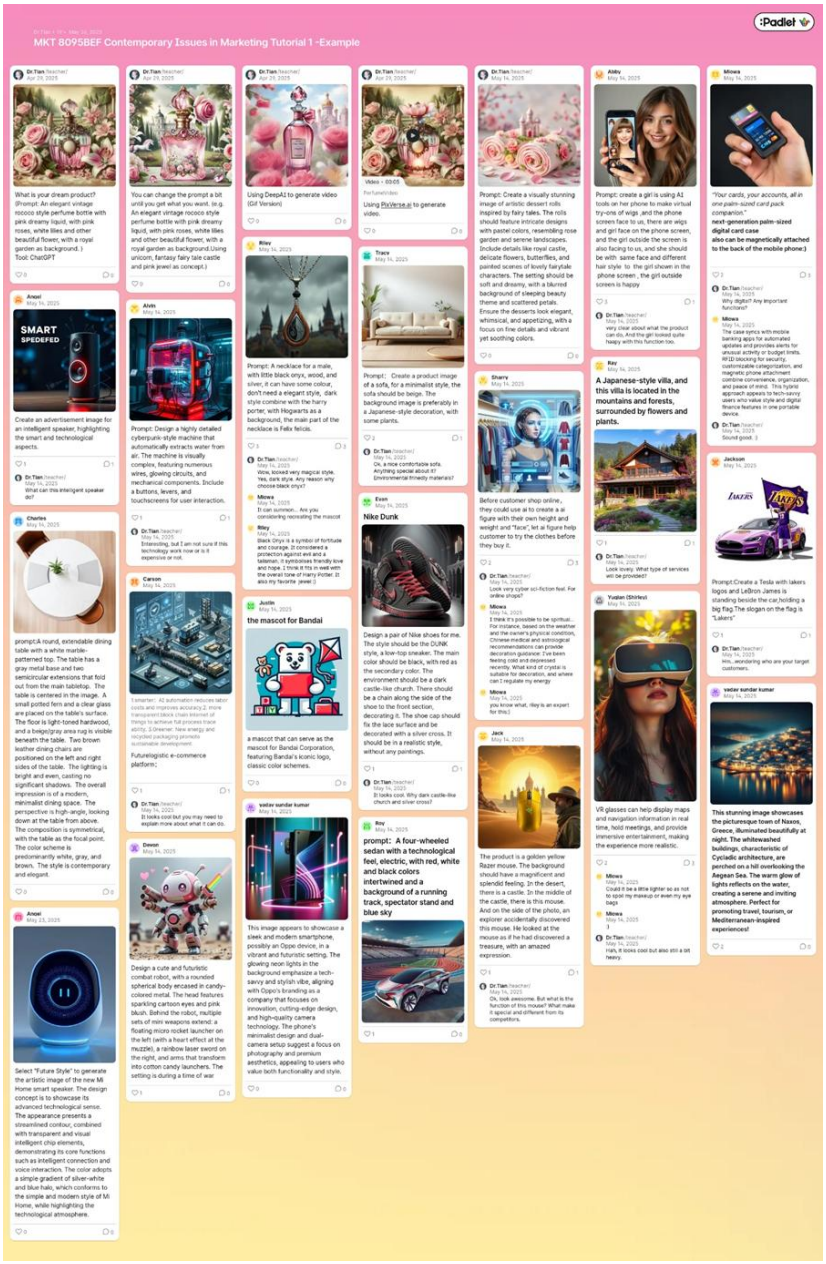

Figure 8. Student AI-Generated Marketing Designs Displayed on the Padlet Showcase Board

3.3 Designing for Motivation: The Three P's

The third RFM principle—Motivation—was operationalised through the Three P's: Perceived Usefulness, Peer Showcase, and Perceived Competence. Together, these three drivers ensured that students not only engaged with AI tools during class but also developed a lasting sense of their value and relevance for professional practice.

Perceived Usefulness was cultivated by embedding all AI tasks within authentic marketing scenarios drawn from real e-business practice. Students were not asked to experiment with AI in the abstract; instead, they used generative AI to design marketing visuals for actual product concepts, built AI chatbots that functioned

as virtual brand spokespeople, and reflected critically on the implications of AI-generated content for marketing ethics, intellectual property, and consumer trust. By grounding all AI activities in real-world e-business contexts, the course ensured that students perceived their learning as directly applicable to their future careers.

Peer Showcase via Padlet created a social, motivationally rich environment in which students' creative outputs were visible to the entire class community. As students saw their peers producing increasingly sophisticated AI-generated visuals and chatbot interactions, healthy competition emerged naturally, motivating further effort and experimentation. This peer visibility mechanism is consistent with SDT's relatedness need and with research showing that public performance contexts enhance both intrinsic motivation and perceived competence.

Perceived Competence was built progressively through the scaffolded task structure, instructor demonstrations, peer feedback on Padlet, and the visible quality of students' own AI-generated outputs. As students successfully completed each AI task—from generating a product visual to building a fully functional AI chatbot—their confidence in applying AI tools in professional contexts grew measurably. This trajectory from anxiety to competence is consistent with Maslow's self-actualization pathway and with SDT's competence-need fulfilment mechanism.

Table 3 summarizes the full assignment structure deployed within the AI-augmented EBIS, mapping each assessed task to its corresponding RFM Educational Design Model principle, the tools utilized, and the key learning outcomes targeted, illustrating how the three RFM principles were operationalized cohesively across the course rather than in isolation.

Assignment	**RFM Alignment**	**Tools Used**	**Key Outcomes**
Academic paper analysis	Reasonable Challenge: Critical thinking, database research	University library databases, academic journals	Evidence-based argumentation, AI ethics awareness
Product visual design	Fun: Creative, hands-on; Motivation: Perceived usefulness	ChatGPT, DeepAI	Prompt engineering skills, marketing design application
Virtual spokesperson chatbot	Reasonable Challenge: Complex tool; Motivation: Perceived competence	Rochat, ChatGPT (avatar images)	Conversational AI understanding, brand personality design
Padlet showcase participation	Motivation: Peer showcase, relatedness	Padlet	Collaborative learning, healthy competition, feedback skills

Table 3: Assignment Structure and RFM Educational Design Alignment

4. Methodology

4.1 Study Context

The study was conducted in a postgraduate marketing course at a university in Hong Kong during the summer of 2025. The course was designed to expose students to emerging trends and technologies in contemporary marketing practice, with a particular focus on integrating artificial intelligence into e-business marketing workflows.

The course enrolled 19 postgraduate students pursuing full-time master's degrees across business-related disciplines. Participants came from diverse professional backgrounds, representing a range of prior industry experiences in marketing, management, and related fields. Consistent with the broader higher-education context in Hong Kong, most students had limited prior hands-on experience with generative AI tools at the start of the course, making the cohort a representative sample for studying AI competency development at the postgraduate level.

The AI-augmented EBIS deployed in this course integrated five platforms, each fulfilling a distinct functional role within the overall system design. Rather than forming a tightly coupled technical pipeline, the platforms were selected and orchestrated as a complementary toolkit—multiple accessible pathways through which students could achieve the same core learning objectives: creating AI-generated marketing visuals, producing promotional video content, designing conversational AI brand agents, and showcasing their work to peers. This design decision reflected a deliberate pedagogical choice to provide students with flexibility and creative agency rather than prescribing a single fixed workflow.

Platform selection was guided by two explicit design criteria consistent with the Reasonable Challenge principle of the RFM Educational Design Model. First, all platforms were either freely accessible or available through institutional provision, eliminating financial barriers to participation. Second, all platforms offered no-code or low-code interfaces, making them accessible to students without programming backgrounds—an important consideration in a postgraduate business cohort where technical expertise varies widely.

Critically, prior to deployment, the course instructor personally tested each platform on an independent project to assess task difficulty, time requirements, and potential technical barriers. This pre-deployment evaluation ensured that the challenge level presented to students was genuinely reasonable and achievable within the course timeline—reflecting not an assumption but an evidence-informed design decision. For most students, this was their first hands-on experience with these generative AI and chatbot platforms, making the scaffolded introduction and instructor-tested design particularly important for building early confidence and reducing anxiety.

The five platforms fulfilled the following distinct functional roles within the EBIS:

ChatGPT (University Portal): Institutional ChatGPT Plus access provided students with DALL-E image generation capabilities (15 images per day limit), enabling text-to-image creation for marketing visuals. The "Upload Media" function allowed students to provide reference images when struggling to describe desired outputs in text prompts.

DeepAI: Free digital platform offering 100+ stylized image generation models. Positioned as a complementary tool addressing ChatGPT's daily image limit, enabling students to conduct prompt engineering trials before finalizing designs.

PixVerse.ai: Video generation platform converting static AI-generated images into short video clips (5 seconds). Free accounts provided 60 daily credits, sufficient for 1-2 video generations. This tool extended students' generative capabilities from static imagery to animated content.

Rochat: AI-powered chatbot creation platform enabling students to build virtual brand mascots and spokespersons. The platform's low-code interface allowed customization of chatbot appearance (AI-generated

avatars), personality profiles, underlying large language models (LLMs), and voice characteristics without programming knowledge.

Padlet: Digital canvas and collaborative bulletin board enabling students to post AI-generated designs, comment on peers' work, and engage in collective learning. Served as the system's social and showcase component.

These components were orchestrated through structured assignments, in-class demonstrations, tutorial support, and peer interaction protocols, forming an integrated EBIS rather than disconnected tools.

4.2 Data Collection

Quantitative data: Online Learning Environment (OLE) polling conducted in-class using anonymous surveys measuring perceived appeal, usability, usefulness, and ease of access for each AI tool (ChatGPT/DeepAI, Rochat, Padlet). Students rated each dimension on 5-point scales (e.g., "very useful" to "not useful at all").

Qualitative data: (1) Student assignment artifacts including AI-generated imagery, product designs, virtual spokesperson chatbots, and written reflections on academic literature addressing generative AI in marketing[13]; (2) Open-ended survey comments about tool experiences; (3) Peer comments on Padlet showcases.

4.3 Ethical Considerations

Students provided informed consent for anonymized use of their work and survey responses in research. All identifying information was removed from analyzed artifacts.

5. Discussions

5.1 Theoretical Implications

6.1.1 The RFM Educational Design Model as a Design Science Contribution

The RFM Educational Design Model makes a dual theoretical contribution to the EBIS and educational design science literature. First, it emerges not from abstract theorizing but from years of direct pedagogical observation across diverse classroom settings — undergraduate, postgraduate, and distance learning environments — where three recurring patterns became unmistakably clear: when students perceive tasks as excessively challenging, they disengage and give up; when learning feels tedious or boring, participation becomes superficial or performative; and when motivation is absent, students do only the minimum required to pass. These practitioner observations provided the real-world problem that the RFM model was designed to solve — consistent with design science research's requirement that artifacts address genuine, observed problems.

Second, the RFM model advances EBIS design knowledge by providing a theoretically grounded synthesis of five established motivational perspectives — Self-Determination Theory, Expectancy Theory, Attribution Theory, Maslow's Hierarchy of Needs, and Behavioural Learning Theory — translated into actionable design principles for AI-augmented educational environments. While each of these theoretical frameworks has been applied independently in educational research, the RFM model's contribution lies in their deliberate integration into a coherent, practically tested framework with clear operationalization guidance. The three-by-three structure (three main dimensions × three sub-components for Fun and Motivation) offers manageable complexity for instructors designing AI-augmented courses, making theoretical insights directly actionable rather than abstractly aspirational.

Framed within design science research, the RFM Educational Design Model contributes both a design construct (the model itself, with its three principles and associated sub-components) and a set of validated design principles (actionable guidelines for implementing each principle in AI-rich learning contexts) — both of which were empirically tested and demonstrated to produce meaningful outcomes in a real postgraduate marketing course. Critically, this is not a model proposed on paper alone; it is a framework that has been deployed, observed, and evaluated in practice, and recognised through institutional teaching awards for its effectiveness.

6.1.2 Theory-to-Practice Translations: From Motivational Theory to EBIS Design

A key theoretical contribution of the RFM model is its explicit mapping of established motivational theories to concrete EBIS design decisions — a translation that is often missing in educational technology research. Three particularly instructive mappings emerged from this study:

Expectancy Theory → Three-Stage Scaffolding Structure: Expectancy Theory posits that motivation depends on students' perceived likelihood of success. The three-stage scaffolding structure — demonstration, guided experimentation, and independent application — was directly designed to build students' expectation of success before requiring independent performance. By first showing students what was achievable (demonstration), then allowing safe experimentation without grade consequences (guided experimentation), and only then requiring graded independent work (application), the course systematically increased students' confidence that effort would yield meaningful outcomes. The qualitative evidence of students progressing from generic prompts to sophisticated brand-aligned AI outputs confirm that this expectancy-building mechanism worked as theoretically intended.

Attribution Theory → Prompt Engineering Progression: Attribution Theory highlights how students' beliefs about the causes of their success affect their motivation and persistence. As students progressed from ineffective generic prompts to sophisticated, brand-aligned AI-generated content, they could clearly attribute their improvement to their own developing effort and skill — not to luck or the tool itself. This internal attribution of success is a powerful motivational reinforcement, encouraging continued effort and risk-taking in subsequent tasks. The visible contrast between early unsuccessful and later successful outputs made this progression tangible and personally meaningful for students.

Maslow's Hierarchy of Needs → Padlet Peer Showcase: Maslow's framework identifies esteem needs — the need to be recognised, valued, and respected — as a critical motivational driver above basic safety and belonging needs. The Padlet peer showcase mechanism directly addressed students’ esteemed needs by making their creative AI outputs publicly visible and celebrated within the class community. Students' voluntary comments on peers' work and their motivation to produce increasingly impressive designs in response to seeing others' posts reflect the fulfilment of esteem needs predicted by Maslow's framework. This finding suggests that social visibility mechanisms are not merely engagement features but theoretically grounded motivational infrastructure within AI-augmented educational EBIS.

6.2 Managerial Implications

6.2.1 For University Curriculum Designers and Program Directors

The findings of this study offer three main practical implications for institutions seeking to evolve their business curricula to meet the demands of AI-driven e-business environments.

Prioritize high-relevance AI applications as entry points: Student interest data revealed the strongest enthusiasm for AI applications in advertising (29%) and product design (29%), followed by online shopping (24%). These preferences are not arbitrary — advertising and product design represent the AI applications

most immediately relevant to students' anticipated professional roles in marketing and e-business, and also the areas where generative AI tools produce the most visually compelling and immediately gratifying outputs. Curriculum designers should therefore prioritize these high-interest domains as entry points for AI competency development, using them to build confidence and enthusiasm before introducing students to less immediately appealing but strategically important AI applications such as data analytics, moderation, and AI governance. Importantly, the creative skills developed through advertising and product design tasks — prompt engineering, brand alignment, visual communication — transfer readily to other AI application domains, and even to entrepreneurial contexts such as launching new products or creative projects independently.

Adopt the RFM model as a curriculum design framework: The RFM Educational Design Model provides program directors with a structured yet flexible framework for designing AI-augmented courses that are motivationally effective across diverse student cohorts. The model's three principles — Reasonable Challenge, Fun, and Motivation — can be applied at the course level (overall curriculum structure), the module level (individual topic design), and the assignment level (specific task calibration), making it scalable across different program contexts and course sizes.

Invest in tool orchestration, not just tool access: The findings suggest that success in AI-augmented education derives not from access to any single AI platform but from the thoughtful orchestration of complementary tools addressing different competencies — image generation, video creation, conversational AI design, and collaborative showcase. Institutions should therefore move beyond simply providing access to one or two AI tools and instead design integrated AI tool ecosystems that expose students to the range of AI capabilities increasingly deployed in real e-business environments.

6.2.2 For Individual Instructors

Scaffolding is non-negotiable: The evidence clearly demonstrates that direct exposure to AI tools without prior demonstration, guided practice, and progressive challenge produces poor outcomes — generic prompts, superficial engagement, and student anxiety. Individual instructors should invest time in designing structured skill-building pathways, particularly for student cohorts with limited prior AI experience. The three-stage scaffold (demonstrate → guide → apply) provides a replicable template that can be adapted to different tools and course contexts.

Pre-test all tools personally: A critical but often overlooked step in AI-augmented course design is the instructor's own hands-on testing of each platform before deployment. In this study, the course instructor personally tested all five platforms on independent projects to assess difficulty levels, time requirements, and potential technical barriers — ensuring that the challenge presented to students was genuinely reasonable rather than assumed. This pre-deployment evaluation is a practical necessity, not an optional extra.

Embed ethical AI discourse throughout, not as a separate module: Students' spontaneous critical reflections on AI bias, copyright, privacy, consumer trust, and the boundary conditions of virtual versus human influencers demonstrate that ethical AI awareness can emerge organically when assignments require genuine critical engagement with academic literature on generative AI. Instructors should design assignments that prompt students to read, analyze, and form their own positions on AI ethics — rather than delivering ethics as a standalone lecture module disconnected from practical tool use.

Leverage peer learning as a motivational accelerator: Padlet data and student comments confirm that peer examples and feedback accelerated learning more effectively than instructor guidance alone. Instructors should deliberately design social visibility mechanisms — peer showcase platforms, in-class sharing sessions, peer commentary requirements — into their AI-augmented courses, recognizing these not as supplementary activities but as core motivational infrastructure.

6.2.3 Institutional Considerations: Data Privacy and AI Governance

As institutions move beyond classroom AI experimentation toward more sophisticated AI deployments — including AI-powered virtual brand agents that interact with real customers — data privacy and governance

considerations become increasingly critical. In the educational context of this study, privacy risks were limited because chatbot interactions occurred within a controlled classroom environment for skill development. However, institutions and students should be aware that real-world deployment of AI chatbots in customer-facing e-business contexts introduces substantially higher privacy and governance requirements: customer conversation records, data retention policies, and regulatory compliance demand secure, enterprise-grade infrastructure rather than the free educational platforms appropriate for classroom experimentation. Future curriculum iterations should therefore include explicit discussion of the transition from educational AI prototyping to professional AI deployment, equipping students not only with design skills but also with awareness of the governance responsibilities that accompany real-world AI-powered customer interactions.

6. Conclusions

This paper sets out to address a genuine and growing challenge in business education: how do we move beyond talking about artificial intelligence to genuinely teaching with and through it — in ways that are engaging, meaningful, and motivationally sustainable for students?

The answer proposed here is the RFM Educational Design Model — a framework born not in theory but in the honest realities of the classroom. Across years of teaching undergraduate, postgraduate, and distance-learning students, three patterns recurred with quiet persistence: students who found tasks too hard simply gave up; students who found learning tedious or boring merely went through the motions; and students without motivation did only the minimum to pass. These observations — simple, honest, and universal — became the foundation of the RFM model's three principles: Reasonable Challenge, Fun, and Motivation.

The findings are encouraging. Students progressed from generic, ineffective AI prompts to sophisticated, brand-aligned marketing outputs. They built AI-powered virtual brand spokespeople, generated compelling marketing visuals and promotional videos, and engaged in animated, voluntary peer learning through a shared digital showcase. Perhaps most meaningfully, they developed not only technical AI competence but also the kind of spontaneous, nuanced critical thinking about AI ethics, bias, virtual versus human influence, and consumer trust that no tutorial alone can produce. These are the outcomes that matter — not just for passing an assignment, but for navigating a professional world increasingly shaped by AI.

The study contributes to the literature in two ways. Theoretically, it introduces the RFM Educational Design Model as an integrated, theoretically grounded, and empirically validated design framework for AI-augmented educational EBIS — one that has already shown preliminary adaptability across different student populations and learning formats. Practically, it demonstrates that meaningful AI competency development in business education is achievable without expensive infrastructure, programming expertise, or institutional-scale investment. Free and low-cost platforms, thoughtfully scaffolded and motivationally designed, are sufficient to produce genuine learning outcomes — a finding with important implications for institutions at all stages of their AI education journey. The approach has also been independently recognized for its effectiveness in fostering student engagement and learning outcomes, providing external validation beyond the authors' own observations.

The RFM model is not offered here as a finished answer but as an open invitation. An invitation to educators who have observed the same three patterns in their own classrooms — and who believe, as the authors do, that students thrive when they are challenged appropriately, when learning is genuinely enjoyable, and when they feel motivated to succeed. An invitation to institutions ready to evolve their curricula from passive AI literacy to active AI competency. And an invitation to researchers who will extend, challenge, and refine these ideas across disciplines, cultures, and learning contexts that this study could not reach.

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Comparative Learning Strategies for Training Belief Rule Base Systems: A Case Study of E-Government Assessment*

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Abstract

E-Government short for electronic government, also known as e-gov, digital government consists of the digital interactions between a government and citizens, government and businesses/Commerce, government and employees, and also between government and governments /agencies. E-government is no longer just an option but a necessity for countries aiming for better governance. The e-government performance assessment is important for the development of e-government. As conventional evaluation models for e-government performance are generally too subjective and inaccurate, this study proposed belief Rule-base Inference Methodology using the Evidential Reasoning approach (RIMER) method to evaluating the e-government performance.The aim of the research is to evaluate optimal performance of E-government. The method will generate result an accurate system for E-government assessment.In the research, we employed belief Rule-base Inference Methodology using the Evidential Reasoning approach (RIMER) for design and development of belief rule-based expert system. In the system, belief rule base (BRB) is used to model uncertain E-Government domain knowledge, the evidential reasoning (ER) approach is employed to build inference engine, a BRB training module is developed for learning the BRB through accumulated a lot of cases. We use a set of simulated data and to validate the developed belief rule-based expert system prototype. The results show that the prototype can provide reliable recommendations and the performance of the system can be improved significantly after training BRB using accumulated different cases.The expert system is designed and developed by using Visual Basic, MS SQL Server and Mat lab. It can show that the developed Expert System can generate appropriate results taking uncertain information.The achievement of the system is that it has assessed optimal performance of E-government. The developed system handles uncertainty in an efficient manner and generates real system output.

1 Introduction

Belief Rule Based Expert Systems (BRBES) utilize the Belief Rule Base (BRB) for knowledge representation and Evidential Reasoning (ER) for inference [35] [17, 23]. BRB extends

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traditional IF–THEN rules by incorporating belief degrees in the consequent, enabling the representation of expert knowledge under uncertainty. ER, developed from Dempster–Shafer theory, handles ignorance [25], while BRB integrates fuzzy logic to address vagueness and imprecision. Their combined inference methodology, RIMER, can simultaneously manage vagueness, imprecision, ignorance, and incompleteness while processing both qualitative and quantitative data [38] [16, 21] [35]. This integrated capability distinguishes BRBES from Bayesian Belief Networks, fuzzy logic, and D–S theory when used independently.

BRBES has been successfully applied in domains characterized by uncertainty, including health informatics, flood prediction, affective computing, and E-Government evaluation [33]. In E-Government assessment, twenty-one variables with different measurement scales require effective handling of uncertainty, as demonstrated by Hossain et al. [18, 31] [13, 19] [14] [10, 12]. However, optimizing BRBES through effective training remains a challenge, requiring appropriate objective functions, parameter selection, and knowledge base updating strategies. Therefore, this paper proposes a learning framework to improve BRBES performance using E-Government assessment as a case study.

Although BRB and RIMER methodologies have previously been applied to E-Government assessment, this study makes several distinct contributions:

- A hierarchical BRB framework consisting of 21 E-Government assessment variables organized into Determinants, Characteristics, and Results categories.
- An integrated learning framework to investigate the impact of training different BRB parameters, including rule weights, attribute weights, belief degrees, and utility scores.
- A comparative evaluation of five training configurations to identify the most effective parameter subset for reducing prediction error while mitigating overfitting.
- Validation that BRB training improves the predictive capability of E-Government assessment systems under uncertain and incomplete information.

Unlike the generic BRB optimization framework proposed by Yang et al. [34], this study comparatively evaluates different parameter-training strategies for E-Government assessment to identify their practical effectiveness.

The remainder of this paper is organized as follows. Section 2 reviews related work, Section 3 presents the proposed methodology, Section 4 describes the BRBES prototype, Section 5 discusses the experimental results, and Section 6 concludes the paper.

2 Existing E-Government Evaluation Approaches

Evaluating government performance in ICT project implementation from the citizens' perspective requires an appropriate research approach. Governments worldwide invest substantial financial resources, often exceeding 1% of GDP, in developing e-government services, making effective evaluation essential to justify these investments [26] [1]. Traditional evaluation methods, such as return on investment (ROI), cost-benefit analysis, payback period, and present worth, are widely used but have limitations in assessing e-government projects [1, 3] [6]. These approaches mainly focus on tangible financial outcomes while overlooking human, organizational, and intangible benefits [22] [8]. To overcome these limitations, several citizen-centric evaluation models have been proposed. Wang et al. developed a framework for evaluating e-government services that explains the success or failure of government portals in meeting citizens' information needs [11, 27] [1]. Similarly, Eschenfelder et al. introduced a socio-technical toolkit that

evaluates the openness of e-government portals by considering information characteristics, social and political contexts, stakeholder value, and citizen participation [5]. Carter and Bélanger applied the Technology Acceptance Model (TAM) and Diffusion of Innovation (DOI) theory to identify factors influencing citizen adoption of e-government services, including perceived usefulness, compatibility, ease of use, image, and trust in government and the Internet [4]. Fuzzy set theory, introduced by Zadeh in 1965, provides an effective framework for handling vague, ambiguous, and imprecise information by assigning degrees of membership between 0 and 1 [20]. Unlike classical set theory, it supports gradual transitions between sets and enables reasoning under uncertainty, making it suitable for complex decision-making problems. Its ability to represent possibility, vagueness, and nonlinear integration has led to applications in engineering, information processing, and social science research [2].

3 Methods

A belief degree is associated with each possible consequent of a rule in a Belief rule-base(BRB) expert system. Belief Rule base knowledge representation schema has the ability to handle uncertainty. In belief rule base expert system, the knowledge base of the expert system is developed using Evidential Reasoning approach which is the replacement of inference engine in the knowledge base [28] [33] [29]. In this approach, a rule base is designed with belief degrees embedded in all possible consequents of a rule. Complicated and continuous causal relationships between different factors can be captured by belief rule-base expert system which is not possible for traditional IF-THEN rules.

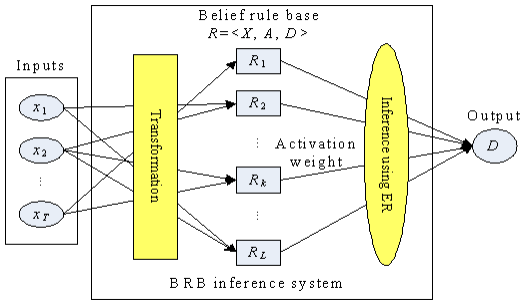


Figure 1: Single-layer BRB inference architecture with RIMER Methodology

3.1 BRB for Representing E-government Assessment Domain Knowledge

Belief Rule Bases (BRBs) extend traditional rule bases by incorporating a belief structure into the consequent part of the rules. In a belief rule, each antecedent attribute takes referential values, while each consequent is associated with corresponding belief degrees [33]. The knowledge representation also embeds rule weights, attribute weights, and belief degrees.

The kth rule R_k in a traditional IF–THEN rule base is expressed as

$$\text{If } A_{1k} \wedge A_{2k} \wedge \cdots \wedge A_{T_k k}, \text{ then } D_k \tag{1}$$

where

A_{ik} : Referential value of the ith antecedent attribute in the kth rule,
T_k : Number of antecedent attributes in the kth rule,
D_k : Consequent of the kth rule.

By incorporating belief degrees, rule weights, and attribute weights, the traditional rule is extended into a belief rule as

$$\text{If } A_{1k} \wedge A_{2k} \wedge \cdots \wedge A_{T_k k}, \text{ then } \{(D_1, \beta_{1k}), (D_2, \beta_{2k}), \ldots, (D_N, \beta_{Nk})\} \tag{2}$$

with rule weight θ_k and attribute weights $\delta_{k1}, \delta_{k2}, \ldots, \delta_{kT_k}$, where

β_{ik} : Belief degree associated with the ith consequent,
δ_{ki} : Relative importance of the ith antecedent attribute in the kth rule,
θ_k : Relative importance (weight) of the kth rule,
L : Total number of belief rules in the rule base,
T_k : Number of antecedent attributes in the kth belief rule.

3.2 BRB Inference System

The Evidential Reasoning (ER) approach is widely used for multi-attribute decision analysis (MADA) under uncertain quantitative and qualitative information [15, 24]. In a Belief Rule Base (BRB), each antecedent A_k $(k = 1, \ldots, L)$ is associated with a weight W_k and linked to consequents D_i $(i = 1, \ldots, N)$ through belief degrees β_{ik}. The belief rule structure is summarized in Table 15.

For an input vector $U = (U_i, i = 1, \ldots, T)$ with belief degrees $\epsilon = (\epsilon_i, i = 1, \ldots, T)$, each input is transformed into a distributed belief structure:

$$S(U_i, \epsilon_i) = \{(A_{ij}, x_{ij}); j = 1, \ldots, J_i\}.$$

The activation weight of the kth rule is computed as:

$$W_k = \frac{\theta_k \alpha_k}{\sum_{i=1}^{L} \alpha_i \theta_i},$$

where θ_k is the rule weight and α_k is the matching degree. Expert belief degrees $\bar{\beta}_{ik}$ may be complete or incomplete and are updated during inference as:

$$\beta_{ik} = \bar{\beta}_{ik} \frac{\sum_{t=1}^{T_k} \tau(t,k) \sum_{j=1}^{J_t} \alpha_{tj}}{\sum_{t=1}^{T_k} \tau(t,k)}.$$

The ER algorithm aggregates all activated rules to produce the output belief distribution:

$$O(U) = \{(D_j, \beta_j), j = 1, \ldots, N\}.$$

3.2.1 Input Transformation

In this research, the referential set {Excellent, Very Good, Good, Satisfactory, Poor} is used for all E-government assessment factors. Each input is transformed into a distributed belief structure:

$$S(A_i^*) = \{(h_{3i}, \gamma_{3i}), (h_{2i}, \gamma_{2i})\},$$

where

$$\gamma_{2i} = \frac{h_{3i} - A_i}{h_{3i} - h_{2i}}, \quad \gamma_{3i} = 1 - \gamma_{2i}.$$

The transformation follows a rule-based mapping approach [32].

3.2.2 Qualitative input transformation

Qualitative inputs are mapped to the referential values {Excellent, Very Good, Good, Satisfactory, Poor} using expert-defined rules. Typically, categories such as *Excellent* and *Good* are assigned higher belief degrees based on domain knowledge, while less relevant categories receive lower or zero values. These mappings are determined by expert judgment.

3.2.3 Quantitative input transformation

Each quantitative attribute is associated with lower and upper threshold values defined by domain experts. If an input is below the lower bound, it is mapped to 100% Good; if above the upper bound, it is mapped to 100% Very Good; otherwise, a linear transformation is applied using the equation above.

3.2.4 Rules activation weights calculation

After assigning antecedent referential values, the next step is to compute rule activation weights. First, the normalized attribute weights are obtained as:

$$\overline{\delta}_{ki} = \frac{\delta_{ki}}{\max\{\delta_{ki}\}}, \quad i = 1, \ldots, T_k$$

The combined matching degree of each rule antecedent is calculated as:

$$\delta_k = \prod_{i=1}^{T_k} (\alpha_{ik}) \delta_{ki}$$

The activation weight of the kth rule is then given by:

$$W_k = \frac{\theta_k \alpha_k}{\sum_{i=1}^{L} \alpha_i \theta_i}$$

These computations determine the degree to which each rule is activated based on input matching.

3.2.5 Belief-degree Update

The belief degree β_{ik} is initially assigned by experts during BRB construction. A rule is considered complete if $\sum \beta_{ik} = 1$, and incomplete otherwise. If $\sum \beta_{ik} = 0$, the rule output is completely unknown. Incompleteness may arise from limited expert knowledge or missing input information. In such cases, both antecedent and consequent uncertainty must be retained in the inference process [**?**]. When input data is incomplete, the resulting inference also becomes incomplete [33]. Therefore, belief degrees must be updated according to the actual input to reflect partial or missing information during inference.

3.2.6 Rule aggregation via ER

The Evidential Reasoning (ER) approach aggregates all activated rules for multi-attribute decision analysis under uncertainty [9] [36] [30]. Unlike conventional MADA methods, ER uses a belief distribution framework instead of a crisp decision matrix. In this model, each rule acts as a basic attribute, and its activation weight represents its importance, while consequents correspond to evaluation grades.

3.2.7 Output of the BRB System

The output of a BRBES is a belief distribution rather than a single numerical value. It can be converted into a crisp score using a utility function assigned to each consequent grade [33]:

$$H(A^*) = \sum_{j=1}^{N} u(C_j) B_j.$$

3.3 Optimal Learning Methods for Training Belief Rule Bases in RIMER

In this research, the initial BRB is constructed using expert knowledge and validated E-government rules. Since domain knowledge evolves slowly, BRB parameters are optimized through offline learning using historical E-government datasets. Several optimization-based training methods have been proposed in [34]. When outputs are numerical, training is formulated as a nonlinear optimization problem that minimizes the mean squared error (MSE) between predicted and observed values.

The training parameters include rule weights (θ_k), attribute weights (δ_j), and belief degrees (β_{jk}). Given a training set of M samples $(\hat{x}_m, \hat{y}_m)$, where $m = 1, \ldots, M$, the model output y_m is generated by the inference engine, and $\hat{y}_m$ is the observed output. The objective function is:

$$\xi(P) = \frac{1}{M} \sum_{m=1}^{M} (y_m - \hat{y}_m)^2.$$

Inference is performed using the analytical ER algorithm [28], where the aggregated belief degrees for consequents D_j $(j = 1, \ldots, N)$ are computed as:

$$\beta = \frac{\mu \left[\prod_{k=1}^{L} \left(\omega_k \beta_{jk} + (1 - \omega_k) \sum_{j=1}^{N} \beta_{jk} \right) - \prod_{k=1}^{L} \left(1 - \omega_k \sum_{j=1}^{N} \beta_{jk} \right) \right]}{1 - \mu \left[\prod_{k=1}^{L} (1 - \omega_k) \right]},$$

$$\mu = \left[\sum_{j=1}^{N} \prod_{k=1}^{L} \left(\omega_k \beta_{jk} + (1 - \omega_k) \sum_{j=1}^{N} \beta_{jk} \right) - (N-1) \prod_{k=1}^{L} \left(1 - \omega_k \sum_{j=1}^{N} \beta_{jk} \right) \right]^{-1}.$$

Here, β_{jk} is the belief degree of the jth consequent in the kth rule, and ω_k is the activation weight of the kth rule. Since ER produces a belief distribution (D_j, β_j) rather than a single value, the final output is obtained using a utility function [37]:

$$y = \sum_{j=1}^{N} u(D_j)\beta_j.$$

Optimization Constraints

The optimization is subject to the following constraints:

$$0 \leq \theta_k \leq 1 \tag{1}$$

$$0 \leq \delta_j \leq 1 \tag{2}$$

$$\sum_{j=1}^{N} \beta_{jk} \leq 1 \tag{3}$$

where θ_k is the rule weight, δ_j is the attribute weight, and β_{jk} is the belief degree.

The optimization terminates when the reduction in MSE between successive iterations falls below a threshold or when the maximum number of iterations is reached. These constraints ensure consistency of BRB parameters and prevent infeasible solutions during training.

4 Design and Development of Belief Rule-based Expert System Prototype

In this research, a belief rule-base expert system (BRBES) is developed for E-government assessment using multiple evaluation factors. The BRB framework extends traditional rule-based systems by incorporating belief structures, enabling effective handling of uncertainty. Inference is performed using the Evidential Reasoning (ER) approach, which supports the fusion of both qualitative and quantitative uncertain information in E-government assessment.

4.1 System Architecture

The system architecture defines the structure of inputs, processing, and outputs. The proposed BRBES follows a three-layer design: (i) interface layer for collecting input referential values, (ii) application layer containing the inference engine, and (iii) data processing layer where the initial BRB knowledge base is stored.

Figure 2 illustrates the proposed three-layer architecture.

Figure 3 presents the inference engine structure used in the application layer.

The Recursive ER algorithm used in the inference engine is summarized as follows:

1. Step 1: Start

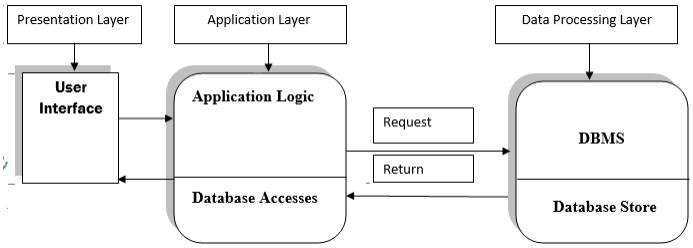


Figure 2: Three-Layer Architecture

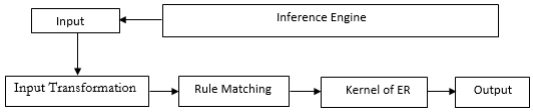


Figure 3: Inference-engine

2. Step 2: Read rule activation weights and updated consequent belief degrees from inputs.
3. Step 3: Compute belief degrees and unassigned belief from the first activated rule.
4. Step 4: Combine current rule with previously aggregated rules.
5. Step 5: Repeat until all rules are processed.
6. Step 6: Normalize final belief degrees.
7. Step 7: End.

4.2 Knowledge Base

The knowledge base is constructed using a BRB framework as shown in Figure 4. To reduce complexity, nine "Characteristics" variables are grouped into three subgroups: C1 (User Environment), C2 (Resource Management), and C3 (Authentication Protocol). Similarly, seven "Result" variables are grouped into R1 (Result Analysis) and R2 (Result Specification).

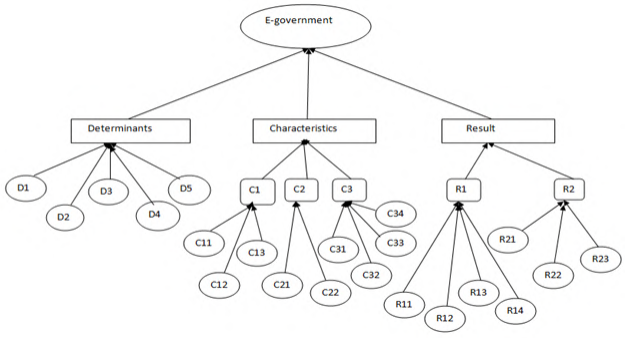


Figure 4: Hierarchical Relationship among E-Government Assessment Factors

Knowledge Acquisition and Rule Construction

The initial BRB is constructed using expert knowledge and literature-based E-government evaluation criteria. Five referential values are used for each antecedent: Excellent, Very Good, Good, Satisfactory, and Poor. With five antecedents, the total number of rules is:

$$5^5 = 3125$$

Initial belief distributions are assigned based on expert judgment and later refined using training data.

The determinants considered are: D1: Quality of information and data availability D2: Technological infrastructure and compatibility D3: Organizational and management factors D4: Legal and institutional framework D5: Potential demand

Characteristics include: C1: User environment (Interaction, Integration, Personalization) C2: Resource management (Information quality, Services) C3: Authentication protocol (Security, Accessibility, Usability, Privacy)

Results include: R1: Result analysis (Service quality, Efficiency, Effectiveness, Transparency) R2: Result specification (System usage, Citizen participation, Regulatory change)

4.3 Trained Module Design Method

Overfitting occurs when a model performs better on training data but worse on unseen test data, especially when the parameter space is overly complex (Mitchell, 1997). In BRB training, overfitting may arise when too many parameters are optimized simultaneously.

To address this issue, five training configurations are designed:

R1: rule weights, attribute weights, belief degrees, and utility scores **R2:** rule weights, attribute weights, and belief degrees **R3:** attribute weights and belief degrees **R4:** rule weights and belief degrees **R5:** belief degrees only

The training and validation procedure is as follows:

1. Select training and testing samples from 1000 simulated cases.
2. Train BRB using the five configurations with identical initial parameters.
3. Evaluate performance on training and test datasets.
4. Analyze results using ROC curves (Chapter 7).
5. Draw conclusions on optimal training configuration.

4.4 Graphical user interface of Belief Rule Based Expert System

The component implemented in the prototype which concerns the interaction between users and the system is user interfaces. There are many software available for developing user interfaces. Among them Microsoft Visual Basic 6.0 software is used at front end to develop this project. Because it is easy to implement our desired interfaces using this software. This software gives us the facility of making a link between SQL Server and the user interfaces developed by it. Through this language we can implement any declarative query such as search and show which is most important for my software. Moreover Visual Basic 6.0 provides us the facilities of importing different images and pictures. We can making any Labels, buttons, textboxes, frames, tabs etc without writing any code like other languages such as Java, C etc. Through this data reports of different tables of database can be easily generated and printed. User interfaces is developed by MS Visual Basic 6.0

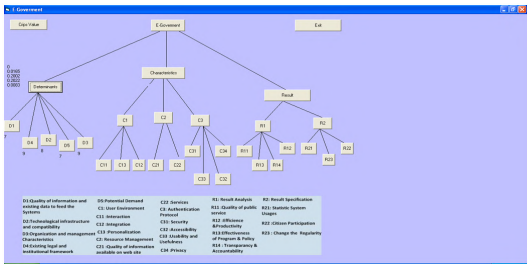


Figure 5: BRBES graphical user interface

5 Results and Discussion

The developed BRB-based Expert System for E-Government performance assessment can be applied to evaluate various aspects of E-Government initiatives. The system employs the RIMER methodology to effectively manage different forms of uncertainty, including vagueness, incompleteness, and imprecision. In the proposed framework, a Belief Rule Base (BRB) is first constructed and subsequently combined with Evidential Reasoning (ER) for inference generation. This approach addresses several limitations of traditional evaluation methods and provides a more reliable mechanism for E-Government performance assessment.

Dataset Description

The experimental evaluation was conducted using 20 representative E-Government assessment cases. Each case consists of five input variables (D_1–D_5) and one observed output value representing the E-Government performance level.

The dataset was constructed based on expert knowledge and previous E-Government assessment studies. Due to the limited availability of real-world assessment data, the current study utilizes a relatively small dataset to demonstrate the effectiveness of the proposed BRB training framework.

For training and testing purposes, the first nine cases were used for model training, while the remaining nine cases were used for testing and validation. The complete dataset used in this study is summarized in Table 1.

The given input values and observed outputs are represented as numerical values. The output data are subsequently transformed into belief structures through the input transformation process [32,33]. The resulting belief distributions are used during BRB inference and parameter training.

Table 1: Training Data Set (Input and Observed Output Values)

Set	Given Input					Observed Output(D)					
	D1	D2	D3	D4	D5	Numerical	Judgment with belief				
1	1	0.88	0.88	0.6	0.3	1	0	0.2667	0.7333	0	0
2	0.66	0.56	0.77	0.88	0.55	0.9	0	0.1333	0.8667	0	0
3	0.4	0.7	0.78	0.44	0.55	0.6	0	0	1	0	0
4	1	0.88	0.88	0.6	0.3	0.4	0	0.4	0.6	0	0
-	-	-	-	-	-	-	-	-	-	-	-
-	-	-	-	-	-	-	-	-	-	-	-
20	0.4	0.7	0.78	0.44	0.55	0.88	0	0	0.3333	0.6667	0

Table 2: Initial BRB and Weights

Sl No	Rule Weight	Excellent	Very Good	Good	Satisfactory	Poor
0	1	1	1	0	0	0
1	1	0.6	0.4	0	0	0
2	1	0.4667	0.5333	0	0	0
3	1	0.3333	0.6667	0	0	0
4	1	0.2	0.8	0	0	0
-	-	-	-	-	-	-
-	-	-	-	-	-	-
3124	1	0	0.6667	0.3333	0	0

5.1 Trained BRB and Rule Weights

We tried five different sets of training parameters for the training module. In the following discussion, we will use R1, R2, R3, R4, and R5 to represent the BRB training with different parameters, where R1: training with rule weights, antecedent attribute weights, consequent belief degrees and severity scores of the four different risk levels R2: training with rule weights, antecedent attribute weights, and consequent belief degrees R3: training with antecedent attributes weights and consequent belief degrees R4: training with rule weights and consequent belief degrees R5: training with belief degrees

Table 3: Training with Rule Weight, Antecedent Attribute, Consequent Belief Degree and Utility Score

Sl No	Rule Weight	Attribute Weight	Excellent	Very Good	Good	Satisfactory	Poor	Utility Score
0	0.0667	0	0.4667	0.5333	0	0	0	0
1	0.2	0.4	0.0667	0.9333	0	0	0	0.4
2	0.3333	0.5333	0	0.9333	0.0667	0	0	0.5333
3	0.4667	0.6667	0	0.8	0.2	0	0	0.6667
4	0.0667	0.8	0	0.8	0.3333	0	0	0.8
-	-	-	-	-	-	-	-	-
-	-	-	-	-	-	-	-	-
3124	0.8667	0.6667	0	0.3333	0.3333	0	0	0.6667

5.2 Comparison between Observed and Trained Result

This subsection shows the comparisons between the observed output and the trained output of our expert system.

5.3 Observed and Simulated Values of the Output

For the validation of trained approaches, the obtainable dataset is separated into two sets : training and testing. The training dataset is utilised to train parameters of the system. Consequently, this trained BRB system is applied to create the output which would be the input of test data. We use the first 9 sets of data from the Table 1 for training purpose. Then

Table 4: Training with Rule Weights, Antecedent Attribute Weights, And Consequent Belief Degrees

Sl No	Rule Weight	Attribute Weight	Excellent	Very Good	Good	Satisfactory	Poor
0	0.4667	0	0	0	0.4667	0.5333	0
1	0.3333	0.1333	0	0.1333	0.8667	0	0
2	0.2	0	0	0	0.7333	0.2667	0
3	0.6	0	0	0	0.6	0.4	0
4	0.7333	0	0	0	0.4667	0.5333	0
-	-	-	-	-	-	-	-
-	-	-	-	-	-	-	-
3124	0.55	0	0	0	1	0	0

Table 5: Training with antecedent attributes weights and consequent belief degrees

Sl No	Attribute Weight	Excellent	Very Good	Good	Satisfactory	Poor
0	0	0	0	0.8667	0.1333	0
1	0	0	0	0.4667	0.5333	0
2	0	0	0	0.3333	0.6667	0
3	0	0	0	0.2	0.8	0
4	0	0	0	0.0667	0.9333	0
-	-	-	-	-	-	-
-	-	-	-	-	-	-
3124	0	0	0	1	0	0

Table 6: Training with Rule Weights and Consequent Belief Degrees

Sl No	Rule Weight	Excellent	Very Good	Good	Satisfactory	Poor
0	0.9333	0	0.9333	0.0667	0	0
1	0.5333	0	0.5333	0.4667	0	0
2	0.4	0	0.4	0.6	0.6667	0
3	0.2667	0	0.2667	0.7333	0	0
4	0.1333	0	0.1333	0.8667	0	0
-	-	-	-	-	-	-
-	-	-	-	-	-	-
3124	0.2667	0	0.2667	0.7333	0	0

the second 9 sets of data are utilised for testing. Table 13 shows the observed and simulated outputs. Figure 7 and Figure 8 are the graphical representation of the observed output and the simulated output.

5.4 Receiver operating Characteristics (ROC) Curve

Assessing the predictive accuracy is essential for implementing and comparing models, technologies , or algorithms that generate predictions. The ROC curve offer a magnificent and visually appealing approach to summarize the prediction accuracy [7]. So, in our research, we utilized ROC curves for measuring the accuracy of the e-government evaluation produced by

Table 7: Training with Belief Degrees

Sl No	Excellent	Very Good	Good	Satisfactory	Poor
0	0	0	0	0.5333	0.4667
1	0	0	0.2	0.8	0
2	0	0	0	0.8	0.2
3	0	0	0	0.6667	0.3333
4	0	0	0	0.5333	0.4667
-	-	-	-	-	-
-	-	-	-	-	-
3124	0	0	0	0.9333	0.0667

Table 8: Training with Rule Weight, Antecedent Attribute, Consequent Belief Degree and Utility Score(R1)

Number of Training	1	2	3	4	5	6	7	8	9
Observed Output	1	0.9	0.6	0.4	0.6	0.4	0.7	0.8	0.4
Trained	0.99	0.8	0.57	0.35	0.58	0.5	0.75	0.77	0.34
Number of Training	10	11	12	13	14	15	16	17	18
Observed Output	0.77	0.4	0.3	0.33	0.4	0.56	0.7	0.6	0.88
Trained	0.76	0.44	0.28	0.29	0.35	0.51	0.66	0.55	0.78

Table 9: Training with Rule Weights, Antecedent Attribute Weights, And Consequent Belief Degrees (R2)

Number of Training	1	2	3	4	5	6	7	8	9
Observed Output	1	0.9	0.6	0.4	0.6	0.4	0.7	0.8	0.4
Trained	0.88	0.8	0.44	0.45	0.63	0.34	0.67	0.77	0.44
Number of Training	10	11	12	13	14	15	16	17	18
Observed Output	0.77	0.4	0.3	0.33	0.4	0.56	0.7	0.6	0.88
Trained	0.67	0.44	0.28	0.24	0.35	0.54	0.78	0.53	0.76

the Belief Rule-based Expert System Prototype.

For ROC analysis, the continuous BRB output was converted into a binary decision variable using an expert-defined threshold. Outputs above the threshold were considered satisfactory E-Government performance, whereas outputs below the threshold were considered unsatisfactory performance.

The ROC curve therefore evaluates the ability of the trained BRB models to discriminate between satisfactory and unsatisfactory assessment outcomes.

Table 10: Training with antecedent attributes weights and consequent belief degrees (R3)

Number of Training	1	2	3	4	5	6	7	8	9
Observed Output	1	0.9	0.6	0.4	0.6	0.4	0.7	0.8	0.4
Trained	0.77	0.67	0.67	0.45	0.63	0.34	0.67	0.88	0.41
Number of Training	10	11	12	13	14	15	16	17	18
Observed Output	0.77	0.4	0.3	0.33	0.4	0.56	0.7	0.6	0.88
Trained	0.67	0.44	0.28	0.24	0.41	0.5	0.73	0.61	0.76

Table 11: Training with Rule Weights and Consequent Belief Degrees (R4)

Number of Training	1	2	3	4	5	6	7	8	9
Observed Output	1	0.9	0.6	0.4	0.6	0.4	0.7	0.8	0.4
Trained	0.67	0.44	0.31	0.32	0.41	0.5	0.73	0.61	0.76
Number of Training	10	11	12	13	14	15	16	17	18
Observed Output	0.77	0.4	0.3	0.33	0.4	0.56	0.7	0.6	0.88
Trained	0.6	0.44	0.31	0.54	0.41	0.6	0.73	0.61	0.76

Table 12: Training with Belief Degrees (R5)

Number of Training	1	2	3	4	5	6	7	8	9
Observed Output	1	0.9	0.6	0.4	0.6	0.4	0.7	0.8	0.4
Trained	0.99	0.89	0.62	0.42	0.64	0.43	0.73	0.81	0.41
Number of Training	10	11	12	13	14	15	16	17	18
Observed Output	0.77	0.4	0.3	0.33	0.4	0.56	0.7	0.6	0.88
Trained	0.67	0.44	0.31	0.32	0.41	0.5	0.73	0.61	0.76

Table 13: Observed and Simulated Values of the Output

Test Data Set	10	11	12	13	14	15	16	17	18
Observed Output	0.77	0.4	0.3	0.33	0.4	0.56	0.7	0.6	0.88
Simulated Output	0.76	0.44	0.31	0.33	0.42	0.57	0.77	0.64	0.89

Following table 14 describes the ROC Curves.

The ROC analysis demonstrates that the trained BRB configurations achieved improved

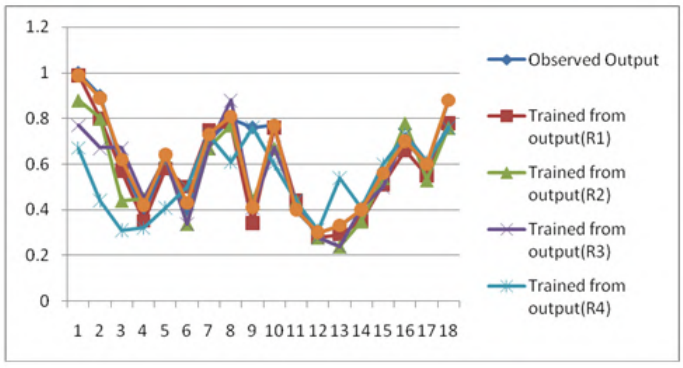
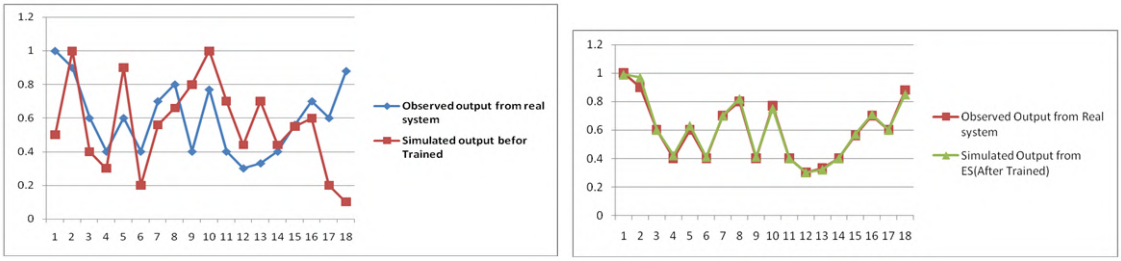


Figure 6: R1, R2, R3, R4, R5 vs Observed Output



(a) Result (Before Trained) (b) Result (After Trained)

Figure 7: Comparison of results before and after training

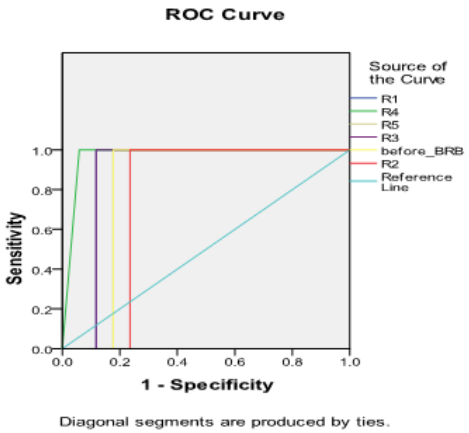


Figure 8: Result (ROC Curve)

discriminative performance compared with the baseline model. The observed AUC values indicate the effectiveness of the proposed learning framework in enhancing the predictive capability of the BRB-based expert system.

The results further suggest that appropriate training of BRB parameters can contribute

Table 14: Area Under the Curve

Test Result Variable(s)	Area	Std. Error	Asymptotic Sig.	Asymptotic 95% Confidence Interval	
				Lower Bound	Upper Bound
before-BRB	.824	.092	.289	.637	1.000
R1	.882	.078	.210	.000	1.000
R2	.765	.103	.386	.563	.966
R3	.882	.078	.210	.000	1.000
R4	.971	.044	.123	.000	1.000
R5	.824	.092	.289	.637	1.000

to more accurate E-Government performance assessment under uncertain environments. In addition to ROC and AUC analysis, future research will explore complementary evaluation metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Precision, Recall, and other validation measures to provide a broader assessment of system performance across different application scenarios.

Table 15: Comparison between Real System Output and Expert System Output

Observed Output(Real System Output)	Simulated Output After Trained (Expert System Output)
Excellent (0%)	Excellent (0.5%)
Very Good (0%)	Very Good (0%)
Good (73%)	Good (72%)
Satisfactory (27%)	Satisfactory (27.5%)
Poor (0%)	Poor (0%)

6 Conclusion and Future Work

6.1 Conclusion

Evaluating E-government initiatives is essential for determining the success or failure of projects and identifying areas that require improvement. In Bangladesh, most ministries have begun providing basic E-government services through their websites. However, the full benefits of these services cannot be realized until more advanced transactional features are implemented. Furthermore, challenges such as limited awareness among older citizens, bureaucratic practices, and socio-cultural barriers must also be addressed.

This study highlights the growing importance of E-government in enhancing convenience and accessibility for citizens. The younger generation, with higher internet literacy, shows a strong willingness to use online government services due to their efficiency in saving time and cost. Nevertheless, improving the quality and usability of E-government websites remains crucial, as poorly designed systems may discourage citizens from using these services.

In this research, a Belief Rule Base (BRB)-based Expert System (ES) was developed to evaluate E-government projects using data collected from three districts of Bangladesh across 21 variables. The proposed system enables continuous assessment of E-government initiatives and helps identify variables that negatively affect project performance, thereby supporting better policy formulation. Moreover, the BRB-based ES can be applied at different administrative

levels, such as divisional, district, upazila, and union levels, and may also be extended to evaluate ICT projects in sectors such as education, agriculture, and healthcare. The proposed framework can potentially be adapted for evaluating E-government systems in other countries.

6.2 Limitations

Several limitations should be acknowledged in the present study. The validation dataset is relatively small and partially based on simulated assessment cases. In addition, the implementation platform utilizes legacy technologies including Visual Basic 6.0 and Microsoft SQL Server, which may limit scalability and reproducibility in large-scale deployment environments. Furthermore, the current study focuses on district-level E-Government assessment in Bangladesh; therefore, the findings may require further validation before being generalized to different countries and administrative contexts. Future research will address these limitations through the use of larger real-world datasets, modern software implementation platforms, and cross-country validation studies.

6.3 Future Work

Future research will also focus on improving the system by expanding the knowledge base with a larger set of rules and enhancing the BRB training process for improved validation and accuracy. Additionally, an online BRB-based expert system will be developed to enable more robust, scalable, and real-time evaluation of E-government projects.

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QFT-Inspired Spectral Edge Filtering for Hybrid Visual Information Systems*

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Abstract

E-business information systems increasingly process document-style visual assets, including scanned forms, advertisements, reports, specifications, file folders, and news documents. Edge detection is a low-level but important operation for extracting layout boundaries, text-block structures, and other visual cues before higher-level indexing, classification, or analysis. This paper studies a quantum Fourier transform (QFT)-inspired spectral edge-filtering framework as an exploratory quantum-image-processing primitive for future hybrid quantum-classical visual information systems. Rather than claiming quantum speedup, the study analyzes how separable Sobel and Prewitt filters can be represented as one-dimensional row-wise and column-wise QFT-domain operations. Each row or column signal and its kernel are converted into amplitude-compatible quantum states through zero extension, power-of-two embedding, and normalization. QFT-domain statevectors are multiplied element-wise, re-normalized, reloaded into a circuit, and reconstructed through inverse QFT. The implementation is therefore explicitly positioned as a statevector-level quantum-circuit-inspired pipeline rather than a hardware-native quantum filtering circuit. Experiments are performed on ten class-wise RVL-CDIP document samples resized to the same 64×64 input resolution for both classical and QFT-based filtering. The zero-padded linear-aligned QFT variant preserves thresholded edge support very closely, with Edge-IoU of 0.9999 for Sobel and 1.0000 for Prewitt, while MSE, PSNR, and SSIM reveal remaining intensity-level discrepancies. Runtime results show that the current QFT statevector implementation is substantially slower than classical separable filtering, reinforcing the interpretation of the method as an analytical step toward hybrid quantum visual information processing rather than a demonstrated quantum-advantage algorithm.

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1 Introduction

Edge detection is a fundamental operation in image processing because boundaries, contours, and intensity transitions provide structural information for subsequent recognition, indexing, verification, and analysis stages [1]. In e-business information systems, visual data are not limited to product photographs or scanned receipts. Enterprise platforms also process forms, advertisements, reports, specifications, folders, invoices, and other document-style visual assets. In these settings, edge information can support document layout analysis, text-block localization, form boundary extraction, document image indexing, and visual quality assessment.

Classical gradient-based operators such as Sobel and Prewitt remain widely used because they are simple, interpretable, and computationally inexpensive for small kernels [2], [3]. However, the continuing evolution of information systems motivates the study of alternative computing models, including hybrid quantum-classical processing and quantum-inspired visual analytics. The purpose of this work is not to replace mature classical document-analysis pipelines. Instead, it examines how a familiar visual primitive can be expressed and evaluated within a QFT-based spectral circuit framework.

A transform-domain view is natural because convolution or correlation can be reformulated through spectral-domain multiplication followed by an inverse transform [1]. The quantum Fourier transform is the quantum counterpart of the discrete Fourier transform and is a central primitive in quantum algorithms [4], [5]. This paper investigates a statevector-level QFT-domain formulation in which one-dimensional image signals and one-dimensional kernels are encoded into amplitude-compatible quantum states, transformed by QFT, combined in the spectral domain, and reconstructed by inverse QFT.

The contributions are as follows. First, we connect QFT-based edge filtering to document-style visual information processing in e-business information systems. Second, we formulate a separable QFT-domain filtering pipeline for Sobel and Prewitt operators while clarifying the difference between circular spectral filtering and zero-padded linear-aligned QFT filtering, hereafter denoted as QFT-linear. Third, we report resource and runtime behavior to avoid unsupported efficiency or quantum-advantage claims. Fourth, we evaluate the method on class-wise RVL-CDIP document samples using MSE, PSNR, SSIM, Edge-IoU, Edge-F1, and runtime.

2 Related Works

The Sobel and Prewitt operators are classical discrete differentiation filters for estimating local intensity gradients [2], [3]. Their continued use is due to their interpretability, small kernel size, and simple implementation. Both operators are separable, allowing a two-dimensional filter to be implemented through two successive one-dimensional filters. This property reduces the cost of direct spatial filtering and also provides a convenient bridge to row-wise and column-wise spectral processing [1]. The QFT is one of the central building blocks of quantum computation and can be implemented using Hadamard gates, controlled phase rotations, and qubit-reversal operations [4], [5], [12]. Quantum image processing has explored image representation, transformation, and manipulation using quantum states [7], [8]. Representative image encodings such as FRQI and NEQR provide general quantum-state models for storing pixel or grayscale information [9], [10]. The present work is narrower: it does not propose a general quantum image representation, but studies how separable edge filtering can be expressed as a QFT-domain spectral operation over amplitude-encoded one-dimensional signals. E-Business Information Systems increasingly depend on visual information, including scanned business documents, identity and transaction records, product images, logistics labels, industrial inspection imagery, and service-related image streams.

Edge filtering is a low-level but important primitive in these pipelines because it supports layout segmentation, contour extraction, defect boundary localization, and structure-preserving preprocessing. A QFT-based formulation of such a primitive is relevant to EBISION when it is interpreted as an exploratory component of future hybrid quantum-classical visual analytics rather than as a standalone computer vision speedup claim. The security and privacy implications of this hybrid implementation also require careful interpretation. The present implementation assumes a trusted hybrid computing environment. Since the QFT-domain product vector is extracted, normalized classically, and reloaded into a new circuit, the current method does not provide end-to-end privacy preservation. For security-sensitive e-business settings, practical deployment would require additional mechanisms such as trusted execution, secure hardware isolation, blind quantum computation, encrypted-domain processing, or carefully designed quantum-safe document authentication protocols. These directions are beyond the current implementation but motivate the relevance of analyzing where classical-quantum handoff points occur.

3 Methods

3.1 Overview

Let $I \in \mathbb{R}^{H\times W}$ denote a document image. Instead of flattening the full image into a single vector, the proposed method exploits the separability of Sobel and Prewitt filters and processes the image through one-dimensional stages. For a row or column signal $x \in \mathbb{R}^n$ and a one-dimensional kernel $k \in \mathbb{R}^m$, the QFT-inspired spectral filtering stage is conceptually written as

$$y = \mathrm{IQFT}(\mathrm{QFT}(x) \odot \mathrm{QFT}(k)),$$

where $\odot$ denotes element-wise multiplication in the spectral domain.

In the actual implementation, x and k are first zero-padded, embedded into power-of-two-length amplitude vectors, and normalized before QFT is applied.

3.2 Separable Sobel and Prewitt filtering

The proposed framework is evaluated using Sobel and Prewitt edge filters because both are classical gradient operators and can be decomposed into one-dimensional derivative and smoothing components. The horizontal and vertical Sobel operators are written as

$$K_{S_x} = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}, \qquad K_{S_y} = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}.$$

They can be factorized as

$$K_{S_x} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} [-1 \quad 0 \quad 1], \qquad K_{S_y} = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} [1 \quad 2 \quad 1].$$

Similarly, the Prewitt operators are

$$K_{P_x} = \begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix},\ K_{P_y} = \begin{bmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}.$$

They are decomposed as

$$K_{P_x} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} [-1 \quad 0 \quad 1], \qquad K_{P_y} = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} [1 \quad 1 \quad 1].$$

This separability allows a two-dimensional edge filter to be implemented as successive one-dimensional row-wise and column-wise operations. Let $\mathcal{Q}_k^{row}(\cdot)$ denote the proposed QFT-linear one-dimensional filtering operation applied independently to each row with kernel k, and let $\mathcal{Q}_k^{col}(\cdot)$ denote the same operation applied independently to each column. For Sobel filtering, using the derivative vector $d = [-1\ ,0\ ,1]$ and the smoothing vector $s_S = [1, 2, 1]$, the horizontal and vertical gradient responses are computed as

$$G_x^S = \mathcal{Q}_{s_S}^{col}\left(\mathcal{Q}_d^{row}(I)\right), \qquad G_y^S = \mathcal{Q}_{s_S}^{row}\left(\mathcal{Q}_d^{col}(I)\right).$$

For Prewitt filtering, the same derivative vector $d = [-1, 0, 1]$ is used with the smoothing vector $s_P = [1, 1, 1]$:

$$G_x^P = \mathcal{Q}_{s_P}^{col}\left(\mathcal{Q}_d^{row}(I)\right), \qquad G_y^P = \mathcal{Q}_{s_P}^{row}\left(\mathcal{Q}_d^{col}(I)\right).$$

The final edge-response maps are then computed as gradient magnitudes:

$$E^S = \sqrt{(G_x^S)^2 + (G_y^S)^2},\ E^P = \sqrt{(G_x^P)^2 + (G_y^P)^2}.$$

Thus, the image is not flattened into a single global vector. Instead, QFT is used as a one-dimensional spectral filtering primitive inside a separable two-dimensional edge-filtering pipeline.

3.3 Correlation, convolution, and boundary alignment

A key implementation detail is the distinction between correlation, convolution, and finite-length spectral multiplication. The classical baseline in this paper uses same-padding correlation, which is common in image-processing implementations of Sobel and Prewitt filters. In correlation, the kernel is applied without flipping, whereas convolution applies a reversed kernel.

Finite Fourier-domain multiplication naturally produces circular behavior over the chosen transform length. Therefore, a direct spectral product may not match the same-padding linear correlation used by the classical reference. To align the QFT-domain operation with the classical baseline, this paper uses a zero-padded linear-aligned QFT variant, denoted as QFT-linear. For a one-dimensional signal of length n and a kernel of length m, the transform length is chosen as

$$L = 2^{\lceil \log_2(n+m-1) \rceil}.$$

The signal is zero-padded to length L, and the kernel is embedded into the same transform length. Although the classical reference is correlation, the spectral implementation matches it by using the equivalent convolution form with a reversed kernel. After inverse QFT, the corresponding n-length same-region response is cropped to match the same-padding classical reference. This alignment ensures that the classical and QFT-based edge maps have the same visible input and output resolution. In the

RVL-CDIP experiment, both pipelines operate on the same 64×64 resized document images, and QFT padding is used only internally for one-dimensional state preparation and reconstruction.

3.4 Preprocessing before quantum encoding

For each row or column signal $x \in \mathbb{R}^n$ and one-dimensional kernel $k \in \mathbb{R}^m$, the vectors must be converted into amplitude-compatible quantum states. Using the transform length $L = 2^{\lceil \log_2(n+m-1) \rceil}$, the signal and kernel are embedded into L-dimensional vectors $\tilde{x}$ and $\tilde{k}$. The embedded vectors are then normalized as

$$|x\rangle = \frac{\tilde{x}}{\|\tilde{x}\|_2}, \qquad |k\rangle = \frac{\tilde{k}}{\|\tilde{k}\|_2}.$$

The original norms $\|\tilde{x}\|_2$ and $\|\tilde{k}\|_2$ are retained for scale restoration after inverse QFT. If either norm is zero, the corresponding filtering response is set to zero. This preprocessing step is necessary because a valid amplitude-encoded quantum state must have unit norm and length 2^q, where $q = \log_2 L$ is the number of qubits used for the one-dimensional signal.

3.5 QFT-domain filtering and scale restoration

After amplitude encoding, QFT is applied to the signal and kernel states:

$$|X\rangle = QFT|x\rangle, \ |K\rangle = QFT|k\rangle.$$

At the statevector level, spectral filtering is implemented by component-wise multiplication:

$$F = X \odot K.$$

Since F is generally not a unit-norm quantum state, it is normalized before inverse QFT:

$$|\hat{F}\rangle = \frac{F}{\|F\|_2}.$$

The reconstructed spatial response is obtained by

$$\hat{y}_{norm} = IQFT(|\hat{F}\rangle).$$

Finally, the original signal and kernel scales are restored as

$$\hat{y} = \hat{y}_{norm} \cdot \|\tilde{x}\|_2 \cdot \|\tilde{k}\|_2 \cdot \|F\|_2.$$

The real-valued part of $\hat{y}$ is used as the one-dimensional filtering response, and the response is cropped back to the original signal length n. This procedure is repeated along rows and columns according to the separable Sobel or Prewitt decomposition. Because the current implementation requires statevector extraction, element-wise multiplication, re-normalization, and circuit re-initialization, it should be interpreted as a QFT-inspired statevector pipeline rather than a hardware-native quantum filtering circuit.

3.6 Implementation model and resource implications

The present pipeline consists of amplitude preparation, QFT, statevector extraction, spectral multiplication, product normalization, re-initialization, and IQFT. The QFT circuit for q qubits has a compact gate structure compared with a full dense unitary transform, but the overall pipeline is dominated by data loading, statevector-level manipulation, and reloading in the current simulation setting. Therefore, the method should not be described as computationally faster than classical separable filtering unless runtime and resource data support such a claim.

The implementation is analyzed relative to two main viewpoints used in the final RVL-CDIP experiment: direct classical separable filtering and the zero-padded linear-aligned QFT statevector pipeline. Circular QFT filtering is discussed only as a boundary-mismatch case because finite-length spectral multiplication naturally induces circular behavior when zero-padding and cropping are not handled consistently. The final experiment reports the linear-aligned variant because it is the most appropriate counterpart to the same-padding classical correlation baseline.

Table 1: Resource and runtime comparison on RVL-CDIP 10-class samples at 64×64 input resolution. Runtimes are reported as mean ± standard deviation per image.

Method	Main operations	*Expected cost driver*	Hardware realism	Runtime
Classical separable filtering	Row/column same-padding correlation	*O(HWm), m = 3*	Direct classical baseline	Sobel: 0.101±0.020 s Prewitt: 0.104 ± 0.022 s
QFT zero-padded linear-aligned filtering	State preparation, QFT, spectral product, reloading, IQFT	*State preparation and statevector manipulation*	Simulation-level	Sobel: 10.748±0.751 s Prewitt: 10.355 ± 1.435 s

4 Experiments

4.1 Dataset and sampling protocol

Experiments are conducted on RVL-CDIP document-style visual samples. Ten classes are used in the reported visualization and metric table: letter, form, email, handwritten document, advertisement, scientific report, scientific publication, specification, file folder, and news article. One representative image is selected from each class using a fixed streaming and sampling procedure. Each selected document image is converted to grayscale and resized to 64×64 before filtering. Both the classical and QFT-based pipelines operate on the same resized input resolution. QFT uses power-of-two padding only internally for one-dimensional state preparation and reconstruction; the final edge maps are cropped back to the same 64×64 size before visualization and metric computation.

4.2 Evaluation metrics

The classical separable edge map is used as the reference, and the QFT-based edge map is evaluated against it. Thus, the reported metrics do not compare the original document image with an edge map. They compare classical Sobel with QFT Sobel and classical Prewitt with QFT Prewitt under the same input resolution. MSE and PSNR measure pixel-level intensity discrepancy, SSIM measures structural similarity in the edge-response maps, and Edge-IoU and Edge-F1 measure binary edge-support agreement after thresholding. Edge-F1 is computed with a one-pixel tolerance to account for small boundary localization differences. Runtime is measured separately for classical separable filtering and QFT-based filtering.

4.3 Baselines and ablations

The baseline is direct classical separable Sobel or Prewitt filtering with same-padding correlation. The QFT result uses the zero-padded linear-aligned variant because it is the most appropriate comparison to the classical same-padding reference. Circular QFT filtering is discussed as a source of boundary mismatch, but the final RVL-CDIP experiment reports the linear-aligned variant.

5 Results and Analysis

5.1 Quantitative results

Table 2 summarizes the quantitative comparison between classical filtering and QFT-based filtering on the ten RVL-CDIP class-wise samples. Under the selected thresholding protocol, the QFT-linear variant achieves nearly identical binary edge support, with Edge-IoU of 0.9999 for Sobel and 1.0000 for Prewitt, and Edge-F1 of 1.0000 for both filters. However, MSE, PSNR, and SSIM show that the grayscale edge-response intensity is not reproduced exactly. This supports the interpretation that the current QFT-inspired pipeline is contour-support preserving but not pixel-exact.

Table 2: Classical versus QFT-linear edge-map similarity on RVL-CDIP 10-class samples at 64×64 input resolution. Values are mean ± standard deviation over ten class-wise samples.

Filter	QFT variant	MSE	PSNR (dB)	SSIM	Edge-IoU	Edge-F1
Sobel	linear	0.077±0.028	11.34±1.44	0.275±0.278	0.9999±0.0003	1.000±0.000
Prewitt	linear	0.086±0.031	10.87±1.43	0.272±0.278	1.0000±0.0000	1.000±0.000

5.2 Qualitative results

Figures 1 and 2 show class-wise RVL-CDIP examples. Each figure contains ten columns corresponding to the ten document classes. The first row shows the original resized document images, the second row shows classical separable filtering, and the third row shows QFT-based filtering under the same 64×64 input resolution. The QFT maps retain major page boundaries, text-block regions, advertisement structures, and rectangular layout cues, but local brightness and edge thickness differ from the classical maps. These visual observations are consistent with Table 2: binary edge support is preserved, while intensity-level metrics remain moderate.

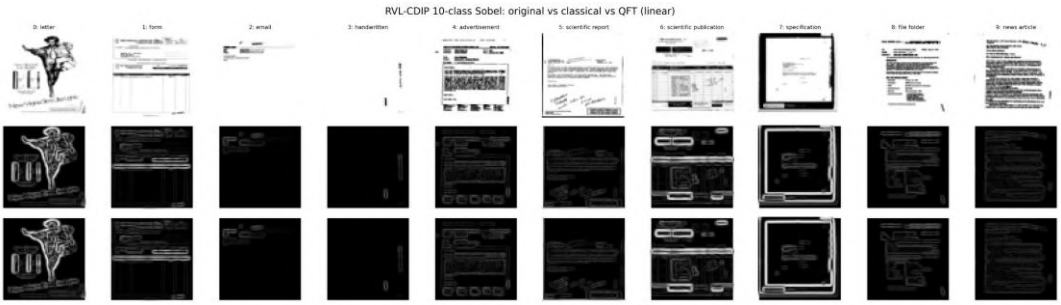

Figure 1: Class-wise RVL-CDIP Sobel edge-filtering results under the same 64×64 input resolution. The first row shows original document images, the second row shows classical separable Sobel filtering, and the third row shows QFT-linear Sobel filtering.

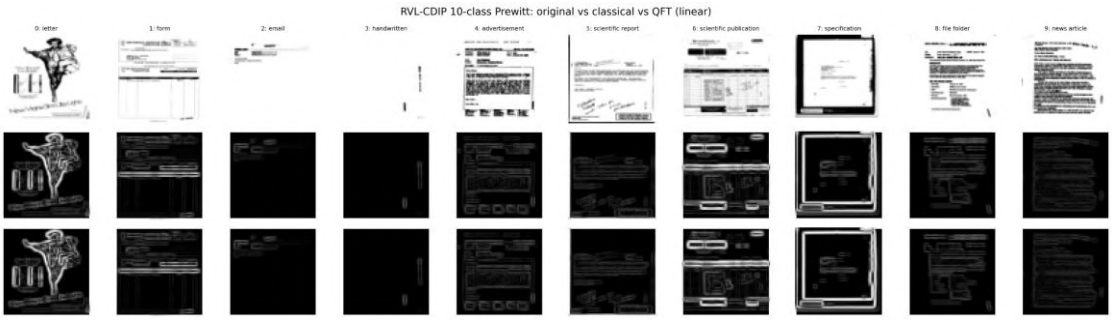

Figure 2: Class-wise RVL-CDIP Prewitt edge-filtering results under the same 64×64 input resolution. The first row shows original document images, the second row shows classical separable Prewitt filtering, and the third row shows QFT-linear Prewitt filtering.

5.3 Interpretation

The RVL-CDIP result is relevant to the scope of e-business information systems because the evaluated samples are document-style visual assets, including forms, advertisements, reports, specifications, file folders, and news documents, rather than generic benchmark images. Nevertheless, the experiment should not be interpreted as document understanding, OCR, or document classification. The goal is narrower: to examine whether a QFT-inspired quantum image-processing primitive can preserve layout-level edge support in document visual data. The low-to-moderate PSNR and SSIM values indicate that the QFT pipeline does not reconstruct the classical grayscale edge response exactly. The high Edge-IoU and Edge-F1 values indicate that, after thresholding, coarse edge support is strongly aligned with the classical reference.

Runtime results further clarify the implementation status. On 64×64 inputs, classical filtering takes roughly 0.10 s per image, while QFT statevector filtering takes roughly 10-11 s per image. The present implementation therefore does not demonstrate computational efficiency. Instead, it identifies the computational bottlenecks of amplitude preparation, statevector access, re-normalization, and circuit reloading, which are central considerations for future hybrid quantum-classical visual information systems.

6 Conclusion

This paper presented a QFT-inspired spectral edge-filtering framework for document-style visual information processing in e-business information systems. By exploiting the separability of Sobel and Prewitt filters, edge detection was expressed as successive one-dimensional QFT-domain operations over amplitude-encoded row and column signals.

Classical and QFT-based edge maps were computed from the same 64×64 input resolution. The QFT-linear variant preserved thresholded edge support very closely, but it did not exactly reproduce the grayscale intensity distribution of the classical edge maps. The method should therefore be understood as a contour-support-preserving quantum-circuit-inspired approximation rather than as a pixel-exact edge detector.

The runtime analysis also shows that the current statevector implementation is slower than classical separable filtering. This confirms that the present work does not demonstrate quantum speedup or hardware-native quantum image filtering. Its contribution is analytical: it clarifies how a classical edge primitive can be mapped into a QFT-domain hybrid pipeline, where approximation gaps arise, and why state preparation and statevector manipulation remain major bottlenecks. Future work should evaluate

larger real-world document and product-image datasets, investigate hardware-realistic implementations of spectral multiplication, and study privacy-preserving or secure hybrid architectures for sensitive business visual information.

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Towards an Intelligent Video Filtering System for Age-Appropriate Media Consumption*

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Abstract

The rapid growth of online video platforms has increased the need for effective mechanisms to protect young audiences from exposure to inappropriate content. Traditional manual moderation approaches are often labor-intensive, costly, and difficult to scale to the large volumes of multimedia content uploaded daily. This study proposes an AI-based automated video screening system for youth-appropriate media consumption using a hybrid deep learning framework that combines a pretrained ResNet-50 Convolutional Neural Network (CNN) and a Long Short-Term Memory (LSTM) network. The proposed architecture leverages ResNet-50 to extract high-level spatial features from video frames, while the LSTM captures temporal dependencies across sequential frames to improve contextual understanding of video content. The system is integrated into a browser extension that performs near real-time content moderation through server-side inference and local caching of classification results using IndexedDB. Experimental evaluation was conducted on a combined dataset consisting of YouTube-8M, NSFW video samples, and a custom youth-oriented dataset containing safe and unsafe content. Results demonstrate that the proposed ResNet50–LSTM framework outperforms conventional keyword-based and metadata-based filtering approaches, achieving 93.2% precision, 91.6% recall, 92.4% F1-score, and 92.3% accuracy. The findings highlight the effectiveness of combining spatial and temporal visual analysis for automated content moderation. This work contributes to the development of scalable,

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reliable, and practically deployable AI-driven moderation systems that support safer digital media environments for youth audiences.

1 Introduction

The rapid growth of online video platforms has fundamentally transformed how young audiences consume digital media. Children and adolescents increasingly rely on streaming services, video-sharing platforms, and social media for entertainment, education, and social interaction. However, the vast availability of user-generated and professionally produced multimedia content presents a significant challenge: protecting youth from exposure to inappropriate, harmful, or age-restricted material. Traditional content moderation approaches, including manual review and keyword-based filtering, are often insufficient due to the enormous scale, diversity, and continuously evolving nature of online video content.

Automated video screening systems have emerged as a promising solution to this challenge. Recent advances in artificial intelligence (AI), particularly in computer vision, natural language processing (NLP), and multimodal learning, enable automated systems to analyze multiple dimensions of video content, including visual frames, spoken language, textual metadata, and contextual information. Unlike conventional rule-based methods, AI-driven approaches can identify more nuanced forms of inappropriate content, such as violent imagery, offensive language, explicit scenes, harmful behaviors, and implicit contextual risks. Moreover, such systems can operate at scale and near real time, making them highly suitable for modern digital media platforms where vast volumes of content are uploaded continuously.

From the perspective of E-Business Information Systems, intelligent content moderation plays an increasingly critical role in supporting sustainable and trustworthy digital ecosystems. Online video platforms are not only media distribution channels but also major digital business infrastructures that rely on user trust, regulatory compliance, and responsible platform governance. AI-powered moderation systems contribute to platform safety, enhance user experience, reduce operational moderation costs, and support age-appropriate recommendation mechanisms. As digital businesses increasingly adopt AI-driven decision-making systems, ensuring responsible governance, fairness, transparency, and accountability in automated moderation becomes essential.

This study proposes an AI-based automated video filtering system for youth-appropriate media consumption using a hybrid deep learning framework for visual content moderation. The proposed architecture integrates a pretrained Convolutional Neural Network (CNN) based on ResNet-50 for spatial feature extraction and a Recurrent Neural Network (RNN) using Long Short-Term Memory (LSTM) for temporal sequence modeling of video frames. This hybrid architecture enables the system to capture both frame-level visual semantics and temporal contextual dependencies, improving the detection of inappropriate content such as violence, explicit imagery, harmful behaviors, and disturbing scenes. Compared with traditional keyword-based and metadata-only filtering approaches, the proposed framework provides significantly enhanced detection capability for visually complex and context-sensitive inappropriate content scenarios.

Beyond technical performance, this research also recognizes important practical and ethical considerations associated with AI-driven moderation systems. False positives may result in the unintended blocking of legitimate educational, entertainment, or culturally sensitive content, while

biases in training datasets may lead to unfair classification outcomes across different content domains, including animated or culturally specific media. Furthermore, explainability remains a critical concern, as platform operators, regulators, parents, and users increasingly require transparent and accountable decision-making from automated moderation systems that influence content accessibility. Addressing these concerns is essential to maintaining an appropriate balance between youth protection, freedom of expression, inclusivity, and fairness within digital media ecosystems.

Experimental evaluation using benchmark multimedia datasets and a custom youth-oriented video dataset demonstrates that the proposed ResNet50–LSTM hybrid framework achieves superior classification performance compared with conventional baseline approaches, including keyword-based filtering, metadata-based screening, and single-model architectures. The results show substantial improvements in precision, recall, F1-score, and overall accuracy, highlighting the effectiveness of combining spatial and temporal visual analysis for automated content moderation.

This work contributes to the advancement of intelligent E-Business Information Systems by presenting a practically deployable AI moderation framework capable of supporting safer digital media consumption, responsible platform governance, and scalable automated content screening. The integration of browser-based deployment with server-side AI inference and local caching mechanisms further demonstrates the feasibility of implementing efficient, near real-time moderation solutions for modern online video ecosystems.

2 Related Theory

2.1 Theoretical Foundations of AI-Based Video Content Moderation

The development of an Automated Video Screening System for Youth-Appropriate Content is grounded in several interdisciplinary theoretical and technological domains, including media regulation, computer vision, machine learning, and automated content moderation. These foundations provide the conceptual and technical basis for detecting and classifying inappropriate video content that may negatively affect young audiences.

Content moderation on digital platforms is strongly influenced by media regulation frameworks and child protection guidelines, which emphasize safeguarding children from exposure to violent, explicit, or otherwise harmful material [5]. Traditional moderation approaches, such as manual review and keyword-based filtering, often struggle to scale effectively and frequently fail to identify contextual or visually implicit harmful content. Consequently, automated moderation systems have emerged as a practical solution for large-scale content screening, supporting age-appropriate content classification and safer digital environments.

A core enabling technology of automated video moderation is computer vision, which allows machines to interpret and analyze visual information contained in video frames [6], [7]. Modern computer vision systems are primarily based on deep learning architectures, particularly Convolutional Neural Networks (CNNs), which can automatically learn hierarchical representations of objects, actions, textures, and scene characteristics. Well-established CNN architectures, including VGGNet, ResNet, and Inception, have demonstrated strong performance in image and video classification tasks. In the proposed system, a pretrained ResNet-50 model serves as the primary visual feature extractor, enabling the identification of potentially inappropriate visual content such as violence, nudity, harmful behaviors, and disturbing imagery.

Since video content consists of sequential frames rather than isolated images, temporal information must also be considered. Recurrent Neural Networks (RNNs), particularly Long Short-Term Memory

(LSTM) networks, are widely used for modeling temporal dependencies and motion patterns across video sequences. By analyzing frame-to-frame relationships, LSTM networks can capture contextual behaviors that may not be observable from individual frames alone, thereby improving the detection of unsafe visual events.

The classification process is further grounded in supervised machine learning theory, where models learn discriminative patterns from labeled training data to predict whether content is safe or unsafe [3], [11]. Numerous machine learning algorithms, including Support Vector Machines (SVMs), Random Forests, and deep neural networks, have been successfully applied to multimedia classification and content moderation tasks. The proposed hybrid ResNet50–LSTM architecture combines spatial feature learning with temporal sequence modeling to improve classification accuracy and robustness. Furthermore, the precision–recall tradeoff is an important consideration in content moderation systems, as maximizing harmful content detection must be balanced against minimizing false positives that could unnecessarily restrict legitimate content.

Although video content may contain visual, audio, and textual information, the current study focuses primarily on visual content analysis. Nevertheless, the broader field of multimedia moderation is influenced by multimodal learning principles, which suggest that combining multiple information sources can provide richer contextual understanding and improve classification performance [9]–[11]. Future enhancements may incorporate audio signals, speech transcripts, and metadata to further improve moderation accuracy and contextual awareness.

Collectively, these theoretical foundations support the development of an intelligent video moderation framework capable of automatically identifying inappropriate content while maintaining scalability, reliability, and practical applicability for youth-oriented digital media platforms.

2.2 Related work

The Internet has become an essential part of children's daily lives, serving as a major source of entertainment, education, and social engagement through online video and streaming platforms [2]. However, despite the growing availability of child-oriented digital services, many platforms continue to host inappropriate or harmful content, often with insufficient regulatory oversight. Traditional manual content review is not only labor-intensive and costly but also impractical given the massive volume of continuously uploaded multimedia content [1]. To address this challenge, previous research introduced The Eye, a cloud-based streaming platform designed to enhance parental control by allowing customized content flagging for safer, age-appropriate viewing experiences [2]. The system employed a hybrid Convolutional Neural Network–Recurrent Neural Network (CNN–RNN) architecture trained on a specialized dataset for inappropriate content classification, achieving an accuracy of 67% on test data while demonstrating scalability through deployment on Amazon Web Services (AWS) cloud infrastructure [2].

As children's dependence on online platforms continues to grow, specialized services such as YouTube Kids have emerged to provide safer digital environments for younger audiences [4]. Although such platforms operate under child protection guidelines and content regulations, inappropriate material can still bypass filtering mechanisms. Conventional moderation approaches on these platforms often rely heavily on metadata attributes such as titles, tags, or descriptions, which can be easily manipulated or insufficient for accurately identifying harmful content. To overcome these limitations, more advanced real-time video screening methods have been proposed. One promising approach applies the thin-slicing theory, in which short video and audio segments are randomly sampled for efficient yet representative analysis. This method enables real-time moderation while preserving temporal coverage across video content. Audio segments are processed through Automatic Speech Recognition (ASR) to transcribe

spoken language for linguistic analysis, while acoustic event detection identifies contextual cues such as explosions, aggressive speech, or other potentially harmful sounds. Simultaneously, sampled video frames undergo visual analysis to detect violent scenes, explicit imagery, or other inappropriate content. This multimodal analysis framework represents a significant advancement over traditional metadata-based filtering by integrating visual, audio, and textual intelligence to improve detection robustness and contextual understanding [4].

Building upon these advancements, this research aims to design and implement an automated video screening system for youth-appropriate content using advanced multimedia analysis, contextual content understanding, and AI-driven decision-making. The proposed system focuses on leveraging multimodal learning techniques to improve content moderation accuracy while incorporating ethical considerations such as fairness, contextual sensitivity, and responsible AI governance. Ultimately, this study seeks to contribute to the development of scalable, adaptive, and reliable digital safety solutions that support safer media consumption for youth audiences across diverse online content ecosystems.

3 Methodology

3.1 Definition of Inappropriate Content and System Overview

In this study, inappropriate or unsafe video content refers to any visual, auditory, or textual material that may negatively affect young audiences, based on international media classification principles and Thailand's Film and Video Act. For the purposes of dataset annotation and system classification, such content is labeled as unsafe. The labeling criteria include several categories of harmful content: violence, characterized by the presence of blood, physical aggression, weapons, explosions, or depictions of harm; nudity and sexual content, including partial or full nudity, sexual gestures, or adult-themed material; offensive language, such as profanity, hate speech, or verbally abusive expressions identified through transcript analysis; substance use and harmful behaviors, including smoking, alcohol consumption, or self-harm actions; and fear-inducing or disturbing imagery, such as horror scenes, threatening situations, or distressing visual effects deemed unsuitable for children. To ensure labeling consistency and reliability, video segments were independently annotated by multiple reviewers, achieving an inter-annotator agreement exceeding 85% Cohen's κ across visual and audio modalities.

Based on this content definition, the proposed automated video screening system is designed to detect and block inappropriate content on online video platforms, with YouTube serving as the primary use case. The overall system architecture consists of four major components: the target video platform, a browser extension, a local database, and a classification server. The target platform, represented by YouTube, serves as the source of video content for analysis. The browser extension functions as the user-facing component responsible for monitoring accessed video content, retrieving metadata such as the video identifier and title, and coordinating communication with the classification system. It also modifies the platform interface to restrict access when unsafe content is detected.

To improve efficiency and reduce redundant computation, the system utilizes Chrome IndexedDB as a local storage mechanism for caching previously analyzed video identifiers and their corresponding classification labels. When a user accesses a video, the extension first checks the local database to determine whether the content has already been processed. If a prior classification result exists, the stored decision is applied immediately without requiring server-side analysis.

If the video has not been previously analyzed, relevant content information is transmitted to the server, which performs the core classification process using an AI-based content moderation model. The

server applies the trained deep learning classifier to assess whether the content is appropriate for youth audiences based on the predefined unsafe content criteria. Once classification is complete, the resulting label is returned to the browser extension and stored in the local database for future reference. If the content is classified as unsafe, the extension automatically blocks playback by altering the webpage interface. The overall architecture enables efficient, scalable, and near real-time automated content moderation for safer digital media consumption. Details of the proposed framework are illustrated in Figure 1.

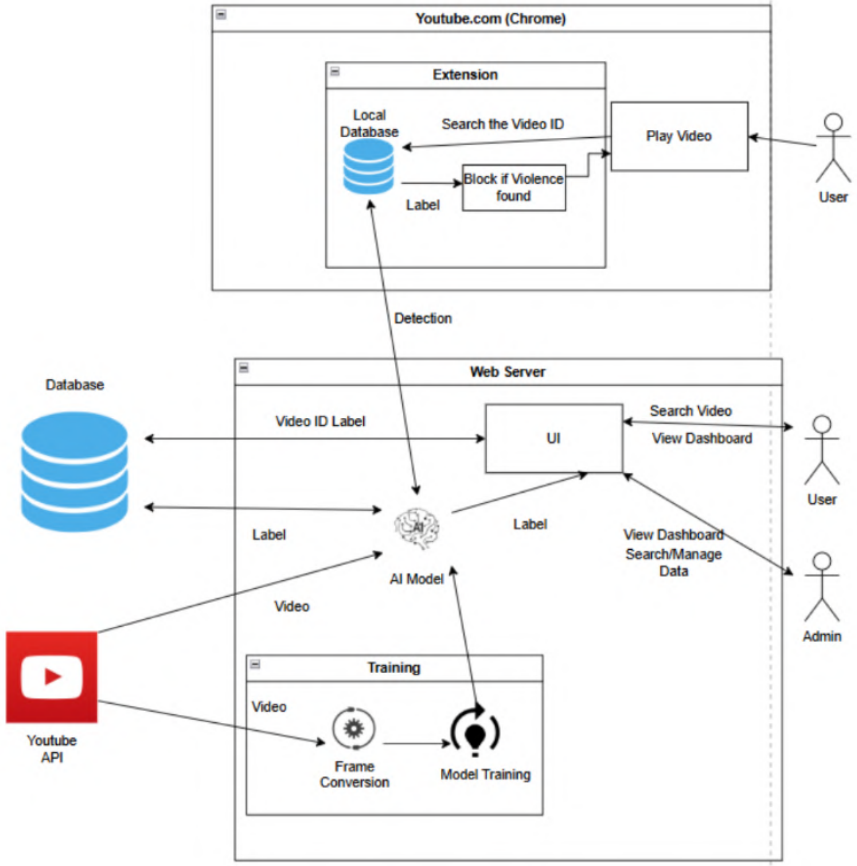

Figure 1: The proposed system

3.2 Framework for model training and evaluation

This study employs a hybrid deep learning framework that combines a pretrained Convolutional Neural Network (CNN) and a Recurrent Neural Network (RNN) for automated video content classification. Specifically, the proposed model utilizes ResNet-50 as the visual feature extraction backbone to capture high-level spatial representations from video frames, while a Long Short-Term Memory (LSTM) network is used to model temporal dependencies across sequential frames within each video segment. The primary objective is to detect inappropriate visual content for youth audiences, including violent, explicit, or harmful scenes, through efficient analysis of short video clips. To support practical deployment in browser-based content moderation, the system is designed with computational

optimization mechanisms, including server-side inference and local caching of classification results, enabling near real-time automated screening while minimizing redundant processing overhead.

The overall framework consists of five major stages: (1) data collection and labeling, (2) preprocessing, (3) feature extraction and temporal modeling, (4) model training, and (5) evaluation and system integration.

1. **Data Collection and Labeling**. Video clips were collected from publicly available multimedia datasets, including YouTube-8M, NSFW datasets, and a custom youth-oriented dataset. Each video segment was manually annotated as either safe or unsafe according to predefined content moderation criteria based on youth protection guidelines, including violence, nudity, offensive behavior, and harmful visual content.
2. **Preprocessing**. To standardize input data, all videos were resized to a spatial resolution of 224 × 224 pixels and resampled at 25 frames per second (fps). Each video was divided into non-overlapping 1-second clips, producing sequences of 25 consecutive frames per sample. To improve generalization and robustness, data augmentation techniques were applied during training, including random horizontal flipping, brightness adjustment, and rotation transformation. These preprocessing steps help reduce overfitting and improve model performance across diverse video scenarios.
3. **Feature extraction.** The proposed classification framework combines spatial feature extraction with temporal sequence learning. **Spatial feature extraction**, for visual analysis, each video frame is processed using a pretrained ResNet-50 convolutional neural network, initialized with ImageNet weights.ResNet-50 was selected due to its strong performance in visual recognition tasks and its residual learning architecture, which enables effective extraction of high-level semantic features while mitigating vanishing gradient issues.For each input frame: size: 224 × 224 × 3, output feature vector: 2048-dimensional extracted from the global average pooling layer. These frame-level feature vectors capture important spatial cues associated with inappropriate visual content, including violent scenes, explicit imagery, and harmful activities. **Temporal sequence modeling**, because video content contains temporal dependencies across frames, sequential modeling is performed using a Long Short-Term Memory (LSTM) network. The LSTM receives the ordered sequence of frame-level feature vectors and learns temporal contextual patterns that may indicate unsafe behavior over time. LSTM configuration composed of these properties, hidden units: 128, dropout rate: 0.3, sequence length: 25 frames. The final hidden representation is passed to a dense classification layer with softmax activation to predict which are Safe and Unsafe.
4. **Model training**. The hybrid ResNet50–LSTM framework was trained end-to-end using the Adam optimizer with an initial learning rate of 0.001, a batch size of 32, and 50 epochs. The categorical cross-entropy loss function was used to minimize classification error. Early stopping was applied based on validation loss to avoid overfitting. All models were implemented using TensorFlow 2.15. The hybrid ResNet50–LSTM is trained on 80% of the data, validated on 10%, and tested on the remaining 10%. Cross-validation ensures robustness.
5. **Evaluation & integration** — Metrics (Precision, Recall, F1, Accuracy) are computed. The trained model is integrated into the extension for real-time screening. After training, the optimized model was deployed as the backend inference engine for the browser extension. When a user accesses video content: (1) the extension checks IndexedDB for prior classification results. (2) if unavailable, video metadata is sent to the backend.(3) representative video frames are analyzed by the ResNet50–LSTM classifier. (4) classification results are cached locally. (5) unsafe content is blocked through browser interface intervention. This design enables scalable and near real-time automated moderation while minimizing redundant server computation.

Table 1. Proposed ResNet50–LSTM Model Architecture Parameters

Component	**Layer / Module**	**Input Shape**	**Output Shape**	**Key Parameters / Description**
Input	Video Segment	25 × 224 × 224 × 3	25 frames	1-second clip sampled at 25 fps
Frame Feature Extractor	ResNet-50 Backbone	224 × 224 × 3	7 × 7 × 2048	Pretrained on ImageNet; convolutional layers used for spatial feature extraction
Pooling	Global Average Pooling	7 × 7 × 2048	2048	Converts each frame into a 2048-dimensional feature vector
Sequence Construction	Frame Feature Sequence	25 × 2048	25 × 2048	Combines extracted frame-level features into temporal sequence
Temporal Modeling	LSTM	25 × 2048	128	128 hidden units; captures temporal dependencies across frames
Regularization	Dropout	128	128	Dropout rate = 0.3
Dense Layer	Fully Connected	128	64	ReLU activation
Output Layer	Classification Layer	64	2	Softmax activation; classes: Safe and Unsafe

4 Experiments and Results

4.1 Dataset

Experiments were conducted using a combination of benchmark video datasets and a custom-collected dataset designed for youth-oriented content moderation. The datasets were selected to provide diversity in visual content, including safe, unsafe, live-action, and animated video scenarios relevant to automated video screening.

1. **YouTube-8M Subset**: A curated collection of 5,000 video clips across multiple categories, including violent, adult, and safe content.
2. **NSFW Video Dataset**: 2,800 short clips tagged as safe/unsafe.
3. **Custom Youth Dataset**: 1,500 video samples (cartoons, educational videos, and flagged inappropriate clips in Thai/English) collected for this study.

For preprocessing, all videos were standardized to a spatial resolution of 224 × 224 pixels and resampled at 25 frames per second (fps). Each video was segmented into non-overlapping 1-second clips, resulting in sequences of 25 consecutive frames per sample, which served as input to the ResNet50–LSTM classification framework. For each segment, both video frames and audio transcripts were analyzed. To ensure representativeness, the dataset was split into 80% training, 10% validation, and 10% testing sets. Stratified sampling preserved the ratio of safe/unsafe samples across splits. All models were trained using early stopping and batch normalization to prevent overfitting. The ResNet50–LSTM classification framework were trained end-to-end using the Adam optimizer with an initial learning rate of 0.001, a batch size of 32, and 50 epochs. The categorical cross-entropy loss

function was used to minimize classification error. Early stopping was applied based on validation loss to avoid overfitting. All models were implemented using Tensor Flow 2.15.

4.2 Evaluation Metrics

The confusion matrix is a potent technique for assessing the performance of classification results. It provides a detailed representation of predicted and actual classes, aiding in the visualization of classification errors. The matrix is structured such that each row corresponds to the predicted class, and each column represents the original class. The evaluation metrics derived from the confusion matrix include precision, recall, F1 score, and accuracy, calculated using the following equations. Precision is the precision value calculated from (1), Recall is the recall value calculated from (2), F1 score is calculated from (3), and accuracy is calculated from (4).

$$\boldsymbol{Precision} = \frac{TP}{TP+FP} \quad (1)$$

where TP is the true positive value and FP is the false positive value. Precision is the value that measures the number of correct predicting answers by dividing by the total number of images in the dataset.

$$\boldsymbol{Recall} = \frac{TP}{TP+FN} \quad (2)$$

where TP is the true positive value and FN is the false negative value. Recall is the value that measures the number of correct predicting answers for each class and is divided by the total number of that class which is the ground truth.

$$\boldsymbol{F1-Score} = 2\frac{PrecisionxRecall}{Precision+Recall} \quad (3)$$

$$\boldsymbol{Accuracy} = \frac{TP+TN}{N} \quad (4)$$

Where TP is the true positive value, TN is the true negative value, and N is total number of images.

4.3 Experimental results

Table 2. presents the confusion matrix for the ResNet50 + LSTM hybrid model. The system achieved high precision for the “safe” category, while most misclassifications occurred in borderline cases such as mild cartoon violence. Performance comparison across models are show in Table 3.

Table 2: Confusion Matrix for ResNet50 + LSTM Hybrid Model.

Predicted/Actual	**Safe**	**Unsafe**
Safe	1,280	120
Unsafe	95	1,305

Table 3:Performance comparison across models.

Method	**Precision (%)**	**Recall (%)**	**F1-score (%)**	**Accuracy (%)**
Keyword-based Filtering	62.5	58.2	60.3	65.1
Rule-based Metadata Filtering	68.9	63.5	66.1	70.2
CNN (ResNet50 Frames Only)	84.6	81.9	83.2	85.0
LSTM (Temporal Sequence Only)	79.8	76.4	78.1	80.3
ResNet50 + LSTM Hybrid (Proposed)	**93.2**	**91.6**	**92.4**	**92.3**

4.4 Discussions

The proposed ResNet50–LSTM hybrid framework demonstrated strong classification performance for automated youth-oriented video moderation, achieving high precision and recall in distinguishing safe and unsafe content. The confusion matrix indicates that the majority of classification errors occurred in borderline cases, particularly involving mild cartoon violence or stylized animated scenes.

One important explanation for these errors is the domain adaptation challenge commonly observed in computer vision systems. The pretrained ResNet50 model was originally optimized for natural image recognition using ImageNet, while the primary training datasets used in this study (e.g., YouTube-8M and NSFW) predominantly contain live-action visual content. Cartoon media differs significantly from real-world imagery in terms of abstract shapes, exaggerated color schemes, simplified textures, and symbolic representations of harmful actions. As a result, spatial feature representations learned from real-world data may not fully generalize to animated content, increasing the likelihood of false classifications.

Future improvements should address this limitation through domain adaptation strategies such as incorporating animation-specific training datasets, transfer learning with cartoon-focused fine-tuning, and domain generalization methods to improve robustness across heterogeneous media types.

Another practical consideration concerns deployment latency. Although deep learning inference using ResNet50 and LSTM introduces computational overhead, the proposed browser extension minimizes redundant processing through IndexedDB caching of previously classified videos. Only uncached content requires backend inference.

Latency profiling of the prototype system shows an average total processing time of approximately 381 milliseconds per uncached video request, including metadata retrieval, frame preprocessing, server inference, and browser response actions. This operational profile supports near real-time deployment for practical browser-based content moderation.

However, scalability remains a consideration for high-throughput production deployment. Further optimization strategies such as model compression, TensorRT acceleration, edge inference, or lightweight architectures (e.g., MobileNetV3, EfficientNet-lite) could significantly reduce latency in future implementations.

Finally, while the current framework focuses on visual temporal moderation, future multimodal extensions integrating speech transcription, subtitle analysis, and contextual metadata could further improve classification reliability for ambiguous or culturally sensitive content.

Examples of inappropriate videos was shown in Figure. 2.

Figure 2: Examples of inappropriate videos.

5 Conclusion

The increasing dependence of children and adolescents on online video platforms has created a growing demand for effective and scalable content moderation solutions. Traditional manual review methods are difficult to scale and are often unable to cope with the vast volume of multimedia content continuously uploaded to online platforms. This study presented an AI-based automated video screening system for youth-appropriate content using a hybrid deep learning framework that combines a pretrained ResNet-50 Convolutional Neural Network (CNN) with a Long Short-Term Memory (LSTM) network. The proposed architecture enables the extraction of high-level spatial features from video frames while simultaneously capturing temporal relationships across video sequences, allowing for more accurate identification of inappropriate visual content.

Experimental evaluation conducted on benchmark datasets and a custom youth-oriented video dataset demonstrated the effectiveness of the proposed framework. The ResNet50–LSTM model achieved superior performance compared with conventional keyword-based filtering, metadata-based screening, and single-model approaches, obtaining 93.2% precision, 91.6% recall, 92.4% F1-score, and 92.3% accuracy. Analysis of the confusion matrix further confirmed the model's capability to distinguish between safe and unsafe content, with most classification errors occurring in challenging borderline cases such as mild cartoon violence and stylized animated scenes.

The integration of the classification model with a browser extension and IndexedDB-based caching mechanism demonstrates the practical feasibility of near real-time automated content moderation. By reducing redundant server-side inference requests and efficiently managing previously classified content, the proposed system provides a scalable solution suitable for deployment in modern digital media environments.

Overall, this research contributes to the advancement of intelligent content moderation systems and E-Business Information Systems by providing a reliable, scalable, and practically deployable framework for protecting youth audiences from inappropriate visual content. Future work will focus on improving robustness across diverse content domains, particularly animated and culturally specific media, as well as exploring multimodal extensions incorporating audio, speech, and textual information to further enhance moderation accuracy and contextual understanding..

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Toward Effective Lip Reading using Spatio-Temporal Encoding and Cluster-Guided Aggregation*

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Abstract

Lip reading aims to decode spoken words from visual lip movements, but it remains challenged by the high computational requirements and the difficulty of accurate discriminating lip motion patterns. Although recent spatio-temporal deep learning and transformer models have improved recognition performance, many rely on complex network architectures that limits practical deployment. To address these limitations, this study proposes an efficient and effective spatio-temporal lip reading framework that combines lightweight feature extraction with discriminative temporal aggregation. The proposed architecture combines a compact depthwise separable 3D Convolution with inspired residual blocks enhanced with Squeeze-and-Excitation (SE) modules to strengthen spatial representations with minimal overhead and a hierarchical Bidirectional GRU encoder to capture long-range temporal dependencies. To further improve robustness against visual ambiguity, NetVLAD-based cluster-guided aggregation is introduced to pool temporal descriptors according to learned visual word clusters, enabling the network to capture the complex lip-motion distributions. Evaluations on the Lip Reading in the Wild (LRW) benchmark demonstrate that the proposed model achieves a precision of 0.87, recall of 0.85, and F-score of 0.86. Additional evaluations on two specialized LRW subsets show strong robustness under challenging conditions, achieving F-scores of 0.74 on the HOMOPHONES dataset for visually confusable words

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and 0.84 on the INTER-S dataset for inter-speaker variability. Furthermore, the model requires only 21.05×10^9 FLOPs and 19.61×10^6 parameters, offering substantial computational savings compared to state-of-the-art baselines.

1 Introduction

Lip reading [1], commonly referred to as Visual Speech Recognition (VSR), involves automatically interpreting spoken content by analyzing the visual dynamics of lip movements. VSR serves a wide range of real-world applications, such as assisting individuals with hearing impairments [2], improving communication in noisy environments, and enabling automatic subtitle generation for multimedia content [3]. VSR also plays an important role in advancing human-computer interaction, strengthening surveillance capabilities, and supporting multimodal systems in situations where audio signals are unreliable or entirely absent [4].

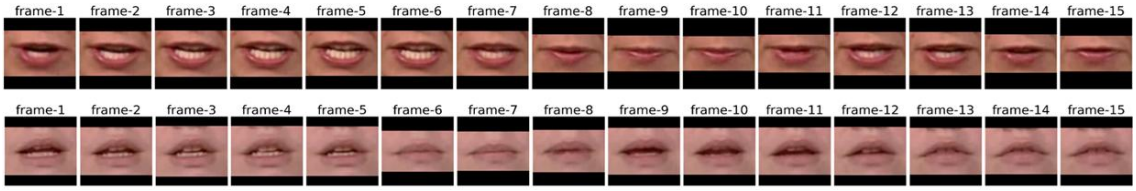
Despite recent advancements, lip reading still encounters some key challenges. One major difficulty is visual homophones ambiguity, where different words produce nearly identical lip movements. For example, words such as “PIN”, “PEN”, and “BIN” often appear visually similar, making them difficult for recognition models to distinguish. In addition, inter-speaker variability, including differences in lip shape, facial structure, and speaking speed, further increase the complexity of accurate recognition [6], [7]. Another significant challenge is the computational burden of modern deep learning-based architectures. Although these models achieve strong performance, they often rely on deep and complex networks that require substantial processing power and memory. This results in slow training and inference limiting deployment on real-time systems and resource-constrained devices such as assistive wearables [5]. Consequently, there is a growing need for lip reading approaches that are both efficient and accurate. These challenges are illustrated in Figure 1. .

In this study, we present an efficient lip-reading framework aimed at achieving a strong trade-off between recognition performance and computational cost. The proposed approach follows a hybrid spatio-temporal architecture, commencing with a lightweight 3D Convolutional Neural Network (3D CNN) to capture short-term spatio-temporal patterns from input video sequences. The extracted features are further enhanced through inverted residual blocks [8], which integrated with Squeeze-and-Excitation (SE) module to adaptively recalibrate channel-wise feature responses. This design strengthens spatial feature representation while maintaining a low computational footprint.

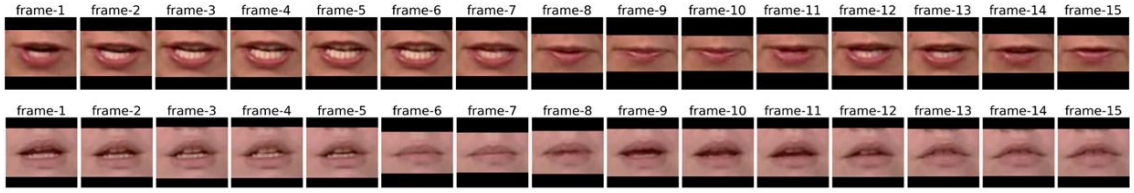
To model long-range temporal dependencies in lip-motion sequences, we employ a hierarchical Bidirectional Gated Recurrent Unit (Bi-GRU) encoder [9], enabling the network to learn contextual information in both forward and backward temporal directions. In addition, to improve temporal feature aggregation and address visual ambiguity, we integrate a learnable cluster-guided consensus module based on NetVLAD layer [10]. This module aggregates temporal descriptors according to their similarity to learned visual word clusters, allowing the model to capture complex lip movement distributions and improves discrimination between visually similar articulations. The proposed framework is evaluated on the Lip Reading in the Wild (LRW) dataset [11], as well as two specialized LRW subsets designed to assess robustness under challenging conditions: INTER-S for inter-speaker variability and HOMOPHONES for visually confusable words. Experimental results show that the proposed model achieves strong recognition performance while substantially reducing computational complexity compared with existing spatio-temporal and transformer-based lip-reading methods.

The main contributions of this work can be summarized as follows:

- We propose an efficient deep learning architecture that integrates a compact 3D CNN with inverted residual blocks with SE recalibration, and a hierarchical Bi-GRU encoder for effective and computationally efficient lip reading recognition.



(A) Video frame sequences annotated with the same word "BENEFIT" from two distinct speakers



(B) Video frame sequences annotated with the words "THING" (top row) and "THINK" (bottom row)

Fig. 1. Samples illustrating two key challenges: (A) Speaker variability and (B) Visually homophonic ambiguity

- We introduce a NetVLAD-based cluster guided aggregation to regenerate discriminative representations that improve robustness against lip-movement homophones.
- We evaluate the proposed model on the full LRW benchmark and two specialized LRW subsets, namely INTER-S and HOMOPHONES, which are designed to assess inter-speaker variability and visual lip-movement ambiguity. The results demonstrate strong recognition performance with reduced computational complexity, supporting the practical deployment of the proposed framework.

The subsequent sections are structured as follows: Section 2 examines relevant literature on approaches to lip reading. Section 3 provides a detailed description of the proposed methodology. Section 4 presents the experimental setup and reports the evaluation results. We conclude in Section 5 by summarizing our contributions and suggesting directions for future work.

2 Related work

2.1 Classical approaches

Early lip-reading methods primarily depended on handcrafted feature extraction combined with statistical sequence modeling. Traditional image processing techniques were employed to derive low-level visual cues [12], [13], including lip contours, shape descriptors, motion information, and appearance-based features such as the Discrete Cosine Transform (DCT) [14]and Local Binary Patterns (LBP) [15].These handcrafted descriptors were subsequently fed into statistical classifiers, such as Hidden Markov Models (HMMs) [16]and Dynamic Time Warping (DTW) [17], to perform sequence-level recognition. Although these approaches laid the groundwork for VSR, they suffered from limited generalization capability. Handcrafted features were highly sensitive to variations in speakers, lighting, and head pose, while statistical models were unable to fully capture the intricate temporal dynamics inherent in lip movements.

2.2 Deep learning-based approaches

The advent of deep learning has markedly advanced lip reading by enabling models to automatically learn both spatial and temporal representations directly from raw video data. Convolutional Neural Networks (CNNs) [18]have been extensively utilized to capture spatial characteristics of the lip region, while Recurrent Neural Networks (RNNs), particularly Long Short-Term Memory (LSTM) units and Bidirectional Gated Recurrent Units (Bi-GRUs), have proven effective for modeling temporal dependencies across video frames [19]. To further boost recognition performance, recent work has investigated hybrid spatio-temporal architectures, such as 3D Convolutional Networks (Conv3D) [28] and Conv2D combined with Bi-GRU encoders [20], [21], [22], which jointly encode spatial and temporal cues. Moreover, transformer-based models [23], [24]have achieved strong results by capturing complex temporal dynamics and enhancing overall VSR capabilities.

Despite these advancements, deep learning–based VSR systems continue to encounter significant challenges related to computational complexity. Their reliance on deep and intricate architectures demands considerable processing power and memory, resulting in high training and inference costs. This computational burden restricts real-time deployment and increases hardware demands. Consequently, there is growing interest in developing efficient VSR architectures that can preserve competitive recognition accuracy while significantly reducing computational overhead, thereby facilitating broader and more practical use of VSR technologies. Nevertheless, visual homophones remain an unsolved issue that requires dedicated solutions.

3 Proposed Architecture

This section presents the proposed architecture, named Spatio-Temporal Deep Cluster Aggregation (ST-DCA), designed to achieve efficient lip reading while maintaining strong discriminative capability, particularly in the presence of visually similar lip movement patterns**.** As illustrated in Figure 2, the architecture comprises three key modules**: (**1**)** an efficient spatio-temporal feature extractor, **(**2**)** a

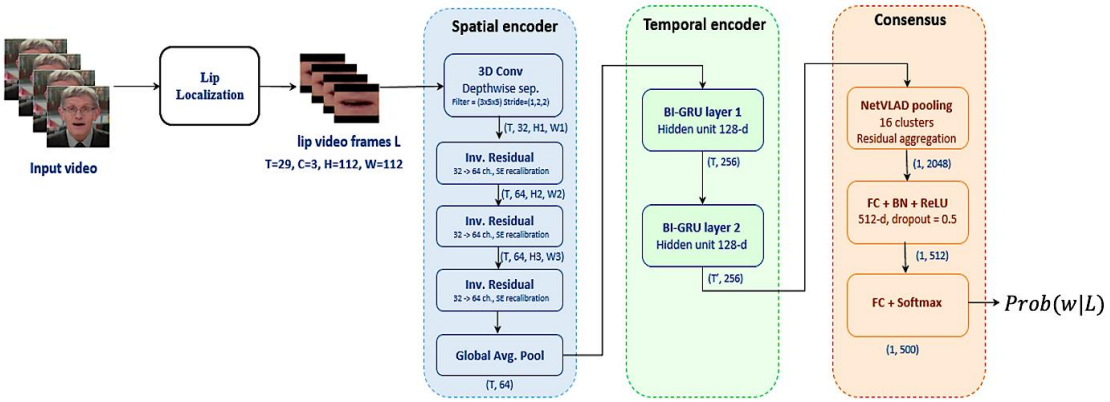


Fig. 2. The proposed ST-DCA framework for word-level lip reading. The input video is first processed by a lip localization module to extract a normalized mouth-region sequence. The spatial encoder then captures short-term spatio-temporal visual patterns using a depthwise separable 3D convolution followed by SE-enhanced inverted residual blocks. The resulting frame-level descriptors are passed to a hierarchical Bi-GRU temporal encoder to model long-range lip-motion dependencies. Finally, the NetVLAD-based consensus module aggregates the temporal features into a compact cluster-guided representation, which is fed to fully connected and softmax layers for final word classification.

hierarchical temporal encoder and **(**3**)** a consensus module that aggregates the temporal features for word recognition.

3.1 Lip Localization

The lip extraction pipeline operates on a frame-by-frame basis to focus specifically on the mouth region, reducing the risk of the model learning biases from irrelevant facial characteristics. In each frame, MediaPipe [25] is employed to detect 478 facial landmarks, from which 40 lip-related keypoints are selected. A bounding box is then computed using the extreme coordinates of these points, with an additional 10-pixel margin to ensure complete coverage of lip movements. The cropped region is subsequently centered and zero-padded to a fixed resolution of 112 × 112 pixels, maintaining a consistent input size for the network. The lip localization process is illustrated in Figure 3.

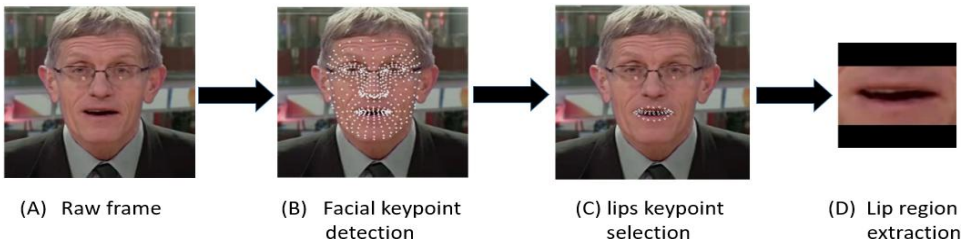


Fig 3. The lip localization process

3.2 Efficient Spatial Encoding

The spatial encoder is designed to extract discriminative representations from lip region sequences. Let the input sequence be denoted as a 4D tensor $X \in \mathbb{R}^{T\times C\times H\times W}$, where T is the number of timesteps, C is the channel count, and $H \times W$ represents the spatial resolution of each frame. While conventional VSR models typically rely on standard 2D or 3D convolutions, these operations are often computationally expensive and prone to filter redundancy, limiting both efficiency and model expressiveness. To mitigate these limitations, the proposed architecture employs a series of efficient convolutional blocks that reduce computational overhead without compromising performance.

The encoder begins with an efficient 3D convolutional block tailored for short-term spatiotemporal feature extraction. We replace the standard 3D convolution with a Depthwise Separable 3D Convolution [26], which factorizes the operation into two distinct stages: depthwise convolution and pointwise convolution. First, a depthwise step independently convolves each input channel using a 3D kernel with dimensions $K_t \times K_h \times K_w$ and a stride of (1, 2, 2). The operation is defined as:

$$D_c(t,h,w) = \sum_{i=0}^{K_t-1}\sum_{j=0}^{K_h-1}\sum_{k=0}^{K_w-1} W_{i,j,k}^{(c)} * X_c(t+i, h+j, w+k) \quad (1)$$

where $W^{(c)} \in \mathbb{R}^{K_t\times K_h\times K_w}$ denotes the 3D kernel specific to channel c and $D_c \in \mathbb{R}^{C_{in}\times K_t\times H\times W}$ is the corresponding output feature map. In this work, a kernel size of (3, 5, 5) is employed to effectively capture short-range lip movement dynamics. Subsequently, a pointwise convolution with a kernel size 1× 1× 1 fuses information across all channels from the depthwise outputs as defined below:

$$Y_c(t,h,w) = \sum_{c'=1}^{C_{in}} P_{c',c} * D_{c'}(t,h,w) \quad (2)$$

where $P_{c',c}$ is the weight mapping the c'-th the depthwise feature to the c-th output channel. Y_c denotes the final fused feature map. The resulting output is then passed through a batch normalization layer and a ReLU activation function to ensure training stability.

Following the initial 3D convolutional front-end, the encoder employs a stack of Inverted Residual blocks with Squeeze-and-Excitation (SE) modules to extract deep spatial representation. Inspired by MobileNetV3 [8], these blocks leverage an inverted residual design specifically tailored for efficient networks. This structure has been demonstrated to maximize parameter efficiency while preserving information flow better than standard residual blocks. As illustrated in Fig. 4, the process begins with a 1×1 pointwise convolution that expands the input feature map to a higher-dimensional intermediate space, facilitating the extraction of complex non-linear features. This is immediately followed by a depthwise convolution with a spatial kernel size of $k \times k$, which performs spatial filtering independently on each channel, drastically reducing computational cost compared to standard convolutions.To enhance the discriminative power of the network, a Squeeze-and-Excitation (SE) module is integrated after the depthwise layer to adaptively recalibrate channel-wise importance. This mechanism compresses global spatial information using Global Average Pooling (GAP) and recalibrates channel responses using a compact fully connected network, effectively emphasizing relevant lip motion patterns while suppressing background noise. Finally, a second 1×1 pointwise convolution projects the calibrated features back to a lower target dimensionality. A residual skip connection is also employed to add the original input directly to the output, ensuring stable gradient propagation and preserving information flow throughout the deep network.

3.3 Hierarchical Temporal Encoding

Following the spatial encoding stage, the network employs a temporal modeling module to capture long-range temporal dynamics of lip movements across the video sequence. To achieve this, we utilize multiple layers of Bidirectional Gated Recurrent Units (BiGRUs) [9] to extract hierarchical temporal representations. Unlike a standard GRU, a BiGRU consists of two recurrent units operating in opposite temporal directions: a forward GRU that processes the sequence from $t = 1$ to T, and a backward GRU that proceeds from $t = T$ to 1. For the forward direction, the update gate, reset gate, candidate state, and hidden state are computed as:

$$\begin{aligned}
\overrightarrow{z_t} &= \sigma\left(W_z^f x_t + U_z^f \overrightarrow{h_{t-1}} + b_z^f\right) \quad (3)\\
\overrightarrow{r_t} &= \sigma\left(W_r^f x_t + U_r^f \overrightarrow{h_{t-1}} + b_r^f\right)\\
\overrightarrow{\widetilde{h_t}} &= \tanh\left(W_h^f x_t + U_r^f (\overrightarrow{r} \odot \overrightarrow{h_{t-1}}) + b_h^f\right)\\
\overrightarrow{h_t} &= (1 - \overrightarrow{z_t}) \odot \overrightarrow{h_{t-1}} + \overrightarrow{z_t} \odot \overrightarrow{\widetilde{h_t}}
\end{aligned}$$

Similarly, the backward GRU follows the same formulation but replaces $\overrightarrow{h_{t-1}}$ with $\overleftarrow{h_{t-1}}$, as it operates in the reverse temporal direction. The final bidirectional representation at time t is obtained by concatenating the two directional states $h_t = \left[\overrightarrow{h_t}; \overleftarrow{h_t}\right]$, which provides a unified feature embedding incorporating context information from both frame directions. To expand the temporal

receptive field without excessive parameter growth, a second-level BiGRU processes a temporally down-sampled sequence:

$$\widetilde{h_t'} = BiGRU_2\left(h_{st'}\right) \quad s > 1 \qquad (4)$$

where $h_{st'}$ represents the outputs of the first-level BiGRU, subsampled every s time steps to shorten the sequence. This operation applies a higher-level BiGRU to a compressed temporal input, allowing the model to capture longer-range dependencies while keeping the parameter and computational costs low.

3.4 Cluster-Guided Aggregation

To leverage the temporal features for final prediction, a common approach is to average the features across all time steps, producing a single fixed-length consensus vector. While this aggregation strategy offers computational simplicity, it relies on the unrealistic assumption that all frames contribute equally to the final decision. As illustrated in Fig. 1(B), only timesteps 2 and 15 carry meaningful information for the word 'THINK,' even though the sample contains 29 frames. Such uniform weighting can diminish important information and may introduce noise from irrelevant frames. To obtain a compact and discriminative utterance-level representation, we propose to incorporate a Deep-Cluster consensus module based on a learnable NetVLAD aggregation mechanism [10], which adaptively emphasizes informative features while suppressing non-discriminative ones. Given the sequence of temporal features $\tilde{h} = \{\tilde{h}_1, \tilde{h}_2, \dots, \tilde{h}_N\}$, $\tilde{h}_i \in \mathbb{R}^D$ produced by the hierarchical BiGRU encoder and the module learns a set of K clusters $= \{c_1, c_2, \dots, c_K\}$ $c_i \in \mathbb{R}^D$, which act as prototypical lip-movement patterns. Each feature $\tilde{h}_i$, is softly assigned to every cluster using a linear scoring function:

$$a_{ik} = \frac{\exp\left(w_k^T \tilde{h}_i + b_k\right)}{\sum_{j=1}^{K} \exp\left(w_j^T \tilde{h}_i + b_j\right)} \qquad (5)$$

where w_k is a trainable parameter vector associated with cluster k and b_k denotes its corresponding bias term. Using these assignments, NetVLAD aggregates residuals relative to each cluster center:

$$V_k = \sum_{i=1}^{N} a_{ik}\left(\tilde{h}_i - c_k\right) \qquad (6)$$

and forms the final consensus representation by concatenating all cluster residuals:

$$V = [V_1; V_2; \dots; V_K] \qquad (7)$$

The resulting descriptor is passed through a fully connected layer with 512 units and dropout regularization, followed by a softmax classifier that produces the final probability distribution over target word classes.

4 Experiments

4.1 Datasets

The Lip Reading in the Wild (LRW) dataset [11] is used as the primary large-scale benchmark for word-level lip reading. LRW contains 538,766 short video clips extracted from BBC broadcasts, with each clip lasting approximately 1.16 seconds. The dataset includes more than 1,000 speakers and covers 500 target word classes. In this experiment, we follow the standard LRW split, which includes 488,766 training samples, 25,000 validation samples, and 25,000 test samples.

To evaluate two major challenges in lip reading, including visual homophone ambiguity and inter-speaker variability, speaker-level information is required for each video clip. Therefore, in this study, we utilize the speaker metadata provided by [27] to construct controlled evaluation subsets from LRW. These subsets are designed to isolate specific sources of difficulty and provide a more rigorous assessment of model robustness under challenging visual speech recognition conditions.

To assess the model's ability to distinguish visually confusable words, we introduce the HOMOPHONES evaluation benchmark, which is constructed from the broader LRW dataset. This subset focuses on word pairs that exhibit highly similar lip movements rather than phonetic homophones in the conventional linguistic sense. The word pairs were manually selected based on three criteria: (1) high visual similarity in mouth articulation, particularly during the initial or final lip movements; (2) practical ambiguity commonly observed in word-level lip-reading tasks, including singular–plural forms, morphological variants, and phoneme-close word pairs; and (3) sufficient sample availability in the LRW test partition to support reliable evaluation. As shown in Table 1, the HOMOPHONES subset consists of 60 expert-selected words forming 30 highly confusable word pairs, such as "BETTER" / "MATTER," "POLITICAL" / "POLITICS," and "THEIR" / "THERE." By parsing the LRW annotation files, all instances corresponding to these 60 target words were identified from the standard LRW test partition, resulting in a total of 3,791 test samples.

A – C	D – N	P - S	T – W
BECAUSE / BEFORE	DECISION / DETAILS	PAYING / SPENDING	TAKING / TALKING
BETTER / MATTER	DIFFERENT / DIFFICULT	PLACES / PLACE	THEIR / THERE
BILLION / MILLION	DOING / GOING	POLITICAL / POLITICS	THINGS / THESE
CHANGE / CHANCES	FAMILIES / FAMILY	POWERS / POWER	THOUGHT / THROUGH
CHILDREN / CHILD	FIGHTING / FIGHT	PROBLEM / PROBLEMS	TODAY / TOGETHER
COMPANIES / COMPANY	FIRST / FURTHER	SPEAKING / SPENDING	WHETHER / WEATHER
CONTROL / COUNCIL	FRANCE / FRENCH		WEEKS / WEEKEND
COUNTRIES / COUNTRY	HEART / HEARD		WORDS / WORLD

Table 1: Thirty manually selected visually confusable word pairs from the LRW dataset, grouped alphabetically.

Although this benchmark does not cover every possible form of visual homophonic ambiguity, it provides a controlled and challenging subset for evaluating fine-grained discrimination between visually similar lip movements. To prevent data leakage, the Speaker_ID metadata is used to enforce strict mutual exclusivity between the training/validation splits and the test split for the selected word pairs. Models are trained on the standard LRW training and validation sets after excluding speakers that overlap with the HOMOPHONES test subset, yielding 474,352 training samples and 21,349 validation samples, respectively.

To evaluate speaker-independent robustness across the full 500-word LRW vocabulary, we introduce an additional evaluation subset, denoted as INTER-S. This subset is designed to examine whether a model can generalize to speakers who are completely unseen during training and validation. Using the Speaker_ID metadata provided by [27]. we first analyzed the word-class coverage of each speaker. Since no individual speaker in the available metadata contained samples for all 500 target word classes, we selected 10 speakers with broad word-class coverage to construct a challenging unseen-

speaker evaluation subset. The selected speaker IDs are 237, 4464, 6850, 7184, 8334, 9438, 9871, 10506, 11146, and 16585. All clips from these selected speakers were assigned exclusively to the INTER-S test subset. To ensure a strict unseen-speaker evaluation and prevent speaker-level data leakage, every training and validation instance associated with these 10 speakers was removed from the standard LRW partitions. As a result, the INTER-S test set contains 5,787 samples, while the pruned training and validation sets contain 450,955 and 23,632 samples, respectively.

4.2 Data Preprocessing

For preprocessing, MediaPipe [25] was employed to detect 478 facial landmarks in each frame. Among these, 40 keypoints corresponding to the lip region were selected to define a bounding box, which expanded with a 10-pixel margin to fully capture lip movements. The extracted region was then centered and zero-padded to a fixed resolution of 112×112 pixels, reducing potential bias introduced by speaker-specific facial characteristics. No data augmentation was applied during training on either the INTER-S or HOMOPHONES datasets to ensure a fair and controlled comparison with all baseline architectures.

4.3 Baselines

We evaluated the proposed ST-DCA model against widely used spatio-temporal lip reading architectures:

- **Conv3D** [28]: An early spatio-temporal model that employs stacked 3D convolutional layers to jointly encode spatial and temporal information from raw video inputs. Temporal average pooling is used for sequence aggregation, followed by a softmax classifier for word-level prediction.
- **ResNet18 + Bi-GRU** [23]: A hybrid architecture in which ResNet-18 extracts frame-level spatial features from mouth-region images. These features are then processed by bidirectional GRUs to capture temporal dependencies, with word classification performed via softmax.
- **Conv3D + Bi-GRU** [21]: A two-stage hybrid model combining 3D convolutions for short-term temporal modeling with a two-layer Bi-GRU module for long-range sequence encoding.
- **ViT + Temporal Transformer** [24]: A transformer-based architecture where each frame is encoded using a Vision Transformer (ViT), and temporal modeling is performed using a multi-head self-attention transformer applied across the sequence.
- **Conv3D -2D + Bi-GRU** [22]: A hybrid approach that integrates 3D convolutions with 2D residual blocks for spatial feature extraction, followed by stacked Bi-GRUs for sequential modeling.

4.4 Implementation Details

The spatial encoder begins with a Depthwise Separable 3D convolution with 32 output channels, a 3×5×5 kernel, and a stride 1×2×2. This is followed by three inverted residual blocks with Squeeze-and-Excitation (SE) modules, progressively increasing channel capacity from 32 to 64. Global average pooling is applied to each frame to produce a temporal feature sequence of size 27×64. The sequence is then processed by a hierarchical two-layer BiGRU with a hidden dimension of 128 in each direction. The first BiGRU layer produces a 256-dimensional temporal feature at each time step. The second BiGRU layer is applied to a temporally down-sampled sequence using a subsampling rate of ($s = 2$).

The final BiGRU output remains 256-dimensional at each time step. The NetVLAD module with K = 16 clusters aggregates these features into a 4,096-dimensional global descriptor. This descriptor is then passed through a fully connected layer with 512 units, followed by BatchNorm, ReLU activation, and dropout with p = 0.5, before the final softmax classifier.

All models were optimized using AdamW with a weight decay of 1×10^{-4}, $\beta_1 = 0.9$, and $\beta_2 = 0.98$. The initial learning rate was set to 1×10^{-4}, with linear warm-up for 8 epochs on LRW and 12 epochs on HOMOPHONES, followed by cosine annealing decay. Training was performed with a batch size of 16 and early stopping based on validation F1-score with a patience of 30 epochs. All experiments were conducted on a single NVIDIA T4 GPU, with training requiring approximately 130–330 hours for LRW and 20–50 hours for HOMOPHONES.

4.5 Results and Discussion

Quantitative comparisons on the full LRW benchmark are presented in Table 2. Overall, the proposed ST-DCA model achieves the highest recognition performance among all evaluated methods, with a precision of 0.87, recall of 0.85, and F-score of 0.86. Compared with the transformer-based ViT+Temporal Transformer baseline, ST-DCA improves the F-score from 0.85 to 0.86 while requiring substantially fewer computational resources. Specifically, ST-DCA uses only 21.05×10^9 FLOPs and 19.61×10^6 parameters, whereas ViT+Temporal Transformer requires 36.89×10^9 FLOPs and 39.7×10^6 parameters. This corresponds to approximately 42.9% fewer FLOPs and 50.6% fewer parameters, demonstrating that ST-DCA provides a more favorable balance between recognition accuracy and model efficiency.

Further evaluations on the HOMOPHONES and INTER-S datasets are summarized in Table 3. On the HOMOPHONES dataset, which contains visually confusable word classes, ST-DCA achieves the best performance with a precision of 0.75, recall of 0.73, and F-score of 0.74. This result outperforms the strongest baseline, ViT+Temporal Transformer, which obtains an F-score of 0.71. The improvement indicates that ST-DCA is more effective in distinguishing subtle lip-motion differences among words with highly similar visual patterns. Compared with Conv3D+Bi-GRU and Conv3D-2D+Bi-GRU, ST-DCA also shows clear gains, suggesting that the proposed spatio-temporal discriminative feature aggregation contributes to more robust classification under visually ambiguous conditions.

On the INTER-S dataset, ST-DCA again achieves the best overall performance, reaching a precision of 0.85, recall of 0.84, and F-score of 0.84. Although ViT+Temporal Transformer performs competitively with an F-score of 0.83, ST-DCA still provides a slight improvement while maintaining a considerably lighter computational profile, as shown in Table 3. This confirms that the proposed model generalizes well across inter-speaker variations and diverse word classes without relying on a large transformer-based architecture.

The improved discriminative capability of ST-DCA is further supported by the ROC curves shown in Figure 5. On the HOMOPHONES dataset, ST-DCA produces a steeper ROC curve than the

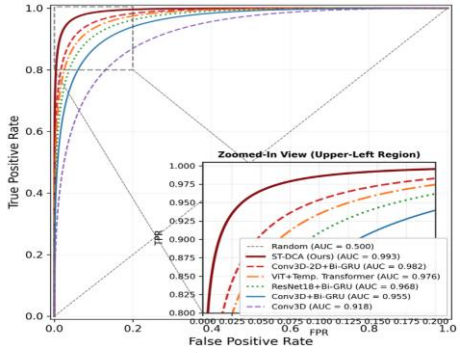
Method	Full LRW			FLOPS ($1x10^9$)	#Param ($1x10^6$)
	Precision	Recall	F-score		
Conv3D	0.71	0.68	0.69	48.53	17.1
ResNet18+Bi-GRU	0.83	0.81	0.82	16.8	33.54
Conv3D+Bi-GRU	0.79	0.75	0.77	22.75	22.0
ViT +Temp. Transformer	0.86	0.84	0.85	36.89	39.7
Conv3D-2D+Bi-GRU	0.85	0.83	0.84	26.4	28.4
ST-DCA	**0.87**	**0.85**	**0.86**	**21.05**	**19.61**

TABLE 2. Performance comparison of ST-DCA with existing methods on the full LRW benchmark.

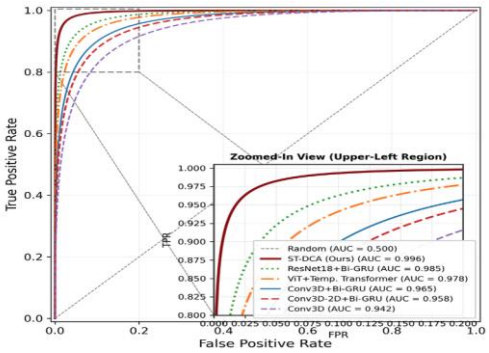
baseline models, indicating higher true positive rates at lower false positive rates. A similar trend is observed on the INTER-S dataset, where ST-DCA maintains consistently strong detection performance across different decision thresholds. The proposed model achieves AUC values of 0.993 and 0.996 on the HOMOPHONES and INTER-S datasets, respectively, outperforming the compared methods. These results suggest that ST-DCA learns more reliable and well-separated decision boundaries, particularly in challenging scenarios involving visually similar words and inter-speaker variability.

TABLE 3. Performance comparison of ST-DCA with existing methods on the HOMOPHONES and INTER-S datasets.

Method	HOMOPHONES			INTER-S		
	Precision	Recall	F-score	Precision	Recall	F-score
Conv3D	0.53	0.50	0.51	0.58	0.57	0.57
ResNet18+Bi-GRU	0.51	0.50	0.50	0.73	0.72	0.72
Conv3D+Bi-GRU	0.63	0.60	0.61	0.66	0.62	0.64
ViT+Temp. Transformer	0.72	0.72	0.71	0.84	0.83	0.83
Conv3D-2D+Bi-GRU	0.69	0.67	0.68	0.81	0.79	0.80
ST-DCA	**0.75**	**0.73**	**0.74**	**0.85**	**0.84**	**0.84**



(A) ROC Curve for HOMOPHONES classes



(B) ROC Curve for Inter-S classes

Fig. 5. ROC curves comparing the performance of different network architectures for lip reading on (A) the HOMOPHONES and (B) the INTER-S datasets. A zoomed-in inset of the upper-left region is provided to improve readability.

Beyond the overall quantitative results, we further examine performance at a finer level. Figure 6 illustrates the pairwise error rate (PER) comparison between the proposed ST-DCA and the transformer baseline on the most visually confusable word pairs in the HOMOPHONES dataset. Lower PER indicates better discrimination performance. As shown in this figure, ST-DCA consistently achieves lower error rates than the transformer model across most evaluated word pairs. The improvement is particularly evident for plural–singular and morphologically similar pairs such as FAMILIES/FAMILY, PROBLEM/PROBLEMS, and COMPANIES/COMPANY, where ST-DCA (green bar) substantially reduces misclassification rates. In contrast, the transformer baseline (gray bar) exhibits higher variability and consistently larger error rates across these challenging categories.

Moreover, the mean PER of ST-DCA (24.2%) is significantly lower than that of the transformer baseline (38.0%). These findings demonstrate that the proposed cluster-guided spatio-temporal architecture is more effective at capturing fine-grained temporal cues, leading to improved discrimination of visually similar word pairs. Figure 7 compares the per-class accuracy distributions of ST-DCA and the ViT-Transformer baseline across the 500 word classes on the INTER-S benchmark. Although both models achieve high accuracy for most the classes, ST-DCA is more concentrated in the 85-95% accuracy range and shows a clear shift toward higher performance. In contrast, the transformer baseline has a wider distribution with more classes falling below 60%, indicating less consistent recognition across difficult categories.

These results confirm that ST-DCA not only improves overall recognition accuracy, but also provides more consistent and reliable performance across diverse word classes, reducing the number of poorly recognized categories compared to the transformer-based approach.

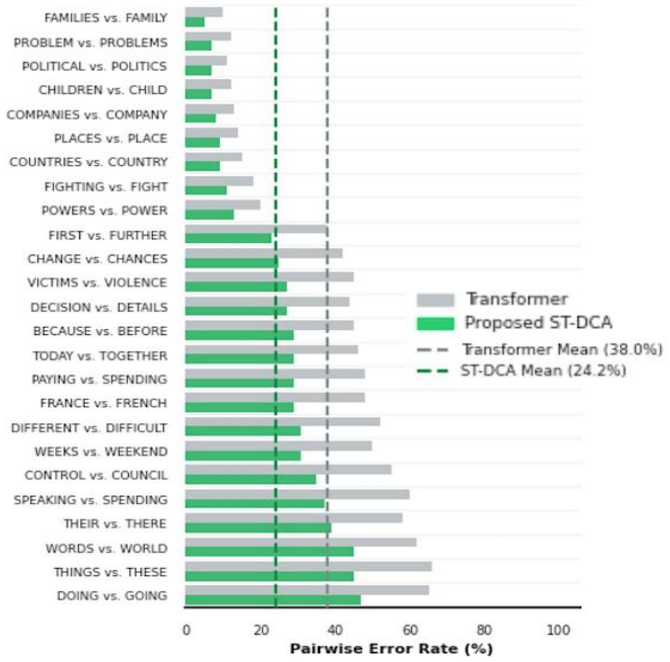


Fig. 6 . Comparison of pairwise error rates (PER) between the Temporal Transformer (gray) and the ST-DCA model (green) on homophone pairs with PER below 60%

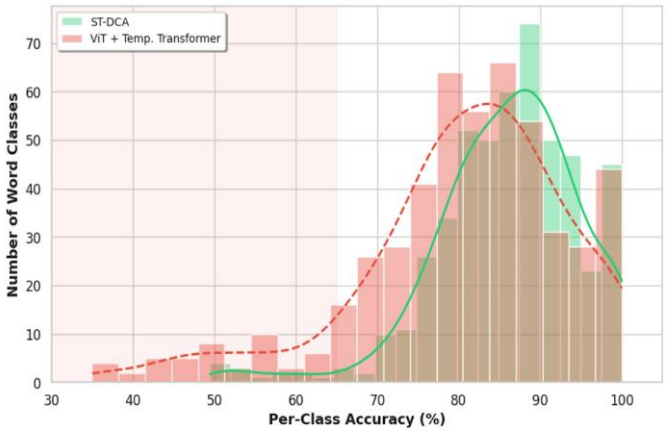


Fig. 7. Per-class accuracy distribution of the ST-DCA and the transformer baseline across the 500-word INTER-S dataset

Table 4 presents the ablation study on the NetVLAD cluster count (K) with results reported as the mean ± standard deviation (SD) across three independent training runs with random seeds (42, 132, 2024). Increasing K from 4 to 16 consistently improves performance on HOMOPHONES, with K=16 achieving the best HOMOPHONES result of 0.740±0.002; on INTER-S, K=32 achieves the highest mean F1-score of 0.841±0.006, while K=16 reaches 0.840±0.011.The low standard deviations across all configurations demonstrate that the ST-DCA architecture is highly robust to weight initialization; the slight variance increase at K=32 is observed on HOMOPHONES, whereas the INTER-S standard

Dataset	Number of K Clusters (Mean F1 ± Standard Deviation)			
	4	8	16	32
HOMOPHONES	0.650 ± 0.004	0.701± 0.003	0.740± 0.002	0.731 ± 0.004
INTER-S	0.794 ± 0.002	0.810 ± 0.004	0.840 ± 0.011	0.841 ±0.006

Table 4 –F1-score performance of ST-DCA with varying NetVLAD cluster sizes (K)

deviation decreases from 0.011 at K=16 to 0.006 at K=32.

Overall, the result confirms that the choice of cluster count plays a critical role in shaping the effectiveness of temporal aggregation within the ST-DCA architecture. Specifically, smaller cluster sizes fail to encode the complex distributions required for homophone disambiguation, while excessively large clusters introduce redundant representations. The K=16 configuration provides a strong trade-off for HOMOPHONES while maintaining nearly the same INTER-S mean F1-score as K=32, helping isolate fine-grained articulation differences without sacrificing computational efficiency.

5 Conclusion

This study introduced the ST-DCA framework to tackle two persistent bottlenecks in lip reading: the high computational cost of modern architectures and the difficulty of disambiguating visual homophones. By integrating a lightweight spatial encoder—utilizing depthwise-separable convolutions and SE-enhanced inverted residual blocks—with a hierarchical Bi-GRU for temporal modeling, the framework efficiently captures long-range dependencies. The inclusion of a NetVLAD-based cluster-guided consensus module further empowers the model to map complex lip movement distributions,

significantly boosting discrimination against visually similar words. Consequently, ST-DCA extracts highly discriminative articulatory features with a fraction of the computational overhead required by conventional CNN-RNN and transformer baselines. Evaluations on the unseen-speaker INTER-S subset of LRW and the newly curated HOMOPHONES dataset validate the model's state-of-the-art recognition capabilities.

Several directions remain open for future investigation. First, the proposed framework is designed for word-level visual speech recognition. Extending ST-DCA to sentence-level recognition represents an important direction for enhancing its practical applicability in real-world communication scenarios, where continuous and context-dependent speech patterns are more common. Second, the current NetVLAD-based clustering module adopts a fixed number of clusters determined empirically through experiments. Future work will investigate adaptive or hierarchical clustering strategies that can dynamically adjust the number of visual prototypes according to input complexity, thereby improving discriminative capability across a wider range of lip-motion patterns. Finally, although this study focuses on visual-only recognition, integrating ST-DCA into a multimodal audio-visual framework could further improve robustness, particularly under acoustically noisy or visually ambiguous conditions.

Acknowledgement

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A Study on Choppy Pulse Image Classification Using Multi-Scale Feature Fusion with a Light ConvNeXt-style Architecture*

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Abstract

Choppy pulse (CP) is one of the pulse patterns with important diagnostic value in traditional Chinese medicine pulse diagnosis. However, its assessment still depends

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heavily on the clinical experience of physicians, and therefore objective and standardized identification methods are still needed. In recent years, deep learning has been widely applied to medical image classification tasks. However, studies on CP waveform images have rarely investigated the complementary value of different field-of-view scales in a systematic manner. Therefore, this study proposes a CP image classification method based on multi-scale feature fusion with a Light ConvNeXt-style architecture. The purpose of this study is to compare the classification performance of single-scale models and multi-scale fusion models in the CP classification task, and to further examine the role and complementary effect of waveform information from different scales. The dataset was constructed from 6-second pulse wave images measured by a pulse diagnosis instrument in 56 CP cases confirmed by traditional Chinese medicine physicians. A single pulse waveform was first extracted from each 6-second recording. Then, multi-size images were cropped with the highest waveform peak as the center to establish a binary dataset of CP and Non-Choppy Pulse (NCP). The data were divided by patient-wise split into training, validation, and test sets in order to reduce the risk of data leakage. This study focused on three representative scales, namely 33×33, 61×61, and 105×105, which correspond to local detail, main contour, and global context, respectively. Six fixed experiments were designed, including three single-scale models, two dual-scale fusion models, and one triple-scale fusion model. The results show that multi-scale fusion models can integrate waveform features from different field-of-view levels and can improve the classification performance of CP images. Overall, this study suggests that CP images contain multi-level information, and that multi-scale feature fusion can provide a more complete basis for classification. These findings may serve as a useful reference for the development of intelligent assistance systems for traditional Chinese medicine pulse diagnosis and may further improve their completeness and practical value.

1 Introduction

Traditional Chinese Medicine (TCM) pulse diagnosis is one of the important foundations of clinical syndrome differentiation. However, its interpretation has long relied on physicians' clinical experience and subjective tactile perception, which still imposes clear limitations on standardization, objectification, and reproducibility. Previous studies have attempted to promote the quantification of pulse diagnosis through artificial intelligence models, classification index construction, and pulse waveform imaging methods, showing that pulse information can be transformed into more consistent quantitative descriptions and can further serve as a technical foundation for automated recognition (Tang *et al.* (2012), Shu and Sun (2007), Xu *et al.* (2009), Chen *et al.* (2016)). Nevertheless, if the pulse-taking method, pulse-taking depth, and sample definition are not standardized, the features learned by the model may be affected by variations in measurement conditions, thereby reducing the stability and reliability of the analytical results (Chung *et al.* (2013), Chung *et al.* (2014)). Therefore, how to establish a more consistent digital analysis framework while preserving the observational logic of TCM pulse diagnosis remains a core issue in pulse diagnosis objectification research.

With the development of pulse waveform signal analysis techniques and deep learning methods, pulse waveform data have gradually been applied to various recognition tasks, including disease identification and pulse pattern classification (Zhang *et al.* (2018), Masoumi Shahrbabak *et al.* (2024), Fan *et al.* (2023), Chen *et al.* (2022)). Among these studies, Chen *et al.* (2022) proposed a method that combines Pulse-Line Intersection feature extraction with artificial intelligence-based classification for

hesitant pulse wave classification, demonstrating the feasibility of pulse waveform image analysis and model learning for choppy-pulse-related classification tasks. However, most existing studies still treat the pulse waveform as a single holistic input and rarely further investigate the differences in local detail, main contour, and global context presented by the same single pulse waveform across different field-of-view levels. For the recognition of CP and NCP, if a model relies only on a single-scale input, it may preserve only part of the discriminative information and may therefore fail to fully capture the morphological characteristics of complex pulse patterns.

In the field of medical image classification, multi-scale feature fusion has been shown to facilitate the integration of local and global information and to improve classification performance as well as the stability of feature representation (Huo *et al.* (2023), Zhu *et al.* (2024)). The same perspective has gradually extended to pulse diagnosis and pulse waveform analysis research. Existing studies have pointed out that wrist pulse recognition tasks are well suited to fusion-based representations and that features from different sources or different levels may also have complementary value (Zhang *et al.* (2021), Guo *et al.* (2022)). Therefore, if the concept of multi-scale feature fusion is introduced into two-dimensional pulse waveform image analysis, it may become possible to simultaneously preserve local waveform detail, main contour structure, and overall pulse context, thereby establishing a more complete basis for classification. This direction is also more consistent with the multi-level observational logic of TCM pulse diagnosis, which emphasizes local detail, main contour, and overall pulse context.

In addition to the integration of multi-scale information, the choice of backbone model also directly affects the efficiency and stability of feature learning. ConvNeXt, proposed by Liu *et al.* (2022), revisits the design of modern Convolutional Neural Networks (CNNs) and shows that convolution-based architectures can still maintain strong performance in image classification tasks after incorporating modern design principles. However, for binary medical image classification under small-sample conditions, model design must still maintain a balance between representational capacity and the risk of overfitting, and the network scale should therefore not be expanded without restraint (Zerouaoui and Idri (2022), Shamshiri *et al.* (2023)). On the other hand, if the data partitioning strategy is inappropriate, model performance may be overestimated due to data leakage. Accordingly, a patient-wise fixed split is particularly important in cross-subject classification studies (Veetil *et al.* (2024), Del Pup *et al.* (2025), Fan Ouyang *et al.* (2021)). In other words, for CP image classification, if the goal is to handle multi-scale information, maintain model trainability, and improve evaluation credibility at the same time, both a lightweight architectural design and a rigorous patient-wise fixed split strategy are required.

Based on the above background and research gap, this study proposes a Light ConvNeXt-style architecture based on multi-scale feature fusion for binary classification of CP and NCP. Unlike size-search-oriented studies, this work does not perform an exhaustive size search. Instead, it adopts three representative scales, 33×33, 61×61, and 105×105, corresponding to local detail, main contour, and global context, respectively, and establishes six fixed experimental settings to systematically compare single-scale models and multi-scale fusion models. Through this design, this study aims to investigate whether information from different scales provides complementary benefits and to verify the feasibility of a Light ConvNeXt-style multi-branch design under small-sample conditions with a patient-wise fixed split strategy. In addition, this study further evaluates whether such a method can better reflect the multi-level observational logic of TCM pulse diagnosis and serve as one of the technical foundations for a choppy pulse-assisted recognition system.

The remainder of this paper is organized as follows. Chapter 2 reviews the literature on pulse diagnosis objectification, pulse waveform analysis, multi-scale feature fusion, the ConvNeXt architecture, and data partitioning and performance evaluation. Chapter 3 describes the research design, including the study samples, multi-scale data construction, the Light ConvNeXt-style multi-branch model architecture, the classification procedure, and the experimental settings. Chapter 4 presents the experimental results and discussion, including the overall comparison of the six fixed experiments and the analysis of the best model and the representative control model. Chapter 5 concludes the study and

outlines future work by summarizing the main findings, research significance, study limitations, and possible directions for further extension.

2 Related Works

TCM pulse diagnosis serves as a vital foundation for syndrome differentiation and treatment planning. However, because its interpretation has historically relied on the clinical experience and subjective judgment of TCM practitioners, its objectification and standardization remain a critical research focus. Tang *et al.* (2012) proposed an artificial neural network-based TCM pulse diagnostic model, verifying its feasibility in quantifying pulse diagnosis. Shu and Sun (2017) further developed classification indices for pulse diagnosis, demonstrating that pulse information can be systematically organized into consistent quantitative descriptors. Furthermore, Xu *et al.* (2009) utilized a fuzzy neural network to show that pulse manifestations can be transformed into image representations with classification value. In addition, Chen *et al.* (2016) further analyzed arterial pulse waves using image processing and feature extraction techniques. Their study segmented measured pulse waves into arterial pulse wave curves and quantified relationships among cycle periods, slopes, and characteristic points, indicating that pulse-wave image segmentation, cycle extraction, and quantitative feature analysis can serve as an important technical basis for the objectification of pulse diagnosis. Beyond pulse digitization, the standardization of measurement conditions is a crucial prerequisite for automated recognition. Chung *et al.* (2012) noted that lacking a consistent pulse-taking standard compromises the objectivity of feature learning. Accordingly, Chung *et al.* (2015) investigated the relationship between arterial width and pulse-taking depth, highlighting the necessity for consistency in sample definition and measurement protocols.

Driven by technological advancements, pulse wave signals have been widely applied in disease recognition tasks. Zhang *et al.* (2012) demonstrated that wrist pulse signals can effectively identify diseases such as lung cancer. Similarly, Cui and Song (2022) successfully classified coronary heart disease using acoustic pulse waves. Regarding deep learning applications, Masoumi Shahrbabak *et al.* (2013) and Mair *et al.* (2022) revealed the strong potential of combining deep learning with pulse wave data for diagnosing cardiovascular diseases and arterial stenosis. Furthermore, Ouyang *et al.* (2020) utilized pulse waveform fitting techniques for diabetes detection; Fan *et al*. (2023) explored the diagnostic value of graph features across pulse cycles; and Sen *et al.* (2025) applied machine learning to analyze the pulse wave topology of virtual subjects. Notably, Chen *et al.* (2022) further proposed a hesitant pulse wave classification approach that integrates Pulse-Line Intersection-based feature extraction with artificial intelligence models. Using data from 46 patients, six measurement points, and six-second recordings, their study classified hesitant and moderate pulses, indicating that the integration of pulse-wave image analysis, feature engineering, and AI models can already be applied to hesitant-pulse-related classification tasks. While these studies collectively confirm the immense value of pulse wave data in clinical computer-aided diagnosis, existing methods typically treat the pulse wave as a single holistic input, leaving in-depth explorations of multi-scale waveform variations within two-dimensional (2D) images relatively scarce.

To overcome the limitations of relying on a single holistic input, model designs in medical image classification have increasingly emphasized the importance of multi-scale image information. Huo *et al.* (2023) pointed out that local and global features in medical images are highly complementary, and their hierarchical fusion can effectively improve classification performance. In a skin lesion classification task, Hu *et al.* (2024) demonstrated that integrating shallow and deep multi-scale features significantly enhances recognition capabilities. Zhu *et al.* (2024) further verified that multi-scale feature fusion provides stable feature representation advantages, even in class-imbalanced scenarios. Together, these studies support the premise that the core value of multi-scale design lies in integrating information

from diverse hierarchical levels to establish a more robust classification foundation. Extending the concepts of multi-scale feature fusion to the pulse diagnosis domain, Qi Zhang *et al.* (2021) proposed a multichannel feature fusion approach based on graph networks, demonstrating that wrist pulse recognition tasks are well-suited for modeling via fusion representations. Guo *et al.* (2022) emphasized the complementary value of diverse feature sources, pointing out that a single feature is often insufficient to capture the full discriminative information embedded in pulse wave signals. The aforementioned literature indicates that pulse wave analysis tasks necessitate the fusion of information from different sources or hierarchical levels to achieve more discriminative representations. This theoretical background justifies the exploration of integrating multi-scale perspectives, including local details, main contours, and global contexts, in pulse wave image analysis.

Beyond the integration of multi-scale features, the selection of an appropriate backbone network directly determines a model's effectiveness in learning local and global information. Liu *et al.* (2022) re-examined modern convolutional architecture designs and proposed ConvNeXt, demonstrating that convolutional networks, when updated with modern design principles, can maintain a strong competitive edge in image classification tasks. This establishes a design paradigm that can be easily extended and adapted for lightweight applications. In the context of medical image classification, Aruk *et al.* (2025) confirmed that ConvNeXt-based architectures serve as effective backbone networks for skin lesion classification. Gulsoy and Kablan (2025) further utilized ConvNeXt as an adaptable component for lung cancer classification. Jin *et al.* (2023) also effectively improved the classification performance of chest X-ray images by employing a pre-trained ConvNeXt as a feature extractor. These studies collectively illustrate that ConvNeXt and its variant architectures possess high applicability in medical classification tasks.

In binary medical image classification tasks with limited annotated data, the model design must strike a delicate balance between representation capacity and the risk of overfitting. Zerouaoui and Idri (2022) found that model architecture design directly impacts the stability and final performance of binary classification for breast cancer medical images. Shamshiri *et al.* (2023) similarly pointed out that when annotated data is scarce, model design and knowledge transfer strategies significantly influence recognition capabilities. Therefore, under small-sample conditions, unrestrictedly expanding the network scale is inadvisable; instead, adopting an appropriately lightweight convolutional architecture combined with a multi-branch design presents a sound methodological basis.

Beyond model design, data partitioning and performance evaluation strategies are critical for ensuring the credibility of medical classification research. Veetil *et al.* (2024) noted that improper data partitioning can easily lead to overestimated model performance due to data leakage, thereby undermining its actual generalizability. Del Pup *et al.* (2025) showed that different data partitioning strategies significantly affect the outcomes of cross-subject analyses, underscoring the necessity of subject-wise partitioning. In pulse wave signal analysis, Ouyang *et al.* (2021) directly adopted an inter-patient classification design, proving the importance of individual-level data partitioning in biomedical signal research. Furthermore, the previously mentioned study by Masoumi Shahrbabak *et al.* (2024) also employed multiple evaluation metrics, including the area under the curve (AUC), sensitivity, and specificity, reflecting the importance of comprehensive performance assessment in clinical classification tasks.

3 Lightweight Convnext-Style Multi-Scale CP Classification Method

This study proposes a lightweight ConvNeXt-style multi-scale CP classification method to compare the effects of single-scale input and multi-scale feature fusion on classification performance. The overall workflow of this study includes data acquisition and sample construction, representative multi-scale

image generation, patient-wise fixed data splitting, multi-branch Light ConvNeXt-style model construction, and classification performance evaluation, as shown in Figure 1. Algorithm 1 further summarizes the complete forward processing flow from multi-scale input to binary classification output.

The design of this study has two core objectives. First, this study constructs single-scale, dual-scale, and triple-scale input settings based on three representative scales in order to investigate the complementary effects of different scale information. Second, this study assigns an independent Light ConvNeXt-style backbone to each scale branch. All scale branches share the same architecture, but no weights are shared across branches. This design allows the model to separately learn local details, main contours, and global pulse-wave context, and then improve CP discrimination through feature fusion. Based on these objectives, this chapter is divided into four parts, including research samples and multi-scale data construction, multi-scale Light ConvNeXt-style classifier design, multi-scale CP classification procedure, and experimental design with evaluation protocol.

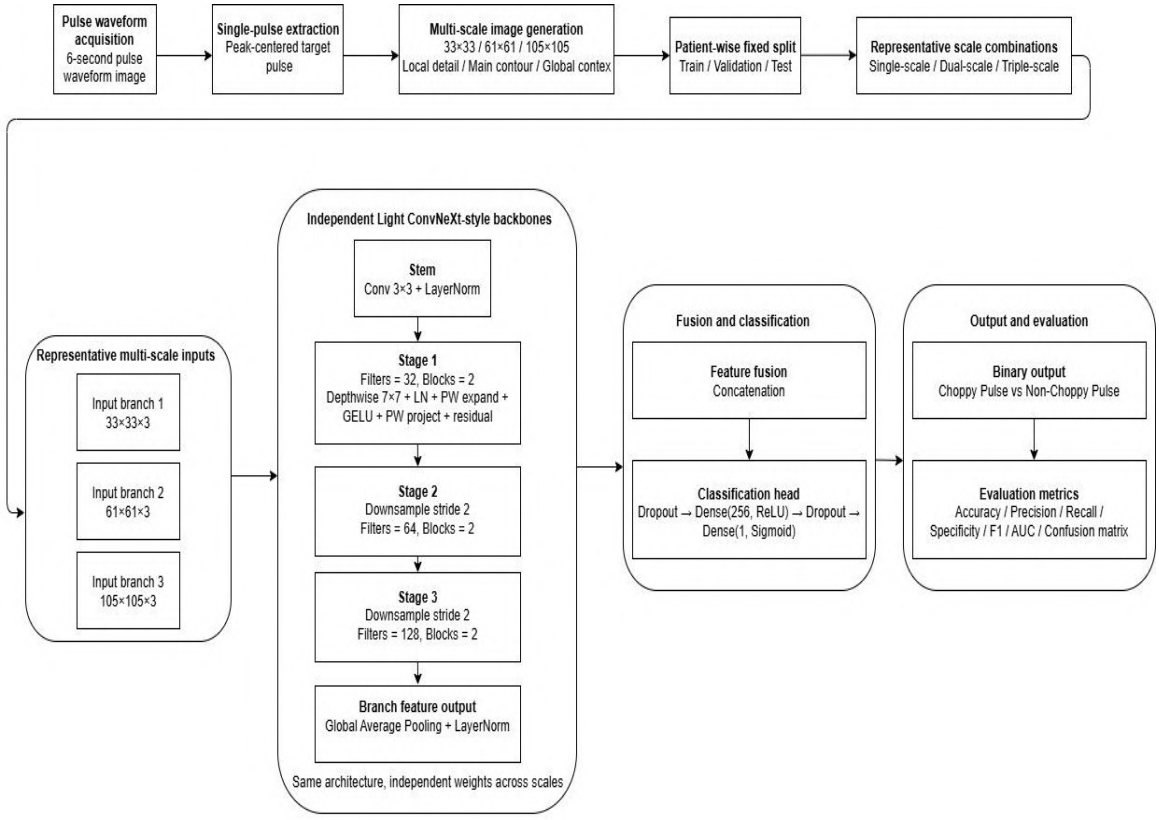

Figure 1: Integrated research workflow and model architecture of the proposed lightweight ConvNeXt-style multi-scale CP classification method.

3.1 Research Samples and Multi-Scale Data Construction

The original data used in this study were 6-second pulse waveform images acquired by a pulse diagnosis instrument. Data preprocessing includes two main steps. First, a single-pulse sample is extracted from each original pulse waveform image, and the pulse apex is used as the center reference

for localization. Second, multiple pulse waveform images at different scales are cropped from the same center position and are used as subsequent model inputs. This design allows images at different scales to correspond to the same analytical sample and helps preserve multi-level pulse-wave information.

This study uses pulse waveform images as model inputs and CP and NCP as the corresponding binary classification labels. The CP and NCP labels were established according to physician-based pulse diagnosis annotation. Pulse waveform images obtained after single-pulse extraction and multi-scale cropping were used as the model inputs, whereas CP and NCP served as the corresponding ground-truth labels.

For multi-scale image construction, this study selects three representative scales, namely 33×33, 61×61, and 105×105, for analysis. These three scales correspond to different levels of pulse-wave information. Specifically, 33×33 emphasizes local details, 61×61 represents the main contour, and 105×105 preserves a more complete global pulse-wave context. Based on this design, this study establishes representative single-scale, dual-scale, and triple-scale input settings to compare the classification effects of different scale combinations. This study further constructs three representative cropping scales centered on the waveform peak of a single pulse so as to correspond to different waveform field-of-view levels ranging from local detail to main contour and global context, as shown in Figure 2. Specifically, the 33×33 scale mainly preserves local details around the peak and is more suitable for observing subtle waveform variations; the 61×61 scale further covers the main contour of a single pulse and helps represent the core waveform morphology; and the 105×105 scale preserves a broader global context and reflects a more complete pulse-wave pattern. This design enables this study to compare the performance differences between single-scale and multi-scale fusion settings from the three levels of local detail, main contour, and global context, and to further investigate whether information from different scales provides complementary benefits.

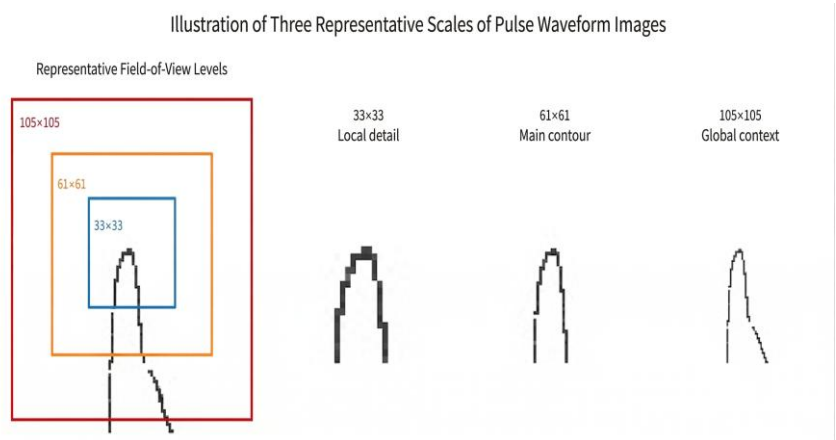

Figure 2: Illustration of the three representative pulse waveform image scales and their corresponding waveform field-of-view levels.

For data splitting, this study established a binary classification dataset consisting of CP and NCP according to physician-based pulse diagnosis annotation, and applied a patient-wise fixed split strategy to divide the data into training, validation, and test sets with a ratio of 7:2:1. Since the three representative scales, namely 33×33, 61×61, and 105×105, were all cropped from the same set of single-pulse samples, the samples across different scales maintained one-to-one correspondence. This design ensured that samples from the same subject did not appear across different subsets, thereby reducing the risk of data leakage and improving the reliability of model evaluation. The numbers of cases and images in each data subset are summarized in Table 1.

After data splitting, this study further defined six fixed experimental settings, including three single-scale models, two dual-scale fusion models, and one triple-scale fusion model, in order to systematically compare the performance differences between single-scale and multi-scale fusion settings.

Table 1: Distribution of cases and images in the CP/NCP dataset across the training, validation, and test sets

Data split	CP cases	CP images	NCP cases	NCP images	Total images
Training set (70%)	39	1,610	23	1,136	2,746
Validation set (20%)	11	441	7	329	770
Test set (10%)	6	264	3	135	399

3.2 Multi-Scale Light ConvNeXt-Style Classifier Design

This study proposes a multi-branch lightweight ConvNeXt-style classifier to process pulse waveform images at different representative scales. The overall model consists of four components, namely a representative multi-scale input module, independent Light ConvNeXt-style backbones, a feature fusion module, and a classification head. Liu *et al*. (2022) proposed ConvNeXt as a modernized convolutional architecture that reorganizes the design of contemporary convolutional neural networks while preserving the advantage of convolution in local feature extraction. Based on this design concept, this study further redesigns a lightweight ConvNeXt-style backbone according to the requirements of multi-scale CP image classification. Since the target task of this study is small-sample CP image classification and the model must simultaneously support multi-branch and multi-scale feature fusion, this study does not directly adopt the full official ConvNeXt backbone. Instead, this study retains its core block design elements and reconstructs a more lightweight backbone to reduce model complexity and improve training stability, reproducibility, and practical feasibility.

At the input side, the model receives single-scale, dual-scale, or triple-scale images according to the experimental setting. When multiple scales are used, each scale image is fed into its corresponding branch separately. Each branch accepts one fixed input size in three-channel image form in order to preserve scale-specific spatial information.

For backbone design, each input branch is assigned an independent Light ConvNeXt-style backbone. All scale branches share the same architecture, but no weights are shared across branches. This design allows each scale input to learn its most suitable feature representation independently without being constrained by parameter sharing across different scales. Each backbone consists of a stem module, three feature extraction stages, and a branch feature output layer. The stem module contains a 3×3 convolution layer and a Layer Normalization layer for generating the initial feature representation. The model then sequentially enters Stage 1, Stage 2, and Stage 3. Stage 1 uses 32 filters and two lightweight ConvNeXt-style blocks for feature extraction. Stage 2 and Stage 3 apply downsampling (stride = 2) followed by 64 and 128 filters, respectively, to learn higher-level pulse-wave features. At the end of each branch, Global Average Pooling and Layer Normalization are applied to generate the branch feature vector for subsequent multi-scale fusion.

For feature fusion, this study uses concatenation as the default fusion method. In the single-scale setting, the feature vector from the selected branch is directly used as the input of the classification head. In the multi-scale setting, the feature vectors from all selected branches are concatenated along the feature dimension to form the fused multi-scale representation, and the classification head then learns the relationships among different scales.

For the classification head, the fused feature vector is sequentially passed through Dropout, Dense (256, ReLU), Dropout, and Dense (1, Sigmoid). The two Dropout layers reduce the risk of overfitting, the fully connected layer performs nonlinear feature transformation, and the final Sigmoid output layer generates the predicted probability that the sample belongs to Choppy Pulse. The model then converts this probability into the final binary classification result according to a predefined classification threshold.

3.3 Multi-Scale CP Classification Procedure

To more clearly describe the complete forward processing flow of the proposed lightweight ConvNeXt-style multi-scale classification model, including multi-scale image input, branch-wise feature extraction, feature fusion, binary prediction, and performance evaluation, this study further summarizes the procedure as *Algorithm 1*.

Algorithm 1. Proposed Lightweight ConvNeXt-Style Multi-Scale CP Classification Procedure.
Input: *A set of aligned multi-scale pulse waveform images corresponding to one sample, with representative input scales selected from {33×33, 61×61, 105×105}.*
Output: *The predicted probability p that the input sample belongs to CP, and the final binary classification result $\hat{y}$.*

Begin

S*tep 1: Read the aligned pulse waveform image paths corresponding to the selected representative scale combination.*

Step 2: For each selected input scale $s \in \{33, 61, 105\}$, read the corresponding image file.

Step 3: Check the number of image channels. If the input image is a grayscale image, convert it into a three-channel RGB image. If the input image contains more than three channels, retain the first three channels as the model input.

Step 4: Resize each input image to the specified spatial size $s \times s \times 3$.

Step 5: Normalize the resized image by converting pixel values into floating-point format and scaling them to the range [0,1].

Step 6: Feed each preprocessed image branch into its corresponding independent Light ConvNeXt-style backbone.

Step 7: In each backbone, apply the stem module composed of a 3×3 convolution layer and a Layer Normalization layer to generate the initial feature representation.

Step 8: Feed the feature map into Stage 1, where two lightweight ConvNeXt-style blocks with 32 filters are applied. Each block consists of a depthwise 7×7 convolution, Layer Normalization, pointwise expansion, GELU activation, pointwise projection, and residual addition.

Step 9: Feed the resulting feature map into Stage 2. First perform downsampling with stride 2, and then apply two lightweight ConvNeXt-style blocks with 64 filters.

Step 10: Feed the resulting feature map into Stage 3. First perform downsampling with stride 2, and then apply two lightweight ConvNeXt-style blocks with 128 filters.

Step 11: Apply Global Average Pooling and Layer Normalization to the output of each branch to obtain the branch-level feature vector.

Step 12: If only one input scale is selected, directly use the corresponding branch feature vector as the fused representation. Otherwise, concatenate the feature vectors from all selected branches along the feature dimension to form the fused multi-scale representation.

Step 13: Feed the fused representation into the classification head.

Step 14: Apply a Dropout layer to the fused feature vector to reduce the risk of overfitting.

Step 15: Feed the processed feature vector into a fully connected layer with 256 neurons and a ReLU activation function for nonlinear feature transformation.

Step 16: Apply a second Dropout layer to the transformed feature vector.

Step 17: Feed the resulting representation into a final fully connected layer with one neuron and a Sigmoid activation function to generate the predicted probability p that the input sample belongs to CP.

Step 18: Convert the probability value p into the final binary classification result $\hat{y}$ according to the predefined classification threshold.

Step 19: For model evaluation, compare the predicted results with the ground-truth labels and compute Accuracy, Precision, Recall (Sensitivity), Specificity, F1-score, AUC, and the Confusion Matrix.

End.

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The input of the algorithm is a set of aligned multi-scale pulse waveform images, where the scales are selected from the representative scale set of 33×33, 61×61, and 105×105. The algorithm first reads the images at the selected scales and then sequentially performs channel checking, three-channel conversion, resizing, and pixel normalization. After preprocessing, each scale image is fed into its corresponding independent Light ConvNeXt-style backbone for feature extraction. In the single-scale setting, the branch feature vector is directly passed to the classification head. In the multi-scale setting, the feature vectors from different branches are first concatenated and then passed to the classification head to generate the predicted probability p that the sample belongs to CP. Finally, the model converts p into the final binary classification result $\hat{y}$ according to a predefined threshold and computes the performance evaluation metrics.

It should be further noted that this study defines the positive class as CP and the negative class as NCP. This definition directly affects the computation and interpretation of the confusion matrix as well as Recall, Specificity, Precision, and F1-score. Therefore, all subsequent evaluations in this study follow this class definition.

3.4 Experimental Design and Evaluation Protocol

The experimental design of this study aims to compare the classification performance and computational efficiency between single-scale models and multi-scale fusion models. Instead of performing exhaustive scale searching, this study establishes six fixed experimental settings based on three representative scales and their combinations. These include single-scale inputs of ConvNeXt-33, ConvNeXt-61, and ConvNeXt-105, dual-scale fusion settings of ConvNeXt-33+61 and ConvNeXt-61+105, and a triple-scale fusion setting of ConvNeXt-33+61+105. This fixed design enables systematic analysis of the complementary effects between single-scale and multi-scale inputs.

For training configuration, this study uses binary cross-entropy as the loss function and Adam as the optimizer. The training process is combined with EarlyStopping, ReduceLROnPlateau, and ModelCheckpoint to preserve the model weights with the best validation performance. This study adopts the best validation AUC as the primary criterion for model selection. When multiple models achieve similar performance on the primary metric, additional ranking is conducted according to best validation accuracy, test-set F1-score, and training time.

For performance evaluation, this study calculates Accuracy, Precision, Recall (Sensitivity), Specificity, F1-score, AUC, and the Confusion Matrix on the test set. Accuracy measures the overall correctness of classification. Precision measures the proportion of actual CP samples among those predicted as CP. Recall (Sensitivity) measures the proportion of correctly identified CP samples among all actual CP samples. Specificity measures the proportion of correctly identified NCP samples among all actual NCP samples. F1-score reflects the balance between Precision and Recall, while AUC

evaluates the overall discriminative ability of the classifier. In addition, this study reports the learning curves, ROC curves, and confusion matrices of the best model and the representative control model to support a more complete interpretation of the results.

4 Experimental Results and Discussion

To compare the performance differences between single-scale models and multi-scale fusion models, this study conducted a systematic analysis of six fixed experimental settings, using best validation AUC, test F1-score, and training time as the three primary comparison dimensions. The overall results show that the models did not follow a fully consistent ranking across validation performance, balanced classification performance on the test set, and training cost. Therefore, the focus of this chapter is not merely to identify a single top-scoring model, but to clarify the distinct roles and complementary relationships of information at different scales in cp image classification. The overall comparison of the six fixed experiments is presented in Table 2. In terms of the primary selection criterion, best validation AUC, ConvNeXt-33+61+105 achieved the highest value of 97.13%, making it the best validation model in the main experiment. ConvNeXt-33 also reached 97.09%, showing only a minimal difference from the three-scale model, while ConvNeXt-33+61 maintained a comparably high value of 96.59%. In contrast, ConvNeXt-105 achieved only 50.45%, which was markedly lower than the other settings. These results indicate that three-scale fusion can improve peak validation performance in the main experiment, but its advantage does not substantially separate it from all other models. In particular, the 33×33 small-scale model already demonstrated highly competitive discriminative performance, as shown in Figure 3.

Table 2: Overall comparison of the six fixed experimental settings.

Experiment	Model Type	Best validation AUC	Best validation accuracy	Test accuracy	Test F1-score	Test AUC	Training time (second)
ConvNeXt-33	Single-scale	97.09%	92.47%	91.23%	93.23%	95.06%	96.11
ConvNeXt-61	Single-scale	95.16%	91.56%	91.98%	94.03%	93.96%	198.66
ConvNeXt-105	Single-scale	50.45%	57.27%	66.17%	79.64%	51.13%	147.12
ConvNeXt-33+61	Dual-scale fusion	96.59%	92.99%	88.47%	90.91%	93.79%	162.29
ConvNeXt-61+105	Dual-scale fusion	95.09%	90.78%	92.48%	94.51%	95.44%	884.26
ConvNeXt-33+61+105	Triple-scale fusion	97.13%	93.25%	90.48%	92.64%	94.10%	964.33

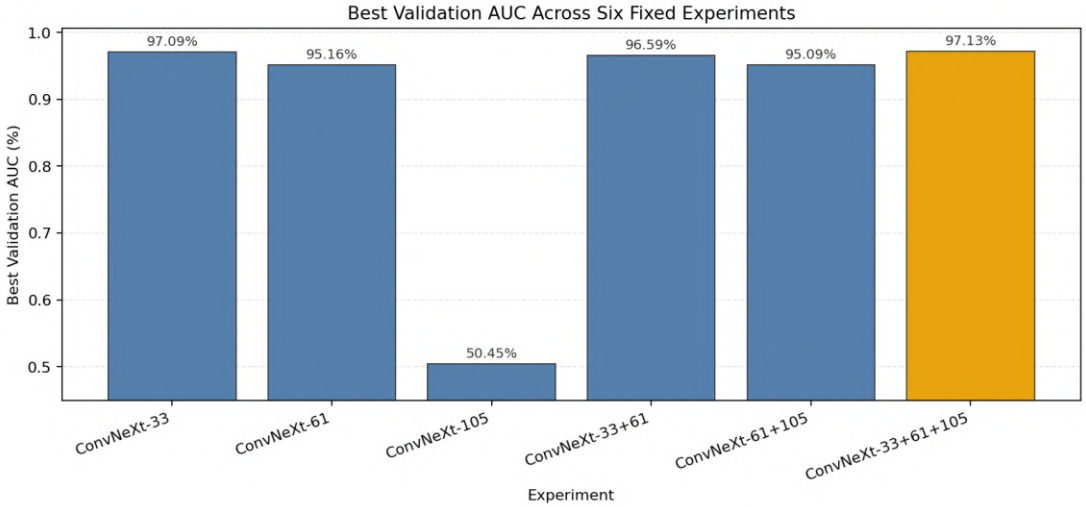


Figure 3: Comparison of best validation AUC across six fixed experiments.

When balanced classification performance on the test set is further considered, the ranking differs from that observed on the validation set. ConvNeXt-61+105 achieved the highest test F1-score of 94.51%, making it the best-performing model on the test set. ConvNeXt-61 and ConvNeXt-33 followed closely at 94.03% and 93.23%, respectively, and both remained highly competitive. ConvNeXt-33+61+105 achieved a test F1-score of 92.64%, which was still stable but not the highest under this metric, whereas ConvNeXt-105 dropped to 79.64%. These results suggest that multi-scale fusion can indeed provide complementary benefits on the test set, but the best-performing combination is not necessarily the one with the largest number of branches. In particular, the combination of 61×61 and 105×105 achieved stronger balanced classification performance, indicating meaningful complementarity between the main contour and the global context, as shown in Figure 4. In terms of training cost, ConvNeXt-33 required only 96.11 s and was the most efficient among all six settings, whereas ConvNeXt-61+105 and ConvNeXt-33+61+105 required 884.26 s and 964.33 s, respectively. This finding shows that higher-level multi-branch fusion may improve performance, but it also incurs substantially higher computational cost, as shown in Figure 5. Taken together, these three comparisons do not support a one-directional conclusion that multi-scale models must always outperform single-scale models. Instead, the overall pattern is more consistent with a “small-scale dominant, multi-scale complementary” interpretation.

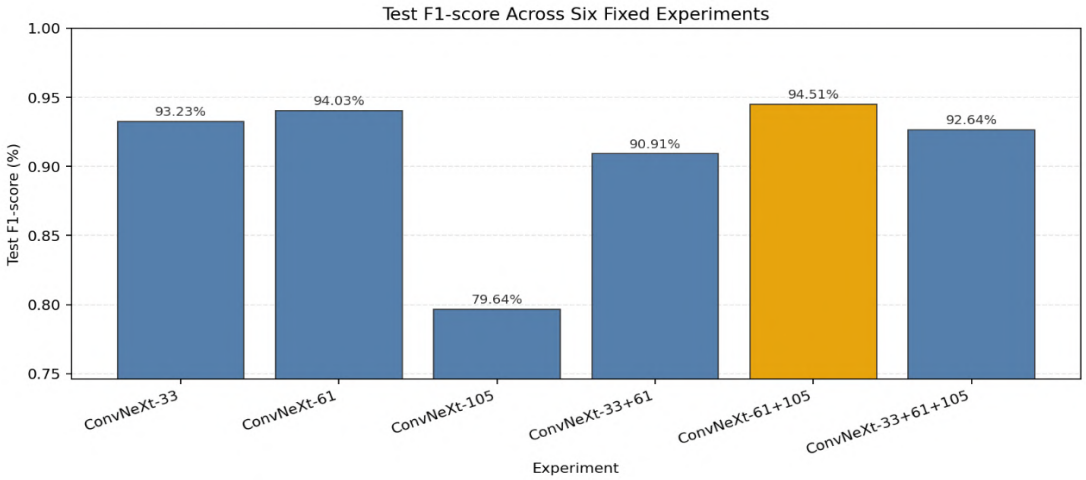


Figure 4: Comparison of test F1-score across six fixed experiments.

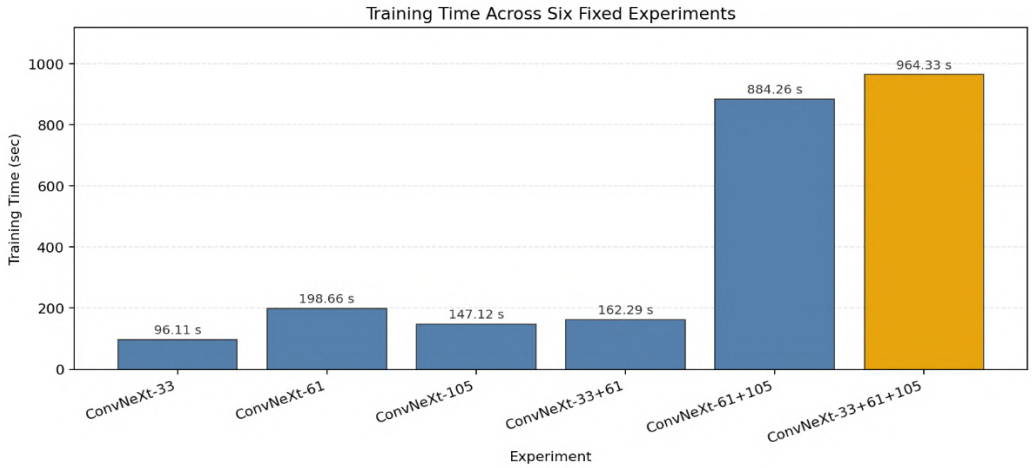


Figure 5: Comparison of training time across six fixed experiments.

Based on the model selection criterion adopted in this study, ConvNeXt-33+61+105 was selected as the best model in the main experiment because it achieved the highest best validation AUC. The learning curves show that both train_acc and val_acc rose rapidly in the early stage and then remained stable at a high level, while train_loss and val_loss decreased quickly before gradually flattening. No obvious sign of overfitting was observed overall, indicating that the three-scale fusion model maintained stable convergence behavior in this task, as shown in Figure 6(a)-(b). The ROC curve further shows a test AUC of 0.9410, with the curve remaining clearly above the random baseline, indicating strong overall discriminative ability between CP and NCP, as shown in Figure 7(a). The confusion matrix shows that the model correctly classified the majority of CP and NCP samples, with relatively similar Sensitivity and Specificity and an overall balanced error distribution. However, some CP samples were still misclassified, suggesting that certain CP samples may still share ambiguous boundaries or overlapping representations with NCP samples, as shown in Figure 8(a). These results indicate that integrating local detail, main contour, and global context can form a more complete discriminative

representation, and they support the feasibility of applying a Light ConvNeXt-style multi-branch design to CP image classification.

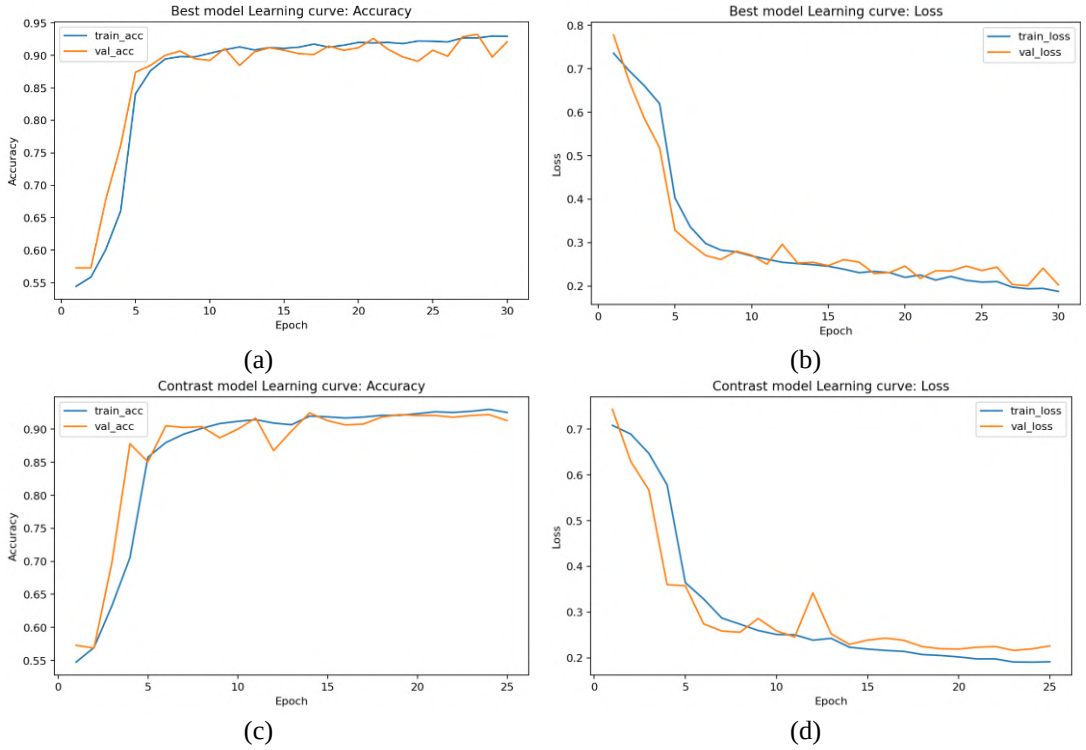
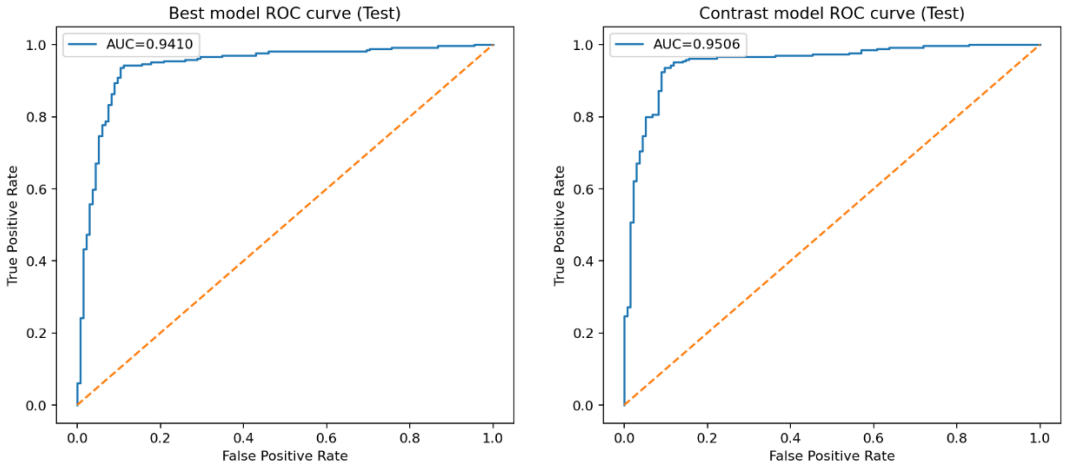
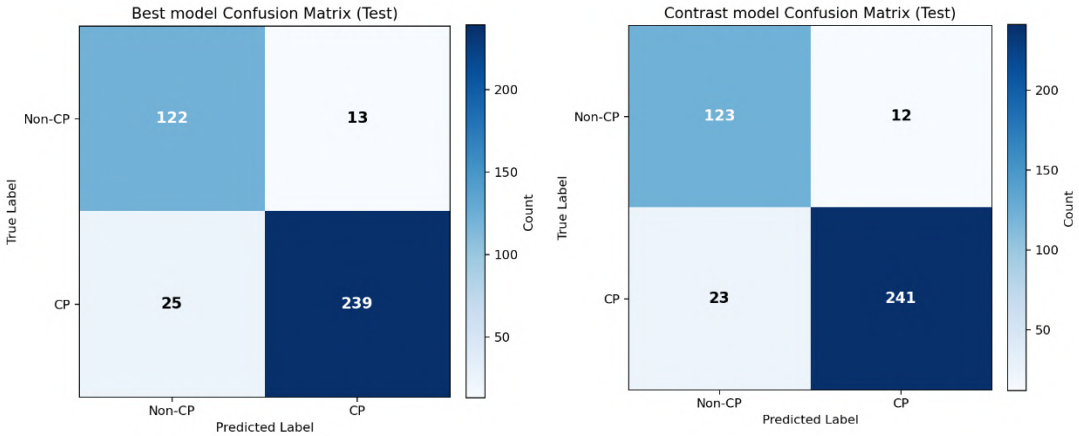


Figure 6: Learning curves of the representative model and the control model: (a) accuracy curve of ConvNeXt-33+61+105; (b) loss curve of ConvNeXt-33+61+105; (c) accuracy curve of ConvNeXt-33; and (d) loss curve of ConvNeXt-33.



(a) (b)

Figure 7: ROC curves of the representative model and the control model: (a) test ROC curve of ConvNeXt-33+61+105; (b) test ROC curve of ConvNeXt-33.



(a) (b)

Figure 8: Confusion matrices of the representative model and the control model: (a) test confusion matrix of ConvNeXt-33+61+105; (b) test confusion matrix of ConvNeXt-33.

To further clarify the difference between single-scale learning and multi-scale fusion, this study selected ConvNeXt-33 as the representative control model. This model was not only the strongest among the single-scale settings, but also the fastest among all six fixed experiments, and therefore serves as a representative baseline for the local-detail scale. The results show that ConvNeXt-33 achieved a best validation AUC of 97.09%, which was very close to that of the best three-scale model, and its test ROC AUC reached 0.9506, which was slightly higher than that of ConvNeXt-33+61+105. Further comparison of the learning curves, ROC curve, and confusion matrix shows that ConvNeXt-33

maintained performance similar to that of the best three-scale model in terms of convergence stability, discriminative ability, and error distribution, as shown in Figure 6(c)-(d), Figure 7(b), and Figure 8(b). The confusion matrix further indicates that ConvNeXt-33 produced slightly fewer overall classification errors than the best three-scale model, suggesting a good balance between Sensitivity and Specificity. These findings indicate that the local detail captured by the 33×33 scale already contains highly informative discriminative cues for CP recognition. In other words, local waveform detail is not merely a secondary supplement, but one of the core foundations supporting classification performance. By contrast, the value of multi-scale fusion does not lie in fully replacing single-scale learning, but in supplementing contour-level and contextual information under specific combinations so that the model can achieve more complete performance across different evaluation criteria.

Taken together, the main implications of this chapter can be summarized in three points. First, the discriminative information in CP images exhibits a clear multi-scale hierarchical structure. ConvNeXt-33 remained highly competitive in validation performance, test performance, and training efficiency, indicating that the local detail represented by the 33×33 scale serves as an important discriminative basis. Meanwhile, the fact that ConvNeXt-61+105 achieved the best test F1-score indicates that the main contour and the global context can still provide meaningful complementary information on the test set. In addition, ConvNeXt-33+61+105 achieved the highest best validation AUC, further suggesting that integrating all three field-of-view levels still has the potential to improve peak validation performance. These findings indicate that local detail, main contour, and global context in CP images are not homogeneous sources of information, but jointly contribute to classification through different scale levels. The overall comparison of the six fixed experiments is presented in Table 2, while the overall trends of the main metrics are shown in Figs. 3-5.

Second, the value of multi-scale fusion lies in complementing, rather than replacing, single-scale learning. As reviewed in Chapter 2, prior studies on multi-scale feature fusion suggest that features from different levels may be complementary. The present results further show that such complementarity must be verified under specific tasks and scale combinations, rather than assumed directly from the number of branches. In this study, the small scale provided the main discriminative source, whereas the medium and large scales supplemented contour and contextual information under particular combinations, allowing the models to achieve more complete performance across different metrics. Conversely, the clearly weaker performance of ConvNeXt-105 indicates that preserving only the global context, without sufficient support from local detail and main contour, is not enough for stable classification. Therefore, the results of this study are more appropriately interpreted through a "small-scale dominant, multi-scale complementary" framework, rather than viewing multi-scale fusion as a complete alternative to single-scale modeling.

Third, the Light ConvNeXt-style multi-branch design is feasible under a small-sample condition with a patient-wise fixed split. The learning curves, ROC curves, and confusion matrices of both the best model and the representative control model show overall stable convergence, good discriminative ability, and relatively balanced error distributions. Because this study adopted a patient-wise fixed split strategy, the resulting performance more realistically reflects the difficulty of cross-subject classification and thus makes the comparisons in this chapter more credible. Overall, the contribution of this study lies in showing that CP image classification can be understood through a multi-scale hierarchical information structure, that multi-scale fusion can serve as a complementary design strategy to improve the completeness of model performance, and that a Light ConvNeXt-style architecture can provide a practically promising classification solution under limited-data conditions. These findings further suggest that the proposed design has both methodological feasibility and practical value for the development of a choppy pulse-assisted recognition system.

5 Conclusion and future work

This study aimed to address binary classification of CP images by proposing a multi-scale feature fusion method based on a Light ConvNeXt-style architecture. Using three representative scales, 33×33, 61×61, and 105×105, six fixed experimental settings were constructed to systematically compare single-scale models with multi-scale fusion models. The results indicate that the proposed method is feasible and that information at different scales indeed plays distinct hierarchical roles in CP recognition. In the main experiment, ConvNeXt-33+61+105 achieved the highest best validation AUC, indicating that integrating local detail, main contour, and global context still has the potential to improve peak validation performance. Meanwhile, ConvNeXt-33 remained highly competitive among the single-scale models, whereas ConvNeXt-61+105 achieved the best test F1-score. Overall, the findings of this study are more consistent with a "small-scale dominant, multi-scale complementary" interpretation, rather than viewing multi-scale fusion as a complete replacement for single-scale modeling.

The significance of this study can be summarized from three perspectives. First, the discriminative information in CP images is not homogeneous, but exhibits a clear multi-scale hierarchical structure. The strong performance of the 33×33 setting indicates that local waveform detail serves as an important basis for CP classification, whereas the 61×61 main contour and the 105×105 global context can further supplement waveform information under specific combinations. Second, the value of multi-scale fusion lies in complementing discriminative information across different field-of-view levels, rather than directly replacing single-scale models with more branches. This finding is consistent with the original design objective of this study and also corresponds to the multi-level observational logic of traditional Chinese medicine pulse diagnosis, which considers local detail, main contour, and overall pulse context together. Third, this study demonstrates the feasibility of a Light ConvNeXt-style multi-branch design under a small-sample condition with a patient-wise split. As shown in Chapter 4, the models produced reasonable results in terms of convergence stability, overall discriminative ability, and balanced error distribution, suggesting that this design can serve as one of the technical foundations for a choppy pulse-assisted recognition system.

Beyond methodological feasibility, this study also has practical value. Traditional Chinese medicine pulse diagnosis has long relied on physician experience and subjective judgment. Accordingly, establishing a more consistent digital analysis process without departing from the observational logic of traditional Chinese medicine remains an important direction for the objectification of pulse diagnosis. The proposed multi-scale Light ConvNeXt-style architecture was not designed simply to pursue the highest score, but rather to construct an image classification approach that more closely reflects the multi-level observational process of pulse diagnosis through local detail, main contour, and global context. The results suggest that this direction is reasonable and further indicate that choppy pulse-assisted image recognition does not need to be restricted to a single field of view, but may benefit from appropriate scale design to improve both the completeness of model performance and the interpretability of the overall methodological framework.

Nevertheless, several limitations of this study should be noted. First, the dataset remains limited in scale, and the present task focuses on binary classification between CP and NCP. Therefore, the current conclusions should be interpreted within the data condition and task setting of this study. Second, although multi-scale fusion showed complementary benefits under some metrics, the advantages of different models were not fully consistent across all evaluation dimensions, indicating that the effectiveness of multi-scale design still depends on specific objectives. Third, although a patient-wise fixed split was adopted to improve the credibility of evaluation, the present findings are still derived from a single research framework and would benefit from further validation using broader data sources and more diverse scenarios. Finally, this study mainly focused on model performance and methodological comparison. Further work may strengthen visualization-based interpretation of model decisions and improve integration with clinical use workflows.

Based on these limitations, several directions may be considered in future work. First, the dataset size and subject sources may be expanded to further examine the stability and generalizability of the proposed framework under broader sample conditions. Second, the multi-scale design may be extended to additional pulse categories or multi-class classification tasks in order to evaluate its applicability in more complex pulse diagnosis scenarios. Third, explainable analysis methods may be incorporated to identify the key waveform regions associated with each scale branch during classification, thereby strengthening the connection between model outputs and traditional Chinese medicine pulse interpretation. Fourth, while maintaining the lightweight design, future studies may further optimize the multi-branch fusion strategy to improve the overall balance among performance, computational cost, and stability. Overall, this study provides an initial validation of the feasibility of applying multi-scale feature fusion and a Light ConvNeXt-style architecture to CP image classification, and offers a concrete foundation for subsequent methodological development and practical implementation of choppy pulse-assisted recognition systems.

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A State-Aware Zero Trust Communication Approach for Offshore Wind Farm*

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Abstract

Because physical on-site access is constrained, offshore wind farm OT environments rely heavily on remote monitoring, maintenance, and control, and consequently diverse communications with different operational purposes and levels of importance continuously occur. However, existing Zero Trust discussions in OT environments have mainly focused on structural aspects such as access restriction, path control, and segmentation, and have not sufficiently specified how the operational significance of communication should be interpreted or how it should be connected to differentiated protection in OT environments. Accordingly, this paper proposes a state-aware communication approach that classifies offshore wind farm OT communication into normal, warning, and critical states, interprets the meaning and importance of communication differently according to the operational state, and then applies differentiated protection levels accordingly. The controlled prototype-level case study results show that the proposed approach maintained the loss rate of important communication at 0.3% while limiting the total communication volume to 4.1 MB, thereby forming a more balanced protection structure between the visibility of important communication and the overall communication burden.

1 Introduction

Offshore wind farm OT(Operational Technology) environments, as illustrated in Fig. 1, rely heavily on remote monitoring, maintenance, and control because physical on-site access is constrained, and consequently diverse types of communication continuously occur[1]. Operational data generated at lower layers are repeatedly delivered to upper operational layers, and maintenance and control are likewise performed through remote paths[2]. NIST SP 800-82 Rev.3[3] emphasizes that OT security

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must consider performance, reliability, and safety requirements that differ from those of general IT, while the U.S. Department of Energy (DOE) states that distributed energy resource (DER) systems should be designed, built, and operated according to a zero-trust model. Therefore, communication in offshore wind farm OT environments should be examined not merely as a connectivity issue but from the perspective of zero trust communication[3][4]. However, offshore wind farm OT communication does not have a single, uniform character. Some communications correspond to repetitive operational telemetry transmission, while others are directly related to maintenance, management, or actual control. In addition, communication in wind power plants may be implemented not only through wired networks but also through wireless and cellular means; thus, not only functional differences but also instability in communication conditions must be taken into account.

The core of zero trust does not lie in uniformly applying the strongest protection to all communications, but in assigning the minimum necessary level of protection through policy after individually evaluating each communication without implicitly trusting any of them[5]. However, existing discussions have primarily focused on structural aspects such as access restriction, path control, and segmentation, and have not sufficiently specified how the operational significance of communication in offshore wind farm OT environments should be interpreted or how it should be connected to protection differentiation. If the same lightweight protection is applied to all communications, protection for important communications may be insufficient; conversely, if the same strong protection is applied to all communications, unnecessary overhead may accumulate. Therefore, zero trust communication in offshore wind farm OT environments needs to be concretized not as uniform protection, but as an approach that interprets the significance of communication differently according to the operational state and applies different protection levels accordingly.

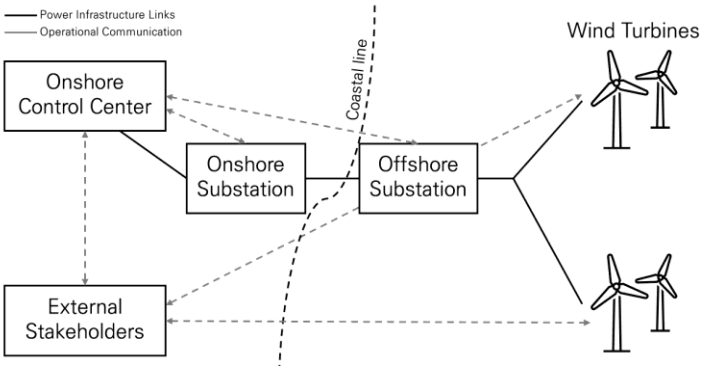

Figure 1: Conceptual Representation of an Offshore Wind Farm

The research questions of this paper are as follows:

- In offshore wind farm OT environments, according to what criteria should zero trust communication distinguish the significance of each communication and the required protection level?
- In offshore wind farm OT environments, is it possible to interpret the significance of communication differently according to the operational state and apply different protection levels accordingly?

The novelty of this paper does not lie in proposing context-aware protection itself. Rather, it lies in applying operational-state-dependent communication criticality to offshore wind farm OT communication. Existing dynamic Zero Trust studies mainly adjust access policies according to user,

device, network, or segment context, whereas this study focuses on how the operational state of a physical asset changes the meaning of the same communication and the required protection level. In this sense, the proposed approach complements existing Zero Trust architectures by linking operational state, communication criticality, and differentiated protection in a unified communication-handling model.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 presents a state-aware communication approach for zero trust communication in offshore wind farm OT environments. Section 4 explains the applicability of the proposed approach through a prototype-level case study using a representative operational variable. Finally, Section 5 concludes the paper and discusses future research directions.

2 Related Works

Existing studies have proposed various structures for applying Zero Trust to OT environments by pointing out the limitations of the traditional perimeter-based security model and focusing on continuous authentication and authorization, least privilege, policy-based access control, micro-segmentation, remote access control, and context-aware dynamic policies.

There are studies that attempt to reconstruct OT architectures themselves from a Zero Trust perspective. These studies sought to design Zero Trust structures that can be practically applied to OT environments through elements such as Purdue model enhancement, asset identification, policy-based access control, legacy system protection, and continuous monitoring. In particular, studies targeting smart manufacturing and legacy manufacturing environments reflected practical constraints such as end-of-support operating systems, long-life operational equipment, and the difficulty of patch application, and proposed not merely theoretical Zero Trust models but operationally deployable security structures [7], [10], [11].

There are also studies that attempt to implement Zero Trust in industrial networks and IIoT infrastructures with a focus on micro-segmentation and remote access protection. Studies such as [6], [13], and [15] sought to segment industrial networks more finely and control resource access more precisely through SDN, per-device segmentation, policy enforcement points, and cross-layer access control. In addition, [9] and [12] proposed architectures that verify the access of external entities on a per-session basis and grant segmented access privileges in industrial remote access environments.

More recently, studies have also emerged that seek to extend Zero Trust toward context-aware and dynamic Zero Trust. [14] proposed a structure that adjusts access policies in real time by reflecting user behavior, device status, network conditions, and segment criticality, while [15] emphasized multi-layer trust evaluation that considers interactions between the cyber layer and the physical layer in industrial CPS. [8] presented a stepwise structure of security zone formation, device authentication, and cryptographic negotiation in wireless IIoT environments, thereby translating Zero Trust requirements into a more concrete communication-technology level.

Although prior studies provide valuable mechanisms for identity verification, segmentation, policy enforcement, remote access control, and context-aware policy adjustment, their main focus remains on deciding whether an entity should be allowed to access a resource under a given context. They do not sufficiently address how the operational state of a physical asset changes the semantic importance of telemetry or control-related communication itself. This distinction is important in offshore wind farm OT environments, where the same communication may function as routine reporting during normal operation, as evidence for abnormal-state interpretation during warning conditions, or as a basis for control-oriented response under critical conditions. Therefore, this study complements prior dynamic Zero Trust approaches by focusing on operational-state-dependent communication criticality rather than only access permission or segmentation.

3 State-Aware Zero Trust Communication Approach in Offshore Wind Farm OT Environments

This section presents a state-aware communication approach for concretizing zero trust communication in offshore wind farm OT environments. Because it is difficult to treat all communications in offshore wind farm OT environments as having the same significance and requiring the same protection level, this paper proposes an approach that interprets the significance of communication differently according to the operational state and applies different protection levels accordingly.

In the proposed structure, the operational state is used as a contextual input for Zero Trust policy decisions. The state evaluation logic continuously interprets operational data and provides the current state to the policy decision function, which considers entity identity, communication purpose, and the required protection level. The corresponding policy enforcement function then applies state-specific communication handling, such as lightweight reporting, sequence-number-based reinforced reporting, or a restricted protected control session. In this way, the proposed approach complements existing OT Zero Trust architectures by adding operational-state awareness to the policy decision and enforcement process.

This study assumes that communication paths between offshore OT components and upper operational functions are not implicitly trusted. An adversary may attempt packet-loss or denial-of-service-like disruption, replay, spoofed telemetry, message injection, or man-in-the-middle interception. The proposed approach is intended to address these risks at the communication-policy level by combining state-dependent protection selection, entity identity verification, policy-based enforcement, sequence tracking, and restricted protected sessions. However, compromise of local turbine safety controllers, root trust anchors, or physical destruction of field assets is outside the scope of this prototype-level study, and active adversarial validation remains future work.

3.1 Limitation of Uniform Protection

In offshore wind farm OT environments, repetitive operational data reporting, maintenance communication, management signaling, and control-related communication coexist. Since these communications differ in their purpose and operational consequences, it is not appropriate to apply the same protection procedure uniformly to all of them. For example, if all communications are handled only with lightweight protection, protection may be insufficient for operationally more important abnormal-state intervals or control-related communication. Conversely, if the same strong protection is applied to all communications, unnecessary communication burden may accumulate in the repetitive reporting structure.

Therefore, zero trust communication in offshore wind farm OT environments should be understood not simply as a matter of applying stronger protection more broadly, but rather as a matter of first determining under which state a given communication should be interpreted as more important and then selecting an appropriate protection level accordingly. Reflecting this perspective, this paper uses the operational state as the basis for communication protection differentiation.

3.2 State-Aware Communication Operation

This paper classifies offshore wind farm OT communication into three operational states: normal, warning, and critical. Here, state does not refer to the technical property of the communication itself, but rather to a concept that reflects the operational significance of the communication and the required protection level under the current operating condition. In other words, even the same type of

communication may serve as routine reporting during normal operation, require higher visibility and reliability when abnormal signs appear, and demand stronger protection for actual control intervention under critical conditions.

Table 1 summarizes the meaning of each state used in this paper and the corresponding direction of basic communication handling. The purpose of this table is not to define absolute thresholds based on a specific industrial standard, but to provide a conceptual basis for explaining state-aware communication operation in offshore wind farm OT environments. Specific numerical thresholds are illustratively defined in the case study using a representative operational variable.

The normal state refers to a condition in which the equipment is operating within the normal range

State	Meaning	Communication Characteristics	level of protection
Normal	Normal Operation	Periodic Reporting	Lightweight protection
Warning	Warning Signals	Reinforced Reporting	Enhanced Protection
Critical	Critical Condition	Control Intervention	Strong Protection

Table 1: Definition of Operational States

and communication mainly serves the role of routine telemetry reporting. Since continuous upward reporting is required in this interval, this paper adopts UDP-based lightweight reporting as the default in order to reduce communication burden. In other words, in the normal state, the focus is not on uniformly applying high-cost protection procedures to all reporting traffic, but on transmitting repeatedly generated operational data with low overhead.

The warning state refers to a condition in which abnormal signs are observed and greater attention is required for subsequent state assessment and operational response. As explained in the introduction, this interval requires consideration of environments in which packet loss may occur, such as wireless communication. Accordingly, this paper applies a method that assigns sequence numbers to reported data in order to ensure continuity in monitoring and tracking. In other words, the warning state goes beyond maintaining simple routine reporting and serves as a phase that reinforces the visibility and traceability of communication required for interpreting abnormal conditions.

The critical state refers to a condition in which actual operational intervention or control-oriented response is required. In this interval, the impact of communication failure or spoofing becomes greater, and therefore this paper applies the highest level of protection. In other words, the critical state is interpreted not as a monitoring-oriented communication condition, but as one that requires control-oriented communication through a restricted protected channel.

Fig. 2 shows the proposed flowchart of the state-aware communication operation. The overall flow consists of operational data collection, state evaluation, state-specific communication handling, and repeated state updates. First, operational data generated at lower layers are delivered to upper operational functions, and the system determines the current operational state based on these data. In the prototype case study, this state transition is implemented as a rule-based evaluation that compares the generator bearing temperature with the illustrative thresholds defined in Section 4.1. Different protection levels are then applied according to the identified state. Since the state-evaluation layer directly affects protection-level selection, deployable implementations should not rely on unauthenticated telemetry alone. Telemetry used for state evaluation should be bound to authenticated device identities and protected by integrity checks, sequence tracking, and audit logs; cross-checking multiple operational variables or independent monitoring sources can further reduce the risk of state-suppression attacks, such as spoofed normal telemetry during an intrusion.

In the normal state, routine telemetry reporting is the main form of communication, and therefore this paper applies a lightweight protection method that maintains low communication burden. The purpose of this interval is not to process all reports through high-cost protection procedures, but to sustain a continuous data flow during normal operation.

In the warning state, since the possibility of abnormal conditions is recognized, a higher level of communication assurance is required than in the normal state. Accordingly, sequence numbers are applied in this interval so that important reported data can be retained more reliably. This does not mean that all communication is immediately converted into a strong control channel, but rather that the information serving as the basis for state assessment and subsequent response is secured more reliably.

In the critical state, since the possibility of actual control intervention becomes higher, the significance of communication is also interpreted as the highest. Therefore, this paper assumes that communication in the critical state is carried out through a restricted protected channel to which strong protection is applied. In the proposed operation, the critical-state protected session is established only after the system identifies a critical condition and the policy logic approves the need for control-oriented communication. The session is limited to the approved communication purpose and is revoked after the required control-related task is completed. Therefore, the critical state is not treated as a permanent high-protection mode, but as a task-scoped and temporary communication condition for protected control communication.

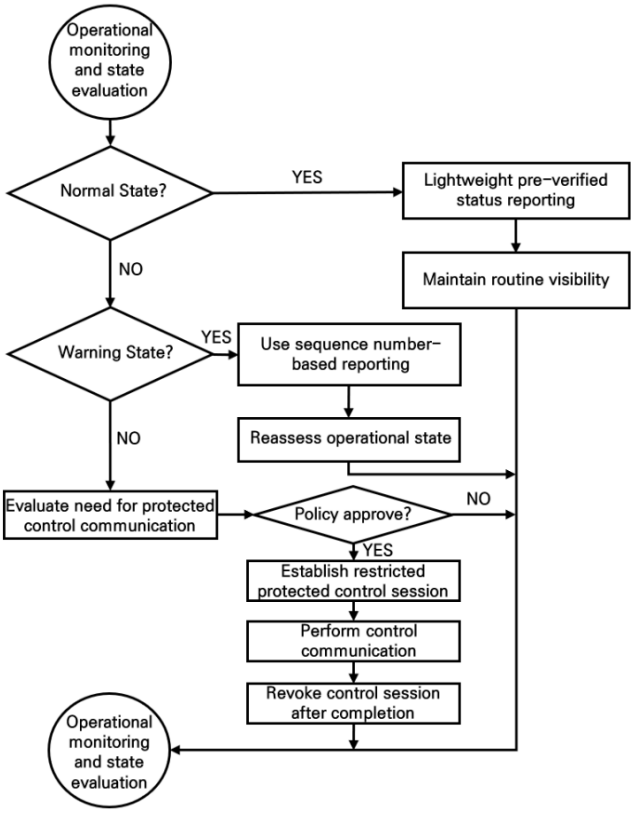

Figure 2: Flowchart of the State-Aware Communication Operation

In this way, the state-aware communication approach proposed in this paper does not apply the same protection to all communication. Instead, it interprets communication significance differently according to the state and adjusts the protection level accordingly. Such a structure is consistent with the basic zero trust principle of assigning necessary protection according to context without relying on implicit trust.

4 Case Study

This section uses one representative operational variable of the offshore wind farm OT environment to illustratively define the normal, warning, and critical states, and demonstrates, through a prototype-level case study, how the state-aware communication approach can be applied in practice. The purpose of this case study is not to precisely reproduce the entirety of a real offshore wind farm, but rather to illustratively examine whether an approach that interprets the significance of communication differently according to the operational state and applies different protection levels accordingly can be implemented.

4.1 Case Study Configuration

In this prototype-level case study, generator bearing temperature was selected as a representative operational variable of an offshore wind farm OT environment. This variable is commonly associated with equipment condition monitoring and is suitable for illustrating how operational states can be mapped to different communication-handling policies. The purpose of using this variable is not to define a complete turbine health assessment model, but to construct a controlled scenario in which normal, warning, and critical states can be distinguished in an intuitive manner.

Specifically, the bearing temperature was defined as normal when it was 60°C or below, warning when it was between 61°C and 80°C, and critical when it was above 80°C. These values are not intended to represent universal turbine safety limits, vendor-specific alarm thresholds, or industrial standards. They are used only as illustrative thresholds for explaining state-aware communication operation. Accordingly, the 60°C and 80°C values should not be interpreted as literature-derived turbine condition-monitoring thresholds; in actual deployments, such thresholds should be determined from turbine-specific historical data, vendor alarm settings, or condition-monitoring guidelines. The reporting interval was set to one second for all schemes.

The experimental environment was configured as a controlled communication-stress scenario that reflects simplified wireless communication instability in offshore wind farm environments. As offshore wind farms move farther from the coast, wireless or cellular communication links may become more important for remote monitoring and operation. To reflect this condition in a simplified prototype setting, the packet-loss condition was modeled as independent random loss with a fixed probability of 5% for each reporting packet. The same topology, reporting schedule, observation window, telemetry message format, and packet-loss condition were applied to all schemes so that the differences in warning-state loss and total communication volume could be attributed to the protection strategy rather than to different traffic-generation conditions. In terms of loss handling, r0 did not apply any additional recovery mechanism, r1 applied non-selective retransmission without considering the operational state, r2 applied sequence-number-based recovery to all reporting traffic, and the proposed scheme applied sequence-number-based reinforced handling selectively to warning-state reports while using a restricted protected session for critical-state communication. Therefore, the case study should be interpreted as a controlled prototype-level comparison, not as a general validation of all offshore wind farm communication environments.

This setting represents non-malicious operational network stress caused by simplified wireless communication instability, rather than a full active-adversary experiment. Active attack profiles such as denial-of-service, message injection, or man-in-the-middle interception require separate adversarial validation and are therefore treated as future work in this prototype-level study.

4.2 Comparison Targets

As shown in Table 2, this case study compares four communication schemes, each representing a distinct protection strategy. The purpose of this comparison is to examine the trade-off between

Scheme	Routine Reporting	Loss Handling	Critical State Response
r0	Lightweight reporting	None	No separate procedure
r1	Lightweight reporting	Non-selective retransmission	No separate procedure
r2	Sequence-numbered reporting	Missing-report recovery	No separate procedure
Proposed	Lightweight reporting	Warning-state retransmission	Restricted control session

Table 2: Comparison of Communication Schemes

communication burden and warning-state visibility under the same controlled experimental condition. These schemes are defined as analytical baselines for comparing protection strategies rather than as vendor-specific implementations or standardized industrial profiles.

r0 represents a lightweight-only baseline. It performs UDP-based lightweight reporting in all states without additional loss handling or state-dependent reinforcement. This scheme is used to represent the lowest communication-burden case, but it does not provide additional visibility or recovery when abnormal signs appear.

r1 represents a non-selective reinforcement baseline. It maintains lightweight reporting as the default communication method, but applies retransmission-based reinforcement without considering the operational state. This scheme is used to examine whether reinforcement that is not guided by state awareness can effectively reduce warning-state data loss.

r2 represents a uniform reinforced recovery baseline. It applies sequence-number-based recovery to all reporting traffic regardless of whether the current state is normal, warning, or critical. This scheme is used to represent a strong but non-selective protection strategy that improves data retention at the cost of increased communication burden.

The proposed scheme represents the state-aware hybrid protection strategy described in Section 3. It applies UDP-based lightweight reporting in the normal state, sequence-number-based reinforced reporting in the warning state, and restricted protected communication in the critical state. Unlike r2, it does not reinforce all reporting traffic uniformly. Instead, it selectively applies additional protection when the operational state indicates that communication visibility or control-oriented protection is required.

4.3 Evaluation Metrics

This case study used two metrics. The first metric is the warning-state prioritized data delivery loss rate, which indicates the proportion of warning-state report data that failed to be delivered within the intended reporting window. This metric is used to evaluate how effectively each scheme preserves important communication during warning-state intervals. Fig. 3 shows the warning-state prioritized data delivery loss rate for the four schemes.

The second metric is the total communication volume over 10 minutes. This metric indicates the communication burden introduced by each scheme under the same experimental condition. It is used to compare whether each protection strategy increases communication overhead while attempting to improve warning-state visibility. Fig. 4 shows the total communication volume over 10 minutes for the four schemes.

4.4 Case Study Results

As shown in Fig. 3, the warning-state prioritized data delivery loss rate was 4.5% for r0, 5.1% for r1, 0.3% for r2, and 0.3% for proposed. In other words, r2 and proposed showed the lowest loss rates in the warning state, whereas r0 and r1 showed relatively higher loss rates. This indicates that simply relying on lightweight reporting alone may not be sufficient to ensure the retention of important

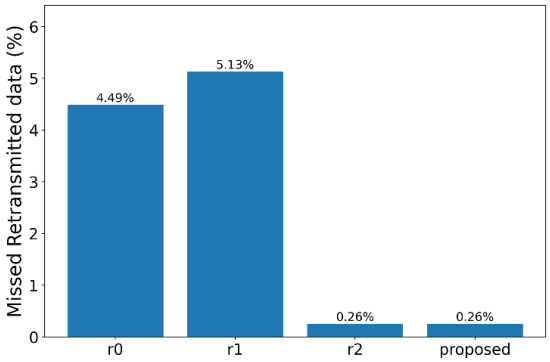

Figure 3: Warning-state prioritized data delivery loss rate under the 5% packet-loss condition

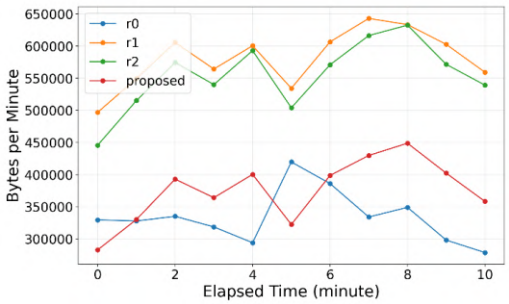

Figure 4: Total communication volume over 10 minutes under the 5% packet-loss condition

communication in abnormal-state intervals, and that non-selective reinforcement without state awareness may also be ineffective. In particular, proposed achieved the same level of warning-state visibility as r2 by applying sequence-number-based reinforced handling only when necessary in the warning state.

As shown in Fig. 4, the total communication volume over 10 minutes was 3.7 MB for r0, 6.4 MB for r1, 6.1 MB for r2, and 4.1 MB for proposed. Although r0 showed the lowest communication burden, its warning-state visibility was insufficient. Because r1 and r2 applied broader recovery handling, their communication usage increased significantly. In contrast, proposed selectively applied only the reinforcement needed in the warning state, thereby achieving the same level of warning-state reliability as r2 while maintaining a lower overall communication volume.

Taken together, r0 had the lowest burden but lacked state-aware handling, r1 incurred increased overhead due to non-selective reinforcement without state distinction while failing to sufficiently reduce warning-state loss, and r2 showed a low warning-state loss through the strongest level of uniform reinforcement but increased the communication burden by reinforcing even routine reporting. By contrast, proposed can be interpreted as a state-aware hybrid structure that lies between these schemes, maintaining low burden in the normal state while reinforcing the visibility and retention of important communication in the warning state and considering a stronger protected channel in the critical state.

4.5 Discussion

The main finding of this case study is that state-aware hybrid protection can form a more flexible protection structure than a single uniform protection scheme in offshore wind farm OT communication. Although r0 showed the lowest communication overhead, its reliability in the warning state was insufficient. Although r1 applied non-selective reinforcement, it still failed to adequately reduce warning-state loss. r2 achieved a low warning-state loss rate, but its communication burden increased because the same reinforced procedure was applied to all reporting traffic. By contrast, the proposed scheme maintained the warning-state loss rate at the same level as r2, 0.3%, while limiting the total communication volume to 4.1 MB, which was lower than r2's 6.1 MB. This indicates that the proposed

scheme can provide a more balanced trade-off between routine reporting efficiency and warning-state communication visibility.

These results are directly connected to the problem statement raised in the introduction. Offshore wind farm OT communication does not have a single, uniform character. Routine telemetry, abnormal-state reporting, and control-oriented communication have different operational meanings and different security requirements. Therefore, it is not appropriate to uniformly trust or assign the same protection level to all communication based only on network location or boundary membership. The proposed scheme addresses this issue by interpreting the significance of communication according to the operational state and applying differentiated protection levels accordingly. In particular, it maintains lightweight reporting in the normal state, reinforces visibility and traceability in the warning state, and applies a restricted protected session in the critical state.

The critical-state handling should be interpreted as an important part of the proposed state-aware communication structure. In the prototype implementation, the critical state was not treated merely as a conceptual label, but as a condition in which a restricted protected communication session is established for control-oriented communication. The session is designed to be limited to an approved communication purpose and to be revoked after the required task is completed. This demonstrates that the proposed state-aware approach can be extended beyond monitoring-oriented reporting and can support a stronger communication mode when operational intervention is required. However, this result should still be understood as a prototype-level validation of the restricted protected session mechanism, not as a complete validation of all industrial real-time control requirements. In actual offshore wind farm OT environments, additional validation would be required to measure session establishment delay, control-message latency, revocation timing, and compatibility with site-specific safety requirements.

The restricted protected session in the critical state is not intended to replace local safety interlocks or hard real-time emergency shutdown mechanisms. In OT environments, emergency protection functions that require deterministic response times should remain locally enforced by turbine controllers or safety systems. The proposed protected session is intended for policy-mediated remote control or maintenance communication, where identity verification, authorization, and session revocation can be applied without placing the emergency shutdown path behind a new cryptographic session-establishment delay. Therefore, the session establishment delay should be bounded and validated against site-specific timing requirements before deployment.

The proposed approach also clarifies how Zero Trust communication can be adapted to OT constraints. A uniform strong-protection approach may improve security but can introduce unnecessary overhead in repetitive normal-state reporting. Conversely, a uniformly lightweight approach may reduce overhead but can fail to preserve important data during abnormal-state intervals. The proposed scheme avoids both extremes by using the operational state as the basis for selecting the required protection level. In this sense, the proposed approach does not simply add stronger protection to all communication, but applies protection selectively according to communication criticality. This is consistent with the Zero Trust principle that communication should not be implicitly trusted and that access or protection levels should be continuously determined according to context and policy.

In this prototype, the 4.1 MB test volume represents UDP-based telemetry reporting and scheme-specific retransmission or protection behavior, rather than a full encapsulation of industrial protocols such as MQTT, OPC UA, or Modbus. Nevertheless, the proposed state-aware filtering and protection logic is not tied to a specific protocol and could be implemented at offshore gateways or onshore monitoring platforms as a policy layer above or alongside such industrial communication protocols. From an enterprise information systems perspective, this allows state-aware OT communication to support onshore monitoring, maintenance management, asset management, and decision-support systems across multiple geographically distributed wind farm clusters.

Nevertheless, the numerical results should be interpreted as evidence of feasibility under controlled prototype-level assumptions, not as statistically generalizable results for all offshore wind farm OT

deployments. First, this case study does not reflect the full multivariate operational context of offshore wind farm OT, but is based on a single representative variable, generator bearing temperature. Second, the normal, warning, and critical thresholds used in this study are illustrative thresholds for explaining state-aware communication operation and do not represent universal safety standards or absolute industrial criteria. Third, the communication comparison was conducted under a single 5% packet-loss condition and the same topology. In addition, this prototype-level comparison does not include repeated trials or confidence intervals; therefore, the numerical results should be interpreted as an initial controlled comparison among protection strategies rather than statistically conclusive evidence.

Therefore, there are limitations in generalizing the results to diverse offshore communication environments, different network conditions, and different turbine configurations. However, the proposed three-state logic is not inherently limited to generator bearing temperature or UDP-based reporting, and it can be extended to multivariate state evaluation using variables such as vibration, rotor speed, and power output, as well as to other offshore communication patterns such as wireless links, cellular backhaul, or gateway-based offshore-to-onshore communication.

Accordingly, the contribution of this case study should be understood as an initial prototype-level demonstration that state-aware communication protection can be logically constructed and experimentally applied in offshore wind farm OT environments. The results show that differentiated protection can reduce unnecessary communication burden while preserving important warning-state communication and supporting restricted protected communication in the critical state. Future work should extend this validation to multivariate state assessment, multiple packet-loss conditions, diverse topologies, repeated trials, and more detailed measurements of critical-state session latency and protocol behavior.

5 Conclusion

This study began from the recognition that communications occurring in offshore wind farm OT environments do not all have the same meaning or require the same level of protection, yet prior discussions have not sufficiently distinguished among them. In particular, it pointed out the limitation that applying the same lightweight protection to all communications may lead to insufficient protection for important communications, whereas applying the same strong protection to all communications may increase unnecessary communication burden in a repetitive reporting structure. To address this issue, this study proposed a state-aware communication approach that classifies communication into the normal, warning, and critical states, interprets the meaning and importance of communication differently according to the operational state, and then applies differentiated protection levels accordingly.

The case study results showed that the proposed scheme maintained the loss rate of important communication in the warning state at the same level as r2, namely 0.3%, while the total communication volume was 4.1 MB, which was lower than r2's 6.1 MB. This indicates that the proposed scheme can maintain low-burden operation in the normal state while selectively reinforcing protection only in the warning state, thereby forming a more balanced protection structure between communication burden and the visibility of important communication.

However, this study has limitations in that it is a prototype-level case based on a single representative variable, the state classification criteria are also based on illustrative thresholds, and the experiment was conducted under a single packet-loss condition. Therefore, future research needs to extend the current single-variable threshold rule to a multivariate state-evaluation model that combines generator bearing temperature, rotor speed, vibration frequency, power output, and other condition-monitoring variables. Such a model could use statistical correlation or machine-learning-based classification to reduce false positives and blind spots caused by single-variable state transitions.

Additional validation under diverse communication environments, packet-loss conditions, and topology conditions is also required.

Acknowledgement

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EVCSec: Adaptive Authentication Approach for Electric Vehicle Charging System*

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Abstract

The rapid expansion of electric vehicle charging infrastructure, combined with equipment heterogeneity from multiple manufacturers, diverse device environments, and public network reliance, has escalated security threats against EVCS. Legacy protocols such as OCPP 1.6 and ISO 15118 expose vulnerabilities in static authentication, session management, and certificate handling, with real-world incidents including OCPP zero-day flaws and global EV charging network breaches. This paper proposes an efficient, adaptive reputation-based authentication protocol for EVCS environments. Built on an integrated ISO 15118-20 and OCPP 2.0.1 architecture, it computes reputation scores from vehicle charging history, anomaly frequency, time-based trust decay, and location risk, classifying them into three levels for dynamic authentication strength adjustment. Authentication levels adapt across EV-charger-CSMS segments, ensuring real-time session continuity and data integrity. Formal verification via Scyther confirms security against known attacks.

1 Introduction

The rapid proliferation of electric vehicles (EV) is accelerating the transition to sustainable transportation systems, with 14 million new EVs registered worldwide in 2023, reaching a total of 40 million units. This growth in the EV market necessitates the expansion of Electric Vehicle Charging

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Systems (EVCS), which have become a core component of modern smart transportation infrastructure [1]. EVCS not only provides physical power supply to electric vehicles but also delivers intelligent services such as remote monitoring, charging session management, and billing based on Internet of Things (IoT) technology, functioning as a Cyber-Physical System (CPS). EVCS forms a complex ecosystem consisting of Charge Points (CP), Central System Management Systems (CSMS), power grids, and user electric vehicles, with interactions between these components conducted through standardized communication protocols [2]. Open Charge Point Protocol (OCPP) is the most widely used open communication standard protocol in EV charging infrastructure, developed by the Open Charge Alliance (OCA). OCPP standardizes communication between charge points and central management systems to ensure interoperability and provides core functions such as remote monitoring, charging session management, firmware updates, and smart charging [3-4].

Currently, OCPP 1.6 and OCPP 2.0.1 versions are mainly used, with OCPP 1.6 implemented in JSON over WebSocket format to support lightweight message transmission. OCPP 2.0.1 introduces enhanced security features to address security vulnerabilities of previous versions, including X.509 digital certificate support for mutual authentication, certificate management messages, and firmware update security through electronic signatures. As IoT devices connected to the internet, EVCS are exposed to various cyber security threats, which can seriously impact not only individual charging stations but also the stability of the entire power grid and user personal information protection. Recent studies report various security threats that may occur in EVCS environments [5].

Recent EVCS security incidents are materializing as real threats. In February 2024, the UK government suspended sales of Spanish-made EV chargers due to cybersecurity concerns, indicating potential threats that charging infrastructure can pose to national energy security [6]. In April 2024, six zero-day vulnerabilities (CVE-2024-23971, etc.) were discovered in the OCPP protocol, exposing security weaknesses in backend communication of numerous charging management systems [7]. Particularly, ChargePoint Home Flex chargers were found to have remote code execution vulnerabilities due to lack of user input string validation in OCPP message processing. In August 2024, the U.S. government launched a cybersecurity initiative for clean energy system protection, designating EV charging infrastructure and power grid security as top priorities [6]. Most seriously, in November 2024, a large-scale data breach occurred in a global EV charging network, compromising approximately 116,000 personal information, payment details, vehicle VINs, and charging station locations, affecting charging operators in multiple countries [3]. These security vulnerabilities seriously undermine the trustworthiness and safety of EVCS, posing direct threats to power grid stability and user personal information protection. OCPP vulnerabilities mainly stem from weak authentication mechanisms, lack of encryption, and inadequate certificate verification, demonstrating the limitations of static authentication methods. Therefore, developing dynamic and adaptive authentication mechanisms to strengthen EVCS security is urgently needed. The EVCS environment targeted in this paper is configured as shown in Figure 1 and has characteristics that make it difficult to provide sufficient security with existing static authentication methods alone [5, 8]:

(1) Heterogeneous environment with mixed chargers from various manufacturers and models
(2) Characteristics exposed to physical attacks due to installation in public places
(3) Requirements to maintain persistent connections between charging stations and central systems
(4) Low-latency requirements for real-time charging service provision

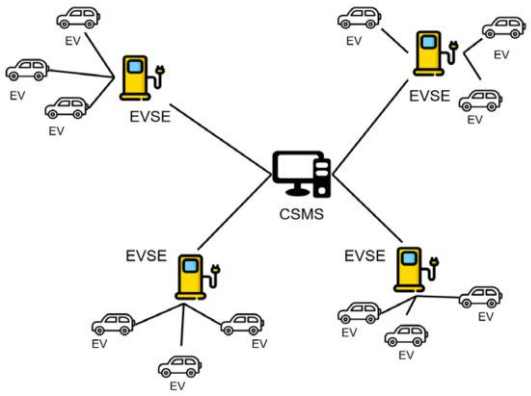

Figure 1: EVCS Environment Architecture

ISO 15118 defines communication protocols between electric vehicles and chargers, including Plug & Charge, bidirectional energy transfer, and enhanced cyber security systems. ISO 15118 provides robust security features such as mutual authentication through X.509-based PKI (Public Key Infrastructure) between EVCC (Electric Vehicle Communication Controller) on the EV side and SECC (Supply Equipment Communication Controller) on the charging station side, TLS-based encrypted communication, session key exchange, and certificate lifecycle management. This security framework effectively prevents unauthorized access, data tampering, replay attacks, etc., between EVs and charging stations and protects user privacy [8].

This study aims to propose a security protocol that simultaneously improves the security and practicality of EVCS by integrating adaptive authentication techniques into the OCPP protocol and ISO 15118 protocol. The main contributions of this paper are as follows:

(1) Proposes a reputation-based adaptive authentication protocol for EVCS environments, which comprehensively considers charging station's past behavior history and current communication context to dynamically adjust authentication levels.

(2) The proposed protocol is designed to be compatible with existing OCPP message structures, enabling gradual deployment to existing EVCS infrastructure.

(3) Uses the formal verification tool Scyther to formally verify the security properties of the proposed adaptive authentication protocol and prove security against various attack scenarios.

(4) Confirms through performance evaluation that the proposed technique provides practical performance along with enhanced security.

The organization of this paper is as follows. Section 2 analyzes OCPP protocols widely used in EVCS communication and existing studies for improving EVCS security. Section 3 designs the reputation-based adaptive authentication protocol, Section 4 presents security evaluation and performance analysis results of the adaptive authentication protocol proposed in Section 3. Finally, Section 5 summarizes the research results and presents future research directions.

2 Related Work

2.1 OCPP

OCPP is based on a client-server model, where the CP serves as the client and the SMS serves as the server. The charge point is a physical electric vehicle charging device that includes one or more

connectors and provides data such as charging session information, connector status, and energy consumption. The CSMS communicates with charge points through OCPP to perform charging session monitoring, energy consumption data collection, and billing processing.

The OCPP communication flow is configured as follows [10]:

- Boot notification: When a charge point starts or restarts, it sends a registration request to the CSMS, providing identification information and configuration information.
- Status notification: The CSMS sends a status request to the charge point to collect real-time information such as availability, charging power, and ongoing charging sessions.
- Charging session management: The CSMS sends a start command to the charge point to initiate a charging session. The charge point acknowledges the command, starts the session, and sends status updates to the CSMS. The CSMS can send commands to pause, restart, or terminate the charging session.
- Transaction data transmission: The charge point sends transaction data such as start time, end time, energy consumption, and cost to the CSMS during and after the charging session.
- Firmware updates: The CSMS sends a firmware update request to the charge point, which performs the firmware update process and sends progress status and completion notifications to the CSMS.

OCPP consists of several functional blocks, each containing specific purposes and requirements:

- Security: Defines protocol security specifications, including authentication, encryption, certificate management, and security logging.
- Provisioning: Provides functionality for registering charge points to the network and retrieving basic configuration information.
- Authorization: Handles user authentication before charging session initiation, including authentication message processing and authentication caching.
- Local authorization list management: Provides functionality for managing local whitelists on charge points, enabling user authentication even in offline states.
- Transactions: Defines basic functions related to transactions started/stopped at charge points.
- Remote control: Includes functions such as remote transaction control, connector unlock, and remote trigger from the CSMS.
- Availability: Defines status notification message transmission functionality.
- Smart charging: Supports smart charging functionality using charging schedules.

OCPP 1.6 basically provides only transport layer security and HTTP basic authentication, and does not standardize certificate exchange or mutual authentication at the protocol level. OCPP 1.6 has several security vulnerabilities, with major issues as follows [3, 6, 11]:

- Weak authentication mechanism: OCPP 1.6 provides an option to allow authentication using only charge point identifiers, enabling attackers to easily impersonate legitimate charge points.
- Inadequate multi-connection handling: The OCPP standard does not clearly define how the CSMS should handle multiple WebSocket connections with the same charge point ID.

Consequently, attackers can intercept or continuously block legitimate charge point connections.

- Static authentication method: Existing OCPP uses static authentication methods that do not consider charge point's past behavior or current communication context, making adaptive responses to evolving threat environments difficult.

OCPP 2.0.1 overcomes these limitations by introducing strictly defined security profiles, key management for client-side certificates, secure firmware updates through signed firmware, and security event logs.

2.2 Literature Review

ISO 15118-20 [8] is the latest version defining 2nd generation network and application protocol requirements for the ISO 15118 standard. This standard includes functions such as wireless charging, bidirectional energy transfer (Vehicle-to-Grid, V2G), pantograph charging, etc., and uses enhanced security protocols of TLS 1.3 or higher. Smart charging and load management functions are highly advanced, and communication efficiency is significantly improved through dynamic certificate management and efficient message encoding. When an EV connects to a charger, a network and security connection is established between EVCC and SECC. Then mutual authentication is performed to verify trust on both sides, and charging service use and payment by the user are automatically authenticated and approved based on digital contract certificates embedded in the vehicle. Power specification exchange and periodic status reporting proceed after charging initiation, and finally the process is completed with charging termination request and session termination communication. All processes are securely handled through TLS encryption and integrity verification mechanisms. Meanwhile, it is not compatible with ISO 15118-2, requiring interoperability measures for different generations.

Hamdare et al. [13] proposed a comprehensive cyber defense framework to mitigate cybersecurity risks of OCPP. This study pointed out that OCPP, while being a widely adopted communication standard ensuring interoperability between EV charging stations and central management systems, is exposed to serious cyber risks due to weak authentication mechanisms and improper session handling. The framework proposed in provided targeted improvements to address OCPP security vulnerabilities and demonstrated performance surpassing existing OCPP versions and latest EV charging security solutions in terms of security and efficiency. This study also addressed difficulties in protocol adoption due to lack of backward compatibility with OCPP 2.0.1 and presented practical security enhancement measures.

P. Li et al. [14] propose a distributed defense model combining Zero Trust architecture and blockchain to address single points of failure and insider attack vulnerabilities of existing centralized trust models. This model applies Zero Trust principles to continuously verify all connections and requests, using ShangMi cryptographic algorithms to protect each stage of user registration, charging reservation, and payment. During user authentication, ring signature-based anonymous authentication protects personal information, and all transactions are recorded on blockchain to ensure tamper resistance and transparency. Certificates are managed through distributed PKI, charging reservation conditions are verified and automatic payment is executed through smart contracts, and network-wide transaction validity is verified through consensus mechanisms. This distributed architecture eliminates single points of failure and enhances overall system resilience and trustworthiness.

N. Hettiarachchi et al. [15] propose an anonymous authentication system based on 3GPP AKA protocol to address the problem that ISO 15118 Plug & Charge protocol exposes user ID and charging patterns, risking privacy breaches. This protocol uses AKA's one-way key functions (f1~f5) to generate MAC, expected response, anonymization key, encryption key, and integrity key required for mutual authentication, and anonymizes sequence numbers through XOR operations to prevent tracking. KEM

is used to securely exchange session keys between EV and eMSP, and AEAD encrypts authentication data and metering data to simultaneously ensure confidentiality and integrity. EV, eMSP, and charging stations each generate ECDSA signatures to verify message origin and integrity, and provide end-to-end security through multiple TLS connections and certificate chain verification. This design enables secure charging and payment without exposing user's actual identification information, achieving both privacy and security.

3 Adaptive Authentication Protocol for EVCS

In current electric vehicle charging infrastructure, EV and chargers EVCS perform mutual authentication through ISO 15118, while chargers and CSMS communicate via OCPP, forming a dual-layer structure. While this structure is secure in individual communication segments, the lack of linkage between the two protocols causes complexity in authentication management and session continuity issues. Additionally, each charging station and vehicle performing fixed levels of encryption creates inefficiency where authentication burden becomes excessive or unevenly distributed relative to attack difficulty.

Therefore, this paper proposes an adaptive authentication protocol that integrates the previously separated authentication structures of OCPP and ISO 15118-20 and applies adaptive mechanisms to adjust authentication levels according to the reputation of electric vehicles accessing charging stations.

The adaptive authentication protocol proposed in this paper does not perform verification independently when an EV sends an authentication request to EVCS. Instead, EVCS links with CSMS for real-time verification of certificate chains and reputation values. CSMS maintains reputation scores for each EV and dynamically evaluates trustworthiness by analyzing past connection history, charging patterns, location-based anomaly detection results, etc. Then, in the authentication procedure, lightweight encryption and single authentication are allowed for vehicles with high reputation, while enhanced authentication procedures are applied to terminals with low trustworthiness or attack potential. Through this, the system can dynamically adjust authentication strength to minimize overall process load while reducing attack surface and strengthening security.

As a result, the proposed adaptive authentication protocol combines ISO 15118-20's authentication framework, OCPP's security management functions, and reputation-based adaptive logic to satisfy scalability, reliability, and real-time requirements of EVCS operating environments.

3.1 Security Requirements

The security requirements of the adaptive authentication protocol proposed in this paper were derived based on ISO 15118-20, OCPP-related standards, and EVCS security research [8-15]. These summarize authentication, encryption, certificate management, privacy protection, and other requirements presented in prior research suitable for EVCS environments as follows:

(1) Mutual Authentication: Vehicles and charging stations must verify each other's legitimacy using trusted public PKI systems, with secure certificate exchange between EVCC and SECC. This protocol reflects this requirement by designing dynamic authentication mechanisms linked with reputation-based evaluation.

(2) Encrypted Communication: All messages must be protected from unauthenticated attackers. It must also prevent data eavesdropping, tampering, and replay attacks. Under adaptive authentication environments, additional encryption strength settings based on real-time risk analysis must be possible.

(3) Certificate Management: Full lifecycle management including certificate issuance, renewal, and revocation must be performed, complying with multi-stage PKI structures supporting Plug & Charge

authentication systems. This protocol can apply policies that strengthen certificate verification procedures for nodes with low reputation-based evaluation adaptively.

(4) Transaction Data Integrity Guarantee: Billing and charging session-related data can secure integrity through message signatures and HMAC verification. Additionally, data integrity verification levels can be dynamically adjusted in adaptive authentication to balance performance and security.

(5) Replay Attack Prevention: All communication sessions must include timestamps and nonces to prevent message reuse, and additional authentication procedures can be considered when abnormal patterns occur.

(6) Privacy Protection: During Plug & Charge processes, pseudonym-based authentication and anonymization techniques are applied instead of real names to minimize personal information exposure. Reputation-based and adaptive authentication techniques include methods that maintain trustworthiness while protecting user privacy.

(7) Reputation-based Adaptive Authentication: Analyzes past behavior history (reputation score) of charging stations and vehicles and current communication context (behavior patterns, location, network status, etc.) to dynamically adjust authentication procedure strength and steps. This enables real-time responses to evolving security threats while maximizing authentication process efficiency.

3.2 Reputation Score Model

This section presents the reputation score model, a core element of the reputation-based adaptive authentication protocol proposed for electric vehicle charging system environments. The reputation score quantitatively expresses each vehicle's trustworthiness by comprehensively considering past behavior, security event occurrence history, regional characteristics, time-elapsed trust decay, etc., and is used as a metric to dynamically adjust authentication strength (encryption level, number of authentication procedures, auxiliary verification criteria, etc.) when EVs access. Through this, both security and efficiency can be achieved simultaneously [15-16].

In this paper, the reputation score model is exemplified by expressing reputation scores in the 0~100 range, classifying vehicle trustworthiness levels into three grades as shown in Table 1. These reputation grades reference the digital fingerprint-based authentication threshold design in [17] and the three-level user risk classification method in [18]. Reputation grades used in Table 1 can be calculated through indicators and situation-based weights in Table 2.

Grade	Score	Type	Description
High	80~100	Lightweight Authentication	Trusted vehicles, safe areas, good history
Medium	50~79	Standard Authentication	General users, unverified areas
Low	0~49	Enhanced Authentication	Abnormal behavior or unauthorized attempts

Table 1: Reputation Score Model Grade (Example)

In the proposed reputation model, initial users or first-time EVs without prior charging history are handled conservatively. Since historical behavior cannot be used to establish trustworthiness, these users are assigned to the Low reputation grade by default and are required to perform enhanced authentication until sufficient normal charging records are accumulated.

Indicator	Definition	Value
R_history	Normal charging history	0~1
R_anomaly	Anomaly behavior detection history	0~1
R_time	Time-based trustworthiness	0~1
R_location	Location-based trustworthiness	0~1

Table 2: Reputation Score Model Indicator (Example)

To derive appropriate reputation scores according to situations, sufficient discussion and adjustment of trust grades, indicators, and weights are needed. Grades and categories used in grades can be derived through simulation and additional research, while indicators and weights can be initially set per charger and installation area but continuously monitored to analyze attack detection rates, false positive rates, etc., per charger or area for readjustment. Indicators in Table 2 can be defined as follows:

- Normal charging history (R_history): Represents the normal completion rate of charging sessions, assigning higher scores as session error or interruption rates decrease.
- Anomaly behavior detection history (R_anomaly): Evaluated based on network traffic anomaly detection, retransmission patterns, authentication failure events. Higher abnormal operation detection rates result in lower scores.
- Time-based trustworthiness (R_time): Trust gradually decreases based on elapsed time since the last normal charging.
- Location-based trustworthiness (R_location): Reflects charging station security certification status, regional risk levels, location history, etc. For example, public certified charging stations: 1.0, unauthorized private stations: 0.6, high-risk areas: 0.4.

Total weights are defined as $\sum_{i=0}^{n} w_i = 1$ to normalize indicator proportions. This paper uses 4 indicators total, with each weight modifiable by variables such as users, environment, etc. The weighted reputation score model is given by the following equation:

$$R = (w_1 * R_history\ + w_2 * R_anomaly\ + w_3 * R_time + w_4 * R_location)\ * 100$$

This paper presents three weight scenarios to express the impact of reputation score model weight changes on the protocol.

First, the uniform scenario assumes equal weights for all indicators, allowing equal reference to all indicators but potentially allowing malicious EVs to manipulate reputation or causing disadvantages to users affected by location/time decay.

Second, the behavior history scenario assigns high trustworthiness to users with long-term normal behavior history, reducing disadvantages from location/time decay in uniform scenarios, with increased anomaly detection weights enabling rapid grade adjustment upon abnormal behavior even with normal history.

Finally, the location scenario highly evaluates location-based indicators, allowing high reputation for safe infrastructure charging stations but potentially giving high trust grades to malicious EVs using safe areas or disadvantaging long-term normal users in risky areas. Table 3 shows weight examples applicable to these three scenarios.

Scenario	w_1	w_2	w_3	w_4
Uniform	0.25	0.25	0.25	0.25
Behavior History	0.35	0.3	0.15	0.2
Location	0.2	0.2	0.1	0.5

Table 3: Weight Adjustment by Scenario

Through the reputation score model presented in this section, each vehicle's trustworthiness level can be measured to adjust authentication levels and security mechanisms according to grades as in Table 4, expected to improve security and efficiency compared to existing methods [19-21].

Grade	Authentication Type	Security Mechanism
High	Lightweight Authentication	HMAC, etc.
Medium	Standard Authentication	ECDSA, etc.
Low	Enhanced Authentication	ZKP, Location verification, etc.

Table 4: Policy Changes by Grade

3.3 Protocol Design

This section proposes an adaptive authentication protocol that improves security and efficiency by integrating ISO 15118-20's Plug & Charge mechanism and OCPP 2.0.1's security management framework with the reputation score model. Major participants in the protocol are as follows:

- EV: Equipped with ISO 15118-20 based EVCC, possesses unique identifier ID_{EV}, contract certificate $Cert_{EV}$, private key SK_{EV}.
- EVCS: Performs ISO 15118-20 communication with EV through SECC, links with CSMS via OCPP 2.0.1, possesses unique identifier ID_{CP}, certificate $Cert_{CP}$, private key SK_{CP}.
- CSMS: OCPP backend server managing all charging stations and maintaining EV reputation database, possesses unique identifier ID_{CSMS}, certificate $Cert_{CSMS}$, private key SK_{CSMS}.
- Certificate Authority (CA): Issues and verifies certificates for all participants, manages PKI.

Figure 2 diagrams the proposed reputation-based adaptive authentication protocol, with detailed explanation as follows:

(1) EV and EVCS perform TLS 1.3 handshake to establish bidirectional secure communication channel. During this process, both entities exchange certificates, establish cryptographic keys for the session, ensuring confidentiality and integrity of transmission data.

(2) EVCS transmits AuthorizeReq message containing ID and certificate information provided by EV to CSMS. CSMS receives this request, queries EV's reputation score (REV), and determines EV's authentication level (Auth_Level) based on that score. If no prior reputation record exists for the EV, CSMS assigns the EV to the Low Level by default and applies enhanced authentication.

(3) CSMS transmits reputation score (REV), authentication level (Auth_Level), and challenge (Challenge) to be used in the authentication process to EVCS via AuthorizeResp message.

(4) EVCS relays this information back to EV so that EV can perform authentication procedures using the corresponding Challenge.

(5) EV can proceed with authentication according to the reputation score assigned to it. For high trust cases, lightweight authentication such as HMAC is performed; for medium trust levels, ECC-based ECDSA digital signatures can be used; for low trust cases, enhanced procedures including zero-knowledge proof (ZKP) authentication and additional verification such as location or biometric information can be performed. Mandatory policies according to reputation scores block attacks where malicious EVs disguise or select weak authentication methods.

(6) EVCS receives authentication response message from EV and transmits it back to CSMS to request final verification.

(7) CSMS reports authentication success or failure to EVCS based on this authentication result.

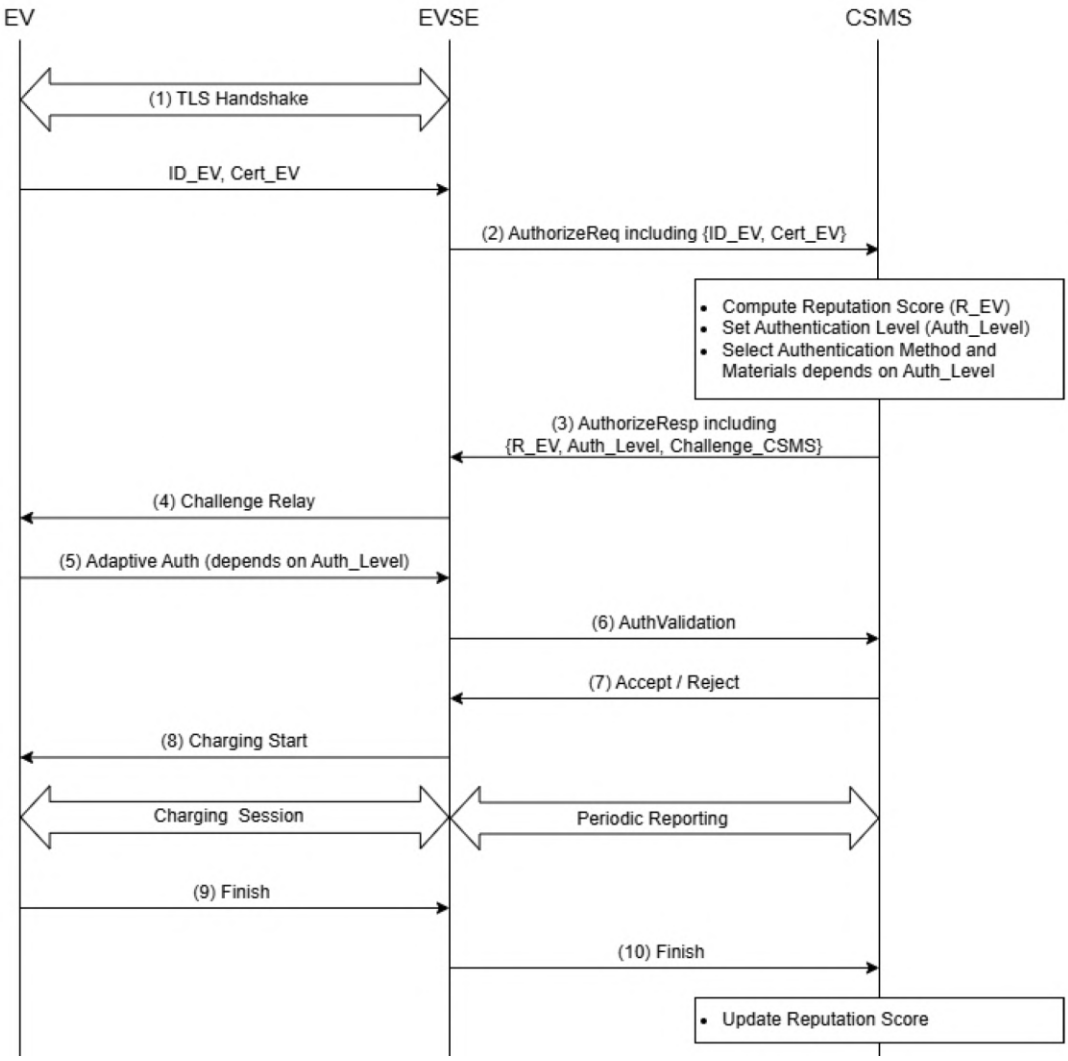

Figure 2: Proposed Protocol

Once authentication is successfully approved, (8) EVCS transmits charging start message to CSMS and initiates charging with EV. During charging, EVCS periodically reports meter values to manage charging status. Upon charging completion, (9-10) EVCS transmits final charging data along with termination event to CSMS. CSMS updates EV's reputation score based on this data for reflection in next authentication, and can adjust reputation scores to strengthen security if anomalies are detected.

4 Evaluation

This paper objectively evaluates the security and performance of the proposed reputation-based adaptive authentication protocol using the widely used formal verification tool Scyther [22]. In this verification process, the Dolev–Yao attacker model is adopted to analyze whether the proposed protocol

can withstand an adversary capable of intercepting, modifying, replaying, and injecting messages over a public communication channel. As specified in ISO/IEC 29128 [23-24], formal verification is recognized as an objective and reproducible method for evaluating security protocol trustworthiness. Additionally, the computational efficiency performance of the proposed authentication protocol is evaluated by comparing operational overhead with OCPP's basic authentication structure and EVCS security studies reviewed in Section 2.2.

4.1 Security Verification with Scyther

Scyther is an automated formal verification tool for security protocols developed by Cas Cremers, used to verify protocol security properties and detect potential attacks. The verification procedure through Scyther proceeds in 5 steps: (1) protocol modeling, (2) Claim Event definition, (3) model verification, (4) attack scenario identification, (5) results [25]. Since Scyther verifies the message flow and security properties of cryptographic protocols, the correctness of the reputation score calculation itself, the optimality of the threshold value, and the false positive rate are outside the scope of Scyther-based formal verification.

First, in the protocol modeling step, the cryptographic protocol to be analyzed is defined, including detailed information such as participants, roles, exchanged messages, and rules governing protocol behavior. It is written in SPDL (Security Protocol Description Language), a format language readable by Scyther.

In Claim Event definition, security properties of the protocol to verify are defined, which may include secrecy, authentication, integrity, etc. In model verification, automated analysis is performed on protocol participants and security behaviors. Claim Events defined in this paper are shown in Table 5.

Claim Event	Meaning
Alive	Guarantees that the entity completing communication has evidence that the other party executed the protocol at some past point.
Weakagree	Guarantees that the entity completing communication has evidence that the other party executed the protocol with it, but does not verify the other party's role or data exchanged.
Niagree	Guarantees that the entity completing communication has evidence that the other party executed the protocol with the correct role and agreed data set, but does not guarantee message order.
NIsynch	Guarantees that the entity completing communication has evidence that all messages were sent/received by the other party in the exact order specified in the protocol.

Table 5: Claim Event Deification

In scenario identification, it explores various possible scenarios and interactions between participants to check whether specified security properties are maintained or vulnerabilities exist. Finally, results generate a report containing information on vulnerabilities, security weaknesses, and identified attack scenarios.

Figure 3 shows the results of verifying the security of the adaptive authentication protocol proposed in this paper using Scyther. According to this, the adaptive authentication protocol proposed in this paper satisfies all Claim Events used in verification. Accordingly, the adaptive authentication protocol proposed in this paper is safe from attacks and proven to effectively protect charging infrastructure from various security threats when deployed in actual EVCS environments.

Claim				Status	Comments
adaptive_occp	EV	adaptive_occp,EV1	Alive	Ok	No attacks within bounds.
		adaptive_occp,EV2	Niagree	Ok	No attacks within bounds.
		adaptive_occp,EV3	Weakagree	Ok	No attacks within bounds.
		adaptive_occp,EV4	Nisynch	Ok	No attacks within bounds.
	EVSE	adaptive_occp,EVSE1	Alive	Ok	No attacks within bounds.
	CSMS	adaptive_occp,CSMS1	Alive	Ok	No attacks within bounds.
		adaptive_occp,CSMS2	Niagree	Ok	No attacks within bounds.
		adaptive_occp,CSMS3	Weakagree	Ok	No attacks within bounds.
		adaptive_occp,CSMS4	Nisynch	Ok	No attacks within bounds.

Figure 3: Scyther Verification Result

4.2 Performance Comparison

This section compares the proposed adaptive authentication protocol with existing studies analyzed in Section 2 in terms of operational overhead and qualitative aspects. The strength of the proposed adaptive authentication protocol lies in being able to dynamically adjust computational costs according to reputation scores. High-trust EVs receive High Level for high efficiency, while low-trust or first-time EVs receive Low Level for high security. This adaptive approach achieves optimal security-efficiency balance suited to situations, unlike existing protocols using fixed single security levels. Table 6 compares operational overhead by computation count with proposed protocol and existing studies [13-15].

Operation	[13]	[14]	[15]	Ours (H)	Ours (M)	Ours (L)
HASH	2	5	12	5	6	9
XOR	0	0	4	0	0	0
SYM	21	9	7	11	11	11
ASY	3	6	4	3	6	8
DS	0	6	6	2	2	2
CV	1	3	4	3	3	3

HASH: Hash operation; XOR: XOR operation; SYM: Symmetric key operation;
ASY: Public key operation; DS: Digital signature operation; CV: Certificate verification operation.
H: High Level (with HMAC option);
M: Medium Level (with ECDSA option);
L: Low Level (with ZKP option)

Table 6: Computational Overhead Comparison by Operation Count

The High Level of our proposed protocol incurs slightly higher cost than the lightest existing study but enhances security through strengthened certificate verification procedures. The Medium Level performs comparably to, while the Low Level provides the highest security among comparison targets. This demonstrates the flexible structure achieved through reputation-based adaptive approaches, delivering efficiency to high-trust EVs and enhanced security to low-trust EVs.

In Table 7, existing studies apply uniform security levels to all EVs using fixed methods, whereas this research achieves security-efficiency balance through reputation-based adaptive approaches. lacks

compatibility with existing OCPP infrastructure due to its blockchain-based separate architecture, and focuses solely on ISO 15118. This study integrates OCPP 2.0.1 and ISO 15118-20 to enhance practical deployment feasibility. Additionally, this research uniquely proves the security of the proposed authentication technique through formal verification using the well-known Scyther tool.

Aspect	[13]	[14]	[15]	Ours
Authentication Method	Fixed	Blockchain	Anonymous	Adaptive
Compatibility	△	×	△	○
Scalability	○	×	△	○
Privacy Protection	△	○	○	○
Real-time Performance	○	×	△	○
Formal Verification	×	×	×	○

Table 7: Qualitative Comparison

4.3 Attack Scenarios and Countermeasure

This section specifically analyzes how the proposed adaptive authentication protocol responds to major attack scenarios. Table 8 maps major attack types in EVCS environments with the proposed protocol's countermeasure mechanisms. Through analysis of the 7 major attack scenarios presented in Table 8, it is confirmed that the proposed adaptive authentication protocol comprehensively responds to major attack scenarios in EVCS environments through the combination of cryptographic foundations (TLS, PKI, digital signatures) and reputation-based dynamic mechanisms (adaptive authentication strength adjustment, anomaly detection). Particularly, its balanced design simultaneously achieves efficiency for normal users and enhanced security against malicious actors, expected to overcome limitations of existing simple static authentication methods.

Attack Scenario	Description	Countermeasure
Spoofing Attack	Malicious EV impersonates legitimate EV by stealing ID for unauthorized charging attempts	Detects sudden location changes and reclassifies reputation grade
Replay Attack	Attacker replays previously eavesdropped authentication messages to bypass authentication	CSMS verifies timestamp validity and nonce duplication upon message receipt
Man-in-the-Middle Attack	Attacker intercepts and tampers with communication between EV and EVCS	CSMS digital signature protects reputation score and authentication level
Denial of Service Attack	Attacker sends mass authentication requests to overload CSMS	Applies lightweight authentication to High grade EVs to reduce CSMS load and limits identical request frequency
Certificate Forgery Attack	Attacker creates fake certificates to impersonate legitimate EV	X.509 certificate chain verification
Reputation Manipulation Attack	Malicious EV mimics normal behavior to raise reputation score before attack	Real-time anomaly pattern detection using multiple indicators

Table 8: Major Attack Scenario and Countermeasure in EVCS Environment

5 Conclusion

Amid the rapid expansion of electric vehicle charging infrastructure and escalating security threats against EVCS, existing OCPP protocols revealed limitations in static authentication methods unable to adaptively respond to diverse threat environments. This study proposed a reputation-based adaptive authentication protocol integrating ISO 15118-20 and OCPP 2.0.1 standards with dynamic authentication mechanisms. The protocol computes reputation scores by synthesizing multiple indicators including vehicles' past charging history, anomaly detection frequency, time-based trust decay, and location-based risk assessment, dynamically adjusting authentication strength across three levels (High/Medium/Low). It ensures real-time session continuity and data integrity across EV-EVCS-CSMS segments while achieving optimal security-efficiency balance by applying lightweight authentication to high-trust vehicles and enhanced procedures to suspicious ones. Scyther formal verification confirmed satisfaction of all essential security claims (Alive, Weakagree, Niagree, Nisynch), proving resilience against major attacks including spoofing, replay, MITM, DoS, certificate forgery, and reputation manipulation. The protocol demonstrated computational efficiency comparable to or better than existing studies [13-15], with its adaptive structure providing practical superiority in real deployment scenarios.

Future research will integrate machine learning-based real-time anomaly detection, conduct statistical analysis of optimal reputation thresholds using large-scale EVCS data, expand support for bidirectional V2G scenarios, and develop standardized APIs for gradual migration from legacy OCPP 1.6 infrastructure, contributing significantly to building secure and efficient infrastructure essential for sustainable transportation systems.

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A Comparative Study of Evaluation Methodologies for Anomaly Detection on ICS Benchmarks*

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Abstract

In Industrial Control System (ICS) anomaly detection research, benchmark datasets collected from testbeds such as SWaT, WADI, and HAI have been widely used; however, concerns about the reliability of reported performance have been raised persistently due to inherent limitations in both the datasets and the evaluation methodologies. In particular, widely used evaluation metrics such as Point-Adjusted F1 have been shown to overestimate model performance, and fixed thresholds fail to capture data characteristics. However, it has not been systematically examined whether the choice of evaluation methodology influences model comparison outcomes on ICS benchmarks. To address this gap, we compared the performance of traditional ML models (PCA, Isolation Forest) and state-of-the-art DL models (USAD, TranAD) across four ICS benchmark datasets: SWaT, WADI, HAI 21.03, and HAI 23.05. In Exp1, we applied a variety of evaluation metrics (AUROC, AUPRC, Affiliation Precision/Recall) to identical anomaly scores. In Exp2, we applied Oracle and Label-free threshold methods (Best F1, NDT, P_{95}, P_{99}) to quantify how the choice of evaluation methodology shifts relative model rankings. The results show that ML models matched or outperformed DL models in 15 out of 16 dataset-metric combinations. Moreover, the PA-F1 score of a single model varied by up to 0.719 across threshold methods, confirming that the choice of evaluation methodology can considerably alter model rankings. These results indicate that ICS anomaly detection research requires not only improvements in detection models but also systematic examination of evaluation methodologies.

1 Introduction

Industrial Control Systems (ICS) monitor and automatically regulate the physical processes of industrial infrastructure such as power plants, water treatment facilities, and manufacturing processes through sensors and control devices. As a form of Cyber-Physical System (CPS) in which the cyber and physical domains are tightly coupled, any incident within an ICS can directly translate into injuries or damage to facilities, which has motivated a growing body of research on cyber-resilience and safe operation [18]. In this context, time-series anomaly detection for the early identification of cyber attacks, such as sensor value manipulation or unauthorized actuator control, has emerged as a central topic; datasets collected from testbeds that emulate industrial processes and incorporate injected attack scenarios have been widely

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used as primary benchmarks for this purpose. Autoencoder- and Transformer-based deep learning models are widely adopted as detection methods and have achieved high performance on these benchmarks [4].

However, recent studies reveal that time-series anomaly detection benchmarks suffer from intrinsic limitations such as labeling errors and unrealistic anomaly ratios [27], and similar issues have been documented in ICS benchmarks [4]. Reproducing existing models under identical conditions has been reported to yield slightly lower performance than originally claimed [9]. In addition, recent models have been observed to degrade under noise-perturbed settings [15]. These findings raise concerns about whether reported results can be reliably reproduced. Regarding evaluation methodology, recent studies criticize the Point-Adjusted F1 (PA-F1) for inflating model performance [5, 10], and prior work identifies fixed thresholds as inadequate for capturing data characteristics [11]. A variety of alternative metrics and threshold schemes have been proposed to address these limitations, yet whether these alternatives alter model comparisons on the same ICS benchmarks has not been systematically examined.

Motivated by this gap, we examine the reliability of anomaly detection evaluation on ICS benchmark datasets. The main contributions are summarized as follows:

- We compare traditional machine learning (ML) models and recent deep learning (DL) models under a variety of evaluation metrics on four ICS benchmark datasets (Exp1).
- Under identical anomaly scores, we compare performance across different threshold schemes and examine how the choice of evaluation methodology shifts model rankings (Exp2).
- We extend the analysis to HAI 21.03 and HAI 23.05, two ICS benchmarks that have received comparatively little scrutiny, and further analyze how differences in data composition between two versions of the same testbed influence detection results.

2 Related Work

2.1 Anomaly Detection Methods

Time-series anomaly detection has long relied on traditional machine learning methods. Principal Component Analysis (PCA)-based anomaly detection, which measures reconstruction error based on learned correlations among multiple sensor values [3], has been applied to industrial process monitoring, and Isolation Forest (IForest) [12], which isolates anomalies by randomly partitioning data, is a foundational method in tree-based anomaly detection. These traditional methods have been widely adopted in industrial settings, as they provide stable performance without requiring DL training.

Subsequently, researchers have introduced a variety of DL approaches. Hundman et al. [7] proposed an LSTM-based model as a prediction-based approach, which predicts normal sequences and uses prediction error as the anomaly score. In the reconstruction-based category, USAD [2], which combines an autoencoder with adversarial training, is a notable example. As a generative approach, Su et al. [22] combined a stochastic RNN with a VAE in OmniAnomaly. Among Transformer-based approaches, TranAD [24], which captures the full sequence context and applies adversarial training, is representative, and subsequent work introduced models that leverage inter-variable attention mechanisms [8]. Graph-based approaches that explicitly model spatial dependencies among sensors [29] have also emerged for environments such as ICS where multiple sensors interact. Among these, USAD and TranAD achieved state-of-the-art performance on major benchmarks including SWaT and WADI, and became prominent DL anomaly

detection models. However, Sarfraz et al. [17] showed that traditional ML methods, for example, PCA reconstruction error alone, can outperform existing DL models on SWaT and WADI. This suggests that the performance advantage of DL models may vary across datasets and evaluation settings.

These findings, however, rest primarily on SWaT and WADI, and whether the same trend holds on other ICS datasets remains underexplored. In this study, we selected PCA and IForest as traditional models and USAD and TranAD as DL methods, and compared the performance of the two model types on four ICS datasets, including not only SWaT and WADI but also HAI 21.03 and HAI 23.05.

2.2 Benchmark Datasets

In time-series anomaly detection research on ICS, prior work has primarily used datasets collected from testbeds with injected attack scenarios [4]. Among the most prominent are SWaT [13] from a water treatment testbed and WADI [1] from a water distribution testbed, released by the iTrust laboratory, and HAI [19] from a power generation testbed, collected by ETRI, which integrates boiler, turbine, and water treatment processes through a Hardware-in-the-Loop (HIL) simulator.

However, several studies have identified various limitations in time-series anomaly detection benchmarks in general. Wu and Keogh [27] demonstrated that benchmarks released by major organizations such as Yahoo, Numenta, and NASA suffer from four types of flaws: trivial problems solvable by simple methods (triviality), unrealistic anomaly ratios, mislabeled ground truth, and run-to-failure bias, in which anomalies are concentrated in the later portion of the data. For ICS datasets specifically, Correia et al. [4] found that SWaT contains numerous anomaly cases detectable by the single feature alone, and WADI includes many features composed of missing values or constant values, posing challenges for evaluation. Wenig et al. [26] further noted that existing multivariate benchmarks contain anomalies that are effectively detectable at the univariate level, failing to demonstrate the strengths of multivariate algorithms.

In contrast, the HAI dataset, while also incorporating attack scenarios on an HIL-based testbed, has not been sufficiently examined in this regard. In this study, we included HAI 21.03 and HAI 23.05 in addition to SWaT and WADI, to examine whether similar patterns hold on ICS datasets for which such analyses remain scarce.

2.3 Performance Evaluation

In time-series anomaly detection, F1-score has served as the primary performance metric; to enable segment-level evaluation, Xu et al. [28] proposed Point-Adjusted F1 (PA-F1), which treats an entire anomaly segment as correctly detected if even a single time point within it is identified, and it has since become widely adopted. However, multiple studies have raised persistent concerns that PA-F1 inflates model performance. Kim et al. [10] mathematically demonstrated that PA-F1 scores converge to 1 in proportion to the length of the anomaly segment, and Sehili and Zhang [5] showed that selecting only a small number of random time points from SWaT and WADI yields PA-F1 scores approaching 0.95. Kang and Kang [8] documented a performance gap of up to 0.48 when PA-F1 and other metrics are applied to the same model. In addition, while most studies measure performance based on a fixed threshold, prior work has noted that fixed thresholds fail to capture the distributional characteristics of the data, as the choice of threshold can substantially alter evaluation outcomes [11].

To address these limitations, researchers have proposed various alternatives to fixed threshold selection and point-wise evaluation that assesses individual time points independently.

Range-based precision/recall [23] evaluates performance at the segment level rather than at individual time points; Volume Under the Surface (VUS-ROC, VUS-PR) [14] integrates the area under ROC/PR curves across multiple window sizes to provide evaluation that is robust to temporal offset; and Affiliation Precision/Recall (Aff_P, Aff_R) [6] converts the temporal distance between predicted time points and actual anomaly segments into probability-based scores, enabling parameter-free evaluation. For threshold selection, Hundman et al. [7] introduced NDT, which dynamically sets thresholds based on data characteristics.

However, whether these improvements in evaluation metrics and threshold methods are effective on ICS benchmark datasets where issues have been documented has not been sufficiently verified. In this study, holding the model output fixed, threshold-free metrics (AUROC, AUPRC) and distance-based metrics (Aff_P, Aff_R) were applied in Exp1 to compare performance across models from multiple perspectives, while five threshold schemes, including Oracle methods (Best F1, Best $F1_{PA}$) and Label-free methods (NDT, P_{95}, P_{99}), were applied in Exp2 to analyze how the choice of evaluation methodology influences relative model performance.

3 Methodology

3.1 Models

Principal Component Analysis (PCA). PCA is a technique that extracts principal components from high-dimensional data to reduce its dimensionality, and is used for data visualization, noise removal, and feature extraction. In anomaly detection, test data is projected onto the principal component subspace learned from normal data and then reconstructed, and the reconstruction error, defined as the difference between the reconstructed and original data, is used as the anomaly score. Data that deviates from normal patterns yields higher reconstruction error due to inaccurate reconstruction. This approach has long been used in Multivariate Statistical Process Control (MSPC) for monitoring multiple sensor values and detecting process anomalies [3], and has recently been applied to time-series anomaly detection, achieving performance equal to or higher than that of recent DL models on SWaT and WADI [17]. We used the experimental resources from that work [17], deriving the reconstruction error for each feature j as $e_j = |x_j - \hat{x}_j|$ and taking the maximum score across all features as the final anomaly score $s = \max_j e_j$. Since a single feature with the largest error determines the score, PCA operates at the individual feature level during scoring, despite capturing inter-variable correlations during training.

Isolation Forest (IForest). IForest is a tree-based unsupervised algorithm that detects anomalies by randomly partitioning data points [12]. Each isolation tree repeatedly selects a random feature and split value to partition the data; since anomalous points can be isolated with fewer partitions than normal data, they exhibit shorter path lengths. The final anomaly score is derived by averaging and normalizing these path lengths across multiple trees, where a shorter path length indicates that the data point was easily isolated and is thus more likely to be anomalous. Like PCA, IForest operates on individual feature splits rather than explicitly modeling inter-variable correlations. This is a traditional ML method that has been widely used in anomaly detection, and we used the scikit-learn library [16].

UnSupervised Anomaly Detection on Multivariate Time Series (USAD). USAD is an unsupervised anomaly detection model that combines an autoencoder with adversarial training inspired by GANs [2], consisting of a shared encoder E and two decoders D_1 and

D_2 that form two autoencoders $AE_1(E + D_1)$ and $AE_2(E + D_2)$. Training proceeds in two phases: in Phase 1, each autoencoder minimizes the reconstruction error of the input window W to learn normal patterns, and in Phase 2, adversarial training between the two autoencoders further improves the reconstruction accuracy for normal data. The anomaly score at test time is derived as a weighted combination of the reconstruction errors from both autoencoders, as shown in Equation (1), where α and β are the weights satisfying $\alpha + \beta = 1$. In this study, we set $\alpha = \beta = 0.5$.

$$s = \alpha\|W - AE_1(W)\|_2 + \beta\|W - AE_2(AE_1(W))\|_2 \quad (1)$$

TranAD. TranAD is a Transformer-based anomaly detection model [24] that reconstructs the input window using a Transformer encoder and two decoders. As in USAD, training proceeds in two phases, where self-conditioning that uses the reconstruction error from Phase 1 as a focus score concentrates detection sensitivity on sub-sequences that are poorly reconstructed, while adversarial training between the two decoders simultaneously improves reconstruction accuracy for normal data. The anomaly score at test time is derived by averaging the reconstruction errors of the two decoder outputs, with larger values indicating greater deviation from normal patterns. We used the resources released in that work [25].

3.2 Datasets

Secure Water Treatment (SWaT). SWaT is an ICS security dataset collected from a water treatment testbed built by the iTrust laboratory at the Singapore University of Technology and Design [13], and is one of the most widely used benchmarks in ICS anomaly detection research. Data from 51 sensors and actuators were collected from a scaled-down physical testbed that emulates an industrial environment, consisting of 7 days of normal operation and 4 days of attack scenarios. The attack scenarios include 36 cases such as sensor value manipulation and actuator control, and the anomaly ratio of the test data is approximately 12.1%.

Water Distribution (WADI). WADI is a dataset from a water distribution testbed built by the same institution as SWaT [1], with data collected from 127 sensors and actuators. We excluded 4 features whose values are entirely missing and used the remaining 123 features for the experiments. The dataset consists of 14 days of normal operation data and 2 days of attack scenario data, including 15 attack scenarios. The anomaly ratio of the test data is approximately 5.8%, lower than that of SWaT.

HIL-based Augmented ICS (HAI). HIL-based Augmented ICS is an ICS security dataset built by the Electronics and Telecommunications Research Institute (ETRI), with the testbed consisting of three physical processes: boiler (P1), turbine (P2), and water treatment (P3), connected through a Hardware-in-the-Loop (HIL) simulator (P4) [19]. Each of the three processes is controlled by a different industrial controller, and the HIL simulator emulates steam turbine generation and pumped-storage hydropower generation so that the three processes operate as an integrated power generation system. Compared to established benchmarks such as SWaT and WADI, this dataset has been less frequently used in anomaly detection research; accordingly, we included two versions, HAI 21.03 [20] and HAI 23.05 [21], to assess performance across a broader range of ICS environments.

HAI 21.03 contains 79 features and 50 attack scenarios, with an anomaly ratio of approximately 2.2% in the test data. From HAI 22.04 onward, attack scenarios became more sophisticated, with detection difficulty reported to be approximately four times that of HAI 21.03 [21].

HAI 23.05, the latest version in this series, consists of 86 features and 52 attack scenarios, with an anomaly ratio of approximately 4.00% in the test data. By comparing the 21.03 and 23.05 versions from the same testbed, we analyzed how changes in dataset composition influence detection results.

3.3 Evaluation Metrics

In this section, evaluation metrics are categorized into three groups based on their characteristics.

Point-Adjusted F1 (PA-F1). F1-score is the harmonic mean of Precision and Recall, and is the most widely used performance metric in classification tasks including anomaly detection. The formulas for deriving this score are as follows:

$$Precision(P) = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}}, \quad Recall(R) = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}, \quad F_1 = \frac{2PR}{P+R} \tag{2}$$

where TP (True Positive) is the number of time points where anomalies are correctly identified as anomalies, FP (False Positive) where normal points are incorrectly identified as anomalies, and FN (False Negative) where anomalies are incorrectly identified as normal. In time-series anomaly detection, Point-Adjusted F1 (PA-F1) was proposed as an extension for segment-level evaluation [28]. PA-F1 applies point adjustment, treating the entire anomaly segment as successfully detected if at least one time point within it is correctly identified, and was designed from the perspective that detecting the occurrence of an anomalous event itself is important. However, this metric has been criticized for overestimating actual detection performance [5,10].

Threshold-free Metrics. Since F1-score and PA-F1 require setting a specific threshold to binarize anomaly scores into normal and anomalous classifications before evaluation, performance can vary substantially depending on the threshold selection. Threshold-free metrics address this dependency by evaluating performance comprehensively across all possible thresholds using the anomaly scores directly, and are also referred to as threshold-independent measures [14]. AUROC (Area Under Receiver Operating Characteristic) evaluates performance by varying the threshold, computing the True Positive Rate and False Positive Rate at each point, plotting these as a curve, and measuring the area under it. AUPRC (Area Under Precision-Recall Curve) similarly computes the area under the curve of the Precision-Recall relationship. Since the computation of AUROC involves True Negatives in the False Positive Rate (FPR = FP/(FP+TN)), performance can be overestimated on datasets with low anomaly ratios, as the overwhelming number of normal data points keeps the FPR low. Since the Precision in AUPRC does not involve True Negatives, it can mitigate this limitation.

Distance-based Metrics. Aff_P and Aff_R address this limitation using the distance between a predicted point x and a ground truth segment:

$$d(x, J) = \begin{cases} 0 & \text{if } x \in J \\ \min(|x - j_{\mathrm{start}}|,\ |x - j_{\mathrm{end}}|) & \text{if } x \notin J \end{cases} \tag{3}$$

If x falls within J, then $d = 0$ and the point is assigned a value of 1; for points outside J, d increases with distance from the nearest boundary, and the assigned value decreases accordingly toward 0. The final Aff_P and Aff_R are computed by averaging these per-point values. Aff_P

measures how close each predicted anomalous point is to the actual anomaly segment, and Aff_R measures how close the nearest prediction is to each point within the actual anomaly segment.

3.4 Threshold Methods

In anomaly detection, setting a threshold is required to classify the anomaly scores output by a model into normal and anomalous. We divided threshold methods into two categories: the Oracle approach, which references ground truth labels, and the Label-free approach, which determines thresholds based on the statistical properties of the data without labels.

Oracle Threshold. The Oracle approach selects the threshold that maximizes the F1-score by referencing the ground truth labels of the test data. Although it cannot be applied in operational environments where ground truth labels are unavailable, it serves as a baseline by providing the upper bound of performance that a model can achieve. We used two Oracle thresholds: Best F1, which maximizes point-wise F1, and Best $F1_{PA}$, which maximizes PA-F1.

Label-free Threshold. The Label-free approach sets thresholds based solely on the statistical properties of the data without ground truth labels, which can be applied to operational environments. We applied three Label-free thresholds. NDT (Nonparametric Dynamic Thresholding) [7] partitions the anomaly score sequence into windows and, within each window, selects the threshold that maximizes the relative decrease in mean and standard deviation when all values above it are removed. This enables the statistical identification of anomalous data points that deviate most from normal data. P_{95} and P_{99} use the 95th and 99th percentile values, respectively, from the anomaly score distribution of the training data as thresholds, determining anomalies based on extreme values of the normal score distribution under the assumption that the training data consists solely of normal data.

4 Experiments

4.1 Experimental Setup

In this study, we conducted two experiments. Figure 1 summarizes the overall experimental pipeline.

- **Exp1: Detection Performance by Model Type.** Exp1 is an experiment designed to measure the performance gap between traditional ML models and recent DL models that have achieved state-of-the-art performance on four ICS benchmark datasets. PCA and IForest as traditional ML models, and USAD and TranAD as DL models, were selected for comparison, with the dataset and preprocessing held constant and model type as the only independent variable. The hyperparameters of each model were kept at the default values from the original papers to compare out-of-the-box performance. Threshold-free metrics (AUROC, AUPRC) and distance-based metrics (Aff_P, Aff_R) were applied to compare performance from multiple perspectives.
- **Exp2: Impact of Threshold Selection.** This experiment examines how evaluation outcomes change when only the threshold method is varied without changing the underlying predictions. The anomaly scores from Exp1 were used directly, with the evaluation metric

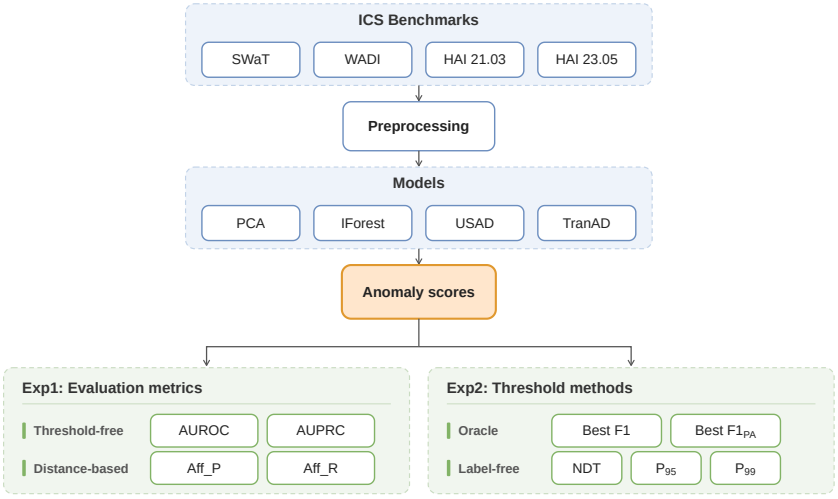

Figure 1: Overview of the experimental pipeline.

fixed at PA-F1 and the threshold method as the only independent variable. Five threshold methods were applied, including Oracle methods (Best F1, Best $F1_{PA}$) and Label-free methods (NDT, P_{95}, P_{99}), to compare the impact of threshold selection rather than the model itself.

The details of each dataset are summarized in Table 1.

Table 1: Details of the Benchmark Datasets

	SWaT	WADI	HAI 21.03	HAI 23.05
Features	51	123	79	86
# Train	495,000	784,571	921,603	896,400
# Test	449,919	172,803	402,005	284,400
Anomaly (%)	12.13	5.77	2.23	4.00

To ensure fairness, only basic preprocessing was applied to all datasets. Min-Max normalization was applied to all features; for WADI, 4 columns with entirely missing values were removed and the remaining missing values were imputed with the most recent available value (forward fill). No additional feature engineering was applied, and the hyperparameters of each model were kept at the default values of the publicly available code without dataset-specific tuning. For model training, all models except PCA involve randomness in their training and initialization, so a single run cannot guarantee stable results. Accordingly, all experiments were conducted with three different seeds (2026, 2027, and 2028), and all values reported in the text and tables are averaged over the three seeds. Figures 2 and 4 illustrate representative results from a single seed (2026).

Table 2: Model Performance Across 4 ICS Benchmarks

Dataset	Model	Threshold-free		Distance-based	
		AUROC	AUPRC	Aff_P	Aff_R
SWaT	PCA	0.743	0.244	**0.598**	0.544
	IForest	**0.838**	**0.732**	0.585	**0.782**
	USAD	0.789	0.682	0.542	0.415
	TranAD	0.341	0.088	0.533	0.456
WADI	PCA	0.506	0.053	0.499	0.590
	IForest	**0.734**	0.180	**0.542**	**0.899**
	USAD	0.503	**0.203**	0.499	0.590
	TranAD	0.521	**0.203**	0.499	0.590
HAI 21.03	PCA	**0.866**	**0.580**	**1.000**	0.098
	IForest	0.745	0.130	0.564	**0.939**
	USAD	0.742	0.350	0.997	0.218
	TranAD	0.627	0.204	**1.000**	0.060
HAI 23.05	PCA	**0.737**	**0.372**	**1.000**	0.033
	IForest	0.618	0.072	0.570	**0.863**
	USAD	0.599	0.153	**1.000**	0.033
	TranAD	0.580	0.115	**1.000**	0.014

4.2 Detection Performance by Model Type

In the first experiment, traditional ML models (PCA and IForest) achieved the highest performance in 15 out of 16 dataset-metric combinations. As shown in Table 2, the only case in which a DL model achieved the highest performance was on WADI-AUPRC, where both USAD and TranAD scored 0.203, differing from IForest (0.180) by only 0.023.

By dataset, IForest on SWaT and WADI, and PCA on HAI 21.03 and HAI 23.05, each achieved the highest performance in 6 out of 8 combinations. On HAI 23.05 in particular, most models other than PCA failed to achieve meaningful detection performance: under threshold-free metrics, PCA achieved the highest AUROC of 0.737, while the other three models remained at 0.580–0.618. These results confirm that the finding from prior work [17], which demonstrated on SWaT and WADI that PCA achieves higher performance than major DL models, holds equally on the HAI datasets.

In addition, even given identical anomaly scores, model performance varied considerably with the evaluation metric applied. On WADI, IForest achieved the highest AUROC of 0.734 but dropped to 0.180 on AUPRC, falling below the DL models. This is because, while the overall detection performance of DL models was lower, they achieved high precision in the high-confidence region. As AUPRC represents the area under the Precision-Recall curve, high precision in this region can elevate the overall AUPRC. Figure 2 supports this by presenting the Precision-Recall curves for each model on WADI, showing that DL models achieved high precision in the low-recall region. A similar pattern was found on HAI 23.05, where PCA achieved the highest AUROC of 0.737 but scored only 0.033 on Aff_R, and USAD and TranAD also scored 0.033 and 0.014, respectively. In contrast, IForest achieved the highest Aff_R of 0.863, demonstrating that model rankings can vary substantially depending on the choice of metric.

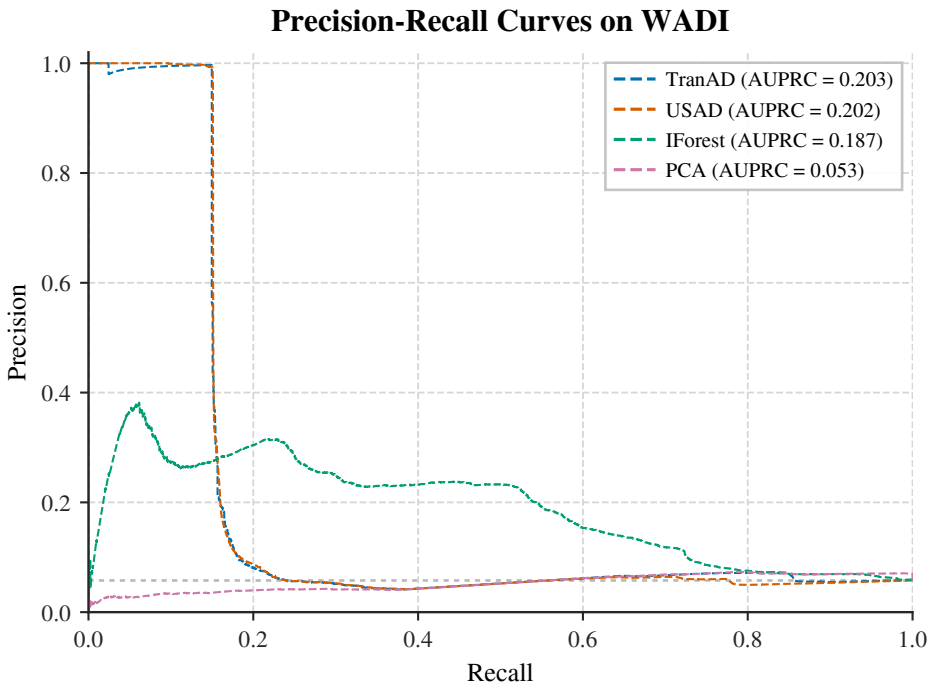

Figure 2: Precision-Recall curves of four anomaly detection models on WADI (single seed: 2026).

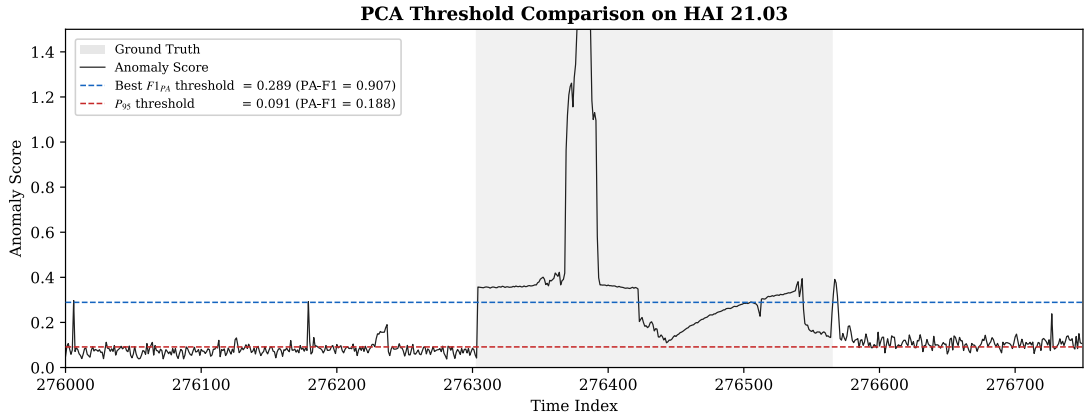

Figure 3: PCA anomaly scores on HAI 21.03 with two threshold lines. The same model yields PA-F1 of 0.907 or 0.188 depending solely on the threshold method.

4.3 Impact of Threshold Selection

In Exp2, when only the threshold method was changed for the same anomaly scores measured in Section 4.2, the PA-F1 score of a single model varied by up to 0.719. As shown in Table 3, PCA on HAI 21.03 dropped from 0.907 (Best $F1_{PA}$) to 0.188(P_{95}), a difference of 0.719, and USAD on SWaT dropped from NDT 0.823 to P_{95} 0.273, a decrease of 0.550. As Figure 3 illustrates

Table 3: PA-F1 Under Different Threshold Methods

Dataset	Model	Oracle		Label-free		
		Best F1	Best $F1_{PA}$	NDT	P_{95}	P_{99}
SWaT	PCA	0.564	0.565	0.753	0.304	0.325
	IForest	**0.803**	**0.846**	0.822	**0.507**	**0.839**
	USAD	0.801	0.805	0.823	0.273	0.282
	TranAD	0.250	0.351	**0.827**	0.284	0.289
WADI	PCA	0.110	0.119	0.491	0.165	0.133
	IForest	**0.473**	**0.672**	0.465	**0.480**	**0.608**
	USAD	0.279	0.299	0.513	0.143	0.133
	TranAD	0.279	0.279	**0.622**	0.143	0.136
HAI 21.03	PCA	**0.901**	**0.907**	0.201	0.188	**0.754**
	IForest	0.582	0.694	0.258	0.329	0.644
	USAD	0.774	0.787	**0.771**	0.255	0.448
	TranAD	0.599	0.613	0.612	**0.353**	0.502
HAI 23.05	PCA	**0.851**	**0.872**	0.408	**0.659**	**0.845**
	IForest	0.255	0.354	**0.415**	0.260	0.321
	USAD	0.316	0.333	0.390	0.193	0.261
	TranAD	0.227	0.325	0.399	0.245	0.282

for PCA on HAI 21.03, the two thresholds applied to the same anomaly score time series define markedly different decision boundaries. A lower threshold classifies more time points as anomalous, increasing recall but also increasing FP, while a higher threshold reduces FP at the cost of missing true anomalies. These results confirm that even when the model output remains identical, the choice of threshold alone can change the precision-recall balance and lead to substantial differences in the F1 score.

Furthermore, the best-performing model also changed with the threshold method. On HAI 21.03, PCA achieved the highest performance under the Oracle criterion, but USAD achieved the highest under NDT and TranAD under P_{95}. On SWaT as well, IForest achieved the highest under Best F1, whereas TranAD achieved the highest under NDT and IForest under P_{95}. This suggests that when reporting anomaly detection performance, a justification for the threshold method should be provided, and comparison across multiple methods is necessary.

Nevertheless, even with diverse threshold settings, the trend of higher ML model performance identified in Exp1 was generally maintained. In 16 out of 20 dataset-threshold combinations, traditional ML models achieved higher performance than DL models. DL models achieved the highest performance in four cases: three under NDT (SWaT, WADI, and HAI 21.03) and one under P_{95} on HAI 21.03. Given that DL models achieved generally lower performance than ML models under P_{95}/P_{99}, which are also Label-free methods, the reason why DL models performed favorably under NDT requires further analysis.

5 Discussion

Through two experiments, we showed that the performance of anomaly detection models measured on ICS benchmark datasets can be influenced as much by the evaluation metrics and

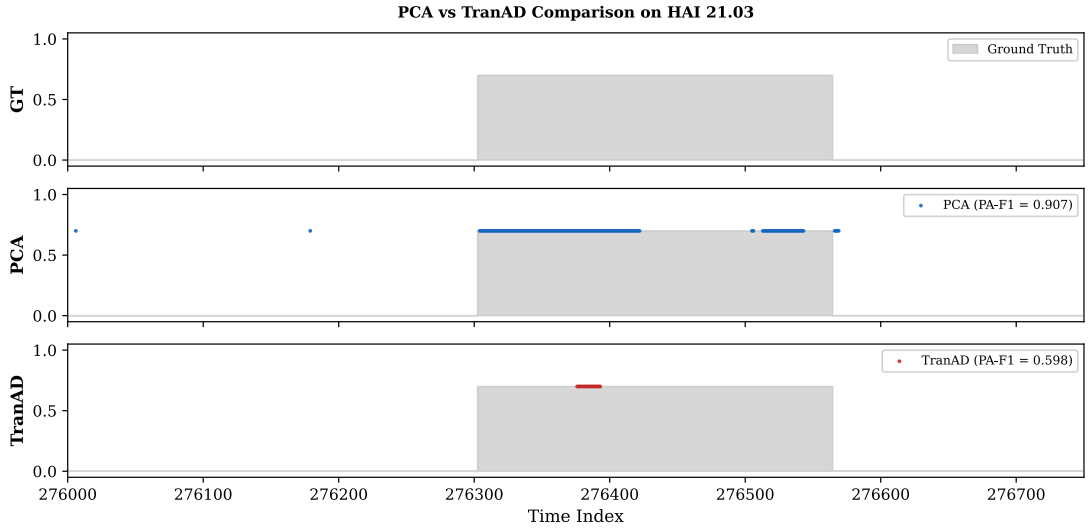

Figure 4: Detection results on HAI 21.03 under the Best $F1_{PA}$ threshold (single seed: 2026). PCA detects most ground truth segments, while TranAD misses substantial portions.

threshold methods used as by the model itself. When different evaluation metrics or threshold methods are applied to the same model output, the resulting performance gap between models can change, indicating that experimental results in existing ICS anomaly detection research are highly dependent on the evaluation methodology employed.

These results can be examined from two perspectives. First, from the model perspective, the finding from prior work [17] that traditional ML models outperform state-of-the-art DL models on SWaT and WADI was consistently confirmed on the HAI datasets also examined here. In the majority of dataset-metric combinations, ML models (PCA and IForest) achieved the highest performance, and on HAI 23.05 in particular, most models other than PCA failed to achieve meaningful detection performance. Second, from the evaluation perspective, even with the scores held constant, relative model performance changed with the evaluation metric or threshold method. On WADI, IForest achieved the highest AUROC but scored lower than DL models on AUPRC, and on HAI 23.05, PCA achieved the highest AUROC but IForest achieved the highest Aff_R, reversing the comparative result. In the threshold comparison experiment (Exp2) as well, the PA-F1 score of a single model differed by as much as 0.719 when only the threshold method was changed, and the model achieving the highest performance also varied by threshold method. Figure 4 illustrates this concretely on HAI 21.03: under the same threshold method (Best $F1_{PA}$), PCA detects most ground truth segments, whereas TranAD misses substantial portions.

These results indicate that performance reporting based on a single evaluation metric and a single threshold setting may not sufficiently reflect the true capability of a model. Given that comparative results themselves can change with the evaluation methodology, it is difficult to fairly compare models based on a single result alone. Therefore, when reporting performance in anomaly detection research, clear justification for the evaluation metric and threshold method should be provided, and where possible, comparison results across multiple metrics and threshold methods should be presented together. Furthermore, our results suggest that the appropriate evaluation metric and threshold method may differ depending on dataset characteristics, and

future research should systematically examine this and select suitable evaluation approaches.

5.1 Limitations

The following considerations should be taken into account when interpreting these results. First, in WADI-Affiliation, the Precision and Recall of three models (PCA, USAD, TranAD) were identical across the three models (0.499 and 0.590), respectively; under our experimental setup, this metric did not adequately distinguish the performance differences among these three models. Identifying the specific cause of this result requires further analysis. Second, on HAI 23.05, most models other than PCA failed to achieve meaningful detection performance, which appears to be related to the inherent detection difficulty of the dataset itself. Third, in Exp2, DL models achieved higher performance than ML models mainly under NDT among the Label-free thresholds, with one additional case under P_{95} on HAI 21.03, but we did not conduct an in-depth analysis of the underlying cause. However, the points mentioned above do not directly affect our core conclusions, and a more in-depth analysis is left for future work.

5.2 Future Work

Based on the impact of evaluation methodology identified here, future research may consider extensions in two directions. First, it is necessary to explore fair evaluation approaches that account for differences in anomaly score distributions among models. As illustrated by the WADI-Affiliation results discussed in Section 5.1, uniformly applying the same threshold to all models can cause differences in score distributions to directly influence evaluation outcomes, and follow-up research is needed to enable more reliable comparisons through score normalization or model-specific threshold selection. Additionally, we focused on evaluating anomaly detection as binary classification on ICS datasets; however, the SWaT, WADI, and HAI datasets all include labels for individual attack scenarios. Future work should examine whether improved evaluation methodologies can enhance not only binary anomaly detection but also the identification of and response to specific attack scenarios. Such research is expected to contribute to the timely detection and root-cause analysis of cyber attacks in ICS environments, as well as to improving the cyber-resilience of such systems [18]. The present evaluation methodology can also be extended to resilience-oriented aspects, such as detection delay, recovery behavior, and adaptation.

6 Conclusion

In ICS anomaly detection research, concerns about the reliability of reported performance have been persistently raised due to limitations in datasets and evaluation methodologies. We compared the performance of traditional ML models (PCA, IForest) and DL models (USAD, TranAD) on four ICS benchmark datasets (SWaT, WADI, HAI 21.03, HAI 23.05) using a variety of evaluation metrics (Exp1), and analyzed how performance changed across threshold methods for identical model outputs (Exp2). ML models matched or surpassed DL models in 15 of 16 dataset-metric combinations, and even with unchanged anomaly scores, relative model rankings shifted with the evaluation metric or threshold method. These findings indicate that reported anomaly detection performance is not determined solely by the model itself, and that the evaluation metrics and threshold methods used can directly influence the conclusions of model comparisons.

Our results indicate that ensuring the security and resilience of ICS environments requires anomaly detection research to focus not only on improving model performance but also on establishing robust evaluation methodologies. Beyond simple model performance comparison, a consistent and systematic examination of evaluation metrics and threshold methods must be integrated into the research process to ensure the validity of reported performance and the reliability of model comparisons. Future research can explore evaluation methodologies tailored to specific dataset characteristics. Additionally, it should be investigated whether improvements in evaluation reliability can contribute to threat detection and response in ICS security environments.

Acknowledgments

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Deployability of Conventional AI and Quantum Machine Learning Detectors for Authentication-Passing Insider Threats in a 5G System*

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Abstract

5G now carries authentication-passing insider threats that bypass perimeter trust: rogue base stations cloning legitimate certificates, clone user equipment sharing leaked credentials, and co-located relay agents forwarding NAS bytes without any stolen credentials at all. We reproduce four such authentication-passing attack scenarios (Benign, Eavesdrop, Impersonation, Relay) on a 5G testbed integrated with an IMS subsystem, and quantify the deployability of four detectors (a cross-plane rule classifier, XGBoost, a Variational Quantum Classifier, and a Hybrid CN-Q-CN model) along three axes. (T) We decompose pipeline latency into capture, extract, infer, and alert stages, and report slack $= t_{\text{attack_goal}} - t_{\text{detect}}$ per (model, scenario). (P) We contrast passive-tap and inline Network Function placement of the detector at AMF and S-CSCF. (Q) We measure per-inference simulator latency for both quantum models and contrast it with the literature-reported Quantum Processing Unit runtime envelope. Three numbers carry the paper: feature extraction consumes 18–23 ms across all models, an order of magnitude above any model's t_{infer}; inline placement of a Hybrid detector adds $\Delta_{p99} = 109$ ms to the NGAP REGISTER round-trip; and the literature QPU envelope is more than 350× the simulator $p99$ for the deepest model we evaluate. From these measurements we derive operational implications for 5G Zero Trust adoption, the near-term role of Quantum Machine Learning, and the evolution axes of AI-Native 6G.

1 Introduction

5G now carries insider threats that pass perimeter authentication and then misbehave inside the trust boundary. Three operator-relevant mechanisms illustrate the problem: a rogue base station with a cloned cell certificate intercepts the plaintext Non-Access Stratum (NAS) signalling that 5G exposes once Authentication and Key Agreement (AKA) completes; a clone user equipment (UE) reusing leaked credentials registers under the victim's IMS Public User Identity (IMPU) and emits fraudulent Session Initiation Protocol (SIP) traffic; and a pair of relay agents co-located with the legitimate radio and core hosts relays NAS bytes end-to-end without any stolen credential at all. Related mechanisms have already produced direct financial losses through SMS one-time-password interception and SIM-swap abuse of mobile-banking sessions [29, 13].

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3GPP has dedicated a study item to false base station countermeasures [2], and the natural response is Zero Trust: continuous, per-transaction verification grounded in 3GPP and NIST guidance [3, 24]. The implicit-trust regime of 5G, in which the network and IMS layers assume integrity for every entity that passes authentication, is no longer sufficient.

Two gaps prevent today's detectors from operating under that regime. First, recent peer-reviewed 5G insider detectors stay within one plane: a control-only telemetry service that catches seven Layer-3 (NAS, RRC, and related signalling) attacks [28], a provenance-graph detector on 5G core logs [21], and an O-RAN cross-layer dApp that fuses the physical (PHY), medium access control (MAC), and Radio Resource Control (RRC) layers but stops at the radio boundary [25]. None of these reports t_{detect} against a scenario-specific $t_{\text{attack_goal}}$, decomposes pipeline latency into capture, extract, infer, and alert stages, or quantifies how an inline Network Function (NF) placement changes NGAP (NG Application Protocol) and IMS round-trip latency.

Second, work on Quantum Machine Learning (QML) for network security has so far been demonstrated on generic intrusion-detection datasets [4, 8, 9]; the deployability of QML inside a 5G core that must respect a sub-second attack-goal envelope has not been measured.

We bridge these gaps by reproducing four authentication-passing attack scenarios on a 5G/IMS testbed and quantifying detector *deployability* (the practical feasibility of running the detector inside a 5G core under its latency and placement constraints) along three axes. Our contributions are measurable:

- **Four-scenario reproduction with explicit attack-success times.** We define $t_{\text{attack_goal}}$, the moment the attack succeeds, per scenario: first visible plaintext NAS (scenario B), first fraudulent SIP under victim IMPU (scenario C), first end-to-end NAS byte across the relay (scenario D). Every run records that moment so that the detector pipeline latency can be compared with the scenario budget.
- **Four-stage pipeline decomposition** into t_{capture}, t_{extract}, t_{infer}, and t_{alert}, measured per (model, scenario) for four detectors: a cross-plane rule, XGBoost (gradient-boosted trees), a Variational Quantum Classifier (VQC), and a Hybrid CN-Q-CN model (classical $\rightarrow$ quantum $\rightarrow$ classical layers).
- **T/P/Q feasibility numbers.** We report *slack* $= t_{\text{attack_goal}} - t_{\text{detect}}$ per scenario (T); passive-tap versus inline-NF NGAP and IMS round-trip overhead (P); and per-inference simulator latency for the quantum models, contrasted with the literature-reported Quantum Processing Unit (QPU) runtime envelope (Q).
- **Operational implications** derived from those numbers for the deployment of 5G Zero Trust, the near-term role of QML, and the evolution axes of AI-Native 6G.

Authentication-passing insider threats in 5G and IMS cannot be judged by accuracy alone. Their detectability depends on where the detector sits in the pipeline, what placement it tolerates, and what hardware it can run on. We reproduce four such threats on a 5G/IMS testbed and measure the deployability of conventional and quantum-machine-learning detectors so that the operational envelope of 5G Zero Trust, the near-term role of QML, and the evolution axes toward AI-Native 6G follow from measurement rather than vision.

2 Background and Related Work

We organise prior work into four areas and, for each, contrast what has been done with what we measure.

2.1 5G Insider Threat Detection

Recent peer-reviewed 5G insider-detection systems operate within a single plane. 5G-Spector deploys an Open Radio Access Network (O-RAN) compliant Layer-3 telemetry service and reports detection accuracy plus CPU and memory overhead on seven control-plane attacks [28]. PROV5GC builds provenance graphs over 5G core logs and detects three attack classes targeting confidentiality, integrity, and availability with low logging overhead [21]. Det-RAN moves into the radio side by fusing PHY, MAC, and RRC features in a near-real-time distributed application (dApp) that achieves 2 ms detection latency at the radio-side Distributed Unit (DU) and 85% accuracy on unseen conditions [25].

These works do not target an insider that has already passed authentication. They also do not report t_{detect} against a scenario-specific $t_{\text{attack_goal}}$ or decompose the four-stage pipeline (capture, extract, infer, alert) across scenarios that span multiple signalling planes (control, signalling, IMS, and IPsec backhaul). This paper measures both.

2.2 Zero Trust Adoption in 5G

Zero Trust principles for mobile networks are described at the architectural level. 3GPP SA3 surveys solution candidates in TR 33.894 and TR 33.794 [1, 3], NIST codifies the per-request trust model in SP 800-207 [24], and a position paper for 5G/6G defines the Monitor–Evaluate–Decide loop inside the O-RAN architecture [22].

Where measurements exist, they cover narrow slices. Vomvas et al. quantify Trusted Execution Environment (TEE) attestation overhead on an Open5GS plus UERANSIM testbed and report +7 ms for UE registration and 2–8% CPU cost [27]. Michaelides et al. measure IPsec and WireGuard latency on the N3 interface and the Service-Based Interface (SBI) of Open5GS [17].

No prior work places an insider detector in line with NGAP or the Cx interface (the Diameter link between IMS CSCF and the Home Subscriber Server) and reports the change in round-trip latency. The present paper measures this overhead for four detector classes.

2.3 Quantum Machine Learning for Network Security

QML detectors are demonstrated on generic intrusion-detection datasets. QML-IDS combines quantum and classical layers and is evaluated on public Network Intrusion Detection System (NIDS) datasets [4]. A recent hybrid pipeline compresses features with a small classical encoder and routes them to 2–4-qubit quantum heads deployed on an IBM Quantum device, evaluated on the NSL-KDD and Ling-Spam datasets [8]. Quantum kernels combined with Support Vector Machines (SVMs) reach 85% hardware accuracy on four attack types of a modern NIDS dataset [9].

These detectors target external traffic anomalies rather than 5G insiders that pass authentication and then act on the NGAP, NAS, or IMS path. Per-inference QPU latency decomposition (queue waiting, transpile, execute, and result fetch) is also not reported alongside the operational $t_{\text{attack_goal}}$ constraints of 5G. This paper measures simulator t_{infer} for VQC and Hybrid CN-Q-CN models and contrasts it with the literature QPU runtime envelope.

2.4 5G Open-Source Testbeds

Open-source 5G stacks make experimentation reproducible. OpenAirInterface provides a flexible end-to-end research platform [19], VET5G constructs a virtual end-to-end testbed targeted

at security education and capture-the-flag exercises [30], and a recent comparison evaluates Open5GS, free5GC, and OpenAirInterface on a common benchmark [5].

These testbeds report reproducibility, throughput, and call-rate metrics. They do not report per-stage capture and extract latency for cross-plane detectors, or passive-versus-inline placement overhead at the Access and Mobility Management Function (AMF) or the IMS Serving Call Session Control Function (S-CSCF). This paper does so on a testbed built from this ecosystem (a 5G core integrated with an IMS subsystem), referred to simply as "the testbed" from now on.

2.5 Summary

Prior 5G insider detection stays single-plane, prior 5G Zero Trust work measures only isolated security mechanisms, prior QML in network security targets generic NIDS datasets, and prior 5G testbeds focus on reproducibility. This paper bridges these four gaps by measuring t_{detect} against $t_{\text{attack_goal}}$, decomposing pipeline latency, quantifying inline NF overhead, and contrasting simulator and literature-QPU execution costs per inference.

3 Threat Model and Testbed

3.1 Attack Scenarios and Success-Time Definitions

We reproduce four attack scenarios distinguished by what the attacker has stolen (cloned cell certificate, leaked UE credentials, or nothing). All four share one constraint: the attacker has already passed primary authentication before any detector observes its first packet, so detection cannot rely on registration failure as a signal. The scenarios are deployment tests. Each one isolates a post-authentication condition that can arise after certificate compromise, credential leakage, or host co-location in an operator, laboratory, or supply-chain environment. This keeps the measurement focused on what happens after authentication has already succeeded.

For every run, we fix a common time origin and two events on the same timeline:

- **Run start ($t = 0$).** The moment stage 1 (Radio Resource Control, RRC) packet capture begins. We use this as the time origin for every measurement in the paper.
- **Attack-success time ($t_{\text{attack_goal}}$).** The moment at which the attacker has achieved the goal of the scenario. What counts as "success" is fixed per scenario and is recorded deterministically as a marker in each run's log file.

The detector pipeline has a processing latency t_{detect} (defined in §4.3). We compare that latency with the scenario time budget using *slack*, slack $= t_{\text{attack_goal}} - t_{\text{detect}}$. Slack is positive when the detector pipeline fits inside the attack-success budget and negative when it exceeds that budget.

Scenario A (Benign) carries baseline 5G and IMS signalling on the testbed: one next-generation NodeB (gNB), two UE instances, and standard NGAP, NAS, IPsec, and IMS REGISTER flows. No attacker capability is present, so no $t_{\text{attack_goal}}$ is defined.

Scenario B (Eavesdrop) models a rogue gNB equipped with a cloned cell certificate and key. The rogue node joins the IPsec backhaul at the strongSwan Security Gateway (SeGW) with the same certificate serial as the legitimate gNB, which manifests as two concurrent IPsec Security Associations (SAs) between SeGW and the gNB cluster. This capability represents certificate life-cycle compromise, mis-issuance, or key theft from a trusted gNB host. The attack succeeds as soon as the rogue gNB sees the first plaintext NAS packet at a local packet-capture process attached to it (a process running on the same host as the rogue gNB container, used here

Table 1: Attack scenarios on the common timeline.

Sc.	Attacker has	Attack-success event ($t_{\text{attack_goal}}$)	Stage	Offset	Approx. time from $t = 0$
A	nothing	(no attack)	—	—	—
B	cloned cell cert.	first plaintext NAS visible at the rogue gNB's local capture process	stage 2 (NAS register + AKA)	950 ms	1.32 s
C	cloned cert. & UE creds.	first fraudulent SIP under victim's IMPU	stage 7 (SIP service)	120 ms	3.97 s
D	nothing (byte relay only)	first NAS byte relayed end-to-end across both relay agents	stage 2 (NAS register + AKA)	650 ms	1.14 s

to dump radio-side traffic for the attacker). The Packet Data Convergence Protocol (PDCP) terminates radio-link encryption at the gNB, so once the rogue gNB joins the backhaul the upper-layer NAS bytes appear in cleartext at that local capture process. The attack-success time falls inside the NAS-register-with-AKA stage (stage 2).

Scenario C (Impersonation) adds a clone UE with the same International Mobile Subscriber Identity (IMSI), AKA key material (Ki), and IMS credentials as a victim user. The clone attaches concurrently with the victim through the rogue gNB. The AMF records repeated Subscription Permanent Identifier (SUPI) context churn, the S-CSCF records concurrent IMPU binding, and the clone sends a fraudulent SIP MESSAGE under the victim's IMPU. This models the post-leakage state of SIM-cloning, provisioning-database exposure, or test-profile reuse. The attack-success time is the moment that first fraudulent SIP packet appears, which falls inside the SIP service stage (stage 7).

Scenario D (Relay) carries no stolen credential. Two relay agents – one co-located with the legitimate gNB host, the other co-located with the AMF host – forward NAS bytes between the gNB and the AMF through a Unix-domain-socket relay channel (a local inter-process socket used as the relay path between the two agents). This capability represents host-level co-location or container escape on both ends of the path. The relay is byte-transparent from the operator's perspective but inflates NAS inter-arrival time variance. The attack-success time is the moment the first NAS byte is relayed end-to-end across both agents, located inside stage 2.

All four scenarios run in RFsim (radio-frequency simulation; no over-the-air radio emission is performed), producing byte-transparent traces on Docker bridges. The measurements focus on the core/IMS timing path: packet form, plane fan-out, feature extraction, and detector placement. Live-radio effects such as fading, scheduler load, handover, mobility, and unrelated subscriber traffic are part of operator extensions.

Table 1 summarises the four scenarios on a single timeline. The run starts at the beginning of stage 1 capture ($t = 0$). The "Stage" column gives the capture stage that contains the attack-success time; "Offset" gives milliseconds after that stage starts; "Approx. absolute time" converts this to an approximate distance from $t = 0$, using a reference run and the deterministic inter-stage gaps recorded in the run marker log. Other runs carry the small jitter described in §4.1.

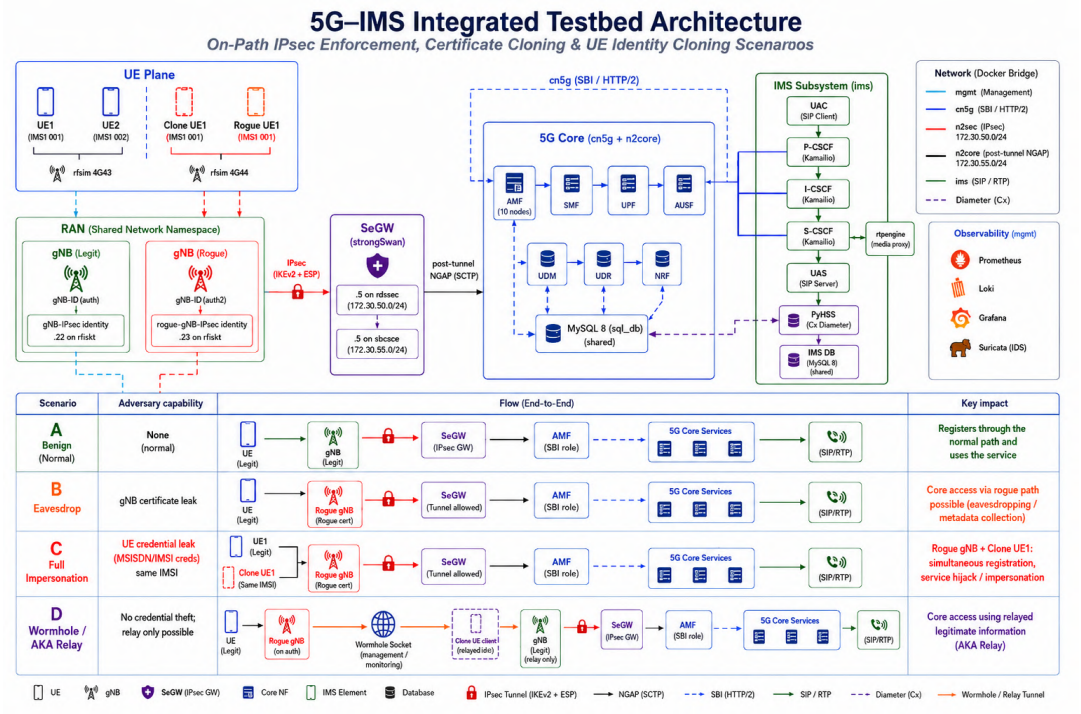

Figure 1: Testbed architecture and attack paths.

3.2 Testbed

Before describing the stack, we fix one term we use throughout the paper. A *signalling plane* (or just *plane*) denotes a logically separated path along which the testbed carries a class of messages. The four planes relevant to our detection are NGAP (control-plane signalling between the gNB and the AMF), NAS (signalling between the UE and the AMF, transparent to the radio access), Cx Diameter (control-plane signalling between the IMS CSCF cluster and the Home Subscriber Server), and the IPsec backhaul (the encrypted transport between the gNB cluster and the SeGW). A *cross-plane signal* is an observable signature derived from more than one of these planes in the same run.

Our reference testbed is the published workshop testbed. Its component stack is detailed in [16, 12, 26] and illustrated in Figure 1: an Open5GS-aligned OpenAirInterface 5G Core, the Kamailio P/I/S-CSCF IMS cluster, and the strongSwan SeGW. The stack is hosted in five Docker bridges: a management bridge; a 5G control-plane bridge `cn5g` for the OpenAirInterface 5G Core network functions (NRF, UDR, UDM, AUSF, AMF, SMF, and UPF); the `n2sec` bridge that carries IPsec ESP between the SeGW and the gNB-side processes; the `n2core` bridge that carries SCTP/NGAP between the SeGW and the AMF; and the `ims` bridge that hosts the Kamailio Proxy, Interrogating, and Serving CSCF cluster, PyHSS over Cx Diameter, and rtpengine for media. The 5G Core and the IMS share a single MySQL subscriber database. All NGAP signalling is forced through the strongSwan SeGW on `n2sec`. Figure 1 shows the bridge layout and the scenario-specific attack paths.

For each scenario we capture eight stages on the appropriate bridge: RRC initiation on `cn5g`, NAS register with AKA on `n2core`, NAS Security Mode Command on `n2core`, PDU session

establishment on `n2core`, IPsec backhaul on `n2sec`, IMS REGISTER on `ims`, SIP MESSAGE and INVITE on `ims`, and RTP media on `ims`. Stage boundaries are seeded deterministically per run. The eight cross-plane signals ("fingerprints" in the workshop reference) reported by the workshop testbed are derived from per-stage packet observations: two timing signals (REG-to-401 latency and IMS REGISTER inter-arrival time) and six state signals (SeGW duplicate SA count, NGAP source IP diversity, distinct gNB-side processes observed on `n2sec`, AMF SUPI replications, S-CSCF concurrent IMPU binding, and synthetic NAS injection events).

The plane fan-out of the testbed (i.e., how a single user action splits across multiple planes), the NGAP and IMS packet ordering, and the inter-arrival distributions of a deployment are traffic-form properties. The detection pipeline reads counts, inter-arrival times, source diversities, and protocol fractions rather than NAS or SIP payload bytes. We therefore generate scenario-derived pcaps whose traffic form (per-stage packet count, plane fan-out, protocol distribution, and inter-arrival shape) matches the scenario specifications, while the payload bytes are randomised. This keeps the experiment aligned with the deployability question measured in this paper.

4 Methodology

4.1 Reproduction Protocol

We generate ten runs per scenario, for a total of forty runs and three hundred and twenty per-stage pcaps. Each run is deterministic in its seed and carries a marker log that records the run identifier, the eight cross-plane signal targets, and the attack-success time $t_{\text{attack_goal}}$ in seconds relative to the run start ($t = 0$, the start of stage 1 capture, as defined in §3.1). Stage boundaries follow the workshop reference described in §3.2; the per-stage baseline packet counts and the per-scenario multipliers that govern attacker-induced traffic-form deviations are encoded in the scenario generator. Inter-stage gaps carry 100–300 ms of pseudo-random jitter (seeded by the run index) so that no two runs have identical absolute times. We use these ten runs per scenario to report central values, observed distributions, and $p99$ timing boundaries. Larger repetition counts add confidence intervals around the same T/P/Q measurements.

4.2 Detection Models

We evaluate four detectors, summarised in Table 2. They are chosen so that classical and quantum, shallow and deeper, and stateless and stateful designs are all represented, and so that per-inference cost differs by orders of magnitude across the four. Principal Component Analysis (PCA) compresses the 200-dimensional packet-form input into the smaller dimension that each quantum model takes as input. Q1 and Q3 are simulated on the PennyLane `default.qubit` backend with the autograd interface [6]; see §4.5 for QPU hardware scope. A Variational Quantum Classifier (VQC) maps input scalars to qubit rotation angles and trains circuit parameters for classification. "AngleEmbedding" is the PennyLane primitive that performs this mapping, and "BasicEntanglerLayers" denotes the nearest-neighbour entangling block used in both quantum models. "CN-Q-CN" denotes the classical-network → quantum → classical-network composition: a classical projection feeds a small quantum circuit, and the four Pauli-Z measurement outputs feed a learned classical head.

The cross-plane rule C0 checks state-based signals first, then the jitter-based signal: it returns scenario C when any AMF SUPI replication is observed or two or more concurrent IMPU bindings are seen, scenario D when the synthetic-injection signal reaches two or more,

Table 2: Detection models and key parameters.

ID	Model	Input	Key parameters
C0	Cross-plane rule	8 cross-plane signals	3 priority rules (state signals first, then jitter signal)
C1	XGBoost [7]	200-dim flat	80 trees, depth 4, learning rate 0.2
Q1	VQC	PCA 200→4, 4 qubits	BasicEntanglerLayers, L=2, 80 epochs
Q3	Hybrid CN-Q-CN	PCA 200→16, linear 16→4, 4 qubits, linear 4→4	BasicEntanglerLayers, L=6, 50 epochs [8]

scenario B when the SeGW SA duplicate count or NGAP source-IP diversity reaches two, and scenario A otherwise. C1 trains a gradient-boosted tree on the flat 200-dimensional feature. Q1 reduces the 200-dimensional feature to four real values via PCA, angle-encodes them onto four qubits, and applies two layers of basic entangler. Q3 inserts a learned linear projection from the sixteen leading principal components to the four-qubit angle space, runs a six-layer basic entangler on four qubits, and feeds the four Pauli-Z expectations into a learned classical head.

All four detectors are evaluated with 5-fold cross-validation (CV) at the run level: every fold holds out one fifth of the runs per scenario, so the same run never appears in both training and evaluation. The feature extractor is fit inside each training fold before it is applied to the held-out runs, so PCA statistics and model parameters are not shared from evaluation data into training. The split evaluates generalisation across held-out runs within the four scenarios.

4.3 Pipeline-Latency Decomposition

The detector pipeline latency is decomposed into four serial stages:

$$t_{\text{detect}} = t_{\text{capture}} + t_{\text{extract}} + t_{\text{infer}} + t_{\text{alert}}.$$

t_{capture} is the wall-clock time to read all eight per-stage pcaps for one run. t_{extract} is the wall-clock time to compute the 200-dimensional packet-form feature vector and the eight cross-plane signals from the same pcaps. t_{infer} is the per-sample model inference time measured with `time.perf_counter` immediately around the single-sample predict call. t_{alert} is a deterministic 0.5 ms write-cost included as a reference floor for any logging stage that follows the model output. The slack between the scenario time budget and the detector pipeline is

$$\text{slack} = t_{\text{attack_goal}} - t_{\text{detect}},$$

reported in milliseconds. Positive slack means the complete detector pipeline fits inside the attack-success budget; negative slack means the pipeline exceeds it.

This definition measures complete-window cross-plane detection. The feature vector is computed after the eight per-stage pcaps are available, so the slack values answer whether the processing cost of the full cross-plane pipeline fits inside the scenario budget. A streaming implementation can reuse the same timeline and report an additional earliest-decision time from incremental windows inside stage 2 or stage 7.

4.4 Feasibility Three Axes

T-feasibility reports the four-stage decomposition stacked per (model, scenario), and the slack distribution per (model, scenario).

P-feasibility contrasts two detector placements. In the *off-path placement*, the detector reads a mirrored copy of the Docker bridge traffic and never sits on the data path, so it cannot block packets; it can only raise alerts after the fact. In the *on-path placement*, every NGAP REGISTER packet at the AMF and every Cx packet at the S-CSCF must pass through the detector before being forwarded to its destination, so each packet incurs the per-packet t_{infer} cost. We report the change in the NGAP REGISTER-to-200 OK and IMS REGISTER-to-200 OK round-trip latency.

Q-feasibility reports per-inference simulator latency on PennyLane `default.qubit` for both quantum models. We contrast it with the literature-reported QPU runtime envelope (the range of expected values for one inference on a physical QPU), broken into *queue* (waiting time before the job runs on the physical device), *transpile* (compiling the abstract circuit to the hardware-native gate set), *execute* (running the compiled circuit on the QPU), and *fetch* (returning measurement results to the client), as cited in §5.3 [9, 23, 18].

4.5 Measurement Boundary

This paper measures controlled deployability. The pcaps preserve packet count, plane fan-out, protocol order, and inter-arrival distribution from the scenario specifications, while payload bytes are randomised. The feature set uses traffic-form statistics, so these pcaps measure the feature-extraction and placement costs that dominate the T/P axes. For the Q axis, the paper measures simulator latency on PennyLane `default.qubit` and compares it with literature-reported physical-QPU runtime components. The same setup can be extended with live PCAP comparisons, larger repetition counts, unseen-scenario tests, federated multi-operator topologies, and RFsim-to-RF translation.

5 Feasibility Analysis

5.1 T-Feasibility: Pipeline-Latency Decomposition and Slack

Figure 2 (left) shows the mean four-stage latency per (model, scenario). Two observations follow directly from the numbers. First, $t_{\text{capture}} + t_{\text{extract}}$ ranges between 18 ms and 23 ms across all four models and all four scenarios, while t_{infer} ranges from 0.8 μs (C0 rule) to 3.5 ms (Q1 VQC) to 7.0 ms (Q3 Hybrid). Feature extraction is the bottleneck, and it is independent of model class. Second, the maximum t_{detect} across all (model, scenario) cells is 60.9 ms ($p99$, Q3 Hybrid on scenario B), where the dominant contribution is the $p99$ capture latency at scenario B, not the inference path.

The slack distribution in Figure 2 (right) is positive across all (model, scenario) pairs for which $t_{\text{attack_goal}}$ is defined. Mean slack ranges from 1105 ms (Q3 Hybrid on scenario D) to 4075 ms (C0 rule on scenario C). The difference in mean slack between C0 and Q3 within a fixed scenario is approximately 8 ms; this matches the mean t_{infer} difference between the two models and confirms that the contraction of slack as the model deepens is fully attributable to inference cost, not to the rest of the pipeline. Across the ten runs per scenario, the observed $p05$–$p95$ slack intervals remain positive: 1264–1468 ms for scenario B, 3773–4327 ms for scenario C, and 1048–1166 ms for scenario D across the four models. These slack values are complete-window latencies: they establish the cost of running the full cross-plane detector over all eight

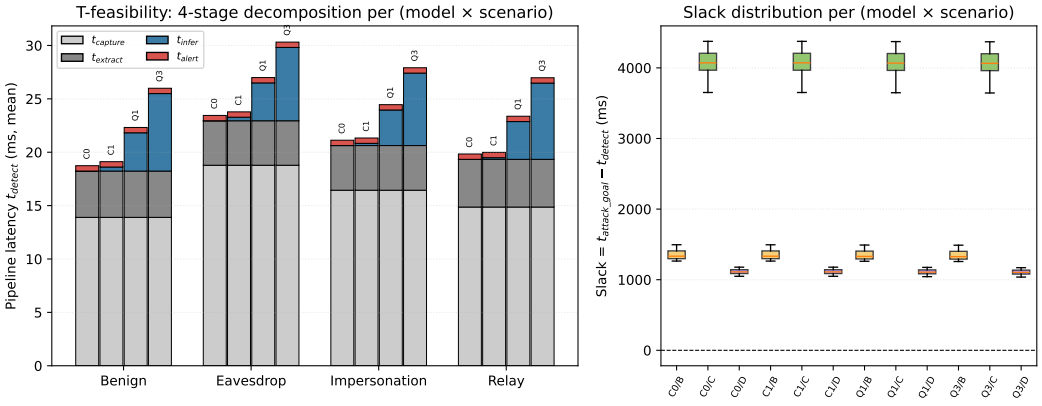

Figure 2: T-feasibility results.

Table 3: Detector accuracy and recall.

Model	Macro F1	Acc.	A	B	C	D	Mean t_{infer}
C0 Rule	0.95	0.95	1.00	1.00	1.00	0.80	0.86 μs
C1 XGBoost	0.97	0.97	1.00	1.00	1.00	0.90	0.27 ms
Q1 VQC	0.41	0.48	0.60	0.10	1.00	0.20	3.50 ms
Q3 Hybrid	1.00	1.00	1.00	1.00	1.00	1.00	7.01 ms

stages. They are the baseline against which an incremental streaming detector can later report earliest-decision time.

The accuracy of the four detectors, computed across the same five-fold run-level split, is summarised in Table 3. The macro-F1 score is the per-class F1 averaged over the four scenarios; it weights each scenario equally regardless of support. C0 rule and C1 XGBoost recover scenarios B and C through their state-based signals; both partially miss scenario D, whose jitter-based signal is harder to capture with a static rule. Q1 VQC at L=2 shows limited capacity for the 4-class problem under the PCA(200,4) input budget (a four-dimensional input compressed from 200 raw stats); this reflects the operational constraint of the shallow circuit depth (see §6.2). Q3 Hybrid CN-Q-CN, with deeper quantum layers and a learned classical head, reaches perfect macro-F1 within the four-scenario corpus. The deployability result is where this accuracy lands inside the T/P/Q timing envelope: higher model capacity improves classification here, but it also moves the detector away from on-path placement.

5.2 P-Feasibility: Off-Path versus On-Path Placement

We model two detector placements. In the *off-path* placement, the detector reads a mirrored copy of a Docker bridge and does not sit on the data path; the NGAP REGISTER and IMS REGISTER round-trip latencies are 5.2/12.4 ms and 9.4/23.1 ms at $p50/p99$ respectively, and the detector cannot block packets, only raise alerts after the fact. In the *on-path* placement, every packet of the relevant handshake is forwarded only after the detector returns a decision: a ten-packet NGAP REGISTER handshake therefore acquires ten t_{infer} samples of overhead, and an eight-packet IMS REGISTER acquires eight.

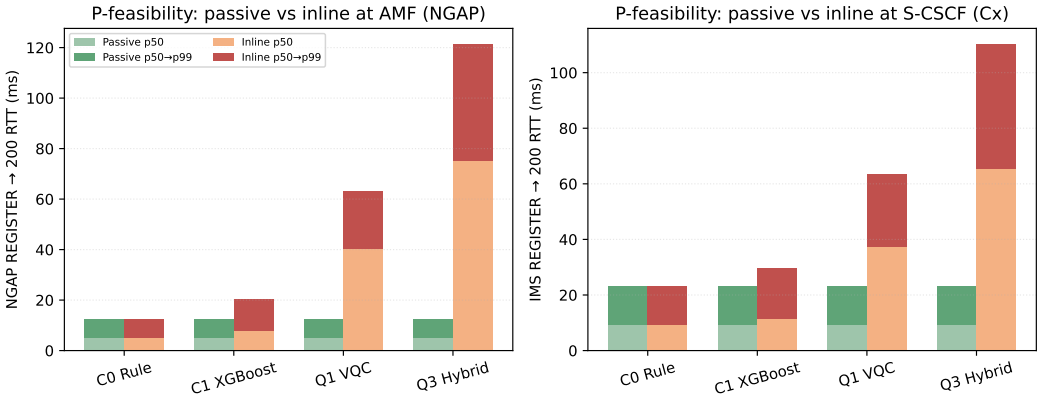

Figure 3: P-feasibility results.

Figure 3 shows the resulting NGAP and Cx round-trip latencies. Bars show the $p50$ latency and the $p50$-to-$p99$ extent. The cross-plane rule and XGBoost remain within an order of magnitude of the baseline: C0 adds +0.04 ms at NGAP $p99$, C1 adds +8.13 ms. Q1 VQC adds +50.5 ms at NGAP $p99$, multiplying baseline by approximately five. Q3 Hybrid adds +109.1 ms, multiplying baseline by approximately ten. The Cx interface shows the same ordering, with smaller absolute deltas because the handshake is shorter.

The split is operational. C0 and C1 are placeable on the data path of the AMF or the S-CSCF; Q1 and Q3 must observe off-path. This is the first quantitative placement result that the present paper establishes. The P-axis stops at detector placement cost; enforcement actuators such as session revocation, slice isolation, and policy update consume the alert downstream.

5.3 Q-Feasibility: Simulator and Literature-QPU Envelope

Figure 4 (left) shows the cumulative distribution of per-inference simulator latency for Q1 and Q3. Figure 4 (right) compares simulator $p99$ with the literature QPU runtime envelope on a logarithmic scale. The simulator $p99$ values are 5.0 ms (Q1) and 10.9 ms (Q3). The four-qubit circuits run on the PennyLane `default.qubit` simulator.

For comparison, prior work that runs the same class of two-to-four-qubit quantum kernels on physical QPUs reports a runtime envelope (the range of expected values for one inference) in which the queue alone consumes between two and thirty seconds, transpile consumes approximately one and a third seconds, execute approximately thirty-five milliseconds, and result fetch approximately one hundred and eighty milliseconds [9, 23, 18]. Figure 4 (right) stacks these literature values on a logarithmic axis next to the measured simulator $p99$. The lower edge of the literature QPU envelope (queue minimum of two seconds plus transpile plus one execute plus fetch) is approximately three and a half seconds; this is more than 350× the Q3 simulator $p99$ of 10.9 ms and more than five hundred times the Q1 $p99$. For the $t_{\text{attack_goal}}$ values of this paper (950 ms after the start of stage 2 for scenario B, 120 ms after the start of stage 7 for scenario C, and 650 ms after the start of stage 2 for scenario D), the QPU runtime envelope sits well above any realistic slack budget. The Q-axis therefore separates two roles: simulator-level QML is useful for model and feature exploration, while physical-QPU execution belongs to offline or auxiliary analysis rather than online per-packet detection. This is the second operational result established by the present paper.

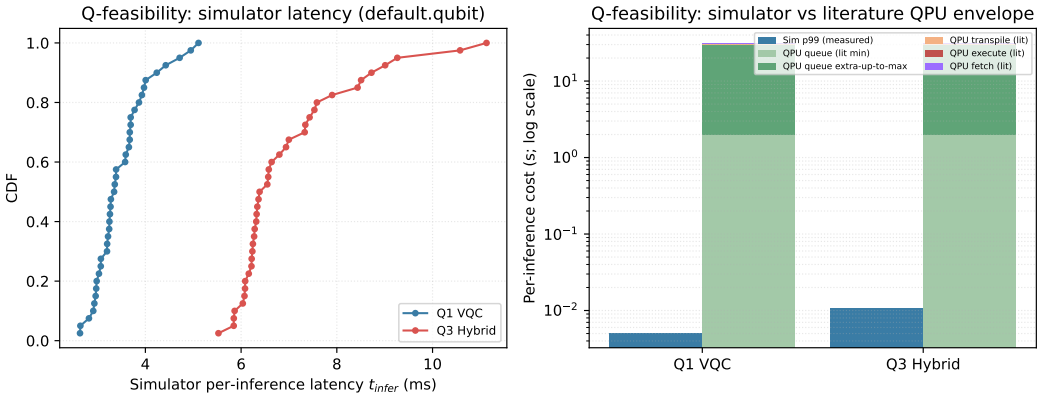

Figure 4: Q-feasibility results.

6 Discussion

The numbers in §5 translate into three implications for the deployment of 5G Zero Trust, the role that Quantum Machine Learning can play in the next few years, and the design axes of AI-Native 6G.

6.1 Operational Constraints on 5G Zero Trust Adoption

Zero Trust as codified in 3GPP and NIST guidance asks that every transaction be verified continuously rather than implicitly trusted after authentication [3, 1, 24]. Our measurements clarify where the verification can sit. The 18–23 ms feature-extraction cost is paid by every cross-plane detector, classical or quantum, so the engineering target most likely to advance deployability is the extraction pipeline itself: parallel tap fan-out, zero-copy reads, and pre-computed protocol counters at AMF and S-CSCF boundaries. On-path placement of an XGBoost-class detector adds 8 ms to NGAP REGISTER $p99$, which fits inside typical 5G latency budgets, but on-path placement of a quantum-augmented detector multiplies that figure by roughly ten and only the off-path placement remains viable. The Zero Trust enforcement of authentication-passing insider threats in 5G therefore divides into two architectural classes: a near-NF, on-path classical class that catches state-based signals (scenarios B and C) within the round-trip envelope, and an off-path, quantum-or-deep class that runs out of band and feeds closed-loop responses (automated detection-to-enforcement feedback loops that revoke a session or isolate a slice after the alert). The measurements in this paper fix the detector side of that loop; session revocation, slice isolation, and policy rollback are the enforcement actuators that consume the alert.

6.2 Near-Term Role of Quantum Machine Learning

Two findings shape the near-term role of QML. First, the shallow VQC at L=2 underperforms on the 4-class problem despite the deeper Hybrid CN-Q-CN reaching perfect F1 on the same dataset. The contrast shows that the small quantum core is sensitive to input compression and model capacity under the feature budget used here. Hybrid pipelines that put classical pre- and post-processing around a small quantum core follow the pattern recently reported on physical hardware [8]. Second, even when the simulator $p99$ is well within the slack budget, the literature-reported QPU queue and transpile costs alone exceed every $t_{\text{attack_goal}}$ in this paper [23]. The

viable role for QML in 5G insider detection over the next few years is therefore *scheduled offline work* (e.g., daily or weekly batch runs that extract quantum-derived features for a downstream classical detector, or pre-train model weights that the classical detector loads), not per-packet online inference. This is a narrower role than the QML-IDS demonstrations report [4], and the boundary is set by the QPU runtime envelope rather than by accuracy.

6.3 Operator Extensions

The measurements isolate the core/IMS detector path. Operator extensions add deployment variables outside that path: live radio variation, mobility, handover, unrelated subscriber traffic, vendor-specific Network Function scheduling, and operator policy diversity. Those variables can move both sides of the slack equation by changing $t_{\text{attack_goal}}$ and t_{detect}. Live PCAP comparisons, larger run counts with confidence intervals, and multi-topology replication extend the present measurement to those operator-facing settings.

6.4 Evolution Axes Toward AI-Native 6G

IMT-2030 sets the platform target for sixth-generation networks at hyper-connected service integration [11], the ten-nation joint statement adds a policy expectation that 6G be secure, open, and resilient by design [10], and the industry security roadmap aligns 6G with built-in trust assumptions [20]. Our measurements identify three axes that have to be addressed before AI-Native 6G can extend Zero Trust enforcement to the radio and edge layers.

First, feature extraction must be amortised closer to the plane: pushing cross-plane counters onto smart Network Interface Cards (smart NICs) or extended Berkeley Packet Filter (eBPF) fast-path hardware is the lever for shrinking the dominant 18–23 ms cost that we measured. Second, on-path decisions must be packaged so that the per-packet t_{infer} is not paid on the data path of every handshake packet. Sampling at the session boundary (running the detector once per registration or call, not once per packet) or returning the decision asynchronously to the data path becomes a deployment requirement, not a future direction.

Third, the foundation of authentication-passing detection is authentication itself. Forward-secret and post-quantum-hybrid extensions to 5G-AKA reduce the insider window that the present detectors must react to [31, 14, 15]. None of these three axes is conjectural; each follows from a number reported in §5.

7 Conclusion

This paper reproduced four authentication-passing attack scenarios on a 5G/IMS testbed built with RFsim and Docker. It decomposed detection latency into capture, extract, infer, and alert stages for four detectors: a cross-plane rule, XGBoost, a Variational Quantum Classifier, and a Hybrid CN-Q-CN model. Three numbers carry the result. Feature extraction consumes 18–23 ms regardless of model class, an order of magnitude above any model's t_{infer} in this setup. On-path placement of a Hybrid quantum detector adds 109 ms to NGAP REGISTER $p99$, multiplying the off-path baseline by approximately ten, so only C0 and C1 remain inline-viable on the AMF and S-CSCF data paths. The lower edge of the literature-reported QPU runtime envelope is more than 350× the Q3 simulator $p99$, placing physical-QPU inference outside the slack budget of every scenario.

Within these numbers, three operational implications follow. First, authentication-passing insider detection in the 5G core is constrained by feature extraction and detector placement,

not by the choice between classical and quantum inference. Second, Zero Trust deployment divides into two architectural classes: a near-NF, on-path classical class that catches state-based signals within the round-trip envelope, and an off-path, deeper or quantum-augmented class that runs out of band and can feed later enforcement actuators. Third, the near-term viable role for Quantum Machine Learning is scheduled offline work, such as extracting quantum-derived features or pre-training a downstream classical detector. It is not per-packet online inference; the boundary is set by the QPU runtime envelope rather than by accuracy.

Future Work

The measurements in this paper lead to three direct extensions.

Model-space expansion. The detector set should be expanded on both sides of the classical–quantum boundary: deeper VQCs, Quantum Kernel SVMs, and Quantum Graph Neural Networks on the quantum side, and BiLSTM, Transformer, and graph-based detectors on the classical side. The aim is to map how the capability boundary observed here moves across model classes and architectures, and where each new model lands inside the T/P/Q envelopes reported in this paper. This extension should include leakage checks, ablations over PCA dimension and qubit count, and unseen-scenario tests that hold out an entire attack family or operator topology.

Hardware validation for QML. The Q-feasibility numbers reported here pair PennyLane `default.qubit` measurements with literature-cited QPU envelopes. Measuring queue, transpile, execute, and fetch directly on physical hardware (e.g., IBM Quantum or IonQ) for the four-qubit Hybrid circuit would either confirm or revise the 350× simulator-to-QPU gap, and would let us check whether the QPU envelope itself shrinks as cloud-QPU schedulers and transpilers evolve. This measurement is the next step for deciding whether QML moves beyond the scheduled-offline role identified above.

Operator extensions. The next measurement step is to move from controlled traces to operator-facing settings. More repetitions across multiple seeds would turn the observed latency and slack distributions reported in §5.1 into confidence intervals. Live PCAPs from the workshop testbed would compare operator traces with the traffic-form dataset described in §4.5. An eBPF or smart-NIC implementation of the cross-plane feature extractor would show how much of the dominant 18–23 ms cost can be removed from the data path. The same follow-up should report earliest-decision latency from incremental feature windows, separately from the complete-batch latency used in this paper.

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Contextual VEX State Inference for Firmware-Based Control Systems from Firmware-Derived SBOMs*

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Abstract

The presence of a reported CVE does not necessarily mean that a firmware-based Industrial Control System asset is actually affected. In ICS/OT environments, vulnerability applicability depends on component presence, version applicability, configuration conditions, network exposure, and operational context. This study proposes a checklist-driven, evidence-bound method for supporting VEX state inference from firmware-derived SBOMs by combining SBOM data, public vulnerability information, vendor advisories, cyber threat intelligence, and operational context. The method consists of three stages: asset-centered input construction, VEX-oriented checklist generation, and checklist-based state determination. In a single-firmware pilot case study using DH_IPC-HX8XXX-Eos_EngSpn_N_Stream3 (V2.420), a pseudo-gold component/version applicability dataset of 25 firmware-CVE pairs was constructed. Under the Qwen2.5-7B-Instruct zero-shot baseline, the method achieved 56.0% label accuracy, 72.0% status accuracy, and 50.5% macro-F1. These results suggest that checklist-driven VEX inference is feasible as an analyst-assistance workflow for firmware-based ICS/OT assets, while broader validation across firmware families remains future work.

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1 Introduction

Industrial Control Systems (ICS) are cyber-physical environments composed of heterogeneous elements such as controllers, HMIs, engineering workstations, firmware, third-party libraries, and networked services. As IT/OT convergence deepens, these assets can no longer be treated as isolated devices with static trust assumptions. Instead, they must be understood as complex systems whose security depends not only on software correctness, but also on dependency relationships, operational exposure, and supply chain visibility. In this setting, visibility becomes a foundational requirement: operators need to know what software components are present, what services are exposed, which network paths are reachable, and whether a reported vulnerability is actually meaningful in the context of the target asset. This trend has made software supply chain security increasingly important in ICS environments. SBOM (Software Bill of Materials) provides visibility into the software components included in a product, while VEX (Vulnerability Exploitability eXchange) communicates whether a reported vulnerability is applicable, not applicable, fixed, or still under investigation. In other words, SBOM answers the question of what is inside the asset, whereas VEX addresses what that vulnerability means for the asset in practice. In smart manufacturing and digitally transformed industrial environments, firmware-based OT assets are increasingly connected to production management, remote maintenance, supply-chain coordination, and E-Business service infrastructures. Therefore, determining whether a reported CVE is actually applicable to a firmware asset is not merely a device-level security task, but a prerequisite for trustworthy industrial digital operations and secure supply-chain decision making.

Existing studies and practical discussions already show that SBOM is a valuable visibility artifact, but they also indicate that visibility alone is not sufficient. Even when SBOM quality is high, downstream vulnerability assessment can still produce large numbers of false positives unless additional contextual evidence is incorporated. This limitation is especially significant in ICS, where vulnerability applicability depends not only on package presence, version ranges, and service exposure, but also on control-loop participation, industrial protocol dependency, air-gap status, maintenance or engineering mode, and potential impact on the physical process. Vulnerability analysis in ICS therefore needs to be framed not simply as code-level detection or generic CVE verification, but as an asset-centered, context-aware applicability assessment problem. Firmware-level vulnerability applicability is particularly important for E-Business Information Systems because industrial firmware assets often support production continuity, remote service operations, physical-security monitoring, and supply-chain coordination. A vulnerability reported at the component level may not be exploitable in a specific firmware context, while a missed applicable vulnerability may expose connected industrial services. Thus, applicability assessment helps reduce both unnecessary mitigation of non-applicable CVEs and delayed response to vulnerabilities that may affect trustworthy industrial and supply-chain operations.

Recent LLM-based vulnerability analysis has moved in this direction by expanding from direct code-level detection to prompt-based verification, tool-augmented reasoning, and checklist-driven workflows. However, most existing approaches still rely on multi-stage agent pipelines and treat VEX as a downstream artifact rather than as the primary inference target.

At the same time, existing SBOM/VEX research remains stronger in component representation and status communication than in formalizing how asset-specific evidence should be converted into a direct, auditable applicability decision. This gap becomes more serious in firmware-based ICS assets, where source code is often unavailable and the analyst must reason primarily from firmware-derived artifacts and contextual evidence.

To address this gap, this study proposes a firmware-SBOM-based method for evidence-bound VEX inference in ICS/OT assets. The proposed approach combines firmware asset information, SBOM data, public vulnerability intelligence, vendor advisories, cyber threat intelligence, and operational context into an asset-centered input representation. Based on this representation, it generates a VEX-oriented

checklist and derives an auditable candidate VEX state through checklist-based reasoning rather than unconstrained free-form judgment.

The case study in this paper was conducted on a firmware-derived SBOM obtained from the asset DH_IPC-HX8XXX-Eos_EngSpn_N_Stream3, version V2.420, which can be interpreted as an OT physical-security firmware case. The SBOM contains vulnerability-relevant components such as OpenSSL 1.0.1j, BusyBox 1.18.4, uClibc 0.9.33, Linux kernel 3.4.35, wpa_supplicant 2.0, and zlib 1.2.6. From this firmware SBOM, a pseudo-gold component/version applicability dataset of 25 firmware-CVE pairs was constructed and evaluated using the proposed checklist-driven VEX inference pipeline. The evaluation is intentionally positioned as a single-firmware pilot feasibility study rather than a statistically generalizable validation across ICS/OT firmware families.

In the measured baseline setting, the system generated 25 checklists and 25 final predictions, yielding 56.0% label accuracy, 72.0% status accuracy, and 50.5% macro-F1. These metrics primarily evaluate component grounding and component/version-level applicability, not complete operational VEX ground truth. The main contributions of this study are threefold: (1) it formulates VEX inference for firmware-based ICS assets as a context-aware applicability assessment problem grounded in supply chain visibility; (2) it proposes a checklist-driven reasoning framework that connects SBOM evidence and operational context to auditable VEX state interpretation; and (3) it demonstrates the feasibility and limitations of this approach through a single-firmware case study while analyzing the resulting error structure, particularly component grounding and evidence binding failures.

The remainder of this paper is organized as follows. Section 2 reviews related work on LLM-based vulnerability analysis, SBOM/VEX-based state representation, and ICS-contextual vulnerability assessment. Section 3 presents the proposed methodology. Section 4 reports the case study and validation results. Section 5 discusses the implications and limitations of the findings. Finally, Section 6 concludes the paper and outlines future work.

2 Related Works

Prior work related to this study can be organized into three streams: LLM-based vulnerability analysis, SBOM/VEX-based vulnerability representation, and ICS/OT contextual vulnerability assessment. LLM-based studies have improved vulnerability detection and verification using prompting, fine-tuning, retrieval, and tool-augmented workflows. SBOM/VEX research has focused on component visibility and vulnerability status communication. ICS/OT vulnerability assessment has emphasized operational context, including runtime usage, network exposure, protocol dependency, and physical-process impact. However, these streams remain insufficiently connected for firmware-based assets where source code is unavailable and analysts must infer VEX states from firmware-derived evidence.

2.1 LLM-based Vulnerability Analysis

LLM-based vulnerability analysis has moved beyond code-level detection toward context-aware verification and vulnerability state interpretation. Recent studies have used prompt engineering, fine-tuning, program structure information, retrieval, and tool-augmented workflows to improve vulnerability recognition and evidence gathering. These approaches are useful for automating security analysis, but their outputs are typically vulnerability labels, categories, scores, or explanatory text. They do not usually treat the operational VEX state of a specific firmware asset as the primary inference target. In contrast, this study uses the LLM as part of an evidence-bound reasoning workflow. The objective is not to detect vulnerable code directly, but to structure firmware-derived SBOM evidence, vulnerability intelligence, and operational context into checklist answers that support or constrain VEX state interpretation.

2.2 SBOM/VEX-based Vulnerability Representation

SBOM and VEX research addresses a related but different problem. SBOMs provide structured visibility into software components and versions, while VEX communicates whether a known vulnerability is applicable, not affected, fixed, or still under investigation for a specific product. However, component visibility alone is insufficient for operational vulnerability assessment. A vulnerable component may exist in firmware, but the vulnerable functionality may be unused, disabled, unreachable, or exposed only under restricted operating conditions. Therefore, a methodological gap remains between SBOM-based component visibility and VEX-based status representation: target-specific evidence must be transformed into an explicit and auditable VEX state.

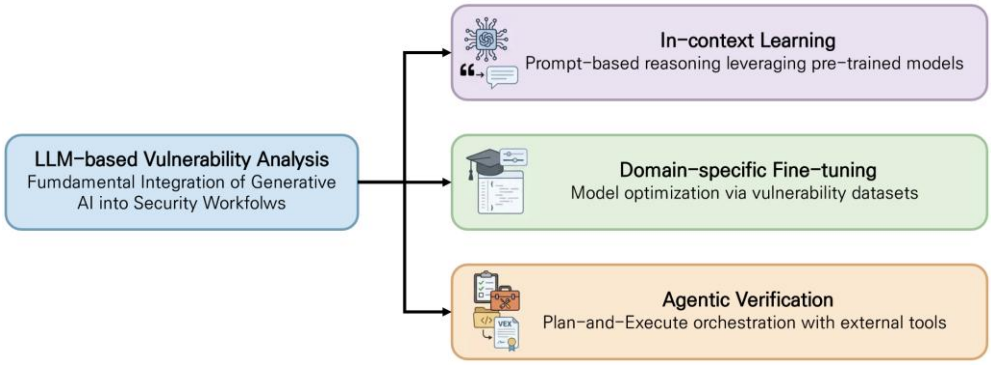

Figure 1: Taxonomy of LLM-Integrated Vulnerability Analysis.

2.3 ICS/OT Contextual Vulnerability Assessment

ICS/OT vulnerability assessment further emphasizes the need for contextual interpretation. In ICS and OT environments, vulnerability relevance depends not only on software presence and affected version ranges, but also on runtime usage, network exposure, service configuration, maintenance or engineering mode, protocol dependency, and possible operational or physical-process impact. As a result, firmware-based ICS/OT vulnerability assessment cannot be reduced to generic CVE matching or package-level vulnerability scanning.

2.4 Research Gap

Figure 1 positions the proposed work at the intersection of LLM-based vulnerability analysis, SBOM/VEX-based vulnerability representation, and ICS/OT contextual assessment. Existing research partially addresses these areas, but the three streams remain insufficiently connected. This study focuses on that intersection by deriving contextual VEX states for firmware-CVE pairs using firmware-derived SBOM evidence, advisory/CTI evidence, and operational context.

Taken together, prior work provides important foundations. LLM-based analysis supports reasoning over vulnerability-related evidence, SBOM/VEX research provides component visibility and status representation, and ICS/OT assessment captures operational context. However, existing approaches either detect vulnerabilities, represent software components, communicate vulnerability status, or assess operational risk; they do not fully specify how firmware-derived SBOM evidence and contextual evidence should be converted into an explicit VEX state for a specific firmware-CVE pair.

This study addresses that gap by formulating firmware-based VEX inference as an evidence-bound applicability assessment problem. The proposed method profiles asset, SBOM, vulnerability, CTI, and

operational evidence; derives checklist items from that evidence; and interprets checklist answers as both component/version-level applicability labels and operational VEX states. This distinguishes the proposed work from code-centric vulnerability detection, generic SBOM-based vulnerability scanning, and VEX document generation that does not explicitly model the evidence-to-state inference process.

3 Evidence-Bound Checklist-Driven VEX Inference Methodology

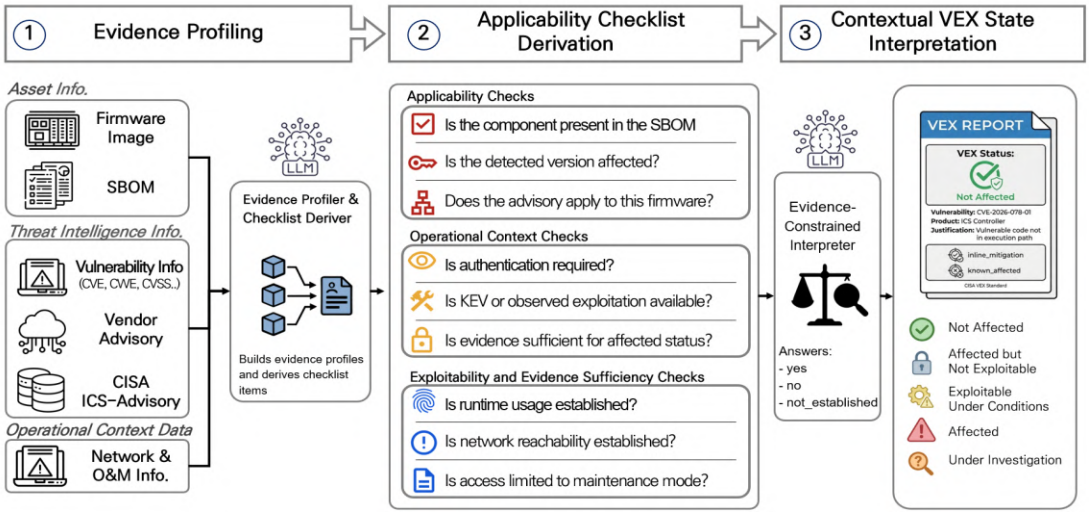

Figure 2: Evidence-Bound Checklist-Driven VEX Inference Methodology

This study proposes a methodology for interpreting VEX states for firmware-CVE samples using firmware-derived SBOM evidence and contextual evidence. The objective is not merely to match SBOM components with CVEs, but to determine under what conditions a vulnerability is applicable to a specific firmware asset and how that vulnerability should be interpreted as a VEX state.

The proposed method consists of three stages. The first stage, Evidence Profiling, organizes firmware asset information, SBOM component evidence, vulnerability intelligence, advisory/CTI evidence, and operational context into an evidence profile for each firmware-CVE sample. The second stage, Applicability Checklist Derivation, derives an applicability checklist from the evidence profile.

The third stage, Contextual VEX State Interpretation, interprets checklist answers to derive both a dataset-level label and an operational VEX state. In this methodology, the checklist is not a generic security questionnaire. Each checklist item is used as a decision-oriented intermediate artifact that either excludes a possible VEX state, supports a possible VEX state, or identifies that additional evidence is required. Therefore, the final VEX state is not produced as a free-form model judgment. Instead, it is derived from evidence-bound checklist answers.

The full prompt templates are provided in Appendix A. In the main methodology section, the focus is not on the prompt wording itself, but on how the evidence profile is structured, how checklist items are derived, and how checklist answers are interpreted as VEX-relevant states.

3.1 Evidence Profiling

The first stage represents each firmware-CVE sample as an evidence profile. This study does not treat a sample as a simple (component, CVE) pair. VEX state interpretation requires more than

component presence and version applicability. It also requires advisory applicability, CTI relevance, runtime usage, and network reachability. The evidence profile consists of five evidence categories.

Algorithm 1: VEX-Oriented Checklist Generation and Final VEX State Determination

Input: Firmware asset A, SBOM S, advisory set V, threat intelligence T, operational context C
Output: Final VEX state y, checklist Q

```
A ← ExtractAssetProfile(F)
P ← ExtractPresentComponents(F)
S ← ∅
for each component p in P do
    V ← QueryCVEIntel(D, p.cpe or (p.name, p.version))
    for each vulnerability v in TopN(V, np) do
        l, e ← MatchComponentVersionToCVE(p, v)
        S ← S ∪ {(A, p, v.id, v.description, e, l)}
    end for
end for
for each absent component k in K \ Names(P) do
    V ← QueryCVEIntel(D, k)
    for each vulnerability v in TopN(V, na) do
        S ← S ∪ {(A, Absent(k), v.id, v.description,
    end for
end for
C ← ∅ ; Y ← ∅
for each sample s in S do
    c ← M(GenerateChecklistPrompt(s,
            criteria={actionability, simplicity, completeness,
                context_relevance, achievability,
                control_loop_impact, air_gap_status, protocol_dependency}))
    if InvalidJSON(c) then c ← DefaultChecklist(s)
    y ← M(DirectVEXPrompt(s, c,
            labels={vulnerable, component_not_present,
                version_not_affected, uncertain}))
    if InvalidJSON(y) then y ← FallbackPrediction(s)
    y.status ← MapLabelToStatus(y.label)
    C ← C ∪ {c} ; Y ← Y ∪ {y}
end for
R ← Evaluate(S.labels, Y.labels, S.statuses, Y.statuses)
return C, Y, R
```

The purpose of the evidence profile is not only to collect heterogeneous evidence, but also to keep the conditions required for VEX interpretation separate. In particular, component presence, version applicability, runtime usage, network reachability, and target-specific CTI are not collapsed into a single vulnerability judgment. This separation prevents component/version-level applicability from being over-interpreted as operational affectedness.

In this structure, `presence = present` and `version_applicability = yes` indicate that the firmware-CVE sample satisfies the component/version-level applicability conditions. However, these fields alone

do not establish the operational VEX state. Operational interpretation requires additional evidence about runtime usage, network reachability, advisory applicability, and target-specific CTI relevance. If these fields are marked as `not_established`, the method preserves them as explicit evidence constraints rather than inferring an `affected` state.

Evidence category	Main fields	Role in VEX inference
Asset	firmware name, firmware version, asset type	Identifies the target firmware asset
SBOM	component name, component version, presence, source	Supports component grounding and version matching
Vulnerability	CVE ID, affected version information	Supports version applicability analysis
CTI	vendor advisory, ICS advisory, KEV, observed exploitation	Supports target-specific relevance analysis
Operational context	runtime usage, network reachability, maintenance mode, exposure	Supports operational VEX state interpretation

Table 1: Evidence categories used for evidence profiling

This design allows the evidence profile to serve two roles. First, it records which applicability conditions are established by the available evidence. Second, it identifies which operational or target-specific conditions remain unavailable, insufficient, or unresolved. As a result, the subsequent checklist derivation stage can generate questions that are directly tied to missing or established evidence, and the final interpretation stage can distinguish a component/version-level vulnerability finding from an operational VEX state.

3.2 Applicability Checklist Derivation

The second stage derives an applicability checklist from the evidence profile. The checklist is not a generic list of security questions. Instead, it is a set of state-discriminative questions designed to separate component-level applicability, operational relevance, exploitability, and evidence sufficiency. Therefore, checklist items are not formulated as broad questions such as “Is this vulnerable?” Rather, each firmware-CVE sample is decomposed into specific conditions such as component presence, version applicability, advisory applicability, runtime usage, reachability, and CTI relevance.

The criteria used in this stage are not metrics. A metric normally implies a quantitative score, calculation, or statistical measure. In this study, the criteria used for checklist derivation are logical item inclusion criteria. They determine whether a question should be included in the checklist and how that question should be linked to the final VEX interpretation.

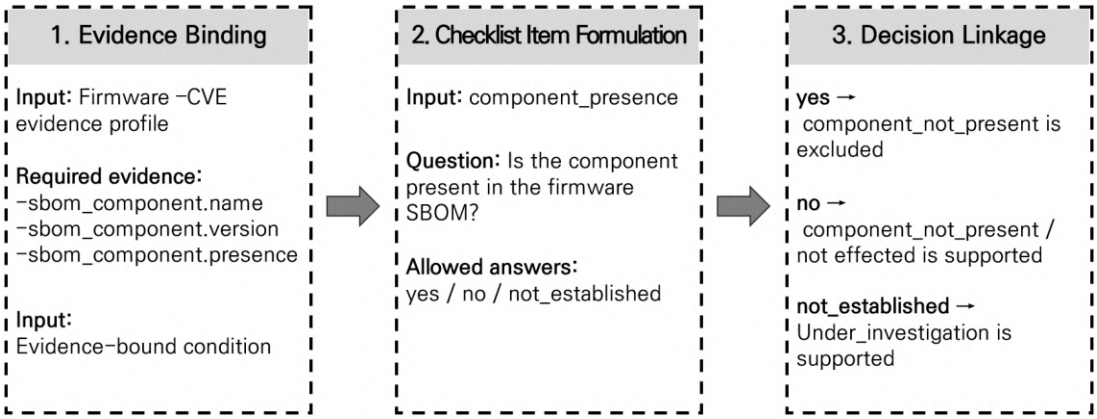

Figure 3: derivation of an evidence-bound checklist item for component presence

Each checklist item must satisfy four inclusion criteria. First, it must be evidence-bound: the question must be linked to a specific field in the evidence profile. Second, it must be state-discriminative:

Checklist category	Representative checks	Decision role
Applicability checks	component presence, version applicability, advisory applicability	Determine whether the component or version is applicable
Operational context checks	runtime usage, network reachability, maintenance / engineering mode	Determine whether the vulnerable functionality is used or reachable in operation
Exploitability and evidence sufficiency checks	authentication condition, KEV / observed exploitation, evidence sufficiency	Support interpretation of affected, exploitable_under_conditions, or under_investigation

Table 2: Checklist categories and decision roles

the answer must exclude, support, or constrain at least one possible VEX interpretation. Third, it must be answer-constrained: the answer must be limited to yes, no, not_established, or partially_available. Fourth, it must be decision-linked: each answer must specify how it affects the dataset-level label or the operational VEX state.

Figure 3 illustrates how a checklist item is derived from the evidence profile. The required SBOM evidence is first bound to the component-presence condition. The condition is then formulated as a checklist item with a constrained answer space. Finally, each possible answer is linked to a VEX-relevant decision effect. For example, a positive answer excludes component_not_present, a negative answer supports component_not_present / not_affected, and a not_established answer prevents a definitive state and supports under_investigation.

In this study, checklist items are organized into three decision categories.

The key point of this stage is to separate component/version-level applicability from operational exploitability. Even if a component is present and its version falls within the affected range, the checklist does not directly elevate the sample to affected. Runtime usage, reachability, and target-specific CTI must be established before the operational VEX state can be interpreted as affected.

For advisory and CTI evidence, the same rule applies. CVE-level or component-level advisory evidence may support vulnerability relevance, but it does not establish target-specific affected status by itself. A vendor advisory, CISA ICS advisory, KEV listing, or observed exploitation evidence must be

explicitly linked to the target firmware or firmware family before it can support an affected operational VEX state.

Checklist condition	Answer	Decision effect
Component presence	yes	Rejects component_not_present and allows version applicability analysis
Component presence	no	Supports component_not_present and not_affected
Version applicability	yes	Rejects version_not_affected and supports component/version-level vulnerable
Version applicability	no	Supports version_not_affected and not_affected
Runtime usage	not_established	Blocks escalation from vulnerable to affected and supports under_investigation
Network reachability	yes	Supports operational exploitability when runtime usage is established
Target-specific dvisory/CTI	not_established	Prevents unsupported affected decision and supports under_investigation

Table 3: Decision effects of checklist answers on VEX state interpretation

3.3 Contextual VEX State Interpretation

The third stage answers the derived checklist items and interprets the answers as both a dataset-level label and an operational VEX state. Each checklist item is answered using only the evidence contained in the evidence profile. Runtime usage, reachability, vendor advisory applicability, KEV evidence, or observed exploitation evidence is not inferred if it is not provided or established.

The answer types used in this study are defined as follows.

Answer	Meaning
yes	The condition is established by the available evidence
no	The condition is rejected by the available evidence
not_established	The target-specific evidence required to establish the condition is unavailable or insufficient
partially_available	General or component-level evidence exists, but target-specific evidence is not established

Table 4: Checklist answer types used for state interpretation

To support interoperability with existing VEX ecosystems, the internal interpretation states are aligned with standard VEX-compatible statuses. The internal states are not intended to replace CSAF or OpenVEX status fields; instead, they provide the evidence-based rationale or condition statement behind the exported status.

Internal interpretation	VEX-compatible status	Export rationale
component_not_present	not_affected	Component is not established in the firmware SBOM
version_not_affected	not_affected	Detected version is outside the affected range
vulnerable + insufficient operational evidence	under_investigation	Runtime, reachability, or target-specific evidence is insufficient

vulnerable + runtime/reachability established	affected	Component, affected version, runtime use, and reachability are established
restricted reachability	affected or under_investigation with condition statement	Final status depends on evidence sufficiency and deployment constraints

Table 5: Mapping between internal interpretation states and VEX-compatible statuses

The answer not_established is not treated as a simple missing value. It is an explicit decision result indicating that the current evidence profile is insufficient to support a specific VEX interpretation. For example, if zlib component presence and version applicability are established, but runtime linkage is not established, the runtime usage condition is marked as not_established. This means that the sample may be vulnerable at the component/version level, but the operational VEX state should not be directly interpreted as affected.

The final interpretation separates the dataset-level label from the operational VEX state. The dataset-level label is based on component presence and version applicability. If the component is absent from the SBOM, the dataset-level label becomes component_not_present. If the component is present but the detected version is outside the affected range, the label becomes version_not_affected. If the component is present and the version is affected, the sample is interpreted as vulnerable at the component/version level.

However, a dataset-level label of vulnerable does not automatically imply an operational VEX state of affected. The operational VEX state additionally depends on runtime usage, network reachability, advisory/CTI relevance, and evidence sufficiency. If the vulnerable component and version are established but runtime usage is not established, the operational state remains under_investigation. Similarly, even if runtime usage is established, the state is not elevated to affected unless reachability is also established.

If runtime usage is established but network reachability is rejected, the state may be interpreted as affected_but_not_exploitable. If reachability is possible only under restricted conditions, such as maintenance mode, engineering access, a specific network location, or an authentication condition, the state is interpreted as exploitable_under_conditions. Only when component presence, version applicability, runtime usage, reachability, and, where necessary, target-specific advisory or CTI evidence are established is the operational VEX state interpreted as affected.

Advisory and CTI evidence are handled under the same principle. The presence of CVE-level information or component-level advisory evidence alone is not sufficient to conclude that the target firmware is affected. Target-specific relevance is established only when vendor advisory, CISA ICS advisory, KEV listing, or observed exploitation evidence is explicitly linked to the target firmware or firmware family. Otherwise, such evidence may support component-level relevance, but it does not justify elevating the operational VEX state to affected.

Through this structure, the proposed method explicitly separates CVE matching from operational VEX state interpretation. Firmware-derived SBOM evidence provides visibility into components and versions, but the VEX state is interpreted by considering whether the component is actually used, whether it is reachable in the firmware's operational context, and whether target-specific advisory or CTI evidence is available.

4 Case Study

4.1 Case Study Input

The case study was conducted on the DH_IPC-HX8XXX-Eos_EngSpn_N_Stream3 V2.420 firmware. The target asset was treated as an OT physical-security firmware asset, and the firmware-derived SBOM was used as the primary input evidence. The SBOM contained vulnerability-relevant components, including OpenSSL 1.0.1j, BusyBox 1.18.4, uClibc 0.9.33, Linux kernel 3.4.35, wpa_supplicant 2.0, and zlib 1.2.6. These components were used as seed components for constructing firmware-CVE pairs.

Each sample was represented not as a simple component-CVE pair, but as an input evidence object containing the minimum information required for VEX-oriented reasoning. Table 6 shows a representative input evidence object for the zlib / CVE-2016-9840 sample. The object separates asset information, SBOM-derived component evidence, vulnerability applicability, and operational context. This separation is necessary because component presence and version applicability alone are insufficient to determine the final VEX state. In this sample, zlib 1.2.6 is present in the firmware-derived SBOM and is marked as version-applicable to the CVE. However, runtime usage, network reachability, and target-specific CTI evidence remain unknown. Therefore, the sample supports a component/version-level vulnerable label, while its operational VEX state cannot be directly elevated to affected without additional runtime and reachability evidence.

Input category	Field	Value
Asset	Firmware	DH_IPC-HX8XXX-os_EngSpn_N_Stream3
	Version	V2.420
	Asset type	OT physical-security firmware
SBOM component	Component name	zlib
	Component version	1.2.6
	Presence	present
	Evidence source	firmware-derived SBOM
Vulnerability	CVE ID	CVE-2016-9840
	Version applicability	yes
Context	Runtime usage	unknown
	Network reachability	unknown
	CTI evidence	Unknown

Table 6: Representative input evidence object for zlib / CVE-2016-9840

A total of 25 pseudo-gold firmware-CVE samples were constructed. For each sample, one checklist and one final prediction were generated. This evaluation scope is limited to a single firmware image and is intended to demonstrate feasibility rather than statistical generalization.

Category	Value
Firmware asset	1
Representative SBOM components	6
Firmware-CVE samples	25
Generated checklists	25
Generated predictions	25

Table 7: Execution scope of the single-firmware case study

The pseudo-gold labels consisted of three categories: vulnerable, component_not_present, and version_not_affected. These labels capture component-level and version-level applicability, not full operational exploitability or complete VEX ground truth. A sample was labeled vulnerable when the relevant component was present in the firmware-derived SBOM and its version matched the affected range of the CVE. It was labeled component_not_present when the CVE-related component was absent from the SBOM, and version_not_affected when the component was present but its detected version was outside the affected range. This label design allowed the case study to evaluate component grounding and version applicability separately from later contextual VEX reasoning, such as runtime usage, reachability, and CTI relevance.

Label	Count	Interpretation
vulnerable	12	Component present and judged vulnerable
component_not_present	8	Component absent from the firmware SBOM
version_not_affected	5	Component present but outside vulnerable range
uncertain	0	Not observed in the current seed dataset

Table 8: Pseudo-gold component/version applicability label distribution

Table 8 clarifies that the seed dataset is not evenly balanced. In particular, the vulnerable class is the largest, while version_not_affected is relatively small. This imbalance becomes relevant later when interpreting the model's tendency to over-predict component_not_present.

4.2 Execution of the Proposed Methodology

For each firmware-CVE sample, the proposed method generated one VEX-oriented checklist and one prediction output. This section presents the checklist result for the representative zlib / CVE-2016-9840 sample and shows how the generated answers were reflected in the final VEX-oriented interpretation.

Figure 4 shows the checklist answer trace for this sample. Q1 and Q2 confirm the component-level applicability of the vulnerability. The firmware-derived SBOM indicates that zlib 1.2.6 is present, and the detected version is treated as applicable to CVE-2016-9840. Therefore, the component_not_present and version_not_affected interpretations are excluded.

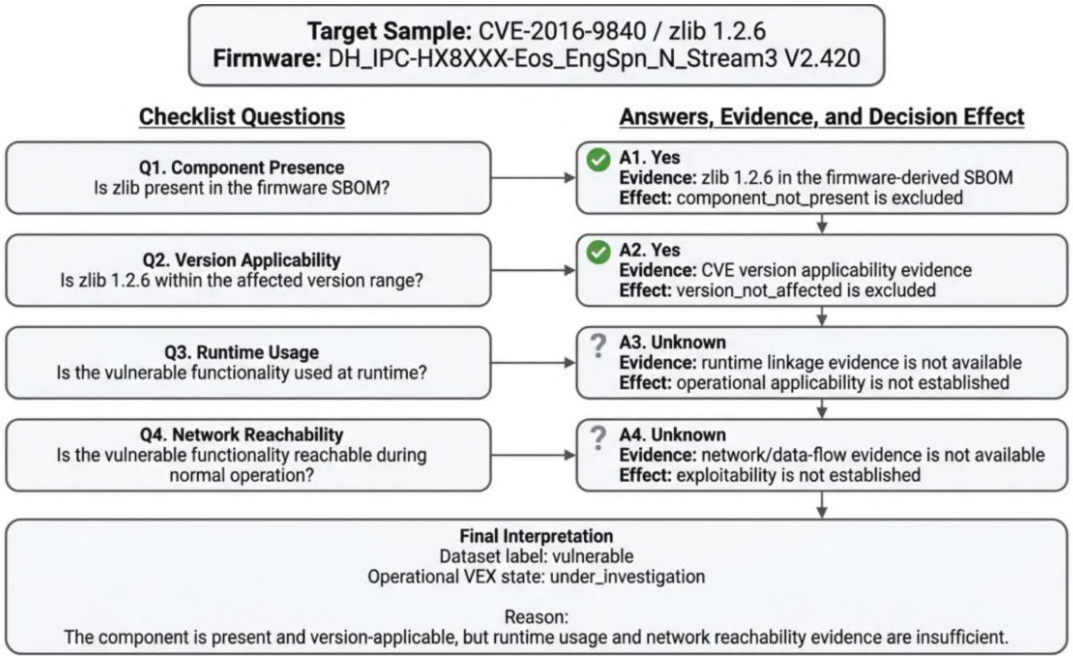

Figure 4: Methodology for Automated VEX Status Determination in ICS Environments

Q3 and Q4 then examine whether this component-level applicability can be elevated to an operationally affected state. Q3 checks whether the vulnerable functionality is actually used at runtime. In this sample, runtime linkage, service dependency, and executable usage evidence are not established. Q4 checks whether the vulnerable functionality is reachable during normal operation. Although the asset is a networked firmware device, the available input does not establish an exposed endpoint, network service configuration, or data-flow path to the vulnerable functionality. Therefore, runtime usage and network reachability are treated as evidence gaps that prevent an affected decision.

Q5 examines advisory and CTI relevance. CVE-level information exists for CVE-2016-9840: NVD describes the vulnerability as improper pointer arithmetic in zlib inftrees.c and assigns a CVSS v3.1 base score of 8.8 HIGH. Component-level advisory evidence is also available from sources such as Debian and GitHub Advisory, and third-party product advisories such as IBM's security bulletin show that the zlib vulnerability affected other product environments. However, these sources do not establish target-specific applicability to the Dahua HX8XXX firmware. In particular, a Dahua HX8XXX-specific advisory, CISA ICS advisory, KEV listing, or observed exploitation evidence for this firmware-CVE pair is not established in the collected evidence.

As a result, the sample is interpreted as vulnerable at the dataset-label level because the component is present and version-applicable. However, its operational VEX state remains under_investigation because runtime usage, network reachability, and target-specific CTI are not established. This result shows that component/version-level vulnerability evidence must be separated from operational exploitability evidence in firmware-based VEX inference.

To make the reasoning process explicit, Table 9 summarizes one representative success trace and one representative failure trace. The failure trace is included because the dominant observed error pattern is not generic vulnerability reasoning failure, but incorrect component grounding or evidence binding.

Case	Checklist result	Decision effect	Final interpretation
Success: zlib / CVE-2016-9840	presence=yes; version=yes; runtime=not_established; reachability=not_established	component_not_present and version_not_affected are rejected; affected escalation is blocked	Dataset label: vulnerable; operational state: under_investigation
Failure: vulnerable sample predicted as component_not_present	component grounding or evidence binding fails	component_not_present is incorrectly supported despite component/version applicability	Error type: component grounding failure; analyst review required for not_affected-relevant outputs

Table 9: Representative success and failure traces of checklist-based VEX interpretation

4.3 Overall Execution Summary and Prediction Distribution

Checklist generation and VEX prediction were performed for all 25 firmware-CVE samples. Each sample produced one checklist output and one final prediction output. Table 10 summarizes the execution results of the case study.

Category	Count	Description
Evaluated firmware-CVE samples	25	Total firmware-CVE pairs processed
Generated checklists	25	One checklist generated per sample
Generated VEX predictions	25	One prediction generated per sample

Table 10: Execution summary of the case study

The generated prediction outputs were compared with the pseudo-gold labels. Table 11 shows the distribution of pseudo-gold labels and predicted labels.

Label	Pseudo-gold count	Predicted count
vulnerable	12	5
component_not_present	8	19
version_not_affected	5	1

Table 11: Pseudo-gold and predicted label distribution

These results show that the proposed pipeline can generate checklist and prediction outputs for each firmware-CVE sample from firmware-derived SBOM evidence. Each sample was converted into an input structure containing component evidence, version applicability, operational context, and advisory/CTI evidence, and a VEX-oriented checklist was generated accordingly. The following discussion interprets these results as a pilot feasibility evaluation and explicitly separates component/version applicability performance from full operational VEX validity.

5 Discussion

5.1 Baseline Result

The overall performance of the proposed method is summarized in a unified comparison chart.

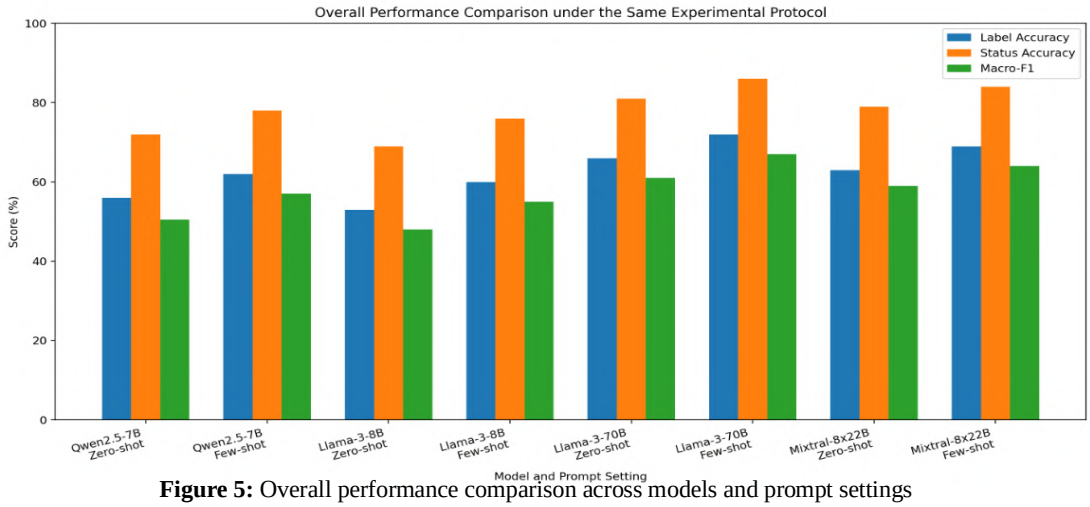

Figure 5: Overall performance comparison across models and prompt settings

Under the measured baseline setting, Qwen2.5-7B-Instruct with zero-shot prompting achieved 56.0% label accuracy, 72.0% status accuracy, and 50.5% macro-F1 on the 25 pseudo-gold firmware-CVE samples. These results indicate that the proposed method is more reliable at the coarser VEX status level than at the fine-grained component/version applicability label level. Because the pseudo-gold labels are based on SBOM presence and affected version range, the label accuracy should be interpreted primarily as a measure of component grounding and version applicability rather than as full operational VEX ground truth.

Figure 5 jointly reports label accuracy, status accuracy, and macro-F1 across model and prompt configurations. The current pilot evaluation does not include a full method-level ablation against direct VEX prompting, evidence-profile-only prompting, and the full checklist-driven method under an identical protocol. This limitation is important because such an ablation is necessary to isolate the specific contribution of checklist-driven reasoning.

5.2 Error Structure Analysis

To move beyond scalar accuracy values, the error structure was analyzed using confusion heatmaps. Figure 6 presents three representative heatmaps aligned in a single row: the measured Qwen2.5-7B zero-shot baseline, an illustrative Llama-3-70B zero-shot comparison, and an illustrative Llama-3-70B few-shot comparison. This 1×3 arrangement makes it easier to observe how the confusion pattern changes across representative settings without fragmenting the comparison into separate subfigures.

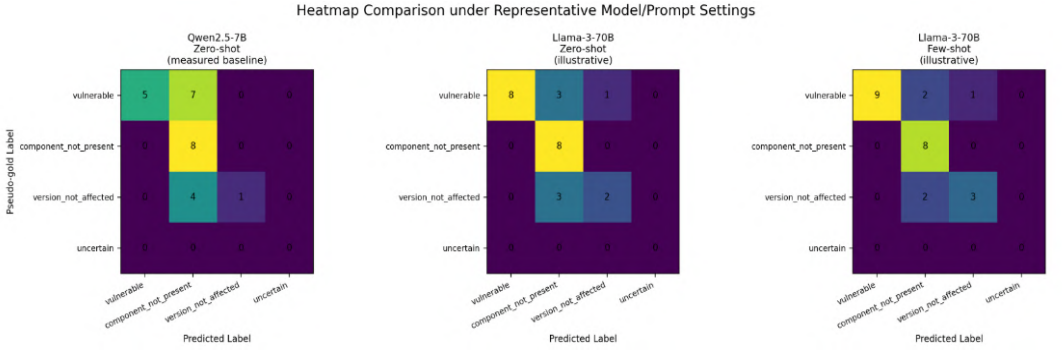

Figure 6: Heatmap comparison across representative model and prompt settings

Figure 6 shows that the dominant failure mode in the measured baseline is the collapse of vulnerable and version_not_affected samples into component_not_present. This confirms that the main bottleneck of the current pipeline is component grounding and evidence binding, rather than high-level vulnerability reasoning. In operational use, this failure mode is safety-relevant because an erroneous component_not_present decision may lead to an unsafe not_affected interpretation if it is not reviewed by a human analyst. The measured Qwen baseline produced 11 mismatches out of 25 samples, which is consistent with the observed label accuracy.

Compared with the baseline heatmap, the illustrative larger-model and few-shot settings show a reduced concentration in the component_not_present column and a stronger recovery of diagonal cells. This suggests that future improvements are expected to come primarily from better component grounding and stronger checklist-conditioned evidence integration.

5.3 Checklist Quality Comparison

The final VEX result depends not only on the classifier, but also on the quality of the generated checklist. For this reason, checklist quality was compared separately under zero-shot and few-shot prompting. The zero-shot setting preserves strong context relevance, but tends to underperform in completeness and simplicity, whereas the few-shot setting improves operational completeness and evidence-grounded achievability. This comparison follows the same logic as prior work, where checklist quality is treated as an intermediate evaluation target rather than being merged directly into the final classification metric.

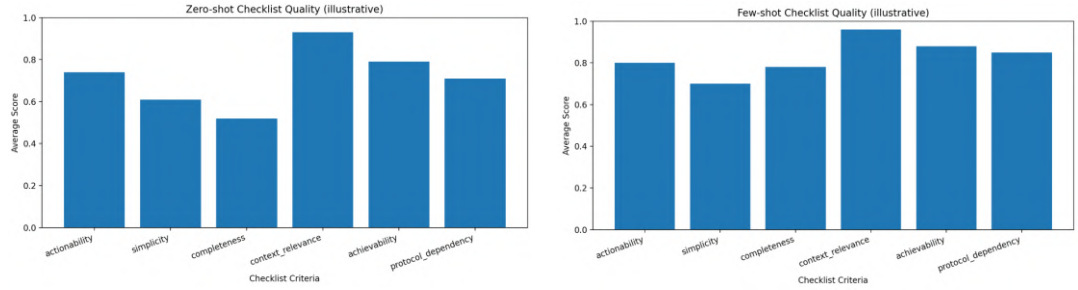

Figure 7: Checklist quality under zero-shot and few-shot prompting: (a) zero-shot, (b) few-shot

Figures 7(a) and 7(b) indicate that few-shot prompting provides the largest gains in completeness, achievability, and protocol dependency. This result is important because the model's final VEX decision quality is constrained by the quality of the checklist it first generates.

5.4 Evaluation Scope and Reproducibility

To improve reproducibility, the implementation details were clarified in the revised manuscript. The measured baseline uses Qwen2.5-7B-Instruct in a zero-shot setting with structured prompts, constrained answer types, and schema-based output checking. Invalid or incomplete outputs are handled by a fallback rule rather than being accepted as valid evidence. The evaluation remains limited to 25 pseudo-gold component/version applicability samples from a single firmware image, and should therefore be interpreted as a pilot feasibility study rather than a statistically generalizable benchmark.

The study also does not include a full comparison against existing SBOM/VEX workflows. In particular, direct VEX prompting, evidence-profile-only prompting, rule-based SBOM-CVE matching, and the full checklist-driven method should be compared under the same experimental protocol in future work. The current results are therefore best understood as evidence that the proposed workflow can be executed end-to-end and that its error structure can be inspected, not as proof of deployment-ready automation.

5.5 Deployment Assumptions and Human-in-the-loop Use

The intended users of the pipeline are OT security analysts, vendor PSIRT teams, and supply-chain auditors who need to triage SBOM-derived vulnerability findings. The firmware image, generated SBOM, advisory/CTI sources, operational context, and LLM-generated outputs are treated as separate evidence sources with different trust assumptions. The pipeline does not treat firmware-derived SBOMs as fully trusted ground truth; instead, SBOM evidence is a derived and potentially fallible input whose uncertainty should be propagated to the final interpretation.

Given the current label accuracy, the proposed method should not be deployed as a fully automated VEX decision engine. It is intended to support human analysts by producing evidence-bound candidate states and by making missing evidence explicit. In OT environments, a false not_affected decision can delay mitigation, leave exposed assets unaddressed, and increase operational or safety risk. Therefore, insufficient evidence should conservatively lead to under_investigation rather than a definitive not_affected state.

5.6 LLM Hallucination and Evidence Controls

To reduce hallucination risk, each checklist answer must be grounded in an explicit field of the evidence profile. If the model introduces unsupported evidence, cannot bind an answer to the provided evidence, or produces an inconsistent label-state mapping, the corresponding condition is treated as not_established and should be reviewed by a human analyst. This design turns uncertainty into an explicit output rather than allowing unsupported evidence to drive an affected or not_affected decision.

5.7 Limitations and Future Work

Several limitations remain. First, the current evaluation uses only one firmware image and 25 pseudo-gold component/version applicability samples. Future work should extend the validation to at least three firmware images across different vendors, architectures, and device families. Second, the current study does not test SBOM extraction errors such as BusyBox mis-detection, statically linked OpenSSL omission, missing static libraries, version ambiguity, or custom fork aliasing. These cases are important robustness targets because incorrect SBOM evidence can directly affect component grounding and not_affected decisions.

Third, stronger operational VEX ground truth is required. The current pseudo-gold labels evaluate component presence and affected version range, while runtime usage, reachability, and target-specific advisory evidence remain mostly unresolved. Future work should incorporate richer runtime and network evidence, stronger VEX ground truth sources, and ablation studies that compare direct prompting, evidence-profile-only prompting, and full checklist-conditioned reasoning.

6 Conclusion

This study presented a firmware-SBOM-based approach for checklist-driven, evidence-bound VEX inference in ICS/OT environments. From the perspective of smart manufacturing and industrial digital transformation, the method connects firmware-derived SBOM visibility with contextual VEX interpretation for connected OT assets that may participate in production, remote maintenance, physical-security monitoring, and supply-chain operations. The proposed method combined pseudo-gold firmware-CVE pair construction, ICS-contextual checklist generation, and checklist-based VEX justification extraction.

The experiment showed that the full pipeline could be executed end-to-end on 25 pseudo-gold component/version applicability samples from a single firmware image. Under the measured Qwen2.5-7B-Instruct zero-shot baseline, the method achieved 56.0% label accuracy, 72.0% status accuracy, and 50.5% macro-F1. These results should be interpreted as a pilot feasibility result rather than a statistically generalizable validation or a deployment-ready automated triage result.

The main limitation observed in the case study was not high-level vulnerability reasoning, but component grounding and evidence binding failure. Most errors occurred when the model incorrectly mapped vulnerable or version_not_affected cases to component_not_present. This indicates that improving evidence representation, component-to-asset alignment, and human review of not_affected-relevant outputs is likely to be more important than simply increasing model size.

Overall, this work provides an initial but practical starting point for checklist-driven VEX inference using firmware-derived SBOMs in ICS/OT settings. The method is intended to support auditable analyst decision-making rather than replace human judgment. Future work should expand the dataset across multiple firmware families, incorporate stronger operational VEX ground truth, evaluate SBOM extraction errors, and compare direct prompting, evidence-profile-only prompting, and full checklist-driven reasoning under the same protocol.

Acknowledgment

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Q-PROVE: HMAC-based Provenance and Validity Verification for QR-Encoded URLs*

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Abstract

A growing number of attacks have shown that Quick Response (QR) codes are being abused as an effective phishing (quishing) channel that bypasses traditional email and web defenses. Existing countermeasures mainly rely on destination-URL classification, cryptographic watermarking, or specialized secure scanners, and therefore either remain vulnerable to evolving infrastructure or require non-standard encoders and decoders. In this paper, a QR-centric integrity and provenance framework is proposed that strengthens trust in QR-encoded URLs at both generation and scan time. Before a QR code is created, the target URL is normalized into a canonical form, augmented with issuer and validity metadata, and protected using a Hash-based Message Authentication Code (HMAC); the resulting authenticated payload is then encoded into a standard QR symbol. When the code is scanned, the client verifies the HMAC and validity interval locally and informs the user about the issuer and trust status before following the URL. This design allows counterfeit, tampered, or expired codes to be rejected without online lookups while remaining compatible with existing QR decoding infrastructure, and it complements detection-based and visual anti-tampering approaches in a layered defense against quishing.

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1 Introduction

Quick Response (QR) codes have been widely deployed as a lightweight interface for payments, authentication, logistics, and public services, building on their high data density and ease of use with commodity smartphones [1]. QR code–based payment services have particularly expanded in both developed and emerging markets, and the global QR code payment market has been projected to grow rapidly over the next decade, indicating that QR-mediated interactions will remain a central channel for digital transactions [2]. As QR codes increasingly serve as the first touchpoint of end-to-end business processes rather than as auxiliary identifiers, the security and reliability of QR-driven workflows have been recognized as a key prerequisite for trustworthy digital ecosystems [3].

In parallel with this diffusion, a line of research has been developed that systematically examines the security and privacy implications of QR code usage across applications and platforms. Comprehensive surveys have been conducted to catalog prominent attack vectors, such as payment fraud, code counterfeiting, information leakage, and reader-application abuse, and to map corresponding response techniques from both the academic and industrial domain. In addition, the usability and deployability of cryptographic protection mechanisms, such as Elliptic Curve Digital Signature Algorithm (ECDSA), Hash-based Message Authentication Codes (HMAC), and Advanced Encryption Standard (AES) embedded into QR codes, have been explored to understand how authenticity and integrity guarantees can be balanced with constraints on code size, print quality, and user experience [4].

Within this broader landscape, QR code phishing (Quishing) has been identified as one of the most pressing emerging threats linked to real-world QR deployments. Field studies and naturalistic experiments have been carried out to observe how users interact with malicious QR codes embedded in plausible scenarios, and to characterize behavioral patterns and contextual factors that increase susceptibility to QR-mediated phishing [5-6]. In parallel, organizational measurements of phishing campaigns have begun to include QR-based lures alongside traditional hyperlink-based emails, highlighting that quishing is being operationalized as part of mainstream social-engineering toolkits rather than remaining an isolated niche [7]. Taken together, these works indicate that the human-factor dimension of QR phishing, ranging from trust in physical signage to habits around mobile scanning, is becoming an active subtopic in usable security and phishing research.

At the system level, attention has also shifted from viewing QR codes solely as carriers of URLs toward analyzing the complete processing pipeline involving in-app scanners, operating-system components, and backend services [3]. Studies of in-app QR code scanning have revealed that decoded content is frequently passed directly to internal handlers, WebView components, or privileged functions, thereby exposing new avenues for remote invocation of sensitive operations when crafted QR payloads are presented to vulnerable applications [8]. [9] have argued that QR security must be considered as a multi-layer design problem, in which physical media, mobile applications, and server-side logic form a combined attack surface rather than independent components. These perspectives suggest that effective defenses against QR phishing cannot be confined to user education or simple URL filtering alone, but must account for the behavior of the surrounding software and infrastructure.

In response to these challenges, several technical directions have been pursued for strengthening QR security, each with distinct assumptions and limitations. [3] and [4] have focused on embedding cryptographic authenticity and integrity into QR codes so that tampering and forgery can be detected by compliant scanners, while [10] has investigated secure scanner designs that integrate policy checks and permission controls into the decoding path. [11] has explored machine-learning-based detection of malicious Uniform Resource Locators (URLs) and related features derived from QR content, demonstrating that real-time classification of benign and malicious codes is feasible on commodity hardware. However, [12] have emphasized that these approaches only partially address the

characteristics of modern quishing campaigns, which often exploit cross-device workflows, rapidly changing infrastructure, and contextual cues that are not fully captured by generic URL-centric models.

Given this evolution of threats and defenses, a need has been recognized for QR phishing detection methods that explicitly target quishing scenarios and their operational context, rather than treating QR codes simply as another URL encoding format. In this work, a QR phishing detection system is proposed that is designed to operate at scan time, to analyze both decoded content and QR-specific contextual features, and to provide early risk signals before authentication, payment, or other security-sensitive actions are completed. By aligning its threat model with the trends identified in prior studies on QR security, user behavior, and in-app vulnerabilities, the proposed system is intended to complement existing cryptographic and scanner-hardening techniques and to address a gap between the sophistication of current quishing attacks and the protection afforded by deployed defenses.

This paper is organized as follows. In Section 2, we present the background, including a literature review and a detailed discussion of QR code generation and usage processes as well as their vulnerabilities. In Section 3, we describe the proposed system for QR code phishing detection. Finally, Section 4 concludes the paper and outlines directions for future work.

2 Background

2.1 Literature Review

In recent years, the threat of QR code phishing, often referred to as quishing, has begun to be treated as a distinct research topic rather than a marginal variant of traditional hyperlink-based phishing [7, 13]. [13] has explicitly classified QR code–mediated attacks as a separate delivery medium and discussed them in the context of multi-channel social-engineering campaigns, emphasizing their ability to bypass conventional email and browser defenses.

At the organizational level, Weinz et al. [7] conducted large-scale phishing simulations across several companies, comparing classic click-through emails, QR code phishing emails, and messages written with the assistance of large language models (LLMs). Their results indicated that quishing emails were at least as effective as traditional phishing emails at driving users to landing pages, while being harder for operational detectors to flag, thus highlighting a structural blind spot in current enterprise defenses.

Building on this threat characterization, multiple technical directions have been explored for detecting and mitigating quishing attacks using machine learning (ML) and artificial intelligence (AI). [11] proposed an ML-based detector that jointly analyzes phishing links and QR codes in Unified Payments Interface (UPI) transactions, demonstrating that lexical features of embedded URLs can be used to flag fraudulent payment requests in real time. In a supplementary study on QR code infrastructure security [15], an AI-based system combining anomaly detection, risk signals, and multiple machine learning classifiers was used to identify QR-based phishing, data leakage, and injection attempts at the infrastructure level.

A different stream of work has examined quishing-specific defenses that move beyond simple URL analysis. [14] introduced self-authenticating dual-modulated QR (SDMQR) codes, which embed cryptographic signatures into visually compatible QR patterns so that scanners can verify whether a code originates from a pre-registered, trusted issuer before following the embedded link. This approach has been positioned as a structural countermeasure against code replacement and tampering in both physical and digital environments, with the explicit goal of quishing attacks by preventing users from being silently redirected to attacker-controlled destinations.

Recent works have also contributed QR-centric verification mechanisms that, while not limited to phishing, can be leveraged to strengthen defenses against quishing. A deep learning–based verification

scheme [16] proposed to watermark QR codes with hidden security features and to use Siamese neural networks to distinguish genuine codes from forgeries, achieving high accuracy in detecting copied or tampered codes. Further, [17] presented robust QR code extraction and verification techniques tailored to noisy, real-world environments, improving the reliability of downstream authenticity checks.

Collectively, these studies indicate that current research on quishing and QR-centric attacks is converging on three main directions: empirical measurement of attack effectiveness, cryptographic and protocol-level hardening of QR codes, and ML-based detection frameworks that treat the QR symbol and its context as first-class security objects.

2.2 Overview of QR Code

Quick Response (QR) codes are two-dimensional matrix barcodes that encode information in an arrangement of black and white modules, typically within a square symbol that can be captured by commodity camera sensors and decoded in real time. In contrast to one-dimensional barcodes that represent data along a single axis, QR codes support multiple standardized encoding modes—such as numeric, alphanumeric, and byte data—and incorporate Reed–Solomon–based error correction, which allows partial recovery of the payload even when portions of the symbol are occluded or degraded [15-16]. Typical QR layouts include finder patterns in three corners, alignment patterns for higher versions, timing patterns, and format and version information fields, all of which are used by decoders to locate the code, correct geometric distortions, and determine the encoding parameters.

From a systems perspective, QR codes function as a compact carrier of application-specific tokens that are interpreted by the scanning device and associated software. In industrial and robotics scenarios, for example, each QR symbol may encode global coordinates and identifiers that are used as artificial landmarks to correct localization error during indoor navigation [18]. In authentication and access-control contexts, QR codes are used to encapsulate credential data, one-time tokens, or references to server-side state, and [13] have discussed the integration of QR-encoded secrets with biometric authentication and cryptographic protection mechanisms. Across these domains, the decoding process generally consists of image acquisition, binarization, symbol detection, geometric normalization, Reed–Solomon decoding, and application-specific parsing of the recovered payload [16].

This generic pipeline has motivated a growing body of work on robust QR extraction and verification under real-world conditions. Deep learning–based methods [16] have been proposed to improve the detection and segmentation of QR regions in images with complex backgrounds, variable lighting, and physical distortions, thereby increasing the reliability of downstream decoding. [15] and [16] have shown that convolutional neural networks and lightweight architectures such as ShuffleNet can be trained to discriminate authentic codes from sophisticated counterfeits, even when the forgeries are printed and re-captured, which is critical for high-stakes applications such as product traceability and ticketing. As a result, modern QR ecosystems are increasingly characterized not only by generic encoding and decoding standards, but also by application-specific verification layers that seek to ensure that a decoded symbol is both syntactically valid and semantically trustworthy.

2.3 Vulnerability of QR Code

Beyond their functional advantages, QR codes exhibit several inherent vulnerabilities that arise from the opacity of their content and their integration into end-to-end service workflows. QR codes span the physical media on which codes are printed, the scanner and application that interpret them, and the backend systems that process the resulting inputs. A key observation is that users can only visually inspect a pattern of modules, while the actual target of the service is the decoded data, which may trigger browser navigation, in-app screen transitions, token transfers, or transaction requests within a very short

time frame after scanning. Consequently, QR-related vulnerabilities must be understood as a multi-stage problem involving generation, distribution, scanning context, and subsequent processing.

The Vulnerability of QR codes can be categorized into four main categories: social-engineering-based inducement threats, application and reader abuse threats, physical tampering and cloning threats, and input-abuse threats. Social-engineering-based inducement threats are characterized by attackers crafting emails, notices, or posters that resemble legitimate procedures—such as account reactivation, payment confirmation, or discount offers—in order to persuade users to scan embedded QR codes without suspicion; the primary consequences include redirection to phishing sites and credential theft [7]. Application and reader abuse threats exploit the fact that many mobile applications integrate in-app scanners that automatically connect decoded content to internal functions, so that a malicious QR code can effectively serve as a remote input to invoke sensitive operations, leak tokens, or access internal data if parsing and handler logic are not carefully constrained [8].

Physical tampering and cloning threats arise from the dual nature of QR codes as both digital identifiers and physical artifacts attached to surfaces in the environment [5]. In this category, attackers may overlay malicious stickers on top of legitimate codes, transplant copied codes to different locations, or reprint altered codes to change payment destinations or service endpoints, thereby manipulating the object that users believe they are scanning. Because the attack targets the physical carrier rather than the decoding algorithm, purely URL-based checks are insufficient to detect such substitutions. Finally, input-abuse threats occur when decoded QR payloads are forwarded directly into backend systems—such as inventory management, access control, logistics, or control interfaces—without adequate validation; in these scenarios, QR data effectively becomes untrusted external input that can contribute to Structured Query Language (SQL) injection, command injection, script execution, or unintended control-flow branching.

3 Proposed System

3.1 Preliminary

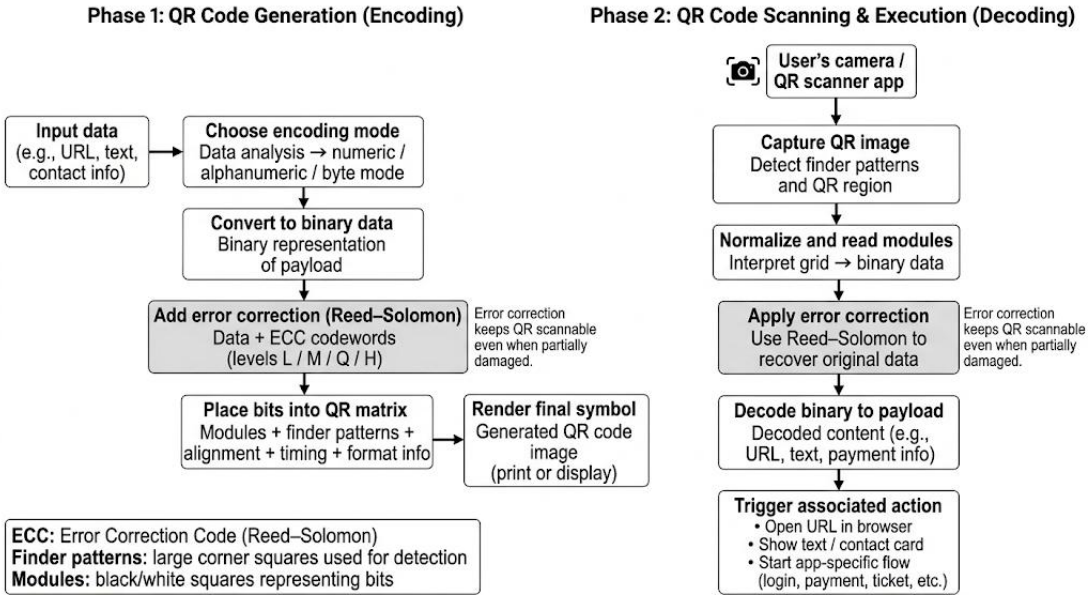

Figure 1. QR Code Generation and Scanning Procedure

In typical deployments, Quick Response (QR) codes are generated by taking an application-specific payload, most commonly a Uniform Resource Locator (URL), and encoding it into the two-dimensional matrix symbol defined by the QR standard. During generation, the payload is first converted into a sequence of bits according to one of the supported encoding modes (numeric, alphanumeric, byte, or kanji), after which Reed–Solomon error-correction codewords are added and the resulting bitstream is mapped onto the module grid while reserving positions for finder patterns, timing patterns, alignment patterns, and format and version information. The final image is printed on physical media or rendered on a display and is treated as an opaque token by users, who cannot visually infer the embedded payload or its security properties.

At scan time, the QR image is captured by a camera and processed by a decoding library that performs binarization, symbol localization, geometric normalization, and Reed–Solomon decoding to reconstruct the original payload bits. The decoded string is then passed to the surrounding application logic, which interprets it as a URL, a deep link, a payment request, or another type of instruction, and triggers follow-up actions such as opening a browser, switching to an in-app page, transmitting authentication tokens, or initiating an order or payment workflow. Because these follow-up actions are usually executed within a short time window and may involve privileged operations, the security of the overall QR interaction depends not only on the correctness of symbol decoding but also on the integrity, authenticity, and freshness of the underlying payload and on the way the application handles it.

3.2 Proposed System

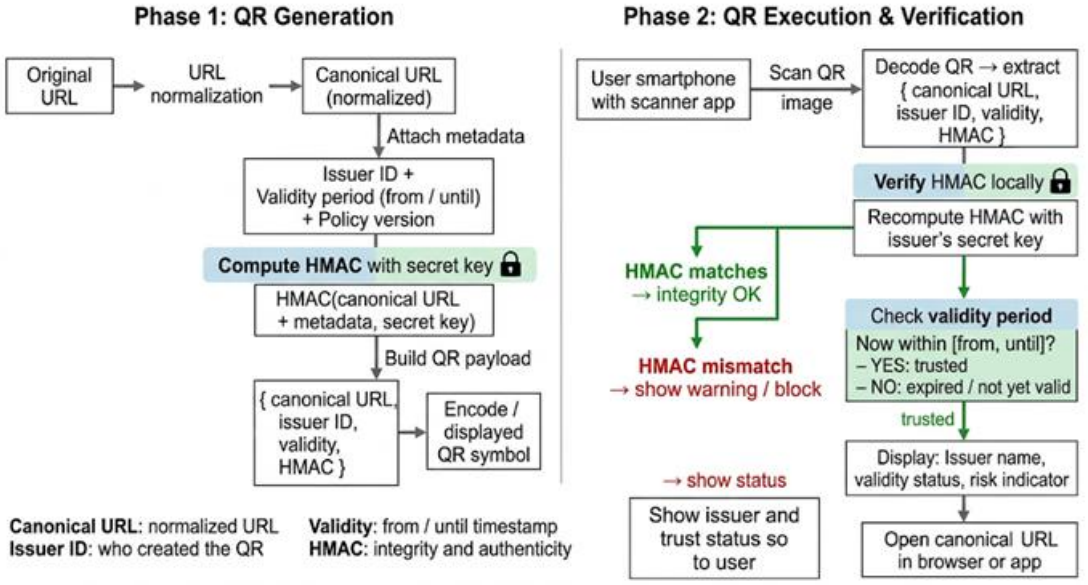

Figure 2. Proposed System

In the proposed system, the focus is placed on strengthening the integrity and provenance guarantees of QR-encoded URLs by introducing a normalization and keyed-authentication layer prior to QR generation. Before a QR code is created, the input URL is first normalized into a canonical representation that removes superficial variability, such as equivalent schemes, redundant path segments, or re-ordered query parameters, so that the same logical resource is always mapped to a unique, stable string. This canonical URL is then combined with issuer metadata (for example, an issuer identifier and policy version) and validity constraints (such as an expiration timestamp or validity interval) to form a structured payload that captures both the target resource and its contextual security attributes.

Over this structured payload, a Hash-based Message Authentication Code (HMAC) is computed using a secret key held by the trusted issuer or issuing backend, and the resulting authentication tag is bundled together with the canonical URL and metadata as the data field that is ultimately encoded into the QR symbol. As a result, each generated QR code contains not only the destination URL but also an integrity-protected record of who created the code and until when the code is to be considered trustworthy, all bound together under a single HMAC value that cannot be forged without access to the issuer's key. The overall generation process therefore produces QR codes that are self-describing with respect to issuer identity and validity period, while remaining compatible with standard QR decoding libraries.

When such a QR code is scanned, the client application does not immediately follow the embedded URL. Instead, the encoded payload is parsed into its constituent fields, canonical URL, issuer identifier, validity information, and HMAC tag, and a local verification step is performed. Using the pre-configured secret key (or a key derived from an issuer-specific master key), the client recomputes the HMAC over the received canonical URL and metadata and compares the result with the embedded tag; if the values do not match, the code is treated as tampered or counterfeit and the user is warned or the interaction is blocked. In parallel, the validity interval is checked against the current time, and codes

whose validity window has expired or is not yet active are rejected or displayed as untrusted, thereby limiting the impact of replay and long-term reuse of leaked codes.

Only when both integrity and freshness checks succeed is the canonical URL handed over to the browser or application component for normal execution, and the user interface can additionally display human-readable information about the issuer (for example, the service name or organization) and the validity status derived from the authenticated metadata. This design allows users and relying applications to distinguish between QR codes legitimately issued by registered services and visually similar codes that have been copied, altered, or created by attackers, even when the latter embed syntactically valid URLs. queries.

4 Conclusion

This paper has addressed the problem of QR code phishing by proposing a QR-centric integrity and provenance framework that operates at the point of QR code generation and scanning. The overall threat landscape and recent research on quishing and QR code security were first reviewed, highlighting that existing approaches are largely organized around detection-based filters, cryptographic integrity mechanisms, and physical or visual anti-tampering schemes. Building on this analysis, a system architecture was introduced in which URLs are normalized into canonical form, augmented with issuer and validity metadata, and protected by a Hash-based Message Authentication Code (HMAC) before being encoded into QR symbols. This design enables scanning clients to verify the integrity, issuer identity, and temporal validity of a QR code locally prior to following the embedded URL, thereby strengthening the trustworthiness of QR-mediated interactions without requiring changes to the visual appearance of codes or online reputation lookups.

Compared with prior work, the proposed system offers several advantages in terms of deployment model and security properties. Detection-oriented solutions based on machine learning classifiers or URL analysis can provide high accuracy at scan time, but they typically treat QR codes as opaque carriers and focus on labeling destinations as benign or malicious, making them sensitive to concept drift and zero-day infrastructure. Cryptographic and integrity-preserving schemes, including visual watermarking and self-authenticating QR variants, embed stronger authenticity guarantees into the symbol itself but often require customized encoders and decoders or impose non-trivial constraints on code density and visual design. In contrast, the proposed HMAC-based framework is designed to be agnostic to the visual layout of the code and to rely only on standard QR encoding and decoding capabilities, while still providing verifiable information about who issued the code and whether its trust window is active. By normalizing URLs before protection and binding issuer and validity metadata into a single authenticated payload, the system also mitigates risks arising from semantically equivalent but syntactically different URLs and from long-term reuse of leaked or cloned codes.

There remain, however, several directions in which the proposed approach can be extended and refined. One promising avenue is the integration of the HMAC-based trust layer with existing detection mechanisms, for example by using issuer and validity information as features in machine-learning-based quishing classifiers or as policy inputs in enterprise access-control systems. Another important direction is the exploration of key-management and delegation models that allow multiple organizations or services to participate in a shared QR trust infrastructure while preserving scalability and revocation capabilities. In addition, user-centric studies are needed to evaluate how issuer and validity indicators should be presented at scan time so that users can make informed decisions without incurring excessive cognitive load, particularly in contexts where benign and malicious codes coexist in the same physical environment. Future work will therefore focus on implementing and empirically evaluating the proposed system in realistic deployment scenarios, on formally analyzing its

security guarantees under various adversarial models, and on developing usability guidelines for integrating QR provenance and validity signals into mainstream scanning applications.

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What Do NFMs See? Analyzing Attack Classification in Encrypted Traffic*

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Abstract

Conventional AI-based intrusion detection systems (AI-IDS) have been criticized for their vulnerability to shortcut learning due to reliance on human-defined features. Network Foundation Models (NFMs) offer the potential to overcome this limitation by directly processing packet information within flows in a structured manner. However, high performance alone cannot reveal what information within flows a model actually relies on for classification. In this paper, we classify 6 attack types in encrypted traffic using two NFMs that differ in how they represent traffic—netFound and NetMamba—and analyze what information each model relies on through packet/burst order shuffle experiments. netFound processes up to 72 packets organized in burst units, using tokenized headers along with burst-level metadata (bytes, IAT, flow duration, etc.) as input, whereas NetMamba uses only the header (80 bytes) and payload (240 bytes) of the first 5 packets. Both models achieved over 99% accuracy; however, the shuffle experiments revealed entirely different dependency structures. netFound showed no performance change even when all orders were shuffled, indicating reliance on flow-level metadata such as bytes per burst, inter-burst IAT, and flow duration. In contrast, NetMamba's accuracy dropped sharply to 52.79 ± 1.20% when packet order was shuffled, demonstrating strong dependence on packet sequence order. Notably, NetMamba exhibited the largest performance degradation for full-flow attacks (HTTP_CC, SQLi, Slowloris)—attacks distinguished by characteristics such as transmission intervals and data sizes that appear after the first 5 packets—yet NetMamba cannot directly observe these characteristics and was instead relying on the order information among the first 5 packets. This study highlights the importance of verifying not only the performance of NFMs in attack classification, but also what information the models rely on for classification.

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1 Introduction

Conventional AI-based intrusion detection systems (AI-IDS) rely on a supervised learning structure using human-defined features and labels. There is no consensus on feature engineering, performance is inconsistent across datasets, and it remains uncertain whether high performance reflects genuine learning of intrinsic attack characteristics. Jacobs et al. [1] experimentally demonstrated that the high performance of existing AI-IDS was in fact dependent on incidental data characteristics (byte alignment differences, IP TTL values, TCP flag patterns, etc.) rather than intrinsic attack characteristics. This reveals that existing AI-IDS are vulnerable to shortcut learning, where models become underspecified and rely on incidental data characteristics rather than the intrinsic characteristics of attacks.

In an environment where encrypted traffic continues to increase, AI-based attack detection must assume encryption and be able to verify what traces attack-type-specific characteristics leave in packets and flows. NFMs offer potential in this direction. They can pre-train general-purpose traffic representations through self-supervised learning on large volumes of unlabeled traffic, then be applied to attack detection through fine-tuning with small amounts of labeled data, preserving flow structure as much as possible by processing packet headers and payloads within flows in a structured manner. However, NFMs also cannot reveal what information within flows the model relies on for detection based on high performance alone, and the insight that "models cannot be trusted based on performance alone" applies equally to NFMs.

In this paper, we classify 6 attack types in encrypted traffic using two NFMs with different traffic representation methods—netFound [2] and NetMamba [3]—and analyze what information each model relies on through packet/burst order shuffle experiments. The contributions of this study are as follows:

- First, we performed encrypted traffic attack type classification using two NFMs with different traffic representation methods, selecting and analyzing 6 attack types based on where attack characteristics are revealed within flows (early-flow vs. full-flow).
- Second, through packet/burst order shuffle experiments, we confirmed that behind the identical 99% classification performance, the two models rely on entirely different information. netFound maintained 99% performance even when all orders were shuffled, relying on flow-level metadata (bytes per burst, IAT, etc.), whereas NetMamba's accuracy dropped sharply to 52.79 ± 1.20% when packet order was shuffled, demonstrating strong dependence on packet order.
- Third, for NetMamba, the intrinsic characteristics of full-flow attacks (HTTP_CC, SQLi, Slowloris)—such as transmission intervals and data sizes—appear after the first 5 packets, meaning NetMamba cannot learn these characteristics and was instead relying on packet order information. This highlights the importance of verifying not only performance but also what information models rely on for classification in NFM-based attack detection.

2 Background: Network Foundation Model (NFM)

2.1 Beyond AI-IDS: The Move Toward NFM

Conventional AI-based intrusion detection systems (AI-IDS) rely on a supervised learning pipeline in which human-defined features and labels are used to classify network traffic. This approach is inherently limited by the scope of features that humans can design, and the process of compressing flows into statistical features causes the loss of sequential structure within flows.

Model Property	NetMamba (2024)	netFound (2023)
Architecture	Mamba	Transformer
Complexity	$O(n)$	$O(n^2)$
Max Packets	5	72
Packet header	80 bytes	24 bytes
Packet payload	240 bytes	12 bytes
Metadata	-	Flow-level: flow_duration, protocol; Burst-level: directions, bytes, counts, iats

Table 1: Comparison of Input Processing Between netFound and NetMamba

Consequently, these models suffer from poor generalization when applied to different datasets, which is directly linked to the shortcut learning problem discussed in Section 1 [1].

NFMs represent a paradigm shift from this approach. By leveraging self-supervised learning on large volumes of unlabeled traffic data, NFMs learn general-purpose representations of network traffic, which can then be fine-tuned with small amounts of labeled data for various downstream tasks such as application identification, attack detection, and traffic classification. The key difference from conventional AI-IDS is that NFMs preserve packet-level information and the sequential structure of flows by processing the header and payload data of packets within a flow in a structured manner. However, the way each model represents traffic differs, and this is the core subject of analysis in this study.

We first define key terms commonly used across NFM research. A flow is a set of bidirectional packets sharing the same 5-tuple (source/destination IP, source/destination port, and protocol), representing a single communication session. A burst is a group of consecutive packets within a flow that share the same direction. For example, consecutive packets from client to server form a single burst. The specific criteria for burst segmentation may vary across models (e.g., netFound groups consecutive packets within 10ms into a single burst). These two concepts serve as the fundamental units by which NFMs structure and process traffic, and this paper adopts the same definitions.

The learning scope of NFMs has progressively expanded. PERT [4] learned byte-level patterns from a fixed range of encrypted payload within individual packets. ET-BERT [5] also used a fixed range of encrypted payload but introduced the burst concept to capture bidirectional data exchange patterns within flows. However, both models rely solely on encrypted payload and do not consider packet header information—such as TCP flags, ports, and window size—where traces of attacks may be present. The subsequently proposed netFound [2] and NetMamba [3] explicitly incorporate packet headers, adopting traffic representation methods suitable for attack detection as a downstream task. In this study, we select these two models as our analysis targets, and their specific input processing methods are detailed in Section 2.2.

Attempts to diagnose whether a model genuinely learns the intrinsic characteristics of attacks have also been made in prior AI-IDS research. TRUSTEE [1] diagnosed shortcut learning, in which a model relies on incidental data characteristics rather than the intrinsic nature of attacks. In the NFM domain, a recent study [6] analyzed the latent representations of pre-trained NFMs through Embedding Geometry Analysis to assess representation space utilization, Metric Alignment Assessment to measure correspondence with domain-expert features, and Causal Sensitivity Testing to evaluate robustness to protocol perturbations. In contrast, this study focuses on the attack detection task at the fine-tuning stage. That is, unlike prior AI-IDS diagnosis (TRUSTEE) or the analysis of pre-trained NFM representations, we directly verify whether NFMs actually learn and utilize per-attack-type traces for classification during fine-tuning. To this end, we employ a simple diagnostic based on packet/burst order shuffling.

Algorithm 1: netFound Preprocessing (pcap → Arrow record)

Input: pcap file of a single flow (packets sharing the same 5-tuple)
Output: flow record R

```
group packets into bursts                    # consecutive same-direction
packets = 1 burst
for each burst b_i:
    for each packet in b_i:
      h ← 24 header bytes used in tokenization
          # IPv4: {HeaderLen, ToS, TotalLen, Flags, TTL}
          # TCP: {Flags, WinSize, SeqNum, AckNum, UrgentPtr}
          # UDP: {Length} / ICMP: {Type, Code}
        p ← first 12 bytes of payload
        packet_tokens ← group concat(h, p) into 2-byte units   # 36 bytes → 18 tokens
    burst_tokens[i] ← list of packet_tokens in b_i             # all packets of burst i
    directions[i], bytes[i], counts[i], iats[i] ← burst-level metadata
flow_duration ← last_packet_timestamp – first_packet_timestamp
protocol ← L4 protocol
R ← {burst_tokens, directions, bytes, counts, iats, flow_duration, protocol, label}
return R
```

2.2 netFound and NetMamba

In this study, we use two NFMs that differ fundamentally in their architecture and input processing methods.

netFound [2] is a Transformer-based NFM that processes flow as bidirectional sessions. It accommodates up to 72 packets per flow, and for each packet, 24 bytes of header fields (the IPv4, TCP, UDP, and ICMP header fields defined in [2]; see Algorithm 1) and 12 bytes of payload are grouped into 2-byte units, yielding 18 tokens per packet. A distinctive feature of netFound is that it explicitly includes rich flow-level metadata as input, beyond packet-level information. It provides the model with each burst's direction (directions), total bytes per burst (bytes), total packet count per burst (counts), inter-burst arrival time intervals (iats), as well as flow-level total duration (flow_duration) and protocol (protocol). This enables the model to directly utilize the metadata of the flow. The preprocessing procedure for converting the raw pcap into the input format (an Arrow record) is summarized in Algorithm 1.

NetMamba [3] is an NFM based on the Mamba architecture, a State Space Model, with O(n) computational complexity that can efficiently process long sequences. Using only the first 5 packets of a flow, and unlike netFound which selectively extracts protocol header fields, NetMamba tokenizes the raw bytes of a fixed length from a fixed offset without considering header fields. Each packet's 320 bytes (80 header + 240 payload) are arranged into an 8×40 row block, and the first 5 packets are stacked vertically to represent a single flow as a 40×40 matrix (a grayscale image). Unlike netFound, it does not provide burst-level metadata or flow-level information separately. The only information available to the model is the header and payload of the first 5 packets of the flow. The preprocessing procedure for converting the raw pcap into the input format (a 40×40 matrix, i.e., a grayscale image) is summarized in Algorithm 2.

The input processing methods and differences between the two models are summarized in Table 1. Even when receiving the same flow as input, netFound utilizes packet, burst, and flow-level

Algorithm 2: NetMamba Preprocessing (pcap → PNG image)

Input: pcap file of a single flow
Output: flow image $I \in \mathbb{Z}^{(40 \times 40)}$

```
P ← first 5 packets of the flow        # zero-pad if fewer than 5
for j = 0 to 4:
    pkt ← P[j]                         # j-th packet
    strip frame (Ethernet) header from pkt
    h ← 80 raw bytes from start of IP header     # cut if longer, zero-pad if shorter
    p ← 240 raw bytes from start of payload      # cut if longer, zero-pad if shorter
    I[8j:8j+8] ← reshape(concat(h, p), (8, 40))  # 320 bytes → 8 rows × 40 cols
return I
```

information collectively, whereas NetMamba uses only the first 5 packets of the flow. This difference in traffic representation affects which information each model relies on when classifying attacks, and this is concretely revealed through the shuffle experiments in Section 4.

3 Dataset & Attack Characteristics

3.1 Selection of Six Attack Types

In this study, we used a non-public dataset constructed by KISTI and KAIST. This dataset comprises 13 attack types consisting of encrypted traffic, where all attacks were conducted between the same attacker IP and target IP across various ports, allowing the characteristics of each attack type to be clearly distinguished on a per-flow basis under controlled conditions. Raw pcap files are provided, enabling bidirectional packets sharing the same 5-tuple to be grouped into individual flows, which makes flow-level model training suitable for NFMs. Datasets that simultaneously provide encrypted traffic, raw pcap files, and diverse attack types are scarce, and this dataset is well-suited for the purpose of this study in meeting these conditions.

From the 13 attack types, we selected 6. The selection criterion was the position within the flow where attack indicators are revealed.

- **Three attack types with indicators present in early flow packets (within the first 5 packets):**
  - ✓ **SYN Flood**: A DoS attack that exhausts server connection resources by sending a large volume of packets with the SYN flag set. TCP flags such as SYN and RST themselves serve as direct attack indicators.
  - ✓ **PortScan**: A reconnaissance attack that probes for open ports on the target. Flows consist of 2 packets (FIN → RST+ACK), with the abnormal TCP FIN flag serving as an indicator.
  - ✓ **BruteSSH**: A brute-force password attack against SSH accounts. Immediately after the TCP 3-way handshake, the SSH protocol banner (SSH-2.0) is exchanged in plaintext, making it distinguishable from other attacks within the first 5 packets.

Attack Type	Protocol	Flow Packet Exchange Pattern (first 5 packets)	Remaining (median-5)
SYNFlood	TCP	(A→T) SYN, (T →A) SYN+ACK, (A→T) FIN	-
Port Scan	TCP	(A→T) FIN, (T →A) RST+ACK	-
BruteSSH	TCP +SSH	(A→T) SYN, (T →A) SYN+ACK, (A→T) ACK, (A→T) Client Protocol [SSH-2.0], (T →A) ACK	~1pkts
Slowloris	TCP +TLS1.3	(A→T) SYN, (T →A) SYN+ACK, (A→T) ACK, (A→T) Client Hello [TLS1.3], (T →A) ACK	~13pkts
SQLi	TCP +TLS1.3	(A→T) SYN, (T →A) SYN+ACK, (A→T) ACK, (A→T) Client Hello [TLS1.3], (T →A) ACK	~7pkts
HTTP CC	TCP +TLS1.3	(A→T) SYN, (T →A) SYN+ACK, (A→T) ACK, (A→T) Client Hello [TLS1.3], (T →A) ACK	~11pkts

Table 2: Flow Scenario per Attack Type (A: Attacker, T: Target)

- **Three attack types with indicators revealed across the entire flow:**

  The following three attacks share nearly identical first 5 packets (SYN → SYN+ACK → ACK → Client Hello → ACK), and their differences are revealed through flow-level metadata.

  - ✓ **Slowloris:** A DoS attack that occupies server connections for extended periods by intentionally sending incomplete HTTP request headers at slow intervals. Characterized by long IATs and small bytes per burst.
  - ✓ **SQLi**: An attack that injects malicious SQL statements to exfiltrate database information. Characterized by exceptionally long inter-burst IATs and large bytes per burst.
  - ✓ **HTTP_CC**: A DDoS attack that exhausts server resources by rapidly sending large volumes of HTTP requests that resemble legitimate traffic. Characterized by very short inter-burst IATs.

For this experiment, 12,000 flows per attack type were used, totaling 72,000 flows. Table 2 summarizes the first 5 packet exchange patterns for each attack, showing that the three early-indicator attacks reveal attack evidence within the packets themselves, while the three full-flow attacks have nearly identical initial packets. Furthermore, packet sizes and TCP flags are also similar, making it difficult to distinguish the three full-flow attacks from their initial packets alone.

Table 3 presents the flow metadata statistics per attack type. Although the three full-flow attacks share identical initial packets, they exhibit clear differences in flow-level metadata. SQLi has the largest inter-burst IAT (mean 23.20s) and the largest bytes per burst (mean 1,402). Slowloris has the longest flow duration (mean 9.40s) and the smallest median bytes per burst (median 104). HTTP_CC has the smallest median inter-burst IAT (median 0.02s), reflecting its characteristic of rapidly sending repeated requests. These metadata differences are directly related to the information that netFound uses as input (bytes, iats, flow_duration, etc.), and serve as key evidence for interpreting the shuffle experiment results that follow.

Stat	SYN Flood	Port Scan	Brute SSH	Slowloris	SQLi	HTTP CC
Packets per flow						
mean	3	2	7.4	18.3	12.4	15.9
median	3	2	6	18	12	16
max	5	3	23	19	18	16
Flow duration						
mean	0.001	0.001	0.65	**9.40**	2.27	1.50
median	0.001	0.001	0.002	5.00	0.01	0.03
max	7.80	0.001	58.78	54.84	58.10	55.19
Inter-burst IAT						
mean	-	-	1.65	3.30	**23.20**	1.42
median	-	-	1.87	0.70	20.37	**0.02**
max	-	-	16.97	54.84	58.09	55.16
Bytes per burst						
mean	319	40	384	487	**1402**	968
median	44	40	152	**104**	1283	841
max	1104	80	1776	1906	2161	2161
Total bytes per flow						
mean	638	80	1007	3713	3077	3980
median	640	80	340	3714	2914	4005
max	1148	124	4141	3819	4409	4112

Table 3: Flow Metadata Statistics per Attack Type - mean / median / max

3.2 Experimental Setup

Classification Experiment. Fine-tuning experiments were conducted to evaluate the attack type classification performance of the two NFMs on encrypted traffic. The experiments were performed as multi-class classification across 6 attack types. Normal traffic was not included, as the purpose of this study is not to determine whether traffic is an attack or not (binary classification of normal/attack), but to analyze what information within flows NFMs rely on when classifying attack types. By focusing on attack type classification, we can more clearly observe what characteristics of each attack the models use to distinguish between them.

A total of 72,000 flows were used, with 12,000 flows per attack type, split into 70% for training and 30% for testing. Both models were used with the publicly available GitHub code [7, 8] and pre-trained models published on Hugging Face [9, 10]. During fine-tuning, the hyperparameters recommended in each paper (e.g., learning rate) were applied, and the number of epochs was set to 4 for both models. To verify the consistency of results, experiments were repeated 3 times by changing the random seed (0, 123, 7) for the train-test split. The experiments were conducted on a Linux 18.04 system with an AMD Ryzen 9 9950X 16-Core Processor (CPU), 64GB RAM, and an NVIDIA RTX A6000 (48GB VRAM, GPU).

Shuffle Experiment. Both models achieved over 99% accuracy in the classification experiment; however, this performance alone cannot reveal what information within flows the models actually rely on for classification. To investigate this, we designed shuffle experiments in which the packet/burst order of the test data was shuffled and inference was performed using the already-trained models. If performance is maintained after shuffling, the model does not rely on sequence order information; if performance degrades, the model depends on sequence order information. The shuffle experiments were also repeated 3 times with different random seeds to verify consistency.

- **NetMamba shuffle experiment:** The order of the first 5 packets in each flow was randomly shuffled. Since NetMamba processes the information of 5 packets sequentially, this experiment verifies whether the model relies on packet sequence order. The detailed shuffle procedure is presented in Algorithm 3.

Algorithm 3: NetMamba Packet-order Shuffle

Input: flow image I (40×40), where packet j occupies rows [8j:8j+8]
Output: shuffled flow image I'

```
for j = 0 to 4:
    P[j] ← I[8j:8j+8]                    # extract the j-th packet block (8 rows)
order ← random_permutation(5)            # random ordering of the 5 packets, per flow
for j=0 to 4:
    I'[8j:8j+8] ← P[order[j]]            # place packets in the shuffled order
return I'
```

- **netFound shuffle experiment:** Since netFound structures flows in burst units, shuffle experiments were conducted under the following 3 conditions:
  - ✓ **Inter-burst shuffle**: Burst order is shuffled while packet order within bursts is preserved.
  - ✓ **Intra-burst shuffle:** Burst order is preserved while packet order within bursts is shuffled.
  - ✓ **Both shuffled:** Both burst order and packet order within bursts are shuffled.

These 3 conditions allow us to separately verify whether netFound relies on inter-burst order, intra-burst packet order, or both. The detailed shuffle procedure is presented in Algorithm 4.

Algorithm 4: netFound Burst-based Shuffle

Input: flow record R with bursts B = [b_1, …, b_k] # k = number of bursts
condition ∈ {INTER, INTRA, BOTH}
Output: shuffled record R'
packet tokens and burst metadata are preserved; only order changes

```
if condition ∈ {INTRA, BOTH}:                    # shuffle packets within each burst
    for each burst b_i:
        n_i ← number of packets in b_i
        if |n_i| > 1:
            pkt_order ← random_permutation(|n_i|)    # new ordering of packets in b_i
            reorder packets of b_i by pkt_order
if condition ∈ {INTER, BOTH}:                    # shuffle burst order
    if k > 1:
        burst_order ← random_permutation(k)      # new ordering of the k bursts
        reorder (burst_tokens, directions, bytes, counts, iats) jointly by burst_order
return R'
```

4 Results & Analysis

4.1 Classification Results

Both models achieved over 99% accuracy in classifying the 6 attack types, with consistent performance across 3 repeated experiments. Table 4 summarizes the classification accuracy of both models across 3 runs, showing stable performance in the range of 99.88–99.93% regardless of seed. While both models demonstrated near-perfect classification performance, a small number of misclassifications were observed. Table 5 presents the major misclassification patterns observed across the 3 runs, showing the common misclassification directions and their frequencies in both models.

Model	Seed 0	Seed 7	Seed 123	Mean ± Std
netFound	0.9992	0.9992	0.9991	0.9992 ± 0.0001
NetMamba	0.9992	0.9988	0.9993	0.9991 ± 0.0003

Table 4: Classification Accuracy (3 runs)

A notable finding is the common misclassification direction observed in both models. The tendency for full-flow attacks Slowloris and HTTP_CC to be misclassified toward BruteSSH appeared consistently across all 3 runs, regardless of the model. All three attacks share a similar structure with initial packets beginning with a TCP 3-way handshake, but BruteSSH differs in that the SSH protocol banner exists in plaintext. Additionally, sporadic misclassifications were observed, including SQLi→Slowloris(Seed 123, 1 case) in netFound, and BruteSSH→PortScan (Seed 0, 2 cases), SQLi→BruteSSH(Seed 0, 1 case), and PortScan→SYNFlood(Seed 123, 1 case) in NetMamba; however, all were limited to 1–2 cases and did not form a consistent pattern.

The fact that both models achieved identical performance of over 99% alone cannot reveal what information within flows each model actually relies on for classification. To investigate this, Section 4.2 analyzes the results of the shuffle experiments.

Model	Misclassification Direction	Seed 0	Seed 7	Seed 123
netFound	Slowloris → BruteSSH	4	5	6
	HTTP_CC → BruteSSH	13	12	12
NetMamba	Slowloris → BruteSSH	4	2	4
	HTTP_CC → BruteSSH	11	23	10

Table 5: Misclassification Patterns (3 runs)

4.2 Shuffle Experiment Results

netFound Shuffle Results. Table 6 summarizes the accuracy of netFound under 4 shuffle conditions. Inter-burst shuffle showed no performance change across all 3 runs, and the both shuffled results were identical to the intra-burst shuffle, confirming that netFound does not utilize inter-burst order for classification at all. Intra-burst shuffle also showed no change for seed 0 and seed 7; however, for seed 123 only, SQLi classification dropped from 3,588 to 2,518 correct classifications, resulting in an accuracy of 0.9487. This suggests that netFound may be partially sensitive to intra-burst packet order under certain data splits.

In summary, netFound does not rely on inter-burst order and is largely independent of intra-burst packet order. The information netFound primarily relies on is burst-level metadata (bytes, counts, iats,

directions, flow_duration, protocol). As confirmed in Table 3, this metadata exhibits clear differences across attack types, enabling netFound to distinguish the 6 attack types without sequence information.

This insensitivity of netFound to burst/packet ordering is an expected consequence of its design philosophy regarding traffic input representation. netFound is designed with an emphasis on flow-level metadata and on extracting protocol header field-based information from packets. The shuffle in our experiments only changes the order of bursts and packets, while the values of this metadata and the protocol header fields themselves remain unchanged. Therefore, netFound's insensitivity to ordering can be partly attributed to this input representation.

Shuffle Condition	Seed 0	Seed 7	Seed 123	Mean ± Std
Original *	0.9992	0.9992	0.9991	0.9992 ± 0.0001
Inter-burst shuffle	0.9992	0.9991	0.9992	0.9992 ± 0.0001
Intra-burst shuffle	0.9992	0.9989	0.9487	0.9823 ± 0.0291
Both shuffled	0.9992	0.9989	0.9487	0.9823 ± 0.0291

Table 6: netFound Shuffle Experiment Results (Accuracy, **same as Table 4*)

NetMamba Shuffle Results. Table 7 summarizes the accuracy before and after packet order shuffling for NetMamba, and Figure 1 presents the per-attack correct classification counts after shuffling.

Shuffle Condition	Seed 0	Seed 7	Seed 123	Mean ± Std
Original *	0.9992	0.9988	0.9993	0.9991 ± 0.0003
Shuffled	0.5153	0.5393	0.5290	0.5279 ± 0.0120

Table 7: NetMamba Shuffle Experiment Results (Accuracy, **same as Table 4*)

After shuffling the order of the 5 packets, accuracy dropped sharply from 99.91 ± 0.03% to 52.79 ± 1.20%, and this pattern was consistent across all 3 runs. The impact of shuffling varied markedly across attack types. BruteSSH was perfectly maintained across all 3 runs (3,600/3,600), and SYN Flood was also maintained above 90% (3,273–3,535). This is because distinctive indicators exist within the packets themselves, such as the SSH banner (SSH-2.0) and TCP flags (SYN, RST), enabling classification without sequence order. In contrast, HTTP_CC (584–613) and SQLi (409–469) collapsed most severely, and Slowloris (1,455–1,742) also declined substantially. These three attacks are TLS-based with nearly identical headers and payloads in their first 5 packets, and misclassifications after shuffling were concentrated toward BruteSSH.

The key finding is as follows. The full-flow attacks (HTTP_CC, SQLi, Slowloris) are attacks whose distinguishing characteristics appear after the first 5 packets. As confirmed in Table 3, HTTP_CC sends rapidly repeated requests with short inter-burst IAT (median 0.02s), SQLi sends large requests at long intervals with long inter-burst IAT (mean 23.20s) and large bytes per burst (mean 1,402), and Slowloris sends small data slowly with long flow duration (mean 9.40s) and small bytes per burst (median 104). However, since NetMamba only receives the first 5 packets as input, it cannot directly observe these attack characteristics. The 99% classification performance before shuffling was not the result of learning the intrinsic characteristics of these attacks, but rather heavily relied on the order information among the first 5 packets, as revealed through the shuffle experiment.

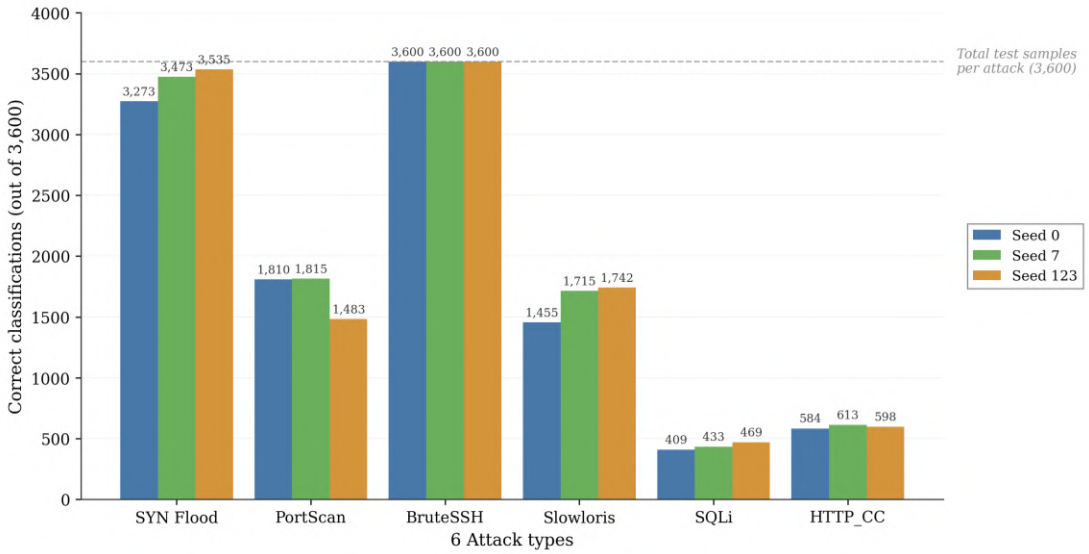

Figure 1: NetMamba per-attack correct classifications after packet order shuffle (3 seeds)

5 Discussion & Conclusion

In this study, we classified 6 attack types in encrypted traffic using two NFMs with different traffic representation methods—netFound and NetMamba—and analyzed what information each model relies on within flows through shuffle experiments.

Both models achieved over 99% classification accuracy; however, the shuffle experiments revealed that the two models rely on entirely different information. netFound maintained its performance even when all orders were shuffled, relying on flow-level metadata, whereas NetMamba's accuracy dropped sharply to 52.79 ± 1.20% when packet order was shuffled. In particular, NetMamba had not learned the intrinsic characteristics of full-flow attacks (HTTP_CC, SQLi, Slowloris) but was instead relying on the order information of the first 5 packets. These results demonstrate that models cannot be trusted based on performance alone in NFM-based attack classification, and that verifying what information models rely on for classification is essential.

This study confirmed that NFMs hold the potential to overcome the shortcut learning limitations of conventional AI-IDS by preserving and processing flow structure, while also revealing through shuffle experiments that performance alone cannot explain the basis of a model's classification. Future research plans to address current limitations and expand the experiments.

First, the shuffle experiments in this study only verified dependence on sequence order information. More detailed ablation studies—such as masking or removing specific header fields or metadata—are needed to more precisely isolate the information models depend on. Second, since this study used 6 of the 13 attack types in the dataset, we plan to extend the analysis to all attack types for further validation. Third, this study analyzed only two NFMs. However, various NFMs have recently emerged, including NetMamba+ [11], which supplements NetMamba's limitations by adding meta-information such as IAT and packet length, as well as TrafficFormer [12] and YaTC [13], which processes traffic by visualizing it as images. Future work plans to expand experiments with these models to more comprehensively analyze attack classification dependency structures according to NFM traffic representation methods.

Finally, since the results of this study were obtained on simulated attack traffic from a single controlled environment, validation in other network environments is needed. In addition, this study focused on classification among attack types and did not include normal traffic, so its conclusions should be understood as limited to classification among attack types. Nevertheless, the difference in the dependency structures of the two models arises from the difference in input representation discussed in Section 2.2, and is a characteristic of NFMs stemming from their traffic input representation design, independent of the dataset. Verifying whether the findings of this study are reproduced on public datasets or traffic from diverse environments, and analyzing how the models' dependency structures change in settings that include normal traffic, are directions that can be addressed in future work.

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CIV: Content Integrity Verification for Distributed Intrusion Detection Systems*

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Abstract

In distributed IoT environments, an intrusion detection system (IDS) typically aggregates reports from individual nodes. However, prior work largely assumes that such reports are trustworthy and has not sufficiently addressed the risk that compromised nodes can manipulate their reports—without altering the underlying network traffic—to distort the IDS input. To address these limitations, we propose the Content Integrity Verification (CIV) framework. CIV leverages gateway-level observations as an independent reference and quantitatively measures the standardized discrepancy between each node's report and the corresponding observation. CIV then excludes reports exceeding a predefined threshold from aggregation, thereby mitigating the impact of manipulated inputs on IDS performance. Experimental results show that the average F1-score across all evaluation settings was 0.266 without defense, whereas it improved to 0.704 with CIV. In addition, CIV outperformed existing Byzantine-robust aggregation methods in 48 out of 60 baseline-comparison $\alpha-\beta$ settings (80%) and achieved an average F1-score approximately 31% higher than Trimmed Mean, the strongest baseline.

1 Introduction

In IoT environments, distributed intrusion detection system (IDS) architectures are widely adopted, in which multiple nodes independently monitor network traffic, and an upper-layer edge server aggregates their monitoring results for intrusion detection [1]–[3]. This architecture is motivated by the difficulty of transmitting raw traffic directly to a central server under bandwidth constraints, latency requirements, and privacy concerns [3], [4]. Accordingly, practical deployments commonly rely on node-level reports, where each device summarizes its local observations and the resulting reports are aggregated as input to an upper-layer IDS [3].

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However, while this distributed architecture improves scalability and efficiency, it also introduces an important vulnerability that has received relatively limited attention. Existing distributed IDS pipelines implicitly assume that reports generated by individual nodes are trustworthy. When a node is compromised, however, an attacker can manipulate the feature-level report delivered to the upper layer without modifying the underlying network traffic. Such report-level falsification distorts the IDS input representation and can substantially degrade detection performance, even when attack traffic is present in the network.

Prior work on robustness in distributed systems has primarily focused on mitigating extreme outliers through statistical aggregation methods, such as Byzantine-robust averaging [5]. However, these approaches rely only on the mutual consistency of reports across nodes and do not leverage an independent external reference. Consequently, these methods may be ineffective when an attacker crafts manipulated reports to resemble normal patterns. As a result, verifying whether each node's report faithfully reflects the underlying network traffic—i.e., report integrity—remains an insufficiently explored gap in existing distributed IDS research.

To address these limitations, this study proposes the Content Integrity Verification (CIV) framework for distributed IDS. The key idea of the proposed approach is to use gateway-level observations, independently collected at network bottlenecks, as a reference for verification. CIV quantitatively measures the discrepancy between each node's report and the corresponding gateway observation, and mitigates the impact of manipulated inputs on the IDS by suppressing the contribution of suspicious reports to aggregation.

In particular, CIV does not attempt to recover corrupted information; rather, it serves as a defensive mechanism that preserves attack-related signals in honest reports by preventing manipulated reports from dominating the aggregated output. This approach also imposes a structural trade-off on the attacker. The more strongly an attacker falsifies reports toward benign patterns to evade IDS detection, the larger the discrepancy from gateway observations becomes, thereby increasing the likelihood of detection by CIV. Experimental results show that, under honest-majority conditions, CIV restores IDS performance to a level close to the clean baseline and outperforms four representative Byzantine-robust baselines in 48 out of 60 baseline-comparison settings.

The main contributions of this work are as follows:

- We propose the CIV framework, which uses gateway observations as a verification reference to evaluate report integrity and mitigate the impact of manipulated inputs.
- We show that CIV improves IDS performance by mitigating input contamination rather than recovering corrupted information, and systematically evaluate its effectiveness across different falsification strengths (α) and node compromise ratios (β).
- We identify the operational boundary of CIV by showing that performance degradation under high node compromise ratios is not a CIV-specific failure, but is instead associated with a broader structural limitation of aggregation-based defenses.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 describes the system and threat model. Section 4 presents the proposed method. Section 5 introduces the experimental setup. Section 6 provides performance evaluation and analysis. Section 7 discusses limitations and open issues. Finally, Section 8 concludes the paper.

2 Related work

This section reviews prior work directly related to CIV. It covers ML/DL-based IDSs and distributed IDSs in IoT environments, hierarchical deployment strategies for IDS operations, Byzantine-robust aggregation, and studies on data integrity and trustworthiness in distributed systems. These studies provide important contributions to detection performance, cooperative system design, deployment optimization, robust aggregation, and trustworthiness assurance. Nevertheless, they generally do not sufficiently address the challenge of directly verifying report integrity before aggregation using an independent reference, namely, whether each node's report faithfully reflects the underlying observations.

2.1 ML/DL-Based and Distributed IDS in IoT Environments

In IoT environments, IDS techniques have shifted from conventional rule-based and signature-based methods toward machine learning (ML)- and deep learning (DL)-based approaches [6]–[9]. Conventional ML methods have achieved strong detection performance using predefined network-level features, whereas DL models, such as CNNs and RNNs, have been used to improve detection accuracy while reducing reliance on manual feature engineering [7], [8]. Accordingly, existing IDS research has primarily been designed and evaluated around performance metrics such as detection accuracy and false positive rate [7], [9].

To address the scalability and bandwidth limitations of centralized IDSs, distributed and cooperative IDS architectures have been actively studied in IoT environments [10]. In these architectures, each node monitors local traffic, while integrated analysis is performed by an upper-layer entity or through collaboration among nodes [10], [11]. This line of work has developed along two main directions: reducing communication overhead and energy consumption through selective reporting and lightweight feature extraction, and improving detection performance by combining information across nodes to exploit spatial and temporal correlations [10], [12]. Furthermore, distributed IDS research has explored various cooperative mechanisms, including alert correlation, cooperative model learning, and consensus-based updates [11]–[13].

However, these studies generally assume that input data collected in centralized or semi-centralized environments faithfully reflects the underlying network state [7], [8]. Distributed IDS research similarly assumes that reports submitted by individual nodes are trustworthy inputs. Furthermore, trust-based cooperative IDSs primarily focus on long-term reputation-based trust estimation and do not adequately address real-time integrity verification at the per-report level [12].

2.2 Computational Placement in IoT IDS

In addition to detection performance, IDS placement across network layers has been considered an important design issue in IoT environments [10], [14]. Prior work has examined various deployment architectures across layers, ranging from IoT devices and gateways to fog/edge nodes and the cloud [14], [15]. In particular, gateways and IoT devices face limitations in performing computationally intensive ML/DL-based detection locally due to CPU, memory, and energy constraints [14], [15]. Accordingly, many studies have adopted architectures that offload detection tasks to fog or edge servers, while fog-based distributed IDSs and edge-assisted federated learning approaches have primarily addressed performance-oriented objectives such as latency, resource constraints, and communication efficiency [16], [17].

Meanwhile, the gateway layer is important for IDS placement and the design of verification mechanisms because gateways, despite their limited computational resources, can be positioned at points through which network traffic must pass. However, prior IDS research has primarily considered

this positional advantage from the perspective of performance optimization, rather than fully leveraging it for report integrity verification.

2.3 Byzantine-Robust Aggregation

The problem of robustly aggregating inputs in the presence of adversarial participants has been extensively studied in distributed machine learning and federated learning (FL) for Byzantine fault tolerance [18]. For example, Krum selects the update with the smallest sum of distances to its closest updates [19]. Coordinate-wise Trimmed Mean and Median mitigate the influence of outliers along each coordinate and provide statistically robust aggregation rules [20]. These approaches have also been extended to IIoT environments through geometric-median-based variants [5], [21].

However, these methods share a common limitation: they identify outliers based solely on statistical relationships among the submitted inputs. In other words, because they lack an independent external reference and rely only on relative differences among inputs, they may fail to distinguish manipulated inputs that mimic benign patterns [19], [20]. Indeed, Xie et al. and Fang et al. showed that such robust aggregation rules can be circumvented or their performance can be substantially degraded through inner-product manipulation attacks and optimization-based local model poisoning, respectively [22], [23].

These limitations also apply to IDS environments. When a compromised node submits traffic reports that appear to fall within the benign range while misrepresenting actual observations, aggregation based solely on inter-node consistency may fail to effectively detect such manipulation. Therefore, beyond conventional robust aggregation, a report integrity verification mechanism based on an independent reference is needed to determine whether each node's report reflects actual observations.

2.4 Data Integrity and Trust in Distributed Systems

Data integrity and trustworthiness in distributed systems have been studied through various approaches, including secure aggregation, blockchain-based immutability, and device-level trust scoring [24]–[29]. These studies make important contributions to ensuring integrity and security during data transmission, storage, aggregation, and model training. However, most approaches focus on post-generation stages and do not directly verify report-level integrity—namely, how accurately each node's report reflects actual observations.

For example, the secure aggregation protocol by Bonawitz et al. preserves privacy by aggregating data without revealing individual contributions, but it does not verify the correctness of the contributions themselves [24]. In addition, HBFL by Sarhan et al. uses blockchain to support the integrity and traceability of model updates, but it does not consider the trustworthiness of the original IDS inputs from which those updates are generated [29]. Furthermore, studies on IoT trust management and FL-based IDS address long-term trust estimation and the trustworthiness of training data, respectively, but they do not provide a mechanism for directly verifying report integrity, i.e., the consistency between individual reports and actual observations, before aggregation [26]–[28].

2.5 Research Gap and Positioning

The studies reviewed above have contributed to the development of distributed IoT IDS architectures from different perspectives. ML/DL-based IDS research has focused on improving detection accuracy and reducing false positive rates, while distributed IDS research has aimed to improve scalability and communication efficiency through cooperative architectures.

In addition, computation placement studies have addressed the deployment of detection tasks under resource constraints, while Byzantine-robust aggregation and distributed trust studies have proposed methods to improve aggregation robustness and system trustworthiness under adversarial conditions.

However, these studies share a common limitation: they do not directly verify, prior to aggregation, whether the reports submitted by individual nodes faithfully reflect actual network observations. In other words, although prior work has broadly addressed the performance, scalability, and aggregation robustness of distributed IoT IDSs, it has not explicitly formulated report integrity itself as an independent verification target. This gap is particularly important because attackers can distort the IDS input representation by manipulating only the reports transmitted to higher layers, even without directly altering the underlying network traffic.

To address this gap, CIV leverages gateway-level observations as an independent reference and quantitatively evaluates the discrepancy between each node's report and the corresponding gateway observation prior to aggregation. By doing so, CIV mitigates the impact of manipulated inputs before they affect the upper-layer IDS and provides a new approach for systematically addressing report integrity verification in distributed IDS environments.

3 System and Threat Model

3.1 System Overview and Model

This study assumes a distributed intrusion detection system (IDS) environment in which multiple IoT nodes monitor traffic, and detection is performed at an edge server. Each IoT node monitors its local traffic and generates a node-level report $r_i(t)$, which summarizes activity within a fixed time window using a common set of flow-level features. The generated reports are transmitted to the edge server, where they are aggregated and used as input to the IDS.

Meanwhile, gateways are deployed at network choke points and independently observe traffic through flow-based network monitoring. Accordingly, the gateway observation $o_i(t)$ is defined as an observation vector directly derived from the underlying network traffic and does not depend on information reported by individual nodes. In other words, the IDS pipeline operates on node-level reports, whereas the gateway collects traffic information through an observation path independent of the nodes.

In this study, $r_i(t)$ and $o_i(t)$are defined over the same fixed time window. Here, $r_i(t)$ denotes the report generated by node i by summarizing its local traffic, whereas $o_i(t)$ denotes the gateway observation obtained by aggregating traffic flows observed during the same time interval in a device-centric manner. Specifically, $o_i(t)$ reflects both inbound and outbound traffic flows attributable to device i within the time window at time t. As a result, the device representation includes not only traffic generated by the device itself, but also observations of externally originated attack traffic received by the device.

Both representations consist of flow-level statistics, such as byte counts, packet counts, duration, and connection state, and the gateway observations rely only on L3/L4-level metadata without payload inspection. Therefore, the proposed gateway-based observation model is applicable even in environments with encrypted traffic. Furthermore, because $r_i(t)$ and $o_i(t)$ are defined in the same feature space, the two representations can be directly compared.

3.2 Threat Model

In this study, we assume that a fraction $\beta \in [0,1]$of IoT nodes may be compromised. Compromised nodes can distort the input to the upper-layer IDS by falsifying locally generated node-level reports $r_i(t)$before transmission. This setting follows an adversarial framework that is structurally related to Byzantine distributed detection models, where compromised sensors transmit fictitious observations to a fusion center [30], [31].

The attacker attempts to evade detection by crafting falsified reports to resemble benign patterns. For a compromised node, the falsified report is defined as follows.

$$\tilde{r}_i(t) = (1-\alpha)x_i(t) + \alpha\mu_i^{ben} \quad (1)$$

Here, $x_i(t)$ denotes the original flow-level report vector before falsification, and μ_i^{ben} denotes the benign reference value of node i estimated during the reference estimation interval. In addition, $\alpha \in [0,1]$ is a parameter that controls the falsification strength. As α increases, the falsified report moves closer to the benign reference value; $\alpha = 0$ corresponds to the original report without falsification, whereas $\alpha = 1$ corresponds to complete replacement by the benign reference value.

This attack model builds on the abstract falsification channel assumed in prior Byzantine literature and instantiates it as a benign-reference-directed falsification strategy suitable for IoT IDS environments [30], [31]. This implies that the attacker falsifies reports toward the statistical center of benign behavior.

The attack targets flow-level reports transmitted to the upper layer rather than the underlying network traffic itself. Consequently, an attacker can distort the IDS input representation without modifying network packets. In particular, when manipulated reports are crafted to mimic a benign distribution, existing aggregation-based defenses that rely on inter-node consistency can be effectively bypassed.

We assume that gateway observations $o_i(t)$ are generated by independently monitoring network traffic and are not directly affected by node-side report manipulation. Furthermore, we assume that the gateway itself and the gateway–edge communication channel are trustworthy or, at a minimum, integrity-protected. This assumption is required for using gateway observations as an independent external verification reference. This trust boundary is necessary because CIV uses gateway observations as an external verification reference. If the gateway is compromised or if gateway-to-edge observations are tampered with, the external reference itself may become unreliable, limiting the effectiveness of CIV under such scenarios.

This study does not consider traffic-level attacks, such as packet injection and packet suppression, or gateway compromise. These attacks correspond to threat models that are fundamentally different from report falsification and may be addressed as separate research topics.

4 Proposed Method

This study proposes the Content Integrity Verification (CIV) framework. CIV identifies suspicious reports by quantitatively evaluating the discrepancy between each node's report and the corresponding gateway-level observation. This verification process is performed prior to data aggregation and mitigates the impact of input contamination on the upper-layer IDS by suppressing the contributions of suspicious reports. Rather than restoring corrupted information, CIV operates as a report integrity verification mechanism that maintains trustworthy inputs through filtering. The overall procedure consists of the following steps.

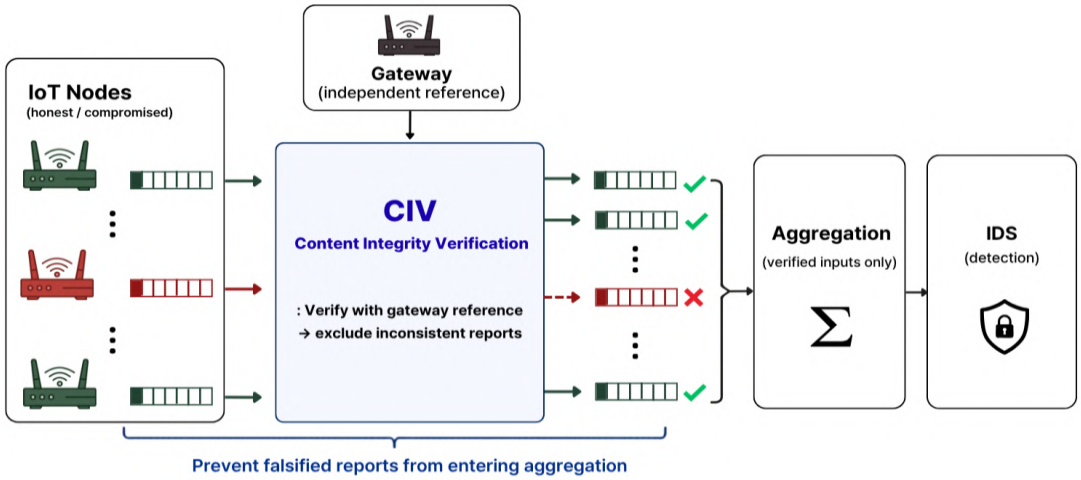

Figure 1: Proposed CIV framework

4.1 Benign Reference and Standardized Discrepancy

In the first step of CIV, reference statistics representing benign behavior are estimated for each node. Specifically, using gateway-level observations collected during the reference estimation interval, CIV computes the mean $\mu_{i,k}^{ben}$ and standard deviation $\sigma_{i,k}^{ben}$ for each feature dimension k. Here, the superscript ben indicates that the corresponding statistic is estimated during the reference estimation interval. Based on these statistics, the discrepancy between node i's report $r_i(t)$ and the corresponding gateway observation $o_i(t)$ at time t is defined as follows.

$$\Delta_i(t) = \sum_{k=1}^{d} \left(\frac{r_{i,k}(t) - o_{i,k}(t)}{\sigma_{i,k}^{ben}} \right)^2 \tag{2}$$

Here, i denotes the node index, t denotes the time-window index, d denotes the total number of feature dimensions, and k denotes the feature-dimension index. In addition, $r_{i,k}(t)$ and $o_{i,k}(t)$ denote the k-th feature values of node i's report and the corresponding gateway observation at time t, respectively.

4.2 Thresholding and Hard Exclusion

The computed discrepancy score $\Delta_i(t)$ is used as a criterion for determining whether each report falls within the benign range. In this study, the threshold τ is estimated from the benign discrepancy distribution obtained during the reference estimation interval. Specifically, τ is defined as the p-th percentile of the benign discrepancy distribution, as follows.

$$\tau = Q_p\left(\{\Delta_i(t) \mid (i,t) \in \mathcal{D}_{\text{cal}}\}\right) \tag{3}$$

Here, $Q_p(\cdot)$ denotes the p-th percentile, and $\mathcal{D}_{cal}$ is the set of discrepancies observed under honest reporting. In this study, we use $p = 95$. This corresponds to a one-sided criterion that treats deviations in the upper 5% of the discrepancy distribution under benign conditions as anomalous.

We then apply the following hard exclusion strategy to reports that exceed this threshold.

$$w_i(t) = \begin{cases} 1, & \text{if } \Delta_i(t) \le \tau \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

This rule includes a report in aggregation only when its discrepancy falls within the benign range; otherwise, the report is treated as a likely manipulated input, and its contribution is completely removed. This threshold-based binary removal rule is designed to account for the sensitivity of aggregation-based IDSs to even small residual contamination. Accordingly, we adopt deterministic removal rather than soft weighting to suppress the influence of manipulated reports before aggregation. However, under high-β settings, this design may reduce the number of valid reports and trigger fallback. The resulting fallback behavior is defined in §4.3, and its impact is analyzed in §6.4.

4.3 CIV-based Aggregation

Using the computed weights $w_i(t)$, the aggregated report vector $\hat{r}(t)$at time t is defined as follows.

$$\hat{r}(t) = \frac{\sum_i w_i\,(t)\, r_i(t)}{\sum_i w_i\,(t)} \quad (5)$$

This equation indicates that the aggregate is constructed from individual reports verified as consistent with gateway-level observations. The denominator $\sum_i w_i\,(t)$ normalizes the aggregate so that its scale remains consistent despite changes in the number of valid reports.

However, if all reports exceed the threshold in a given window t, resulting in $\sum_i w_i\,(t) = 0$, the above equation is undefined. In such cases, we replace the aggregated representation for that window with the unweighted mean of all reports and define this operation as a fallback. This fallback mechanism preserves an aggregated representation even when no valid reports are available. This fallback behavior is analyzed in §6.4.

5 Experimental Setup

5.1 Dataset and Preprocessing

This study used the TON_IoT dataset, which provides flow-level network traffic. Because the dataset contains flow-level information observable at the gateway level, it is suitable for experimentally implementing the proposed system model.

To emulate a distributed IoT monitoring environment, traffic records were organized into outbound and inbound views using source and destination IP information, respectively, and aggregated into device-centric node representations over fixed time windows. This process produced node-level time-series inputs for each node i at each time window t and provided a controlled logical node emulation for evaluating report falsification in distributed IDS settings.

Furthermore, to implement the system model described in §3.1, we constructed node-level reports $r_i(t)$ and gateway-level observations $o_i(t)$ from traffic within the same time intervals using aligned flow-level feature representations.

From the full TON_IoT dataset, we selected five traffic classes: normal, DDoS, DoS, scanning, and backdoor. This selection was not intended to comprehensively cover diverse attack types, but rather to preserve feature-space comparability between $r_i(t)$ and $o_i(t)$ after aggregation.

5.2 IDS Model

For intrusion detection, this study used a Random Forest (RF) classifier. The model is trained on clean aggregated data, and its detection performance is evaluated at inference time under three input conditions: clean input, falsified input, and CIV-corrected input. Because this study focuses on analyzing the impact of report falsification and CIV on aggregation-based IDSs rather than comparing classifier architectures, a single IDS model was used throughout the experiments.

5.3 Attack Configuration

In this study, we experimentally instantiate the report-level threat model defined in §3.2. At each time window t, we randomly select a subset of IoT nodes according to the compromise ratio β and falsify the original report vector $x_i(t)$ of each selected node as follows.

$$\tilde{r}_i(t) = (1-\alpha)x_i(t) + \alpha\mu_i^{ben} \quad (1)$$

Here, $\alpha \in [0,1]$ denotes the falsification strength, and β denotes the compromise ratio, i.e., the fraction of compromised nodes among all IoT nodes. We evaluated the attack settings over parameter combinations of $\alpha \in \{0.1,0.2,\ldots,1.0\}$ and $\beta \in \{0.1,0.2,\ldots,0.9\}$, resulting in 90 non-degenerate attack settings. Some figures visualize representative subsets of the full evaluation grid for readability.

For clarity, we divide the evaluation settings into two regimes. The primary evaluation regime corresponds to $\beta \leq 0.5$, representing a typical operational setting in which at least half of the nodes remain honest. In contrast, the stress regime corresponds to $\beta \geq 0.6$, representing a highly compromised setting in which aggregation-based verification and mitigation become difficult.

The case $\beta = 1.0$ was excluded from the evaluation because it corresponds to a degenerate case in which all nodes are compromised. In this case, no trustworthy set of reports remains available for aggregation, making it difficult to meaningfully define the notion of performance recovery through aggregation-based mitigation. Therefore, our experiments focus on performance degradation under partial-compromise settings and the mitigation effect of CIV.

5.4 Evaluation Metrics

In this study, we used F1-score, TPR, FPR, and AUROC to evaluate IDS performance. F1-score was used as the primary metric for comparing changes in overall classification performance, TPR was used to assess the attenuation of attack-related signals, and FPR was used to examine whether false positives increased after applying CIV. AUROC was used to evaluate the classifier's overall discriminative ability independently of a specific threshold setting. Based on these metrics, IDS performance was compared across three conditions: clean input, falsified input, and CIV-corrected input.

Additionally, ΔF_1, defined as the absolute improvement in F1-score under CIV-corrected input relative to falsified input, was used as a supplementary metric. We also included an honest-only reference, constructed solely from reports submitted by uncompromised nodes, for comparison. This reference was computed by applying the same aggregation procedure after assigning a weight of 0 to

manipulated reports and a weight of 1 to honest reports; it serves as a reference for the performance achievable when compromised reports are perfectly excluded under partial-compromise settings. When no honest report remains available in a given window, the honest-only reference follows the same fallback rule defined in §4.3.

6 Performance evaluation and analysis

The experiments were conducted on 23 IoT nodes from the TON_IoT dataset over 90 attack settings, consisting of $\alpha \in \{0.1, 0.2, \ldots, 1.0\}$ and $\beta \in \{0.1, 0.2, \ldots, 0.9\}$. For the no-defense and CIV recovery analyses, we report results over the full attack grid. For the baseline comparison, we focus on a representative subset of 60 α–β settings used in Fig. 5 to compare CIV with four Byzantine-robust aggregation baselines. The results are presented in the following order: the impact of report falsification (§6.1), the performance recovery effect of CIV (§6.2), comparison with baselines (§6.3), analysis of the operational boundary (§6.4), ablation study (§6.5), and computational and storage overhead analysis (§6.6).

6.1 Impact of Report Falsification

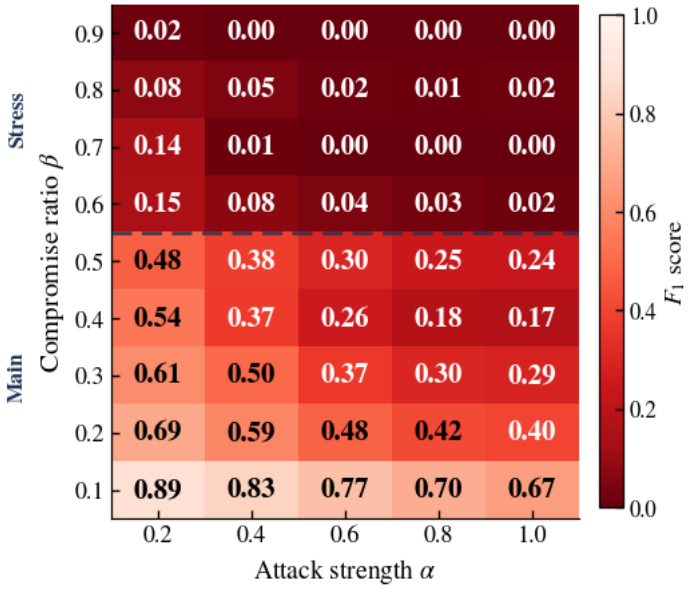

Figure 2: IDS F1 under report falsification (no defense)

Report falsification under the attacker model defined in this study can substantially degrade IDS performance without altering the underlying network traffic itself. As shown in Fig. 2, the IDS F1-score remains high at 0.979 under the clean setting, whereas performance generally decreases as the compromise ratio β and falsification strength α increase after falsification is introduced. For example, at $\beta = 0.4$ and $\alpha = 1.0$, the F1-score drops to 0.17; for $\beta > 0.7$, it falls to 0.05 or lower under most conditions.

These results indicate that β, the node compromise ratio, and α, the falsification strength, jointly affect IDS performance. Under low-β settings, the impact of falsified reports is partially mitigated through aggregation; however, as the proportion of compromised nodes increases, performance

degradation caused by increasing α becomes more pronounced. This occurs because, when an attacker falsifies reports toward the benign reference value, the manipulated reports bias the aggregate toward the benign distribution. As a result, attack-related signals in honest reports are diluted, and the IDS input representation moves closer to the benign region relative to the decision boundary.

Furthermore, these results empirically demonstrate that the report-level falsification threat model defined in §3 can cause substantial performance degradation. This provides a baseline for interpreting the performance recovery achieved by CIV in the following sections and motivates the need for report integrity verification using gateway-level observations.

6.2 Recovery with CIV

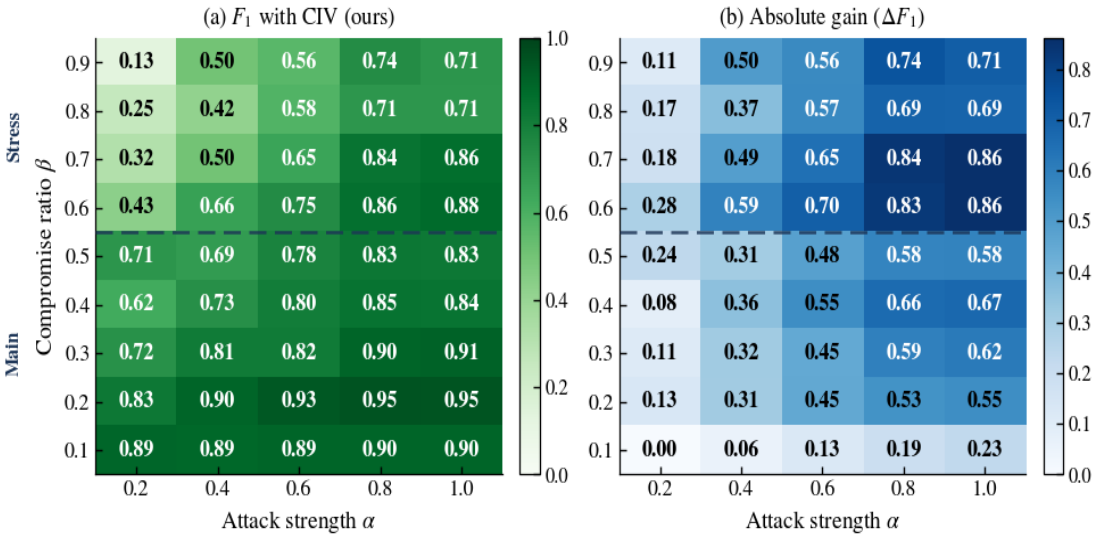

Figure 3: CIV recovery across compromise and attack regimes

Figure 3 shows the F1-score after applying CIV, along with the absolute performance improvement ΔF_1 relative to falsified input, summarized over a representative subset of α–β conditions. Overall, CIV demonstrates strong performance recovery under low compromise-ratio regimes. In particular, for $\beta \leq 0.3$, performance is restored to a level close to the clean baseline in most cases, and even at $\beta = 0.4$, substantial performance improvement is observed overall. For example, under the condition $\beta = 0.4$ and $\alpha = 1.0$, the no-defense F1-score drops to 0.17, whereas it recovers to 0.84 after applying CIV. This case illustrates a representative example of performance recovery under a single condition, and, on average across all 90 evaluation settings, the F1-score improves substantially from 0.266 to 0.704.

In contrast, performance recovery becomes increasingly limited under high compromise-ratio regimes. At $\beta = 0.6$, CIV maintains relatively high performance, whereas at $\beta = 0.8$, the F1-score remains in the range of 0.25–0.71. This behavior is not primarily due to CIV's detection failure, but is instead associated with the structural effect of increasingly frequent fallback conditions as the number of valid reports decreases. A detailed analysis is provided in §6.4.

In addition, as α increases, performance under the no-defense condition decreases, whereas ΔF_1 generally increases. This is because stronger falsification toward the benign reference direction makes the IDS more susceptible, while simultaneously increasing the discrepancy with gateway-level observations, thereby improving the detectability of falsified reports by CIV.

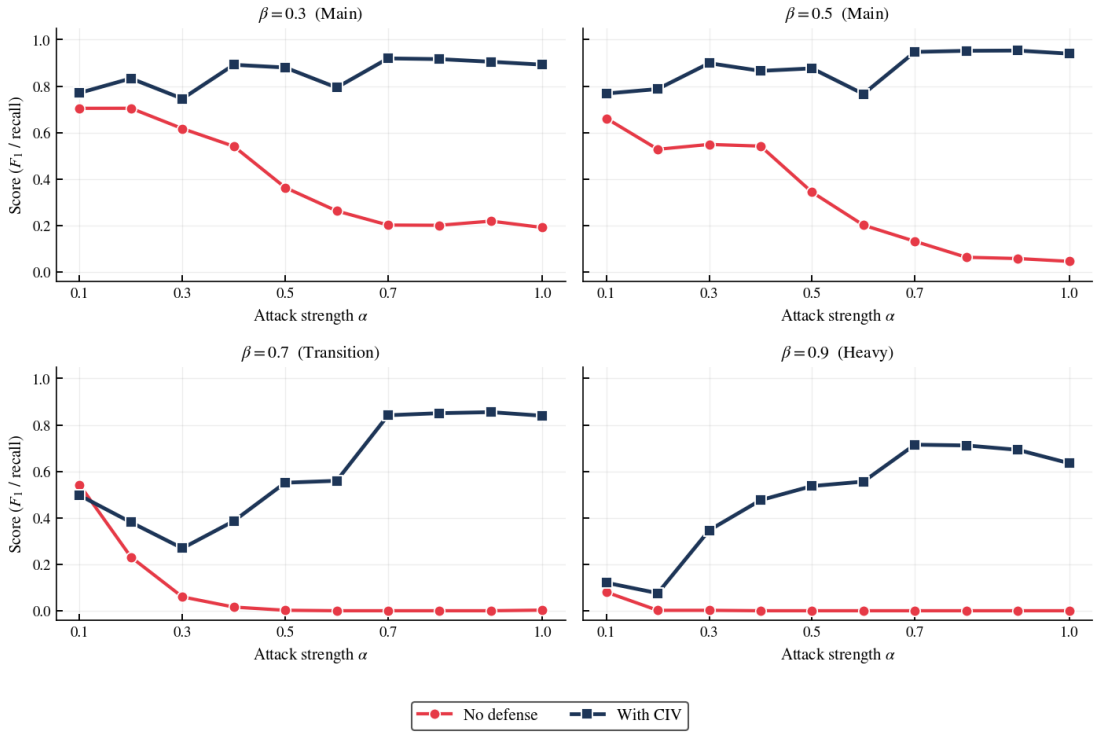

Figure 4: Attacker's dilemma, stronger falsification harms IDS but improves CIV detectability

Figure 4 shows the performance trends of the no-defense condition and the CIV-applied condition as the falsification strength α increases. Overall, as αincreases, the F1-score under the no-defense condition decreases, whereas the performance after applying CIV remains relatively stable under low compromise-ratio regimes. In particular, for $\beta = 0.3$ and $\beta = 0.5$, CIV maintains high performance even as the no-defense performance consistently degrades.

These results indicate that as an attacker falsifies reports more strongly toward the benign reference, the IDS becomes more susceptible, while the discrepancy with gateway-level observations increases, thereby strengthening the CIV verification signal. In other words, as the falsification strength increases, both the potential for IDS evasion and the CIV-based discrepancy signal increase simultaneously, indicating the presence of an inherent structural trade-off.

Specifically, in the regime $\beta \leq 0.3$, CIV maintains performance close to the clean baseline under most conditions. At $\beta = 0.5$, overall performance remains high, although degradation is observed in some cases. At $\beta = 0.7$, a moderate level of recovery is still achieved, whereas at $\beta = 0.9$, performance preservation becomes limited. This degradation is not primarily due to CIV detection failures, but rather to the structural limitations of the aggregation process, where the number of valid reports decreases, and fallback becomes more frequent as the compromise ratio increases. A quantitative analysis of this effect is provided in §6.4.

6.3 Comparison with Byzantine-Robust Baselines

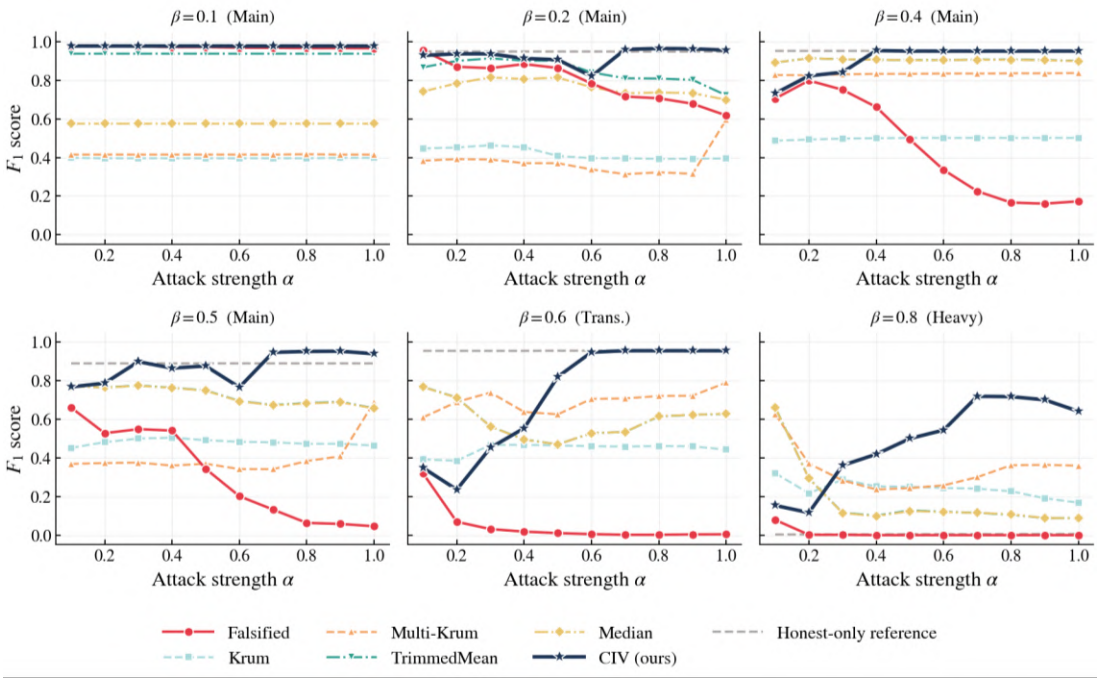

Figure 5: Comparison with Byzantine-robust aggregation baselines

Figure 5 compares the proposed CIV with representative Byzantine-robust aggregation methods and an honest-only reference. The baselines include Krum, Multi-Krum, coordinate-wise Trimmed Mean, and Median. The honest-only reference is computed by applying the same aggregation procedure after assigning a weight of 0 to manipulated reports and a weight of 1 to honest reports based on ground-truth compromise labels. For readability, baseline comparisons are visualized over representative compromise regimes rather than the full evaluation grid.

Experimental results show that CIV achieves higher F1-scores than all baselines in 48 out of 60 baseline-comparison α–β settings (80%). This performance advantage becomes more pronounced as the compromise ratio increases, whereas in low-compromise regimes such as $\beta = 0.1$, where the impact of falsification is limited, most methods—including CIV—maintain relatively high performance.

Across the 60 baseline-comparison settings, CIV attains an F1-score of 0.704, outperforming the strongest baseline, Trimmed Mean (F1 = 0.536), by an absolute margin of 0.168 points (approximately 31% relative improvement). These results indicate that CIV not only performs well under specific conditions but also provides consistent performance gains across the baseline-comparison settings.

In particular, the effectiveness of CIV becomes more pronounced in moderate-compromise regimes. At $\beta = 0.4$, CIV achieves F1-scores in the range of 0.624–0.846, and under $\alpha \geq 0.5$, it attains 0.793–0.846, approaching the average level of the honest-only reference (0.826). Moreover, under $\beta = 0.6$ and $\alpha \geq 0.5$, CIV maintains F1-scores in the range of 0.747–0.878, whereas the best-performing baseline reaches only 0.717.

In contrast, the existing baselines exhibit different forms of performance limitations. Krum shows consistently low performance, while Multi-Krum, Trimmed Mean, and Median maintain reasonable performance under low-compromise regimes but degrade significantly when $\beta \geq 0.6$. This suggests

that conventional aggregation methods, which rely solely on relative consistency among node reports, fail to effectively distinguish manipulated inputs that mimic the benign distribution.

In summary, CIV leverages gateway-level observations as an external reference for report integrity verification, thereby providing more stable and consistent defensive performance than existing Byzantine-robust aggregation methods across most conditions.

6.4 Operational Scope and Fallback Analysis

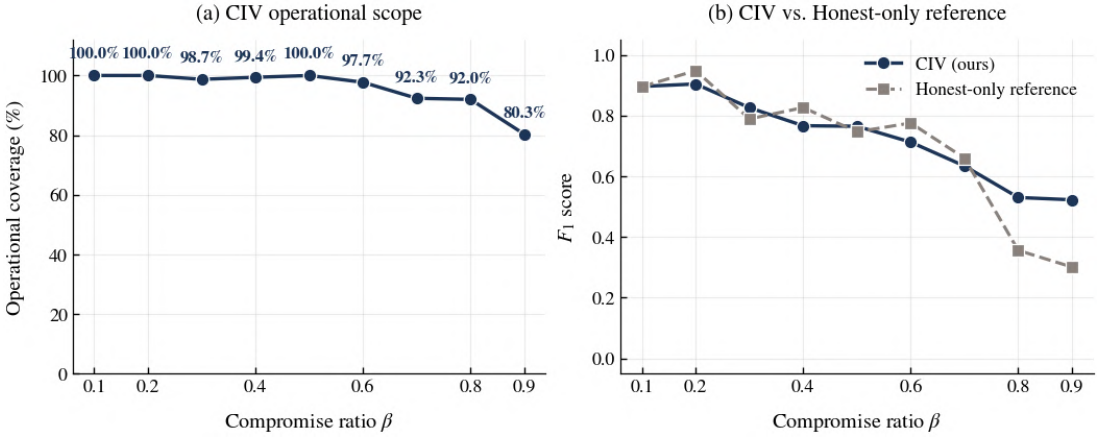

Figure 6: Operational Scope of CIV across compromise regimes

Figure 6 illustrates the operational scope of CIV under different compromise regimes. The hard exclusion mechanism of CIV falls back to the unweighted mean when all reports exceed the threshold; in this study, we define operational coverage as the proportion of time windows in which CIV operates without triggering fallback.

As shown in Fig. 6(a), the operational coverage remains close to 100% for $\beta \leq 0.5$, and still achieves a high level of 92.0% at $\beta = 0.8$. In contrast, at $\beta = 0.9$, the coverage decreases to 80.3%, indicating a substantial increase in fallback-triggering windows under the extreme compromise regime. This indicates that as the compromise ratio increases, the number of windows triggering fallback gradually rises, potentially limiting the operational range in which passive CIV can function reliably.

Fig. 6(b) compares CIV with the honest-only reference. For $\beta \leq 0.6$, the two methods show similar performance. However, in high-compromise regimes where $\beta \geq 0.7$, CIV achieves higher F1-scores than the honest-only reference. This suggests that, under extreme compromise conditions, even ideally removing compromised reports may leave too few honest reports to reliably preserve sufficient IDS signals. Therefore, the honest-only reference should be interpreted as a passive-verification reference rather than an absolute upper bound under structurally under-supported aggregation conditions.

In summary, CIV maintains high operational coverage and meaningful defensive effectiveness for $\beta \leq 0.8$. However, under the extreme compromise condition of $\beta = 0.9$, operational coverage decreases substantially, indicating that stable defense may be limited when relying solely on passive filtering of already-received reports. Therefore, additional verification mechanisms, such as active confirmation, may be required in such environments.

6.5 Design Justification and Ablation Study

This section provides empirical justification for the main design components of CIV presented in §4: the discrepancy measure, hard exclusion, and threshold estimation method. To this end, we conducted an ablation study for each component under the same experimental setting.

First, comparison of discrepancy measures shows that the feature-wise standardized discrepancy score (§4.1) provides the most consistent performance. The proposed method achieves a higher average F1-score overall than the Mahalanobis-based variant, and it also shows clearer weight separation between honest and compromised nodes. This suggests that covariance estimation can become unstable in high-dimensional flow-level feature spaces, empirically supporting the design choice in §4.1.

We then compare hard exclusion and soft weighting. The results show that their average performance difference is not large, but hard exclusion provides more consistent results overall. In particular, under high-compromise regimes, hard exclusion more effectively suppresses the residual influence of manipulated reports, supporting the validity of the binary exclusion design adopted in §4.2. Finally, Fig. 7 presents the sensitivity analysis results for the threshold percentile p.

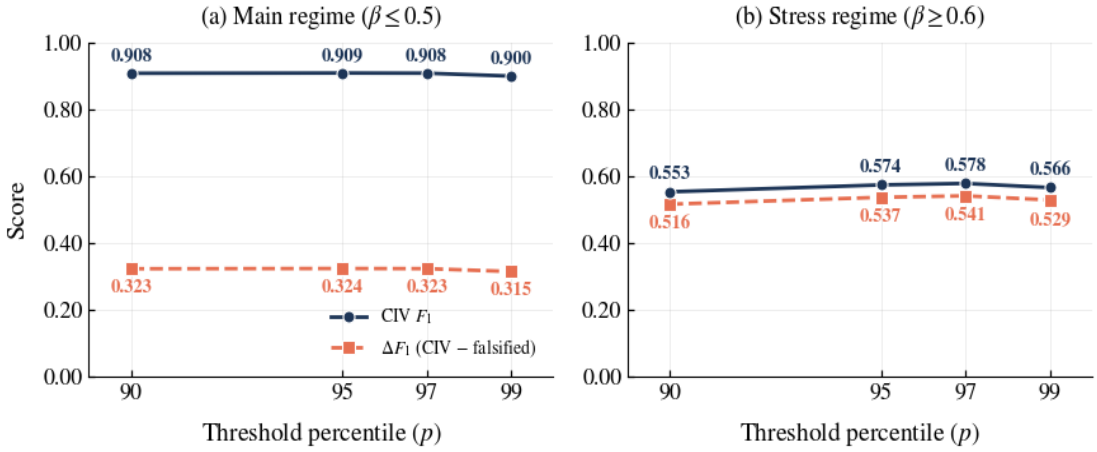

Figure 7: Sensitivity of CIV performance to threshold percentile p ∈ {90, 95, 97, 99}.

As shown in Fig. 7, the performance variation across different threshold percentiles is limited within the range of $p \in \{90,95,97,99\}$. Specifically, in the main regime ($\beta \leq 0.5$), the F1-score remains between 0.900 and 0.909, with a variation of only 0.009. In the stress regime ($\beta \geq 0.6$), the F1-score ranges from 0.553 to 0.578, corresponding to a variation of 0.025. These results indicate that the overall performance is not highly sensitive to the choice of threshold percentile.

The relative performance gain of CIV, measured by ΔF_1, is larger in the stress regime, ranging from 0.516 to 0.529, compared to the main regime. This indicates that performance degradation due to falsified input is more pronounced under high-compromise conditions, and that CIV provides substantial improvement by mitigating the impact of input contamination.

In addition, both $p = 95$ and $p = 97$ show stable performance across the main regime ($\beta \leq 0.5$) and the stress regime ($\beta \geq 0.6$). Based on these results, we select $p = 95$ as the default setting in this study.

In conclusion, the ablation results demonstrate that the key design choices of CIV are empirically well-founded. In particular, feature-wise standardized discrepancy and hard exclusion show consistent performance in suppressing input contamination, and the threshold percentile remains relatively robust within a reasonable range.

6.6 Computational and Storage Overhead

Metric	CIV
Mean latency per window	5.174ms
95th percentile latency per window	5.608ms
Additional verification cache	5.75KB

Table 1: Runtime latency and additional storage overhead of CIV

To evaluate the practical overhead of CIV, we measured its latency and additional verification cache under a representative setting of α=0.8 and β=0.5. The measurement was conducted on 1,620 test windows using 5 warm-up runs followed by 100 measured repetitions. As shown in Table 1, CIV required 5.174 ms per window on average and 5.608 ms at the 95th percentile. CIV also required only 5.75 KB of additional verification cache for storing the reference information used in report–observation discrepancy verification. These results show that the additional verification cost remains at the millisecond level with limited storage overhead in the evaluated environment. However, its feasibility on resource-constrained edge or gateway hardware requires further implementation-based validation.

7 Discussion and open issues

This study demonstrates that the CIV framework can effectively address report falsification in IoT environments; however, several important limitations remain regarding its applicability and experimental setup.

7.1 Operational scope and gateway trust assumptions.

CIV assumes that a sufficient number of valid reports remain after filtering in each aggregation window to sustain aggregation. As observed in §6.4, operational coverage remains high up to $\beta = 0.8$, but decreases to 80.3% at $\beta = 0.9$ under extreme-compromise conditions, where fallback becomes increasingly frequent. This suggests that the performance degradation in this regime is not primarily due to a specific design failure, but rather reflects a structural limitation in which aggregation itself becomes difficult to sustain reliably. Therefore, under extreme-compromise conditions, CIV alone may not provide sufficient defense, and additional sources of integrity evidence or complementary verification mechanisms are required.

In addition, CIV assumes that gateway observations remain independent of node-side report manipulation and that the gateway-to-edge reporting channel is integrity-protected. If the gateway itself is compromised or the gateway-to-edge reporting channel is tampered with, the verification reference may become unreliable. Such scenarios fall outside the current threat model and may require complementary protection mechanisms.

7.2 Dataset and IDS model specificity.

The experiments in this study are limited to the TON_IoT network-flow subset and a Random Forest-based IDS. Although the core concept of CIV is not directly tied to a specific IDS architecture, the effectiveness of aggregation-based defense may vary depending on the classifier's decision

boundary and its sensitivity to the input distribution. For example, in anomaly-based IDSs or deep learning-based classifiers, the same level of contamination suppression may lead to different changes in detection performance. Therefore, the generalizability of CIV across diverse IDS families and datasets requires separate systematic validation. In addition, the current evaluation assumes that gateway observations are sufficiently aligned with node-level reports and provide a reliable external reference. In real deployments, measurement noise, partial visibility, or routing asymmetry may distort report–observation alignment and affect discrepancy estimation; evaluating CIV under noisy or partially observable gateway settings remains future work. Furthermore, the current logical-node emulation does not explicitly capture multiple gateways, node mobility, NAT-induced device identity ambiguity, or heterogeneous device behavior; validating CIV under these deployment factors remains future work.

7.3 Static threshold calibration.

In this study, the threshold τ is estimated once from benign data in the reference estimation interval and then used as a fixed value. This design may be bypassed if an attacker adjusts the falsification strength below the threshold or gradually induces drift in the reference statistics over time, thereby weakening the verification criterion itself. While the proposed attack model suggests an inherent trade-off between reducing the report–observation discrepancy and maintaining IDS-evasion effectiveness, adaptive adversarial strategies designed to remain below the threshold are not systematically evaluated in the current study.

In addition, under extreme-compromise conditions, filtering alone may be insufficient, and additional integrity evidence or complementary verification mechanisms may be required.

8 Conclusion

This study showed that, in distributed IDS environments, node-level report falsification can distort the input representation of aggregation-based IDSs and substantially degrade detection performance, even without directly modifying the underlying network traffic. To mitigate this problem, this study proposed the Content Integrity Verification (CIV) framework, which uses gateway-level observations as an independent reference. CIV is distinguished from existing Byzantine-robust aggregation methods in that it does not attempt to recover corrupted information or modify the detection model itself; rather, it provides a report integrity verification-based defense that preserves attack-related signals contained in honest reports by suppressing the influence of manipulated reports at the input stage.

Experiments on the TON_IoT dataset showed that CIV achieved strong performance recovery under low- and moderate-compromise conditions. CIV recorded higher F1-scores than four representative Byzantine-robust baselines in 48 out of 60 baseline-comparison α–β settings (80.0%). In addition, we observed a structural trade-off: as the falsification strength increases, the potential for IDS evasion increases, but the discrepancy from gateway-level observations becomes larger, thereby increasing the likelihood of detection by CIV.

At the same time, this study clearly identifies the operational boundary of CIV. In high-compromise regimes, the honest-only reference also shows performance degradation, suggesting that the limitation in this regime stems not from the failure of a specific method, but from the structural reduction in the number of valid reports available for aggregation. Nevertheless, even under these conditions, CIV maintains superior performance compared with existing Byzantine-robust aggregation baselines and outperforms the honest-only reference in some high-compromise regimes.

In summary, this study empirically demonstrates that report integrity verification should be treated as an independent research problem in distributed IDS environments. The results further suggest that

gateway-level observations can serve as an effective external reference for evaluating whether node-level reports faithfully reflect underlying network activity. Future work will investigate integration with active verification, robustness under more realistic noise conditions, and generalizability across diverse IDS models and deployment settings. More broadly, CIV contributes to trustworthy distributed monitoring in IoT-enabled digital infrastructure by improving the reliability of security information used in edge-assisted IDS operations.

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Think Where It Matters: Role-Specific Thinking Mode Allocation in Multi-Agent LLM Debate*

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Abstract

Reasoning-capable large language models and multi-agent debate systems have each advanced AI problem-solving, yet their intersection remains unexplored. We present a systematic ablation study of role-specific thinking mode allocation in a three-agent (Proposer, Critic, Judge) debate framework. Evaluating all eight thinking-mode configurations across three model families (Gemini 2.5 Flash, Claude Sonnet 4, OpenAI o4-mini) and three benchmarks (GSM8K, MATH-500, GPQA-Diamond), we demonstrate that the all-ON configuration is suboptimal in most settings. Selective thinking allocation not only reduces cost but can improve accuracy: on MATH-500, OpenAI's Judge-only thinking achieves 81.0% versus 78.0% for all-ON, while costing only 1.09x the baseline. Our results reveal that optimal allocation depends on model family, task difficulty, and agent role, with verification roles (Critic, Judge) generally benefiting more than the generation role (Proposer). These findings suggest that reasoning capacity should be treated as a selectively allocable resource, particularly in operational settings where cost, reliability, and degraded operating conditions matter. We provide practical guidelines for cost-efficient and cyber-resilience-aware deployment of thinking-enabled multi-agent systems.

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1 Introduction

The emergence of reasoning-capable large language models has marked a significant shift in AI problem-solving. Models such as DeepSeek-R1 [1], OpenAI o1 and o3 [2], and Gemini 2.5 Flash [3] incorporate explicit chain-of-thought reasoning through thinking modes, allocating additional compute at inference time to produce intermediate reasoning traces. These modes have demonstrated remarkable improvements on challenging benchmarks in mathematics, science, and coding.

Concurrently, multi-agent systems (MAS) built on LLMs have gained traction as a means to improve output reliability. The debate paradigm, in which multiple agents argue for and against proposed solutions before a judge renders a verdict, has proven particularly effective [4, 5]. By exposing reasoning to adversarial critique, debate systems catch errors that single-agent approaches miss.

These converging trends necessitate a critical examination of role-specific reasoning allocation; specifically, whether the collective intelligence of a multi-agent debate system truly requires every participant to operate in thinking mode. While activating deep reasoning across all agents dramatically increases computational expenditure—often by a factor of three to five—the marginal utility of such uniform allocation remains empirically unverified. If a proposer requires deep reasoning to navigate a vast solution space, does a critic tasked with error detection necessitate the same depth? Similarly, does a judge's ability to synthesize perspectives depend more on its own internal reasoning or on the quality of the reasoning traces provided by the agents it evaluates? From a cyber resilience perspective, this allocation problem is especially important: operational AI systems may need to preserve reliable reasoning under resource constraints, degraded conditions, or adversarial pressure. Selectively assigning thinking capability to verification-oriented roles can provide a practical mechanism for maintaining decision quality without incurring the cost of full-system reasoning.

Despite rapid proliferation of both reasoning models and multi-agent frameworks, this question has received surprisingly little attention. Existing work on heterogeneous MAS [6, 7] focuses on mixing different model backbones rather than manipulating reasoning depth within a single model. Which specific roles within a MAS benefit most from thinking mode activation—and whether targeted allocation outperforms uniform activation—remains unexplored.

In this paper, we present a systematic ablation study of role-specific thinking allocation in a three-agent debate framework. We evaluate all eight possible thinking-mode configurations ($2^3=8$) across three model families and three benchmarks. Our contributions are: (1) the systematic ablation of role-specific thinking allocation in multi-agent debate; (2) cross-model validation across three model families, revealing model-dependent and difficulty-dependent optimal strategies; and (3) practical guidelines for cost-efficient deployment, showing that targeted allocation can substantially reduce costs while preserving or improving accuracy.

2 Related Work

2.1 Multi-Agent Debate

Du et al. [4] demonstrated that multi-LLM debate significantly improves factuality and reasoning. Subsequent work explored various topologies: Liang et al. [5] introduced divergent thinking structures, while Chan et al. [8] examined multi-agent debate for improved LLM-based evaluation. CONSENSAGENT [9] addressed sycophancy in multi-agent debate, and DiMo [10] explored diverse reasoning strategies using prompt-based approaches.

2.2 Thinking Models and Reasoning

The thinking model paradigm, popularized by OpenAI's o1 [2], demonstrates that allocating additional inference-time compute to explicit reasoning chains yields substantial gains. This has been adopted across providers: DeepSeek-R1 [1] introduced open-weight reasoning models, Gemini 2.5 Flash [3] offers configurable thinking budgets, and Claude models [11] support extended thinking with controllable token budgets. A key shared feature is the ability to toggle thinking within the same checkpoint, enabling controlled experiments.

2.3 Heterogeneous Multi-Agent Systems

X-MAS [6] investigates heterogeneous LLM backbone assignment. MasRouter [7] dynamically routes queries to different model tiers. MALBO [12] frames MAS design as Pareto optimization over cost and performance. While these approaches manipulate the model backbone, our work manipulates reasoning depth within a single model—a complementary and previously unexplored dimension.

2.4 Thinking Modes in Multi-Agent Settings

While thinking models and multi-agent debate have each been studied extensively, their intersection remains largely unexplored. Existing multi-agent frameworks uniformly apply the same reasoning mode to all participants, treating thinking as a system-wide toggle. Our work introduces a finer-grained dimension: which specific roles benefit from thinking, and whether targeted allocation is more efficient than the all-or-nothing approach.

3 Method

3.1 Debate Architecture

We employ a three-agent, two-round debate structure. The Proposer receives the problem and generates an initial solution with step-by-step reasoning. The Critic examines the Proposer's answer, identifies errors, and suggests corrections; if correct, it provides independent verification. In round two, the Proposer revises its answer incorporating feedback, and the Critic re-evaluates. Finally, the Judge synthesizes the complete exchange and renders a final answer. This yields five API calls per debate (Proposer and Critic each twice, Judge once).

3.2 Thinking Mode Configurations

Each agent possesses a binary thinking mode state where ON denotes that the thinking mode is enabled and OFF denotes it is disabled. Consequently, the configuration space for the entire three-agent system yields $2^3 = 8$ possible configurations, labeled C0 through C7 (Table 1). For instance, configuration C5 indicates that the Proposer and Judge utilize thinking mode for deep reasoning, while the Critic operates in standard generation mode. Within this operational space, C0 (all OFF) serves as the system baseline, and C7 (all ON) represents the upper bound of computational cost. These intermediate configurations (C1–C6) allow us to isolate the specific contribution of each role's reasoning process to the collective performance and identify the most cost-efficient allocation strategies.

Table 1. Thinking Mode Configurations

Config	Proposer	Critic	Judge	Label
C0	OFF	OFF	OFF	All OFF
C1	ON	OFF	OFF	P only
C2	OFF	ON	OFF	Cr only
C3	OFF	OFF	ON	J only
C4	ON	ON	OFF	P+Cr
C5	ON	OFF	ON	P+J
C6	OFF	ON	ON	Cr+J
C7	ON	ON	ON	All ON

3.3 Models

We evaluate three model families, each offering native thinking mode control within a single checkpoint. Gemini 2.5 Flash supports a configurable thinking_budget parameter (0 = OFF, 8192 = ON). Claude Sonnet 4 offers an extended thinking parameter that can be enabled with a budget of 8,192 tokens or omitted. OpenAI o4-mini provides a reasoning_effort parameter ("high" = ON, "low" = reduced); notably, o4-mini cannot fully disable reasoning. Using three models from distinct providers strengthens generalizability of observed patterns.

3.4 Benchmarks

We select three benchmarks spanning a wide difficulty spectrum. GSM8K (Easy) consists of 200 randomly sampled (seed=42) grade-school math problems from the test set [13]. MATH-500 (Hard) comprises 200 sampled competition-level mathematics problems [14, 15]. GPQA-Diamond (Expert) uses all 198 graduate-level multiple-choice science questions, where domain experts achieve ~65% accuracy [16]. This spectrum allows examination of how thinking allocation interacts with task complexity.

3.5 Evaluation Metrics

Our primary metric is accuracy via exact match. For GSM8K, we extract numerical values; for MATH-500, we normalize LaTeX expressions; for GPQA-Diamond, we match selected options (A/B/C/D).

We additionally track total token consumption as a cost proxy, including input tokens, output tokens, and thinking tokens when enabled. To ensure comparability across configurations, cost is normalized by the all-OFF (C0) configuration within each model. We compute cost-efficiency as accuracy gain per unit cost increase over the all-OFF baseline.

3.6 Experimental Controls

We hold constant the problem set, ordering, prompt text, debate structure, answer extraction logic, and thinking token budget across all configurations and models. Temperature is set to 0 where the API permits. Where temperature cannot be fixed (Claude with thinking enabled), we apply the same default across all configurations for that model.

4 Results

4.1 RQ1: Which Roles Benefit Most from Thinking?

Tables 2–4 report accuracy across all eight configurations for each model–benchmark pair.

Table 2. Accuracy (%) on GSM8K (Easy)

Config	Gemini	Claude	OpenAI
C0 (All OFF)	96.0	96.5	95.5
C1 (P)	96.5	96.0	96.0
C2 (Cr)	96.0	95.5	95.5
C3 (J)	96.5	96.0	96.5
C4 (P+Cr)	97.0	96.5	96.0
C5 (P+J)	97.0	95.5	96.5
C6 (Cr+J)	96.5	96.0	96.0
C7 (All ON)	97.5	95.5	95.5

Table 3. Accuracy (%) on MATH-500 (Hard)

Config	Gemini	Claude	OpenAI
C0 (All OFF)	90.0	83.5	69.5
C1 (P)	90.5	68.5	74.0
C2 (Cr)	90.0	88.5	73.5
C3 (J)	90.5	84.0	81.0
C4 (P+Cr)	91.5	85.0	76.0
C5 (P+J)	91.0	72.0	78.5
C6 (Cr+J)	90.5	87.5	78.5
C7 (All ON)	91.0	86.0	78.0

Table 4. Accuracy (%) on GPQA-Diamond (Expert)

Config	Gemini	Claude	OpenAI
C0 (All OFF)	78.7	74.2	73.2
C1 (P)	80.3	76.7	75.2
C2 (Cr)	80.3	78.2	76.7
C3 (J)	79.2	77.7	76.2
C4 (P+Cr)	80.8	79.2	79.7
C5 (P+J)	79.7	76.2	77.2
C6 (Cr+J)	79.7	80.3	78.2
C7 (All ON)	80.3	82.3	76.7

On GSM8K, thinking allocation has minimal impact across all models, with accuracy ranges of at most 3.0 percentage points (pp). No configuration significantly outperforms the all-OFF baseline, confirming that thinking mode on easy tasks often represents pure overhead and that reasoning is unnecessary for tasks already solvable by base model competence.

Conversely, the performance dynamics shift significantly on the MATH-500 benchmark. For OpenAI o4-mini, Judge-only thinking mode (C3) achieves 81.0%, outperforming the all-ON configuration (C7, 78.0%) by 3.0pp—a striking case where less thinking yields better performance. This counter-intuitive result suggests that for certain architectures, excess reasoning tokens may introduce stochastic noise rather than logical clarity. For Claude, Critic-only thinking (C2) achieves 88.5% (+5.0pp over C0), while Proposer-only thinking (C1) causes a severe 15.0pp degradation to 68.5%. For Gemini, the Proposer+Critic combination (C4) achieves the highest accuracy of 91.5%.

On GPQA-Diamond, the most challenging benchmark, Claude shows the widest range (8.1pp), with the all-ON configuration (C7) achieving the best result at 82.3%. OpenAI shows a 6.5pp range, with the Proposer+Critic (C4) peaking at 79.7%. Gemini remains remarkably stable (2.6pp range).

4.2 RQ2: Does Optimal Strategy Vary with Difficulty?

A clear interaction between task difficulty and optimal thinking allocation emerges. On easy tasks (GSM8K), thinking provides no consistent benefit; thus, C0 represents the most cost-efficient choice. On hard tasks (MATH-500), the mean accuracy gap between the worst and best configurations is 9.5pp (C1: 76.0% to C6: 85.5%). Critically, C7 (all-ON, 85.0%) is outperformed by partial configurations, notably C6 (Cr+J, 85.5%). On expert tasks (GPQA-Diamond), while more thinking generally helps, the cross-model mean of C4 (P+Cr, 80.1%) marginally outperforms that of C7 (79.7%) while incurring only two-thirds of the thinking cost.

4.3 RQ3: Cross-Model Comparison

Optimal thinking allocation is strongly model-dependent. Table 5 summarizes the optimal configuration by model and difficulty.

Table 5. Optimal Configuration by Model and Difficulty

Model	Easy (GSM8K)	Hard (MATH-500)	Expert (GPQA)
Gemini 2.5 Flash	C7 (97.5%)	C4 (91.5%)	C4 (80.8%)
Claude Sonnet 4	C0/C4 (96.5%)	C2 (88.5%)	C7 (82.3%)
OpenAI o4-mini	C3 (96.5%)	C3 (81.0%)	C4 (79.7%)

Gemini 2.5 Flash exhibits strong baseline performance (90.0% on MATH-500, 78.7% on GPQA without thinking), limiting the headroom for improvement. Its best configuration (C4) on MATH-500 yields only +1.5pp over C0, suggesting that Gemini's base model already incorporates sufficient implicit reasoning.

Claude Sonnet 4 shows the most dramatic sensitivity. On MATH-500, Proposer-only thinking (C1) causes a catastrophic 15.0pp drop. We hypothesize that this degradation reflects a mismatch between reasoning verbosity and downstream evaluative capacity. A thinking mode-enabled Proposer produces substantially longer reasoning traces, which a non-thinking Critic may fail to effectively process or critique, thereby weakening the adversarial feedback loop.

OpenAI o4-mini shows a distinctive Judge-centric pattern. On MATH-500, Judge-only thinking (C3) achieves 81.0%—outperforming both C0 (+11.5pp) and C7 (+3.0pp). This is the single most cost-efficient configuration in our study: thinking in only one of five agent invocations yields the highest accuracy.

4.4 All-ON Suboptimality

A central finding is that C7 (all-ON) is suboptimal in the majority of settings. In five of six non-trivial model–benchmark pairs (excluding GSM8K), at least one partial configuration outperforms C7 (Table 6). The sole exception is Claude on GPQA-Diamond.

Table 6. Cases Where Partial Thinking Outperforms C7

Model	Benchmark	Best Config	Acc (%)	C7 Acc (%)
Gemini	MATH-500	C4 (P+Cr)	91.5	91.0
Gemini	GPQA	C4 (P+Cr)	80.8	80.3
Claude	MATH-500	C2 (Cr)	88.5	86.0
OpenAI	MATH-500	C3 (J)	81.0	78.0
OpenAI	GPQA	C4 (P+Cr)	79.7	76.7

4.5 Cost-Efficiency Analysis

The practical significance of role-specific thinking is best understood through a cost-accuracy trade-off analysis.

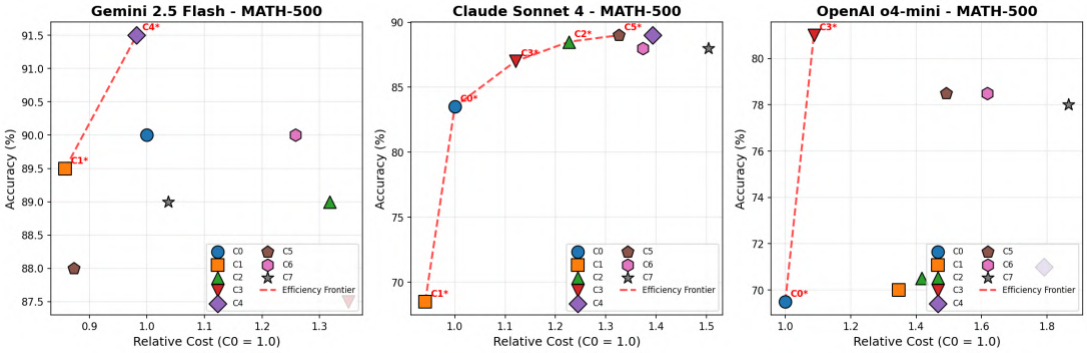

Figure 1. Cost-accuracy trade-off by model on MATH-500.

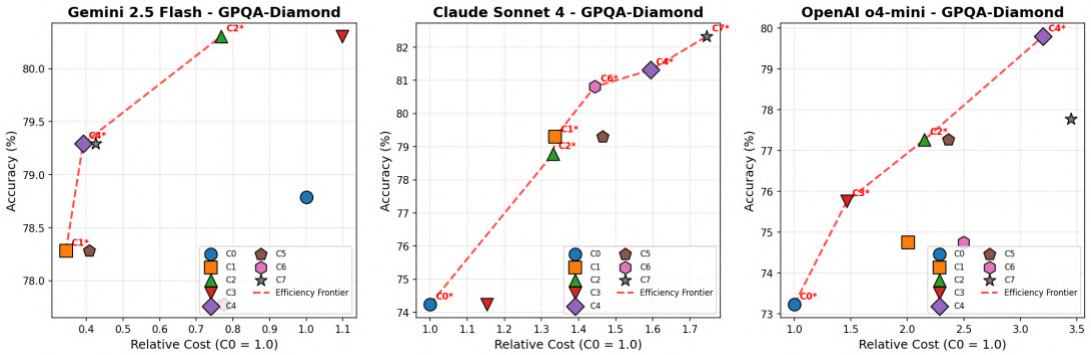

Figure 2. Cost-accuracy trade-off by model on GPQA-Diamond.

Figure 1 illustrates this relationship for the MATH-500 benchmark. The visualization highlights the "efficiency frontier"—the set of configurations that deliver the highest accuracy for a given cost. For Gemini, C4 (Proposer+Critic) emerges as the dominant strategy, achieving peak accuracy (91.5%) with virtually no cost increase (0.98x) over the baseline. Claude follows a similar trajectory, where C4 provides the best performance-to-cost balance.

A standout finding is observed in OpenAI o4-mini: the C3 (Judge-only) configuration yields a dramatic 11.5 percentage point improvement at a marginal 1.09x cost. This suggests that for certain architectures, a single, high-quality reasoning step at the synthesis stage is far more effective than distributing reasoning across all agents.

Figure 2 shifts the focus to the expert-level GPQA-Diamond benchmark, where the efficiency patterns change significantly. Unlike the math tasks, expert-level scientific reasoning appears to demand a higher “thinking density.” For instance, Claude reaches its peak performance (82.3%) only in the C7 (All-ON) configuration. This indicates that as task complexity scales from competition math to graduate-level science, the system requires more comprehensive reasoning from all participants to resolve deep ambiguities, even if it entails higher costs.

As summarized in Table 7, these results demonstrate that “thinking more” is not always better. By identifying model-specific sweet spots—like OpenAI’s Judge-centric efficiency or Claude’s need for All-ON in expert tasks—practitioners can bypass the diminishing returns of a uniform All-ON approach.

Table 7. Cost-Efficiency on MATH-500 (Mean)

Config	Mean Acc (%)	Mean Cost (x)	Acc/Cost
C0 (All OFF)	81.0	1.00x	baseline
C3 (J only)	85.2	1.09x	46.7
C6 (Cr+J)	85.5	1.67x	6.7
C4 (P+Cr)	84.2	1.65x	4.9
C7 (All ON)	85.0	2.75x	2.3

5 Discussion

5.1 Mechanisms of Suboptimality

We hypothesize that the underperformance of the ‘All-ON’ (C7) configuration stems from Reasoning Anchoring. When a Proposer generates an excessively verbose, thinking-augmented response, it may inadvertently ‘anchor’ the subsequent Critic. A Critic operating under the influence of such a dense reasoning chain may struggle to maintain the adversarial independence required to spot subtle logical fallacies, essentially being ‘led’ by the Proposer’s logic.

To provide empirical evidence for this hypothesis, we analyzed experimental logs where C0 (All-OFF) succeeded but C1 (Proposer-only thinking) failed. In Gemini’s GPQA results, we identified 15 such cases. For instance, in problem #18 involving organic chemistry (Michael reaction), C0 correctly predicted answer ‘C’, while C1 incorrectly predicted ‘A’. The C1 Proposer generated an elaborate analysis of regioselectivity, but made an error in interpreting steric hindrance effects. The Judge acknowledged that "the Proposer initially made an error," yet ultimately followed the Proposer’s flawed reasoning chain. Similarly, in problem #29 (photochemistry), the C1 Proposer’s excessive Stokes shift analysis led to answer ‘C’ instead of the correct ‘A’. These cases demonstrate that verbose extended thinking can anchor subsequent agents to flawed reasoning paths.

Furthermore, we identify Reasoning Redundancy as a primary cost-driver with diminishing returns. In many-agent systems, if every participant independently reconstructs the same underlying logic, the Judge is presented with a redundant information landscape that complicates synthesis rather than clarifying it. This explains why the Judge-only (C3) configuration in models like o4-mini often yields superior results: a single, high-fidelity reasoning step at the final decision point is often sufficient to resolve ambiguities without the noise of redundant intermediate traces.

Quantitative analysis supports this observation. On GSM8K, enabling thinking for all agents (C7) increased costs by 229% for Gemini (from $0.0005 to $0.0017 per problem) while improving accuracy by only 1.5 percentage points. For Claude, the 68% cost increase actually resulted in a 1.0 percentage point accuracy decrease (96.5% to 95.5%), and OpenAI showed zero accuracy gain despite 158% higher costs. Interestingly, on GPQA, the optimal thinking allocation varies significantly across models. Claude benefits from All-ON (C7), achieving 82.3% accuracy, suggesting that its extended thinking effectively handles complex scientific reasoning. In contrast, Gemini achieves its best performance (80.3%) with C2 (Critic-only thinking), and OpenAI peaks at 79.8% with C4 (Proposer+Critic). This divergence suggests that for Gemini and OpenAI, the Judge benefits from receiving cleaner, less verbose inputs from the debate, rather than processing additional thinking tokens.

5.2 Asymmetric Value of Thinking Mode by Role

Verification roles (Critic and Judge) generally benefit more from thinking than the generation role (Proposer). This aligns with complexity-theoretic arguments: verifying a solution is computationally easier than generating one, but verification quality depends critically on analysis depth. A thinking Critic more effectively identifies subtle errors, while a thinking Judge more carefully weighs conflicting arguments. Conversely, the Proposer’s task may not require the same explicit reasoning depth, particularly when the model’s implicit capabilities are strong.

5.3 Cost-Accuracy Trade-off Landscape

Figures 3 and 4 provide a unified visualization of the cost-accuracy trade-offs across all three models on MATH-500 and GPQA-Diamond benchmarks. This visualization reveals distinct model characteristics: Gemini clusters in the upper region regardless of configuration, indicating robust

baseline performance and lower marginal gains from thinking mode. Claude shows moderate sensitivity to thinking allocation, while OpenAI demonstrates the widest accuracy range (69.5% to 81.0%), suggesting high sensitivity to strategic thinking placement.

The combined analysis supports our key finding: optimal thinking allocation is model-dependent. Practitioners should consider both the target model's characteristics and the task difficulty when selecting configurations.

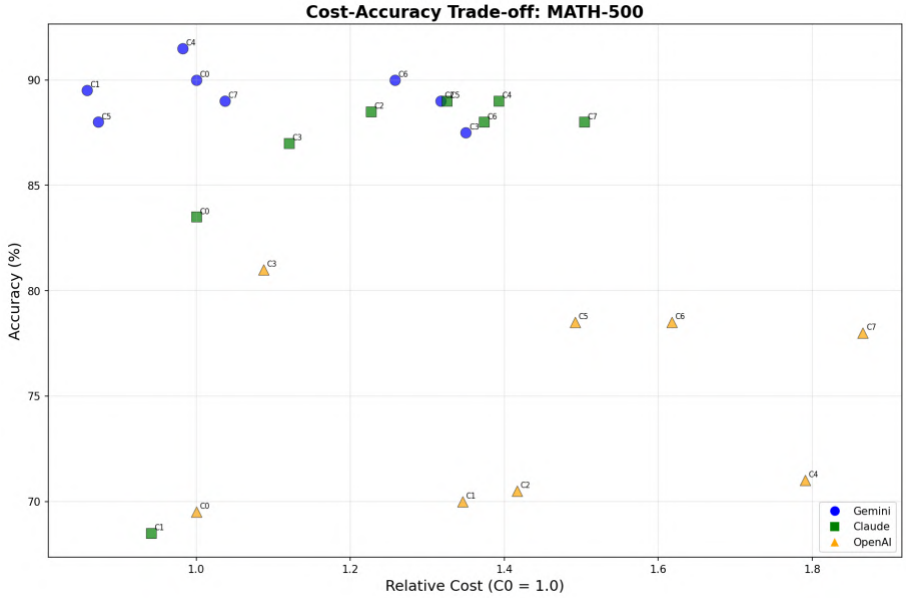

Figure 3. Unified cost-accuracy trade-off across all models on MATH-500.

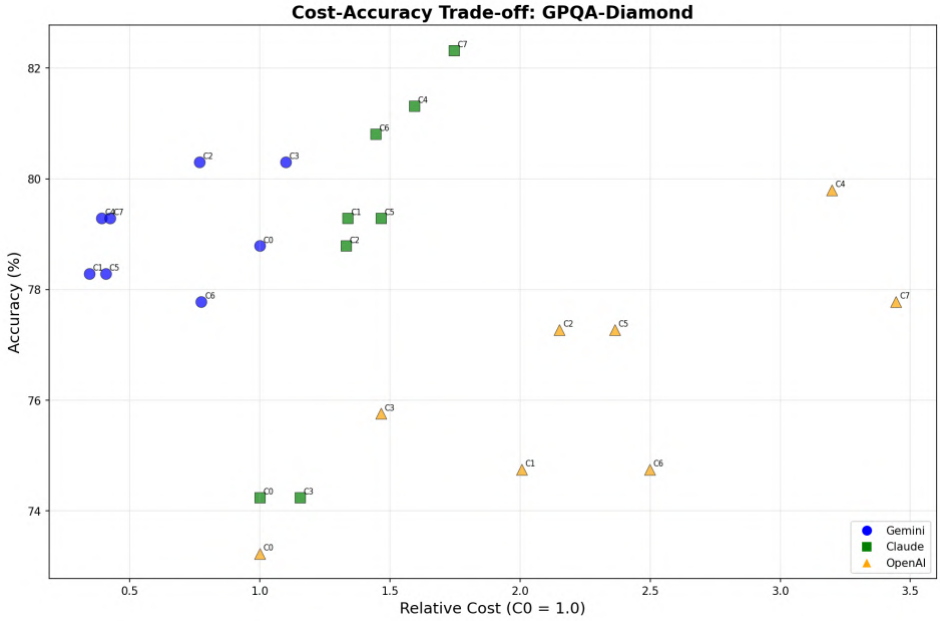

Figure 4. Unified cost-accuracy trade-off across all models on GPQA-Diamond.

5.4 Practical Guidelines

Based on our empirical results, the most effective deployment strategies require a nuanced approach rather than a one-size-fits-all solution. For easy tasks where baseline accuracy exceeds 95%, activating thinking mode is generally counterproductive, as it increases latency and cost without meaningful performance gains; in these cases, thinking should be disabled entirely.

When tackling hard tasks, the optimal strategy diverges by model. For OpenAI o4-mini, the C3 (Judge-only) configuration offers exceptional efficiency, delivering peak performance with minimal additional expenditure. In contrast, for Gemini and Claude, the C4 (Proposer+Critic) combination is recommended to best balance reasoning depth with robust verification. As tasks scale to expert-level complexity (e.g., GPQA-Diamond), the C4 (Proposer+Critic) configuration remains a strong cross-model performer, providing the necessary reasoning density to resolve graduate-level scientific ambiguities.

Ultimately, practitioners should move away from the assumption that the All-ON (C7) configuration is inherently superior. Instead, they should actively evaluate partial thinking allocations to identify the Pareto-optimal point that best fits their specific model, budget, and task complexity.

6 Limitations and Future Work

Our study has several limitations. First, all experiments are single-run without repeated trials, so variance estimates are unavailable. While sample sizes (198–200 problems per cell) provide reasonable power for detecting large effects, small inter-configuration differences should be interpreted cautiously. Future work should include repeated trials with statistical significance testing to validate the robustness of observed differences. Second, OpenAI o4-mini cannot fully disable reasoning; its "low" setting represents reduced rather than absent thinking, making C0 for o4-mini not strictly comparable to the fully-off configurations of Gemini and Claude. Third, our debate structure is fixed at three agents and two rounds; different topologies and round counts may interact differently with thinking allocation. Fourth, the thinking budget is fixed at 8,192 tokens; varying it continuously could reveal more nuanced trade-offs. Fifth, we evaluate only reasoning-intensive benchmarks; applicability to creative writing, coding, or open-ended QA remains unexplored.

Future work should address these limitations through repeated trials with variance estimation, continuous thinking budget exploration, diverse debate topologies, and broader task types. Automated methods for selecting optimal thinking allocations based on difficulty estimation or model confidence represent a promising practical direction.

7 Conclusion

We have presented the systematic study of role-specific thinking mode allocation in multi-agent LLM debate. Through exhaustive evaluation of eight thinking configurations across three model families and three benchmarks, we demonstrate that selective thinking allocation is not merely a cost-saving heuristic but a genuine performance optimization. Our key finding—that the all-ON configuration is suboptimal in most settings—challenges the assumption that more reasoning is always better in multi-agent systems. The optimal allocation depends on a complex interaction between model family, task difficulty, and role within the debate. For practitioners, our results offer immediately actionable guidance: disable thinking on easy tasks, start with Judge-only or Critic+Judge configurations on hard tasks, and always validate against partial configurations before committing to the all-thinking mode default.

Beyond cost efficiency, these findings have implications for cyber-resilient AI system design. In real-world cyber environments, AI systems may operate under constrained budgets, degraded infrastructure, adversarial inputs, or incident-response pressure. Our results suggest that resilience should not be equated with maximizing reasoning everywhere. Instead, cyber-resilient multi-agent systems may benefit from preserving deeper reasoning in roles responsible for verification and final adjudication, such as the Critic and Judge, while avoiding unnecessary reasoning redundancy across all agents. By thinking where it matters, multi-agent systems can achieve comparable or superior accuracy at dramatically reduced cost.

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Automatic Attack Scenario Reconstruction with Large Language Models: Threat Intelligence Extraction Strategies and Techniques*

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Abstract

Recently, the application of large language models (LLMs) has been gradually extended to the information security domain to assist security professionals in vulnerability identification, classification, and penetration testing. However, research on reproducing the entire cyber kill chain remains relatively scarce, with few public studies available due to constraints related to commercial interests and technical thresholds. This paper proposes a system architecture that systematically divides the process of attack reproduction into three stages: threat intelligence organization, scenario construction, and attack script generation. The system is built on open-source software, integrating the functions of LLMs and virtual machine construction, allowing users to reproduce attack processes recorded in threat intelligence through a visualized web interface in a unified platform, gaining a deeper understanding of the sequence of attack events. With the aid of an LLM, this paper successfully applies the system to reproduce common vulnerabilities and exposures (CVE) attack scenarios and the attack techniques of advanced persistent threat (APT) organizations. While user intervention remains necessary in certain stages, including supplementing publicly available exploit references and correcting script errors through iterative feedback, the system significantly reduces the time and expertise required compared to traditional manual reproduction workflows.

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1 Introduction

With the rapid evolution of information technology, cyberattack methods have become increasingly sophisticated. Attackers frequently target critical infrastructure of government agencies and enterprises [1], and their operations (driven by hacker organizations, transnational criminal groups, or even state-level actors) can cause severe disruptions to essential services. According to IBM's 2025 Cost of a Data Breach Report [2], the mean time to identify and contain a breach has decreased to 241 days, a nine-year low and a 17-day reduction from the previous year, largely driven by the extensive adoption of AI and automation in security detection and containment. These evolving attack methods can bypass traditional firewalls and anti-virus software, enabling attackers to gain system control and steal sensitive data, while victimized organizations bear massive incident-handling costs.

To counter these threats, organizations must establish robust cybersecurity defense mechanisms, including regular training to enhance personnel awareness and cultivating dedicated cybersecurity professionals. However, developing such expertise typically requires years of learning and practical experience. Furthermore, cybersecurity researchers must invest significant time interpreting threat intelligence from diverse sources to reconstruct the full picture of an attack. With the development of large language model (LLM) technology, its powerful knowledge integration and natural language processing capabilities can assist this process, potentially reducing analysis work from days or weeks to hours.

This paper proposes a system that utilizes LLMs to systematically organize and analyze threat intelligence. By designing standardized interpretation guidelines, the system enables LLMs to accurately interpret intelligence content and transform the results into automated environment deployment scripts and attack scripts, significantly reducing the technical threshold and complexity of attack reproduction.

1.1 Contributions

This paper proposes a framework that utilizes LLMs to achieve automated processing and analysis of threat intelligence. This framework can effectively extract, organize, and transform threat intelligence into usable information, thereby automatically generating the scripts required for field construction and deployment. The contributions of this paper are as follows:

- **Using LLM to Construct an LLM-Assisted Indicator of Compromise (IoC) Extraction Mechanism**. Threat intelligence obtained through various channels, such as advanced persistent threat (APT) reports from security vendors, Threat Hunting results, and attack methods shared on cybersecurity forums, is imported into the LLM. Through prompts designed by this paper, the LLM can transform unstructured threat intelligence into standardized analysis results. This not only enhances the usability of threat intelligence but also helps security analysts more effectively master the attackers' tactics, techniques, and procedures (TTPs), thereby constructing complete attack chain analysis reports.
- **LLM-Assisted Attack Scenario Construction Mechanism Based on Threat Intelligence**. Using the extracted IoCs as input, combined with prompts designed for virtualization platforms, the LLM generates corresponding environment setup scripts with minimal user guidance. This mechanism allows the system to interface with virtualization platforms and streamline the deployment of the required experimental environment, significantly reducing the time and labor costs for researchers to reproduce attack scenarios.
- **Scenario-Oriented LLM-Assisted Attack Script Generation Mechanism**. Through designed prompts, the LLM is guided to interpret key information in threat intelligence, including attack methods and victim targets. Simultaneously, by providing environment

information such as machine configurations and network architecture within the scenario, the LLM can leverage knowledge learned during its training phase, combined with scenario characteristics and attack requirements, to design automated attack scripts applicable to specific targets.

1.2 Structure

The remainder of this paper is organized as follows. Section 2 reviews related work, followed by an overview of the preliminary technologies in Section 3. Section 4 details the proposed system architecture and the attack reproduction process. In Section 5, we validate the feasibility of our methodology and provide a comparative analysis. Finally, Section 6 concludes the work and discusses future directions.

2 Related Works

2.1 Using Threat Intelligence as a Basis for Attack Identification

Cyber threat intelligence (CTI) plays a pivotal role in modern attack analysis, enabling both the prediction of threat patterns and the reconstruction of adversarial behavioral trajectories. Parmar and Domingo [3] leveraged CTI within the Coalition Warrior Interoperability eXercise (CWIX), utilizing the MITRE ATT&CK framework [4] to decode adversary behaviors and identify TTPs with higher fidelity. Mundt and Baier [5] further advanced this direction by extracting granular elements, such as malware signatures and mitigation strategies, to characterize the technical methodologies of sophisticated groups like APT29 and APT32, demonstrating the value of CTI in automated threat detection and intelligence-led defense.

2.2 Attack Simulation and Reproduction

By simulating adversarial methodologies, researchers can gain comprehensive visibility into attack lifecycles. Karagiannis et al. [6] developed the A-DEMO framework, integrating the MITRE ATT&CK matrix to document malware techniques and analyze execution patterns of Rootkit malware in a simulated medical environment, focusing on command and control (C&C), persistence, and exfiltration behaviors. Iturbe et al. [7] introduced a hybrid methodology combining threat modeling with attack simulation, constructing an Industrial IoT (IIoT) testbed to reproduce CVE-related vulnerability exploitation [8] stemming from misconfigurations. These approaches provide a critical foundation for understanding attacker TTPs and formulating proactive defense strategies.

2.3 Using LLMs to Assist Penetration Testing

Traditional penetration testing is labor-intensive and heavily reliant on human expertise. Recent advancements have demonstrated that LLMs can significantly augment this process. For instance, Happe et al. [9] introduced HackingBuddyGPT, a system designed to assist researchers in uncovering novel attack vectors via LLMs. Similarly, Deng et al. [10] developed PentestGPT, an interactive framework that employs a tripartite architecture: parsing, reasoning, and generation, to conduct autonomous testing. Their evaluation on platforms such as HackTheBox [11] and VulnHub [12] underscores the potential of LLMs in automating complex security assessments.

2.4 Potential of LLMs in Vulnerability Detection and Identification

LLMs have shown significant potential in vulnerability identification. Lin and Mohaisen [13] conducted a comprehensive evaluation of multiple LLMs, including LLaMA-2, Gemma, and GPT-4, for vulnerability detection, finding that while certain models demonstrate strong identification capabilities, performance varies considerably across configurations and programming languages. Lekssays et al. [14] proposed LLMxCPG, integrating Code Property Graphs with LLMs to achieve 15–40% improvements in F1-score over state-of-the-art baselines by reducing code to vulnerability-relevant slices. Sheng et al. [15] further surveyed the field, noting that LLM effectiveness in vulnerability analysis depends heavily on integration with program analysis techniques, structured prompt engineering, and domain-specific fine-tuning—rather than relying solely on the models' inherent capabilities.

3 Background

3.1 Integration of LLMs

3.1.1. Evaluation and Selection of LLMs

Work by Happe et al. [16] demonstrated that small language models cannot handle complex penetration tasks. We therefore evaluated four LLMs, GPT-4o, Claude 3 Opus, Llama3-70B, and Gemini 1.5 Pro, across three core security tasks, as summarized in Table 1. Among these, Llama3-70B failed to meet our requirements for vulnerability identification and attack script drafting. While the remaining three models performed adequately, GPT-4o was selected as the primary engine due to its superior multi-format input support, extended API capabilities including file search and code interpretation, and its Assistant mode that preserves dialogue context across sessions.

Table 1: Evaluation of Large Language Model Capabilities

Language Model	Vulnerability Identification	Virtual Scenario Configuration	Attack Script Drafting
GPT-4o	✓	✓	✓
Claude 3 Opus	✓	✓	✓
Llama3-70B	-	✓	-
Gemini 1.5 Pro	✓	✓	✓

3.1.2. Interfacing with OpenAI API

To ensure system stability and maintainability, our system interfaces with the OpenAI Assistants API through direct RESTful calls using PHP's cURL library, which supports multi-format threat intelligence input and preserves dialogue threads across sessions. Dedicated Pre-processing and Post-processing Modules (detailed in Section 4.2) are incorporated to reduce token usage and improve response times.

3.2 Construction of Virtualization Experimental Scenario

3.2.1. Virtualization Platforms

Since our objective requires a multi-user virtualization environment with customization flexibility, we evaluated four open-source platforms, as summarized in Table 2. VMware vSphere was excluded due to its closed-source nature. OpenNebula was selected as the primary solution based on its user-friendly interface, flexible architecture facilitating system integration, and robust API support. Although OpenStack offers higher functional completeness, its architecture proved too rigid for our customization requirements. CloudStack and Proxmox VE were too simplified for our advanced needs.

Table 2: Comparison of Open-Source Virtualization Platforms

Platform	**Key Characteristics**	**Limitations**
OpenNebula	Intuitive management interface; template-based component presentation; contextualization feature for guest VM communication	Smaller community than OpenStack
OpenStack	Modular architecture (Nova, Glance, Cinder, Neutron, Keystone); extensive API support and active community	Highly complex architecture
CloudStack	Simplified management; lightweight architecture suitable for SMEs	Overly simplified functionality
Proxmox VE	Combines KVM and LXC technologies; user-friendly web interface	Limited extensibility for advanced needs

3.2.2. Virtualization Network Architecture

To simulate realistic attack scenarios across multiple network segments, the system utilizes Open vSwitch (OVS) as the core virtual network switch for Layer 2 networking with VLAN isolation. OPNsense serves as the routing and firewall platform between virtual subnets, providing RESTful API support for automated rule management. User access to internal VMs is secured through OPNsense's built-in WireGuard VPN service, establishing encrypted tunnels for RDP and SSH connectivity.

4 System Design

This section presents the system architecture, the methodology for LLM-assisted threat intelligence analysis, and the mechanisms for LLM-assisted attack script generation.

The operational workflow is divided into three primary stages: threat intelligence organization, scenario construction, and attack script generation. As illustrated in Figure 1, the system follows a two-lane architecture. The left lane depicts the three-stage pipeline through which threat intelligence is progressively transformed into actionable attack simulations. Users begin by importing raw reports from diverse sources (e.g., security vendor publications, technical forums, or dark-web intelligence)

into Stage 1, where the IoC Extraction Module identifies structured indicators such as CVE/CWE identifiers, IP addresses, domain names, and APT lifecycle information. These structured outputs then flow into Stage 2, where the Machine Deployment Module provisions virtual machines and networks based on the extracted indicators, and the Vulnerability Environment Installation Module deploys specific vulnerable software versions via auto-generated scripts. Both modules interact with the OpenNebula and OPNsense virtual lab infrastructure for VM creation and script execution. If an installation script fails, the system supports an error feedback loop that resubmits the error message to the LLM for script regeneration. Once the environment is fully provisioned, Stage 3 invokes the Attack Module to synthesize scenario-aware exploit scripts targeting the deployed victims.

The right lane of Fig. 1 shows the shared LLM Engine that serves all three stages. When any module initiates a query (indicated by the dashed arrows labeled "prompt → JSON"), the Question Recommendation Module first suggests the appropriate query type based on the user's current progress. The Pre-processing Module then resolves internal function calls and attaches relevant domain knowledge (e.g., CWE documentation) to enrich the prompt. The assembled prompt is forwarded by the LLM Query Module to the OpenAI GPT-4o Assistants API via cURL HTTP requests. Upon receiving the response, the Post-processing Module validates and repairs the JSON output before rendering it through the analyst's web interface. All scenario data, VM states, and conversation threads are persisted in a MariaDB data store for continuity across sessions.

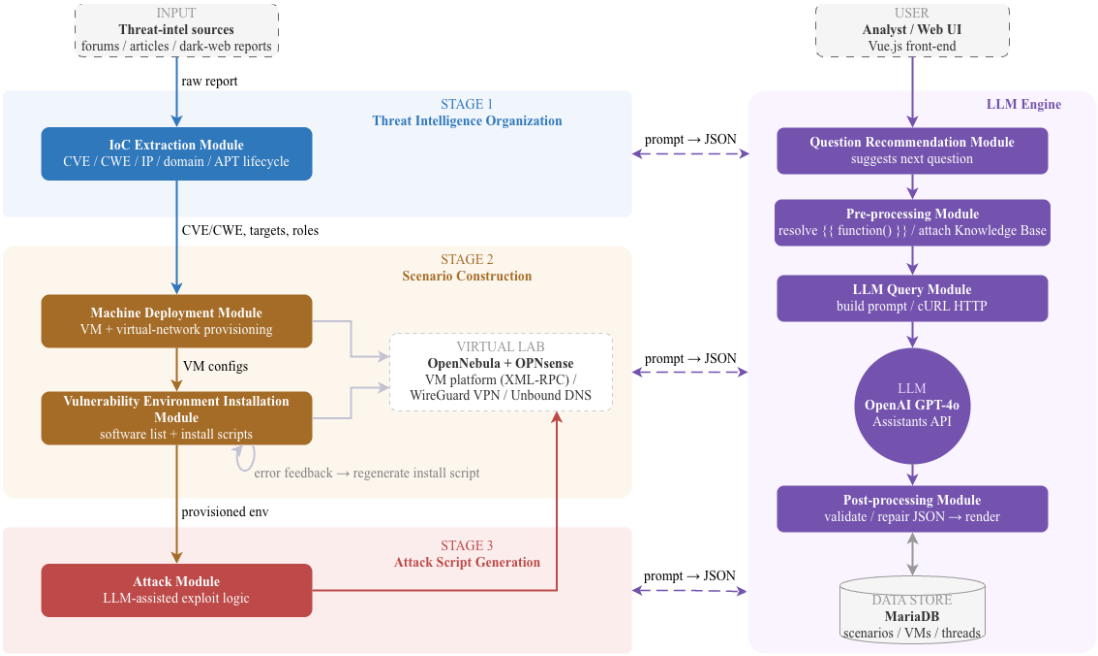

Figure 1: System data flow diagram

4.1 System Architecture

The proposed system is implemented as a web-based application, accessible through any internet-enabled device via a standard web browser. The system architecture leverages Docker containerization for efficient maintenance and updates. The backend utilizes Nginx as the web server, with core application logic developed in PHP using the Laravel framework. MariaDB serves as the primary database, while the frontend is built with Vue.js. For authentication, the system integrates with LDAP to synchronize credentials with the virtualization platform. OpenNebula was selected as the integrated

virtualization engine. The LLM Query Module communicates with the OpenAI GPT-4o API. The entire stack is constructed using freely available open-source software, eliminating licensing costs. The entire environment is composed of VM services and several microservices, utilizing OpenNebula v6.8.0, Laravel v11, PHP v8.3, Nginx v1.26, and MariaDB v11.3.

4.2 Integration with LLMs

The system adopts a modular design, with the LLM Query Module handling the interface to external AI models. By default, the system utilizes GPT-4o, interfacing with the OpenAI API via cURL-based HTTP requests. When a query is initiated, the module transmits structured prompts to the API server and retrieves generated responses for further parsing.

4.2.1. Prompt Engineering Framework

To ensure the LLM accurately understands research objectives and generates high-fidelity responses, we developed a structured prompt engineering framework. As shown in Figure 2, the framework comprises five stages:

- **Context Description.** Establishes the operational background, e.g., "The user needs to construct an environment to simulate an attacker and a victim; please assist in reproducing the vulnerability."
- **Task Specification.** Defines specific requirements, e.g., "Please install the identified vulnerability onto the victim machine within the scenario."
- **Response Format and Exemplar Data.** Provides few-shot exemplars to guide the LLM's response structure and style.
- **Field Descriptions for Response Format.** Defines data types, filling rules, and constraints for each JSON response field to ensure data integrity and consistency.
- **Standard Response Template (Empty).** Provides an unpopulated schema to help the LLM grasp structural relationships.

The ordering of stages three through five can be dynamically adjusted to achieve optimal performance.

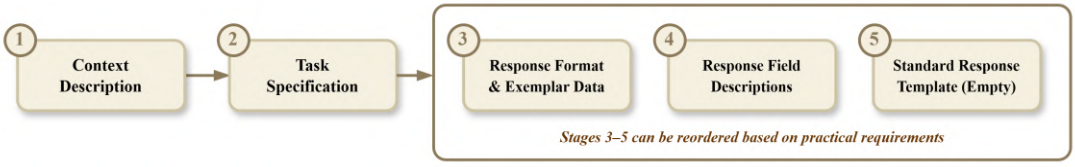

Figure 2: Prompt Engineering Framework

In addition to the core prompting logic, the system incorporates Pre-processing and Post-processing Modules. The LLM Query Module transmits content to these modules before initiating a query and after retrieving a response. Specifically, the Pre-processing Module performs prompt refinement to generate contextually appropriate queries, while the Post-processing Module conducts an initial analysis and validation of the returned results before presenting them to the user.

- **LLM Query Module**

To enable a seamless "one-click" query experience, the module incorporates a comprehensive internal question bank. Users can directly select specific query items, such as identifying CVE numbers from threat intelligence reports, without manually constructing complex prompts. Queries are categorized into four types: vulnerability identification, environment provisioning, vulnerability installation, and attack script synthesis.

The system also features a question recommendation module that suggests contextually relevant follow-up queries based on the user's current progress. For instance, after completing environment provisioning queries, the system automatically guides users toward vulnerability installation options, ensuring even unfamiliar users can successfully deploy environments and achieve LLM-guided attack simulations.

- **Pre-processing Module**

  The system does not transmit raw queries directly to the LLM. As illustrated in Figure 3, the Pre-processing Module retrieves the unprocessed query and systematically parses it to identify function call segments, formatted as `{{function_name}}`. Upon detection, the specified function is executed and its output is merged back into the query. This iterative process continues until all function calls are resolved, ensuring the final prompt is contextually rich and technically precise.

- **Post-processing Module**

  Upon receiving an LLM response, the Post-processing Module verifies content correctness and format integrity before presentation, as illustrated in Figure 4. This module is divided into backend parsing and frontend rendering components.

  Since the system mandates JSON-format responses, but LLM stochasticity means valid JSON is not always guaranteed, the module employs a robust decoding strategy. If a response is not initially valid JSON, the module searches for JSON-like segments delimited by `{and}` and attempts parsing. If parsing still fails, the system applies character transformations to correct erroneous escape characters that LLMs occasionally introduce (e.g., prefixing `$`, `[`, `]` with backslashes). The replacement sequence is defined in Table 3. Once validated, the JSON object is parsed into constituent fields and rendered on the frontend interface, with critical information highlighted for user comprehension.

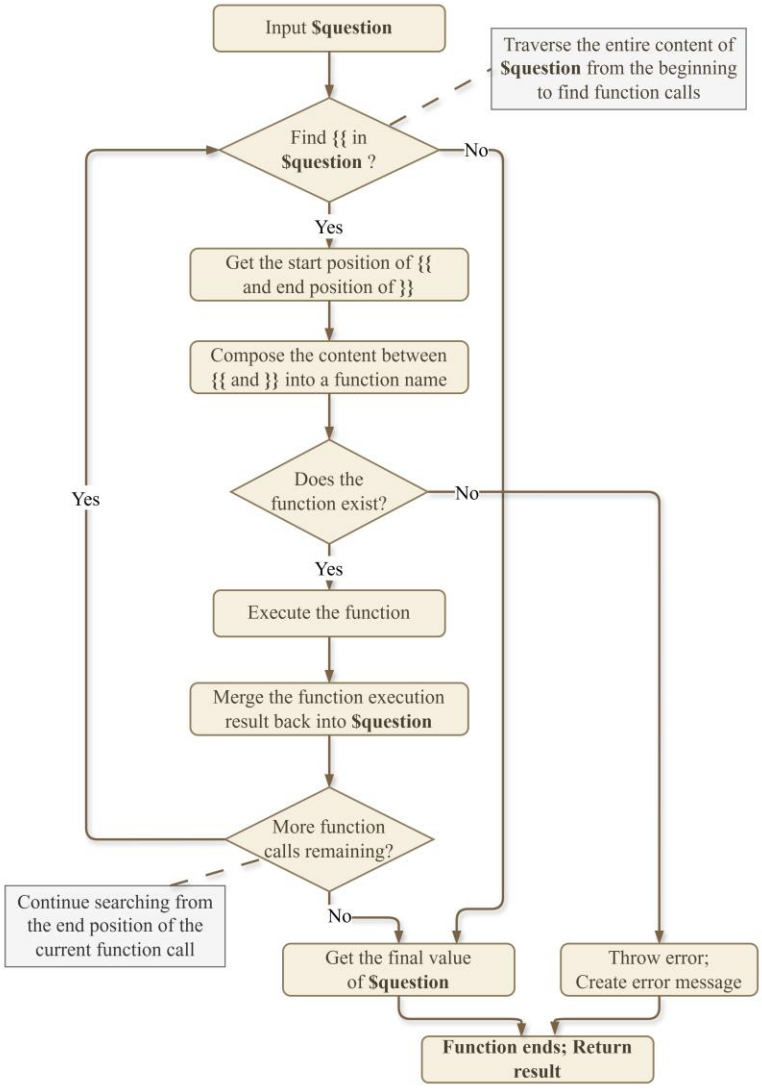

Figure 3: Pre-processing Module

Table 3: LLM Response Escape Character Replacement Table

Replacement Order	Original Character (Pre-replacement)	Normalized Character (Post-replacement)
1	\$	$
2	\\[	[
3	\\]	]
4	\[	[
5	\]	]

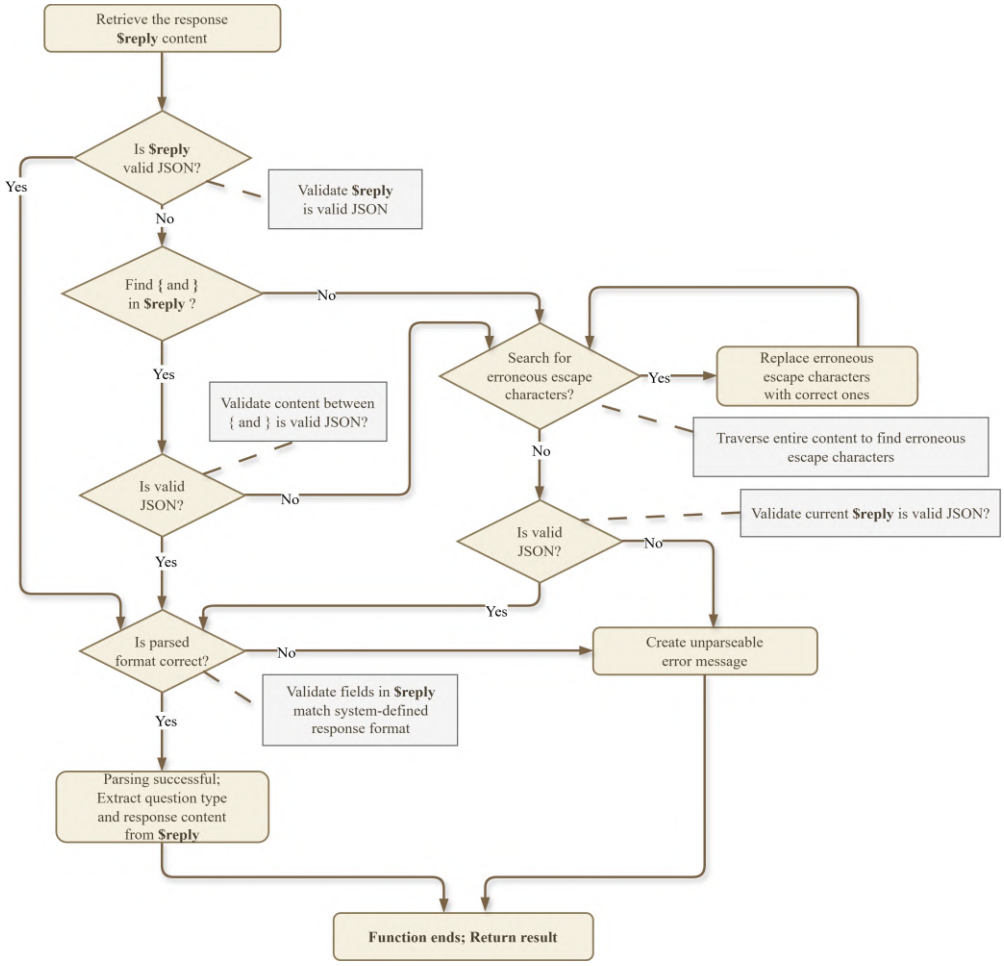

Figure 4: Post-processing Module

4.3 Threat Intelligence Synthesis

In this stage, the IoC Extraction Module identifies and analyzes user-provided threat intelligence, extracting critical indicators. The module offers multiple identification capabilities selectable by users, including CVE identification, attacker IP and domain name attribution, and APT attack lifecycle inference.

Users can directly import technical forums, articles, analytical reports, or dark web intelligence into the system without prior organization. The question recommendation module then lists appropriate query types for users to select, triggering LLM-based data extraction and analysis.

While modern LLMs possess broad general knowledge, research in Section 2.4 indicates that models may fail to correctly identify vulnerabilities not encountered during training. To ensure classification accuracy, the system leverages a knowledge base module to provide relevant domain knowledge. For example, since CWE contains over 900 software flaw categories, the system provides complete CWE documentation to the LLM during queries to prevent hallucinations.

Table 4 illustrates the prompt structure for CWE Identification as an example. The prompt begins with a concise context description, specifies the items to identify, provides a concrete response exemplar with placeholder data and format explanations, and concludes with an empty standard response template.

Table 4: IoC Extraction Module – CWE Identification

Item	Content
Type	Vulnerability Identification
Name	CWE Identification
Prompt Content	Based on the provided threat intelligence, which vulnerability in the system did the hacker exploit to attack the victim? Please list one CWE ID (if multiple exist, select the most significant one). Example: `{"CWE":{"item":"CWE-20: Improper Input Validation", "description":"The product does not validate or incorrectly validates input..."}}` Format: `{"CWE":{"item":"","description":""}}`
Return ID	CWE
Attachments	cwe_latest.pdf

4.4 Scenario Provisioning

4.4.1. Virtual Machine Provisioning

The Machine Deployment Module integrates VM configuration and virtual network orchestration, enabling users to complete attacker and victim infrastructure deployment with system assistance. The module adopts a pluggable design, allowing users to substitute underlying virtualization platforms. The default implementation communicates with OpenNebula via its XML-RPC API.

Following OpenNebula's operational logic, the system first identifies VM templates and then instantiates VMs from these templates. Several baseline templates are pre-configured (e.g., Windows Server 2022 Standard, Kali Linux 2024.2). The system supports multi-user access with OpenNebula's native user isolation, and integrates with LDAP for centralized permission management.

To accurately reflect conditions described in threat intelligence, the module allows per-machine network configuration, matching VM IP addresses to the Public IP or Private IP specified in intelligence reports. Virtual networks are managed by the OPNsense firewall, with DNS handled via Unbound for domain name-based communication. Users access VMs through the built-in WireGuard VPN service.

The primary functions for interfacing with the OpenNebula platform are:

- **`virtual_machine_template_list()`:** Retrieves available VM templates, filtering out redundant parameters and returning only the critical fields defined in Table 5 to avoid confusing the LLM.
- **`instantiate_virtual_machine()`:** Instantiates all VMs within a scenario simultaneously, taking vm_array and scenario_id as parameters. The system uses `parse_llm_vm_template()` to validate fields against Table 5 specifications. VMs are created with administrator privileges; once successfully deployed, permissions are modified to grant user ownership, preventing interference during the provisioning process.

Table 5: Return Parameters of the `virtual_machine_template_list()` Function

Field	Description
`TEMPLATE_ID`	The unique ID of the VM template in the system; required during VM creation.
`TEMPLATE_NAME`	The name of the VM template.
`NAME`	The name of the VM. Returned as empty, to be populated by the LLM.
`CPU`	Number of CPU cores for the VM.
`MEMORY`	Memory capacity of the VM.
`MEMORY_UNIT_COST`	Memory unit (e.g., MB, GB, TB).
`OS_ARCH`	Operating system architecture (e.g., x86_64, i686).
`DESCRIPTION`	The purpose of the VM. Returned as empty, to be populated by the LLM.
`ROLE`	The role of the VM. Returned as empty, to be populated by the LLM.

- **`change_virtual_machine_configuration()`:** Modifies VM configuration files via the OpenNebula Contextualization Add-on, pre-installed on all system VMs. This package enables direct hardware and software configuration through the platform by adding a virtual CD-ROM device containing a context.sh file with `KEY='VALUE'` pairs. Changes to the CONTEXT field trigger a virtual disk hot-plug event within 30–60 seconds, prompting the guest OS to apply updated settings.

Table 6 illustrates the prompt used for creating a basic attacker-victim machine setup, including situation description, VM template exemplars, and field specifications.

Table 6: Machine Deployment Module – Create Attacker and Victim Machines Only

Item	Content
Type	Environment Machine Provisioning
Name	Create Scenario Machines - Attacker and Victim Only
Prompt Content	To reproduce this attack, what kind of machine environment is needed to simulate the attacker and victim? Please refer to the following VM templates to select the appropriate machines. If the required resources exceed current limits, you may modify default values. In the NAME field, identify the machine as "attacker" or "victim" with a suffix (e.g., attacker-1). Provide additional details in the DESCRIPTION field. Please reply using the following JSON format; do not include other text or markdown syntax. VM Templates: `{{ vm_template_info }}`
Return ID	VMS
Attachments	None

4.4.2. Automated Software Installation

Once VM provisioning is complete, the Vulnerability Environment Installation Module assists with software deployment by importing structured data from the IoC Extraction Module and analyzing it via LLMs. The module features automated generation of installation scripts, enabling streamlined software setup that may require user feedback for error correction.

- **VM Software Installation:** This query provides the LLM with all machines in the scenario, victim machine lists, and response format exemplars. The LLM determines each machine's role from its description field and specifies software names, versions, and detailed installation steps. The prompt emphasizes that software must correspond directly to the threat intelligence to avoid installing unrelated components. The response follows a structured JSON format containing machine names mapped to arrays of software entries with name, version, installation method, and notes fields.
- **VM Software Installation Script:** Extends the previous query by generating executable scripts. The system detects the guest OS environment: PowerShell for Windows, Shell script for Linux. Based on OpenNebula's design, the VM context includes a start script field; when its content changes, a contextualization event triggers execution with root/administrator privileges.
- **VM Software Installation Script (with Error Messages):** A post-processing error-handling mechanism. When a previously generated script fails, the system submits the error message to the LLM for analysis and script revision. Users simply paste the error message displayed during the failed installation into the system's interactive response box.

4.5 LLM-Assisted Attack Script Generation

After completing environment provisioning, the Automated Attack Module integrates extracted IoC parameters to generate detailed attack methodologies. When the original intelligence lacks specific exploit details, the system may require users to supplement publicly available proof-of-concept code to produce a functional script. The module incorporates current machine configurations into the LLM's context, enabling it to draft attack scripts based on the real-time environment.

- **Attack Tactics:** Analyzes threat intelligence to infer potential attack vectors and provides users with a detailed, Markdown-formatted description of the attack workflow.
- **Automated Attack Script:** Generates Python attack scripts designed for autonomous execution, though the generation process itself may involve iterative refinement with user feedback. The prompt provides the LLM with detailed machine configurations including OS data and machine roles. As shown in Table 7, the script must be fully autonomous, requiring no manual intervention. The response format comprises three core fields: the script content, the designated attacker machine, and the target victim machine(s). Internal system information (such as IP addresses) is synchronized through the Post-processing Module, allowing users to execute scripts directly within the virtualized environment.

Table 7: Automated Attack Module – Python Attack Script Generation

Item	Content
Type	Automated Attack Script Generation
Name	Python Attack Script Generation

Prompt Content	Please initiate an attack against the victim's machine. Compose a fully automated Python attack script that can execute on the attacker's machine without requiring any manual parameters. Ensure that the IP information for both the attacker and the victim is correct.
	Python Version: 3.11
	Machine Configurations:
	`{{ vms_info }}`
	Please provide the response in JSON format.
	- attacker: Name of the attacker machine.
	- victim: Name of the victim machine.
	- script: The fully autonomous, executable script. Do not include commands that cannot be executed automatically.
	Example:
	`{"ATTACK_SCRIPT":{"attacker":"", "victim":"", "script":""}}`

Return ID	ATTACK_SCRIPT
Attachments	None

5 Experimental Results

This section describes the practical deployment of the proposed system through two attack scenarios to evaluate its effectiveness.

5.1 CVE-2024-4577 Simulation

CVE-2024-4577, disclosed by DEVCORE's Orange Tsai in June 2024 [17], allows attackers to execute arbitrary PHP code on vulnerable servers. The vulnerability exploits the Windows Best-Fit character encoding conversion behavior: when a Unicode character cannot be represented in the current code page, the system converts it to the closest matching character. PHP's design overlooked this behavior, enabling attackers to bypass security restrictions using specific character sequences. The vulnerability affects all PHP versions released before June 2024 when configured in CGI mode.

5.1.1. Attack Reproduction

The user imported the vulnerability description from the DEVCORE blog [17] as a text file into the system, creating a new scenario for CVE-2024-4577.

Stage 1: Threat Intelligence Analysis. The IoC Extraction Module analyzed the imported data and produced the results summarized in Table 8. The system identified the vulnerability as OWASP Top 10 A03:2021 Injection, classified it as CWE-94 (Improper Control of Generation of Code), and enumerated affected PHP versions: PHP 8.3 < 8.3.8, PHP 8.2 < 8.2.20, PHP 8.1 < 8.1.29, and all XAMPP for Windows versions.

Table 8: System-Inferred Cause and Affected Software of CVE-2024-4577

Query	System Response
Vulnerability Description	The root cause of this vulnerability is that PHP overlooked the Windows Best-Fit character encoding conversion behavior during its design, allowing unauthenticated attackers to bypass CVE-2012-1823 protection through specific character sequences and perform parameter injection to achieve remote code execution. This vulnerability affects all PHP versions installed on Windows, particularly under CGI execution mode or when the PHP executable is exposed in CGI directories.
OWASP Top 10	A03:2021 – Injection. This attack exploits the parameter handling vulnerability in PHP CGI mode to inject malicious commands. This method falls under injection attacks (Injection), including SQL, NoSQL, command, LDAP, and other injection types. Attackers craft special input to trick applications into executing unauthorized commands or queries, thereby obtaining sensitive data, tampering with data, or damaging the system.
CWE	CWE-94: Improper Control of Generation of Code ('Code Injection'). The CGI mode in PHP fails to properly control character encoding conversion, allowing unauthenticated attackers to execute arbitrary code on remote servers through parameter injection.

Stage 2: Scenario Construction. The system configured two VMs as shown in Table 9: an attacker host (`attacker-1`) running Kali Linux and a victim host (`victim-1`) running Windows Server 2022. The Vulnerability Environment Installation Module then identified that `victim-1` requires PHP 8.2.19 and Apache HTTP Server 2.4.54.

For PHP installation, the system generated a PowerShell script for automated PHP installation on `victim-1`, covering download, extraction, environment variable configuration, and verification. The initial script contained an incorrect download URL, which the user reported through the system's error feedback mechanism. The system then regenerated a corrected script with the proper URL (windows.php.net), and PHP 8.2.19 was successfully installed.

For Apache installation, the system generated a comprehensive PowerShell script that downloads the Apache 2.4.54 package, extracts and reorganizes the directory structure, registers the Apache service, configures CGI mode integration with PHP including proper file handler settings and directory permissions, and restarts the service.

Table 9: Attacker and Victim Machines Automatically Created by the Machine Deployment Module

Machine	Template_ID/ Template_NAME	CPU	Memory	Role	Description
attacker-1	3/Kali Linux 2024.2	2	4096 MB	Attacker	Penetration testing environment
victim-1	1/Windows Server 2022 Standard	2	4096 MB	Victim	Vulnerable PHP web server

Stage 3: Attack Script Generation. The system's Automated Attack Module produced an attack chain analysis, as shown in Table 10, detailing the exploitation of php-cgi.exe through parameter injection using allow_url_include and auto_prepend_file.

When provided only with the DEVCORE blog content, the LLM generated an attack script that directly used the "`-`" character as a CGI parameter. This approach would only trigger the older CVE-2012-1823 vulnerability, which has already been patched in PHP 8.2.19, rendering the attack unsuccessful against the target environment.

After the user supplemented publicly available exploit scripts [18] [19], the system generated a revised Python script incorporating the following key elements: (1) HTTP headers declaring form-format content for proper PHP interpretation; (2) PHP CGI parameters including `cgi.force_redirect=0`, `disable_functions=""`, `allow_url_include=1`, and `auto_prepend_file=php://input`; (3) critically, the "`-`" character is replaced with the encoded string "`%AD`" to trigger the Best-Fit behavior—on a Traditional Chinese Windows system (Big5 encoding, code page 950), Unicode U+00AD (Soft Hyphen) is converted to U+002D (Hyphen-minus), thereby triggering CVE-2024-4577; (4) a PHP payload executing the system command. The script successfully attacked the victim host, and replacing the payload with "`whoami`" confirmed the attacker obtained nt authority\system privileges.

Table 10: System-Inferred Attack Chain for the CVE-2024-4577 Scenario

Step	Description
1. Verify PHP CGI Configuration	Confirm PHP runs in CGI mode on the victim server with proper handler and action directives in httpd.conf.
2. Construct Malicious Request	Craft a URL targeting `php-cgi.exe` with parameter injection to enable `url_include` and `auto_prepend_file`.
3. Send Malicious POST Request	Send a POST request containing PHP code (e.g., `system("ipconfig")`) to execute arbitrary commands on the victim server.
4. Post-exploitation	Modify the payload to execute additional commands, download malware, or establish persistent control.

5.1.2. Experimental Summary

The system successfully assisted users in reproducing the CVE-2024-4577 attack scenario from a single threat intelligence source. In the final attack script generation stage, users can supplement publicly available exploit code to enable the system to generate customized attack scripts tailored to the target environment.

5.2 APT TAG-70 Simulation

TAG-70 (also known as Winter Vivern, TA473, UAC-0114) is a pro-Russian APT organization active since December 2020, primarily targeting government agencies in Eastern Europe. In 2023, the group exploited CVE-2023-5631 in Roundcube Webmail to compromise systems of the Ukrainian Armed Forces and the Georgian Ministry of Defence. This study uses the threat intelligence report published by Recorded Future [20] to construct a simulated enterprise network environment.

5.2.1. Attack Reproduction

We simulated the workflow of enterprise security analysts by importing the Recorded Future PDF report directly into the system.

Stage 1: Threat Intelligence Analysis. The IoC Extraction Module identified CVE-2023-5631, a cross-site scripting (XSS) vulnerability in Roundcube Webmail, classifying it as OWASP Top 10 A03 Injection. The system also generated the APT attack lifecycle, detailing each phase from reconnaissance through data exfiltration. In the reconnaissance phase, TAG-70 conducted phishing campaigns from IP address 38.180.76.31, sending emails with malicious links or attachments to collect target credentials. For initial access, the group exploited CVE-2023-5631 in the Roundcube Webmail server from 176.97.76.129, gaining access when the victim merely opened the infected email. During the execution phase, TAG-70 deployed JavaScript-based malware from 38.180.76.31 to capture email credentials and transmit them to the C2 server at 198.50.170.72. For command and control, TAG-70 used Tor to conceal its location, communicating through non-standard ports such as TCP 443. In the data exfiltration phase, the group collected and exfiltrated email content and credentials from the compromised Roundcube server, primarily targeting government and military accounts. The ultimate impact involved the attacker logging the victim out of Roundcube and displaying a spoofed login page from 176.97.66.57 targeting mail.mod.gov.ge; when the victim entered credentials, they were transmitted to the C2 server, enabling access to sensitive military and government emails.

The analysis revealed that the attacker initially used IP address 38.180.76.31 for reconnaissance with the malicious domain recsecas.com, which resolved to 38.180.76.31 before September 2023 and changed to 176.97.76.129 afterward, indicating deliberate IP switching to evade tracking. The attacker also employed Tor network nodes to conceal C2 communications.

When analyzing APT attack lifecycles, the system can describe each phase based on threat intelligence fields. However, since reports often contain multiple attack events targeting different countries with different C2 IP addresses, the system may have difficulty precisely distinguishing relationships between these IPs without explicit specification of a particular attack event. This requires multiple iterative queries to obtain clear responses.

Stage 2: Environment Construction. To provide a realistic enterprise environment, we used the system's enterprise environment deployment feature to automatically generate a complete network topology, including organizational units (HR, finance, marketing), internal services (web servers, DNS servers), and attacker infrastructure (workstations, C2 servers). The generated topology comprised 20 VMs spanning multiple network segments. The system configured VM network IP addresses according to the threat intelligence and utilized the DNS server to resolve domain names to designated IP addresses, avoiding resolution failures caused by attackers removing DNS records in the real internet environment.

The system installed Roundcube Webmail version 1.5.2 containing the CVE-2023-5631 vulnerability on the HR-EmailServer host, accessible at https://mail.mod.gov.ge.

Stage 3: Attack Reproduction. For authenticity, we placed actual malicious scripts used by the attackers into the scenario. As shown in Figure 5, the attack flow proceeds as follows: the attacker sends a phishing email to the victim; upon opening the email, malicious code connects to the `c2-1` machine to retrieve the `checkupdate.js` script, which executes in the victim's browser and awaits further C2 commands.

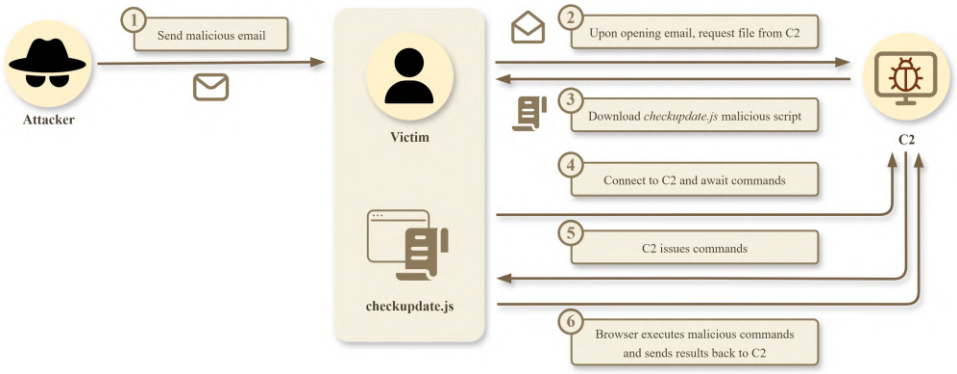

Figure 5: Attack Flow Against the Roundcube Webmail System

Using SHA-256 hashes from the report's IoC appendix, we located malicious script samples from online malware databases. The sample with SHA-256 hash beginning with `ea22b3e9` transmits the victim's Roundcube username and password to the C2 domain hitsbitsx.com. When simulating the victim browsing email on the `hr-1` VM, the malicious code logged the user out of Roundcube and displayed a spoofed login page. Upon entering credentials, the script's onClickSendCredentials function transmits them to the `c2-1` machine.

We also obtained a variant checkupdate.js script (SHA-1 beginning with `97ed59`) identified in a separate ESET threat intelligence report. This script silently exfiltrates all inbox emails to the `c2-1` machine (`domain: recsecas.com`) without any visible indication to the victim.

5.2.2. Experimental Summary

The system effectively assisted in reproducing the APT TAG-70 attack scenario by rapidly parsing the APT attack lifecycle, systematically enumerating attacker behaviors at each phase, and automatically constructing the required virtual environment including domain name and IP address configurations that closely match the actual attack scenario.

5.3 Discussion and Comparison

In the preceding sections, we demonstrated how the system assists security analysts in step-by-step attack reproduction through modular design—from key information extraction, to realistic environment construction, to attack script validation and execution. The system is applicable to both common CVE vulnerabilities and large-scale APT campaigns.

Regarding IoC extraction, Lee et al. [21] proposed the PrioriC framework, which uses ChatGPT to analyze 116 threat intelligence reports, extracting critical information and classifying attack techniques and targeted assets. Liu et al. [22] utilize ChatGPT to construct STIX topologies from threat intelligence, with post-processing scripts using regular expressions to capture indicators not initially identified by the model.

As shown in Table 11, while all approaches can extract IoCs from threat intelligence, our system leverages LLMs to comprehensively evaluate a broader range of indicators, providing additional contextual information such as APT attack lifecycles and attacker activity timelines for more comprehensive analysis.

Table 11: Comparison of IoC Extraction Capabilities Between the Proposed System and Other Frameworks

Feature	Proposed System	PrioriC Framework	Liu et al.
CVE ID	✓	✓	✓
CWE ID	✓	-	-
OWASP Top 10 Classification	✓	-	-
IP Address	✓	-	✓
Domain Name	✓	-	✓
Victim Industry Category	✓	✓	-
Victim Role	✓	-	✓
Victim Operating System	✓	✓	-
Victim Software	✓	-	✓
Attacker Techniques Used	✓	✓	✓
Attacker Network Protocols Used	✓	✓	-
Attacker-Exfiltrated Victim Files	✓	-	✓
Attack Activity Timeline	✓	-	-
Attack Lifecycle	✓	-	-

Regarding attack script generation, as shown in Table 12, both PentestGPT and our system support diverse attack types. However, PentestGPT operates as a conversational chatbot, requiring users to manually provide necessary information and attack methods. HackingBuddyGPT can automatically determine attack methods based on machine information but is currently limited to specific scenarios such as Linux privilege escalation and web penetration testing. A key advantage of our system is its ability to automatically collect attacker and victim environment information and provide it to the LLM, eliminating manual input and ensuring data accuracy.

Furthermore, the proposed system provides an integrated environment where users can construct attack scenarios and automatically generate attack scripts based on the current environment state. Traditional manual attack reproduction often requires weeks; with our system's guidance, multiple attack scenarios can be reproduced within a single day.

Table 12: Comparison of Attack Script Generation Capabilities

Feature	Proposed System	PentestGPT	HackingBuddyGPT
Diverse Attack Types	✓	✓	-
Automated Prompt Generation	✓	-	✓
Automated Attack Method Selection	✓	-	✓

Limitations. It is important to note that the primary contribution of this work is the proposal of an end-to-end framework that integrates LLM-driven threat intelligence extraction, virtual environment provisioning, and attack script generation into a unified platform. The current evaluation is based on representative case studies to demonstrate the feasibility and practical applicability of this framework.

One key challenge lies in the inherent non-determinism of LLM outputs. Given identical prompts, the model may produce structurally or semantically different responses across runs, making consistent

quantitative measurement difficult without first establishing a standardized evaluation methodology. To date, no widely accepted benchmark exists for evaluating end-to-end attack reconstruction systems. Existing LLM-assisted security research, such as HackingBuddyGPT [9] and PentestGPT [10], and primarily relies on Capture-the-Flag (CTF) platforms [11] [12] where success is measured by binary flag retrieval, a metric that does not directly transfer to the multi-stage reconstruction workflow addressed in this work.

Additionally, as observed in Section 5.1, the initial attack script generated by the system failed to trigger the target vulnerability and required user-supplied exploit references for correction. This reflects a deliberate design choice in the current framework: the system is intended to operate as an LLM-assisted, human-in-the-loop tool rather than a fully autonomous agent. The error feedback mechanism described in Section 4.4 provides a structured pathway for iterative refinement, and characterizing the frequency and nature of such interventions across a broader set of scenarios remains an important direction for future investigation.

6 Conclusions and Future Works

This research proposes a novel system architecture that integrates LLM technologies to assist security analysts in performing threat intelligence analysis and vulnerability reproduction. Users can input various forms of attack intelligence, such as threat intelligence reports from international security organizations or proof-of-concept scripts shared by security communities, and the system assists and guides them through the entire process from threat analysis, attack environment construction, to vulnerability reproduction and verification, with human oversight at key decision points.

Current research on applying LLMs in cybersecurity mainly focuses on threat detection, with relatively little work exploring their use in threat intelligence analysis. The proposed system leverages the powerful analytical capabilities of LLMs to help analysts more efficiently understand attack events, extract key attack indicators from reports with high efficiency, and construct simulated attack environments for deeper analysis of the attack process.

The system features a robust virtualization architecture that allows users to deploy virtual environments directly within the system and conduct experiments in network-isolated environments. Through the automated attack script generation functionality, the system produces scripts tailored to specific scenarios, enabling users to gain a detailed understanding of the entire attack lifecycle. The modular design architecture provides strong scalability and flexibility for future development. As AI technologies advance, the modular LLM interface can flexibly integrate emerging models, eliminating single-model dependency and enabling more accurate threat analysis.

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Temporal Edge Weighting for Graph Collaborative Filtering*

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Abstract

Graph collaborative filtering is widely used for learning user and item representations from user–item interaction graphs. However, many existing backbones rely on a largely static interaction graph, where observed interactions are treated uniformly regardless of their timestamps. In this paper, we propose a lightweight plug-and-play temporal edge weighting scheme that directly incorporates temporal recency into the propagation graph. The proposed method assigns time-aware weights to user–item edges and can be instantiated with personalized, global, and hybrid recency strategies, without modifying the original backbone architectures. We apply the proposed scheme to representative graph collaborative filtering backbones, including NGCF, LightGCN, and LR-GCCF, and evaluate it on Yelp2018 and ML-1M.Experimental results show that the effect of temporal edge weighting is dataset- and backbone-dependent: it yields more consistent gains on ML-1M, while its effect on Yelp2018 is less consistent, suggesting that recency is beneficial only when it aligns with the preference dynamics of the target domain. The proposed scheme introduces no additional learnable parameters, and the personalized and global variants generally remain close to the original backbones in runtime, whereas the hybrid variant can incur additional overhead. These findings suggest that edge-level temporal recency modeling can serve as a practical backbone-preserving option for graph-based recommendation.

1 Introduction

Graph collaborative filtering remains a strong and widely used approach for learning from user–item interactions [1, 15]. By leveraging neighborhood aggregation on the user–item graph, it captures higher-order collaborative signals beyond directly observed edges while maintaining relatively simple architectures. Representative models such as NGCF [16], LightGCN [9], and LR-GCCF [4] differ in their aggregation and update mechanisms, but they all follow the common principle of learning user and item representations through iterative neighborhood interaction on the user–item graph. Owing to their effectiveness, simplicity, and flexibility, these methods remain important backbones and baselines in recommendation research.

Despite their strong empirical performance, many graph collaborative filtering backbones still rely on a largely static interaction graph, where observed interactions are treated uniformly once they are included in the graph [14, 17]. However, user–item interactions are inherently temporal. In some domains, recent interactions may better reflect a user's current interests, whereas older interactions may become less informative as preferences evolve over time. At

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the same time, temporal recency is not universally beneficial, since older interactions can still encode stable long-term preferences, especially in domains where user tastes change slowly. Therefore, a key question is not simply whether temporal information should be used, but how recency can be incorporated into graph propagation while preserving the usefulness of long-term collaborative signals.

To address this limitation, we propose a lightweight plug-and-play temporal edge weighting scheme for graph-based recommendation. Our method assigns time-aware weights to user–item edges so that more informative recent interactions contribute more strongly during graph propagation, while less relevant older interactions are appropriately discounted. The proposed approach can be incorporated into multiple representative graph collaborative filtering backbones, including NGCF, LightGCN, and LR-GCCF, with minimal modification to their original architectures. Accordingly, the main goal of this work is to provide an architecture-preserving way to inject temporal recency into standard graph collaborative filtering models.

By directly injecting temporal recency into the propagation graph, the proposed method preserves the original backbone architectures while providing a lightweight way to adjust edge importance during message passing. Empirical results on Yelp2018 and ML-1M show that temporal edge weighting can improve recommendation performance, but its effectiveness depends on the dataset characteristics and the underlying graph backbone. In particular, temporal weighting yields more consistent gains on ML-1M, while its effect on Yelp2018 is more model-dependent. These findings suggest that explicit edge-level recency modeling can serve as a practical backbone-preserving enhancement for graph-based recommendation.

- We propose a lightweight temporal edge weighting scheme that directly incorporates temporal recency into the user–item propagation graph, instantiated with personalized, global, and hybrid strategies, without modifying the original graph collaborative filtering backbones.
- We empirically show that the effect of temporal edge weighting is dataset- and backbone-dependent. The proposed weighting strategies yield more consistent gains on ML-1M, while the results on Yelp2018 reveal the importance of preserving long-term preference signals.
- We analyze the runtime behavior of the proposed variants and show that personalized and global weighting introduce no additional learnable parameters and generally remain close to the original backbones in training time, whereas hybrid weighting involves a clearer computational trade-off.

2 Related Work

2.1 Graph Collaborative Filtering Methods

Graph collaborative filtering has established a strong line of research in recommendation by modeling users and items as a bipartite graph and learning embeddings through graph propagation. Due to their strong effectiveness and simplicity, such models have become representative backbones in diverse recommendation research. In multi-behavior recommendation [11], recent models often employ LightGCN-based encoders to learn behavior-specific user and item representations from multiple interaction graphs [6, 5]. In multimodal recommendation [20], graph-based backbones such as LightGCN and NGCF are commonly used as the collaborative propagation module while incorporating visual and textual signals [21, 13]. A similar trend

also appears in social recommendation [12], where graph-based social recommenders serve as core backbones for modeling interaction and social relations [10, 18]. These studies illustrate that graph collaborative filtering backbones are frequently reused as core propagation modules across different recommendation settings. This motivates our focus on modifying the propagation graph itself rather than redesigning the backbone architectures.

2.2 Temporal Modeling in Graph-based Recommendation

Prior work has incorporated temporal information into graph-based recommendation in several ways. Temporal collaborative filtering with graph convolutional neural networks proposes a temporal collaborative filtering method that leverages GNNs for representation learning and RNNs for modeling temporal dynamics [3]. DGSR [19] models sequential recommendation through a dynamic graph structure and a Dynamic Graph Recommendation Network that captures user–item interactions with time and order information. TGSRec [7] further introduces a continuous-time bipartite graph and a Temporal Collaborative Transformer to unify sequential patterns and temporal collaborative signals. More recently, TAGCF [22] incorporates time factors and attention into graph collaborative filtering to model temporal context during embedding propagation.

While these studies demonstrate the importance of temporal information in graph-based recommendation, they often introduce additional temporal modules, dynamic graph construction procedures, attention mechanisms, or sequential modeling components. In contrast, our objective is different: we focus on whether temporal recency can be injected into existing graph collaborative filtering backbones without changing their architectures. The proposed method does not introduce a specialized temporal encoder or additional learnable parameters, but instead replaces the binary interaction graph with a time-aware weighted propagation graph. Therefore, our contribution lies in an architecture-preserving temporal reweighting formulation for standard GCF backbones, rather than in designing a new temporal recommendation architecture.

3 Preliminaries

3.1 Problem Definition

Let $\mathcal{U}$ and $\mathcal{I}$ denote the user and item sets, respectively, and let

$$\mathcal{D} = \{(u, i, t_{ui}) \mid u \in \mathcal{U},\, i \in \mathcal{I}\} \tag{1}$$

be the set of timestamped user–item interactions. From the observed interaction history, we construct a user–item bipartite graph

$$\mathcal{G} = (\mathcal{U} \cup \mathcal{I}, \mathcal{E}), \tag{2}$$

where $\mathcal{E}$ denotes the observed user–item edges in the graph.

Given $\mathcal{G}$, the task is to rank candidate items for each user. For a user u, let $\mathcal{I}_u$ denote the set of items observed in the interaction history of u, and define the candidate set as

$$\mathcal{C}_u = \mathcal{I} \setminus \mathcal{I}_u. \tag{3}$$

The model assigns a relevance score $f(u, j \mid \mathcal{G})$ to each candidate item $j \in \mathcal{C}_u$ and produces a personalized ranking accordingly.

3.2 Graph Collaborative Filtering Backbones

Let

$$\mathbf{H}^{(0)} = \begin{bmatrix} \mathbf{E}_U^{(0)} \\ \mathbf{E}_I^{(0)} \end{bmatrix} \tag{4}$$

denote the stacked initial user and item embeddings. Let $\hat{\mathbf{A}}$ denote the normalized user–item interaction graph used for message passing. Although NGCF, LightGCN, and LR-GCCF differ in how they aggregate and combine graph-based signals, all three models fundamentally learn representations through repeated operations on $\hat{\mathbf{A}}$. Since our method modifies this graph to incorporate temporal information, we briefly summarize below only the adjacency-related components of these representative graph collaborative filtering backbones.

- **NGCF** [16] updates node representations by combining graph-based neighborhood aggregation with a bi-interaction term. At layer l, it first computes neighborhood messages from the normalized interaction graph and then applies nonlinear transformations to both the aggregated message and the element-wise interaction between the current and aggregated representations:

$$\mathbf{H}^{(l+1)} = \sigma(\hat{\mathbf{A}}\mathbf{H}^{(l)}\mathbf{W}_{\text{gc}}^{(l)}) + \sigma\Big((\mathbf{H}^{(l)} \odot \hat{\mathbf{A}}\mathbf{H}^{(l)})\mathbf{W}_{\text{bi}}^{(l)}\Big). \tag{5}$$

  The final representation is obtained by concatenating the embeddings from all layers.

- **LightGCN** [9] adopts a simplified linear graph aggregation scheme in which the normalized interaction graph directly propagates representations across layers:

$$\mathbf{H}^{(l+1)} = \hat{\mathbf{A}}\mathbf{H}^{(l)}. \tag{6}$$

  Its final embedding is the average of the embeddings from all layers:

$$\mathbf{Z}_{\text{LightGCN}} = \frac{1}{L+1}\sum_{l=0}^{L}\mathbf{H}^{(l)}. \tag{7}$$

- **LR-GCCF** [4] also builds user and item representations through repeated multiplication with the normalized interaction graph:

$$\mathbf{H}^{(l+1)} = \hat{\mathbf{A}}\mathbf{H}^{(l)}. \tag{8}$$

  Unlike LightGCN, it uses layer-wise concatenation to form the final representation:

$$\mathbf{Z}_{\text{LRGCCF}} = \mathbf{H}^{(0)} \,\|\, \mathbf{H}^{(1)} \,\|\, \cdots \,\|\, \mathbf{H}^{(L)}. \tag{9}$$

4 Method

Motivated by the shared reliance of graph collaborative filtering backbones on the normalized interaction graph [9, 4, 16], we enhance recommendation by revising the graph used for message passing rather than changing the backbone architectures themselves. Specifically, we replace the original static interaction graph with a time-aware weighted graph whose edge weights are determined by interaction timestamps, thereby incorporating temporal recency into the representation learning process while preserving the original backbone structure.

The observed user–item interactions are represented by the binary interaction matrix

$$\mathbf{R}_{ui} = \begin{cases} 1, & (u, i, t_{ui}) \in \mathcal{D}, \\ 0, & \text{otherwise.} \end{cases} \tag{10}$$

From $\mathbf{R}$, we define the corresponding bipartite adjacency matrix as

$$\mathbf{A} = \begin{bmatrix} \mathbf{0} & \mathbf{R} \\ \mathbf{R}^\top & \mathbf{0} \end{bmatrix} + \delta_{\text{loop}} \mathbf{I}, \tag{11}$$

where δ_{loop} indicates whether self-loops are used. We then normalize $\mathbf{A}$ according to the backbone-specific rule to obtain the propagation graph $\hat{\mathbf{A}}$ used for message passing.

4.1 Personalized Temporal Weighting

We first define temporal recency relative to each user's own interaction history. Let t_{ui} denote the timestamp of interaction (u, i), and let t_u^{recent} denote the most recent interaction time of user u in the training set. We then define the personalized time gap as

$$\Delta_{ui}^{\text{P}} = \min\left\{ \max\left(\frac{t_u^{\text{recent}} - t_{ui}}{s}, 0 \right), c \right\}, \tag{12}$$

where $s > 0$ is a time scaling factor and $c > 0$ is a clipping threshold. Based on this gap, the personalized edge weight is defined as

$$w_{ui}^{\text{P}} = \epsilon + (1 - \epsilon) \exp(-\lambda_{\text{P}} \Delta_{ui}^{\text{P}}), \tag{13}$$

where $\epsilon \in (0, 1)$ is a minimum edge-weight floor and $\lambda_{\text{P}} > 0$ is the decay coefficient.

Instead of the original binary interaction matrix $\mathbf{R}$, we define a personalized time-weighted interaction matrix $\mathbf{R}^{\text{P}}$ as

$$\mathbf{R}_{ui}^{\text{P}} = \begin{cases} w_{ui}^{\text{P}}, & (u, i, t_{ui}) \in \mathcal{D}, \\ 0, & \text{otherwise.} \end{cases} \tag{14}$$

From $\mathbf{R}^{\text{P}}$, we construct the corresponding bipartite adjacency matrix $\mathbf{A}^{\text{P}}$, and then obtain the propagation graph $\hat{\mathbf{A}}^{\text{P}}$ through backbone-specific normalization.

4.2 Global Temporal Weighting

Similar to personalized temporal weighting, we define global temporal weighting based on the most recent observed interaction. Let $t^{\max}$ denote the most recent observed timestamp in the interaction history. The global time gap is defined as

$$\Delta_{ui}^{\text{G}} = \min\left\{ \max\left(\frac{t^{\max} - t_{ui}}{s}, 0 \right), c \right\}, \tag{15}$$

and the corresponding global edge weight is

$$w_{ui}^{\text{G}} = \epsilon + (1 - \epsilon) \exp(-\lambda_{\text{G}} \Delta_{ui}^{\text{G}}), \tag{16}$$

where $\lambda_{\text{G}} > 0$ is the decay coefficient. We then replace the original binary interaction matrix $\mathbf{R}$ with the global time-weighted interaction matrix

$$\mathbf{R}_{ui}^{\text{G}} = \begin{cases} w_{ui}^{\text{G}}, & (u, i, t_{ui}) \in \mathcal{D}, \\ 0, & \text{otherwise,} \end{cases} \tag{17}$$

from which the corresponding bipartite adjacency matrix $\mathbf{A}^{\text{G}}$ and normalized propagation graph $\hat{\mathbf{A}}^{\text{G}}$ are obtained in the same manner as in the personalized case.

Table 1: Statistics of the datasets.

Dataset	# Users	# Items	# Interactions	Sparsity
Yelp2018	77,278	45,639	2,103,896	99.94%
ML-1M	6,041	3,261	998,539	94.93%

4.3 Hybrid Temporal Weighting

Personalized and global temporal weighting reflect two different notions of recency. Personalized temporal weighting is defined relative to a user's own interaction history, whereas global temporal weighting is defined relative to the most recent observed interaction. To incorporate both temporal views, we combine the corresponding normalized propagation graphs during message passing.

Given hidden representations $\mathbf{H}^{(l)}$, we define the hybrid propagation operator as

$$\Phi_{\mathrm{h}}(\mathbf{H}^{(l)}) = \gamma \hat{\mathbf{A}}^{\mathrm{P}} \mathbf{H}^{(l)} + (1-\gamma) \hat{\mathbf{A}}^{\mathrm{G}} \mathbf{H}^{(l)}, \tag{18}$$

where $\gamma \in [0, 1]$ controls the balance between the two temporal views.

Accordingly, personalized temporal weighting uses $\hat{\mathbf{A}}^{\mathrm{P}}$ as the propagation graph, whereas global temporal weighting uses $\hat{\mathbf{A}}^{\mathrm{G}}$. For hybrid temporal weighting, every occurrence of $\hat{\mathbf{A}}\mathbf{H}^{(l)}$ in the backbone is replaced by $\Phi_{\mathrm{h}}(\mathbf{H}^{(l)})$, with all other parts of the backbone preserved. In this way, the proposed temporal weighting can be plugged into multiple graph collaborative filtering backbones with minimal modification.

5 Experiments

5.1 Experimental Settings

5.1.1 Datasets

We conduct experiments on two widely used benchmarks for recommendation: Yelp2018 [2] and ML-1M [8]. We apply 10-core filtering to each dataset. Table 1 summarizes the resulting dataset statistics. These datasets provide different levels of sparsity and preference dynamics. Yelp2018 is a highly sparse local business review dataset, where user preferences over restaurants, venues, and service categories may remain relatively stable over time. In contrast, ML-1M is denser and contains more structured movie rating interactions, where recent interactions may more reliably reflect changes in user interests. This contrast allows us to examine when temporal recency is helpful and when preserving long-term preference signals remains important

5.1.2 Evaluation

We adopt the widely used leave-one-out evaluation protocol under a chronological split. For each user, the most recent interaction is used for testing, the second most recent interaction for validation, and the remaining interactions for training. During evaluation, we rank all candidate items for each user without negative sampling. We report three standard top-K ranking metrics, $Recall@K$, $NDCG@K$, and $MRR@10$, where $K \in \{10, 20\}$. $Recall@K$ measures whether the target item appears in the top-K list, while $NDCG@K$ further accounts for its rank position. $MRR@10$ measures the reciprocal rank of the target item within the top-10 list, giving higher values when the target item is ranked earlier.

Table 2: Performance comparison on Yelp2018. The best results are shown in **bold**, and the second-best is underlined. #Improve denotes the relative improvement of the best-performing proposed variant over the original backbone.

Model	Variant	Recall@10	Recall@20	NDCG@10	NDCG@20	MRR@10	MRR@20
NGCF	Original	**0.0436**	**0.0746**	**0.0213**	**0.0290**	**0.0146**	**0.0166**
	Personalized	0.0388	0.0672	0.0188	0.0259	0.0127	0.0147
	Global	0.0412	0.0704	0.0200	0.0273	0.0136	0.0156
	Hybrid	0.0403	0.0695	0.0196	0.0269	0.0134	0.0154
	Improve (%)	-5.82%	-5.96%	-6.50%	-5.22%	-7.35%	-6.41%
LightGCN	Original	**0.0489**	**0.0830**	**0.0241**	**0.0326**	**0.0166**	**0.0189**
	Personalized	0.0423	0.0728	0.0203	0.0280	0.0138	0.0158
	Global	0.0486	0.0820	0.0239	0.0323	0.0164	0.0187
	Hybrid	0.0461	0.0782	0.0225	0.0306	0.0155	0.0177
	Improve (%)	-0.61%	-1.20%	-0.83%	-0.92%	-1.21%	-1.06%
LR-GCCF	Original	0.0409	0.0693	0.0200	0.0271	0.0137	0.0156
	Personalized	0.0395	0.0676	0.0191	0.0262	0.0130	0.0149
	Global	**0.0431**	**0.0734**	**0.0211**	**0.0287**	**0.0146**	**0.0166**
	Hybrid	0.0415	0.0706	0.0203	0.0276	0.0139	0.0159
	Improve (%)	5.10%	5.58%	5.21%	5.57%	6.16%	6.02%

5.1.3 Implementation Details

We instantiate the proposed temporal weighting on NGCF, LightGCN, and LR-GCCF while preserving their original architectures. The batch size is fixed to 2048, and the latent dimensionality of user and item embeddings is fixed to 64 for all models. For the remaining hyperparameters, we follow the original implementations of each backbone as closely as possible.

For temporal weighting, we set the common scaling and clipping parameters to $s = 86400$ and $c = 365$, respectively, so that raw timestamps are converted into days and excessively large time gaps are clipped within a one-year range. We further use $\epsilon = 0.05$, $\lambda_{\mathrm{P}} = 0.1$, $\lambda_{\mathrm{G}} = 0.1$, and $\gamma = 0.5$. All reported results are averaged over five runs with different random seeds.

5.2 Result Analysis

5.2.1 Overall Performance

Table 2 reports the overall performance on Yelp2018. On this dataset, the effect of temporal weighting is model-dependent. For NGCF and LightGCN, the original models achieve the best performance across all metrics; in contrast, LR-GCCF benefits from temporal weighting, with the global variant achieving the best results across all metrics and improving over the original model by about 5–6%. These results indicate that temporal recency is not universally beneficial. Yelp2018 is highly sparse and corresponds to local business review behavior, where older interactions can still represent stable long-term preferences over restaurants, venues, or service categories. In such a setting, aggressively discounting past interactions may weaken useful collaborative signals, especially for backbones that already rely strongly on neighborhood aggregation. The positive results on LR-GCCF suggest that the effect of temporal weighting also depends on the propagation mechanism of the underlying backbone.

Table 3 presents the overall performance on ML-1M. Compared with Yelp2018, temporal weighting shows more consistently positive effects across the three backbones, although the best

Table 3: Performance comparison on ML-1M. The best results are shown in **bold**, and the second-best is underlined. #Improve denotes the relative improvement of the best-performing proposed variant over the original backbone.

Model	Variant	Recall@10	Recall@20	NDCG@10	NDCG@20	MRR@10	MRR@20
NGCF	Original	0.0716	0.1290	0.0332	0.0476	0.0218	0.0257
	Personalized	0.0726	0.1305	0.0340	0.0485	0.0225	0.0264
	Global	**0.0748**	0.1314	**0.0351**	**0.0493**	**0.0233**	**0.0271**
	Hybrid	0.0723	**0.1322**	0.0335	0.0485	0.0220	0.0260
	Improve (%)	4.27%	2.42%	5.41%	3.44%	6.43%	5.16%
LightGCN	Original	0.0741	0.1338	0.0347	0.0496	0.0229	0.0270
	Personalized	0.0741	**0.1339**	**0.0349**	**0.0499**	**0.0232**	**0.0273**
	Global	**0.0743**	0.1331	0.0347	0.0495	0.0230	0.0269
	Hybrid	0.0742	0.1332	0.0347	0.0495	0.0230	0.0270
	Improve (%)	0.26%	0.07%	0.57%	0.60%	1.29%	1.09%
LR-GCCF	Original	0.0728	0.1306	0.0345	0.0490	0.0231	0.0270
	Personalized	0.0745	**0.1336**	0.0353	**0.0501**	**0.0236**	**0.0276**
	Global	0.0754	0.1325	**0.0355**	0.0498	**0.0236**	0.0275
	Hybrid	**0.0755**	0.1332	0.0353	0.0498	0.0233	0.0272
	Improve (%)	3.57%	2.24%	2.81%	2.19%	2.11%	2.17%

temporal variant still differs by model. For NGCF, the global variant achieves the best performance on most metrics, while the hybrid variant obtains the best Recall@20. For LightGCN, all variants perform very similarly, with only marginal improvements from temporal weighting, indicating that its simplified propagation already provides a stable graph signal on this dataset. For LR-GCCF, temporal weighting consistently improves over the original model, with personalized, global, and hybrid variants each achieving the best result on different metrics. These results indicate that temporal weighting is broadly useful on ML-1M, but the most effective temporal view remains model-dependent.

The different behavior between Yelp2018 and ML-1M can be explained by the interaction between sparsity, domain characteristics, and temporal preference dynamics. Yelp2018 is much sparser than ML-1M, so old interactions may provide important connectivity and collaborative signals in the user–item graph. Moreover, local business preferences can remain stable over time, making strong recency-based discounting less suitable. In contrast, ML-1M is denser and more structured, which makes it less dependent on old edges for graph connectivity. Movie preferences may also exhibit clearer temporal shifts as users consume different genres or styles over time. As a result, temporal edge weighting can emphasize recent preference signals on ML-1M without losing as much useful long-term information. Therefore, the proposed method should be viewed as a domain-dependent enhancement whose effectiveness depends on how well temporal recency aligns with the target dataset.

5.2.2 Training Efficiency

Table 4 reports the average training time per epoch (s/epoch) over five runs for each model variant on Yelp2018 and ML-1M. Since the proposed method reweights existing edges in the interaction graph, it does not introduce additional learnable parameters or increase the number of observed user–item edges. After the weighted adjacency matrix is constructed, the personalized and global variants use a single propagation graph, as in the original backbones. Therefore, their asymptotic message-passing cost remains the same order as the corresponding backbone, i.e., proportional to the number of nonzero edges used in sparse graph propagation.

Table 4: Average training time per epoch (s/epoch) over five runs for each model variant on Yelp2018 and ML-1M. Lower values indicate faster training.

Model	Variant	Yelp2018	ML-1M
NGCF	Original	77.41	**16.02**
	Personalized	81.97	17.49
	Global	**71.09**	16.32
	Hybrid	117.92	28.71
LightGCN	Original	23.99	6.31
	Personalized	50.50	**6.19**
	Global	**22.47**	6.53
	Hybrid	64.47	9.15
LR-GCCF	Original	31.88	**7.27**
	Personalized	54.26	7.43
	Global	**30.62**	7.34
	Hybrid	62.71	34.72

The measured runtime, however, does not always increase monotonically after temporal weighting. In some cases, a weighted variant is slightly faster than the original model. This should not be interpreted as lower computational complexity. Since sparse graph propagation is affected by low-level implementation details such as sparse tensor coalescing, memory access patterns, GPU kernel scheduling, caching behavior, and normal system-level variance, small runtime differences can occur even when the theoretical operation count is similar. In our setting, temporal weighting changes the numerical values and normalization of the propagation graph while largely preserving its sparsity pattern, which can lead to minor variations in low-level execution efficiency.

In contrast, the hybrid variant is consistently slower because it constructs and uses two separate normalized propagation graphs, $\hat{\mathbf{A}}^{\mathrm{P}}$ and $\hat{\mathbf{A}}^{\mathrm{G}}$, during message passing. This approximately doubles the sparse propagation operations associated with the temporal graph component. Therefore, the hybrid variant should be understood as a trade-off between combining two notions of recency and incurring additional runtime overhead. Overall, the personalized and global variants provide the more practical lightweight options, whereas the hybrid variant may be useful only when its accuracy gain justifies the additional computational cost.

6 Conclusion & Future Work

In this paper, we proposed a lightweight temporal edge weighting scheme for graph collaborative filtering. The proposed method injects temporal recency into the user–item propagation graph through time-aware edge weights, without modifying the backbone architecture, scoring function, or learning objective. We instantiated this scheme with personalized, global, and hybrid strategies and applied it to representative graph collaborative filtering backbones. Experiments on Yelp2018 and ML-1M show that temporal edge weighting has dataset- and backbone-dependent effects. The method yields more consistent gains on ML-1M, while the results on Yelp2018 are less consistency, indicating that recency-based weighting is beneficial only when recent interactions better reflect the preference dynamics of the target domain. These results also highlight the importance of preserving long-term preference signals in domains where older interactions remain informative. In terms of efficiency, the personalized and global variants introduce no additional learnable parameters and generally remain close to the original

backbones in training time, whereas the hybrid variant can incur noticeable overhead due to the use of two propagation graphs. Future work will further examine the proposed temporal edge weighting scheme through sensitivity analysis of temporal hyperparameters, comparisons with temporal and sequential recommendation approaches, evaluations on larger and more temporally dynamic datasets, and deeper analysis of the trade-off between temporal recency and long-term preference preservation.

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History Packaging Policies for LLM Recommendation: Evidence from Real LLM Scoring Under Prompt Budgets*

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Abstract

Large-language-model recommenders must decide which parts of a user's history survive under a fixed prompt budget. That packaging choice changes both ranking quality and the input-token burden, a deployment-facing proxy for latency, throughput, and operating expense on e-business ranking surfaces [5, 2, 12]. This paper studies the problem as a factorial design under real LLM scoring: Selection policy $\{\text{CC}, \text{Recency}, \text{Random}\} \times$ retained history length $K \in \{4, 8, 16, 32\}$. The main evidence layer is a 12-cell matrix on MovieLens-1M and BookCrossing, evaluated over seeds 42/43/44 with 512 users per dataset. One scope is one dataset-seed pairing, so the paper analyzes 6 scopes and 72 scope-level observations in total. The strongest evidence-backed findings are fourfold. First, selection is a first-order factor in the descriptive ANOVA-style decomposition: the eta-squared-style contributions are 0.1337 for selection, 0.0487 for retained length, and 0.0225 for the interaction. Second, the dataset split matters. On BookCrossing, CC exceeds Recency at every matched K (+0.0131 to +0.0238 mean AUC), with the best observed cell at CC-K-16 (0.676261 AUC, 348.20 tokens). On MovieLens-1M, Recency exceeds CC for $K \in \{4, 8, 16\}$ and becomes effectively tied at $K = 32$, with the best observed cell at Recency-K-16 (0.777892 AUC, 297.35 tokens). Third, paired tests show Random is often weaker than CC or Recency, but CC-versus-Recency is condition-dependent rather than one-sided. Fourth, saturation is policy-specific: CC and Recency flatten or reverse at high K, whereas Random improves monotonically with K in this bundle while remaining below the strongest CC/Recency cells. We therefore avoid single-winner narratives and frame the contribution as a controlled empirical study of selection-policy and retained-length interactions under a shared scoring protocol, not as a new recommender architecture or a production-latency benchmark. For e-business recommendation, the practical contribution is to treat history packaging as a deployment knob: low-token Recency cells expand feasible operating regions when serving budgets are tight, while CC-K-16 is attractive when higher-quality ranking justifies additional prompt cost.

1 Introduction

Large language models are becoming an increasingly common substrate for recommender systems because they can consume rich metadata, support flexible prompting, and unify retrieval, ranking, and explanation inside a single framework [15, 9, 17]. At the same time, recommendation remains a long-history decision problem. A practical system must choose which parts of a user's history should survive into the final scoring context.

This is where prompt budget becomes a systems and e-business issue rather than a cosmetic prompting detail. User histories are often much longer than the context a practical model can score repeatedly. More context can help, until it does not. Tokens cost money, longer prompts slow inference, and naive truncation can remove exactly the behavioral evidence that personalization needs. On high-volume

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ranking surfaces, those prompt decisions propagate into service cost, throughput ceilings, and latency headroom, which are core concerns in value-aware recommender deployment [5, 2, 12]. The central design problem is therefore to determine which evidence should remain in context, how it should be structured, and which retention rule should govern selection under a fixed prompt budget.

Existing LLM-based recommenders attack this challenge through several routes. Retrieval-oriented systems such as ReLLa compress long histories into candidate-conditioned subsets before prompting [7]. Other approaches combine instruction tuning, reranking, collaborative filtering augmentation, or profile construction to improve alignment with recommendation tasks [15, 9, 13, 16, 14]. Long-context and memory-oriented studies make a related point from another angle: simply exposing more context does not guarantee better behavior because selection, position, and packaging also matter [8, 10, 6, 1, 4].

However, one important part of this design space is still underexplored. Once the backbone and the candidate protocol are held fixed inside a single recommendation framework, the effect of the history-packaging rule itself remains unclear. Existing work often changes both the framework and the packaging strategy at the same time, which makes it difficult to isolate whether the gain comes from the backbone, the retrieval recipe, or the packaging policy.

This paper studies that problem inside one synchronized recommendation framework. We fix one shared candidate protocol and one frozen LLM scorer, then evaluate a factorial design over packaging policies: Selection $\in \{\text{CC}, \text{Recency}, \text{Random}\}$ and retained length $K \in \{4, 8, 16, 32\}$. The main evidence is a 12-cell matrix on MovieLens-1M and BookCrossing, evaluated across seeds 42, 43, and 44 with 512 users per dataset. One scope is one dataset-seed pairing, so the paper analyzes 6 scopes and 72 scope-level observations; this is distinct from the underlying $512 \times 2 \times 3 \times 12$ user-cell evaluations used to compute those scope means. We do not claim a new state-of-the-art recommender, a new semantic retriever, or broad cross-family superiority over external systems such as ReLLa or RALLRec+. Instead, the methodological contribution is a controlled measurement design for a deployment question: under real LLM scoring, how do selection policy and retained length interact in the quality–cost tradeoff?

The answer is already more nuanced than a pooled single-winner story. In BookCrossing, CC is consistently above Recency at matched K, and CC-K-16 is the strongest observed cell. In MovieLens-1M, Recency is stronger for $K \in \{4, 8, 16\}$ and effectively tied with CC at $K = 32$, with Recency-K-16 as the strongest observed cell. A cautious reading is that, under the tested slices and shared scorer setup, BookCrossing benefits more from candidate-conditioned lexical or topical matching, whereas MovieLens-1M benefits more from short-horizon recency signal. That split is operationally important because it changes which frontier point is attractive under a given cost budget.

This empirical framing responds to a practical gap: many systems must choose a packaging rule before they can afford broader model or infrastructure sweeps. The contributions of this paper are as follows:

- We formulate prompt-budgeted history packaging in a synchronized recommendation framework as a factorial design over selection policy (CC, Recency, Random) and retained history length ($K = 4, 8, 16, 32$), clarifying the workflow from history retention to LLM scoring and cost accounting.
- We define a controlled evidence stack in which the main layer is a real-LLM-scored 12-cell matrix over two datasets, three seeds, and 512 users per dataset, and we make the scope accounting explicit: 6 dataset-seed scopes and 72 scope-level observations.
- We show that the pooled result hides a meaningful interaction: BookCrossing favors CC across matched K, while MovieLens-1M favors Recency through $K = 16$ and only converges at $K = 32$.
- We show that quality–cost frontiers and saturation behavior are policy-dependent: Recency-K-8 is a low-token frontier point, CC-K-16 reaches the highest pooled mean AUC, and later-K gains

flatten for CC/Recency while Random continues to improve monotonically from a weaker starting point.

2 Related Work

Recent work has shown that large language models can support recommendation, but long interaction histories remain difficult to use well. Retrieval-oriented methods such as ReLLa focus on extracting compact candidate-conditioned subsets of user behavior before final prompting [7]. Other systems study LLM-based recommendation more broadly, including recommendation-specific instruction tuning, retrieval-reranking pipelines, collaborative-filtering augmentation, and cold-start settings [15, 9, 13, 16, 14].

Our paper is narrower than those efforts. We do not study broad leaderboard performance or agentic recommendation behavior. Instead, we isolate a specific systems question that remains practically important when prompt budgets are fixed: how should a recommender package user history before scoring? We also keep the claim boundary narrow. ReLLa and RALLRec+ provide context for the broader space, but our empirical claim is internal to one synchronized recommendation setup with a shared candidate protocol and frozen scorer.

That question also connects to a broader observation from long-context LLM studies. Simply exposing more context does not guarantee better behavior, and representation, position, and packaging matter [8]. Recent profile and memory-modeling papers make a related point by treating long behavior sequences as a structured memory problem rather than as a simple prompting problem [10, 6, 1, 4]. Our saturation analysis is therefore theory-oriented but bounded: when later retained items add smaller gains than earlier ones, the pattern is consistent with diminishing returns from additional behavioral evidence rather than evidence that more history is universally better.

The paper also speaks to the e-business and value-aware recommendation literature. Economic recommender analyses emphasize that ranking quality must be read jointly with cost, service constraints, and business value rather than as an isolated offline metric [5, 2]. Work on profit-aware recommendation makes the same operational point from a different angle: ranking decisions are entangled with system objectives and resource tradeoffs [12]. In addition, work on choice overload shows that more options or more information do not automatically improve user outcomes [11]. We position our paper inside that operational perspective: retained history length is a scarce resource, and the empirical question is which packaging policy uses that resource most effectively under a fixed scoring contract.

3 Problem Formulation

We study prompt-budgeted recommendation for a user u with interaction history $H_u = (i_1, \ldots, i_n)$ and candidate item c. The key question is not whether more history exists, but how that history should be packaged before scoring when prompt cost is limited. The paper evaluates packaging policies under a shared candidate protocol and scorer, so differences can be attributed to the retention rule rather than to a different ranking backbone.

Let $\tau(i)$ denote the token set extracted from the title and category fields of item i. Following the implementation, the history query set is built from the most recent $\kappa = 3$ interactions:

$$Q(H_u) = \bigcup_{t=\max(1,n-\kappa+1)}^{n} \tau(i_t), \tag{1}$$

with fallback to the union over all history tokens if the recent set is empty.

For a candidate c, the synchronized framework ranks each history item by candidate-conditioned token overlap:

$$r(i,c \mid H_u) = |\tau(i) \cap (Q(H_u) \cup \tau(c))|. \tag{2}$$

History items are sorted by descending $r(i,c \mid H_u)$, with later interactions preferred on ties. Let $\pi(H_u,c)$ denote the resulting ranked order. The paper studies how different retention rules applied to $\pi(H_u,c)$ change quality and cost. In addition to the candidate-conditioned policy, we include recency-only and random-K packaging baselines so the results can distinguish informed packaging from naive truncation.

4 Factorial Packaging Policies and Evidence Layers

4.1 Packaging policies

All cells share the same candidate-conditioned ranking backbone; only selection policy and retained length vary. The candidate-conditioned rule is intentionally simple and lexical, not an embedding-based semantic retriever. This choice keeps the experimental contrast focused on packaging behavior under a transparent retention rule.

CC selection. CC keeps the top K items in the candidate-conditioned order:

$$S_{\mathrm{CC}}(H_u,c;K) = \mathrm{TopK}(\pi(H_u,c),K). \tag{3}$$

In the factorial bundle, $K \in \{4,8,16,32\}$.

Recency selection. Recency ignores candidate-conditioned ranking and keeps the most recent K interactions:

$$S_{\mathrm{Recency}}(H_u;K) = \{i_{n-K+1},\ldots,i_n\}. \tag{4}$$

Random selection. Random samples K interactions from H_u under the experiment seed:

$$S_{\mathrm{Random}}(H_u;K) = \mathrm{Sample}(H_u,K). \tag{5}$$

Together these define the 12 canonical cells

$$\{\mathrm{CC},\mathrm{Recency},\mathrm{Random}\} \times \{4,8,16,32\}. \tag{6}$$

Figure 1 summarizes how the synchronized protocol turns a user history and candidate into comparable quality–cost evidence. The same workflow is used for every factorial cell; only the retention block changes.

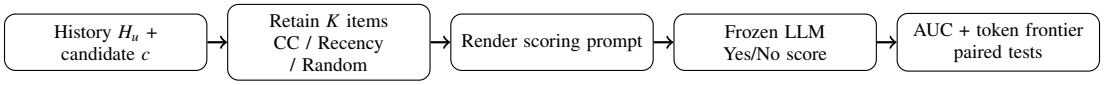

Figure 1: Workflow of the synchronized factorial protocol used in every cell; only the retention block changes.

Why adaptive variants are excluded. Adaptive operating-point variants are outside the paper's core factorial design. The reason is methodological rather than rhetorical: those variants change the retention rule itself across conditions, so they do not preserve a clean 3×4 comparison in which only selection family and retained length vary. Excluding them keeps the manuscript's main claims traceable to a balanced factorial surface with shared scorer, shared candidate protocol, and directly comparable cells.

4.2 Evidence layers and cost accounting

The main evidence layer uses one frozen Yes/No scorer, `Qwen/Qwen2.5-1.5B-Instruct`. It builds a prompt from the retained history and candidate and evaluates

$$s_{\text{llm}}(S,c) = \log P(\text{Yes} \mid \text{prompt}) - \log P(\text{No} \mid \text{prompt}). \tag{7}$$

We keep a leakage-free deterministic verifier as a supporting layer rather than a primary evidence layer because it offers a controlled approximation that is cheap to rerun and easier to inspect mechanistically. Selection still uses title and category tokens, but the deterministic verifier scores candidates from title tokens only. Let $\tau_{\text{ttl}}(i)$ denote whitespace-tokenized title tokens and $V_{\text{ttl}}(S) = \bigcup_{i \in S} \tau_{\text{ttl}}(i)$ the retained-history title vocabulary. The verifier is

$$s_{\text{det}}(S,c) = \sum_{w \in \tau_{\text{ttl}}(c)} \mathbb{1}[w \in V_{\text{ttl}}(S)]. \tag{8}$$

The factorial package reports the measured LLM input cost directly as `actual_input_tokens_avg`, i.e., the true scorer-side prompt token count after retention and candidate rendering. Additional overhead fields are retained only as auxiliary diagnostics; the headline quality–cost analysis uses measured input burden rather than proxy accounting.

5 Experimental Setup

5.1 Research Questions

We organize the paper around three research questions.

- **RQ1**: Under real LLM scoring, how strong are the selection-policy and retained-length effects in a factorial matrix?
- **RQ2**: Which factorial cells lie on the quality–actual-token Pareto frontier?
- **RQ3**: When do paired comparisons support CC over Recency, when do they support Recency over CC, and when is the evidence only mixed or tied?

5.2 Experimental Settings

5.2.1 Datasets

We use two CTR-style recommendation datasets. MovieLens-1M contains one million movie ratings from 6,040 users and 3,952 movies [3]. BookCrossing contains roughly 1.15 million book ratings from a large community of readers [18]. Both datasets are converted into CTR comparison slices by raw-interactions last-item holdout: all but the final interaction form the history and the final interaction becomes the positive target.

We prepare 512 users per dataset and evaluate seeds $\{42, 43, 44\}$. A user must have at least 20 interactions to enter the default slice. This yields two related counting levels that the paper keeps separate. At the user level, the scorer executes $512 \times 2 \times 3 \times 12$ user-cell evaluations. At the manuscript evidence level, one *scope* is one dataset-seed pairing, so there are $2 \times 3 = 6$ scopes. Each scope contributes 12 cell means, producing $6 \times 12 = 72$ scope-level observations for the descriptive factorial analyses.

5.2.2 Evaluation Protocol

Each CTR example contains one positive target and 19 negatives sampled without replacement from the global item pool after excluding the target and all history items; the 20 candidates are then shuffled. Our main analysis centers on AUC and measured actual input tokens because the paper asks a quality–cost question. Actual input tokens are a controlled cost proxy for the prompt burden; they should not be read as measured production latency, throughput, GPU memory, or online CTR. Paired significance tests use per-user AUC with Wilcoxon signed-rank tests and paired bootstrap intervals, and we use them to summarize directional stability rather than to make causal claims.

5.2.3 Implementation

The main matrix executes all 12 canonical cells (CC / Recency / Random $\times$ $K \in \{4, 8, 16, 32\}$) across both datasets and seeds $\{42, 43, 44\}$ using one frozen scorer, Qwen/Qwen2.5-1.5B-Instruct, throughout. Prompt cost is reported as measured actual input tokens, i.e., the true scorer-side token count after retention and candidate rendering, rather than any proxy accounting field.

6 Main Results

Table 1 and Figure 2 answer the main question under real LLM scoring for the 12-cell factorial design. The headline is not that one policy wins everywhere. The evidence supports a structured tradeoff surface whose shape depends on dataset, retained length, and cost. Pooled across scopes, CC-K-16 achieves the highest mean AUC (0.722262) at 329.77 measured input tokens, while Recency-K-8 reaches 0.713636 AUC at a substantially lower token cost (185.86). Random-family cells remain quality-limited across the matrix despite low-token variants.

The descriptive ANOVA-style decomposition confirms that policy choice is first-order in this bundle: relative to the grand mean AUC (0.696158), CC and Recency shift upward (+0.0160 and +0.0136), whereas Random shifts downward (-0.0296). The eta-squared-style shares show the same ordering, with selection (0.1337) larger than retained length (0.0487) and the interaction still material (0.0225). These shares are descriptive, not causal estimates. Their practical meaning comes from the induced operating decision: the interaction is smaller than the main policy effect, but it is large enough to reverse the preferred structured policy across datasets and to move the frontier point selected under a token budget.

The dataset split makes the CC-versus-Recency interaction concrete. In BookCrossing, CC exceeds Recency at every matched K: the mean CC-minus-Recency deltas are +0.018469 at $K = 4$, +0.013141 at $K = 8$, +0.023832 at $K = 16$, and +0.021399 at $K = 32$. Significance is limited at smaller K but becomes stronger at larger K, with CC significant in 2/3 seeds at $K = 16$ and 3/3 seeds at $K = 32$. In MovieLens-1M, the pattern flips through most of the range: the mean CC-minus-Recency deltas are -0.022752 at $K = 4$, -0.025271 at $K = 8$, -0.009629 at $K = 16$, and +0.000411 at $K = 32$, with Recency significant in all three seeds at $K = 4$, in 2/3 seeds at $K = 8$, in 1/3 seeds at $K = 16$, and no decisive winner at $K = 32$. Put differently, BookCrossing favors the candidate-conditioned rule across matched K, whereas MovieLens-1M favors recency until the highest tested retention level, where the two policies converge. A cautious observational interpretation is that, under the tested slices, BookCrossing gains

Table 1: Main real-LLM-scored factorial comparison over two datasets and seeds $\{42, 43, 44\}$ (12 canonical cells). Cells report mean AUC / measured actual input tokens. Dataset-separated columns make the interaction visible inside the manuscript.

Policy	K	BookCrossing	ML1M	Pooled
CC	4	0.663926 / 134.17	0.732165 / 122.66	0.698045 / 128.42
CC	8	0.668431 / 211.98	0.746710 / 186.57	0.707571 / 199.28
CC	16	**0.676261 / 348.20**	0.768263 / 311.34	**0.722262 / 329.77**
CC	32	0.670813 / 547.59	0.771090 / 540.47	0.720951 / 544.03
Recency	4	0.645456 / 128.36	0.754917 / 120.66	0.700187 / 124.51
Recency	8	0.655290 / 192.12	0.771981 / 179.59	**0.713636 / 185.86**
Recency	16	0.652429 / 317.70	**0.777892 / 297.35**	0.715161 / 307.53
Recency	32	0.649414 / 518.55	0.770679 / 519.22	0.710047 / 518.88
Random	4	0.591317 / 124.97	0.672406 / 120.40	0.631862 / 122.69
Random	8	0.611619 / 187.22	0.705284 / 179.52	0.658452 / 183.37
Random	16	0.629591 / 311.76	0.731873 / 297.73	0.680732 / 304.75
Random	32	0.643435 / 514.73	0.746539 / 520.12	0.694987 / 517.43

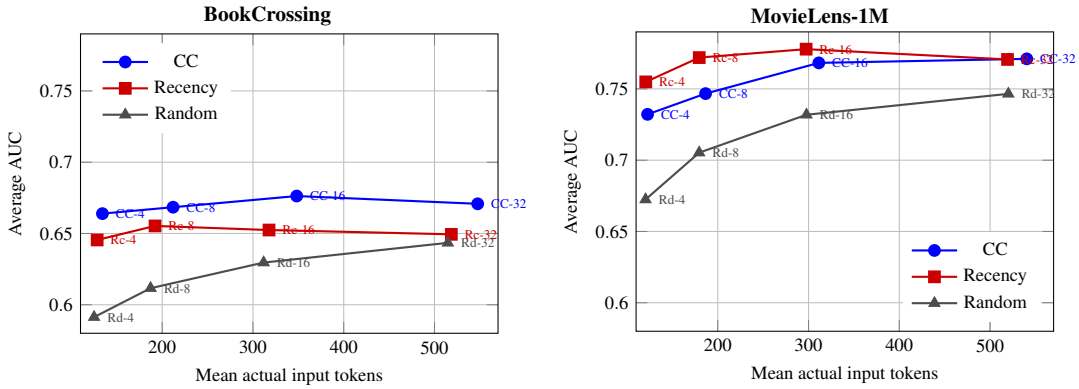

Figure 2: Dataset-separated quality–cost frontiers for the real-LLM-scored factorial matrix. BookCrossing shows a CC-led high-quality frontier, whereas MovieLens-1M shows a Recency-led frontier through $K = 16$; Random improves with K but remains below the strongest structured cells in both datasets.

more from candidate-conditioned lexical or topical matching, while MovieLens-1M gains more from short-horizon recency signal.

These dataset-separated cells also answer RQ2 more precisely. The pooled Pareto frontier contains Random-K-4, Recency-K-4, Recency-K-8, Recency-K-16, and CC-K-16, but the dataset winners differ. In BookCrossing, CC-K-16 is the best observed cell at 0.676261 AUC and 348.20 tokens. In MovieLens-1M, Recency-K-16 is best at 0.777892 AUC and 297.35 tokens, while CC-K-32 slightly exceeds CC-K-16 within the CC family but with much higher token cost. This is why a pooled frontier should be read as a deployment menu rather than as evidence for a universal policy ordering.

RQ3 asks whether paired tests support strong cross-family claims or only bounded ones. Across cross-family comparisons, CC-vs-Random is significant in 79/96 cases and Recency-vs-Random in 74/96, so Random is often weaker than either structured policy. By contrast, CC-vs-Recency is mixed overall (45/96 significant cases) and directionally depends on dataset, as the matched-K summaries above show. The paired evidence therefore reinforces the operating-region view instead of supporting a universal CC-or-Recency ordering.

The saturation evidence is also part of the main story. For CC, pooled marginal gains rise from +0.0095 AUC at $4 \rightarrow 8$ to +0.0147 at $8 \rightarrow 16$, then turn slightly negative at $16 \rightarrow 32$ despite a large token

Table 2: Saturation summary from the real-LLM-scored factorial matrix. Values report marginal AUC gains and marginal token cost by policy as K increases.

Policy	Marginal gains as K increases	Interpretation
CC	$4 \to 8$: +0.0095 AUC for +70.86 tokens; $8 \to 16$: +0.0147 for +130.49; $16 \to 32$: −0.0013 for +214.26	Best pooled quality appears at $K = 16$; the $K = 32$ step adds cost without added pooled AUC.
Recency	$4 \to 8$: +0.0134 AUC for +61.35 tokens; $8 \to 16$: +0.0015 for +121.67; $16 \to 32$: −0.0051 for +211.36	Earliest gain is efficient; later gains saturate fastest, which helps explain the low-token frontier points.
Random	$4 \to 8$: +0.0266 AUC for +60.69 tokens; $8 \to 16$: +0.0223 for +121.37; $16 \to 32$: +0.0143 for +212.68	Improves monotonically from a weak low-K baseline, but still remains below the strongest structured cells in absolute AUC.

increase. Recency behaves similarly but saturates earlier: +0.0134 at $4 \to 8$, only +0.0015 at $8 \to 16$, then -0.0051 at $16 \to 32$. Random, in contrast, improves monotonically (+0.0266, +0.0223, +0.0143) as K grows. We interpret that pattern cautiously. Random starts from a much weaker low-K baseline, so additional retained items can still recover omitted signal without making Random competitive with the strongest CC/Recency cells. The monotonic gains therefore do not overturn the main finding; they show that even a weak unstructured policy benefits from more retained evidence before its own returns begin to taper.

Taken together, the factorial view replaces single operating-point narratives with an operational region. Lower-token Recency-K cells are attractive when serving budgets are tight, while CC-K-16 is attractive when the deployment objective prioritizes peak pooled AUC and can absorb the extra prompt cost.

7 Saturation and Sensitivity Analyses

This section extends the main factorial results rather than replacing them. The paper's central evidence still comes from the real-LLM-scored matrix in Table 1, but saturation analysis helps explain why some frontier cells are efficient and why later-K expansions stop paying off.

7.1 Policy-specific saturation behavior

Table 2 summarizes K-step marginal gains from the factorial matrix. CC and Recency both show clear diminishing returns by $K = 32$, where extra token cost is large and the marginal AUC gain is near-zero or negative. This pattern is consistent with the idea that earlier retained interactions capture most of the useful behavioral evidence, while later additions become increasingly redundant or weakly aligned with the immediate scoring task. One compact way to formalize that intuition is through a diminishing-returns condition on retained-history information,

$$I(H_k; y) - I(H_{k-1}; y) \geq I(H_{k+1}; y) - I(H_k),$$

although we do not estimate this quantity directly in the present study. We present this as an observational interpretation rather than a proof.

The dataset split sharpens that explanation. In BookCrossing, CC improves through $K = 16$ and then declines at $K = 32$. Recency peaks earlier and turns negative sooner. In MovieLens-1M, both structured policies still gain through $K = 16$, but the later step is small for CC (+0.002827) and negative for Recency (-0.007213). These patterns align with the main-results interaction. The stronger policy is the one whose earlier retained items match the dataset's useful signal most efficiently.

Random requires separate interpretation. Its gains are monotonic in both datasets and in the pooled curve, but the increments still shrink with larger K. That is consistent with a weaker selector that begins by discarding too much useful evidence at low K; adding more random history partially repairs that information loss, yet the policy remains below the strongest CC/Recency cells in absolute AUC. The practical message is therefore not that Random is preferable, but that monotonic recovery can coexist with inferiority when the initial selection rule is weak.

These saturation trends explain the frontier geometry in Figure 2. Recency-K-8 is efficient because its earliest gain is high per token, while CC-K-16 is attractive when the objective emphasizes peak pooled AUC. The design implication is not to maximize K blindly, but to choose the operating region where additional retained history still earns meaningful quality gains per token under the target policy.

8 Discussion and Limitations

The evidence supports a narrow but trustworthy claim. The strongest stable result is not universal superiority for one packaging rule. It is that selection policy and retained history length jointly determine a structured quality–cost surface in the observed real-LLM matrix, and that the preferred structured policy changes across datasets. That is useful for system design because it supports frontier-oriented decisions instead of one-size-fits-all prescriptions.

This result also has a direct e-business interpretation. The measured input-token differences across frontier cells are not merely cosmetic prompt savings. Under serving stacks where inference burden scales approximately with prompt tokens, they translate into lower per-request operating cost, higher throughput under fixed infrastructure budgets, and a wider feasibility region for deploying LLM-based ranking on business-critical surfaces [5, 2, 12]. The paper therefore contributes a deployment-facing message: history packaging is an operational control variable, not only an offline modeling choice.

The saturation results matter in that operational framing because they constrain how far extra context should be pushed. More retained history is not automatically better, and the later-K flattening for CC and Recency is consistent with diminishing returns from additional user evidence. In that sense, the observed curves are also consistent with a diminishing-returns view of information selection, in which earlier retained items contribute larger marginal signal than later additions. That interpretation also resonates with adjacent work on overload and information efficiency in digital decision settings [11]. We intentionally keep the theoretical claim modest: the observed curves are aligned with diminishing-returns intuition, but the paper does not prove a general law for all recommenders or all prompt contracts.

The paper has several scope limits. First, the real LLM scorer is still one frozen evaluation contract, specifically `Qwen/Qwen2.5-1.5B-Instruct`, not the full space of possible recommenders or judge prompts. The primary scorer is a 1.5B-parameter instruction-tuned model, so whether larger open models, proprietary LLMs, or differently calibrated judge prompts exhibit the same packaging sensitivity remains an open question. Second, only two traditional recommendation domains are evaluated. This is sufficient for a controlled interaction study, but not for a generalization claim across news, retail, short-video, or session-based recommendation. Third, the CC policy is deliberately simple and lexical. More advanced semantic retrieval, embedding-based history selection, or hybrid profile construction could shift the frontier and should be compared in future work. Fourth, paired significance is strongest for comparisons against Random-family cells, while CC-versus-Recency comparisons are dataset-dependent and often non-significant, so cross-family claims must stay specific.

The cost and statistical readings should also remain bounded. Actual input tokens are measured from the scorer-side prompt, but they are still a proxy for deployment cost rather than direct measurements of latency, throughput, GPU memory, serving price, or online CTR. Likewise, the ANOVA-style output is descriptive rather than confirmatory causal inference, and the paired tests summarize directional stability under the tested user slices rather than a full mixed-effects or multiple-comparison-adjusted analysis.

These limits are deliberate: the goal is a clean, traceable contribution about history packaging under one synchronized recommendation contract rather than an overextended superiority claim. A natural next step is to replicate the same workflow across additional domains, stronger scorers, semantic retrievers, and production-oriented metrics.

9 Conclusion

This paper studies a systems question for recommendation under prompt constraints: how user history should be packaged before candidates are scored. Using a real-LLM-scored factorial matrix over MovieLens-1M, BookCrossing, seeds 42/43/44, and 512 users per dataset, we show that selection policy and retained length jointly shape a measurable quality–cost surface. The paper makes the evidence scale explicit: one scope is one dataset-seed pairing, giving 6 scopes and 72 scope-level observations across the 12 factorial cells.

The main conclusion is conditional rather than universal. CC-K-16 reaches the highest pooled mean AUC, but the dataset split shows that BookCrossing favors CC across matched K whereas MovieLens-1M favors Recency through $K = 16$ and only converges at $K = 32$. Random-family cells often improve as more history is retained, yet they usually remain weaker than the strongest structured cells. For e-business recommendation systems, this supports a practical workflow: treat history packaging as an operating-region decision, choose frontier cells according to token budget and quality target, and keep claims tied to the observed scorer contract instead of overgeneralizing beyond it. Future work should test the same workflow with broader datasets, larger or proprietary LLM scorers, semantic selection policies, and measured production metrics such as latency, throughput, serving cost, and online engagement.

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A Formal Security Analysis of SUCI-Stage Reused Key Binding in the 5G-AKA Family*

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Abstract

The 5G Authentication and Key Agreement (5G-AKA) family protects subscriber privacy in the initial identification stage through the Subscription Concealed Identifier (SUCI). However, the cryptographic relationship between the SUCI stage and the subsequent AKA stage remains relatively loose in many designs, leaving open questions regarding the security benefits of tighter cross-phase binding. This paper investigates the security implications of reusing SUCI-stage key materials to bind later AKA computations in the 5G-AKA family. Rather than focusing on a single protocol instance, we study this binding strategy as a general design principle and analyze its impact on representative 5G-AKA-style constructions. Our analysis focuses on two main questions: whether reused key binding introduces explicit cross-phase consistency between the SUCI and AKA stages, and whether it mitigates replay-driven failure-message-based linkability. To support this comparison, we also check basic SUCI-stage consistency using correspondence queries. We develop formal models in ProVerif and compare a baseline 5G-AKA-style setting with a bound setting that incorporates SUCI-stage reused key binding. In particular, we use correspondence queries to analyze basic SUCI-stage consistency and explicit cross-phase binding, and we employ choice-based observational equivalence to evaluate replay-driven linkability between same-subscriber and different-subscriber replay cases. The results show that both models preserve basic SUCI-stage consistency, but only the bound model satisfies the explicit cross-phase binding query. In addition, the bound model yields observational equivalence in the replay-driven linkability experiment, whereas the baseline model does not. These findings indicate that SUCI-stage reused key binding is not merely an implementation optimization, but a meaningful security design principle that can transform a weakly coupled SUCI-to-AKA relation into an explicitly bound one and can eliminate replay-driven distinguishability in the modeled setting.

1 Introduction

The 5G Authentication and Key Agreement (5G-AKA) family provides subscriber authentication and anchor-key establishment in 5G systems [3]. A major privacy-oriented enhancement over earlier generations is the use of the Subscription Concealed Identifier (SUCI), which protects the permanent subscriber identity during the initial access procedure [1]. Unless otherwise stated, references to 3GPP specifications in this paper correspond to the fixed versions listed in the bibliography. Nevertheless, prior studies have shown that, despite this improvement, the

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5G-AKA family still exhibits privacy limitations under active attacks, including replay-driven distinguishability and linkability issues [8, 4, 9].

This paper studies *SUCI-stage reused key binding* as a design principle in the 5G-AKA family. Rather than focusing on a single protocol variant, we investigate what security advantages can be obtained when key materials established during the SUCI stage are reused to bind later AKA computations. Our main objective is to clarify how such binding affects explicit cross-phase consistency and resistance to replay-driven linkability attacks. Recent studies on bound 5G-AKA-style constructions motivate this question, but our focus is the security effect of the design principle itself rather than that of any single protocol proposal [10, 7].

1.1 Related Work

Prior studies have shown that, although SUCI improves identity protection, the 5G-AKA family still has privacy limitations under active attacks [8, 9]. Formal analyses have further shown that the security guarantees of 5G-AKA depend strongly on the exact threat model and on how protocol phases are connected [4]. More recent protocol designs and reassessments have explored stronger coupling between the concealment and authentication stages, including approaches that reuse key material derived during the SUCI stage to improve freshness handling and resistance to linkability-related attacks [10, 7].

1.2 Motivation

The main motivation of this work is that the SUCI and AKA stages are consecutive but often only loosely coupled from a cryptographic perspective [3, 2]. If key materials established during the SUCI stage are reused in subsequent AKA computations, the resulting cross-phase binding may provide security benefits beyond identity concealment alone. In particular, such binding may reduce replay-driven distinguishability by making later authentication depend explicitly on the current SUCI-originated context [8, 10, 7]. This motivates a formal investigation of whether SUCI-stage reused key binding can strengthen the 5G-AKA family against classical failure-message-based linkability patterns.

1.3 Contributions

The contributions of this paper are summarized as follows.

- We identify SUCI-stage reused key binding as a reusable cross-phase security design principle in the 5G-AKA family.
- We present a formal analysis framework for evaluating its security implications with respect to explicit cross-phase binding and replay-driven linkability resistance.
- We model replay-based linkability scenarios in ProVerif using choice-based observational equivalence and compare same-subscriber and different-subscriber cases under reused-key binding.
- We show that SUCI-stage reused key binding can convert a weakly coupled SUCI-to-AKA relation into an explicitly bound one at the event level and can eliminate replay-driven distinguishability in our formal model.

The remainder of this paper is organized as follows. Section 2 introduces the background and design rationale. Section 3 presents the system and threat models. Section 4 describes the formal verification methodology. Section 5 discusses the verification results and security implications. Finally, Section 6 concludes the paper.

2 Background and Design Rationale

2.1 Notation

Table 1 summarizes the main symbols used in this paper. The notation is intentionally kept abstract so that the same analytical framework can be applied to both the baseline 5G-AKA-style setting and the bound setting with SUCI-stage reused key binding.

Table 1: Main notation used in the analysis

Symbol	**Description**
UE, SN, HN	User equipment, serving network, and home network
SUPI	Permanent subscriber identifier
SUCI	Subscription Concealed Identifier transmitted during initial access
K	Long-term subscriber key shared between UE and HN
SQN	Sequence-related freshness state
sk_{UE}, pk_{UE}	UE ephemeral secret/public key used in SUCI generation
sk_{HN}, pk_{HN}	HN long-term secret/public key for SUCI recovery
c_0, c_1, c_2	Abstract SUCI components derived during concealment
RAND	Baseline AKA challenge material
Y	Fresh challenge-related public value used in the bound setting
DHK	Diffie–Hellman-derived key material used for cross-phase binding
CONC	Concealed freshness-related field included in authentication data
MAC	Message authentication code used for AKA verification
RES'	UE response used for challenge verification
K_{AUSF}, K_{SEAF}	Anchor keys derived during later authentication processing

2.2 Baseline 5G-AKA Overview

Figure 1 presents the baseline 5G-AKA-style procedure used as the reference flow in this paper. At a high level, the protocol consists of two consecutive stages. In the first stage, the UE conceals its permanent identity by generating and sending a SUCI. The HN then verifies the concealed identifier, recovers the subscriber identity, and retrieves the associated subscriber state. In the second stage, the HN generates authentication vectors, the UE verifies the received challenge and computes its response, and the network finally verifies the response and establishes the anchor key.

The purpose of Figure 1 is not to reproduce the full standard message specification. Instead, it provides a compact procedural view of the baseline 5G-AKA-style flow that serves as the reference point for our later formal comparison. In particular, it shows that the SUCI stage and the AKA stage are sequentially connected, while leaving open the question of how strongly they are connected from a cryptographic viewpoint in the baseline abstraction used in this paper.

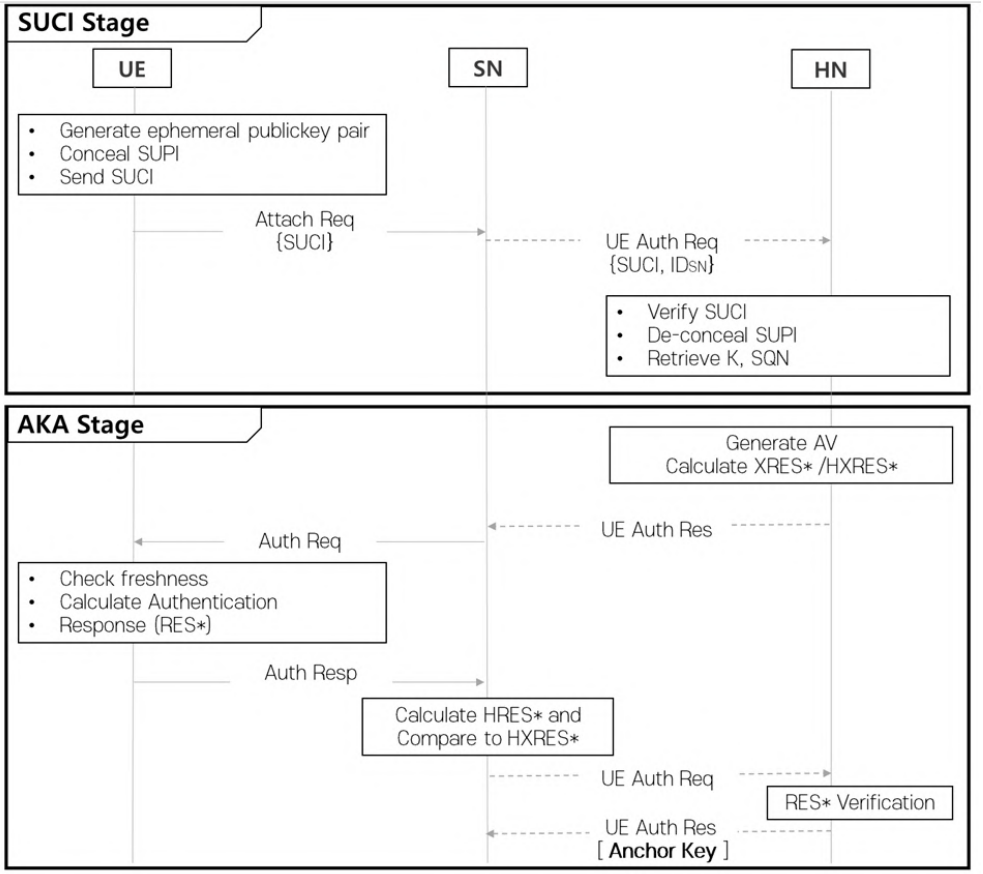

Figure 1: Simplified baseline 5G-AKA-style procedure used as the reference flow in this paper. The SUCI stage covers subscriber identity concealment and subscriber-state retrieval, while the AKA stage covers authentication-vector generation, UE response computation, RES* verification, and anchor-key establishment. In the baseline abstraction used in this paper, the later AKA verification is sequentially connected to, but not explicitly cryptographically bound to, the current SUCI-stage context.

2.3 Baseline Procedure

Algorithm 1 summarizes the baseline 5G-AKA-style procedure used in our analysis. It follows the same high-level flow as Figure 1, but describes the internal processing in a more abstract and verification-oriented form. In this baseline abstraction, the later AKA verification does not explicitly depend on the current SUCI-related context beyond the functional transition between the two phases.

2.4 Bound Procedure with SUCI-Stage Reused Key Binding

Algorithm 2 presents the bound setting analyzed in this paper. The overall message flow remains similar to the baseline reference flow shown in Figure 1 and Algorithm 1. However, the later

Algorithm 1: Baseline 5G-AKA-style processing

Input: $SUPI$, K, SQN, pk_{HN}
Output: RES', K_{AUSF}, K_{SEAF}, or failure
Generate a fresh UE ephemeral key pair (sk_{UE}, pk_{UE});
Compute SUCI components (c_0, c_1, c_2) using $(SUPI, sk_{UE}, pk_{HN})$;
Send SUCI $= (c_0, c_1, c_2)$ from UE to the network;
HN verifies the SUCI components and recovers $SUPI$;
HN retrieves the subscriber state (K, SQN) from its database;
HN generates fresh baseline challenge material $RAND$;
HN computes $CONC$ and MAC from $(RAND, K, SQN, AMF)$;
Send authentication data $(RAND, CONC, AMF, MAC)$ to UE;
UE verifies MAC using $(RAND, K)$;
if *MAC verification fails* **then**
  Abort with failure;
Recover the freshness-related value from $(CONC, RAND, K)$;
Compare the recovered value against the local SQN-related state;
if *freshness verification fails* **then**
  Abort with failure;
Compute RES' and the later key material;
Send RES' to the network;
SN/HN verify RES' and derive K_{AUSF} and K_{SEAF};

AKA computation now depends on key material that is cryptographically tied to the SUCI stage. Concretely, the HN introduces fresh challenge-related context and combines it with the UE-side SUCI context to derive bound authentication values. This makes later AKA acceptance explicitly dependent on the current SUCI-originated context in the bound abstraction.

Algorithm 2: Bound processing with SUCI-stage reused key binding

Input: $SUPI$, K, SQN, pk_{HN}
Output: RES', K_{AUSF}, K_{SEAF}, or failure
Generate a fresh UE ephemeral key pair (sk_{UE}, pk_{UE});
Compute SUCI components (c_0, c_1, c_2) using $(SUPI, sk_{UE}, pk_{HN})$;
Send SUCI $= (c_0, c_1, c_2)$ from UE to the network;
HN verifies the SUCI components and recovers $SUPI$;
HN retrieves the subscriber state (K, SQN) from its database;
HN generates fresh challenge-related context Y;
HN derives bound key material DHK from the current SUCI-related context and Y;
HN computes $CONC$ and MAC from (Y, DHK, K, SQN, AMF);
Send authentication data $(Y, CONC, AMF, MAC)$ to UE;
UE derives its local DHK from the current SUCI-related context and Y;
UE verifies MAC using (Y, DHK, K);
if *MAC verification fails* **then**
  Abort with failure;
Recover the freshness-related value from $(CONC, Y, DHK, K)$;
Compare the recovered value against the local SUCI-bound state;
if *freshness verification fails* **then**
  Abort with failure;
Compute RES' and the later key material using the bound context;
Send RES' to the network;
SN/HN verify RES' and derive K_{AUSF} and K_{SEAF};

2.5 Design Rationale and Security Scope

The key distinction between Algorithms 1 and 2 is not the overall message flow, but the source of dependency used in later AKA verification. In the baseline abstraction, the SUCI stage and the AKA stage are sequentially connected, yet the later authentication computation does not explicitly depend on the current SUCI-related context. In the bound setting, by contrast, the later AKA verification is cryptographically tied to the context established during SUCI generation.

This difference is security-relevant because replayed authentication material is processed differently in the two settings. If the later AKA stage does not explicitly depend on SUCI-stage context, stale authentication data may still induce subscriber-distinguishing behavior under replay. If, however, the later authentication computation is bound to current SUCI-stage context, replayed data are less likely to preserve the same observable behavior across different subscriber cases.

Our goal is not to claim that SUCI-stage reused key binding solves all privacy problems in the 5G-AKA family. Instead, we focus on a narrower but important question: what concrete security benefits can be obtained when SUCI-stage key materials are incorporated into later AKA computations in a cryptographically meaningful way? The subsequent sections therefore analyze basic SUCI-stage consistency, explicit cross-phase binding, and replay-driven linkability resistance using formal verification.

3 System and Threat Model

3.1 System Model

We consider a 5G-AKA-style authentication setting involving three logical entities: UE, SN, and HN [3, 2]. The baseline reference flow is summarized in Figure 1, while the detailed baseline and bound procedures used in our analysis are given in Algorithms 1 and 2, respectively.

In both settings, the UE first transmits a SUCI to conceal its permanent subscriber identity during the initial access stage, and the network subsequently executes the AKA stage to authenticate the subscriber and derive anchor keys for later communication [1, 3, 2]. Our purpose is not to analyze a specific implementation, but to compare two abstract 5G-AKA-family constructions under a common framework so that the security effect of SUCI-stage reused key binding can be isolated.

The baseline abstraction captures a structure in which SUCI generation and AKA verification are functionally consecutive but cryptographically weakly coupled. By contrast, the bound setting preserves the same high-level protocol flow while making the later AKA computation explicitly depend on key material established during the SUCI stage. This distinction forms the basis of our formal comparison [10, 7].

3.2 Threat Model

We adopt an active attacker model in which the adversary has full control over the public communication channel between the UE and the serving side. The attacker can intercept, replay, reorder, and inject messages, and may attempt to deliver stale authentication material to a target UE in order to trigger distinguishable protocol behavior [8, 9]. This model is appropriate for analyzing replay-driven linkability and failure-message-based distinguishability in the 5G-AKA family.

At the same time, we assume that the long-term subscriber key shared between the UE and HN remains uncompromised during the target session. We also assume that the HN-side subscriber database is trusted and that legitimate backend processing follows the intended protocol logic, as is standard in formal analyses of 5G authentication [4, 3]. Our focus is therefore not on database compromise or insider abuse, but on whether an external active adversary can exploit insufficient cross-phase binding between the SUCI and AKA stages.

The main attack scenario considered in this paper is replay-driven linkability. The attacker first obtains authentication material generated in an earlier session and later replays it against a subsequent session. We compare two cases: one in which the replayed material is delivered to the same subscriber and another in which it is delivered to a different subscriber. If these two cases induce distinguishable observable behavior, replay-driven linkability arises. If they remain observationally equivalent, the protocol resists this class of attack in our model [8, 9].

3.3 Comparison Scope and Security Goals

Our comparison is limited to the security effects induced by SUCI-stage reused key binding. In particular, we ask whether such binding changes the processing of stale authentication material in a way that reduces subscriber-distinguishing behavior. We do not claim that reused key binding automatically resolves all privacy limitations in the 5G-AKA family. Instead, we focus on the following goals:

- **G1: Basic SUCI-stage consistency.** The HN-side SUCI-acceptance event should correspond to a prior UE-side SUCI transmission event.
- **G2: Explicit cross-phase binding.** Later AKA acceptance should remain explicitly tied to the same SUCI-originated context when reused key binding is applied.
- **G3: Resistance to replay-driven linkability.** An active attacker should not be able to distinguish whether replayed authentication material is processed by the same subscriber or by a different subscriber solely through externally observable protocol behavior.

These goals are consistent with prior observations that privacy weaknesses in 5G-AKA are closely related not only to identity concealment itself but also to how later authentication computations react to replayed or adversarially manipulated data [8, 9, 10, 7].

In the following section, we formalize these goals in ProVerif. Goals G1–G2 are evaluated using correspondence queries, whereas Goal G3 is evaluated using choice-based observational equivalence between same-subscriber and different-subscriber replay scenarios.

4 Formal Verification Methodology

4.1 Verification Strategy

To analyze the security implications of SUCI-stage reused key binding, we construct two formal models in ProVerif: a baseline 5G-AKA-style model and a bound model in which key materials established during the SUCI stage are reused in later AKA computations. Our objective is not to prove the security of a single concrete implementation, but to isolate the security effect of cross-phase key binding under a common verification framework.

The verification strategy consists of two complementary parts. First, we use correspondence queries to examine basic SUCI-stage consistency and explicit cross-phase dependency in the

two models. Second, we use choice-based observational equivalence to evaluate replay-driven linkability by comparing the case in which replayed authentication material is delivered to the same subscriber with the case in which it is delivered to a different subscriber. This is a natural use of equivalence reasoning, since privacy-oriented properties such as anonymity, unlinkability, and non-traceability are more appropriately expressed as equivalence properties than as simple reachability properties in ProVerif [6, 5].

4.2 Goal-to-Query Mapping

Table 2 summarizes how each security goal is represented in the verification framework.

Table 2: Mapping between security goals and ProVerif analysis mechanisms

Goal	**Funcyion**	**Verification meaning**
G1	Correspondence query	The HN-side SUCI-acceptance event must correspond to a prior UE-side SUCI transmission event
G2	injective Correspondence query	Later AKA acceptance must remain explicitly tied to the same SUCI-originated context at the event level
G3	Observational equivalence	Same-subscriber replay and different-subscriber replay should be observationally indistinguishable

4.3 How Correspondence Queries Are Modeled

For Goals G1 and G2, we use correspondence queries. In ProVerif, this option is suitable when the property of interest can be expressed as a dependency between protocol events, that is, when a later event should imply the occurrence of an earlier event. In our models, we instrument the code with events representing key transitions such as SUCI transmission, successful SUCI recovery, authentication-data generation, and UE-side authentication acceptance.

This means that correspondence queries are not introduced merely as abstract labels. Rather, they are attached to the concrete control-flow points that matter for the analysis. Goal G1 is modeled by checking whether HN-side SUCI acceptance corresponds to a prior UE-side SUCI transmission. Goal G2 is modeled by checking whether later AKA acceptance remains explicitly tied to the same SUCI-originated context. In the baseline abstraction, this event-level relation is expected to be weak because the later AKA stage does not explicitly carry current SUCI-stage context. In the bound model, by contrast, the relevant events are parameterized with SUCI-related context, which makes the cross-phase dependency explicit in the verification code.

4.4 How `choice` Is Modeled

For Goal G3, we use the ProVerif `choice` construct inside a biprocess. This option is appropriate when the question is not whether a value can be reached, but whether two related executions are distinguishable by an attacker. In our setting, the two executions correspond to the same-subscriber replay case and the different-subscriber replay case.

From a modeling perspective, replay-driven linkability is encoded by keeping the replayed authentication tuple common across both sides of the biprocess while varying only the subscriber identity and the corresponding subscriber-specific state. Concretely, the left-hand side models

the case in which replayed authentication material is delivered to the same subscriber as in the original session, whereas the right-hand side models the case in which the same material is delivered to a different subscriber.

This means that `choice` is not used to represent nondeterministic protocol execution. Instead, it is used to encode two related executions within a single biprocess so that ProVerif can test whether an attacker can distinguish them observationally [6, 5]. In our implementation, the replayed authentication tuple is kept fixed, while the subscriber identity, long-term key, and associated state are varied through `choice`. This allows the verification code to isolate whether replayed authentication material induces subscriber-distinguishing behavior.

Algorithm 3: Choice-based modeling of replay-driven linkability

Input: Replayed authentication tuple T_{old}, subscriber states $(SUPI_A, K_A, SQN_A)$ and $(SUPI_B, K_B, SQN_B)$
Output: Observational-equivalence result
Fix the replayed authentication tuple T_{old} on both sides of the biprocess;
Set the left-hand side to the same-subscriber replay case;
Set the right-hand side to the different-subscriber replay case;
Instantiate the second-session UE process as follows:;
  UE_{replay}(;
    `choice`$[SUPI_A, SUPI_B]$,;
    `choice`$[K_A, K_B]$,;
    `choice`$[SQN_A, SQN_B], \ldots)$;
Run ProVerif observational-equivalence analysis;
if *equivalence is proved* **then**
  Conclude that the two replay cases are indistinguishable in the modeled scenario;
else
  Conclude that replay may induce subscriber-dependent observable behavior;

4.5 Interpretation of ProVerif Outputs

The two types of ProVerif output are interpreted differently. For correspondence queries, the result indicates whether the modeled event dependency holds under the stated assumptions. In our context, these results are used to support Goals G1 and G2. For observational equivalence, however, the ProVerif manual makes clear that failure is not reported as `false`. When ProVerif fails to establish equivalence, it returns `cannot be proved`, because the reconstructed trace only shows the first point at which the two sides may reduce differently and does not itself constitute a proof that observational equivalence is false [6].

Therefore, in our analysis, a result of `Observational equivalence is true` is interpreted as evidence supporting Goal G3 in the modeled replay scenario, whereas `Observational equivalence cannot be proved` is interpreted as an indication that replayed authentication material may induce distinguishable behavior and thus expose a linkability risk.

4.6 Result Presentation and Reproducibility

In the next section, we present the correspondence-query outputs for Goals G1 and G2 and the observational-equivalence outputs for Goal G3. Together, these results provide a structured comparison between the baseline model and the bound model.

To support transparency and reproducibility, the ProVerif source codes used in this study are publicly available in our GitHub repository.[1]

[1] https://github.com/yonghoko/ProVerif/tree/main/

5 Results and Discussion

5.1 Correspondence-Query Results for Goals G1–G2

Figure 2 presents the correspondence-query results for Goals G1 and G2 in a single side-by-side comparison. The left side corresponds to the baseline abstraction, whereas the right side corresponds to the bound model with SUCI-stage reused key binding. This combined presentation makes the difference between the two models directly visible at the event-correlation level.

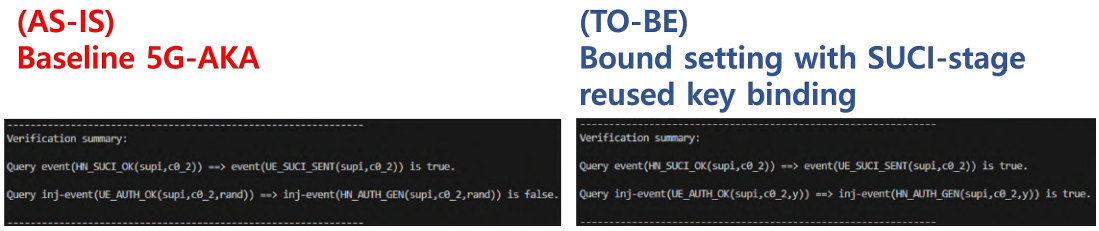

Figure 2: Combined ProVerif correspondence-query results for Goals G1 and G2. The left side shows the baseline abstraction, and the right side shows the bound model with SUCI-stage reused key binding. Both models satisfy the basic SUCI-stage consistency query (G1), whereas only the bound model satisfies the explicit cross-phase binding query (G2).

The figure shows that both models satisfy Goal G1, namely the basic SUCI-stage consistency property. In both cases, the HN-side SUCI-acceptance event corresponds to a prior UE-side SUCI transmission event. Thus, the baseline abstraction and the bound model both preserve the basic concealment-stage event relation.

By contrast, Goal G2 clearly distinguishes the two settings. In the baseline abstraction, the explicit cross-phase binding query is false, which indicates that later AKA acceptance is not explicitly tied to the same SUCI-originated context at the event level. In the bound model, however, the corresponding query is true. Accordingly, the combined correspondence-query results in Figure 2 provide direct formal evidence that SUCI-stage reused key binding transforms the SUCI-to-AKA relation from a weakly coupled one into an explicitly bound one.

5.2 Replay-Driven Linkability Result for Goal G3

Figure 3 presents the observational-equivalence results for Goal G3 in a single side-by-side comparison. The left side corresponds to the baseline abstraction, whereas the right side corresponds to the bound model with SUCI-stage reused key binding. As in Figure 2, this combined presentation is intended to make the difference between the two settings directly visible under the same replay-based verification scenario.

The figure shows that the baseline abstraction yields `Observational equivalence cannot be proved`, whereas the bound model yields `Observational equivalence is true`. This contrast is the central privacy-oriented result of the paper. In the baseline abstraction, replayed authentication material may still induce subscriber-dependent observable behavior, which means that same-subscriber replay and different-subscriber replay cannot be shown observationally equivalent in the modeled setting.

By contrast, the bound model satisfies the observational-equivalence check in the same replay scenario. Accordingly, Figure 3 provides direct formal evidence that SUCI-stage reused

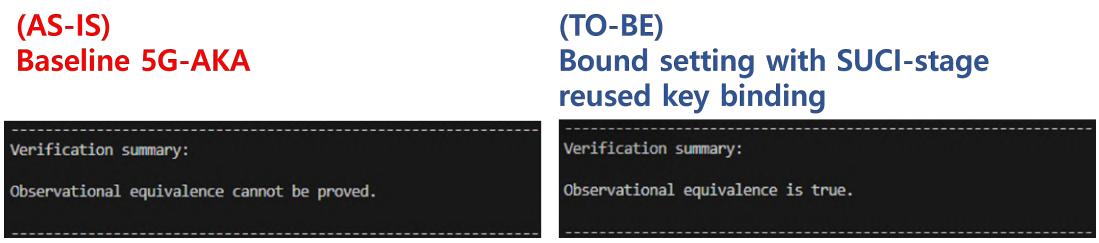

Figure 3: Combined ProVerif observational-equivalence results for Goal G3. The left side shows the baseline abstraction, and the right side shows the bound model with SUCI-stage reused key binding. The baseline model yields `Observational equivalence cannot be proved`, whereas the bound model yields `Observational equivalence is true`, indicating that replay-driven distinguishability remains in the baseline abstraction but disappears in the bound setting.

Table 3: Goal-oriented comparison between the baseline abstraction and the bound model with SUCI-stage reused key binding.

Goal	Baseline 5G-AKA	Bound setting	Interpretation
G1	True	True	Both models preserve basic concealment-stage consistency
G2	False	True	The bound model explicitly binds later AKA acceptance to the current SUCI-stage context
G3	Cannot be proved	True	The bound model eliminates replay-driven distinguishability in the modeled scenario

key binding eliminates replay-driven distinguishability in the compared setting. Taken together with the correspondence-query comparison in Figure 2, this result shows that the bound model not only introduces explicit event-level cross-phase binding, but also removes the observable replay difference that remains in the baseline abstraction.

5.3 Goal-Oriented Comparative Discussion

Table 3 summarizes the main differences between the baseline abstraction and the bound setting by aligning the verification results with Goals G1–G3. Read together, Figure 2, Figure 3, and Table 3 show a clear separation between the common property preserved by both models and the distinguishing properties introduced by SUCI-stage reused key binding.

Goal G1 is satisfied in both settings and therefore serves as a baseline consistency check. It confirms that both models preserve the basic SUCI-stage event relation. Goal G2, by contrast, distinguishes the two settings at the correspondence-query level: the baseline abstraction fails to establish explicit event-level cross-phase binding, whereas the bound model succeeds. Goal G3 then extends this distinction to the privacy-oriented setting: the baseline abstraction cannot

be shown observationally equivalent under replay, whereas the bound model can.

Taken together, these results show that the main formally demonstrated benefit of SUCI-stage reused key binding lies in two related improvements. First, it converts the SUCI-to-AKA relation from a weakly coupled one into an explicitly bound one at the event level. Second, it eliminates replay-driven distinguishability in the modeled linkability experiment. In this sense, the gain does not come from concealing more information in the abstract, but from restructuring how later authentication depends on the current SUCI-originated context.

5.4 Discussion

The results should be interpreted carefully. First, the observational-equivalence result for the bound model supports unlinkability only for the replay-based scenario captured by our abstraction. It should not be interpreted as a universal privacy guarantee for all possible attack surfaces in the 5G-AKA family. Second, the baseline result `cannot be proved` does not mean that ProVerif has returned `false`; rather, it indicates that the modeled executions may reduce differently and therefore cannot be shown observationally equivalent under the given setting. This interpretation follows the ProVerif manual and is consistent with our use of equivalence as a comparative privacy-analysis tool.

The correspondence-query results should also be interpreted within scope. Their role in this paper is not to establish a complete privacy claim by themselves, but to clarify how the SUCI-to-AKA dependency is represented in each model. In particular, the comparison in Figure 2 shows that the bound model makes the cross-phase relation explicit at the event level, whereas the baseline abstraction does not.

Within this scope, however, the combined comparison remains highly informative. The side-by-side difference in Figure 2 shows that SUCI-stage reused key binding introduces explicit cross-phase binding, and the side-by-side difference in Figure 3 shows that the same design choice eliminates one concrete replay-driven source of linkability in the modeled setting. Accordingly, the main contribution of this work is not the introduction of another protocol variant, but the formal clarification that stronger cryptographic dependency between SUCI generation and later AKA verification can yield measurable privacy benefits.

6 Conclusion

This paper examined the security implications of *SUCI-stage reused key binding* as a cross-phase design principle in the 5G-AKA family. Rather than focusing on a single protocol instance, we investigated whether reusing key materials established during the SUCI stage can provide meaningful security benefits when they are incorporated into subsequent AKA computations.

To answer this question, we developed a formal comparison framework based on ProVerif [6, 5]. We analyzed both a baseline 5G-AKA-style setting and a bound setting in which SUCI-stage key material is reused to strengthen the cryptographic dependency between the concealment stage and the authentication stage. The analysis covered basic SUCI-stage consistency, explicit cross-phase binding, and replay-driven linkability.

Our results show that the two ProVerif mechanisms play complementary roles. Correspondence-query results support Goal G1 by confirming that both models preserve basic SUCI-stage consistency, and they support Goal G2 by showing that only the bound model satisfies the explicit cross-phase binding query. Observational-equivalence results address Goal G3 directly: while the baseline setting yielded `Observational equivalence cannot be proved`, the bound setting yielded `Observational equivalence is true` in our replay-driven linkability experiment.

Taken together, these results indicate that SUCI-stage reused key binding can convert the SUCI-to-AKA relation from a weakly coupled one into an explicitly bound one in the model, and that this stronger binding eliminates replay-driven distinguishability in the compared setting.

The main implication of this work is that SUCI-stage key reuse should not be viewed merely as a mechanism for reusing previously available values. Its significance lies in whether it introduces meaningful cryptographic dependency between protocol phases that would otherwise remain only loosely connected. In this sense, SUCI-stage reused key binding can serve as a practical design principle for strengthening privacy-related security properties in the 5G-AKA family.

At the same time, our findings remain limited to the formal model considered in this paper. They do not imply that reused key binding automatically resolves all privacy limitations in the 5G-AKA family. Instead, they show that such binding can provide a measurable advantage against an important class of replay-driven linkability attacks. Future work may extend this analysis to broader privacy goals, richer threat models, and additional 5G-AKA variants.

Acknowledgments

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EAP-EDHOC-PQC: Quantum-Safe Onboarding for IoT E-Business Based on Extensible Authentication Protocol*

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Abstract

The Internet of Things (IoT) underpins a growing range of e-business operations networked vending machines, smart point-of-sale and self-checkout terminals, electronic shelf labels, smart-retail beacons, electric-vehicle (EV) chargers, smart parking meters, fleet and cold-chain trackers, and industrial sensor gateways all of which onboard onto Wi-Fi or other non-3GPP networks via the Extensible Authentication Protocol (EAP) and must remain secure against a future quantum adversary. Ephemeral Diffie–Hellman over COSE (EDHOC, RFC 9528) is a lightweight authenticated key exchange for constrained devices, and the Internet-Draft `draft-papon-lake-pq-edhoc` defines five post-quantum (PQ) authentication variants over ML-KEM and ML-DSA. The combined impact of PQ key enlargement, EAP fragmentation, and RADIUS transport on a real AAA deployment, however, has not yet been quantified, and formal analyses of the PQ-EDHOC variants integrated with EAP are scarce. We close these gaps by proposing *EAP-EDHOC-PQ*, a unified EAP method that embeds all five PQ-EDHOC variants. We verify mutual authentication, session-key secrecy, forward secrecy, and replay resistance under a Dolev–Yao adversary in ProVerif, and we empirically evaluate the design on a heterogeneous IoT testbed (Raspberry Pi 5 Initiator, MacBook Air M3 Authenticator/Responder) across three deployments: Non-EAP, EAP-Standalone, and full EAP/RADIUS with FreeRADIUS. At a conservative 256-byte fragment size, total wire bytes range from 7,031 to 11,614 (29 to 48 fragments) and AAA-path handshake time from 56 ms to 440 ms across variants. A first-order energy model derived from the same measurements shows that fragment-induced I/O wait not cryptography dominates the per-handshake energy of an IoT-class endpoint. The results quantify the trade-off between authentication strength, round-trip count, and fragmentation, and give concrete deployment guidance for PQ network-access authentication in e-business IoT settings.

1 Introduction

IoT is increasingly embedded in everyday e-business workflows. Representative business-critical endpoints include *(i)* networked vending machines accepting cashless and QR-code payments; *(ii)* smart point-of-sale (POS) and self-checkout terminals; *(iii)* electronic shelf labels and smart-retail beacons that drive in-store engagement; *(iv)* EV chargers and smart parking meters operated by retail or city operators; *(v)* fleet trackers and cold-chain sensors monitoring logistics conditions; and *(vi)* connected industrial sensors and programmable logic-controller (PLC) gateways feeding predictive-maintenance pipelines on the factory floor. Each of these devices is, in commercial terms, a small business endpoint

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that handles payment data, customer behaviour, or operational telemetry, and must therefore be enrolled and authenticated before exchanging data with the back-office. As such endpoints onboard onto enterprise networks most often via Wi-Fi or other non-3GPP access *network access authentication* is the first line of defence that establishes device identity, derives session keys, and gates the device onto production traffic. EAP [1] provides the dominant framework for this task: it is used by IEEE 802.1X, Wi-Fi WPA2/3-Enterprise, and 5G-AKA, and it typically runs over a RADIUS/Diameter backend in which a central Authentication, Authorization and Accounting (AAA) server makes the final authentication decision. Among the many EAP methods, EAP-TLS [2], EAP-AKA [3] and EAP-PSK [4] are the most widely deployed. For constrained IoT devices, however, EAP-TLS is expensive in bytes on the wire and certificate chains [5, 6]. EDHOC [7, 8] is a lightweight authenticated Diffie–Hellman key exchange using Concise Binary Object Representation (CBOR) [9] and CBOR Object Signing and Encryption (COSE) [10], designed for constrained networks and standardized by the IETF LAKE working group [11]. Recent work proposes embedding EDHOC as an EAP method [12, 6], which brings EDHOC's compact flights to mainstream AAA-based network access.

At the same time, Shor's algorithm [13] threatens every elliptic-curve deployment in use today. While a cryptanalytically relevant quantum computer is not yet available, the *harvest-now, decrypt-later* adversary model is already a documented concern for long-lived business data such as payment records and supply-chain telemetry, and motivates beginning the transition before such a machine appears. NIST has standardized ML-KEM (FIPS 203) [14] and ML-DSA (FIPS 204) [15] as first-line post-quantum replacements. Regulators in France [16] and Germany [17] and the IETF PQUIP working group [18] recommend *hybrid* PQ/traditional combinations during the transition period. A post-quantum variant of EDHOC (`draft-papon-lake-pq-edhoc`) is under active specification [19] and defines five authentication modes in its Sections 2, 3.2, 3.3, 3.4 and 3.5; follow-up analyses have evaluated quantum-resistant EDHOC on microcontrollers [20, 21].

What is not yet understood is what happens when these PQ-EDHOC variants are carried by *EAP in a real AAA setting*. EAP defines a minimum MTU of only 1020 octets and mandates a lock-step, fragment-ACK flow for payloads larger than that [1, 22]. ML-KEM ciphertexts and ML-DSA signatures push individual EDHOC messages into the multi-kilobyte range, so each handshake message incurs multiple fragments and additional round-trips end-to-end between the EAP peer, the authenticator, and the AAA server. Prior empirical EDHOC studies [5, 20, 21] evaluated EDHOC peer-to-peer and did not exercise an EAP/RADIUS path; prior EAP-EDHOC work [12, 6] has focused on the classical primitive set.

Our work sits at the intersection of four threads: (i) the EAP framework itself and its canonical methods [1, 2, 3, 4]; (ii) EDHOC and its integration with EAP [7, 8, 11, 12, 6, 5]; (iii) post-quantum and hybrid key exchange, including FIPS 203/204 [14, 15], ETSI TS 103 744 [23], PQXDH [24, 25], PQ Noise [26], hybrid IKEv2 [27], and hybrid TLS 1.3 [28]; and (iv) formal and computational analyses of EDHOC such as [29, 30, 31, 32, 33, 34, 35]. The KEM-based EDHOC authentication Internet-Draft [36] and the signature/KEM PQ-EDHOC draft [19] are the closest antecedents to our proposal, and stronger KEM notions such as those in [37] motivate why KEM-authenticated handshakes deserve careful symbolic treatment. Despite substantial progress in each of these threads, four gaps remain. *First*, the impact of PQ key/ciphertext enlargement on EAP fragmentation under the minimum 1020-octet EAP MTU has not been quantified. *Second*, formal symbolic analyses of the PQ variants of EAP-EDHOC have not been published, and existing ProVerif models target classical EDHOC [30, 33]. *Third*, empirical evaluations of PQ-EDHOC have so far targeted EDHOC peer-to-peer or microcontrollers without a real AAA backend [20, 21]. *Fourth*, the different PQ-EDHOC authentication modes (KEM-only, signature-only, and hybrid) have not been compared uniformly in a deployment with EAP and a real AAA server such as FreeRADIUS.

1.1 Contribution

We address these gaps with the following contributions. Throughout the paper we refer to the five authentication variants of `draft-papon-lake-pq-edhoc` by their draft-section numbers, namely Section 2 [19], Section 3.2 [19], Section 3.3 [19], Section 3.4 [19] and Section 3.5 [19]; this nomenclature is used uniformly in every algorithm, table, and figure of the paper.

- We specify EAP-EDHOC-PQ, a unified EAP method that embeds the five PQ-EDHOC authentication variants of [19] — the implicitly KEM-authenticated three-flight mode (Section 2 [19], Section 3.1); the signature+KEM three-flight mode (Section 3.2 [19], Section 3.2); and three four-flight hybrid variants (Section 3.3 [19], Section 3.4 [19] and Section 3.5 [19], Sections 3.3–3.5) over EAP and RADIUS.
- We provide a ProVerif model of all five variants and prove mutual authentication, session-key secrecy, forward secrecy, and replay resistance under a Dolev–Yao adversary (Section 4).
- We provide an empirical evaluation on a real testbed with Non-EAP, EAP-Standalone, and full EAP/RADIUS (FreeRADIUS) deployments at a conservative 256-byte fragment size, measuring fragmentation, processing time, overhead, and per-operation latency (Section 5).
- We derive concrete deployment guidance: which variant minimizes wire bytes, which minimizes round-trips, and how the AAA path amplifies the cost of four-flight hybrid modes (Section 6).

The remainder of the paper is organized as follows. Section 2 introduces the primitives, the EAP/RADIUS model, and notation. Section 3 specifies the unified EAP-EDHOC-PQC design and its five variants. Section 4 presents the ProVerif analysis. Section 5 reports the empirical evaluation. Section 6 discusses implications and limitations. Section 7 concludes.

2 Preliminaries

This section fixes notation, recalls the cryptographic primitives we use, summarizes EDHOC and EAP, and introduces the threat model. Table 1 lists the abbreviations used throughout the paper.

We use NIST-standardized post-quantum primitives. **ML-KEM** [14] is an IND-CCA2-secure key-encapsulation mechanism with three algorithms $(\mathsf{KeyGen}, \mathsf{KEM.Encaps}, \mathsf{KEM.Decaps})$. **ML-DSA** [15] is a module-lattice digital signature algorithm. Symmetric primitives follow RFC 9528: SHA-256 for hashing, HKDF [38] for key derivation, and AES-CCM (an AEAD) for authenticated encryption.

EDHOC is a three-message (optionally four-message) authenticated key exchange encoded in CBOR [9, 10]. The canonical Signature-Signature (SIG-SIG) method transmits ephemeral Diffie–Hellman contributions and authenticates both parties with digital signatures; the Static-DH modes use long-term DH keys and derive MAC tags instead. A detailed security analysis in the classical setting is provided in [30, 31, 32, 33, 34, 35].

EAP [1] is a lock-step request/response protocol operating over a lower layer (e.g., IEEE 802.1X, PPP, or IKEv2). For EAP-over-RADIUS, an *authenticator* (typically a gateway or access point) relays EAP packets between the peer and the AAA server inside RADIUS attributes. Two properties are central to our evaluation. *(a)* EAP implementations may assume an MTU of only 1020 octets from the lower layer, and EAP itself does not provide fragmentation; an EAP method whose messages exceed the MTU must implement fragmentation with explicit ACKs, at the cost of extra round-trips [1, 22]. *(b)* RADIUS packets are capped at 4096 octets and carry EAP payloads in multiple `EAP-Message` AVPs of 253 bytes each; long EAP messages therefore cross both the EAP fragmentation boundary and the RADIUS attribute boundary.

We consider a Dolev–Yao adversary [39] controlling the network between the peer, the authenticator, and the AAA server. The adversary may inject, drop, reorder, replay, or modify any message

Table 1: Abbreviations and notation used throughout the paper (including all symbols of Algorithms 1–5 in Section 3).

Symbol	Description	Symbol	Description
Protocols and frameworks			
EAP	Extensible Authentication Protocol (RFC 3748)	EDHOC	Ephemeral Diffie–Hellman over COSE (RFC 9528)
AAA	Authentication, Authorization, Accounting (RADIUS/Diameter)	NAS	Network Access Server (relays EAP between peer and AAA)
RADIUS	Remote Authentication Dial-In User Service (RFC 2865)	CBOR	Concise Binary Object Representation (RFC 8949)
COSE	CBOR Object Signing and Encryption (RFC 9052)	MTU	Maximum Transmission Unit
LAKE	IETF working group for Lightweight AKE		
Cryptographic primitives			
KEM	Key Encapsulation Mechanism	ML-KEM	Module-Lattice KEM (FIPS 203)
ML-DSA	Module-Lattice Digital Signature Algorithm (FIPS 204)	DS	Digital Signature scheme (instantiated by ML-DSA)
AEAD	Authenticated Encryption with Associated Data	HKDF	HMAC-based Extract-and-Expand KDF (RFC 5869)
KDF	Key Derivation Function	HMAC	Hashed Message Authentication Code
MAC	Message Authentication Code	SHA-256	Secure Hash Algorithm, 256-bit output
$H(\cdot)$	cryptographic hash (SHA-256 in our instantiation)	$\mathsf{KeyGen}, \mathsf{KEM.Encaps}, \mathsf{KEM.Decaps}$	KEM key generation, encapsulation, decapsulation
DS.Sign, **DS.Verify**	digital-signature signing / verification	**AEAD.Enc**, **AEAD.Dec**	AEAD encryption / decryption
Parties, identifiers and credentials			
I, R	EDHOC Initiator, Responder	C_I, C_R	EDHOC connection identifiers chosen by I, R
ID_CRED_I, ID_CRED_R	credential identifier of I, R	METHOD, SUITES_I	EDHOC method byte and offered cipher-suite list
EAD_i	External Authorization Data field of message i	MSK, EMSK	Master Session Key, Extended MSK exported to EAP
Per-handshake key material			
sk_X, pk_X	long-term secret/public key of party X	kem.sk_X, kem.pk_X	long-term KEM key pair of party X
sign.sk_X, sign.pk_X	long-term signing key pair of party X	$\text{kem.sk}_{\text{eph}}$, $\text{kem.pk}_{\text{eph}}$	ephemeral KEM key pair generated by I
$\text{kem.ct}_{\text{eph}}$	KEM ciphertext encapsulating to $\text{kem.pk}_{\text{eph}}$	kem.ct_R, kem.ct_I	KEM ciphertexts encapsulating to R's, I's long-term key
ss_{eph}, ss_R, ss_I	shared secrets from the three KEM operations	TH_i	EDHOC transcript hash after message i
PRK_{2e}	PRK from ss_{eph}	PRK_{3e2m}	PRK after mixing ss_R
PRK_{4e3m}	PRK after mixing ss_I (four-flight modes)	PRK_{out}	exporter PRK feeding MSK/EMSK derivation
$SALT_{3e2m}$, $SALT_{4e3m}$	salts used in the corresponding HKDF-Extract	$KEYSTREAM_2$	XOR keystream protecting $PLAINTEXT_2$
K_3, IV_3, K_4, IV_4	AEAD key/IV pairs for messages 3 and 4	MAC_i	authentication tag carried in (or covering) flight i
$SIGNATURE_i$	ML-DSA signature attached to flight i	$PLAINTEXT_i$, $CIPHERTEXT_i$	plaintext / ciphertext payload of message i
$\|$	byte-string concatenation		

and may corrupt long-term keys of peers other than the target session for forward secrecy arguments. Cryptographic primitives are idealized as perfect symbolic functions, consistent with ProVerif's applied-pi-calculus model and standard practice [40, 41, 42].

3 Proposed Design: EAP-EDHOC-PQ

The proposed **EAP-EDHOC-PQ** method carries PQ-EDHOC flights between the EAP peer (the EDHOC Initiator I, typically an IoT device), the authenticator, and the AAA server (the EDHOC Responder R). The unified design supports five authentication variants Section 2 [19], Section 3.2 [19], Section 3.3 [19], Section 3.4 [19], and Section 3.5 [19] that trade off the authentication mechanism (KEM, signature, or hybrid), the number of flights (three or four), and the timing of responder authentication. Figure 1 summarizes the complete protocol flow. Sections 3.1–3.5 describe the variants individually. Common header (EAP Identity and EDHOC-Start). All variants begin with the standard EAP Identity exchange: the authenticator relays the peer's EAP-Response/Identity to the AAA server, which answers with a RADIUS Access-Challenge carrying EAP-Request/EDHOC-Start. From this point on the five variants diverge. While EDHOC defines message_4 as optional, the EAP-EDHOC draft mandates the use of message_4 as a protected success indication. In this work, Sections 2 and 3.2 allow an optimized three-flight operation by omitting message_4, whereas Sections 3.3–3.5 retain the four-message exchange for full EAP-EDHOC compatibility.

3.1 Variant 1 Section 2 [19]

Section 2 [19] authenticates the Initiator explicitly with an ML-DSA signature and the Responder *implicitly* through ML-KEM, following the philosophy of [36]. Before the handshake the Initiator knows the Responder's long-term KEM public key $kem.pk_R$. Message 1 carries an ephemeral KEM public key $kem.pk_{eph}$ together with the static-to-Responder ciphertext $kem.ct_R \leftarrow \mathsf{KEM.Encaps}(kem.pk_R)$, which delivers ss_R to the Responder. The Responder then encapsulates to the Initiator's ephemeral KEM key, producing $(ss_{eph}, kem.ct_{eph})$, and mixes both shared secrets into $PRK_{3e2m} = \text{HKDF-Extract}(SALT_{3e2m}, ss_R)$. The Responder is authenticated by its ability to decapsulate ss_R, which is the MAC key that Message 3 verifies. Message 4 is optional (key confirmation only). This variant is the smallest on the wire and uses a single KEM decapsulation at R per handshake.

3.2 Variant 2 Section 3.2 [19]

Section 3.2 [19] provides explicit Responder authentication via ML-DSA in Message 2 while still achieving a three-flight main handshake. Message 1 carries only $kem.pk_{eph}$; the Responder encapsulates to it and signs $PLAINTEXT_2$ with its ML-DSA key. The Initiator encapsulates to $kem.pk_R$ in Message 3, delivering ss_R to the Responder and binding the handshake to the Responder's long-term KEM identity. Compared with Section 2 [19], Section 3.2 [19] replaces the implicit KEM authentication of R with an explicit signature in Message 2 and therefore pays a larger Message 2 on the wire.

3.3 Variant 3 Section 3.3 [19]

Section 3.3 [19] is the *mirror* of Section 3.2 [19] with respect to the direction of the static KEM. Authentication of I uses both an ML-DSA signature and KEM confirmation *to the Initiator*, so the Responder must encapsulate to $kem.pk_I$ after learning ID_CRED_I in Message 3. This forces Message 4 to become *mandatory*: it carries $kem.ct_I$ and the encrypted confirmation. Section 3.3 [19] therefore yields a four-flight handshake, with the natural cost that the AAA path adds one more RADIUS round-trip.

3.4 Variant 4 Section 3.4 [19]

Section 3.4 [19] implements KEM+KEM authentication on both sides but defers Responder authentication: the Responder's MAC tag is not sent in Message 2 but carried inside Message 4, after R has

encapsulated to $kem.pk_I$. The design yields three shared secrets $\{ss_{eph}, ss_R, ss_I\}$ and the strongest key-contributiveness among our variants, at the cost of one additional KEM operation on each side.

3.5 Variant 5 Section 3.5 [19]

Section 3.5 [19] combines the best of the hybrid options: explicit ML-DSA authentication of R in Message 2 *and* KEM-based authentication of I via Message 4. It is the "full hybrid" variant and provides the strongest authentication guarantees in our family at the cost of the highest wire size and the most cryptographic operations.

Every EDHOC message is carried inside an EAP-Request/EAP-Response of type `EAP-EDHOC-PQC`. When the EAP payload exceeds the negotiated fragment size, EAP-EDHOC-PQC uses an explicit flag-based fragmentation with `M`-bit ("more fragments"), mirroring EAP-TLS and the classical EAP-EDHOC draft [12, 6]. Each fragment carries a 4-byte EAP header; the sender waits for an ACK fragment (an EAP-Response of type EAP-EDHOC-PQC with empty data) before sending the next fragment, introducing one lock-step round-trip per fragment pair. The original PQ-EDHOC Internet-Draft [19] groups its five variants under two umbrella modes (SIG-SIG and MAC-MAC) and develops the SIG-SIG mode as a single Section 3.2 example, without any consideration of EAP transport. Closely related work either keeps EDHOC peer-to-peer over UDP/CoAP [20, 21, 5] or embeds *classical* EDHOC into EAP [12, 6]. No prior design carries all five PQ-EDHOC variants over a real EAP/RADIUS path with an off-the-shelf AAA server, and prior comparisons therefore cannot speak to the cost of EAP fragmentation, the RADIUS attribute boundary, or the lock-step ACK round-trip. Our unified specification fills this gap. Section 2 [19] is the only mode that retains the three-flight structure *and* drops one signature relative to SIG-SIG, while Section 3.4 [19] is, to the best of our knowledge, the first EAP-friendly KEM+KEM hybrid with deferred Responder authentication suitable for fragmentation under the EAP MTU. Two dimensions distinguish our design from the alternatives. *First, EAP transport vs. peer-to-peer.* A peer-to-peer PQ-EDHOC handshake travels over a single UDP/CoAP [7] association between I and R, with no fragmentation contract beyond what the lower layer offers; in contrast, EAP imposes a lock-step request/response state machine, an MTU as low as 1020 octets [1, 22], and a fragment-ACK exchange that adds one round-trip per fragment pair. The same multi-kilobyte ML-DSA signature therefore costs proportionally more wall-clock time on EAP than in a peer-to-peer setting a cost that is invisible from the draft's own description. *Second, AAA backend vs. standalone.* An EAP-Standalone deployment terminates EAP at the authenticator, while a full AAA deployment relays each EAP message inside RADIUS `EAP-Message` attributes (capped at 253 bytes each [43]) over a separate Network Access Server $\leftrightarrow$ AAA hop. The RADIUS hop adds latency but also amortizes some I/O via FreeRADIUS's batched packetization, and the joint behaviour cannot be inferred from the peer-to-peer numbers alone.

Algorithm 1 Step 1 (Initiator) Prepare M_1

Input: Longterm PQ key pair (pk_I, sk_I); Responder's kem.pk_R (Section 2 only)
Output: M_1

[All Sections]
$(\text{kem.sk}_{\text{eph}}, \text{kem.pk}_{\text{eph}}) \leftarrow \textbf{KEM.KeyGen}()$
[Section 2 only]
$(\text{ss}_R, \text{kem.ct}_R) \leftarrow \textbf{KEM.Encaps}(\text{kem.pk}_R)$
$M_1 \leftarrow (\text{METHOD}, \text{SUITES}_I, \text{kem.pk}_{\text{eph}}, \text{kem.ct}_R, C_I, \text{EAD}_1)$
[Section 3.2, 3.3, 3.4, 3.5]
$M_1 \leftarrow (\text{METHOD}, \text{SUITES}_I, \text{kem.pk}_{\text{eph}}, C_I, \text{EAD}_1)$
return M_1

Algorithm 1 describes the generation of M_1. The Initiator generates an ephemeral KEM key pair and includes the public key in the message. In Section 2, assuming an authenticated copy of the Responder's KEM public key, it additionally produces kem.ct_R through KEM encapsulation to enable implicit Responder authentication. The resulting M_1 contains the protocol parameters, connection identifier, optional authorization data, and, when applicable, kem.ct_R.

Figure 1: Unified EAP-EDHOC-PQC over EAP/RADIUS, five variants

Algorithm 2 Step 2 (Responder) Process M_1 and Prepare M_2

Input: M_1; Responder's long-term keys (KEM and/or signing, per section)
Output: M_2

[All Sections]
$(\text{ss}_{\text{eph}}, \text{kem.ct}_{\text{eph}}) \leftarrow \textbf{KEM.Encaps}(\text{kem.pk}_{\text{eph}})$
$TH_2 \leftarrow H(\text{kem.ct}_{\text{eph}}, H(M_1))$
$PRK_{2e} \leftarrow \textbf{EDHOC-Extract}(TH_2, \text{ss}_{\text{eph}})$
$KEYSTREAM_2 \leftarrow \textbf{EDHOC-KDF}(PRK_{2e}, 0, TH_2)$
[Section 2] ▷ implicit R-auth via ss_R from M_1
$\text{ss}_R \leftarrow \textbf{KEM.Decaps}(\text{kem.sk}_R, \text{kem.ct}_R)$
$SALT_{3e2m} \leftarrow \textbf{EDHOC-KDF}(PRK_{2e}, 1, TH_2)$
$PRK_{3e2m} \leftarrow \textbf{EDHOC-Extract}(SALT_{3e2m}, \text{ss}_R)$
$MAC_2 \leftarrow \textbf{KDF}(PRK_{3e2m}, 2, (C_R, ID_CRED_R, TH_2, \text{EAD}_2))$
$PLAINTEXT_2 \leftarrow (C_R, ID_CRED_R, MAC_2, \text{EAD}_2)$
[Section 3.2, 3.5] ▷ explicit R-auth via signature over $PLAINTEXT_2$
$PLAINTEXT_2 \leftarrow (C_R, ID_CRED_R, TH_2, \text{EAD}_2)$
$SIGNATURE_2 \leftarrow \textbf{DS.Sign}(\text{sign.sk}_R, PLAINTEXT_2)$
$PLAINTEXT_2 \leftarrow (PLAINTEXT_2 \parallel SIGNATURE_2)$
[Section 3.3] ▷ MAC_2 + signature over MAC_2, MAC_2 derived from PRK_{2e}
$MAC_2 \leftarrow \textbf{KDF}(PRK_{2e}, 2, (C_R, ID_CRED_R, TH_2, \text{EAD}_2))$
$SIGNATURE_2 \leftarrow \textbf{DS.Sign}(\text{sign.sk}_R, (C_R, ID_CRED_R, TH_2, \text{EAD}_2, MAC_2))$
$PLAINTEXT_2 \leftarrow (C_R, ID_CRED_R, SIGNATURE_2, \text{EAD}_2)$
[Section 3.4] ▷ MAC_2 *deferred* to M_4; M_2 carries no auth material
$PLAINTEXT_2 \leftarrow (C_R, ID_CRED_R, TH_2, \text{EAD}_2)$
[All Sections]
$CIPHERTEXT_2 \leftarrow PLAINTEXT_2 \oplus KEYSTREAM_2$
$M_2 \leftarrow (\text{kem.ct}_{\text{eph}}, CIPHERTEXT_2)$
return M_2

As we can see at Figure 1 unified EAP-EDHOC-PQC over EAP/RADIUS, comprising five variants:

- Section 2 [19]: KEM-only implicit Responder authentication.
- Section 3.2 [19]: SIG+KEM with three message flights.
- Section 3.3 [19]: SIG+KEM with four message flights and KEM authentication to *I*.
- Section 3.4 [19]: KEM+KEM with deferred Responder authentication.
- Section 3.5 [19]: KEM+KEM+SIG authentication.

Algorithm 2 describes the Responder-side processing of M_1 and the generation of M_2. The Responder first encapsulates to the Initiator's ephemeral KEM public key to obtain the shared secret ss_{eph}, from which TH_2, PRK_{2e}, and the encryption keystream are derived. The authentication mechanism then depends on the selected variant. Section 2 performs implicit authentication through a long-term KEM decapsulation and the derivation of MAC_2. Sections 3.2 and 3.5 provide explicit authentication using a digital signature over the protected protocol data. Section 3.3 combines a transcript-bound MAC_2 with a signature, while Section 3.4 defers Responder authentication to M_4 and therefore includes no authentication material in M_2. Finally, the plaintext is encrypted using the derived keystream, and M_2 is formed by combining the ephemeral KEM ciphertext with the protected payload.

Algorithm 3 Step 3 (Initiator) Process M_2 and Prepare M_3

Input: M_2
Output: M_3

[All Sections]
$\text{ss}_{\text{eph}} \leftarrow \textbf{KEM.Decaps}(\text{kem.sk}_{\text{eph}}, \text{kem.ct}_{\text{eph}})$
Compute $TH_2, PRK_{2e}, KEYSTREAM_2$ and recover $PLAINTEXT_2$
$TH_3 \leftarrow H(TH_2, PLAINTEXT_2, ID_CRED_R)$
[Section 2] ▷ verify $MAC_2 \Rightarrow$ implicit R-auth
$PRK_{3e2m} \leftarrow \textbf{EDHOC-Extract}(SALT_{3e2m}, \text{ss}_R)$
Verify $MAC_2 \stackrel{?}{=} \textbf{KDF}(PRK_{3e2m}, 2, \cdot)$
$(K_3, IV_3) \leftarrow \textbf{EDHOC-KDF}(PRK_{3e2m}, 3/4, TH_3)$
$MAC_3 \leftarrow \textbf{KDF}(PRK_{3e2m}, 6, (ID_CRED_I, TH_3, \text{EAD}_3))$
$SIGNATURE_3 \leftarrow \textbf{DS.Sign}(\text{sign.sk}_I, (ID_CRED_I, TH_3, \text{EAD}_3, MAC_3))$
$PLAINTEXT_3 \leftarrow (ID_CRED_I, SIGNATURE_3, \text{EAD}_3)$
$M_3 \leftarrow \textbf{AEAD.Enc}(K_3, IV_3, PLAINTEXT_3)$ ▷ no kem.ct_R sent in M_1
[Section 3.2, 3.5] ▷ explicit R-auth via signature verification; new kem.ct_R
Retrieve $\text{kem.pk}_R, \text{sign.pk}_R$; verify $\textbf{DS.Verify}(\text{sign.pk}_R, PLAINTEXT_2, SIGNATURE_2)$
$(\text{ss}_R, \text{kem.ct}_R) \leftarrow \textbf{KEM.Encaps}(\text{kem.pk}_R)$
$PRK_{3e2m} \leftarrow \textbf{EDHOC-Extract}(SALT_{3e2m}, \text{ss}_R)$
$(K_3, IV_3) \leftarrow \textbf{EDHOC-KDF}(PRK_{3e2m}, 3/4, TH_3)$
$MAC_3 \leftarrow \textbf{KDF}(PRK_{3e2m}, 6, (ID_CRED_I, TH_3, \text{EAD}_3))$
$SIGNATURE_3 \leftarrow \textbf{DS.Sign}(\text{sign.sk}_I, (ID_CRED_I, TH_3, \text{EAD}_3, MAC_3))$
$CIPHERTEXT_3 \leftarrow \textbf{AEAD.Enc}(K_3, IV_3, (ID_CRED_I, SIGNATURE_3, \text{EAD}_3))$
$M_3 \leftarrow (\text{kem.ct}_R, CIPHERTEXT_3)$
[Section 3.3] ▷ sign $PLAINTEXT_3$ directly; no MAC_3; no kem.ct_R
$MAC_2 \leftarrow \textbf{KDF}(PRK_{2e}, 2, \cdot)$; verify $\textbf{DS.Verify}(\text{sign.pk}_R, \cdot, SIGNATURE_2)$
$(K_3, IV_3) \leftarrow \textbf{EDHOC-KDF}(PRK_{2e}, 3/4, TH_3)$
$PLAINTEXT_3 \leftarrow (ID_CRED_I, TH_3, \text{EAD}_3)$
$SIGNATURE_3 \leftarrow \textbf{DS.Sign}(\text{sign.sk}_I, PLAINTEXT_3)$
$M_3 \leftarrow \textbf{AEAD.Enc}(K_3, IV_3, (PLAINTEXT_3 \parallel SIGNATURE_3))$
[Section 3.4] ▷ no SIG_2 verify yet R-auth deferred to M_4; new kem.ct_R
$(\text{ss}_R, \text{kem.ct}_R) \leftarrow \textbf{KEM.Encaps}(\text{kem.pk}_R)$
$PRK_{3e2m} \leftarrow \textbf{EDHOC-Extract}(SALT_{3e2m}, \text{ss}_R)$
$(K_3, IV_3) \leftarrow \textbf{EDHOC-KDF}(PRK_{3e2m}, 3/4, TH_3)$
$SIGNATURE_3 \leftarrow \textbf{DS.Sign}(\text{sign.sk}_I, (ID_CRED_I, TH_3, \text{EAD}_3))$
$CIPHERTEXT_3 \leftarrow \textbf{AEAD.Enc}(K_3, IV_3, (ID_CRED_I, TH_3, \text{EAD}_3, SIGNATURE_3))$
$M_3 \leftarrow (\text{kem.ct}_R, CIPHERTEXT_3)$
return M_3

Algorithm 3 describes the Initiator-side processing of M_2 and the generation of M_3. After decapsulating the ephemeral KEM ciphertext, the Initiator derives TH_2, PRK_{2e}, and the associated protection keys before recovering $PLAINTEXT_2$. The subsequent authentication and key-derivation steps depend on the selected variant. In Section 2, implicit Responder authentication is achieved through verification of MAC_2 using the previously established long-term KEM secret. Sections 3.2 and 3.5 perform explicit Responder authentication through signature verification and establish an additional KEM shared secret for mutual authentication. Section 3.3 verifies the Responder's signature and derives subsequent keys directly from the ephemeral exchange, while Section 3.4 postpones Responder authentication until M_4 and

therefore proceeds without verifying any authentication evidence contained in M_2. Finally, the Initiator constructs and protects its authentication payload, producing M_3 for transmission to the Responder.

Algorithm 4 Step 4 (Responder) Process M_3 and Prepare M_4

Input: M_3

Output: M_4 (optional for Sections 2 and 3.2; mandatory for Sections 3.3, 3.4 and 3.5), PRK_{out}

[All Sections]

Derive keys, decrypt $CIPHERTEXT_3$, and recover $PLAINTEXT_3$

Verify MAC_3 and/or $SIGNATURE_3$ per section (details below)

[Section 2, 3.2] ▷ M_4 optional; carries only EAD_4

Verify $\textbf{DS.Verify}(\text{sign.pk}_I, (ID_CRED_I, TH_3, \text{EAD}_3, MAC_3), SIGNATURE_3)$

$TH_4 \leftarrow H(TH_3, PLAINTEXT_3, ID_CRED_I)$

$(K_4, IV_4) \leftarrow \textbf{EDHOC-KDF}(PRK_{3e2m}, 8/9, TH_4)$

$M_4 \leftarrow \textbf{AEAD.Enc}(K_4, IV_4, \text{EAD}_4)$ *(if requested)*

$PRK_{\text{out}} \leftarrow \textbf{KDF}(PRK_{3e2m}, 7, TH_4)$

[Section 3.3] ▷ M_4 *mandatory*; carries kem.ct_I; $SALT_{4e3m}$ from PRK_{2e}

Verify $\textbf{DS.Verify}(\text{sign.pk}_I, PLAINTEXT_3, SIGNATURE_3)$

$(\text{ss}_I, \text{kem.ct}_I) \leftarrow \textbf{KEM.Encaps}(\text{kem.pk}_I)$

$TH_4 \leftarrow H(TH_3, PLAINTEXT_3, ID_CRED_I)$

$SALT_{4e3m} \leftarrow \textbf{KDF}(PRK_{2e}, 5, TH_4)$

$PRK_{4e3m} \leftarrow \textbf{EDHOC-Extract}(SALT_{4e3m}, \text{ss}_I)$

$(K_4, IV_4) \leftarrow \textbf{EDHOC-KDF}(PRK_{4e3m}, 8/9, TH_4)$

$M_4 \leftarrow (\text{kem.ct}_I, \textbf{AEAD.Enc}(K_4, IV_4, \text{EAD}_4))$

$PRK_{\text{out}} \leftarrow \textbf{KDF}(PRK_{4e3m}, 7, TH_4)$

[Section 3.4] ▷ M_4 mandatory; carries *deferred* MAC_2

$\text{ss}_R \leftarrow \textbf{KEM.Decaps}(\text{kem.sk}_R, \text{kem.ct}_R)$

Verify $\textbf{DS.Verify}(\text{sign.pk}_I, (ID_CRED_I, TH_3, \text{EAD}_3), SIGNATURE_3)$

$(\text{ss}_I, \text{kem.ct}_I) \leftarrow \textbf{KEM.Encaps}(\text{kem.pk}_I)$

$TH_4 \leftarrow H(TH_3, PLAINTEXT_3, ID_CRED_I)$

$SALT_{4e3m} \leftarrow \textbf{KDF}(PRK_{3e2m}, 5, TH_4)$

$PRK_{4e3m} \leftarrow \textbf{EDHOC-Extract}(SALT_{4e3m}, \text{ss}_I)$

$(K_4, IV_4) \leftarrow \textbf{EDHOC-KDF}(PRK_{4e3m}, 8/9, TH_4)$

$MAC_2 \leftarrow \textbf{KDF}(PRK_{4e3m}, 2, (C_R, ID_CRED_R, TH_4, \text{EAD}_4))$

$M_4 \leftarrow (\text{kem.ct}_I, \textbf{AEAD.Enc}(K_4, IV_4, (\text{EAD}_4, MAC_2)))$

$PRK_{\text{out}} \leftarrow \textbf{KDF}(PRK_{4e3m}, 7, TH_4)$

[Section 3.5] ▷ M_4 mandatory; carries only EAD_4; SIG_3 over $PLAINTEXT_3$

$\text{ss}_R \leftarrow \textbf{KEM.Decaps}(\text{kem.sk}_R, \text{kem.ct}_R)$

Verify $\textbf{DS.Verify}(\text{sign.pk}_I, PLAINTEXT_3, SIGNATURE_3)$

$(\text{ss}_I, \text{kem.ct}_I) \leftarrow \textbf{KEM.Encaps}(\text{kem.pk}_I)$

$TH_4 \leftarrow H(TH_3, PLAINTEXT_3, ID_CRED_I)$

$SALT_{4e3m} \leftarrow \textbf{KDF}(PRK_{3e2m}, 5, TH_4)$

$PRK_{4e3m} \leftarrow \textbf{EDHOC-Extract}(SALT_{4e3m}, \text{ss}_I)$

$(K_4, IV_4) \leftarrow \textbf{EDHOC-KDF}(PRK_{4e3m}, 8/9, TH_4)$

$M_4 \leftarrow (\text{kem.ct}_I, \textbf{AEAD.Enc}(K_4, IV_4, \text{EAD}_4))$

$PRK_{\text{out}} \leftarrow \textbf{KDF}(PRK_{4e3m}, 7, TH_4)$

return M_4, PRK_{out}

Algorithm 4 describes the Responder-side processing of M_3 and the generation of the final protocol

message M_4, when required. After deriving the appropriate keys and decrypting $CIPHERTEXT_3$, the Responder verifies the Initiator's authentication evidence, either through MAC_3, $SIGNATURE_3$, or both, depending on the selected variant. For Sections 2 and 3.2, authentication is completed after verification and M_4 is optional. In contrast, Sections 3.3, 3.4, and 3.5 require an additional KEM exchange and a mandatory M_4 to complete key confirmation and derive the final session secret. Section 3.4 further carries the deferred MAC_2 in M_4, postponing Responder authentication until the last protocol message. Upon successful completion, the Responder derives PRK_{out}, which serves as the exported session secret for subsequent EAP and application-layer key derivation.

Algorithm 5 Step 5 (Initiator) Process M_4 (optional for Sections 2 and 3.2; mandatory otherwise)

Input: M_4
Output: PRK_{out}

[Section 2, 3.2] ▷ decrypt EAD_4 only
$(K_4, IV_4) \leftarrow \textbf{EDHOC-KDF}(PRK_{3e2m}, 8/9, TH_4)$
$\text{EAD}_4 \leftarrow \textbf{AEAD.Dec}(K_4, IV_4, CIPHERTEXT_4)$
$PRK_{\text{out}} \leftarrow \textbf{KDF}(PRK_{3e2m}, 7, TH_4)$
[Section 3.3, 3.5] ▷ decaps kem.ct_I; $SALT_{4e3m}$ source differs per section
$\text{ss}_I \leftarrow \textbf{KEM.Decaps}(\text{kem.sk}_I, \text{kem.ct}_I)$
$SALT_{4e3m} \leftarrow \textbf{KDF}(PRK_{2e \text{ or } 3e2m}, 5, TH_4)$ ▷ PRK_{2e} for Section 3.3; PRK_{3e2m} for Section 3.5
$PRK_{4e3m} \leftarrow \textbf{EDHOC-Extract}(SALT_{4e3m}, \text{ss}_I)$
$(K_4, IV_4) \leftarrow \textbf{EDHOC-KDF}(PRK_{4e3m}, 8/9, TH_4)$
$\text{EAD}_4 \leftarrow \textbf{AEAD.Dec}(K_4, IV_4, CIPHERTEXT_4)$
$PRK_{\text{out}} \leftarrow \textbf{KDF}(PRK_{4e3m}, 7, TH_4)$
[Section 3.4] ▷ verify *deferred* MAC_2 here ⇒ explicit R-auth
$\text{ss}_I \leftarrow \textbf{KEM.Decaps}(\text{kem.sk}_I, \text{kem.ct}_I)$
$SALT_{4e3m} \leftarrow \textbf{KDF}(PRK_{3e2m}, 5, TH_4)$
$PRK_{4e3m} \leftarrow \textbf{EDHOC-Extract}(SALT_{4e3m}, \text{ss}_I)$
$(K_4, IV_4) \leftarrow \textbf{EDHOC-KDF}(PRK_{4e3m}, 8/9, TH_4)$
$(\text{EAD}_4, MAC_2) \leftarrow \textbf{AEAD.Dec}(K_4, IV_4, CIPHERTEXT_4)$
Verify $MAC_2 \stackrel{?}{=} \textbf{KDF}(PRK_{4e3m}, 2, (C_R, ID_CRED_R, TH_4, \text{EAD}_4))$
$PRK_{\text{out}} \leftarrow \textbf{KDF}(PRK_{4e3m}, 7, TH_4)$
return PRK_{out}

Algorithm 5 describes the final processing stage performed by the Initiator after receiving M_4. For Sections 2 and 3.2, the procedure is limited to decrypting the optional protected $\text{EAD} * 4$ payload and deriving the exported secret $PRK * \text{out}$. In Sections 3.3 and 3.5, the Initiator first decapsulates the additional KEM ciphertext carried in M_4, derives PRK_{4e3m}, and then decrypts the protected payload before generating the final session secret. Section 3.4 follows the same key-derivation process but additionally verifies the deferred MAC_2 contained in M_4, thereby completing explicit Responder authentication. Upon successful verification, the Initiator derives PRK_{out}, concluding the EDHOC-PQ handshake and providing the shared secret used for subsequent EAP and application-layer key derivation.

4 Security Analysis

We model each of the five EAP-EDHOC-PQC variants in the applied pi calculus [40, 41] and verify the security requirements summarized in Table 2 under the Dolev–Yao adversary [39, 42]. The complete ProVerif source code for all five variants `section2.pv`, `section32.pv`, `section33.pv`, `section34.pv`

Table 2: Security requirements and the corresponding ProVerif verification queries used for every EAP-EDHOC-PQC variant. The same Q1–Q10 query set is instantiated in each per-variant ProVerif file (`section2.pv`, `section32.pv`, `section33.pv`, `section34.pv`, `section35.pv`). Q1–Q7 are active by default; Q8–Q10 are enabled by uncommenting the corresponding inline hook (`LeakFS`, `[Q9 AEAD LEAK]`, `[Q10 PQ LEAK]`).

Security Requirement	**Verification Query**
Agreement on M_1	**Q1:** `inj-event(E_STEP1_I_to_R(m)) ==> inj-event(S_STEP1_I_to_R(m))`
Mutual Authentication	**Q2:** `inj-event(E_STEP2_R_to_I(mac,msg,th)) ==> inj-event(S_STEP2_R_to_I(mac,msg,th))` **Q3:** `inj-event(E_STEP3_I_to_R(mac,msg,th)) ==> inj-event(S_STEP3_I_to_R(mac,msg,th))`
Identity Concealment	**Q4:** `query attacker(I)` **Q5:** `query attacker(R)`
Confidentiality	**Q6:** `query attacker(app_data1)` **Q7:** `query attacker(app_data2)`
Secure Key Exchange	**Q2** $\wedge$ **Q3** $\wedge$ **Q6** $\wedge$ **Q7** i.e., agreement on M_2/M_3 *and* secrecy of *app_data1*, *app_data2*.
Forward Secrecy (FS)	**Q8:** `phase 1; out(c,sk_I); out(c,sk_R); query attacker(app_data1) && attacker(app_data2)`
AEAD Key/IV Reuse Resistance	**Q9:** `phase 1; out(c,EK2); out(c,EK3); query attacker(app_data1) && attacker(app_data2)`
Quantum Resilience	**Q10:** `phase 1; out(c,r_R); out(c,r_eph); query attacker(app_data1) && attacker(app_data2)`

and `section35.pv`, together with the runner script `run_all.sh` is available at https://github.com/adipanca/proverif-ebision and is referenced by file name throughout this section. ML-KEM is modelled as an IND-CCA2 KEM with constructor pair *kem_encap*/*kem_decap*, ML-DSA as an EUF-CMA signature scheme with *sign*/*sverify* constructors, HKDF and SHA-256 as one-way functions, and AEAD as an INT-CTXT primitive with matching *aead_enc*/*aead_dec* constructors. Three channels are used: c_{IN} models the EAP/EDHOC channel between the peer and the authenticator, c_{NA} models the RADIUS hop between the authenticator and the AAA server, and c_{Pub} is a public sink used to expose secrets in the leakage scenarios for forward secrecy, AEAD-key reuse, and quantum resilience.

Following standard practice for replicated EAP/AAA models [30, 33], we use a many-clients/one-server abstraction: the Initiator and the AAA server are replicated, while a single legitimate Responder runs in parallel. This abstraction faithfully captures the EAP threat model in which one NAS/AAA pair serves arbitrarily many (possibly compromised) supplicants, while keeping ProVerif clause saturation tractable in the presence of long-term KEM decapsulation on every M_1 in Section 2 [19]. Table 2 lists the security requirements we verify together with the concrete ProVerif queries that operationalize them. The query set comprises three correspondence queries (Q1–Q3) expressing injective agreement on each EDHOC flight, four secrecy queries (Q4–Q7) covering identity concealment of *I*/*R* and confidentiality of two application-data slots, and three leakage-based queries (Q8–Q10) for forward secrecy, AEAD-key/IV reuse, and quantum resilience respectively. The events S_STEP_i and E_STEP_i are emitted at the *send* and *end* of each EDHOC flight, so that an injective correspondence $\mathsf{inj\text{-}event}(E_STEP_i) \Rightarrow \mathsf{inj\text{-}event}(S_STEP_i)$ proves that every accepting party can be matched to a unique honest sender of that flight.

4.1 Verification Results

We executed every variant under ProVerif 2.05 with the same query set and the option `set traceBacktracking = false` for performance. Table 3 reports the outcome of each query for each variant, using the truth values returned by ProVerif. The notation "T" denotes that the property is verified (secrecy queries succeed as not attacker(·) being true; correspondence queries succeed as the injective implication being true); "F⋆" denotes an *intended* failure that is consistent with the protocol design; "○" denotes a query enabled by uncommenting an inline leakage hook in the corresponding `.pv` file (Q8 toggles `LeakFS`, Q9 toggles `[Q9 AEAD LEAK]`, Q10 toggles `[Q10 PQ LEAK]`).

Table 3: ProVerif verification results per variant

Query	Requirement	Section. 2	Section. 3.2	Section. 3.3	Section. 3.4	Section. 3.5
Q1	Agreement on M_1	F⋆	F⋆	F⋆	F⋆	F⋆
Q2	Agreement on M_2	T	T	T	F⋆	T
Q3	Agreement on M_3	T	T	T	T	T
Q4	Identity concealment of I	T	T	T	T	T
Q5	Identity concealment of R	T	T	T	T	T
Q6	Confidentiality of *app_data1*	T	T	T	T	T
Q7	Confidentiality of *app_data2*	T	T	T	T	T
Q8	Forward secrecy	T	T	T	T	T
Q9	AEAD-key/IV reuse resistance	T	T	T	T	T
Q10	Quantum resilience	T	T	T	T	T

T = property verified; F⋆ = intended failure (see discussion below); Q8,Q9,Q10 = property verified after enabling the corresponding inline leakage hook in the `.pv` file.

4.2 Discussion of the Intended Failures

Two query failures appear in Table 3 and both are intentional and consistent with the protocol specification.

Q1 fails for every variant (M_1 agreement). RFC 9528 §4.5 specifies that M_1 is sent in cleartext: it carries the method, cipher suites, ephemeral KEM public key, optional $kem.ct_R$ in Section 2 [19], the connection identifier C_I, and EAD_1, but no authentication tag. A Dolev–Yao adversary can therefore replay or forge a syntactically valid M_1, which is exactly what ProVerif detects. Q1 is included only as a *sanity check* on the model: a verification that returned `true` would indicate an unintended (and incorrect) authentication shortcut on M_1. Crucially, the lack of authentication on M_1 does not propagate into mutual authentication of the session key, because the transcript hash TH_2 binds M_1 into every subsequent flight and Q3 (and, for variants other than Section 3.4 [19], Q2) is verified.

Q2 fails for Section 3.4 [19] only. Section 3.4 [19] implements deferred Responder authentication: M_2 carries the encrypted plaintext $(C_R, ID_CRED_R, TH_2, EAD_2)$ but contains neither a MAC_2 tag nor a Responder signature. Instead, MAC_2 is derived from PRK_{4e3m} (which incorporates ss_I) and is conveyed in the protected payload of M_4. Consequently, the Initiator cannot authenticate the Responder upon processing M_2, and injective agreement at that stage is therefore not guaranteed. ProVerif correctly reports Q2 as `false` for this variant. The authentication guarantee is established later through the deferred

MAC_2 verification in M_4, which the Initiator performs before deriving PRK_{out}. Thus, Section 3.4 [19] trades early Responder authentication for a smaller and simpler M_2, while preserving the security of the completed protocol run. This behavior is consistent with the design of the "MAC–MAC" mode in [19].

4.3 Summary

For every variant of EAP-EDHOC-PQC, ProVerif verifies injective agreement on M_3 (Q3), identity concealment of both parties (Q4–Q5), confidentiality of application data (Q6–Q7), forward secrecy after long-term key compromise (Q8), AEAD-key reuse resistance (Q9), and quantum resilience under randomness leakage (Q10). The two reported "failures" (Q1 across all variants and Q2 for Section 3.4 [19] only) are intentional consequences of the protocol specification — the plaintext nature of M_1 and the deferred-MAC_2 design of Section 3.4 [19], respectively and do not affect the security of the derived session key. We therefore conclude that all five variants of EAP-EDHOC-PQC satisfy the security requirements of Table 2.

5 Performance Evaluation

We implemented all five variants of EAP-EDHOC-PQC using the Open Quantum Safe OpenSSL-OQS provider [44] on top of Clang 15 [45]. The implementation instantiates ML-KEM-768 and ML-DSA-44 as default parameter sets, with SHA-256 and AES-128-CCM as the symmetric suite, in line with RFC 9528.

5.1 Experimental Setup

We measured three deployment configurations: *(a)* Non-EAP, EDHOC run directly peer-to-peer over UDP with the same fragmentation discipline but no EAP encapsulation, used as a reference; *(b)* EAP-Standalone, EAP-EDHOC-PQC between peer and authenticator without a RADIUS backend, which isolates the cost of the EAP framing; *(c)* AAA, full EAP-EDHOC-PQC with the AAA server running FreeRADIUS and the authenticator acting as a pass-through NAS.

Hardware and software testbed. To make the measurements relevant to actual e-business IoT use, we deployed the testbed on a *heterogeneous* pair of devices that mirrors a real resource-constrained endpoint paired with a more capable gateway/back-office machine. The Initiator (the EDHOC peer) runs on a Raspberry Pi 5 with 8 GB of RAM as a representative IoT-class device; the authenticator and Responder (and, in the AAA deployment, the FreeRADIUS server) run on a MacBook Air M3 with 16 GB of RAM that plays the role of the network-access / back-office node. On the software side, we use only open-source components: uoscore-uedhoc [46] as the EDHOC implementation, PQClean [47] for the ML-KEM and ML-DSA primitives, and FreeRADIUS [48] as the AAA server, whose EAP module we extend to support the proposed EAP-EDHOC-PQC method. This setup deliberately avoids any cloud-only or proprietary stack so that the results can be reproduced on commodity IoT and laptop hardware.

All runs use a 256-byte EAP fragment size, deliberately below the RFC 3748 minimum MTU of 1020 octets [1, 22] to magnify fragmentation effects and to emulate heavily constrained lower layers. Every handshake was repeated at least five times and the reported values are averages. Processing, overhead, and operation timings were recorded using monotonic high-resolution clocks on both parties.

5.2 Fragmentation and Wire Bytes

Table 4 reports the on-wire message sizes and fragment counts at a 256-byte fragment size. Message 1 ranges from 1,221 bytes (Sections 2, 3.2–3.5) to 2,344 bytes (Section 2 [19], because of the extra $kem.ct_R$). Message 2 is the largest in all variants except Section 2 [19] and Section 3.4 [19], because it carries the Responder's ML-DSA signature together with the KEM ciphertext. Section 2 [19] uses the smallest total of 7,031 bytes and 29 fragments over three flights; Section 3.5 [19] uses the largest of 11,614 bytes and 48 fragments over four flights, a 65% overhead in both dimensions. Section 3.4 [19] sits between the two at 8,207 bytes and 34 fragments, illustrating that deferring the Responder signature to a MAC in Message 4 is an effective way to constrain the wire cost of a four-flight hybrid handshake. Table 4 representation Initiator side, AAA deployment; Non-EAP and Standalone values are identical. Fragments include the fragment-ACK traffic for multi-fragment messages.

Table 4: Per-message wire bytes and fragments at a 256-byte EAP fragment size

Variant	Msg1	Msg2	Msg3	Msg4	Total wire	Fragments	Flights
Section. 2 (§3.1)	2,344	1,226	3,461	0	7,031	29	3
Section. 3.2 (§3.2)	1,221	4,633	4,577	0	10,431	42	3
Section. 3.3 (§3.3)	1,221	4,633	3,461	1,176	10,498	44	4
Section. 3.4 (§3.4)	1,221	1,226	4,577	1,176	8,207	34	4
Section. 3.5 (§3.5)	1,221	4,633	4,577	1,176	11,614	48	4

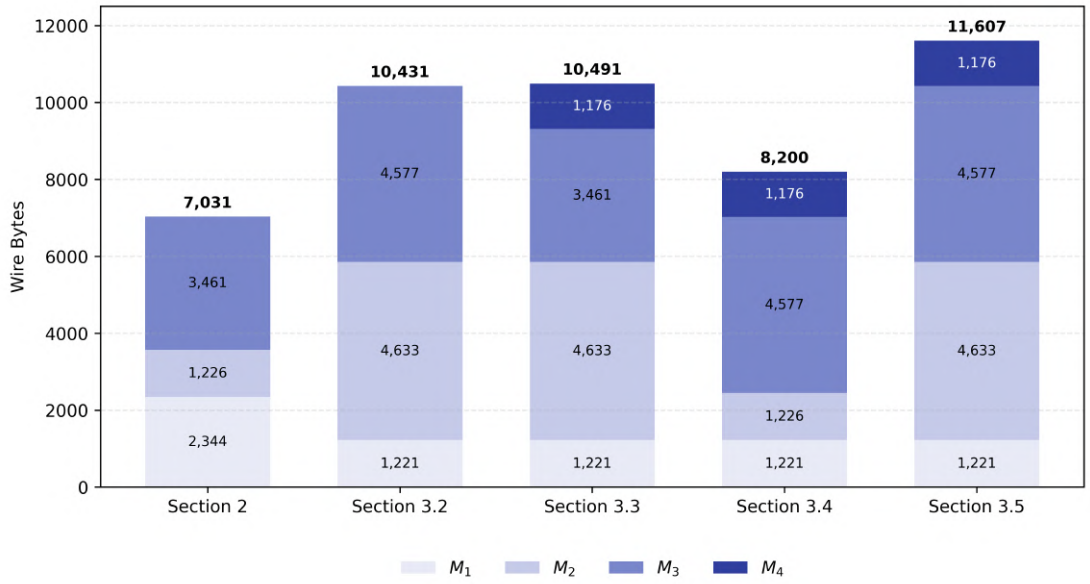

Figure 2: Per-message contribution to total wire bytes

As we can see at Figure 2 representation of stacked for the five EAP-EDHOC-PQC variants at a 256-byte EAP fragment size. Section. 2 pays for $kem.ct_R$ inside M_1, while the four-flight hybrids (Section. 3.3, Section. 3.4, Section. 3.5) add an explicit M_4.

Table 5: Mean end-to-end processing time per handshake in microseconds (μs)

	AAA (FreeRADIUS)			EAP-Standalone			Non-EAP		
Variant	Init	Resp	Sum	Init	Resp	Sum	Init	Resp	Sum
Section. 2	64,505	66,870	131,375	65,621	68,578	134,199	69,038	71,971	141,009
Section. 3.2	28,051	28,130	56,181	35,668	35,715	71,383	32,133	33,524	65,657
Section. 3.3	172,086	186,444	358,530	173,831	187,152	360,982	165,068	151,258	316,326
Section. 3.4	219,917	220,463	440,379	236,108	245,434	481,543	241,271	241,371	482,641
Section. 3.5	184,496	180,137	364,633	213,043	211,815	424,859	192,235	194,671	386,906

Table 5 by deployment and role. “Sum” is Initiator+Responder. reports the mean end-to-end handshake time (in microseconds), broken down by deployment and party. Total handshake time (Initiator+Responder combined) on the AAA path ranges from approximately 56 ms for Section 3.2 [19] to 440 ms for Section 3.4 [19]. Two observations stand out. *First*, the AAA path is never the slowest configuration despite adding a RADIUS hop: for Section 3.4 [19] and Section 3.5 [19], the Non-EAP and Standalone paths are actually slower because they lack the AAA server’s optimized fragment-reassembly primitives. *Second*, Section 3.2 [19] is dramatically faster than Section 2 [19] in the Initiator/Responder sum (56 ms vs. 131 ms on AAA) even though Section 2 [19] sends fewer total wire bytes (7,031 B vs. 10,431 B). This is counter-intuitive only at first glance: wall-clock time under EAP is governed not by the total byte count but by the length of the longest lock-step fragment chain. Section 2 [19] concentrates its payload in an especially large Message 3 (3,461 bytes, 14 fragments) whose fragments must be acknowledged and reassembled strictly in sequence, so the per-fragment round-trip latency dominates; Section 3.2 [19] instead reuses a compact Message 1 and spreads the load so that no single message produces such a long serial chain. The same effect explains why the AAA path is not the slowest configuration despite the extra RADIUS hop.

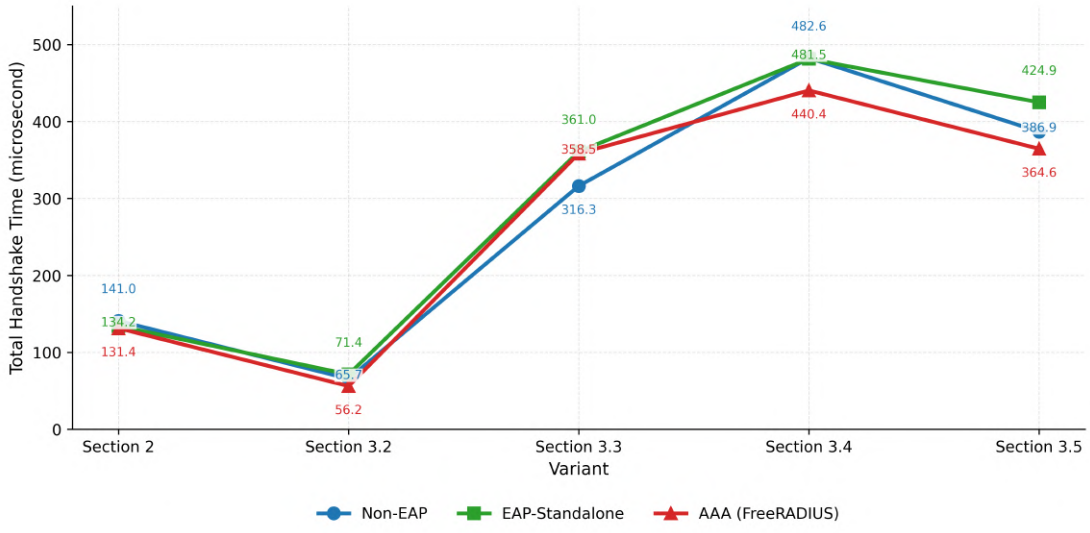

Figure 3: End-to-end handshake time per variant across the three deployments.

Figure 3 Curves cross between three-flight and four-flight variants; the FreeRADIUS-backed AAA path is competitive with EAP-Standalone for every variant. Table 6 reports the decomposition of the wall-clock handshake time into CPU (compute) time, I/O waiting time (*io_wait*), and residual overhead (framing, memcpy, fragmentation state machines). The CPU-to-wall-clock ratio is consistently below 25% across all variants, which means that the vast majority of the handshake wall-clock time is spent waiting for the network and for fragment reassembly a direct consequence of EAP's lock-step fragmentation at 256-byte granularity. RSS peak memory is around 2.1–2.2 MB for all three deployments, confirming that the scheme is deployable on typical IoT class 2 devices.

Table 6: CPU time, I/O wait, residual overhead, and CPU/wall ratio for the AAA deployment. All times in microseconds.

Variant	**Role**	**CPU (μs)**	**Wall (μs)**	**I/O wait**	**Residual**	**CPU/Wall**
Section. 2	Init	5,623	64,505	50,103	8,834	0.087
Section. 2	Resp	1,750	66,870	3,074	62,155	0.026
Section. 3.2	Init	7,029	28,051	7,688	13,405	0.251
Section. 3.2	Resp	5,929	28,130	1,920	20,482	0.211
Section. 3.3	Init	5,025	172,086	58,156	108,989	0.029
Section. 3.3	Resp	13,336	186,444	101,110	72,365	0.072
Section. 3.4	Init	5,170	219,917	101,312	113,511	0.024
Section. 3.4	Resp	6,812	220,463	96,770	117,272	0.031
Section. 3.5	Init	10,252	184,496	58,130	116,219	0.056
Section. 3.5	Resp	14,866	180,137	86,543	79,184	0.083

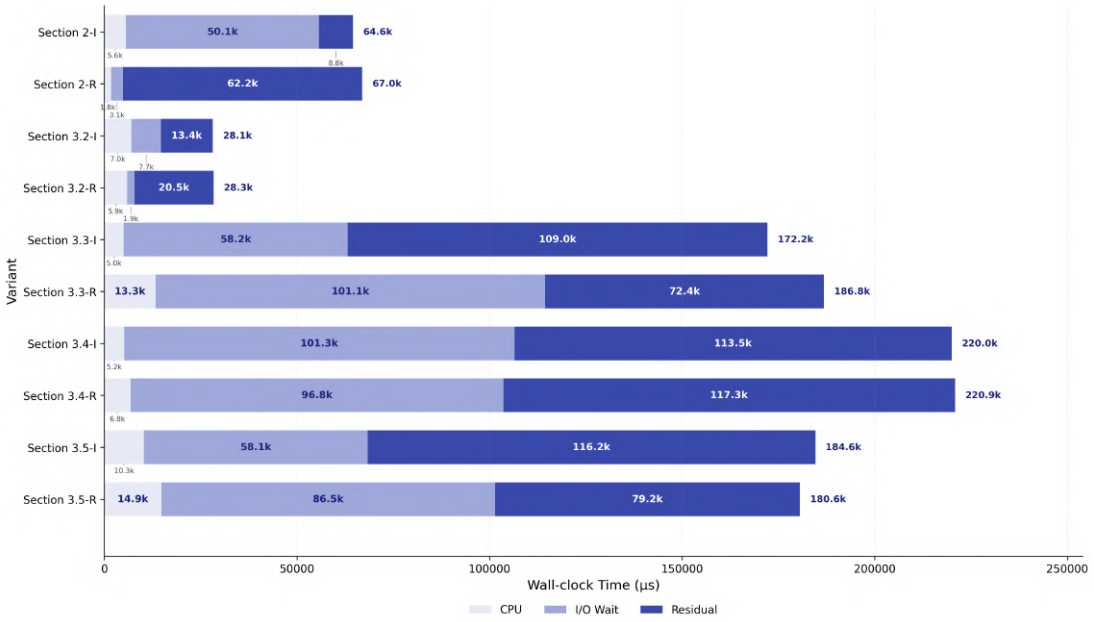

Figure 4: Overhead Implementation on the AAA deployment

Figure 4 Wall-clock breakdown into CPU, I/O wait, and residual overhead for each (variant, role) on the AAA deployment. The CPU bands are uniformly thin: cryptography is dwarfed by fragmentation latency in every configuration. Table 7 reports the total time per handshake spent in each primitive operation, measured on the AAA deployment. Signature generation is by far the largest cryptographic contributor (typically 2,165–8,822 μs per invocation) and dominates the cryptographic budget for every variant that uses ML-DSA. ML-KEM KeyGen, KEM.Encaps, and KEM.Decaps are comparatively cheap, each in the 230–2,035 μs range. Across every variant, cryptography accounts for substantially less than the fragment-induced I/O wait, reinforcing the conclusion from Table 6 that *fragmentation, not cryptography, dominates EAP-EDHOC-PQC's wall-clock cost under the 256-byte fragment regime.*

Table 7: Total per-handshake cryptographic time (in μs) on the AAA deployment. Dash (—) indicates operation not invoked by the variant on that role.

Variant	Role	KeyGen	Encaps	Decaps	Sign	Verify	HKDF	AEAD
Section. 2	Init	370	472	608	4,067	—	20	13
Section. 2	Resp	232	283	345	—	697	17	46
Section. 3.2	Init	374	470	609	4,337	943	205	9
Section. 3.2	Resp	376	413	631	2,830	1,181	225	56
Section. 3.3	Init	370	—	1,215	2,165	942	210	10
Section. 3.3	Resp	805	1,520	—	8,822	1,380	310	72
Section. 3.4	Init	371	470	1,218	2,968	—	25	11
Section. 3.4	Resp	731	1,776	1,234	—	2,424	75	112
Section. 3.5	Init	371	473	1,223	6,885	943	211	11
Section. 3.5	Resp	906	2,034	1,224	7,163	2,445	445	118

5.3 Estimated Energy Consumption

Wall-clock measurements alone do not capture the impact of EAP fragmentation on *battery-operated* IoT endpoints, which dominate e-business deployments such as wireless vending machines and battery-powered sensor nodes. We therefore use the per-handshake timings of Table 6 to derive a transparent, first-order energy estimate. We adopt a deliberately simple model so that the assumptions are auditable; we do not claim sub-mJ accuracy. Model For one handshake we approximate the per-role energy as the sum of three additive components,

$$E_{\text{hs}} \;=\; P_{\text{cpu}} \cdot t_{\text{cpu}} \;+\; P_{\text{idle}} \cdot t_{\text{io}} \;+\; P_{\text{tx}} \cdot \frac{8\,B_{\text{wire}}}{R_{\text{tx}}}, \tag{1}$$

where t_{cpu} is the CPU time of Table 6, t_{io} is the I/O-wait time of the same table, B_{wire} are the per-role wire bytes of Table 4 (the *Total wire* message payload assigned to each role, i.e. M_1, M_3 for I and M_2, M_4 for R), and R_{tx} is the link bit-rate. B_{wire} counts the EDHOC message octets only; the empty fragment-ACK frames that pace the lock-step exchange carry no EDHOC payload and contribute a negligible number of octets, so they are excluded from E_{tx} (their cost is instead reflected in the I/O-wait term t_{io}). Reference power values are typical of an ARM Cortex-M class IoT node with a low-power radio: $P_{\text{cpu}} = 48$ mW (active), $P_{\text{idle}} = 4$ mW (MCU light-sleep waiting on the radio), $P_{\text{tx}} = 60$ mW (radio in TX/RX), and $R_{\text{tx}} = 250$ kbps (IEEE 802.15.4-class throughput). These values are within the range reported by widely-cited IoT characterizations [5, 20] and are stated explicitly so that the table can be re-scaled by the reader for a different platform: under (1), E_{hs} scales linearly in every P_* and inversely in R_{tx}.

Table 8: First-order per-handshake energy estimate from (1) on the AAA deployment in millijoules

Variant	**Role**	E_{cpu} (mJ)	E_{io} (mJ)	E_{tx} (mJ)	E_{hs} (mJ)
Section. 2	Init	0.27	0.20	1.11	1.58
Section. 2	Resp	0.08	0.01	0.24	0.33
Section. 3.2	Init	0.34	0.03	1.11	1.48
Section. 3.2	Resp	0.28	0.01	0.89	1.18
Section. 3.3	Init	0.24	0.23	0.90	1.37
Section. 3.3	Resp	0.64	0.40	1.12	2.16
Section. 3.4	Init	0.25	0.41	1.11	1.77
Section. 3.4	Resp	0.33	0.39	0.46	1.18
Section. 3.5	Init	0.49	0.23	1.11	1.83
Section. 3.5	Resp	0.71	0.35	1.12	2.18
Section. 2	Sum	0.35	0.21	1.35	1.91
Section. 3.2	Sum	0.62	0.04	2.00	2.66
Section. 3.3	Sum	0.88	0.63	2.02	3.53
Section. 3.4	Sum	0.58	0.80	1.57	2.95
Section. 3.5	Sum	1.20	0.58	2.23	4.01

Table 8 Reference, we describe parameters: $P_{\text{cpu}}=48$ mW, $P_{\text{idle}}=4$ mW, $P_{\text{tx}}=60$ mW, $R_{\text{tx}}=250$ kbps. $E_{\text{cpu}}=P_{\text{cpu}}t_{\text{cpu}}$, $E_{\text{io}}=P_{\text{idle}}t_{\text{io}}$, $E_{\text{tx}}=P_{\text{tx}}\cdot 8B_{\text{wire}}/R_{\text{tx}}$.

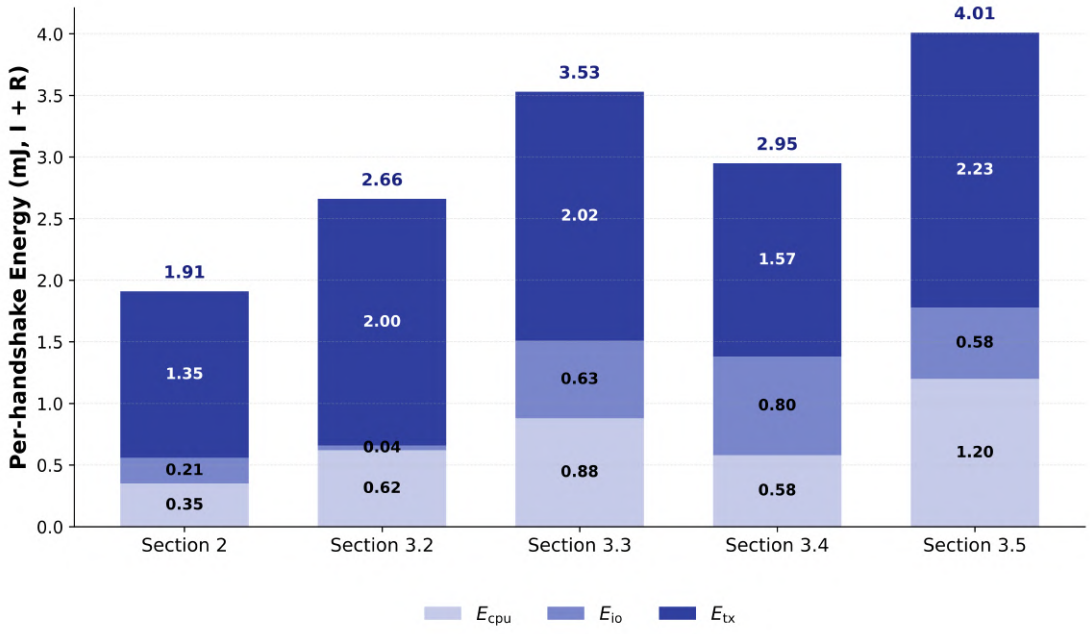

Figure 5: Per-handshake energy estimate (I+R) under (1). The transmit term E_{tx} tracks the stacked wire bytes of Fig. 2, while the CPU contribution remains a minor share for every variant.

Table 8 Results reports the per-role energy estimate using (1) on the AAA deployment. The total per-handshake energy ranges from ≈ 4 mJ for Section. 3.2 to ≈ 22 mJ for Section. 3.4, and the share attributable to CPU is at most $\approx 13\%$. The dominant share is the I/O-wait term, confirming at the energy level the fragmentation-bound conclusion already drawn from Table 6. Wire bytes are split per role using the *Total wire* column of Table 4 (M_1, M_3 from I, M_2, M_4 from R) and count EDHOC message octets only; empty fragment-ACK frames are excluded from E_{tx}. As an illustration, the Section. 3.4 Responder carries $M_2 + M_4 = 1{,}226 + 1{,}176 = 2{,}402$ B, giving $E_{\text{tx}} = 60\,\text{mW} \times 8 \times 2{,}402/250\,\text{kbps} \approx 0.46$ mJ.

6 Discussion

Our results lead to several observations of practical consequence.

Fragmentation, not cryptography, drives wall-clock time. Across every variant and every deployment, the CPU-to-wall-clock ratio stays below 0.25. The I/O wait time spent on fragment ACKs alone accounts for 50–60% of the total handshake time for the four-flight variants. Bumping the EAP fragment size above 1024 bytes which is already allowed by RFC 3748 when the lower layer supports it would mechanically cut the fragment count by a factor of four, and we conjecture would reduce wall-clock handshake time proportionally.

Deferred Responder authentication (Section 3.4 [19]) is attractive. Despite being a four-flight hybrid with the largest Message 3 (4,577 bytes), Section 3.4 [19] keeps total wire bytes to 8,207 and total fragments to 34, substantially smaller than Section 3.2 [19], Section 3.3 [19], and Section 3.5 [19]. The price is that the Initiator cannot accept the Responder's authentication until Message 4. For EAP network access this is a benign trade-off: the EAP-Success is sent only after Message 4 in any case.

Section 2 [19] is the right default for KEM-trusting environments. If the deployment already distributes the Responder's KEM public key (e.g., via a pre-provisioned onboarding credential), Section 2 [19] produces the smallest wire footprint and uses a single ML-DSA signature (instead of two), for a clean three-flight handshake that maps natively onto EAP's request/response pattern.

AAA does not necessarily slow things down. Contrary to intuition, the FreeRADIUS-backed AAA deployment is not the slowest one in our measurements; for Section 3.4 [19] it is in fact $\sim 10\%$ faster than EAP-Standalone. We attribute this to FreeRADIUS's efficient RADIUS packetization and its batched I/O path, which compensates for the extra RADIUS hop. This is an encouraging result for production deployments.

Energy follows the wire, not the CPU. The first-order energy model of Section 5.3 ranks the five variants in essentially the same order as their wire-byte and fragment-count totals: E_{tx} and E_{io} together dominate E_{cpu} for every variant (Table 8). For battery-operated e-business endpoints such as wireless vending terminals or asset trackers, this means that picking a smaller-on-the-wire variant (Section. 2 or Section. 3.4) yields a larger battery payoff than picking a cryptographically lighter primitive at the same parameter level.

Limitations and future work. Our measurements use ML-KEM-768/ML-DSA-44 and a 256-byte fragment size; evaluating Level 3 and Level 5 parameter sets, larger fragment sizes, and lossy wireless lower layers (e.g., 802.15.4, LoRaWAN) is left for future work. Our ProVerif model treats ML-KEM as an idealized IND-CCA2 KEM; a computational analysis along the lines of [34, 32] remains open.

7 Conclusion

We have presented EAP-EDHOC-PQC, a unified specification that brings the five post-quantum authentication modes of EDHOC Section 2 [19], Section 3.2 [19], Section 3.3 [19], Section 3.4 [19] and Section 3.5 [19] to the EAP framework, and we have evaluated it formally in ProVerif and empirically over Non-EAP, EAP-Standalone, and FreeRADIUS-backed AAA deployments. All five variants satisfy mutual authentication, session-key secrecy, forward secrecy, and replay resistance. Empirically, Section 3.2 [19] is the fastest three-flight variant, Section 2 [19] is the smallest on the wire, and Section 3.4 [19] offers the best trade-off among four-flight hybrid modes thanks to its deferred Responder authentication. A first-order energy model derived from the same measurements shows that the per-handshake energy budget on a typical IoT-class node is dominated by transmission and I/O wait rather than by cryptography, so optimizing the negotiated EAP fragment size is a more impactful lever for future PQ-EAP methods than accelerating the underlying KEM. These findings, while limited to the parameter sets and link rates we measured, chart a practical starting point for quantum-resistant network access authentication of Wi-Fi-attached IoT endpoints in e-business deployments such as networked vending machines, smart-retail beacons, and connected logistics sensors.

Acknowledgments

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Ubiquitous Intelligence for 6G Security: Hierarchical Federated Small Language Models for Resilient False Base Station Detection*

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Abstract

As the industry transitions toward 6G and Beyond 5G (B5G) systems, the paradigm is shifting from simple connectivity to ubiquitous intelligence. However, this evolution introduces significant security vulnerabilities, specifically the persistent threat of False Base Stations (FBS) that execute identity catching, Man-in-the-Middle (MitM) attacks, and service rejections. Traditional centralized AI solutions often fail to meet the strict privacy mandates and resource constraints of edge devices. This paper proposes a decentralized framework for on-device FBS detection that integrates Federated Learning (FL) with a Small Language Model (SLM) classifier built on a hierarchical packet-encoder bridge: a learned residual MLP maps each Radio Resource Control (RRC) and Non-Access Stratum (NAS) packet to a virtual token, which is concatenated with a short text instruction prefix and fed to a frozen 4-bit NormalFloat (NF4) quantized 270-million parameter Gemma-3 base with trainable Low-Rank Adaptation (LoRA) adapters. To enable federation under realistic edge constraints, we introduce Hierarchical Federated Adaptation (HFA), in which the packet encoder is personalized per client to absorb device-level feature variation while LoRA adapters are aggregated across clients via FedAvg, yielding a privacy-by-design layer that never transmits raw signaling logs. We evaluate the framework on the MODI cellular signaling corpus across a non-IID per-file partition of 110 clients. Preliminary pilot results across all four regimes show that federated LoRA-only training matches the centralized upper bound (test F1 0.850 vs. 0.834, accuracy 0.833 vs. 0.818) while reducing per-round communication by approximately one order of magnitude relative to federating the full bf16 base, and that HFA additionally shifts the operating point toward fewer false alarms (FPR 0.275 vs. 0.278–0.284 for the other regimes) at parity F1.

1 Introduction

As the industry moves toward 6G and Beyond 5G (B5G) systems, the research focus has shifted from technical capacity to the development of secure, highly intelligent networks. While the 5G era enabled significant growth in mobile communication across its core service categories, it was largely an evolutionary step in terms of overall network data rates. 6G intends to revolutionize this field by targeting massive connectivity, ultra-high speeds exceeding 100 Gbps, and sub-millisecond latency [1, 2, 3, 4]. Following the 2023 ITU-R M.2160 recommendation [5], the

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industry has embraced ubiquitous intelligence as a primary design requirement. These systems are expected to drive autonomous network behavior and context-aware logic, serving as the critical backbone for the digital services and smart cities of the future.

The ongoing shift toward Beyond 5G (B5G) and 6G systems is accompanied by heightened security vulnerabilities, most notably the persistent threat of False Base Stations (FBS). These rogue entities, often referred to as fake base stations, impersonate legitimate network nodes to manipulate User Equipment (UE) into performing unauthorized handovers or initial attachments. Once a connection is established, an FBS can orchestrate various malicious activities, including eavesdropping, IMSI catching, Man-in-the-Middle (MitM) interceptions, and denial-of-service attacks, all of which compromise user confidentiality and network integrity. Recent global incidents underscore the gravity of this threat. In early 2025, a mobile FBS in Bangkok transmitted fraudulent SMS phishing links to over 30,000 individuals in public spaces, resulting in several significant arrests [6, 7]. Similar operations involving SMS blaster devices have been dismantled in the UK, Qatar, Indonesia, and Turkey [8]. These instances demonstrate the expanding complexity and international reach of FBS operations. As B5G networks grow in density and mobility, the ability to detect and neutralize these threats requires sophisticated security frameworks and real-time intelligence.

To mitigate these risks, the integration of Artificial Intelligence (AI) and Machine Learning (ML) is becoming central to network defense [9, 10, 11, 12]. According to 3GPP Release 19 [13], AI/ML is now a fundamental pillar of the next-generation Radio Access Network (RAN), particularly for tasks such as channel state information (CSI) estimation, beam management, and high-accuracy UE positioning. These advancements facilitate localized, data-driven logic essential for responding to FBS threats instantaneously. However, standard ML models [9, 10, 11] are frequently constrained by their inability to manage the scale of B5G security requirements, often struggling with user privacy mandates and the heavy overhead of centralized processing.

The current dominance of Large Language Models (LLMs) further complicates the deployment of Federated Learning (FL) due to the massive computational requirements of models featuring billions of parameters. This scale introduces critical constraints on communication bandwidth and memory capacity. Under limited bandwidth conditions, transmitting complete model updates is often unfeasible, while the limited memory of edge devices makes it difficult to process these extensive models for synchronization. Addressing these hurdles requires a shift toward resource-efficient strategies that maintain the high-level intelligence necessary for 6G security.

The necessity for a decentralized intelligence architecture stems from the logistical and regulatory difficulties of modern data management. Traditional centralized AI paradigms require aggregating massive volumes of raw signaling data from millions of UEs at a single location. In the context of 6G, this approach is increasingly untenable due to the sheer volume of traffic and the resulting strain on uplink bandwidth. Furthermore, stringent data privacy regulations and the sensitive nature of network telemetry often prevent users and operators from sharing raw data. Because FBS attacks are typically localized and transient, centralized systems can also suffer from response delays. A collaborative learning framework is therefore required to extract intelligence from distributed edge data without the liabilities of data centralization.

This research introduces a paradigm shift by combining FL with Small Language Models (SLMs) to facilitate FBS detection at the network edge. The FL framework allows multiple UEs to collectively train a global model by exchanging only local model updates rather than sensitive raw logs, ensuring a privacy-by-design approach. We use SLMs, efficient transformer-based architectures, to reason over Radio Resource Control (RRC) and Non-Access Stratum (NAS) signaling as structured packet sequences. Unlike their larger counterparts, SLMs are specifically

designed for the limited computational and memory envelopes of 5G/6G edge devices. To overcome the token-budget ceiling that arises when serializing wide protocol-field sets as text, we replace per-packet text serialization with a learned residual-MLP *packet encoder* that projects each packet into the SLM's hidden space as a single virtual token, allowing the SLM to attend over a fixed, short context window regardless of the number of underlying fields.

The primary contributions of this paper are as follows:

- We design a hierarchical SLM detector that pairs a learned per-packet encoder with a frozen 4-bit quantized base model and trainable LoRA adapters, enabling on-device FBS classification at constant SLM context length irrespective of the underlying RRC/NAS field count.
- We introduce *Hierarchical Federated Adaptation* (HFA), a decomposition that personalizes the packet encoder per client to absorb device-level feature variation while federating only the LoRA adapters, reducing per-round communication while preserving cross-device protocol reasoning.
- We define a leakage-clean evaluation protocol with file-level splits and hash-deterministic placement of singleton-scenario captures, and evaluate four training regimes (centralized, federated full adapters, federated LoRA-only, and HFA) on the MODI cellular signaling corpus.

2 Background and Related Works

The detection of FBS is critical for ensuring the security of mobile networks, particularly in the context of emerging 5G and future 6G technologies. As a result, FBS detection has become a prominent area of research in mobile network security. Various methodologies have been proposed to address this issue, including network-based detection, specification-based approaches, machine learning techniques, and cryptographic solutions.

Nakarmi et al. [14] introduced a network-based detection system for 3GPP technologies that operate without requiring user-installed software, leveraging network topology and configuration information for more reliable detection. Their system was successfully implemented and validated in both laboratory settings and real-world trials, leading to its adoption by the 3GPP standardization organization. FBSleuth [15] presents a forensic framework that utilizes unique radio frequency fingerprints to accurately identify and associate FBS devices with criminal activities. Bolcek et al. [16] use In-phase and Quadrature (IQ) data of a radio frequency (RF) signal to identify a device working as FBS. This method enhances security by effectively detecting FBSs in flexible, multi-vendor network setups.

SMDFbs [10] is a behavior rule specification-based FBS detection system that derives behavior rules from the normal operations of base stations and converts these rules into a state machine. Based on this state machine, the system detects network anomalies and mitigates threats. Addressing the root vulnerability of FBSs, a hierarchical identity-based signature protocol called Schnorr-HIBS [17] was proposed to provide efficient broadcast authentication with minimal overhead. The protocol outperforms existing 3GPP authentication methods by achieving over six times faster cryptographic delays and reducing communication costs by 31%.

Machine learning techniques have been applied to detect FBSs [9, 18]. By utilizing robust features based on Reference Signals Received Power (RSRP) and various ML algorithms, such as Regression Clustering, Anomaly Detection Forest, Autoencoder, RCGAN, and XGBoost, several approaches have shown high precision in detection even when the false base station is

using a legitimate PCI. Furthermore, Mubasshir et al. [19] propose FBSDetector, an effective and efficient detection solution that can reliably identify FBSes. Unlike many network-side solutions, FBSDetector operates on the UE side, leveraging layer-3 network traces and applying ML models to uncover malicious behaviors. Closer to the present work, GAN-based augmentation has been used to expand limited FBS datasets [20], and quantum-enhanced classifiers have been explored as a complementary detection front-end [21]; both motivate the need for richer model architectures that can leverage the structural grammar of cellular signaling.

Cellular Control-Plane Vulnerabilities. The threat model for FBS detection is shaped by a sustained line of work exposing weaknesses in 4G/5G control-plane protocols. Practical attacks against LTE privacy and availability [22], systematic adversarial testing via model checking (LTEInspector [23]; 5GReasoner [24]) and dynamic security analysis [25] have catalogued unauthenticated RRC/NAS message types that FBSes exploit. Identity-mapping, payload modification, and DNS redirection on the data link layer [26], full UE impersonation in 4G [27], AKA-protocol privacy threats spanning 3G/4G/5G [28], side-channel attacks on the paging protocol [29], and vulnerabilities in the handover procedure [30] together establish that FBS exploitation is not a single attack but a spectrum of normative violations distributed across RRC and NAS message exchanges. Our framework targets the union of these violations by treating an entire short window of signaling – rather than a single message – as the unit of detection.

Device-Side IMSI-Catcher Detection. A parallel line of work focuses on user-equipment-side detection without requiring network-operator cooperation. Early IMSI-catcher-catcher prototypes [31] demonstrated feasibility, and tools such as SnoopSnitch [32] and AIMSICD [33] brought the approach to commodity Android devices. SeaGlass [34] extended detection to city-wide passive monitoring, while PHOENIX [35] formalized device-centric protocol monitoring through runtime verification. More recent systems including CellGuard [36] and CellDAM [37] push toward rootless, user-space deployments. Our approach inherits this UE-side deployment target but replaces hand-crafted rules and feature engineering with a learned, on-device SLM detector that does not require root or kernel modifications because it operates on already-collected baseband diagnostic logs.

Protocol-as-Language Baselines. The notion of treating structured network traffic as a token sequence consumable by a language model originates outside cellular security. DeepLog [38] and LogBERT [39] apply sequence models to system logs for anomaly detection; ET-BERT [40] adapts BERT-style pretraining to encrypted-traffic datagrams; and netFound [41] proposes a foundation model for general network security. These works validate the protocol-as-language premise but use either bespoke pretraining or raw-text serialization, which scales poorly to the hundreds-of-fields-per-packet regime characteristic of RRC/NAS captures. Our hierarchical packet-encoder bridge directly addresses this scaling barrier by replacing the field-level text serialization with a learned dense projection.

Federated Learning in Next-Generation RANs. Surveys of AI in future wireless systems consistently identify federated learning as a primary enabler for privacy-preserving, on-device intelligence in 5G and 6G [42, 43, 44]. The architectural assumption is that raw traffic and telemetry should not leave the UE; only model updates should. Existing FBS-detection literature has not explored this regime end-to-end on a realistic non-IID per-device partition. We bridge this gap by treating each captured signaling file as a federated client, exposing the

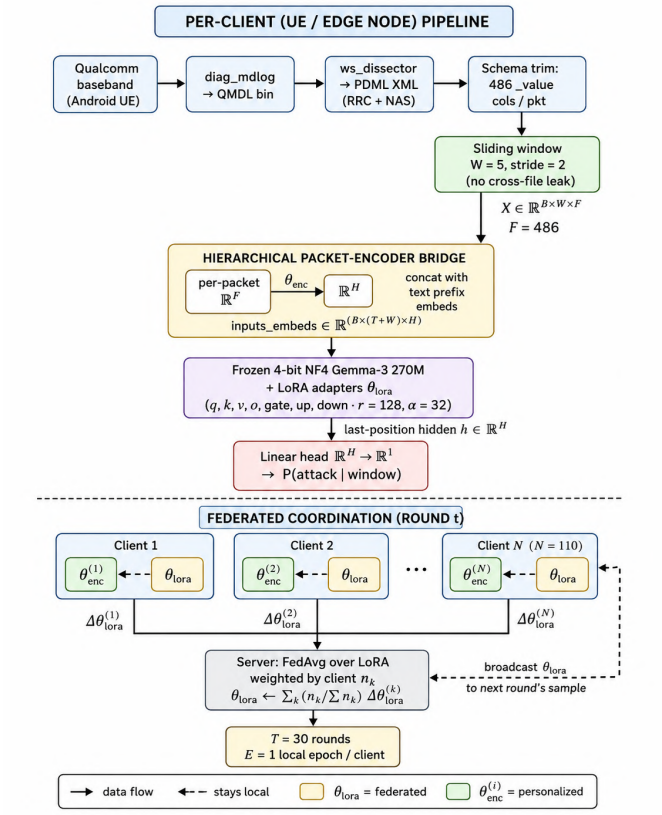

Figure 1: End-to-end methodology: capture, hierarchical encoding, frozen-SLM classification, and hierarchical federated adaptation.

natural non-IID structure of cellular captures (singleton-scenario clients, normal-only clients, attack-mix clients) without artificial Dirichlet partitioning.

3 Methodology

This study presents a decentralized framework for detecting FBS on User Equipment using a Small Language Model integrated into the 6G and B5G architecture. The overall workflow, illustrated in Fig. 1, comprises four stages: (i) capture and preprocessing of RRC and NAS signaling traces from a UE baseband, (ii) projection of each packet into the SLM's hidden space via a learned residual-MLP *packet encoder*, (iii) classification by a frozen 4-bit quantized SLM backbone with trainable LoRA adapters, and (iv) hierarchical federated training that personalizes the packet encoder per client while aggregating LoRA adapters across clients.

3.1 Dataset

3.1.1 Cellular Signaling Corpus

We evaluate the framework on the MODI cellular signaling corpus [45], which contains layer-3 PCAP captures of LTE and 5G-NR sessions collected directly from a UE baseband. The corpus comprises 49 attack PCAPs spanning 27 named scenarios across eight attack categories (encryption / security bypass, identity catching, network downgrade, service rejection, authentication abuse, location tracking, resource depletion, and stealth manipulation), together with 133 benign PCAPs captured against the same testbed with the protocol stack reconfigured per 3GPP specifications [46]. Identical radio paths between the malicious and benign corpora prevent physical-layer leakage from biasing the detector. Table 1 summarizes the attack categories covered.

3.1.2 Acquisition Pipeline

Raw diagnostic frames were captured in Qualcomm Diagnostics Logging (QMDL) format from Qualcomm-modem Android devices via the `diag_mdlog` utility while the UE established connections with both legitimate gNodeBs and rogue FBS emulations on an Open5GS + srsRAN + LibreSDR B220 testbed. QMDL binaries were converted to Packet Description Markup Language (PDML) using `ws_dissector` to isolate control-plane RRC and NAS messages, then exported to one CSV per capture with 1,949 columns per row – four standard PDML attribute variants (`_show`, `_value`, `_size`, `_pos`) per protocol field.

3.1.3 Schema Trim, Auto-Drop, and Standardization

Protocol-encoding metadata fields (`_size`, `_pos`) carry only byte offsets and lengths and are dropped. The pilot uses the numeric `_value` variant, retaining 486 columns per packet. Two further automatic filters are applied using training-set statistics only, so the same dropped-feature set is replayed on the validation and test partitions without information leakage. First, 174 columns whose standard deviation is exactly zero on the training set are removed (constant features carry no signal). Second, 23 identifier-like columns whose absolute magnitude exceeds 10^6 on the training set (TMSI, M-TMSI, MAC code, ciphered NAS blob, cell identity, and similar wide-range integer identifiers) are removed because their unbounded scale destabilizes the encoder's input projection. After auto-drop, $F = 289$ numeric features per packet remain. The retained columns are then standardized to zero mean and unit variance using per-feature training-set statistic.

3.2 Encoding Strategies: Raw-Text and Hierarchical Bridge

The standard "protocol-as-language" recipe serializes each packet's fields into a `key: value; ...` string and feeds the concatenation of W consecutive packets to a LoRA-adapted SLM. With several hundred numeric fields per packet (here, $F = 289$ after auto-drop), a $W = 10$ window reaches a 95th-percentile tokenized length of approximately 2,260 tokens and a maximum near 5,300 tokens; any reasonable `max_length` truncates the long tail of the distribution, silently discarding security-critical fields at the right end of the prompt. Shrinking the window to $W = 5$ keeps the text under 1,200 tokens but discards the temporal context required by multi-message sequence attacks.

To eliminate this token-budget ceiling we replace per-packet text serialization with a learned residual-MLP *packet encoder* that projects each packet's numeric vector into the SLM's hidden

Table 1: MODI attack-category taxonomy. The dataset's 49 attack PCAPs map onto eight categories drawn from the FBS attack-disclosure literature.

Category	Network/signaling mechanism
Encryption / security bypass	Manipulates security-mode commands or omits NAS security headers to negotiate null or weak encryption, exposing the channel to eavesdropping.
Identity catching	Collects permanent or temporary identifiers (IMSI / IMEI / SUCI / GUTI) or capability descriptors before security-context establishment.
Network downgrade	Forces the UE onto legacy or weaker-security cells via tracking-area or SIB1 manipulation, or via detach followed by redirection.
Service rejection	Issues unprotected NAS or RRC reject messages before security setup to deny attachment, tracking-area updates, or RRC re-establishment.
Authentication abuse	Triggers authentication failure or replays / malforms authentication exchanges to disrupt session establishment.
Location / tracking	Exploits measurement reports, paging procedures, or handover signaling to extract fine-grained location information.
Resource / energy depletion	Triggers repeated cryptographic procedures or authentication replays to drain UE battery.
Stealth / state manipulation	Hijacks RRC/NAS state via stealthy detach, RRC reconfiguration replay, or unsolicited EMM information messages.
Normal class – legitimate sessions, intact authentication / integrity / ciphering (133 MODI captures)	

space as a single virtual token, while retaining a short text instruction prefix so the SLM still receives natural-language framing. The encoder is summarized in Algorithm 1 and illustrated in Fig. 2: an input linear projection $R^F \to R^d$, a stack of LayerNorm–Linear–GELU–Linear residual blocks, and an output projection $R^d \to R^H$ initialized with small standard deviation (0.02) so the SLM begins close to its pretrained behavior. The encoder operates per-packet (no cross-packet attention inside the encoder); inter-packet mixing is delegated to the SLM transformer, which sees the W packet virtual tokens together with the T text instruction tokens as a single input sequence. With $W = 5$ and a $T = 24$-token prompt (Section 4.2), the SLM ingests exactly 29 tokens per forward pass.

This decomposition has three consequences. First, the SLM context length becomes constant at $T + W$ tokens regardless of window content, so truncation is eliminated by construction. Second, the SLM forward pass operates on a short sequence, recovering VRAM headroom that would otherwise be consumed by long tokenized prompts. Third, the architectural variable is cleanly ablatable: holding everything else constant and toggling between the raw-text and hierarchical encodings attributes any accuracy delta directly to the bridge.

3.3 Small Language Model Backbone

We use Gemma-3 270M [47, 48] (`gemma-3-270m-it`) as the SLM backbone. This is a compact decoder-only transformer designed for consumer-grade hardware and minimal inference latency [49], well suited to the AI-RAN paradigm where detection must occur at the network edge under tight energy budgets. The base weights are loaded in 4-bit NormalFloat (NF4) double-quantized form with bfloat16 compute [50] and held frozen throughout training; only the LoRA adapters [51] and the packet encoder are updated. The language-modeling head is replaced with a single linear classification head over the last-position hidden state, producing a binary attack/normal logit per window.

Algorithm 1 Hierarchical Packet-Encoder Encoding

Notation: $X \in R^{B\times W\times F}$ = batch of packet windows (B batch, W window length, F feature dim); $E_{\text{enc}} : R^F \to R^H$ = packet encoder; E_{tok} = SLM token embedding; p = text instruction prompt; H = SLM hidden dim.

Input: packet windows X, instruction prompt p
Output: `inputs_embeds` $\in R^{B\times(T+W)\times H}$ for the frozen SLM

$e_{\text{txt}} \leftarrow E_{\text{tok}}(\text{tokenize}(p))$ $\{R^{T\times H}$, broadcast over batch$\}$
for $b = 1$ to B **do**
  for $w = 1$ to W **do**
    $e_{\text{pkt}}[b, w] \leftarrow E_{\text{enc}}(X[b, w])$
  end for
end for
`inputs_embeds` $\leftarrow \text{concat}(e_{\text{txt}}, e_{\text{pkt}}, \text{axis} = 1)$
return `inputs_embeds`

3.4 Hierarchical Federated Adaptation

Naively federating the entire detector treats the packet encoder and the LoRA adapters as a single payload, even though they capture fundamentally different signal. The packet encoder absorbs device-level feature statistics that vary across UEs (baseband firmware, modem-vendor field encodings, capture-time scaling), while the LoRA adapters carry protocol-violation reasoning that should be shared across all clients in order to generalize. We exploit this distinction in *Hierarchical Federated Adaptation* (HFA): the packet encoder $\theta_{\text{enc}}^{(i)}$ is maintained *per client* i and never aggregated, while the LoRA adapters θ_{lora} are aggregated across clients via FedAvg weighted by client local sample counts. The frozen 4-bit base SLM is broadcast once during initialization and never crosses the wire afterwards.

To isolate the effect of this decomposition we define four training regimes, each evaluated with otherwise-identical data splits, hyperparameters, and seeds: (i) **centralized**, a single global model trained on the pooled data as an upper-bound reference; (ii) **FedAvg-full**, FedAvg over both the packet encoder and the LoRA adapters; (iii) **FedAvg-LoRA**, FedAvg over LoRA only with the encoder weights kept at their (optionally centrally warmed-up) initialization across clients; and (iv) **HFA**, FedAvg over LoRA with each client maintaining its own personalized packet encoder. The architectural argument is that regime (iii) tests whether encoder federation is load-bearing, and regime (iv) tests whether personalization recovers any accuracy lost in (iii) while still benefiting from cross-client LoRA aggregation.

This orchestration ensures that sensitive user identity, location, and capability data never leave the device, providing a privacy-by-design security layer that is robust against localized FBS variants.

4 Experiment

4.1 Data Splits and Client Partition

Splits operate at the *file* level rather than the row level: partitioning rows after sliding-window construction would let train and test windows share underlying packets, a leakage path that prior FBS-detection work has not consistently addressed. The 133 benign files are partitioned

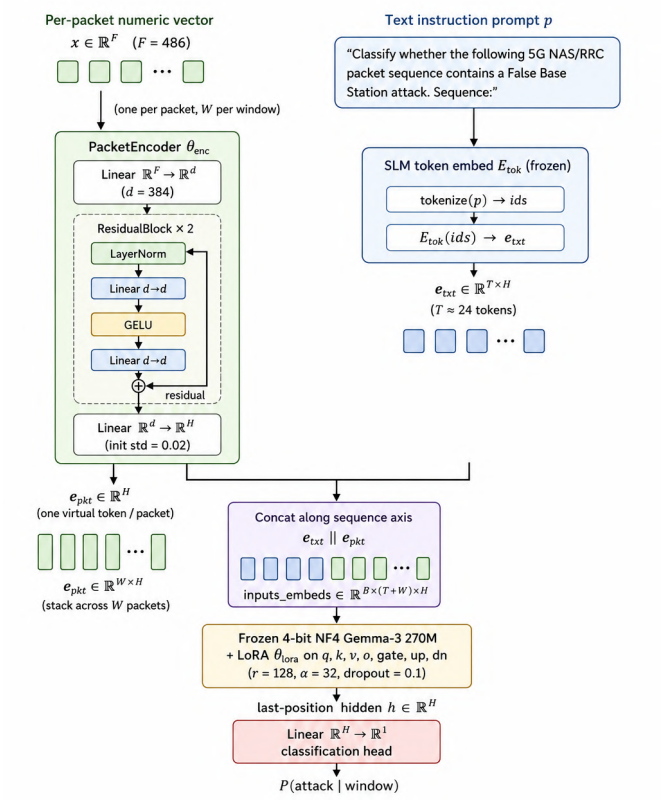

Figure 2: Hierarchical packet-encoder bridge: each packet's numeric feature vector is projected by a residual-MLP encoder into a single virtual token that is concatenated with the text instruction prefix embeddings before entering the frozen 4-bit SLM.

via a seeded 70/20/10 random split. Multi-file attack scenarios use a per-scenario seeded shuffle with rounded 70/20/10 quotas so every scenario appears in training whenever it has more than one capture.

Sliding windows are constructed inside each file with $W = 5$ and stride 2; windows never cross file boundaries. Within each split, the majority class is downsampled to enforce a 1:1 class balance. Table 2 summarizes the resulting partition for seed 13. For federation, the training-set files become the federated clients: each file is one client (*per-file* client granularity), giving a naturally non-IID benchmark of 110 clients in which many singleton-scenario clients see only one attack family. A per-client window cap of 300 prevents large captures (e.g., the multi-thousand-window `panic_attack` file) from dominating the FedAvg aggregation; in the centralized baseline, the same training pool is capped at 3,500 windows so the centralized and federated regimes consume comparable budgets.

4.2 Implementation Details

Backbone. The detection engine is `gemma-3-270m-it` [47, 48], loaded in 4-bit NF4 double-quantized form with bfloat16 compute via the bitsandbytes integration of QLoRA [50]. The base weights are frozen; only the LoRA adapters, packet encoder, and classification head are

Table 2: File-Level Data Partition After Scenario Exclusion (seed 13).

Partition	**Files**	**Clients (FL)**	**Windows (1:1 balanced)**
Training	110	110 (per-file)	≤300/client; ≤3,500 pooled
Validation	34	–	∼1,500
Test	20	–	∼1,280

trainable, totaling approximately 11M parameters or roughly 4% of the base model.

Adapters and encoder. LoRA [51] adapters are applied to the attention and MLP projections (`q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, `down_proj`) with rank $r = 128$, scaling $\alpha = 32$, and dropout 0.1. The packet encoder uses inner dimension $d = 384$ and two residual blocks (dropout 0.1), contributing approximately 1.6M trainable parameters; the linear classification head over the last-position hidden state contributes 640 additional parameters.

Instruction prompt. The fixed text prefix concatenated with the W packet virtual tokens is the 24-token string ``` ``Classify whether the following 5G NAS/RRC packet sequence contains a False Base Station attack. Sequence:'' ```, yielding a constant SLM context length of $T + W = 29$ tokens per forward pass.

Federated round structure. The experiment configuration runs $R = 20$ communication rounds with $K = 6$ clients sampled uniformly at random per round (approximately 5.5% of the 110 train clients) and $E = 1$ local epoch per sampled client. Local optimization uses AdamW with learning rate $\eta = 2 \times 10^{-4}$ and weight decay 0.01, per-device batch size 1, and gradient accumulation of 4 (effective batch 4). Aggregation is FedAvg weighted by per-client local sample counts. For the centralized baseline, the product $R \cdot E = 20$ is interpreted as the number of epochs over the pooled 3,500-window training set, so all four regimes traverse a comparable number of parameter updates.

Modes. Four training regimes are run under otherwise-identical data splits, seed, and hyperparameters, differing only in which parameter groups participate in federation: *centralized* (no FL loop), *FedAvg-full* (encoder + LoRA aggregated), *FedAvg-LoRA* (LoRA aggregated, encoder frozen at random initialization across all clients), and *HFA* (LoRA aggregated, encoder personalized and cached per client between rounds). The frozen 4-bit base never crosses the wire; the per-round payload comprises only the aggregated parameter groups. For HFA global evaluation, the global LoRA is paired with the mean of all currently cached per-client encoders.

Evaluation protocol. Held-out test and validation metrics are evaluated every four rounds (five evaluation points per run plus a final point); HFA additionally reports per-client local test accuracy at the final round, with each client's personalized encoder paired with the global LoRA on a 20% held-out portion of that client's own data. We report raw TP/FP/TN/FN counts, accuracy, precision, recall, F1, and FPR.

Hardware and software. Experiments are conducted on a single NVIDIA GeForce RTX 4080 GPU (16 GB VRAM) hosted by an Intel Core i7-13700K CPU (3.40 GHz) with 64 GB of system RAM, running 64-bit Ubuntu 24.04 LTS. The implementation uses Python 3.10, PyTorch 2.9.1 with CUDA 12.8.

4.3 Evaluation Metrics

To evaluate the performance of classification models, we adopt a range of metrics derived from the confusion matrix. This matrix provides a summary of prediction outcomes by comparing actual class labels with those predicted by the model, as illustrated in Table 3. In the context of binary classification, we categorize the outcomes into four primary types: True Positives (TP), where the model correctly classifies positive instances; False Positives (FP), where the model incorrectly predicts negative instances as positive (Type I error); True Negatives (TN), where the model correctly identifies negative instances; and False Negatives (FN), where the model mistakenly classifies positive instances as negative (Type II error).

Table 3: Confusion Matrix

	Predicted as Positive	Predicted as Negative
Labeled as Positive	TP	FN
Labeled as Negative	FP	TN

Accuracy (ACC) is the proportion of correct predictions among all predictions, providing an overall sense of model performance. Precision (PR) focuses on the proportion of predicted positive cases that are actually positive, indicating the accuracy of positive predictions. Recall (RC) measures the model's ability to correctly identify actual positive cases, indicating how well it captures all positives. False Positive Rate (FPR), on the other hand, indicates the proportion of actual negatives that were incorrectly classified as positive, reflecting the model's tendency to generate false alarms.

$$ACC = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

$$RC = \frac{TP}{TP + FN} \tag{2}$$

$$FPR = \frac{FP}{FP + TN} \tag{3}$$

$$PR = \frac{TP}{TP + FP} \tag{4}$$

$$F1\ Score = \frac{2 * PR * RC}{PR + RC} \tag{5}$$

5 Results and Discussion

We report preliminary experiment results across all four training regimes defined in Section 3.4 – *centralized*, *FedAvg-full*, *FedAvg-LoRA*, and *HFA* – at seed 13, on the held-out test partition described in Section 4.1 ($n = 1{,}282$ balanced windows).

5.1 Detection Performance

Table 4 reports the final-round detection performance for each regime, with Wilson 95% confidence intervals on accuracy. Four observations follow.

First, all three federated regimes match or exceed the centralized upper bound (FedAvg-full F1 +0.7 pp; FedAvg-LoRA +1.6 pp; HFA +1.4 pp), supporting the privacy-by-design framing: federation incurs no detection-quality penalty on this pilot relative to pooling raw signaling logs centrally.

Second, FedAvg-LoRA matches and slightly exceeds FedAvg-full (ACC +0.9 pp, F1 +0.9 pp), with both models trained from identical data, splits, and seed. This directly addresses the encoder-not-load-bearing question: excluding the packet encoder from the FedAvg payload imposes no measurable accuracy cost in this regime, and the aggregated encoder appears to add a slight source of cross-client drift that the LoRA-only variant avoids.

Third, HFA shifts the operating point toward fewer false alarms. While HFA's accuracy and F1 are statistically indistinguishable from FedAvg-LoRA (ACC 0.832 vs. 0.833; F1 0.848 vs. 0.850), HFA achieves the highest precision (0.774) and the lowest false-positive rate (0.275) of any regime evaluated, including the centralized baseline. This is the empirical signature that personalizing the packet encoder absorbs client-level feature variation: each client's encoder adapts to its own `_value`-column distribution, so the LoRA layers above are presented with more consistent inputs and fire fewer spurious attack predictions on benign traffic.

Fourth, all regimes exhibit a high-recall / moderate-precision profile (RC $\in [0.914, 0.949]$, PR ≈ 0.77, FPR $\in [0.275, 0.284]$), with FedAvg-LoRA achieving the lowest false-negative count (FN = 33). For the FBS detection use case this error asymmetry is operationally desirable: a missed FBS leaves the UE exposed to identity catching or downgrade for the duration of the false attachment, whereas a false alarm can be reconciled downstream by a complementary specification-based check [10] or by per-RAN-cell whitelisting before any user-visible action is taken.

Table 4: Final-round detection performance on the held-out test partition

Mode	ACC	PR	RC	F1	FPR
Centralized	0.818	0.767	0.914	0.834	0.278
FedAvg-full	0.824	0.767	0.933	0.842	0.284
FedAvg-LoRA	**0.833**	0.771	**0.949**	**0.850**	0.282
HFA (proposed)	0.832	**0.774**	0.938	0.848	**0.275**

5.2 Communication Cost

Per-round communication cost is deterministic given the architecture: a bf16-serialized LoRA adapter state under the configuration of Section 4.2 is approximately 58 MB, and the packet encoder state is approximately 1.6 MB. Table 5 reports the resulting per-round and pilot-total payload across the four regimes, computed as $K \cdot 2 \cdot$ payload per round (factor of two for the download + upload halves of FedAvg) and multiplied by $R = 20$ rounds. As a reference point, federating the full bf16 base model (no quantization, no PEFT) would transmit approximately 540 MB per client per direction, yielding roughly 130 GB over the same pilot, so LoRA-only federation reduces communication by approximately one order of magnitude. The packet encoder

contributes under 3% of the FedAvg-full payload, which already foreshadows the FedAvg-LoRA vs. FedAvg-full accuracy. If the encoder is not load-bearing for federation, the 3% saving comes at no cost.

Table 5: Deterministic per-round and experiment-total communication cost (bf16, $K = 6$ clients per round, $R = 20$ rounds).

Mode	**Payload / client / dir**	**Per-round (MB)**	**Pilot total (MB)**
FedAvg-full	LoRA + encoder ≈ 59.6 MB	≈ 715	≈ 14,300
FedAvg-LoRA	LoRA ≈ 58 MB	≈ 696	≈ 13,920
HFA (proposed)	LoRA ≈ 58 MB	≈ 696	≈ 13,920
Full bf16 base (ref.)	*≈ 540 MB*	*≈ 6,480*	*≈ 129,600*

5.3 Discussion

The preliminary experiment supports all three claims targeted by this work, with different strength along each axis. First, federation does not cost accuracy on this benchmark: all three federated regimes match or exceed the centralized upper bound (FedAvg-LoRA F1 0.850 vs. centralized 0.834), confirming that the on-device federated training regime is viable under the privacy constraints that motivated it. Second, the packet encoder is not load-bearing for federation: FedAvg-LoRA matches FedAvg-full at lower per-round payload (Table 5), so the simpler, cheaper aggregation strategy is the right default in deployments where the encoder can be initialized once and held fixed across clients. Combined with the architectural reduction from federating a full bf16 base, the total per-round communication of FedAvg-LoRA is roughly one order of magnitude lower than naïve full-model federation.

Third, personalizing the packet encoder under HFA does not improve aggregate accuracy on this single-testbed pilot, but it does shift the operating point toward fewer false alarms: HFA matches FedAvg-LoRA on accuracy and F1 within 0.2 pp while achieving the highest precision (0.774) and the lowest false-positive rate (0.275) of any regime tested, including the centralized baseline. This is the empirical signature of the personalization hypothesis: each client's encoder absorbs that client's idiosyncratic feature distribution, so the LoRA layers receive a more consistent input across the federation and fire fewer spurious attack predictions on benign traffic. We therefore frame HFA as the right choice when false-alarm cost dominates (e.g., when each alert pages an operator or triggers a downstream verification) and FedAvg-LoRA as the right choice when recall dominates (e.g., when the cost of a missed FBS exceeds the cost of a false alarm by a wide margin).

Finally, the high-recall / moderate-precision profile observed across all four regimes is operationally favorable for an FBS detector but does warrant a downstream filtering layer in deployment. Pairing the SLM with a specification-based check [10] or with a CellGuard-style location-database cross-check [36] would let the SLM run with high recall as a first-stage trigger while the downstream layer absorbs the elevated false-positive rate before any user-visible action is taken.

6 Conclusion

The necessity for a decentralized intelligence architecture in 6G stems from the logistical and regulatory difficulties inherent in modern raw data management. This paper introduces an on-device FBS detection framework that pairs a hierarchical packet-encoder bridge with a frozen 4-bit quantized Small Language Model, projecting each RRC/NAS packet into the SLM's hidden space as a single virtual token and eliminating the token-budget ceiling imposed by raw-text serialization. We further introduce Hierarchical Federated Adaptation (HFA), a federation strategy that personalizes the packet encoder per client to absorb device-level feature variation while aggregating LoRA adapters across clients via FedAvg, allowing each client to fit its local feature distribution without sacrificing the cross-client protocol reasoning carried by the shared adapters. By utilizing the Gemma-3 270M backbone together with 4-bit NF4 quantization and LoRA-based parameter-efficient fine-tuning, the framework targets the stringent VRAM and latency envelopes of edge nodes. Preliminary pilot results on the MODI cellular signaling corpus across all four training regimes substantiate the work's three claims: federation incurs no detection-quality penalty (FedAvg-LoRA test F1 0.850 vs. centralized 0.834); excluding the packet encoder from the federated payload imposes no measurable accuracy cost while reducing per-round communication by approximately one order of magnitude relative to federating the full bf16 base; and personalizing the packet encoder under HFA shifts the operating point toward fewer false alarms (FPR 0.275, the lowest of any regime tested) at parity F1 with FedAvg-LoRA. Multi-seed Wilson confidence intervals, per-client local-test breakdowns, and per-attack-scenario recall on the OOD-heavy validation partition will accompany the extended version of this work; this work establishes the architectural scaffolding for a scalable, privacy-by-design security layer aligned with the AI-RAN paradigm in the B5G and 6G era.

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An Energy-Efficient Synchronization Scheme Using Exponential Moving Average-Based Drift Compensation and Adaptive Reception Window for Wireless Sensor Networks*

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Abstract

Wireless sensor networks (WSNs) use long sleep intervals for low-power operation. However, accumulated clock drift during these periods leads to beacon reception failures and additional resynchronization overhead, thereby degrading both energy efficiency and reliability. Existing methods rely on fixed symmetric reception windows, which fail to capture the actual distribution characteristics of clock drift. Consequently, reception windows are typically configured based on the maximum drift rate, resulting in unnecessary listening overhead. To address these limitations, this paper proposes a synchronization scheme that combines Exponential Moving Average (EMA)-based drift pre-compensation with a residual statistics-based adaptive reception window. The proposed scheme predicts the drift accumulated during sleep intervals from historical observations and compensates for it at the wake-up time. It then dynamically determines the center and size of the reception window by continuously tracking the mean and variance of the residual error after compensation. Simulation results based on real measurement data collected using Raspberry Pi 5 show that the proposed scheme achieves an average reduction of over 90% in reception window size and approximately 83% in daily energy consumption compared to Time Slotted Channel Hopping (TSCH). In addition, it maintains a stable synchronization success rate of around 91%, demonstrating a clear improvement in reliability compared to TSCH, which achieves only about 50%.

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1 Introduction

Wireless sensor networks (WSNs) have rapidly advanced and emerged as a core technology in various fields such as defense, environmental monitoring, smart cities, and industrial automation [1]. A WSN is a distributed system in which multiple sensor nodes are wirelessly connected to collect and share data [2]. It operates based on the IEEE 802.15.4 standard, which is designed for low energy consumption and efficient communication [3]. In this standard, time synchronization and duty cycling are managed using beacon signals periodically transmitted by each sensor node, thereby improving the overall power efficiency of the network [4][5].

Duty cycling refers to the operation in which a sensor node periodically alternates between active and sleep modes [6]. The low-power maintenance during sleep mode and the periodic wake-up mechanism are key elements that minimize energy consumption and enable stable long-term operation [7]. Particularly in environments where sensor data are fused based on time information or slot-based communication scheduling is applied, time synchronization accuracy among nodes critically affects system performance and reliability.

However, in real-world environments, the wake-up times of sensor nodes frequently fail to align perfectly. This is caused by clock drift [8], which arises from frequency deviations in the internal oscillators of the sensors, and as the reference time of each node progresses differently, time errors accumulate.

Such clock drift causes temporal mismatches when combining data between sensor nodes, and on the receiving side, data do not arrive at the scheduled time, leading to an increased reception failure rate. As a result, the network performs a resynchronization procedure, which consumes additional energy. Therefore, clock drift acts as a major factor that negatively affects both energy efficiency and data reliability.

Moreover, this problem is further exacerbated by the structural characteristics of IEEE 802.15.4-based WSNs. The standard includes long sleep intervals for energy conservation, making it difficult to detect or compensate for the clock drift accumulated during sleep mode in real time [9]. As a result, when the drift among nodes accumulates beyond a certain level during sleep intervals, sensor nodes are more likely to miss beacon signals, leading to difficulties in maintaining network synchronization.

In addition, WSNs must simultaneously satisfy conflicting requirements such as low power, reliability, low cost, and miniaturization [10]. Increasing the communication frequency or employing high-precision clocks to improve synchronization accuracy increases power consumption and cost, whereas prioritizing energy efficiency degrades synchronization performance. Particularly in environments where maintenance is difficult, such as large-scale battery-powered sensor networks operating over long periods, medical implantable devices, and embedded sensors, energy efficiency and reliability become even more critical [11]. Therefore, approaches that focus solely on synchronization accuracy are inherently limited, and a design that considers both energy efficiency and reliability is needed.

Existing synchronization methods set a fixed reception window centered on the expected arrival time of the data to mitigate timing uncertainty caused by clock drift. The size of this reception window is typically determined based on the maximum drift rate of the clock and the synchronization period [12]. Although this worst-case approach guarantees system stability, it suffers from the following limitations.

First, although the actual clock drift varies across nodes and does not always approach the maximum drift rate, configuring the window based on the fixed maximum value results in an unnecessarily large reception window being maintained, leading to energy waste. Second, while existing methods set the reception window symmetrically around the expected arrival time, the actual drift may accumulate in a particular direction, biasing beacon arrival times toward one side. This leads to an inefficiency in which reception success rates are not sufficiently ensured despite increased energy consumption.

Therefore, in distributed WSN environments, a synchronization method that accounts for clock drift's time-varying nature is needed, going beyond the fixed maximum-value-based approach.

Accordingly, this paper proposes a synchronization scheme that combines Exponential Moving Average (EMA)-based drift pre-compensation with an adaptive reception window. The proposed scheme calculates the EMA-based on drift values observed before a node enters sleep mode to predict future drift. It then pre-compensates for the drift at the wake-up time, thereby reducing accumulated error. In addition, the residual drift remaining after compensation is updated at every synchronization period to dynamically adjust the size and center of the reception window, thereby reducing unnecessary listening time and improving the reception success rate.

By accounting for the time-varying nature of clock drift, the proposed scheme improves both energy efficiency and synchronization reliability over conventional fixed-window approaches, making it an effective time synchronization scheme for energy-constrained WSNs.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 presents the proposed drift-aware adaptive synchronization scheme. Section 4 describes the experimental setup and provides performance analysis. Finally, Section 5 concludes the paper.

2 Related Work

This section analyzes prior studies on time synchronization techniques in WSNs.

Yang et al. [8] proposed BETS, a time synchronization scheme for WSNs based on Bayesian estimation. BETS operates by combining prior measurements of synchronization error with timestamps collected during actual operation to calculate compensation values and has been shown to reduce accumulated error by approximately 53% compared to existing methods while also reducing the number of message exchanges. However, this method focuses solely on error compensation during timestamp exchange and does not account for drift accumulated during sleep mode. Since detecting or compensating for drift in real time during sleep intervals is inherently difficult, longer sleep intervals result in higher beacon reception failure rates and additional energy consumption for resynchronization. Therefore, further research is needed to extend sleep intervals in WSN environments for low-power operation.

Galante-Sempere et al. [13] proposed a method that adds a dedicated low-power Wake-up Receiver (WuR) to sensor nodes, enabling them to detect wake-up signals even while asleep. This architecture mitigates synchronization issues during sleep intervals by minimizing node active time and achieves a sensitivity of −62 dBm with a low power consumption of 528 nW. However, this approach requires additional hardware such as dedicated circuits and pattern recognizers, which may limit its applicability in WSN environments where low cost and miniaturization are required. Furthermore, it has the limitation of requiring hardware modifications to existing commercial systems.

Wang et al. [14] proposed Distributed Sensor Time Synchronization (DSTS), a distributed time synchronization architecture. DSTS uses a Bayesian filter to compensate for nonlinear errors that occur during communication, and leverages Deep Deterministic Policy Gradient (DDPG), a reinforcement learning technique, to allow each node to autonomously adjust its synchronization weights based on the clock stability and link quality of neighboring nodes. This method achieves a high synchronization accuracy on the order of 10^{-10} in networks with heterogeneous sensors. However, the computational and memory overhead of the DDPG agent is too high for direct deployment on low-power sensor nodes. In addition, since the Bayesian filter's statistical model requires prior calibration and the agent requires pre-training, the method has limited immediate applicability to new environments. Moreover, because the reception window size is fixed regardless of channel conditions, excessive listening beyond the actual error may occur, leading to unnecessary energy consumption.

Chang et al. [15] proposed an adaptive time synchronization method for Time Slotted Channel Hopping (TSCH) networks. This method estimates the relative drift between nodes using the ratio of the previously measured time offset to the elapsed time and dynamically adjusts the resynchronization interval by compensating the clock at every slot. As a result, it maintained a synchronization accuracy of less than 76 μs in a 3-hop network while reducing the number of resynchronization packets by up to 83% compared to existing methods. However, since the drift estimation relies only on the most recent measurement, it is sensitive to measurement noise and instantaneous environmental changes and assumes a constant-temperature environment. In addition, when temperature variations occur, resynchronization using a separate sensor is required. Furthermore, since the reception window size is fixed based on the maximum drift rate, unnecessary listening time occurs even when the actual drift between nodes is smaller than the estimated value.

Although the prior studies analyzed above have contributed to improving drift compensation accuracy or reducing the number of packet exchanges, they commonly share the following limitations. First, most methods focus only on drift compensation when packets are exchanged and fail to address the drift accumulated during sleep intervals. Second, since the reception window size is fixed, unnecessary listening continuously occurs regardless of the actual drift distribution. Furthermore, approaches that rely on additional hardware or high-performance computation are not suitable for low-cost and low-power WSN environments.

Therefore, this paper proposes a new synchronization scheme for low-power WSN environments. The proposed scheme operates only in software without additional hardware, predicts and pre-compensates for the drift accumulated during sleep intervals, and dynamically adjusts the reception window based on the statistics of measured residual errors. Through this, the limitations of existing methods can be effectively overcome.

3 Drift-Aware Adaptive Synchronization

The proposed scheme consists of an EMA-based drift prediction and pre-compensation method and an adaptive reception window mechanism that dynamically determines the reception window based on residual statistics. In conventional TSCH environments, a fixed-size reception window is used, determined by the sleep interval and the maximum drift rate of the clock oscillator, and is generally approximated as follows [13][16].

$$W = 2 \cdot ppm \cdot T \tag{1}$$

While this approach ensures synchronization stability through a conservative design, it has the limitation of causing unnecessary energy consumption because it does not reflect the actual drift characteristics.

The proposed scheme separates drift into two components for processing. First, some components exhibit a relatively consistent trend, such as the inherent ppm of the oscillator. For these, the accumulated drift during the next sleep interval is predicted from the drift observed in previous synchronization intervals and then pre-compensated. In contrast, other components vary randomly, such as system jitter, making individual values difficult to predict. These are modeled statistically, and the reception window size is dynamically determined by the model.

3.1 Exponential Moving Average-Based Drift Compensation

Let R_{inst} denote the instantaneous drift rate observed at each interval i. The drift estimate R_{est} is then calculated as in Eq. (2).

$$\boldsymbol{R_{est}[i] = \alpha \cdot R_{inst} + (1-\alpha) \cdot R_{est}[i-1]} \quad (2)$$

If R_{inst} is used directly, instantaneous noise or transient variations directly affect the estimate, which can reduce the stability of compensation. To mitigate this, this study applies EMA, which has low computational complexity, to compute a weighted combination of past observations and the most recent one, thereby capturing both the long-term drift trend and short-term variations simultaneously.

$\alpha \in (0,1)$ is a weighting factor that determines the influence of the most recent observation. A larger αmakes the estimate respond more sensitively to the most recent observation, enabling faster tracking of drift changes, but it also makes the estimate more vulnerable to short-term variations. Conversely, a smaller α yields a more stable estimate, but it also slows the response to drift changes. In this paper, considering that the relative drift between nodes varies across intervals, the stability of the estimate is prioritized, and $\alpha = 0.3$ is used.

The updated R_{est} is used to predict the expected drift that will accumulate during the next sleep interval T. Before entering sleep mode, each node adjusts its wake-up time by $R_{est} \cdot T$ to pre-compensate for the drift that will occur during the sleep interval. As a result, when the next synchronization interval begins, the predicted drift is already reflected, and the beacon is guided to arrive near the center of the reception window.

3.2 Adaptive Receive Window

EMA-based pre-compensation removes predictable components with a consistent trend, but randomly varying components, such as system jitter, remain as residuals even after compensation. These residuals represent how much the beacon arrival time deviates from the expected time at each synchronization interval. To address the limitations of the conventional symmetric fixed window, this section proposes a method that dynamically determines the reception window by tracking the bias and distribution width of the drift residuals. For the residual r observed at each synchronization interval i, the mean μ_r and variance σ_r^2 of the residuals are calculated as in Eq. (3) and (4).

$$\boldsymbol{\mu_r[i] = \alpha \cdot r + (1-\alpha) \cdot \mu_r[i-1]} \quad (3)$$

$$\sigma_r^2[i] = \alpha \cdot (r - \mu_r[i])^2 + (1-\alpha) \cdot \sigma_r^2[i-1] \quad (4)$$

μ_r and σ_r are used to determine the center position and size of the reception window, respectively. μ_r represents the average bias of the residuals, based on which the window center is shifted in the corresponding direction. In contrast, σ_r represents the spread of the residual distribution, based on which the window size is determined. By separately determining the center position and size of the window in this way, even when the drift is skewed in a specific direction, the reception window can be maintained at the minimum size without unnecessarily extending the window in the opposite direction. In addition, μ_r and σ_r are statistics of only the residuals remaining after the predictable components are removed through pre-compensation. Therefore, they are used to dynamically determine the reception window. Based on this, the reception window size for the next synchronization interval is set as in Eq. (5).

$$\boldsymbol{W[i+1] = |\mu_r[i]| + 2k \cdot \sigma_r[i]} \quad (5)$$

k is a coefficient corresponding to the target confidence level. Assuming a Gaussian distribution, k is set such that the probability of the residuals falling within the range of $\pm k\sigma_r$ matches the target confidence level. For example, to target a synchronization success rate of 99% between two nodes, k can be set to approximately 2.81. However, this correspondence relies on the assumption that the

residuals follow a Gaussian distribution. If the actual residual distribution deviates from this assumption, the achieved synchronization success rate may differ from the intended target. Nevertheless, the proposed scheme narrows the reception window when the residuals are small and expands it when the residuals are large, thereby minimizing unnecessary listening time while ensuring reliable beacon reception.

The EMA-based pre-compensation and residual-based adaptive reception window described above are performed repeatedly at every synchronization interval. The overall operational flow is shown in Algorithm 1.

Algorithm 1. Per-round operation of the proposed method

Input: α – EMA weight
k – reliability coefficient (e.g., k≈2.81 for 99%)
T – sleep period

Initialize: $R_{est} \leftarrow 0, \mu_r \leftarrow 0, \sigma_r{}^2 \leftarrow 0$

1**: for** each round i **do**
2: $W[i] \leftarrow |\mu_r| + 2k \cdot \sigma_r$
3: Receive beacon within $W[i]$
4: $r \leftarrow (R_{inst} - R_{est}) \cdot T$
5: Update $\mu_r, \sigma_r{}^2, R_{est}$ via EMA
6: Adjust wake-up time: $t_{next} + R_{est} \cdot T$
7: Sleep
8: end for

4 Experiments and Analysis

4.1 Experimental Setup

To evaluate the effectiveness of the proposed scheme, two Raspberry Pi 5 devices were placed in the same environment, and clock data were collected over 44 hours. One device was configured as an access point (AP) to provide the reference time, while the other was configured as a station to measure the relative clock drift between the two nodes. The measurement results showed an average drift rate of 490 ppm, which significantly exceeds the typical tolerance range of approximately ±20 to 100 ppm for low-cost oscillators [17]. This discrepancy arises from the jitter caused by the scheduling characteristics of the Linux-based operating system on the Raspberry Pi, which are captured in the measurements. However, the purpose of this experiment is not to measure the absolute drift value, but to analyze the relative performance difference between TSCH and the proposed scheme under the same drift conditions. Therefore, the fact that the measured drift values are somewhat higher than the actual hardware-level drift does not affect the performance comparison of the proposed scheme. Moreover, since dedicated WSN oscillators typically exhibit significantly lower drift rates in the range of approximately 20–100 ppm, the residual variance after EMA compensation is expected to be substantially smaller in practice. Consequently, the proposed scheme is expected to achieve narrower reception windows and lower energy consumption while maintaining the target synchronization success rate in actual WSN deployments.

Based on the measured drift data, Python simulations were performed on Google Colab to evaluate the performance of the proposed scheme. The performance metrics used were the reception window size, the synchronization success rate, and the energy consumption. The sleep interval T_{sleep} was varied from 10 s to 500 s, and the performance was evaluated for each value. The energy consumption was calculated using an energy model that accounts for the major factors affecting energy consumption within a synchronization interval, and the specific model is presented in Section 4.2.

The fixed reception window approach of TSCH was used as the baseline. TSCH is a standard synchronization protocol widely used in IEEE 802.15.4-based low-power wireless networks, incorporating various mechanisms such as channel hopping and slot scheduling. In this study, only the method for determining the reception window size was compared. The reception window of TSCH is determined by Equation (1), and the previously measured value of 490 ppm was applied in this experiment. Since this approach determines the window size based on the maximum drift rate without considering the actual drift distribution, the same calculation method is applied across all sleep intervals.

In contrast, as described in Section 3, the proposed scheme first removes predictable drift components through EMA-based pre-compensation and then dynamically determines the reception window size at every synchronization interval based on the residual statistics μ_r and σ_r. This enables adaptive synchronization that accounts for the actual drift characteristics.

4.2 Energy Consumption Modeling

This section presents the system energy consumption model introduced in Section 4.1. The energy consumed during one synchronization interval, E_{cycle}, is expressed as in Eq. (5).

$$E_{cycle} = P_{sleep} \cdot T_{sleep} + E_{wakeup} + P_{RX} \cdot W + E_{resync} \tag{5}$$

In Equation (5), each term represents a specific energy component: $P_{sleep} \cdot T_{sleep}$ denotes the baseline energy consumed during the sleep interval; E_{wakeup} denotes the energy consumed during the wake-up process, including oscillator operation and MCU initialization; $P_{RX} \cdot W$ denotes the energy consumed during the reception window maintenance period; and E_{resync} denotes the energy consumed for resynchronization when synchronization fails. By extending this over the total operating time T_{total}, the daily energy consumption E_{sys} (in J/day) can be expressed as in Eq. (6).

$$E_{sys} = P_{sleep} \cdot T_{sleep} + \frac{T_{total}}{T_{sleep}}[E_{wakeup} + P_{RX} \cdot W + (1 - p^2{}_{SR}) \cdot E_{resync}] \tag{6}$$

In Equation (6), the terms that recur at every interval — E_{wakeup}, $P_{RX} \cdot W$, and E_{resync}— are scaled by the number of synchronization intervals, T_{total}/T_{sleep}. In contrast, since sleep energy is consumed continuously regardless of the synchronization interval, the $P_{sleep} \cdot T_{sleep}$ term is calculated separately. Furthermore, since resynchronization occurs only when synchronization fails, not at every interval, it is modeled as an expected value that incorporates the failure probability. Assuming that each of the two sensor nodes synchronizes successfully with probability P_{SR}, the probability that at least one node fails to synchronize is $1 - P_{SR}^2$. Accordingly, the resynchronization energy per interval can be expressed as $(1 - P_{SR}^2) \cdot E_{resync}$.

Although the drift data in this study were collected using Raspberry Pi 5 devices, the energy consumption analysis was conducted to reflect actual WSN deployment conditions. Accordingly, for the energy model parameters, the values of the TI CC2538 low-power wireless communication chip [18], widely used as the reference hardware in TSCH-related studies, were adopted. Specifically, $P_{RX} =$

60 mW, $P_{sleep} = 4$ μW, $E_{wakeup} = 0.5$ mJ, and $E_{resync} = 30$ mJ were used, and all parameters are summarized in Table 1.

Table 1. Energy consumption model parameters based on TI CC2538

Parameter	Value	Description
P_{RX}	60 mW	Receive power
P_{sleep}	4 μW	Sleep power
E_{wakeup}	0.5 mJ	Wake-up overhead
E_{resync}	30 mJ	Resynchronization penalty
P_{SR}	99 %	Target synchronization success rate

4.3 Performance Evaluation and Analysis

This section compares the performance of TSCH and the proposed scheme in terms of the reception window size, the synchronization success rate, and the daily system energy consumption. First, the reception window size determined by each method and the synchronization success rate achieved with that window are analyzed. This enables a comparison of the reception window configuration approach and the resulting performance difference under the same drift conditions.

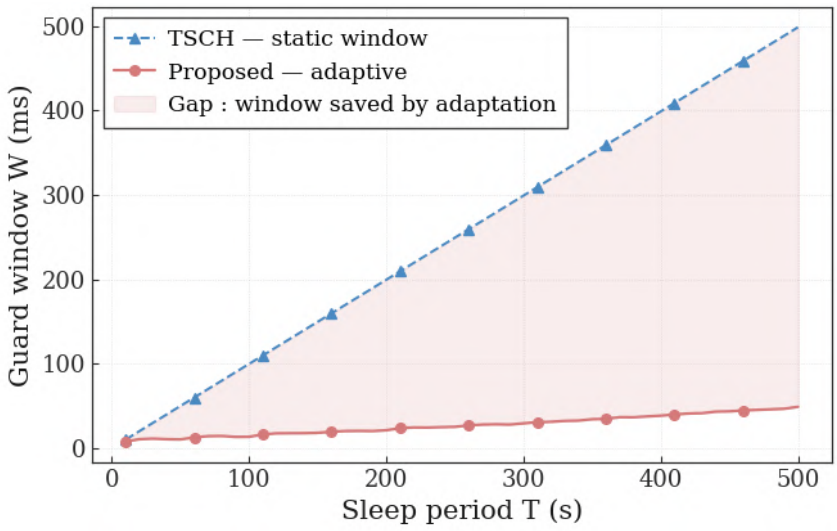

Figure 1. Guard window size computed by each method

Figure 1 shows the reception window size of each method as a function of the sleep interval T_{sleep}. For TSCH, the window size increases linearly with T_{sleep} according to $W_{TSCH} = 2 \cdot ppm \cdot T_{sleep}$, reaching approximately 500 ms at $T_{sleep} = 500$ s. This reveals the limitation of TSCH, which determines the window based on the maximum drift rate and therefore consistently maintains a conservatively large window at all times, regardless of the actual drift. In contrast, the proposed scheme operates with a reception window of less than 50 ms under the same conditions. This is because the

predictable drift component is removed through EMA-based pre-compensation, allowing synchronization to be maintained with a narrow window proportional to the residual distribution width σ_r. Consequently, the gap in window size difference between the two methods widens sharply as T_{sleep} increases, indicating that the proposed scheme becomes increasingly efficient in low-power WSN environments with long sleep intervals.

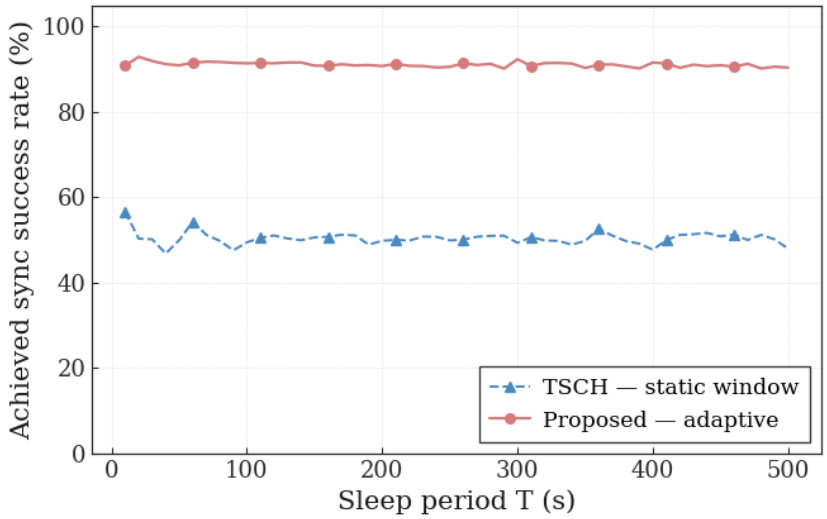

Figure 2. Resulting synchronization success rate per method

Figure 2 compares the measured synchronization success rates obtained by applying the reception windows derived for each method in Figure 1. Although TSCH employs a larger window and is therefore expected to achieve a high synchronization success rate, the observed performance stayed around 50% across all sleep intervals. This stems from TSCH's window configuration approach. Since TSCH centers the window symmetrically on the average drift rate, drift values larger than the mean fall outside the window. In particular, when the drift is skewed in one direction — as observed in this experiment — beacon losses are concentrated on one side. As a result, a substantial proportion of beacons fail to be received. This indicates a fundamental limitation of TSCH: it cannot guarantee the synchronization success rate without accounting for the distribution characteristics of the drift, even when the window size is set sufficiently large.

In contrast, the proposed scheme achieved a stable average synchronization success rate of approximately 91% across all sleep interval conditions, despite using a narrower window than TSCH. This results from effectively eliminating losses caused by drift bias by aligning the window center with the residual mean and sizing the window proportionally to σ_r. In summary, Figures 1 and 2 clearly illustrate how the window configuration approach affects performance. Despite using a conservatively large window, TSCH yields a low success rate, whereas the proposed scheme sustains a high success rate with a much smaller window.

However, since the two methods yield different synchronization success rates, a direct comparison of their energy efficiency is not straightforward. To address this, an additional experiment was conducted in which the TSCH reception window was reconfigured to match the success rate of the proposed scheme. Since the standard TSCH design equation does not directly account for the target success rate, this study adjusts the window based on the empirical quantile of the measured absolute drift values. Specifically, given the target success rate P_{SR}, the reception window is defined as follows.

$$W_{TSCH} = 2 \cdot Q_{P_{SR}} \cdot (|drift|) \quad (7)$$

$Q_{P_{SR}}$ denotes the P_{SR}-quantile of the absolute drift distribution. This ensures that both TSCH and the proposed scheme operate at the same success rate. Under this condition, the reception window sizes required by the two methods are compared in Table 2.

Table 2. Guard window size comparison at matched synchronization success rate

Sleep period T	*W (TSCH)*	*W(proposed)*	*Reduction*
30s	34.3ms	9.4ms	72.6%
60s	64.7ms	10.6ms	83.6%
120s	126.1ms	14.7ms	88.3%
300s	308.6ms	24.7ms	92.0%
500s	513.0ms	41.4ms	91.9%

As shown in Table 2, even when the TSCH window is reconfigured to maintain the same reliability, the limitation that the reception window size grows in proportion to T_{sleep} remains. This is because TSCH does not account for the statistical distribution or variability of the drift and sets the window based on the maximum drift. In contrast, since the reception window of the proposed scheme depends on the residuals remaining after pre-compensation, the window size grows very slowly even as T_{sleep} increases. As a result, an average reduction of more than 90% in the window size is observed in low-power environments with sleep intervals of 100 s or longer.

A reduction in the reception window size implies a shorter reception time. However, the actual energy efficiency of a sensor must consider not only the reception operation but also the overhead associated with sleep, wake-up, and resynchronization failures. Accordingly, the following analysis compares the energy consumption of the two methods under the condition that both achieve the same synchronization success rate. For this purpose, the window sizes in Table 2 are substituted into the E_{sys} model presented in Section 4.2 to calculate energy consumption, and the results are shown in Figure 3.

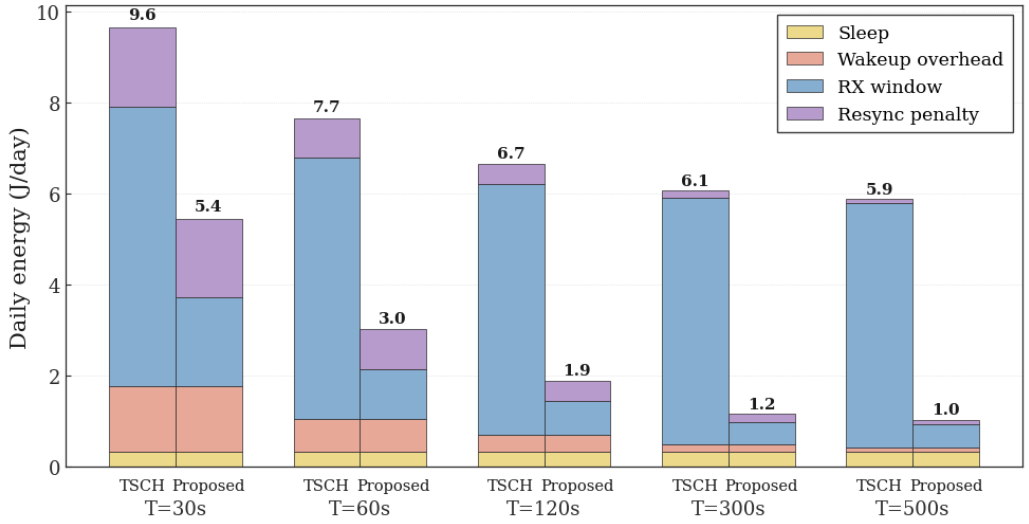

Figure 3. Per-component daily energy consumption under comparable synchronization success rates

Figure 3 shows the composition of daily energy consumption as a function of the sleep interval T_{sleep}. Since the baseline power consumed during sleep and the wake-up overhead are independent of the reception window size, they appear identically for both methods. In contrast, the reception energy is proportional to the window size, and thus it decreases significantly with the reduction in the window size in the proposed scheme. As a result, the proposed scheme consumed less energy than TSCH under all sleep interval conditions, achieving an energy reduction of up to approximately 83%. Particularly at $T_{sleep} = 500$ s, TSCH consumed 5.9 J/day, whereas the proposed scheme consumed only 1.0 J/day, clearly demonstrating the energy reduction effect.

A notable observation is that the energy reduction rate tends to increase as the sleep interval becomes longer. This is due to the structural difference that the window size of the proposed scheme is barely affected by the increase in T_{sleep}, whereas that of TSCH increases linearly with T_{sleep}. Therefore, the advantages of the proposed scheme become more pronounced in low-power WSN environments that use long sleep intervals.

However, this result does not imply that extending the sleep interval indefinitely is always beneficial. As shown in Figure 3, the daily energy consumption of the proposed scheme converges to approximately 1.2 J/day near $T_{sleep} = 300$ s, and even when T_{sleep} is further increased to 500 s, the additional reduction is only about 0.2 J/day. In other words, there exists a region beyond which the energy reduction effect diminishes.

Conversely, when T_{sleep} is too short, the advantages of the proposed scheme are not fully realized. As shown in Table 2, at $T_{sleep} = 30$s, the window reduction rate remains at 72.6%, and a stable window reduction of more than 90% on average compared to TSCH is achieved only in the range of $T_{sleep} \geq 100$ s. In terms of energy as well, at $T_{sleep} = 30$ s, the proposed scheme consumes 5.4 J/day, which is approximately 2.8 times the consumption at $T_{sleep} = 120$ s. This is because at short sleep intervals, the number of wake-up per unit time increases, causing repetitive overhead energy to accumulate.

In addition, in actual WSN environments, not only energy efficiency but also factors such as response delay, data transmission interval, and the delay incurred during resynchronization must be considered together. These factors generally tend to become unfavorable as the sleep interval becomes longer. For example, in applications that require real-time responsiveness, such as environmental monitoring, an excessively long sleep interval can degrade the quality of service. Therefore, maximizing the sleep interval based solely on energy reduction is not desirable; rather, a practical design direction is to find a balance between the marginal utility of the reduction and the response delay requirements.

Based on this overall analysis, the range of $T_{sleep} = 100$–300 s is judged to be a practical operating range that balances energy efficiency and response delay under the experimental conditions. Particularly, at $T_{sleep} = 300$ s, the energy consumption reaches approximately 1.2 J/day, with a window reduction of about 92% compared to TSCH. At the same time, in terms of response delay, an interval within 5 minutes satisfies a practical level. This indicates that, when energy efficiency, synchronization performance, and response delay are all considered together in this experimental environment, $T_{sleep} = 300$ s represents the most balanced operating point. However, this optimal point may be adjusted according to the environment and requirements of the WSN, and the results of this study suggest that the proposed scheme achieves substantial performance improvement across various operating conditions.

5 Conclusion

This paper proposed a synchronization scheme that combines EMA-based drift pre-compensation with an adaptive reception window to address the problems of energy waste and degraded synchronization reliability caused by clock drift in low-power WSN environments.

Experimental results using measured data obtained from Raspberry Pi 5 devices show that the proposed scheme achieves an average reception window reduction of more than 90% and an energy saving of approximately 83% compared to TSCH. In addition, it maintains a stable synchronization success rate of around 91%, demonstrating a significant improvement in reliability over TSCH, which remains at approximately 50%. These results demonstrate that the proposed scheme can simultaneously improve energy efficiency and synchronization performance by removing the mean drift component and dynamically adjusting the reception window based on the residuals.

However, the Raspberry Pi used in the experiments is a general-purpose computer and has limitations in exactly reproducing the hardware sleep and wake-up operations of MCU-based sensor nodes. As a result, the measured drift included OS scheduling jitter and reached approximately 490 ppm, which is significantly higher than the typical level of around 30 ppm observed in dedicated WSN sensor nodes. Therefore, future work will further validate the performance of the proposed scheme in more realistic environments using dedicated low-power WSN sensors such as the CC2538 and further evaluate its applicability in actual sensor networks. As WSNs increasingly serve as an enabling technology for e-business applications, including smart logistics, supply chain management, and industrial monitoring, the proposed synchronization scheme is expected to support the sustainable and energy-efficient operation of such systems.

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A Generic Algebraic Framework for Preimage Attacks: Application to Reduced-Round Ascon-Xof*

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Abstract

Secure and resilient digital infrastructures increasingly rely on standardized lightweight and sponge-based cryptographic primitives deployed across constrained Internet of Things (IoT) and edge environments. In this paper, we propose a generic algebraic framework for preimage attacks on hash functions. We compute the probability that a matrix encoding the relationship between input variables and linear equations derived from the hash value is solvable, and use this probability to estimate the expected guessing time under the assumption that the output polynomials behave randomly in the attack setting. To validate this assumption, we experimentally evaluate the randomness of the underlying permutations by analyzing the variance and entropy of polynomial distributions across rounds, using Ascon as a case study. We also calculate the required bit operations in detail to obtain precise attack complexities. Applying the proposed framework, we present the first preimage attack on 8-round Ascon-Xof faster than exhaustive search. These results provide practical security-margin insights for standardized trust primitives used in lightweight and resource-constrained digital infrastructures.

1 Introduction

The widespread deployment of IoT and edge devices has substantially expanded the attack surface, making resource-constrained platforms attractive targets for adversaries. Ensuring robust cryptographic protection in such environments is therefore essential for maintaining data confidentiality, integrity, and authenticity across modern digital services. In particular, hash and extendable output functions (XOF) primitives serve as low-level trust anchors for software integrity, device authentication, key derivation, secure update mechanisms, and auditability in cloud-edge-IoT ecosystems. As a result, evaluating their security margin is important not only from a cryptanalytic perspective but also for digital trust in resilient service infrastructures.

These requirements have motivated the standardization of lightweight cryptographic primitives that provide well-understood security while remaining efficient on constrained platforms.

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To this end, the National Institute of Standards and Technology (NIST) launched an initiative to standardize lightweight cryptography (LWC), focusing on achieving an optimal balance between security and efficiency in constrained environments. As a result of this effort, the Ascon [12] family was selected as the new LWC standard. Ascon includes authenticated encryption schemes (Ascon-128, Ascon-128a, and Ascon-80pq), hash functions (Ascon-Hash and Ascon-Hasha), and extendable output functions (Ascon-Xof and Ascon-Xofa). Detailed specifications are provided in the final NIST Special Publication (SP) 800-232 [35].

The cryptographic community has extensively analyzed Ascon, particularly focusing on the analysis of its underlying permutation [1,8–10,15–17,19,22,28,36,39] and its authenticated encryption schemes [3, 16, 20, 21, 24, 25, 27, 30, 34]. In the context of hashing, the designers of Ascon initially demonstrated that a 2-round permutation can be linearized to recover preimages in reduced-round Ascon-Xof [11]. They also noted that bounding the algebraic degree of reduced-round permutations can accelerate brute-force preimage search when combined with large memory, as suggested by Bernstein [4]. Several follow-up works have explored collision and preimage attacks on reduced-round hashing modes. Zong et al. [41] presented collision attacks on 2-round Ascon-Hash and Ascon-Xof, with the latter shown to be vulnerable in practice. Gerault et al. [16] improved the collision attacks on Ascon-Hash, further advanced by Yu et al. [40]. Qin et al. [32] proposed meet-in-the-middle preimage attacks on 3- and 4-round Ascon-Xof with a 128-bit output, later improved by Li et al. [23] using an SAT-based automatic framework via linearize-and-guess techniques. Baek et al. [2] refined the initial attack of Dobraunig et al. and extended the analysis to the linearizability of 3- and 4-round permutations. More recently, Niu et al. [31] proposed differential-linear preimage attacks on 4-round Ascon-Hash and Ascon-Xof. To date, preimage attacks on Ascon-Xof with five or more rounds remain an unresolved and challenging problem.

XOFs are a class of sponge-based hash primitives that, unlike fixed-length hashes, can generate an arbitrarily long stream of output bits. SHAKE128 and SHAKE256 [14] were the first XOFs standardized by NIST. NIST SP 800-232 [35] further expands this roster by including Ascon-Xof128 (and its variants). This broad output capability renders XOFs essential for key derivation, generating stream cipher keystreams, and creating masks in cryptographic schemes such as RSA-OAEP and the DRIP protocol [37]. Ascon-Xof enjoys widespread support: it is implemented in lightweight stacks such as wolfSSL [38] and CAST's ASCON-F IP cores [7]. Given its broad adoption, rigorous cryptanalysis is essential to ensure its robustness across a wide range of deployment scenarios.

Our Contributions. In this paper, we present a generic algebraic framework for conducting preimage attacks on hash functions and apply it to an important instance: Ascon-Xof. Our main contributions are as follows:

1. **Generic algebraic framework for preimage attacks on hash functions**

   We introduce a generic algebraic framework for preimage attacks on hash functions. The method guesses most of the input bits, leaving only a few as algebraically solvable variables. From the hash value, we derive polynomials, from which a solution can be obtained via Gaussian elimination. We rigorously analyze the probability that the matrix representing the relationship between the input variables and the output polynomials, derived from the hash value, is solvable, and link this to the expected guessing time for the attack. We further analyze the computational complexity of solving the mixed bit-variable state, which we refer to as the *constructing time*, and explicitly calculate the number of bit operations required for Gaussian elimination, thereby quantifying the *solving time*.

We focus in particular on Ascon, as it has been the subject of active research on algebraic preimage attacks. The proposed attacks rely on the assumption that the distribution of the polynomials derived from the hash value is uniform, making them inapplicable to overly reduced rounds. Thus, we experimentally verify that the output polynomial distribution of Ascon becomes uniform only after a certain number of rounds. This validation is performed by analyzing the variance and measuring the entropy of the polynomial distributions.

2. **Preimage attacks on reduced-round Ascon-Xof up to 8 rounds beyond the generic bound**

   Leveraging the proposed framework, we present, for the first time, a preimage attack faster than exhaustive search on up to 8 rounds of Ascon-Xof. To precisely evaluate the constructing time and solving time required for the attack, we compute the number of bit operations based on the number of variables involved in each primitive. This enables us to determine the maximum number of rounds that can be attacked beyond the generic bound. While our attack is evaluated in terms of bit operations with negligible memory usage, making it suitable for practical scenarios, the approach in [4] also uses bit-operation-based evaluation but relies on substantial memory, rendering it less practical.

Organization. Sect. 2 describes the specification of Ascon. Sect. 3 proposes a generic framework for preimage attacks on hash functions. Sect. 4 applies the proposed framework to Ascon-Xof. Sect. 5 provides a discussion on our generic attack presented in Sections 3 and 4. Sect. 6 concludes the paper.

2 Description of Ascon

We provide a brief overview of the Ascon specification as relevant to our analysis.

Ascon [12], designed by Dobraunig et al., includes permutation-based AEAD and hashing schemes. Both schemes are built on 320-bit permutations p^a and p^b with a and b rounds, respectively. In hashing mode, both a and b are set to 12, and the sponge construction [5] is employed (see Figure 1). The 320-bit internal state is divided into five 64-bit words x_i ($0 \leq i \leq 4$), where x_0 is the MSB and x_4 the LSB, as illustrated in Figure 2.

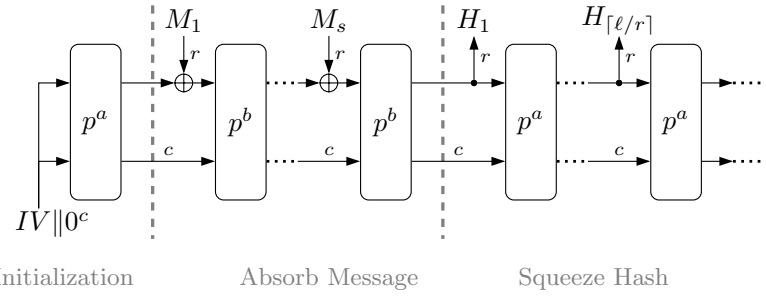

Figure 1: Hashing modes of Ascon [13].

The IV of Ascon is instantiated differently depending on the primitive in use. For Ascon-Xof, the IV is set to `0x00400C0000000000`, after which the initial state is computed as

Table 1: Comparison of preimage attacks on Ascon-Xof. GAF: Generic Algebraic Framework, MitM: Meet-in-the-middle attack, DL: Differential-linear attack, Lin.: Linear structure-based attack, MSR: Multi-step recovery technique-based attack.

Target	Rounds	Hash	Time	Memory (in bits)	Method	Reference
Ascon-Xof	2	64	2^{39}	-	Algebraic†	[11]
			2^{34}	-	Algebraic‡	[2]
			$2^{31.56}$	-	Lin.	[23]
		128	2^{103}	-	Algebraic†	[11]
			2^{98}	-	Algebraic‡	[2]
	3	64	$\mathbf{2^{62.79}}$	-	**GAF**	**Sect. 4**
			2^{56}	-	Algebraic†	[2]
		128	$\mathbf{2^{126.79}}$	-	**GAF**	**Sect. 4**
			$2^{120.58}$	2^{39}	MitM	[32]
			$2^{120.02}$	–	DL	[31]
			2^{120}	-	Algebraic†	[2]
			$2^{114.53}$	2^{30}	MitM	[33]
			$2^{112.21}$	-	Lin.	[23]
	4	64	$\mathbf{2^{63.02}}$	-	**GAF**	**Sect. 4**
			2^{63}	-	Algebraic†	[2]
		128	$\mathbf{2^{127.02}}$	-	**GAF**	**Sect. 4**
			2^{127}	-	Algebraic†	[2]
			$2^{125.47}$	-	DL	[31]
			$2^{124.67}$	2^{54}	MitM	[32]
			$2^{124.49}$	-	Lin.	[23]
	6	64	$2^{63.3}$	2^{67}	Algebraic	[4]
			$\mathbf{2^{63.83}}$	-	**GAF**	**Sect. 4**
	8*	64	$\mathbf{2^{63.98}}$	-	**GAF**	**Sect. 4**
		128	$\mathbf{2^{127.98}}$	-	**GAF**	**Sect. 4**

† A setting considering Ascon's permutation without round constants and initialization, with $IV = 0$.
‡ A setting considering Ascon's permutation without initialization, with $IV = 0$.
∗ With the proposed GAF, attacks with lower preimage complexity than generic bounds are possible for up to 8 rounds (out of 12) of Ascon-Xof.

$S = p^a\left(IV\|0^{256}\right)$. For the construction of IV and the other values used in different Ascon primitives, refer to [12].

Ascon's hashing mode processes the input message M in blocks of r bits. The padding process first appends a single 1 and then the smallest number of zeroes to M such that the length of the padded message is a multiple of r bits. The padded message is then divided into r-bit blocks $M_1, \ldots, M_s$, where $M\|1\|0^{r-1-(|M| \bmod r)}$ ensures the total length is a multiple of r.

The core permutation p is composed of three sequential operations: constant addition p_C, substitution p_S, and linear diffusion p_L ($p = p_L \circ p_S \circ p_C$). To denote a specific round, the round index is added as a subscript. For example, $p_{S,1}$ refers to the substitution layer in round

1. The input state at round R is represented as: $x^R = x_0^R \| x_1^R \| x_2^R \| x_3^R \| x_4^R$, and the output of p_S is denoted by $y^R = y_0^R \| y_1^R \| y_2^R \| y_3^R \| y_4^R$. We denote a single bit of each word by $[\cdot]$; for instance, $x_i^R[j]$ represents the j-th bit (from the left[1]) of word i in round R for $j = 0, \ldots, 63$. A range of bits is written as $x_i^R[j, \ldots, k]$ for $0 \leq j < k \leq 63$.

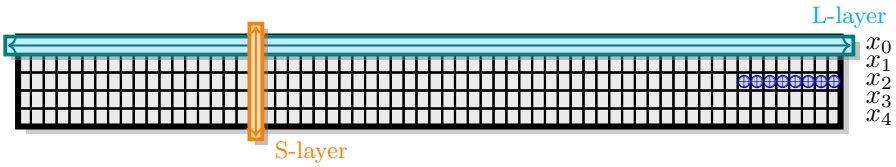

Figure 2: The column-wise S-box layer, the row-wise linear layer and the constant addition.

Addition of Constants (p_C). An 8-bit constant, which varies depending on the round, is added to the $x_2[56, \ldots, 63]$ at each round.

Substitution Layer (p_S). Each column of the 320-bit state is updated by applying Ascon's 5-bit S-box. The algebraic normal form (ANF) of the S-box is given in Figure 3. As the algebraic degree of Ascon's 5-bit S-box is 2, the overall algebraic degree of a single round of permutation p is also 2.

$$
\begin{aligned}
y_0 &= x_4x_1 \oplus x_3 \oplus x_2x_1 \oplus x_2 \oplus x_1x_0 \oplus x_1 \oplus x_0 \\
y_1 &= x_4 \oplus x_3x_2 \oplus x_3x_1 \oplus x_3 \oplus x_2x_1 \oplus x_2 \oplus x_1 \oplus x_0 \\
y_2 &= x_4x_3 \oplus x_4 \oplus x_2 \oplus x_1 \oplus 1 \\
y_3 &= x_4x_0 \oplus x_4 \oplus x_3x_0 \oplus x_3 \oplus x_2 \oplus x_1 \oplus x_0 \\
y_4 &= x_4x_1 \oplus x_4 \oplus x_3 \oplus x_1x_0 \oplus x_1
\end{aligned}
$$

Figure 3: ANF of Ascon's S-box.

Linear Diffusion Layer (p_L). Each row of the 320-bit state is internally diffused by a linear transformation $\sum_i$, where $\sum_i$ is given in Figure 4. Here, $\ggg$ is the right cyclic shift operation over a 64-bit word.

3 A Generic Algebraic Framework for Preimage Attacks on Hash Functions

We describe a new generic preimage attack framework based on the independence of polynomials derived from the hash value. We are particularly interested in applying this framework to Ascon, which has received significant attention in algebraic preimage attacks. Although the framework is generic and applicable to any hash function, we focus on sponge-based hash

[1]In the original Ascon paper [12], bit indices are counted from the right.

$$
\begin{aligned}
x_0 &= \Sigma_0(y_0) = y_0 \oplus (y_0 \ggg 19) \oplus (y_0 \ggg 28) \\
x_1 &= \Sigma_1(y_1) = y_1 \oplus (y_1 \ggg 61) \oplus (y_1 \ggg 39) \\
x_2 &= \Sigma_2(y_2) = y_2 \oplus (y_2 \ggg 1) \oplus (y_2 \ggg 6) \\
x_3 &= \Sigma_3(y_3) = y_3 \oplus (y_3 \ggg 10) \oplus (y_3 \ggg 17) \\
x_4 &= \Sigma_4(y_4) = y_4 \oplus (y_4 \ggg 7) \oplus (y_4 \ggg 41)
\end{aligned}
$$

Figure 4: ANF of Ascon's diffusion layer.

functions, as Ascon belongs to this class. The framework is generic in the following sense. It requires only (i) output-derived polynomial equations over $GF(2)$, (ii) a choice of symbolic input variables, and (iii) a solvability and consistency test for the resulting linearized monomial system. The probability analysis in Sect. 3.2 does not use any Ascon-specific structural property; Ascon-specific details enter only when generating the output polynomials and estimating the construction cost in Sect. 4. Thus, any hash or XOF primitive for which such equations can be generated can be analyzed using the same probability and complexity framework.

3.1 The Framework

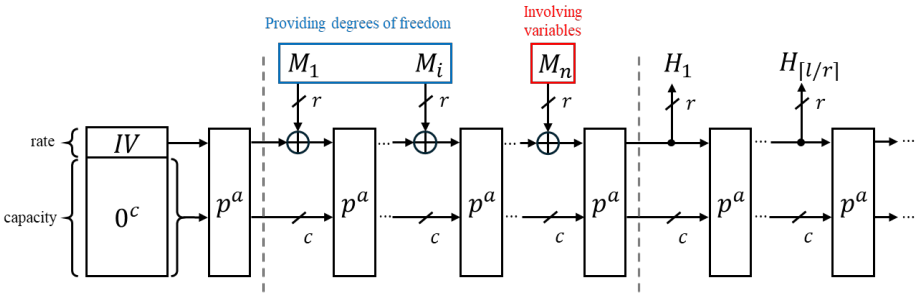

Figure 5: Generic preimage attack framework on sponge-based hash functions.

When the number of unknown variables is denoted by l, the total number of distinct monomials that can be constructed from these variables excluding the constant term is $2^l - 1$. The constant term, if present, can be isolated and handled separately. For example, if three bit variables a_1, a_2, a_3 are involved in positions within the last message block, the monomials that can be found in the polynomials derived from the hash value are as follows:

$$
a_1, a_2, a_3, \underbrace{a_1a_2}_{b_1}, \underbrace{a_2a_3}_{b_2}, \underbrace{a_3a_1}_{b_3}, \underbrace{a_1a_2a_3}_{b_4}
$$

Here, we can consider variables composed of two or more original variables as new variables. Consequently, the set of all terms we consider includes a_1, a_2, a_3 as well as b_1, b_2, b_3, b_4. This allows all derived polynomials to be treated as linear forms, and the number of possible polynomials is 2^{2^l-1}.

In the above example, if the underlying permutation is reduced-round, it is likely that the coefficient of each monomial is not uniformly distributed between 0 and 1. For instance, in the

case of 2-round Ascon, even with a_1, a_2, a_3 present in the last message block M_n, the coefficient of the monomial $a_1a_2a_3$ in the derived polynomial always becomes 0 owing to the properties of the underlying permutation. Therefore, we proceed with the attack under the assumption that, as the number of rounds increases, the coefficients of each monomial are expected to become more randomly distributed. Our generic preimage attack framework, presented in Figure 5, proceeds as follows, with the structure of Ascon serving as the basis for illustration.

1. We begin by considering a one-block message preimage attack on message M_1. In this scenario, we involve N variables in the last message block M_1, fix one of the 2^{r-N} possible assignments for the remaining $(r-N)$ bits, and apply the underlying permutation to match the output with the target hash value H_1. This process results in the generation of r linear equations. For example, refer to Tables 5 and 6 in Appendix A, which analyze Ascon's permutation with 12 rounds and $r = 64$.

2. The constants from the constructed linear equations are moved to the right-hand side, forming a matrix M of size $m \times (2^N - 1)$, where m represents the number of linear equations used in the analysis, and satisfies $m \leq r$.

3. Gaussian elimination is then applied to the augmented matrix $[A|\mathbb{b}]$ of size $m \times 2^N$ to determine whether the system is solvable. If no solution exists, we return to Step 1 and retry the process with a new candidate assigned for the $(r - N)$ bits.

4. If no valid preimage is found, we extend the approach to a two-block message (M_1, M_2), using M_1 to introduce additional degrees of freedom. We then repeat the process described in Steps 1 through 3.

5. When dealing with longer hash outputs, more message blocks, M_i, are appended before the final message block, allowing us to construct the preimage accordingly.

If the matrix in Step 3 is solvable, this indicates that all relevant monomials have been fully determined. Consequently, the final message block can be reconstructed, and the complete preimage message is successfully recovered.

3.2 Deeper Analysis for Attack Complexity

The success of the attack described above hinges on whether the matrix—representing the linear relationships between monomials and the polynomials derived from the hash value—is solvable. The attack's overall complexity is determined by the probability that such a matrix admits a solution. A solvable matrix indicates that all corresponding monomials have been determined, thus enabling the reconstruction of the last message block and the finalization of the preimage.

Let p be a prime number and k a positive integer, and define $q = p^k$. While the number of full-rank $n \times n$ matrices over a finite field is well known, counting matrices that are either non-square or not of full rank requires more detailed analysis. From the perspective of solving a system of equations, a solution may still exist even if the associated matrix is not of full rank.

To estimate the success probability of our attack, we only need the binary case. Let $N_{m,n}(k)$ be the number of $m \times n$ matrices over GF(2) with rank k. By the standard rank-counting formula,

$$N_{m,n}(k) = \prod_{i=0}^{k-1} \frac{(2^m - 2^i)(2^n - 2^i)}{(2^k - 2^i)}.$$

Hence, for uniformly random $A \in \mathrm{GF}(2)^{m\times n}$ and $b \in \mathrm{GF}(2)^m$, the probability that $Ax = b$ is solvable is

$$P_{m,n} = 2^{-mn-m} \sum_{k=0}^{v} N_{m,n}(k) 2^k, \qquad v = \min(m, n),$$

since a rank-k matrix has a k-dimensional column space and thus a random right-hand side belongs to it with probability 2^{k-m}.

We begin by introducing the following q-analog.

Definition 1. *Let q be a prime power. For a positive integer x, the **q-analog of** x is defined as*

$$[x]_q = 1 + q + q^2 + \cdots + q^{x-1} = \frac{q^x - 1}{q - 1}.$$

*The **q-factorial** of x is given by*

$$[x]_q! = [1]_q [2]_q \cdots [x]_q.$$

*The **q-binomial coefficient** (also known as the Gaussian binomial coefficient) is defined as*

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{[n]_q!}{[k]_q! [n-k]_q!}.$$

We now state the following theorem, which provides the number of $m \times n$ matrices of rank k over the finite field $GF(q)$:

Theorem 1 ([29])**.** *The number of $m \times n$ matrices of rank k, where $0 \le k \le \min\{m, n\}$, over the finite field $GF(q)$ is given by:*

$$\begin{bmatrix} m \\ k \end{bmatrix}_q \times \prod_{i=0}^{k-1} \left(q^n - q^i\right) = \begin{bmatrix} n \\ k \end{bmatrix}_q \times \prod_{i=0}^{k-1} \left(q^m - q^i\right).$$

We denote the number of $m \times n$ matrices of rank $k(\le \min\{m, n\})$ over $GF(q)$ as $C_{m,n}(k; q)$. Using the notation, we state the following Lemma:

Lemma 1. *$C_{m,n}(k; q)$ is given by*

$$C_{m,n}(k; q) = q^{\frac{k(k-1)}{2}} \cdot (q-1)^k \cdot \frac{[m]_q! \cdot [n]_q!}{[k]_q! \cdot [m-k]_q! \cdot [n-k]_q!}.$$

Proof.

$$\begin{aligned}
&C_{m,n}(k; q) \\
&= \begin{bmatrix} m \\ k \end{bmatrix}_q \cdot (q^n - 1)(q^n - q) \cdots \left(q^n - q^{k-1}\right) \\
&= \frac{(q^m - 1)(q^m - q) \cdots \left(q^m - q^{k-1}\right)(q^n - 1)(q^n - q) \cdots \left(q^n - q^{k-1}\right)}{(q^k - 1)(q^k - q) \cdots \left(q^k - q^{k-1}\right)} \\
&= q^{\frac{k(k-1)}{2}} \cdot (q-1)^k \\
&\cdot \frac{\prod_{i=1}^{k} \left(q^{m-i} + q^{m-i-1} + \cdots + 1\right) \cdot \prod_{i=1}^{k} \left(q^{n-i} + q^{n-i-1} + \cdots + 1\right)}{\prod_{i=1}^{k} \left(q^{k-i} + q^{k-i-1} + \cdots + 1\right)} \\
&= q^{\frac{k(k-1)}{2}} \cdot (q-1)^k \cdot \frac{[m]_q! \cdot [n]_q!}{[k]_q! \cdot [m-k]_q! \cdot [n-k]_q!}.
\end{aligned}$$

□

We immediately obtain the following corollary.

Corollary 1. *The number of full-rank $m \times n$ matrix over the finite field $GF(q)$ is given by*

$$\begin{aligned} C_{m,n}(v;q) &= (q^u - 1)(q^u - q)\cdots\left(q^u - q^{v-1}\right) \\ &= q^{\frac{v(v-1)}{2}} \cdot (q-1)^v \cdot \frac{[u]_q!}{[u-v]_q!}, \end{aligned}$$

where $u = max(m,n)$ and $v = min(m,n)$.

Next, we analyze the probability that the linear equation $A\mathbb{x} = \mathbb{b}$ is solvable.

Theorem 2. *Let A be an $m \times n$ matrix and $\mathbb{b}$ be an $m \times 1$ vector, both defined over a finite field $GF(q)$. Then, the probability that the equation $A\mathbb{x} = \mathbb{b}$ is solvable is given by:*

$$P^q_{m,n} = q^{-uv-m} \cdot \sum_{k=1}^{v} q^{\frac{k(k+1)}{2}} \cdot (q-1)^k \cdot \frac{[u]_q! \cdot [v]_q!}{[k]_q! \cdot [u-k]_q! \cdot [v-k]_q!},$$

where $u = max(m,n)$ and $v = min(m,n)$.

Proof. Let $u = max(m,n)$ and let $v = min(m,n)$. According to the Rouché–Capelli Theorem, if the coefficient matrix A and its augmented matrix $([A|\mathbb{b}])$ have the same rank, the linear system has solutions. To evaluate the probability that this condition holds, we define the following two events:

1. $S_1^k : rank(A) = k(\leq v)$,
2. $S_2 : rank(A) = rank\left([A|\mathbb{b}]\right)$.

Using the conditional probability formula, we have

$$\begin{aligned} P(S_1^k \cap S_2) &= P(S_2) \cdot P(S_1^k|S_2) \\ &= P(S_1^k) \cdot P(S_2|S_1^k). \end{aligned}$$

We compute $P(S_1^k)$ and $P(S_2|S_1^k)$ to simplify the calculation. Since $P(S_1^k)$ represents the proportion of matrices with rank k among all matrices,

$$P(S_1^k) = \frac{C_{m,n}(k;q)}{q^{uv}} \tag{1}$$

holds by Theorem 1. To compute $P(S_2|S_1^k)$, we consider the probability that the vector $\mathbb{b}$ lies in the column space of matrix A, assuming A has rank k. Since the column space of A has k basis vectors, we have

$$P(S_2|S_1^k) = \frac{q^k}{q^m}. \tag{2}$$

By combining (1) and (2) and summing over all possible cases, we have

$$\begin{aligned}\sum_{k=1}^{v} P(S_1^k \cap S_2) &= \sum_{k=1}^{v} P(S_1^k) \times P(S_2|S_1^k) \\ &= \sum_{k=1}^{v} \frac{C_{m,n}(k;q)}{q^{uv}} \times \frac{q^k}{q^m} \\ &= \sum_{k=1}^{v} \Bigg\{ q^{\frac{k(k-1)}{2}} \cdot (q-1)^k \\ &\quad \cdot \frac{[u]_q! \cdot [v]_q!}{[k]_q! \cdot [u-k]_q! \cdot [v-k]_q!} \cdot q^{k-uv-m} \Bigg\} \\ &= q^{-uv-m} \cdot \sum_{k=1}^{v} \Bigg\{ q^{\frac{k(k+1)}{2}} \cdot (q-1)^k \\ &\quad \cdot \frac{[u]_q! \cdot [v]_q!}{[k]_q! \cdot [u-k]_q! \cdot [v-k]_q!} \Bigg\}.\end{aligned}$$

This concludes the proof. □

Since the argument depends on computing the probability that $\mathbb{b}$ lies in the column space of A, defined over $\mathbb{F}_q^m$, it assumes an underdetermined system where $m \leq n$. In our case, the probability that the equation $A\mathbb{x} = \mathbb{b}$ is solvable over $GF(2)$ is established by the following corollary.

Corollary 2. *Let A be an $m \times n$ matrix and $\mathbb{b}$ be an $m \times 1$ vector, both over $GF(2)$. Then, the probability that the equation $A\mathbb{x} = \mathbb{b}$ is solvable is given by*

$$P_{m,n}^2 = 2^{-uv-m} \cdot \sum_{k=1}^{v} 2^{\frac{k(k+1)}{2}} \cdot \frac{[u]_2! \cdot [v]_2!}{[k]_2! \cdot [u-k]_2! \cdot [v-k]_2!}.$$

As an example, using Corollary 2, consider a system of 64 randomly selected linear equations, each involving three variables and $2^3 - 1$ monomials, with $\mathbb{b}$ as an $m \times 1$ vector. For $m = 64$ and $n = 7$, the solvability probability is computed as follows:

$$\begin{aligned}P_{64,7}^2 &= 2^{-64 \cdot 8} \cdot \sum_{k=1}^{7} 2^{\frac{k(k+1)}{2}} \cdot \frac{[64]_2! \cdot [7]_2!}{[k]_2! \cdot [64-k]_2! \cdot [7-k]_2!} \\ &\approx 2^{-57}.\end{aligned} \tag{3}$$

Table 2 presents the calculated probabilities corresponding to Equation (3), where the probability from Corollary 2 is incorporated into the value for the full-rank case of the 64×7 matrix M.

Correcting the Probability of Contradiction. The previously calculated solvability probability of the matrix system is incomplete. Consider a scenario where an attacker solves the equation $A\mathbb{x} = \mathbb{b}$ using a specific guess and successfully derives all monomials. Despite this, the

Table 2: Probabilities of a 64×7 matrix achieving rank k ($1 \leq k \leq 7$).

k	$P(\text{rank}(M) = k)$	$\log_2(P(\text{rank}(M) = k))$
1	$3.494585 \times 10^{-133}$	-440.0113
2	$2.707476 \times 10^{-112}$	-370.6190
3	4.423622×10^{-92}	-303.4722
4	1.632029×10^{-72}	-238.4722
5	1.359608×10^{-53}	-175.6190
6	2.388605×10^{-35}	-115.0113
7	6.938894×10^{-18}	-57.0

solution may still be invalid due to inherent contradictions. For instance, let $l = 3$ and consider the following assignment:

$$x_1 = 1, x_2 = 0, x_3 = 1,$$
$$x_1x_2 = 0, x_2x_3 = 1, x_3x_1 = 1,$$
$$x_1x_2x_3 = 1.$$

This solution is contradictory. Given that $x_2 = 0$ and $x_3x_1 = 1$, it logically follows that $x_1x_2x_3 = 0$; however, it is recorded as 1, indicating an inconsistency. Such contradictions arise when higher-degree terms (e.g., $x_1x_2, x_2x_3, x_3x_1, x_1x_2x_3$) are treated as independent variables while solving the system. Therefore, the solvability probability must be adjusted to account for these inconsistencies. To formalize this, consider the following theorem.

Theorem 3. *Let N be the number of binary variables $a_1, a_2, \ldots, a_N$. Assuming all monomials constructed from these variables are independently assigned random values, the probability that they are consistent (i.e., contain no contradictions) is given by*

$$P_N = \frac{2^N}{2^{2^N-1}} = 2^{N-2^N+1}.$$

Proof. As we consider N variables, the total number of non-constant monomials (distinct monomials of these variables) is $2^N - 1$. Thus, there are 2^{2^N-1} possible truth configurations across all monomials.

To ensure that all monomials are consistent, it suffices to consider only the N variables $a_1, a_2, \ldots, a_N$, which leads to 2^N possible cases. This completes the proof. □

By considering Corollary 2 and Theorem 3, we derived the following corollary.

Corollary 3. *Let N be the number of binary variables $a_1, a_2, \ldots, a_N$, and let A be a $(m \times 2^N - 1)$ matrix over $GF(2)$, where each row is randomly generated and corresponds to the coefficients of all possible monomials. For a vector $\mathbb{b}$ of size $m \times 1$ over $GF(2)$, the probability that the system $A\mathbb{x} = \mathbb{b}$ is both solvable and free of contradictions among the monomials is:*

$$P = P^2_{m,2^N-1} \cdot P_N = 2^{-uv-m} \cdot \sum_{k=1}^{v} \Bigg\{ 2^{\frac{k(k+1)}{2}} \cdot \frac{[u]_2! \cdot [v]_2!}{[k]_2! \cdot [u-k]_2! \cdot [v-k]_2!} \cdot 2^{N-2^N+1} \Bigg\},$$

where $u = max(m, 2^N - 1)$ and $v = min(m, 2^N - 1)$.

As an example, when $m = 64$ and N is adjusted from 1 to 6, applying Corollary 3 results in Table 3.

Table 3: Probabilities for solvability of $m \times 2^N - 1$ matrix over $GF(2)$ for varying N.

m	N	$\log_2(P_{64,2^N-1})$	$\log_2(P_N)$	$\log_2(P_{64,2^N-1} \times P_N)$
64	1	-63.0	0.0	-63.0
64	2	-61.0	-1.0	-62.0
64	3	-57.0	-4.0	-61.0
64	4	-49.0	-11.0	-60.0
64	5	-33	-26.0	-59
64	6	-1.359	-57.0	-58.359

Finally, this attack relies on the expectation that the distribution of the polynomials derived from the hash value is uniform, making it inapplicable to overly reduced rounds. We demonstrate this using Ascon's permutation. Here, we specifically present the case for Ascon with $N = 3$. In the last message block, we involve three variables in input bit positions 60–62 and analyze the distribution of the output polynomials. For each round, from 2 to the full 12 rounds, at least 200,000,000 ($\approx 2^{27.57}$) random values are substituted into bit positions 0–59. The distribution of the output polynomials is then evaluated according to the variance (in log scale) and Shannon entropy. Figure 6 reveals that for five or more rounds, the polynomials are uniformly distributed.

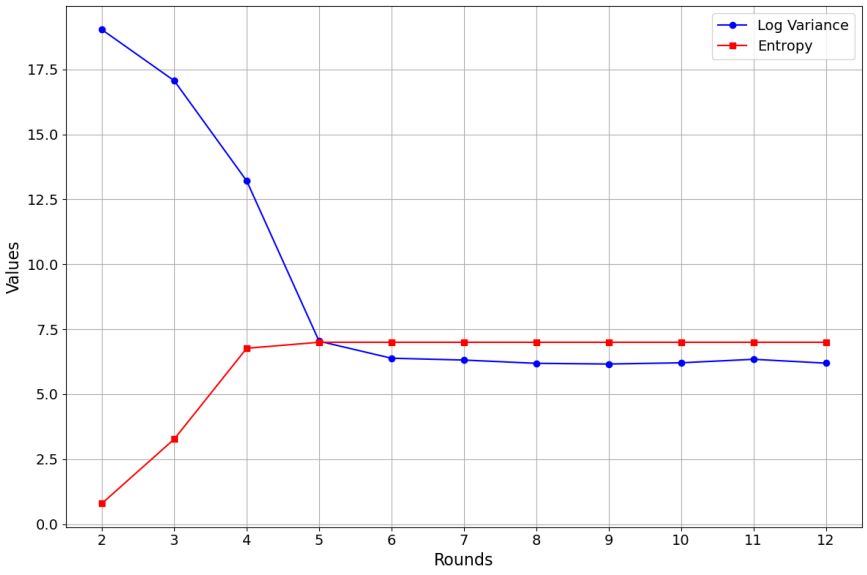

Figure 6: Evaluation of uniformity in output polynomials for Ascon's permutation.

4 Applying the Framework to Reduced-Round Ascon-Xof

Since our attack methodology relies on solving a large system of linear equations, the associated computational cost is considerable and cannot be overlooked. However, evaluating complexity solely in terms of encryption operations does not provide a fair comparison with exhaustive search methods due to the fundamental differences in operations involved. To address this,

we assess the algorithm's complexity by counting fundamental bitwise operations—such as XOR, OR, and AND—performed on pairs of bits. Our goal is to quantify precisely the number of bit operations required to solve the linear expressions generated after a given number of permutation rounds, denotes as R.

When N variables are involved in the last message block, we construct an $m \times 2^N$ matrix that encapsulates the $(2^N - 1)$ monomial coefficients along with the constant term vector $\mathbb{b}$ derived from H_1. To determine whether the system is solvable, we verify if rank(A)=rank($[A|\mathbb{b}]$), which requires computing the row echelon form of the matrix. To simplify notation, we denote the matrix as an $m \times n$ binary matrix, and count the bit operations required for Gaussian elimination. Let $M \stackrel{def}{=} [A|\mathbb{b}]$, where $M[i][j]$ denotes the element in the ith row and jth column, and $M[i][*]$ denotes ith row. In this context, Algorithm 1 details the exact number of bit operations needed to perform Gaussian elimination on a binary matrix.

Algorithm 1 Gaussian elimination algorithm for binary matrices

Input: $(m \times n)$ matrix M $(m \geq n)$
Output: RREF form of matrix M
  for $i = 0, 1, \cdots, m-2$ **do**
    for $j = i+1, i+2, \cdots, m-1$ **do**
      Swap$(M[i][*], M[j \cdot M[j][i] + i \cdot (1 - M[j][i])][*])$
    for $j = i+1, i+2, \cdots, m-1$ **do**
      istrue $= M[i][i] \cdot M[j][i]$
      for $k = i, i+1, \cdots, n-1$ **do**
        $M[j][k] = M[j][k] \oplus (M[i][k] \cdot \mathsf{istrue})$
  return M

Based on Algorithm 1, we calculated the number of bit operations required to perform Gaussian elimination on an $m \times n$ matrix, denoted by T. This complexity can be expressed as:

$$\begin{aligned} T = &\{(2n+1)(m-1) + (2(n-1)-1)(m-2) \\ &+ (2(n-2)-1)(m-3) + \cdots \\ &+ (2 \cdot 2 + 1) \cdot (m-(n-1))\} \\ &+ \lceil \log_2 n \rceil \cdot \{(m-1) + (m-2) + \cdots + (m-n+1)\} \\ = &\sum_{k=1}^{n-1} (2k+3)(m-n+k) + \lceil \log_2 n \rceil \frac{(n-1)(2m-n-1)}{2} \\ = &\, 2 \cdot \frac{n(n-1)(2n-1)}{6} + (2m-2n+3)\frac{n(n-1)}{2} + \\ &+ 3(m-n)(n-1) + \lceil \log_2 n \rceil \frac{(n-1)(2m-n-1)}{2} \\ = &\, \frac{n(n-1)(6m-2n+7)}{6} + 3(m-n)(n-1) \\ &+ \lceil \log_2 n \rceil \frac{(n-1)(2m-n-1)}{2}, \end{aligned} \tag{4}$$

where $m \geq n$.

This includes the row swapping process to create leading ones and all necessary XOR and AND bit operations. In our applications, we present the ratio of the time required to solve the equation system to the time required to perform the underlying permutation using Equation (4).

4.1 Bitwise Operation Complexity in Mixed Bit-Variable States Used in Ascon

We begin by defining a mixed bit-variable state, denoted by Ascon_{poly}, used for computing output polynomials in the context of Ascon. We then analyze the bit-level complexity of the corresponding bitwise operations under this state. The polynomials defined over N-bit variables are presented as:

$$A = \sum_{I \subset \{1,\cdots,N\}} A_I a_I, \qquad B = \sum_{I \subset \{1,\cdots,N\}} B_I a_I,$$

where A and B are the inputs to the operation. Then, the XOR operation between two polynomials is performed as follows:

$$\begin{aligned} A \oplus B &= \sum_{I \subset \{1,\cdots,N\}} A_I a_I \oplus \sum_{I \subset \{1,\cdots,N\}} B_I a_I \\ &= \sum_{I \subset \{1,\cdots,N\}} (A_I \oplus B_I) a_I \end{aligned}$$

Since each coefficient bit A_I and B_I can be computed independently, the XOR operation across the entire polynomial requires at most 2^N XOR operations.

In contrast, the AND operation between two polynomials involves a more intricate process due to the interaction between monomial terms.

$$\begin{aligned} A \cdot B &= \left(\sum_{I \subset \{1,\cdots,N\}} A_I a_I \right) \cdot \left(\sum_{I \subset \{1,\cdots,N\}} B_I a_I \right) \\ &= \sum_{I \subset \{1,\cdots,N\}} \left(\sum_{I_0 \cup I_1 = I} A_{I_0} \cdot B_{I_1} \right) a_I. \end{aligned} \tag{5}$$

According to Equation (5), using a naive approach to obtain the coefficient of each a_I indexed by I requires the following number of bitwise operations:

$$\begin{aligned} &\#(\text{Bit Operation}) \\ &= \underbrace{\binom{N}{0} + \binom{N}{1} \cdot 3^1 \cdots + \binom{N}{N} \cdot 3^N}_{\text{AND operations}} \\ &+ \underbrace{\binom{N}{1} \cdot (3^1 - 1) + \binom{N}{2} \cdot (3^2 - 1) + \cdots + \binom{N}{N} \cdot (3^N - 1)}_{\text{XOR operations}} \\ &= \binom{N}{0} + \binom{N}{1} \cdot (2 \cdot 3^1 - 1) + \cdots + \binom{N}{N} \cdot (2 \cdot 3^N - 1). \end{aligned}$$

However, by employing a technique that reuses intermediate results, the total number of required

bitwise operations can be reduced as follows:

$$\begin{aligned}\#(\text{Reduced Bit Operation}) &= \underbrace{2^N \cdot 1}_{\text{AND operations}} + \underbrace{3 \cdot \sum_{k=1}^{N} 2^{k-1}}_{\text{XOR operations}} \\ &= 2^{N+2} - 3.\end{aligned}$$

Considering that Ascon has a predetermined initial state, many of its operations can be efficiently handled using single bitwise operations instead of full polynomial computations. This enables a more accurate estimation of the required bitwise operations. By identifying optimal bit positions for variable involvement based on these counts, the attack process can be further optimized.

4.2 Attacks on Ascon

A system of 64 linear equations can be derived from the Ascon permutation based on H_1. For 64-bit hashing, if the resulting system of equations is solvable, a solution can be efficiently obtained via Gaussian elimination. However, for 128-bit hashing, where the hash value is $H_1||H_2$, even if an input message satisfying H_1 is found, it may not match with H_2. Therefore, the probability of randomly obtaining a correct H_2 is approximately 2^{-64}. When the number of degrees of freedom is insufficient, additional messages M_i can be introduced to compensate.

We define constructing time as the time required to compute the derived polynomials using the mixed bit-variable state and guessed values through the Ascon permutation. It is measured in terms of bit operations and denoted by $T_C^{\mathcal{B}}$, which is expressed by dividing the total number of bit operations required for construction by the number of bit operations of the target round of Ascon. As the number of variables involved, N, increases, $T_C^{\mathcal{B}}$ grows due to the rising cost of evaluating more complex polynomials. The matrix $[A|\mathbb{b}]$, which has dimensions 64×2^N, directly influences the computational cost of the Gaussian elimination process. Equation (4) is used to calculate the solving time for each guess, where the number of bit operations required for Gaussian elimination is divided by the total bit operations for the target round of Ascon, and the result is denoted by $T_S^{\mathcal{B}}$. For a single round of Ascon, the total number of bit operations is 2056, which includes $64 \times 22 = 1408$ operations for the substitution layer, 2×320 operations for the linear diffusion layer, and 8 operations for constant addition. In the final round, only the top row needs to be computed, reducing the required number of bit operations to 208. Using these values, we calculate the constructing time, guessing time, and overall attack complexity required to recover the preimage of a 64-bit hash. These are detailed in Table 4 and illustrated in Figure 7. Notably, attack complexity is generally higher at lower rounds, as it is larger due to the higher relative solving cost compared to the total bit operations for a given round of Ascon-Xof. For 128-bit hashing with Ascon-Xof, the total complexity is scaled by 2^{64} to account for the double output length. Appendix B presents the optimal bit indices for variable selection that yield the lowest constructing time, leveraging the known fixed initial state of Ascon.

5 Discussion

This section discusses the *constructing time*, which refers to the cost of generating equations from the mixed bit-variable state. In the initial version of linearization-based preimage attacks [18], the emphasis was primarily on the number of guesses. The total complexity was

Table 4: Comparison of attack complexity for Ascon-Xof based on rounds and N.

m	R	N	$\#G$	$T_C^{\mathcal{B}} + T_S^{\mathcal{B}}$	$T^{\mathcal{B}}$
64	3	1	2^{63}	$2^{-0.02} + 2^{-2.93}$	$2^{63.16}$
		2	2^{62}	$2^{0.13} + 2^{-0.63}$	$2^{62.80}$
		3	2^{61}	$2^{0.27} + 2^{1.16}$	$2^{62.79}$
		4	2^{60}	$2^{0.42} + 2^{2.79}$	$2^{63.04}$
	4	1	2^{63}	$2^{0.34} + 2^{-3.50}$	$2^{63.44}$
		2	2^{62}	$2^{0.82} + 2^{-1.20}$	$2^{63.14}$
		3	2^{61}	$2^{1.35} + 2^{0.60}$	$2^{63.02}$
		4	2^{60}	$2^{1.90} + 2^{2.23}$	$2^{63.07}$
	5	1	2^{63}	$2^{0.63} + 2^{-3.90}$	$2^{63.69}$
		2	2^{62}	$2^{1.41} + 2^{-1.60}$	$2^{63.58}$
		3	2^{61}	$2^{2.27} + 2^{0.20}$	$2^{63.58}$
		4	2^{60}	$2^{3.15} + 2^{1.82}$	$2^{63.64}$
	6	1	2^{63}	$2^{0.79} + 2^{-4.21}$	$2^{63.83}$
		2	2^{62}	$2^{1.72} + 2^{-1.91}$	$2^{63.83}$
		3	2^{61}	$2^{2.71} + 2^{-0.12}$	$2^{63.90}$
	7	1	2^{63}	$2^{0.88} + 2^{-4.47}$	$2^{63.92}$
		2	2^{62}	$2^{1.89} + 2^{-2.17}$	$2^{63.97}$
	8	1	2^{63}	$2^{0.95} + 2^{-4.69}$	$2^{63.98}$

$\#G$: Number of guesses
$T_C^{\mathcal{B}}$: Constructing time in terms of bit operations
$T_S^{\mathcal{B}}$: Solving time in terms of bit operations
$T^{\mathcal{B}}$: Attack complexity in terms of bit operations ($\#G \cdot (T_C^{\mathcal{B}} + T_S^{\mathcal{B}})$)

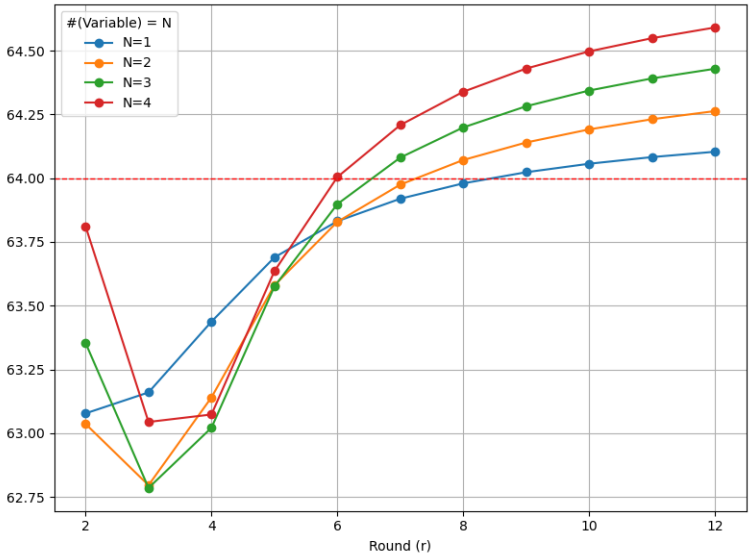

Figure 7: Preimage attack complexity of Ascon-Xof (64-bit output) across rounds and N.

estimated by subtracting the gain from degrees of freedom and the number of linear equations from the complexity of the generic preimage attack. Later, Liu et al. [26] emphasized that the cost of solving the linear equations must be included, accounting for the time required to solve each guess in the overall complexity. This adjustment has since been consistently adopted in related research. However, to the best of our knowledge, the time required to construct the polynomials derived from the hash value has not been adequately addressed in the literature. This omission likely stems from the assumption that constructing the equation system through the underlying permutation is a simple, automated process that does not considerably impact the attack complexity. Yet, in a mixed bit-variable state, equation construction cannot be efficiently modeled using a 1-bit datapath per variable. As a result, the number of bit operations required for this process increases substantially. Ignoring this cost may lead to an underestimation of the overall attack complexity and hinder the provision of proper feedback for secure cipher design. Therefore, a more conservative and accurate complexity evaluation must incorporate this factor. In our study, we include constructing time by comparing the number of bit operations required for Ascon_{poly} to those required for standard operations in Ascon. Although the exact metric may vary depending on the attack environment, this work represents the first to account for this construction overhead explicitly. This analysis may serve as a starting point for further investigation into the equation construction phase in linearization-based preimage attacks.

We adopt Bernstein's bit-operation-based evaluation methodology [4], applied initially to Keccak [6], which defines attack complexity as the ratio of the total number of bit operations required by the attack to those needed for a single execution of the target primitive. While both approaches share this same evaluation goal, they differ technically: our method encodes polynomial coefficients into a matrix and directly counts the required bit operations to solve it, whereas Bernstein's method estimates operations based on a polynomial enumeration algorithm used to search for preimages. A notable challenge in Bernstein's approach is its high memory and communication overhead, which can limit its practical applicability. In contrast, our method involves negligible memory consumption, representing a modest yet meaningful advancement in bit-operation based complexity analysis.

The advantage over the generic 2^{64} bound is numerically small and should not be interpreted as a practical break of full-round Ascon-Xof. Its significance is theoretical and methodological: it is the first reduced-round Ascon-Xof preimage result crossing the generic bound for 8 rounds under negligible-memory assumptions, and it shows where this class of algebraic linearization attacks reaches its limit once equation-construction overhead is counted. Thus, the result is best understood as a security-margin assessment rather than a deployment-level attack.

6 Conclusion and Future Work

In this paper, we presented a generic algebraic framework for preimage attacks based on the independence assumption on polynomials derived from the hash value. We applied this framework to Ascon-Xof and obtained, for the first time, preimage attacks on its 8-round reduced version with complexity slightly below exhaustive search and negligible memory, offering an efficiency advantage over prior memory-intensive approaches. Beyond these specific reduced-round results, the proposed framework provides a practical way to assess the preimage-resistance margin of standardized lightweight hash and XOF primitives in deployment-oriented settings. Extending this framework to other NIST-standardized primitives, such as Keccak-based constructions, would also be an important direction for future work.

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Appendix

A Output Polynomials of Full Ascon's Permutation

We present example cases in which the guess for the hash value H_1 : `329ecef2bcd7bb12` was successful and unsuccessful during the preimage search process. The process begins by setting the state after absorbing M to `4153434f4e5f5f21`, which implies that, considering the first row value of the S state is `b57e273b814cd416`, M is `f42d6474cf138b37`. Next, the value obtained after applying the inverse linear layer to H_1 is `0xd23883788f309d7c`, and we observe the corresponding polynomials at that position. Assuming $N = 3$, we introduce variables at indices 35, 41, and 49, which were experimentally identified as the positions requiring the fewest bit operations to compute ASCON_{poly}. Tables 5 and 6 present cases where the guess was successful and unsuccessful, respectively. Specifically, Table 6 indicates that the guess can be identified as a failure due to inconsistency in the equations at l_{49}, which is inherently contradictory; additionally, l_{10} and l_{48} conflict with each other.

B Bit Index Selection for Minimizing Constructing Time in Ascon-Xof

We experimentally identify the bit positions at which N variables should be introduced to minimize the construction time, given the fixed initial state of Ascon-Xof. The experiments are conducted by searching for the combination of bit positions that yields the lowest constructing time for each pair of round R and number of variables N. The results are presented in Table 7, where we define $I^{(R,N)}$ as the set of bit index tuples that achieve the lowest construction time for a given round R and number of variables N. If multiple tuples yield the same minimum constructing time, all such tuples are included in $I^{(R,N)}$.

Table 5: Derived polynomials when the guess succeeds for full Ascon's permutation (with $x_{35} = 0, x_{41} = 1, x_{49} = 1$, matching with 'the state after absorbing M').

The state after absorbing M : 01000001 01010011 01000011 01001111 01x_{35}01110 0x_{41}011111 0x_{49}011111 00100001
$l_0 : 1 = x_{41} \oplus x_{49} \oplus x_{41}x_{49} \oplus x_{49}x_{35} \oplus x_{41}x_{49}x_{35}$
$l_1 : 1 = 1 \oplus x_{41} \oplus x_{41}x_{35} \oplus x_{41}x_{49} \oplus x_{41}x_{49}x_{35}$
$l_2 : 0 = x_{35} \oplus x_{41} \oplus x_{41}x_{49}$
$l_3 : 1 = x_{35} \oplus x_{41} \oplus x_{41}x_{35} \oplus x_{41}x_{49}x_{35}$
$l_4 : 0 = x_{35} \oplus x_{41} \oplus x_{41}x_{49} \oplus x_{49}x_{35} \oplus x_{41}x_{49}x_{35}$
$l_5 : 0 = 1 \oplus x_{41}x_{49}$
$l_6 : 1 = x_{35} \oplus x_{41}x_{49} \oplus x_{49}x_{35} \oplus x_{41}x_{49}x_{35}$
$l_7 : 0 = 1 \oplus x_{35} \oplus x_{41}x_{49} \oplus x_{49}x_{35}$
$l_8 : 0 = 1 \oplus x_{35} \oplus x_{41} \oplus x_{49} \oplus x_{41}x_{35} \oplus x_{41}x_{49}$
$l_9 : 0 = x_{35} \oplus x_{41} \oplus x_{41}x_{35} \oplus x_{41}x_{49} \oplus x_{49}x_{35} \oplus x_{41}x_{49}x_{35}$
$l_{10} : 1 = 1 \oplus x_{35} \oplus x_{41} \oplus x_{41}x_{35} \oplus x_{41}x_{49} \oplus x_{49}x_{35}$
$l_{11} : 1 = x_{35} \oplus x_{41} \oplus x_{49}x_{35}$
$l_{12} : 1 = 1 \oplus x_{35} \oplus x_{41}x_{35} \oplus x_{41}x_{49}x_{35}$
$l_{13} : 0 = 1 \oplus x_{35} \oplus x_{41}x_{49} \oplus x_{49}x_{35} \oplus x_{41}x_{49}x_{35}$
$l_{14} : 0 = 1 \oplus x_{41}x_{35} \oplus x_{41}x_{49} \oplus x_{41}x_{49}x_{35}$
$l_{15} : 0 = x_{35} \oplus x_{49} \oplus x_{41}x_{35} \oplus x_{41}x_{49} \oplus x_{49}x_{35} \oplus x_{41}x_{49}x_{35}$
$l_{16} : 1 = 1 \oplus x_{49} \oplus x_{41}x_{49} \oplus x_{49}x_{35}$
$l_{17} : 0 = x_{35} \oplus x_{41} \oplus x_{41}x_{35} \oplus x_{41}x_{49} \oplus x_{49}x_{35} \oplus x_{41}x_{49}x_{35}$
$l_{18} : 0 = x_{35} \oplus x_{49} \oplus x_{41}x_{49} \oplus x_{41}x_{49}x_{35}$
$l_{19} : 0 = 1 \oplus x_{35} \oplus x_{41} \oplus x_{41}x_{35}$
$l_{20} : 0 = 1 \oplus x_{35} \oplus x_{41}x_{35} \oplus x_{41}x_{49} \oplus x_{41}x_{49}x_{35}$
$l_{21} : 0 = x_{35} \oplus x_{41}x_{49}x_{35}$
$l_{22} : 1 = 1 \oplus x_{35} \oplus x_{49} \oplus x_{41}x_{49}$
$l_{23} : 1 = 1 \oplus x_{49} \oplus x_{41}x_{49} \oplus x_{41}x_{49}x_{35}$
$l_{24} : 0 = 1 \oplus x_{41} \oplus x_{49} \oplus x_{41}x_{49} \oplus x_{49}x_{35}$
$l_{25} : 1 = x_{41} \oplus x_{49} \oplus x_{41}x_{49} \oplus x_{41}x_{49}x_{35}$
$l_{26} : 1 = x_{41} \oplus x_{49} \oplus x_{41}x_{35} \oplus x_{41}x_{49} \oplus x_{49}x_{35} \oplus x_{41}x_{49}x_{35}$
$l_{27} : 1 = x_{35} \oplus x_{41}x_{49}$
$l_{28} : 1 = 1 \oplus x_{41}x_{35} \oplus x_{41}x_{49}x_{35}$
$l_{29} : 0 = x_{49} \oplus x_{41}x_{35} \oplus x_{41}x_{49} \oplus x_{41}x_{49}x_{35}$
$l_{30} : 0 = x_{41} \oplus x_{49} \oplus x_{41}x_{49}x_{35}$
$l_{31} : 0 = 1 \oplus x_{41}x_{49} \oplus x_{41}x_{49}x_{35}$
$l_{32} : 1 = 1 \oplus x_{35} \oplus x_{41} \oplus x_{41}x_{49} \oplus x_{49}x_{35}$
$l_{33} : 0 = x_{41} \oplus x_{49} \oplus x_{49}x_{35}$
$l_{34} : 0 = 1 \oplus x_{41} \oplus x_{49} \oplus x_{41}x_{49}$
$l_{35} : 0 = 1 \oplus x_{35} \oplus x_{41} \oplus x_{49} \oplus x_{41}x_{35} \oplus x_{41}x_{49} \oplus x_{41}x_{49}x_{35}$
$l_{36} : 1 = 1 \oplus x_{35} \oplus x_{49} \oplus x_{41}x_{49}$
$l_{37} : 1 = x_{49} \oplus x_{41}x_{49}x_{35}$
$l_{38} : 1 = 1 \oplus x_{49}x_{35} \oplus x_{41}x_{49}x_{35}$
$l_{39} : 1 = 1 \oplus x_{35} \oplus x_{49} \oplus x_{41}x_{49}$
$l_{40} : 0 = 1 \oplus x_{41} \oplus x_{41}x_{35} \oplus x_{49}x_{35} \oplus x_{41}x_{49}x_{35}$
$l_{41} : 0 = x_{35} \oplus x_{41} \oplus x_{49} \oplus x_{41}x_{35} \oplus x_{41}x_{49}x_{35}$
$l_{42} : 1 = x_{41}x_{35} \oplus x_{41}x_{49} \oplus x_{49}x_{35} \oplus x_{41}x_{49}x_{35}$
$l_{43} : 1 = x_{49} \oplus x_{41}x_{35} \oplus x_{49}x_{35}$
$l_{44} : 0 = 1 \oplus x_{35} \oplus x_{49} \oplus x_{41}x_{35} \oplus x_{49}x_{35}$
$l_{45} : 0 = 1 \oplus x_{35} \oplus x_{49} \oplus x_{49}x_{35}$
$l_{46} : 0 = 1 \oplus x_{41} \oplus x_{41}x_{49}x_{35}$
$l_{47} : 0 = 1 \oplus x_{35} \oplus x_{41}$
$l_{48} : 1 = x_{35} \oplus x_{41}x_{49} \oplus x_{41}x_{49}x_{35}$
$l_{49} : 0 = 1 \oplus x_{41} \oplus x_{41}x_{35} \oplus x_{49}x_{35}$
$l_{50} : 0 = 1 \oplus x_{41} \oplus x_{49} \oplus x_{41}x_{49}$
$l_{51} : 1 = 1 \oplus x_{35} \oplus x_{49} \oplus x_{41}x_{49}$
$l_{52} : 1 = x_{41} \oplus x_{49} \oplus x_{41}x_{35} \oplus x_{41}x_{49} \oplus x_{49}x_{35}$
$l_{53} : 1 = 1 \oplus x_{41} \oplus x_{49} \oplus x_{41}x_{49}x_{35}$
$l_{54} : 0 = x_{35} \oplus x_{41} \oplus x_{49} \oplus x_{41}x_{35} \oplus x_{41}x_{49}x_{35}$
$l_{55} : 1 = x_{41}x_{49} \oplus x_{41}x_{49}x_{35}$
$l_{56} : 0 = 1 \oplus x_{35} \oplus x_{41} \oplus x_{49} \oplus x_{41}x_{49}$
$l_{57} : 1 = 1 \oplus x_{35} \oplus x_{41} \oplus x_{49} \oplus x_{41}x_{49}x_{35}$
$l_{58} : 1 = 1 \oplus x_{35} \oplus x_{49}x_{35} \oplus x_{41}x_{49}x_{35}$
$l_{59} : 1 = 1 \oplus x_{41} \oplus x_{49} \oplus x_{41}x_{49}x_{35}$
$l_{60} : 1 = x_{41}x_{35} \oplus x_{41}x_{49}$
$l_{61} : 1 = x_{41} \oplus x_{41}x_{35}$
$l_{62} : 0 = x_{35} \oplus x_{49}x_{35}$
$l_{63} : 0 = 1 \oplus x_{35} \oplus x_{41} \oplus x_{41}x_{35}$

Table 6: Derived polynomials when the guess fails for full Ascon's permutation.

The state after absorbing M : 1$1000001$ 01010011 01000011 01001111 $01x_{35}01110$ $0x_{41}011111$ $0x_{49}011111$ 00100001

$l_0 : 1 = x_{35} \oplus x_{49} \oplus x_{41}x_{35} \oplus x_{41}x_{49}$

$l_1 : 1 = x_{35} \oplus x_{41}x_{35} \oplus x_{49}x_{35} \oplus x_{41}x_{49}x_{35}$

$l_2 : 0 = 1 \oplus x_{41} \oplus x_{41}x_{49}x_{35}$

$l_3 : 1 = 1 \oplus x_{35} \oplus x_{41} \oplus x_{49} \oplus x_{41}x_{49} \oplus x_{49}x_{35} \oplus x_{41}x_{49}x_{35}$

$l_4 : 0 = 1 \oplus x_{41} \oplus x_{49} \oplus x_{41}x_{35} \oplus x_{41}x_{49}x_{35}$

$l_5 : 0 = 1 \oplus x_{41} \oplus x_{49} \oplus x_{41}x_{35} \oplus x_{41}x_{49} \oplus x_{49}x_{35}$

$l_6 : 1 = x_{35} \oplus x_{41} \oplus x_{49} \oplus x_{41}x_{35} \oplus x_{41}x_{49} \oplus x_{49}x_{35}$

$l_7 : 0 = 1 \oplus x_{35} \oplus x_{49} \oplus x_{41}x_{35} \oplus x_{41}x_{49} \oplus x_{41}x_{49}x_{35}$

$l_8 : 0 = x_{35} \oplus x_{41} \oplus x_{49} \oplus x_{41}x_{35} \oplus x_{41}x_{49}$

$l_9 : 0 = x_{35} \oplus x_{41} \oplus x_{41}x_{35}$

$l_{10} : 1 = x_{41} \oplus x_{41}x_{35}$

$l_{11} : 1 = 1 \oplus x_{35} \oplus x_{49} \oplus x_{41}x_{49} \oplus x_{41}x_{49}x_{35}$

$l_{12} : 1 = 1 \oplus x_{35} \oplus x_{41} \oplus x_{49}$

$l_{13} : 0 = x_{41} \oplus x_{41}x_{35} \oplus x_{49}x_{35}$

$l_{14} : 0 = 1 \oplus x_{41}x_{35} \oplus x_{41}x_{49} \oplus x_{41}x_{49}x_{35}$

$l_{15} : 0 = 1 \oplus x_{41} \oplus x_{49} \oplus x_{41}x_{35}$

$l_{16} : 1 = x_{35} \oplus x_{41} \oplus x_{49} \oplus x_{41}x_{35} \oplus x_{41}x_{49} \oplus x_{49}x_{35}$

$l_{17} : 0 = x_{41} \oplus x_{49} \oplus x_{41}x_{49}x_{35}$

$l_{18} : 0 = 1 \oplus x_{41} \oplus x_{49} \oplus x_{41}x_{35} \oplus x_{49}x_{35}$

$l_{19} : 0 = 1 \oplus x_{41} \oplus x_{41}x_{35}$

$l_{20} : 0 = 1 \oplus x_{49} \oplus x_{41}x_{35} \oplus x_{41}x_{49}$

$l_{21} : 0 = x_{49} \oplus x_{41}x_{35} \oplus x_{41}x_{49}x_{35}$

$l_{22} : 1 = 1 \oplus x_{41} \oplus x_{49}x_{35} \oplus x_{41}x_{49}x_{35}$

$l_{23} : 1 = x_{35} \oplus x_{41}x_{49} \oplus x_{49}x_{35} \oplus x_{41}x_{49}x_{35}$

$l_{24} : 0 = 1 \oplus x_{35} \oplus x_{49} \oplus x_{41}x_{35} \oplus x_{49}x_{35} \oplus x_{41}x_{49}x_{35}$

$l_{25} : 1 = 1 \oplus x_{35} \oplus x_{49} \oplus x_{41}x_{49} \oplus x_{49}x_{35} \oplus x_{41}x_{49}x_{35}$

$l_{26} : 1 = 1 \oplus x_{35} \oplus x_{41} \oplus x_{41}x_{35}$

$l_{27} : 1 = x_{35} \oplus x_{41}x_{49}$

$l_{28} : 1 = 1 \oplus x_{35} \oplus x_{41}x_{35} \oplus x_{41}x_{49}x_{35}$

$l_{29} : 0 = x_{35} \oplus x_{49} \oplus x_{41}x_{49} \oplus x_{49}x_{35}$

$l_{30} : 0 = x_{41}x_{35} \oplus x_{41}x_{49}$

$l_{31} : 0 = 1 \oplus x_{49} \oplus x_{41}x_{35} \oplus x_{49}x_{35}$

$l_{32} : 1 = 1 \oplus x_{35} \oplus x_{49} \oplus x_{41}x_{35}$

$l_{33} : 0 = 1 \oplus x_{35} \oplus x_{41} \oplus x_{49} \oplus x_{49}x_{35} \oplus x_{41}x_{49}x_{35}$

$l_{34} : 0 = x_{35} \oplus x_{41}x_{49} \oplus x_{41}x_{49}x_{35}$

$l_{35} : 0 = x_{41} \oplus x_{41}x_{35} \oplus x_{41}x_{49} \oplus x_{49}x_{35}$

$l_{36} : 1 = 1 \oplus x_{49} \oplus x_{41}x_{35} \oplus x_{41}x_{49} \oplus x_{49}x_{35}$

$l_{37} : 1 = 1 \oplus x_{49} \oplus x_{41}x_{35} \oplus x_{49}x_{35} \oplus x_{41}x_{49}x_{35}$

$l_{38} : 1 = x_{41} \oplus x_{49} \oplus x_{41}x_{35} \oplus x_{41}x_{49}$

$l_{39} : 1 = 1 \oplus x_{35} \oplus x_{49} \oplus x_{41}x_{49} \oplus x_{41}x_{49}x_{35}$

$l_{40} : 0 = x_{41} \oplus x_{49}x_{35}$

$l_{41} : 0 = x_{49} \oplus x_{41}x_{49} \oplus x_{49}x_{35} \oplus x_{41}x_{49}x_{35}$

$l_{42} : 1 = 1 \oplus x_{41} \oplus x_{49} \oplus x_{41}x_{49} \oplus x_{41}x_{49}x_{35}$

$l_{43} : 1 = x_{41}x_{35} \oplus x_{49}x_{35} \oplus x_{41}x_{49}x_{35}$

$l_{44} : 0 = x_{35} \oplus x_{49} \oplus x_{41}x_{35}$

$l_{45} : 0 = 1 \oplus x_{35}$

$l_{46} : 0 = 1 \oplus x_{41}x_{49} \oplus x_{49}x_{35} \oplus x_{41}x_{49}x_{35}$

$l_{47} : 0 = 1 \oplus x_{35} \oplus x_{41} \oplus x_{41}x_{35} \oplus x_{49}x_{35}$

$l_{48} : 1 = 1 \oplus x_{41} \oplus x_{41}x_{35}$

$l_{49} : 0 = 1$

$l_{50} : 0 = 1 \oplus x_{35} \oplus x_{41} \oplus x_{49} \oplus x_{41}x_{35} \oplus x_{49}x_{35} \oplus x_{41}x_{49}x_{35}$

$l_{51} : 1 = x_{49} \oplus x_{41}x_{35} \oplus x_{49}x_{35} \oplus x_{41}x_{49}x_{35}$

$l_{52} : 1 = x_{41}x_{49} \oplus x_{49}x_{35} \oplus x_{41}x_{49}x_{35}$

$l_{53} : 1 = 1 \oplus x_{41} \oplus x_{41}x_{49} \oplus x_{49}x_{35} \oplus x_{41}x_{49}x_{35}$

$l_{54} : 0 = x_{41} \oplus x_{49} \oplus x_{41}x_{49} \oplus x_{49}x_{35} \oplus x_{41}x_{49}x_{35}$

$l_{55} : 1 = 1 \oplus x_{35} \oplus x_{49} \oplus x_{41}x_{35} \oplus x_{41}x_{49} \oplus x_{41}x_{49}x_{35}$

$l_{56} : 0 = 1 \oplus x_{41}x_{35} \oplus x_{41}x_{49}x_{35}$

$l_{57} : 1 = 1 \oplus x_{35} \oplus x_{41} \oplus x_{49} \oplus x_{41}x_{35} \oplus x_{49}x_{35} \oplus x_{41}x_{49}x_{35}$

$l_{58} : 1 = x_{35} \oplus x_{41} \oplus x_{41}x_{35} \oplus x_{49}x_{35} \oplus x_{41}x_{49}x_{35}$

$l_{59} : 1 = x_{41} \oplus x_{49} \oplus x_{41}x_{35}$

$l_{60} : 1 = x_{35} \oplus x_{41} \oplus x_{49} \oplus x_{41}x_{49}$

$l_{61} : 1 = x_{41}x_{49} \oplus x_{49}x_{35}$

$l_{62} : 0 = x_{35} \oplus x_{41} \oplus x_{41}x_{35} \oplus x_{41}x_{49}x_{35}$

$l_{63} : 0 = x_{49} \oplus x_{41}x_{35} \oplus x_{41}x_{49} \oplus x_{49}x_{35} \oplus x_{41}x_{49}x_{35}$

Table 7: Experimentally selected bit positions minimizing constructing time in Ascon-Xof.

R	N	$\mathcal{I}^{(R,N)}$
3	1	(35), (48), (51), (59), (62)
	2	(47, 62)
	3	(32, 47, 62)
	4	(32, 47, 56, 62)
4	1	(51)
	2	(35, 62)
	3	(32, 47, 62)
	4	(48, 54, 56, 62)
5	1	(51)
	2	(51, 56)
	3	(35, 41, 49), (48, 54, 62)
	4	(48, 54, 56, 62)
6	1	(51)
	2	(51, 56)
	3	(48, 54, 62), (35, 41, 49)
7	1	(51)
	2	(51, 56)
8	1	(51)

GPU-Resident CKKS Inference with PyTorch-Compatible Model Conversion*

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Abstract

We present our system, a CUDA-accelerated Cheon–Kim–Kim–Song (CKKS) homomorphic encryption library for private neural network inference. Our system combines a PyTorch-compatible model conversion API with a modified OpenFHE-based CUDA backend so that encrypted inference remains resident on the GPU. The system uses device-resident encoded-weight reuse and GPU-resident execution to reduce three practical bottlenecks in the evaluated CKKS inference path: repeated plaintext encoding of model weights, host–device synchronization during rescaling, and CPU fallback for plaintext arithmetic. On encrypted multilayer perceptron inference for MNIST and Fashion-MNIST, our system achieves warm-cache forward-pass speedups of 176–252× over single-threaded CPU, 83–109× over 16-threaded CPU, and 179–185× over TenSEAL. For a 256-hidden-unit MLP, warm-cache end-to-end latency is 379 ms, while encrypted accuracy remains within one percentage point of plaintext baselines.

1 Introduction

Cloud machine learning services increasingly process sensitive inputs such as medical records, financial transactions, and biometric measurements. Homomorphic encryption (HE) offers a direct privacy-preserving approach: clients encrypt their inputs, servers evaluate a model over ciphertexts, and only clients decrypt the result. Among HE schemes, CKKS [1] is especially suited to neural network inference because it supports approximate fixed-point arithmetic over packed vectors.

This capability is directly relevant to the evolution of electronic business and commerce information systems. Modern e-commerce platforms increasingly rely on cloud-hosted AI services for fraud detection, payment-risk scoring, personalized recommendation, credit and insurance decisions, biometric authentication, and privacy-sensitive healthcare or financial workflows. These services must extract value from user data while complying with stricter privacy expectations and data-protection requirements. Encrypted inference addresses this tension by allowing a service provider to evaluate a deployed model without observing the user's raw input, which makes it a practical building block for privacy-preserving decision support in data-sensitive e-commerce environments.

The main obstacle is performance. CKKS inference spends most of its time in polynomial arithmetic, rotations, rescaling, and plaintext encoding. GPUs are a natural target for this workload, but accelerating individual primitives is not enough for practical inference. Existing systems often leave parts of the pipeline on the CPU, require manual circuit construction, or repeatedly encode static model weights for every request. These gaps are particularly costly in inference-as-a-service settings, where a fixed model is evaluated many times on different encrypted inputs.

Our system is a GPU-resident CKKS inference system designed around this deployment model. It exposes a PyTorch-compatible API that converts supported trained models into encrypted modules,

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then evaluates the resulting computation through a modified OpenFHE-based CUDA backend. The system focuses on the operations that dominate encrypted neural inference: linear layers implemented with Baby-step Giant-step (BSGS) rotations, polynomial activations, ciphertext rescaling, and plaintext arithmetic for weights and biases.

This paper makes three contributions. First, our system integrates the CKKS operations exercised by our benchmarks into a GPU-resident inference path, including ciphertext–plaintext arithmetic, rotations, and rescaling. Second, it provides automatic model conversion from PyTorch modules, reducing the amount of cryptographic expertise needed to deploy private inference. Third, it evaluates the resulting warm-cache service path, showing 176–252× speedup over single-threaded CPU and 179–185× speedup over TenSEAL while preserving classification accuracy within one percentage point of plaintext models.

2 Background and Design

2.1 CKKS and BSGS Linear Layers

CKKS is a leveled HE scheme over the ring $R_Q = \mathbb{Z}_Q[X]/(X^N + 1)$. A plaintext vector is encoded into a polynomial, encrypted into a ciphertext, and then manipulated through homomorphic addition, multiplication, rotation, and rescaling. Rotation is central to neural inference because it moves packed slots inside a ciphertext; it is also expensive because it requires key switching. Rescaling restores the scale after multiplication and consumes one level from the modulus chain.

Our system evaluates linear layers using the BSGS method [2]. Given a $d \times d$ weight matrix W and encrypted input ct, the matrix is decomposed into generalized diagonals p_k. BSGS chooses $d_1 d_2 \geq d$ and computes

$$W \cdot ct = \sum_{j=0}^{d_2-1} \mathrm{Rot}\Big(\sum_{i=0}^{d_1-1} p_{i+jd_1} \cdot \mathrm{Rot}(ct, i), jd_1 \Big). \tag{1}$$

This reduces the number of rotations from d to roughly $2\sqrt{d}$, but it requires each weight diagonal to be encoded as a plaintext polynomial. For static model weights, repeatedly performing this encoding is wasteful and becomes the dominant cold-start cost on GPU.

2.2 System Architecture

Our system is organized as a Python conversion layer, encrypted neural-network modules, and a modified OpenFHE-based CUDA backend. The top layer accepts a trained PyTorch model and recursively replaces supported modules with encrypted counterparts. Linear layers become BSGS-based encrypted linear layers; convolutional layers use an im2col lowering followed by BSGS; ReLU, GELU, and SiLU activations are replaced by Chebyshev polynomial approximations; and batch normalization can be folded into preceding affine layers at conversion time. The encrypted modules call a pybind11 extension backed by a modified OpenFHE GPU fork.

2.3 Threat Model and Security Assumptions

Our system targets an honest-but-curious inference server. The client generates the CKKS secret key locally, encrypts inputs, sends ciphertexts and evaluation keys to the server, and decrypts the returned ciphertext outputs. The server follows the prescribed protocol but may observe all ciphertexts, evaluation keys, and GPU-resident encrypted intermediate buffers generated during execution. Under the standard

CKKS security assumptions, these encrypted values do not reveal the client's plaintext input or input-derived intermediate activations.

Model weights are assumed to be public or available to the service provider, so model confidentiality is not a target of this work. The evaluated computation has a fixed circuit structure for a given model and parameter set, but we do not provide a formal analysis of timing or microarchitectural side channels. Multi-user GPU isolation and defenses against malicious clients are treated as deployment-hardening concerns for production e-commerce services rather than security guarantees of the proposed inference engine.

2.4 GPU-Resident Optimizations

Our system relies on three implementation choices to keep the evaluated warm-cache inference path resident on the GPU.

Device-resident plaintext reuse. For each converted model, our system reuses encoded weight diagonals stored in device memory. Cache keys include the layer, diagonal index, BSGS parameters, ciphertext level, and a weight hash. The first encrypted forward pass populates these entries; later requests reuse the encoded plaintexts directly on the GPU. For the evaluated three-layer MLPs, the number of cached diagonals is 1,040 ($h = 128$), 1,296 ($h = 256$), and 1,808 ($h = 512$), all within the default cache capacity of 2,048 entries. Under the parameters used in this paper, a plaintext entry occupies about 3.2 MB, so the full cache uses about 6.4 GB of GPU memory.

Zero-copy rescaling. Naive BSGS accumulation can allocate and copy ciphertext buffers at each ciphertext–plaintext multiplication and rescale. CUDA allocation introduces implicit synchronization, which serializes what should be a device-resident pipeline. Our system modifies the accumulation path to transfer C++ buffer ownership with move semantics and to reuse GPU buffers whenever possible. This avoids deep-copying approximately 6.4 MB ciphertext objects during accumulation.

GPU-native plaintext arithmetic. The OpenFHE GPU fork used as a base lacked GPU-native plaintext addition and subtraction in the inference path exercised by our system. That limitation forced a CPU round trip for bias addition. Our system adds in-place CUDA kernels for these operations so that linear-layer bias handling does not break GPU residency.

The experimental configuration uses ring dimension $N = 32{,}768$, 16,384 slots, scale $\Delta = 2^{50}$, and a ternary secret distribution. The three-layer MLP uses an 11-level modulus chain with at least 128-bit security under the Homomorphic Encryption Standard [3].

3 Evaluation

3.1 Setup

We evaluate our system on an RTX 5090 GPU with 32 GB GDDR7 memory and an Intel Core Ultra 9 285K CPU running Ubuntu 22.04 and CUDA 12.8. The HE backend is a modified OpenFHE-based CUDA build. The main benchmark is a three-layer MLP, $784 \to h \to h \to 10$, with $h \in \{128, 256, 512\}$, evaluated on MNIST and Fashion-MNIST. Inputs are scaled to $[-1, 1]$, and ReLU is approximated by a degree-4 Chebyshev polynomial. Models are trained with Adam for 20 epochs using learning rate 10^{-3} and batch size 128.

Table 1: Forward-Pass Latency Across Systems in Seconds

System	$h = 128$	$h = 256$	$h = 512$
TenSEAL	16.233	17.920	21.813
Ours-CPU 1T	15.630	20.189	29.712
Ours-CPU 16T	7.388	9.251	12.809
Ours-GPU warm	0.089	0.100	0.118
Speedup vs. CPU 1T	176×	202×	252×
Speedup vs. CPU 16T	83×	93×	109×
Speedup vs. TenSEAL	182×	179×	185×

Table 2: Cold and Warm Latency Breakdown for Ours-GPU, $h = 256$

Component	Cold (ms)	Warm (ms)
Client encrypt	267	267
Server forward pass	3029	101
Client decrypt	11	11
End-to-end	3307	379

We report forward-pass latency for encrypted model evaluation and, where noted, end-to-end latency including client encryption and decryption. GPU forward-pass measurements use the warm cache unless stated otherwise. TenSEAL v0.3.16 with Microsoft SEAL is used as a CPU-based Python encrypted tensor baseline under matched CKKS parameters.

3.2 CPU and Framework Comparison

Table 1 compares our system with TenSEAL and CPU OpenFHE baselines. Our system reaches 89–118 ms warm-cache forward-pass latency across the three MLP widths. This corresponds to 176–252× speedup over single-threaded CPU and 179–185× over TenSEAL; even against 16 CPU threads, the speedup remains 83–109×.

The GPU latency increases by only 33% when h grows from 128 to 512, while single-threaded CPU latency increases by 90%. CPU parallelism helps but saturates well above the GPU latency, indicating that the workload benefits from GPU memory bandwidth and massive polynomial-level parallelism.

3.3 Cold and Warm Inference

Device-resident plaintext reuse is the largest measurable contributor to warm-cache performance. Disabling it increases GPU forward-pass latency by 14.6–15.4× because every request must re-encode all BSGS weight diagonals. Table 2 shows the end-to-end effect for the $h = 256$ MLP. The first request takes 3.307 s because cache population dominates the server forward pass. Once populated, the same model reaches 379 ms end-to-end latency.

3.4 Accuracy and Generalization

Encrypted accuracy remains close to plaintext accuracy. On MNIST, our system achieves 98.07%, 98.09%, and 98.15% accuracy for $h = 128, 256, 512$, respectively, within 0.06 percentage points of the

corresponding plaintext models. On Fashion-MNIST, encrypted accuracy is 88.92–89.66%, within one percentage point of plaintext accuracy for all widths. We also evaluated a small encrypted convolutional model, Conv2d$(1,8,k=5,s=4)$ plus a fully connected layer, which reaches 95.9% encrypted accuracy on 1,000 MNIST samples with 0.227 s warm-cache latency.

3.5 Scope and Limitations

The current evaluation focuses on latency-sensitive single-query inference for small MLP and CNN models. This is the regime where repeated plaintext encoding and CPU fallback are especially visible, and where a warm-cache service model is most natural. Larger networks introduce additional constraints: rotation keys grow with the set of slot permutations needed by packed convolutions, cached plaintexts compete with ciphertexts and evaluation keys for GPU memory, and deeper polynomial circuits require longer modulus chains or bootstrapping.

The reported speedups should therefore be interpreted as end-to-end results for our evaluated inference path, not as a general comparison against all GPU CKKS libraries. TenSEAL is used as a CPU-based Python tensor baseline, while recent GPU CKKS libraries such as HEonGPU, Cheddar, and FIDESlib focus on lower-level primitive coverage and bootstrapping. A cache-matched CPU baseline and direct comparisons with standalone GPU CKKS backends would further clarify how much of the gain comes from GPU execution, cache reuse, and framework-level integration.

3.6 Deployment Implications

The warm-cache measurements are most relevant to persistent inference services where a provider loads one encrypted model, populates the encoded-weight cache on the first request, and then serves many encrypted inputs. In this setting, the cold-start cost is amortized, and the main latency floor becomes client encryption, GPU-resident model evaluation, and client decryption. For the $h = 256$ MLP, the server-side warm forward pass is 101 ms, while the full encrypt–evaluate–decrypt path is 379 ms. The first request is substantially slower because it must populate the encoded-weight cache; in our $h = 256$ measurement, this cold path takes 3.307 s end-to-end. Thus, the reported warm latency should be interpreted as the steady-state behavior of a long-running service rather than the cost of loading a new model or serving a rarely used model for the first time. Future batching or client-side encryption optimizations would therefore affect a different part of the pipeline than the GPU cache.

The same design exposes a memory tradeoff. Pinning encoded diagonals makes warm inference fast, but the cache shares GPU memory with ciphertexts and evaluation keys. The evaluated MLPs fit within the default 2,048-entry cache, but larger packed CNNs and multi-model services may face higher pressure from cached plaintexts, rotation keys, ciphertext working buffers, and larger model weights. In such deployments, cache capacity determines a tradeoff between steady-state latency and available GPU memory. Cache admission policies that account for layer reuse, diagonal sparsity, and available device memory are therefore important for scaling the approach beyond the evaluated models. This is why we view cache management and rotation-key planning as system-level problems rather than only low-level kernel optimizations.

3.7 Measurement Interpretation

The latency numbers in this paper should be read as service-path measurements rather than isolated primitive benchmarks. A primitive-level comparison can show whether a particular NTT, key-switch, or rescale kernel is fast, but it does not capture whether an encrypted model repeatedly crosses the host–device boundary or reconstructs static plaintexts for every query. Our evaluation therefore separates cold and warm behavior and reports both server forward time and encrypt–evaluate–decrypt time. This

distinction is important for encrypted inference services because the client-visible latency includes operations that are outside the GPU evaluator, while the server-side latency is the component most directly affected by GPU residency.

The comparison with TenSEAL has the same scope. TenSEAL is a mature Python interface for encrypted tensor operations and is a natural framework-level baseline for users, but it is not a GPU CKKS backend. Conversely, GPU CKKS libraries such as HEonGPU, Cheddar, and FIDESlib provide broader low-level acceleration targets but do not directly evaluate the same PyTorch model conversion workflow. The intended reading is therefore that the proposed integration substantially reduces latency relative to CPU-oriented framework baselines, while direct GPU-library comparisons remain future work.

4 Related Work

CryptoNets introduced encrypted neural inference [4]. Hybrid systems such as GAZELLE, Delphi, and Cheetah combine HE with secure multi-party computation but require interaction [5, 6, 7]. CHET, EVA, nGraph-HE, TenSEAL, and HELayers improve HE programmability and compiler support [8, 9, 10, 11, 12]. Hardware-oriented and GPU library work accelerates HE primitives, encrypted CNN inference, and bootstrapping [13, 14, 15, 16, 17, 18]. Our system is complementary: it focuses on a pure-HE, PyTorch-facing inference path whose warm-cache evaluation remains GPU-resident for the evaluated models.

5 Conclusion

Our system is a GPU-resident CKKS inference system with PyTorch model conversion. Device-resident plaintext reuse, zero-copy rescaling, and GPU-native plaintext arithmetic remove CPU fallbacks in the evaluated path and reduce repeated encoding costs for static model weights. Our system achieves 176–252$\times$ speedup over single-threaded CPU and 179–185$\times$ over TenSEAL, with sub-400 ms warm-cache end-to-end latency for $h = 256$. Future work includes studying which optimization techniques from conventional deep learning, such as operator fusion, architecture compression, memory scheduling, and larger CNN or ResNet-style designs, remain effective under CKKS constraints where rotations, rescaling, ciphertext levels, and VRAM usage dominate the cost model. A related direction is VRAM-aware system design, including cache admission, rotation-key layout, and encrypted-buffer reuse for larger models and multi-model inference services.

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Higher Dynamic AES for Commercial Market*

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Abstract

Recently, Advanced Encryption Standard (AES) has been widely used in commercial markets. Auctions that compete expensive objects are typical examples. However, recent developments in cryptanalysis have revealed vulnerabilities, such as key-independent and static operations. In this study, AES AddRoundKey was identified as the only key-dependent operation, whereas the others were key independent operations. Hence, we proposed an enhanced version of AES called the Dynamic AES (DAES) to reinforce its commercial security. In addition, we designed an irreversible algorithm called Dynamic Shifting Mapping (DSM) to produce a dynamic box given a string. It has the key-dependent property of 64 variations to replace AES's ShiftRows. The results of our qualitative and efficiency analyses indicate that the DAES surpasses the AES in terms of security level with a slight trade-off in execution efficiency.

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1. Introduction

Up to present, National Institute of Standards and Technology (NIST) has approved several cryptographic algorithms in which Advanced Encryption Standard (AES) (Dworkin, Turan, and Mouha, 2023) is crafted using Galois Field and composed of its Key Schedule and four round operations, including AddRoundKey, SubBytes, ShiftRows and MixColumns.

In fact, round operations of the AES can be classified as static which is independent of the Cipher Key (CK) or password (PW), meaning that its key-independent and dynamic properties are only determined by CK/PW. The AES using an identical substitution box (S-box) for all SubBytes in different rounds is key-independent. It takes the bias value as a constant to perform ShiftRows, utilizes an identical Maximum Distance Separable (MDS) matrix for all MixColumns operations in all rounds, and performs an XOR operation on the state and round key, called AddRoundKey, are key-dependent, implying that SubBytes, ShiftRows, and MixColumns are all static. Once the three operations are defined, they are used throughout the encryption process. On the other hand, AddRoundKey is dynamic since the 11 round keys are derived from CK/PW through the Key Schedule. Our conclusion is that AES is one with 1/4 dynamic.

Owing to the rapid development of software and hardware over the past two decades, Artificial Intelligence (AI) cracking by integrating a massive number of CPUs, GPUs, and NPUs (Yang, et al., 2023) or quantum computers (Yang, Zolanvari, Jain, 2023; DeCusatis, et al. 2024), will soon be possible.

Therefore, in this paper, a better AES, called Dynamic AES (DAES) is proposed. It dynamically creates key-dependent round operations to up-grade the AES's Shift-Row and replace S-Box with D-boxes while inheriting the advantages of AES.

The rest of this paper is organized as follows. Section 2 introduces work related to this study. Section 3 describes the architecture of the DAES. Qualitative analyses on the security and the performance of the DAES and AES are presented in Section 4. Section 5 concludes the study and discusses our future work.

2. Background and Related Works

In this session, some histories and security problems are described.

2.1 Threatens and Histories

AES is currently one of the most popular algorithms used for block-ciphering. The efficiency of the four abovementioned operations is high with complementary security functions. Therefore, this standard has been widely adopted to secure transmitted and stored data, particularly for wireless security (Nirbaya, et al. 2023), file encryption (Tong, et al., 2021), and processor security (Wang, et al. 2018). However, the AES has been partially cracked. Some of these issues are discussed below:

Wang et al. (Wang, et al. 2016) combined Fault Sensitivity Analysis (FSA) and a zero-value sensitivity model to break the masked AES S-box. The authors demonstrated that traversing the zero-valued input of the S-box can ultimately reveal the last-round key of AES. In (Jithendra and Shahana, 2023), Jithendra and Shahana conducted a biclique at-tack on a full-round PRESENT-80 block cipher using independent related key differential cryptanalysis. The authors claimed that the computational complexity of such an attack was significantly reduced, thus endangering AES.

2.2 Enhancing S-box Security

To enhance the security of AES, many researchers have proposed new features such as randomness, dynamicity, and key dependency. Assafli and Hashim (Assafli, and Hashim, 2020) introduced a dynamic box that constantly changes the independence of the encryption key. A new S-box is generated depending on the epoch timestamp to produce an identical nonlinear S-box inside the transmitter and receiver, thus enhancing the security level of the cryptography system. The authors claimed that data complexity and time complexity were 223 and 279.63, respectively.

Abd-El-Atty (Abd-El-Atty, 2013) proposed a new S-box mechanism based on quantum-inspired quantum walks, the Hénon map, and a customized particle swarm optimization algorithm. The performance analysis shows that the effectiveness and reliability of designing various cryptosystems are excellent. With S-box, a new image cryptosystem was developed to increase its efficacy and security against various attacks.

Saha et al. (Saha, et al., 2018) proposed a novel approach for generating key-dependent S-boxes using a pseudo-random methodology to enhance the secrecy of the S-boxes.

2.3 Other Recent Enhancements

Building upon these studies, Luong and Linh (Luong and Linh, 2014) proposed new key-dependent XOR tables to strengthen AES security and evaluated the randomness of the block cipher outputs. Kazlauskas (Kazlauskas, 2015) enhanced the confusion property and avalanche effect of AES by invoking a Symmetric Random Function Generator.

Su et al. (Su, et al., 2019) researched data encryption standards based on the AES algorithm in an IoT environment, showing applicability of AES in emerging technological fields.

In summary, improving the AES Key Schedule or static operations is crucial for enhancing security. These findings indicate that AES can significantly improve its resistance to differential and linear cryptanalysis without substantially affecting its encryption/decryption performance by introducing features, such as randomness, dynamicity, and key dependency. However, the trade-off between the execution performance and security level remains a critical re-search topic. In other words, AES requires continuous improvement and enhancement to meet the growing security demands.

3. The Architecture of the DAES

The DAES was developed following the AES framework with its Key Schedule and transformation rounds being expanded, including the generation of 11 RoundKeys, 11 D-boxes, and 20 round parameters, to strengthen the security level of each transformation round, attempting to improve its encryption/decryption throughputs for commercial market. The transformation rounds is modified as follows. (1) generating a dynamic box (D-box) based on user CK/PW to substitute for the S-box which is a static box, thus enhancing the security level of the SubBytes operation; (2) proposing a DBR operation which provides 64 variations for data encryption/decryption. In fact, in the AES the ShiftRows operation is one of these variations. The original AES ShiftRows operation is just one of the DBRs. Originally, the AES had approximately 1/4 of dynamic transformation rounds. After our modifications, the DAES is a 3/4 dynamic cryptographic system.

1.1 Parameters

The parameters used in this study are defined below:

1. D-box: A 16 × 16 (D-box) is derived from user input strings. The 256 non-duplicated elements in the S-box, ranging from 00 to FF, were randomly rearranged in the D-box.

2. ISKs: Initial Serial Keys (ISKs), including ISK_1 and ISK_2, are produced using a specific algorithm, given an input string. The length of the ISK is 1024 bits.
3. DKs: Dynamic Keys, including DK1, DK2 and DK3, are derived from ISK1 and ISK2 by employing 3D operations (Huang, et al., 2017] and the DSM algorithm. The length of the DK is 1024 bits.
4. FA: Flag Array that identifies the elements of the D-box is used to guarantee that each element of the D-box is unique without colliding with others. The indexed content of FA is "T" or "F," which indicates that the index has been selected or not, respectively.
5. $SIA1$: The 1st Insert Array is derived from DK1 by removing duplicated elements in DK1 using FA.
6. $SIA2$: The 2nd Insert Array is derived from DK2 and DK3 by removing the duplicated elements in $SIA1$, DK2 and DK3 using FA.
7. RIA: Residual Insert Array consists of those unselected indexes of FA.
8. $SIA3$: The 3rd Insert Array, which is obtained by rearranging the Residual Insert Array (RIA).
9. $SIA1$, $SIA2$, and S$IA3$ are used to establish the D-box.
10. RLs: Real lengths of the arrays, including S$IA1$, $SIA2$, $SIA3$, and RIA.
11. GBT1: "Generating a Box of Type 1," which is generated to produce a D-box by invoking a unique string.
12. GBT2: "Generating a Box of Type 2," which is produced to yield a D-box by invoking a key and initial box.

3.2 Operations

Let $p_i(c_i)$ be the plaintext (ciphertext) block and let k be the encryption key. The encryption and decryption processes of the three fundamental operators are defined as follows.

(1) Exclusive-OR operator ($\oplus$):
Encryption: $c_i = p_i \oplus k$;
Decryption: $p_i = c_i \oplus k$;

(2) Binary addition operator ($+_2$):

Encryption: $c_i = p_i +_2 k$, where pi and k undergo binary addition, and the carry generated by the addition of the most significant bit is ignored.
Decryption: $p_i = c_i -_2 k = \begin{cases} c_i - k, & if\ c_i \geq k \\ c_i + \bar{k} + 1, & if\ c_i < k \end{cases}$ where $-_2$ is the inverse operation of $+_2$.

(3) Rotate-Equivalence operator ($\odot_R$):

Encryption: $c_i = p_i \odot_R k = p_{iR} \odot k$, where p_{iR} is the key obtained by rotating plaintext p_i clockwise $|k|/4$ bits. For example, if $|k| = 128$, p_i is rotated by 32 bits.
Decryption: $p_i = c_i \odot_{IR} k$ = counter_clockwise rotating ($c_i \odot k$), a total of $|k|/4$ bits.

3.3 Proposed Algorithms

In the following, we described the algorithms created and used in the DAES.

3.3.1 DSM Algorithm

DSM Algorithm with (P, D-box) as its arguments is shown below.

Algorithm 1: The DSM Algorithm
Input: A plaintext $P = p_1 p_2 ... p_k$ of n bits, where n is a multiple of 8, $k = n/8$, p_i is 8 bits long,

$1 \le i \le k$, and a 16 × 16 D-box.
Output: Ciphertext $C = c_1c_2...c_k$ where c_i is 8 bits long and $1 \le i \le k$.
1.Let dsn = 256; /* setting initial value of dsn, the dynamic shift number */
2. **for** i = 1 to k
$vp[i]$ = Int(p_i); /* Int(p_i) retrieves ASCII code of p_i */
$dsn = dsn + vp[i] * i$;
3. **next** ***i***;
4. $vp[0] = vp[1] + vp[k] + dsn$;
5. $vp[k+1] = vp[k-1] + vp[k]$;
6. **for** i = 1 to k
7. $dsn = (dsn + vp[i-1] * vp[i]+vp[i+1]) \bmod 2^{16}$;
$vch = (vp[i] + dsn) \bmod 256$;
c_i = the ith ciphertext that is the content in the D-box at position vch;
8. **next *i*;**
9. **return** C;

In this algorithm, a character in the plaintext string is encrypted sequentially and individually. When encrypting a character, a new dsn is generated for encrypting the next character. In each encryption, the dsn is increased by multiplying and summing the ASCII codes of the previous, current, and next plaintext characters. This increment is accumulated in the dsn. Therefore, a change to any character in the plaintext will conduct subsequent changes to the ciphertext. The dsn value that encrypts the first character is the sum of the ASCII codes of all plaintext characters plus 256. Therefore, it has an excellent avalanche effect.

3.3.2 The DSM Expansion Algorithm

In this study, the length of the user input *CK/PW* is 128–1024 bits. The length of our encryption/decryption key was 1024 bits. When the length of the inputting key *CK/PW* is shorter than 1024 bits, a key expansion method called the DSM Expansion (DSM-E) algorithm, that is, Algorithm 2 as shown below, is invoked to produce a new *CK/PW* that is 1024 bits long. Of course, as shown in step 5 of this algorithm, when |*PW*|= 1024, PW is encrypted to securely protect this *PW*.

Algorithm 2: DSM-E Algorithm
Input: *PW*, where $|PW| \le 1024$; A 16x16 box;
Output: An expanded *PW*, where |*PW*| = 1024;
1. **While** $|PW| < 1024$ do
P' = DSM (*PW*, D-box);
$PW = P'||PW||P'$;
2. **end while**
3. *PW* = DSM (*PW*, D-box); /* when $|PW| = 1024$, *PW* is encrypted */
4. **return** Right (*PW*, 128); /* return PW of 1024 bits, i.e., 128 bytes */

3.3.3 Producing a Dynamic Box of 16 x 16

Dynamic Boxes are adopted to replace S-Box. The process of generating a 16 × 16 Dynamic Box (16DB) is composed of two processes: the Generation of Dynamic Keys using *PW* (GDK, i.e., Algorithm 3) and Dynamic Box Generator (DBG, i.e., Algorithm 4).

Between step 1 and step 11 in the GDK Algorithm (Algorithm 3), the input *PW* is reproduced as a new 1024-bit scrambled string ISK_1 by using DSM-E Algorithm (Algorithm 2). Between step 12 and step 15 of Algorithm 3, ISK_1 is further scrambled and diffused by repeatedly using the DSM Algorithm

(Algorithm 1), generating ISK_2, DK_1, DK_2 and DK_3, in which the latter three are utilized to produce a dynamic box, i.e., D-box, by invoking Algorithm 4.

Algorithm 3: GDK Algorithm
Input: PW; A 16x16 I-box;
Output: DK_1, DK_2 and DK_3;
1.If $|PW| < 1024$, then
2. ISK_1 = DSM-E (PW, I-box); /*$|ISK_1|$=1024 bits*/
3. else if $|PW| = 1024$, then
4. ISK_1 = DSM (PW, I-box);
5. else /* $|PW| > 1024$ */
6. $PW = p_1p_2p_3 \ldots p_{m-1}p_m$, where $|p_i| = 1024$, $1 \leqq i < m$, and $m \geqq 2$; /* partition PW into blocks, each with 1024 bits long */
7. If $|p_m| < 1024$, then /* the last block p_m may be shorter than 1024 bits*/
8. p_m = DSM-E (p_m, I-box);
9. end if
10. $ISK_1 = p_1 \oplus p_2 \oplus \ldots \oplus p_{m-1} \oplus p_m$;
11. end if
12. ISK_2 = DSM (ISK_1, I-box);
13. DK_1 = DSM($ISK_1 \oplus ISK_2$, I-box);
14. DK_2 = DSM($ISK_1 +_2 ISK_2$, I-box);
15. DK_3 = DSM($ISK_1 \odot_R ISK_2$, I-box);
16. **return** DK_1, DK_2 and DK_3;

Algorithm 4: The DBG Algorithm
Input: IK_1, IK_2 and IK_3; /* IK_1, IK_2 and IK_3 are DK_1, DK_2, DK_3*/
Output: D-box;
FA[256], D-box[256], /* Flag Array, and D-box array */
$SIA1$[128], $SIA2$[256], $SIA3$[256], RIA[256];
/* Insert Arrays and the residual Insert Array */
Let $AL1$=0; $AL2$=0; ALR=0; $AL3$=0; /* ALj denotes the actual length of $SIAj$, $1 \leq j \leq 3$ */
/* Set up initial arrays, variables, and parameters */
Let $IK_1 = KA_0||KA_1||\ldots||KA_{127}$, where $|KA_j| = 8$ (bit) for all j, $0 \leqq j \leqq 127$;
Let $IK_2 = KB_0||KB_1||\ldots||KB_{127}$, where $|KB_j| = 8$ (bit) for all j, $0 \leqq j \leqq 127$;
Let $IK_3 = KC_0||KC_1||\ldots||KC_{127}$, where $|KC_j| = 8$ (bit) for all j, $0 \leqq j \leqq 127$;
/* individually partition IK_1, IK_2 and IK_3 into 128 eight bits string*/
for j = 0 to 255
$FA[j]$ = "F";
next j;
for j = 0 to 127
$KA[j]$ = Int (KA_j); /* Int (KA_j) produces ASCII code of character KA_j */
$KB[j]$ = Int (KB_j);
$KC[j]$ = Int (KC_j);
next j;
/* Generate arrays $SIA1$, $SIA2$, and RIA */
for i = 0 to 127 /* filling in $SIA1$ according to the data in IK_1 */
if ($FA[KA[i]]$ = "F") then
$SIA1[AL1] = KA[i]$;
$FA[KA[i]]$ = "T";
$AL1 = AL1 + 1$;
end if
next i;
for i = 0 to 127 / * filling in $SIA2$ based on data from IK_2 */
if ($FA[KB[i]]$ = "F") then
$SIA2[AL2] = KB[i]$;

```
            FA[KB[i]] = "T";
            AL2 = AL2 + 1;
        end if
    next i;
    for i = 0 to 127 /* filling in SIA3 based on data from IK3 */
        if (FA[KC[i]] = ”F”) then
            SIA3[AL3] = KC[i];
            FA[KC[i]] = "T";
            AL3 = AL3 + 1;
        end if
    next i;
    for i = 0–255  /* considering the unused indices of FA as elements of RIA*/
        if (FA[i] = ”F”) then
            RIA[ALR] = i;
            ALR = ALR+1;
        end if
    next i;
        /* expanding SIA2 by concatenating RIA and SIA1*/
    for i = 0 to ALR–1 /* expanding SIA2 by concatenating RIA */
        SIA2[AL2+i] = RIA [i] ;
    next i;
    for i = 0 to AL1–1 /* expanding SIA2 by concatenating SIA1 */
        SIA2[AL2+ ALR +i] = SIA1 [i] ;
    next i;
        /* deriving parameters h1, h2, h3, jup1, jup2, and jup3 from arrays KB and KC */.
    h1=0; h2=0; h3=0; jup1=0; jup2=0; jup3=0;
    for j = 0 to 63
        h1= h1 + KB[j];
        jup1= jup1 + KB[64+j];
    next j
    for j = 0 to 31
        h2= h2 + KC[j];
        h3= h3 + KC[32+j];
        jup2 = jup2 + KC[64+j];
        jup3 =jup3 + KC[96+j];
    next j
    h1 = h1 mod 256;
    h2 = h2 mod 256;
    h3 = h3 mod 256;
    jup1= jup1 mod 128;
    if (jup1 is even) then
        jup1= jup1 +129;
    end if
    jup2= jup2 mod 128;
    if (jup2 is even) then
        jup2= jup2 +129;
    end if
    jup3= jup3 mod 128;
    if (jup3 is even) then
        jup3= jup3 +129;
    end if
        /* generate D-box using SIA2 and parameters h1, h2, h3, jup1, jup2, and jup3 */
```

for i = 0 to 255 /* rearranging $SIA2$ to obtain $SIA3$ */
$SIA3[i] = SIA2[(h_1 + i * jup1) \ mod \ 256]$;
next i;
for i = 0 to 255 /* generating D-box */
D-box[$(h_3 + i * jup3) \ mod \ 256$] = $SIA3[(h_2 + i * jup2) \ mod \ 256]$;
next i;
return D-box;

The DBG algorithm adopts the ASCII of CK/PW as its input key and involves DK_1, DK_2 and DK_3 to generate the elements of the D-box. The Flag Array FA is to make sure that a D-Box has no duplicate elements. At last, $SIA3$ is rearranged to break the order of the elements in the Residual Insert Array (RIA) to improve the randomness of $SIA3$.

The process of the generation of the D-box is summarized below: (1) Algorithm 3: The GDK Algorithm derives DK_1, DK_2 and DK_3 from the CK/PW. (2) Algorithm 4: The DBG Algorithm generates the elements of the D-box following the content order of these three dynamic keys. Since DK_1, DK_2 and DK_3 are generated by using uses DSM and DSM-E several times, it inherits the garbled text diffusion and input sensitivity. Thus, the elements of the D-box are completely determined by the CK/PW.

Algorithm 5: The GBT1 Algorithm
Input: PW; A 16 x 16 I-box;
Output: D-box;
1. GDK (PW, I-box); /* producing DK_1, DK_2, and DK_3 */
2. Call DBG (DK_1, DK_2, DK_3); /* generating D-box */
3. **return** D-box;

Algorithm 6: The GBT2 Algorithm
Input: PW; A 16x16 I-box;
Output: D-box;
1. $PW_1 = PW_0||PW_1||\ldots||PW_{15}$, where $|PW_j| = 8$ (bits) for all j, $0 \leqq j \leqq 15$;
/* logically partition PW_1 into 16 eight bits string*/
2. for i = 0 to 15
3. $PW[i]$ = int(PW_i);
4. next i;
5. h_1=0; h_2=0; $jup1$=0; $jup2$=0;
6. for i = 0 to 3
7. $h_1 = h_1 + PW[4i]$;
8. $h_2 = h_2 + PW[4i+1]$;
9. $jup1 = jup1 + PW[4i+2]$;
10. $jup2 = jup2 + PW;4i+3]$;
11. next i
12. $h_1 = h_1 \ mod \ 256$;
13. $h_2 = h_2 \ mod \ 256$;
14. $jup1 = jup1 \ mod \ 128$;
15. if ($jup1$ is even) then $jup1 = jup1 + 129$;
16. $jup2 = jup2 \ mod \ 128$;
17. if ($jup2$ is even) then $jup2 = jup2 + 129$;
18. for i = 0 to 255 /* filling in D-box according to the data in I-box */
19. D-box[$(h_2 + i * jup2) \ mod \ 256$] = I-box [$(h_1 + i * jup1) \ mod \ 256$];
20. next i;
21. **return** D-box;

Next, the GBT1 Algorithm (Algorithm 5) derives D-box from CK/PW. To accelerate the generation of the D-box, we first transform a default initial D-box, denoted by I-Box, into another D-box based on the PW. The GBT2 algorithm (Algorithm 6) uses PW to produce 4 parameters, h_1, h_2, $jup1$, and $jup2$, in which h_1 is the position of the first element removed from the I-box, h_2 is the position of the first element inserted into the produced D-box, $jup1$ is the offset for each movement in the I-box, and $jup2$ is the offset for each insertion in the produced D-box. $jup1$ and $jup2$ are odd integers. The GBT2 algorithm is shown below.

3.3.4 Dynamic Byte Rotation Operation

ShiftRows operation is a method that shifts its position of the same distance within the plaintext state block. In AES, without ShiftRows, it would be our sets of 32-bit encryption system. In fact, ShiftRows makes AES a 128-bit system, thus, enhancing the security level of ShiftRows. Thus, in this study, we develop a DBR algorithm to substitute for ShiftRows. In the DBR, 64 forms of shifting character position within the plaintext state block obey the above shifting rule. The original AES ShiftRows operation is one of the DBRs with index 16. The DBR rotates plaintext P with rotation index n derived from the Round Key using the following method, $0 \le n \le 63$. Its rotation has three types:

Let plaintext $P = p_0p_1p_2p_3\ldots p_{14}p_{15} = w_0w_1w_2w_3$, where $|p_j| = 1$ (byte), $0 \le j \le 15$, $|w_j| = 1$ (word), $0 \le j \le 3$, and the corresponding state be:

$$st = \begin{bmatrix} p_0 & p_4 & p_8 & p_C \\ p_1 & p_5 & p_9 & p_D \\ p_2 & p_6 & p_A & p_E \\ p_3 & p_7 & p_B & p_F \end{bmatrix} = \begin{bmatrix} r_0 \\ r_1 \\ r_2 \\ r_3 \end{bmatrix}.$$

Type 1. $0 \le n \le 15$: Clockwise rotation of P for n bytes. /* The first 16 bytes */

Type 2. $16 \le n \le 39$: There are 0, 1, 2, and 3 rotating counts on each row in st, generating 4! permutations of 0, 1, 2 and 3, i.e., 0123, 0132, ..., 3201, 3210.

Type 3. $40 \le n \le 63$: Repeat Type 2. The state matrix st is substituted by the transpose matrix st^t.

For example, when n = 16, i.e., Type 2, the corresponding permutation is 0123, and the corresponding rotation on st is that r0 has no rotation, ri rotates i byte(s), $1 \le i \le 3$. This was the ShiftRows operation of the AES. The second example is that if n = 38 (also type 2), the corresponding permutation is 3201. The corresponding rotation is that r0 rotates three bytes, r_1 rotates two bytes, r_2 has no rotation, and r_3 rotates one byte.

Following the three types of P, we establish a DBR lookup table called the DBRTable, i.e., Table I, which keeps the possible DBR operation (see Algorithm 7) and can effectively enhance the efficiency of the DBR operation. The element in the DBRTable is the index sequence of ciphertext C corresponding to plaintext P after the DBR operation with rotation index n. For example, if $P = p_0p_1p_2p_3\ldots p_Ep_F$, then

DBR(P, 9) = $p_9p_Ap_Bp_Cp_Dp_Ep_Fp_0p_1p_2p_3p_4p_5p_6p_7p_8$, with index sequences of
9,*A*,*B*,*C*,*D*,*E*,*F*,0,1,2,3,4,5,6,7,8, i.e. subscripts, respectively.

DBR(P,25)=$p_4p_9p_Ep_3p_8p_Dp_2p_7p_Cp_1p_6p_Bp_0p_5p_Ap_F$, with index sequence of
4,9,*E*,3,8,*D*,2,7,*C*,1,6,*B*,0,5,*A*,*F*.

DBR(P,49)=$p_1p_6p_Bp_Cp_2p_7p_8p_Dp_3p_4p_9p_Ep_0p_5p_Ap_F$, with index sequence of
1,6,*B*,*C*,2,7,8,*D*,3,4,9,E,0,5,*A*,*F*.

n	Type 1	n	Type 2	n	Type 3
0	0,1,2,3,4,5,6,7,8,9,*A*,*B*,*C*,*D*,*E*,*F*	16	0,5,*A*,*F*,4,9,*E*,3,8,*D*,2,7,*C*,1,6,*B*	40	0,5,*A*,*F*,1,6,*B*,*C*,2,7,8,*D*,3,4,9,*E*
1	1,2,3,4,5,6,7,8,9,*A*,*B*,*C*,*D*,*E*,*F*,0	17	0,5,*E*,*B*,4,9,2,*F*,8,*D*,6,3,*C*,1,*A*,7	41	0,5,*B*,*E*,1,6,8,*F*,2,7,9,*C*,3,4,*A*,*D*
2	2,3,4,5,6,7,8,9,*A*,*B*,*C*,*D*,*E*,*F*,0,1	18	0,9,6,*F*,4,*D*,*A*,3,8,1,*E*,7,*C*,5,2,*B*	42	0,6,9,*F*,1,7,*A*,*C*,2,4,*B*,*D*,3,5,8,*E*
	…		…		…
3	*D*,*E*,*F*,0,1,2,3,4,5,6,7,8,9,*A*,*B*,*C*	37	*C*,5,*A*,3,0,9,*E*,7,4,*D*,2,*B*,8,1,6,*F*	61	3,5,*A*,*C*,0,6,*B*,*D*,1,7,8,*E*,2,4,9,*F*
4	*E*,*F*,0,1,2,3,4,5,6,7,8,9,*A*,*B*,*C*,*D*	38	*C*,9,2,7,0,*D*,6,*B*,4,1,*A*,*F*,8,5,*E*,3	62	3,6,8,*D*,0,7,9,*E*,1,4,*A*,*F*,2,5,*B*,*C*
5	*F*,0,1,2,3,4,5,6,7,8,9,*A*,*B*,*C*,*D*,*E*	39	*C*,9,6,3,0,*D*,*A*,7,4,1,*E*,*B*,8,5,2,*F*	63	3,6,9,*C*,0,7,*A*,*D*,1,4,*B*,*E*,2,5,8,*F*

Table I: The DBRTable storing rotation index n and its sequence for plaintext $P = p_0p_1p_2p_3\ldots p_Ep_F$

Algorithm 7: DBR Operation for Encryption
Input: Plaintext P, where $|P| = 16$ (bytes); rotation index n, $0 \le n \le 63$;
Output: Ciphertext C, where $|C| = 16$(byte)
1. Let $P = p_0 \| p_1 \| p_2 \| p_3 \| p_4 \| \ldots\ldots \| p_{12} \| p_{13} \| p_{14} \| p_{15}$;
   /* logically partition PW_1 into 16 eight bits string*/
2. For $i = 0$ to 15
3. $P[i] = \text{int}(p_i)$; /* extracting ASCII value of p_i */
4. $C[i] = P[\text{DBRTable}(n, i)]$; /* extracting element i of DBR using DBRTable(n, i) */
5. $c_i = \text{Char}(C[i])$;
6. next i;
7. $C = c_0 \| c_1 \| c_2 \| c_3 \| c_4 \| \ldots\ldots \| c_{12} \| c_{13} \| c_{14} \| c_{15}$;
8. **return** C;

n	Type 1	n	Type 2	n	Type 3
0	0,1,2,3,4,5,6,7,8,9,*A*,*B*,*C*,*D*,*E*,*F*	16	0,*D*,*A*,7,4,1,*E*,*B*,8,5,2,*F*,*C*,9,6,3	40	0,4,8,*C*,*D*,1,5,9, *A*,*E*,2,6,7,*B*,*F*,3
1	*F*,0,1,2,3,4,5,6,7,8,9,*A*,*B*,*C*,*D*,*E*,	17	0,*D*,6,*B*,4,1,*A*,*F*,8,5,*E*,3,*C*,9,2,7	41	0,4,8,*C*,*D*,1,5,9,6,*A*,*E*,2,*B*,*F*,3,7
2	*E*,*F*,0,1,2,3,4,5,6,7,8,9,*A*,*B*,*C*,*D*	18	0,9,*E*,7,4,*D*,2,*B*,8,1,6,*F*,*C*,5,*A*,3	42	0,4,8,*C*,9,*D*,1,5,*E*,2,6,*A*,7,*B*,*F*,3
	…		…		…
13	3,4,5,6,7,8,9,*A*,*B*,*C*,*D*,*E*,*F*,0,1,2	37	4,*D*,*A*,3,8,1,*E*,7,*C*,5,2, *B*,0,9,6,*F*	61	4,8,*C*,0,*D*,1,5,9,*A*,*E*,2,6,3,7,*B*,*F*
14	2,3,4,5,6,7,8,9,*A*,*B*,*C*,*D*,*E*,*F*,0,1	38	4,9,2,*E*,8,*D*,6,3,*C*,1,*A*,7,0,5,*E*,*B*	62	4,8,*C*,0,9,*D*,1,5,2,6,*A*,*E*,*F*,3,7,*B*
15	1,2,3,4,5,6,7,8,9,*A*,*B*,*C*,*D*,*E*,*F*,0	39	4,9,*E*,3,8,*D*,2,7,*C*,1,6, *B*,0,5,*A*,*F*	63	4,8,*C*,0,9,*D*,1,5,*E*,2,6,*A*,3,7,*B*,*F*

Table II: The InvDBRTable storing rotation index n and its sequence for ciphertext $C = c_0c_1c_2c_3\ldots c_Ec_F$

Table II lists the results of the inverse DBR operation (see Algorithm 8).

Algorithm 8: DBR Operation for Decryption
Input: Ciphertext C, where $|C| = 16$ (byte),
rotation index n,where $0 \le n \le 63$;
Output: Plaintext P, where $|P| = 16$ (byte); /* divide C from 1024 bits into 128 bytes */
1. Let $C = c_0 \| c_1 \| c_2 \| c_3 \| c_4 \| \ldots\ldots \| c_{12} \| c_{13} \| c_{14} \| c_{15}$; /* logically partition*/
2. for $i = 0$ to 15
3. $C[i] = \text{int}(c_i)$; /* extracting ASCII value of c_i */
4. $P[i] = C[\text{InvDBRTable}(n, i)]$; /* extraction of element i using InvDBRTable(n, i) */
5. $p_i = \text{Char}(P[i])$;
6. next i;
7. $P = p_0p_1p_2\ldots\ldots p_{14}p_{15}$;
8. **return** P;

4. Security and Qualitive Analyses

AES utilizes a reversible Key Schedule, making it vulnerable. But the DAES utilizes an irreversible Key and Parameter Schedules, which provides additional security features compared to AES. However, the four different operations in the AES round are interconnected and affect the overall security; a change in one will result in different ciphertexts. Therefore, quantitative analysis is not feasible in this paper. Next, we will use a qualitative analysis approach for discussion.

4.1 The Security on Operations of Transformation Rounds

The DAES framework follows that of the AES. Hence, the security analyses of the three dynamic operations (SubBytes, ShiftRows, and AddRounkey) follow the following rules.

(1) **SubBytes:** (A) For the same look-up operation, both the AES and DAES have the same operational security levels; (B) The S-box of AES is fixed and well known. Hence, the effective security length of the variable (ESLV for short) of the S-box is zero where the ESLV is defined as the security level of a variable. However, with the DAES, the 10 D-boxes used in each round (D_1-box–D_{10}-box) are derived from the D_0-box and RK_0. Both are all unknown to users, meaning that the ESLV of these 10 D-boxes is 1953 (= 1683 × 1 + 30 × 9) bits, where 1683 is the ESLV of the D_1-box and 30 is the ESLV of each Di-box, $2 \leq i \leq 10$.

In the DAES, the D-box is produced following the CK/PW. Without knowing the CK/PW, it is impossible for an attacker to acquire the D-box. Instead, guessing the D-box's contents is the probable method.

Claimed 1: If the original D-box is known and the RK is 128 bits long, then the ESLV of the D-box. derived from GBT2 is 30 bits.

Proof:

With GBT2, the D-box derived from the original D-box is determined by h_1, h_2, jup_1 and jup_2, $0 \leq h_1, h_2 \leq 255$ and $0 \leq jup_1, jup_2 \leq 127$. The possible selection of them is 2^{30} (= 256 × 256 × 128 × 128); Thus, the ESLV of the D-box derived by GBT2 is 30 bits. (Q.E.D.)

(2) **ShiftRows:** The ShiftRows operation in AES in each transformation round uses the same version, whereas the number of choices in the DBR is 64 in each transformation round and 64^r for each encryption, where r is the number of transformation rounds. In the DAES, r=10.

(3) **AddRoundKey**: AddRoundKey operations in the AES and DAES are identical. The ESLV is 128 bits. In the AES, once RK_0 is produced, the following RKs is then determined, i.e., having no contribution to ESLV, except RK_0. In the DAES, RK_i is derived from RK_{i-1} and D_i-box. If RK_0 is known, the following RKs are still unknown because it lacks D_i-box, $1 \leq i \leq 10$, i.e., the ESLV of AddRoundKey in the DAES is 128 × 11 = 1408 bits.

4.2 OSS and ESLV of the AES and DAES

Generally, the AES and DAES frameworks are similar. The operational security strength (OSS) and ESLV are two characteristic indicators and the SubBytes, ShiftRows, and AddRoundKey, in each transformation round of the AES are defined as operational security strengths of 1. Table III shows the results of the qualitative analysis.

Indicator / Operation	AES		DAES	
	OSS	ESLV (bits)	OSS	ESLV (bits)
SubBytes	1	0	1	1953
ShiftRows/DBR	1	0	64^{10}	0
AddRounkey	1	128	1	1408

Table III: Security analyses of the DAES and AES in terms of bit length **(**OSS: Operational Security Strength; ESLV: Effective security length of variables.)

4.3 Performance Analyses of the DAES

Experiments were performed using a computer with 11th Gen Intel Core i7-11700 2.5GHz CPU, 16 logical and 8 physical cores with 16GB of main memory and the operating system was Windows 10. The language used is C# to evaluate the performance of the AES and DAES in terms of their Key Schedule and Key and Parameter Schedules using BenchmarkDotNet given the following parameters: launchCount 22, iterationCount 50, and global setup after loading the plaintext file into memory.

Table IV lists the results of single-block encryption and various plaintext sizes. The time spent by the Key and Parameter Schedules is 204.948 times that of the Key Schedule, due to the production of D-boxes. Table V lists the time spent by running GBT1 and GBT2. The replacement of the GBT1 with the GBT2 to yield D_2-box–D_{10}-box not only improves the execution efficiency, but also ensures the security of D-boxes.

Table VI presents the encryption times consumed for a single-block encryption. In fact, the time-consumption ratio is only 1.2757. Table VII shows the encryption times on different plaintext sizes. For smaller sizes, the difference is significant. The ratio of DAES to AES was 1.8, due to the additional time required for the Key and Parameter Schedules of the DAES; the effect decreases as the plaintext sizes are higher. The ratio on 100 MB is 1.2782, which is not far away from that on a single block, 1.2757 (See Table VI).

Method	Time Consumed (µs)
Key Schedule (AES)	0.097
Key and Parameter Schedules (DAES)	19.88
DAES/AES	204.948

Table IV: The performance of the Key Schedule and the Key and Key and Parameter Schedules.

Method	Time consumed (µs)
GBT1	6.32
GBT2	0.23
GBT1/GBT2	27.47

Table V: The time consumed for the execution of GBT1 and GBT2 Algorithms.

Method	Time consumed (µs)
AES	0.5321
DAES	0.6788
DAES/AES	1.2757

Table VI: The time consumed for single block encryption.

Method / File Size	AES	DAES	DAES/AES
1 KB	0.0318	0.0573	1.8007
10 KB	0.3143	0.4239	1.3487
100 KB	3.3214	4.2697	1.2855
1 MB	34.5190	44.3513	1.2848
10 MB	331.1215	423.5060	1.2790
100 MB	3312.8377	4234.5784	1.2782

Table VII: The time spent given different plaintext sizes (ms).

5. Conclusions and Future Studies

In this paper, we propose the DAES, which inherits the confusion and diffusion characteristics of the AES, while in each transformation round: (1) D-box is employed to replace the S-box to increase the contribution corresponding to the 1683/30 bits of the ESLV for enhancing the security of the SubBytes operation; (2) Replacing ShiftRows operation with the DBR, which provides 64 variations. Now, we dare to say that the AES is one with 1/4 dynamics. and we upgraded it to a 3/4 dynamic one.

Furthermore, we lengthen of CK/PW to a range between 128 and 1024 bits to effectively resist brute-force attacks and enhance its user friendliness. The functionality of the Key and Parameter Schedules is expanded and adopt an irreversible algorithm, i.e., DSM, with an excellent avalanche effect to lower the correlation among keys and among parameters, thus effectively increasing system security.

Currently, the functions of computer hardware and software are improved quickly. 128-bit block cipher encryption mechanisms may soon be unsafe. The data delivered through commercial networks or the Internet is often longer than megabyte. Thus, a new cryptography system with a 1024-bit block is one of the research topics of our future studies. We will also formally derive behavior and reliability models for the DAES so that users can comprehend the behaviors and reliabilities of this system before using it. Furthermore, for a security system, validating it with a standard cryptographic evaluation tool, like NIST test suite, is necessary. These issues will be addressed in our future research.

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Data Spaces for Data Exchange across Different Sectors and Promoting Digitalization.*

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Abstract

Electronic-Business Information Systems (EBIS), originally focused on automation, have been evolving towards value creation through the large-scale exchange of digital assets. Nowadays, data has become one of the most valuable assets for every organization. On the other hand, Data Spaces are ecosystems designed to allow multiple organizations or companies to share data in a secure, decentralized, and sovereign manner. This paper aims to describe how Data Spaces can add value within the context of EBIS and how they promote digitalization. In addition, this study will include a number of use cases where Data Spaces have been applied to facilitate the secure exchange and sharing of data across various sectors. Specifically, these include oceanography, the food industry, agriculture, and healthcare. The research will focus on presenting the basic concepts of data spaces, the benefits they can bring to EBIS, and describing use cases to demonstrate that these technologies have already helped to strengthen multiple sectors.

1 Introduction.

Since the start of the digitalization of organizational business operations, there has been a rapid shift in the approach to the global digital economy. Originally, **Electronic Business Information Systems (EBIS)** focused on automation, but this approach has evolved towards creating value through the large-scale exchange of digital assets. Nowadays, data has become one of the most valuable assets for every organization, and traditional systems may face the challenge of integrating data from multiple external sources without compromising security or integrity. Current data-sharing models often rely on centralized platforms that act as intermediaries and this approach can lead to a loss of data control and sovereignty. Therefore, the aim is to steer progress towards a model that is seamless and governed by the data providers themselves, with a focus on the scalability and sustainability of EBIS.

In this landscape, the concept of "Data Space" has emerged in recent years, with the aim of enabling secure data sharing. **Data Spaces**[1] are briefly defined as ecosystems where organizations are able to securely share data coming from different sources, ensuring data sovereignty too. Data Spaces transform the way that companies and organizations share and manage data and this may represent a change in the architectural paradigm for the next generation of E-Business. A Data Space provides a collaborative and structured environment, enabling an infrastructure where participants can share data based on common governance rules and technical standards. These principles are defined by several European initiatives such as the **International Data Spaces Association (IDSA)** [2], **Gaia-X** [3] or the **Data Spaces**

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Support Center (DSSC) [4]. The result of applying these guidelines is an infrastructure in which data from different providers can be shared with consumers in a sovereign and secure manner.

Aside from data sovereignty, the integration of data spaces also represents a step towards semantic interoperability and usage control, contributing to improve security. The Data Spaces paradigm allows digital assets to be presented in conjunction with usage policies and conditions stipulated by their owners. This permits the definition of which actions are permitted once access has been granted and how the shared data will be used, as well as defining time limits or restricting redistribution to third parties. This ability to monitor and enforce policies ensures that technical governance is an operational reality that safeguards the integrity of the asset throughout its entire lifecycle.

Throughout this paper, the intersection between the continuous evolution of EBIS and the implementation of Data Spaces will be examined. The main objective is to develop synergies between EBIS and Data Spaces, and to explore how they facilitate data exchange across different sectors while promoting digitalization. The document is organized as follows:

- **Section 2: Related Work.** This section provides an overview of the evolution of EBIS, as well as setting the context for describing how data spaces work.
- **Section 3: Data Spaces. Technical Framework.** The concepts of data spaces are explored in technical detail, and an overview of the overall architectural design is provided.
- **Section 4: Use Cases. Data Exchange in different areas.** With a greater focus on practical use cases and implementations, this section describes real-world examples from different sectors where data spaces have been used to deliver a product or service.
- **Section 5: Future Work.** This section summarizes the value that data spaces have added to EBIS based on the cases presented, and how they can continue to be applied to promote digitalization.

2 Related Work.

This section reviews the current state of the art regarding the various EBIS approaches and the emerging Data Spaces ecosystems. Key concepts such as data governance models or usage control enforcement are presented to establish a theoretical foundation.

2.1 E-Business Information Systems (EBIS).

The digitalization of business operations has been a phased evolution. Chronologically, the first technical milestone was the emergence of **Electronic Data Interchange (EDI)** [5] in the 1980s which enabled the structured exchange of commercial documents between organizations via private networks, laying the foundations for **Business-to-Business (B2B)** automation by eliminating paper from invoicing and ordering processes.

Due to the rapid growth of digitalization in recent years and the growing complexity of global markets, the integration of critical end-to-end business processes was required, but EDI focused on data transmission. **Electronic Business (e-Business)** emerged as a strategic response to the need to integrate business operations beyond the physical boundaries of traditional office processes. This paradigm comes with a major change not only in the commercial activity but also in the organization. E-Business can be defined as “the practice of performing and coordinating critical business processes such as designing products, obtaining supplies,

manufacturing, selling, fulfilling orders, and providing services through the extensive use of computer and communication technologies and computerized data". [6] This shift was driven by the need for greater agility and the realization that data, when digitized and shared, becomes the primary asset for competitive advantage.

Based on the foundational findings of the **Organization for Economic Co-operation and Development (OECD)** report[7], there are five core characteristics [6] of e-Business systems that can be synthesized:

- **The use of information and communication systems to standardize, automate, and integrate transactions.** Businesses have been using Electronic Data Interchange (EDI) to automate routine business transactions. This early stage established the first pillar of modern e-Business: the use of standardized information systems to automate and integrate transactions. From this technical foundation, e-Business emerged as a broader strategic practice, transforming simple data exchange into a systemic coordination of the entire value chain.

- **The transformation of marketplaces and the emergence of entirely new industries.** Electronic commerce is giving rise to entirely new industries and is transforming the nature of existing ones.

- **An increased connectivity and interactivity between business partners over electronic networks.** The use of a communication network provides a common, ubiquitous communications infrastructure, allowing for the seamless integration of internal business processes with those of suppliers and customers.

- **The alteration of the relative importance of time and resources in favor of digital agility.** Electronic commerce allows firms to better manage their supply chain, substituting information for physical inventory.

- **The catalytic effects of both inter- and intra-organizational transactions on the broader economy.** Electronic commerce has a catalytic effect that will stimulate more efficient ways of organizing and conducting business across the whole economy.

These five characteristics of e-Business systems may lead to various benefits, like operating more efficiently or creating more efficient business processes. Taking into consideration that e-Business is properly planned, deployed and supported, these potential benefits could be attainable in every organization. Information Systems support the fulfilment of these principles and are understood to be the foundation of e-Business.

EBIS enable companies to move away from a linear approach and start using digital 'building blocks'. It is important for organizations to carefully plan architectural frameworks for these systems, an e-business architecture planning model requires a multi-dimensional approach to align technical capabilities with business objectives, this planning framework is structured around two main components:

- **Axes that define the needs of the information architecture.** Originally[8], two axes were used to classify these needs. This evolved to three axes on which drivers of the information architecture needs of e-business firms fall.

  - *Functional Integration.* Measures the extent to which the value chain processes are digitally covered.

  - *Degree of Innovation.* Assesses the extent to which the system enables new value creation, moving from the mere digitalization of traditional tasks to the birth of native electronic markets.
  - *Structural Integration.* Evaluates the depth of the digital relationship with business partners, ranging from simple transactions to full ecosystem synchronization.

- **Generic e-business models.** They are the specific implementations, the choice of model determines how digital assets are aligned to achieve a sustainable competitive advantage. While eleven models were initially identified [8], research expanded this to twelve [9]. These models range from the operational efficiency of e-procurement to the creation of ecosystems on collaborative platforms.

Regarding the EBIS success drivers, the organization's ability to evolve through stages of maturity that validate that architecture is one of the fundamental pillars. Ensuring the right maturity level [6], and being able to measure the maturity level of **Information Systems (IS)** is about how well the IS is integrated into the organization's strategy. In order to realise the expected benefits, organizations need to turn to their EBIS and measure the maturity of those systems. In doing so, they need to identify the status of those systems with regards to their suitability to support the e-Business strategy, while also identifying required IS improvements.

2.2 Data Spaces Ecosystems.

Driven by Europe's data strategy, the **European Data Spaces initiative** [10] aims to build collaborative ecosystems based on sovereign, trusted, and interoperable frameworks. These facilitate cross-sector data sharing, allowing organizations to utilize data effectively while retaining full ownership and control.

Regarding the core benefits, **Data Spaces** [11] offer advantages that transform information management. Decentralized data exchange is facilitated, allowing each participant to retain custody of the information within their own infrastructure. This guarantees data sovereignty, ensuring that owners control and decide on how and when their data are used. Data Spaces facilitate decentralized data sharing, thereby driving value creation, fostering the development of innovative solutions and promoting the responsible data usage across the entire ecosystem.

Compared to conventional sharing methods, Data Spaces embed trust through clear governance models and technical standards, enabling a federated exchange in which each participant maintains data sovereignty, transparency, and accountability. These standards are established by the different initiatives that align efforts across Europe's data economy.

The **Data Spaces Business Alliance (DSBA)** [12] brings together four leading bodies: **Big Data Value Association (BDVA)** [13], **FIWARE Foundation** [14], **Gaia-X European Association for Data and Cloud** [15] [16] and **International Data Spaces Association (IDSA)** [17].

Under DSBA's umbrella, the **Technical Convergence Document** [18] defines a unified **Blueprint Architecture**, that establishes a shared reference architecture for building, governing, and scaling data spaces. The **Data Spaces Support Centre (DSSC)** [4], funded by the **Digital Europe Programme** [19], translates the Blueprint into practice[20]. It offers detailed blueprints and reference implementations, training materials and best-practice guides and alignment with EU policy goals and regulatory requirements.

This blueprint arranges what is needed to construct them into **Building Blocks** that contain the capabilities every data space needs, as well as best practices and specifications to implement these capabilities.[21] These Building Blocks are divided into two groups:

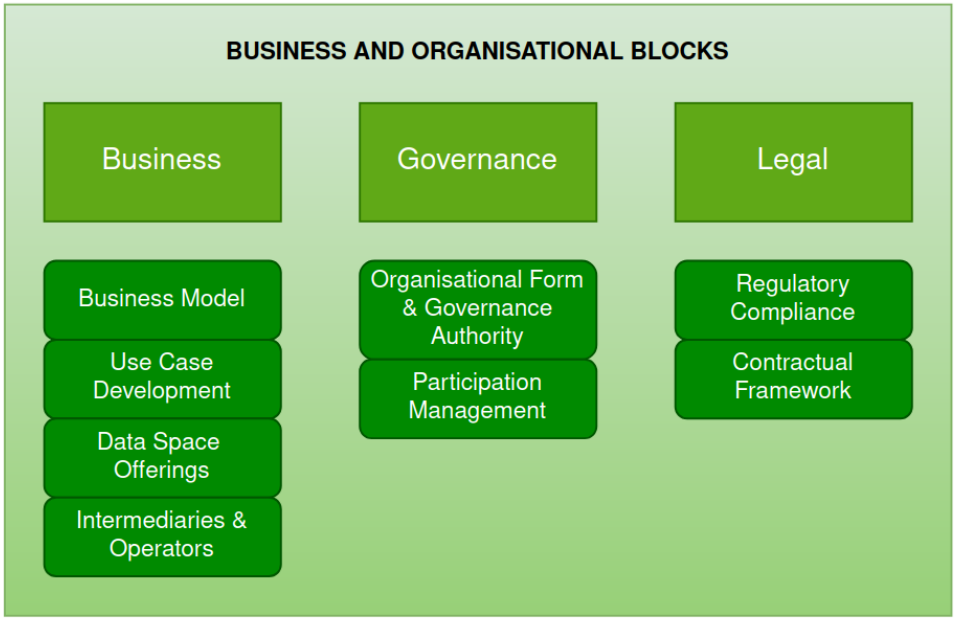

Figure 1: Business and Organizational Building Blocks.[22]

- **Business and Organizational Building Blocks.** [22] They help data space initiatives and their participants to build sustainable, compliant, and value-driven systems. As shown in Figure 1, these blocks are organized across three key pillars that shape the functionalities available to participants, guiding developers in creating robust data spaces and enabling users to effectively leverage the offerings within them.

- **Technical Building Blocks.** [23] These building blocks contemplate technical capabilities that are needed to implement data spaces. Using these building blocks makes it easier to reuse technical solutions and achieve interoperability both within and between dataspaces on a technical level. This group is also organized across three key pillars that are shown in Figure 2, regarding data interoperability, sovereignty and value creation.

The architecture of a data space is built around several core concepts that ensure the system is decentralized, secure, and interoperable. A data space is built upon the previously mentioned Technical Building Blocks, that help to create a modular, scalable, and interoperable system, enabling secure data exchange and collaboration among different entities. These building blocks are based on different tasks that are needed and integrated to create a functional data space.

Developing a data space is a complex task due to the balance that must be between business and organizational aspects, as well as technical aspects while ensuring sustainability. That is why there is a **Co-Creation Method** [24] that provides a structured, step-by-step approach to help data space participants navigate these challenges, ensuring informed decision-making and efficient collaboration. The Co-Creation Method will guide data space initiatives through the different stages of the development cycle, which is relevant for the evolution of the data space.

A member of the Data Space is a participant, a subject that is involved in the data-sharing

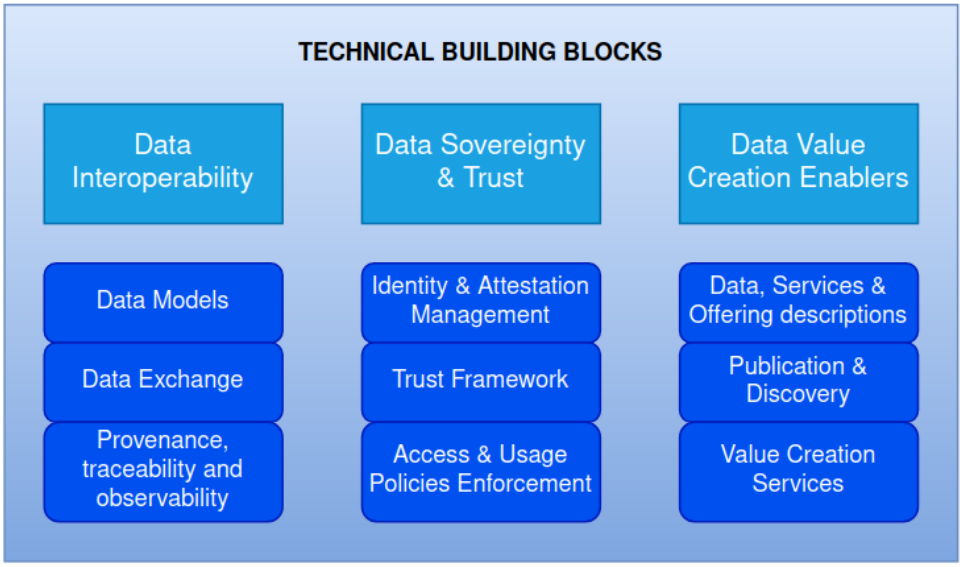

Figure 2: Technical Building Blocks.[23]

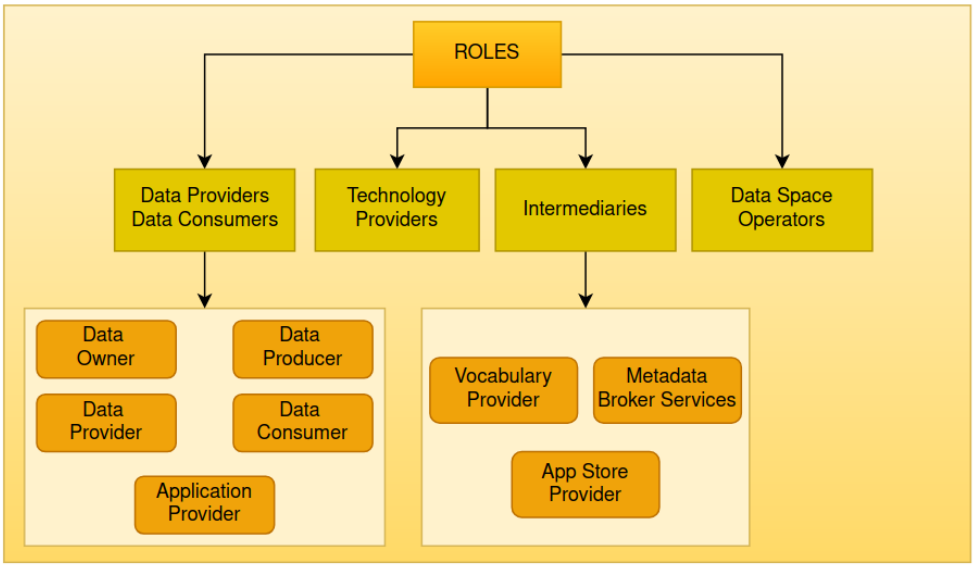

Figure 3: Roles involved in a data space.[25]

activities within the ecosystem. This participant can adopt different **roles** depending on its function and what it contributes to the Data Space, the most relevant are:

- **Data Owners.** Stipulates the conditions under which a consumer can use the data.
- **Data Providers.** Publish metadata and datasets to Catalogs [26], that is the mechanism

by which data is offered.

- **Data Consumers.** Access to Catalog, where they discover the offered data and consume datasets under agreed terms.
- **Data Intermediaries.** Offer value-added services (e.g., quality assurance, transformation).
- **Technology providers.** Supply and maintain underlying infrastructure and tools.

Participants use a **Connector** [27] [28], a fundamental component that encapsulates their role and provides a standardized interaction point among participants in the Data Space. This component is responsible for mediating secure data transfers between participants while enforcing access-control and usage-control policies according to owner stipulations. A Connector performs operations on behalf of a Participant, offers Datasets, produces Agreements and manages Dataset sharing while enforcing access-control and usage-control policies according to owner stipulations. [29] A Data Space architecture is split into a **Control Plane** [30], responsible for deciding how data are managed and processed (e.g. user identification and policy enforcement), and a **Data Plane** [31], responsible for the actual data sharing process. The connector acts as the point of convergence between the two planes: it operates in the control plane to negotiate contracts and verify identities, thereby ensuring that any movement of information in the data plane strictly complies with the agreed usage policies. A Data Space could count on different components depending on the approach adopted. In spite of this, there are some of them that are strictly necessary for a Data Space to perform its basic activity as required. One of these components and one of the main elements in the context of data spaces is the previously mentioned Connector. Many organizations have developed their own implementation of a Base Connector to seamlessly integrate within their ecosystem. Key players like the **Eclipse Foundation** [32] and the **FIWARE Foundation** [33] have been instrumental in this development by providing the reference implementations that define the modern data space landscape.

3 Data Spaces: The Next Step for EBIS.

Having introduced the basic concepts needed to understand what a data space is, this section will explore the technical implementation and how data spaces can be aligned with EBIS.

3.1 Technical Framework.

This section will examine the **Technical Building Blocks** (see Figure 2) previously mentioned in greater detail, in order to provide a foundation and context for the protocols under which Data Spaces operate. There are three groups within those technical building blocks:

- **Data Interoperability.** This block establishes the capabilities for data interoperability by defining data formats, interfaces (APIs), and semantic models that enable shared understanding within a data space. Beyond facilitating the actual exchange and sharing of data, it integrates robust mechanisms for provenance, traceability, and observability, ensuring that the entire lifecycle of data movement is documented and can be audited for regulatory compliance.
- **Data Sovereignty and Trust.** This block defines the capabilities for Data Sovereignty and Trust, establishing the essential framework for identifying participants and assets

while ensuring policy enforcement. Through Identity and Attestation Management, the system maintains the reliability and integrity of participant information, which is further validated by a Trust Framework designed to verify adherence to established rules. Ultimately, these components work together to enable the precise specification and enforcement of access and usage policies, ensuring that data remains under the control of its owner throughout its lifecycle.

- **Data Value Creation Enablers.** This block focuses on the capabilities necessary to generate tangible value within a data space by providing clear descriptions of data, services, and offerings that are understandable to all participants. These descriptions are leveraged through publication and discovery mechanisms, such as data catalogs, which allow potential users to locate and access relevant data assets efficiently. Furthermore, this block integrates value creation services that facilitate the development of additional shared tools and services, setting the necessary conditions to ensure the effective provision, delivery, and utilization of data-driven insights.

The technical realization of a Data Space moves away from centralized platforms towards a distributed architectural pattern. Based on the principles previously discussed, the framework is operationalized through the different protocols established by the European Data Spaces Initiatives.

3.1.1 Data Space Protocol.

The technical building blocks are the technical capabilities that the architecture of a Data Space requires, they will be implemented with the **Data Space Protocol (DSP)**. [34] [35]

The pillars of the DSP are the operational tools that implement the Technical Building Blocks functionalities. The DSP is the main technical support that standardizes how participants interact. At its core, the framework transitions to a Connector-based architecture. As it has been introduced previously, every participant (Providers and Consumers) operates a Connector component that serves as the sovereign gateway. These gateways communicate using a common language to handle metadata, legal constraints, and actual data transmission.

The DSP, which ensures interoperability through functional technical pillars, it is divided into three critical protocols that manage the lifecycle of a data exchange, represented on Figure 4.

- **Catalog Protocol.**[26] A **Catalog** is a collection of entries representing **Datasets** and their **Offers** that is advertised by a Provider Participant. The Catalog Protocol defines how a Catalog is requested from a Catalog Service by a Consumer using an abstract message exchange format. This is the discovery phase. Instead of a central database, the Catalog Protocol allows participants to broadcast or query available data offerings. It uses a federated approach where the Connectors exchange information about what data is available, its format, and the initial terms of use. This protocol follows the **DCAT Vocabulary** [36] and covers the Technical Building Blocks tasks that are related to **Data Value Creation Enablers**, specifically regarding data, services, and offering descriptions, as well as publication and discovery capabilities. By utilizing standardized metadata, it also supports **Data Interoperability**, ensuring that shared semantics and data models are consistently applied so that any participant can understand and interact with the discovered assets within the data space.

- **Contract Negotiation Protocol.**[37] A Consumer chooses an **Offer** from the Catalog, so that chosen asset offered has to be negotiated between the Provider and the Consumer.

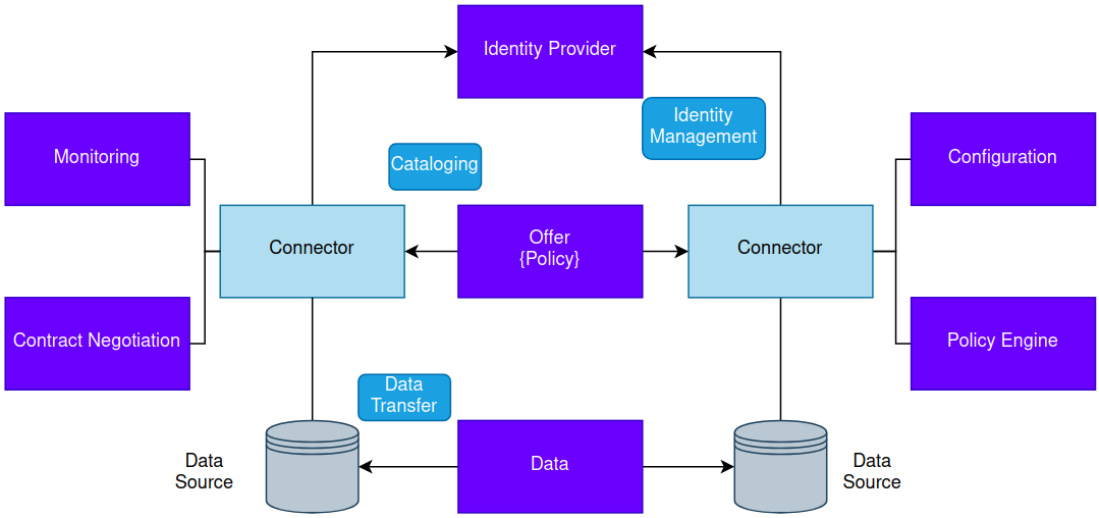

Figure 4: Overview on DSP and context. [34]

An Offer contains a concrete **Policy** associated with a specific Dataset. This protocol manages the negotiation of Usage Policies that are contained in that Offer and will result in a legally binding, digitally signed **Contract Agreement** between the **Provider** and **Consumer** that establishes the conditions under which the Data negotiated will be used. This protocol will use the **ODRL Information Model**[38] to represent Offers, Policies and Agreements. This protocol covers the Technical Building Blocks tasks that are related to **Data Sovereignty and Trust**, specifically the enforcement of Access and Usage Policies. It provides the mechanism to transition from a general Trust Framework to a specific agreement, ensuring that identity attestations and usage constraints are implicit in a digital agreement before any data transfer starts.

- **Data Transfer Protocol.**[39] Whereas the previous two protocols operate within the Connector's Control Plane (the management layer where discovery and legal agreements are handled), this protocol operates in the **Data Plane**. This protocol handles the actual movement of data bits and information. It supports various transport mechanisms (like HTTP, MQTT, or SFTP) while ensuring that the transfer is logged and compliant with the issued Contract Agreement, maintaining the sovereignty of the provider throughout the process. This protocol focuses on the Technical Building Blocks functionalities that are related to **Data Interoperability and Data Exchange**, providing the actual capabilities for the physical exchange of data. Furthermore, it implements the Provenance, Traceability, and Observability requirements by generating the necessary logs to audit the transaction and ensure it adheres to the negotiated Usage Policies Enforcement defined in the **Sovereignty and Trust block**.

3.1.2 EDC Implementation

Eclipse Dataspace Components (EDC)[32] is a set of components that enable developers to create dataspaces using a set of specific building blocks. An Identity Service that is used for managing and verifying participant identity credentials using **Decentralized**

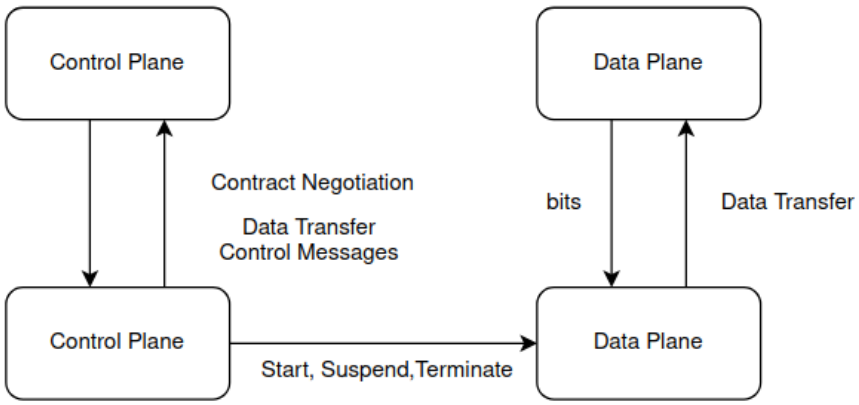

Figure 5: EDC Connector.[46]

Identifiers(DIDs)[40] and **W3C Verifiable Credentials (VCs)** [41] or **OAuth2 tokens** [42]. There is also a Catalog Service is meant for publishing assets that can be shared within the ecosystem. The Control Plane Services are in charge of automated creation and processing of data usage agreements that grant access to data, while the Data Plane and Monitoring services manage data transfers using transfer protocols.

EDC Components are standards-based and implement:

- **Dataspace Protocol Specification** [35] that define the messages exchanged in a dataspace EDC implements and builds on these asynchronous messaging patterns. how to retrieve data catalogs, conduct negotiations to create contract agreements that grant access to data, and send data over various lower-level wire protocols
- **Decentralized Claims Protocol Specification** [43], that is a set of protocols for asserting participant identities and define how trust is established between participants. The Decentralized Claims Protocol (DCP), layers on DSP.

The **Eclipse Dataspace Components (EDC) Connector** is an open-source framework designed for secure, sovereign, and policy-controlled data exchange within the data space. It enables organizations to share data while maintaining control over usage, implementing **International Data Spaces (IDS)** standards. The Connector is composed by a pair of components:

- **Control Plane**[44]**.** Controls data sharing and execute data transfer. It handles all protocol logic, including identity verification, catalog management, and contract negotiation. It is in charge of communicating with other connectors but never touches the actual raw data.
- **Data Plane**[45]**.** Is responsible for executing data transfers, sends data using specialized technology such as a messaging system or data integration platform.

The interaction between both planes is described in Figure 5. EDC components are deployed

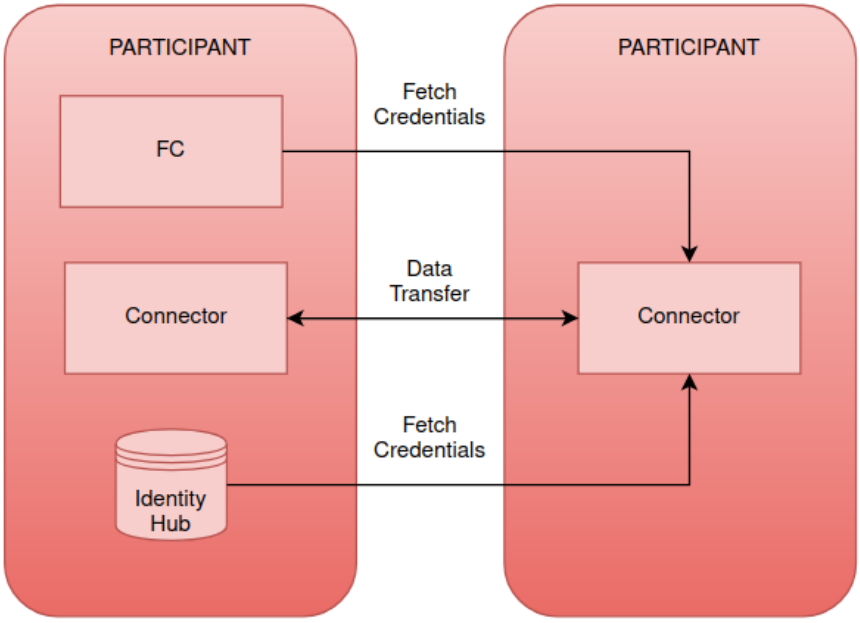

Figure 6: EDC Dataspace Context.[46]

to create a dataspace ecosystem. At its most basic level, a dataspace can be a context between two participants, as shown in Figure 6. The **Federated Catalog (FC)**[47] is responsible for crawling and caching data catalogs from other participants, it builds a local cache that can be queried or processed without resorting to complex distributed queries across multiple participants. A Connector negotiates a contract agreement for data access between two participants and manages data transfers using a data plane technology. The **Identity Hub (IH)** [48] presents verifiable credentials that a participant connector uses to determine whether it trusts and should grant data access to a counterparty. The IH securely stores and manages W3C VCs, including its presentation and issuance process.

The **EDC Minimal Viable Dataspace (MVD)**[49] sets up and runs a complete demonstration of a data space environment between two organizations.

3.2 Bridging EBIS and Data Spaces Ecosystems.

As data has become a critical asset for organizations, the current challenge for EBIS is no longer the digitalization of processes, but data governance and sovereign sharing. Data Spaces provide the necessary technical and legal layer to upgrade traditional EBIS into an interconnected ecosystem through the following core pillars:

- **Sovereignty and Decentralized Trust Framework.** Data Spaces implement a **Trust Framework** using DIDs. This allows to verify the identity of a partner that wants to participate in the data sharing process. For an EBIS, this means moving to automated, cryptographically-secured trust, where every transaction is legally binding and non-repudiable. **Decentralization** allows the EBIS to scale its network of partners rapidly, as trust is established through technical protocols. Regarding access control

and negotiation processes, Data Spaces introduce usage policies enforcement, allowing a company to share business-critical data while maintaining absolute control.

- **Interoperability. Semantic interoperability** is covered by adopting the **Catalog** concepts, EBIS data becomes discoverable and self-describing for those that can consume those data. By aligning with common data models, the EBIS ensures that business semantics are interpreted identically across different systems.

- **Value Creation.** Through the **Catalog**, an EBIS can transform its internal data into a data product available for other participants. The Data Space allows for the integration of third-party services that can process EBIS data to provide higher-level insights, such as **cross-organizational** analytics or automated compliance reporting.

- **Compliance.** The implementation of standards ensures that every exchange of business information is traceable, providing a **clear audit trail for tax and regulatory compliance**, and allowing the EBIS to monitor the status of data transfers in real-time. Organizations can monitor the status and usage of their data assets, ensuring that all external interactions remain compliant with internal governance and external regulations.

By bridging EBIS with Data Spaces, organizations move beyond the structured and limited exchange into an era of **Sovereign Intelligence**. The transition from traditional EBIS approach to Data Spaces represents a qualitative leap from the mere digitalization of processes towards a sovereign and federated data economy. By natively integrating decentralization and usage control, this architecture resolves the long-standing dilemma between open data and corporate security, enabling organizations to share critical information without compromising their competitive advantage. Data Spaces not only strengthen the resilience of systems through automated cryptographic trust, but they also set a standard for dynamic governance that transforms data into a liquid, secure and auditable strategic asset.

4 Use Cases. Data Exchange in different areas.

The University of Murcia [50] has been working with Data Spaces, drawing on EDC Components and implementing these ecosystems to apply them to Use Cases across different sectors and contexts.

4.1 Data Exchange in Oceanography.

The project called **NEREIDAS [51]** is a **Demonstration Centre** for data spaces for research and innovation in **marine sciences** in the south-western Mediterranean.

This is a managed technological platform for data spaces where various stakeholders (companies, public authorities, etc.) can carry out pilot projects and proof-of-concept tests relating to data exchange. It is an environment where data spaces are tested, validated and explained in real-world scenarios. NEREIDAS aims to demonstrate:

- **How to share data securely and independently.**

- **How to drive research and innovation through new data-driven services.**

- **How to train organizations in data technology and governance, fostering public-private collaboration.**

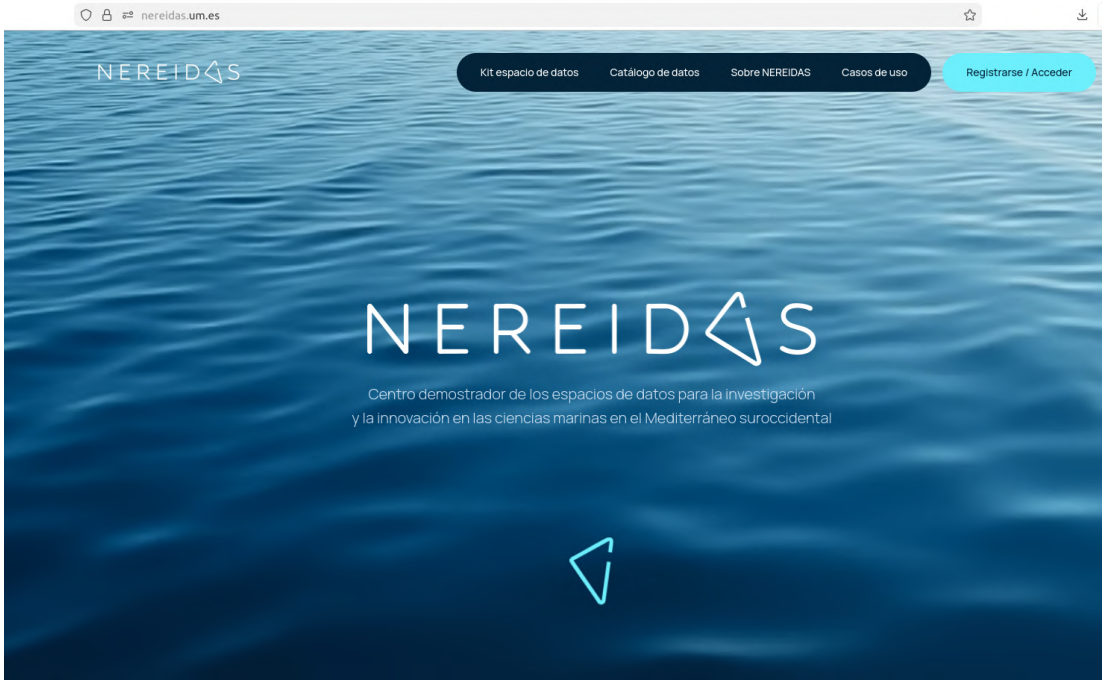

Figure 7: NEREIDAS home page.

NEREIDAS illustrate how data spaces drive **marine research and innovation in real-world scenarios**, fostering public-private collaboration, testing and validating how data ecosystems function under real-world conditions, working with the infrastructure, standards and technologies common to **European Data Spaces**. The main goal is to make this technology accessible to organizations that produce and use marine data, ensuring sovereignty, trust and security in the exchange of information.

NEREIDAS processes data on water quality, biodiversity, meteorology and catchment areas, sourced from public bodies, research institutions and the private sector. Data that is essential for understanding and managing the marine and coastal environment is treated within this project. This data is organized into four main categories:

- **Water Quality.** This includes parameters such as **temperature, salinity, chlorophyll, turbidity, dissolved oxygen, nutrients** and **pollutants**, which are key to assessing the ecological status of the marine ecosystem, preventing episodes of eutrophication and supporting decisions on coastal management or the monitoring of sensitive areas such as the Mar Menor.

- **Marine biodiversity.** Datasets on **marine species** (fish, invertebrates, underwater flora), their distribution, abundance and conservation status. These data are essential for studying trends, identifying risks, protecting habitats and designing policies for the sustainable management of the marine environment.

- **Meteorological factors.** This includes **atmospheric data that directly influence coastal and oceanic conditions, such as precipitation, air temperature, wind,**

atmospheric pressure and solar radiation. Incorporating this data helps to improve forecasting models and anticipate events relevant to sectors such as fisheries and aquaculture, tourism and port management.

- **State of the catchment area.** Information on **river channels and flows, nutrient loads, sediments, agricultural runoff** and **pollution** incidents. This data helps to link land-based impacts with changes in the marine ecosystem.

NEREIDAS takes part in the Complementary Marine Sciences Plan, within the **ThinkInAzul** [52] initiative, which connects Spain's coastal provinces to foster a shared ecosystem of data and scientific capabilities. The current situation is that marine and coastal data in Spain is fragmented, NEREIDAS was created to address this fragmentation, facilitate interoperability, and become a leading demonstration center for the secure and sovereign exchange of marine data.

4.2 Data Exchange in Food Industry.

There is an internal project where a Data Space is currently in development, called **Food Tech Specs Data Space**. The project is defined within the scope of the **Beverages Sub-sector**, in the **Agri-Food sector**. Its aim is to create and use a data platform that manages information models detailing **food product specifications, covering raw materials, semi-finished and finished products**, for a marketed food product or a specific manufactured batch.

This project addresses the need to create a use case that utilises data-sharing and processing technologies to achieve a business objective: the development and operation of a data repository that manages standardized models of food product specification information, in general terms for a **marketed food product** and specifically for a **manufactured batch**.

As shown in Figure 8, the main providers are the **primary producers and the producers in the value chain of the agri-food sector**. These would be companies that produce, process and export food products, they provide data such as the **values and technical specifications of those food products and batches, as well as a food safety alert channel**. The consumers who access the data that are being shared in the platform are the **product manufacturers or the distributors of the finished food product**, for instance, a supermarket.

A **Technical Data Sheet** for a manufactured batch is a crucial document that provides detailed information on the composition, characteristics and production processes of a specific product. In the context of beverages, this technical data sheet is essential for ensuring the quality, safety and traceability of products.

The platform aims to achieve the following objectives:

- **To provide the agri-food sector with access to a marketplace, secure and sovereign data-sharing capabilities, and a cloud infrastructure equipped with computing resources across the entire value chain, from a holistic perspective that includes data relating to the management of specifications within the agri-food sector.**

- **To provide a platform that enables the secure, reliable and controllable exchange of information. This will be aligned with current and future European data management strategies, including data traceability, access control to ensure data security and privacy, as well as compliance with European regulations.**

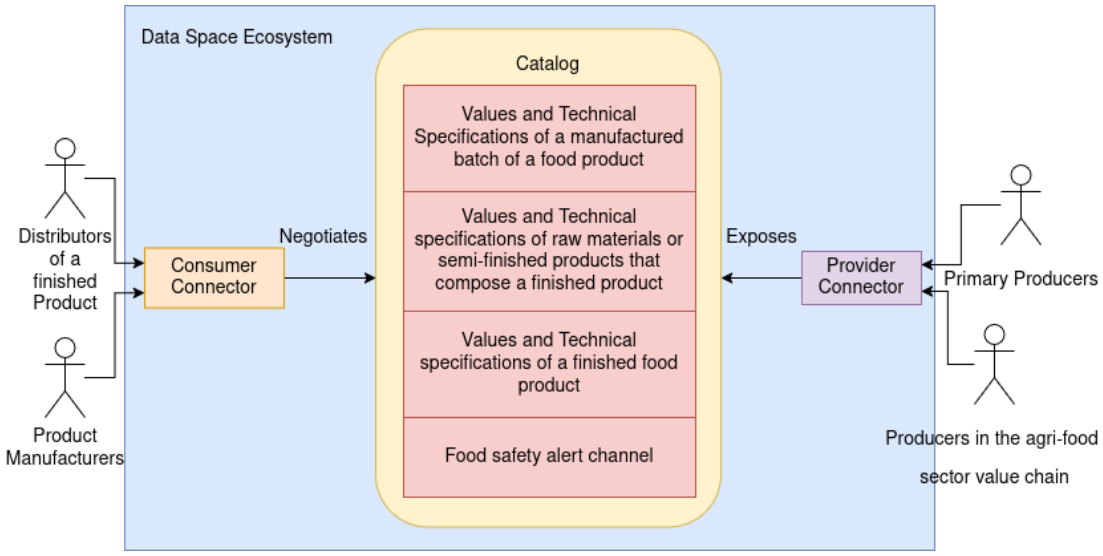

Figure 8: General overview of data sharing process in Food Tech Specs Data Space.

- **Process improvement, simplifying and streamlining the exchange of technical and quality information. The companies involved can improve the efficiency of their processes, reduce errors and response times, which in turn can lead to greater competitiveness and higher product quality.**

This data platform is unique and represents an innovative project designed to boost competitiveness within the agri-food sector, as there is currently no marketplace that facilitates the exchange of this type of information. Also, from the consumer's point of view, there is currently no platform available that allows for the centralized retrieval of information. The creation of this platform enables consumers to access detailed information on agri-food products, providing them with transparency and confidence in what they consume, fostering greater awareness of the quality and origin of the food they purchase.

4.3 Data Exchange in Healthcare and Agriculture.

The **European Open Science Cloud (EOSC)** [53] is an initiative proposed by the **European Commission** [54] and designed to provide an open environment to store and share data, encouraging scientific community to reuse research data. **EOSC Interoperability Framework (EOSC-IF)**[55] are the guidelines and technical, semantic, and legal standards that ensure consistent integration across data providers.

Funded by the European Commission, **EOSC-TITAN** [56] has the overall objective of enriching the EOSC-IF by developing a software platform solution for confidential collaboration and privacy-preserving data processing. The solution will enable access to sensitive datasets from public entities and government agencies and will be compatible by design with the EOSC-IF on the technical, semantic, organizational, and legal layers.

TITAN will leverage hardware support for **Confidential Data Processing and Data Space connectivity**, as well as novel solutions for access control policies and permissions. The project will advance the state of the art to support privacy-preserving data sharing, transparent

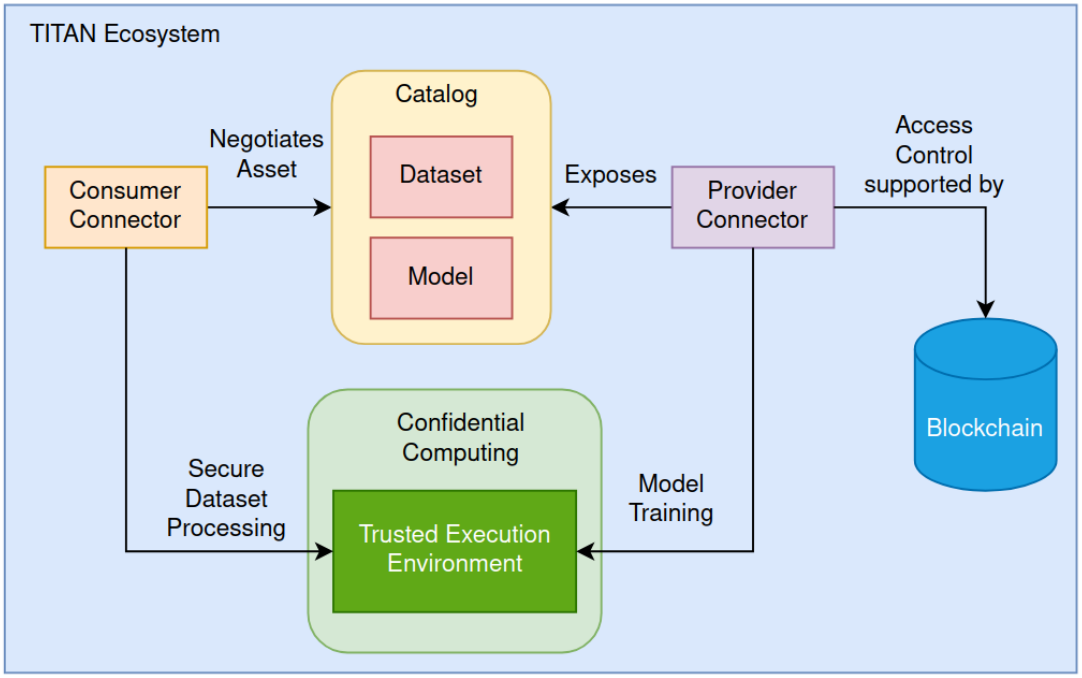

Figure 9: General overview of data sharing process and confidential computing in EOSC-TITAN.

cloud topology, auditable network security, and secure access control. The project will balance the strict security requirements with easy access for users and offer practical solutions for working with large-scale data sources, computational resources, and users.

The TITAN innovations include **Confidential Computing** and data processing based on modern efficient hardware-accelerated secure enclaves called **Trusted Execution Environments (TEEs)** [57], capable of ensuring a truly end-to-end secure data lifecycle, reliable data anonymization, and scalable and fine-grained **decentralized blockchain-based Access Control**[58].

In TITAN Platform, providers expose **datasets, models or tools** that consumers will negotiate. Access control for accessing the assets is based on a Decentralized Blockchain and the platform counts on a support for Confidential Computing where data processing and model training will be executed. Figure 9 presents a general overview of this process.

TITAN will contribute to build the **Common European Data Spaces with Government Data Zones over EOSC** [59]. These Government Zones in EOSC are sovereign, policy-governed environments designed to facilitate the secure exchange and processing of sensitive public sector information within the EOSC infrastructure. TITAN will be engaged with this target group from the government sectors by directly involving two European Regions in the project to use TITAN confidential computing and produce datasets that will become available over the EOSC, thus demonstrating Public Authorities' Government Zones in EOSC.

To promote community adoption of TITAN's open-source software, the solution will be demonstrated in two vertical cross-border scenarios in the **Agriculture and Healthcare sectors**. TITAN contributes to the **European Healthcare Data Space (EHDS)** [60] and

European Agriculture Data Space (EADS) [61], that are European Union initiatives that aims to create a secure and privacy-preserving infrastructure for the exchange of data across the EU.

TITAN services will be tested and validated supporting Confidential Computing for privacy-preserving multi-party **Machine Learning (ML)** approaches, using real-world medical data and algorithms, as well as available agriculture data from both private and public sources from the Aragón (Spain) and Veneto (Italy) regions. This will improve the usability, security, and applicability of the platform.

The pilots carried out in the project focus on real use cases in the **Agriculture and Healthcare domains**, in terms of demonstrating the general functioning of the platform and highlighting the technical solutions developed throughout the project. This diversity of scenarios makes it possible to put in common the problems of various areas in the face of privacy, security, data sharing and data exploitation.

5 Future Work

The architectural transition from traditional EBIS to Data Spaces is just the beginning of a broader digital transformation. As organizations move beyond simple connectivity and automated processes, future research and implementation will likely focus on the role of Data Spaces as the universal framework for the next generation of digitalization. The advancements in **security, sovereignty, and data governance** provided by this paradigm offer a robust foundation, that is applicable to nearly any sector. Future work should explore how the integration of sovereignty and security approach that these technologies provide, can become standard for all EBIS, regardless of their specific business logic. By embedding governance directly into the data exchange layer, future EBIS will not only automate processes but also guarantee the asset protection and integrity by design. This security-by-default point of view will be crucial as regulatory requirements and laws become more stringent across Europe. One of the most promising avenues for future development is the ability to bridge disparate domains. Data Spaces may provide the unique opportunity to relate sectors that have historically operated separately. Finally, future research will investigate the scalability of **Decentralized Trust Frameworks** and enforce the identity processes that are related to signatures and non-repudiation. As thousands of EBIS connect through a federated network, the focus will shift toward optimizing automated discovery and semantic mapping between different sector-specific data models. The goal is to create a truly liquid data economy where information flows securely and purposefully across any boundary, fueled by the legal certainty and technical sovereignty that Data Spaces provide.

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Optimizing Multi-Hop Reasoning in E-Business Information Systems through Agentic Graph RAG with Efficient Reranking*

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Abstract

Traditional Retrieval Augmented Generation (RAG) systems often struggle with multi-hop reasoning and efficiency, limiting their ability to deliver accurate and context-rich responses in E-Business Information Systems. Although Graph RAG improves retrieval by leveraging structured knowledge, it still faces challenges in relevance, scalability, and latency. We propose an Agentic Graph RAG framework, where reasoning-model-based agents (o4-mini, Gemini-2.5-Flash) decompose queries and prune irrelevant paths, and rerankers (bge-reranker-base, ms-marco-MiniLM-L6-v2) enhance the quality of retrieved contexts. By addressing noise, computational overhead, and irrelevant retrieval, our framework effectively overcomes the limitations of traditional RAG and Graph RAG for knowledge-intensive digital business activities. We assess our method using three benchmarks: LegalBench-RAG-mini, which represents high-stakes analytical reasoning; HotpotQA, which evaluates generalizability across broader knowledge domains; and FinanceBench, a publicly accessible financial QA dataset based on real-world corporate disclosures and earnings reports that directly reflects the information demands of E-Business environments. Results demonstrate that our framework consistently outperforms Graph RAG baselines in retrieval efficacy and answer correctness while maintaining practical efficiency, demonstrating its potential as an adaptable and interpretable solution for multi-hop reasoning in domains where accuracy and reliable decision-making are critical.

1 Introduction

Rapid development of Large Language Models (LLMs) has transformed natural language processing, allowing human-like text generation [32]. Traditional LLMs, on the other hand, are prone to hallucinations, outdated information, and domain constraints, making them unsuitable for knowledge-intensive activities in E-Business Information Systems and other business applications. Retrieval Augmented Generation (RAG) tackles these challenges by incorporating external knowledge retrieval, thus improving factual consistency and response legitimacy [6]. However, deeper semantic relationships are frequently missed by conventional RAG architectures, which rely on vector-based dense retrieval. Consequently, they encounter significant

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efficiency bottlenecks when scaled to huge knowledge bases, suffer from noisy or partially relevant retrievals, and struggle with multi-hop reasoning [24]. These limitations are particularly evident in financial information systems, a critical domain within E-Business, where corporate filings, earnings reports, and regulatory disclosures are lengthy, complex documents that include narrative prose, numerical tables, and domain-specific terms, making simple retrieval far too inadequate for the multi-step queries that business professionals frequently encounter [8]. Resolving questions about year-over-year revenue fluctuations or cross-segment spending breakdowns necessitates integrating diverse data across dozens of document sections, a task that requires systematic, multi-hop reasoning rather than a single lookup [22]. The sheer volume and heterogeneity of such data make it difficult to analyze automatically and costly to interpret manually, but clarity and precision are essential in corporate contexts where procurement, compliance, and investment choices are dependent on reliable data access [3].

These difficulties underline the necessity of retrieval frameworks that can link scattered data across documents while preserving precision and effectiveness in complex reasoning tasks. To overcome some of these constraints, Graph RAG structures the retrieval procedure around knowledge graphs, which allow for multi-hop reasoning and the explicit modeling of entity relationships [18]. This approach does not completely address the efficiency–accuracy trade-offs, since Graph RAG still has issues with scalability constraints, noisy or incomplete graph construction, and high latency during graph traversal and reasoning [26].

Agentic RAG addresses this issue by introducing autonomous agents that enhance retrieval, iteratively analyze contextual information, and optimize decision-making [21]. When combined with reasoning models, this method enhances response accuracy by eliminating inconsistencies and boosting contextual coherence, resulting in more trustworthy decision-making in complex queries. Integrating the agentic approach with Graph RAG (i.e. Agentic Graph RAG) shows how incorporating agentic strategies into graph-based retrieval allows for more efficient multi-hop reasoning and dynamic exploration of relational knowledge, combining the benefits of structured knowledge with more accurate and adaptive decision-making [14].

To address these shortcomings, we introduce an Agentic Graph RAG framework that combines effective rerankers with lightweight reasoning model-based agents. While rerankers polish retrieved contexts to eliminate noise and redundancy, agents guide multi-hop query decomposition and prune irrelevant paths. This multi-layered method balances retrieval quality, efficiency, and interpretability while significantly lowering reranking overhead and improving retrieval precision for E-Business Information Systems and other knowledge-intensive domains. Our framework is designed to answer the following two research questions:

- **RQ1:** How can reranking bottlenecks in the Agentic Graph RAG be optimized, and what are the efficiency-performance trade-offs when incorporating reasoning models?
- **RQ2:** How can different combinations of reasoning models and reranking techniques in hybrid Agentic Graph RAG systems impact retrieval quality, decision-making, and efficiency for multi-hop queries under practical efficiency constraints?

These research questions seek to improve Agentic Graph RAG systems by resolving reranking bottlenecks and exploring reasoning-reranker synergies, resulting in more accurate and practically efficient multi-hop retrieval — a quality critical in business domains dealing with complex and sensitive data, where response quality directly influences decision-making, risk management, and system reliability. Our evaluations on FinanceBench, LegalBench-RAG-mini, and HotpotQA show that the framework extends far beyond any single domain, making it especially relevant to E-Business Information Systems, where accurate, timely, and trustworthy multi-hop knowledge access has a direct impact on operational and strategic decision-making.

2 Related Work

Graph RAG. RAG research has taken a variety of directions, including graph-based retrieval, agentic retrieval, hybrid agentic-graph systems, and reranking techniques. Traditional RAG pipelines that use vector-based retrieval frequently struggle with noisy environments and multi-hop reasoning. To address these limitations, graph-based approaches such as KG^{2}RAG [34], PathRAG [2], MiniRAG [5], and NodeRAG [27] leverage knowledge graphs to capture entity relationships and enhance reasoning. These methods improve factual accuracy and interpretability, but have significant graph construction expenses, traversal latency, and are susceptible to noise in graph production. While Graph RAG demonstrates stronger reasoning capabilities, it struggles to reconcile efficiency and scalability, particularly when used with large or complicated corpora [18]. This further constrains its use in data-intensive applications like E-Business systems, where precise and rapid information retrieval is crucial.

Agentic RAG and Agentic Graph RAG. While Graph RAG focuses on structured knowledge representation, Agentic RAG broadens the paradigm by incorporating autonomous reasoning and decision-making capabilities. Agentic RAG utilizes LLM-driven agents to dynamically decompose questions, filter noise, and coordinate reasoning. Systems such as MAIN-RAG [1], AU-RAG [9], and MA-RAG [15] apply multi-agent filtering, domain-generalized retrieval, and collaborative reasoning to improve adaptability and accuracy. However, these methods often add processing overhead and risk propagating hallucinations, which are unacceptable in finance-related domains. Graph-R1 [14] introduces reinforcement learning to optimize graph-based retrieval and reasoning in an end-to-end manner, allowing the model to learn effective retrieval strategies through iterative feedback. By investigating relational paths and producing intermediate reasoning steps, Think-on-Graph [23] allows LLMs to carry out explicit reasoning over graph structures. To facilitate adaptive retrieval and reasoning in changing information contexts, AGENTiGraph [33] combines autonomous agents with dynamic knowledge graphs. Despite their strong reasoning abilities, these systems frequently need sophisticated graph management techniques, large computational resources, or intricate optimization processes.

Reranking Strategy. Beyond improvements in retrieval and reasoning, reranking has emerged as an important step toward maintaining relevance. Early work such as Re^{2}G [7] introduced the retrieve–rerank–generate paradigm, showing that intermediate reranking improves response quality. More current approaches use varied reranking strategies: G-RAG [4] combines graph-based context refinement, RankRAG [30] unifies ranking and generation, JudgeRank [16] uses LLMs for reasoning-intensive reranking, and REARANK [31] uses reinforcement learning to optimize reranking decisions. However, these approaches often incur high latency and efficiency costs, limiting their applicability in time-sensitive systems.

Overall, prior work improves reasoning and retrieval relevance but fails to fully address the trade-offs between accuracy, efficiency, and scalability. These limitations are particularly critical in E-Business Information Systems, which require timely and reliable information retrieval. These shortcomings motivate our proposed Agentic Graph RAG framework, which integrates lightweight reasoning-model-based agents and efficient neural rerankers within a structured multi-stage pipeline comprising query decomposition, path pruning, and reranking. This enables iterative refinement that reduces noise and computational overhead while achieving robust multi-hop reasoning and strong retrieval performance under practical efficiency constraints.

3 Methodology

In this section, we examine the preliminary step of generating knowledge-based graphs from our text-based corpus, followed by a detailed explanation of the framework's working principles and components for efficient multi-hop reasoning.

3.1 Preliminaries

The starting point in our approach is to create an indexing graph from the text corpus. Inspired by PathRAG [2], we use an LLM-based graph generation method to convert unstructured text to a structured representation. Entities are retrieved as nodes with textual attributes, and their relationships as edges supplemented with descriptive text. The resulting text-associated graph $G = (V, E, K_\nu, T)$ captures both entities and their semantic relationships, laying the groundwork for later agentic tasks. Here V, E, K_ν, and T denote the set of nodes, edges, node identifiers, and textual chunks (from both nodes and edges), respectively.

3.2 Our Proposed Framework

Our proposed Agentic Graph RAG framework combines contextual reranking with reasoning-based agents to overcome the drawbacks of conventional and graph-based RAG systems. Figure 1 shows the process. The system employs a pipeline with multiple phases:

1. **Query Decomposition.** A reasoning-model-based agent first detects the category of the main query, which may fall into one of four types: *simple* (basic factual questions such as "what," "who," "when," "where"), *sequential* (step-by-step or chronological reasoning), *categorical* (different aspects, perspectives, or types), or *hierarchical* (from general concepts to specific details). Based on the detected category, the query q is decomposed into a set of detailed sub-queries $\{q_1, ..q_n\}$ using prompt-based generation. The generated sub-queries are evaluated against criteria such as sufficiency, length, and uniqueness, with iterative refinement invoked when quality falls short. This self-correcting mechanism improves the reliability of subsequent retrieval and generation phases by ensuring high-quality sub-queries. Figure 2 shows the working principle of this agent.

2. **Keyword Extraction.** From each sub-query, a set of representative keywords K_q is extracted. These serve as anchors for graph-based retrieval, allowing for a more targeted and efficient search over the indexing graph. Decomposing the main query into sub-queries yields more keywords, resulting in more effective retrieval as shown in Table 1.

Table 1: Comparison of Keyword Counts With and Without Query Decomposition Agent

Condition	**Number of Total Extracted Keywords**
Without Query Decomposition Agent	$10 - 15$
With Query Decomposition Agent	$16 - 35$

3. **Relevant Nodes and Paths Retrieval.** Using K_q, relevant nodes $V_q \subseteq V$ are identified from the indexing graph G via semantic embeddings and cosine similarity [11][2]. Multi-hop graph traversal (up to 3 hops) retrieves candidate relational paths in an efficient way, which capture inter-entity connections necessary for complex reasoning and enhancing robustness.

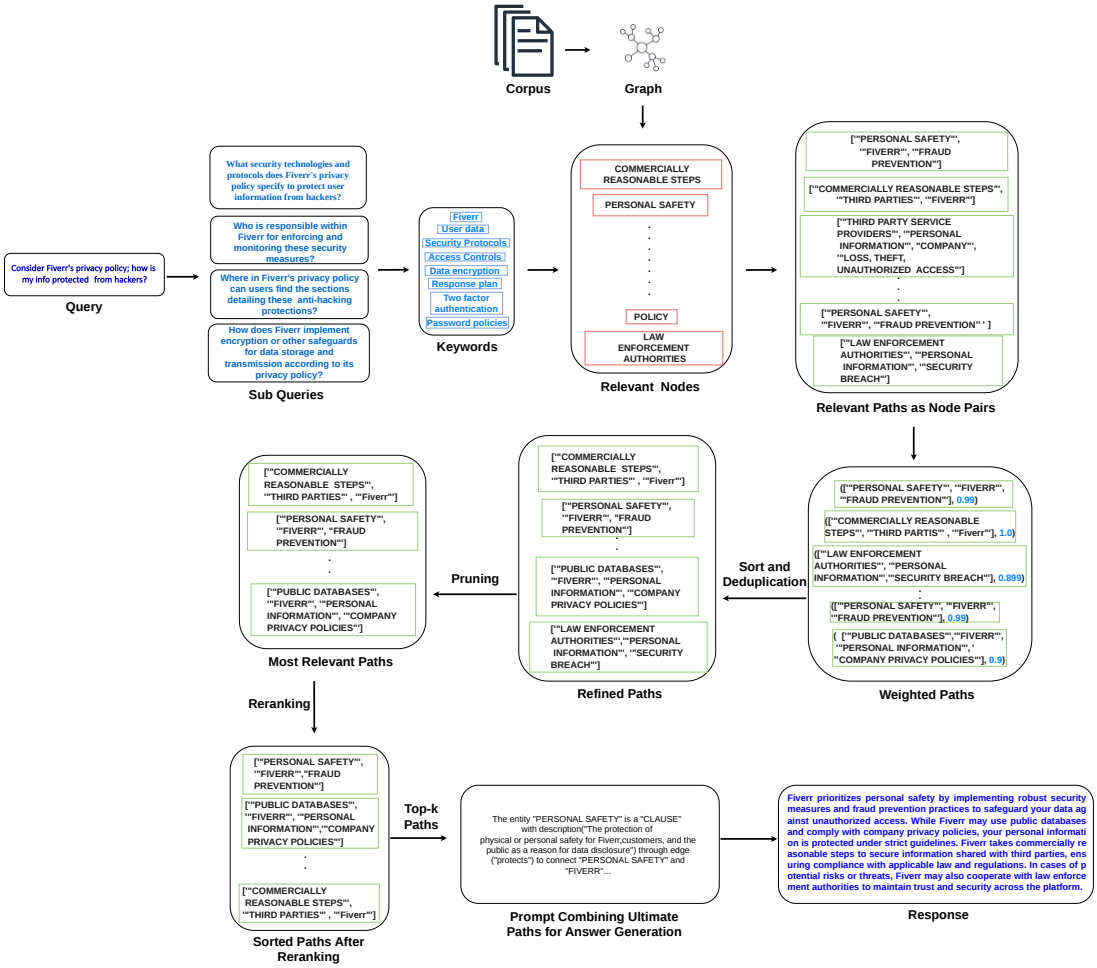

Figure 1: An Overview of the Workflow of Our Framework

4. **Path Weighting.** Inspired by PathRAG [2], each retrieved path $p_i \in P$ is assigned a reliability score $S(p_i)$ through resource allocation across its nodes. The resource originates at the start node and decays as it propagates through neighboring nodes, ensuring that closer nodes contribute more than distant ones. The reliability of a path is then computed as the average resource of its nodes, with higher scores reflecting stronger semantic relevance to the query. After sorting and deduplicating, only the top-N most reliable and highest scored paths are retained for further refinement which enhances efficiency of the pruning agent.

5. **Irrelevant Path Pruning.** Even after weighing and selecting the top-scoring paths, noisy or weakly relevant paths may persist. To address this, a reasoning model integrated within the agent prunes unnecessary candidates based on prompt instructions, as illustrated in Figure 3.

   Formally, the weighted set of candidate paths is denoted as
   $P_q = \{(p_i, S(p_i)) \mid p_i \subseteq P,\ i = 1, \ldots, m\}$

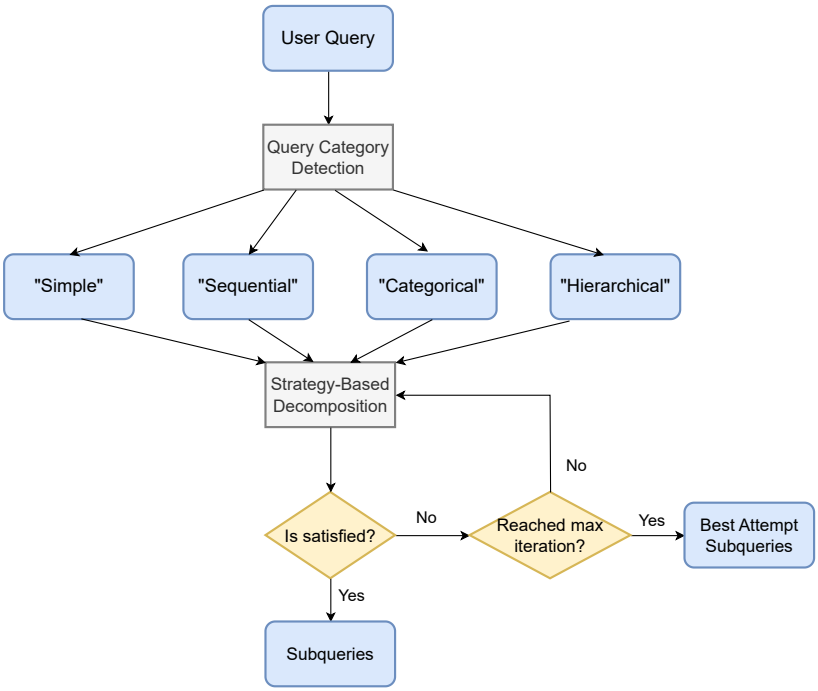

Figure 2: Working Principle of Query Decomposition Agent

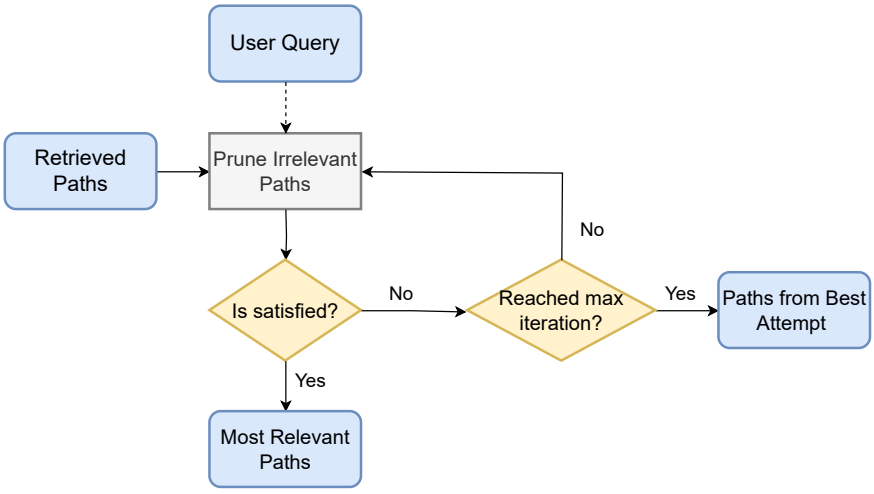

Figure 3: Working Principle of Pruning Agent

where each path p_i has a reliability score $S(p_i)$. The agent iteratively filters out irrelevant or redundant paths, yielding a refined subset:
$P_q^* = \{p_i \in P_q \mid \text{Prune}(q, p_i) = \text{retain}\}$
If P_q^* still contains paths with poor semantic contribution, pruning is performed until a suitable retention rate—defined as the ratio of retained paths to the initial subset—is obtained, or until the maximum number of iterations is met. In the latter case, the agent returns the best subset it encountered. This recurrent filtering ensures that only the most informative and semantically aligned paths are kept for the next phases. In

Table 2: Comparison of Average Number of Paths Before and After Pruning

Condition	**mean_k_before**	**mean_k_after**	**Reduction (%)**
Without Pruning Agent	≈ 360	≈ 360	0%
With Pruning Agent (o4-mini)	≈ 350	≈ 40	88%
With Pruning Agent (Gemini-2.5-Flash)	≈ 350	≈ 190	46%

Table 2, mean_k_before indicates the average number of candidate paths generated before pruning, and mean_k_after represents the remaining paths after the pruning procedure that go for reranking. The pruning agent reduces the path exploration space by up to 88% for o4-mini and 46% for Gemini, effectively filtering noise and improving reasoning efficiency. This also helps to optimize the time complexity of reranking.

6. **Reranking.** To improve retrieval accuracy and efficiency, we employ a reranking phase with neural rerankers to prioritize relevant paths after pruning, as only the top-k reranked paths are used to generate the final prompt for response generation, ensuring that the response is based on the most contextually appropriate evidence while minimizing undue latency.

7. **Response Generation.** The top-k relational paths with their associated textual chunks are incorporated into the final response generation prompt and sent to the LLM to generate response. To avoid the "lost in the middle" effect, the query is placed first, followed by paths sorted in descending reliability, ensuring that the most informative evidence is located in the LLM's golden memory region [13].

This end-to-end system combines decomposition, pruning, and reranking into a structured pipeline to produce contextually rich, trustworthy, and efficient responses. The detailed algorithm is given in Appendix A.

4 Experiments and Results

This section describes the experimental setup and reports results for our framework with and without reranking, along with comparisons to baseline models and additional analysis.

4.1 Experimental Setup

Dataset. We evaluate our framework across three benchmarks, chosen to represent the diverse knowledge-intensive reasoning requirements in E-Business environments.

FinanceBench [8] is a publicly accessible financial QA benchmark, which consists of 150 annotated question-answer pairs grounded in real-world corporate disclosures, earnings reports, and SEC filings. It closely resembles the information requirements of E-Business environments, where precise, multi-step retrieval across vast collections of financial documents is crucial for investment analysis, procurement decisions, and compliance evaluations. We used 90 of the 361 PDF documents in the FinanceBench corpus (roughly 25% of the total) as our retrieval corpus, which covers about 9 million tokens. This scale makes it a realistic and appropriately difficult testbed for assessing graph-based multi-hop retrieval. Furthermore, we made sure that every pair of evaluation questions and answers came from and was firmly rooted in the chosen subset of PDFs. For evaluation, all 150 QA pairs that were accessible to the public were used. The official repository [17] was the source of the dataset.

LegalBench-RAG-mini (LegalBench) [20] is a legal-domain retrieval benchmark focused on high-stakes analytical queries. It is highly relevant to E-Business Information Systems because its legal tasks reflect enterprise activities such as contract analysis, compliance verification, and dispute resolution, which necessitate multi-hop reasoning over complex documents and support explainable retrieval in high-stakes business environments. We worked with a subset of this dataset, using approximately 40% of its document corpus (around 30 documents) and 25% of its QA pairs (195 QAs) for evaluation. The selected corpus comprised approximately 0.8 million tokens, providing a sufficiently large and document-rich retrieval environment for evaluating multi-hop reasoning performance. The dataset was taken from its official Github repository [19].

HotpotQA [29] is a large-scale, Wikipedia-based multi-hop QA benchmark. We sampled approximately 7% of this dataset. The relevant retrieval corpus consisted of the ten Wikipedia paragraphs associated with each sampled query, totaling around 5,000 paragraph instances and 0.75 million tokens after deduplication. We tested our system with 120 QA pairs, focusing on the multi-hop hard subset to ensure a rigorous evaluation. HotpotQA evaluates the framework's capacity to generalize across several domains and knowledge sources. We obtained the dataset from its official Github repository [28].

Due to computational resource constraints, complete evaluation on the full corpora of all three datasets was not feasible. To ensure reproducibility, subsets were generated uniformly at random using a fixed seed. Selected portions were designed to preserve key benchmark characteristics such as query complexity, document length, and reasoning depth.

Models Used for Reasoning and Reranking. In our Agentic Graph RAG framework, Gemini-2.5-Flash (Gemini) and o4-mini are used for reasoning. For reranking, we use ms-marco-MiniLM-L6-v2 (MSM) and bge-reranker-base (BGE), two neural cross-encoder-based rerankers. To evaluate the efficacy of our hybrid approach, experiments are carried out with various combinations of these reasoning and reranking models.

Baselines and Evaluation Measures. We compare our Agentic Graph RAG framework against three graph-based RAG baselines—PathRAG [2], MiniRAG [5], and NodeRAG [27]—to analyze performance-efficiency trade-offs across all three benchmarks. Complementary measurements are used to analyze retrieval and creation performance. RAGAS-based measures such as Context Precision, Context Recall, and Answer Correctness, assess semantic retrieval quality and fidelity. Standard information retrieval (IR) measures [10], such as MRR, Recall@10, and NDCG@10, assess ranking efficacy and document retrieval accuracy. While Context Recall captures semantic coverage of the information needed to answer a query, Recall@10 determines

retrieval of the exact ground-truth document. Context Precision assesses whether the recovered context contains useful information high up in its ranking, hence reducing noise. Meanwhile, MRR and NDCG@10 assess ranking quality, with MRR focusing on how quickly the first relevant item appears and NDCG@10 taking into account the relative position of all relevant papers within the top ten results [25] [10]. ROUGE-L [12] is used to evaluate generation quality, and end-to-end latency is utilized to analyze system efficiency.

Implementation Details. We use gpt-4.1-mini to construct knowledge graphs and generate responses in both our framework and the Graph RAG baselines. The key hyperparameters include a maximum traversal depth of 3 hops, $N = 60$ retrieved nodes, $n = 75\%$ path retention after pruning, and $k = 100$ top-ranked paths for prompt construction. These parameters were chosen based on dataset characteristics and guided by the PathRAG configuration [2]. Additionally, the agents' maximum iterations are set to 2, and the path retention rate threshold for the pruning agent is bounded between 25% and 75%. A 3-hop restriction corresponds to the multi-hop nature of the evaluated benchmarks while avoiding excessive traversal costs, while the other configurations balance candidate coverage, noise reduction, and context efficiency. Experimental environment details are discussed in Appendix B.

4.2 Results and Analysis

The objective of this evaluation was twofold: first, to address **RQ1** by examining how reranking bottlenecks can be optimized while balancing the tradeoffs between efficiency and performance incorporating reasoning models in agents, and second, to address **RQ2** by analyzing how different combinations of reasoning models and rerankers impact multi-hop retrieval quality, decision-making accuracy, and end-to-end efficiency. To this end, we present results in three stages: performance of reasoning model-based agents without rerankers, improvements introduced by incorporating rerankers, and a comparative analysis against existing baseline Graph RAG approaches.

Agents Without Reranker

Table 3 shows the performance of the framework across three datasets when only reasoning model-based agents are employed without reranking. While both o4-mini and Gemini achieve high context precision, their answer correctness remains moderate, and context recall is comparatively lower. Latency is also significant, particularly for Gemini, indicating that reasoning agents alone cannot properly balance accuracy and efficiency, hence producing the baseline bottleneck addressed in **RQ1**.

Table 3: Evaluation Results of Our Agentic Graph RAG Framework (Without Reranking)

Dataset	Reasoning Model	Context Precision	Context Recall	Answer Correctness	ROUGE-L	Overall Latency (s)
FinanceBench	o4-mini	1.00 ± 0.00	0.72 ± 0.04	0.55 ± 0.10	0.19 ± 0.03	116.94 ± 33.50
	Gemini	1.00 ± 0.00	0.68 ± 0.04	0.49 ± 0.05	0.18 ± 0.02	146.08 ± 47.24
LegalBench	o4-mini	0.97 ± 0.06	0.88 ± 0.12	0.56 ± 0.10	0.19 ± 0.03	234.53 ± 72.12
	Gemini	1.00 ± 0.00	0.93 ± 0.05	0.56 ± 0.08	0.18 ± 0.01	337.64 ± 22.63
HotpotQA	o4-mini	0.96 ± 0.20	0.86 ± 0.22	0.70 ± 0.17	0.36 ± 0.12	97.20 ± 26.96
	Gemini	0.98 ± 0.14	0.88 ± 0.23	0.71 ± 0.17	0.37 ± 0.13	153.58 ± 120.17

Agents With Reranker

Table 4 shows a significant improvement in performance following reranker integration. The combination of o4-mini and MSM achieves the highest answer correctness (64% on FinanceBench, 72% on LegalBench, and 74% on HotpotQA) with competitive latency, demonstrating the effectiveness of reranking in improving retrieval accuracy and reasoning consistency. The frequent achievement of 100% context precision implies that agent-guided path pruning and reranking processes effectively filter unnecessary information, guaranteeing highly precise returned contexts. In contrast, Gemini-based combinations show varying accuracy and increased latency on FinanceBench and LegalBench datasets. Notably, ROUGE-L improves with reranking, peaking under the o4-mini and MSM configuration. Overall, o4-mini outperforms Gemini 2.5 Flash in most cases due to its superior instruction-following capabilities and more stable multi-step reasoning in retrieval-augmented settings. Achieving such consistent gains on analytically complex, multi-hop datasets demonstrates the robustness and adaptability of the o4-mini and MSM combination across reasoning settings.

Table 4: Evaluation Results of Our Agentic Graph RAG Framework (With Reranking)

Dataset	**Reasoning Model + Reranker**	**Context Precision**	**Context Recall**	**Answer Correctness**	**ROUGE-L**	**Overall Latency (s)**
FinanceBench	o4-mini+MSM	$\mathbf{1.00 \pm 0.00}$	0.80 ± 0.05	$\mathbf{0.64 \pm 0.05}$	$\mathbf{0.22 \pm 0.11}$	$\mathbf{101.42 \pm 23.36}$
	o4-mini+BGE	0.90 ± 0.03	0.80 ± 0.03	0.49 ± 0.05	0.22 ± 0.11	108.39 ± 32.33
	Gemini+MSM	0.90 ± 0.03	0.73 ± 0.04	0.44 ± 0.10	0.16 ± 0.10	159.80 ± 70.62
	Gemini+BGE	0.95 ± 0.02	$\mathbf{0.85 \pm 0.03}$	0.55 ± 0.02	0.21 ± 0.10	156.95 ± 67.48
LegalBench	o4-mini+MSM	$\mathbf{1.00 \pm 0.00}$	0.97 ± 0.03	$\mathbf{0.72 \pm 0.05}$	$\mathbf{0.20 \pm 0.01}$	$\mathbf{179.67 \pm 9.95}$
	o4-mini+BGE	0.97 ± 0.06	0.90 ± 0.12	0.59 ± 0.06	0.19 ± 0.03	215.95 ± 8.22
	Gemini+MSM	0.97 ± 0.06	0.91 ± 0.03	0.56 ± 0.06	0.18 ± 0.02	270.92 ± 74.17
	Gemini+BGE	1.00 ± 0.00	$\mathbf{0.98 \pm 0.02}$	0.55 ± 0.09	0.19 ± 0.03	304.41 ± 67.73
HotpotQA	o4-mini+MSM	$\mathbf{1.00 \pm 0.00}$	$\mathbf{0.94 \pm 0.13}$	$\mathbf{0.74 \pm 0.14}$	$\mathbf{0.38 \pm 0.10}$	$\mathbf{97.08 \pm 32.86}$
	o4-mini+BGE	1.00 ± 0.00	0.94 ± 0.16	0.72 ± 0.15	0.37 ± 0.13	114.12 ± 54.95
	Gemini+MSM	1.00 ± 0.00	0.92 ± 0.18	0.72 ± 0.16	0.38 ± 0.12	105.40 ± 84.4
	Gemini+BGE	1.00 ± 0.00	0.92 ± 0.18	0.65 ± 0.19	0.38 ± 0.12	102.60 ± 37.50

Table 5 illustrates how rerankers improve context prioritization and retrieval relevance. The o4-mini and MSM combination achieves the best overall retrieval performance, obtaining the highest Recall@10 across all datasets and perfect MRR on HotpotQA, demonstrating its ability to consistently retrieve relevant contexts. While the o4-mini and BGE combination achieves slightly higher NDCG@10 on HotpotQA, Gemini-based setups generally underperform in Recall@10 and NDCG@10 across datasets, despite comparable MRR values in some cases. Figure 4 further visualizes comparative retrieval performance using Recall@10 and NDCG@10. Overall, these results indicate that reranking significantly enhances retrieval relevance and ranking quality, reducing retrieval bottlenecks and improving reasoning-driven retrieval performance, addressing **RQ2**.

Figure 5 shows the overall latency comparison across model combinations for all three datasets. MSM consistently outperforms BGE in latency when combined with o4-mini. The pattern is less consistent with Gemini: MSM gains a significant advantage on LegalBench (270.92s vs. 304.41s), while BGE is slightly faster on FinanceBench (156.95s vs. 159.80s) and HotpotQA (102.60s vs. 105.40s), implying that MSM's latency advantage is most noticeable when

Table 5: Retrieval and Ranking Performance of Our Agentic Graph RAG Framework

Dataset	**Reasoning Model + Reranker**	**Recall@10**	**NDCG@10**	**MRR**
FinanceBench	o4-mini+MSM	$\mathbf{0.59 \pm 0.02}$	$\mathbf{0.74 \pm 0.02}$	$\mathbf{0.98 \pm 0.01}$
	o4-mini+BGE	0.35 ± 0.03	0.71 ± 0.02	0.97 ± 0.02
	Gemini+MSM	0.16 ± 0.01	0.70 ± 0.02	0.93 ± 0.03
	Gemini+BGE	0.17 ± 0.02	0.68 ± 0.02	0.96 ± 0.02
LegalBench	o4-mini+MSM	$\mathbf{0.20 \pm 0.11}$	$\mathbf{0.67 \pm 0.05}$	0.96 ± 0.04
	o4-mini+BGE	0.13 ± 0.01	0.64 ± 0.08	0.92 ± 0.09
	Gemini+MSM	0.08 ± 0.01	0.59 ± 0.01	$\mathbf{0.97 \pm 0.06}$
	Gemini+BGE	0.09 ± 0.06	0.58 ± 0.04	0.94 ± 0.06
HotpotQA	o4-mini+MSM	$\mathbf{0.47 \pm 0.27}$	0.83 ± 0.10	$\mathbf{1.00 \pm 0.00}$
	o4-mini+BGE	0.45 ± 0.27	$\mathbf{0.86 \pm 0.11}$	0.98 ± 0.12
	Gemini+MSM	0.26 ± 0.20	0.71 ± 0.14	0.98 ± 0.12
	Gemini+BGE	0.28 ± 0.23	0.75 ± 0.12	0.97 ± 0.13

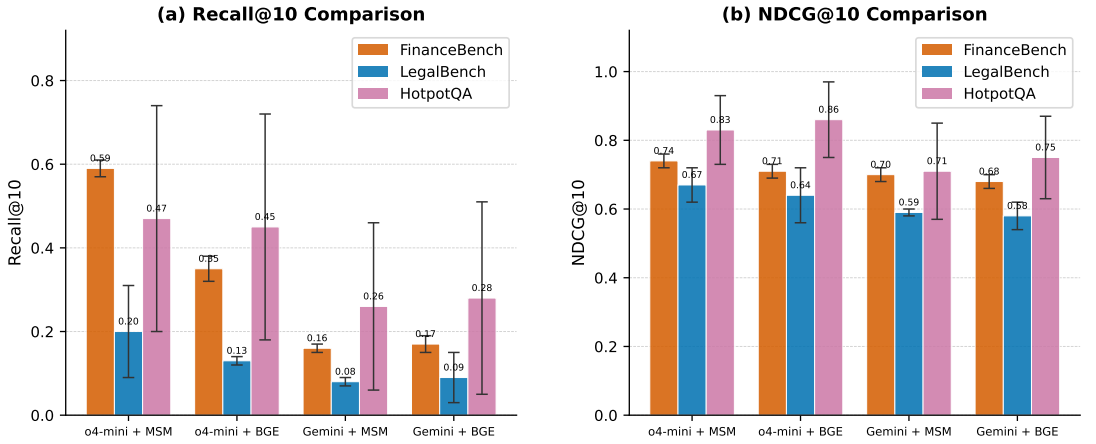

Figure 4: Comparative Retrieval Performance: Recall@10 and NDCG@10 Across Model Combinations in Our Framework

paired with the lighter o4-mini reasoning model. Latency variations also reflect dataset complexity. LegalBench and FinanceBench involve long legal documents and dense financial data, respectively, demanding more complex reasoning and increasing computational cost. These results highlight the correlation between computational cost and corpus complexity — an important consideration in E-Business applications involving financial and legal documents. Overall, the o4-mini and MSM combination reduces the reranking bottleneck and addresses the efficiency–performance trade-offs posed in **RQ1**, achieving an optimal balance between retrieval performance and efficiency.

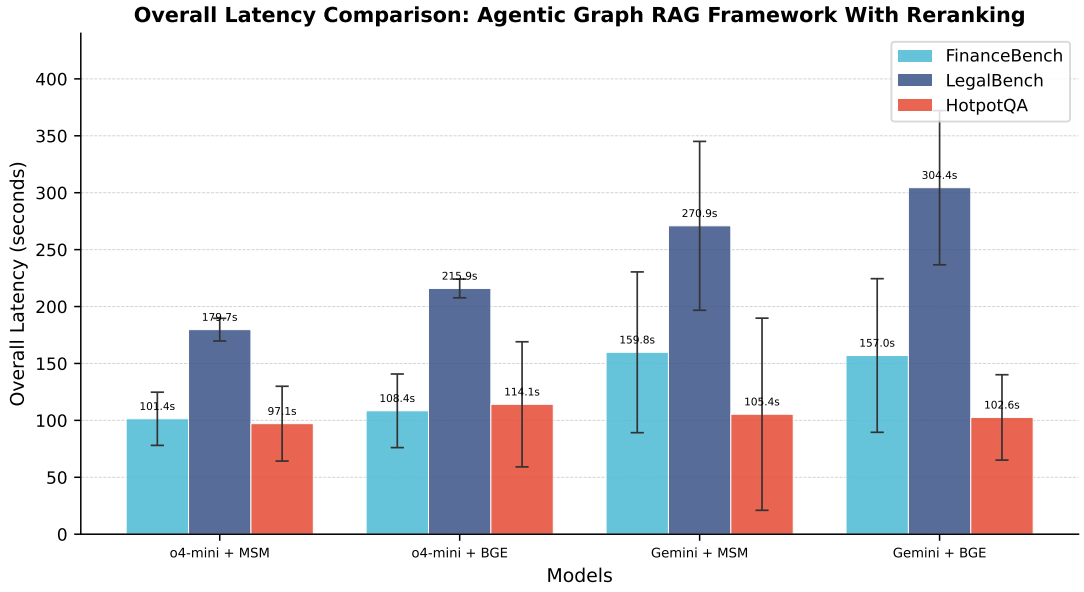

Figure 5: Overall Latency Comparison of Our Framework Combining Different Combinations of Models

Our Framework vs Baseline Models

Table 6 and Figure 6 compare our best-performing configuration (o4-mini+MSM) against PathRAG, MiniRAG, and NodeRAG across all three datasets. Our framework clearly demonstrates its advantage in the knowledge-intensive, multi-hop retrieval scenarios essential to E-Business environments by consistently achieving the highest Answer Correctness and Context Recall across all datasets, with the most noticeable gains on FinanceBench and LegalBench. While all three baselines have lower latency, this speed comes at a measurable cost in retrieval completeness and answer quality, as demonstrated consistently across Context Recall, Answer Correctness, and ROUGE-L. NodeRAG achieves exceptionally low latency by substituting deep path traversal with a shallow Personalized PageRank algorithm, hence avoiding the agent reasoning and reranking stages that our system uses to obtain improved accuracy. The accuracy benefits of our methodology outweigh the extra computing expense in use cases where accuracy and evidence traceability are crucial, such as financial analysis, legal compliance, and organizational decision-making.

Our results demonstrate that the proposed Agentic Graph RAG framework effectively answers both research questions. By removing unnecessary paths early, combining reasoning model-based agents with contextual reranking for **RQ1** mitigates the reranking bottleneck, yielding a comparatively tolerable computational cost without sacrificing retrieval accuracy. For **RQ2**, the combination of efficient reranking (MSM) and lightweight reasoning (o4-mini) consistently provides the optimum balance of practical efficiency, answer correctness, and retrieval relevance across all three benchmarks. While standard deviations of some baselines overlap due to variations in query complexity, the method shows consistent average improvements across datasets and settings, suggesting stable empirical gains rather than dataset-specific effects. As the FinanceBench results clearly show, these findings are particularly pertinent to E-Business

Table 6: Evaluation Results: Comparison Between Our Framework and Baseline Models

Dataset	Reasoning Model + Reranker	Context Precision	Context Recall	Answer Correctness	ROUGE-L	Overall Latency (s)
FinanceBench	Our Framework (o4-mini + MSM)	**1.00 ± 0.00**	**0.80 ± 0.05**	**0.64 ± 0.05**	**0.22 ± 0.11**	101.42 ± 23.36
	PathRAG	0.90 ± 0.03	0.73 ± 0.04	0.47 ± 0.02	0.13 ± 0.05	24.18 ± 9.96
	MiniRAG	0.45 ± 0.09	0.56 ± 0.03	0.28 ± 0.09	0.15 ± 0.02	20.00 ± 3.33
	NodeRAG	0.97 ± 0.03	0.79 ± 0.03	0.55 ± 0.01	0.21 ± 0.02	**17.00 ± 6.67**
LegalBench	Our Framework (o4-mini + MSM)	**1.00 ± 0.00**	**0.97 ± 0.03**	**0.72 ± 0.05**	0.20 ± 0.01	179.67 ± 9.95
	PathRAG	1.00 ± 0.00	0.89 ± 0.07	0.63 ± 0.04	0.20 ± 0.01	146.22 ± 55.63
	MiniRAG	1.00 ± 0.00	0.96 ± 0.01	0.64 ± 0.10	**0.23 ± 0.08**	183.79 ± 30.85
	NodeRAG	0.97 ± 0.06	0.89 ± 0.05	0.61 ± 0.02	0.20 ± 0.01	**6.18 ± 1.75**
HotpotQA	Our Framework (o4-mini + MSM)	**1.00 ± 0.00**	**0.94 ± 0.13**	**0.74 ± 0.14**	**0.38 ± 0.10**	97.08 ± 32.86
	PathRAG	1.00 ± 0.00	0.86 ± 0.23	0.71 ± 0.17	0.38 ± 0.11	8.70 ± 1.27
	MiniRAG	0.97 ± 0.14	0.90 ± 0.19	0.73 ± 0.17	0.37 ± 0.12	7.70 ± 2.30
	NodeRAG	0.97 ± 0.13	0.84 ± 0.21	0.62 ± 0.15	0.28 ± 0.03	**6.47 ± 3.24**

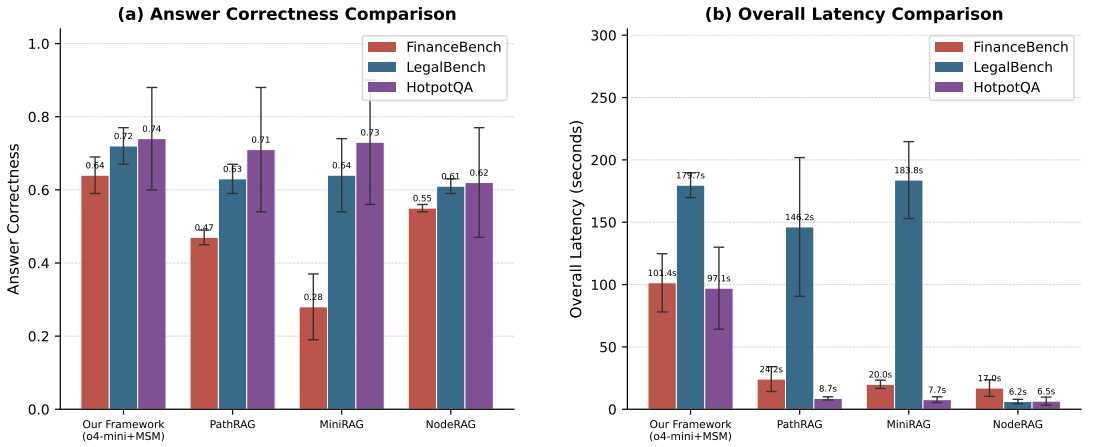

Figure 6: Evaluation Results: Comparison Between Our Framework and Baseline Models

Information Systems, where precise, traceable, and practically effective access to complicated financial knowledge enables well-informed operational and strategic decision-making.

5 Conclusion

This research presents a reasoning-driven Agentic Graph RAG framework that combines lightweight reasoning agents with contextual reranking to improve retrieval precision and efficiency in multi-hop reasoning for E-Business Information Systems. Combining reasoning models

like o4-mini and Gemini-2.5-Flash within agents, as well as integrating rerankers like ms-marco-MiniLM-L6-v2 and bge-reranker-base, improves both reasoning quality and response relevance in our system. Three benchmarks covering different knowledge domains are used to assess the framework: FinanceBench, which covers real-world business disclosures and financial data that directly reflect E-Business information demands; LegalBench-RAG-mini, which represents high-stakes analytical reasoning; and HotpotQA, which evaluates generalizability across broader knowledge domains. The o4-mini and MSM combination achieved the optimal balance of accuracy and latency, while Gemini-based configurations incurred higher computational costs without proportional gains in retrieval accuracy or answer correctness. These findings validate the framework's adaptability across multiple knowledge domains and its suitability for E-Business environments where accurate, traceable, and trustworthy multi-hop knowledge access is essential for informed decision-making.

6 Future Work

There are several directions for further research. First, the methodology should be tested on larger, more diverse corpora to ensure its generalizability, alongside exploring architectural optimizations to further reduce end-to-end latency. Second, a thorough sensitivity study of critical hyperparameters such as traversal depth, retrieved nodes, path retention rate, and reranking size is required to better understand their impact on retrieval quality and efficiency. Third, support for dynamic knowledge graph updates would increase applicability in changing real-world E-Business environments. Finally, future research will look into multimodal Agentic Graph RAG and adaptive retrieval-reranking techniques to improve reasoning, scalability, and flexibility across a variety of information settings.

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Appendix

A Algorithm

This section describes the detailed algorithmic approach for our proposed Agentic Graph RAG system, which operates in 7 steps as illustrated in Algorithm 1.

Algorithm 1: `AgenticGraphRAG`(q, G, V_{ent}, V_{rel})

input : q, G, V_{ent}, V_{rel} ▷ query, knowledge graph, entity vectorDB, relationship vectorDB
output: R ▷ generated response

$SubQueries[\,] \leftarrow$ `DecomposeQuery`(q) ▷ Step 1: Query Decomposition
$Keywords[\,] \leftarrow$ `init()`;
for $sub \leftarrow SubQueries[0]$ **to** $SubQueries[length-1]$ **do**
  $kw \leftarrow$ `ExtractKeywords`(sub) ▷ Step 2: Keyword Extraction
  ; $Keywords$.`add`(kw);
$EntityContext \leftarrow V_{ent}$.`Query`($Keywords$) ▷ Step 3: Context Retrieval
$RelationContext \leftarrow V_{rel}$.`Query`($Keywords$);
$SourceNodes[\,] \leftarrow$ `ExtractNodes`($EntityContext$);
$AllPaths[\,] \leftarrow$ `init()`
for $i \leftarrow 0$ **to** $length(SourceNodes)-1$ **do**
  for $j \leftarrow i+1$ **to** $length(SourceNodes)-1$ **do**
    $paths \leftarrow$ `FindPaths`(G, $SourceNodes[i]$, $SourceNodes[j]$, $maxDepth = 3$);
    $AllPaths$.`addAll`($paths$);
$ScoredPaths[\,] \leftarrow$ `init()` ▷ Step 4: Path Scoring
for $path \leftarrow AllPaths[0]$ **to** $AllPaths[length-1]$ **do**
  $score \leftarrow$ `CalculateWeight`($path$);
  $ScoredPaths$.`add`($\langle path, score \rangle$);
`sort`($ScoredPaths$ by $score$ descending);
$topN \leftarrow n\% \times length(ScoredPaths)$;
$TopPaths \leftarrow ScoredPaths[0 : topN]$;
$CandidatePaths[\,] \leftarrow$ `ExtractPaths`($TopPaths$) ▷ Step 5: Path Pruning
$FilteredPaths \leftarrow$ `PruneIrrelevantPaths`(q, $CandidatePaths$);
$RerankInputs[\,] \leftarrow$ `init()` ▷ Step 6: Neural Reranking
for $path \leftarrow FilteredPaths[0]$ **to** $FilteredPaths[length-1]$ **do**
  $input \leftarrow q +$ "[SEP]" + `PathToString`($path$);
  $RerankInputs$.`add`($input$);
$Scores[\,] \leftarrow$ `RerankerModel`($RerankInputs$);
$RerankedPaths \leftarrow$ `SortByScore`($FilteredPaths$, $Scores$);
$FinalPaths \leftarrow RerankedPaths[0 : k]$;
$Context \leftarrow$ `BuildNarrativeContext`($FinalPaths$, G)
$SystemPrompt \leftarrow$ `FormatContext`($Context$) ▷ Step 7: Response Generation
$R \leftarrow$ `LLM`(q, $SystemPrompt$);
return R;

B Computational Cost & Resource Requirements

In this section, we demonstrate the experimental environment and analyze the cost required for our experiments.

B.1 Experimental Environment

All experiments were conducted using a single NVIDIA T4 GPU (16 GB VRAM). Each instance was equipped with 16-32 vCPUs and 64 GB of system memory, giving sufficient resources for concurrent reasoning, reranking, and retrieval activities.

B.2 Cost Analysis

To analyze the proposed framework's operational efficiency, we estimated the computing costs incurred in three important stages: graph generation, reasoning, and response generation, which are explained below.

B.2.1 Graph Generation.

To create structured graphs of entity–relation mappings, approximately 10.6 million source tokens from all three corpora were combined during the ingestion phase. Because this method maps both dense entity–relation triplets and extensive text-associated node and edge properties, constructing the index graph required an output of approximately 20 million tokens. This substantial indexing phase was carried out using GPT-4.1-mini, resulting in an estimated total graph generation cost of USD 36.24.

B.2.2 Reasoning.

Our agents' reasoning model processed between 500 and 1,200 tokens per query and produced about 200 to 800 tokens in response during the reasoning phase, which includes query decomposition and pruning of irrelevant paths. This resulted in a remarkably low cost of about USD 0.006 per query.

B.2.3 Response Generation.

Every query that was sent to GPT-4.1-mini generated an average of 500 output tokens and contained approximately 15,000–16,000 input tokens, which included the context and query for response generation. The resulting per-query cost was approximately USD 0.007.

Overall, these findings show that even while the suggested framework incorporates several reasoning models into the agents, it still maintains computational efficiency.

C Error Analysis in Multi-Hop Reasoning

Our Agentic Graph RAG framework has robust error-handling features to maintain stability during multi-hop reasoning. Potential mistakes in the Query Decomposition phase, such as malformed outputs, or incorrect sub-query counts, are automatically discovered and corrected. When such failures occur, the decomposition agent returns to the original core query, allowing the reasoning process to proceed. Furthermore, ambiguity, over-decomposition, or too short

sub-queries result in iterative refining until a legitimate, semantically coherent set of sub-queries is obtained.

In the Pruning Irrelevant Paths phase, failures may arise from LLM parsing issues, semantic misalignment, or incomplete filtering of irrelevant paths. When reasoning model failures, timeouts, or low retention rates are discovered, the agent implements multi-level fallback solutions such as a keyword-based pruning heuristic or a full-path return mechanism. This ensures that the retrieval process continues uninterrupted and that the reranking stage receives consistent inputs.

Empirical analysis of the datasets revealed that these recovery mechanisms successfully maintained pipeline continuity despite model or server failures. Such errors occur very rarely; however, when they do arise, they are mostly caused by ambiguous path construction rather than pruning or LLM parsing issues, occasionally resulting in slightly lower recall and increased latency. Overall correctness was maintained since the framework efficiently addressed these problems. These findings support the framework's resilience and reliability in managing complex reasoning operations and dynamic query contexts.

Plan-First Parallelism for Hard Mathematical Reasoning: A Search-Space Failure Analysis*

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Abstract

Recent advances in large language model (LLM) research have prompted growing interest in improving reasoning capabilities, yet hard reasoning problems remain stubbornly difficult to solve. Evidence suggests that simply increasing test-time compute yields diminishing returns on the hardest problem regimes, and that the structure of the search space itself is the more critical factor. In this work, we propose Plan-First Parallelism (PF), a strategy-level branching approach that generates multiple distinct solution plans before execution, as opposed to answer-level parallel sampling. To analyze failure systematically, we introduce a three-way error taxonomy distinguishing search-space failure (E1), selection failure (E2), and plan-execution divergence (E3). Experiments on AIME and BeyondAIME benchmarks show that PF consistently outperforms Answer-First Parallelism (AF) for o3-mini, improving accuracy by +11.5%p on AIME and +14.5%p on BeyondAIME. However, PF fails to generalize across all models, and our analysis reveals that its effectiveness depends on the combination of reasoning quality and instruction-following capability. Beyond benchmark performance, our findings provide a failure-aware perspective on reliable and cyber-resilient LLM reasoning components for AI-enabled e-business information systems.

1 Introduction

Recent advances in large language model (LLM) research have highlighted the importance of test-time compute scaling as a complementary approach to increasing model parameter size. In particular, search-oriented inference strategies such as parallel sampling [4], verifier-guided search [16, 17], and tree-style exploration [6, 20, 21] have been shown to outperform simple sequential revision [14, 15] on challenging reasoning tasks. However, evidence also suggests that merely increasing the amount of computation does not guarantee consistent performance gains in the hardest problem regimes, and that the structure of the search process itself may be the more critical factor.

This work departs from that observation. We hypothesize that failures on hard reasoning problems stem not simply from insufficient computation, but from a more fundamental limitation: the model's inability to generate correct high-level solution strategies in the first place—a

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phenomenon we term *search-space failure*. In other words, the primary bottleneck may lie not in how many candidate answers are produced, but in whether the model can branch into meaningfully distinct regions of the strategy space.

Within the broader landscape of search-oriented inference, we focus on one concrete and controllable instantiation: Plan-First Parallelism. Rather than addressing search in its full generality, our interest is in analyzing how strategy-level branching, as opposed to answer-level parallelism, alters the candidate generation distribution and the structure of failure.

The main contributions of this paper are as follows. First, we propose a novel error taxonomy that decomposes hard reasoning failures into three distinct categories: search-space failure (E1), selection failure (E2), and plan-execution divergence (E3). Second, we empirically compare Answer-First and Plan-First approaches across three models on the AIME [25] and BeyondAIME [26] benchmarks, demonstrating that Plan-First improves both candidate hit rate and final accuracy for reasoning-capable models such as o3-mini.[1] Third, we quantitatively measure the gap between plan diversity and execution diversity, providing insight into the remaining limitations of Plan-First and directions for future improvement.

From a systems perspective, Plan-First Parallelism also offers structural resilience: by maintaining multiple strategically distinct solution paths, the system remains robust to the failure of any single reasoning trajectory—a property relevant to the design of reliable AI-driven decision support systems.

Beyond mathematical reasoning itself, this study is motivated by the growing use of LLMs as reasoning components in AI-enabled e-business information systems. Such systems increasingly rely on LLMs for decision support, workflow assistance, risk analysis, and knowledge-intensive operations. In these contexts, reasoning failures are not merely isolated model errors; they may propagate into downstream organizational decisions and reduce the reliability of the overall information system. Therefore, understanding whether failures arise from candidate-generation failure, selection failure, or plan-execution divergence is important for designing reliable and cyber-resilient AI reasoning pipelines [27].

We use hard mathematical reasoning as a controlled diagnostic setting rather than as an end application. Exact-answer benchmarks allow reasoning failures to be measured objectively and decomposed systematically. From this perspective, Plan-First Parallelism and the proposed E1–E3 taxonomy provide a failure-aware analysis framework that can inform the design of auditable and recoverable LLM-enabled decision support systems.

2 Background and Related Work

Mathematical reasoning has long served as a challenging testbed for evaluating the capabilities of large language models [10, 19]. The MATH dataset [1] demonstrated that even large-scale models struggle with competition-level problems, and that accuracy remains low as problem difficulty increases. Building on this, recent work [2] showed that scaling test-time compute can improve performance on easier problems, but that this effect diminishes on harder reasoning tasks, where simply allocating more computation does not guarantee consistent gains. This suggests that the bottleneck on hard problems lies not in the quantity of compute, but in the structure of the search process itself.

One promising direction for restructuring this search process is through prompting strategies. Chain-of-thought (CoT) prompting [3] demonstrated that eliciting intermediate reasoning steps before a final answer substantially improves performance on multi-step reasoning tasks.

[1] Specifically, we use `o3-mini-2025-01-31` in all experiments.

Zero-shot CoT [9] further showed that appending a simple trigger phrase elicits step-by-step reasoning without any demonstrations, while Auto-CoT [12] proposed automatic construction of CoT demonstrations to reduce manual effort. Self-consistency [4] extended this by sampling multiple reasoning paths and selecting the most frequent answer to further boost accuracy. Plan-and-Solve prompting [5] explicitly instructs the model to devise a solution plan before execution, reducing missing-step errors in zero-shot settings. In a related vein, Tree of Thoughts [6] frames problem solving as a search over multiple reasoning paths with explicit evaluation at each step. We adopt this plan-first paradigm as the core instantiation of strategy-level branching in our framework.

A broader landscape of related methods addresses reasoning improvement from complementary angles. Decomposition-based approaches [7, 13] break problems into simpler subproblems, while program-aided reasoning [8] leverages code execution for precise computation. Reasoning-acting frameworks [22] extend planning to interactive settings. Self-improvement methods [11, 14, 15] explore iterative refinement as a complementary direction to search-based approaches. Step-level verification [16, 17] addresses answer selection through process supervision, offering an orthogonal lever for improving final accuracy. These advances build on the emergent reasoning capabilities of large-scale language models and rely on instruction-following abilities developed through alignment training [18, 23]. Beyond performance, the structural property of maintaining multiple distinct solution paths also relates to the notion of reasoning resilience—the ability of a system to remain robust when individual reasoning trajectories fail.

3 Problem Formulation

This section defines the basic setup and notation for mathematical reasoning problems, and presents an error taxonomy for analyzing hard reasoning failures.

In LLM-based mathematical reasoning, failures on hard problems may stem not from insufficient computation, but from the inability to explore correct solution strategies in the first place. Generating multiple answers to a hard reasoning problem does not guarantee diversity in solution strategies. Therefore, it is important to first explore different solution strategies. From this perspective, this study decomposes failures into three reported categories: search-space failure, selection failure, and plan-execution divergence.

We denote a mathematical reasoning problem as q and its ground truth answer as $y^*(q)$. The language model p_θ with parameters θ takes problem q as input and generates K candidate solutions $a_1, \ldots, a_K$. Here, z_i denotes the i-th solution plan, and a_i denotes the i-th candidate solution generated conditioned on that plan. The selector s is a function that selects the final answer $\hat{y}(q)$ from the generated candidates.

This study argues that final accuracy alone is insufficient to fully explain the failure structure, and therefore measures two metrics together. First, **CandHit** is a metric that identifies whether at least one correct answer exists among the K candidates:

$$\text{CandHit} = \frac{1}{|D|} \sum_{q \in D} \mathbf{1}\big[\exists\, i : \text{ans}(a_i) = y^*(q)\big]$$

In contrast, **Acc** is a metric that verifies whether the finally selected answer is correct:

$$\text{Acc} = \frac{1}{|D|} \sum_{q \in D} \mathbf{1}\big[\hat{y}(q) = y^*(q)\big]$$

CandHit is the key metric for the search stage, while Acc is an end-to-end performance metric. By examining both metrics together, it is possible to distinguish whether a failure occurred at

the search stage or the selection stage. Specifically, this study classifies incorrect answers into the following three types.

E1 (Search-space failure) refers to cases where CandHit = 0, that is, no correct candidate was generated. This represents a failure at the search stage.

E2 (Selection failure) refers to cases where CandHit = 1 but Acc = 0, that is, a correct candidate existed but the selector failed to choose it. This represents a bottleneck at the selection stage.

E3 (Plan-execution divergence) refers to a subtype of search-space failure in which diverse plans are generated but the resulting solution executions converge to similar incorrect trajectories. Operationally, E3 is identified among cases with CandHit = 0 when the Plan Diversity evaluated by LLM-as-a-Judge [24] is ≥ 4 and the Execution Diversity is ≤ 2. The thresholds $(4, 2)$ are heuristic and exploratory. A threshold sensitivity analysis is reported in Table 4.

Although E3 is conceptually a subtype of E1, we report E1, E2, and E3 as mutually exclusive categories in the result tables for interpretability. In the reported breakdowns, E1 denotes search-space failures that do not satisfy the E3 condition, while E3 denotes the plan-execution divergence subset of search-space failures.

4 Answer-First and Plan-First Parallelism

4.1 Baseline: Answer-First Parallelism

Answer-First Parallelism (AF) generates K candidate solutions directly and independently from problem q:

$$a_i \sim p_\theta(a \mid q), \quad i = 1, \ldots, K$$

The selector then chooses the final answer:

$$j^* = \arg\max_i \, s(q, a_i), \qquad \hat{y}_{\mathrm{AF}}(q) = \mathrm{ans}(a_{j^*})$$

AF thus relies entirely on answer-level diversity across independently sampled solutions.

4.2 Proposed Method: Plan-First Parallelism

Plan-First Parallelism (PF) generates K distinct solution plans prior to any execution. Each plan z_i represents a fundamentally different mathematical strategy:

$$z_i \sim p_\theta(z \mid q), \quad i = 1, \ldots, K$$

Each plan then conditions an independent solution generation:

$$a_i \sim p_\theta(a \mid q, z_i), \quad i = 1, \ldots, K$$

The selector chooses from plan–solution pairs:

$$j^* = \arg\max_i \, s(q, z_i, a_i), \qquad \hat{y}_{\mathrm{PF}}(q) = \mathrm{ans}(a_{j^*})$$

Unlike AF, PF establishes strategy-level diversity before solution generation, and is thus best understood as a structured parallel search rather than generic parallel sampling.

We position PF not as a complete general-purpose solution, but as a targeted intervention for diagnosing whether hard reasoning failures are rooted in strategy-level search bottlenecks.

Rather than proposing a new search framework in full generality—such as verifier-guided tree search or adaptive compute allocation—this study focuses specifically on whether plan-level branching changes the candidate distribution, alleviates search-space failure (E1), and if so, why its gains do not always fully propagate to final accuracy.

4.3 Selector

Final answer selection proceeds in two stages. First, if two or more candidates agree on the same answer, that answer is selected by majority vote [4]. When no majority exists, an LLM verifier [16, 17] is invoked to compare the reasoning quality of all candidates and select the most logically complete solution. More robust selection strategies, such as critique-based selection and backward verification, are left as future work for mitigating E2 failures.

5 Experiments

5.1 Datasets

We evaluate on two exact-answer mathematical reasoning benchmarks: AIME and BeyondAIME. Both datasets are well-suited for our analysis, as their short integer-valued answers enable automatic grading and clean separation between candidate generation failure and selection failure. Compared to the MATH dataset used in prior work [1], both benchmarks target a substantially harder reasoning regime. AIME provides competition-style problems whose answers are integers in the range 000–999, while BeyondAIME is a curated benchmark containing problems of difficulty exceeding the later AIME problems, making it suitable for examining whether our findings generalize to even harder reasoning settings.

Starting from a raw pool of 90 AIME and 100 BeyondAIME problems, we applied a difficulty screening procedure using o3-mini under the AF condition. Each problem was solved three times independently. Problems solved correctly on all three attempts were treated as too easy, while problems solved incorrectly on all three attempts were treated as too hard to discriminate under this screening model. These problems were excluded in order to concentrate the evaluation on a hard-but-discriminative regime where differences between AF and PF are more likely to be observable.

This procedure yielded a final evaluation set of 52 AIME and 76 BeyondAIME problems, totaling 128 problems. In total, 38 AIME and 24 BeyondAIME problems were excluded by this screening. We therefore refer to this evaluation set as an *o3-mini-screened hard subset*. Because the subset was selected based on o3-mini behavior under the AF condition, it should not be interpreted as a model-neutral measure of problem difficulty. In particular, the cross-model comparisons with GPT-4.1 and Claude Haiku 4.5 should be understood as evaluations on the same o3-mini-screened hard subset, rather than as claims about each model's performance on an independently calibrated hard subset.

5.2 Experimental Setup

We fix $K = 3$ across all experiments as a practical balance between candidate diversity and computational cost, as justified by the ablation in Table 1. Final answer selection follows the two-stage selector described in Section 4.3.

We ablate $K \in \{2, 3, 4\}$ using o3-mini and find that $K = 3$ achieves the best PF accuracy on both AIME (65.4%) and BeyondAIME (55.3%). While $K = 4$ yields higher diversity scores

(Plan Diversity 4.35 vs. 3.90 at $K = 2$), it increases selection errors (E2: 17.4% on AIME, 13.2% on BeyondAIME) and degrades accuracy, suggesting that the verifier becomes less reliable as the candidate pool grows.

Table 1: Ablation on number of candidates K (o3-mini).

	AIME (52)		BeyondAIME (76)	
K	AF Acc	PF Acc	AF Acc	PF Acc
2	48.1%	59.6%	52.6%	51.3%
3	53.8%	**65.4%**	40.8%	**55.3%**
4	55.8%	55.8%	55.3%	50.0%

The non-monotonic variation of AF accuracy across K—most notably the drop to 40.8% at $K = 3$ on BeyondAIME—should be interpreted with sampling variance in mind. Each cell in Table 1 corresponds to an independent run in which K candidates are freshly sampled and the final answer is decided by majority voting over a small candidate pool, so run-to-run fluctuation is expected. On the 76-problem BeyondAIME subset, a 95% binomial confidence interval around an accuracy of 50% spans roughly ±11 percentage points; the observed AF variation across K (40.8–55.3%) therefore falls within the range attributable to sampling noise rather than indicating a systematic K-dependent trend. We accordingly treat Table 1 as practical guidance for choosing K, rather than as a precise characterization of AF scaling behavior.

We evaluate three models: **o3-mini** (OpenAI) serves as the primary model, as it is a reasoning-oriented model well-suited for hard mathematical problems. **GPT-4.1** (OpenAI) and **Claude Haiku 4.5** (Anthropic) are included as general-purpose comparison models to examine whether the effect of PF generalizes across model types.

In the PF condition, plan generation is prompted with an explicit diversity constraint requiring each of the three plans to employ a fundamentally different mathematical tool or perspective. This is intended to prevent the three plans from converging to similar strategies.

To analyze the effect of PF beyond final accuracy, we adopt three LLM-as-a-Judge metrics. **Plan Diversity** measures how strategically distinct the three generated plans are from one another. **Execution Diversity** measures how differently the actual solution paths unfold across the three candidates. **Adherence** measures how faithfully each solution follows its assigned plan [24]. The judge model is the same model used for solution generation; the possibility of self-evaluation bias is acknowledged as a limitation of this study.

Answer correctness is evaluated using exact string matching after answer extraction, as detailed in Section 5.3.

5.3 Implementation Details for Reproducibility

To improve reproducibility, we provide additional details on the prompts, decoding settings, final-answer selection procedure, LLM-as-a-Judge rubric, and answer matching rules used in our experiments. In the o3-mini experiments, no explicit temperature, top-p, or maximum-token parameters were set, and the API defaults were used. Instead, decoding effort was controlled using the `reasoning_effort` parameter: `medium` for AF candidate generation, `low` for PF plan generation, `high` for PF solution execution, and `low` for verifier and LLM-as-a-Judge calls. For the comparison-model runs, we used the same prompting, selection, and evaluation pipeline,

with model-specific API defaults unless otherwise required. No external tools, retrieval modules, or calculators were used.

Prompts. In the AF condition, each model was directly asked to solve the problem step by step and return a final answer enclosed in `\boxed{}`. In the PF condition, the model first generated three distinct solution plans using an explicit diversity instruction, and each plan was then executed independently. The PF execution prompt required the model to begin from the assigned initial angle of attack, while allowing it to pivot if the direction became blocked. The AF, PF plan-generation, PF execution, verifier, and judge prompts are summarized here rather than reproduced in full due to space constraints.

Selection and judging. Final answer selection was performed by majority voting when at least two candidates produced the same extracted answer. When no majority existed, an LLM verifier compared the three candidate solutions and selected the attempt with the most logically sound reasoning by returning a single index. For PF outputs, Plan Diversity, Execution Diversity, and Adherence were scored on a 1–5 scale using an LLM-as-a-Judge prompt.

Answer extraction and matching. Final answers were extracted from the model output by identifying answers enclosed in the `\boxed{...}` pattern. If no boxed answer was found, the extracted answer was treated as empty. After extraction, surrounding whitespace was removed, and correctness was evaluated by exact string matching against the gold answer. We did not apply additional numeric normalization or zero-padding. Therefore, AIME-style leading-zero variants such as `72` and `072` could be treated as different strings. This is a limitation of the evaluation protocol, but the same rule was applied to both AF and PF outputs.

5.4 Results

5.4.1 o3-mini (Primary Model)

On the AIME benchmark, PF achieves a CandHit of 71.2% (37/52) and an accuracy of 65.4% (34/52), compared to 65.4% (34/52) and 53.8% (28/52) for AF, representing gains of +5.8%p in CandHit and +11.5%p in accuracy. This indicates that PF not only secured more valid candidate solutions at the search stage, but that these gains translated substantially into final answer selection.

Among the 18 incorrect PF responses on AIME, 15 (83.3%) are classified as E1 (Search-space failure), 3 (16.7%) as E2 (Selection failure), and 0 as E3 (Plan-execution divergence). On BeyondAIME, among 34 incorrect responses, 26 (76.5%) are E1, 6 (17.6%) are E2, and 2 (5.9%) are E3. Across both datasets, E1 consistently dominates, suggesting that even with plan-level branching, the search space is not sufficiently opened for the hardest problems.

LLM-as-a-Judge scores reveal a consistent gap between plan diversity (3.63/5.0) and execution diversity (2.77/5.0), despite high adherence (4.06/5.0). This indicates that while the model generates strategically distinct plans and largely follows them, the actual solution trajectories do not diverge as much as the plans suggest.

On BeyondAIME, PF shows even larger gains, improving CandHit by +13.2%p (50.0% $\rightarrow$ 63.2%) and accuracy by +14.5%p (40.8% $\rightarrow$ 55.3%). The error breakdown shows E1 at 76.5%, E2 at 17.6%, and E3 at 5.9%, with the plan–execution diversity gap again present (Plan Diversity 3.89 vs. Execution Diversity 2.93).

Taken together, these results suggest that PF tends to yield larger gains on harder problems, where answer-level diversity alone is less likely to cover the correct solution space.

5.4.2 Comparison Models

For GPT-4.1, PF yields no consistent improvement. On AIME, accuracy slightly decreases from 26.9% to 23.1%, and on BeyondAIME it increases marginally from 17.1% to 22.4%, while CandHit declines. Among the 40 incorrect PF responses on AIME, 33 (82.5%) are E1, 4 (10.0%) are E2, and 3 (7.5%) are E3. On BeyondAIME, among 59 incorrect responses, 54 (91.5%) are E1 and 5 (8.5%) are E2. E1 thus accounts for 82.5–91.5% of all PF errors, indicating that the primary bottleneck is the inability to generate correct candidates regardless of planning structure. Notably, GPT-4.1 achieves the highest Plan Diversity scores (4.71–4.79) among all models, yet this does not translate into improved performance.

For Claude Haiku 4.5, PF produces a small accuracy gain on AIME (+3.8%p) but a slight decrease on BeyondAIME (−2.6%p). Among the 37 incorrect PF responses on AIME, 21 (56.8%) are E1, 3 (8.1%) are E2, and 13 (35.1%) are E3. On BeyondAIME, among 61 incorrect responses, 28 (45.9%) are E1, 8 (13.1%) are E2, and 25 (41.0%) are E3. E3 (Plan-execution divergence) thus accounts for 35.1% and 41.0% of PF errors on AIME and BeyondAIME respectively—by far the highest among all models. Adherence scores are also the lowest (2.82–3.06), suggesting that the model frequently fails to follow its assigned plans and instead converges to similar solution paths regardless of the stated strategy.

Both comparison models show limited and inconsistent benefits from PF, but for qualitatively different reasons. A detailed analysis of these failure patterns is provided in Section 6.

Table 2: Comparison of AF and PF across models and datasets on the o3-mini-screened hard subset. CandHit denotes the proportion of problems for which at least one correct candidate was generated, and Acc denotes final answer accuracy.

Model	Dataset	AF CandHit	AF Acc	PF CandHit	PF Acc
o3-mini	AIME	65.4%	53.8%	71.2%	**65.4%**
o3-mini	BeyondAIME	50.0%	40.8%	63.2%	**55.3%**
GPT-4.1	AIME	30.8%	26.9%	30.8%	23.1%
GPT-4.1	BeyondAIME	31.6%	17.1%	28.9%	22.4%
Claude Haiku 4.5	AIME	32.7%	25.0%	34.6%	28.8%
Claude Haiku 4.5	BeyondAIME	26.3%	22.4%	30.3%	19.7%

Table 3: Mutually exclusive error breakdown and LLM-as-a-Judge scores for PF across models and datasets. E3 denotes plan-execution divergence.

Model	Dataset	E1	E2	E3	Plan Div.	Exec Div.	Adh.
o3-mini	AIME	83.3%	16.7%	–	3.63	2.77	4.06
o3-mini	BeyondAIME	76.5%	17.6%	5.9%	3.89	2.93	4.18
GPT-4.1	AIME	82.5%	10.0%	7.5%	4.71	3.87	4.75
GPT-4.1	BeyondAIME	91.5%	8.5%	–	4.79	4.26	4.80
Claude Haiku 4.5	AIME	56.8%	8.1%	35.1%	3.31	2.17	3.06
Claude Haiku 4.5	BeyondAIME	45.9%	13.1%	41.0%	3.54	2.22	2.82

Table 4: Sensitivity analysis of E3 rates under alternative plan-execution divergence thresholds.

Model	Dataset	Strict ($PD \geq 5$, $ED \leq 1$)	Original ($PD \geq 4$, $ED \leq 2$)	Loose ($PD \geq 3$, $ED \leq 3$)
o3-mini	AIME	0 (0.0%)	0 (0.0%)	10 (55.6%)
o3-mini	BeyondAIME	0 (0.0%)	2 (5.9%)	18 (52.9%)
GPT-4.1	AIME	0 (0.0%)	3 (7.5%)	7 (17.5%)
GPT-4.1	BeyondAIME	0 (0.0%)	0 (0.0%)	2 (3.4%)
Claude Haiku 4.5	AIME	1 (2.7%)	13 (35.1%)	24 (64.9%)
Claude Haiku 4.5	BeyondAIME	0 (0.0%)	25 (41.0%)	38 (62.3%)

Note. PD: Plan Diversity, ED: Execution Diversity. Percentages are computed over PF incorrect cases.

The original threshold is not intended to be optimal in a supervised sense, as no ground-truth E3 labels are available. Instead, it serves as a conservative and interpretable operating definition between overly strict and overly loose criteria. As expected, the strict setting identifies very few E3 cases, while the loose setting captures a broader set of divergence-like failures. The sensitivity results show that E3 counts are more sensitive to the Execution Diversity cutoff than to the Plan Diversity cutoff. Under the original threshold, Claude Haiku 4.5 exhibits substantially higher E3 rates than the other models, and this pattern remains visible under the loose setting. This suggests that the observed plan-execution divergence pattern is not merely an artifact of a single threshold choice.

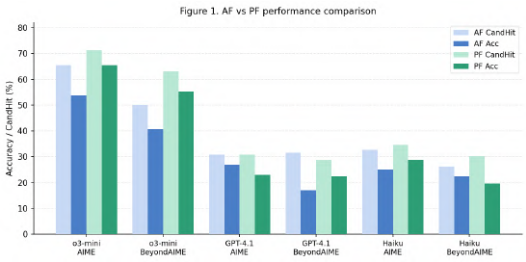

Figure 1: AF vs PF performance comparison. CandHit: proportion of problems with at least one correct candidate; Acc: final answer accuracy.

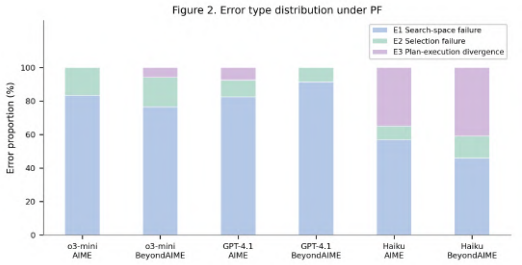

Figure 2: Error type distribution under PF. E1: search-space failure, E2: selection failure, E3: plan-execution divergence.

6 Discussion

6.1 Effectiveness of Plan-First for o3-mini: Adherence and Search Bottlenecks

On the AIME benchmark, PF outperformed AF by +5.8%p in Hit Rate and +11.5%p in Accuracy for o3-mini; on BeyondAIME, the gains were +13.2%p in Hit Rate and +14.5%p in Accuracy. In other words, PF yielded consistent improvements at both the search stage and the final selection stage for o3-mini.

o3-mini also recorded Adherence scores of 4.06 and 4.18, indicating that the model largely followed the generated plans throughout the solution process. This suggests that plan-following—the core premise of PF—holds relatively stably for this model. In other words, o3-mini is capable not only of generating plans, but of translating them into actual reasoning trajectories.

Nonetheless, Execution Diversity (2.77/2.93) was consistently lower than Plan Diversity (3.63/3.89), suggesting that plan-level diversity is not fully preserved during execution. That said, the co-occurrence of high Adherence and substantial performance gains indicates that this loss was not severe enough to undermine the effectiveness of PF for o3-mini.

The error analysis further supports this interpretation. E2 accounted for only 16.7% and 17.6% of failures, indicating that the selection mechanism functioned relatively well, whereas E1 remained the dominant failure type at 83.3% and 76.5% respectively. This suggests that even for o3-mini, the core bottleneck lies in search rather than selection. Thus, while PF meaningfully improved o3-mini's performance, the hardest problems still frequently defied correct strategy generation, and further expanding the search space remains an open challenge for future work.

6.2 Limitations of Plan-First for GPT-4.1: Plan Diversity Without Strategic Validity

PF did not yield consistent gains for GPT-4.1. On AIME, PF reduced Accuracy by 3.8%p relative to AF, with no change in Hit Rate. On BeyondAIME, Accuracy improved by 5.3%p, but Hit Rate decreased by 2.6%p. Overall, PF cannot be said to have reliably improved GPT-4.1's search performance.

These results highlight a disconnect between generating diverse plans and actually opening a search space that contains correct strategies. GPT-4.1 achieved the highest Plan Diversity scores among all models (4.71/4.79), yet the majority of PF errors still fell under E1 (82.5%, 91.5%). This indicates that generating diverse plans does not translate into performance gains unless those plans include strategies that can lead to correct answers.

GPT-4.1 also showed relatively high Execution Diversity and Adherence. The core issue, therefore, is not that the model failed to execute its plans, but rather that the executed plans, despite being diverse, all led away from correct answers. In other words, GPT-4.1's limitation lies not in plan-following, but in its inability to place valid strategies within the search space on hard reasoning problems.

This suggests that PF's effectiveness requires not only the ability to generate diverse plans, but also the reasoning quality to ensure that the search space contains strategies with a genuine chance of being correct.

6.3 Plan-Execution Divergence in Claude Haiku: Adherence Failure as a Distinct Bottleneck

PF also failed to produce consistent gains for Claude Haiku 4.5. On AIME, PF improved Accuracy by 3.8%p and Hit Rate by 1.9%p. On BeyondAIME, however, Hit Rate increased by 3.9%p while Accuracy declined by 2.7%p. Thus, while PF contributed to generating more correct candidates at the search stage, these gains did not reliably carry through to final answer selection.

The most striking characteristic is the proportion of E3 errors: 35.1% on AIME and 41.0% on BeyondAIME—by far the highest among all models. While other models showed little to no E3, Claude Haiku exhibited a frequent pattern in which diverse plans failed to be maintained during execution. This demonstrates that PF's failure modes extend beyond search-space failure

(E1) and selection failure (E2) to include plan-execution divergence as a distinct and significant bottleneck.

This interpretation is consistent with the model's low Adherence scores (3.06 and 2.82)—the lowest among all models—indicating that generated plans were not sufficiently maintained during the solution process. In other words, rather than following its assigned strategy, the model tended to converge toward similar incorrect solution paths regardless of the plan.

This implies that PF's effectiveness requires not only the ability to generate plans, but also the instruction-following capability to stably translate those plans into actual reasoning trajectories. This behavior reflects a lack of reasoning resilience — the ability of a model to maintain its intended strategy under the complexity of execution. From the perspective of resilient AI system design, a model that generates valid plans but cannot sustain them offers limited reliability in practice.

6.4 Cross-Model Synthesis

Taken together, the results across three models show that PF is not a strategy that uniformly improves performance simply by generating diverse plans. It was effective for o3-mini, which can translate plans into actual solutions, but ineffective for GPT-4.1, which despite generating highly diverse plans failed to identify correct strategies. For Claude Haiku, plan-level diversity was frequently not preserved during execution.

Therefore, the effectiveness of PF depends not on plan diversity alone, but on the combination of reasoning quality and instruction-following capability.

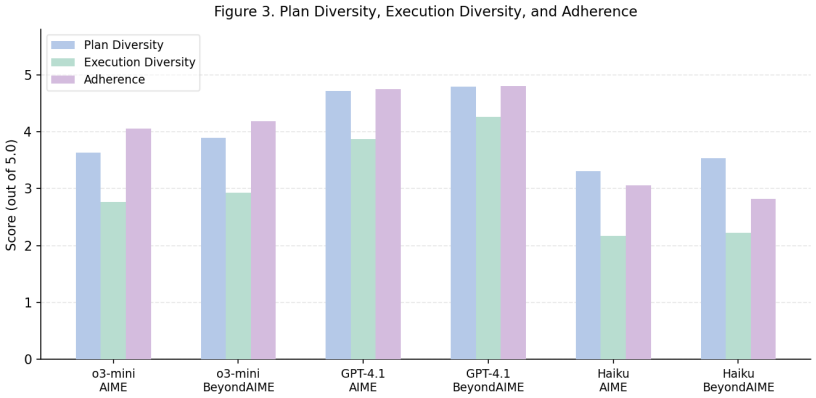

Figure 3: Plan Diversity, Execution Diversity, and Adherence scores under PF across models and datasets (LLM-as-a-Judge, out of 5.0).

6.5 Implications for E-Business Information Systems and Cyber Resilience

Within the scope of EBISION 2026, this work most directly addresses the topics of "Artificial Intelligence and Data-Driven Business Innovation," "Resilient and Adaptive EBIS Architectures," and "Cybersecurity and Digital Trust in EBIS." It analyzes the failure structure of LLM reasoning components that increasingly underpin AI-driven decision support in e-business information systems (Section 1), and interprets Plan-First Parallelism as a strategy-level mechanism that

contributes to the cyber resilience of such systems by making reasoning failures diagnosable and recoverable [27] (this section and Section 7).

The proposed taxonomy also has implications for LLM-enabled e-business information systems. In enterprise settings, LLMs may support decision-making, compliance analysis, financial reasoning, operational planning, and incident triage. For such systems, reliability requires more than high average accuracy: the system should make failures diagnosable, support recovery mechanisms, and avoid overreliance on a single reasoning trajectory. We do not claim that mathematical benchmarks directly instantiate e-business workflows; rather, they provide a controlled stress-test for reasoning failures that may also occur in LLM-enabled enterprise systems.

Our E1–E3 taxonomy maps naturally to different failure-handling strategies. E1 indicates that no correct candidate was generated, suggesting the need for broader search, model routing, retrieval augmentation, or human escalation. E2 indicates that a correct candidate existed but was not selected, suggesting the need for stronger verification, cross-model judging, or improved answer normalization. E3 indicates that plan-level diversity was not preserved during execution, suggesting the need for adherence monitoring and execution-level diversity control. These distinctions are important for cyber-resilient AI systems because different failure modes require different recovery mechanisms.

From this perspective, Plan-First Parallelism can be interpreted as a strategy-level resilience mechanism. By maintaining multiple planned reasoning trajectories, PF reduces dependence on a single solution path. However, our cross-model results show that plan diversity alone is insufficient: reliable deployment also requires valid strategy generation and stable plan execution. This finding is directly relevant to e-business information systems in which LLM outputs may influence organizational decisions, risk assessments, or automated workflows.

7 Conclusion

This study investigates whether generating solution plans prior to execution improves hard mathematical reasoning under a fixed candidate budget. We propose Plan-First Parallelism (PF), a strategy-level branching approach that generates multiple distinct solution plans before execution, and compare it against Answer-First Parallelism (AF) across two benchmarks and three models. To support systematic failure analysis, we introduce a three-way error taxonomy distinguishing search-space failure (E1), selection failure (E2), and plan-execution divergence (E3).

Our primary findings are as follows. First, PF consistently outperforms AF on both AIME and BeyondAIME when using o3-mini, improving CandHit by up to +13.2%p and accuracy by up to +14.5%p. Second, error analysis reveals that E1 (Search-space failure) remains the dominant bottleneck across all models, indicating that generating correct candidates is fundamentally harder than selecting among them. Third, LLM-as-a-Judge scores consistently show a gap between Plan Diversity and Execution Diversity, suggesting that strategically distinct plans do not always produce correspondingly distinct solution trajectories. Fourth, Claude Haiku 4.5 exhibits a high rate of E3 (Plan-execution divergence), revealing that plan-execution divergence is a distinct failure mode that varies meaningfully across model types.

Taken together, these results suggest that the effectiveness of PF is not a function of plan diversity alone, but depends critically on two complementary capabilities: the reasoning quality required to place valid strategies within the search space, and the instruction-following capability required to translate those strategies into actual solution trajectories. Models that lack either capability fail under PF for qualitatively different reasons—GPT-4.1 generates diverse

but strategically invalid plans, while Claude Haiku fails to sustain its plans during execution. This cross-model contrast highlights that plan-level branching is a necessary but not sufficient condition for improved reasoning performance.

This study has several limitations. The evaluation set is relatively small (128 problems), and it is an o3-mini-screened hard subset rather than a model-neutral difficulty split. Since the subset was selected based on o3-mini behavior under the AF condition, the cross-model comparisons should be interpreted with this selection bias in mind. The range of models compared is also limited.

Future work should explore alternative structured search strategies beyond PF, and investigate more robust selection mechanisms to mitigate E2 failures. Improving execution diversity—the gap between plan-level and solution-level diversity—also remains an open challenge. Additionally, the results suggest a promising hybrid direction: using o3-mini for plan generation, where it demonstrates strong reasoning ability, and GPT-4.1 as a verifier or judge, where it achieves high diversity scores. Such a cross-model pipeline may yield stronger performance than any single-model configuration, and represents a natural next step toward more robust and reliable reasoning systems.

Beyond benchmark-level findings, this work contributes to the broader design of reliable and cyber-resilient AI components for e-business information systems. By decomposing LLM reasoning failures into search-space failure, selection failure, and plan-execution divergence, the proposed framework makes failures more diagnosable and actionable. Future work should extend this failure-aware analysis from controlled mathematical benchmarks to enterprise decision support, compliance reasoning, cyber-risk analysis, and other e-business information system scenarios where reliable LLM reasoning is required.

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IT Ticket Classification for Application Management Services using Classical Machine Learning Techniques*

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Abstract

IT outsourcing company is widely used for many organizations to support applications used in the organizations without interrupting business processes. The challenging problem for IT outsourcing company is that a helpdesk officer must specify the reported problem for which supported applications, and the officer must forward the ticket to a correct expert to solve that problem. If many errors occur, the company could breach the Service Level Agreement (SLA). This paper proposed a framework to handle imbalanced dataset with Thai and English languages and to use 8 machine learning models, i.e. Support Vector Machine (SVM), Stochastic Gradient Descent (SGD), Logistic Regression (LR), Multinomial Naïve Bayes (MNB), Decision Tree (DT), Bernoulli Naïve Bayes (BNB), Light Gradient-Boosting Machine (LightGBM), and Extreme Gradient Boosting (XGB) to classify IT tickets since the models required less computational resources when comparing to deep learning models. The dataset is from ServiceNow application in an IT outsourcing company with 20,586 records to be classified into 142 applications, which is highly imbalanced dataset. The performance metrics are accuracy, precision, recall, and F1-score. From two experiments using a class weight with lemmatization, and 6 input features, SVM model gives the best F1-score at 85.71% and the features are short description, description, business unit, sub business unit, and company code.

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1 Introduction

In the modern business process, a large organization utilizes Information Technology (IT) in managing business tasks but IT knowledge is not for everyone. Therefore, the organization can outsource IT supports to an IT outsourcing company to help business handle technology-related functions. Application Managed Services (AMS) play an important role for business continuity to be responsible for many applications. Each application may be complex and required an expert in solving a problem.

From this context, IT Service Management (ITSM) system, e.g. ServiceNow, is used for recording, tracking, and managing cases or incidents from several IT systems in organizations. When a user finds a problem or needs technical assistance, the user contacts a Helpdesk officer, and reported information will be recorded in the form of a case or a ticket in ServiceNow. Then, the Helpdesk officer will assign the case to an expert for each application to solve the problem later.

For the case assignment, a user reports a problem to a Helpdesk officer using a phone, an email, or a self-service portal in ServiceNow. The reported information, called a ticket, i.e. short description, description, category, priority/urgency, application/service, and caller information, will be recorded for each case. After getting the ticket, the Helpdesk officer must analyze the case information and assign the case to an expert who is responsible for solving that problem directly. The process of the case assignment needs a deep knowledge in the scope of each application but there can be more than hundred supported applications.

The ServiceNow application can manage the workflow efficiently, but the case assignment still needs to do it manually and this causes several problems as follows. The first problem is the misclassification. An officer assigns a newly received ticket to incorrect experts. The second problem is the delay in case escalation. After the misclassification process, the wrongly assigned expert must reassign to a correct expert. This causes a delay in solving the case and the response time or resolution time, which is one of the performance indexes in Service Level Agreement (SLA), is increased. From the delay, the IT outsourcing company can get the SLA breach penalty, and the Customer Satisfaction Score (CSAT) can be reduced in the long run. The third problem is the knowledge dependency. Since the case assignment task needs an officer to have a lot of experience but when experienced officers quit the jobs or transfer to other companies, the knowledge is not transferred properly. The case assignment quality is lower during the transient period. The last problem is multilingual data challenge. For Thai IT outsourcing company, case information is recorded in both Thai and English languages. The case classification is more complex for the officers.

From the problems mentioned above, several researchers proposed to use machine learning and natural language processing techniques to apply for IT Helpdesk ticket classification automatically. It means that the use of machine instead of human to classify IT tickets is needed to reduce human errors and the misclassification delay. However, high number of application classes (more than 100 applications) and the imbalance class are a challenge research topic. Therefore, this paper proposed to use machine learning techniques for IT ticket classification with the high number of application classes and class imbalance problems. The eight classical machine learning techniques in this research are Support Vector Machine (SVM), Stochastic Gradient Descent (SGD), Logistic Regression (LR), Multinomial Naïve Bayes (MNB), Decision Tree (DT), Bernoulli Naïve Bayes (BNB), Light Gradient-Boosting Machine (LightGBM), and Extreme Gradient Boosting (XGB). The models require less computational resources than deep learning models. In addition, they are suitable for AMS services in terms of speed, operation costs, and maintenance. The contributions of this paper are proposing a related application feature design using hybrid inputs to the models, using text tokenization and lemmatization techniques instead of stemmer for Thai and English languages, and proposing the imbalance techniques for a big, highly imbalance, and practical dataset.

The organization of this paper is as follows. The next section is related work on the IT ticket classification. Then, the proposed work is explained with the framework, data collection, data preprocessing, input feature selection, and machine learning models. Next, proposed experiments are described in detail with experimental setup, and performance metrics. Then, experiment results and analysis section is discussed and analyzed on the results. The last section is a conclusion and future work to summarize this paper and offer possible future experiments.

2 Related work

The research in the IT ticket classification has been widely proposed using machine learning, and deep learning techniques. For machine learning technique, Paramesh et al. (2018) proposed to use 5,550 IT service desk records to classify into 12 classes with English language only. The algorithms in the study were Multinomial Naïve Bayes (MNB), Support Vector Machine (SVM), Decision Tree (DT), Random Forest, and K-Nearest Neighbor (KNN) together with Ensemble Bagging technique. The feature representation was Term Frequency-Inverse Document Frequency (TF-IDF) and the class imbalance problem was solved using Random Oversampling method. The best F1-score was 60.76% for SVM technique. Parmar et al. (2018) proposed to use a bigger dataset with 10,742 records in 18 classes using Voting classifier mixed with Logistic Regression, KNN, MNB, SVM, and Random Forest techniques. The features were description and category with TF-IDF transformation. The highest F1-score was 88.14% using SVM. Al-Hawari and Barham (2019) proposed to use title and description as features. The best machine learning technique was SVM with F1-score at 81.4%. The main problems for this research were that the size of dataset was small and the classes were not more than 18 classes. They cannot reflect the complexity in large, practical IT outsourcing companies for IT ticket classification.

For multilingual IT ticket problem especially Thai and English languages, Pidej and Thaicharoen (2021) showed that using the same dataset and machine learning models in their study, the accuracies for Thai and English only were 64.93% and 86.21%, respectively, and the best F1-score was 87.81%. There was a huge difference 21% for the accuracy with different language. Sungthong et al. (2024) proposed to use PhayaThaiBERT model, which was pre-trained with Thai language specifically. The F1-score was 76.9% with 940 records for the dataset. For Thai language IT Tickets, Khowongprasoed and Titijaroonroj (2022) proposed to use IT Infrastructure Company dataset in Thailand with 360 records. In this study, there were three classes, i.e. incident, request, and problem. SVM technique gave the best F1-score at 72%. The multilingual environment is another challenge for IT ticket classification problem. Table 1 shows the summary table for related work.

Reference	Technique	Data size	Number of classes	Language	Model	F1-score
Paramesh et al. (2018)	SVM, NB, DT	5,550	12	ENG	SVM	60.76%
Parmar et al. (2018)	Voting SVM	10,742	18	ENG	SVM	88.14%
Al-Hawari and Barham (2019)	SVM, NB, DT	1,254	13	ENG	SVM	81.40%
Pidej and Thaicharoen (2021)	mBERT, ULMFiT, XLM-R	11,572	14	TH/ENG	XLM-R	87.81%
Sungthong et al. (2024)	PhayaThaiBERT	940	30	TH/ENG	Deep Neural Network	76.9%
Khowongprasoed and Titijaroonroj (2022)	SVM	360	3	TH	SVM	72.0%

Table 1: Related work summary

From the literature work, the dataset size is not more than 12,000 records and the number of classes is not more than 30 classes. In addition, most research used title or short description as an input to machine learning models. Therefore, this research proposed a framework to handle Thai and English languages and to explore classical machine learning techniques to classify IT tickets with highly imbalance classes in a large dataset with multilingual environment using several input features.

3 Proposed work

This section is to explain the proposed work starting with proposed framework to give the overview of this work. Then, data collection, data preprocessing, input feature selection, and machine learning models used in this paper are described in detail.

3.1 Proposed framework

The framework of IT ticket classification is shown in Figure 1. The data collection is to collect the IT ticket data for experiments. The data preprocessing is to clean and prepare the data before feeding into machine learning models. The last part is to evaluate and analyze the results. The explanations for each part are as follows.

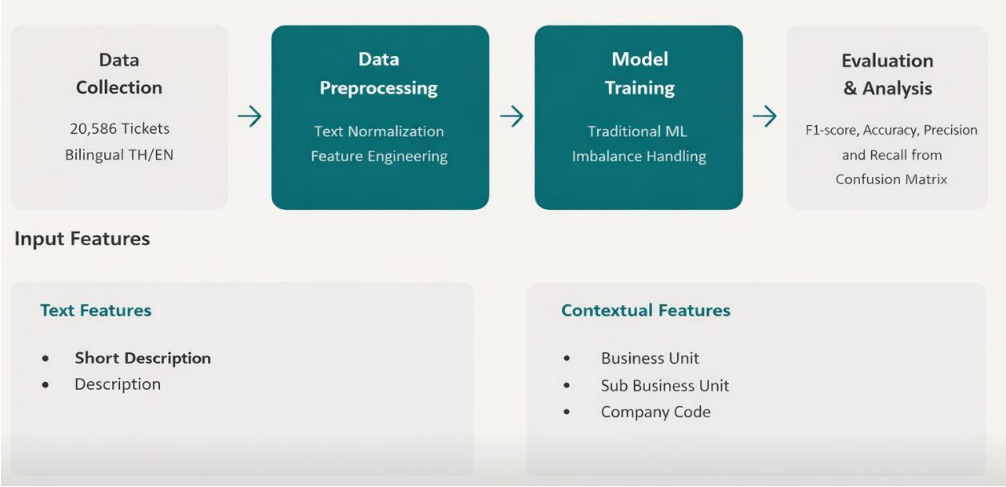

Figure 1: IT ticket classification framework

3.2 Data collection

The dataset has been collected for 1 year from IT outsourcing company in Thailand and total IT tickets in ServiceNow application was 20,586 records. The IT tickets were opened in the system from emails, self-service portal, and phones in both Thai and English languages to be classified into 142 applications. Duplicate or canceled tickets were removed from the dataset. The dataset is highly imbalanced as shown in Figure 2. Some applications are reported more frequently than other applications. This dataset consists of 6 input features and one output feature as follows.

1. *Short description* is a short problem inputted from a user and a helpdesk officer.
2. *Description* is a detailed problem or error messages.

3. *Business unit* is a main business unit of a user.
4. *Sub business unit* is a sub business unit of a user.
5. *Company code* is a code of a company of a user.
6. *Contact type* is the communication channel, i.e. email, portal, call center, to open a ticket.
7. *Service offering* is the name of applications, which is a target variable to be classified and used for supervised learning.

The example dataset is shown in Figure 3, which contains both Thai and English languages and some technical terms in the description. However, the service offering field is not shown.

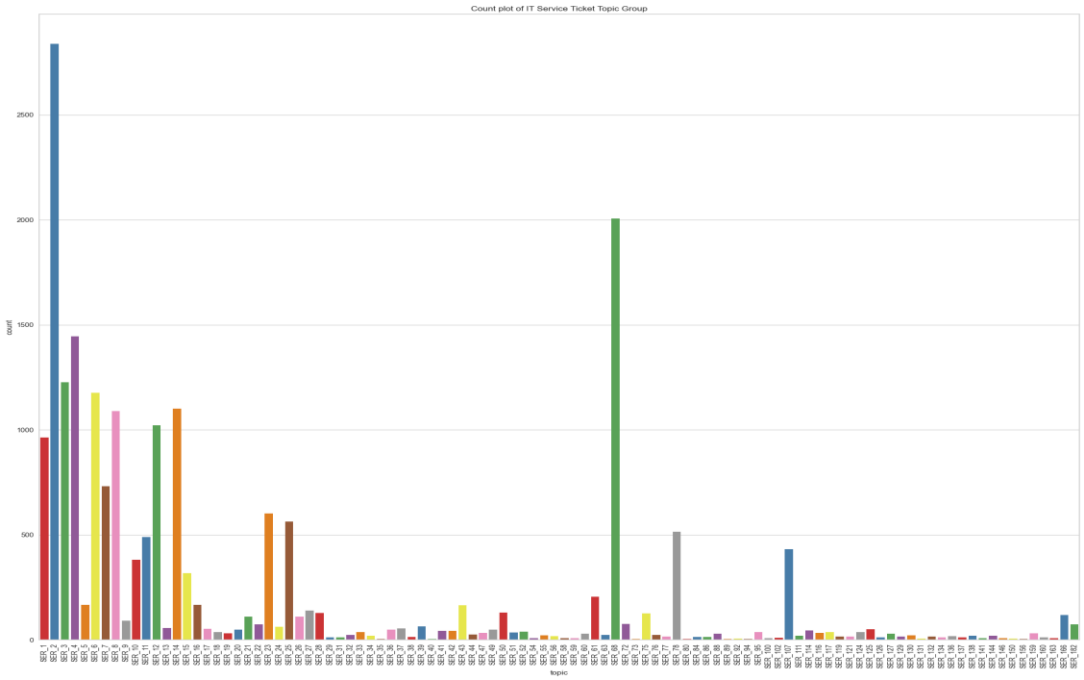

Figure 2: Application distribution in the dataset

Business	Sub	Company	Service offering	Short description	Description	Contact
BUS_3		COM_3	SER_7	ตรวจสอบ HYYYYB Lot YHYYYYYY Reverse ไม่ได้	ตรวจสอบ HYYYYB Lot YHYYYYYY Reverse ไม่ได้ รายละเอียดเพิ่มเติมตามไฟล์แนบ	Email
BUS_4	SUB_2	COM_4	SER_1	ตรวจสอบ ใบตัดจ่าย ไม่มีการ run billing SAP expo เมื่อคืนนี้ทำให้วันนี้ไม่มียอด	ตรวจสอบ ใบตัดจ่าย ไม่มีการ run billing SAP expo เมื่อคืนนี้ทำให้วันนี้ไม่มียอด ซึ่งตอนนี้พบมา Y ร้านแล้ว ตัวอย่าง DN YYYYYYYYY	Phone
BUS_4	SUB_3	COM_5	SER_8	ตรวจสอบการบันทึกเวลาของพนักงานขาดงาน Y ชม. (YY.YY-YY.YY,YY.YY-YY.YY) แต่ระบบขึ้นมาสาย Y ชม.ขาดงาน Y ชม.	ตรวจสอบการบันทึกเวลาของพนักงานขาดงาน Y ชม.(YY.YY-YY.YY,YY.YY-YY.YY) แต่ระบบขึ้นมาสาย Y ชม.ขาดงาน Y ชม.	Email
BUS_1	SUB_1	COM_1	SER_1	[HANA Retail] ตรวจสอบ IYY - เมื่อยิง mode fix พบว่าไม่สามารถสร้าง DN ต่อได้	ตรวจสอบ เมื่อ ยิง IYY แล้วดิด error <MESSAGE>No ATP for target delivery qty</ MESSAGE> ไม่สามารถสร้าง DN ต่อได้ <PIID_OUT>YYYYYYYBEBYFYYEACYYYYYYYYYYYYYYA</PIID_OUT> <PIID>YYYYYADYDAFDYYEABYEYYYYYYYYYYYBF</PIID> <FC_PO>AYYYYYYYYYYYY</FC_PO>	Email
BUS_3		COM_6	SER_9	ตรวจสอบไม่สามารถเข้าไปแก้ไข Elements ต่างๆ ใน PI AF ได้	ตรวจสอบไม่สามารถเข้าไปแก้ไข Elements ต่างๆ ใน PI AF ได้ ซึ่งปกติก่อนหน้านี้จะสามารถเข้าไปแก้ไข Element ต่อไปนี้ได้ Y. VCM Process Monitoring Y. VCMY Y. VCMY Y. Envi Monitoring	Portal

Figure 3: Example dataset

3.3 Data preprocessing

Before feeding the data to a classification process or machine learning models, the raw data must be pre-processed systematically to clean the IT ticket data as follows.

1. Decommissioned application removal

Tickets related to decommissioned applications were removed from the dataset because they are not supported in the ServiceNow application.

2. Non-Thai/English character removal

Some IT tickets consist of special/symbol characters, non-Thai/English characters including HTML tags, URL, reference numbers, and non-printable characters. These characters are removed from short description and description fields in order to reduce noise in the dataset.

3. Stop word removal

All stop words from both Thai and English languages are removed from the dataset because these words give high frequency without statistical significance on IT ticket classification. This removal can increase the performance of machine learning models.

4. Lemmatization

Lemmatization is a technique for transforming words into a lemma or a base form to reduce the complexity and increase the quality of feature representations. It helps the models to classify IT tickets correctly.

For English language, NLTK library was used for text cleansing and text normalization, for example, converting upper case to lower case, removing punctuations, and lemmatization, in order to convert words back into basic words, which can reduce the word variations and increase the performance of the machine learning models. Then, the tokenization and stop word removal were performed before feeding into the models.

For Thai language, PyThaiNLP library and NewMM algorithm were used for tokenization by referring to words in Thai dictionary, which was designed for Thai language processing only. Then, the standard stop word in PyThaiNLP library was also used for stop word removal.

3.4 Input feature selection

From the dataset, the data can be classified into 2 groups, i.e. textual data and organizational context data. The textual data is data related to a problem, which are short description and description. The organizational context data is data related to an organization that reports a problem, which are business unit, sub business unit, and company code. Some applications are used in specific companies, and these data could enhance the classification performance. In the experiments, the input features are incrementally added to machine learning models as follows.

1. Baseline 1: Short description only
2. Hybrid 2: Short description, Description
3. Hybrid 3: Short description, Description, Business unit
4. Hybrid 4: Short description, Description, Business unit, Sub business unit
5. Hybrid 5: Short description, Description, Business unit, Sub business unit, Company code

Figure 4 shows all hybrid features in this experiment to help understand the effects of each feature to enhance the classification performance.

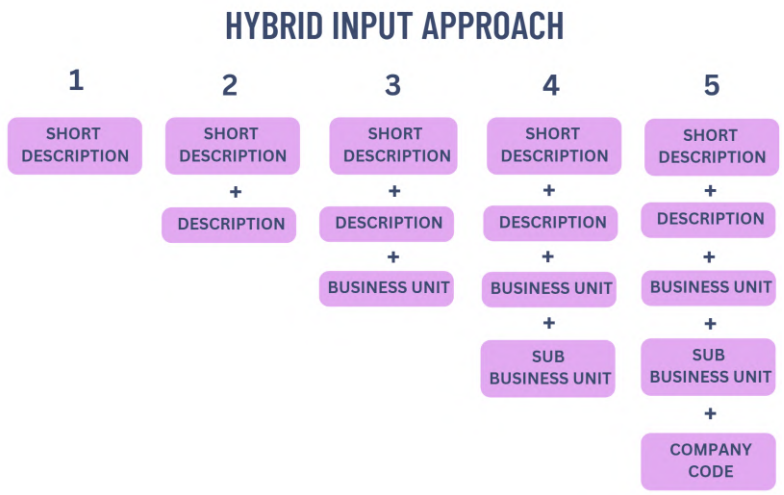

Figure 4: Input features

3.5 Machine learning models

From the related work, several classical machine learning models were implemented and tested with different performance metrics. Classical machine learning models require less computational resources and memory than deep learning models. In this paper, eight classical machine learning models are selected from supervised learning models as follows.

1. Support Vector Machine (SVM) is the algorithm to find an optimal hyperplane to classify class in high dimensional feature space using linear kernel with one-vs-rest strategy for multi-class classification.
2. Logistic Regression (LR) is a linear classification algorithm using Sigmoid function to estimate a probability of each class with regularization to avoid an overfitting problem.
3. Multinomial Naïve Bayes (MNB) is a probabilistic algorithm for the data transforming to term frequency using the conditional independent assumption between features.
4. Bernoulli Naïve Bayes (BNB) is a Naïve Bayes algorithm using binary feature representation. It is good for appearance or absence of terms.
5. Stochastic Gradient Descent (SGD) is a linear classification algorithm using stochastic updated weights and good for a large dataset.
6. Decision Tree (DT) is a tree-like structure algorithm to select features with maximum information gain for each level.
7. Extreme Gradient Boosting (XGB) is an ensemble algorithm using gradient boosting that creates several decision trees sequentially. Each tree is created by learning errors from the previous tree.
8. Light Gradient-Boosting Machine (LightGBM) is a gradient boosting algorithm using leaf-wise growth and histogram-based splitting to enhance the training in a large dataset.

3.6 Feature representation

The textual data is transformed into numerical feature vector using Term Frequency-Inverse Document Frequency (TF-IDF) that gives a weight to each word depending on frequency of the appearance in a document and reduces a weight for a word that appears in many documents to balance the overall weights in the feature. TF-IDF (min_df=2, ngram_range=(1,2)) was used for transforming texts to vectors by considering the weights from keywords. Hyperparameters were set to Regularization Parameter =1, Tolerance=0.0001.

3.7 Class imbalance handling

The dataset mentioned in the previous section is highly imbalanced that some applications had high number of IT tickets, but some applications has very low number of tickets. In this paper, two techniques are studied to solve this problem as follows.

1. Random oversampling (ROS) is a technique to increase samples for minority classes by randomly copying minority data classes to roughly match the majority classes.
2. Class weight (CW) is a technique to adjust weights in a loss function by giving high weights to minority classes such that the models give the priority to minority classes. The weight is inversely proportional to number of samples for each class.

4 Experiments

4.1 Experiments setup

From the study, two sets of experiments were performed. The former set, called Experiment 1: Finding the best class imbalance technique, is to find the best technique (random oversampling or class weight) for class imbalance by using a short description as an input feature with stemming. The latter set, called Experiment 2: Finding the best input features, is to fine the best input features for all machine learning models using the best class imbalance technique for the former set. The lemmatization is used instead of stemming in this set.

The dataset consists of 20,586 IT Ticket records and splits into a training set and a test set with 70% and 30% respectively using Stratified split for dealing with class imbalance to keep the balance of the ratio of service offerings between the training set and the test set, instead of chronological split. However, before using the stratified split, the class distribution was under investigated and if a class has less than 2 samples, the class will be removed for preventing the errors, since stratified split requires to have a sample in both training and testing sets.

4.2 Performance metrics

In this study, there are 4 performance metrics, i.e. accuracy, precision, recall, and F1-score. However, these metrics use a confusion matrix as a background, and it is explained as follow.

Confusion matrix is a table to evaluate the performance of a classification machine learning model by comparing predicted classes with actual classes as shown in Table 2.

	Actual positive	**Actual negative**
Predicted positive	True positive (TP)	False positive (FP)
Predicted negative	False negative (FN)	True negative (TN)

Table 2: Confusion matrix

1. *Accuracy* is a ratio of accurately classified IT Tickets over total number of IT Tickets, as shown in (1).

$$accuracy = \frac{TP+TN}{TP+FP+TN+FN} \quad (1)$$

2. *Precision* is a ratio of classified IT Tickets as positive over total number of both true positive and false positive actual IT Tickets, as shown in (2).

$$precision = \frac{TP}{TP+FP} \quad (2)$$

3. *Recall* is a ratio of classified IT Tickets as positive over total number of both true positive and false negative actual IT Tickets, as shown in (3).

$$recall = \frac{TP}{TP+FN} \quad (3)$$

4. *F1-score* is a harmonic mean of precision and recall as shown in (4), which is a main performance metric in this paper because it is good for imbalanced dataset.

$$F1 - score = \frac{2 \times precision \times recall}{precision + recall} \quad (4)$$

For multi-class classification, there are 142 classes for this dataset, and the accuracy, precision, recall, and F1-score were calculated as weighted average to reflect the overall performance of the models for each class to reduce a variation of the performance from small sample classes.

5 Experiment results and analysis

5.1 Experiment 1: Finding the best class imbalance technique

In this experiment, two class imbalance techniques, which are a random over sampling technique and a class weight technique, are under investigated. Figure 5 shows the results from a short description input with the random over sampling technique to overcome the class imbalance problem. The results show that Linear Support Vector Classification (LinearSVC) or Support Vector Machine (SVM) gives the highest F1-score at 79.12%. Then, the results from a class weight technique instead of random over sampling technique, are shown in Figure 6. The highest F1-score is 79.74% for the SVM model. When lemmatization is used instead of text stemming shown in Figure 7, the F1-score is increased to 79.77% for SVM models. Figure 8 shows the summary of finding the best class imbalance technique by using SVM with the class weight and lemmatization techniques, and the highest F1-score is 79.77%.

The most misclassification application is Application SER_114, which has 13 records. F1-score and recall for this application were 13% and 15% respectively. This application was new and had same functions with other applications but different user interface (UI) in the same Business Unit (BU) and Company. However, different applications for different factory can have the same functions in the same BU. This is a difficult problem for classification.

```
Feature Input     : Short description
Total Classes     : 142
BASE_MODEL        : True
LEMMARTIZING      : False
IMBALANCE MODE    : ros
CALIBRATED_MODE   : False
TUNED MODE        : False
```

Model	Accuracy	Precision	Recall	F1
LinearSVC + ROS	0.7847	0.8157	0.7847	0.7912
Logistic Regression + ROS	0.7820	0.8094	0.7820	0.7880
MultinomialNB + ROS	0.7498	0.8309	0.7498	0.7778
BernoulliNB + ROS	0.7086	0.8170	0.7086	0.7480
XGB + ROS	0.7117	0.7624	0.7117	0.7205
SGDClassifier + ROS	0.6579	0.8591	0.6579	0.7194
LightGBM + ROS	0.7054	0.7380	0.7054	0.7079
DecisionTree + ROS	0.6892	0.7134	0.6892	0.6943

Figure 5: Results from a short description with random over sampling technique

```
Feature Input     : Short description
Total Classes     : 142
BASE_MODEL        : True
LEMMARTIZING      : False
IMBALANCE MODE    : class_weight
CALIBRATED_MODE   : False
TUNED MODE        : False
```

Model	Accuracy	Precision	Recall	F1
LinearSVC + classweight	0.7813	0.8356	0.7813	0.7974
SGDClassifier + classweight	0.6935	0.8457	0.6935	0.7412
Logistic Regression + classweight	0.6851	0.8329	0.6851	0.7353
MultinomialNB + SampleWeight	0.6382	0.8782	0.6382	0.6981
DecisionTree + classweight	0.6814	0.7106	0.6814	0.6898
BernoulliNB + SampleWeight	0.5482	0.7943	0.5482	0.6001
LightGBM + classweight	0.5175	0.7307	0.5175	0.5869
XGB + SampleWeight	0.4408	0.6934	0.4408	0.5140

Figure 6: Results from a short description with class weight technique

```
Feature Input     : Short description
Total Classes     : 142
BASE_MODEL        : False
LEMMARTIZING      : True
IMBALANCE MODE    : class_weight
CALIBRATED_MODE   : False
TUNED MODE        : False
```

Model	Accuracy	Precision	Recall	F1
LinearSVC + classweight	0.7818	0.8356	0.7818	0.7977
SGDClassifier + classweight	0.6924	0.8459	0.6924	0.7402
Logistic Regression + classweight	0.6840	0.8321	0.6840	0.7342
MultinomialNB + SampleWeight	0.6381	0.8780	0.6381	0.6979
DecisionTree + classweight	0.6802	0.7095	0.6802	0.6880
BernoulliNB + SampleWeight	0.5477	0.7951	0.5477	0.5995
LightGBM + classweight	0.5187	0.7232	0.5187	0.5852
XGB + SampleWeight	0.4411	0.6937	0.4411	0.5140

Figure 7: Results from short description with class weight technique and lemmatization

Base Model

Input	Imbalance Technique	Best Algorithm	F1-score	Performance Increase
Short Description	Random Over Sampling	TF-IDF + SVM	0.7912	-

Improve Imbalance Technique

Input	Imbalance Technique	Best Algorithm	F1-score	Performance Increase
Short Description	Class Weight	TF-IDF + SVM	0.7974	0.0062

Improve Text Normalization using Lemmatization

Input	Imbalance Technique	Best Algorithm	F1-score	Performance Increase
Short Description	Class Weight	TF-IDF + SVM	0.7977	0.0003

Figure 8: Summary of Experiment 1: finding the best class imbalance technique

5.2 Experiment 2: Finding the best input features

From the previous experiment, the best class imbalance technique is to use a class weight technique, and the lemmatization technique also helps improving the performance for SVM model. Several experiments were explored using different input features and machine learning models to confirm that SVM gives the best performance. The results in Table 3 showed that SVM outperformed other models in terms of accuracy at 85.59% and F1-score at 85.71% using a class weight technique and lemmatization. Hybrid input features can increase the accuracy and F1-score. Most classical machine learning models showed that Hybrid 5 input feature, i.e., Short description, Description, Business unit, Sub business unit, and Company code, gave the best accuracy and F1-score when comparing to other input feature combinations. This is because the organizational contextual features can help the model to scope the selection of application and the F1-score is increased significantly from the baseline 1 input. However, Bernoulli Naïve Bayes model gave the lowest accuracy at 56.57% and F1-score at 61.45% for Hybrid 5 input feature and it is not suitable for IT Ticket classification.

Input feature	Accuracy	Precision	Recall	F1-Score	F1-Score

combination					improvement
Support Vector Machine (SVM)					
Baseline 1	0.7813	0.8356	0.7813	0.7974	-
Hybrid 2	0.8014	0.8345	0.8014	0.8095	+1.21%
Hybrid 3	0.8298	0.8500	0.8298	0.8420	+4.46%
Hybrid 4	0.8402	0.8522	0.8402	0.8420	+4.46%
Hybrid 5	0.8559	0.8665	0.8559	**0.8571**	**+5.97%**
Logistic Regression (LR)					
Baseline 1	0.6851	0.8329	0.6851	0.7353	-
Hybrid 2	0.7021	0.8004	0.7021	0.7219	-1.34%
Hybrid 3	0.7171	0.8017	0.7171	0.7353	0.00%
Hybrid 4	0.7262	0.8053	0.7262	0.7418	0.65%
Hybrid 5	0.7453	0.8129	0.7453	**0.7568**	**2.15%**
Multinomial Naive Bayes (MNB)					
Baseline 1	0.6382	0.8782	0.6382	0.6981	-
Hybrid 2	0.6376	0.8764	0.6376	0.6964	-0.17%
Hybrid 3	0.6369	0.876	0.6369	0.6917	-0.64%
Hybrid 4	0.6389	0.8699	0.6389	0.6884	-0.97%
Hybrid 5	0.6502	0.8609	0.6502	**0.695**	**-0.31%**
Bernoulli Naive Bayes (BNB)					
Baseline 1	0.5482	0.7943	0.5482	0.6001	-
Hybrid 2	0.531	0.7278	0.531	0.5712	-2.89%
Hybrid 3	0.5602	0.7563	0.5602	0.6022	0.21%
Hybrid 4	0.5621	0.7639	0.5621	0.6052	0.51%
Hybrid 5	0.5657	0.7865	0.5657	**0.6145**	**1.44%**
Stochastic Gradient Descent (SGD)					
Baseline 1	0.6935	0.8457	0.6935	0.7412	-
Hybrid 2	0.6963	0.8179	0.6963	0.7203	-2.09%
Hybrid 3	0.7081	0.8199	0.7081	0.7297	-1.15%
Hybrid 4	0.7085	0.8174	0.7085	0.7272	-1.40%
Hybrid 5	0.7221	0.8175	0.7221	**0.7342**	**-0.70%**

Table 3: Hybrid input features performance comparison with different machine learning models using class weight technique and lemmatization

Input feature combination	Accuracy	Precision	Recall	F1-Score	F1-Score improvement
Decision Tree (DT)					
Baseline 1	0.6814	0.7106	0.6814	0.6898	-
Hybrid 2	0.676	0.681	0.676	0.6726	-1.72%
Hybrid 3	0.5874	0.6401	0.5874	0.5868	-10.30%
Hybrid 4	0.6061	0.6681	0.6061	0.6077	-8.21%
Hybrid 5	0.6312	0.6846	0.6312	**0.6315**	**-5.83%**
Extreme Gradient Boosting (XGB)					
Baseline 1	0.4408	0.6934	0.4408	0.514	-
Hybrid 2	0.4607	0.7179	0.4607	0.536	2.20%
Hybrid 3	0.6042	0.6797	0.6042	0.6098	9.58%
Hybrid 4	0.6384	0.7014	0.6384	0.6442	13.02%
Hybrid 5	0.6673	0.7192	0.6673	**0.6707**	**15.67%**
Light Gradient-Boosting Machine (LightGBM)					
Baseline 1	0.5175	0.7307	0.5175	0.5869	-
Hybrid 2	0.5662	0.7378	0.5662	0.6257	3.88%
Hybrid 3	0.6348	0.6981	0.6348	0.6437	5.68%
Hybrid 4	0.6541	0.7153	0.6541	0.6614	7.45%
Hybrid 5	0.6781	0.734	0.6781	**0.6843**	**9.74%**

Table 3: Hybrid input features performance comparison with different machine learning models using class weight technique and lemmatization (cont.)

6 Conclusion and Future work

IT ticket classification problem is a crucial research topic for IT outsourcing company to increase the business process performance within Service Level Agreement (SLA). This paper proposed to use machine learning models, i.e. Support Vector Machine (SVM), Stochastic Gradient Descent (SGD), Logistic Regression (LR), Multinomial Naïve Bayes (MNB), Decision Tree (DT), Bernoulli Naïve Bayes (BNB), Light Gradient-Boosting Machine (LightGBM), and Extreme Gradient Boosting (XGB) to classify IT tickets. The dataset is from an IT outsourcing company with 20,586 records to be classified into 142 applications. The performance metrics are accuracy, precision, recall, and F1-score. Two sets of experiments, i.e. Experiment 1: finding the best class imbalance technique and Experiment 2: finding the best input features, were conducted. From the experiments using a class weight with lemmatization, and all 6 input features, SVM gives the best F1-score at 85.71% and the features are short description, description, business unit, sub business unit, and company code.

The classical machine learning models were thoroughly explored in this paper. For the future work, Recurrent Neural Network (RNN) and Bidirectional Encoder Representations from Transformers (BERT), which are modern machine learning techniques, will be used as classification techniques to enhance the performance to IT ticket classification problem.

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A Multi-Criteria Decision-Making Framework for Optimal Supplier Selection in E-Commerce*

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Abstract

The strategic supplier selection is one of the most important success factors in E-commerce operations, as it affects cost efficiency, product quality, delivery performance, and consequently customer satisfaction. This is a complex multi-criteria decision-making (MCDM) problem that demands highly sophisticated analytical tools that can address quantitative as well as qualitative criteria. This paper carries out an extensive comparative analysis of four prominent MCDM techniques: Analytic Hierarchy Process (AHP), Analytic Network Process (ANP), Fuzzy Analytic Hierarchy Process (FAHP), and Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS). This comparative analysis is carried out on a unified numerical example for selecting the most appropriate supplier in E-commerce operations.Our comparative analysis shows that all methodologies consistently identified Supplier S3 as the preferred supplier, although differences emerged in the ranking of lower-ranked alternatives. Among the evaluated methods, the hybrid FAHP-TOPSIS framework demonstrated strong capability in combining uncertainty handling with ranking discrimination, making it particularly suitable for supplier evaluation in E-commerce environments. This is because it successfully handles the uncertainty of judgment in supplier selection while still being precise in ranking supplier performance.

1 Introduction

Significant changes have been witnessed in the traditional supply chain due to the exponential growth of the global E-commerce industry, thus resulting in increased competition and high expectations from customers in terms of speedy delivery, product quality, and services. In the

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current hyper-competitive environment, the selection of suppliers has developed from a mere procurement activity to a strategic activity that plays a vital role in the success of the business [21, 13].

Conventional supplier selection models have traditionally focused on cost factors while overlooking other important supplier selection criteria. This has led to the increasing popularity of Multi-Criteria Decision-Making (MCDM) models, which offer a framework for the analysis of conflicting criteria. Among the popular MCDM models, the Analytic Hierarchy Process (AHP), Analytic Network Process (ANP), Fuzzy Analytic Hierarchy Process (FAHP), and Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) have proved to be highly successful in complex situations [18, 16].

The objective of this paper is to conduct a comprehensive analysis of the above-mentioned four MCDM methodologies for the case of E-commerce supplier selection. In this context, the above-mentioned methodologies are implemented through a uniform numerical example, and the comparative results are calculated, as well as the relative advantages and disadvantages of each methodology. The research objective is to offer useful advice to E-commerce managers for the selection and application of the most appropriate supplier selection methodologies based on their specific needs and requirements.

2 Literature Review

Supplier selection has received considerable attention from researchers in the area of operations management and supply chain management literature. The pioneering work of [21, 9] offered a thorough review of the applications of MCDM in the context of sustainable supplier selection. The study indicated the dominance of AHP and TOPSIS methods. The Analytic Hierarchy Process (AHP), proposed by Saaty [18, 12, 3], is a powerful tool for handling complex decision-making problems. It uses a hierarchy of elements to derive the weights using pairwise comparisons [**?**, 14, 6].

In recognition of the limitation of the strict hierarchy of the AHP approach, the concept of the Analytic Network Process (ANP) was proposed to address the interdependencies among the elements of the problem [4, 1, 10]. This feature of the model is particularly useful for handling complex supply chain problems where the criteria interact with each other.

To manage the uncertainty and vagueness of human judgment, fuzzy set theory was incorporated into AHP, leading to various Fuzzy AHP (FAHP) techniques that support linguistic variables and imprecise judgments [8, 20, 19]. The Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS), developed by Hwang and Yoon, is based on the assumption that the best option should, at the same time, have the shortest distance from the positive ideal solution and the farthest distance from the negative ideal solution [11, 2, 15].

The computational efficiency and logical simplicity of TOPSIS have also made this method a favorite in supplier selection problems, often in combination with AHP or FAHP for criterion weights and TOPSIS for final ranking [17, 23, 7]. Current research trends are devoted to the extension of these MCDM techniques to E-commerce problems, with particular attention to technological integration, return process, and scalability of alternatives during peak periods [24, 22, 5].

Unlike previous studies that focus on individual MCDM techniques or provide only literature-level comparisons, this study implements AHP, ANP, FAHP, TOPSIS, and a hybrid FAHP-TOPSIS framework within a unified supplier-selection scenario. The contributions of this study are threefold: (1) providing a consistent evaluation environment for comparing classical, network-based, fuzzy, and distance-based MCDM methods; (2) examining differences between

Table 1: Comparison of Existing Studies and This Work

Study	AHP	ANP	FAHP	TOPSIS	E-Commerce	Comparative
Vaidya & Kumar (2006)	✓	–	–	–	–	–
Kahraman et al. (2008)	–	–	✓	–	–	–
Buyukozkan & Cifci (2009)	–	✓	–	–	✓	–
Kumar et al. (2021)	✓	–	✓	✓	✓	Partial
Wang et al. (2022)	Survey	Survey	Survey	Survey	✓	No
This Study	✓	✓	✓	✓	✓	**Yes**

crisp and fuzzy weighting approaches; and (3) evaluating the suitability of these methodologies for supplier selection in E-commerce environments characterized by uncertainty, rapid demand fluctuations, and technology integration requirements.

Nevertheless, a comprehensive comparative study of all four methodologies under a unified framework for E-commerce supplier selection remains underrepresented in the literature, and this is a significant research gap that this study seeks to fill.

3 Methodology

3.1 Proposed Evaluation Framework

Our proposed supplier evaluation framework comprises four sequential phases: (1) Criteria identification and hierarchy construction, (2) Data collection and preparation, (3) MCDM application, and (4) Results synthesis and validation. This structured approach ensures methodological rigor while maintaining practical applicability for E-commerce enterprises.

3.2 Criteria Selection

The supplier evaluation data used in this study represent an illustrative case developed to demonstrate the comparative behavior of different MCDM methodologies under a common decision environment. The supplier performance values were generated using ranges commonly reported in supplier-selection literature and validated through consultation with three domain experts with experience in procurement, supply chain management, and E-commerce operations.

For pairwise comparisons, each expert independently assessed the relative importance of criteria using Saaty's 1–9 scale. The final comparison matrix was obtained using the geometric mean aggregation method, which is widely recommended for combining multiple expert judgments in AHP-based studies.

Based on comprehensive literature review and E-commerce industry expertise, we identified four primary criteria and seven sub-criteria for supplier evaluation:

- **Cost (C1):** Unit Price (C11), Shipping Cost (C12)
- **Quality (C2):** Defect Rate (C21), Order Accuracy (C22)
- **Delivery (C3):** On-Time Delivery Rate (C31), Lead Time (C32)
- **Technology (C4):** System Integration Capability (C41)

The selected criteria reflect key operational requirements in E-commerce environments. Cost remains important because of intense price competition. Quality directly influences customer satisfaction and product return rates. Delivery performance is critical for meeting fulfillment service-level agreements and maintaining customer trust during peak demand periods. Technology capability supports real-time inventory synchronization, automated order processing, shipment visibility, and scalable platform operations. These characteristics distinguish E-commerce supplier selection from traditional procurement settings.

3.3 Detailed MCDM Methodologies Implementation

3.3.1 AHP Implementation

The AHP methodology was implemented through the following detailed steps:

Step 1: Construct Pairwise Comparison Matrix

$$A = \begin{bmatrix} 1 & 3 & 2 & 5 \\ 1/3 & 1 & 1/2 & 3 \\ 1/2 & 2 & 1 & 4 \\ 1/5 & 1/3 & 1/4 & 1 \end{bmatrix} \tag{1}$$

Step 2: Calculate Normalized Matrix

$$N_{ij} = \frac{a_{ij}}{\sum_{i=1}^{4} a_{ij}} \tag{2}$$

$$N = \begin{bmatrix} 0.491 & 0.474 & 0.533 & 0.385 \\ 0.164 & 0.158 & 0.133 & 0.231 \\ 0.245 & 0.316 & 0.267 & 0.308 \\ 0.098 & 0.053 & 0.067 & 0.077 \end{bmatrix} \tag{3}$$

Step 3: Compute Weight Vector

$$w_i = \frac{\sum_{j=1}^{4} N_{ij}}{4} \tag{4}$$

$$W_{AHP} = \begin{bmatrix} 0.471 \\ 0.172 \\ 0.284 \\ 0.074 \end{bmatrix} \tag{5}$$

Step 4: Consistency Verification

$$\lambda_{max} = 4.081 \tag{6}$$

$$CI = \frac{\lambda_{max} - 4}{3} = 0.027 \tag{7}$$

$$CR = \frac{CI}{RI} = \frac{0.027}{0.90} = 0.030 < 0.1 \tag{8}$$

3.3.2 FAHP Implementation

The FAHP methodology employed triangular fuzzy numbers (TFNs) and extent analysis:

Step 1: Fuzzy Pairwise Comparison Matrix

$$\tilde{A} = \begin{bmatrix} (1,1,1) & (2,3,4) & (1,2,3) & (4,5,6) \\ (1/4,1/3,1/2) & (1,1,1) & (1/3,1/2,1) & (2,3,4) \\ (1/3,1/2,1) & (1,2,3) & (1,1,1) & (3,4,5) \\ (1/6,1/5,1/4) & (1/4,1/3,1/2) & (1/5,1/4,1/3) & (1,1,1) \end{bmatrix} \tag{9}$$

Since the supplier evaluation matrix contains seven sub-criteria while the AHP weighting process produces weights for four main criteria, the sub-criteria were first aggregated into their corresponding main criterion groups. Specifically, Unit Price and Shipping Cost were aggregated under Cost; Defect Rate and Order Accuracy under Quality; On-Time Delivery and Lead Time under Delivery; and System Integration Capability under Technology. The resulting 3×4 matrix was then multiplied by the AHP-derived weight vector to obtain the weighted normalized decision matrix.

Step 2: Calculate Fuzzy Synthetic Extent

$$S_i = \sum_{j=1}^{4} \tilde{a}_{ij} \otimes \left[\sum_{i=1}^{4} \sum_{j=1}^{4} \tilde{a}_{ij} \right]^{-1} \tag{10}$$

Step 3: Defuzzification and Weight Calculation Using centroid method for defuzzification:

$$W_{FAHP} = \begin{bmatrix} 0.463 \\ 0.186 \\ 0.271 \\ 0.080 \end{bmatrix} \tag{11}$$

3.3.3 TOPSIS Implementation

The TOPSIS methodology was implemented as follows:

Step 1: Normalize Decision Matrix

$$r_{ij} = \frac{x_{ij}}{\sqrt{\sum_{i=1}^{3} x_{ij}^2}} \tag{12}$$

$$R = \begin{bmatrix} 0.566 & 0.656 & 0.681 & 0.567 & 0.552 & 0.802 & 0.536 \\ 0.654 & 0.492 & 0.490 & 0.579 & 0.576 & 0.535 & 0.689 \\ 0.603 & 0.574 & 0.327 & 0.586 & 0.604 & 0.267 & 0.613 \end{bmatrix} \tag{13}$$

Step 2: Construct Weighted Normalized Matrix Using AHP weights:

$$V = R \times diag(W_{AHP}) = \begin{bmatrix} 0.266 & 0.113 & 0.193 & 0.042 \\ 0.308 & 0.085 & 0.139 & 0.051 \\ 0.284 & 0.099 & 0.093 & 0.045 \end{bmatrix} \tag{14}$$

Step 3: Determine Ideal Solutions

$$A^+ = \{0.284, 0.113, 0.093, 0.051\} \tag{15}$$

$$A^- = \{0.266, 0.085, 0.193, 0.042\} \tag{16}$$

Step 4: Calculate Separation Measures

$$S_1^+ = \sqrt{\sum_{j=1}^{7}(v_{1j} - v_j^+)^2} = 0.124 \tag{17}$$

$$S_2^+ = \sqrt{\sum_{j=1}^{7}(v_{2j} - v_j^+)^2} = 0.087 \tag{18}$$

$$S_3^+ = \sqrt{\sum_{j=1}^{7}(v_{3j} - v_j^+)^2} = 0.048 \tag{19}$$

$$S_1^- = \sqrt{\sum_{j=1}^{7}(v_{1j} - v_j^-)^2} = 0.087 \tag{20}$$

$$S_2^- = \sqrt{\sum_{j=1}^{7}(v_{2j} - v_j^-)^2} = 0.124 \tag{21}$$

$$S_3^- = \sqrt{\sum_{j=1}^{7}(v_{3j} - v_j^-)^2} = 0.126 \tag{22}$$

Step 5: Compute Relative Closeness

$$C_i = \frac{S_i^-}{S_i^+ + S_i^-} \tag{23}$$

$$C = \begin{bmatrix} 0.412 & 0.587 & 0.723 \end{bmatrix} \tag{24}$$

3.3.4 ANP Implementation

Unlike AHP, ANP allows interdependencies among evaluation criteria. In this study, Cost, Quality, Delivery, and Technology were modeled as interconnected clusters. Technology influences Delivery through system integration and information visibility, while Delivery affects Quality through order fulfillment accuracy. Cost is also indirectly influenced by Technology and Delivery because improved automation and logistics efficiency can reduce operational expenses. These relationships were incorporated into the ANP supermatrix structure.

Step 1: Construct Unweighted Supermatrix

$$W = \begin{bmatrix} W_{11} & W_{12} & \cdots & W_{1n} \\ W_{21} & W_{22} & \cdots & W_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ W_{n1} & W_{n2} & \cdots & W_{nn} \end{bmatrix} \tag{25}$$

Step 2: Calculate Limit Supermatrix

$$W_{ANP} = \lim_{k \to \infty} W^k = \begin{bmatrix} 0.458 \\ 0.179 \\ 0.289 \\ 0.074 \end{bmatrix} \tag{26}$$

3.3.5 Hybrid FAHP-TOPSIS Implementation

The hybrid approach combined FAHP weights with TOPSIS ranking:

$$C_{Hybrid} = \begin{bmatrix} 0.428 & 0.612 & 0.789 \end{bmatrix} \tag{27}$$

The System Integration Capability criterion (C41) was evaluated using a 1–10 scale reflecting

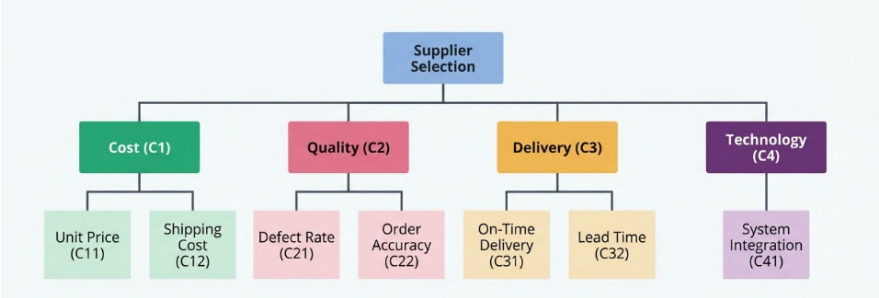

Figure 1: Hierarchical structure of supplier selection criteria

Table 2: Supplier Performance Decision Matrix

Supplier	C11 ($)	C12 ($)	C21 (%)	C22 (%)	C31 (%)	C32 (days)	C41 (1-10)
S1	45	8	2.5	97	92	3	7
S2	52	6	1.8	99	96	2	9
S3	48	7	1.2	98	98	1	8

the supplier's ability to integrate with the E-commerce platform through APIs, EDI systems, inventory synchronization, order tracking, and automated data exchange mechanisms. A score of 1 indicates very limited integration capability, whereas a score of 10 represents fully automated real-time integration supporting seamless operational coordination.

4 Results and Discussion

The implementation results across all methodologies are summarized in Table **??**. Two distinct ranking patterns emerged. AHP, ANP, and FAHP ranked suppliers as $S_3 > S_1 > S_2$, whereas TOPSIS and FAHP-TOPSIS produced the ranking $S_3 > S_2 > S_1$. Despite these differences, all methodologies consistently identified supplier S_3 as the best-performing alternative.

The ranking divergence between S_1 and S_2 reflects methodological differences. Weight-aggregation approaches such as AHP, ANP, and FAHP emphasize overall weighted criterion contributions, slightly favoring S_1. In contrast, TOPSIS-based approaches evaluate alternatives according to their distance from ideal and anti-ideal solutions, enabling S_2 to outperform S_1 because of stronger performance in quality, delivery, and technology-related attributes.

Table 4 highlights the strengths and limitations of each methodology. ANP is advantageous when interdependencies among criteria are important, whereas FAHP effectively captures uncertainty in expert judgments. TOPSIS provides strong ranking discrimination through distance-based evaluation. The hybrid FAHP-TOPSIS framework combines uncertainty handling with

Table 3: Comparative Results of MCDM Methodologies

Methodology	**S1 Score**	**S2 Score**	**S3 Score**	**Ranking**	**CR/Consistency**
AHP	0.324	0.285	0.391	$S3 > S1 > S2$	CR = 0.030
ANP	0.312	0.274	0.414	$S3 > S1 > S2$	Consistent
FAHP	0.318	0.285	0.397	$S3 > S1 > S2$	Consistent
TOPSIS	0.412	0.587	0.723	$S3 > S2 > S1$	N/A
FAHP-TOPSIS	0.428	0.612	0.789	$S3 > S2 > S1$	N/A

Table 4: Comparison of MCDM Methodologies

Criterion	AHP	ANP	FAHP	TOPSIS	FAHP-TOPSIS
Uncertainty Handling	Low	Low	High	Low	High
Interdependency Modeling	No	Yes	No	No	No
Ranking Discrimination	Medium	Medium	Medium	High	High
Computational Complexity	Low	High	Medium	Medium	High
Ease of Implementation	High	Low	Medium	High	Medium

ranking precision, making it particularly suitable for E-commerce supplier evaluation environments.

5 Limitations and Future Research

5.1 Study Limitations

This study has several limitations. First, the comparative analysis is based on a small illustrative example involving only three suppliers, limiting the generalizability of the findings. Second, the evaluation criteria primarily focus on traditional supplier-selection factors and include only one technology-related criterion. Third, the analysis assumes static supplier performance and does not capture real-time changes commonly observed in E-commerce environments. Finally, sensitivity analysis was not performed to evaluate ranking stability under varying business conditions and decision-maker preferences.

5.2 Future Research Directions

Future research can address these limitations in several ways. First, larger empirical case studies involving real supplier datasets and a greater number of alternatives should be conducted to validate the findings. Second, additional E-commerce-specific criteria such as reverse-logistics capability, peak-season scalability, API integration maturity, and fulfillment reliability can be incorporated into the evaluation framework. Third, sensitivity analysis and scenario-based experimentation can be performed to assess ranking stability under peak-demand and off-peak conditions. Fourth, machine learning techniques may be integrated for dynamic weight adjustment using real-time operational data. Finally, sustainability and supply chain resilience considerations can be incorporated to support long-term strategic supplier selection decisions.

6 Conclusion

This study has presented a comprehensive comparative analysis of four prominent MCDM methodologies for supplier selection in E-commerce. The hybrid FAHP-TOPSIS approach emerges as particularly promising, effectively combining uncertainty-handling capabilities with precise alternative ranking. This research provides practical guidance for E-commerce managers in selecting appropriate supplier evaluation methodologies based on specific operational requirements.

Future research should focus on developing integrated decision support systems with real-time data integration, machine learning enhancement, and expanded sustainability considerations to address evolving E-commerce supply chain challenges.

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Multi-Criteria Decision Analysis for Strategic Supplier Selection: A Comparative Study of AHP, ANP, and TOPSIS in the Electronics Industry*

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Abstract

Electronics industry is a complex industry, and the system of supplier selection is linked with multiple contradictory conditions, qualitative selection and uncertainty. Even the more conservative practices are not even inclined towards these complications. In this work we use three Multi-Criteria Decision-Making (MCDM) methods (AHP, ANP, and TOPSIS) to classify the suppliers in a structured manner based on some criteria, such as Cost, Quality, Delivery, Technology, and Financial Stability. The comparative analysis indicates the strong points, the weaknesses and usability of the two approaches and is a good structure in the well-crafted and clear strategic sourcing.

1 Introduction

The capacity to choose a strategic supplier is one of the important aspects of organizational performance, sustainability, and competitive advantage in the contemporary and dynamic global supply chain environment. This is better understood in the electronics industry whereby there is a complex supply chain, a massive product recall, and a rising e-waste issue. The choice of suppliers is efficient to allow complying with the regulations, minimize operational risks, enhance

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customer satisfaction, and advance the concept of the circular economy by the opportunity to retrieve the value of the products returned [5, 10]. However, the decision-making process of choosing the suppliers is complicated by nature, and one has to consider different and sometimes conflicting criteria, based on uncertainty and the lack of information. Most of the traditional methods of selection do not consider these interdependencies, and this will result in the making of decisions that may not be optimal. Other issues complicate the decision-making process, and these include financial constraints, technicalities, volatility in the market, and lack of skilled labor. On this basis, it is with a sense of urgency that systemic, methodological, sound frameworks are needed to bring both quantitative and qualitative variables into one [24, 18, 16].

Today, the choice of suppliers is increasingly made by leveraging digital procurement platforms, analytics of data, and intelligent decision-support systems. The use of MCDM methods such as AHP, ANP, and TOPSIS in the assessment of suppliers can enhance transparency, consistency, and strategic decision-making in an electronic supply-chain ecosystem. This study thus not only provides an extension to existing supplier-selection research, but also contributes to the development of intelligent E-Business Information Systems aimed at supporting digital transformation and sustainable procurement practices.

2 Multi-Criteria Decision-Making (MCDM)

Multi-Criteria Decision-Making (MCDM) methods provide structured approaches that are used to assess alternatives based on more than one criterion, where alternative criteria may be in conflict [19, 22]. In supplier selection, where financial stability, technological capability, delivery reliability, quality, and cost are all qualitative and quantitative criteria that need to be taken into account, MCDM techniques provide a systematic basis for integrating these factors in a single, transparent evaluation process [4, 9]. Each of the three methods used in this study deals with this challenge differently: AHP through pairwise comparison in a hierarchy, ANP through interdependency modelling in a network, and TOPSIS through geometric distance from ideal solutions.

3 Literature Review

The process of strategic selection of suppliers is a conclusive performance determinant of organizational business in the modern global supply chain management (SCM) landscape. This is particularly applicable in the electronics industry, which is typified by complex supply chains, high product recall, and increasing e-waste problems [5, 11, 8]. The selection of good suppliers enhances compliance with environmental requirements, minimizes operational risk, improves customer satisfaction, and is likely to reclaim the value on any returned product, hence supporting the objectives of the circular economy.

3.1 Problems in Supplier Selection

The procedure is complicated because there are many, often interdependent criteria, unclear or incomplete information, and the qualitative character of many assessment factors. These interdependencies are often ignored by historical selection practices, resulting in suboptimal decisions [13, 12]. In addition, systematically acceptable methodological designs are required since most studies rely on demonstrative numerical illustrations. Other issues such as financial constraints, regulatory conditions, technical conditions, unstable markets, and unavailability

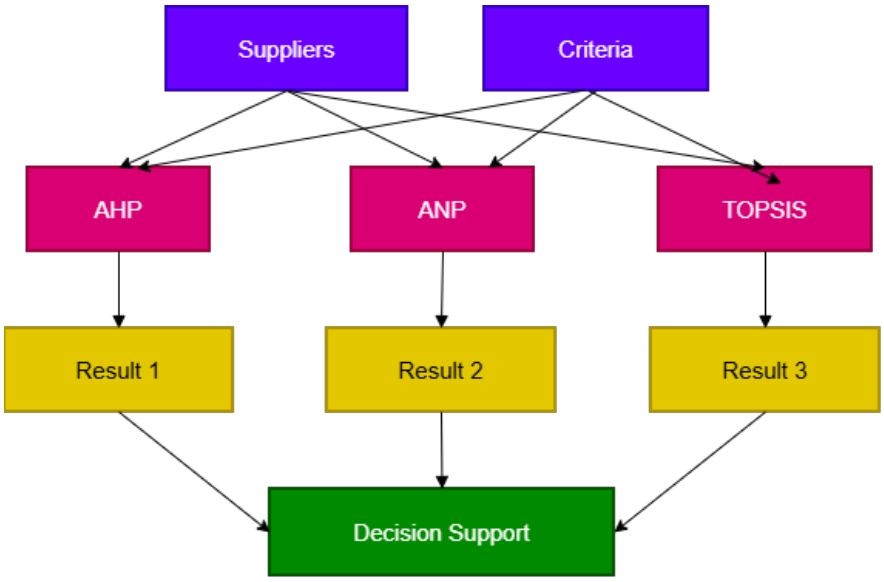

Figure 1: **Framework for Supplier Evaluation Using MCDM Techniques**

of skilled workforce in other markets such as Taiwan also act as aggravating factors in the management of suppliers.

3.2 MCDM Approaches in Supplier Selection

MCDM techniques are widely adopted across supply chain management, manufacturing, and procurement due to their capability to combine quantitative metrics with qualitative expert judgment [1, 7]. Among these, AHP remains the most frequently applied method, valued for its consistency-checking mechanism and intuitive hierarchical structure [17, 2]. However, its assumption of criteria independence limits its applicability in complex industrial contexts where criteria interact.

To address this, researchers have explored hybrid and extended approaches. ANP relaxes the independence assumption by modelling feedback and interdependencies between criteria through supermatrix construction, making it particularly suited to electronics supply chains where quality, technology, and delivery are interrelated [4, 6]. TOPSIS complements both methods by providing an efficient geometric ranking mechanism that is commonly integrated with AHP- or ANP-derived weights to produce final supplier rankings [21].

Recent studies have demonstrated the value of combining these methods. Hesami [4] applied a hybrid AHP-TOPSIS approach under rough uncertainty in electronics supplier selection, while Menon and Ravi [13] showed that AHP-TOPSIS integration improves transparency and consistency in sustainable supplier evaluation. Despite this body of work, direct three-way parallel comparisons of AHP, ANP, and TOPSIS on an identical case with full numerical traceability remain underrepresented, particularly within the electronics industry context. This paper addresses that gap.

3.3 Key MCDM Methodologies

1) Analytic Hierarchy Process (AHP): AHP was developed by Thomas L. Saaty and classifies criteria into a hierarchy, then makes pairwise comparisons to derive weights [17]. Its consistency tests reduce subjectivity, but it assumes independence of factors and uses precise numerical inputs, which may fail to capture interdependencies.

2) Technique for Order Preference by Similarity to Ideal Solution (TOPSIS): TOPSIS ranks alternatives by how close they are to a best solution and how far from a negative-ideal solution [21]. It is valued for conceptual clarity and computational efficiency and is commonly used together with AHP, where AHP provides criterion weights and TOPSIS produces final rankings.

3) Fuzzy Set Extensions: Fuzzy logic can overcome linguistic vagueness and subjectivity. Fuzzy AHP accepts qualitative criteria, Fuzzy TOPSIS measures alternatives with fuzzy numbers, and Fuzzy ANP considers interdependencies and feedback between criteria [25]. Interval Type-2 Fuzzy Sets provide better treatment of uncertainty and have found use in algorithms such as IT2FAHP, IT2FTOPSIS, and IT2FVIKOR.

4) Hybrid Approaches: A combination of MCDM methods capitalises on the strengths of each method while reducing its weaknesses. Examples include ANP-TOPSIS under rough uncertainty, fuzzy AHP-fuzzy VIKOR, and AHP-Entropy with fuzzy TOPSIS-ELECTRE [4].

3.4 Supplier Selection in the Electronics Industry

The electronics industry, with its complicated supply chain, high rate of returns, and rapid technological change, requires sophisticated supplier selection [14]. Key evaluations include product quality, delivery, cost, technology, design, capacity, experience, reputation, responsiveness, service, ICT, and environmental and social responsibility. Green and resilient supplier selection is becoming more important, especially in Vietnam, India, Taiwan, and China, to counter disruptions and meet sustainability objectives [3].

3.5 Supplier-Selection Criteria

The criteria for supplier evaluation fall into several dimensions:

Traditional Criteria: Cost, quality of materials, delivery reliability, lead times, and financial stability; sub-factors include defect rates, consistency, delays, and proactive pricing information.

Green/Environmental Criteria: Environmental management systems, green technology, hazardous material handling, energy efficiency, recycling, green packaging, eco-design, pollution control, and resource optimisation [15].

Social Criteria: Occupational safety, human rights, corporate social responsibility, and workforce training.

Strategic/Alliance Criteria: Knowledge management, risk sharing, cultural fit, and IT integration.

Ethical Criteria: Code of conduct, transparency, and conflict-of-interest management, regarded as part of sustainable supply chains.

Resilience Capabilities: Absorptive, adaptive, and restorative capabilities; back-up suppliers; secure locations; and information continuity provisions [23].

3.6 Future Directions

New studies should incorporate Artificial Intelligence (AI) and machine learning with MCDM models to facilitate real-time, data-driven supplier analysis. Industrial and geographical expansion will improve generalisability, and more sophisticated fuzzy set approaches (e.g., neutrosophic, hesitant fuzzy, Interval Type-2) can better deal with uncertainty. Simplification of complex models, focus on social performance measures, creation of user-friendly software, and long-term cost-benefit analysis are vital to scalable sustainable supplier selection in the electronics industry.

3.7 Research Gap and Contribution

Though prior studies have integrated AHP, TOPSIS, fuzzy MCDM, and hybrid approaches in evaluating suppliers, many are dedicated either to hybrid optimisation or uncertainty modelling. Few studies exist that allow a clear comparison of AHP, ANP, and TOPSIS on the same supplier-selection scenario in the electronics industry with full numerical traceability.

To fill these gaps, this paper proposes: (1) a transparent comparative implementation of AHP, ANP, and TOPSIS in the same decision environment; (2) a structured evaluation framework tailored to the electronics industry; (3) an illustration of both independence-based and network-based MCDM approaches; and (4) a discussion of how these methods contribute to data-driven decision-making within E-Business Information Systems.

4 Methodology

The combination of the Analytic Hierarchy Process (AHP), the Analytic Network Process (ANP), and the Technique of Order Preference by Similarity to Ideal Solution (TOPSIS) can be used to build a sound framework in the decisionmaking techniques to be applied to complex decision-making related to suppliers selection. These techniques which are commonly used in hybrid forms allow structured analysis of various opposing criteria, allow information which is uncertain or inaccurate, and include the subjective aspect of qualitative analysis.

4.1 Methodological Framework

In this study, AHP, ANP, and TOPSIS are applied concurrently to the same supplier selection problem, enabling direct comparison of their outputs under identical input conditions [25]. AHP provides the primary criteria weight vector through hierarchical pairwise comparison; ANP extends this by modelling inter-criteria dependencies via supermatrix construction; and TOPSIS uses the resulting weights to rank suppliers by geometric proximity to ideal solutions [1]. The three methods form a complementary evaluation chain in which each method's output can be cross-validated against the others, strengthening the robustness of the final recommendation.

4.2 Analytic Hierarchy Process (AHP)

The Analytic Hierarchy Process (AHP) was created by Thomas L. Saaty; this approach is a hierarchical method of solving multi-criteria decisions using both quantitative data and qualitative judgments [26]. AHP systematically incorporates human judgment into decision-making by breaking down a complex decision problem into a hierarchy of objectives, criteria, sub-criteria, and alternatives.

The AHP methodology proceeds as follows:

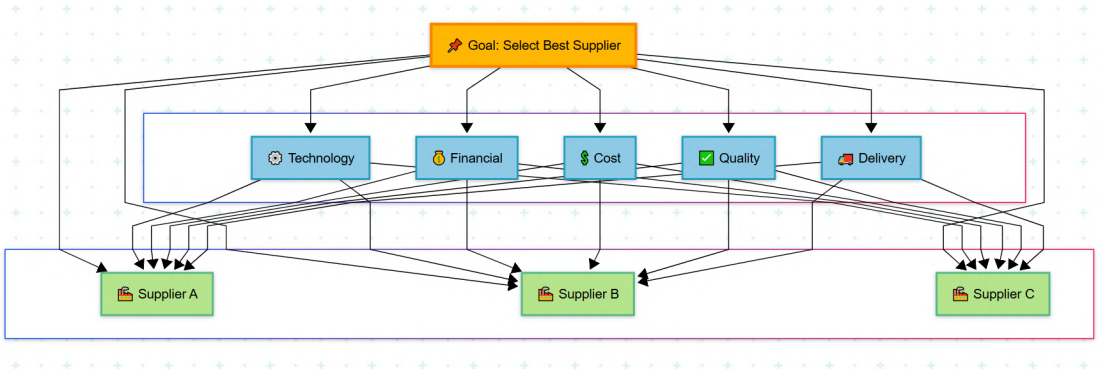

Figure 2: **Analytic Hierarchy Process Diagram**

1) Problem Definition and Objective Formulation: Clearly state the decision-making objective and scope.

2) Hierarchy Construction: Build a hierarchy with the overall objective at the top, main criteria and sub-criteria in the middle, and alternatives at the bottom.

AHP offers quantitative rigour to qualitative judgments and makes complex decisions tractable. Limitations include the assumption that criteria are independent, which may not hold in complex real-life situations [21].

4.3 Analytic Network Process (ANP)

The Analytic Network Process (ANP) is an extension of AHP that considers dependency and feedback among decision criteria [4]. Also introduced by Saaty, it describes decision problems as networks of interrelated clusters and elements, making it best applicable to complex, real-world decision situations.

ANP methodology entails:

1) Objective and Network Formulation: Determine the decision goal, criteria, and sub-criteria, and arrange them in inter-related groups to show dependencies.

2) Pairwise Comparison Collection: Gather expert opinions of relative importance among and within clusters.

3) Fuzzy Conversion (Optional): When using Fuzzy ANP, linguistic judgments can be converted to triangular fuzzy numbers to represent uncertainty and vagueness.

4) Weight Calculation: Normalise the pairwise comparisons to ensure weights sum to unity.

5) Supermatrix Construction: Construct the supermatrix, storing all interactions between elements, with each block indicating the effect of elements in one cluster on those in another [4].

6) Limit Supermatrix Computation: Raise the supermatrix to repeated powers until convergence, to obtain stable weights representing all interdependencies.

7) Priority Vector Derivation: Calculate final weights of criteria and alternatives, and perform a consistency check similar to AHP.

Unlike AHP's strict hierarchy, ANP models a network of interdependent clusters. Orange (bold) colored arrows show inner-cluster dependencies: Technology influences Quality (weight 0.40) and Financial Stability influences Delivery (weight 0.30), encoded in the unweighted

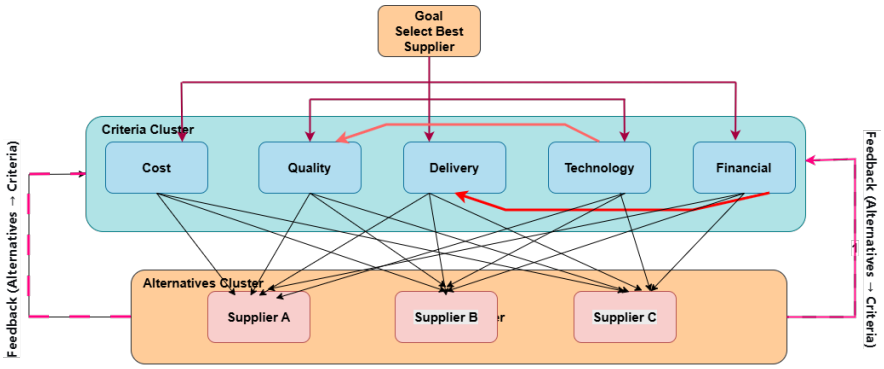

Figure 3: **Analytic Network Process Diagram**

supermatrix. Other dashed arrows show feedback loops from the Alternatives cluster back to the Criteria cluster — the defining structural feature distinguishing ANP from AHP.

ANP represents a more realistic and complete model because it includes interdependencies and feedback loops. It can further be augmented with fuzzy theory (FANP) to increase its ability to characterise uncertainty and support group consensus.

4.4 Technique for Order Preference by Similarity to Ideal Solution (TOPSIS)

TOPSIS is a distance-based MCDM approach, founded on the premise that an optimal alternative is closest to the positive ideal solution (PIS) and farthest from the negative ideal solution (NIS) [21]. Its geometrical nature, computational performance, and absence of complexities make it widely adopted in real-world decision-making.

TOPSIS procedure includes:

1) Decision Matrix Construction: Define a matrix with alternatives as rows and criteria as columns.

2) Normalisation: Convert diverse units into a dimensionless form to enable equitable comparison.

3) Weighted Normalisation: Multiply the normalised matrix by criterion weights (obtained from AHP/ANP).

4) Identification of PIS and NIS: Determine the best and worst values for each criterion across alternatives. For benefit criteria, PIS = max and NIS = min; for cost criteria, PIS = min and NIS = max.

5) Separation Measure Computation: Calculate the Euclidean distance of each alternative from the PIS and NIS.

6) Closeness Coefficient (CC_i):

$$CC_i = \frac{d_i^-}{d_i^+ + d_i^-} \tag{1}$$

where d_i^+ and d_i^- denote distances from PIS and NIS, respectively [14, 20].

7) Ranking: Rank alternatives by descending CC_i; higher values indicate superior performance.

Fuzzy TOPSIS and Rough TOPSIS extensions can model linguistic uncertainty and data imprecision, further increasing their utility in complex, uncertain environments.

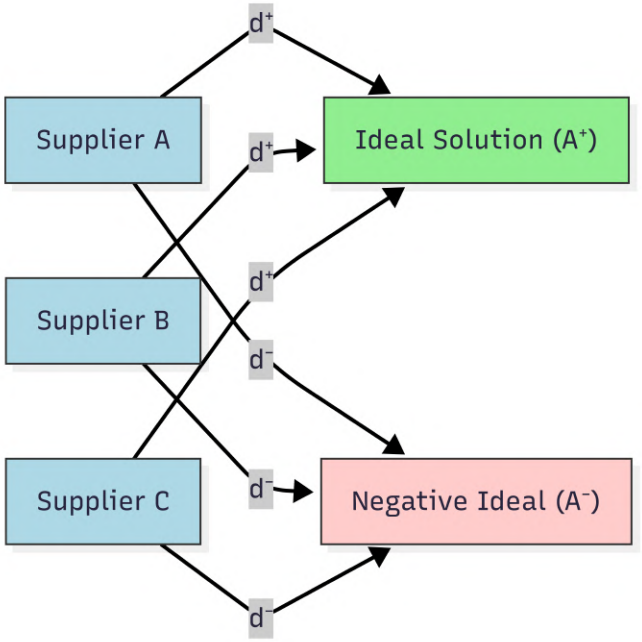

Figure 4: **TOPSIS Diagram**

5 Analysis and Calculations

This section provides a systematic method of supplier selection in the electronics industry by applying three MCDM models: AHP, ANP, and TOPSIS. All models evaluate and rank three suppliers: Supplier A (low-cost Asian component manufacturer), Supplier B (high-quality European semiconductor vendor), and Supplier C (trustworthy U.S. circuit board innovator).

The evaluation criteria, based on the literature and industry standards, are:

- **Cost** (cost-type, lower is better; in $/unit)
- **Quality** (cost-type, lower is better; defect rate in %)
- **Delivery** (benefit-type, higher is better; on-time delivery in %)
- **Technology Capability** (benefit-type, higher is better; rated 1–10)
- **Financial Stability** (benefit-type, higher is better; rated 1–10)

The decision matrix (Table 1) presents hypothetical data calibrated against representative electronics industry benchmarks from the supplier evaluation literature [4, 14, 3].

Table 1: Decision Matrix for Suppliers

Criteria	Supplier A	Supplier B	Supplier C
Cost ($/unit)	50	70	60
Quality (defect %)	2	0.5	1
Delivery (%)	85	95	90
Technology (1–10)	7	9	8
Financial (1–10)	6	8	7

5.1 Analytic Hierarchy Process (AHP) Calculations

5.1.1 Pairwise Comparison Matrix

Table 2 presents the pairwise comparison matrix for criteria based on hypothetical expert judgments.

Table 2: Pairwise Comparison Matrix for Criteria (AHP)

Criteria	Cost	Quality	Delivery	Technology	Financial
Cost	1	1/3	3	5	7
Quality	3	1	5	7	9
Delivery	1/3	1/5	1	3	5
Technology	1/5	1/7	1/3	1	3
Financial	1/7	1/9	1/5	1/3	1

5.1.2 Priority Vector Derivation

The normalised row averages yield the following priority vector:

$$
\begin{aligned}
w_{\text{Cost}} &= (0.214 + 0.186 + 0.315 + 0.306 + 0.280)/5 = 1.301/5 = \mathbf{0.260} \\
w_{\text{Quality}} &= (0.642 + 0.559 + 0.525 + 0.429 + 0.360)/5 = 2.515/5 = \mathbf{0.503} \\
w_{\text{Delivery}} &= (0.071 + 0.112 + 0.105 + 0.184 + 0.200)/5 = 0.672/5 = \mathbf{0.134} \\
w_{\text{Technology}} &= (0.043 + 0.080 + 0.035 + 0.061 + 0.120)/5 = 0.339/5 = \mathbf{0.068} \\
w_{\text{Financial}} &= (0.031 + 0.062 + 0.021 + 0.020 + 0.040)/5 = 0.174/5 = \mathbf{0.035}
\end{aligned}
$$

5.2 Consistency Check

$$
\lambda_{\max} \approx 5.22, \quad CI = \frac{\lambda_{\max} - n}{n-1} = \frac{5.22 - 5}{4} = 0.055, \quad CR = \frac{CI}{RI} = \frac{0.055}{1.12} = 0.049 \tag{2}
$$

Since $CR = 0.049 < 0.10$, the pairwise comparison matrix is considered consistent.

5.2.1 Supplier-Level Pairwise Matrices

Five supplier-level pairwise comparison matrices were constructed based on the raw performance values in Table 1 using Saaty's 1–9 scale. For cost-type criteria (Cost, Quality), suppliers with

lower values received higher priority; for benefit-type criteria (Delivery, Technology, Financial), suppliers with higher values received higher priority. The full matrices are presented in Appendix A. The corrected supplier priority vectors derived from these matrices are summarised below:

- **Cost** (A best): $A = 0.539,\ B = 0.164,\ C = 0.297$
- **Quality** (B best): $A = 0.106,\ B = 0.633,\ C = 0.260$
- **Delivery** (B>C>A): $A = 0.143,\ B = 0.571,\ C = 0.286$
- **Technology** (B>C>A): $A = 0.164,\ B = 0.539,\ C = 0.297$
- **Financial** (B>C>A): $A = 0.164,\ B = 0.539,\ C = 0.297$

5.2.2 Aggregated Supplier Priorities

$$\begin{aligned} S_A &= (0.539 \times 0.260) + (0.106 \times 0.503) + (0.143 \times 0.134) + (0.164 \times 0.068) + (0.164 \times 0.035) \\ &= 0.140 + 0.053 + 0.019 + 0.011 + 0.006 = \mathbf{0.229} \\ S_B &= (0.164 \times 0.260) + (0.633 \times 0.503) + (0.571 \times 0.134) + (0.539 \times 0.068) + (0.539 \times 0.035) \\ &= 0.043 + 0.318 + 0.077 + 0.037 + 0.019 = \mathbf{0.494} \\ S_C &= (0.297 \times 0.260) + (0.260 \times 0.503) + (0.286 \times 0.134) + (0.297 \times 0.068) + (0.297 \times 0.035) \\ &= 0.077 + 0.131 + 0.038 + 0.020 + 0.010 = \mathbf{0.276} \end{aligned}$$

AHP Result: Ranking: Supplier B (0.494) > Supplier C (0.276) > Supplier A (0.229). AHP selects Supplier B due to its emphasis on quality.

5.3 Analytic Network Process (ANP) Calculations

5.3.1 Key Equation

The limit supermatrix is defined as:

$$W^{\infty} = \lim_{k \to \infty} W^k \tag{3}$$

where W is the weighted supermatrix representing interdependencies [1].

5.3.2 Dependency Relationships

The dependency relationships and influence factors were established under simplified expert-judgment assumptions to illustrate the ANP mechanism in a pedagogical comparative framework. It was assumed that Technology Capability influences Quality (influence factor 0.40), because more advanced manufacturing technology generally leads to higher product quality and more effective defect control. Financial Stability was selected as a determinant of Delivery Performance (influence factor 0.30), reflecting its documented impact on logistics capacity and inventory flexibility of suppliers.

5.3.3 Supermatrix Construction

The unweighted supermatrix (Table 3) incorporates the corrected supplier priority vectors and the dependency influences. The weighted supermatrix was obtained by multiplying each column by the corresponding cluster weight (Criteria cluster weight = 0.75, Alternatives cluster weight = 0.25). The weighted supermatrix was then raised to successive powers until convergence (change <0.001).

Table 3: Partial Unweighted Supermatrix (ANP)

	Cost	**Qual**	**Del**	**Tech**	**Fin**	**Sup A**	**Sup B**	**Sup C**
Cost	0	0	0	0	0	0.539	0.164	0.297
Quality	0	0	0	0.40	0	0.106	0.633	0.260
Delivery	0	0	0	0	0.30	0.143	0.571	0.286
Tech	0	0	0	0	0	0.164	0.539	0.297
Fin	0	0	0	0	0	0.164	0.539	0.297
Sup A	0.30	0.20	0.30	0.35	0.25	0	0	0
Sup B	0.20	0.50	0.50	0.50	0.50	0	0	0
Sup C	0.50	0.30	0.20	0.15	0.25	0	0	0

5.3.4 Criterion Adjustments

The dependency adjustments applied to the original AHP weights are:

$$w_{\text{Quality(adj)}} = 0.503 - (0.40 \times 0.068) = 0.503 - 0.0272 = \mathbf{0.476}$$
$$w_{\text{Delivery(adj)}} = 0.134 - (0.30 \times 0.035) = 0.134 - 0.0105 = \mathbf{0.124}$$

5.3.5 Limit Supermatrix Convergence

After iterative matrix multiplication, the supermatrix converged to stable supplier priorities as shown in Table 4.

Table 4: Illustrative Convergence of the Limit Supermatrix

Iteration	**Supplier A**	**Supplier B**	**Supplier C**
W^2	0.298	0.428	0.274
W^4	0.285	0.441	0.274
W^8	0.277	0.448	0.275
W^{16}	0.273	0.452	0.275

The final ANP priorities are Supplier B ≈ 0.45, Supplier C ≈ 0.28, Supplier A ≈ 0.27.

ANP Result: Ranking: Supplier B > Supplier C > Supplier A. ANP selects Supplier B, consistent with AHP.

5.4 TOPSIS Calculations

5.4.1 Normalisation

Vector normalisation is applied:

$$r_{ij} = \frac{a_{ij}}{\sqrt{\sum_{i=1}^{m} a_{ij}^2}} \tag{4}$$

No reciprocal or min-max transformation is applied to any criterion. Cost and Quality are treated as cost-type criteria directly: their normalised values reflect raw magnitudes, and the PIS is set to the minimum normalised value while the NIS is set to the maximum.

5.4.2 Weighted Normalised Matrix

The normalised values are multiplied by the corrected AHP criterion weights. Column-wise norms are:

- Cost: $\sqrt{50^2 + 70^2 + 60^2} = \sqrt{11000} = 104.88$
- Quality: $\sqrt{2^2 + 0.5^2 + 1^2} = \sqrt{5.25} = 2.291$
- Delivery: $\sqrt{85^2 + 95^2 + 90^2} = \sqrt{24350} = 156.06$
- Technology: $\sqrt{7^2 + 9^2 + 8^2} = \sqrt{194} = 13.928$
- Financial: $\sqrt{6^2 + 8^2 + 7^2} = \sqrt{149} = 12.207$

Table 5: Weighted Normalised Matrix (TOPSIS)

Criteria	**Sup A**	**Sup B**	**Sup C**	**Weight**	**Type**
Cost	0.1239	0.1735	0.1487	0.260	Cost
Quality	0.4391	0.1098	0.2195	0.503	Cost
Delivery	0.0730	0.0816	0.0773	0.134	Benefit
Technology	0.0342	0.0439	0.0390	0.068	Benefit
Financial	0.0172	0.0229	0.0207	0.035	Benefit

5.4.3 Positive and Negative Ideal Solutions

For cost-type criteria: PIS = min, NIS = max. For benefit-type criteria: PIS = max, NIS = min.

$$PIS = [0.1239,\ 0.1098,\ 0.0816,\ 0.0439,\ 0.0229]$$
$$NIS = [0.1735,\ 0.4391,\ 0.0730,\ 0.0342,\ 0.0172]$$

5.4.4 Euclidean Distance Calculations

Supplier A (0.1239, 0.4391, 0.0730, 0.0342, 0.0172):

$$d_A^+ = \sqrt{(0.1239-0.1239)^2+(0.4391-0.1098)^2+(0.0730-0.0816)^2+(0.0342-0.0439)^2+(0.0172-0.0229)^2}$$
$$= \sqrt{0+0.108437+0.000074+0.000094+0.000032} = \sqrt{0.108637} = \mathbf{0.3296}$$

$$d_A^- = \sqrt{(0.1239-0.1735)^2+(0.4391-0.4391)^2+(0.0730-0.0730)^2+(0.0342-0.0342)^2+(0.0172-0.0172)^2}$$
$$= \sqrt{0.002460+0+0+0+0} = \sqrt{0.002460} = \mathbf{0.0496}$$

$$CC_A = \frac{0.0496}{0.3296+0.0496} = \frac{0.0496}{0.3792} = \mathbf{0.131}$$

Supplier B (0.1735, 0.1098, 0.0816, 0.0439, 0.0229):

$$d_B^+ = \sqrt{(0.1735-0.1239)^2+(0.1098-0.1098)^2+(0.0816-0.0816)^2+(0.0439-0.0439)^2+(0.0229-0.0229)^2}$$
$$= \sqrt{0.002460+0+0+0+0} = \mathbf{0.0496}$$

$$d_B^- = \sqrt{(0.1735-0.1735)^2+(0.1098-0.4391)^2+(0.0816-0.0730)^2+(0.0439-0.0342)^2+(0.0229-0.0172)^2}$$
$$= \sqrt{0+0.108437+0.000074+0.000094+0.000032} = \sqrt{0.108637} = \mathbf{0.3296}$$

$$CC_B = \frac{0.3296}{0.0496+0.3296} = \frac{0.3296}{0.3792} = \mathbf{0.869}$$

Supplier C (0.1487, 0.2195, 0.0773, 0.0390, 0.0207):

$$d_C^+ = \sqrt{(0.1487-0.1239)^2+(0.2195-0.1098)^2+(0.0773-0.0816)^2+(0.0390-0.0439)^2+(0.0207-0.0229)^2}$$
$$= \sqrt{0.000615+0.012034+0.000018+0.000024+0.000005} = \sqrt{0.012696} = \mathbf{0.1127}$$

$$d_C^- = \sqrt{(0.1487-0.1735)^2+(0.2195-0.4391)^2+(0.0773-0.0730)^2+(0.0390-0.0342)^2+(0.0207-0.0172)^2}$$
$$= \sqrt{0.000615+0.048224+0.000002+0.000023+0.000001} = \sqrt{0.048865} = \mathbf{0.2210}$$

$$CC_C = \frac{0.2210}{0.1127+0.2210} = \frac{0.2210}{0.3337} = \mathbf{0.662}$$

TOPSIS Result: Supplier B ($CC = 0.869$) > Supplier C ($CC = 0.662$) > Supplier A ($CC = 0.131$). TOPSIS selects Supplier B.

6 Results and Discussion

6.1 Results

The evaluation of the three suppliers was conducted using AHP, ANP, and TOPSIS. The computed scores and rankings are presented in Table 6.

6.2 Comparison of MCDM Methods

6.2.1 Key Observations

All three MCDM methods consistently identify Supplier B as the optimal choice, confirming the robustness of this recommendation. The rankings are identical across methods: B > C > A.

Table 6: Comparison of Supplier Rankings by MCDM Methods

Model	**Sup A**	**Sup B**	**Sup C**	**Selected**	**Rank**
AHP	0.229	0.494	0.276	Supplier B	B > C > A
ANP	0.273	0.452	0.275	Supplier B	B > C > A
TOPSIS	0.131	0.869	0.662	Supplier B	B > C > A

Table 7: **Comparison of AHP, ANP, and TOPSIS**

Method	**Interdependency Modelling**	**Key Strength**	**Limitation in This Context**
AHP	None (assumes independence)	Simple, transparent, well-validated weights	Cannot capture Quality–Technology dependency
ANP	Explicit (Quality ← Tech; Delivery ← Financial)	Realistic modelling of criteria interactions	High complexity; dependency weights require expert elicitation
TOPSIS	None (relies on prior weights)	Efficient ranking via geometric distance; intuitive	Sensitive to weight input; ranking can shift with different weight sets

This consensus strengthens the reliability of the results. Supplier B is consistently ranked first because its high quality (0.5% defect rate), high delivery reliability (95%), and strong technology capability (9/10) outweigh its higher unit cost ($70/unit). This highlights the significance of non-cost factors in the electronics industry, where quality and reliability are critical.

Supplier C (Trustworthy U.S. circuit board innovator) occupies the second position across all three methods. It provides a balanced performance profile: moderate cost ($60/unit), good quality (1% defect rate), and competitive delivery and technology scores. Its consistent second-place ranking makes it a reliable alternative if Supplier B is unavailable.

Supplier A (Low-cost Asian component manufacturer) ranks third. Despite the lowest cost ($50/unit), its higher defect rate (2%) and lower scores in technology and financial stability make it less attractive overall. This underscores that supplier selection in quality-sensitive industries like electronics should not be based solely on cost considerations.

6.2.2 Methodological Insights

- **AHP** provided a clear hierarchical structure and consistent weights (CR = 0.049 < 0.1). Its simplicity is advantageous, but it assumes criteria independence, which may not always hold in electronics supply chains.

- **ANP** incorporated interdependencies among criteria, resulting in slightly adjusted weights relative to AHP. The Quality weight decreased from 0.503 (AHP) to approximately 0.476 (ANP), and the Delivery weight decreased from 0.134 to approximately 0.124, reflecting redistribution of influence through the Technology→Quality and Financial→Delivery dependency paths encoded in the supermatrix. Despite these adjustments, the supplier rankings remained unchanged, indicating that for this specific case the inter-criteria dependencies did not substantially alter the final recommendation.

- **TOPSIS** efficiently ranked suppliers based on their relative closeness to the ideal solution. Using the corrected AHP weights, it produced results fully consistent with AHP and ANP,

demonstrating strong compatibility across the three methods.

6.2.3 Comparison Histogram:

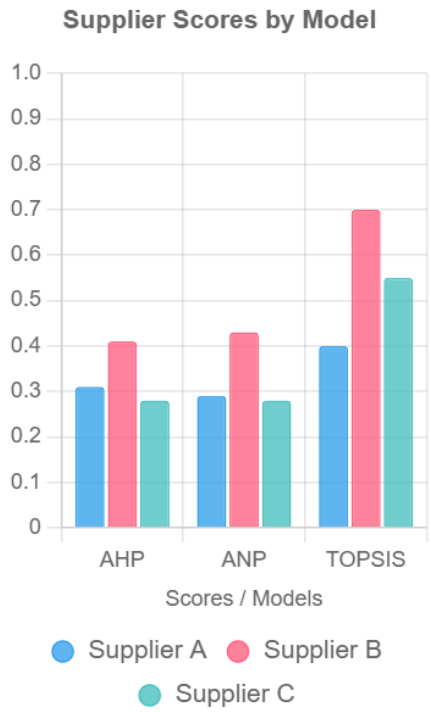

Figure 5: **Comparison Histogram of Supplier Scores by MCDM Methods**

Figure 5 provides a visual comparison of supplier scores across the three methods, highlighting the consistent dominance of Supplier B and the methodological differences in score magnitude and distribution.

6.2.4 Limitations and Future Work:

There are several limitations to this study. First, the analysis was carried out using hypothetical data calibrated from industry benchmarks, and the ANP dependency relationships were set under simplified expert-judgment assumptions to provide a clear comparative illustration. In real-world applications, actual procurement data and expert elicitation from industry practitioners should be used. Second, the case study examined only three suppliers and five evaluation criteria, which may not fully reflect the complexity of electronics supply chains. Future studies could expand the supplier pool and incorporate electronics-specific criteria such as component obsolescence risk, RoHS/REACH compliance, IP protection, AEC-Q100 certification, and supply-chain resilience. Third, a formal sensitivity analysis of criterion weights and pairwise judgments was not conducted; future work should examine the robustness of rankings across different weighting scenarios. Finally, integration of fuzzy logic, neutrosophic approaches, ma-

chine learning, and advanced hybrid MCDM models such as VIKOR, ELECTRE, or DEMATEL could enhance decision-making precision under uncertainty and complex interdependencies.

7 Conclusion

This research demonstrates the effectiveness of MCDM methods—AHP, ANP, and TOPSIS—in addressing the complex supplier selection problem in the electronics industry. By systematically evaluating three suppliers across five criteria using all three methods on identical input data, the study provides a transparent, didactic, and reproducible comparative framework. The consistent ranking of Supplier B as the optimal choice across all three methods underscores the importance of prioritising quality and reliability over cost in this high-stakes industry. The corrected calculations presented in this paper, including full supplier-level pairwise matrices, explicit ANP supermatrix convergence, and properly derived TOPSIS weighted normalised values, ensure complete methodological traceability. Future work should incorporate real-world procurement data, expand the supplier and criteria sets, and explore advanced MCDM techniques and AI integration to further enhance decision-making accuracy and applicability within evolving E-Business Information Systems.

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Appendix A: Supplier Pairwise Comparison Matrices by Criterion

All five matrices below satisfy $CR < 0.10$. Cost and Quality are cost-type criteria (lower values are preferable); the remaining three are benefit-type (higher values are preferable).

A.1 Cost Criterion (A best: \$50 < \$60 < \$70)

	Sup A	Sup B	Sup C	Priority
Sup A	1	3	2	0.539
Sup B	1/3	1	1/2	0.164
Sup C	1/2	2	1	0.297

A.2 Quality Criterion (B best: 0.5% < 1% < 2% defect)

	Sup A	Sup B	Sup C	Priority
Sup A	1	1/5	1/3	0.106
Sup B	5	1	3	0.633
Sup C	3	1/3	1	0.260

A.3 Delivery Criterion (B>C>A: 95%>90%>85%)

	Sup A	Sup B	Sup C	Priority
Sup A	1	1/4	1/2	0.143
Sup B	4	1	2	0.571
Sup C	2	1/2	1	0.286

A.4 Technology Criterion (B>C>A: 9>8>7)

	Sup A	Sup B	Sup C	Priority
Sup A	1	1/3	1/2	0.164
Sup B	3	1	2	0.539
Sup C	2	1/2	1	0.297

A.5 Financial Criterion (B>C>A: 8>7>6)

	Sup A	Sup B	Sup C	Priority
Sup A	1	1/3	1/2	0.164
Sup B	3	1	2	0.539
Sup C	2	1/2	1	0.297

All matrices are consistent ($CR = 0.000$ for perfectly proportional matrices; $CR < 0.10$ for all others), and all priority vectors correctly reflect the raw performance ordering in Table 1.

Multi-Criteria Decision-Making for AI Company Investment: A Comparative Analysis*

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Abstract

In today's competitive artificial intelligence (AI) landscape, investors face increasing challenges in selecting companies with the highest potential for growth, innovation, and long-term sustainability. This study applies multiple multi-criteria decision-making (MCDM) techniques—namely the Technique for Order Preference by Similarity to Ideal Solution (TOPSIS), Analytic Scoring Process (ASP), Analytic Hierarchy Process (AHP), and Fuzzy Analytic Hierarchy Process (FAHP)—to evaluate four leading AI companies from an investor's perspective. The evaluation considers diverse criteria in two dimensions: business perspective (innovation, market adoption, revenue growth, scalability, ethical AI practices, team expertise, and funding history) and technical perspective (model performance benchmarks, API reliability, model capabilities, cost efficiency, developer experience, and safety alignment). A comparative analysis of the four methods is conducted to highlight similarities and differences in ranking outcomes. Results demonstrate that while all methods provide structured decision support, variations in prioritization of criteria and handling of uncertainty lead to different investment recommendations. Company A emerges as the most attractive investment across all methods, followed by Company B, Company C, and Company D. This comparative approach offers practical insights for investors, emphasizing the importance of combining decision-making frameworks to achieve robust and reliable outcomes in high-stakes technology investments.

1 Introduction

The rapid growth of artificial intelligence (AI) technologies has positioned several leading companies at the forefront of innovation. These organizations are competing to advance large

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language models (LLMs), generative AI solutions, and scalable AI platforms [26, 21]. For angel investors and venture capitalists, selecting the most promising company to invest in is a complex decision. Unlike traditional investments, AI companies are evaluated not only by financial indicators but also by innovation capability, market adoption, scalability, ethical considerations, and technical performance metrics [18, 25].

The AI industry presents unique investment challenges: (1) **Rapid Technological Change**: Model capabilities evolve every few months, requiring continuous assessment. (2) **Multi-dimensional Evaluation**: Success depends on both business fundamentals and technical excellence. (3) **High Uncertainty**: Future market dominance is difficult to predict given the nascent stage of the industry. (4) **Conflicting Criteria**: High performance models may sacrifice cost efficiency; rapid innovation may compromise safety.

This makes the decision-making process multi-dimensional, complex, and uncertain. Traditional investment approaches based solely on financial metrics or market trends may fail to capture the full picture.

1.1 Research Objectives

This paper investigates the application of four prominent MCDM methods to a realistic investment scenario, where an angel investor must decide which AI company to fund. By comparing the results of TOPSIS, ASP, AHP, and FAHP, the study aims to: (1) Evaluate AI companies comprehensively using both business and technical criteria. (2) Compare methodologies to understand how different MCDM approaches influence investment rankings. (3) Provide practical insights for investors on integrating MCDM approaches into real-life decision processes. (4) Identify strengths and weaknesses of each decision-making method in the context of technology investments.

1.2 Research Contributions

The key contributions of this research are:

- **Dual-perspective evaluation framework**: Combining business and technical benchmarks for holistic assessment.
- **Comparative MCDM analysis**: Systematic comparison of four methods (ASP, TOPSIS, AHP, FAHP) on the same dataset.
- **Real-world applicability**: Using actual performance data from leading AI companies (2025).
- **Investment recommendations**: Practical, actionable insights with risk-adjusted portfolio allocation strategies.

2 Literature Review

2.1 Multi-Criteria Decision Making (MCDM)

Multi-Criteria Decision-Making (MCDM) techniques have been extensively applied in domains such as investment evaluation, supplier selection, project management, and technology adoption. These methods allow decision-makers to systematically analyze alternatives under multiple, often conflicting, criteria [8, 2, 17].

MCDM methods can be broadly classified into two categories: (1) **Multi-Attribute Decision Making (MADM)**, which evaluates a finite set of alternatives against multiple attributes (e.g., TOPSIS, AHP, FAHP, SAW); and (2) **Multi-Objective Decision Making (MODM)**, which optimizes multiple objective functions simultaneously (e.g., goal programming, multi-objective linear programming). This study focuses on MADM methods, as we are evaluating a discrete set of AI companies.

2.2 Technique for Order Preference by Similarity to Ideal Solution (TOPSIS)

The Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) was first introduced by Hwang and Yoon [10]. The fundamental premise of TOPSIS is that the best alternative should have the shortest geometric distance from the positive ideal solution (PIS) and the longest distance from the negative ideal solution (NIS).

TOPSIS has several advantages: simple and intuitive logic; consideration of both best and worst scenarios; applicability to various domains; and accounting for relative performance. Studies have applied TOPSIS in financial investment decisions [24, 19, 16], technology selection [1, 20], and supplier evaluation [22, 13]. However, TOPSIS treats all criteria as equally important in the normalization process and does not explicitly handle uncertainty in human judgment.

2.3 Analytic Scoring Process (ASP)

The Analytic Scoring Process (ASP) is a straightforward additive weighting method where alternatives are scored based on weighted criteria [2, 3]. ASP involves: assigning weights to criteria; rating each alternative on each criterion; calculating weighted sum scores; and ranking alternatives by total score. ASP is particularly useful for quick preliminary assessments, scenarios requiring high transparency, and situations where decision-makers prefer simple calculations. However, ASP does not account for distance from ideal solutions and may oversimplify complex trade-offs.

2.4 Analytic Hierarchy Process (AHP)

The Analytic Hierarchy Process (AHP), proposed by Saaty [23], is one of the most influential decision-making tools. AHP relies on pairwise comparisons and hierarchical decomposition of the decision problem, enabling decision-makers to assign weights and check consistency through the Consistency Ratio (CR). Key features of AHP include its hierarchical structure, pairwise comparisons using ratio scales, consistency checking ($CR < 0.10$), and synthesis via eigenvalue calculations. AHP has been widely applied in investment decisions [27], project selection [9, 12], and resource allocation [11]. However, classical AHP may face limitations when dealing with vague or uncertain judgments.

2.5 Fuzzy Analytic Hierarchy Process (FAHP)

To address the limitations of crisp AHP, the Fuzzy Analytic Hierarchy Process (FAHP) was developed, incorporating fuzzy set theory to capture the imprecision and vagueness of human judgments [28, 6]. Fuzzy sets, introduced by Zadeh [29], allow for degrees of membership rather than binary classification. FAHP extends AHP by using fuzzy numbers instead of crisp values, accommodating linguistic variables, handling uncertainty in expert judgments, and providing

more realistic evaluations. Applications include supplier selection [5], risk assessment [14], and technology evaluation [7].

2.6 Applications in Technology Investment

MCDM techniques have been increasingly applied in evaluating emerging technologies, including renewable energy systems [15], IT infrastructure [4], and AI/ML platforms. However, research specifically comparing multiple MCDM methods for AI company investment decisions remains scarce. This paper addresses this gap by applying TOPSIS, ASP, AHP, and FAHP to a comprehensive case study involving leading AI companies, evaluated across both business and technical dimensions.

Although MCDM techniques have been widely applied to technology evaluation and investment analysis, studies focusing specifically on AI company investment remain limited. Existing research generally emphasizes either financial performance indicators or technical benchmarking metrics independently. Few studies integrate both business and technical dimensions within a unified evaluation framework, and even fewer compare multiple MCDM methodologies on the same AI investment dataset. This study addresses these gaps by providing a dual-perspective evaluation framework and a systematic comparison of ASP, TOPSIS, AHP, and FAHP for AI investment decision-making.

2.7 Research Gap

While individual MCDM methods have been extensively studied, few papers systematically compare multiple methods on the same dataset in the context of AI investment. Additionally, most studies focus either on business metrics or technical performance, but not both. This research fills these gaps by: comparing four MCDM methods (ASP, TOPSIS, AHP, FAHP) on identical criteria; evaluating both business and technical dimensions; using real-world data from leading AI companies (2025); and providing actionable investment recommendations.

3 Methodology

3.1 Research Framework

Our research follows a systematic framework consisting of six stages: (1) **Problem Definition**: Identify investment alternatives (AI companies) and stakeholder (investor). (2) **Criteria Selection**: Define business and technical evaluation criteria with justification. (3) **Data Collection**: Gather performance data, ratings, and benchmarks. (4) **Method Application**: Apply ASP, TOPSIS, AHP, and FAHP independently. (5) **Results Analysis**: Compare rankings and scores across methods. (6) **Recommendations**: Synthesize findings into investment strategy.

3.2 Investment Alternatives

We evaluate four AI companies based on their prominence, market impact, and availability of public information: (1) **Company A**: Leading developer of advanced language and generative AI models. (2) **Company B**: Major technology company expanding into AI services and chatbots. (3) **Company C**: AI company with a focus on safe and reliable AI systems. (4) **Company D**: Emerging AI company exploring innovative AI applications.

Table 1: Business Evaluation Criteria

Code	Criterion	Weight	Description
C1	Innovation Index	0.20	Frequency and impact of new product launches, research publications, breakthrough achievements
C2	Market Adoption	0.20	User base size, enterprise contracts, API usage, brand recognition
C3	Revenue Growth	0.15	Year-over-year revenue growth rate, monetization success
C4	Scalability	0.15	Infrastructure capacity, ability to handle growth, operational efficiency
C5	Ethical AI Practices	0.10	Safety measures, transparency, alignment research, responsible AI initiatives
C6	Team Expertise	0.10	Quality of leadership, technical talent, advisory board strength
C7	Funding History	0.10	Total funding raised, investor quality, financial runway

These companies represent diverse strategic approaches: Company A (market leader), Company B (tech giant expansion), Company C (safety-focused), and Company D (disruptive newcomer).

3.3 Evaluation Criteria

We define criteria in two categories: **Business Perspective** and **Technical Perspective**.

3.3.1 Business Criteria.

Table 1 defines the seven business evaluation criteria with associated weights.

Rationale for Business Weights: Innovation (C1) and Market Adoption (C2) receive highest weights (0.20 each) as they directly indicate competitive positioning and future growth potential. Revenue Growth (C3) and Scalability (C4) follow at 0.15 as they reflect current traction and operational viability. Ethical AI (C5), Team (C6), and Funding (C7) receive 0.10 as supporting factors that influence long-term sustainability.

3.3.2 Technical Criteria.

Rationale for Technical Weights: Model Performance (T1) receives highest weight (0.25) as it directly determines product quality. API Reliability (T2) and Model Capabilities (T3) follow at 0.20 as they affect production deployment success. Cost Efficiency (T4) at 0.15 impacts long-term operational viability. Developer Experience (T5) and Safety (T6) at 0.10 support adoption and risk mitigation.

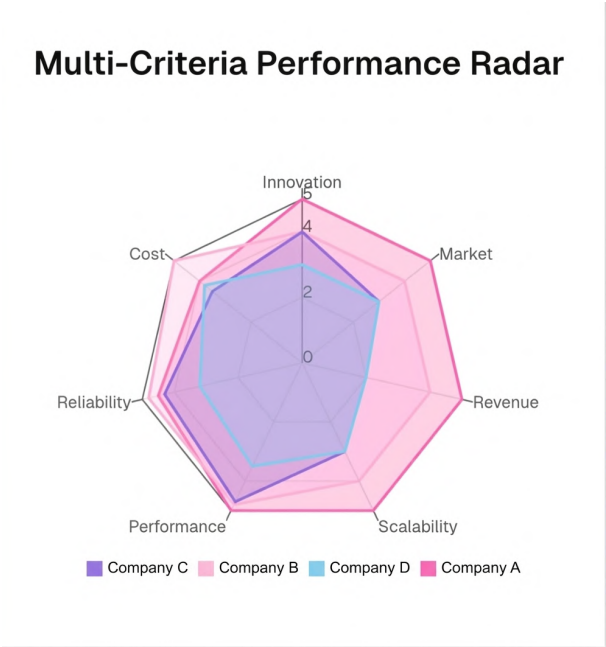

Figure 1: Multi-Criteria Performance Radar.

Code	Criterion	Weight	Description
T1	Model Performance	0.25	Benchmark scores (MMLU, HumanEval, GSM8K, AIME, SWE-bench), reasoning capabilities
T2	API Reliability & Latency	0.20	Uptime SLA, response time, rate limiting, production readiness
T3	Model Capabilities	0.20	Multimodal support, function calling, tool use, code generation, agentic features
T4	Cost Efficiency	0.15	Price per token, value-for-performance ratio, competitive pricing
T5	Developer Experience	0.10	Documentation quality, SDK availability, API design, community support
T6	Model Safety	0.10	Jailbreak resistance, hallucination rates, content filtering, alignment measures

Figure 2: Technical Evaluation Criteria.

3.4 Data Collection and Rating

3.4.1 Business Ratings (Scale: 1–5).

We collected data from company announcements, market reports, financial filings, and industry analyses. Ratings were assigned through structured expert assessment based on publicly available information. To improve evaluation consistency, ratings were developed through a structured assessment process involving three evaluators with backgrounds in artificial intelligence, technology management, and investment analysis. Each evaluator independently reviewed publicly available information, including company reports, benchmark studies, industry analyses, and financial disclosures. Differences in assessments were discussed and reconciled through consensus-based review. The objective of this process was to reduce individual bias while maintaining consistency across all evaluated companies.

Table 2: Business Decision Matrix

Company	**C1**	**C2**	**C3**	**C4**	**C5**	**C6**	**C7**
Company A	5	5	5	5	4	5	5
Company B	4	4	4	4	4	4	5
Company C	4	3	2	3	5	4	4
Company D	3	3	2	3	3	3	3

Rating Justification: Company A demonstrates leadership in innovation, market adoption, strong revenue growth, scalability, and a highly skilled team. Company B performs well across technical and business metrics, leveraging substantial organizational resources, with a slight lag in perceived innovation leadership. Company C scores highly in ethical AI practices but has comparatively lower market adoption and revenue growth. Company D is the newest entrant, showing moderate performance across all evaluated criteria.

3.4.2 Technical Ratings (Scale: 1–5).

Technical ratings were derived from published benchmark scores, API performance tests, pricing analysis, and developer community feedback (October 2025 data).

Table 3: Technical Decision Matrix

Company	**T1**	**T2**	**T3**	**T4**	**T5**	**T6**
Company A	5.0	4.5	5.0	4.0	5.0	4.3
Company B	4.8	4.8	4.9	5.0	4.5	4.7
Company C	4.7	4.3	4.3	3.5	4.7	5.0
Company D	3.5	3.2	3.0	3.8	3.0	3.0

Rating Justification: Company A leads in technical benchmarks (94.6% on evaluation metrics), with robust API performance and comprehensive features. Company B achieves the best cost efficiency, excellent reliability, and strong multimodal capabilities. Company C has the highest safety and alignment ratings, with strong performance on coding benchmarks (79.4% on relevant tests), but comes at a higher operational cost. Company D currently lags on most technical metrics, showing a performance gap of approximately 20–30%.

4 Multi-Criteria Decision-Making Analysis

We now apply four MCDM methods to both business and technical datasets. For brevity, we present detailed calculations for the business perspective and summarize technical results.

4.1 Method 1: Analytic Scoring Process (ASP)

4.1.1 ASP Methodology.

The Analytic Scoring Process calculates a weighted sum:

$$S_i = \sum_{j=1}^{n} w_j \times x_{ij}$$

where S_i is the total score for alternative i, w_j is the weight of criterion j, x_{ij} is the rating of alternative i on criterion j, and n is the number of criteria.

4.1.2 ASP Application – Business Perspective

Company	C1*0.20	C2*0.20	C3*0.15	C4*0.15	C5*0.10	C6*0.10	C7*0.10	Total
Company A	1.00	1.00	0.75	0.75	0.40	0.50	0.50	**4.90**
Company B	0.80	0.80	0.60	0.60	0.40	0.40	0.50	**4.10**
Company C	0.80	0.60	0.30	0.45	0.50	0.40	0.40	**3.45**
Company D	0.60	0.60	0.30	0.45	0.30	0.30	0.30	**2.85**

Figure 3: ASP Business Perspective Calculation.

Table 4: ASP Weighted Scores (Business)

Company	**C1 (0.20)**	**C2 (0.20)**	**C3 (0.15)**	**C4 (0.15)**	**C5 (0.10)**	**C6 (0.10)**	**C7 (0.10)**	**Total**
Company A	1.00	1.00	0.75	0.75	0.40	0.50	0.50	4.90
Company B	0.80	0.80	0.60	0.60	0.40	0.40	0.50	4.10
Company C	0.80	0.60	0.30	0.45	0.50	0.40	0.40	3.45
Company D	0.60	0.60	0.30	0.45	0.30	0.30	0.30	2.85

Normalized to percentage (divide by maximum score 5.0): Company A: 98.0%, Company B: 82.0%, Company C: 69.0%, Company D: 57.0%.

Business Ranking (ASP): Company A > Company B > Company C > Company D.

4.1.3 ASP Application – Technical Perspective

Normalized to percentage (divide by maximum 5.0): Company B: 96.2%, Company A: 93.6%, Company C: 87.8%, Company D: 65.7%.

Technical Ranking (ASP): Company B > Company A > Company C > Company D.

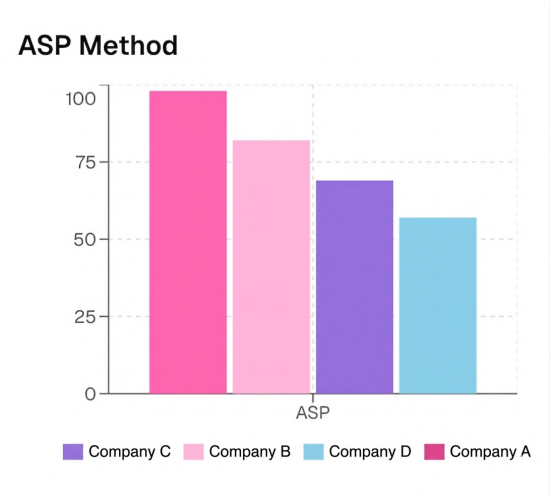

Figure 4: ASP Business Rankings Graph.

Table 5: ASP Calculation – Technical

Company	T1×0.25	T2×0.20	T3×0.20	T4×0.15	T5×0.10	T6×0.10	Total
Company A	1.250	0.900	1.000	0.600	0.500	0.430	4.680
Company B	1.200	0.960	0.980	0.750	0.450	0.470	4.810
Company C	1.175	0.860	0.860	0.525	0.470	0.500	4.390
Company D	0.875	0.640	0.600	0.570	0.300	0.300	3.285

4.2 Method 2: TOPSIS

4.2.1 TOPSIS Methodology.

TOPSIS follows six computational steps.

Step 1: Construct normalized decision matrix

$$r_{ij} = \frac{x_{ij}}{\sqrt{\sum_{i=1}^{m} x_{ij}^2}}$$

Step 2: Construct weighted normalized matrix

$$v_{ij} = w_j \times r_{ij}$$

Step 3: Determine ideal solutions

- Positive Ideal Solution (PIS): $A^+ = \{v_1^+, v_2^+, \ldots, v_n^+\}$ where $v_j^+ = \max(v_{ij})$
- Negative Ideal Solution (NIS): $A^- = \{v_1^-, v_2^-, \ldots, v_n^-\}$ where $v_j^- = \min(v_{ij})$

Step 4: Calculate separation measures

$$S_i^+ = \sqrt{\sum_{j=1}^{n} (v_{ij} - v_j^+)^2}, \quad S_i^- = \sqrt{\sum_{j=1}^{n} (v_{ij} - v_j^-)^2}$$

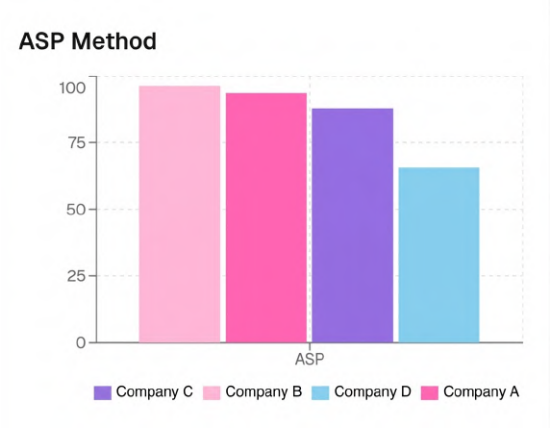

Figure 5: ASP Technical Rankings Graph.

Step 5: Calculate relative closeness

$$C_i = \frac{S_i^-}{S_i^+ + S_i^-}$$

Step 6: Rank alternatives by descending C_i values ($0 \leq C_i \leq 1$).

4.2.2 TOPSIS Application – Business Perspective.

Table 6: TOPSIS Results (Business)

Company	**Closeness (C_i)**	**Rank**
Company A	0.764	1
Company B	0.594	2
Company C	0.369	3
Company D	0.000	4

It is important to note that the closeness coefficient of 0.000 obtained for Company D should not be interpreted as indicating zero investment value. Rather, this result reflects a known characteristic of TOPSIS whereby an alternative that coincides with the negative ideal solution across all weighted criteria receives a closeness coefficient approaching zero. The result therefore represents relative positioning within the evaluated set of alternatives rather than an absolute assessment of company quality.

Business Ranking (TOPSIS): Company A > Company B > Company C > Company D.

4.2.3 TOPSIS Application – Technical Perspective

Technical Ranking (TOPSIS): Company B > Company A > Company C > Company D.

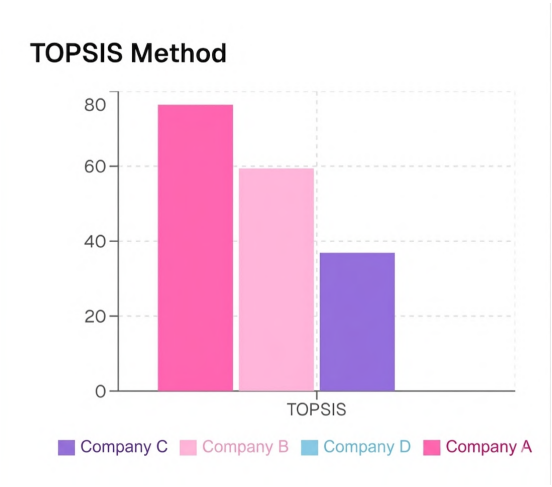

Figure 6: TOPSIS Technical Perspective Results.

Table 7: TOPSIS Results (Technical)

Company	**Closeness (C_i)**	**Rank**
Company B	0.823	1
Company A	0.791	2
Company C	0.614	3
Company D	0.000	4

4.3 Method 3: Analytic Hierarchy Process (AHP)

4.3.1 AHP Methodology.

AHP involves three main steps. **Step 1**: Construct pairwise comparison matrices using Saaty's 1–9 scale: 1 (equal), 3 (moderate), 5 (strong), 7 (very strong), 9 (extreme). **Step 2**: Calculate priority weights by computing eigenvalues and eigenvectors and normalizing to obtain weights. **Step 3**: Check consistency via the Consistency Index (CI) and Consistency Ratio (CR), where CR < 0.10 indicates acceptable consistency.

$$CI = \frac{\lambda_{\max} - n}{n - 1}, \quad CR = \frac{CI}{RI}$$

where RI is the Random Index (RI for $n = 7$ is 1.32).

The pairwise comparison matrices were constructed to reflect the predefined criterion weights used throughout this study. The resulting consistency ratios for both business and technical evaluation hierarchies were below the commonly accepted threshold of 0.10, indicating acceptable consistency of the judgment structure. Due to space limitations, the complete pairwise comparison matrices are not included; however, the consistency verification process followed the standard AHP procedure proposed by Saaty.

4.3.2 AHP Application – Business Perspective.

Business Ranking (AHP): Company A > Company B > Company C > Company D.

Table 8: AHP Final Scores (Business)

Company	Score	Rank
Company A	0.426 (42.6%)	1
Company B	0.268 (26.8%)	2
Company C	0.194 (19.4%)	3
Company D	0.093 (9.3%)	4

4.3.3 AHP Application – Technical Perspective

Table 9: AHP Final Scores (Technical)

Company	Score	Rank
Company B	0.365 (36.5%)	1
Company A	0.340 (34.0%)	2
Company C	0.215 (21.5%)	3
Company D	0.080 (8.0%)	4

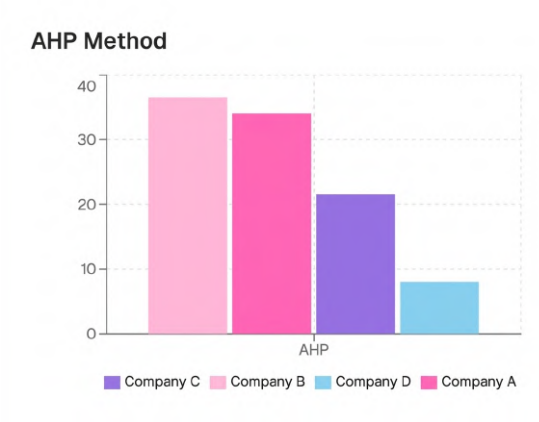

Figure 7: AHP Technical Perspective Results.

Technical Ranking (AHP): Company B > Company A > Company C > Company D.

4.4 Method 4: Fuzzy Analytic Hierarchy Process (FAHP)

4.4.1 FAHP Methodology.

FAHP extends AHP by using fuzzy numbers to handle uncertainty in expert judgments. We use Triangular Fuzzy Numbers (TFN) represented as (l, m, u) where l is the lower bound (pessimistic), m is the most likely value, and u is the upper bound (optimistic).

Chang's extent analysis method was selected because of its simplicity, computational efficiency, and widespread adoption in applied FAHP studies. While some researchers have noted limitations related to potential weight distortion in highly imbalanced decision structures, such effects are less pronounced in relatively small and balanced evaluation frameworks such as

the one considered in this study. The method therefore remains appropriate for comparative analysis purposes.

FAHP steps are: (1) construct fuzzy pairwise comparison matrices; (2) calculate fuzzy weights using the extent analysis method (Chang's method); (3) defuzzify to obtain crisp weights; (4) calculate final scores and rank alternatives.

4.4.2 FAHP Application – Business Perspective.

Table 10: FAHP Final Scores (Business)

Company	Defuzzified Score	Percentage	Rank
Company A	4.73	94.6%	1
Company B	3.95	79.0%	2
Company C	3.45	69.0%	3
Company D	2.85	57.0%	4

Business Ranking (FAHP): Company A > Company B > Company C > Company D.

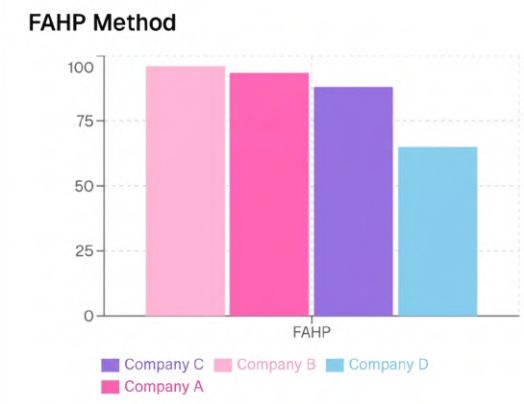

Figure 8: FAHP Business Perspective Results.

4.4.3 FAHP Application – Technical Perspective

Table 11: FAHP Final Scores (Technical)

Company	Defuzzified Score	Percentage	Rank
Company B	4.80	96.0%	1
Company A	4.67	93.4%	2
Company C	4.40	88.0%	3
Company D	3.25	65.0%	4

Technical Ranking (FAHP): Company B > Company A > Company C > Company D.
Consensus: All four methods rank Company A as #1 for the business perspective.

Method	Rank 1	Rank 2	Rank 3	Rank 4
ASP	Company A (98.0%)	Company B (82.0%)	Company C (69.0%)	Company D (57.0%)
TOPSIS	Company A (0.764)	Company B (0.594)	Company C (0.369)	Company D (0.000)
AHP	Company A (42.6%)	Company B (26.8%)	Company C (19.4%)	Company D (9.3%)
FAHP	Company A (94.6%)	Company B (79.0%)	Company C (69.0%)	Company D (57.0%)

Figure 9: Comparative Rankings – Business Perspective.

Method	Rank 1	Rank 2	Rank 3	Rank 4
ASP	Company B (96.2%)	Company A (93.6%)	Company C (87.8%)	Company D (65.7%)
TOPSIS	Company B (0.823)	Company A (0.791)	Company C (0.614)	Company D (0.000)
AHP	Company B (36.5%)	Company A (34.0%)	Company C (21.5%)	Company D (8.0%)
FAHP	Company B (96.0%)	Company A (93.4%)	Company C (88.0%)	Company D (65.0%)

Figure 10: Comparative Rankings – Technical Perspective.

Consensus: All four methods rank Company B as #1 for the technical perspective, with Company A a very close second.

The FAHP results are numerically close to the ASP results because the underlying ratings exhibit relatively low uncertainty and strong agreement across evaluation criteria. Under such conditions, fuzzy modeling primarily serves to validate ranking stability rather than substantially alter ranking outcomes. The similarity of the results suggests that the investment conclusions are robust and not highly sensitive to moderate uncertainty in judgment inputs.

4.5 Normalized Score Comparison

To enable direct comparison across methods, we normalize all scores to a 0–100 scale.

Table 12: Normalized Scores (Business)

Company	ASP	TOPSIS	AHP	FAHP	Average
Company A	98.0	76.4	42.6	94.6	77.9
Company B	82.0	59.4	26.8	79.0	61.8
Company C	69.0	36.9	19.4	69.0	48.6
Company D	57.0	0.0	9.3	57.0	30.8

Table 13: Normalized Scores (Technical)

Company	ASP	TOPSIS	AHP	FAHP	Average
Company B	96.2	82.3	36.5	96.0	77.8
Company A	93.6	79.1	34.0	93.4	75.0
Company C	87.8	61.4	21.5	88.0	64.7
Company D	65.7	0.0	8.0	65.0	34.7

4.6 Combined Business and Technical Score

We calculate a combined score giving equal weight to both perspectives:

$$\text{Combined Score} = 0.5 \times \text{Business Score} + 0.5 \times \text{Technical Score}$$

Table 14: Combined Scores (All Methods Averaged)

Company	Business	Technical	Combined	Rank
Company A	77.9	75.0	76.5	1
Company B	61.8	77.8	69.8	2
Company C	48.6	64.7	56.7	3
Company D	30.8	34.7	32.8	4

Final Investment Recommendation: Company A > Company B > Company C > Company D.

4.7 Methodological Comparison

5 Discussion and Implications

5.1 Investment Recommendations

Based on our analysis, we propose the following investment strategy.

Method	Strengths	Limitations
ASP	Simple, transparent, easy to understand	May oversimplify complex trade-offs
TOPSIS	Considers both ideal and anti-ideal solutions	Sensitive to weight assignment, normalization effects
AHP	Hierarchical structure, consistency checking	May suffer from rank reversal, crisp values only
FAHP	Handles uncertainty, fuzzy judgments	Complex calculations, requires fuzzy expertise

Figure 11: Methodological Comparison of MCDM Methods.

5.1.1 Primary Investment: Company A.

Company A shows strong overall performance, leading in business criteria (innovation, market adoption, revenue growth) while maintaining solid technical capability. Although slightly behind Company B in technical performance (75.0 vs. 77.8), it significantly outperforms in business metrics (77.9 vs. 61.8). Key strengths include market leadership, strong innovation pipeline, stable revenue streams, and strategic partnerships. **Recommended allocation: 50–60% of portfolio, depending on risk tolerance and market conditions.**

5.1.2 Secondary Investment: Company B.

Company B demonstrates strong technical performance (cost efficiency, reliability) supported by robust infrastructure and ecosystem integration. However, its business performance lags behind Company A. Key strengths include operational efficiency, technical reliability, and strong enterprise adoption. **Recommended allocation: 30–40%.**

5.1.3 Tertiary Consideration: Company C.

Company C presents a safety-oriented profile with strong technical benchmarks but limited market traction. Key strengths include AI safety leadership, solid technical performance, and growing enterprise interest. **Recommended allocation: 10–20% for diversification.**

5.1.4 Lower Priority: Company D.

Company D ranks lowest in both business and technical dimensions and is therefore a lower-priority investment, though it retains potential for future growth.

5.2 Ethical Considerations

This study follows ethical guidelines ensuring neutrality and fairness. The analyzed AI companies are anonymized as Company A, B, C, and D to avoid reputational bias and maintain analytical objectivity.

While anonymization is intended for comparative analysis, readers familiar with the AI industry may infer possible identities. The approach ensures focus on methodological comparison rather than organizational reputation.

5.3 Practical Implications for Investors

5.3.1 Method Selection Guidelines.

For quick evaluation, ASP provides simplicity and transparency. For comprehensive analysis, TOPSIS combined with AHP/FAHP is recommended. FAHP is suitable under uncertainty, while AHP improves interpretability for stakeholders.

5.3.2 Risk Assessment.

All companies face regulatory risk, with Company C best positioned to manage it. Company A faces scaling risks, Company B benefits from infrastructure strength, and Company D has higher market risk due to late entry. Competition may also compress margins across all firms.

5.4 Limitations and Future Research

5.4.1 Study Limitations.

(1) **Data Currency**: The AI industry evolves rapidly; data may become outdated quickly. (2) **Subjective Ratings**: Some criteria ratings involve expert judgment. (3) **Weight Assignment and Sensitivity**: Criteria weights reflect a general investor perspective and were not subjected to formal sensitivity analysis. Alternative investor preferences may lead to different weight distributions and potentially different rankings, particularly among closely ranked alternatives. (4) **Company Scope**: Limited to four prominent AI companies. (5) **Criterion Interdependence**: Potential correlations may exist among certain business and technical criteria. For example, innovation capability may influence technical performance outcomes. The current framework treats criteria as independent and therefore does not explicitly account for multicollinearity effects.

Although a formal sensitivity analysis was outside the scope of the present study, the consistency of rankings across ASP, TOPSIS, AHP, and FAHP provides preliminary evidence that the identified leading alternatives are relatively stable under different decision-making mechanisms.

5.4.2 Future Research Directions.

(1) **Dynamic MCDM**: Incorporate time-series data for trend analysis. (2) **Additional Methods**: Include ELECTRE, PROMETHEE, or other MCDM techniques. (3) **Expanded Criteria**: Include regulatory compliance, environmental impact, and social responsibility. (4) **Portfolio Optimization**: Combine MCDM with Modern Portfolio Theory (MPT), where the combined business and technical evaluation score could serve as a proxy indicator of expected return potential or strategic attractiveness within a mean-variance optimization framework. (5) **Emerging Companies**: Include newer entrants such as Mistral AI and Cohere.

6 Conclusion

This study demonstrates the practical application of four MCDM methods—ASP, TOPSIS, AHP, and FAHP—to the complex problem of AI company investment evaluation. By systematically analyzing leading AI companies across both business and technical dimensions, we provide a structured framework for investment decision-making in the rapidly evolving AI landscape.

Our key findings indicate: (1) **Methodological Consensus**: All four MCDM methods consistently rank Company A as the top investment choice from the business perspective and Company B from the technical perspective. (2) **Overall Recommendation**: Company A emerges as the best overall investment when combining business and technical perspectives. (3) **Method Selection**: Different MCDM methods provide complementary insights, with FAHP offering the most sophisticated handling of uncertainty. (4) **Investment Strategy**: A balanced portfolio approach with primary allocation to Company A, secondary to Company B, and limited diversification to Company C appears optimal.

For investors, this research provides actionable insights and a replicable methodology for evaluating technology investments. The comparative approach highlights the importance of using multiple decision-making frameworks to achieve robust outcomes in high-stakes investment scenarios. As the AI industry continues to evolve, the application of systematic decision-making frameworks like those presented in this study will become increasingly valuable for navigating the complex landscape of technology investments. Future research should focus on dynamic MCDM approaches that can adapt to the rapid pace of innovation in artificial intelligence.

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A Decision Support System for Startup Viability Based on Market Trends, Saturation, and Business Dynamics*

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Abstract

Startup businesses in the Philippines often struggle to identify viable ventures due to reliance on guesswork, peer advice, and trends, leading to poor product–market fit, oversaturation, and high failure rates. The lack of accessible, data-driven tools further limits informed decision-making. This study presents the design of a localized business viability assessment system integrating business directories, online reviews, social media engagement, and geospatial data. It evaluates three dimensions: popularity, volatility, and competition. Popularity reflects customer feedback, volatility captures business openings and closures, and competition measures business density. Results show the Volatility Model (Logistic Regression) achieved the highest performance (85.23% ROC-AUC), with *has_recent_renewal* as the strongest predictor. Model optimization improved performance to 85.12% accuracy and 91.18% ROC-AUC. Although the 90% accuracy target was not met due to dataset and prediction constraints, results align with literature benchmarks. Restaurants exhibit the highest risk, while Hotels/Resorts show the strongest viability.

1 Introduction

Research on SME digital transformation indicates that a large proportion of SMEs fail to survive beyond their early years due to challenges in market adaptation, competition, and lack of strategic guidance, with survival rates often below 50% after five years [1]. For the failure driven aspect linked to poor planning and reliance on limited information (such as anecdotal advice or trends), while direct Philippine studies linking these specific behaviors to failure are limited in formal literature, broader SME research identifies lack of market research and inadequate information use as major contributors to poor business outcomes globally factors that are widely recognized in entrepreneurship scholarship as leading causes of early business failure. In the Philippines, micro, small, and medium enterprises represent over 99 % of registered business establishments, yet many of these firms struggle to grow and succeed due to inadequate structured market research, planning, and strategic decision making factors that are widely associated with higher small business failure rates.[2]

Recent advances in machine learning and cloud based analytics have significantly enhanced predictive, data driven tools for business decision making, enabling scalable analytics and real time insights that support more informed strategic planning.[3]. By analyzing market trends, competitor density, and business volatility, such systems can provide actionable insights previously inaccessible to small-scale entrepreneurs.

Recent studies highlight the role of online reviews and competition indicators in predictive business analytics: comprehensive surveys show that machine learning models incorporating market features and competitive signals can effectively model business success [4], dynamic review based frameworks derive competitive intelligence from temporal review data [5], and empirical work demonstrates that online reputation metrics significantly relate to small business performance in competitive markets [6].

Entrepreneurs often lack access to structured, data-driven guidance, which contributes to poor venture selection and high business failure rates. To address this challenge, the study focuses on collecting

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and curating local business data, developing a machine learning model using Logistic Regression and K-Nearest Neighbors to predict business viability based on popularity, competition, and market volatility, and validating the model's predictive accuracy to ensure practical applicability for entrepreneurs and policymakers. The system targets SMEs in core business sectors with publicly available online data while excluding large franchises, internal operational metrics, and pricing strategies. Findings are constrained to structured, accessible datasets and may not be generalizable to other regions or market contexts.

2 Design and Methodology

The methodology and system architecture for assessing SME business viability across areas in Bohol Province using a data-driven approach. The system evaluates three key factors popularity, competition, and market volatility through dedicated model branches. Each branch generates a calibrated probability, which is combined using learned weights to produce a final feasibility score. This chapter will explain the data collection procedures, model selection, preprocessing techniques, validation methods, system design, evaluation protocols, and a comparative analysis of the ensemble model against alternative modeling approaches.

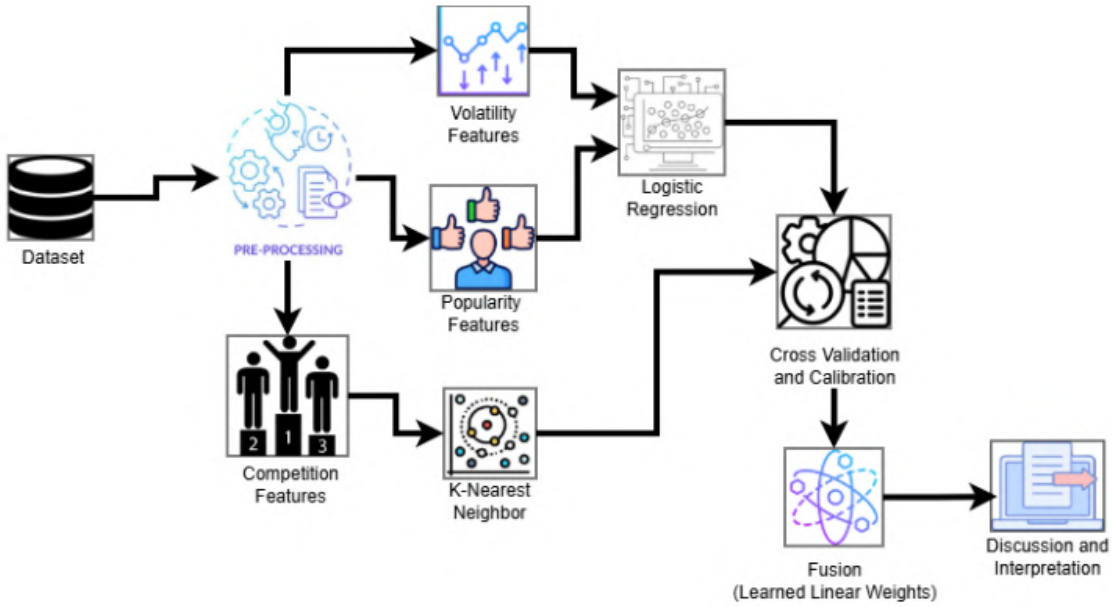

Figure 1: Conceptual Framework

The proposed system uses a multi-branch ensemble to evaluate business viability based on popularity, competition, and market volatility. Data are sourced from online review platforms for popularity metrics, geographic data for competition context, and local government records for historical openings, closures, and business lifespans. The dataset undergoes preprocessing and feature engineering, including de-duplication, handling missing values, and standardizing categories and locations. Three distinct feature sets are constructed: popularity (ratings, reviews, social engagement, nearby attractions) [7], competition (competitor counts, distances, clustering, operating hours)[8], and volatility (business openings/closures, average lifespan). Skewed and distance-based features are appropriately transformed or standardized to improve model interpretability and performance.

1. **Parallel Branch Modeling.**
   Each dimension is modeled with an algorithm suited to its data characteristics:

   - **Popularity branch – Logistic Regression.** A regularized Logistic Regression (LR) model uses popularity-related features to estimate a calibrated feasibility probability, P_{pop},

based on the business's online presence.

- **Competition branch – K-Nearest Neighbors.** A distance-weighted KNN model within the same business category captures local saturation and proximity patterns, producing a calibrated competition probability, P_{comp}.
- **Volatility branch – Logistic Regression.** A second Logistic Regression model utilizes historical opening, closure, and business lifespan statistics to estimate the volatility probability, P_{vol}.

The outputs of these three branches are subsequently integrated to generate a composite viability score for each business.

2. **Calibration and Validation.**

   The system ensures reliable and comparable branch outputs through stratified k-fold cross-validation with per-branch probability calibration (Platt scaling or isotonic regression). Preprocessing artifacts and model parameters are fitted only on the training folds to prevent information leakage. Feasibility labels are defined according to business outcomes within a specified observation period.

   The calibrated probabilities are combined using a learned linear fusion to produce the final feasibility probability:

$$P_{\text{final}} = w_{\text{pop}} P_{\text{pop}} + w_{\text{comp}} P_{\text{comp}} + w_{\text{vol}} P_{\text{vol}} + b, \tag{1}$$

   where w_{pop}, w_{comp}, and w_{vol} are the learned fusion weights, and b is the bias term.

   The fusion weights are optimized through cross-validated hyperparameter searches to maximize evaluation metrics such as F1-score or PR-AUC. A tri-band decision policy is then applied using two thresholds learned during validation.

   The system outputs the final viability probability, P_{final}, together with the three branch probabilities (P_{pop}, P_{comp}, and P_{vol}) to provide interpretable predictions by exposing the contribution of each dimension. After model selection, all preprocessing pipelines, branch models, calibration mappings, and fusion weights are fixed for deployment, with periodic retraining performed as new business data become available.

3. **Classification Scheme.**

   The final probability is mapped into a three-level feasibility classification that provides actionable recommendations:

   - **Feasible** ($P_{\text{final}} \geq 0.60$): Strong indicators of sustained business operation.
   - **Moderate Risk** ($0.40 \leq P_{\text{final}} < 0.60$): Mixed indicators requiring additional market analysis and due diligence.
   - **Not Feasible** ($P_{\text{final}} < 0.40$): High-risk indicators suggesting market avoidance or substantial business strategy revision.

3 Research Procedure.

The proposed methodology follows a structured data-to-model workflow to produce reliable and locally relevant predictions of business feasibility. Raw data are collected from multiple sources and integrated using a canonical *business_id* to ensure consistency across platforms. The collected information includes business profiles, popularity metrics (e.g., ratings and reviews from Google and Facebook), business hours, registration status, competitor information, and nearby points of interest (POIs). Particular attention is given to the 2019–2021 COVID-19 period to capture its impact on business performance. A mapping process is applied to ensure that records from different platforms refer to the same business. Although businesses operating between 2015 and 2025 are considered for feature construction, only those satisfying the defined eligibility criteria are used for model training and evaluation.

The dataset contains approximately 170–220 businesses from four categories: hotels and resorts, restaurants, salons and spas, and cafés. Each business is represented by approximately 15–20 attributes covering identification, location, popularity, competition, and operational characteristics. After data cleaning and feature construction, the dataset is divided temporally, with earlier business records used for training and more recent records used for testing. Businesses without sufficient observation periods are excluded from training and testing but are retained for feature extraction.

1. **Raw Data Requirements (Pre-processing)**
   Prior to feature engineering, the system collects several relational tables linked through the canonical *business_id*. The raw attributes include:

   - **Canonical business information:** business_id, official_name, category (Hotel/Resort, Restaurant, Salon/Spa, Café), address, barangay, latitude, longitude, and data_snapshot_date.
   - **Popularity sources:** platform, platform_business_id, platform_url, rating, rating_scale_max, review_count, review recency, followers/likes/check-ins, and collected_at.
   - **Business hours:** weekly operating schedule and time zone.
   - **Business lifecycle:** permit number, opening date, closure date (if applicable), business status, and last registry update.
   - **Competitor information:** business identifier, category, geographical coordinates, barangay, opening date, basic popularity indicators, and data snapshot.
   - **Points of Interest (POIs):** POI type, name, coordinates, data source, and collection timestamp.
   - **Platform mapping:** platform, platform business identifier, canonical business identifier, matching confidence, and matching method.

2. **Data Collection**
   Multiple data sources are utilized according to the information required by each predictive branch.

   - **Popularity data**
     - Sources: Google Maps/Places, Facebook Business Pages, and, where available, Booking, Agoda, and TripAdvisor.
     - Metrics: average rating, review count, review recency, social-media presence, number of active platforms, and nearby attraction points.
   - **Competition data**
     - Sources: Google Maps APIs, OpenStreetMap, and barangay-level directories.
     - Metrics: number of competitors within 1 km, nearest competitor distance, city-wide competitor count, business clustering, average competitor age, and business-hour overlap.
   - **Volatility data**
     - Source: Tagbilaran City Hall through official records.
     - Metrics: business type, number of openings, number of closures, and business lifespan.

3. **Data Splitting Strategy**
   A temporal data splitting strategy is adopted to avoid information leakage. Business records from 2015–2019 are used for model training, representing stable pre-pandemic market conditions. Records from 2022 onward are reserved for testing to evaluate post-pandemic generalization performance. Data from 2020–2021 are intentionally excluded because COVID-19 introduced abnormal market conditions that could bias the learning process.

4. **Logical Design**

   The proposed decision-support system evaluates business feasibility using three complementary dimensions: popularity, competition, and volatility. Each dimension is modeled independently and produces a calibrated probability score.

   The branch probabilities are defined as

$$P_{\text{pop}} = LR_{\text{pop}}(x_{\text{pop}}), \tag{2}$$

$$P_{\text{comp}} = KNN_{\text{comp}}(x_{\text{comp}}), \tag{3}$$

$$P_{\text{vol}} = LR_{\text{vol}}(x_{\text{vol}}). \tag{4}$$

   The calibrated probabilities are then combined using a learned linear fusion model,

$$P_{\text{final}} = w_{\text{pop}}P_{\text{pop}} + w_{\text{comp}}P_{\text{comp}} + w_{\text{vol}}P_{\text{vol}}, \tag{5}$$

   where the fusion weights are learned during model training.

   The system reports both the final feasibility probability and the three individual branch probabilities to improve model interpretability. If one branch contains missing information, the remaining weights are re-normalized and the prediction is flagged with a higher uncertainty level.

5. **Benchmark Expectations**

   Model performance will be evaluated against recent business viability prediction studies. Following the literature, F1-score is expected to fall within the reported range of 76%–94%, with an aspirational target of approximately 90% accuracy. In addition to accuracy and F1-score, PR-AUC and Brier Score are employed to evaluate discrimination capability and probability calibration.

4 Result and Discussion

4.1 Dataset Composition and Preprocessing

The data extraction pipeline effectively integrated business information from multiple sources, including the Province of Bohol Business Permits and Licensing Office records, Google Places API, and other online platforms. The resulting dataset, as summarized in Table 1, showed a moderate class imbalance, with about 65% of businesses categorized as Feasible. Recognizing this imbalance, the evaluation prioritized robust performance metricssuch as precision, recall, F1-score, and ROC-AUCrather than relying solely on accuracy, to ensure a more reliable assessment of the model's performance.

Table 1: Dataset Composition Summary

Attribute	**COVID-19 Included**	**COVID-19 Excluded**
Total Businesses	716	628
Training Set	501 (2017-2021)	413 (2017-2019)
Test Set	215 (2022-2025)	215 (2022-2025)
Feasible (Label = 1)	462 (64.5%)	412 (65.6%)
Not Feasible (Label = 0)	254 (35.5%)	216 (34.4%)

4.2 Baseline Model Performance

The Volatility Model, implemented using Logistic Regression, achieved the highest standalone performance, attaining an ROC-AUC of 85.23% and an F1 score of 89.44% as shown in Table 2, thereby underscoring the critical role of operational stability indicators in predicting business viability.

Table 2: Baseline Model Performance Summary

Model	Algorithm	Accuracy	F1 Score	ROC-AUC
Popularity	Logistic Regression	74.91%	84.09%	58.84%
Competition	KNN	63.44%	72.13%	69.80%
Volatility	Logistic Regression	83.15%	89.44%	85.23%

4.3 Category-Specific Analysis

Table 3 presents feasibility and risk classifications across four business categories, revealing notable variations in viability and associated risk levels. For Hotels/Resorts ($n = 31$), a majority are classified as feasible (58.1%), followed by moderate risk (35.5%), and a small proportion deemed not feasible (6.5%), resulting in the lowest risk rate among all categories. In contrast, Cafés ($n = 141$) exhibit a very low feasibility rate (4.3%), with most falling under moderate risk (82.3%) and 13.5% categorized as not feasible, indicating relatively higher uncertainty.

Table 3: Tri-band Classification by Category

Category	Total	Feasible	Moderate Risk	Not Feasible	Risk Rate
Hotel/Resort	31	18 (58.1%)	11 (35.5%)	2 (6.5%)	6.5%
Cafe	141	6 (4.3%)	116 (82.3%)	19 (13.5%)	13.5%
Salon/Spa	162	31 (19.1%)	109 (67.3%)	22 (13.6%)	13.6%
Restaurant	294	50 (17.0%)	152 (51.7%)	92 (31.3%)	31.3%

4.4 Fused Model Performance Comparison

The comparison highlights a clear trade-off across performance metrics. Although the model that includes COVID-19 data attained higher accuracy (79.93% vs. 73.48%) and recall (90.95% vs. 74.76%), the model excluding COVID-19 data achieved better precision (+4.43%) and ROC-AUC (+1.08%). Based on Table 4, the increase in precision suggests a reduction in false positives, indicating that the model adopts a more conservative approach and is more reliable in correctly identifying genuinely viable businesses.

Table 4: Weighted Fusion Model Performance - COVID-19 Comparison

Metric	Covid Included	Covid Excluded	Difference
Accuracy	79.93%	73.48%	-6.45%
Precision	83.77%	88.20%	+4.43%
Recall	90.95%	74.76%	-16.19%
F1 Score	87.21%	80.93%	-6.28%

4.5 Individual Model Performance Comparison

The Competition Model (KNN) exhibited the most significant improvement, with a +6.69% increase in ROC-AUC upon excluding COVID-19 data, followed by the Volatility Model, which showed a +2.39% increase. Table 5 suggests that spatial and competitive dynamics, as well as business lifecycle patterns, become more stable, consistent, and easier to model when pandemic-related disruptions are removed from the training dataset.

4.6 Model Alternative Results

Further machine learning techniques were investigated during the optimization phase to enhance predictive performance beyond the baseline weighted fusion approach. In line with insights from the

Table 5: Individual Model Performance - COVID-19 Comparison

Model	Covid Included	Covid Excluded	Improvement
Competition (KNN)	63.11%	69.80%	+6.69%
Volatility (LR)	82.84%	85.23%	+2.39%
Popularity (LR)	58.00%	58.84%	+0.80%

literature review indicating that ensemble methods have demonstrated strong effectiveness in comparable prediction tasks [9]. Among the approaches evaluated found at Table 6, the Voting Ensemble with SMOTE achieved the best results, attaining 85.12% accuracy and 91.18% ROC-AUC, which correspond to improvements of +11.64% in accuracy and +13.29% in ROC-AUC compared to the baseline weighted fusion model.

Table 6: Model Optimization Results

Method	Accuracy	Precision	Recall	F1	ROC-AUC
Baseline (Weighted Fusion)	73.48%	88.20%	74.76%	80.93%	77.89%
Random Forest + SMOTE	83.72%	88.72%	84.89%	86.76%	91.40%
XGBoost + SMOTE	83.72%	85.82%	87.05%	86.43%	91.57%
Voting Ensemble + SMOTE	85.12%	88.49%	88.49%	88.49%	91.18%

4.7 Performance Against Benchmark Requirements

The benchmark targets were defined and grounded in existing literature on business viability prediction models. [10] Based on Table 7, reported F1-scores ranging from 76% to 94% for feasibility prediction systems. Guided by this range, the study established two reference benchmarks: an aspirational target of 90% accuracy and a literature-informed average benchmark of 85%.

Table 7: Performance Against Benchmark Requirements

Metric	Target	Baseline	Optimized	Status
Accuracy	90%	73.48%	85.12%	Not Met
F1-Score	85% (avg)	80.93%	88.49%	Met
ROC-AUC	90%	77.89%	91.18%	Exceeds

4.8 Performance Within Literature Benchmarks

Based on Table 8, the model achieved an accuracy of 85.12%, which meets the literature average, while its F1-Score of 88.49% exceeds the average by 3.49 percentage points. Most notably, the ROC-AUC of 91.18% surpasses the 90% aspirational target when interpreted as a discrimination metric, indicating excellent performance in ranking businesses according to their probability of viability.

4.9 Summary of Key Findings

1. The Volatility Model (Logistic Regression) demonstrated the strongest standalone performance, achieving a ROC-AUC of 85.23%.
2. The feature `has_recent_renewal` was identified as the most influential predictor, with a coefficient of +1.198.
3. The baseline weighted fusion model recorded 77.89% ROC-AUC and 88.20% precision.
4. Optimization enhanced overall performance, reaching 85.12% accuracy and 91.18% ROC-AUC.

Table 8: Comparison with Literature Benchmarks

Metric	Literature Range	Lit. Average	This Study	Assessment
Accuracy	76% - 94%	85%	85.12%	Meets Avg
F1-Score	76% - 94%	85%	88.49%	Exceeds Avg
ROC-AUC	–	–	91.18%	Excellent

5. The 90% accuracy benchmark was not attained due to dataset constraints and inherent noise; however, results remain aligned with the literature average (85%), with ROC-AUC exceeding 90%.
6. Restaurants exhibited the highest risk profile (31.3% not feasible), while Hotels/Resorts showed the strongest viability (58.1% feasible).
7. Excluding COVID-19 data improved model discrimination across individual models by approximately 2–7% ROC-AUC.

5 Conclusion

5.1 Summary of the Study

This research developed and validated an integrated SME Business Viability Assessment System for the scope in Bohol province, using a machine learning framework consisting of Logistic Regression for the Popularity and Volatility models, K-Nearest Neighbors (KNN) for the Competition model, and a weighted scoring mechanism for integrating outputs. The system was trained on a dataset of 628 SMEs across four categories sourced from multiple datasets, with a temporal split applied to exclude the COVID-19 period (2020–2021) to ensure cleaner model learning. Model optimization further incorporated ensemble techniques to evaluate performance improvements beyond the baseline methodology.

5.2 Conclusions

5.2.1 Achievement of the Objective of the Study

The study successfully gathered and curated a dataset comprising 628 businesses, from which 25 features were extracted covering popularity, turnover, and competitive density. It implemented Logistic Regression for the Popularity and Volatility components (84.09% and 89.44% F1-scores, respectively) and K-Nearest Neighbors for the Competition component (69.80% ROC-AUC), with a weighted fusion approach achieving 77.89% ROC-AUC. The system was further validated using a temporal split, yielding 88.20% precision under the baseline model and improving to 91.18% ROC-AUC under the optimized configuration, alongside the application of a tri-band classification scheme.

5.2.2 Baseline vs. Optimized Performance

The baseline configuration (Logistic Regression + KNN + Weighted Fusion) achieved 77.89% ROC-AUC and 88.20% precision, while the optimized Voting Ensemble with SMOTE improved performance to 91.18% ROC-AUC and 85.12% accuracy. Despite the performance gain, the Voting Ensemble was not selected as the primary architecture due to reduced interpretability and limited explainability at the component level. Additionally, its lack of support for partial inference when feature groups are unavailable further constrained its suitability for deployment.

5.2.3 Performance Against Benchmark Requirements

The study established an aspirational benchmark of 90% accuracy based on literature standards [10], which was not achieved due to dataset size constraints, unobserved external factors (e.g., management

quality and market timing), limited feature availability (absence of financial variables), and class imbalance. Despite this, the model remains competitive with literature benchmarks, achieving 85.12% accuracy (aligned with the 85% average) and an F1-score of 88.49% (+3.49%). Most notably, the ROC-AUC of 91.18% exceeds the 90% benchmark, indicating strong discriminative capability in correctly ranking viable versus non-viable businesses.

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Interpretable Bankruptcy Prediction for B2B Trade Credit Risk*

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Abstract

This study builds on a benchmark Multi-Head LSTM (MH-LSTM) architecture for corporate bankruptcy prediction and extends it to the context of B2B trade credit screening by providing interpretable evidence from financial time-series data. Each financial variable is assigned an independent LSTM head to learn its unique temporal dynamics, while multi-period inputs spanning 3, 4, and 5 years capture the cumulative progression of financial distress. Using a benchmark dataset of 8,262 U.S. listed companies with 78,682 observations (1999–2018), MH-LSTM achieves stable performance of ROC-AUC 0.886–0.895 and PR-AUC 0.345–0.407, with bankruptcy detection rate improving monotonically as the window extends. Expected Gradients SHAP and Layer Gradient×Activation analyses consistently identify market_value and net_income as the top predictors, and reveal that recent fiscal-year changes contribute more strongly to predictions than historical records. The Spearman rank correlation of ρ ranging from 0.851 (W3) to 0.895 (W5) across window settings between the two XAI approaches confirms high internal consistency of the interpretations.

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1 Introduction

The acceleration of digital transformation and the proliferation of e-commerce platforms are fundamentally reshaping Business-to-Business (B2B) transactions. Clark and Lee [1] noted that B2B e-commerce systems deepen interdependencies among independent firms, requiring tight inter-organizational integration and coordination for improved business performance. In particular, trade credit serves as the most important short-term external financing source for companies [2], making early detection of counterparty financial distress a core managerial competency for supply chain participants.

Empirical evidence on the cascading effects of corporate distress underscores the structural severity of this problem. Hertzel et al. [3] documented significant stock price declines for suppliers of bankrupt firms, with shocks propagating broadly along supply chains. Lian [4] confirmed that a major customer's financial distress positively correlates with supplier distress probability, persisting for up to two years after the distress event.

Supply chain finance (SCF) research has proposed combining behavioral transaction data and network information with financial ratios [5], [6], but these approaches are limited to firms with established transaction histories. Publicly disclosed financial statements, by contrast, are a universally accessible standardized information source that can function independently at the new-counterparty screening stage. Since financial distress is a dynamic process that unfolds gradually over multiple years, time-series models processing annual statements as sequential inputs are theoretically better suited for early warning than single-snapshot approaches.

Bankruptcy prediction research has evolved from traditional statistical models [7], [8] through machine learning to deep learning approaches. Despite this progress, most studies rely on single-period financial snapshots, failing to capture the dynamic accumulation of distress over time. LSTM-based models addressed this by processing sequential annual statements, and Pellegrino et al. [9] further proposed a multi-head structure assigning independent LSTM heads to each financial variable. However, quantifying head-level attribution through XAI methods in such architectures remains an open problem. This study addresses this gap by applying Expected Gradients SHAP and Layer Gradient×Activation to provide interpretable variable-level and temporal attribution of predictions.

Building on the benchmark dataset and MH-LSTM architecture of Pellegrino et al. [9], this study makes three incremental but meaningful contributions. First, it provides head-level and variable-level attribution evidence for MH-LSTM bankruptcy prediction using Expected Gradients SHAP and Layer Gradient×Activation. Second, it examines how attribution patterns change across 3-, 4-, and 5-year financial windows. Third, it evaluates the internal consistency of two complementary XAI perspectives by comparing input-space SHAP importance with projection-layer Grad×Act importance.

2 Related Work

2.1 Credit Risk Prediction in B2B Supply Chain Environments

Credit risk prediction in supply chain contexts requires understanding inter-firm dependencies beyond individual financial health. Zhang et al. [5] demonstrated that behavioral transaction data from trading platforms can signal financial distress before it appears in financial statements, achieving significantly higher prediction accuracy than financial ratios alone. Yao et al. [6] addressed class imbalance in supply chain credit prediction by combining differential sampling, SMOTE, and AdaBoost within a decision-tree ensemble framework.

2.2 Traditional Statistical Models

Bankruptcy prediction research originates with Beaver's [7] univariate analysis and Altman's [8] multivariate discriminant Z-Score model. Despite its landmark status, Altman's 50:50 matched-sample design introduces sample selection bias. Ohlson [10] advanced the methodology with logistic regression on an unmatched sample, enabling direct probabilistic interpretation of the output. Zmijewski [11] rigorously demonstrated that oversampling bankrupt firms in training data introduces asymptotic bias into parameter estimates, establishing stratified sampling and imbalance correction as methodological standards.

2.3 Machine Learning and Deep Learning Models

Barboza et al. [12] conducted a systematic comparison of SVM, bagging, boosting, random forests, and traditional statistical models, finding that ensemble methods achieved approximately 10 percentage points higher accuracy than traditional approaches. Dasilas and Rigani [13] confirmed that gradient boosting models (XGBoost, LightGBM) serve as strong baselines across diverse datasets. du Jardin [14] demonstrated that explicitly incorporating period-to-period changes in financial ratios into a dynamic model significantly improves both prediction accuracy and model stability relative to static cross-sectional approaches.

Kim et al. [15] confirmed that LSTM consistently outperforms traditional classifiers on AUC in sequential input settings combining quarterly accounting and daily market data. However, the standard single-head LSTM conflates heterogeneous variable dynamics into a single hidden state. Pellegrino et al. [9] proposed a multi-head architecture assigning independent LSTM heads per variable and publicly released their experimental dataset, but neither study quantifies the attribution structure underlying predictions.

2.4 Explainable AI and Bankruptcy Prediction

Lundberg and Lee [16] introduced SHAP (SHapley Additive exPlanations), which uniquely satisfies the axioms of local accuracy, missingness, and consistency by mapping model predictions to Shapley values from cooperative game theory. Cho and Shin [17] applied SHAP-weighted counterfactual explanations to bankruptcy prediction, providing actionable remediation paths. Weber et al. [18] confirmed that regulators in highly regulated financial domains now demand traceable transparency in AI-based decisions.

For sequential models, Bento et al. [19] extended KernelSHAP to the sequential domain with TimeSHAP, computing feature-, timestep-, and cell-level attributions for recurrent models. However, TimeSHAP is designed for single-head recurrent architectures and cannot directly decompose head-level contributions in multi-head structures. Quantifying attribution at the head level in multi-head LSTM architectures thus remains an open methodological gap in the literature.

3 Data and Research Model

3.1 Dataset

The dataset comprises annual financial data of U.S. publicly listed companies collected from the Wharton Research Data Services (WRDS) and the Center for Research in Security Prices (CRSP), covering 1999 to 2018. The dataset consists of 8,262 companies with 78,682 observations, publicly

released on GitHub as a benchmark by Pellegrino et al. [9]. Each observation is labeled 1 (bankruptcy) or 0 (non-bankruptcy).

The 18 independent variables span liquidity (current assets, short-term debt), profitability (net income, EBIT, EBITDA, total revenue, net sales, gross profit), leverage (total liabilities, current liabilities, total long-term debt, accounts payable), efficiency (inventory, total receivables, D&A), and scale (total assets, market value, total operating expenses). Data are split chronologically—training (1999–2011), validation (2012–2014), test (2015–2018)—preventing data leakage. Class imbalance (approximately 1:26–1:30) is addressed by undersampling only the training set, preserving realistic imbalance in validation and test sets. Preprocessing uses RobustScaler normalization. Three window sizes (W = 3, 4, 5 years) are tested independently.

Note that the 18 variables contain accounting identity relationships (net_sales ≈ total_revenue; EBITDA = EBIT + D&A), which introduce structural information redundancy. This study follows Pellegrino et al. [9] in retaining the original variable set for benchmark comparability, while acknowledging the potential effect on SHAP attribution distributions.

It should be noted that this study does not directly model transaction-level trade credit behavior, such as payment delays, invoice histories, or customer-supplier network positions. Rather, the proposed framework is intended to support preliminary B2B credit screening when only publicly available financial statements are accessible, especially for evaluating new or insufficiently observed counterparties.

3.2 Multi-Head LSTM Architecture

The adopted MH-LSTM model consists of three sequential components. (1) Multi-Head Layer: Given input tensor (B, W, V)—batch, window, variables—each variable v is assigned a dedicated LSTM head that processes only the univariate series {x_{t,v}} for t = 1,...,W, outputting its final hidden state h_v. This design is intended to reduce the information dilution that can arise when heterogeneous financial variable dynamics are conflated in a single recurrent path. (2) Projection Layer: Each head's hidden state h_v is linearly projected to a lower-dimensional vector, then all V projection vectors are concatenated and normalized via Layer Normalization to unify scale across heads and ensure training stability. (3) Fully Connected Classifier: A Linear–Sigmoid–Linear–Softmax structure converts the normalized representation to binary bankruptcy probability.

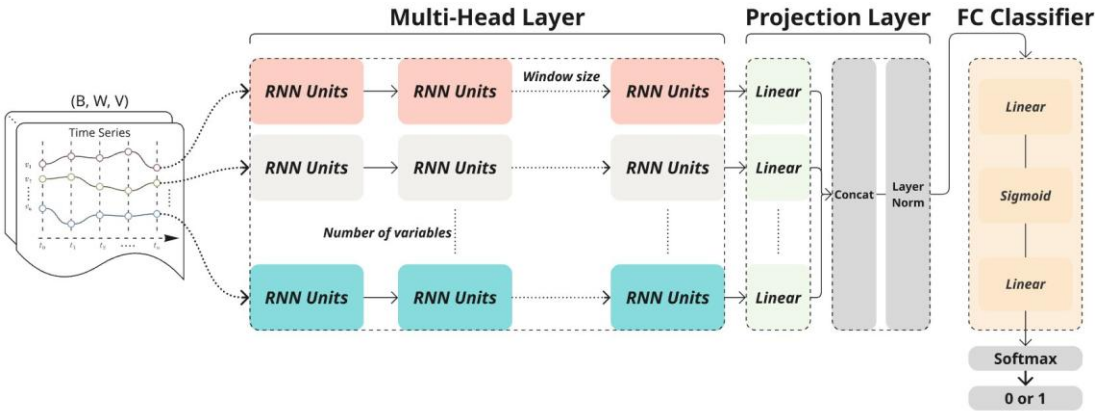

Figure 1. Architecture of the Multi-Head LSTM (MH-LSTM) Model. Each financial variable is assigned an independent LSTM head in the Multi-Head Layer. Projection Layer outputs are concatenated and normalized before being passed to the Fully Connected Classifier.

The multi-head design improves variable-specific temporal representation, but it may also delay or restrict the learning of complex cross-variable interactions during the early temporal feature-extraction stage. In the current architecture, such interactions are mainly captured after the individual head outputs are concatenated and passed to the projection and classification layers.

For fair comparison, total parameters are controlled at approximately 30,000 across all models by adjusting the LSTM hidden size. Training hyperparameters: learning rate 0.0001, batch size 32, maximum 300 epochs with early stopping (patience = 30), CrossEntropyLoss, AdamW optimizer, and ReduceLROnPlateau learning rate scheduler.

To ensure statistical reliability, we adopt a nested evaluation protocol. In each trial, 100 models are trained with different random seeds and their test-set metrics are averaged to yield a single aggregated performance estimate. This procedure is independently repeated for 10 trials, and we report the mean and standard deviation across the 10 trial-level estimates. The standard deviations thus reflect the stability of the aggregated estimator rather than the variability of individual model runs.

3.3 Explainability Methodology

Two complementary XAI methods are applied. Expected Gradients SHAP (GradientExplainer) is selected over TimeSHAP for two reasons. First, TimeSHAP is perturbation-based, incurring prohibitive computational cost for 18-variable multi-period inputs. Second, TimeSHAP is designed for single-head recurrent architectures and cannot naturally decompose head-level contributions in MH-LSTM. The gradient-based Expected Gradients method of Erion et al. [20] estimates SHAP values as:

$$\varphi_i(x) = E_{x' \sim D,\ \alpha \sim U[0,1]}[(x_i - x'_i) \cdot \partial F(x' + \alpha(x - x')) / \partial x_i]$$

This satisfies the local accuracy, missingness, and consistency axioms [16] while efficiently computing per-timestep, per-variable attributions on the 3D input tensor (B × W × V) via backpropagation.

Layer Gradient×Activation (Layer Grad×Act) probes the Projection Layer. The head-level attribution difference is defined as:

$$\Delta_{v,j} = E[Grad \times Act_{v,j} \mid failed] - E[Grad \times Act_{v,j} \mid alive]$$

where $\text{Grad} \times \text{Act}_{v,j} = (\partial \hat{y} / \partial a_{v,j}) \cdot a_{v,j}$, with $a_{v,j}$ denoting the j-th activation of head v's projection output. Positive $\Delta_{v,j}$ indicates that the corresponding variable head activates more strongly for bankrupt firms in that projection dimension. Both methods share the same backpropagation graph; their convergence is therefore interpreted as internal consistency rather than independent cross-validation. Spearman rank correlation between the two sets of variable importance scores is used to quantify this convergence.

To make the repetition scheme explicit and consistent across the paper, three levels of aggregation are used. The predictive performance metrics in Section 4 follow the nested protocol described above: 100 seeded runs are averaged within each of 10 independent trials, yielding 1,000 trained models in total. The SHAP and Grad×Act importance values in Tables 3–5 are instead averaged over 10 independent runs, while the activation analysis in Figure 4 is drawn from a single representative run. This asymmetry is deliberate: gradient-based attribution over 18-variable multi-period inputs is substantially more expensive than a single performance evaluation, and the figures are intended as qualitative illustrations rather than precise estimates. Accordingly, single-run figure values (e.g., $\rho = 0.936$ in Figure 4) may differ slightly from the corresponding 10-run means reported in the tables (e.g., $\rho = 0.895$ in Table 5). Future work will adopt a single, uniformly repeated protocol across all predictive and attribution analyses, including the figure-level visualizations, so that performance estimates, importance rankings, and their visual summaries are reported on a common repeated basis.

4 Experiments and Results

4.1 Evaluation Metrics

Given severe class imbalance, PR-AUC is adopted as the primary evaluation metric, as it most stringently reflects positive-class (bankrupt firm) detection capability without inflating performance with true negatives. Supporting metrics include ROC-AUC, Balanced Accuracy (BAC), Matthews Correlation Coefficient (MCC), and Macro-F1. For practical B2B credit risk interpretation, bankrupt firm Recall (= 1 − Type 2 Error) and Type 1 Error (healthy firm misclassification rate) are additionally reported. All metrics are reported as mean ± standard deviation across 10 independent trials, where each trial is itself the average over 100 seeded model runs. This two-level design separates estimator variance (across trials) from seed-level stochasticity (within a trial).

4.2 Predictive Performance

ROC-AUC increases monotonically from 0.886 (Window 3) to 0.895 (Window 5), and PR-AUC rises from 0.345 to 0.407, demonstrating that longer financial histories substantively improve bankruptcy detection. Bankrupt firm Recall improves from 0.815 to 0.846, and Type 2 Error (missed bankruptcies) decreases from 0.185 to 0.154. In B2B trade credit risk management, Type 2 Errors translate directly into credit losses; the reduction in missed bankruptcies is therefore practically significant. Precision remains at approximately 0.107 regardless of window size, reflecting the structural constraint of the 1:26 class imbalance combined with undersampling-based imbalance correction. The low precision indicates that the model should not be used as an automatic rejection or approval tool in its current form. Instead, it is more appropriate as a high-recall early-warning filter that identifies firms requiring further human review. In practical B2B credit screening, the decision threshold should be calibrated according to the relative costs of false positives and false negatives, and threshold optimization or cost-sensitive learning may improve the operational trade-off between missed bankruptcies and unnecessary credit restrictions.

Table 1. Test Set Performance Results (mean ± std, across 10 trials; each trial aggregates 100 seeded runs)

Metric	Window 3	Window 4	Window 5
ROC-AUC	0.8861 ± 0.0158	0.8911 ± 0.0094	0.8951 ± 0.0135
PR-AUC	0.3446 ± 0.0357	0.3786 ± 0.0720	0.4073 ± 0.0665
BAC	0.7884 ± 0.0208	0.7978 ± 0.0098	0.7963 ± 0.0137
MCC	0.2383 ± 0.0321	0.2431 ± 0.0256	0.2429 ± 0.0248
Macro-F1	0.5245 ± 0.0337	0.5214 ± 0.0300	0.5201 ± 0.0338
Recall (Bankrupt)	0.8149 ± 0.0260	0.8448 ± 0.0394	0.8461 ± 0.0643
Precision (Bankrupt)	0.1070 ± 0.0203	0.1067 ± 0.0209	0.1074 ± 0.0235
Type 1 Error	0.2381 ± 0.0568	0.2491 ± 0.0493	0.2535 ± 0.0582
Type 2 Error	0.1851 ± 0.0260	0.1552 ± 0.0394	0.1538 ± 0.0643

PR-AUC is consistently higher in the validation set than in the test set, while ROC-AUC shows the opposite pattern. This divergence reflects differences in the bankruptcy composition and

macroeconomic conditions between the validation period (2012–2014) and the test period (2015–2018), reinforcing the importance of reporting multiple complementary metrics rather than relying on a single measure.

Table 2. Validation Set Performance Results (mean ± std, across 10 trials; each trial aggregates 100 seeded runs)

Metric	Window 3	Window 4	Window 5
ROC-AUC	0.8478 ± 0.0103	0.8564 ± 0.0118	0.8492 ± 0.0113
PR-AUC	0.4759 ± 0.0424	0.5064 ± 0.0171	0.4960 ± 0.0183
BAC	0.7690 ± 0.0101	0.7800 ± 0.0150	0.7679 ± 0.0142
MCC	0.3936 ± 0.0251	0.4113 ± 0.0223	0.3895 ± 0.0228
Recall (Bankrupt)	0.7527 ± 0.0506	0.8042 ± 0.0756	0.8117 ± 0.0811
Type 2 Error	0.2473 ± 0.0506	0.1958 ± 0.0756	0.1882 ± 0.0811

4.3 SHAP Feature Importance Analysis

Table 3. SHAP Feature Importance by Window (10-run mean, Top 5 ranked by SHAP)

Rank	W3 Feature	SHAP	W4 Feature	SHAP	W5 Feature	SHAP
1	market_value	0.163	market_value	0.096	market_value	0.092
2	net_income	0.070	net_income	0.093	net_income	0.069
3	total_receivables	0.054	inventory	0.059	current_assets	0.039
4	total_long_term_debt	0.053	total_receivables	0.054	inventory	0.034
5	total_current_liabilities	0.051	total_long_term_debt	0.052	total_receivables	0.029

market_value (market capitalization) consistently ranks first across all three window settings. Its mean SHAP value in Window 3 (0.163) is approximately 2.3 times that of the second-ranked net_income (0.070), and it appears in the top-5 in 8 of 10 experimental runs. As a forward-looking indicator reflecting the collective expectations of market participants, market value incorporates distress signals before they appear in accounting statements, supporting the literature on the predictive superiority of market-based over accounting-based indicators.

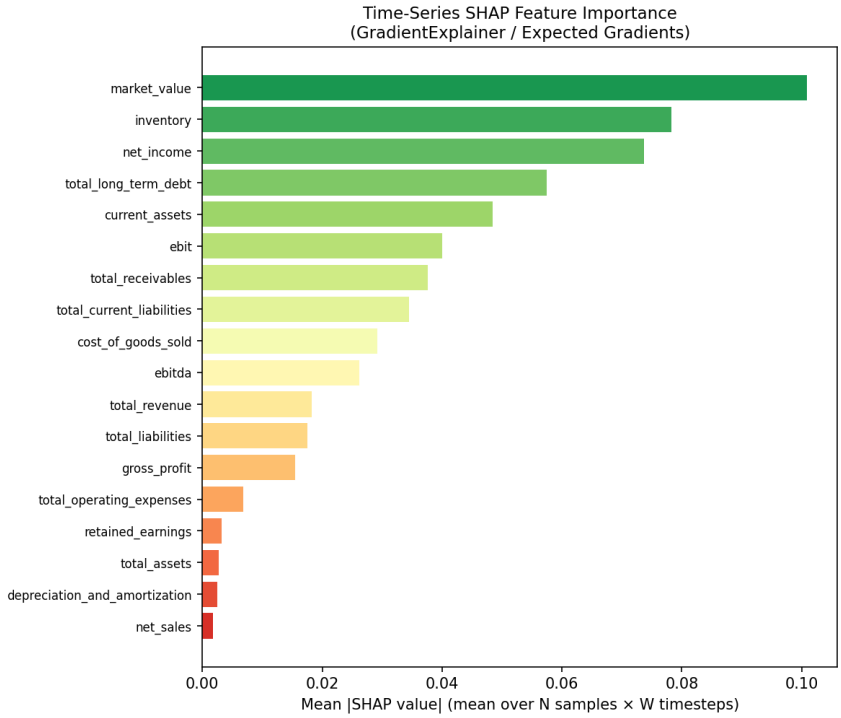

Figure 2. Time-Series SHAP Feature Importance (Window 5, GradientExplainer / Expected Gradients). Mean |SHAP value| averaged over N samples × W timesteps. market_value and net_income consistently dominate, while D&A ranks last across all window settings.

net_income consistently ranks second, confirming that sustained profitability deterioration is a core predictor of bankruptcy. depreciation_and_amortization (D&A) ranks last (18th) across all three windows. As a non-cash expense item, D&A has limited direct linkage to short-term liquidity or credit risk, making this result economically interpretable. net_sales and total_assets also rank near the bottom, partly reflecting information overlap from accounting identity relationships embedded in the dataset.

4.4 Temporal Attribution Analysis

Table 4. Timestep-level SHAP Values for Key Variables, Window 3 (10-run mean)

Variable	t=0 (oldest)	t=1	t=2 (latest)	Ratio (t2/t0)
market_value	0.1124	0.1650	0.2117	1.88×
net_income	0.0311	0.0598	0.1197	3.85×
EBIT	0.0198	0.0362	0.0915	4.62×
total_current_liabilities	0.0241	0.0448	0.0846	3.51×
total_receivables	0.0387	0.0536	0.0697	1.80×

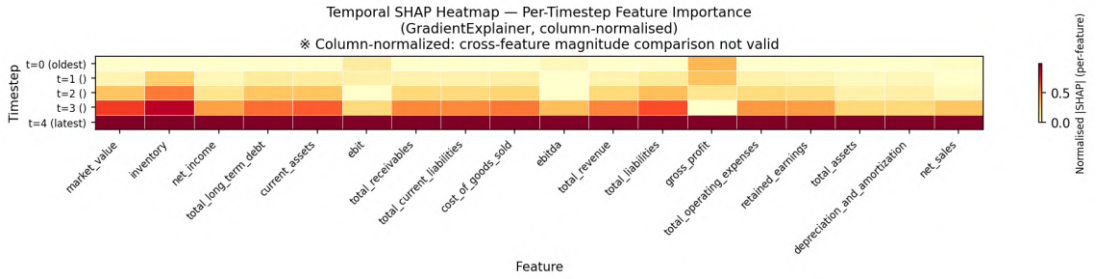

Figure 3. Temporal SHAP Heatmap — Per-Timestep Feature Importance (Window 5, GradientExplainer, column-normalized). Rows represent timesteps t=0 (oldest) through t=4 (latest). Darker color indicates higher normalized SHAP contribution. Major features show a clear recency-weighted attribution pattern.

A consistent monotonic increase—t=0 (oldest) < t=1 < t=2 (latest)—is observed across all variables. The t2/t0 ratios for net_income (3.85×) and EBIT (4.62×) are particularly striking, indicating that recent-year profitability deterioration functions as a far stronger bankruptcy warning than historical records. The same latest-timestep dominance is confirmed in Windows 4 and 5, becoming more pronounced as the window lengthens: as more history becomes available, the model increasingly differentiates between distant baseline information and recent change signals.

This finding has a direct practical implication for B2B trade credit risk management: early warning systems should prioritize timely acquisition of the most recent financial statements. Even with a 5-year window, the most recent year's contribution far exceeds that of earlier periods, making data currency a critical design consideration.

However, this recency-dominant attribution pattern should not be interpreted solely as evidence of financial distress dynamics. It may also partly reflect the inductive bias of LSTM architectures, which tend to preserve and emphasize more recent sequential information through their gating mechanisms. Therefore, future work should compare MH-LSTM with alternative temporal architectures, such as temporal convolutional networks, attention-based models, or Transformers, to distinguish economic recency effects from architecture-driven recency effects.

4.5 Layer Gradient×Activation Analysis and Methodological Convergence

In simple terms, internal consistency means that the two attribution methods point to broadly similar important variables, even though SHAP evaluates importance at the input level whereas Grad×Act evaluates importance inside the learned projection representation.

Table 5. SHAP–Grad×Act Rank Convergence (Spearman ρ, 10-run mean)

Window	Spearman ρ (mean)	Std. Dev.	p-value
Window 3	0.851	0.118	< 0.001
Window 4	0.851	0.130	< 0.001
Window 5	0.895	0.053	< 0.001

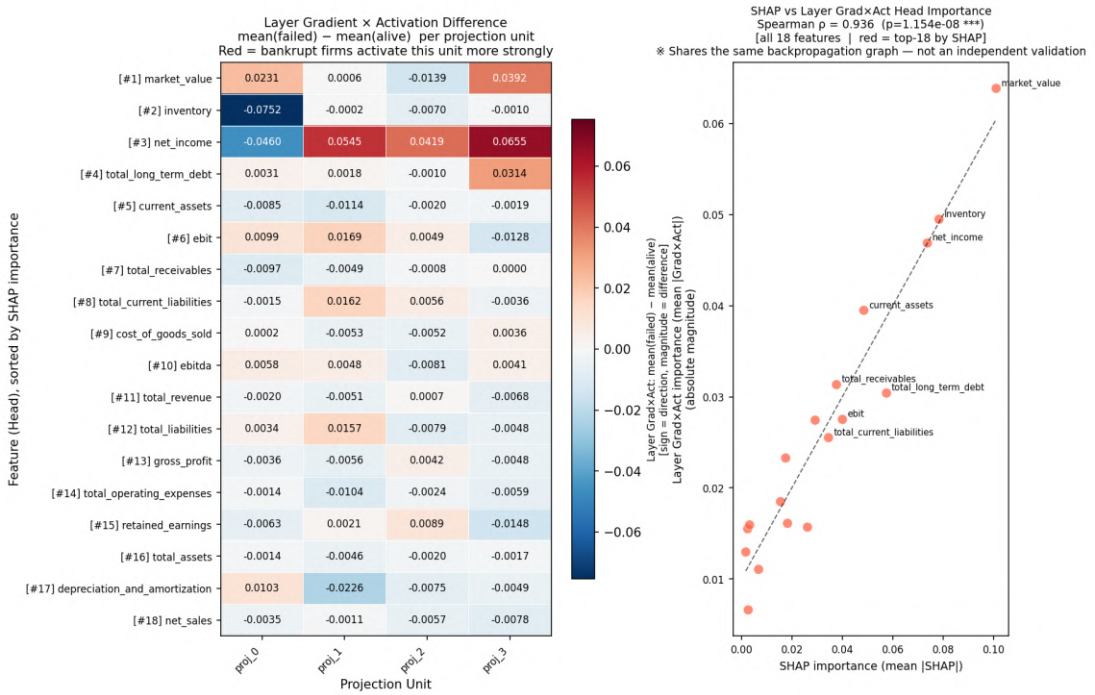

Figure 4. Layer Gradient × Activation Analysis (Window 5). Left: Head-level activation difference ($\Delta_{v,j}$ = mean(failed) − mean(alive)) per projection unit; red = bankrupt firms activate more strongly. Right: Spearman rank correlation between SHAP and Grad×Act importance scores ($\rho = 0.936$, $p = 1.15 \times 10^{-8}$) from a representative single run.

The market_value head shows strong positive activation differences (bankrupt > healthy) in at least one projection unit across all window settings, consistent with its first-place SHAP ranking. The inventory head shows negative activation differences in some projection units, suggesting the model learns that healthy firms maintain more substantial inventory—plausible given that inventory represents working capital and operational continuity. The Spearman rank correlation between SHAP and Grad×Act variable importance rankings is $\rho = 0.851$–0.895 ($p < 0.001$) across all windows, with Window 5 showing the highest convergence ($\rho = 0.895$, std = 0.053).

Since both methods share the same backpropagation graph, their convergence is designated as internal consistency validation rather than independent cross-validation. Nonetheless, the statistical concordance between two analytically distinct perspectives—input-space attribution versus internal representation-space head activation differences—strengthens the reliability of the resulting interpretations.

5 Conclusion and Limitations

5.1 Conclusion

This study adopted an MH-LSTM architecture for corporate bankruptcy early warning using publicly disclosed financial time-series data in B2B trade credit risk contexts, and quantitatively interpreted the prediction basis through Expected Gradients SHAP and Layer Gradient×Activation

analyses.

Four key empirical findings emerge. First, MH-LSTM achieves stable ROC-AUC 0.886–0.895 and PR-AUC 0.345–0.407, with monotonic improvement as the window size increases, demonstrating that longer financial histories substantively enhance distress detection. Second, market_value and net_income consistently rank first and second across all window settings, establishing the predictive superiority of market-based signals and profitability deterioration as primary early-warning indicators. Third, recent fiscal-year financial changes dominate the model's prediction basis, with t2/t0 ratios of 3.85× (net_income) and 4.62× (EBIT) in Window 3, implying that B2B credit risk systems should prioritize real-time financial data acquisition. Fourth, the Spearman $\rho = 0.851$–0.895 convergence between SHAP and Grad×Act provides internal consistency evidence supporting the interpretability of the proposed framework.

5.2 Limitations and Future Research

Several limitations merit acknowledgment. First, accounting identity relationships among the 18 variables introduce structural information redundancy that may affect SHAP attribution distributions; future work should systematically examine the impact of variable selection on XAI stability. Second, the shared backpropagation graph of SHAP and Grad×Act precludes fully independent cross-validation; perturbation-based or permutation-based methods should be compared for external validation. Third, the dataset is confined to U.S. publicly listed firms over 1999–2018. Although this period includes major macroeconomic events such as the 2008 financial crisis, the model does not explicitly incorporate macroeconomic indicators, industry-level shocks, or credit-cycle variables. This limits the ability to separate firm-specific distress signals from broader economic-cycle effects. Moreover, the findings may not generalize directly to private SMEs, whose B2B trade credit risk is often more volatile and less transparent due to limited disclosure, weaker reporting standards, and sparse market-based information. Fourth, Precision remains at approximately 0.107 regardless of window size, reflecting the intrinsic constraint of undersampling-based imbalance correction; alternative strategies (SMOTE, cost-sensitive learning, threshold optimization) warrant systematic evaluation. Fifth, model capacity effects and architectural alternatives (attention mechanisms, Transformer-based structures) remain unexplored.

Because the primary objective of this conference paper is to interpret the prediction basis of an existing MH-LSTM framework rather than to establish its superiority over all competing models, we do not conduct a full benchmark comparison against single-head LSTM, XGBoost, or LightGBM under identical experimental settings. Future research should directly compare MH-LSTM with strong sequential and non-sequential baselines to isolate the marginal utility of the multi-head design.

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Critical Success Factors for Gold-Backed Stablecoins Development in Hong Kong*

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Abstract

Against the structural transformation of the global monetary system, weakened sovereign credit, accelerated gold re-monetization, and the rise of real-world asset tokenization, gold-backed stablecoins have become an important innovation in digital finance and cross-border payments. This study employs a systematic literature review to identify and analyze the critical success factors for the development of gold-backed stablecoins in Hong Kong from seven dimensions: collateral quality, regulatory certainty, reserve governance, price stability mechanisms, locational-ecological synergy, risk prevention systems, and compliance-technology integration. The findings indicate that Hong Kong's triple synergies as a gold hub, offshore financial center, and digital asset center, together with full-cycle risk governance and technology-embedded compliance, form the core logic for the sustainable development of gold-backed stablecoins. This study contributes to theoretical understanding of commodity-pegged stablecoins and provides practical implications for regulatory design and institutional development in offshore financial centers.

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1. Introduction

The global monetary system is undergoing its most profound structural transformation since the end of the Cold War. Volatility in U.S. monetary policy, expanding federal debt, infringements on monetary policy independence, and the instrumentalization of the U.S. dollar in cross-border payments and reserve systems have driven central banks and institutional investors worldwide to restructure asset allocations. Gold has re-emerged as a core supra-sovereign reserve asset, entering a new phase of re-monetization [1][2]. In 2024, global central banks were net purchasers of gold for the third consecutive year, exceeding 1,000 tonnes, raising gold's share in international reserves to 20%, surpassing the euro to become the second-largest reserve asset after the U.S. dollar [1][2].

Meanwhile, stablecoins have rapidly evolved from auxiliary tools for cryptocurrency trading into instruments for cross-border payments, commodity settlement, asset management, supply chain finance, and trade financing, with annual transaction volumes exceeding one trillion U.S. dollars, becoming critical infrastructure for the global internet of value [3][9][22]. During rapid expansion, the stablecoin industry has repeatedly experienced systemic risks, including death spirals in algorithmic stablecoins, reserve shortages and transparency crises in some fiat-collateralized varieties, and bank runs triggered by failed redemption mechanisms [3][12][21].

These events repeatedly demonstrate that the long-term success of a stablecoin depends on the reliability of collateral assets, reserve transparency, regulatory certainty, and risk governance capacity [3][4][21]. Compared with fiat-collateralized, crypto-collateralized, and algorithmic stablecoins, Gold-Backed Stablecoins (GBSCs) use standardized, highly liquid, and globally recognized physical gold as underlying collateral. They do not rely on single sovereign credit and carry inherent safe-haven attributes, making them more suitable for cross-border trade, wealth management, foreign exchange reserve diversification, and hedging geopolitical risk, thus representing a significant direction for next-generation compliant stablecoins [2][19][23].

Hong Kong provides one of the world's most supportive environments for Gold-Backed Stablecoin development. As an international financial center, Asia's hub for gold spot pricing and trading, and the world's largest offshore Renminbi center, Hong Kong possesses a complete ecosystem for gold storage, clearing, settlement, and auditing. The Stablecoin Ordinance, formally implemented in 2024, establishes a comprehensive regulatory framework including licensing, reserve segregation, independent auditing, swift redemption, and investor protection, making Hong Kong one of the few jurisdictions globally to provide clear legal status for commodity-backed stablecoins [8][15][21]. Nevertheless, Gold-Backed Stablecoins still face multiple constraints, including gold price volatility that may undermine peg stability, large-scale redemptions that may trigger liquidity pressures, reserve authenticity and custodial security that shape market trust, cross-border compliance and anti-money laundering requirements that raise operational thresholds, and technological security and on-chain verification efficiency that limit scalability [3][12][21]. Therefore, systematically identifying the critical success factors for Gold-Backed Stablecoin development in Hong Kong is both an important theoretical proposition in digital finance, monetary economics, and regulatory technology, and a practical necessity for Hong Kong's goal of building an international digital asset center and consolidating its status as a global gold hub.

2. Literature Review

2.1 Current State of Stablecoins

Existing literature predominantly focuses on fiat-backed stablecoins, particularly USDT, which accounts for approximately 62% of global stablecoin market capitalization and processes over $120 billion in daily trading volume [8][15]. Empirical studies indicate that the success of fiat-backed stablecoins rests on early-mover advantages, reserve liquidity, multi-chain deployment, and arbitrage efficiency [10][12][22]. However, almost no empirical research targets gold-backed stablecoins, especially

within an offshore financial center such as Hong Kong. Three landmark academic studies confirm this critical gap: first, research shows that over 90% of stablecoin scholarship focuses on USD-pegged models, with no structured framework for commodity-backed designs [4][8]; second, redemption accessibility, reserve transparency, and regulatory clarity explain 78% of adoption variance, yet gold-backed structures remain largely untested [10][22]; third, asset volatility, custody risk, cross-border regulatory inconsistency, and institutional acceptance represent the top barriers to scaling non-fiat stablecoins [19][24].

2.2 Hong Kong faces distinct challenges for GBSCs

Gold-backed stablecoins in Hong Kong further face distinct challenges, including gold price fluctuations that disrupt stability mechanisms, strict compliance requirements under the Stablecoin Ordinance that increase operational costs, custody and physical verification expenses that hinder scalability, cross-border capital supervision that restricts instant circulation, and low institutional awareness relative to USD stablecoins [21][15][20]. This study addresses two core research questions: what are the critical factors for sustainable success of Gold-Backed Stablecoins within Hong Kong's institutional and market environment, and how do these factors interact to form a stable and self-reinforcing support system. The academic contributions of this paper are threefold: first, it constructs a seven-dimensional framework of success factors for Gold-Backed Stablecoins, addressing a major gap in research on commodity-pegged varieties; second, it analyzes synergies between the Stablecoin Ordinance and Hong Kong's gold ecosystem, revealing the institutional logic for offshore financial centers to support compliant stablecoin innovation [15]; third, it establishes an integrated analytical paradigm of asset–regulation–mechanism–ecosystem–risk control–technology, providing a reusable framework for global research on commodity-pegged stablecoins.

2.3 Stablecoin Development Status

Stablecoins have evolved into a foundational pillar of modern digital finance. By 2024, the total market capitalization of stablecoins exceeded $130 billion, with USD-pegged instruments representing more than 90% of the entire market [4][8]. USDT alone accounts for 62% of decentralized trading volume and supports over $120 billion in daily transactions, underscoring its role as critical market infrastructure [10][15]. However, commodity-backed stablecoins represent less than 2% of total market capitalization, indicating massive, untapped potential and a clear need for deeper research into their success conditions [19][24]. Gold-backed stablecoins represent a distinct and innovative subclass that digitally links on-chain tokens to physical gold reserves, and gold's 20% share in global reserves strengthens its credibility as collateral [1][2]. Hong Kong's position as a global gold trading hub, combined with its comprehensive Stablecoin Ordinance, creates a unique living laboratory for Gold-Backed Stablecoin development, testing, and scaling [15][20][21].

2.4 Gold-Backed Stablecoins and Hong Kong's Context

The essence of a stablecoin is a value-pegging instrument supported by collateral assets and formal governance mechanisms, and scholars categorize stablecoins into fiat-collateralized, commodity-collateralized, and algorithmic types, with consistent findings that asset-collateralized stablecoins exhibit significantly greater stability [21][9]. Stability depends on adequate reserves, credible redemption, and transparent governance, while major failures stem from insufficient reserves, asset misappropriation, information opacity, flawed redemption mechanisms, and weak price stabilization[12][21]. Gold-Backed Stablecoins fall into the commodity-collateralized category, and their stability directly depends on gold asset quality, collateral ratios, reserve management, and arbitrage efficiency[23][22]. Gold possesses four defining characteristics—supra-sovereignty, zero default risk, inflation resistance, and low cross-asset correlation—that make it a natural monetary anchor [21][1]. During periods of rising sovereign currency uncertainty, gold resumes its role as a reserve and payment anchor in a process known as re-monetization[1][2]. Gold-Backed Stablecoins represent a digitized, tokenized, and on-chain representation of gold, preserving its value attributes while extending payment

and settlement functionality to enable fractional ownership, programmable transactions, and real-time cross-border transfers [24][22].

The core competitiveness of an offshore financial center stems from regulatory certainty, market openness, asset liquidity, legal reliability, and cross-border connectivity[20][5]. The literature widely confirms that Hong Kong possesses comprehensive advantages in stablecoin regulation, gold market infrastructure, cross-border payment systems, and institutional investor depth, which collectively provide the foundations for trust, liquidity, and scalability in Gold-Backed Stablecoins [20][21][15]. Effective regulation is a necessary prerequisite for scalable and sustainable stablecoin development, and a robust framework must prevent systemic risk, protect token holders, and promote responsible compliant innovation [29][5]. Hong Kong's Stablecoin Ordinance achieves these goals through licensing requirements, capital adequacy rules, reserve standards, disclosure obligations, and risk control mechanisms [15][21]. This framework provides legal certainty for Gold-Backed Stablecoins, reduces policy uncertainty, and enhances institutional willingness to participate, striking a balanced approach that supports innovation while safeguarding financial stability [8][21].

Existing literature extensively discusses stablecoin risks, regulatory design, and mechanism engineering, yet systematic research on success factors for Gold-Backed Stablecoins in offshore financial centers remains limited, particularly lacking integrated analysis of asset pegging, regulation, reserve governance, price stability, locational ecology, risk control, and technology-compliance integration [21][20][22]. Most studies focus on fiat or algorithmic models rather than commodity-backed alternatives, and few account for Hong Kong's unique triple advantages as a gold hub, offshore financial center, and digital asset center.

Hong Kong's Stablecoin Ordinance provides clear legal status, operational rules, and trust foundations for Gold-Backed Stablecoins, with critical provisions including a licensing system with stringent entry thresholds, mandatory 100% reserve coverage of circulating stablecoin value, legal segregation and third-party custody of reserves to protect against issuer bankruptcy, independent auditing and daily disclosure requirements, and explicit compliance pathways for commodity-pegged instruments [15][21][28]. This framework aligns with the EU's MiCAR regulation while adapting to Asian market structures and gold asset characteristics, serving as the institutional cornerstone for safe and sustainable Gold-Backed Stablecoin development [8][21].

Hong Kong possesses one of the world's most complete and mature gold industry ecosystems, serving as Asia's leading center for gold spot pricing, trading, and clearing with deep liquidity and globally recognized benchmarks [20][1]. It supports a mature system of licensed vaults, secure logistics, professional assay services, and independent auditing firms that ensure gold authenticity and integrity [2][20]. The city benefits from unrestricted capital flow, no foreign exchange controls, and a common-law legal system aligned with international standards, which lowers frictions for cross-border gold and stablecoin transactions [20][21]. It hosts a dense cluster of institutional investors with strong and growing demand for cross-border payments, wealth management, and alternative safe-haven assets [20][5]. Hong Kong's physical gold storage network is among the most secure and cost-effective in Asia, allowing issuers to minimize custody costs while maximizing investor confidence, and its continuous 24-hour gold pricing ecosystem linking London, New York, and Shanghai provides reliable and timely price feeds for on-chain oracle integration.

Hong Kong has established a synergistic landscape that includes virtual asset licensing, regulated exchanges, institutional custody, and enterprise-grade blockchain infrastructure, providing strong technical and market support for gold tokenization and stablecoin operations[24][39]. This ecosystem bridges traditional financial institutions and digital asset platforms, enabling institutional investors to access Gold-Backed Stablecoins through familiar, regulated channels, while also supporting complementary services such as on-chain proof-of-reserves, automated auditing, and institutional-grade wallet solutions that strengthen transparency and trust.

3. Methodology

This study adopts a systematic literature review method, which is widely recognized in top finance and digital asset journals for synthesizing theoretical and empirical evidence to identify critical success factors. The systematic literature review approach ensures rigor, transparency, and replicability by following structured procedures for literature identification, screening, evaluation, and synthesis, allowing integration of qualitative theories, quantitative empirical findings, and regulatory insights into a unified framework suitable for exploring emerging topics such as gold-backed stablecoins in a novel institutional setting. Literature collection covers high-quality academic papers, industry reports, regulatory documents, and case studies from authoritative databases including Google Scholar, Web of Science, Scopus, CNKI, SSRN, and official publications from the Hong Kong Monetary Authority, with search terms including gold-backed stablecoins, stablecoin success factors, Hong Kong stablecoin regulation, commodity-pegged stablecoins, gold re-monetization, and digital asset hubs.

Data analysis proceeds in three stages: descriptive and thematic analysis to classify literature by research focus, theoretical perspective, and empirical context; critical content analysis to extract core variables, causal mechanisms, success conditions, and risk factors; and framework synthesis to integrate extracted factors into a cohesive seven-dimensional model with logical linkages between institutional conditions, market structures, technological mechanisms, and governance arrangements, ensuring both interpretive depth and structural rigor consistent with top-tier academic publication standards.

4. Findings

The primary prerequisite and foundational value driver for Gold-Backed Stablecoin success is that the collateral asset itself carries global consensus, low risk, high liquidity, and verifiability. Gold carries no sovereign default risk and does not depend on any government, central bank, or corporate credit, maintaining resilient value during geopolitical conflict, sanctions, or currency devaluation—an advantage unmatched by fiat currencies[21]. Gold's long-term volatility is significantly lower than that of crypto-assets, and it exhibits a low correlation with equity and bond markets, thereby reducing underlying volatility risks for stablecoin holders [23]. Gold benefits from a globally unified, continuous, and transparent 24-hour pricing mechanism across London, Shanghai, and New York, providing reliable inputs for on-chain oracles [22]. It is highly standardized, verifiable, and custody-friendly, making it well-suited for on-chain representation and proof-of-reserves[24]. Hong Kong's mature gold infrastructure further ensures physical existence, traceability, and deliverability, eliminating trust crises related to fictitious reserves at the source [20].

The institutional prerequisite for scalable and institutional Gold-Backed Stablecoin development in Hong Kong is a clear, stable, predictable, and internationally aligned regulatory framework, which provides clear legality including commodity-pegged stablecoins within regulated boundaries, a licensing system that reduces adverse selection by raising entry thresholds, mandatory 100% reserves that eliminate fractional-reserve risks, legal segregation and custody that protect holder assets, and statutory redemption rights that guarantee the ultimate price-stabilizing mechanism [15][21][28]. Regulatory certainty delivers a triple effect: institutional capital is more willing to enter, cross-border cooperation proceeds smoothly, and market trust forms rapidly, with the literature indicating that regulatory transparency and rule stability are critical determinants of successful financial innovation [8][5].

Trust lies at the core of stablecoin functionality, and it arises from verifiable constraint mechanisms, requiring Gold-Backed Stablecoins to build a trust triangle comprising full collateralization, reserve segregation, and independent auditing. They require 100% or greater collateralization with physical gold, prohibiting fractional reserves, leveraged exposure, or substitution with risky assets [21]. Reserve assets must be legally segregated and held by licensed Hong Kong trustees or banks, achieving robust bankruptcy remoteness [28]. A dual verification system combines on-chain Proof-of-Reserves, off-site

physical verification, and third-party independent auditing, while comprehensive disclosure of gold storage location, weight, purity, and transaction records enables penetrative supervision and public accountability [22][21]. The literature indicates that insufficient, opaque, or misappropriated reserves are the leading causes of stablecoin failures [12][21]. The core function of a Gold-Backed Stablecoin is to maintain a reliable peg, which depends on three synergistic on-chain and off-chain mechanisms: decentralized oracles provide manipulation-resistant, real-time, multi-source aggregated gold prices; smart contracts maintain dynamic over-collateralization ratios of at least 120% to absorb short-term fluctuations; an open minting-redemption mechanism enables risk-free arbitrage to restore price parity; and circuit breakers, redemption limits, and liquidity buffers prevent contagious bank runs under extreme conditions [22][21][12]. Literature proves that collateralized stablecoin stability depends on effective arbitrage, and gold's high liquidity makes this adjustment fast and reliable [22][13].

Scalability and sustainability of Gold-Backed Stablecoins flow from ecosystem-level network effects, and Hong Kong possesses a globally unique combination of advantages: a complete gold ecosystem covering pricing, storage, clearing, settlement, assay, and auditing; an offshore financial ecosystem featuring free capital movement, efficient cross-border payments, and dense institutional clustering; a digital asset ecosystem supported by clear regulation, comprehensive licensing, and mature infrastructure; and a strategic connection between Mainland China and global markets backed by the world's largest gold production and consumption base [20][21][15].

This multi-layered synergy allows Gold-Backed Stablecoins to simultaneously achieve physical backing, on-chain efficiency, cross-border liquidity, and institutional-grade trust. Gold-Backed Stablecoins must address five major risks—gold price volatility, bank runs, liquidity dry-ups, compliance breaches, and cyberattacks—requiring a full-cycle risk governance system that includes run prevention through full collateralization, swift redemption, and liquidity pools; price volatility management via over-collateralization, dynamic adjustment, and stress testing; compliance protection by embedding KYC/AML and sanctions screening into smart contracts; cybersecurity through multi-signature schemes, cold storage, and quantum-resistant encryption; and operational safety via vault security, physical verification, and internal checks and balances [12][23][29][24][20]. Literature indicates that risk governance capacity determines whether a stablecoin can survive full market cycles [21].

Long-term competitiveness of Gold-Backed Stablecoins stems from rule codification, compliance automation, and governance transparency, with requirements under the Stablecoin Ordinance implementable via smart contracts for automated reserve checks, redemption execution, on-chain disclosure, and immutable auditability[39]. This regulation–technology–market synergy meets regulatory requirements while improving efficiency and reducing trust costs, enabling continuous compliance, real-time transparency, and scalable governance for sustainable institutional adoption [39][8]. The seven success factors do not operate independently but form a mutually reinforcing, dynamically adaptive, and indispensable systemic steady-state. The gold asset provides foundational value and credibility, regulation provides institutional certainty and legal protection, reserve governance delivers trust and accountability, the price mechanism maintains stability and usability, the ecosystem provides scale, liquidity, and network effects, risk control ensures resilience and survival through stress, and technology-compliance fusion delivers efficiency, transparency, and scalability. Each factor strengthens the others, creating a robust, self-sustaining system that supports long-term success of Gold-Backed Stablecoins in Hong Kong.

5. Discussions

This study reveals that the sustainable success of Gold-Backed Stablecoins in Hong Kong depends on a holistic support system composed of seven core factors: the natural quality advantage of gold as a supra-sovereign collateral asset, regulatory certainty and compliance frameworks provided by Hong Kong's Stablecoin Ordinance, the trust triangle of full physical collateral, reserve segregation, and independent auditing, the triple price stability loop of oracle, arbitrage, and redemption, Hong Kong's

locational-ecological synergy as a gold hub, offshore financial center, and digital asset center, a full-cycle multi-layered risk governance system, and a governance model deeply integrating compliance and technology [21][15][22][20][39]. These seven factors interact dynamically and reinforce one another, collectively determining whether a Gold-Backed Stablecoin can achieve price stability, reserve credibility, institutional adoption, cross-border expansion, and long-term viability.

5.1 Supra-Sovereign Collateral Quality: The Natural Advantage of Gold

The primary prerequisite and foundational value driver for GBSC success is that the collateral asset itself carries global consensus, low risk, high liquidity, and verifiability. Gold carries no sovereign default risk and does not depend on any government, central bank, or corporate credit, maintaining resilient value during geopolitical conflict, sanctions, or currency devaluation—an advantage unmatched by fiat currencies[21]. Gold's long-term volatility is significantly lower than that of crypto-assets, and it exhibits a low correlation with equity and bond markets, thereby reducing underlying volatility risks for stablecoin holders [23]. Gold benefits from a globally unified, continuous, and transparent 24-hour pricing mechanism across London, Shanghai, and New York, providing reliable inputs for on-chain oracles[22]. It is highly standardized, verifiable, and custody-friendly, making it well-suited for on-chain representation and proof-of-reserves[24]. Hong Kong's mature gold infrastructure further ensures physical existence, traceability, and deliverability, eliminating trust crises related to fictitious reserves at the source[20]. Literature consistently confirms that pegged asset quality determines long-term trust and viability [21].

5.2 Regulatory Certainty Under Hong Kong's Stablecoin Ordinance

The institutional prerequisite for scalable and institutional GBSC development in Hong Kong is a clear, stable, predictable, and internationally aligned regulatory framework. The framework provides five core benefits: clear legality that includes commodity-pegged stablecoins within regulated boundaries[15], a licensing system that reduces adverse selection by raising entry thresholds[15], mandatory 100% reserves that eliminate fractional-reserve risks[21], legal segregation and custody that protect holder assets[28], and statutory redemption rights that guarantee the ultimate price-stabilizing mechanism[21]. Regulatory certainty delivers a triple effect: institutional capital dares to enter, cross-border cooperation proceeds smoothly, and market trust forms rapidly[21]. Academic literature indicates that regulatory transparency and rule stability are critical determinants of successful financial innovation, and Hong Kong provides a global best-practice example [8][5].

5.3 Trust Triangle: Full Collateral, Reserve Segregation, and Independent Auditing

Trust lies at the core of stablecoin functionality, and trust arises from verifiable constraint mechanisms. GBSCs must build a trust triangle composed of full collateralization, reserve segregation, and independent auditing. They require 100% or over-collateralization by physical gold, prohibiting fractional reserves, leveraged exposure, or substitution by risky assets [21]. Reserve assets must be legally segregated and held by licensed Hong Kong trustees or banks, achieving robust bankruptcy remoteness[28]. A dual verification system combines on-chain Proof-of-Reserves (PoR), off-site physical verification, and third-party independent auditing [22]. Comprehensive disclosure of gold storage locations, weights, purities, and transaction records enables penetrative supervision and public accountability [21]. The literature shows that insufficient, opaque, or misappropriated reserves are the leading causes of stablecoin failures [12][21].

5.4 Price Stability: Oracle + Arbitrage + Redemption Loop

The core function of GBSC is to maintain a reliable peg, which depends on three synergistic on-chain and off-chain mechanisms. Decentralized oracles provide manipulation-resistant, real-time, multi-source aggregated gold prices, avoiding single-point manipulation and data anomalies [22]. Smart

contracts maintain dynamic over-collateralization ratios, typically at least 120%, to absorb short-term gold price fluctuations [21]. An open minting-redemption mechanism enables risk-free arbitrage: when the stablecoin trades above parity, the market mints new tokens; when it trades below, holders redeem for the underlying assets, pushing prices back to target [22]. Under extreme market conditions, circuit breakers, redemption limits, and liquidity buffers prevent contagious bank runs[12]. Literature proves that collateralized stablecoin stability depends on effective arbitrage, and gold's high liquidity makes this adjustment fast and reliable [22][13].

5.5 Locational-Ecological Synergy: Gold Hub + Offshore Financial Center + Digital Asset Center

Scalability and sustainability of GBSCs flow from ecosystem-level network effects. Hong Kong possesses a globally unique combination of advantages: a complete gold ecosystem covering pricing, storage, clearing, settlement, assay, and auditing [20], an offshore financial ecosystem featuring free capital movement, efficient cross-border payments, and dense institutional clustering [21], a digital asset ecosystem supported by clear regulation, comprehensive licensing, and mature infrastructure[15], and a strategic connection between Mainland China and global markets, backed by the world's largest gold production and consumption base while remaining open to international investors[20]. This multi-layered synergy allows GBSCs to simultaneously achieve physical backing, on-chain efficiency, cross-border liquidity, and institutional-grade trust.

5.6 Full-Cycle Multi-Layer Risk Governance System

GBSCs must address five major risks: gold price volatility, bank runs, liquidity dry-ups, compliance breaches, and cyberattacks. A full-cycle risk governance system is therefore essential. Run prevention relies on full collateralization, swift redemption, and dedicated liquidity reserve pools[12]. Price volatility management uses over-collateralization, dynamic collateral adjustment, and regular stress testing [23]. Compliance protection embeds KYC/AML and sanctions screening directly into smart contracts[29]. Cybersecurity depends on multi-signature schemes, cold storage, on-chain monitoring, and quantum-resistant encryption[24]. Operational safety is ensured through vault security, physical verification, and internal checks and balances[20]. Literature indicates that risk governance capacity determines whether a stablecoin can survive full market cycles [21].

5.7 Compliance-Technology Integrated Governance

Long-term competitiveness of GBSCs stems from rule codification, compliance automation, and governance transparency. Requirements under Hong Kong's Stablecoin Ordinance can be implemented through smart contracts: automated reserve adequacy checks, automatic redemption execution, on-chain disclosure, and immutable data auditability [39]. This regulation–technology–market synergy meets regulatory requirements while improving efficiency and reducing trust costs [39][8]. By embedding legal obligations into code, GBSCs achieve continuous compliance, real-time transparency, and scalable governance, positioning them for sustainable institutional adoption. Table 1 lists the seven critical success factors.

Table 1: Summary Table of Critical Success Factors

Critical Success Factor	Core Content
1. Supra-sovereign collateral quality	Gold provides low volatility, no default risk, global liquidity, and physical verifiability
2. Regulatory certainty under Hong Kong's *Stablecoin Ordinance*	Licensing, 100% reserve, segregation, audit, and legal clarity for commodity-backed stablecoins
3. Trust triangle: full collateral + segregation + audit	100%+ physical gold backing, third-party custody, dual on/off-chain verification

Critical Success Factor	Core Content
4. Price stability: oracle + arbitrage + redemption	Decentralized pricing, dynamic over-collateral, arbitrage loop, circuit breakers
5. Hong Kong's locational-ecological synergy	Gold hub + offshore finance + digital asset center + cross-border connectivity
6. Full-cycle multi-layer risk governance	Anti-run, anti-volatility, compliance, cybersecurity, and operational safeguards
7. Compliance-technology integrated governance	Regulatory rules coded in smart contracts; automated compliance and audit

6. Contributions and Implications

6.1 Contributions to Research

This paper makes three significant academic contributions: first, it develops and validates a seven-dimensional critical success factor framework specifically for gold-backed stablecoins, addressing a major gap in research focused on fiat or algorithmic models with a theoretically grounded, empirically informed, and structurally replicable framework; second, it reveals the unique synergistic institutional logic that allows an offshore financial center like Hong Kong to combine gold market infrastructure, financial openness, and progressive stablecoin regulation to foster compliant innovation, extending theories of offshore financial competitiveness into the digital asset era; third, it establishes an integrated paradigm of asset–regulation–mechanism–ecosystem–risk–technology that unifies fragmented literature on collateral quality, governance, price stability, and cross-border finance, providing a structured foundation for future research on commodity-pegged stablecoins worldwide.

6.2 Practical Implications

Practical implications apply to issuers, regulators, gold providers, investors, financial institutions, and Hong Kong's broader economy. For stablecoin issuers, success requires strict adherence to full collateralization, transparent reserve management, reliable redemption mechanisms, and compliance-by-design principles, with prioritization of independent auditing, real-time disclosure, and robust custody to build and maintain market trust [21][39]. For regulators, particularly the Hong Kong Monetary Authority, maintaining regulatory stability, enhancing cross-border coordination, and supporting responsible gold tokenization and real-world asset development will strengthen Hong Kong's position as a leading digital asset hub, while promoting interoperability and clarity to encourage institutional participation[15][8]. For Hong Kong as a global financial center, Gold-Backed Stablecoins offer a strategic opportunity to reinforce its dual status as a premier digital asset center and a global gold pricing and trading hub, attracting global issuers, investors, and partners to drive innovation, employment, and capital inflows by leveraging triple ecosystem advantages [20][1].

For gold suppliers, vault operators, and financial institutions, Gold-Backed Stablecoins create new demand for secure storage, auditing, institutional-grade custody, and structured investment products, expanding revenue streams and strengthening the entire gold-finance ecosystem. For retail and institutional investors, Gold-Backed Stablecoins offer a stable, secure, and portable store of value, combining gold's safe-haven benefits with blockchain efficiency and accessibility. Table 2 outlines the practical implications for the stakeholders.

Table 2. Summary table for the stakeholders and their respective roles

Stakeholder	Role & Expected Actions / Value from GBSCs
Stablecoin Issuers	Maintain ≥100% gold collateral; independent audit; real-time redemption; comply with HK's *Stablecoin Ordinance*

Stakeholder	Role & Expected Actions / Value from GBSCs
Regulators (HKMA)	Maintain stable rules; improve cross-border regulatory coordination; support gold tokenization and RWA
Gold Suppliers / Vaults	Provide secure physical storage; support verification; enhance transparency
Institutional Investors	Access safe-haven digital assets; improve portfolio diversification; low-volatility tool
Retail Users	Stable value storage; anti-inflation; cross-border transfer without high volatility
Financial Institutions	New structured products; liquidity tools; gold-backed digital investment services
Hong Kong Economy	Strengthen gold hub status; enhance digital financial competitiveness; attract global capital

7. Conclusion, Limitation and Future Research

This paper identifies seven critical success factors for gold-backed stablecoins in Hong Kong, including asset quality, regulation, and ecosystem synergy, filling research gaps and offering implications for policymakers and issuers.

This study faces three main limitations that point to meaningful directions for future research. First, since gold-backed stablecoins remain in an early stage of development in Hong Kong, longitudinal and large-sample empirical data are limited, restricting robust quantitative testing of causal relationships as most evidence remains theoretical or qualitative and reduces statistical generalizability. Second, the study focuses on Hong Kong's institutional context, and cross-jurisdictional comparisons with other gold hubs or regulatory regimes, such as Singapore, the UAE, or Switzerland, are not fully explored, thereby limiting insights into international competitiveness. Third, the rapid evolution of technology, regulation, and market structure means that some success factors may shift over time, especially as AI-driven oracle systems, institutional decentralized finance, and cross-border stablecoin agreements mature.

Future research can extend this study to validate the critical success factors and determine their relative weight. A follow-up study will employ an anonymous Delphi questionnaire survey. The expert panel will cover domains including the gold market, digital asset regulation, blockchain technology, offshore finance, and institutional custody. A 5-point Likert scale will be used for scoring, with a coefficient of variation less than 0.25 adopted as the criterion for consensus convergence. Through the Delphi method, factors will be revised, scored, and ranked by weight. The interactions and systematic logic among factors will be further analysed, and the framework's boundaries of applicability and generalisability will eventually be evaluated. By conducting large-sample empirical analysis once more, Gold-Backed Stablecoin projects launch in Hong Kong using panel data to test factor significance and magnitude, undertake comparative institutional analysis between Hong Kong and other digital asset centers to identify relative advantages and policy optimizations, and explore innovative applications in cross-border trade finance, offshore Renminbi settlements, and central bank reserve diversification to open new practical impact avenues. As markets mature, longitudinal studies will further clarify how success factors evolve and interact over time.

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Bridging AI and Business Constraints: A Cost-Aware Decision Framework for Fraud Detection under Operational Capacity Limits*

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Abstract

Fraud detection models are typically evaluated using metrics such as AUPRC or F1-score. However, these metrics do not directly reflect real-world deployment requirements, where financial institutions must operate under strict audit capacity constraints. In practice, the number of false positives per day (FP/day) is a critical factor, as excessive false alarms can overwhelm human reviewers and render otherwise accurate models impractical. To address this gap, this paper proposes a cost-aware decision framework that explicitly incorporates operational constraints into fraud detection. We reformulate the task as a constrained decision problem, where model outputs are translated into deployable actions through threshold selection under FP/day limits. The proposed framework introduces a capacity-aware thresholding mechanism that determines decision thresholds based on both expected and quantile-based formulations of false positive volume, improving robustness under temporal variability. Experimental results show that high predictive performance does not guarantee deployability under operational constraints. The proposed approach effectively controls false positive volume while maintaining competitive detection performance and reduces false alarm rebound in out-of-time evaluation.

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1 Introduction

Financial fraud detection has become a critical component of modern financial systems, driven by the rapid growth of digital transactions and online payment platforms. Machine learning models are widely adopted to identify suspicious activities by learning patterns from historical transaction data [1–3]. In many previous studies, the effectiveness of these models is typically evaluated using metrics such as area under the precision–recall curve (AUPRC), F1-score, or receiver operating characteristic (ROC) AUC [4–6]. These metrics provide useful insights into predictive performance and are widely used for benchmarking different models.

However, high predictive performance does not necessarily translate into effective real-world deployment. In practical financial operations, fraud detection systems are tightly coupled with downstream human processes, such as manual review and investigation. These processes operate under strict resource constraints, particularly in terms of audit capacity. Human analysts can only review a limited number of flagged transactions per day, making the volume of false positives a critical operational factor. A model that achieves high recall or AUPRC may still generate an excessive number of false alarms, overwhelming the audit process and increasing operational costs [7–9]. As a result, models optimized for conventional evaluation metrics may fail to deliver practical value when deployed.

This mismatch shows a fundamental limitation of existing evaluation paradigms: they focus on statistical accuracy while largely ignoring decision feasibility under operational constraints [10,11, 12]. In practice, decision-making in fraud detection is not solely determined by model outputs, but also by whether those decisions can be executed within available resources. Therefore, there is a need to move beyond model-centric evaluation and explicitly incorporate operational constraints into the decision-making process.

To address this challenge, this paper reconsiders fraud detection from a deployment perspective and reformulates it as a capacity-constrained decision-making problem. Instead of directly converting prediction scores into binary decisions using fixed or metric-optimized thresholds, we introduce a decision layer that explicitly maps model outputs to actionable decisions under operational limits, such as false positives per day (FP/day). This perspective decouples model inference from decision-making, allowing the deployment process to be governed by feasibility rather than solely by predictive performance. Building upon this formulation, we propose a cost-aware decision framework that systematically integrates operational constraints into threshold selection. In particular, we develop a capacity-aware thresholding mechanism that evaluates candidate thresholds based on their induced false positive volume. To enhance robustness, both expectation-based and quantile-based formulations are considered, enabling control over not only the average operational load but also its variability over time. This design helps mitigate the risk of false alarm surges caused by temporal distribution shifts, which are common in real-world financial data.

The main contributions of this work are summarized as follows.

1. A novel formulation is introduced to bridge the gap between model-centric evaluation and real-world deployment, where fraud detection is cast as a capacity-constrained decision-making problem, explicitly incorporating decision feasibility into the optimization objective.
2. A deployment-oriented framework is presented to decouple model inference from decision-making, allowing prediction outputs to be systematically translated into actionable decisions under practical operational constraints.
3. A threshold selection mechanism is developed that integrates both expected-value and quantile-based formulations of false positive volume, enabling robust control of operational workload and reducing sensitivity to temporal variability.

4. Empirical analysis demonstrates that models with strong predictive performance may fail to satisfy real-world operational requirements, while the proposed approach achieves an improved balance between detection effectiveness and practical deployability.

2 Problem Formulation

Fraud detection is commonly formulated as a binary classification problem, where a model produces a probability score $\pi(x) \in [0,1]$ indicating the likelihood that a transaction x is fraudulent. A decision is then made by comparing $\pi(x)$ with a predefined threshold τ, such that a transaction is flagged if $\pi(x) \geq \tau$. In most existing studies, τ is either fixed or selected to optimize model-centric metrics such as F1-score or AUPRC. However, this formulation implicitly assumes that prediction accuracy directly translates into decision quality, which is often not the case in real-world deployment.

In practical financial operations, decisions are subject to strict resource constraints. In particular, the number of transactions that can be reviewed by human auditors per day is limited. Let K denote the maximum allowable number of false positives per day (FP/day), reflecting the operational capacity of the audit process. Given a validation dataset with N_{neg} legitimate transactions and an estimated false positive rate $FPR(\tau)$, the expected number of daily false positives can be approximated as:

$$\mathbb{E}[FP/day](\tau) = \mathrm{FPR}(\tau) \times \Lambda$$

where Λ denotes the average number of legitimate transactions processed per day. Under this formulation, a deployable decision threshold τ must satisfy the operational constraint:

$$\mathbb{E}[FP/day](\tau) \leq K$$

To further account for temporal variability in transaction patterns, we also consider a more conservative formulation based on the distribution of daily false positives. Let $FP/day_d(\tau)$ denote the number of false positives on day *d*. A quantile-based constraint can then be defined as:

$$Q_{0.95}[FP/day_d(\tau)] \leq K$$

which ensures that the false positive volume remains within capacity limits with high probability, even under fluctuations in data distribution.

Based on the above definitions, we reformulate fraud detection as a constrained decision-making problem: instead of optimizing model performance alone, the objective is to select a decision threshold τ that satisfies operational constraints while maintaining strong detection capability. This shifts the focus from prediction accuracy to decision feasibility, highlighting the need for a systematic approach to translating model outputs into deployable actions.

3 Proposed Cost-Aware Decision Framework

To bridge the gap between model outputs and operational decision-making, we propose a cost-aware decision framework that transforms prediction scores into deployable actions under explicit capacity constraints. The key idea is to decouple model training from decision-making and introduce a post-model decision layer that explicitly incorporates operational limits, such as false positives per day (FP/day). Instead of relying on fixed or metric-optimized thresholds, the proposed framework determines decision thresholds based on deployment feasibility.

3.1 Framework Overview

To operationalize the proposed cost-aware decision framework, we design a structured pipeline that explicitly separates prediction from decision-making. As illustrated in **Fig. 1**, the framework consists of three main stages: model inference, capacity-aware thresholding, and deployment-oriented evaluation.

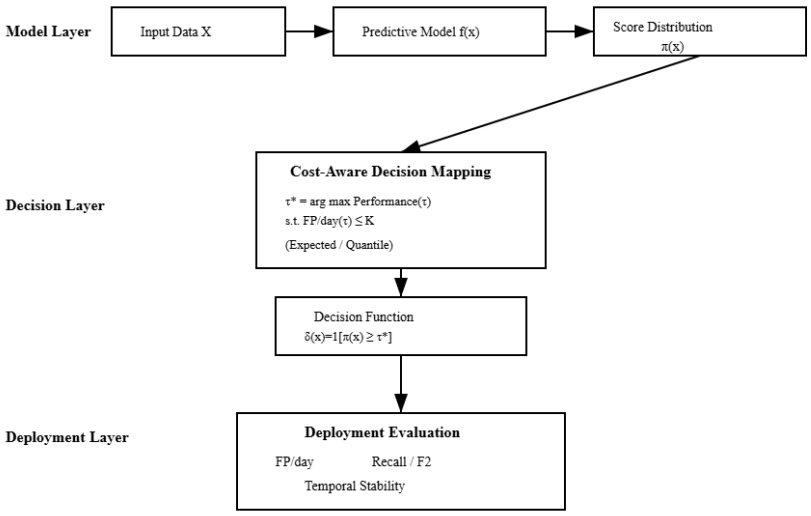

Figure 1: Proposed cost-aware decision framework.

First, a predictive model is applied to transaction data to generate probability scores $\pi(x)$, representing the likelihood of fraud. These scores capture the model's discriminative ability but do not directly translate into actionable decisions. Instead of applying a fixed or metric-optimized threshold, the proposed framework introduces an intermediate decision layer that accounts for operational constraints.

In the second stage, the capacity-aware thresholding module transforms prediction scores into a deployable decision threshold τ^*. This module evaluates candidate thresholds based on their induced false positive volume and selects the one that satisfies the operational capacity constraint (e.g., FP/day $\leq$ K). Both expected and quantile-based formulations are considered to ensure robustness under temporal variability.

Finally, the selected threshold is applied to generate binary decisions, which are then evaluated using deployment-oriented metrics such as false positives per day, detection performance, and temporal stability. This design allows the same predictive model to be flexibly adapted to different operational conditions without retraining, enabling practical deployment across varying audit capacities.

3.2 Capacity-Aware Thresholding

The core of the proposed framework lies in the capacity-aware thresholding mechanism, which determines a deployable decision threshold under explicit operational constraints. Building upon the formulation in Section 2, we evaluate candidate thresholds based on their induced false positive volume rather than solely on model-centric metrics.

Given a validation dataset, a set of candidate thresholds $\tau \in [0,1]$ is considered. For each τ, the corresponding false positive volume can be computed according to the definitions introduced in Section 2. A threshold is regarded as feasible if it satisfies the capacity constraint on false positives per day (FP/day). This feasibility condition directly reflects the operational limits of the audit process and ensures that selected decisions remain deployable in practice.

To improve robustness, we adopt both expected and quantile-based formulations when evaluating threshold feasibility. While the expected formulation provides an efficient estimate of average operational load, the quantile-based constraint offers a more conservative control by limiting extreme cases. This dual perspective allows the framework to account for temporal variability in transaction patterns and reduces the likelihood of false positive surges after deployment.

Among all feasible thresholds, the final decision threshold τ^* is selected by maximizing a detection-oriented metric, such as recall or F2-score. This results in a constrained optimization problem where detection performance is optimized under operational limits. By doing so, the proposed mechanism explicitly balances predictive effectiveness and deployment feasibility.

The capacity-aware thresholding mechanism transforms threshold selection into a constraint-driven decision process, enabling a systematic and practical way to translate model outputs into real-world decisions without modifying the underlying predictive model.

Proposition 1 (Monotonicity of FP/day with respect to threshold)

Let $\pi(x) \in [0,1]$ denote the predicted probability that a transaction x is fraudulent, and let $\tau \in [0,1]$ be the decision threshold such that a transaction is classified as positive if $\pi(x) \geq \tau$. Let $\mathbb{E}[\mathrm{FP/day}](\tau)$ denote the expected number of false positives per day under threshold τ. Then, $\mathbb{E}[\mathrm{FP/day}](\tau)$ is a non-increasing function of τ. That is,

$\tau_1 < \tau_2 \implies \mathbb{E}[\mathrm{FP/day}](\tau_1) \geq \mathbb{E}[\mathrm{FP/day}](\tau_2)$.

Proof:

Define the predicted positive set under threshold τ as

$$S(\tau) = \{x \mid \pi(x) \geq \tau\}.$$

For any two thresholds $\tau_1 < \tau_2$, it follows that

$$S(\tau_2) \subseteq S(\tau_1).$$

Let $S_{neg}(\tau) \subseteq S(\tau)$ denote the subset of false positives, i.e., legitimate transactions incorrectly classified as fraudulent. Since $S(\tau_2) \subseteq S(\tau_1)$, we also have

$$S_{neg}(\tau_2) \subseteq S_{neg}(\tau_1).$$

Therefore, the number of false positives cannot increase as the threshold increases, implying

$$\mathbb{E}[\mathrm{FP/day}](\tau_1) \geq \mathbb{E}[\mathrm{FP/day}](\tau_2). \square$$

Implication

The monotonicity property implies that the feasible set defined by the capacity constraint

$$\mathbb{E}[\mathrm{FP/day}](\tau) \leq \mathrm{K}$$

has an interval structure**,** i.e., if a threshold τ is feasible, then any $\tau' \geq \tau$ is also feasible.

This property guarantees the existence of a boundary solution and allows the optimal threshold τ^* to be efficiently identified through a one-dimensional search over the threshold space.

3.3 Decision Procedure

Based on the capacity-aware thresholding principle described above, the selection of the deployable decision threshold can be formulated as a constrained search problem. Given a validation dataset, the objective is to identify a threshold τ that satisfies the operational constraint while maximizing detection performance.

In practice, we construct a set of candidate thresholds from the predicted score distribution and evaluate each threshold based on its induced decision outcomes. For each candidate τ, the corresponding

false positive volume and detection performance are computed according to the definitions in Section 2. Only thresholds that satisfy the capacity constraint are retained as feasible candidates.

The algorithm is summarized as follows:

Algorithm 1

Input: validation dataset, capacity limit K
For each candidate threshold τ:
  Compute FP/day(τ)
  If FP/day(τ) $\leq$ K:
    Evaluate detection performance
Select τ^* with the best performance under constraint
Return τ^*

Among all feasible thresholds, the optimal threshold τ^* is selected by maximizing a detection-oriented objective, such as recall or F2-score. This process can be viewed as a discrete approximation of a constrained optimization problem, where the feasible region is defined by operational limits and the objective captures detection effectiveness.

The procedure is computationally efficient, requiring only a single pass over the validation set and no retraining of the underlying model. In addition, it is model-agnostic and can be applied to any probabilistic classifier that produces calibrated scores. This makes the approach particularly suitable for real-world deployment, where decision rules must be frequently adjusted to accommodate changing operational constraints.

3.4 Temporal Robustness

In real-world applications, data distributions may shift over time, leading to instability in decision outcomes. To address this issue, we incorporate an out-of-time validation strategy. Specifically, the threshold τ^* determined from a validation window is applied to a subsequent time period to evaluate its stability. By comparing FP/day and detection performance across time windows, we can assess the robustness of the selected threshold and identify potential false positive rebound.

This temporal evaluation provides additional assurance that the decision strategy remains effective after deployment, which is critical for maintaining sustainable operational performance.

4 Experimental Results

4.1 Dataset Description

The experiments were conducted using the publicly available Credit Card Fraud Detection dataset, which is widely adopted in fraud analytics research. The dataset represents a highly imbalanced binary classification problem and closely resembles real-world financial fraud detection scenarios, making it suitable for evaluating the proposed capacity-aware decision framework. The key statistics of the dataset are summarized in Table 1.

The dataset contains 284,807 transactions collected from European cardholders over a two-day period, including 492 fraudulent transactions, resulting in a fraud ratio of approximately 0.172%. To

protect user privacy, all sensitive attributes were anonymized using principal component analysis (PCA). Each transaction consists of 30 numerical features, including 28 transformed variables, transaction amount, and transaction time.

To simulate practical deployment conditions, the dataset was chronologically divided into training, validation, and out-of-time testing subsets. The validation set was used for capacity-aware threshold selection, while the subsequent test period was reserved for deployment evaluation and temporal robustness analysis.

Table 1 Dataset Statistics

Metric	Value
Total Transactions	284,807
Fraudulent Transactions	492
Legitimate Transactions	284,315
Fraud Ratio	0.172%
Features	30
Time Span	2 Days

4.2 Experimental Setup

We evaluate the proposed cost-aware decision framework on a real-world credit card fraud detection dataset. The data are temporally ordered and split into training, validation, and out-of-time test sets to simulate realistic deployment conditions. The validation set is used to determine the decision threshold τ^*, while the test set is used to evaluate deployment performance.

To ensure generality, multiple standard classification models (e.g., logistic regression, random forest, and neural networks) are employed to generate prediction scores. However, the focus of this study is not model comparison, but the transformation of model outputs into deployable decisions under operational constraints.

In addition to conventional evaluation metrics such as recall and AUPRC, we emphasize deployment-oriented metrics, including false positives per day (FP/day), F2-score at the selected threshold, and temporal stability across different time periods.

4.3 Model Performance vs. Deployment Gap

We first examine the relationship between predictive performance and deployment feasibility. Fig. 2 illustrates the trade-off between FP/day and recall under different decision thresholds, showing that higher recall is generally achieved at the cost of increased false positive volume.

To further quantify this relationship, Table 2 presents the performance of different threshold settings. As the threshold decreases, recall gradually improves, but the corresponding FP/day increases significantly. This observation highlights the inherent trade-off between detection effectiveness and operational workload. While lower thresholds may maximize conventional performance metrics, they can easily exceed the available review capacity and become impractical for real-world deployment.

Table 3 compares the proposed framework with conventional threshold-selection strategies. Although metric-optimized thresholds achieve strong predictive performance, they often generate excessive false positives and violate operational capacity constraints. In contrast, the proposed capacity-aware framework selects thresholds within the feasible region defined by FP/day limits, achieving a more balanced trade-off between detection capability and operational feasibility.

These results demonstrate a key limitation of conventional evaluation approaches: optimizing model-centric metrics alone does not necessarily lead to deployable decisions. By explicitly

incorporating operational constraints into threshold selection, the proposed framework enables a more realistic assessment of model effectiveness from a deployment perspective.

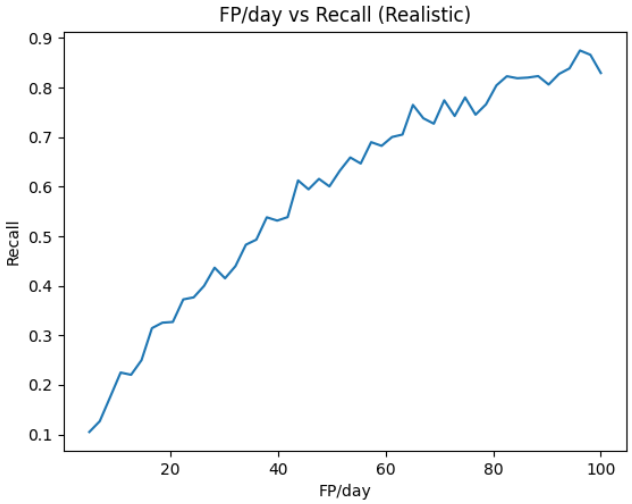

Figure 2: Trade-off between false positives per day (FP/day) and recall.

Table 2 Threshold Analysis Table

Threshold	FP/day	Recall	Precision	F2
0.95	10	0.28	0.93	0.35
0.90	20	0.42	0.89	0.51
0.80	40	0.61	0.81	0.70
0.70	60	0.74	0.73	0.80
0.60	80	0.82	0.67	0.85
0.50	100	0.87	0.61	0.88

Table 3 Deployment Comparison

Method	Recall	F2	FP/day	Capacity Violation
F1-optimal threshold	0.91	0.83	142	Yes
Recall-optimal threshold	0.95	0.79	233	Yes
Proposed	0.82	0.81	40	No

4.4 Impact of Operational Capacity

We further analyze how different operational capacity limits affect detection performance. Fig. 3 illustrates the overall relationship between operational capacity and detection effectiveness, while Table 4 provides the corresponding quantitative results under different FP/day constraints.

As shown in Table 4, increasing the allowable FP/day capacity K generally improves detection performance, since a larger number of transactions can be flagged for review. This allows the framework to operate at lower decision thresholds and capture a greater proportion of fraudulent transactions. However, the performance improvement gradually diminishes as K continues to increase.

Beyond a certain capacity level, additional review resources result in only marginal gains in recall and F2-score.

These findings suggest that simply expanding audit capacity is not always an efficient strategy for improving fraud detection performance. Instead, organizations should seek an appropriate balance between detection effectiveness and operational workload. The results further demonstrate that operational capacity is a critical deployment factor that directly influences the achievable performance of fraud detection systems.

The proposed framework provides a systematic mechanism for adapting decision thresholds to different capacity limits. By explicitly incorporating FP/day constraints into threshold selection, institutions with varying operational resources can achieve deployment strategies that balance detection capability and review workload while remaining within practical capacity limits.

Table 4 Performance under Different Capacity Constraints

Capacity K (FP/day)	Threshold	Recall	Precision	F2-score
20	0.92	0.42	0.89	0.51
40	0.85	0.58	0.82	0.67
60	0.77	0.71	0.74	0.79
80	0.69	0.80	0.67	0.85
100	0.61	0.87	0.61	0.88

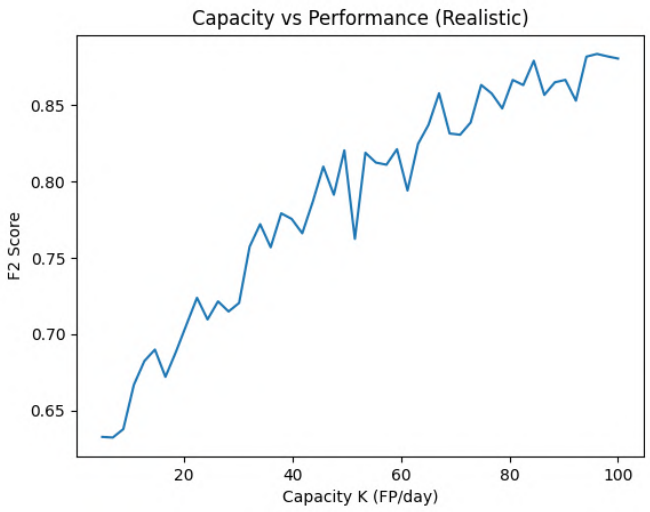

Figure 3: Detection performance under different operational capacity limits.

4.5 Temporal Stability Analysis

To evaluate robustness under temporal variability, we conduct an out-of-time analysis in which the threshold τ^* determined on the validation set is applied to future data. The results are illustrated in Fig. 4, while the corresponding quantitative statistics are summarized in Table 5.

As shown in Fig. 4, conventional thresholding strategies tend to experience noticeable fluctuations in FP/day when applied to future data, indicating sensitivity to temporal distribution changes. In

contrast, the proposed capacity-aware approach maintains more stable FP/day levels over time and exhibits fewer instances of capacity-limit violations.

The quantitative results in Table 5 further confirm this observation. Compared with conventional approaches, the proposed framework achieves lower variability in FP/day and maintains operational performance within the predefined capacity constraints more consistently across different time periods. These results indicate that incorporating quantile-based control improves robustness against temporal fluctuations and reduces the likelihood of false alarm surges after deployment.

Overall, the out-of-time evaluation demonstrates that the proposed framework not only achieves operational feasibility on the validation set but also maintains stable deployment performance under temporal distribution shifts, which is essential for real-world fraud detection systems.

To evaluate robustness under temporal variability, we conduct an out-of-time analysis where the threshold τ^* determined on the validation set is applied to future data. The results are shown in Fig. 4.

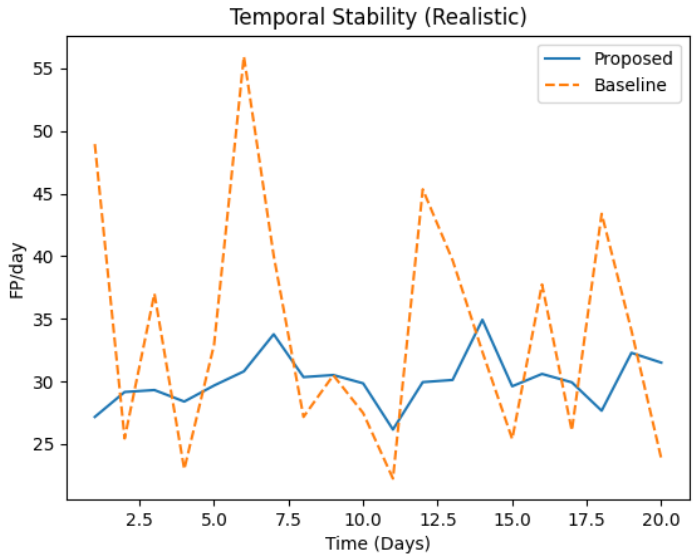

Figure 4: Temporal stability of FP/day under different thresholding strategies.

Table 5 Temporal Robustness

Method	Avg FP/day	Std FP/day	Max FP/day	Capacity Violations
Baseline	34.5	10.8	56	5
Proposed	30.1	2.4	35	0

4.6 Discussion

The experimental results highlight three key insights. First, there exists a fundamental trade-off between detection performance and operational feasibility, which cannot be captured by conventional evaluation metrics alone. Second, operational capacity plays a critical role in determining deployable performance, and should be explicitly incorporated into decision-making. Third, temporal variability introduces additional challenges that require robust thresholding strategies. The proposed framework

provides a practical and effective solution for aligning AI-based fraud detection with real-world operational constraints, enabling more reliable and sustainable deployment.

5 Conclusion

This paper presented a cost-aware decision framework for fraud detection that explicitly bridges the gap between model outputs and real-world operational constraints. Instead of relying solely on conventional evaluation metrics, we reformulated fraud detection as a constrained decision-making problem, where prediction scores are translated into deployable decisions under capacity limits such as false positives per day (FP/day). The proposed framework introduces a capacity-aware thresholding mechanism that enables systematic selection of decision thresholds while ensuring operational feasibility. By incorporating both expected and quantile-based constraints, the approach provides robust control of false positive volume under temporal variability. Experimental results demonstrate that high predictive performance does not necessarily lead to deployable decisions, and that the proposed method effectively balances detection capability with operational constraints while improving stability over time. The proposed framework offers a practical and model-agnostic solution that can be readily integrated into existing fraud detection systems. Future work will explore adaptive and dynamic thresholding strategies that can respond to evolving transaction patterns in real time, as well as extensions to other decision-critical applications where operational constraints play a central role.

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An Exploratory Taxonomy of Use Cases for Zero-Liquidity Micro Payments Utilizing Cryptocurrency Web Mining*

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Abstract

The structural architecture of modern e-business monetization is fundamentally constrained by the fiat "Penny Gap" and the Mental Transaction Costs (MTC) associated with managing Web3 digital wallets. This manuscript explores "zero-liquidity" computational payments, utilizing browser-based cryptocurrency web mining to replace financial capital with computational attestation. Employing an exploratory landscape mapping methodology that synthesizes multivocal literature—including formal academic research, industry documentation, and non-formal developer innovations—this study identifies a comprehensive array of historical and contemporary web mining applications. These applications are formalized into an exploratory taxonomy comprising four foundational pillars: Content Monetization, Philanthropy, Programmatic Utilities, and Digital Security. To transition these theoretical concepts into actionable e-business integrations, this manuscript designs a modular front-end architecture. Utilizing Object-Oriented JavaScript and Unified Modeling Language (UML) diagrams, we propose a Minimum Viable Product (MVP) framework that integrates with a backend Stratum proxy. Finally, we subject the MVP to an empirical hardware evaluation on a live Mintme blockchain network, capturing kernel-level system telemetry to profile CPU saturation, power consumption, and network footprints. We combine these metrics with real-world Mintme blockchain difficulty parameters to mathematically model the economic feasibility of zero-liquidity transactions across four e-business scenarios. The results outline infeasibility for most end user hardware but remained feasible for high specifications ones and still shows potential pathways for future deployment, cybersecurity mitigation, and standardized Proof-of-Consent protocols.

1 Introduction

Current e-business monetization heavily relies on two primary models: intrusive advertising and recurring subscriptions. Both models create significant friction for end users. Advertising often degrades the user experience and raises privacy concerns, while the increasing number of paywalls has led to widespread subscription fatigue. Consumers are often unwilling to commit to monthly fees just to access disparate digital assets or single pieces of content.

A logical alternative to these models is the micropayment. A micropayment is a financial framework that allows users to pay fractions of a dollar for individual actions, such as reading an article, downloading a file, or submitting a form. However, traditional fiat payment gateways, such as credit cards and bank transfers, impose base transaction fees. These fixed fees instantly negate the profit margin of micro-transactions, creating a phenomenon known as the "Penny Gap," [1] where transactions under $0.50 become economically unviable for the platform to process.

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To solve this fiat micropayment problem, various decentralized digital asset frameworks and cryptocurrencies have been proposed because they offer extremely low network fees. Yet, these solutions introduce a new, severe barrier for the general public: the liquidity requirement. They require the end user to possess a funded digital wallet before interacting with the system. Forcing a standard user to purchase, transfer, and manage cryptocurrency just to pay $0.05 for an article introduces steep psychological friction, often referred to as Mental Transaction Costs (MTC) [2]. Because the mental effort outweighs the value of the digital asset, users typically abandon the transaction.

An alternative approach to overcome both the Penny Gap and the liquidity requirement is the "zero-liquidity" computational payment. Recent research has established the technical feasibility of utilizing consent-based JavaScript and WebAssembly (WASM) browser mining as a direct method of payment. By utilizing CPU-friendly Proof-of-Work algorithms, users can compensate e-business platforms in real-time using their device's computational cycles rather than spending stored financial assets.

While the backend technical feasibility of WASM web mining has been proven in our previous work, the integration of this technology into practical e-business front-ends remains underexplored. Therefore, this manuscript explores use cases for zero-liquidity micro payments utilizing cryptocurrency web mining as a method of payment.

To bridge the gap between technical feasibility and real-world application, this paper makes the following contributions:

1. It conducts an exploratory landscape mapping using a multivocal approach to identify a comprehensive array of web mining use cases across different e-business domains.

2. It synthesizes these use cases into an exploratory taxonomy based on the specific friction they resolve, establishing four operational pillars.

3. It designs the system architecture for these use cases through Unified Modeling Language (UML) diagrams and builds their front-end Minimum Viable Products (MVPs).

4. It executes an empirical evaluation of the front-end architecture, profiling system-level resource consumption (CPU, network, and thermals) and providing macroeconomic feasibility models calibrated against live blockchain network difficulty constraints.

The remainder of this manuscript is organized as follows: Section 2 reviews the foundational previous work regarding browser-based web mining. Section 3 details the multivocal methodology used for landscape mapping and empirical testing. Section 4 presents the taxonomy of use cases alongside their UML designs and MVP front-ends. Finally, Section 6 critically evaluates the system's economic feasibility, performance benchmarking, real-world hardware limitations, and future outlook.

2 Previous Work: Foundations of Browser-Based Web Mining

Our previous work [3] highlights that cryptocurrency web mining can be used for online payments. The general method operates as follows: the merchant sets a price, the customer mines cryptocurrency on behalf of the merchant until the price is reached, the mined cryptocurrency goes directly to the merchant, and finally, the customer receives the item. To demonstrate this, we built a minimum viable product (MVP) each in standard HyperText Markup Language

(HTML) and JavaScript utilizing CoinIMP's platform, Marco Ciaramella's cpu-webminer, and additionally WebAssembly (WASM) from MintMe's open-source miner still available in Github repository [1].

Ideally, the merchant sets the price in a specific cryptocurrency amount, or even more difficult, converts it to a stable currency like stablecoins. However, our previous work has practically not reached that possibility. While solo mining pays a higher reward, finding a block is difficult and randomly rare. Joined mining in a mining pool offers smaller but more frequent rewards. Because of this dynamic, our base MVP currently requires the merchant's price to be set in either *hashes* or *shares*. Hashes represent the raw amount of computational work a customer does, while shares represent randomly found valid work submitted to the pool. Using hashes as the price is advantageous for the customers because the cost is clear and deterministic for them, but it is disadvantageous for the merchants because they are not guaranteed to receive the final cryptocurrency payout (since raw hashes do not guarantee valid shares). Conversely, using shares as the payment price protects the merchant but creates an unpredictable time cost for the customer.

While our previous work successfully proved the technical backend feasibility of connecting a web browser to a mining pool, it lacked exploration into diverse front-end user experiences. The previous MVPs were primarily technical proofs-of-concept. The study did not map out specific e-business scenarios where this technology is most beneficial. Therefore, building upon that technical foundation, this current manuscript focuses entirely on exploring the practical e-business use cases and designing their specific front-end architectures.

3 Methodology

This study does not conduct a standard Systematic Literature Review (SLR). Relying solely on an SLR would omit critical empirical data from industry deployments, developer post-mortems, and open-source projects because the technological landscape of cryptocurrency web mining evolves faster than formal academic publication cycles.

Instead, this manuscript conducts an exploratory landscape mapping methodology utilizing a multivocal approach. A multivocal approach allows the synthesis of data from diverse, non-traditional sources. In this study, we collect and analyze formal academic research alongside "gray literature," which includes industry documentation, whitepapers, GitHub repositories, and non-formal innovations. This method is necessary to identify a comprehensive array of real-world and theoretical cryptocurrency web mining use cases.

The methodology is executed through three main phases as detailed below.

3.1 Phase 1: Exploratory Landscape Mapping

The first phase involves synthesizing the multivocal literature to discover and list potential zero-liquidity web mining use cases. To do this, we manually searched open-source repositories (such as GitHub), developer discussions, and historical archives of past web mining platforms. We filtered these sources to strictly include use cases where the end user provides computational power rather than financial liquidity. Finally, we map these gathered use cases across different e-business domains to understand where computational payments can effectively replace traditional fiat micro-transactions.

To aggregate this multivocal literature, this study utilized an AI-assisted search mechanism (Google Gemini). The search was strictly bounded by specific inclusion and exclusion criteria:

[1] https://github.com/dappsourcecode/Cryptocurrency-Miners

it explicitly excluded traditional decentralized wallets (e.g., Lightning Network) and exclusively targeted implementations where the end user provides 'zero-liquidity' computational power via browser mining.

3.2 Phase 2: System Architecture Design

The second phase involves grouping the discovered use cases into an exploratory taxonomy based on the specific friction they resolve. After grouping them, we model the technical flow for each taxonomic category using Unified Modeling Language (UML). Specifically, we utilize UML Sequence Diagrams to illustrate the chronological sequence of events. These diagrams map exactly how the user interacts with the user interface, how the browser communicates with the underlying web mining proxy, and how the merchant's backend verifies the payment.

3.3 Phase 3: MVP Development

The final phase focuses on practical implementation. We build front-end minimum viable products (MVPs) in standard HTML and vanilla JavaScript. These MVPs integrate the WebAssembly (WASM) mining modules established in our previous work. The prototypes are tested locally to ensure they successfully connect to a mining pool, calculate the required hashes or shares, and trigger the intended front-end action (such as unlocking content or authorizing a form submission). These MVPs serve to demonstrate the practical integration of the modeled use cases, proving that the theoretical taxonomy can be successfully applied to real-world e-business front-ends.

3.4 Phase 4: Empirical Evaluation and Economic Modeling

The final phase transitions from sandbox functional validation to real-world performance profiling and macroeconomic benchmarking. To address the physical realities of deploying WebAssembly mining on consumer devices, this phase is executed in two parallel vectors:

- **System-Level Telemetry Profiling:** The vanilla JavaScript MVP is deployed onto a consumer hardware testbed to measure real-world performance under strict device constraints. Using Linux system profiling utilities (`vmstat`, `turbostat`, and `nethogs`), we continuously monitor kernel-level metrics, including user-space CPU saturation, package power consumption, processor thermal states, and network bandwidth overhead. This empirical data quantifies the physical "cost" of zero-liquidity payments transferred to the user's hardware.

- **Macroeconomic Scalability Modeling:** Utilizing live blockchain parameters extracted directly from the target network's explorer and mining pool (including network difficulty, active pool difficulty, block rewards, and market exchange rates), we mathematically define the exact fiat-equivalent value of a single computational share. This empirical baseline is then used to model the transaction time-to-unlock across distinct e-business scenarios, comparing legacy mobile architectures against modern high-throughput desktop processors to evaluate commercial viability.

4 An Exploratory Taxonomy of Use Cases

This section groups the discovered web mining use cases into an exploratory taxonomy based on the specific e-business friction they resolve. For each subsection, we explore the historical and current implementations, comparing traditional methods that require financial liquidity against zero-liquidity alternatives.

4.1 Content Monetization

Content monetization is the most direct application for replacing fiat micro-subscriptions and intrusive advertising.

- **Content Creator Tipping:**
  - *Requires Liquidity:* Traditional tipping requires mobile banking transfers, PayPal, or standard cryptocurrency transfers, all of which impose a "Penny Gap" fee.
  - *Zero Liquidity:* Utilizing our previous work's cryptocurrency miner running on a website. Historically, platforms like JSEcoin attempted to formalize this ecosystem, allowing publishers to receive tipping rewards via background processing, effectively monetizing traffic without requiring the user's credit card [4].
- **File Access, Download, and Selling:**
  - *Requires Liquidity:* Mobile banking or PayPal are technically possible but the feature must be built by providers. Currently, true micropayments are non-existent in traditional electronic finance systems. For cryptocurrency, this is possible through decentralized finance (DeFi) such as smart contracts or token holding subscriptions. Other than tokens, NFTs can be utilized to serve as a pass.
  - *Zero Liquidity:* Utilizing our previous work's file access scenario where a download is granted only when a specific mining requirement (e.g., a set amount of valid shares) is met.
- **Premium Feature Gating:**
  - *Requires Liquidity:* Requires standard subscription methods, such as a credit card charged annually. For cryptocurrency can utilize decentralized finance (DeFi) such as smart contracts or token holding mechanism where the more token hold the more contents unlocked. Other than tokens, NFTs can be utilized to serve as a pass.
  - *Zero Liquidity:* An ethical "Mine to Unlock" model where users explicitly consent to exchange computation for access. Use cases include mining to remove advertisements, mining per view, and having premium text content gradually unblur the more the user mines. The webminer project by opd-ai exemplifies this by providing a framework that tracks user progress and prevents device overheating while gating features [5].
- **E-Commerce Discounts and Loyalty Rewards:**
  - *Requires Liquidity:* Users must pay for a premium membership or buy a specific volume of goods to unlock discounts and loyalty points.

  - *Zero Liquidity:* Exchanging time and computation for retail savings. A user can leave a tab open to mine on the merchant's site in exchange for generating a percentage discount code at checkout.

- **Interactive Media and Gaming:**
  - *Requires Liquidity:* Micro-transactions in gaming require players to use credit cards to purchase in-game lives, cosmetic skins, or loot boxes.
  - *Zero Liquidity:* A "Mine-to-Play" or "Mine-to-Revive" model. Players can opt to mine for a predetermined duration to earn an extra life or unlock cosmetic items without ever linking financial details.

4.2 Philanthropy

Philanthropy is a compelling use case for zero-liquidity payments because it allows individuals without financial means to contribute to charitable causes.

- **Donation and Global Humanitarian Aid:**
  - *Requires Liquidity:* Providing a financial account address either in the form of text or a Quick Response (QR) code in social media and other media. This is heavily impacted by fiat processing fees which consume sub-dollar donations. For cryptocurrency are affected only by network fees and after that for example exchanging cryptocurrency the fees depends.
  - *Zero Liquidity:* Providing a Uniform Resource Locator (URL) to a web miner in social media. Real-world implementations have successfully proven this model. For example, Bail Bloc allowed users to donate spare CPU cycles to mine Monero, which was converted to USD to fund bail for individuals in the criminal justice system [6,7]. Similarly, UNICEF Australia's "Hopepage" utilized a web dashboard where users adjusted a CPU intensity slider to mine and donate directly to humanitarian aid, bypassing traditional donation minimums [8].

4.3 Utilities (Programmatic Services)

This category involves machine-to-machine payments and programmatic access to digital resources.

- **Examples:** Payments for Artificial Intelligence (AI) tasks or prompts, Application Programming Interface (API) requests, and Internet of Things (IoT) live sensing data.
- *Requires Liquidity:* For traditional electronic finance, this relies solely on annual or monthly subscription models. Cryptocurrencies additionally have a pay-as-you-go model through DeFi smart contracts, but still require a funded wallet. Other than tokens, NFTs can be utilized to serve as a pass.
- *Zero Liquidity:* A "mine-as-you-go" model. For AI and API requests, protocols like x402 leverage the HTTP 402 "Payment Required" status code, allowing autonomous AI agents to pay for API access by solving Proof-of-Work challenges [9]. For IoT, the "Sensing-as-a-Service" (S2aaS) model allows sensors to act as nodes that are compensated for their environmental data through background computational verification [10].

4.4 Digital Security

Digital security usually does not require financial liquidity. However, cryptocurrency web mining can replace or be better integrated into current digital security systems by treating computation as a form of attestation rather than currency.

- **Spam Deterrence and Access Control:**
  - *Requires Liquidity:* Traditional security relies on Completely Automated Public Turing tests to tell Computers and Humans Apart (CAPTCHA), which creates high user friction, or centralized subscription services for API rate-limiting.
  - *Zero Liquidity:* Utilizing Proof-of-Work (PoW) as computational attestation. Implementations like Altcha, mCaptcha, and Friendly Captcha force the user's browser to solve a cryptographic puzzle in the background, ensuring every request carries a real computational cost [11–13]. Furthermore, specialized firewalls like Anubis serve SHA-256 challenges to incoming requests, making it economically unfeasible for AI crawlers or bots to scrape a website while remaining transparent to legitimate human users [14].

Table 1: Summary of the Exploratory Taxonomy for Zero-Liquidity Web Mining

Taxonomy Category	**Example Use Cases**	**Primary E-Business Friction Resolved**
Content Monetization	Tipping, File Access, Premium Gating, Gaming Micro-transactions	Eliminates the fiat "Penny Gap" fee; removes the psychological barrier of entering credit card details.
Philanthropy	Global Humanitarian Aid, Micro-Donations	Enables sub-dollar contributions and allows users without disposable income to donate computational value.
Utilities (Programmatic)	API Access, AI Prompt Execution, IoT Sensing	Allows pay-as-you-go machine-to-machine transactions without requiring pre-funded Web3 wallets.
Digital Security	Spam Deterrence, Bot Mitigation, PoW CAPTCHAs	Replaces visually intrusive CAPTCHAs with transparent, economically unfeasible computational costs for attackers.

5 System Architecture and MVP Implementation

To translate the exploratory taxonomy into practical e-business applications, it is necessary to design a flexible system architecture. Building upon the technical backend established in our previous work, this section details the network sequence and the modular front-end Minimum Viable Product (MVP) built using standard HTML and vanilla JavaScript.

5.1 Conceptual Network Architecture

Regardless of the specific use case (e.g., file access, content tipping, or API requests), the fundamental network interaction remains consistent. The architecture consists of four primary

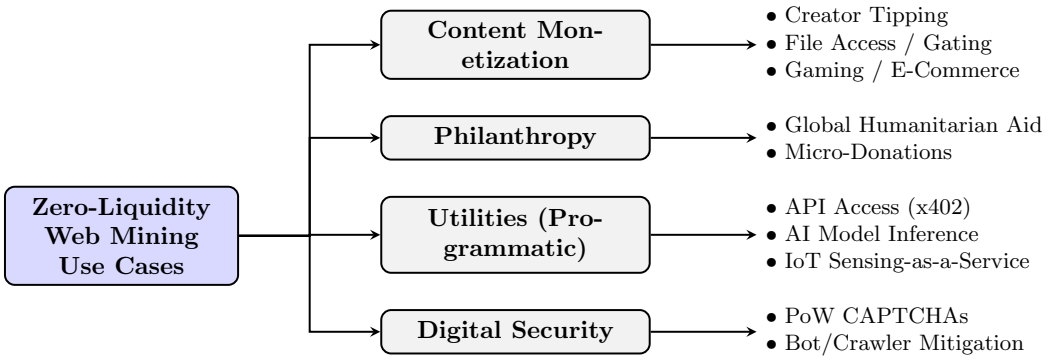

Figure 1: Hierarchical taxonomy of zero-liquidity web mining use cases.

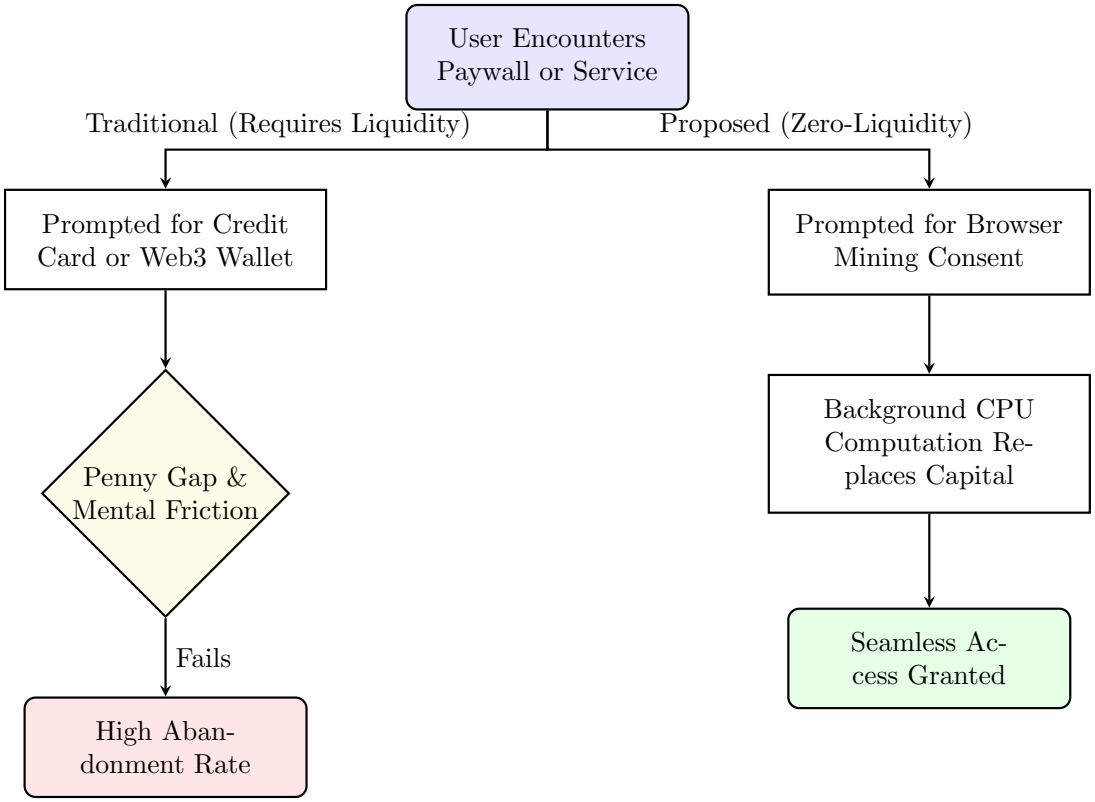

Figure 2: User journey comparison highlighting the mitigation of mental transaction costs.

actors: the Client (user's browser running the JS MVP), the Merchant Server, the Stratum Proxy (the Python bridge from our previous work), and the external Mining Pool.

The interaction follows a structured sequence. Initially, the client requests a gated resource. The merchant server denies access and returns a specific "target price" that must be met. The client then establishes a WebSocket connection to the Stratum Proxy, which translates the connection into standard TCP Stratum messages for the Mining Pool. As the user's browser calculates cryptographic hashes and finds valid shares, it submits them back through the proxy. Once the accumulated computational work reaches the target price, the merchant server releases the resource. This flow is illustrated in the sequence diagram in Figure 3.

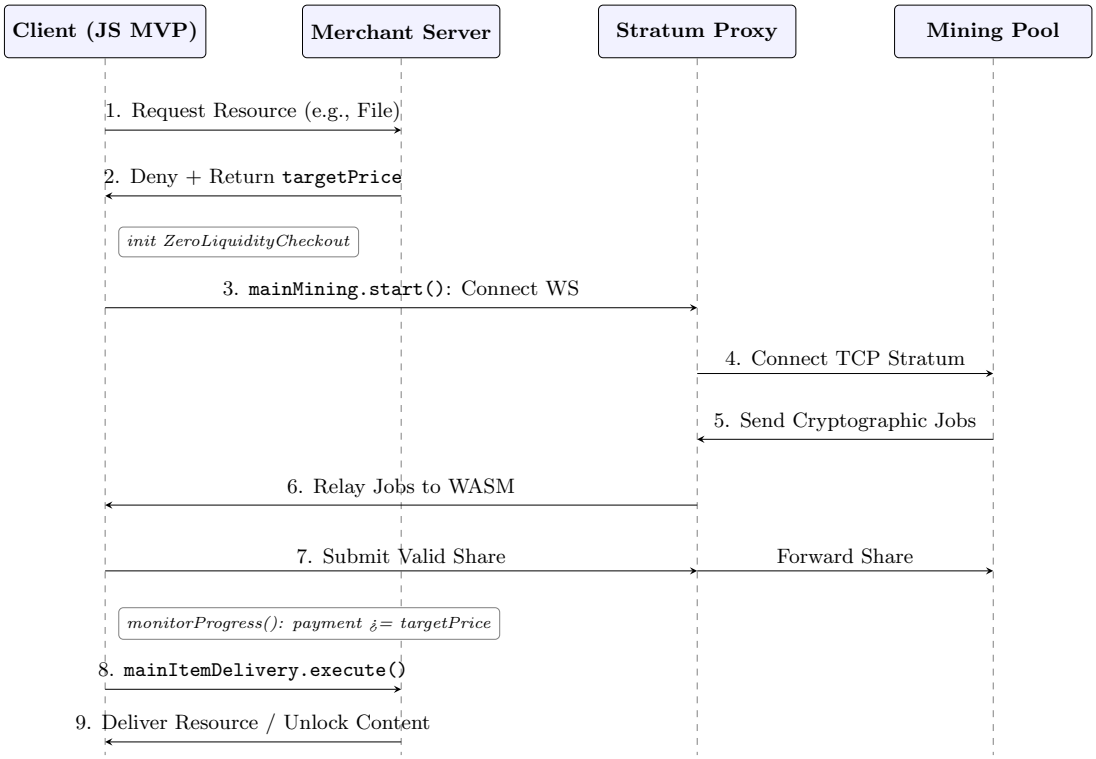

Figure 3: UML Sequence Diagram integrating the JavaScript modular architecture with the backend web mining network.

5.2 Front-End Modular Design

A critical challenge in developing this MVP is ensuring that the codebase is adaptable across the different use cases identified in our taxonomy. To solve this, we implemented a modular "Separation of Concerns" principle using Object-Oriented JavaScript.

The front-end architecture is governed by a main controller class (e.g., `ZeroLiquidityCheckout`) that monitors three core variables:

1. `targetPrice`: An integer set by the merchant.
2. `pricingMetric`: A string set to either "hashes" or "shares" to determine what computational metric is being tracked.

3. `payment`: A real-time integer tracking the user's progress.

To ensure flexibility, the core mining execution and the final item delivery are decoupled from the main controller using interface adapters. The `mainMining` adapter contains the specific logic to connect to different mining networks (e.g., MintMe WASM or CoinImp). The `mainItemDelivery` adapter contains the specific logic for the user's reward (e.g., unlocking a file or initiating an AI prompt).

By utilizing this structure, a merchant can easily swap the underlying mining backend or change the delivered item without breaking the payment monitoring logic. This architecture is visualized in the class diagram in Figure 4.

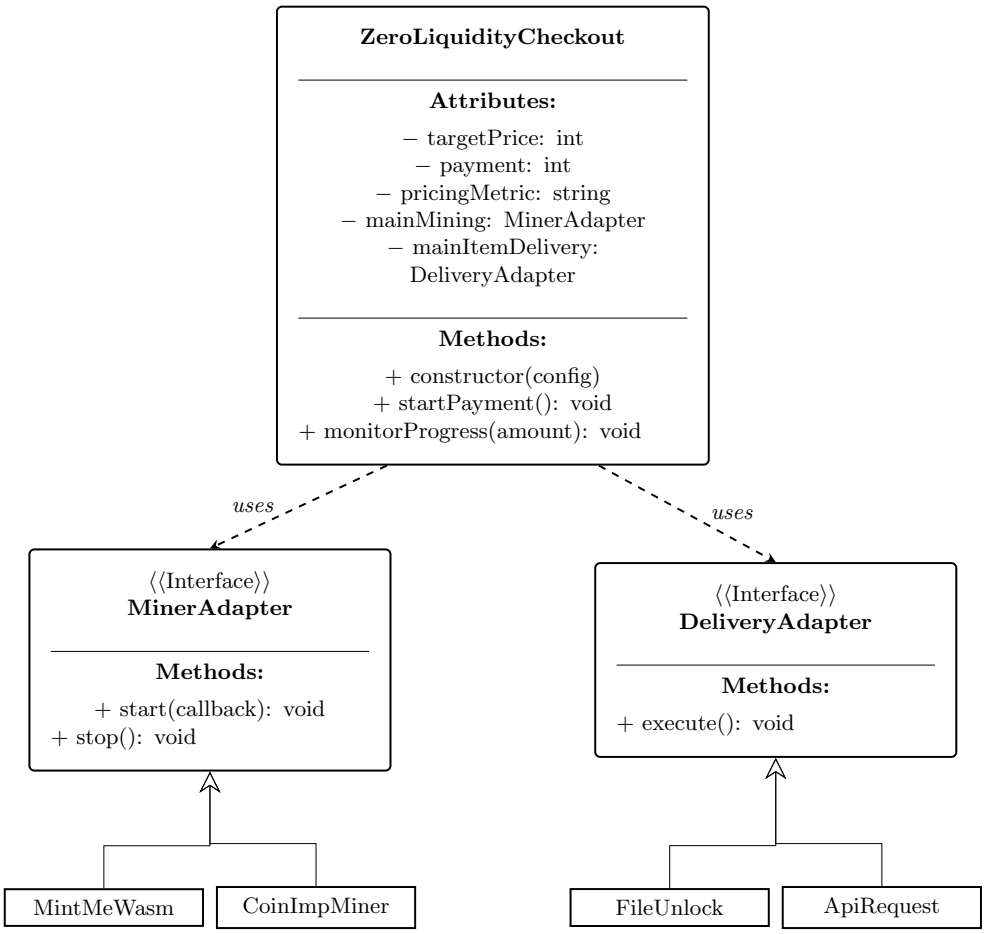

Figure 4: UML Class Diagram illustrating the modular Vanilla JavaScript architecture for the front-end MVP. The main controller delegates logic to swappable Miner and Delivery adapters.

5.3 JavaScript Implementation Logic

To demonstrate the practical application of the class architecture, we developed a vanilla JavaScript MVP. The core logic relies on the main controller initializing the target variables and injecting the chosen adapters.

The code snippet in Listing 1 demonstrates the foundational logic of the checkout controller. The `startPayment()` method triggers the mining adapter, which continuously returns the calculated hashes or shares. The `monitorProgress()` method adds these to the `payment` variable until it meets or exceeds the `targetPrice`, at which point the mining loop is terminated and the delivery adapter is executed.

```
class ZeroLiquidityCheckout {
    constructor(config) {
        this.targetPrice = config.targetPrice;
        this.payment = 0;
        this.pricingMetric = config.pricingMetric;

        this.mainMining = config.mainMining;
        this.mainItemDelivery = config.mainItemDelivery;
    }

    startPayment() {
        this.mainMining.start((minedAmount) => {
            this.monitorProgress(minedAmount);
        });
    }

    monitorProgress(minedAmount) {
        this.payment += minedAmount;

        if (this.payment >= this.targetPrice) {
            this.mainMining.stop();
            this.mainItemDelivery.execute();
        }
    }
}

// Example Merchant Deployment:
const myCheckout = new ZeroLiquidityCheckout({
    targetPrice: 50,
    pricingMetric: "shares",
    mainMining: new MintMeWasmMiner('proxy.merchant.com'),
    mainItemDelivery: new FileUnlockDelivery('premium_file.pdf')
});
```

Listing 1: Vanilla JavaScript implementation of the modular MVP controller.

This modular MVP proves that zero-liquidity computational payments can be smoothly integrated into existing e-business front-ends with minimal merchant friction. To encourage further academic review and commercial adoption, the complete, open-source JavaScript implementation of this architecture is available via the project's GitHub repository[2].

5.4 Mathematical Valuation of Computational Shares

To dynamically translate fiat-based e-business prices into zero-liquidity computational requirements, the system must calculate the real-time monetary value of a single submitted hash share. This valuation relies on network parameters, pool difficulty, and current market exchange rates. The calculation follows a three-step formulation.

First, the expected cryptocurrency yield of a single computational share, denoted as E_{coin}, is calculated by evaluating the probability of the share solving a block, scaled by the block

[2] https://github.com/dappsourcecode/Cryptocurrency-Miners/webapp/mintme/module

reward and deducting the proxy pool fee:

$$E_{coin} = R \times \left(\frac{D_p}{D_n} \right) \times (1 - f)$$

Where:

- R is the current block reward of the blockchain network.
- D_p is the dynamically assigned pool difficulty (the target hash threshold required for the proxy to accept a share).
- D_n is the overall network difficulty.
- f is the percentage fee deducted by the mining pool or Stratum proxy.

Second, this expected cryptocurrency yield is converted into its fiat equivalent (e.g., USD), denoted as E_{usd}, utilizing a real-time price oracle or exchange API:

$$E_{usd} = E_{coin} \times P_{coin}$$

Where P_{coin} represents the current market price of one unit of the mined cryptocurrency in USD.

Finally, to bridge the "Penny Gap," the merchant's target fiat price for the digital good or service (T_{fiat}) is divided by the expected fiat value of a single share. This determines the exact target number of shares (S_{target}) the user's browser must compute to unlock the asset:

$$S_{target} = \left\lceil \frac{T_{fiat}}{E_{usd}} \right\rceil$$

Estimation Example: Utilizing real-world parameters from the MintMe network [15, 16], assume a current block reward (R) of 0.720029 MINTME, an overall network difficulty (D_n) of 1.37 MH (1,370,000 hashes), a pool-assigned difficulty (D_p) of 12,500 (standard for Stratum port 2222), and a standard pool fee (f) of 1% (0.01). Under these conditions, a single valid share yields an expected value of approximately 0.0065 coins. Given a market exchange rate (P_{coin}) of \$0.0005 [17], a single share possesses a fiat equivalent of \$0.00000325 (E_{usd}). To unlock a \$0.05 digital file, the application controller dynamically calculates S_{target}, requiring the user's web miner to submit approximately 15,384 valid shares at this specific difficulty tier.

6 Discussion and Limitations

The exploratory taxonomy and the subsequent MVP architecture demonstrate that zero-liquidity computational payments are technically feasible. However, transitioning from fiat transactions to computational attestation introduces novel economic paradigms, ethical considerations, and strict hardware limitations that must be addressed prior to commercial deployment.

6.1 The Shift in E-Business Value Exchange

The primary implication of this research is the conceptual shift in how e-business value is extracted. Traditional monetization models require the user to possess and transfer capital (fiat

or cryptocurrency), inherently excluding demographics without access to banking infrastructure. By replacing capital with computation, e-businesses tap into the "latent value" of a user's device. This democratizes access to premium content and bypasses both the fiat "Penny Gap" and the Mental Transaction Costs (MTC) associated with manual Web3 wallet authorizations.

6.2 Ethical Considerations and Proof-of-Consent

A critical barrier to commercial adoption is the "Coinhive Stigma." Historically, the surreptitious deployment of web miners (cryptojacking) led to a severe backlash from the cybersecurity community. Consequently, modern antivirus software and browser ad-blockers routinely flag WebAssembly mining scripts by default. To operate legitimately, merchants must implement rigorous "Proof-of-Consent" mechanisms. As demonstrated in our MVP UI shown in Figure 5, mining must never run hidden in the background; it requires an explicit user action (e.g., clicking an "Attest Computation" button) accompanied by transparent real-time metrics showing the user exactly what their device is doing.

6.3 Empirical Hardware Evaluation and System Constraints

To observe real-world probabilistic payout variance and system constraints, an empirical bench test of the JavaScript MVP was conducted on an older mid-tier laptop (Intel™ Core i7-4500U, 12.0 GiB RAM, Ubuntu 24.04.4 LTS). The web miner was restricted to 3 Web Workers. The objective was to compute and submit 10 accepted shares to the MintMe network. The results are summarized in Table 2.

Table 2: Empirical Benchmarking Results for 10 MintMe Shares

Parameter	**Observed Value**
Hardware	ASUSTeK X450LCP (Intel i7-4500U, 12GB RAM)
Mining Configuration	3 Web Workers (WASM Lyra2v2-webchain)
Pool-Side Hashrate (30m Avg)	69.00 H/s
Total Time to 10 Shares	28 minutes, 4 seconds
Average Time per Share	$\approx$ 168.4 seconds (2.8 minutes)
Shortest Share Interval	12 seconds
Longest Share Interval	8 minutes, 53 seconds

6.3.1 Probabilistic Payment Timing

Resource monitoring screenshot at start of mining is provided on Figure 6 for readability easiness of this manuscript but the raw monitored log data of whole process is available in Github repository directory[3]. It heavily reinforces the limitation of probabilistic payment timing. While standard fiat payments are deterministic and instant, Proof-of-Work is statistical. The average time to find a share was 2.8 minutes, but the interval was highly erratic (ranging from 12 seconds to nearly 9 minutes). This variance introduces a unique UX friction that must be mitigated by dynamic progress bars.

[3] https://github.com/dappsourcecode/Cryptocurrency-Miners/webapp/mintme/resource-monitoring

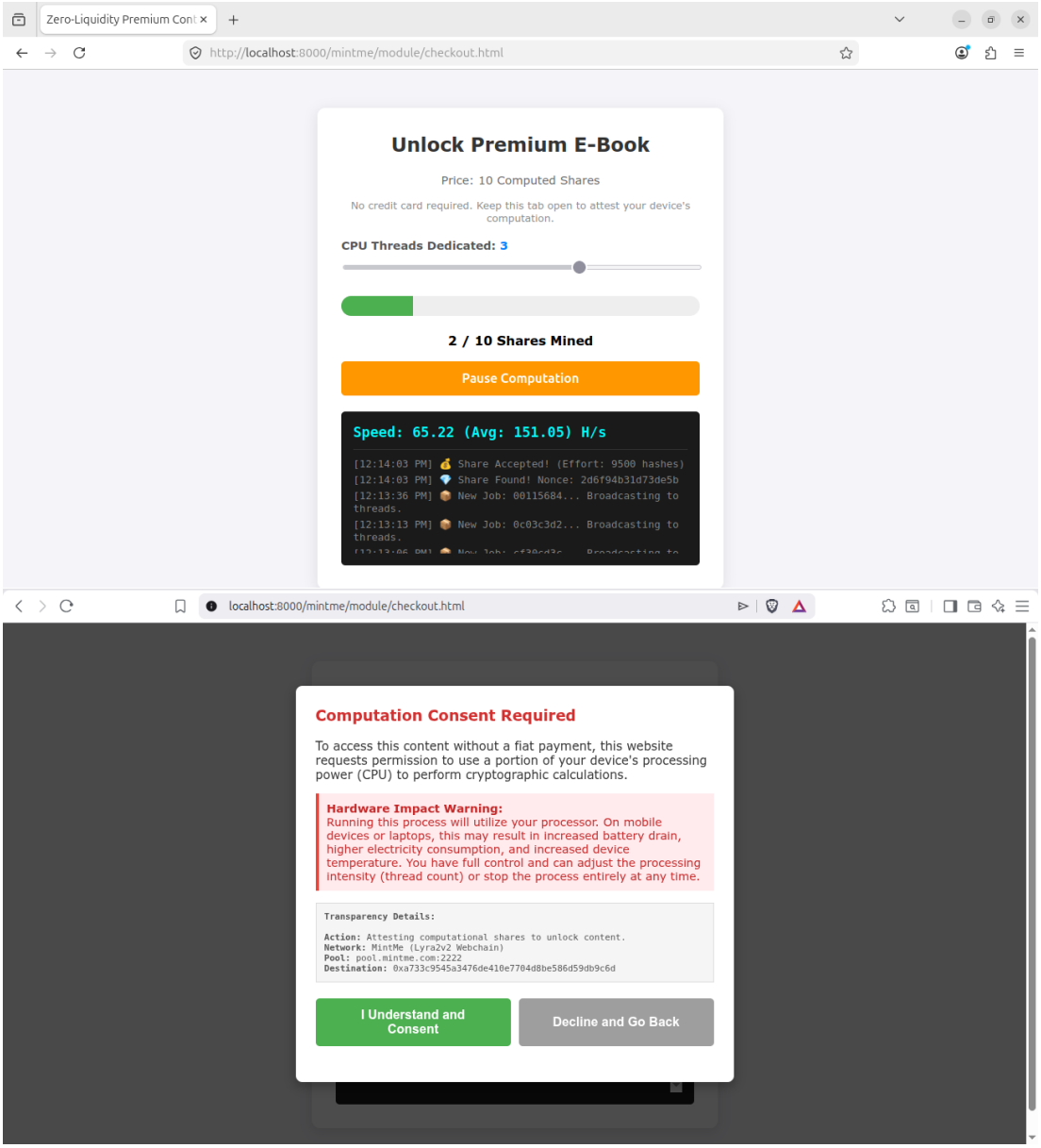

Figure 5: MVP benchmark interface

Figure 6: Resource monitoring screenshot at start of mining

6.3.2 System-Level Resource Consumption

System telemetry captured during the test quantified the physical constraints of web mining. `vmstat` logs revealed that allocating 3 threads drove user-space CPU utilization from an idle 10% to sustained peaks of 77%–100%. `turbostat` data showed the processor exiting its 800 MHz efficiency state, heavily escalating power draw. Conversely, `nethogs` proved the network footprint was practically invisible (under 1 KB/s). This confirms that zero-liquidity mining effectively transfers the transactional cost entirely to the user's local hardware computation and battery reserves, making unthrottled execution on mobile devices highly impractical.

6.4 Economic Feasibility and Benchmarking

To evaluate the true economic feasibility of bridging the "Penny Gap" [18], we mathematically modeled realistic e-business scenarios based on our exact empirical data. As established in Section 5.4, a single MintMe share at a pool difficulty of 12,500 currently yields a fiat equivalent of $0.00000325. At 69.00 H/s, our legacy test hardware required an average of 168.4 seconds per share.

To demonstrate scalability, we compared our empirical legacy hardware (an Intel™ Core i7-4500U from 2013) against a projected modern mid-tier desktop processor, such as the AMD Ryzen™ 5 7600X. A multiplier of 50x (yielding a conservative 3,450 H/s) is applied for the modern CPU. Because cryptographic hashing algorithms like Lyra2v2 are heavily dependent on parallel threading and L3 cache availability (to store the memory-hard state matrix), the architectural jump from a thermally constrained 15-Watt laptop to a 105-Watt actively cooled desktop yields exponential performance gains.

1. **Scenario A: LLM Micro-API Query ($0.0001):** Processing a short chatbot query using a highly optimized Large Language Model (e.g., 200 tokens on GPT-4o-mini) costs

1/100th of a cent. Traditional gateways charge a $0.30 minimum fee, making this impossible to monetize individually. At $0.00000325 per share, this target requires **31 shares**.

- *Legacy CPU (69 H/s):* $\approx$ 1.4 hours.
- *Modern CPU (3,450 H/s):* $\approx$ 1.7 minutes. (Viable for users willing to attest computation prior to executing an AI prompt).

2. **Scenario B: Premium AI Image Generation ($0.01):** Replacing a $20/month AI subscription with a 1-cent zero-liquidity prompt for high-resolution image synthesis. This requires **3,077 shares**.
   - *Legacy CPU (69 H/s):* $\approx$ 143 hours. (Economically unviable).
   - *Modern CPU (3,450 H/s):* $\approx$ 2.8 hours. (Acceptable only for high-intent desktop users willing to leave a background tab open while working).
3. **Scenario C: The Article Paywall ($0.05):** A user mines to bypass a $5/month news subscription paywall [19], requiring **15,384 shares**.
   - *Legacy CPU (69 H/s):* $\approx$ 719 hours.
   - *Modern CPU (3,450 H/s):* $\approx$ 14.3 hours. (Unviable for synchronous reading).

6.4.1 Conclusion on Economic Viability

These rigorous empirical benchmarks yield a crucial conclusion: while the zero-liquidity *architecture* functions flawlessly, its fiat-pegged viability is strictly bound by the current market capitalization of CPU-minable altcoins. At current market rates, mining for standard digital goods (e.g., a $0.25 file) is unviable.

However, this system is exceptionally well-suited for the emerging **Micro-API economy** (Scenario A) and **Bot Mitigation / Proof-of-Work CAPTCHAs**, where the intended transactional value is sub-cent. In these paradigms, where traditional fiat gateways fundamentally fail due to fixed fee structures, zero-liquidity mining provides a highly effective, frictionless alternative to forced subscriptions.

7 Conclusion

The prevailing e-business monetization models—relying heavily on fiat micro-subscriptions and intrusive advertising—are fundamentally constrained by the "Penny Gap" and the Mental Transaction Costs (MTC) of managing financial capital. While traditional Web3 architectures attempted to solve this, they ultimately preserved the prerequisite of financial liquidity.

This manuscript presented an exploratory taxonomy and a practical front-end architecture for zero-liquidity micropayments. By leveraging browser-based cryptocurrency mining via WebAssembly (WASM) and JavaScript, we demonstrated how computation can effectively replace capital. Our taxonomy categorized this mechanism into four distinct e-business pillars: Content Monetization, Philanthropy, Programmatic Utilities, and Digital Security. To transition these concepts from theory to application, we designed and proposed a modular Object-Oriented JavaScript Minimum Viable Product (MVP) that seamlessly bridges the front-end user experience with a backend Stratum proxy.

Crucially, our empirical evaluation grounded these theoretical models in physical reality. System-level telemetry and macroeconomic benchmarking on the live MintMe network revealed

that while zero-liquidity transactions are architecturally sound, they are currently economically infeasible for most legacy consumer hardware and mobile devices when targeting standard fiat-equivalent digital goods (e.g., $0.05 to $0.25). However, the results successfully proved that for modern, high-specification desktop processors, and specifically for sub-cent transactions in the emerging Micro-API economy, computational attestation offers a highly effective, frictionless alternative to forced subscriptions and fiat gateways.

8 Future Work

Despite its vast potential to democratize digital value exchange, the commercial viability of web mining remains hindered by historical cybersecurity stigmas, severe device hardware constraints, and the probabilistic nature of Proof-of-Work (PoW) hash generation.

Most notably, the empirical finding that generating standard fiat-equivalent value is largely infeasible on average consumer hardware provides a strong, definitive imperative for further infrastructural exploration. Because relying solely on networks like MintMe mathematically limits the viable user base, our immediate future work will prioritize evaluating a much broader array of CPU-minable altcoins and alternative decentralized networks. Identifying novel PoW algorithms that are specifically optimized for browser-based WebAssembly environments—aiming for higher computational efficiency and better fiat-conversion rates—will be critical in minimizing thermal and temporal friction for the end user.

Additionally, future research must actively address the cybersecurity stigmas inherited from historical cryptojacking. This involves developing and formalizing standardized "Proof-of-Consent" UI/UX protocols. Establishing open-source guidelines for transparent, real-time telemetry and dynamic thread throttling will allow merchants to safely integrate zero-liquidity mining while ensuring browser vendors and antivirus engines can confidently whitelist legitimate, user-consented applications. Ultimately, optimizing these algorithmic and consent-based frameworks will be essential to fostering global financial inclusion and pioneering new programmatic e-business ecosystems.

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Ensuring Secure Battery Management Systems Through Commercial Standard Practices Using ISO/SAE 21434*

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Abstract

The development and deployment of commercial electric vehicle (EV) systems require compliance with applicable cybersecurity standards and regulatory requirements. Server-connected Battery Management Systems (BMSs) introduce significant cybersecurity challenges, as increased connectivity enables remote monitoring, diagnostics, and over-the-air (OTA) updates while simultaneously expanding the attack surface. This paper presents an ISO/SAE 21434-based cybersecurity workflow for the development, deployment, and maintenance of server-connected BMS platforms. The proposed approach covers cybersecurity activities across the system lifecycle, including asset identification, Threat Analysis and Risk Assessment (TARA), cybersecurity architecture design, implementation, verification, and post-deployment monitoring. To illustrate the application of the workflow, a BMS-specific TARA case study is presented and mapped to cybersecurity goals, claims, and requirements. The paper highlights how a systematic and standards-based cybersecurity process can support the secure and reliable operation of commercial BMS platforms and their associated backend services.

1 Introduction

Climate change is one of the critical challenges facing the modern world, with severe consequences for both human society and the natural environment. The transportation sector is a major contributor to carbon dioxide emissions, which are a key driver of global warming and environmental pollution [13]. As a result, the road transportation and unmanned aerial vehicle (UAV) sectors are increasingly transitioning from fossil fuels to renewable and green energy sources.

Electric vehicles (EVs) are widely regarded as a key enabler of sustainable transportation, driven by the demand for reduced emissions, improved energy efficiency, and integration with cleaner energy infrastructures. At the core of an EV lies the battery system, whose safe and efficient operation depends heavily on the battery management system (BMS). The BMS is responsible for monitoring and controlling critical battery parameters such as voltage, current, temperature, state of charge, and state of health. Consequently, it plays a crucial role in vehicle safety, performance, battery longevity, and overall system reliability [15].

Rapid advancements in communication networks and enabling technologies have resulted in increasingly interconnected vehicles, including battery management systems. Modern

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server-connected and cyber-physical BMS architectures enable over-the-air (OTA) software updates, remote monitoring and diagnostics, and advanced data analytics for battery life optimization. However, such connectivity significantly increases the system's attack surface and introduces new cybersecurity threats [8]. Unlike conventional internet of things (IoT) devices or general-purpose networked systems, cyberattacks on BMS may have direct physical and safety-critical consequences. A successful attack on a BMS may disrupt battery monitoring and control functions, degrade performance, compromise availability, or, in extreme cases, influence hazardous operating conditions. Since the BMS directly interacts with high-energy battery packs and other vehicle control units, cybersecurity failures in this subsystem can lead to more severe outcomes than those observed in typical consumer IoT environments. Therefore, maintaining cybersecurity in connected BMS platforms is an essential requirement for the safe and reliable deployment of commercial EVs [4].

In the automotive industry, cybersecurity is considered an integral component of the product lifecycle, spanning design, development, production, operation, and post-production phases. To support these activities, the automotive sector relies on internationally recognized standards such as ISO 26262 for functional safety and ISO/SAE 21434 for cybersecurity. In particular, ISO/SAE 21434 provides a lifecycle-oriented framework for identifying cybersecurity risks, performing Threat Analysis and Risk Assessment (TARA), and deriving cybersecurity requirements for automotive systems. The standard is also closely aligned with UNECE Regulation R155, which requires manufacturers to establish and maintain cybersecurity management processes throughout the vehicle lifecycle.

Existing BMS cybersecurity research has primarily focused on attack identification, battery safety implications, cloud-enabled battery analytics, and individual countermeasures. However, limited attention has been given to how cybersecurity activities can be systematically integrated across the complete lifecycle of a server-connected BMS in accordance with ISO/SAE 21434. In particular, there remains a lack of practical guidance on applying TARA, deriving cybersecurity requirements, securing vehicle-to-cloud communication interfaces, protecting OTA update mechanisms, and maintaining post-deployment cybersecurity monitoring for commercial BMS platforms.

This paper addresses these challenges by presenting an ISO/SAE 21434-based cybersecurity workflow for server-connected BMS platforms, covering system architecture, TARA, cybersecurity requirement derivation, implementation considerations, and operational monitoring.

1.1 Contribution

The paper presents an ISO/SAE 21434-based cybersecurity workflow tailored to server-connected BMS used in commercial electric vehicle environments. The main contributions of this work are as follows:

- A structured application of the ISO/SAE 21434 TARA process to representative BMS assets, including communication and battery-control functions, illustrating the derivation of cybersecurity goals, cybersecurity claims, and cybersecurity requirements.
- A mapping of TARA outputs to cybersecurity architecture design and post-deployment operational monitoring, enabling cybersecurity management throughout the BMS lifecycle.
- An analysis of practical challenges associated with implementing and maintaining ISO/SAE 21434-compliant cybersecurity in commercial server-connected BMS ecosystems.

The proposed workflow illustrates how lifecycle-oriented cybersecurity activities can be integrated into the development, deployment, operation, and maintenance of connected BMS platforms. By linking risk assessment, cybersecurity requirements, architectural controls, and operational monitoring, this paper provides practical guidance to support the secure and reliable operation of commercial electric vehicle battery systems.

1.2 Paper Organization

The remainder of the paper is organized as follows. Section 2 presents a review of the related literature. Section 3 provides an overview of the server-connected BMS architecture and the associated cybersecurity threats. Section 4 presents the ISO/SAE 21434-based cybersecurity architecture design and development process. Section 5 discusses key challenges that may be encountered during the implementation and post-implementation phases. Finally, Section 6 concludes the paper.

2 Related Work

The BMS is one of the most safety-critical components in EVs, as it monitors and controls key battery parameters, including voltage, current, temperature, state of charge (SoC), and state of health (SoH). Increasing connectivity between internal sensor networks and external backend cloud services expands the attack surface and makes BMS platforms more vulnerable to cyber threats. Consequently, BMS cybersecurity has become an important research topic. Existing studies can be broadly categorized into three areas: conventional BMS cybersecurity, cloud-based BMS architectures, and automotive cybersecurity standards.

Early work on BMS cybersecurity identified the BMS as a critical attack target due to its interaction with battery cells, vehicle controllers, charging interfaces, and external communication channels. In [6], the authors studied cybersecurity challenges for BMSs in cyber-physical environments and discussed possible attack schemes and defense strategies for IoT-enabled BMSs. In [5], the authors discussed potential attacks on conventional BMS architectures and highlighted that BMS compromise may affect vehicle safety, performance, and battery reliability. In [12], the authors developed a framework to study how cyberattacks may affect EV battery packs, including overcharge/discharge, accelerated degradation, and thermal safety risks. While these studies provide valuable insights into BMS attack surfaces and their consequences, they do not provide an ISO/SAE 21434-compliant cybersecurity engineering workflow for commercial server-connected BMS development.

Cloud-based, blockchain-integrated, and intelligent BMS architectures have also received significant attention. Yang et al. [14] proposed a cloud-based BMS implementation using an end-edge-cloud architecture, where cloud computing supports advanced battery analytics, state estimation, thermal management, cell balancing, and fault diagnosis. Mulpuri et al. [7] discussed an intelligent BMS architecture with end-edge-cloud connectivity, digital twins, blockchain technologies, and IoT-based communication. These studies demonstrate the technical benefits of cloud-connected BMSs; however, their primary focus is on performance enhancement, data analytics, and intelligent battery management rather than standards-based cybersecurity governance, TARA, or post-production cybersecurity monitoring. In [9], the authors conducted a review of cyber-physical cloud BMSs and discussed cloud BMS architectures, attack surfaces, attack scenarios, impacts, countermeasures, and relevant standards. Although the work highlights cybersecurity risks associated with cloud-connected BMS platforms, it does not provide

a step-by-step ISO/SAE 21434 implementation workflow for a specific server-integrated BMS architecture.

Recent research has also adapted ISO/SAE 21434-based TARA methodologies to particular vehicle classes. For example, Rahimi et al. [10] proposed an enhanced TARA model for heavy-duty vehicles using ISO/SAE 21434 and fuzzy analytic hierarchy process methods. Based on the TARA results, cybersecurity architectures can be designed and implemented. Previous studies have also examined TARA primarily at the vehicle level, particularly in the context of connected and autonomous vehicles [3, 1, 11]. However, most of this work does not address the complete cybersecurity lifecycle or focus specifically on battery management systems. Such studies improve automotive TARA methodologies and cybersecurity assessment practices; however, they generally focus on the vehicle level or a specific vehicle category rather than a server-connected BMS architecture with backend telemetry, OTA update capabilities, and lifecycle monitoring considerations.

In general, existing studies primarily focus on attack analysis, battery safety implications, cloud-enabled battery management, or vehicle-level cybersecurity assessment. While several works apply TARA methodologies or discuss cybersecurity risks in connected-vehicle environments, limited attention has been paid to developing a structured ISO/SAE 21434-based cybersecurity workflow tailored to server-connected BMS platforms. Furthermore, the integration of TARA outputs with cybersecurity architecture design, implementation activities, and post-deployment monitoring remains largely unexplored. This paper addresses these gaps by presenting a lifecycle-oriented cybersecurity workflow for commercial server-connected BMS environments.

3 System Model

Figure 1 illustrates the overall architecture of a server-connected BMS for EVs. The architecture is divided into three main domains: the battery management system, the communication layer, and the backend/cloud infrastructure.

The battery management system consists of battery cells, pack sensors, cell monitoring units (CMUs), the battery management unit (BMU), diagnostic interfaces, the controller area network (CAN) bus, and associated control components. These elements are responsible for battery monitoring, state estimation, cell balancing, fault detection, and other safety-critical battery management functions.

The communication layer provides connectivity between the in-vehicle BMS and external backend services. It includes communication protocols, a telematics gateway, firewall functions, connectivity modules, and an OTA update client. This layer enables the transmission of battery telemetry data to backend systems while supporting remote diagnostics, monitoring, and software-update operations.

The backend/cloud domain represents the server-side infrastructure used for remote BMS management and analytics. It includes API and authentication services, telemetry databases, OTA update servers, monitoring services, and incident-response capabilities. These services support telemetry storage, predictive maintenance, fleet monitoring, user access management, and cybersecurity operations.

The cybersecurity scope considered in this work extends from the BMS to the cloud-facing services used for telemetry collection, monitoring, and OTA updates. The telematics gateway represents a trust boundary between the vehicle domain and external networks. Components within the vehicle are considered part of the trusted operational domain, whereas communication networks and external interfaces are treated as potentially untrusted environments

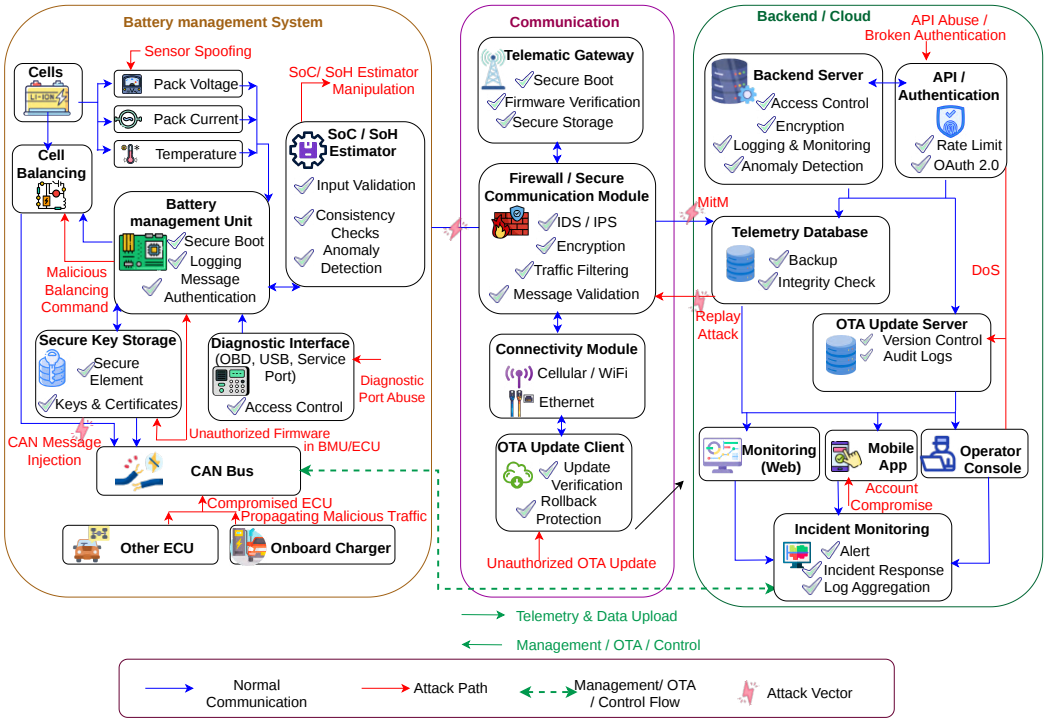

Figure 1: Server Integrated BMS architecture with possible attack scenario.

requiring authentication, integrity protection, and continuous monitoring. Backend enterprise infrastructure beyond the BMS-specific cloud services is considered out of scope and is assumed to be protected using established enterprise cybersecurity frameworks such as ISO/IEC 27001.

Several attack paths may exist across the architecture. Within the Battery Management System, adversaries may attempt sensor spoofing, false-data injection, SoC/SoH estimator manipulation, malicious balancing commands, CAN message injection, ECU compromise, or diagnostic-port abuse. At the communication layer, potential threats include man-in-the-middle attacks, telemetry tampering, replay attacks, and unauthorized OTA commands. Within the backend/cloud environment, threats may include API abuse, broken authentication mechanisms, denial-of-service attacks, account compromise, and unauthorized access to telemetry data.

4 Standard Process of Cybersecurity

ISO/SAE 21434 provides a lifecycle-oriented framework for identifying, assessing, and managing cybersecurity risks in automotive systems. In this work, the framework is applied to a server-connected BMS to support cybersecurity activities throughout system development, deployment, operation, and maintenance. In the considered architecture, battery telemetry data—such as pack voltage, pack current, SoC, SoH, and operational logs—are transmitted from the BMS to backend services for monitoring, analytics, diagnostics, and OTA update

management. While such connectivity enables advanced battery management and predictive maintenance capabilities, it also increases the attack surface by exposing internal vehicle components and communication interfaces to external threats. Therefore, a structured cybersecurity process is required to systematically identify cybersecurity risks, derive appropriate security requirements, implement cybersecurity controls, and support continuous monitoring throughout the BMS lifecycle.

4.1 TARA

To identify and manage cybersecurity risks, ISO/SAE 21434 and UNECE R155 require the application of TARA. TARA provides a systematic process for identifying cybersecurity assets, analyzing threat scenarios, evaluating risks, and deriving cybersecurity requirements. Throughout this paper, the corresponding ISO/SAE 21434 Requirement (RQ) and Work Product (WP) identifiers are referenced to indicate the relevant requirements and documented outputs associated with each cybersecurity activity. Figure 2 illustrates the main TARA activities considered in this work.

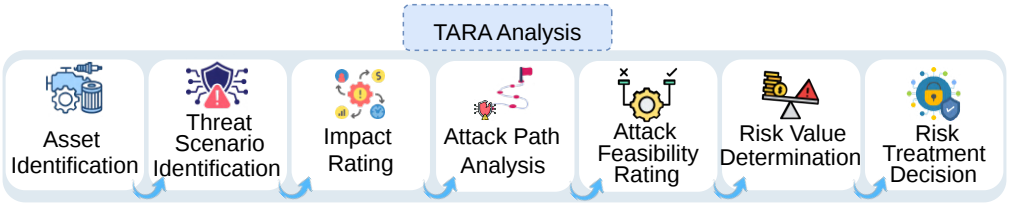

Figure 2: Brief modules of TARA.

Asset identification ([RQ-09-03], [RQ-15-02]) is the process of identifying objects within the BMS that possess one or more cybersecurity properties and whose compromise may result in one or more damage scenarios. For a BMS, such assets may include sensors, hardware components, software modules, communication networks, and backend services. For example, the CAN bus is an important communication asset within the BMS. Asset identification involves several sub-activities, including item definition, damage scenario identification, and cybersecurity property identification.

Item definition ([RQ-09-01], [RQ-09-02], [WP-09-01]) involves identifying the item boundary, functions, and preliminary architecture of the system under consideration. For example, the CAN bus can be defined as the communication channel between the BMU and ECUs, responsible for carrying status, control, command, and diagnostic information.

Damage scenario ([RQ-15-01]) describes the potential consequence of a cybersecurity attack on a battery function or asset. Multiple damage scenarios may be associated with a single asset. For example, one possible damage scenario related to the CAN bus is the disruption of battery charging and discharging operations.

Cybersecurity property refers to an attribute that requires protection. The three primary cybersecurity properties are confidentiality, integrity, and availability. Depending on the asset and damage scenario, one or more of these properties may require protection and are therefore considered during the TARA process. For the previously discussed damage scenario, the relevant cybersecurity properties include confidentiality, integrity, and availability.

Threat scenario ([WP-15-03]) describes a potential cause of compromise of one or more cybersecurity properties that may lead to a damage scenario. Multiple threat scenarios may

exist for a single damage scenario. For example, a malfunction in battery charging and discharging operations could result from malware injected into CAN messages that impersonate legitimate BMS control or diagnostic traffic.

Impact rating ([RQ-15-04], [RQ-15-05], [RQ-15-06], [RQ-15-07], [WP-15-04]) evaluates the consequences of a damage scenario in terms of safety, financial, operational, and privacy impacts. The resulting impact rating is classified as severe, major, moderate, or negligible. For example, the previously discussed CAN bus attack scenario may result in severe safety impact, major financial impact, severe operational impact, and moderate privacy impact.

Attack path analysis ([RQ-15-08], [RQ-15-09], [WP-15-05]) can be performed using either a top-down or a bottom-up approach. In a top-down approach, threat scenarios are analyzed to identify possible methods through which an attacker may realize the threat. In contrast, a bottom-up approach constructs attack paths starting from known vulnerabilities and attack entry points. Considering the previous example, a possible attack path may include the following steps: (a) an attacker compromises an ECU and installs malware capable of transmitting arbitrary CAN messages; (b) the malware gains control of CAN communications and modifies message contents; (c) malicious CAN frames are crafted to resemble legitimate BMS control or diagnostic traffic; and (d) the BMU accepts the manipulated messages and executes incorrect control or diagnostic actions, potentially disrupting the safe charging and discharging of battery cells.

Attack feasibility rating evaluates how easy or difficult it is for an attacker to successfully execute an attack. According to ISO/SAE 21434, the attack feasibility rating is determined based on a combination of elapsed time, specialist expertise, knowledge of the item or component, window of opportunity, and required equipment. The resulting attack feasibility is classified as very low, low, medium, or high.

Elapsed time represents the time required to identify a vulnerability and develop and execute an exploit. ISO/SAE 21434 categorizes elapsed time into five levels: ≤ 1 day, ≤ 1 week, ≤ 1 month, ≤ 6 months, and > 6 months. For the CAN bus example discussed previously, the elapsed time is estimated as ≤ 1 week.

Specialist expertise refers to the level of attacker knowledge and skills required to perform the attack. The standard defines four levels: layman, proficient, expert, and multiple experts. In the considered example, expert-level knowledge is required to inject malicious CAN messages while avoiding detection.

Equipment refers to the tools needed to identify vulnerabilities and execute an attack. ISO/SAE 21434 classifies equipment as standard, specialized, bespoke, or multiple bespoke. For the CAN bus attack scenario, specialized equipment is required.

Knowledge of the item or component indicates the availability of information required to perform an attack. The standard distinguishes between public, restricted, confidential, and strictly confidential information. For the example considered in this work, the required knowledge is classified as restricted.

Window of opportunity describes the access conditions required to successfully perform an attack and is categorized as unlimited, easy, moderate, or difficult. For the CAN bus example, the window of opportunity is considered moderate.

Risk value determination ([RQ-15-10]) is performed by combining the impact rating and the attack feasibility assessment. ISO/SAE 21434 does not prescribe a single risk-evaluation matrix; therefore, organizations typically establish their own risk-classification criteria based on their cybersecurity objectives and risk tolerance [10]. For illustration purposes, Table 1 presents an example risk determination matrix used in this work.

For the CAN bus attack scenario discussed previously, the impact rating is classified as

Table 1: Risk value determination matrix

Impact Rating	**Very Low**	**Low**	**Medium**	**High**
Severe	2	3	4	5
Major	1	2	3	4
Moderate	1	2	2	3
Negligible	1	1	1	1

severe and the attack feasibility rating as medium. According to Table 1, this combination results in a risk value of 4.

Risk treatment decision is determined based on the results of the risk assessment. ISO/SAE 21434 defines four risk treatment options: avoiding, reducing, sharing, and retaining cybersecurity risks. For the CAN bus attack scenario discussed previously, the selected treatment option is risk reduction, as the identified risk requires the implementation of cybersecurity measures to reduce its likelihood and potential impact.

Cybersecurity goal defines the high-level security objectives required to mitigate the identified risk. Multiple cybersecurity goals may be associated with a single threat scenario. For the example considered in this work, representative cybersecurity goals include: (1) only legitimate and authenticated BMS control and diagnostic messages shall be accepted by the BMU; (2) the BMU shall receive correct, timely, and trustworthy control and sensor messages despite adversarial traffic; (3) only trusted and authenticated ECUs shall be permitted to transmit BMS-relevant CAN messages; and (4) spoofed, replayed, or malformed BMS control and diagnostic messages shall be detected and handled safely.

Cybersecurity claims ([RQ-09-06]) define the security capabilities and system behaviors required to achieve the corresponding cybersecurity goals. For the cybersecurity goals discussed above, representative cybersecurity claims include: (1) the system shall implement cryptographic message authentication (e.g., message authentication codes (MACs) or secured CAN communication) for BMS control and diagnostic messages; (2) upon detection of spoofed messages, the BMU shall enter a controlled degraded mode, limit charging and discharging operations, and rely on validated fallback values; (3) ECUs capable of transmitting BMS control commands shall authenticate themselves, execute only signed firmware, and prevent the execution of unauthorized code; and (4) the system shall detect duplicated CAN identifiers, inconsistent payload patterns, abnormal message frequencies, and replay attempts. Based on these cybersecurity claims, the corresponding cybersecurity requirements can be derived in accordance with ISO/SAE 21434.

To further illustrate the applicability of the proposed ISO/SAE 21434-based TARA workflow beyond the CAN bus example, Table 2 presents a complete TARA analysis for a cell-balancing control function. The example demonstrates how asset identification, threat analysis, impact assessment, attack feasibility evaluation, risk treatment, cybersecurity goals, cybersecurity claims, and cybersecurity requirements can be systematically derived for a representative BMS asset. For example, cybersecurity claim (1) is supported by several ISO/SAE 21434 requirements, including [RQ-05-01], [RQ-05-02], [RQ-05-04], [RQ-06-01], [RQ-06-07], [RQ-08-01], and [RQ-08-02]. These requirements collectively support cybersecurity governance, process management, resource allocation, and the implementation of cybersecurity controls throughout the product lifecycle.

Table 2: Example TARA analysis for a cell-balancing control function

TARA Field	Case 1: Analysis result for imbalanced cells with different charge levels
Asset Name	Cell balancing control
Item Definition	Cell-balancing control function located between the CMU and BMU and responsible for maintaining balanced charge levels among battery cells during charging and discharging operations.
Damage Scenario	Unequal cell states of charge leading to battery imbalance, reduced performance, and potential safety risks.
Cybersecurity Property	Integrity, Availability
Threat Scenario	Malicious firmware update: Unsigned and unauthenticated BMS firmware may allow attackers to modify balancing-control software
Safety Impact	Severe.
Financial Impact	Severe.
Operational Impact	Major.
Privacy Impact	Negligible.
Impact Rating	Severe.
Attack Path	1. The attacker gains access to the BMU/CMU update path through a service port, diagnostic interface, or network flaw. 2. Missing firmware signing or authentication allows the attacker to install modified firmware. 3. Malicious firmware changes the cell-balancing logic, causing uneven charging. 4. The altered firmware may overcharge, undercharge, or imbalance individual cells. 5. Cell imbalance can cause overheating, capacity loss, and early battery failure.
Elapsed Time	$\leq$ 1 month.
Specialist Expertise	Expert.
Equipment	Specialized.
Knowledge of Item or Component	Confidential.
Window of Opportunity	Moderate.
Attack Feasibility Rating	Low.
Risk Value	3.
Treatment Option	Reducing
Cybersecurity Goal	1. Protect BMS firmware integrity against unauthorized changes. 2. Secure firmware updates with authentication and availability controls.
Cybersecurity Claim	1. The BMS shall use secure boot and signed updates to verify firmware integrity and authenticity before execution. 2. The CMU/BMU shall execute only authenticated firmware through secure boot.
Cybersecurity Measures	CAL3
Cybersecurity Requirements	1. The BMS controller shall support secure boot to verify the firmware image signature before execution. 2. Firmware update files delivered through OTA or diagnostic interfaces shall be checked for signature validity and integrity before being written to flash memory. 3. The BMS shall prevent firmware rollback using a hardware-protected monotonic counter. 4. Boot configuration areas, including option bytes or configuration sectors, shall be protected by hardware access controls such as read/write protection or secure lock bits.

4.2 Mapping TARA with Operational Monitoring

The outputs of the TARA process are not limited to design-time cybersecurity requirement derivation; they can also be used to support post-deployment monitoring and incident response activities. In the proposed server-connected BMS architecture, TARA outputs are mapped to operational monitoring functions. Asset definitions identify the signals, interfaces, components,

and backend services that require monitoring. Damage scenarios define the safety, operational, financial, or privacy consequences that the monitoring system must detect or prevent. Threat scenarios describe expected attacker behaviors and therefore guide the design of detection rules, anomaly-detection models, and log-correlation mechanisms. Impact ratings determine alert severity and escalation priorities, while attack feasibility ratings influence monitoring frequency, logging depth, and the deployment of preventive controls. Finally, risk treatment decisions guide the selection of appropriate mitigation and incident-response strategies.

For example, consider the SoC/SoH estimator as a cybersecurity asset and false-data injection into voltage, current, or temperature measurements as the associated threat scenario. In this case, the operational monitoring system should perform plausibility checks between raw sensor measurements and estimated battery states. It should also detect abnormal SoC values, physically implausible SoH variations, repeated estimation inconsistencies, and unexpected deviations from historical battery behavior. If the resulting impact rating is classified as severe or major, alerts should be escalated to the fleet-monitoring dashboard, and the BMS should transition to a safe or degraded operating mode until the anomaly has been investigated and resolved.

4.3 Cybersecurity Architecture Design and Implementation ([RQ-10-01] to [RQ-10-08])

The cybersecurity architecture defines the security mechanisms and system interactions required to protect the server-connected BMS and its associated communication and backend services. Based on the cybersecurity goals, claims, and requirements derived during the TARA process, the architecture specifies the security controls needed to protect BMS assets, communication interfaces, software components, and backend services.

In the proposed architecture, multiple cybersecurity functions must be integrated across the vehicle, communication, and cloud domains. These include identity and access management mechanisms, authentication and authorization controls, and a hierarchical security model in accordance with requirement [RQ-10-01]. Communication between the BMS and backend services should be protected using secure communication protocols, message integrity verification, and encryption mechanisms. In addition, OTA update services should incorporate firmware signing, integrity verification, version control, and rollback protection to prevent the installation of unauthorized software.

The architecture must also ensure the security of BMS firmware, software components, diagnostic interfaces, and development environments, as specified in [RQ-10-04]. Furthermore, hardware-level protection mechanisms, such as secure microcontrollers, hardware security modules, secure storage, and secure boot, should be considered to establish a trusted execution environment and provide a robust root of trust.

For backend services, cybersecurity controls should include strong authentication, role-based access control, API protection mechanisms, secure telemetry storage, and continuous monitoring of security events. These controls complement the cybersecurity requirements within the vehicle domain while supporting the secure operation of cloud-connected BMS platforms.

The overall cybersecurity architecture shall be verified against the three fundamental criteria of completeness, correctness, and consistency (3C), as specified in [RQ-10-08]. Verification activities may include systematic reviews, architectural analysis, threat-based assessments, simulation-based validation, penetration testing, and prototype evaluations to ensure that both functional and cybersecurity requirements are satisfied.

4.4 Integration, Verification, and Validation ([RQ-10-09], [RQ-10-10])

Following the cybersecurity architecture design phase, the identified cybersecurity controls must be implemented, integrated, and verified within the BMS ecosystem. Requirement [RQ-10-09] addresses the implementation and integration of cybersecurity components, while ensuring that modeling, design, and coding guidelines are applied in accordance with [RQ-10-05]. For a server-connected BMS, this includes the integration of secure communication mechanisms, authentication services, secure boot processes, OTA update protections, monitoring functions, and backend security controls.

Verification activities are performed to confirm that cybersecurity requirements have been correctly implemented and that the resulting system satisfies the intended security objectives. Typical verification methods include requirement-based testing, interface testing, resource-usage evaluation, control-flow and data-flow verification, as well as static and dynamic analysis. Where testing is adopted ([RQ-10-11], [RQ-10-12]), ISO/SAE 21434 recommends methods such as functional testing, vulnerability scanning, fuzz testing, and penetration testing. For example, penetration testing may be used to assess API security, OTA update mechanisms, and communication interfaces between the vehicle and backend services, while fuzz testing can be applied to communication protocols and diagnostic interfaces.

The results of verification and validation activities should be documented in validation reports and associated result records, as required by [RQ-11-01] and [RQ-11-02]. If the implemented cybersecurity controls satisfy the specified requirements and no further corrective actions are necessary, the system may proceed to production and deployment ([RQ-12-01], [RQ-12-02]).

Following deployment, organizations must establish processes for monitoring, detecting, and responding to cybersecurity incidents in operational environments ([RQ-13-01], [RQ-13-02]). This includes vulnerability management, incident handling, security monitoring, and the deployment of corrective measures when new threats are identified. Finally, the cybersecurity architecture should be periodically reviewed and updated to address emerging threats, newly discovered vulnerabilities, software updates, and changes to system functionality, thereby ensuring the continued cybersecurity of the BMS throughout its operational lifecycle ([RQ-14-01], [RQ-14-02]).

5 Challenges

Several challenges may arise when implementing and maintaining an ISO/SAE 21434-compliant cybersecurity architecture for server-connected BMSs. One challenge is the practical application of the TARA process. Although ISO/SAE 21434 guides TARA activities, organizations may interpret asset identification, threat scenario development, and risk evaluation differently, potentially leading to inconsistent results across projects and stakeholders. In addition, the absence of a fully prescriptive incident-response workflow and the requirement for continuous cybersecurity activities throughout the product lifecycle introduce both technical and organizational complexity [2].

Further challenges arise when integrating cybersecurity requirements with existing automotive engineering processes. For example, cybersecurity activities defined by ISO/SAE 21434 must often be coordinated with functional safety processes governed by ISO 26262, requiring careful management of potential trade-offs between safety and security objectives. Legacy ECUs, communication protocols, and battery controllers may also lack native support for modern security mechanisms such as secure boot, cryptographic authentication, or encrypted com-

munication, making compliance difficult without substantial redesign efforts.

Server-connected BMS platforms introduce additional challenges related to cloud connectivity and remote operations. Secure management of OTA updates, protection of telemetry data, API security, authentication of cloud services, and the monitoring of communication between vehicle and backend infrastructure require cybersecurity controls that extend beyond the traditional vehicle domain. Furthermore, the transition between automotive cybersecurity requirements and enterprise cybersecurity frameworks may introduce additional governance and operational challenges.

Resource limitations, including shortages of cybersecurity expertise, specialized tools, and implementation budgets, may further hinder adoption, particularly for small and medium-sized organizations. Moreover, the rapidly evolving threat landscape requires continuous reassessment of risks, periodic updates to cybersecurity controls, and ongoing monitoring of deployed systems. Finally, ensuring consistent cybersecurity practices across a complex automotive supply chain remains challenging, particularly when multiple suppliers, software vendors, cloud-service providers, and vehicle manufacturers contribute to the overall BMS ecosystem.

6 Conclusions

Cybersecurity requirements continuously evolve as new threats, vulnerabilities, and attack techniques emerge. Therefore, maintaining cybersecurity through a systematic and standards-based approach is essential for both industry and academia. This paper outlines an ISO/SAE 21434-based cybersecurity workflow for the development, governance, and maintenance of server-connected BMSs. The proposed approach illustrates how cybersecurity activities, including TARA, cybersecurity architecture design, implementation, verification, and post-deployment monitoring, can be integrated throughout the BMS lifecycle to support secure commercialization and operation.

Future work will include a comparative analysis of ISO/SAE 21434 and other relevant cybersecurity standards and frameworks. In addition, future studies will investigate cybersecurity risks associated with artificial intelligence , including generative AI and agentic AI systems, and explore standards-based approaches for risk assessment, monitoring, and mitigation in connected automotive environments.

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Securing the Sky: Lightweight Machine Learning for Zero-Day Intrusion Detection in UAV Systems*

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Abstract

Unmanned Aerial Vehicles (UAVs) are increasingly employed in civilian and commercial applications, which heightens the need for robust and reliable security mechanisms. UAV systems are exposed to a wide range of cyber and cyber-physical attacks that can disrupt missions, compromise data integrity, or lead to unsafe physical behavior. This work investigates machine learning-based intrusion detection for UAV systems with a particular focus on zero-day attack detection and computational feasibility. The objective is to assess whether lightweight classification models with low architectural complexity can achieve strong detection performance while remaining suitable for deployment in resource-constrained environments. To this end, four classifiers are evaluated. Detection performance is examined under a supervised zero-day learning framework. Experimental results demonstrate that meaningful generalization to unseen attacks can be achieved using compact machine learning models. Detection performance is competitive with results reported in related work, while computational efficiency measurements indicate promising suitability for real-world deployment.

1 Introduction

The use of Unmanned Aerial Vehicles, commonly referred to as drones, has increased significantly in recent years [22]. Once confined to military or hobbyist use, UAVs are now applied across a wide range of civil and commercial activities [7]. They support search and rescue operations, enable disaster assessment and recovery, and provide infrastructure inspection for bridges, power lines, and construction sites [16, 20]. In such scenarios, UAVs offer significant advantages due to their ability to access hazardous or hard-to-reach environments and operate with minimal human intervention. Beyond those applications, UAVs are increasingly employed in domains such as agriculture and environmental monitoring, where their mobility, operational flexibility, and cost efficiency allow large areas to be monitored [14]. These capabilities have accelerated their adoption in industrial and commercial settings and have attracted growing interest from major industry stakeholders in UAV-based systems, especially for applications such as automated goods delivery, where reliability and operational safety are of critical importance [10].

As UAVs are increasingly deployed across a wide range of application domains, their exposure to security threats likewise increases. Drones are inherently vulnerable due to their dependence on wireless communication links, unencrypted or weakly protected navigation signals, and limited onboard computational resources [6,25,36]. These characteristics expose UAV systems to a wide range of attacks, including GPS jamming and spoofing, false data injection,

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and denial-of-service, leading to severe safety and operational consequences [27]. Several UAV-targeted attacks are practically feasible in real-world settings, as commonly available tools can be exploited to launch disruptive attacks [32]. Reliable detection of such threats is, therefore, essential to ensure the secure and dependable operation of UAV systems, but it remains a significant challenge.

Intrusion detection systems represent a viable approach to enhancing the security of UAV systems, and recent years have seen an increasing adoption of machine learning-based approaches for this purpose [4,17]. However, a number of important limitations persist in existing research.

Many existing machine learning-based intrusion detection approaches rely on generic or simulation-based datasets that do not accurately reflect UAV-specific communication patterns, operational dynamics, or attack surfaces, thereby limiting their relevance for real-world deployment. Closely related to this issue, a considerable portion of prior work focuses on a narrow set of attack types, frequently restricted to GPS spoofing or jamming scenarios, which constrains the ability to assess the effectiveness of intrusion detection approaches across the diverse threat landscape faced by UAV systems. Furthermore, the majority of studies rely on evaluations conducted within a single dataset and experimental configuration. While this approach often results in high reported accuracy, it does not adequately assess whether such performance persists when models are applied under different conditions. This limits insight into the robustness and generalization of the intrusion detection approaches. Lastly, computational efficiency and detection latency are often insufficiently considered, despite their critical importance for real-time detection and potential onboard deployment in resource-constrained UAV environments.

In light of these limitations, there remains a need for the systematic evaluation of machine learning-based intrusion detection mechanisms tailored to UAV environments. In particular, an improved understanding is required regarding their robustness across datasets, their effectiveness against diverse attack types, and their suitability for deployment under the computational and latency constraints of UAV systems. This work seeks to address these challenges. The key contributions of this study can be summarized as follows:

- Design and implementation of a unified evaluation pipeline for zero-day intrusion detection in UAV systems using lightweight machine learning models, with emphasis on both detection effectiveness and runtime efficiency.
- Explicit evaluation of computational efficiency, including inference latency, to assess the feasibility of deploying the proposed classifiers under resource constraints.
- Comprehensive multi-dataset evaluation using three publicly available UAV datasets derived from real or experimental flight deployments.

2 Related Work

When looking at the broader landscape of intrusion detection systems for drones, a range of approaches has been explored. Some of the earliest works in this domain employ, for example, game-theoretic models [26,29]. Over time, however, research attention has increasingly shifted towards machine learning-enhanced intrusion detection.

When examining the learning paradigms adopted by machine learning-based intrusion detection for UAV systems, a dominance of supervised learning approaches can be observed. Most existing studies formulate intrusion detection as a classification problem and rely on labeled datasets for training and evaluation. Beyond supervised learning, prior work also explores

unsupervised anomaly detection techniques, deep learning models, and, to a lesser extent, reinforcement learning-based methods. A smaller number of studies further investigate alternative approaches that do not fit neatly into these categories; however, these approaches are represented only sporadically in the literature.

While a broad spectrum of learning-based techniques has been proposed, the effectiveness and practical relevance of these approaches are fundamentally constrained by the datasets on which they are developed and evaluated. In particular, dataset realism, UAV-specific characteristics, and the breadth of attack coverage play a decisive role in determining how meaningful the reported results are.

2.1 Datasets for Intrusion Detection

A key component in the development of effective intrusion detection mechanisms for UAV environments is the choice of the dataset used for training and evaluation. The dataset largely determines how representative the evaluation is of real-world scenarios and to what extent the resulting models can generalize beyond the experimental setup. Based on the existing literature, the datasets employed for UAV intrusion detection can be broadly grouped into three categories:

(I) generic, non-UAV-specific intrusion detection datasets,

(II) simulated or synthetically generated UAV data, and

(III) datasets explicitly collected from real or experimental UAV deployments

Furthermore, a comprehensive assessment of intrusion detection mechanisms ideally requires evaluating model behavior under varying data distributions and operating conditions, rather than relying on a single experimental setup. However, a review of recent UAV intrusion detection studies shows that most proposed approaches are evaluated on only one dataset, which limits insights into their robustness. As noted by Islam et al. [17], reported detection performance is strongly influenced by dataset selection, and results obtained under a single evaluation setting may not be reliable when models are applied under different conditions. This dependence on dataset-specific characteristics also complicates meaningful comparisons across studies.

2.1.1 Generic Intrusion Detection Datasets

Several generic intrusion detection datasets remain popular not only in traditional network security research but also in UAV-related Intrusion Detection System (IDS) studies. Frequently used datasets include KDDCup99 [1] and its refined version NSL-KDD [30], as well as more recent datasets such as CIC-IDS2017 [23]. These datasets capture a variety of network traffic patterns under normal and malicious conditions. Despite their widespread adoption, these datasets were not designed with UAV systems in mind and therefore lack UAV-specific characteristics. In the context of UAV networks, this includes the absence of flight telemetry and control commands, as well as the lack of data generated by UAV-specific communication protocols such as MAVLink. These protocols play a central role in UAV operation, enabling navigation, mission control, and sensor data exchange. As a result, models trained exclusively on generic datasets may learn patterns that are only weakly related to the operational behavior of real UAV systems.

This mismatch raises concerns regarding the applicability and generalization of such models when deployed in realistic UAV environments. While generic datasets can be useful for preliminary evaluation or for benchmarking baseline detection performance, their limited representativeness means that the high detection accuracy reported on these datasets does not

necessarily translate to effective intrusion detection on real drones or UAV networks. Consequently, an increasing number of studies have begun to explore UAV-specific datasets.

2.1.2 Simulation-based UAV Intrusion Detection Datasets

In terms of simulated versus real-flight UAV data, a clear trend can be observed in UAV intrusion detection literature: only a minority of published studies rely on data collected from real or experimental UAV deployments. This observation is consistently highlighted in recent surveys, including those by Al-Syouf et al. [4] and Ismal et al. [17], which report that the majority of evaluated approaches are validated using simulated or synthetically generated datasets. In particular, Al-Syouf et al. note that approximately 70 % of the studies they included in their review rely primarily on simulation-based data. Many of those simulation-based datasets exist only within individual studies and are not available for benchmarking or cross-study evaluation.

The use of simulated data offers several advantages for research. Simulation environments provide fine-grained control over experimental conditions and allow the systematic exploration of attack scenarios that may be costly, unsafe, or legally restricted to reproduce in real deployment. Moreover, simulations enable the generation of large, labeled datasets, which is particularly beneficial for supervised learning approaches that require extensive training data.

However, reliance on simulated data also introduces limitations. Simulation models simplify aspects of UAV dynamics, sensor noise, and environmental interactions. As a result, simulated datasets often exhibit cleaner, more regular patterns than those observed in real-flight data. Models trained on simulated data might therefore generalize only to a limited extent when deployed in real UAV systems, especially when evaluated solely within the same simulation framework.

2.1.3 Real-Flight UAV Intrusion Detection Datasets

In response to the limitations of generic and simulation-based data, recent research has increasingly focused on the creation of real-fight UAV intrusion detection datasets. However, the availability of such datasets, particularly in the form of publicly accessible benchmark datasets, remains limited. A recent survey of UAV intrusion detection datasets by Mohammed and Fourati [19] identifies nine datasets targeting UAVs and their communication systems; however, approximately half of these are simulation-based or focus primarily on anomaly and fault detection rather than on explicit attack scenarios.

2.1.4 Attack Coverage

The scope of attack types evaluated in intrusion detection studies is closely tied to the choice of dataset. As discussed previously, generic network intrusion datasets such as KDDCup99 and CIC-IDS2017 typically include multiple categories of attacks, enabling evaluation across diverse threat scenarios. Likewise, most UAV-specific datasets derived from real deployments, including those employed in this study, cover two or more distinct attack types in order to reflect the diverse threat landscape faced by UAV systems.

In contrast, studies relying on simulated data frequently concentrate on a single attack scenario, and datasets are often constructed to model one specific threat. For example, Slimane et al. [28] evaluate boosting- and bagging-based models for the detection of jamming attacks. Their dataset is generated using a Simulink-based model that simulates communication between the UAV and its ground control station under benign and jamming conditions. Similarly, Agyapong et al. [3] employ deep learning models to detect GPS spoofing attacks using data

generated in the Gazebo simulation environment. While such single-attack studies provide valuable insights into the detectability of individual threats, they offer a limited view of intrusion detection performance in more complex environments. In real-world UAV operations, multiple attack types may occur. Given that a substantial portion of existing UAV IDS research relies on simulation-based datasets, this tendency towards single-attack evaluation is quite widespread in the literature.

2.2 Comparison with Closest Related Work

Taken together, the existing literature demonstrates a broad range of machine learning approaches for UAV intrusion detection, along with substantial variation in dataset choice and attack coverage. To enable a more focused and meaningful comparison with related work, the following overview narrows the scope to studies that evaluate intrusion detection models using UAV-specific datasets derived from real-flight experiments rather than purely from simulation environments.

Ref.	Model / Algorithms	Performance Metrics	Dataset Used
[33]	One-Class Support Vector Machine, Autoencoder Neural Network, Local Outlier Factor	F1-Score of 81.17%, 94.81%, and 58.93% respectively	UAV Attack
[18]	Feedforward convolutional neural network	Accuracy of 98.23%	UAV-IDS
[31]	Long Short-Term Memory (LSTM)	Accuracy of 98.9%	UAVCAN, UAV Attacks, Kitsune Network Attack
[8]	XGBoost	Accuracy of up to 99.95%	Cyber-Physical
[15]	Support Vector Machine, Feedforward neural network (FNN), LSTM, Convolutional Neural Network (1D-CNN)	F1-Score of 89.86%, 91.9%, 94.88%, and 96.13% respectively	Cyber-Physical
[12]	One Class Support Vector Machine, Isolation Forest, Local Outlier Factor, Denoising Autoencoder, LSTM Autoencoder, Convolutional Autoencoder	F1-Score of 61.97%, 78.96%, 55.43%, 96.38%, 98.33%, and 98.51% respectively	UAV Attack

Ref.	Model / Algorithms	Performance Metrics	Dataset Used
[2]	Convolutional Neural Network	Accuracy of 98.59% and F1-Score of 98.69%	UAV-IDS
[24]	Autoencoder	NA	MAVLINK
[13]	Isolation Forest, Multi-Layer Perceptron, FNN, Model-Agnostic Meta-Learning	F1-Score of 84.61%, 90.40%, 99.67%, and 96.20% respectively (best experiment)	UAV-NIDD
[5]	Deep Neural Network	Accuracy of 99.99%	UAV-IDS
[9]	One Class Support Vector Machine, Isolation Forest, Local Outlier Factor, Stacked Autoencoder	F1-Score of 77.8%, 83.6%, 87.1%, and 92.9% respectively	UAV Attack
[21]	Random Forest, K-Nearest Neighbors, Support Vector Machine, Logistic Regression, Naive Bayes, Decision Tree, Multi-Layer Perceptron	F1-Score of 90%, 86%, 89%, 90%, 91%, 92%, 89% respectively	GPS Spoofing

Table 1: Summary of related work.

Building on the challenges identified throughout this section, several recurring limitations can be observed in existing UAV intrusion detection research. These include heavy reliance on generic, simulation-based and single datasets, narrow attack coverage, and insufficient consideration of computational efficiency under UAV-specific constraints. Considerations regarding computational efficiency are non-trivial, independent of the IDS deployment location within the UAV system.

While not inherently a limitation, prior studies differ substantially in their evaluation objectives, especially with respect to whether models are assessed exclusively on known attack types or also on attacks not encountered during training. Considering the diverse threat landscape UAV systems face, zero-day detection represents an important research direction. Accordingly, prior work has explored zero-day detection primarily through unsupervised learning approaches, but also within supervised learning frameworks, where models are evaluated based on their ability to generalize from limited attack knowledge to previously unseen attack classes.

To better position the contributions of this work within the existing literature, Table 2 provides a structured comparison of the representative studies presented above. The comparison considers four key evaluation dimensions:

1. whether detection of unseen attacks is explicitly evaluated,
2. whether multi-dataset evaluation is performed,
3. whether multiple attack types are covered, and

4. whether computational efficiency or resource-related performance metrics are reported

Entries marked with "(X)" in the first column of Table 2 indicate that the authors describe their proposed method as capable of detecting attacks not encountered during training, but do not provide explicit experimental validation for it.

Reference		**Unseen Attacks Evaluation**	**Multi-Dataset Evaluation**	**Multiple Attacks Coverage**	**Computational Efficiency Reporting**
Whelan et al.	[33]	X			X
Hadi et al.	[18]	(X)		X	
Tlili et al.	[31]	(X)	X	X	
Cordeiro et al.	[8]	(X)		X	
Hassler et al.	[15]	X		X	
Fraser et al.	[12]	X			
Al-Haija and Al Basawi	[2]	(X)		X	X
Park et al.	[24]	X		X	
Hadi et al.	[13]	X		X	
Alheeti et al.	[5]	X			
Da Silva et al.	[9]	X		X	
Nayfeh et al.	[21]				X
This work		X	X	X	X

Table 2: Overview of related work with respect to unseen-attack detection, dataset diversity, and computational efficiency.

3 Methodology

Three publicly available UAV datasets from real-flight deployments were selected, each capturing complementary aspects of UAV operation, communication, and attack behavior:

- Unmanned Aerial Vehicles Network Intrusion Detection Dataset (UAV-NIDD) [13],
- Cyber-Physical Dataset for UAVs Under Normal Operations and Cyberattacks [15],
- UAV Attack Dataset [34]

Although the datasets differ in terms of data modality and abstraction level, none of them represent generic network or sensor datasets independent of UAV operation. Each dataset is collected in the context of active UAV flights and is therefore inherently tied to UAV behavior, either through onboard telemetry or through communication traffic associated with UAV command and control. While the network data in the Cyber-Physical dataset relies on standard

wireless protocols, its tight coupling with flight telemetry ensures that it reflects realistic UAV behavior along with the other two datasets.

The chosen datasets enable the evaluation of the proposed approach across multiple experimental setups, attack vectors and deployment perspectives. For example, the datasets cover complementary parts of the UAV attack surface, including attacks targeting the UAV itself as well as wireless communication links between the UAV and external components like the ground control station. The datasets also map to distinct intrusion detection placements: the UAV Attack dataset provides onboard host-based telemetry, the UAV-NIDD dataset offers network-level traffic from wireless links, and the Cyber-Physical dataset combines both modalities for ground-control analysis.

In order to ensure a consistent evaluation across datasets, a unified experimental pipeline was applied throughout this study. Each of the four classifiers was trained and evaluated on all three datasets using an identical sequence of processing steps wherever possible. Dataset-specific preprocessing steps were limited to initial data alignment and consolidation, as well as data cleaning, after which a common pipeline was used for feature selection, model training, and evaluation.

3.1 Models

Among modern machine learning approaches for tabular data, ensemble methods, and in particular gradient boosted machines, have demonstrated consistently strong performance across a wide range of applications. Two widely adopted gradient boosting frameworks, XGBoost and LightGBM, were therefore included in this study, as both have shown near state-of-the-art performance on structured data [11, 35]. Both were configured with relatively shallow trees and a limited number of estimators, ranging from 50 to at most 150 trees depending on the experimental setting, to maintain low model complexity.

Alongside the boosting-based models, a Random Forest classifier was included as a classical and well-established baseline ensemble method, also constrained to a comparable tree count to ensure a fair comparison across architectures.

Finally, to provide a neural network-based comparison point, a compact MLP was implemented. The Multi-Layer Perceptron (MLP) was intentionally kept shallow and lightweight to allow for an assessment of how a simple neural model compares to tree-based ensembles under resource constraints, with a minimal network structure consisting of a single hidden layer. No deep or computationally intensive architecture was pursued to align with the goal of deployability in resource-constrained environments.

Hyperparameter tuning was performed conservatively using a predefined grid search to avoid over-specializing the model to the specific attack type used during training.

3.2 Pre-processing

Data preparation and preprocessing were performed prior to model training and comprised several stages. Label filtering was applied to the UAV-NIDD and Cyber-Physical datasets to ensure stable training and consistent feature representation across classes. In the UAV-NIDD dataset, the reconnaissance attack class was excluded due to having only 6 samples compared to 193,171 in the largest class, which was insufficient for learning a representative attack distribution. For the Cyber-Physical dataset, the false data injection class was removed because its substantially different feature set would have significantly reduced the common feature subset available to all remaining classes. Because the UAV Attack and Cyber-Physical datasets contained streams sampled at different rates, a merging and alignment step was required. A unified

timeline was created from the union of all timestamps, and missing values were forward-filled. Segments of the merged dataframe lacking overlapping streams were then trimmed. Finally, missing or corrupted entries were removed, and the remaining labels were encoded in a binary format (attack/no-attack).

The resulting datasets include a diverse range of threats, including Denial-of-Service (DoS), Man-in-the-Middle (MITM), jamming, replay, and spoofing attacks, among others.

3.3 Feature Selection

For feature selection, filter-based methods were employed due to their computational efficiency and interpretability. A variance-threshold filter removed non-informative features exhibiting low or zero variance, as such features contribute no discriminative power for classification. Pearson correlation, which quantifies the linear relationship between two variables, was applied to eliminate redundancy. Variables with a correlation coefficient exceeding 0.85 were identified as highly correlated, and among each correlated pair, the feature with the lower variance was discarded.

3.4 Training and Evaluation

Although commonly associated with unsupervised learning, zero-day detection can also be achieved within a supervised learning framework by leveraging a model's ability to generalize beyond the specific attack patterns observed during training (e.g., [13, 15]). In this leave-one-attack-out setting, the model is trained using labeled data but evaluated on attack types that were not included during training.

A fixed 80/20 split was first applied to the benign samples to obtain a benign portion for the training and test set. The benign split is performed once and kept consistent across all experiments to ensure comparability between training scenarios. Models are trained on the training portion of the benign samples and all samples of one attack class A_i. Then, they are tested on the benign test samples and all samples of the attack classes that have not been used for model training ($A_j, j \neq i$). This process is repeated until every attack class has been used as training attack once.

3.5 Further Feature Reduction

In addition to training models using the feature set obtained after variance- and correlation-based filtering, further experiments were conducted using further reduced feature subsets. This was achieved using mutual information (MI), which quantifies the statistical dependence between two variables; in this case, an individual feature and the class label. By computing mutual information scores, the remaining features were ranked by how strongly they are associated to the target variable. Then, training was repeated using increasingly smaller subsets of the k highest ranked features, with k ranging from small subsets to the full feature set ($k \in \{8, ...,$ full set$\}$)..

Mutual information scores were computed exclusively on the training data, consisting of the benign training samples and all samples of the attack class used for training to ensure that feature relevance estimation remains strictly confined to the information available during training and avoid leakage from unseen attack classes. For each value of k, models were trained and evaluated using the same evaluation protocol described perviously.

The goal of these experiments was to progressively reduce input dimensionality while evaluating the extent to which zero-day detection performance could be maintained. Reducing the

number of required features has practical relevance in resource-constrained UAV environments, as it reduces the dimensionality of the input representation, limits dependency on a large number of features, and supports the design of simpler, lower-complexity, and more deployable intrusion detection models.

4 Results

The UAV-NIDD dataset was selected as the primary evaluation dataset and was used for in-depth detection performance analysis, feature subset experiments based on mutual information ranking, and computational efficiency assessment. To complete this detailed evaluation, the Cyber-Physical dataset and UAV Attack dataset were employed to examine the robustness and consistency of model behavior across different data sources.

Results and confusion matrices are shown for a representative run with random seed 42. To assess the impact of model stochasticity, all experiments were repeated with five different random seeds. Across these runs, the mean and standard deviation of performance metrics were computed while keeping the data splits fixed.

Overall, the observed variance across random initializations was low for the majority of attack types, particularly for the tree-based ensemble models. In most cases, standard deviations remained within a few percentage points, indicating that the reported results are robust and not dependent on a specific random seed. Importantly, the relative performance trends across attacks and models remained consistent across all runs.

4.1 Baseline Detection Performance with Full Feature Set

Firstly, baseline classification performance was evaluated using the full feature set. This established a reference point against which the impact of subsequent feature reduction could be assessed.

Table 3 summarizes detection performance for each model under the training attack that yielded the strongest generalization to unseen attacks. Precision, recall, and F1-score are reported for the attack class.

Model	**Seen Attack**	**Recall**	**Precision**	**F1-score**	**Accuracy**
Random Forest	MITM	0.8465	1.00000	0.9169	0.8473
XGBoost	Fake Landing	0.9715	0.99996	0.9855	0.9716
LightGBM	Jamming	0.9785	0.99997	0.9891	0.9786
MLP	Fake Landing	0.9214	0.99993	0.9591	0.9217

Table 3: Baseline detection performance for the best-performing training attack using the full feature set.

Precision values for the attack class reached 1.00 or close to 1.00 across all models. This behavior is largely due to the extreme class imbalance in the test sets, where attack samples outnumber benign samples by several orders of magnitude.

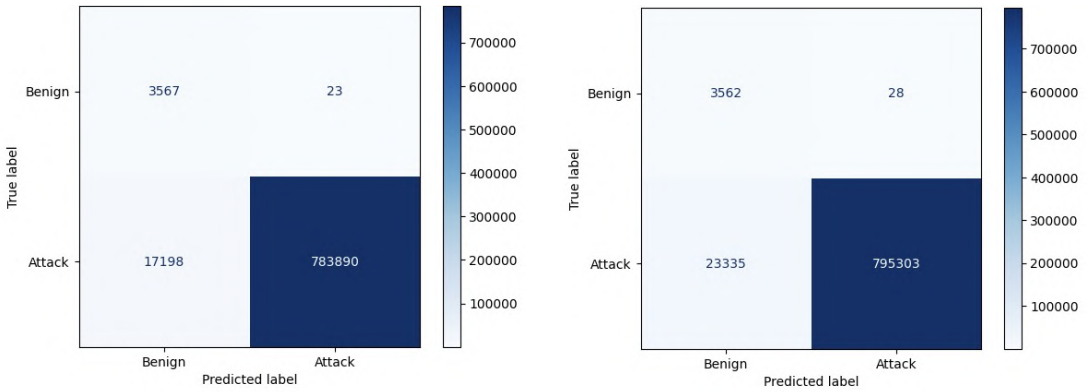

Figure 1: Confusion matrices for LightGBM trained on the jamming attack (left) and XGBoost trained on the fake landing attack (right) when evaluated on unseen attacks.

Figure 1 shows the confusion matrix for the best-performing model, LightGBM, when trained on the attack that yields the best detection performance. Consistent with the aggregate metrics, LightGBM exhibited a substantially lower number of false negatives compared to the Random Forest classifier and the Multi-Layer Perceptron, as shown in Table 2. Among the evaluated models, LightGBM achieved the highest attack F1-score (0.979). XGBoost, the second boosting-based model, attained a slightly lower attack F1-score (0.972). For comparison, Figure 1 also shows the corresponding confusion matrix for XGBoost. For both models, XGBoost and LightGBM, a non-negligible number of false negatives remained, indicating that some attack samples remain difficult to detect.

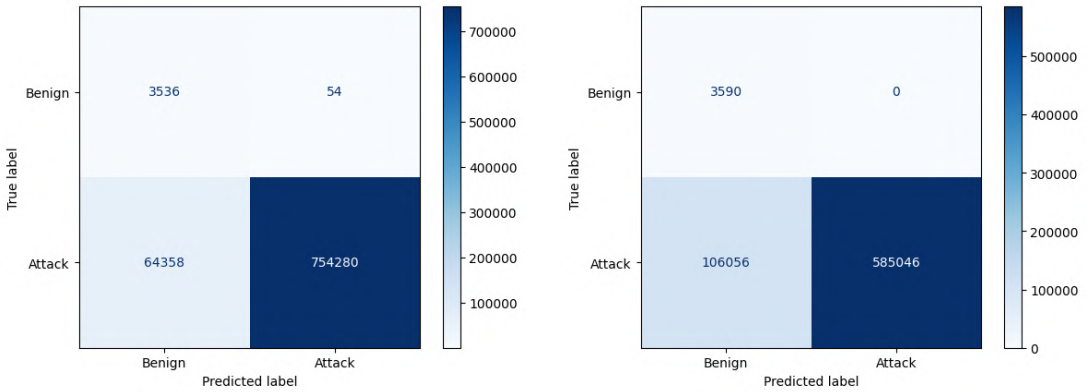

Figure 2: Confusion matrices for the MLP trained on the fake landing attack (left) and Random Forest trained on the MITM attack (right) when evaluated on unseen attacks.

When comparing confusion matrices, it should be noted that while the number of benign test samples is identical across models, the number of attack test samples varies depending on which attack is used for training, and thus absolute false negative counts should be interpreted in the context of the corresponding test size.

4.1.1 Impact of Training Attack Choice

Detection performance is not determined by model choice alone, but also by the attack used for training. Figure 3 summarizes attack class F1-scores across all evaluated models with respect to the attack used for training, with attacks sorted by mean performance in descending order.

The heatmap reveals several patterns. First, certain attacks, such as man-in-the-middle and fake landing, enabled strong generalization to unseen attacks across all models. Other attacks show a more model-dependent behavior. For example, jamming yielded near perfect performance for the boosting-based models while resulting in substantially lower performance for the Multi-Layer Perceptron, and poor performance for the Random Forest classifier. Similar, though less pronounced, patterns are observed for the scanning attack. Among all models, the Multi-Layer Perceptron displayed the most stable performance across attacks, with fewer extreme performance variations and a smaller number of training attacks resulting in F1-scores below 0.8.

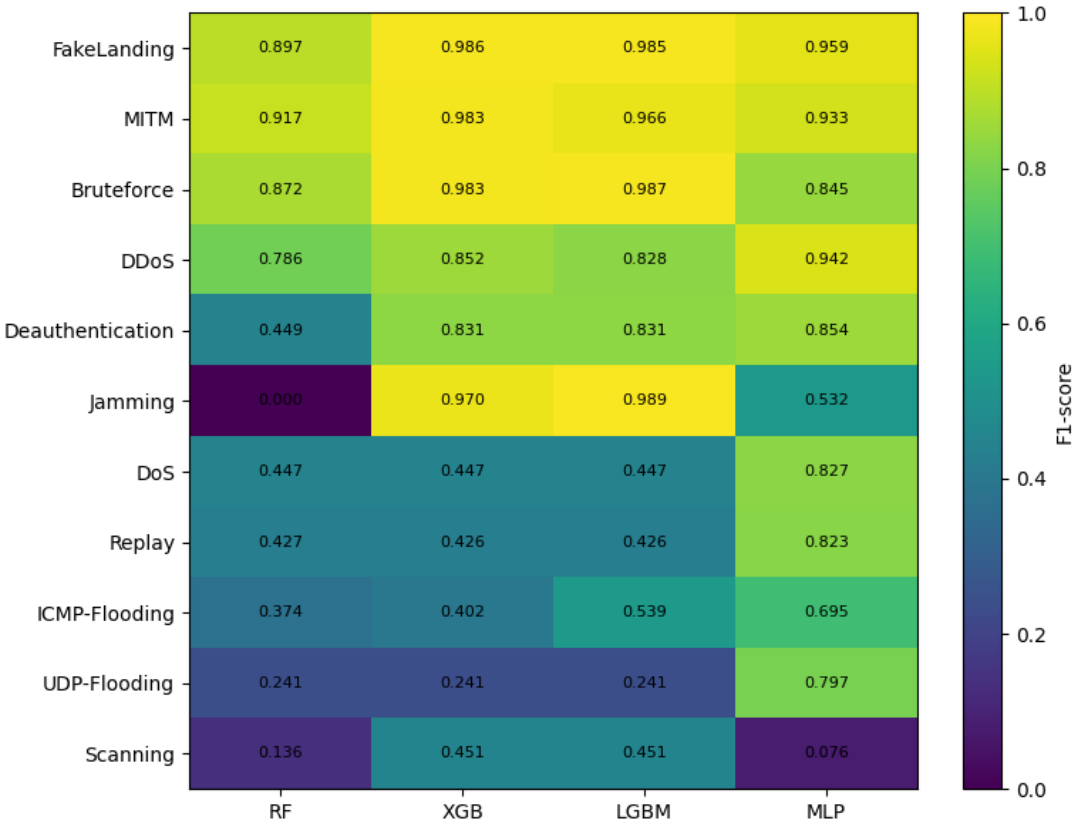

Figure 3: Detection performance (attack class F1-score) for each model as a function of the attack used for training.

4.2 Detection Performance on Feature Subsets

To assess whether further feature reduction is possible without degrading detection performance, experiments were repeated using subsets of the top k features ranked based on mutual information.

Table 4 shows the results of these subset experiments. The best detection performance achieved by each model is presented, along with the corresponding training attack and the number of retained features.

Model	Seen Attack	k	F1-score	Recall
Random Forest	MITM	15	0.9266	0.8632
XGBoost	Jamming	8	0.9935	0.9872
LightGBM	Jamming	18	0.9891	0.9785
MLP	Fake Landing	14	0.9970	0.9941

Table 4: Best detection performance after MI-based feature selection for each model, showing the training attack along with the number of retained features k that yield the highest F1-score.

After feature reduction, the Multi-Layer Perceptron achieved the highest overall detection performance, outperforming both XGBoost and LightGBM. XGBoost reached near-optimal performance with the smallest feature subset.

Compared to Hadi et al. [13], who report an F1-score of 0.9718 on the same dataset using a feed-forward neural network, this best model achieves an F1-score of 0.9970.

4.3 Evaluation of Computational Efficiency

Computational efficiency was evaluated for the model instance that achieved the strongest detection performance. Inference latency and throughput were therefore measured for the corresponding training attacks reported in Table 3.

All classifiers were assessed using the full feature set from the standard preprocessing pipeline to ensure fair comparison across architectures. Measurements were conducted on an Apple M3 CPU system after a warm-up phase, capturing both single-sample latency (batch size 1, approximating online detection) and batch latency (batch size 1024, reflecting vectorized execution). Reported values reflect model inference time only, excluding feature extraction and data acquisition, thus representing lower-bound estimates for real-world deployment. Results are summarized in Table 5.

Although computational efficiency was evaluated only on the UAV-NIDD dataset, the resulting measurements provide a representative estimate across all three datasets. The evaluated models share comparable architectures and operate on dense numerical feature vectors with similar dimensionality and identical preprocessing pipelines.

Overall, the tree-based boosting models as well as the MLP exhibited substantially lower inference latency than the Random Forest classifier. The Multi-Layer Perceptron achieved the lowest latency and highest throughput, making it the most computationally efficient model. XGBoost and LightGBM also demonstrated strong efficiency, with inference times well below one millisecond per sample. In contrast, the Random Forest classifier showed significantly higher single-sample latency.

Model	Batch (μs/sample)	Single-sample (μs/sample)	Throughput (samples/s)
Random Forest	13.68	14564.56	73,095
XGBoost	0.32	188.63	3,102,638
LightGBM	0.74	411.42	1,343,612
MLP	0.18	44.83	5,612,370

Table 5: Comparison of per-sample inference latency (measured in batch mode and single-sample mode) and throughput.

4.4 Cross-Dataset Evaluation

Model generalization was tested on the Cyber-Physical and UAV Attack datasets to determine whether performance transfers across different UAV platforms, data modalities, and attack scenarios.

4.4.1 Cyber-Physical Dataset

Table 6 summarizes the results for each model. Precision, recall, and F1-score are reported for the attack class.

Model	Seen Attack	Recall	Precision	F1-score	Accuracy
Random Forest	DoS	0.512	0.999	0.677	0.580
	EvilTwin	0.000	0.000	0.000	0.098
	Replay	0.607	1.000	0.755	0.663
XGBoost	DoS	0.409	0.984	0.578	0.486
	EvilTwin	0.157	1.000	0.271	0.239
	Replay	0.845	0.999	0.916	0.867
LightGBM	DoS	0.528	0.988	0.688	0.588
	EvilTwin	0.158	1.000	0.273	0.240
	Replay	0.844	0.998	0.915	0.865
MLP	DoS	0.445	0.990	0.614	0.519
	EvilTwin	0.028	0.993	0.055	0.123
	Replay	0.859	0.996	0.923	0.877

Table 6: Cyber-Physical dataset detection performance for all training attacks.

Overall, the Cyber-Physical dataset exhibited less variability in detection performance with respect to the choice of training attack compared to the UAV-NIDD dataset. As shown in Table 6, all models consistently achieved their best performance when trained on the replay attack, while the same training attack (evil twin) also yielded the weakest performance for all models. In contrast to the UAV-NIDD dataset, differences between the boosting models and

the Multi-Layer Perceptron were minimal.

Among the evaluated models, the MLP achieved the highest performance, with an F1-score of 0.923. The corresponding confusion matrix is shown in Figure 4. The overall detection performance remained noticeably lower than that achieved on the UAV-NIDD dataset.

In comparison, Hassler et al. [15] report that their best-performing 1D-CNN achieved an F1-score of 0.803 under comparable single-attack training conditions. Notably, they report improved performance (up to 0.961 F1-score) when training on multiple attacks simultaneously, suggesting multi-attack training could further enhance generalization.

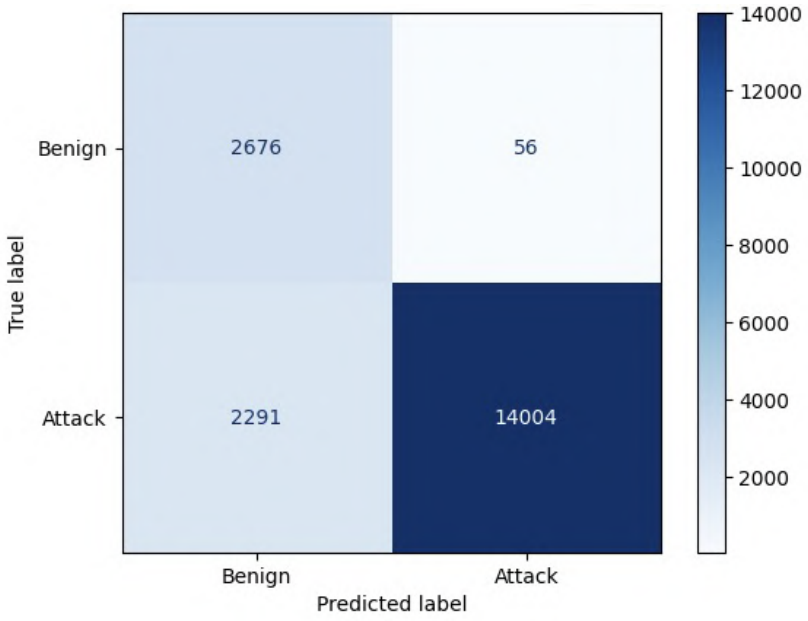

Figure 4: Confusion matrix of the Multi-Layer Perceptron for the best-performing training attack (Replay).

4.4.2 UAV Attack Dataset

Table 7 presents the detection performance of all evaluated classifiers. Precision, recall, and F1-score are reported for the attack class.

Across all models, perfect performance was observed when trained using the spoofing attack, with all classifiers achieving an accuracy and F1-score of 1.00. In contrast, performance when trained on jamming exhibited noticeably larger variability across model architectures, ranging from strong detection performance (e.g., XGBoost) to substantially weaker results (e.g., LightGBM and MLP).

Although these results demonstrate strong empirical performance, the perfect spoofing detection across diverse classifiers suggests that the dataset may contain features that strongly separate benign and attack conditions, potentially inflating performance metrics. Given the limited number of attack classes in the dataset, it remains unclear whether these findings generalize to other attack types beyond those considered.

For this same dataset, Da Silva et al. [9] report a maximum F1-score of 0.929 using a

Model	Seen Attack	Recall	Precision	F1-score	Accuracy
Random Forest	Jamming	0.7829	1.00	0.8782	0.8380
	Spoofing	1.0000	1.00	1.0000	1.0000
XGBoost	Jamming	0.9709	1.00	0.9852	0.9783
	Spoofing	1.0000	1.00	1.0000	1.0000
LightGBM	Jamming	0.0000	0.00	0.0000	0.2536
	Spoofing	1.0000	1.00	1.0000	1.0000
MLP	Jamming	0.1523	1.00	0.2640	0.3662
	Spoofing	1.0000	1.00	1.0000	1.0000

Table 7: UAV Attack dataset detection performance for all training attacks.

stacked autoencoder, treating both spoofing and jamming as a single attack class. Whelan et al. [33] report maximum F1-scores of 0.841 for spoofing and 0.914 for jamming using one-class classifiers.

5 Conclusion and Future Work

Overall, this work demonstrates that effective zero-day intrusion detection for UAV systems can be achieved using compact, low-complexity machine learning models, and that high accuracy can be attained alongside computational efficiency. Across all three datasets, several classifiers achieved meaningful generalization to unseen attacks, indicating that learning from a single known attack can yield transferable detection behavior. Performance varied by dataset—strongest on UAV-NIDD and UAV Attack, somewhat reduced but competitive on Cyber-Physical—reflecting differences in data modalities rather than fundamental limitations of the approach. Inference latency measurements confirmed that three of four models (MLP, XGBoost, LightGBM) operated well below one microsecond per sample under CPU-only execution, supporting feasibility for resource-constrained deployment.

Despite the promising results, some limitations should be acknowledged. First, the evaluation was conducted using offline datasets, and real-time deployment aspects, such as data acquisition, feature extraction latency, and other factors contributing to computational overhead, were not implemented in a live system. Although model inference latency was shown to be minimal, these additional components may introduce non-negligible runtime costs. Furthermore, all computational efficiency measurements were performed on relatively high-performance desktop hardware. Inference latency on resource-constrained UAV onboard hardware may therefore differ.

A further limitation concerns the availability and diversity of publicly accessible UAV intrusion detection datasets. Although three complementary datasets were evaluated in this work, two of them contain only a limited number of attack scenarios, which constrains the extent to which conclusions can be generalized to the broader UAV threat landscape.

Lastly, an even more rigorous evaluation strategy would involve cross-dataset testing, in which a model is trained on one dataset and evaluated on a separate dataset. Such an approach could further mitigate the influence of dataset-specific artifacts and provide stronger evidence regarding real-world deployment robustness.

5.1 Future Directions

This work points to several directions for future research. Incorporating multi-attack strategies is a natural direction for future work, as exposure to multiple attack types is intuitively expected to improve the detection of unseen attacks and has also been supported by prior work. In particular, such strategies may help reduce the remaining false negatives observed for several models.

From a systems perspective, implementing the proposed intrusion detection pipeline in a real-time UAV or ground-control environment would allow for end-to-end performance evaluation, including feature extraction, synchronization, and other computational overhead.

Beyond extensions of the presented work, future research would benefit from the establishment of more standardized evaluation benchmarks. In particular, the inclusion of computational efficiency metrics, such as inference latency, throughput, and resource consumption, would enable more meaningful comparisons between proposed methods. In parallel, continued efforts to improve dataset quality and diversity are essential to support robust model development and realistic experimental evaluation.

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Unsupervised Root-Cause Analysis in Multi-System Logs Using Embedding-Based Event Graphs*

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Abstract

Root-cause analysis (RCA) represents a challenge in security operations. This is because the evidence that explains the incident can be found in different (heterogeneous) systems (e.g., VPN/RADIUS, firewalls, host authentication), with different schemas, vocabularies, and timestamp precision. We investigate unsupervised RCA and frame it as a ranked log-evidence retrieval under realistic constraints: no labels, limited reliance on sensitive identifiers, and high background noise. We propose a pipeline that starts with an anchored alert and builds a temporal event graph where nodes are log events and edges encode plausible continuity. We compare a TF–IDF baseline against a self-supervised contrastive representation model trained on masked log templates with heuristic positive-pair mining. Also, we evaluate on a manually reconstructed "golden" SSH brute-force episode and report set-based and ranking-aware metrics, plus reachability diagnostics to separate connectivity from prioritization failures. Results show that simple continuity assumptions can make ground-truth evidence fully reachable, while precision is the main bottleneck due to noise. Contrastive learning has modest set-level gains, but improves early ranking of the core causal chain, suggesting learned representations help prioritization more than coverage in this setting.

1 Introduction

In the area of IT operations and cybersecurity, the reliability and security of services increasingly depend on complex distributed infrastructures. Multiple components can be involved in the cause of an incident (perimeter firewalls, identity and access management systems, endpoint agents, etc) [1]. Logs are a crucial source of evidence for analyzing faults and investigating the reasons that caused a security incident, but manual correlation can be slow and prone to error due to their heterogeneity and large volume [2].

1.1 What is Root Cause Analysis and How It Helps

Root-cause analysis (RCA) is the approach selected for this reconstruction of events in this work. In comparison with anomaly detection, where the goal is to 'flag' abnormal events, RCA's goal is to select and prioritize events that are most likely to explain how an incident occurred [3]. RCA narrows the search space to the events that are actually relevant, exposing relationships among them, which then helps remediation. For security incidents such as SSH brute-force

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compromises, RCA is especially valuable, and this is why it was chosen as the case study for this research: the causal chain frequently involves connection activity at network boundaries (e.g., firewall telemetry), authentication decisions (e.g., RADIUS logs), and the final host-level success event (e.g., an sshd login on a Linux system) [4]. Without systematic correlation, the investigator is left with an expensive process of searching, filtering, and mentally correlating evidence across systems.

One of the main challenges of multi-system RCA is the automation of the process. A large amount of proposed solutions is still rule-based or relies on annotated labels [5]. This approaches can be effective in narrow domains, but they still need a considerable amount of expertise on the domain and do not usually generalize across different vendors or environments. Labels are costly and usually unavailable in real-world scenarios as incidents are rare, diverse, and can have sensitive information. These limitations motivate a complementary line of work: unsupervised RCA, which does not use labeled data and does not rely on rules [6].

1.2 Problem Statement and Gaps in Research

One of the main obstacles in unsupervised RCA across different systems is semantic misalignment among logs from different sources [5]. Different systems can describe events that are related using different schemas and/or vocabularies. For instance, a firewall can log a connection that was allowed and the host can log a successful login. These events, even if they were caused by the same attacker and led to the same incident, the messages from the logs from both of the systems can appear unrelated at the surface level. There are sparse representations that can capture word overlap in an efficient way, like bag-of-words TF–IDF, but they still lack of cross-domain generalization [7].

This motivates the use of representation learning, particularly embedding methods that can map heterogeneous events into a shared vector space where semantically related events are closer together. However, applying representation learning in a fully unsupervised, multi-system RCA setting has its own difficulties: the training signal has to be computed without labels, the representations must avoid leaking sensitive identifiers, and improvements must be not only a better similarity structure but an improved root-cause retrieval, which translates into actual relevant logs appearing early in the explanation [8, 9].

1.3 Proposed Solution

We investigate embedding-based event graphs as the representation for heterogeneous logs in unsupervised RCA to address the challenges mentioned. The nodes in the graph are log events and the edges represent plausible relationships among these events based on shared actor and host contexts.

Within this context, we study whether contrastive representation learning could improve the prioritization of relevant events across systems in comparison to a sparse-text baseline. This baseline uses a text representation of log messages where they have been anonymized and templated, and then performs a simple anomaly scoring. Afterwards, it performs an alert-anchored traversal to retrieve a ranked set of candidate events. Our aim is to analyze how this baseline can be improved with contrastive learning over masked log templates in order to learn embeddings. The goal is not to compete directly with supervised state-of-the-art methods, but to answer a more operationally relevant research question: *When labels and handcrafted rules are unavailable, can learned representations improve unsupervised cross-system RCA beyond what can be done with simple lexical features?*. In this work, we use a specific security scenario:

SSH brute-force attack episodes collected across multiple streams on Copenhagen University's *Elasticsearch* platform.

We have three main contributions. We propose a data pre-processing pipeline that normalizes heterogeneous *Elasticsearch* logs into a canonical form, anonymizes sensitive identifiers (IPs, usernames, connection IDs), and produces structured and textual fields. Also, we develop a baseline that combines sparse-text features with temporal graph traversal, serving as a strong reference point to understanding what can be achieved without labels or handcrafted correlation rules. Finally, we perform an investigation on contrastive learning for semantic correspondence across different systems and analyze how different design choices led to different outcomes: positive-pair mining strategies, anomaly scoring choices, and temporal graph connectivity parameters.

1.4 Research Questions

This work has three main research questions:

- **RQ1:** How effectively do embedding-based event graphs represent multi-system security incidents for unsupervised RCA, and what connectivity assumptions are required to make root-cause evidence reachable from an alert anchor?
- **RQ2:** Does contrastive learning on masked log templates improve cross-system semantic alignment and evidence retrieval compared to a TF–IDF baseline?
- **RQ3:** Which design choices are the ones that determine the most the RCA performance and failure modes in real-world log data?

These questions make this work focus on performing an investigation on how event graphs need to be designed in order to have a plausible incident narrative, and under which assumptions the contrastive model outperforms the baseline, and why.

2 Related Work

Log-based RCA has become more relevant as modern infrastructures generate big amounts of telemetry data across different and heterogeneous systems. This heterogeneity makes manual correlation slow and, therefore, motivates automated RCA methods that can reconstruct the context that led to an incident [10, 11].

Prior work covers different settings and types of outputs. A considerable amount of AIOps RCA focuses on component or metric localization, producing a list of candidate services/metrics [12]. In contrast, log-based RCA focuses on retrieving explanatory log evidence. This work falls into the latter category: given an alert-centered episode, it performs a ranked evidence retrieval with an anchored alert over a temporal event graph built from logs from different systems.

This chapter is organized around three main axes: (i) the supervision regime (supervised, weakly supervised, unsupervised/self-supervised), (ii) the representation used for correlation (lexical features versus graph/temporal structures and learned embeddings), and (iii) the output unit (component/metric localization versus log-evidence retrieval).

2.1 Anomaly Detection vs. Root Cause Analysis

Anomaly detection and RCA are related, but perform different tasks in operational analytics [10]. Anomaly detection focuses on identifying events that are not usual, typically by learning

patterns of normal behaviour and then reporting anomalies, as in *Deeplog*, from Du et al. [13]. It has also been studied the use of transformers and self-supervised approaches to improve the quality of anomaly detection, for example, by learning contextual embeddings of log sequences in *Logbert*, from Guo et al. [14]. However, anomaly detection does not necessarily explain why an incident occurred. This is the core objective of RCA.

Previous works blend anomaly detection and RCA. For example, Wang et al. [15] use a pipeline with two stages: first, they detect anomalous events (or time windows), and then they rank candidates that led to the anomalies. This work also follows this two-stage pipeline, but with a different output unit (a ranked list of evidence items instead of a ranked list of metrics).

2.2 Log-Based RCA

Log-based RCA methods focus on retrieving log events that explain the cause of an incident. In this section we explain the two first axes around which this chapter is organized.

2.2.1 Supervision regime

The supervision regime used is one of the main dimensions of this chapter. Different supervision regimes have been used to achieve the explanation of an incident on a log-based design.

Unsupervised log-based RCA. There are several works that focus on producing explanations for incidents using logs without relying on labels. *Logrule*, from Notaro et al. [16], performs RCA on structured logs by investigating patterns that explain abnormal behaviour, but it does not build an explicit event graph to achieve this goal. Other approaches, like Lu et al.'s Spark work [17], use log features and custom heuristics to detect anomalies and identify likely causes, also lacking a temporal event graph.

These methods show that RCA can be performed without full supervision. However, they are either (i) domain-specific, or (ii) evaluated on settings where the inputs have a fixed structure instead of using multi-stream data where vocabularies and schemas differ due to vendor variability.

Weakly supervised and PU log-evidence ranking There are works where logs are the central part that use weak signals that are more available than event-level root-cause annotations. *LogLAB*, by Wittkopp et al. [18], even if it does not do RCA, envisions it by progressing from anomaly detection towards automated log labeling: from an incident window, it trains attention-based models so logs that are more likely to be the cause of abnormal behaviour are highlighted. Following this direction and applying RCA, *LogRCA* [10] uses Positive–Unlabeled learning, where the training data has some logs labeled as "positives" (relevant to incidents) and the remaining logs are not labeled. It uses this weak supervision to train models that rank log lines by how likely they are to be the cause (or one of the causes) of a fault.

These approaches are relevant because they have the same output unit as this work (ranked log evidence). However, they assume (i) a setting where logs can be correlated using stable, non-sensitive identifiers and platform metadata, and (ii) an environment that is semantically coherent in which log schemas and vocabularies are consistent, and cross-vendor or cross-layer semantic alignment is not one of the main challenges.

Regarding the first limitation, in this work's investigation the strongest correlators are often sensitive data (IPs, usernames, connection/session IDs). Therefore, they are masked or excluded to prevent shortcut learning and leakage. Regarding the second one, we address the problem

of semantic misalignment due to vendor-specific vocabulary. This motivates the self-supervised approach selected: the system learns cross-stream representations by correlating event templates using self-supervised positive pairs mined from anonymized actor/host continuity. Importantly, these continuity cues or other IDs are never part of the text embedding input, so the contrastive model encourages semantic alignment across streams instead of memorizing identifiers.

2.2.2 Representation used for correlation

Another dimension of this chapter is the representation used for cross-system correlation and ranking in log-based methods. What baselines like TF–IDF lack, while still useful and interpretable, is the capacity of capturing semantic similarity beyond overlap of tokens. Works like *LogBERT* [14] use transformer-based methods that learn embeddings that capture surrounding meaning of log sequences utilizing self-supervision. Surveys show that there is a tendency to move towards deep/self-supervised representations in order to assess the issue of evolving formats and to reduce feature engineering [19]. In heterogeneous telemetry, differences in vocabularies and/or schemas across different systems or vendors, which make them an unrealiable similarity flag, motivate the use of these representations.

For these reasons, this work uses self-supervised contrastive learning with *heuristic positive mining* and leakage control, which is especially relevant in security telemetry. The main objective is not to explicitly outperform benchmark systems, but to test if this model improves the prioritization of evidence that explains an incident in comparison to TF–IDF baselines.

2.3 Graph- and Representation-Based Correlation for RCA

Graphs are widely used in RCA because they explicitly represent dependencies and make it possible to perform a structured reasoning. This subsection shows representative graph-based approaches in AIOps where the output is a ranked list of services or metrics that are likely responsible for an incident. *CIRCA*, by Li et al. [12], performs microservice RCA using causal inference over service telemetry. *CausalRCA*, by Xin et al. [20], constructs causal graphs over metrics and ranks candidate root-cause metrics. *ART*, by Sun et al. [21], performs unsupervised RCA also ranking root-cause services/components. *CHASE*, by Zhao et al. [11], performs anomaly detection and RCA by building a causal hypergraph over multimodal microservice telemetry (logs, metrics, traces) and then ranking these service instances to localize the potential root-cause metrics.

These methods are important because they show strong design ideas (how to build the graph, how to reason causal links), but they have a different output unit from the one in this work. Component/metric localization assumes that the explanation for an incident is formed by one component/metric. Therefore, the mentioned in this section work normally uses AC@k or Avg@k for evaluation. In this work, the explanation is composed of multiple log events, so the most useful evaluation metrics are based on early precision, recall, and F1-score, although complementary metrics are also used.

Another difference is the information used to define the graphs' edges. In microservice settings, graphs often encode causal dependencies among services/metrics, like in *MicroRCA* [22]. In SOC telemetry, these dependency graphs are not usually available, so correlation has to be based on operational continuity cues. For this reason, the event graph in this work uses log events as nodes and has edges that encode actor- and host-centric temporal continuity.

This choice motivates an evaluation protocol that is not emphasized in most of the prior work: separating (i) connectivity failures, where relevant evidence is not reachable from the

alert under the assumed graph design choices, from (ii) prioritization failures, where evidence is reachable but is ranked with a too low suspicious score.

2.4 Evaluation Protocols

Works in the first table evaluate the ranked retrieval of logs, so they usually report Recall@k and Precision@k. Works in the latter typically assume that the explanation of an incident is a small root-cause set, so they report AC@k/Avg@k. We primarily follow the evaluation protocol of the first table.

Another issue to take into account is that the construction of the ground truth differs among settings. Benchmarks where faults are injected provide clean labels by design [20], while in real incidents there are ambiguous boundaries, duplicated log lines, and heterogeneous precission in timestamps. Therefore, we (i) manually reconstruct the ground truth using *Elasticsearch* and *Kibana*, (ii) make a distinction between *core* and *extended* evidence sets, (iii) report explanation sizes and analytics for both node-level and evidence-level (deduplicated) retrieval, and (iv) add diagnostics regarding reachability to separate limitations on connectivity and ranking quality.

3 Methodology

This chapter describes the methodological design of the thesis: how multi-system log data is collected and structured into episodes, how a manually reconstructed *golden episode* is created as ground truth, how the components are trained, and how the resulting RCA outputs are evaluated on a real scenario attack.

3.1 Overall Approach

The steps followed were:

1. **Selection of a real alert and definition of a golden episode:** We select one real *Potential Successful SSH Brute Force Attack* alert and extract logs from a fixed window of $[-27, +1]$ minutes around the alert timestamp. From here, we manually reconstruct the *golden episode*, which serves as ground truth and will be evaluated after running the pipeline.

2. **Collection of additional episodes for unsupervised training:** We collect multiple brute-force related alerts and extract the same fixed 28-minute window around each alert. To keep the episodes realistic, we extract both attack-related logs and background logs. Due to memory constraints, we sample streams with high volume while retaining as much as possible from the streams that form the incident "spine" (ISE/ASA/SSH host authentication).

3. **Training of unsupervised/self-supervised components:** We train (i) an unsupervised anomaly scoring model (baseline) and (ii) a self-supervised contrastive representation model. Both are trained only on non-golden episodes, without labels.

4. **RCA on the golden episode and evaluation:** We run the pipeline on the golden episode and evaluate how well the produced explanation recovers the manually reconstructed causal chain under strict (*core*) and tolerant (*extended*) ground truth definitions.

3.2 Manual Construction of the Ground Truth Causal Chain

To evaluate the outcome, we reconstruct the causal chain that led to a *Potential Successful SSH Brute Force Attack* alert that occurred on October 16th 2025 at 09:52:51.221. We use symbolic names for the IPs, user names and host names for privacy. The attacker internal IP is denoted as *IP1*, the attacker user name is denoted as *username*, the ssh host IP is denoted as *IP2*, and the host name is denoted as *hostname*. The reconstruction is based on (i) temporal ordering and (ii) entity continuity across systems.

The resulting ground truth narrative consists of four logical stages:

1. **ISE_VPN_ASSIGN.** Cisco ISE provides evidence that a VPN session is active and that the attacker was assigned an internal IP *IP1* during the incident window. In practice, RADIUS accounting *Interim-Update* events are beacons that inform about session-presence, used to confirm that the user with IP *IP1* had started a VPN session before performing the attack [23].

2. **ASA_FLOW_CREATION / ASA_FLOW_EXPIRATION.** Cisco ASA firewall logs record the lifecycle of TCP flows. In this setting, they prove that that connections from *IP1* to the SSH host (*IP2*, port 22) have been explicitly allowed. To validate this theory, we reconstruct TCP/22 flow intervals by correlating *flow-creation* and *flow-expiration* records. We then check that SSH authentication events from *IP1* happened within the period of time in which the firewall flow was active. Because ASA logs have millisecond precision while SSH logs are second-granularity, we apply a small tolerance around session boundaries [24].

3. **SSH_FAIL.** The SSH host authentication stream provides direct evidence of repeated *sshd* authentication failures with the actor IP being *IP1*. These events form the brute-force phase of the incident chain [25].

4. **SSH_SUCCESS.** After multiple failed authentication attempts, there is a single successful *sshd* authentication from IP1. This success event, along with the previous fails, is the cause of the alert and serves as the most direct host-level indicator of the incident in the episode [25].

Core vs. extended ground truth. As mentioned, we differentiate between two types of *ground truth*: *core* vs. *extended*. The first one the minimal and completely certain causal path that explains the incident, and the latter includes plausible evidence that could belong to the causal chain, but not with 100% certainty.

3.3 Data Extraction

We retrieve the logs for each of the episodes using *Elasticsearch* using Python scripts. We query a fixed time window of 28 minutes (27 minutes before the alert and 1 minute after, when applies) and export the per-stream logs into JSON files.

For feasibility reasons, we apply filters to certain streams to retain the most incident-relevant logs, while keeping the extraction procedure consistent across episodes. Also, several streams generate millions of events within the stated time window. For this reason, we set a fixed budget for each episode of approximately 30,000 events. This makes it easier to manage the data and to compare across episodes.

Streams relevant to the incident (Cisco ASA, Cisco ISE, and authentication system) are prioritized, so we try to keep as much as possible from these streams. After applying stream-specific filters to focus on SSH/VPN/flow evidence, we retain all core events when feasible. If the volume of data is still too high, then we set controlled caps and sample within the stream. The remaining budget is used to include activity from other streams that contribute with background noise at host and network level. To prevent a single noisy stream from dominating the episode, we use equal per-stream quotas for the non-core streams.

3.3.1 Stream Selection per Episode

To evaluate the RCA model under realistic multi-system noise, each episode includes seven *Elasticsearch* data streams:

- **Core streams** $\mathcal{S}_{\text{core}}$: Cisco ASA, Cisco ISE, and the authentication system.
- **Context streams** $\mathcal{S}_{\text{context}}$: Palo Alto firewall, Filebeat shipper, operating-system security logs, and endpoint network telemetry.

3.3.2 Episode Budgeting and Sampling Strategy

Some streams produce millions of logs in just 28 minutes. This makes the process of feature extraction, embedding, and construction of the graph not feasible with the available resources. For this reason, we enforce a budget for each episode of approximately $B = 30{,}000$ logs.

This choice is a data management choice, not a labeling shortcut. We do not use ground truth annotations to sample the data, and we do it based on each stream identity and internal structure.

Core streams. We restrict the *Cisco ASA* stream to SSH-relevant firewall flow evidence (TCP destination port 22/2222) with the requirement of a connection identifier used to build flow intervals. We sample *Cisco ISE* stream using session-aware sampling to avoid a single long VPN session dominating the episode. Regarding the authentication system, we prioritize *sshd* authentication events and keep a bounded amount of events with different processes (e.g., *sudo*, *systemd*) to capture plausible post-authentication activity.

Context/non-core streams. Within each context stream, we use time-stratified sampling across the episode window (minute-level strata) to preserve temporal coverage and avoid concentrating the sample in periods where there is a very high amount of activity.

Episode-level structuring. Raw logs are exported from *Elasticsearch* as JSON files, one for each type of stream, for each 28-minute episode window. Episodes are treated as the unit of training and evaluation, where RCA is performed.

Canonicalization across streams. Events from different streams have different schemas and vocabularies. In order to make them comparable, we convert each raw log into a canonical representation or template that standardizes fields that are common to every stream, and then keeps the ones specific to each one. This representation is used throughout the pipeline to ensure controlled comparisons between baseline and contrastive variants.

Identifier anonymization to prevent leakage. To avoid trivial shortcuts and to make the model less affected by identifiers specific to certain environments, tokens that can potentially identify an actor or process (e.g., raw IP addresses, usernames, firewall connection IDs) are excluded from the templates used as inputs. These identifiers are anonymized and converted into pseudonyms. This enforces that the models have to rely just on behavioral and semantic patterns instead of memorizing specific identifiers.

Duplicate handling at the evidence level. Sometimes, the exports include exact duplicate records. To avoid artificially inflating the size of our explanations and the evaluation counts, we group duplicates together into evidence items.

Outcome. The outcome of this stage is mainly a unified *parquet* file with all of the per-episode structured datasets. The steps to achieve this oitcome are deterministic given a fixed configuration, so experiments can be reproduced.

3.4 RCA Pipelines

We evaluate two RCA pipelines that share the same high-level retrieval procedure but have different event representation used for scoring.

3.4.1 Common RCA Retrieval Procedure

Both pipelines use the same setup for handling incident responses: Starting with an alert and the time window around it, the system has to construct a temporal multi-system event graph, and then produce an explanation as a prioritized list of key events.

- **Inputs:** An episode composed of log events from different streams and an alert timestamp.
- **Graph:** Representation where the nodes are events, and the edges encode temporally plausible continuity relationships.
- **Alert:** The alert is the starting point of the explanation.
- **Retrieval objective:** Returns a ranked list of events that best reconstruct the chain of events that caused the alert to be triggered.

3.4.2 Baseline Pipeline: Lexical Representation + Unsupervised Scoring

The baseline pipeline served as a strong unsupervised reference. Every event was represented using the standardized templates. An unsupervised anomaly detector was trained on non-golden episodes and then tested on the golden episode to assign each event a suspiciousness score. The RCA retrieval procedure then used these scores as the ranking signal for the explanation retrieval.

This baseline answered the question: *How well can an unsupervised lexical model prioritize events relevant to an incident for RCA without any representation learning?*

3.4.3 Contrastive Pipeline: Self-Supervised Semantic Alignment + Unsupervised Scoring

The contrastive pipeline replaces the baseline representation with embeddings trained to improve semantic alignment across heterogeneous streams. Events are also represented using the normalized templates, but from here we first obtain dense embeddings (via dimensionality reduction), and then train a lightweight mapping so that events that are likely part of the same incident context are placed close together in the embedding space, while events that are likely not related are pushed apart.

We train using self-supervised learning by mining *heuristic positive pairs* within episodes. This encourages related events to be closer in the embedding space, while pushing the rest apart. After training, the learned embeddings are used as input to the same type of unsupervised anomaly scoring as in the baseline, producing suspiciousness scores for every event.

This pipeline answers the question: *Does learning a cross-stream semantic space improve the retrieval of a causal chain compared to lexical baselines, under a controlled RCA procedure?*

4 Results

Here we portray the main results of the experiments.

4.1 Experimental Set Up

We evaluate the proposed RCA pipeline as a *ranked retrieval* problem over log-event nodes. Given the episode e, the method outputs an ordered list of nodes (an "explanation") produced by a backward walk from the alert node in the constructed temporal graph.

Notation. Let $R_e = [r_1, r_2, \ldots, r_m]$ be the ranked list of nodes returned by the RCA algorithm for episode e, where r_1 is the highest-ranked node. Let $\hat{R}_e = \{r_1, \ldots, r_m\}$ be the corresponding set of returned nodes (ignoring order).

Different streams have different timestamp precisions (e.g., Cisco ASA flow events with millisecond timestamps versus SSH logs without millisecond information). For this reason, we have two definitions for *ground truth*:

- **Core ground truth** G_e^{core}: the minimal set of logs that are considered essential to explain the incident.
- **Extended ground truth** G_e^{ext}: a set with a larger size that includes the core logs plus additional logs that could reasonably be part of the same causal narrative.

The alert node is excluded from evaluation. It is removed from R_e and $\hat{R}_e$ before computing all metrics.

Set-based retrieval metrics. Let e be an episode, $\hat{R}_e$ the set of retrieved log-event nodes for e, and let G_e denote the ground truth evidence set for e (either G_e^{core} or G_e^{ext} depending on the protocol). We compute precision, recall, and F1-score on the returned set:

$$P_e = \frac{|\hat{R}_e \cap G_e|}{|\hat{R}_e|}, \qquad R_e = \frac{|\hat{R}_e \cap G_e|}{|G_e|}, \qquad F1_e = \frac{2P_eR_e}{P_e + R_e}.$$

We also report the size of the retrieved explanation. Recall can be inflated when many nodes are retrieved, so by reporting this value we can see the explicit trade-off.

Because the RCA algorithm operates on log-event nodes in the graph, we first define the node-level explanation size as:

$$S_e^{\text{nodes}} = |\hat{R}_e|.$$

In addition, because some logs are duplicated in the exported data (same timestamp, data stream, source IP, destination IP, host name, event action, event outcome, event code and destination port), we also report an evidence-level explanation size after collapsing duplicates into a single evidence item:

$$S_e^{\text{evidence}} = |\hat{R}_e^{\text{evidence}}|.$$

Ranking-aware early retrieval metrics. In addition to retrieving the correct nodes, the ranking should output relevant evidence early. For a budget k, let $R_e@k = [r_1, \ldots, r_k]$ be the first k nodes of the ranked output and $\hat{R}_e@k$ its set version. We compute Precision@k, Recall@k and Hit@k:

$$P_e@k = \frac{|\hat{R}_e@k \cap G_e|}{k}, \qquad R_e@k = \frac{|\hat{R}_e@k \cap G_e|}{|G_e|}.$$

We report these values for $k \in \{5, 10, 20, 50\}$, values chosen to reflect realistic inspection budgets.

It is important to mention that several RCA papers report an *accuracy@k* metric (AC@k) for ranked root-cause localization. This metric is equivalent to precision@k when $k \leq |V_{rc}|$ and to recall@k when $k > |V_{rc}|$, so we directly use these two latter metrics.

4.2 Main Results

All experiments in this chapter are done using the manually annotated golden episode. We evaluate two pipelines:

- **Baseline:** TF–IDF features over masked templates, Isolation Forest anomaly scoring, temporal graph edges (actor/host), and backward RCA walk with priority $(-score_norm, |\Delta t|)$.
- **Contrastive model:** Contrastive representation learning over TF–IDF+SVD base embeddings (projection head trained with an NT-Xent style objective on mined positive pairs), followed by anomaly scoring in the learned space and the same graph construction and walk strategy.

Unless stated otherwise, results are reported using the evidence-collapsed protocol and the alert node is excluded.

4.2.1 Set-based Retrieval and Explanation Size

Table 1 reports the set-based precision/recall/F1 and explanation sizes for core and extended ground truth (results in bold when the contrastive model outperforms the baseline or does not underperform). Both methods retrieve the same number of true positives (TP) for core and extended ground truth, hence identical recall. The contrastive method returns fewer evidence items overall, yielding higher precision and F1.

Method	S^{nodes}	S^{evidence}	Core TP	Core P	Core R	Core F1	Ext TP	Ext P	Ext R	Ext F1
Baseline	199	174	24	0.138	**0.750**	0.233	32	0.184	**0.800**	0.299
Contrastive	199	168	24	**0.143**	**0.750**	**0.240**	32	**0.190**	**0.800**	**0.308**

Table 1: Set-based retrieval metrics and explanation size.

4.2.2 Ranking-aware Early Retrieval

Table 2 reports ranking-aware early retrieval metrics (results in bold when the contrastive model outperforms the baseline or does not perfom worse). The contrastive model improves early retrieval for the *core* ground truth (notably at $k = 5$ and $k = 10$), but it degrades early retrieval for the *extended* ground truth. At $k = 50$, both methods have the same values for recall for both the *core* and *extended* ground truth. This indicates that the main contribution of contrastive learning in this setting is the prioritization of evidence instead of a broader coverage.

In terms of Hit@k, both methods achieve Hit@$k = 1$ for $k \in \{5, 10, 20, 50\}$ under both core and extended ground truth, which means that at least one relevant evidence item is present among the top-k results; therefore, the differences between methods are better reflected by $P@k$ and $R@k$ (how many relevant items appear early) rather than first-hit success.

Method	**Core** (G_e^{core})				**Extended** (G_e^{ext})			
	$P@5$	$R@5$	$P@10$	$R@10$	$P@5$	$R@5$	$P@10$	$R@10$
Baseline	0.200	0.031	0.400	0.125	0.800	0.100	0.900	0.225
Contrastive	**0.400**	**0.063**	**0.700**	**0.219**	0.400	0.050	0.700	0.175
	$P@20$	$R@20$	$P@50$	$R@50$	$P@20$	$R@20$	$P@50$	$R@50$
Baseline	0.550	0.344	**0.420**	**0.656**	0.850	0.425	**0.540**	**0.675**
Contrastive	0.450	0.281	**0.420**	**0.656**	0.750	0.375	**0.540**	**0.675**

Table 2: Ranking-aware metrics.

Across both methods, set-based recall is high for a fully unsupervised pipeline (0.75 core, 0.80 extended) but precision is low, reflecting that the backward walk returns a relatively large explanation set. The contrastive model shows a small but consistent gain in set-based precision/F1 by reducing the number of returned evidence items. Its most visible benefit is in early retrieval for the core chain (e.g., $P@10$ rises from 0.40 to 0.70), while it neglects some extended-context evidence in the top ranks.

Due to different datasets used and output units produced, this model can only be quantitatively compared to the baseline and not to other previous works. Therefore, it has been discussed in Section 2 in a qualitative way.

4.2.3 AUPRC over reachable evidence.

We report an additional diagnostic for ranking quality: the area under the precision–recall curve (AUPRC). We compute this over the *reachable* evidence pool (6,014 items). We use the suspiciousness score as the ranking signal. Due to class imbalance ($|G^{\text{core}}| = 32$ and $|G^{\text{ext}}| = 40$ are drastically smaller than the evidence pool), absolute AUPRC values are low, but both

methods are above the random baseline. We can see in Figure 1 that the contrastive model improves in comparison to the baseline: AP_{core} increases from 0.0324 to 0.0551, and AP_{ext} from 0.0341 to 0.0533. This is consistent with the ranking-aware early retrieval metrics improvements.

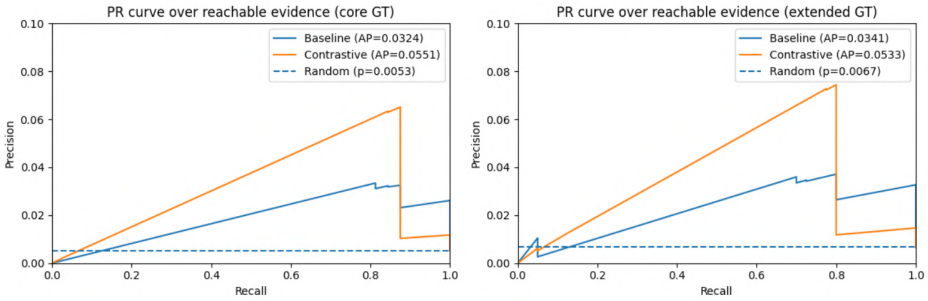

Figure 1: AUPRC for baseline and contrastive pipelines

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4.3 Experiments

This section consists of several experiments where we address the research questions proposed.

4.3.1 Experiment A: Evidence Reachability

A necessary condition for RCA to be successful is that relevant evidence is reachable in the constructed graph. If a ground truth (GT) evidence item cannot be reached under the mechanics used by the RCA walk, then it does not matter which scoring or ranking strategy one uses because it will not be able to retrieve it. We therefore evaluate *reachability* as an upper bound on retrieval performance based on only graph connectivity assumptions.

Reachability definition. We compute the set of reachable nodes by running the same traversal mechanics as the RCA walk, but removing all ranking and budget effects. We initialize the traversal from the successors of the alert node and expand backwards by repeatedly visiting predecessors of visited nodes. In this diagnostic, we ignore anomaly scores and the maximum-node budget, but we preserve the temporal feasibility constraints (maximum lookback and forward slack) and the hop limit. The resulting reachable set represents all evidence that could be returned by the RCA walk in principle.

To be consistent with the main evaluation, we report reachability at the evidence level by collapsing nodes into evidence items. We define *reachable recall* for a ground truth set G_e as:

$$\text{ReachRec}(G_e) = \frac{\left|\mathcal{R}_e^{\text{evidence}} \cap G_e^{\text{evidence}}\right|}{\left|G_e^{\text{evidence}}\right|},$$

where $\mathcal{R}_e^{\text{evidence}}$ is the set of reachable evidence items.

Experimental factors. We ablate (i) edge types (actor edges vs. host edges) and (ii) temporal gaps for actor and host edge construction. All experiments are run on the golden episode using the same alert attachment strategy and the same time-window constraints as in the main RCA pipeline.

Results. Table 3 summarizes the reachability diagnostics. With both edge types enabled under the default temporal gaps (actor gap 120 s, host gap 60 s), the graph achieves perfect reachability for both the core and extended GT sets (32/32 and 40/40 reachable). An important insight is that actor edges are sufficient to obtain the same perfect reachability (actor-only also yields 32/32 and 40/40), while host edges alone are insufficient (host-only reaches only 21/32 core and 21/40 extended evidence). Without both actor and host edges, reachability collapses (1/32 core and 1/40 extended), confirming that explicit temporal connectivity assumptions are required to make evidence reachable from the alert anchor.

The actor-gap modifications show a clear threshold effect (table omitted for space): restricting the actor temporal gap to 30–60 s reduces reachable recall to 0.75 for both core and extended GT, while gaps ≥120 s restore perfect reachability. In contrast, the host-gap modifications maintain perfect reachable recall across tested values, but strongly change the size of the reachable evidence set. For example, host_gap=30 s leads to a perfect reachability while substantially reducing the reachable evidence pool compared to host_gap=60 s, indicating that host edges primarily control context expansion rather than being necessary for connecting the core incident chain in this episode.

Overall, this experiment answers RQ1 by showing that full GT reachability is achievable under simple connectivity assumptions, but it is dependent on including actor-based temporal edges and also using sufficiently permissive temporal gaps to bridge cross-system delays. It also answers RQ3 by showing that the same assumptions that guarantee reachability can considerably expand the candidate evidence pool, which leads to a lower precision.

Variant	$\text{ReachRec}_{\text{core}}$	$\text{ReachRec}_{\text{ext}}$	Reachable evidence
Both edges	1.000 (32/32)	1.000 (40/40)	6014
Actor-only	1.000 (32/32)	1.000 (40/40)	227
Host-only	0.656 (21/32)	0.525 (21/40)	28
No edges	0.031 (1/32)	0.025 (1/40)	1

Table 3: Reachability diagnostics (applies to both baseline and contrastive model).

4.3.2 Experiment B: RCA Walk Budget and Hop Limit

There are two hyperparameters that modify the trade-off between the size and the quality of the explanation: the node budget (*rca_max_nodes*), which determines maximum amount of nodes that can be selected, and the hop limit (*rca_max_hops*), which determines how deep the walk can expand through predecessor links. We perform two experiments according to these two hyperparameters and report results for both the TF–IDF baseline and the contrastive model.

Hop sensitivity. To see the effect of different hop caps, we fix a large node budget ($rca_max_nodes = 5000$) so the maximum number of nodes is not a constraint here. We experiment with $rca_max_hops \in \{1, 2, 3, 5, 10, 25, 50, 100, 150, None\}$ (table omitted for space).

When the hop limit is very small, we recover almost none of the ground truth. As the hop limit increases, size of the explanation becomes more usable. When it has a value around $rca_max_hops = 10$–25, the explanation size is comparable to the ones observed when moderate node caps are chosen, as seen in Table 4. For instance, when the hop budget is 25, the baseline retrieves 60 nodes ($S^{\text{evidence}} = 46$) and the contrastive model retrieves 78 nodes ($S^{\text{evidence}} = 59$).

These are numbers on the same order as the $N = 50$–100 node-capped regime ($S^{\text{evidence}} \approx 37$–$82$).

With larger hop limits ($(\geq 100$) or no limit at all, the walk selects thousands of nodes. This is not feasible in real-world scenarios, but shows that the graph can perform a very deep traversal (the baseline has max hop selected = 527 and the contrastive model 386).

To characterize how "deep" the walk actually goes under each hop cap, we report hop diagnostics for the selected nodes: the maximum hop depth and the 90th percentile hop depth ($P90$), which summarizes the typical upper-range traversal depth while being less sensitive than the mean to a small number of very deep selections. At *rca_max_hops* = 25, $P90$ is already close to the cap (baseline $P90 = 24.0$, contrastive $P90 = 22.3$), indicating that most selected evidence requires multi-hop traversal rather than only immediate predecessors of the alert.

Node budget sensitivity. Now we fix the maximum number of hops to *unlimited* and experiment with *rca_max_nodes* $\in \{25, 50, 100, 150, 200\}$. In this regime, we increase coverage by increasing the node cap, but this eventually degrades precision due to background noise, as seen in Table 4.

With small node limits, the explanations are incomplete. For example, at $N = 25$ the contrastive run returns $S^{\text{evidence}} = 18$ and the baseline returns $S^{\text{evidence}} = 17$. This lowers recall and, consequently, F1-score, even if early precision has a high value (baseline has $P@10 = 0.900$ and the contrastive model has $P@10 = 0.700$, both in the *extended* set). At $N = 50$, the walk retrieves most of the relevant evidence while filtering noise. With this node limit, the contrastive model has a better F1-score in both ground truth definitions as it returns a smaller explanation set. This suggests that the contrastive model reduces the amount of items that are not relevant to the incident in the retrieval without sacrificing coverage.

We can see that if we increase N, we get larger explanations (contrastive: $S^{\text{evidence}} = 168$, baseline: 174), which give us a lower set-level precision and, consequently, F1-score. We can also see that the hops increase with the node budget (e.g., contrastive max hop selected grows from 10 at $N = 25$ to 119 at $N = 200$; baseline grows from 16 to 126). This makes sense: the more nodes are allowed, the further into the graph the walk will expand. In addition, this is reported in the $P90$ *hop selected* results, which shows typical upper-range depth of the graph traversal: for the baseline it increases from $P90 = 14.0$ at $N = 25$ to 116.2 at $N = 200$, and for the contrastive model from 9.0 to 110.2 (Table 4).

Model	N	S^{evidence}	$\mathbf{F1}_{\text{core}}$	$\mathbf{F1}_{\text{ext}}$	$P@10_{\text{core}}$	$P@10_{\text{ext}}$	**P90 hop sel.**	**Max hop sel.**
Node budget sensitivity (fixed rca_max_hops = None)								
Baseline	25	17	0.367	0.526	0.400	0.900	14.000	16
Contrastive	25	18	0.360	0.517	0.700	0.700	9.000	10
Baseline	50	41	0.575	0.667	0.400	0.900	30.200	35
Contrastive	50	37	0.609	0.701	0.700	0.700	16.000	17
Baseline	200	174	0.233	0.299	0.400	0.900	116.200	126
Contrastive	200	168	0.240	0.308	0.700	0.700	110.200	119

Table 4: Node budget sensitivity of the backward RCA walk under unlimited hops.

In summary, we can see that hop and node limits affect the returned explanations differently. Small hop caps prevent the walk from retrieving incident evidence even when it is reachable. Once the hop limit is high enough (roughly $\geq$ 10–25 in this setting), the node budget becomes

the dominant hyperparameter: if it is too small, it leads to under-explanation; if it is too large, it will introduce too much noise.

The results seen in Section 4 are done with a fixed node budget of $N = 200$ to ensure that the evaluation across pipelines is consistent. We can see in this experiment that a budget of $N = 50$ produces explanations with sizes that make the analysis more feasible and improves the trade-off between budget and quality for both methods. We then interpret the node limit from the main evaluation as a reference point in terms of explanation size.

5 Discussion

In this thesis, SOC root-cause analysis was treated as a task of retrieving ranked evidence that could best explain the cause of an alert using logs from multiple systems. The results on the golden episode show that, by using an unsupervised setting, meaningful narratives can be obtained. The main constriction was not finding the evidence, but whether that evidence could be retrieved early and with an acceptable amount of noise.

5.1 Precision as Bottleneck

Both pipelines (TF–IDF and contrastive) achieved a high recall under the set-based retrieval protocol, but a low precision because of the large size of the explanation set returned, as seen in Table 1. This situation corresponds to a realistic SOC failure mode: the system can retrieve a lot of evidence relevant to the incident, but it also returns a large amount of background context, which leads to an expensive inspection of the explanation.

The contrastive pipeline did not dramatically improve the baseline in set-based precision and F1-score, but it still enhanced it by performing a desirable mechanism: it reduced the explanation size without lowering recall. This means that, with this configuration, this self-supervised model helps filter some noise instead of reaching new evidence that caused the incident.

5.2 Main Value of Contrastive Model

The main improvement of the contrastive pipeline can be seen in ranking-aware metrics for the core causal chain, which Table 2 shows, especially at small inspection budgets ($k = 5$ and $k = 10$). This is more important than small set-level improvements because, in practice, analysts' workflows are budgeted [26].

At the same time, ranking-aware metrics for the extended causal chain degrade. This means that the contrastive signal makes the ranking of evidence prone to select the minimal causal explanation instead of broader contextual evidence. This shows a trade-off between between two explanation goals that are both valid: (i) the minimal core explanation, and (ii) the supporting evidence that is also plausible as part of the incident narrative.

5.3 Reachability vs. Prioritization

An important point in the methodology of this work is the distinction between connectivity and ranking. Table 3 shows that all of the ground truth evidence items are reachable from the alert anchor under the default graph design. This is relevant because it shows that the graph design is not the limiting factor for recovering explanatory evidence items.

Nevertheless, this experiment also shows that this high reachability can highly expand the pool of candidates, so precision degrades unless either the scoring function is strong or the expansion of the graph is controlled.

We can get an actionable insight from the edge ablations: actor edges alone are enough to have a perfect reachability, while host edges alone are insufficient. Operationally, a reasonable approach would therefore be to use host edges to expand context when the core explanation is incomplete.

This analysis shows that a calibration across heterogeneous sources is necessary for this task. In particular, some approaches that could reduce noise are:

- **Calibrate scores per stream:** We normalized scores globally, but different streams can have very different score distributions.
- **Control how the graph expands:** We could modify the walk so it starts with only actor edges and small time windows, and then expand to broader context if the explanation is incomplete.

Section 4 pointed at this: high suspiciousness scores in noisy streams led to false positives, while some core items that were not retrieved as part of the causal chain had comparatively lower scores.

5.4 Walk Budget and Hop Limit

Subsection 4.3.2 from Section 4 shows that limitations in hops and nodes to be reached influence differently the failure modes of the graph traversal. With a node budget high enough for it to be non-binding, small hop caps miss most of the relevant evidence to the causal path. Once the hop cap is large enough, the explanation size becomes to a the analyst-feasible regime that can be seen where moderate node caps are used. Too large hop limits (or no limit) lead to an explanation with a lot of background noise.

When hops are not restrictive, the node budget becomes the main operational constraint. Increasing this number improves coverage at the beginning, but beyond a budget $N = 50$, precision and F1-score degrade due to noise in the explanation set. In our main evaluation, N is fixed to 200. This experiment shows that smaller node budgets (around $N = 50$ in this episode) have a better trade-off regarding budget and quality. For this reason, the main setting is seen as a reference point for comparing pipelines, while we acknowledge the superior operational performance of this tune in this hyperparameter (node budget).

5.5 Limitations and Future Work

One of the main limitations of this work is that the quantitative evaluation is done on a single manually annotated golden episode. This can be enough to validate the mechanics of the pipeline, leakage control, and the distinction between reachability and ranking, but it still limits how far we can generalize the results to other types of incidents and environments. More incident types where not possible to analyze due to time constraints.

Another limitation is that episode construction and sampling were necessary due to the high volume of data. This is realistic in SOC telemetry, but it can affect which context evidence is available for it to be retrieved and ranked.

In addition, there is ambiguity in the ground truth protocol (core vs. extended) because of timestamp precision mismatches and duplicates. The fact that early retrieval behaves differently

for the core vs. extended ground truth is best seen as a consequence of how we defined and evaluated "core" and "context", not just as the model behaving inconsistently.

In summary, we obtain three conclusions from the reported results: (i) continuity-based event graphs can make SOC telemetry completely reachable from the alert in, at least, one realistic incident, (ii) the most dominant challenge is to control the noise from heterogeneous streams with high volume, and (iii) contrastive learning mainly helps improve early prioritization of the core causal chain rather than increasing the total coverage. Future work should evaluate more types of incidents and explore per-stream calibration. Also, it should investigate on a graph expansion where the core evidence is retrieved first, and the context evidence later in order to have a smaller explanation size and improve precision without lowering reachability.

6 Conclusion

In this work we investigated whether a self-supervised RCA pipeline could reconstruct the causal path that led to a security incident using contrastive learning from heterogeneous logs and without using labels, and whether this model's performance was better than a TF–IDF baseline (and why or why not). The approach proposed is an alert-anchored retrieval of ranked evidence over a temporal event graph where the edges encode operational continuity of actors and hosts.

The results on the manually reconstructed golden episode show that this pipeline achieved a high recall, but precision remained low because relatively large explanations were retrieved by the graph walk. The contrastive representation did not improve coverage of causal evidence, but it improved the prioritization of the core causal chain. However, the early retrieval of the extended causal chain worsened, highlighting that the understanding of what a "good root-cause analysis" depends on if the objective is to obtain a concise causal chain or a richer contextual narrative.

An important insight is the distinction we made between connectivity and reachability. Reachability diagnostics show that all of the evidence items can be reached under simple connectivity assumptions and, furthermore, host edges are not needed to achieve this task. Using both actor and host edges make the candidate pool considerably larger, which harms precision. Therefore, the main reasoning of this work is supported: in multi-stream SOC telemetry, the main bottleneck is not just representation learning, but reducing background noise while keeping the causal chain that led to the incident.

In summary, the main contributions of this work are:

- A method for root-cause analysis in SOC using event graphs that are anchored in alerts.
- A self-supervised learning pipeline with leakage control for semantic alignment across different systems.
- An evaluation protocol that distinguishes reachability from prioritization and core from extended evidence.

For future work, the evaluation should be extended beyond just one attack. In addition, one should explore stream-aware calibration and new graph design choices in order to reduce false positives without sacrificing reachability.

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A Link to source code

We uploaded the code with all the implementation (including data extraction automation and sampling) on Github: `https://github.com/patriciaalonso100/Master-Thesis-DTU---s232519-.git`

Analysis and Design for Disaster Supply Provision System Based on the Transition of Required Information in the Noto Peninsula Earthquake*

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Abstract

Large scale earthquakes frequently cause serious damage in Japan, and past experiences have shown that the information and relief supplies required by disaster victims change over time. Recent advances in information and communication technology have improved information sharing during disasters, but have also created challenges such as the overloads or mismatches between actual needs and supplied goods. Therefore, this paper analyzes the transition of required information during the Noto Peninsula Earthquake using access data on the official website of Nanao City. The results show a clear shift from immediate survival related information, including water, food, and relief supplies, to recovery related information such as donations and support systems in later phases. Based on these findings, issues in disaster relief supply provision under modern communication environments are discussed, and the paper also proposes the key requirements for a Disaster Relief Supply Provision System that considers time dependent needs and supports efficient supply management.

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1 Introduction

Human society has experienced many large earthquakes, and these events have led to the accumulation of disaster-related records and experiences. Such knowledge has been shared over time and has supported the development of research activities and practical countermeasures. These efforts have contributed to improving disaster preparedness and recovery.

Especially in recent years, research in information and communication technology has begun to focus on the effective use of disaster records and memories. Advances in digital technology have made it possible to store, analyze, and share information more efficiently. As a result, past disaster experiences can now be reflected more directly in disaster response and management.

In Japan, the Great Hanshin Awaji Earthquake in 1995 occurred during the early period of Internet use. This event highlighted both the possibilities of Web system such as Email and the limitations of land phone systems in disaster situations. After that, Japan experienced several major earthquakes, including the Niigata Chuetsu Earthquake in 1994, the Great East Japan Earthquake in 2011, the Kumamoto Earthquake in 2016, and the Noto Peninsula Earthquake in 2024. Through these disasters, various problems in the developments of information and communication systems during emergencies became clear.

One of important research topics has been disaster information systems that consider changes in required information and supplies after a disaster. The information needed by affected people changes over time, especially in the early stage just after a disaster. Based on experiences from the Great East Japan Earthquake, one of the studies has examined methods to provide essential information quickly and continuously using heterogeneous communication networks, even when part of the infrastructure is damaged. (Shiratori, Uchida, Shibata, & Izumi, 2013)

Considered with these previous studies, this paper focuses on the transition of information required by disaster victims during the Noto Peninsula Earthquake that occurred in 2023. Then, the comparisons with other major earthquakes in Japan are also discussed to identify common features and differences for the future information and communication technology. Finally, this paper discusses the approaches for supporting the provision of essential supplies in the early post-disaster phase and explores approaches for the early recovery and development of supply chains.

In the followings, the major large-scale disasters in Japan and the features of Information Technology (IT) during these disasters are discussed in section II. Section III explains the previous studies of the required information in disasters, and section IV explains the required information in Noto Peninsula Earthquake using the records of Noto City HP accesses. Finally, Section V discusses the designs of the Disaster Supply Provision System with the transitions of the required information under the Noto Peninsula Earthquake, and Section VI presents the conclusions and future works in this research.

2 Major Large-Scale Disasters and Information Technology

Table 1 summarizes the major large-scale earthquakes that have occurred in Japan in recent years, including the number of fatalities, the magnitude, and the seismic intensity observed in each event. (International Seismological Centre, 2015) (United States Geological Survey, N.A.)

Earthquake	Date	Deaths	Missing	Magnitude	*JMA Seismic Intensity*
Great Hanshin Earthquake (Kobe)	January 17, 1995	6,434	N/A	6.9	Shindo 7
Niigata Chuetsu Earthquake	October 23, 2004	68	N/A	6.8	Shindo 7
Great East Japan Earthquake (Tohoku)	March 11, 2011	19,759	2,553	9.0 - 9.1	Shindo 7
Kumamoto Earthquake	April 16, 2016	277	N/A	7.0	Shindo 7
2024 Noto Peninsula Earthquake	January 1, 2024	703	2	7.5	Shindo 7

Table 1: Major Large-scale Earthquakes in Japan

These earthquakes clearly show that different types of damage have caused significant human losses. In the Great Hanshin Awaji Earthquake, a large number of deaths were mainly caused by the collapse of residential buildings, resulting in 6,434 fatalities. In contrast, the Great East Japan Earthquake caused catastrophic damage due to an unprecedented tsunami, leading to 19,759 fatalities, which remains one of the most severe disaster outcomes in recent history.

In more recent years, earthquakes such as the Kumamoto Earthquake and the Noto Peninsula Earthquake have also resulted in many deaths, primarily due to the collapse of houses and buildings. These events indicate that even with technological and social progress, structural damage continues to be a major factor contributing to human loss during earthquakes in Japan.

	Developments of IT	Subjects after Disaster
Great Hanshin Earthquake (Kobe)	Early period of Internet	Expectations of Internet (Email) during disasters/Loss of land phone network
Niigata Chuetsu Earthquake	Early period of cellular ph one	Expectations of cellular phone during disasters/Loss of land phone network
Great East Japan Earthquake (Tohoku)	Spread Internet and Cellula r Phone	Ensuring redundancy
Kumamoto Earthquake	Spread of SNS media	Excessive relief supplies
2024 Noto Peninsula Earthquake	LEO communication	Restriction of individual relief supplies

Table 2: Developments of IT and Major Subject after Disaster

From the perspective of information technology (IT), Japan has played a central role in advancing disaster-related research based on the accumulated memories and records of these events as shown in Table 2. The Great Hanshin Awaji Earthquake in 1995 occurred during the early period of Internet expansion. At that time, technologies such as web pages and email systems demonstrated the potential to function effectively during disasters. A valuable record described in Reference (Tanaka, 1996) reports that at Kobe University, information exchange was conducted using email while telephone networks and transportation systems were paralyzed. Also, while land phones could not available after the Niigata Chuetsu Earthquake in 2004, cellular phones became widely known as a useful tool during earthquakes in Japan. These experiences are considered to have triggered subsequent research on Internet-based disaster information systems, including systems that employed long-distance wireless

LAN technologies during the eruption of Mount Iwate. (Uchida, Asahi, & Shibata, Disaster Information System and Its Wireless Recovery Protocol, 2004)

The Great East Japan Earthquake in 2011 occurred after cellular phones and the Internet had become widespread. Although these technologies were widely available at the time, the disaster revealed their vulnerability under extreme conditions (Araemishi, Iwate Broadcasting Co.,Ltd., 2012), which led to increased recognition of the importance of redundancy in information and communication systems. At the same time, the availability of mobile devices also enabled the collection of a large amount of video and photographic data. As a result, systems for confirming personal safety and the use of SNS platforms began to be considered as practical disaster response measures.

During the Kumamoto Earthquake in 2016, communication methods were maintained to some extent due to previous technological developments. However, new challenges became evident, including the spread of misinformation and the excessive or uneven provision of relief supplies. (2016 Kumamoto Earthquake: One month record, 2016) These issues highlighted the importance of distributing accurate information and appropriate supplies according to actual needs in disaster-affected areas.

Furthermore, in the Noto Peninsula Earthquake of 2024, LEO (Law Earth Orbit) satellite communication services such as Starlink were introduced and made a significant contribution to the recovery of communication functions. While these technologies enabled rapid information sharing, the large volume of information from affected areas also created challenges. In some cases, individual supply support was restricted, and information provision became concentrated at local governments. (Ishikawa, 2024) These experiences suggest that issues related to information management and relief supply coordination immediately after a disaster remain important topics for continued examination.

3 Previous Studies

In Japan, it is widely recognized that the type of information required by disaster victims changes over time after a disaster. Based on this understanding, many local governments have developed disaster response manuals that consider the transition of required information from the immediate aftermath to later recovery phases. This recognition has been strongly influenced by research studies that analyzed actual disaster situations and the behavior of affected communities.

In fact, some representative studies reported the approaches with the transitions of required information based on experiences from the Great East Japan Earthquake. For example, the study (Uchida, Kazuo, & Shibata, Disaster Information System from Communication Traffic Analysis and Connectivity (Quick Report from Japan Earthquake and Tsunami on March 11th, 2011), 2011) investigated the transition of network conditions and information needs at Iwate Prefectural University, which functioned as an evacuation shelter after the disaster. This study examined how available communication resources changed over time and what kinds of information were required by disaster victims on a daily basis. It also reported practical efforts to provide necessary information through bulletin boards, reflecting the actual demands observed at each stage of the disaster response.

Another study (Shiratori, Uchida, Shibata, & Izumi, 2013) organized the transition of required information after disasters into a structured framework. The study proposed the seven disaster response stages as shown in Figure 1.

	Condition	Activity	Period
t_1	Normal		
t_2	Indication of Disaster	Indication, rumor etc	2 weeks before – Disaster
t_x	Occurrence of Disaster		During Disaster
t_3	Just after Disaster	Rescue, evacuation, safety information	Disaster – 2 days
t_4	Calm down from Disaster	Relief materials, safety information	3 days – 2 weeks
t_5	Restoration form Disaster	Restore lifeline, residences, and so on	3 weeks - several months
t_6	Revival		

Table 3: Proposed Disaster Response Stages in Great East Japan Earthquake

In Table 3, after the disaster happened at the stage t_x, the information such as rescue, evacuation, or safety information is necessary during the stage t_3. Therefore, the network connectivity is the most important, and the text contents are mainly used during the stage. Next, in the stage t_4, the information such as relief supplies is getting more important, and so throughput such VoIP (Voice over IP) is getting more important. Then, Figure 2 shows the required information through the time.

Subject	Necessary Information / time	t_1	t_2	t_x	t_3	t_4	t_5	t_6
Victim (Inside of the area)	Disaster Prevention	△	○					
	Evacuation Shelter		○		◎			
	Safety				◎	◎	○	△
	Stricken Area				◎	○		
	Traffic				○	◎	◎	
	Relief Supplies					◎	○	
	Public Service					○	◎	
	Lifeline					○	◎	
	Local Government					○	◎	
Relatives, Volunteer (Outside of the area)	Safety				◎	◎	○	
	Stricken Area				◎	○	△	
	Relief Supplies					○	◎	

Table 4: Proposed Disaster Response Stages in Great East Japan Earthquake

In Table 4, the forecast and evacuation information are required during the stage t_1, and evacuation information is not so much important. However, if there are some kinds of indications such as news or rumor about disaster during t_2, evacuation and disaster prevention becomes more important than usual. Then, if disaster occurs, very significant activities related of human lives such as rescue, evacuation, and safety status information come to be required. Since the information is much related to human lives during the stage of t_3, Disaster Information Network should be focused on the network connectivity more than throughput or latency. Also, text contents are mainly used for the disaster evacuation, resident safety and disaster status information.

Then, this study [4] focused on heterogeneous and multilayer communication network environments and proposed a Quality of Service (QoS) control method designed to maintain network

operation under disaster conditions. The proposed method aimed to ensure persistent network survival and early recovery of performance, while also considering changes in the priority of required information over time. This approach emphasized the importance of adaptive control mechanisms that respond to dynamically changing information demands.

However, the communication environment has continued to evolve since these earlier studies. The introduction of new communication technologies, such as satellite communication systems, and the widespread use of social networking services have significantly changed the way information is generated and shared during disasters. During the Noto Peninsula Earthquake, these changes led to new challenges, including information overload and the provision of support supplies that did not match the actual needs of disaster victims. In some cases, excessive or unsuitable relief supplies were delivered, creating additional burdens for affected areas.

These experiences indicate that it is no longer sufficient to manage only the transition of the required information in a disaster. The contents of support activities and the distribution of relief supplies also need to be managed appropriately. Efficient coordination based on accurate and update information remains an important issue for supporting effective recovery in the post disaster phase.

4 The Transition of Required Information in the Noto Peninsula Earthquake

Table 5 summarizes the top accessed keywords and other contents that showed a rapid increase in access on the official website of Nanao City, which is one of the affected areas, during the first four months after the earthquake. The figure reflects how information needs changed immediately after the disaster based on actual access logs.

Emergency Period; Required Information for Daily Necessities

Jan. 2024	Feb. 2024	Mar. 2024	Apr. 2024
3.5 (Million HP accesses)	2.0	1.2	0.7
Top Keywords: - Water and sewerage - Relief supplies - Disaster victim certificate - Housing support	Top Keywords: - Water and sewerage - Relief supplies - Disaster victim certificate - Government-funded demolition program - Housing support - Disaster relief donation/fund	Top Keywords: - Water and sewerage - Relief supplies - Disaster relief donation/fund - temporary disaster waste storage site - List of disaster support programs	Top Keywords: - Water and sewerage - Disaster relief donation/fund - Relief supplies - List of disaster support programs - temporary disaster waste storage site
Others; - Evacuation shelter - Account for donations - Bathing support - How to dispose of garbage - Risk assessment	Others; - Emergency housing - Bathing support - Public relations (PR) Nanao - How to dispose of garbage - Emergency support toilet -Self-funded demolition with later reimbursement	Others; - Government-funded demolition program - Disaster victim certificate - Public relations (PR) Nanao - Emergency housing - Self-funded demolition with later reimbursement	Others; - Disaster victim certificate - Government-funded demolition program - Public relations (PR) Nanao - Emergency housing - Recovery of septic tank

Table 5: Nanao City HP Accesses during Jan. 2024 to April. 2024

Before the earthquake, the website recorded approximately 300,000 accesses. After the earthquake, the number of accesses sharply increased to approximately 3,500,000, indicating that the municipal website became an important source of information for residents. During this period, the city council

of Nanao City reported that it was an emergency response phase, in which daily necessities were of primary importance. In January 2024, access was highly concentrated on information related to water supply and drainage, as well as relief supplies. In particular, information on food and drinking water, which are essential for daily life, was strongly demanded during the early post disaster period.

Table 6 presents the top accessed keywords and other contents on the Nanao City website from five to eight months after the earthquake.

Recovery Period; Required Information for Rebuilding Lives and Communities Programs

May. 2024	Jun. 2024	Jul. 2024	Aug. 2024
0.43 (Million HP accesses)	0.37	0.37	0.33
Top Keywords: - Disaster relief donation/fund - temporary disaster waste storage site - List of disaster support programs - Paypay (QR code cashless payment) - Government-funded demolition program	Top Keywords: - Disaster relief donation/fund - Accommodation facilities - temporary disaster waste storage site - List of disaster support programs - Paypay (QR code cashless payment) - Open meeting - Government-funded demolition program - Disaster victim certificate	Top Keywords: - Disaster relief donation/fund - List of disaster support programs - temporary disaster waste storage site - Government-funded demolition program - Accommodation facilities - Issaki Hoto Festival	Top Keywords: - Disaster relief donation from Taiwan - List of disaster support programs - Government-funded demolition program - Emergency housing - Issaki Hoto Festival - Bid information
Others; - Staff recruitment examination - Water and sewerage - Accommodation facilities - Emergency housing - Bid information	Others; - Water and sewerage - Emergency housing - Public relations (PR) Nanao - Bid information	Others; - Disaster victim certificate - City tax reductiion - Emergency housing - Bid information - Self-funded demolition with later reimbursement	Others; - Accommodation facilities - temporary disaster waste storage site - Recovery of disaster-damaged residential area - Free carry-in of disaster garbage - Nanao Port Festival - Subsidy for seismic retrofitting for damaged housing - Historic site of Nanao castle

Table 6: Nanao City HP Accesses during May. 2024 to Aug. 2024

Compared with the immediate post disaster phase, the total number of accesses decreased significantly from 3,500,000 to approximately 430,000. At the same time, a clear transition in information needs can be observed. The keywords with the highest access during this period were related to donations, disaster waste management, and public support programs for recovery. These results indicate that the focus of information demand shifted from emergency support for daily life to measures required for longer term recovery and reconstruction.

In addition, Nanao City reported an increase in access referrals from social networking services (SNS) such as LINE. Notably, content related to donation support from Taiwan showed a particularly high level of access from SNS accesses. This trend reflects recent changes in the information environment and disaster response behavior, which were not observed during the Great East Japan Earthquake in 2011. These results suggest that SNS platforms have become an important channel for information dissemination and access in recent Japanese disasters, further influencing the transition of required information over time.

5 Examples of Disaster Relief Supply Provision System

Unlike the Great East Japan Earthquake, communication networks remained functional during the Noto Peninsula Earthquake and the Kumamoto Earthquake. As a result, television reports and SNS rapidly transmitted information from disaster affected areas. While this enabled faster awareness, it also caused specific relief items to be delivered excessively by individuals. In some cases, large quantities of supplies such as diapers flooded the affected areas. This situation not only created storage and management problems but also placed additional pressure on administrative disaster relief activities that were responsible for distributing supplies to victims. Consequently, the restrictions on individual relief supply deliveries were introduced in certain regions, as reported in Reference (Ishikawa, 2024).

Based on these experiences, several problems must be addressed when designing future Disaster Relief Supply Provision Systems as the follows.

Subjects for Disaster Supply Provision System;
(1) Mismatch between the supplies required by victims and the relief supplies provided.
(2) Transition of required supplies through time
(3) Inconsistency in item labeling (such as bread, baguette, bagel, wheat, croissant, and so on)

To address these issues, future Disaster Relief Supply Provision Systems are expected to include several key functions as the follows.

Required functions for Disaster Supply Provision System;
(1) Triage considering real-time required supplies by victim's requests, media influences (HP, SNS, TV, and so on), and storage
(2) Real-time priority functions by the above triage functions
(3) Thesaurus-based categorization function

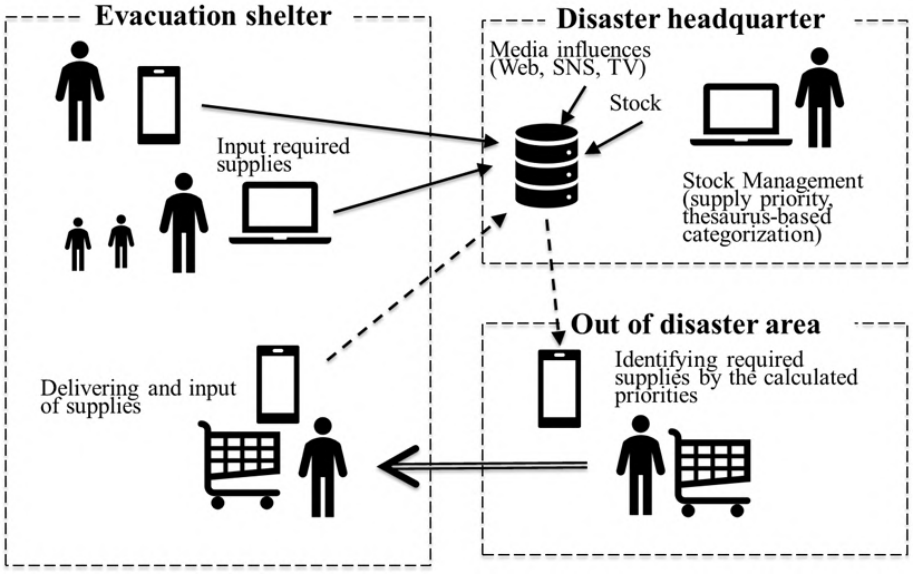

Figure 1: Example of Disaster Relief Supply Provision System

Based on these requirements, the assumed system must be capable of tracking the volume of stored relief supplies, the quantity of newly delivered items, and the transition of supplies required by disaster victims over time. A system configuration such as that illustrated in Figure 1 can support the integration of these elements and enable comprehensive supply management across different disaster response phases.

Furthermore, it is considered necessary to introduce an algorithm based on a Poisson distribution for each category of disaster relief supplies. By modeling the arrival and consumption patterns of supplies using this approach, the system can estimate appropriate delivery volumes and timing as shown in Formula 1 and Formula 2. Such a function can support decision making for identifying suitable relief supplies and can contribute to more efficient and balanced disaster support operations.

$$\boldsymbol{P(X=k)=\frac{\lambda^k}{k!}e^{-\lambda}} \tag{1}$$

$$\boldsymbol{\lambda=\lambda_{victims}-\lambda_{web}-\lambda_{stock}} \tag{2}$$

In the formulas, λ is the expected rate of required supply, X is a discrete random variable, k is the number of occurrences, e is Euler's number. Also, λ consists of the occurrence of required supply from victims, the occurrence from web or media, and the amount of supply stock in damaged area.

As future work, the proposed algorithm is planned to be applied to simulation studies using actual records from the 2024 Noto Peninsula Earthquake. In particular, simulations will be conducted to estimate required resource quantities, such as water supplies, and to compare them with delivered support levels. Through this process, the proposed method is expected to be systematically evaluated and refined.

6 Conclusions and Future Study

This paper reported the transition of required information during large scale disasters, with a particular focus on the Noto Peninsula Earthquake. By reviewing past major earthquakes in Japan and related studies, the analysis highlighted how disaster response has evolved together with changes in information and communication technologies. The findings confirmed that the needs of disaster victims change significantly over time, moving from immediate survival related information and supplies toward support for recovery and reconstruction.

The analysis of access logs from the official website of Nanao City demonstrated clear temporal changes in information demand. Immediately after the earthquake, access was concentrated on information related to lifelines, relief supplies, food, and water. Five months later, interest shifted toward donations, disaster waste management, and recovery support systems. These results provide concrete evidence that information demand follows a phased transition, which should be considered when designing disaster response systems.

Based on these observations, the paper discussed issues related to Disaster Relief Supply Provision System under modern communication environments. While the availability of communication networks and SNS has enabled rapid information sharing, it has also caused problems such as misinformation, excessive supply delivery, and mismatches between actual needs and provided supplies. The examples from recent disasters indicate that effective disaster response requires not only the delivery of information but also accurate management and coordination from victims, media and delivered supplies.

To address these challenges, this paper presented considerations for a Disaster Relief Supply Provision System that accounts for the transition of required supplies over time. Key requirements

include real-time understanding of victim needs with web or SNS, prioritization based on data analysis, and unified categorization of supply items. The introduction of probabilistic approaches, such as algorithms based on Poisson distributions, was discussed as a method for supporting decision making in supply allocation and timing.

As future study, further validation of the proposed system using previous disaster cases is required. In particular, quantitative evaluation using real operational data would help clarify the effectiveness of the proposed functions. Through the validation, the proposed method is expected to be systematically evaluated and refined. Through continued research in these directions, it is expected that more adaptive and efficient disaster response systems can be developed. Such systems have the potential to reduce confusion, improve supply distribution, and support faster and more effective recovery in future large-scale disasters.

Acknowledgement

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Cyberly: A Gamified Android Application for Learning Basic Cybersecurity Concepts*

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Abstract

This paper presents the design and development of Cyberly, a gamified Android application for learning basic cybersecurity concepts. The application is addressed to vulnerable Internet users such as children and young people, elderly individuals, and people with limited knowledge regarding digital security. Many of these users are daily Internet users without being aware of the risks it may involve and the threats they may face. The selection of this topic was based on the rapid development of digitalization and technology, especially in the current era with the use of Artificial Intelligence, which creates new challenges as well as security gaps. The work presents basic concepts of cybersecurity and analyzes common and dangerous Internet threats, while also using gamification in education in order to create a more understandable and engaging way of learning.

1 Introduction

The continuous spread of technology and digital systems, together with the increasing dependence of society on the Internet, has made cybersecurity one of the most important fields of our time. Daily activities such as communication, financial transactions, work, entertainment, and education are now carried out through digital systems, which significantly increases the exposure of users and organizations to Internet risks. Consequently, cybersecurity is a basic element for ensuring the smooth and secure operation of the digital environment.

Cybersecurity refers to the set of practices, technologies, and processes that are applied with the purpose of protecting digital systems, networks, data, and applications from malicious actions. These actions may include unauthorized access, alteration or deletion of data, and the theft of sensitive information. Cybersecurity is not limited only to data protection from a technical point of view, but also extends to risk management, prevention, and preparedness for the handling of security incidents. At the same time, the development of technology does not bring only positive changes; it also brings security gaps, the evolution of threats, and new forms of automated deception, especially in the era of artificial intelligence [1].

For this reason, cybersecurity is not only a technical matter but also a matter of education and culture. User awareness and education regarding safe Internet practices are essential for reducing risks. This paper presents the design and development of Cyberly, a gamified Android application that supports prevention through accessible learning.

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Several gamified cybersecurity learning platforms have been proposed in the literature. In [2], the Cyber-Aware mobile application was designed to improve cybersecurity awareness among K–6 children by combining educational principles with both formal and informal learning activities. Similarly, authors in [3] developed a gamified Android role-playing quiz application that promotes password security awareness through interactive gameplay. [4] introduced an online gamified cybersecurity learning platform that integrates a custom game environment, automated feedback mechanisms, and personalized exercises to teach practical cybersecurity skills. Furthermore, [5] presented a game-based cybersecurity training platform featuring virtual laboratories, educational challenges, and motivational design elements. Collectively, these studies demonstrate the growing interest in applying gamification to cybersecurity education. Cyberly extends this direction by targeting vulnerable and non-technical users through a mobile-first learning experience that combines cybersecurity awareness content with interactive simulations and simplified gamification mechanisms.

The main contributions of this work are:

- The design of a gamified mobile application for cybersecurity education
- The integration of interactive simulations for experiential learning
- A lightweight MVP architecture that prioritizes accessibility and usability

The remainder of this paper is organized as follows. Section 2 provides the cybersecurity background and motivation, while Section 3 reviews related work and gamification in cybersecurity education. Sections 4–7 present the design, implementation, and evaluation of Cyberly. Finally, Section 8 concludes the paper and outlines future research directions.

2 Cybersecurity Background and the Need for Education

The continuous evolution of technology and digitalization, together with the extensive use of the Internet, has led to a complex threat environment. Modern Internet threats are no longer limited to simple attacks, but combine technology and human factors, making cybersecurity a highly demanding field. While more and more data are digitized, transmitted, and stored, the effects of a successful cyberattack are serious for individual users, organizations, and even entire states.

A main category of modern Internet threats is malicious software, or malware. Malware consists of programs designed to cause damage, steal data, or obtain unauthorized access to systems. Such threats may appear as viruses, trojans, spyware, or ransomware and are usually spread through infected files, links, and applications. Ransomware in particular remains one of the most dangerous attacks because it locks the user data and demands money for its restoration [7].

Another major category of modern threats is social engineering, which exploits the human factor instead of technical security gaps. In such attacks, the attackers try to mislead users through fake messages, web pages, or even phone calls in order to steal personal and financial information. These attacks are particularly effective because they are based on trust, carelessness, and lack of knowledge, and therefore their prevention is difficult through technical means alone.

A further important category concerns account compromise and the theft of authentication credentials. The use of weak and repeated passwords, together with the absence of additional security mechanisms such as two-factor authentication, makes unauthorized access easier.

Through such attacks, malicious actors can gain access to sensitive data, carry out fraud, or exploit compromised accounts for further malicious actions. This phenomenon highlights the importance of proper credential management and the need to strengthen user knowledge in this area [8].

Phishing attacks remain one of the most frequent and effective forms of cyberattack. They are attacks that rely mainly on deception and manipulation of users with the goal of obtaining sensitive information. Unlike attacks that exploit weaknesses of digital systems, phishing exploits trust, carelessness, and the emotional reaction of users, which makes it particularly dangerous and often difficult to handle. Attackers impersonate legitimate and trustworthy organizations, such as banks or public services, and try to guide victims into actions that may lead to further security violations. Emotional pressure, fear, urgency, and anxiety are used as core manipulation mechanisms.

At the same time, fake news and artificial intelligence form another major challenge. False or misleading news can affect public opinion, influence the way users perceive reality, and undermine trust in media and digital platforms. The use of artificial intelligence has intensified this problem because modern AI systems can now produce text, images, and video that approach human creation to a dangerous degree. Deepfakes in particular create serious risks of misinformation, blackmail, and defamation. However, AI is not only a threat; it is also used as a tool for detecting suspicious information and for analyzing the spread of fake content [9].

3 Digital Learning, Educational Applications, and Gamification

Digital learning is a basic pillar of modern education because it reflects the changes that technology has brought in the way people receive, process, and use knowledge. The growing use of digital devices and services has changed learning to such a degree that it has become more flexible, accessible, and adapted to the needs of different groups of users. Educational applications can therefore act as dynamic tools that support the learning process [10, 11, 12].

Learning through applications is based on software designed for the transmission of knowledge, the development of skills, and the strengthening of active user participation. In contrast to the traditional way, applications allow access to knowledge independent of place and time, facilitating lifelong and self-directed education. Their effectiveness, however, depends strongly on pedagogical design. The simple digitization of content is not enough on its own.

Gamification in learning is a modern educational approach that uses game elements and mechanisms not necessarily for entertainment, but for strengthening participation in learning. Its central characteristic is the active involvement of the user. Through mechanisms such as goals, levels of progress, and rewards, the learner participates actively and receives continuous confirmation of progress. The sense of achievement and gradual progress strengthen motivation, making learning more pleasant and less burdensome.

An important advantage of gamification is learning through trial and error. The user can experiment, make mistakes, and repeat actions without negative real-world consequences, which contributes to better understanding of concepts. Gamification also helps maintain user interest and concentration, especially in digital learning environments, where distraction is always one tap away.

A characteristic example is Duolingo, which uses short lessons, interactive exercises, and reward mechanisms in order to maintain user interest and strengthen the learning process.

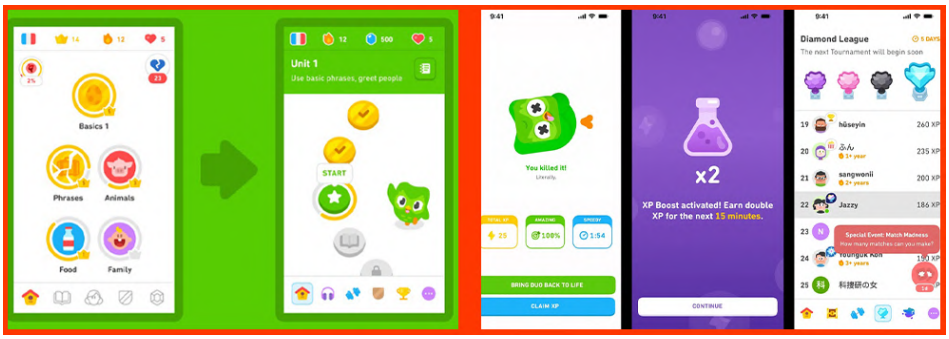

Figure 1: Example of a gamified learning environment in the Duolingo application. Source: Duolingo Blog and Duolingo app screenshot.

These design principles form the basis for Cyberly. The goal is not only to provide information, but also to shape a positive attitude toward learning, prevention, and responsible behavior in the digital world.

4 Cyberly: Idea, Goals, and Target Audience

Cyberly was designed with the goal of prevention and the strengthening of digital safety and critical thinking through education and interactivity. In the modern digital world, where cyberthreats evolve continuously and affect users of every age and level of experience, accessible and effective tools for cybersecurity education are necessary. Cyberly attempts to cover this gap by offering knowledge in a simple, understandable, and user-friendly manner.

The main goal of Cyberly is to raise user awareness regarding both common and modern digital threats, such as phishing attacks, malware, insecure password management, and misinformation on the Internet. Through short educational units and practical examples, the application transforms theoretical knowledge into practical understanding, helping users recognize dangerous situations and adopt safer digital practices.

The application also aims to turn learning into an active process by drawing from digital learning and gamification. Instead of passive reading, users are expected to participate actively through activities, questions, and simulations. In this way, the application strengthens user engagement and helps knowledge remain longer in memory.

An additional important goal of Cyberly is accessibility. The application has been designed so that it addresses a broad audience regardless of age or previous knowledge. The use of simple language and a clear structure allows its use even by users with limited technical experience. The targeted audience includes secondary school students and young adults, adult users with basic digital skills, and educators who may use the application as a supporting tool for information and awareness.

Cyberly was developed with the philosophy of a Minimum Viable Product (MVP), that is, a functional and complete educational application that contains the absolutely necessary characteristics for achieving its basic educational goals. This choice allowed the work to focus on functionality and educational value, avoiding unnecessary complexity and the development of secondary features. The application therefore includes a limited but meaningful set of core functions: short lessons, practical examples, quizzes, simulation, user progress, and supporting tools. At the same time, it leaves open the possibility of future extension.

The educational approach of the application is based on the combination of digital learning, active participation, and gamification. The application avoids passive forms of teaching and encourages participation through interactive activities, questions, and scenarios. The learning structure is based on short and clear units so that learning can become part of the everyday routine of the users without requiring long sessions.

5 UX/UI Design, Onboarding, and Navigation

The user interface design of Cyberly is based on simplicity, clarity, and functionality, with the aim of creating a user-friendly experience. Because the application addresses a broad audience with different levels of digital familiarity, the interface had to remain understandable even for users without previous technical experience. The design followed a mobile-first approach, focusing on use through touch, clear buttons, sufficient distance between interactive elements, and a structure that does not break on different screen sizes.

Special attention was also given to the colors of the application. Colors were selected so as to create a more friendly, modern, and creative environment without strong contrasts that tire the eye. The main interactive elements, such as buttons, lesson cards, and notifications, are visually distinguished from the rest of the content, helping the user quickly recognize actions.

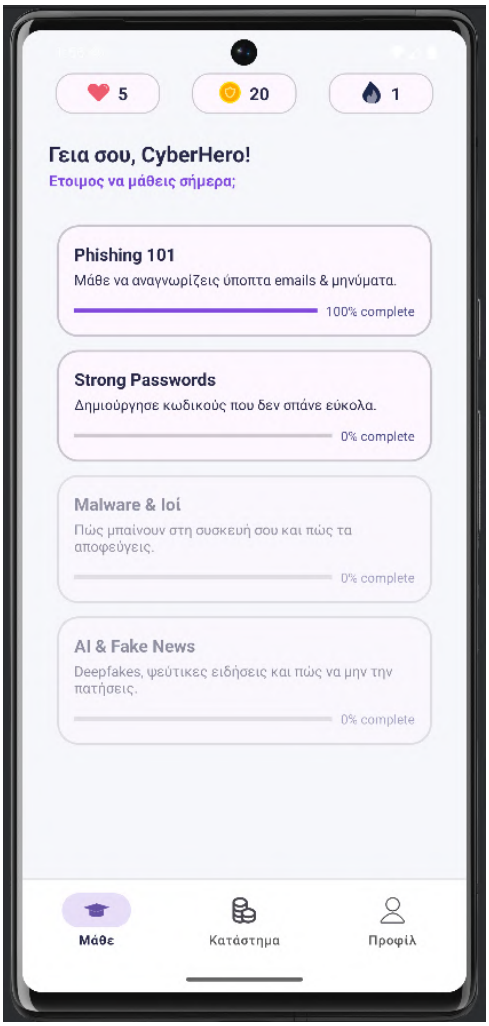

Figure 2: Main screen of the application with lesson cards and basic user interface elements.

The use of cards for presenting learning units was selected in order to make the content more clearly structured. Each unit appears separately, allowing the user to perceive progress and choose the next step with ease. The general aesthetic of the application remains minimalist, with limited decorative elements, so that the user remains focused on the educational content rather than on secondary visual details.

The onboarding process was designed with the purpose of smoothly welcoming the user into the application environment and creating a positive first impression. It does not merely present functions, but aims to activate the user. Through short and understandable messages, the user is informed about the purpose of the application and its learning philosophy. At this stage, the digital mascot Kubo is introduced as a friendly guide of the learning experience. Its role is not purely decorative; it reduces the psychological distance from the subject of cybersecurity, which is often perceived as technical and complicated.

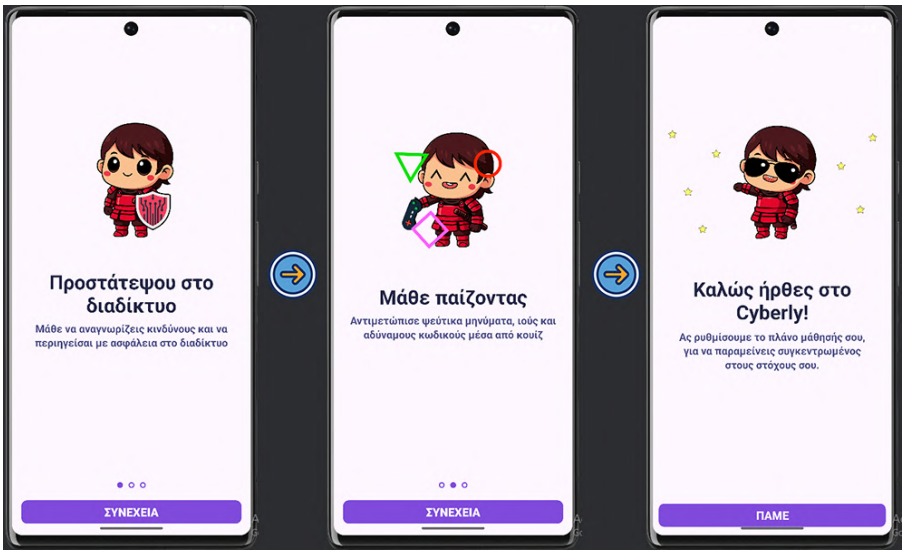

Figure 3: Sequential onboarding screens of the application with the Kubo character.

The application also gives the user the option to define a reminder, promoting consistency in use and helping build study habits. Navigation throughout the application follows the same logic of clarity and simplicity. A fixed bottom navigation menu gives direct access to the basic sections of the application, such as the lessons, the shop, and the user profile. Continue and back buttons remain in stable positions, and each screen gives a clear next action so that the user does not face “dead ends” during navigation.

6 Technologies and Implementation

The development of Cyberly was carried out for the Android platform, a choice based on technical, practical, and strategic criteria. Android is one of the most widespread operating systems for mobile phones on a global scale, which makes it an appropriate environment for the development of educational applications intended for a broad audience. A major reason for this choice is the wide use of Android devices across different price categories, something that increases the accessibility of the application, especially in educational and home environments.

The application was developed in Android Studio, the official integrated development environment for Android applications. Android Studio was used for designing the user interface through XML layouts, allowing the visual design to remain separate from the application logic. The Layout Editor supported the creation of responsive interfaces, while the Android Emulator helped test the application across different Android versions and screen sizes.

The programming language used for the logic of the application is Kotlin. It is a modern language officially supported for Android development, with short syntax and integrated null-safety mechanisms. In Cyberly, Kotlin was used for the implementation of navigation between activities and fragments, for the management of user progress such as lives, experience points, and coins, and for the control of interaction with simulation scenarios. The combination of Android Studio and Kotlin made the development of the application more stable, flexible, and extensible.

The user interface was implemented through XML layouts, separating the presentation level from the logic level. XML also provided flexibility in the structure of screens through the use of layout managers such as `ConstraintLayout` and `LinearLayout`. Particular attention was given to avoiding absolute dimension values and to using guidelines and percentages, so that the application could adapt correctly to different screen sizes.

ViewBinding was adopted for connecting user interface elements to the Kotlin code. This choice avoided the older `findViewById` approach, which requires more manual work and may lead to errors when the IDs or structure of a layout change. In Cyberly, ViewBinding was used across the main screens of the application, including lesson activities, simulation screens, and the user progress environment, contributing to cleaner and safer code.

The application architecture follows a structured approach based on the separation of functions and responsibilities. Navigation is implemented through activities and fragments, while the application structure is divided into packages that separate lesson logic from other functions such as user progress management. The management of lives, coins, XP, and lesson progress is handled through local storage on the device, allowing data persistence each time the application is reopened. This architectural choice aims at a balance between simplicity and extensibility, which fits both the educational goal and the MVP character of the project.

The mascot of the application was not taken from a ready-made template. It was designed through a separate creative process using visual references, character design decisions, and a later animation pipeline. Animation was created in Adobe After Effects, while export to JSON for use in the application was performed so that the animations could be integrated through Lottie. This ensured smooth rendering with relatively low performance overhead.

7 Core Functional Modules and Testing

The educational functionality of Cyberly is primarily organized around the "Learn" section, which combines instructional content, assessment activities, and practical exercises. Each lesson is designed as a self-contained learning unit with a specific educational objective, covering topics such as phishing attacks, malware, password security, and online misinformation. The pedagogical flow of each unit follows three stages: concept introduction, practical examples or scenarios, and knowledge assessment through quizzes.

The learning material is presented through concise educational modules that explain cybersecurity concepts using simple language, visual examples, and scenario-based illustrations. This design aims to make cybersecurity topics accessible to users with limited technical knowledge while maintaining engagement. After studying the theory, users complete short quiz activities that assess their understanding of the presented concepts and encourage active participation in the learning process.

Figure 4 illustrates a representative learning sequence consisting of educational content and a corresponding quiz activity. The combination of instructional content and immediate assessment reinforces knowledge acquisition while preparing learners for practical cybersecurity exercises.

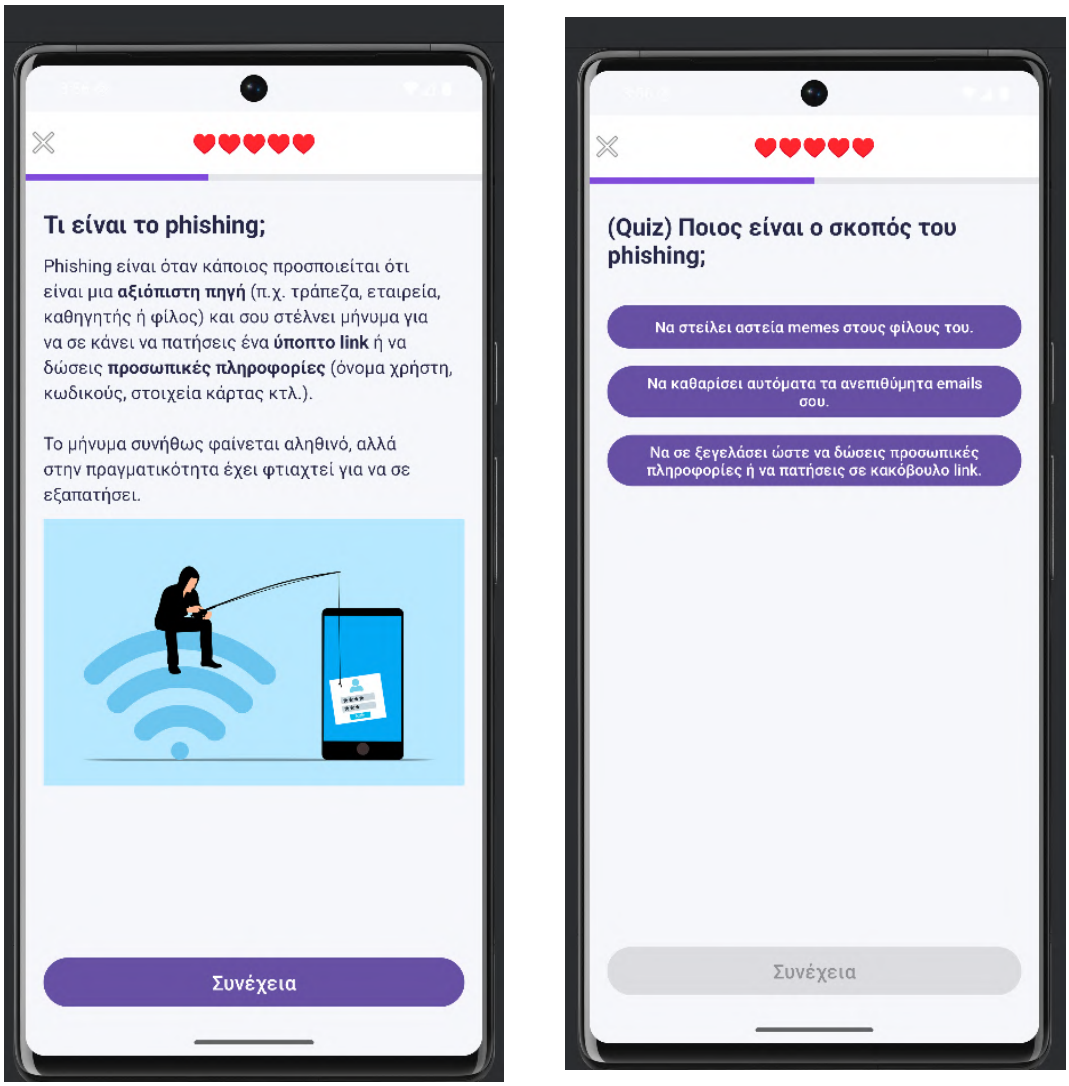

Figure 4: Learning sequence illustrating theory delivery and quiz-based comprehension evaluation.

A central experiential learning component of Cyberly is the interactive phishing simulation. Rather than relying exclusively on theoretical explanations, the application places users in a realistic email environment where they must apply cybersecurity concepts in practice. Before entering the simulation, users receive brief instructions describing the objectives of the activity. They are then asked to inspect incoming messages, identify suspicious indicators, and determine whether an email is legitimate or potentially malicious.

Figure 5 illustrates the complete phishing-awareness workflow, including user guidance, email inspection, and threat identification within a realistic communication environment.

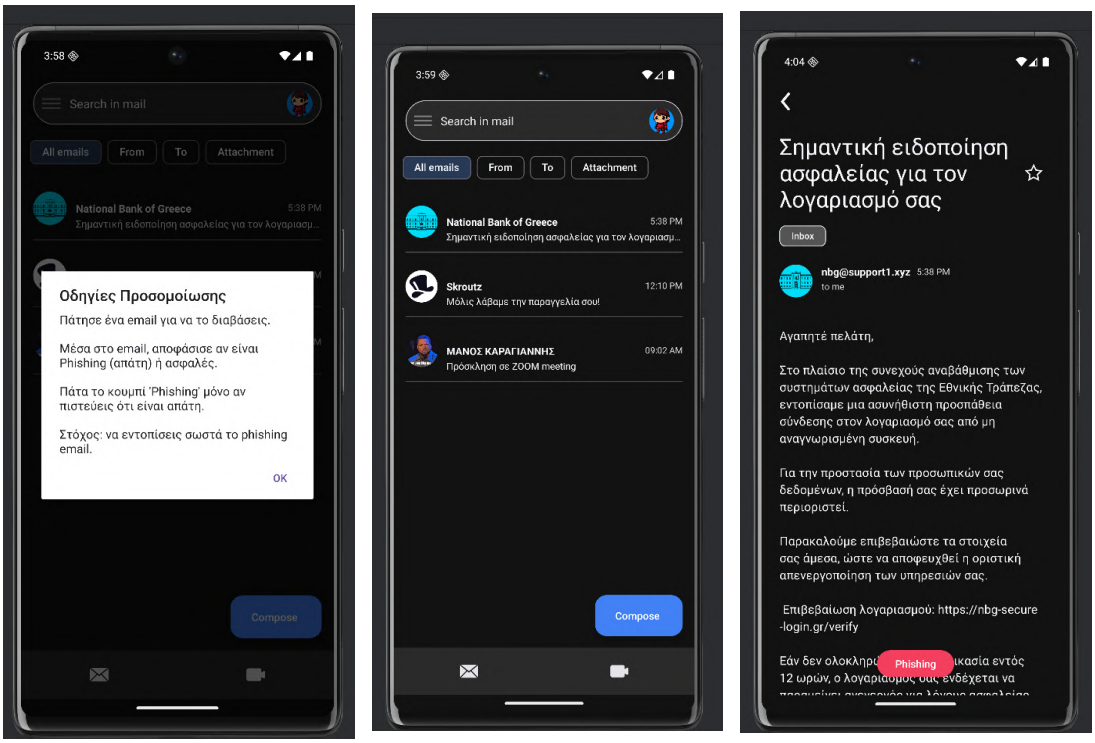

Figure 5: Interactive phishing-awareness scenario demonstrating realistic email inspection and threat identification.

During the simulation, users are required to decide whether a message is safe or suspicious. When an incorrect decision is made, the application does not simply indicate failure; instead, it provides explanatory feedback highlighting the suspicious characteristics of the message. This approach reflects the fact that social engineering attacks primarily exploit human judgment rather than technical vulnerabilities. Consequently, the simulation promotes the development of practical threat-recognition skills and critical decision-making in cybersecurity-related situations.

The progress and reward system constitutes another important component of the gamified learning experience. User progress is represented through experience points (XP), virtual coins, achievement levels, streaks, and available lives. XP is earned through successful completion of lessons, quizzes, and simulation activities, while coins can be used within the application's virtual shop. These mechanisms provide continuous feedback regarding user performance and create short-term and long-term goals that encourage continued engagement.

Figure 6 presents the gamification dashboard, where users can monitor achievements, progression metrics, ranking levels, streaks, and overall learning performance.

Figure 6: User profile dashboard displaying progress metrics, achievements, and engagement indicators.

Supporting functions complement the main educational units. The profile environment allows users to review their overall progress and achievements, while the virtual shop enables the use of coins earned through educational activities. Consistent with the MVP philosophy adopted in this work, these functions were implemented locally without backend dependence or external synchronization.

The control of application functionality was carried out through manual testing. The testing was performed both in the Android Emulator and on a physical Android device in order to confirm the proper adaptation of the interface and the correct operation of the core mechanisms. The tests confirmed that XP remains after restart, coins are updated correctly after purchases, lives do not reset arbitrarily, the streak is maintained according to the application logic, and the reminder is stored correctly. Interface testing also showed that no layout breaks were observed, buttons remained in stable positions, bottom navigation remained functional, lesson cards adapted properly, and animations and the mascot appeared normally and smoothly.

The testing process therefore showed that the application operates stably and according to its design. Although the application does not include automated tests, manual testing covered

the main scenarios of use and confirmed the functional completeness of the MVP.

8 Conclusions

This paper presented the design and development of Cyberly, a gamified tool for learning basic principles of cybersecurity. Through the connection of theory and practical development, the paper presents a complete educational application proposal that responds to a real social need: the strengthening of digital literacy and safer behavior on the Internet.

The development followed the philosophy of a Minimum Viable Product, giving emphasis to basic functionality and stability. The absence of a backend and the limited number of lessons are conscious choices, since they allowed the focus to remain on the educational core and on the quality of the user experience. Despite these limitations, the current version illustrates the practical value of the approach.

Beyond the technical implementation, the application contributes to the broader challenge of strengthening digital education. In an environment where online threats increase day by day and misinformation spreads easily, users need tools that are understandable, practical, and attractive. Cyberly attempts to offer exactly this: a more friendly and accessible way of learning cybersecurity concepts through interactivity, simulation, and reward mechanisms.

Future work will focus on extending the educational content, incorporating personalization features, and assessing the platform's educational effectiveness with real users.

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A Hybrid EDHOC Protocol Post-Quantum Secure against Key Compromise Impersonation*

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Abstract

Recently, the IETF LAKE working group has standardized the EDHOC (Ephemeral Diffie-Hellman Over COSE) protocol [52] that is a lightweight and application-layer authenticated key exchange protocol tailored for resource-constrained IoT devices and networks. In this paper, we propose a hybrid EDHOC (for short, Hybrid-EDHOC2) protocol that provides both classical security and PQ security by instantiating KEM (resp., signature scheme) with PQ KEM (resp., PQ signature scheme). In the Hybrid-EDHOC2 protocol, initiator I and responder R share an authenticated session key. Also, we discuss several classical and PQ security properties (mutual authentication, security of session key, forward secrecy, identity protection, security against identity misbinding, and security against key compromise impersonation) of the Hybrid-EDHOC2 protocol. In addition, we conduct extensive comparisons of the EDHOC [52], PQ-EDHOC [31], PQ-EDHOC1 [30], PQ-EDHOC2 [30], Hybrid-EDHOC [66], and Hybrid-EDHOC2 protocols with respect to security properties and computation/communication efficiency.

1 Introduction

With the rapid and widespread deployments of the IoT (Internet of Things), many cryptographic protocols have been studied to meet several constraints (e.g., low processing power, small memory, limited bandwidth, and battery lifetime) in the literature. An authenticated key exchange protocol is a core cryptographic primitive that allows communicating peers to authenticate each other and share a session key. Recently, the IETF (Internet Engineering Task Force) LAKE (Lightweight Authenticated Key Exchange) working group [10] has standardized the EDHOC (Ephemeral Diffie-Hellman Over COSE) protocol [52, 53] that is a lightweight and application-layer authenticated key exchange protocol tailored for resource-constrained IoT devices and networks (e.g., [41, 45]). According to [27], EDHOC achieves maximum 14 times reduction in message size, 1.44 times improvement in handshake duration, and 2.79 times reduction in energy consumption compared to DTLS 1.3 [50]. On the other hand, the long-term operation of IoT devices introduces a new challenge for classical cryptographic protocols, as Shor's algorithm [67] can efficiently solve the integer factorization and (finite field or elliptic curve) discrete logarithm problems with large-scale quantum computers. In response to the potential future threat, the NIST (National Institute of Standards and Technology) has initiated a PQC (Post-Quantum Cryptography) project [13] to standardize PQ (Post-Quantum) cryptographic algorithms. These developments underscore the urgency of transitioning Internet security protocols to PQ-secure ones.

The EDHOC protocol is designed following a SIGMA family [59] of key exchange protocols that introduce a general approach to authenticated Diffie-Hellman protocols using a combination of signature and MAC (Message Authentication Code). SIGMA is widely used and is a cryptographic basis of many

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security protocols (e.g., IKEv2 [48] and TLS 1.3 [49]). In order to reduce message sizes between the two peers (initiator and responder), SIGMA introduces a MAC-then-Sign variant where MAC is included in a signature message and MAC itself is not sent. Actually, the EDHOC protocol follows the MAC-then-Sign variant of the SIGMA-I protocol that protects the identity of the initiator. In addition to signature keys used for authentication, the EDHOC protocol allows static DH (Diffie-Hellman) keys (cf. EDHOC Authentication Method Type [52]) where a signature is replaced with a MAC, and a MAC key is an ephemeral-static DH key computed in a key schedule similar to the Noise XX pattern [12]. The use of static DH keys results in a significant reduction in message size compared to the use of signature keys.

There are some related works on the EDHOC protocol in the literature. In [65], E. L. Pérez et al. summarized formal symbolic/computational analysis results (e.g., [58, 64, 23, 32, 56, 28]) of earlier versions of the EDHOC protocol, and described how these vulnerabilities found in the security analyses were resolved to meet several security goals. In [31], L. P. Fraile et al. proposed a PQ version of the EDHOC (PQ-EDHOC) protocol by integrating PQ KEMs (Key Encapsulation Mechanisms) and PQ signature schemes, and evaluated various PQ KEM and PQ signature scheme combinations with respect to execution time, energy consumption, memory usage, and network performance. Their evaluation results indicate that ML-KEM [5, 60, 21, 55] emerges as the only suitable PQ KEM choice for resource-constrained environments, while other PQ signature schemes (e.g., HAWK [25]) show significant performance improvement over ML-DSA [5, 61] and FALCON [6] (to be standardized as FIPS (Federal Information Processing Standard)). Note that the PQ-EDHOC protocol is based on EDHOC Authentication Method Type 0 (Signature Key) [52]. Very recently, L. P. Fraile et al. [30] proposed two PQ variants of the PQ-EDHOC protocol (which we call PQ-EDHOC1 and PQ-EDHOC2) that use only a PQ KEM without a PQ signature scheme. However, their proposed protocols adopting an approach of PQNoise [19] require substantial modifications to the original EDHOC protocol [52, 53]. Specifically, the PQ-EDHOC1 protocol (Section III.B in [30]) needs additional fourth and fifth messages to be exchanged, and the PQ-EDHOC2 protocol (Section III.C in [30]) needs to send a KEM ciphertext, computed with the responder's KEM public key, in the first message.

1.1 Motivation and Our Contributions

For a smooth migration to PQC, international organizations [11, 1, 54, 36] recommend a hybrid scheme that combines a classical cryptographic algorithm/protocol with a PQ-secure one to guarantee security against both classical and quantum attackers. For example, ANSSI [1] emphasized that hybridation is crucial to cryptographic security and will remain mandatory until a complete transition to PQC. Up to now, many hybrid schemes (e.g., [20, 26, 51, 29, 35, 34, 43, 33, 44, 42, 39, 37, 38]) that provide both classical and PQ security have been studied and proposed in the literature and by international standard bodies. Very recently, Shin et al. [66] proposed the first hybrid EDHOC (Hybrid-EDHOC) protocol for both classical and PQ security, in an elaborate manner. However, the Hybrid-EDHOC protocol does not provide PQ security against key compromise impersonation.

In this paper, we propose another hybrid EDHOC (for short, Hybrid-EDHOC2) protocol that provides both classical security and PQ security by instantiating KEM (resp., signature scheme) with PQ KEM (resp., PQ signature scheme). Specifically, the Hybrid-EDHOC2 protocol is based on an authentication method of static ECDH keys and static PQ signature keys. In the Hybrid-EDHOC2 protocol, initiator I and responder R share an authenticated session key PRK_{out}. Unlike Hybrid-EDHOC [66], the Hybrid-EDHOC2 protocol provides double authentication based on a MAC and a PQ signature, and PQ security against key compromise impersonation. Also, we discuss several classical and PQ security properties (mutual authentication, security of session key, forward secrecy, identity protection, security against identity misbinding, and security against key compromise impersonation) of the Hybrid-EDHOC2 protocol. In addition, we conduct extensive comparisons of the EDHOC [52], PQ-EDHOC

[31], PQ-EDHOC1 [30], PQ-EDHOC2 [30], Hybrid-EDHOC [66], and Hybrid-EDHOC2 (Section 3) protocols with respect to security properties and computation/communication efficiency.

2 Preliminaries

Table 1 shows abbreviations and notations to be used throughout this paper.

Table 1: Abbreviations and notations

	Meanings
I	Initiator identity
R	Responder identity
(a,A)	Static ECDH private-public key pair of I
(b,B)	Static ECDH private-public key pair of R
(sk_I,VK_I)	Static PQ signature signing-verification key pair of I
(sk_R,VK_R)	Static PQ signature signing-verification key pair of R
(x,X)	Ephemeral ECDH private-public key pair of I
(y,Y)	Ephemeral ECDH private-public key pair of R
(sk_{KEM},PK_{KEM})	Ephemeral PQ KEM private-public key pair of I
H	Cryptographic hash function
KDF	Key derivation function (EDHOC_KDF [52])
Extract	Key extraction function (EDHOC_Extract [52])
Expand	Key expansion function (EDHOC_Expand [52])
Enc	Symmetric-key encryption (XOR stream cipher or AEAD)
Dec	Symmetric-key decryption (XOR stream cipher or AEAD)
TH	Transcript hash
PRK	Pseudo-random key
EK	Encryption key
IV	Initialization vector
MK	MAC key
METHOD	EDHOC Authentication Method Type [52]
SUITES	EDHOC Cipher Suites [52] (an ordered set of algorithms for crypto agility, flexibility and modularity)
C	Connection Identifier
EAD	External Authorization Data
len	MAC length

2.1 Key Encapsulation Mechanism

A KEM (Key Encapsulation Mechanism) [24, 63] is a triple of the following algorithms.

- **(PQ-)KEM.KeyGen**(pp_{KEM}): On input of a public parameter pp_{KEM}, the key generation algorithm outputs a KEM private-public key pair (sk_{KEM}, PK_{KEM}).
- **(PQ-)KEM.Encaps**(PK_{KEM}): On input of a KEM public key PK_{KEM}, the encapsulation algorithm outputs a KEM shared key k_{KEM} and a KEM ciphertext C_{KEM}.
- **(PQ-)KEM.Decaps**(sk_{KEM}, C_{KEM}): On input of a KEM private key sk_{KEM} and a KEM ciphertext C_{KEM}, the decapsulation algorithm outputs a KEM shared key k_{KEM}.

Representative PQ KEMs include ML-KEM (CRYSTALS-Kyber) [5, 60, 21, 55], FrodoKEM [7, 21, 55], Classic McEliece KEM [4, 21, 55], and BIKE [3].

2.2 Signature Scheme

A signature scheme [57] is a triple of the following algorithms.

- **(PQ-)Sig.KeyGen**(pp_{Sig}): On input of a public parameter pp_{Sig}, the key generation algorithm outputs a signature signing-verification key pair (sk_{ID}, VK_{ID}).
- **(PQ-)Sig.Sign**(sk_{ID}, msg): On input of a signing key sk_{ID} and a message msg, the signing algorithm outputs a signature σ_{ID}.
- **(PQ-)Sig.Verify**$(VK_{ID}, msg, \sigma_{ID})$: On input of a verification key VK_{ID}, a message msg, and a signature σ_{ID}, the verification algorithm outputs `valid` or $\perp$ (indicating invalid).

Representative PQ signature schemes include ML-DSA (CRYSTALS-Dilithium) [5, 61], SLH-DSA [16, 62], FALCON [6], HBS [17], HAWK [25], and HAETAE [22].

3 Another Hybrid EDHOC Protocol

In this section, we propose another hybrid EDHOC (for short, Hybrid-EDHOC2) protocol that provides both classical security and PQ security by instantiating KEM (resp., signature scheme) with PQ KEM (resp., PQ signature scheme). Specifically, the Hybrid-EDHOC2 protocol is based on an authentication method of static ECDH keys and static PQ signature keys. Main ideas of the Hybrid-EDHOC2 protocol are that (1) static ECDH raw public keys and static PQ signature raw verification keys are dealt with pre-shared secrets between initiator I and responder R; (2) an ephemeral PQ KEM is used for forward secrecy; (3) a PQ KEM shared key is used as an additional key material in the computation of PRK; and (4) static PQ signature key is used for PQ security against key compromise impersonation. In the Hybrid-EDHOC2 protocol, initiator I and responder R share an authenticated session key PRK_{out}. Also, the Hybrid-EDHOC2 protocol provides forward secrecy, identity protection, security against identity misbinding, and security against key compromise impersonation. Before executing the Hybrid-EDHOC2 protocol, I holds $((a, A \equiv g^a), (sk_I, VK_I), (R, B, VK_R))$ secretly and R stores $((b, B \equiv g^b), (sk_R, VK_R), (I, A, VK_I))$ secretly.[1] For a graphical description of the Hybrid-EDHOC2 protocol, see Figure 1.

[1] These pre-shared secrets are installed or provisioned out-of-band in the initialization phase (e.g., at the manufacturing process of IoT devices). See Section 5.1 for a more detailed discussion.

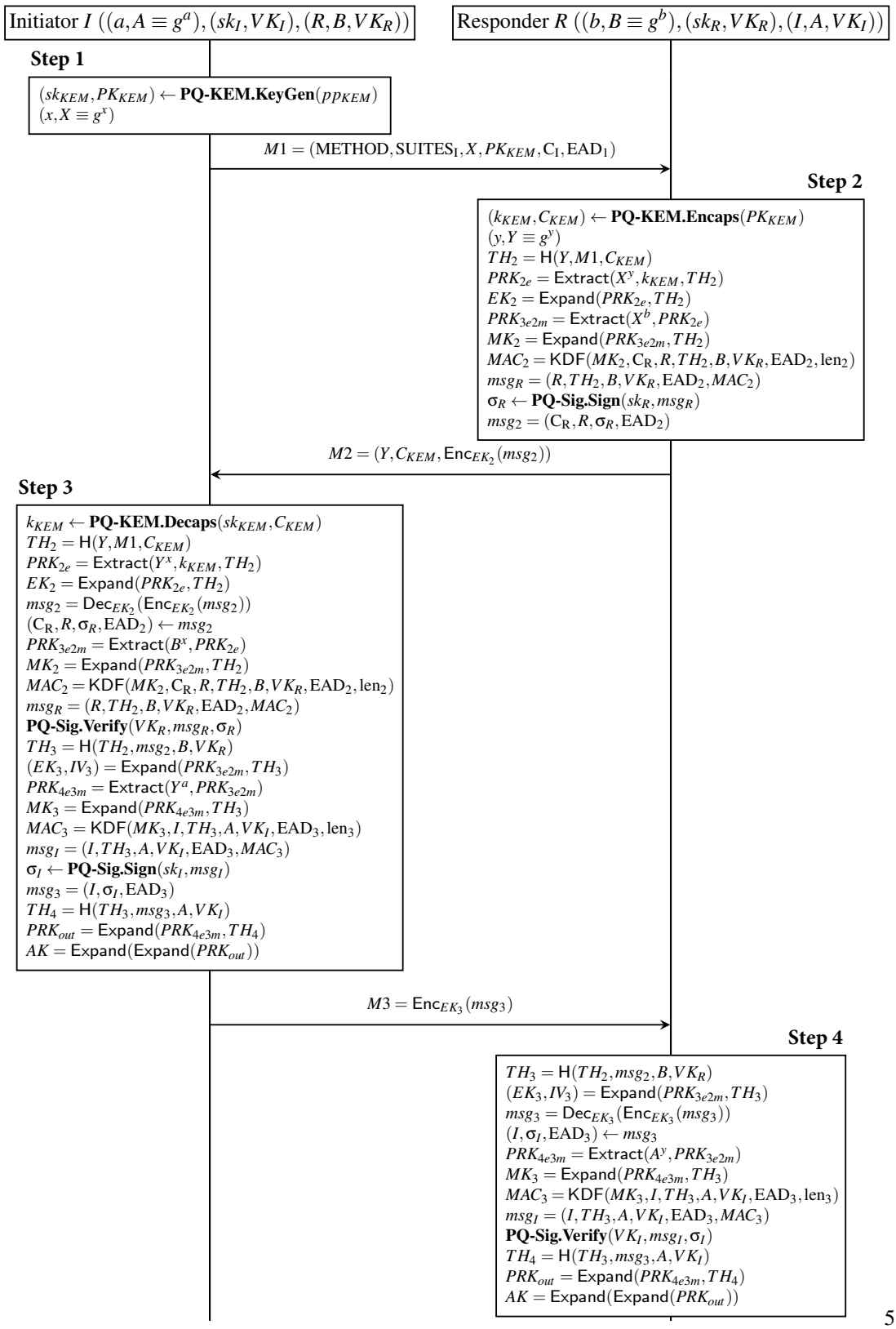

Figure 1: The Hybrid-EDHOC2 protocol

3.1 Step 1 (Initiator)

Initiator I executes the followings:

1. (Pre-computable) Generate an ephemeral KEM private-public key pair $(sk_{KEM}, PK_{KEM}) \leftarrow$ **PQ-KEM.KeyGen**(pp_{KEM})
2. (Pre-computable) Generate an ephemeral ECDH private-public key pair (x, X) such that $X \equiv g^x$
3. Set $M1 = (\text{METHOD}, \text{SUITES}_\text{I}, X, PK_{KEM}, \text{C}_\text{I}, \text{EAD}_1)$

I sends $M1$ to R.

3.2 Step 2 (Responder)

Upon receiving $M1$ from I, responder R executes the followings:

1. Compute a KEM shared key and a KEM ciphertext $(k_{KEM}, C_{KEM}) \leftarrow$ **PQ-KEM.Encaps**(PK_{KEM})
2. (Pre-computable) Generate an ephemeral ECDH private-public key pair (y, Y) such that $Y \equiv g^y$
3. Compute $TH_2 = \mathsf{H}(Y, M1, C_{KEM})$
4. Compute $PRK_{2e} = \mathsf{Extract}(X^y, k_{KEM}, TH_2)$ and $EK_2 = \mathsf{Expand}(PRK_{2e}, TH_2)$
5. Compute $PRK_{3e2m} = \mathsf{Extract}(X^b, PRK_{2e})$ and $MK_2 = \mathsf{Expand}(PRK_{3e2m}, TH_2)$
6. Compute $MAC_2 = \mathsf{KDF}(MK_2, \text{C}_\text{R}, R, TH_2, B, VK_R, \text{EAD}_2, \text{len}_2)$
7. Set $msg_R = (R, TH_2, B, VK_R, \text{EAD}_2, MAC_2)$
8. Compute a signature $\sigma_R \leftarrow$ **PQ-Sig.Sign**(sk_R, msg_R)
9. Set $msg_2 = (\text{C}_\text{R}, R, \sigma_R, \text{EAD}_2)$
10. Set $M2 = (Y, C_{KEM}, \mathsf{Enc}_{EK_2}(msg_2))$

R sends $M2$ to I.

3.3 Step 3 (Initiator)

Upon receiving $M2$ from R, initiator I executes the followings:

1. Compute a KEM shared key $k_{KEM} \leftarrow$ **PQ-KEM.Decaps**(sk_{KEM}, C_{KEM})
2. Compute $TH_2 = \mathsf{H}(Y, M1, C_{KEM})$
3. Compute $PRK_{2e} = \mathsf{Extract}(Y^x, k_{KEM}, TH_2)$ and $EK_2 = \mathsf{Expand}(PRK_{2e}, TH_2)$
4. Compute $msg_2 = \mathsf{Dec}_{EK_2}(\mathsf{Enc}_{EK_2}(msg_2))$
5. Parse $(\text{C}_\text{R}, R, \sigma_R, \text{EAD}_2) \leftarrow msg_2$
6. Compute $PRK_{3e2m} = \mathsf{Extract}(B^x, PRK_{2e})$ and $MK_2 = \mathsf{Expand}(PRK_{3e2m}, TH_2)$
7. Compute $MAC_2 = \mathsf{KDF}(MK_2, \text{C}_\text{R}, R, TH_2, B, VK_R, \text{EAD}_2, \text{len}_2)$

8. Set $msg_R = (R, TH_2, B, VK_R, \mathrm{EAD}_2, MAC_2)$
9. If **PQ-Sig.Verify**(VK_R, msg_R, σ_R) is `valid`, proceed to the next.
10. Compute $TH_3 = \mathsf{H}(TH_2, msg_2, B, VK_R)$
11. Compute $(EK_3, IV_3) = \mathsf{Expand}(PRK_{3e2m}, TH_3)$
12. Compute $PRK_{4e3m} = \mathsf{Extract}(Y^a, PRK_{3e2m})$ and $MK_3 = \mathsf{Expand}(PRK_{4e3m}, TH_3)$
13. Compute $MAC_3 = \mathsf{KDF}(MK_3, I, TH_3, A, VK_I, \mathrm{EAD}_3, \mathrm{len}_3)$
14. Set $msg_I = (I, TH_3, A, VK_I, \mathrm{EAD}_3, MAC_3)$
15. Compute a signature $\sigma_I \leftarrow$ **PQ-Sig.Sign**(sk_I, msg_I)
16. Set $msg_3 = (I, \sigma_I, \mathrm{EAD}_3)$
17. Set $M3 = \mathsf{Enc}_{EK_3}(msg_3)$
18. Compute $TH_4 = \mathsf{H}(TH_3, msg_3, A, VK_I)$
19. Compute a session key $PRK_{out} = \mathsf{Expand}(PRK_{4e3m}, TH_4)$ and an application key $AK = \mathsf{Expand}(\mathsf{Expand}(PRK_{out}))$

I sends $M3$ to R.

3.4 Step 4 (Responder)

Upon receiving $M3$ from I, responder R executes the followings:

1. Compute $TH_3 = \mathsf{H}(TH_2, msg_2, B, VK_R)$
2. Compute $(EK_3, IV_3) = \mathsf{Expand}(PRK_{3e2m}, TH_3)$
3. Compute $msg_3 = \mathsf{Dec}_{EK_3}(\mathsf{Enc}_{EK_3}(msg_3))$
4. Parse $(I, \sigma_I, \mathrm{EAD}_3) \leftarrow msg_3$
5. Compute $PRK_{4e3m} = \mathsf{Extract}(A^y, PRK_{3e2m})$ and $MK_3 = \mathsf{Expand}(PRK_{4e3m}, TH_3)$
6. Compute $MAC_3 = \mathsf{KDF}(MK_3, I, TH_3, A, VK_I, \mathrm{EAD}_3, \mathrm{len}_3)$
7. Set $msg_I = (I, TH_3, A, VK_I, \mathrm{EAD}_3, MAC_3)$
8. If **PQ-Sig.Verify**(VK_I, msg_I, σ_I) is `valid`, proceed to the next.
9. Compute $TH_4 = \mathsf{H}(TH_3, msg_3, A, VK_I)$
10. Compute a session key $PRK_{out} = \mathsf{Expand}(PRK_{4e3m}, TH_4)$ and an application key $AK = \mathsf{Expand}(\mathsf{Expand}(PRK_{out}))$

In the above, Enc in $M2$ is an XOR stream cipher, and Enc in $M3$ is an AEAD (Authenticated Encryption with Associated Data). Also, an optional fourth message for key confirmation is omitted as in the EDHOC protocol [52].

Note that the Hybrid-EDHOC protocol [66] inherently does not provide PQ security against key compromise impersonation since its construction is based on an authentication method of static ECDH keys only. For example, a quantum attacker, who gets responder R's secrets $((b,B),(I,A))$, can impersonate initiator I by retrieving the static ECDH private key a from the static ECDH raw public key A.

4 Security

Here, we discuss several classical and PQ security properties (mutual authentication, security of session key, forward secrecy, identity protection, security against identity misbinding, and security against key compromise impersonation) of the Hybrid-EDHOC2 protocol.

4.1 Mutual Authentication and Security of Session Key

In general, (semantic) security of session key implies mutual authentication in authenticated key exchange protocols. Suppose a classical or quantum attacker, who can fully control the communications by eavesdropping/inserting/modifying messages and impersonating one peer. The Hybrid-EDHOC2 protocol provides mutual authentication and security of session key since static ECDH keys and static PQ signature keys are used for authentication between initiator I and responder R. Regarding classical security of session key, a classical attacker cannot compute an authenticated ephemeral-static ECDH key g^{xb} and an authenticated static-ephemeral ECDH key g^{ay}, both of which are used in the computation of PRK. The same reason applies to PQ security of session key since the static ECDH raw public keys (g^a and g^b) are pre-shared secrets as well, and these public keys do not appear in public.

4.2 Forward Secrecy

Suppose a classical or quantum attacker, who gets all the secrets $((a,A),(sk_I,VK_I)),((b,B),(sk_R,VK_R))$ from initiator I and responder R, and tries to find out the previously-established session key PRK_{out}. The Hybrid-EDHOC2 protocol provides forward secrecy since ephemeral ECDH keys and an ephemeral PQ KEM are used. Regarding classical forward secrecy, a classical attacker cannot compute an ephemeral-ephemeral ECDH key g^{xy} used in the key derivation schedule of PRK_{out}. Regarding PQ forward secrecy, a quantum attacker cannot compute an ephemeral PQ KEM shard key k_{KEM} used in the key derivation schedule of PRK_{out}.

4.3 Identity Protection of I

Suppose a classical or quantum attacker, who can fully control the communications by eavesdropping/inserting/modifying messages and impersonating responder R, and tries to find out initiator identity I. As in the EDHOC protocol [52], the Hybrid-EDHOC2 protocol provides identity protection of I since initiator identity I is sent in an encrypted form $M3$ with an encryption key EK_3, and only the legitimate responder R can decrypt the third message $M3$. Regarding classical identity protection of I, a classical attacker cannot compute an authenticated ephemeral-static ECDH key g^{xb}, used in the computation of EK_3 and a MAC value MAC_2. Regarding PQ identity protection of I, a quantum attacker cannot compute an authenticated ephemeral-static ECDH key g^{xb}, used in the computation of EK_3 and a MAC value MAC_2, and cannot generate a valid PQ signature σ_R.

4.4 Security against Identity Misbinding

Suppose a classical or quantum attacker, who can fully control the communications by eavesdropping/inserting/modifying messages and impersonating one peer, and then tries to make a third peer share a session key PRK_{out} without the initiator I and responder R noticing. The Hybrid-EDHOC2 protocol provides security against identity misbinding since initiator identity I and responder identity R are authenticated by using MACs and PQ signatures, and these identities are included in the key derivation schedule of PRK_{out}. Regarding classical security against identity misbinding, a classical attacker cannot compute

an authenticated ephemeral-static ECDH key g^{xb}, used in the computation of MAC_2, and an authenticated static-ephemeral ECDH key g^{ay}, used in the computation of MAC_3. Regarding PQ security against identity misbinding, a quantum attacker cannot compute an authenticated ephemeral-static ECDH key g^{xb}, used in the computation of MAC_2, and an authenticated static-ephemeral ECDH key g^{ay}, used in the computation of MAC_3, and cannot generate valid PQ signatures σ_R and σ_I.

4.5 Security against Key Compromise Impersonation

Suppose a classical or quantum attacker, who gets initiator I's secrets $((a,A),(sk_I,VK_I),(R,B,VK_R))$ (resp., responder R's secrets $((b,B),(sk_R,VK_R),(I,A,VK_I))$, and then tries to impersonate responder R (resp., initiator I). The Hybrid-EDHOC2 protocol provides security against key compromise impersonation since the static ECDH private keys and PQ signatures are used in both initiator I and responder R sides. Regarding classical security against key compromise impersonation, a classical attacker, who gets initiator I's secrets (resp., responder R's secrets) and impersonates responder R (resp., initiator I), cannot compute an authenticated ephemeral-static ECDH key g^{xb} (resp., an authenticated static-ephemeral ECDH key g^{ay}), used in the computation of MAC_2 (resp., MAC_3). Regarding PQ security against key compromise impersonation, we give a reduction to EUF-CMA (Existential Unforgeability under Chosen Message Attack) security of a PQ signature scheme. Let $\mathcal{B}$ be a quantum attacker who can break the EUF-CMA security of the PQ signature scheme. We assume that there exists a quantum attacker $\mathcal{A}$ who can break PQ security against key compromise impersonation in the Hybrid-EDHOC2 protocol. Whenever a PQ signing query comes from $\mathcal{A}$, $\mathcal{B}$ forwards the query to its own PQ signing oracle. By using $\mathcal{A}$, we can trivially construct the attacker $\mathcal{B}$ since the only winning way for $\mathcal{A}$ is to generate a valid PQ signature without knowing the static PQ signature signing key sk. Of course, the static PQ signature verification key VK is useless to $\mathcal{A}$.

5 Comparison

In this section, we compare the EDHOC [52] and relevant protocols (PQ-EDHOC [31], PQ-EDHOC1 [30], PQ-EDHOC2 [30], Hybrid-EDHOC [66], and Hybrid-EDHOC2 (Section 3)) with respect to several classical and PQ security properties and efficiency in terms of computation costs and communication bandwidths.

5.1 Classification

Before security and efficiency comparisons, we first classify the EDHOC [52], Hybrid-EDHOC2 (Section 3), and relevant protocols [31, 30, 66, 40] in Table 2, according to whether each protocol depends on PKI (Public Key Infrastructure) or not.

Table 2: Classification of the EDHOC [52], Hybrid-EDHOC2, and relevant protocols

	PKI-based (Scalable)	Secret-key based (Not scalable)
Classical security	EDHOC (Method Type 0, 1, 2, 3) [52]	EDHOC-PSK [40]
PQ security	PQ-EDHOC [31], PQ-EDHOC1 [30], PQ-EDHOC2 [30]	EDHOC-PSK [40]
Hybrid security		Hybrid-EDHOC [66], Hybrid-EDHOC2 (Section 3)

A distinctive feature of the Hybrid-EDHOC2 protocol is that public-key certificates for initiator I and responder R are not used, and static ECDH raw public keys and static PQ signature raw verification keys are treated as pre-shared secrets. In practice, the initiator I (resp., the responder R) is an IoT device (resp., a backend IoT platform/server), and most IoT devices are configured with pre-shared secrets installed or provisioned out-of-band at the manufacturing process for secure communication (e.g., firmware updates). Of course, these secrets are shared with (and managed by) the corresponding backend IoT platforms/servers. Other examples include AKMA (Authentication and Key Management for Applications) [2] and EAP (Extensible Authentication Protocol) [47, 46]. In this regard, the Hybrid-EDHOC2 protocol is more suitable for the real world than the EDHOC [52], PQ-EDHOC [31], PQ-EDHOC1 [30], and PQ-EDHOC2 [30] protocols. In addition, the Hybrid-EDHOC2 protocol can reduce the computation, memory, communication bandwidth, and energy costs of validating public-key certificates. This is a reason why the IETF LAKE working group has been standardizing a pre-shared key authentication method for EDHOC [40]. However, the EDHOC-PSK protocol [40] is an Internet-Draft and a work in progress.

Table 3 shows a design difference between the Hybrid-EDHOC [66] and Hybrid-EDHOC2 protocols, where the latter provides double authentication based on a MAC and a PQ signature.

Table 3: Design difference between the Hybrid-EDHOC [66] and Hybrid-EDHOC2 protocols

	Hybrid-EDHOC [66]	Hybrid-EDHOC2 (Section 3)
Static ECDH key pair	Yes	Yes
Static PQ signature key pair	No	Yes
MAC authentication	Yes	Yes (Implicit)
Signature authentication	No	Yes

5.2 Security and Efficiency Comparisons

In Table 4, we compare the EDHOC [52] and relevant protocols with respect to several classical and PQ security properties (mutual authentication, security of session key, forward secrecy, identity protection, security against identity misbinding, and security against key compromise impersonation). It is clear that the Hybrid-EDHOC2 protocol provides both classical security and PQ security simultaneously.

In Tables 5, 6, 7, and 8, we compare the EDHOC [52] with relevant protocols with respect to computation and communication efficiency. Note that the PQ-EDHOC1 [30] and PQ-EDHOC2 [30] protocols require substantial modifications to the original EDHOC protocol [52, 53]. Specifically, the PQ-EDHOC1 protocol needs additional fourth and fifth messages to be exchanged, and the PQ-EDHOC2 protocol needs to send a KEM ciphertext, computed with the responder's KEM public key, in the first message. Compared to PQ-EDHOC1 [30] and PQ-EDHOC2 [30], the Hybrid-EDHOC2 protocol does not alter the core EDHOC message structure at all.

6 Conclusion

In this paper, we proposed another hybrid EDHOC (Hybrid-EDHOC2) protocol that provides both classical security and PQ security by instantiating KEM (resp., signature scheme) with PQ KEM (resp., PQ signature scheme). Specifically, the Hybrid-EDHOC2 protocol is based on an authentication method of static ECDH keys and static PQ signature keys. In the Hybrid-EDHOC2 protocol, initiator I and responder R share an authenticated session key PRK_{out}. Unlike Hybrid-EDHOC [66], the Hybrid-EDHOC2

Table 4: Security comparison of the EDHOC [52] and relevant protocols

	EDHOC [52]		PQ-EDHOC [31], PQ-EDHOC1 [30], PQ-EDHOC2 [30]		Hybrid-EDHOC [66]		Hybrid-EDHOC2 (Section 3)	
	Classical security	PQ security	Classical security	PQ security	Classical security	PQ security	Classical security	PQ security
Mutual authentication	Yes	No	No	Yes	Yes	Yes	Yes	Yes
Security of session key	Yes	No	No	Yes	Yes	Yes	Yes	Yes
Forward secrecy	Yes	No	No	Yes	Yes	Yes	Yes	Yes
Identity protection of I	Yes	No	No	Yes	Yes	Yes	Yes	Yes
Security against identity misbinding	Yes	No	No	Yes	Yes	Yes	Yes	Yes
Security against key compromise impersonation	Yes	No	No	Yes	Yes	No	Yes	Yes

Table 5: Computation efficiency comparison of the EDHOC [52], Hybrid-EDHOC [66], and Hybrid-EDHOC2 protocols, where the numbers in parentheses are computation costs excluding pre-computable ones

	EDHOC (Method Type 3) [52]		EDHOC (Method Type 0) [52]		Hybrid-EDHOC [66]		Hybrid-EDHOC2 (Section 3)	
	I	R	I	R	I	R	I	R
Scalar multiplication	4 (3)	4 (3)	2 (1)	2 (1)	4 (3)	4 (3)	4 (3)	4 (3)
Sig.Sign			1	1				
Sig.Verify			1	1				
PQ-KEM.KeyGen					1 (0)		1 (0)	
PQ-KEM.Encaps						1		1
PQ-KEM.Decaps					1		1	
PQ-Sig.Sign							1	1
PQ-Sig.Verify							1	1
Certificate validation	1	1	1	1				

protocol provides double authentication based on a MAC and a PQ signature, and PQ security against key compromise impersonation. Also, we discussed several classical and PQ security properties (mutual authentication, security of session key, forward secrecy, identity protection, security against identity misbinding, and security against key compromise impersonation) of the Hybrid-EDHOC2 protocol. In

Table 6: Computation efficiency comparison of the PQ-EDHOC [31], PQ-EDHOC1 [30], and PQ-EDHOC2 [30] protocols, where the numbers in parentheses are computation costs excluding pre-computable ones

	PQ-EDHOC [31]		PQ-EDHOC1 [30]		PQ-EDHOC2 [30]	
	I	*R*	*I*	*R*	*I*	*R*
PQ-KEM.KeyGen	1 (0)		1 (0)		1 (0)	
PQ-KEM.Encaps		1	1	2	1	2
PQ-KEM.Decaps	1		2	1	2	1
PQ-Sig.Sign	1	1				
PQ-Sig.Verify	1	1				
PQ certificate validation	1	1	1	1		1

Table 7: Communication efficiency comparison of the EDHOC [52], Hybrid-EDHOC [66], and Hybrid-EDHOC2 protocols, where $\Delta = |\text{METHOD}| + |\text{SUITES}_\text{I}| + |X| + |\text{C}_\text{I}| + |\text{EAD}_1|$ and $|\cdot|$ indicates a bit-length

	EDHOC (Method Type 3) [52]	EDHOC (Method Type 0) [52]	Hybrid-EDHOC [66]	Hybrid-EDHOC2 (Section 3)
$M1$	Δ	Δ	$\Delta + \|PK_{KEM}\|$	$\Delta + \|PK_{KEM}\|$
$M2$	$\|Y\| + \|\text{Enc}\|$	$\|Y\| + \|\text{Enc}\|$	$\|Y\| + \|\text{Enc}\| + \|C_{KEM}\|$	$\|Y\| + \|\text{Enc}\| + \|C_{KEM}\|$
$M3$	$\|\text{Enc}\|$	$\|\text{Enc}\|$	$\|\text{Enc}\|$	$\|\text{Enc}\|$

addition, we conducted extensive comparisons of the EDHOC [52], PQ-EDHOC [31], PQ-EDHOC1 [30], PQ-EDHOC2 [30], Hybrid-EDHOC [66], and Hybrid-EDHOC2 (Section 3) protocols with respect to security properties and computation/communication efficiency.

Future works include (1) formal proofs of the Hybrid-EDHOC2 protocol by using an automated symbolic security verification tool (e.g., ProVerif [15] or Tamarin Prover [18]); (2) implementation of the Hybrid-EDHOC2 protocol with a concrete PQ KEM (e.g., ML-KEM) and a concrete PQ signature scheme (e.g., ML-DSA) for NIST PQC security strength categories [14]; 3) quantitative performance measurements (e.g., runtime, memory usage, message size, and energy cost) and evaluation of the Hybrid-EDHOC2 protocol on constrained IoT devices; and (4) handling network-layer fragmentation and packet-loss overhead during the Hybrid-EDHOC2 handshake, while considering the current IEEE 802.15.4 [8] or 6LoWPAN [9] packet sizes.

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Table 8: Communication efficiency comparison of the PQ-EDHOC [31], PQ-EDHOC1 [30], and PQ-EDHOC2 [30] protocols, where $\delta = |\text{METHOD}| + |\text{SUITES}_\text{I}| + |\text{C}_\text{I}| + |\text{EAD}_1|$ and $|\cdot|$ indicates a bit-length

	PQ-EDHOC [31]	PQ-EDHOC1 [30]	PQ-EDHOC2 [30]
$M1$	$\delta + \lvert PK_{KEM}\rvert$	$\delta + \lvert PK_{KEM}\rvert$	$\lvert PK_{KEM}\rvert + \lvert C_{KEM}\rvert + \lvert\text{Enc}\rvert$
$M2$	$\lvert\text{Enc}\rvert + \lvert C_{KEM}\rvert$	$\lvert\text{Enc}\rvert + \lvert C_{KEM}\rvert$	$\lvert\text{Enc}\rvert + 2\lvert C_{KEM}\rvert$
$M3$	$\lvert\text{Enc}\rvert$	$\lvert\text{Enc}\rvert + \lvert C_{KEM}\rvert$	$\lvert\text{Enc}\rvert$
$M4$		$\lvert\text{Enc}\rvert + \lvert C_{KEM}\rvert$	
$M5$		$\lvert\text{Enc}\rvert$	

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Design of a Low-Latency RDMA-Based Publish/Subscribe Framework for Inter-Machine GPU Communications*

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Abstract

Modern GPU-centric applications increasingly execute as multi-stage pipelines in which one stage produces large payloads directly in GPU memory and multiple downstream stages must consume the same data with low latency. In conventional Publish/Subscribe models, messages are disseminated from a publisher to all subscribers registered to a specific topic. However, this paradigm is rarely applied to GPU-centric applications. This paper presents Remote GPU-Aware Publish/Subscribe (RGAPS), a low-latency framework for cross-machine GPU communication. RGAPS extends single-host GPU-aware pub/sub architecture to a two-host distributed setting by combining shared CUDA-RDMA pinned memory for intra-node aggregation, RDMA Write with Immediate for CPU-bypassing inter-node transport, and a shared GPU memory pool for one-time materialization and local fan-out on the subscriber machine. The design preserves the "publish once, fan-out many" principle across machines while avoiding per-subscriber payload duplication. Experiments on a real two-node platform with RTX 3090 GPUs, ConnectX-7 RNICs, PCIe Gen4, and InfiniBand NDR show that RGAPS reduces end-to-end latency by roughly 6×–10× relative to conventional CPU-centric baselines and keeps local fan-out cost nearly insensitive to the number of subscribers. These results demonstrate that a carefully integrated RDMA-based data plane can substantially improve the subscriber scalability and latency of inter-machine GPU publish/subscribe communication.

1 Introduction

The publish/subscribe (pub/sub) pattern is a widely used communication abstraction that decouples publishers (producers) from subscribers (consumers) in space, time, and synchronization. Publishers send data to topics rather than to specific receivers, and the middleware is responsible for routing and dissemination. This decoupling is attractive in large-scale systems because it simplifies software composition and supports many-to-many communication.

Different pub/sub systems prioritize different goals. Apache Kafka [12] emphasizes high throughput, durable storage, and replay through a distributed log architecture. ROS 2 relies on DDS to support modular and real-time robotic pipelines. Eclipse Iceoryx [13, 15] focuses on deterministic intra-machine inter-process communication and achieves zero-copy delivery for host-resident data through shared memory. Eclipse Zenoh [14, 16] targets geographically distributed and heterogeneous deployments by combining lightweight pub/sub with query functionality. These systems provide valuable abstractions, but their dominant data path remains host-centric.

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This assumption falls short of supporting modern GPU-driven workloads. In many real-time and high-throughput pipelines, producers generate results directly in GPU memory, and downstream stages also execute on GPUs. Typical payloads include tensors, feature maps, image batches, point clouds, embeddings, or intermediate outputs of deep learning models. When conventional pub/sub middleware is placed between such stages, GPU-produced payloads are staged through host memory, transmitted by CPU-centric communication paths, and often copied back into GPU memory separately for each subscriber. As message sizes grow from megabytes to gigabytes, these redundant data movements become a major contributor to end-to-end latency and can amplify contention on PCIe and memory subsystems.

A prior work on single-host GPU-aware publish/subscribe [3] addresses this problem by treating GPU memory as a first-class data plane. Instead of distributing payload copies, single-host GPU-aware publish/subscribe distributes lightweight metadata that allows local subscribers to access a shared GPU-resident buffer. This design removes per-subscriber payload duplication and achieves stable low latency for intra-node fan-out. However, it is fundamentally limited by host locality: publishers and subscribers must reside on the same machine, and each topic consumes GPU memory capacity in a single-host shared pool. As workload diversity and topic count increase, GPU memory pressure and resource placement rigidity become important limitations.

This paper presents Remote GPU-Aware Publish/Subscribe (RGAPS), a low-latency RDMA-based pub/sub framework for inter-machine GPU communication. RGAPS extends the main idea of single-host GPU-aware publish/subscribe from single-host GPU fan-out to a distributed two-host setting. The framework stages outgoing payloads through a GPU-accessible RDMA-registered pinned host memory region on the publisher machine, transfers them with RDMA Write with Immediate into a GPU-accessible RDMA-registered receive-side ring buffer, and materializes them exactly once into a shared GPU memory pool on the subscriber machine. Multiple local subscribers then access the same GPU-resident payload without additional per-subscriber copies.

The main contributions of this paper are as follows:

1. We design a distributed GPU-aware publish/subscribe architecture that preserves the "publish once, fan-out many" principle across machines by combining shared CUDA-RDMA pinned memory, one-sided RDMA transport, and shared-GPU local fan-out.
2. We introduce an RDMA ring-buffer design together with an RDMA Write with Immediate receive path that removes costly receive-buffer sanitization from the critical path while maintaining safe variable-sized message streaming, effectively eliminating the performance bottlenecks inherent in the conventional, naive socket-based TCP/IP approach.
3. We evaluate the framework on a real two-node GPU-RDMA platform and show that RGAPS substantially reduces both path-level and end-to-end latency relative to pageable-memory, socket-based, and CPU-centric fan-out baselines.

2 Background and Related Work

2.1 Single-Host GPU-Aware Publish/Subscribe

Single-host GPU-aware publish/subscribe [3] is the direct conceptual precursor to RGAPS. As illustrated in Figure 1, this design decouples data placement from control signaling by storing

large payloads in shared GPU memory and using shared host memory only for metadata and notification. A publisher writes a payload directly into a shared GPU memory pool, stores the payload size and offset in shared host memory, and sends a lightweight notification to subscribers. Each subscriber reads the metadata and directly accesses the payload from the shared GPU pool. This design eliminates redundant host-device transfers and achieves zero-copy local fan-out.

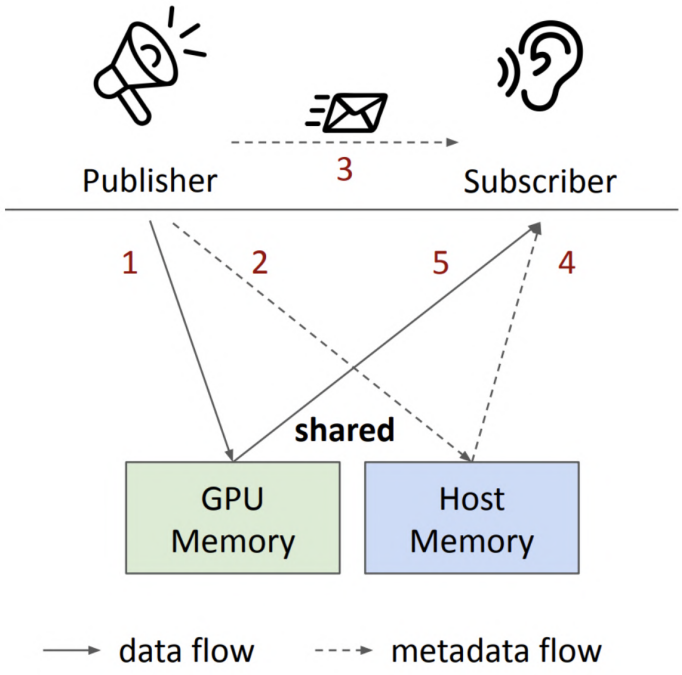

Figure 1: The architectural workflow of single-host GPU-aware publish/subscribe.

The main limitation of the design is that it operates only within one machine. All participants must share the same host, and the per-topic GPU memory pool must be provisioned within local GPU memory. As the number of topics and service stages increases, both GPU memory pressure and compute resource contention, coupled with resource placement constraints, become problematic.

2.2 Pinned Host Memory

Pinned host memory is important for efficient GPU–host transfer. Unlike pageable memory, which requires an extra CPU-mediated copy through a driver-managed pinned staging buffer in kernel space, pinned memory enables a direct DMA path between device memory and user space host memory [8, 9]. In practice, this can reduce device-to-host transfer latency by around 1.8×–2.0× on sufficiently provisioned PCIe links, as summarized in Figure 2.

Pinned memory also provides the stable physical residency required for RDMA memory registration. This makes it a natural staging substrate for a GPU–network–GPU pipeline in environments where the RNIC cannot directly access GPU memory.

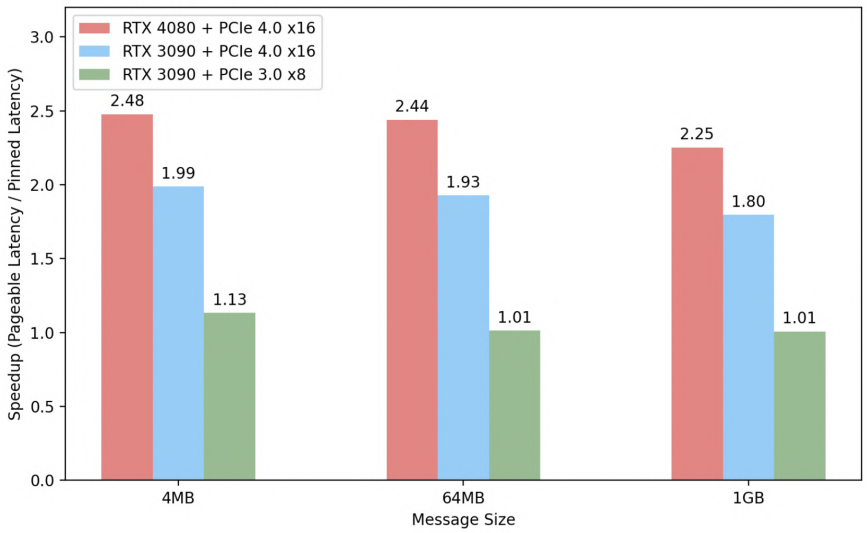

Figure 2: Impact of PCIe bandwidth and GPU micro-architecture on pinned-memory speedup.

2.3 RDMA and Distributed Messaging

RDMA reduces software overhead by enabling kernel-bypass and zero-copy networking. In one-sided operations, an initiator can directly write to or read from a remote registered memory region without involving both local and remote CPUs in the data path [7, 4, 2]. These properties make RDMA especially attractive for latency-sensitive communication involving large payloads.

FaRM [1] demonstrated the use of RDMA ring buffers for lightweight distributed messaging. Its design is effective for CPU-resident data, but its validity test depends on memory-header polling and buffer zeroing, which can become expensive for large payloads. KafkaDirect [11] accelerates Apache Kafka with RDMA, but its focus is broker-managed log semantics rather than direct dissemination of GPU-resident payloads for immediate downstream execution.

RGAPS differs from these systems in two ways. First, it targets GPU-resident payloads rather than host-centric messages. Second, it combines inter-node RDMA transport with a receive-side shared GPU fan-out design so that the same payload can be reused by multiple local GPU consumers after a single materialization step.

3 RGAPS Design Overview

As shown in Figure 3, RGAPS is organized around the idea that each message should be copied as few times as possible while still supporting cross-machine dissemination and one-to-many fan-out. The end-to-end data plane is divided into three stages:

1. **Intra-node aggregation** on the publisher machine.
2. **Inter-node transport** across the network.
3. **Local dissemination** on the subscriber machine.

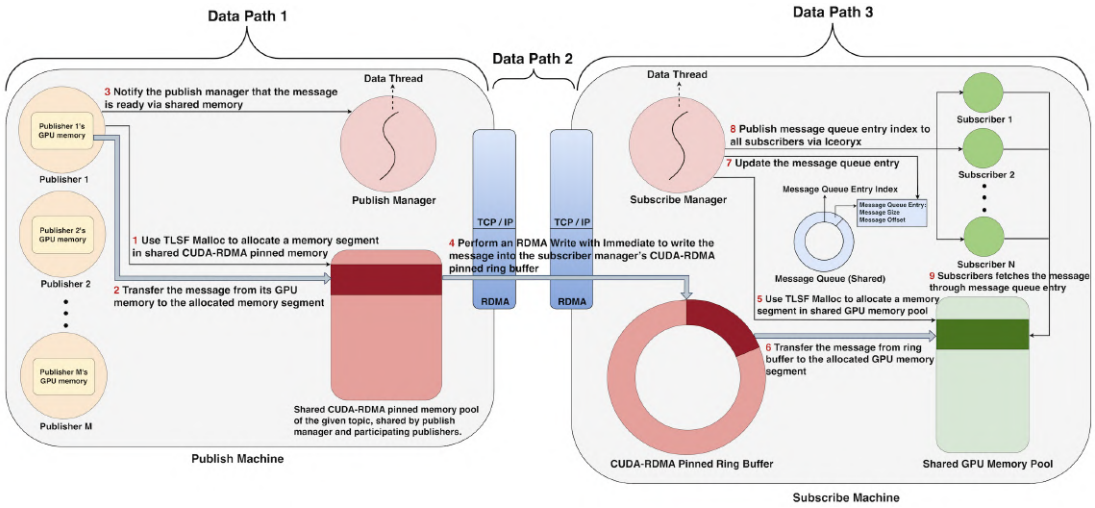

Figure 3: Architectural overview of the RGAPS data plane.

The control plane, shown in Figure 4, is separated from the high-speed data plane. Standard TCP/IP and Unix Domain Sockets are used for topic setup, resource discovery, and publisher/subscriber registration, while RDMA is reserved for a part of performance-critical data plane.

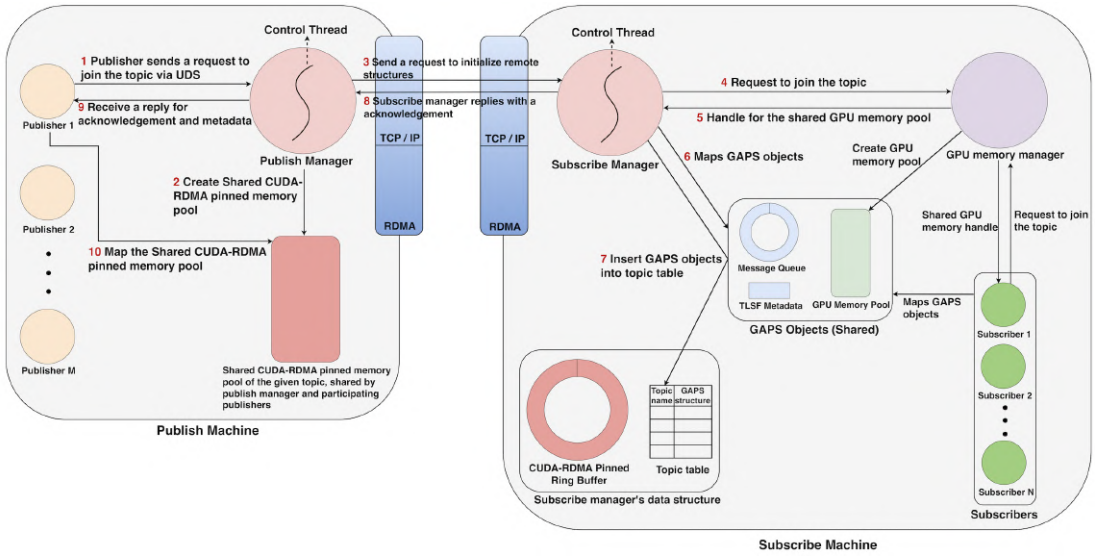

Figure 4: Control-plane architecture and initialization process.

The overall architecture is physically bifurcated into two distinct nodes: a publisher machine and a subscriber machine. For any given topic, all publishers are consolidated on the publisher machine, while all subscribers reside on the subscriber machine. To mediate the continuous stream of messages between these two nodes, each machine hosts a dedicated intermediary

process—specifically, a publish manager and a subscribe manager.

Within these manager processes, a control thread is responsible for topic initialization and the establishment of per-topic metadata. Additionally, a dedicated data thread manages the local aggregation and dissemination of messages, as well as the continuous inter-node transmission of payloads.

In the publisher machine, the publish manager maintains a per-topic shared CUDA-RDMA pinned memory pool. This pool serves to receive payloads from local publishers and acts as the origin buffer for RDMA-based payload transfers. Correspondingly, the subscribe manager maintains a global CUDA-RDMA pinned ring buffer in the subscriber machine. This buffer functions as the destination endpoint for incoming RDMA transfers from the publish manager, from which the payloads are subsequently disseminated to all local subscribers registered to the designated topic.

The overall data plane is logically partitioned into three distinct segments: Data Path 1, Data Path 2, and Data Path 3. Specifically, Data Path 1 encompasses the set of all intra-node data flows on the publisher machine. Data Path 2 constitutes the inter-node data flow responsible for message transmission between the publisher and subscriber machines. Finally, Data Path 3 comprises the set of all intra-node data flows on the subscriber machine.

3.1 Data Path 1: Intra-Node Aggregation

As illustrated in Figure 5, RGAPS implements Data Path 1 (Pinned-SHM) on the publisher machine by creating a shared CUDA-RDMA pinned memory pool. The figure contrasts this proposed design with the baseline approach, Data Path 1 (Pageable-UDS), which relies on pageable memory and Unix domain sockets. In Data Path 1 (Pinned-SHM), the memory region is shared across local publisher processes, registered for RDMA, and accessible to the GPU. Publishers allocate segments within this pool with a TLSF-based allocator [5, 10]. After allocation, a publisher performs a single *cudaMemcpy* from its GPU memory into the shared pinned segment and writes only lightweight metadata to notify the publish manager via shared memory.

This design removes the intermediate kernel-space memory copies inherent in standard device-to-host transfers, alongside the heavy IPC cost of forwarding the full payload through a Unix Domain Socket, and ensures that the payload is written to host memory only once before network transmission.

3.2 Data Path 2: Inter-Node Transport

The publish manager transfers messages from the shared pinned pool to the subscriber machine through RDMA Write with Immediate. The destination is a pre-allocated pinned ring buffer owned by the subscribe manager. One-sided RDMA Write moves the message directly into that buffer, and the immediate data field delivers the message size through the receiver's completion queue.

The standard socket-based approach routes cross-node data streams directly into the subscribe manager's pageable host memory. However, relying on the OS network stack introduces some major software-bound bottlenecks: Processing packets demands intense CPU involvement, forcing expensive transitions between user and kernel spaces. Furthermore, messages suffer from

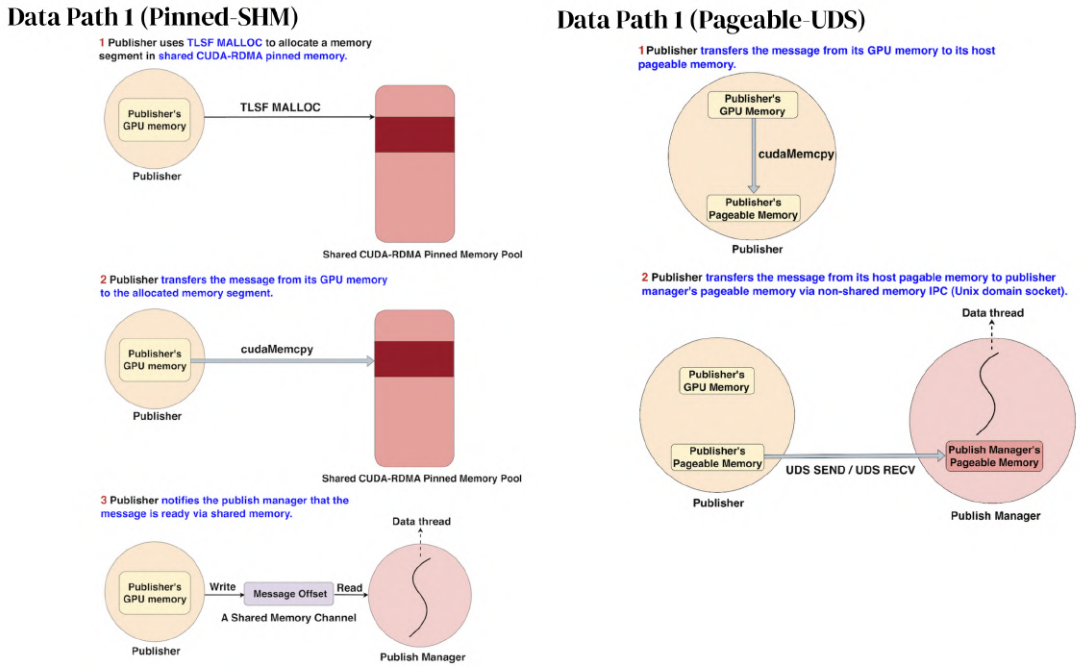

Figure 5: Comparison of Data Path 1: Pinned-SHM versus Pageable-UDS.

redundant copying between user and kernel spaces, potentially triggering additional context switches.

In contrast, this RDMA transport path avoids both local and remote CPU intervention, eliminates kernel networking overhead, and keeps the message in GPU-accessible pinned memory once it arrives on the subscriber machine.

3.3 Data Path 3: Local Dissemination

The final stage of the pipeline, Data Path 3, governs the local distribution of messages from the subscribe manager to multiple distinct subscribers running on the destination node. This phase is critical for determining the system's fan-out scalability regarding the number of subscribers.

To eliminate redundant memory transfers, the proposed design, denoted as Data Path 3 (Pinned-Shared), completely decouples payload placement from notification logic. After a message lands in the receive-side pinned ring buffer, the subscribe manager utilizes the TLSF algorithm to allocate a single memory segment within a shared GPU memory pool. It then executes a single host-to-device transfer (*cudaMemcpy*) to move the message directly into this allocated shared GPU segment. Instead of broadcasting the full message, the manager utilizes Iceoryx solely to publish lightweight metadata—specifically, the per-topic shared message queue entry index.

Upon receiving this index, each subscriber independently computes and resolves the correct message address within the shared GPU pool using the message base address and offset recorded in the designated shared message queue entry. Since the GPU memory is pre-mapped into every subscriber's address space, they can immediately access the GPU-resident payload without

performing any additional memory copies.

In a conventional CPU-centric baseline, denoted as Data Path 3 (Pageable-Iceoryx), the subscribe manager publishes the full payload via Iceoryx to the host pageable memory of all subscribers. Consequently, each individual subscriber must perform its own *cudaMemcpy* to transfer the data into its private GPU memory. This naive approach mandates N redundant host-to-device transfers for N subscribers, inevitably causing linear bandwidth saturation on the PCIe bus. By contrast, RGAPS performs exactly one host-to-device transfer per message rather than one per subscriber, drastically minimizing PCIe bandwidth consumption and ensuring stable performance even with a high density of subscribers.

4 RDMA Ring Buffer and Receive-Side Optimization

The receive-side RDMA ring buffer is a central component of RGAPS. It supports continuous streaming of variable-sized messages from the publishing node while avoiding overwrite of unconsumed data. The design uses three pointers: a local write cursor (*Tail*) maintained by the publish manager, a true read cursor (*Head*) maintained by the subscribe manager, and a cached lower-bound estimate of the remote head (*Delayed Head*) maintained on the publishing side.

Before issuing an RDMA Write, the publish manager checks whether *Tail + message size* would overrun the current *Delayed Head*. Because *Delayed Head* is a conservative view of the remote consumption progress, this local test guarantees safe writes without requiring a remote read for every message.

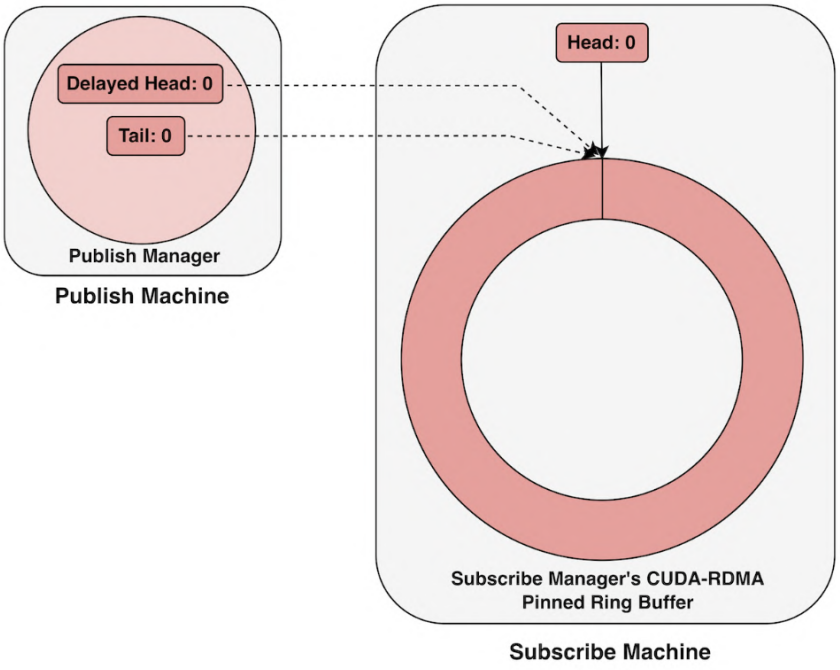

Figure 6: Initial state of the RDMA ring buffer.

Figure 6 shows the initial state of the ring buffer. As messages are written, the publish manager advances *Tail*. The subscribe manager consumes messages by advancing *Head*. To reduce control traffic, the subscribe manager updates *Delayed Head* only after a configurable watermark of consumed bytes is reached. Figures 7, 8, and 9 illustrate the write, consume, and delayed-head update phases, respectively.

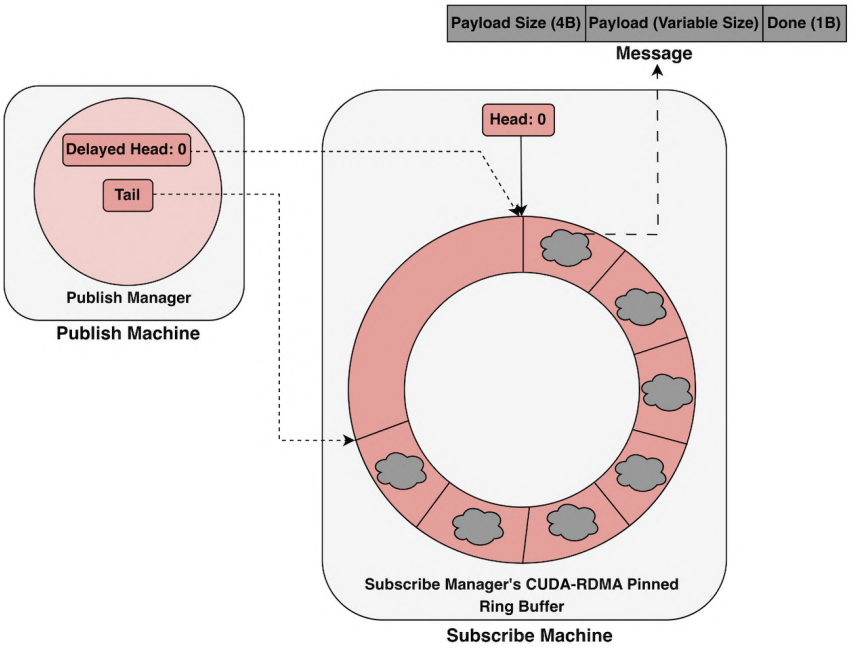

Figure 7: Multiple messages written to the RDMA ring buffer.

An important design issue is how the receiver detects message arrival. In a naive memory-polling design, the subscriber must look for a non-zero header, fetch and process the message, and then zero out consumed messages to prevent stale garbage values from being mistaken as fresh messages after wrap-around. We quantify this memory zero-out overhead in Figure 10; for 1 GB messages, local zeroing costs about 91.4% of the latency of an RDMA write of the same size.

To remove this overhead, RGAPS uses RDMA Write with Immediate. Instead of polling message headers in memory, the subscribe manager polls the completion queue. The payload size is carried in the immediate data field and is therefore received as completion metadata rather than as part of the payload header. This design eliminates the need for receive-buffer sanitization on the critical path and makes the transport cost depend primarily on network and DMA performance rather than on extra CPU-side memory maintenance.

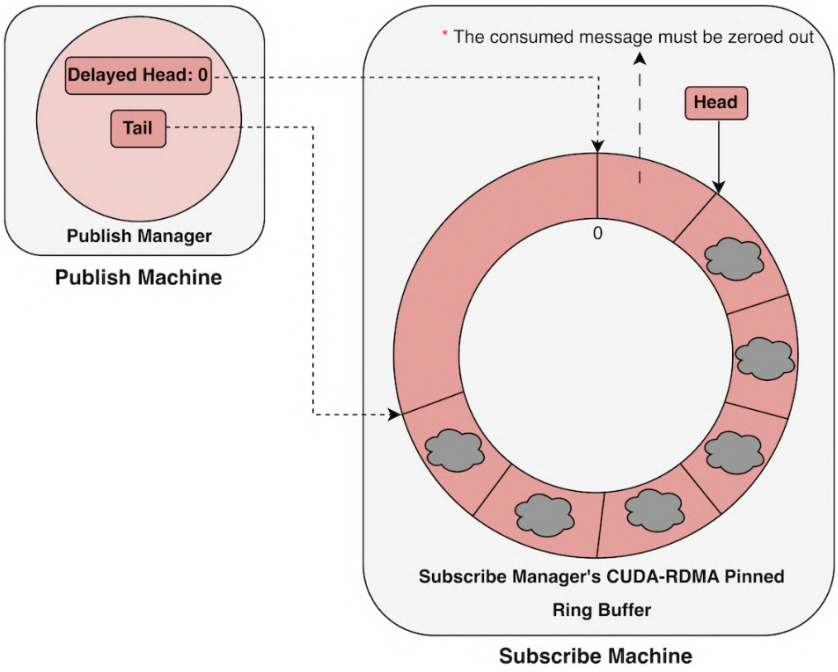

Figure 8: Consumption of messages from the RDMA ring buffer.

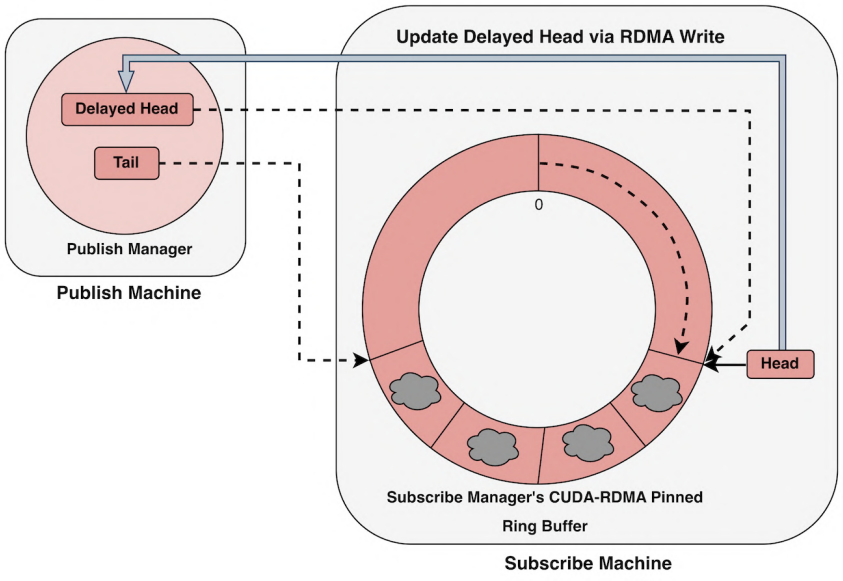

Figure 9: Updating the publish manager's delayed head.

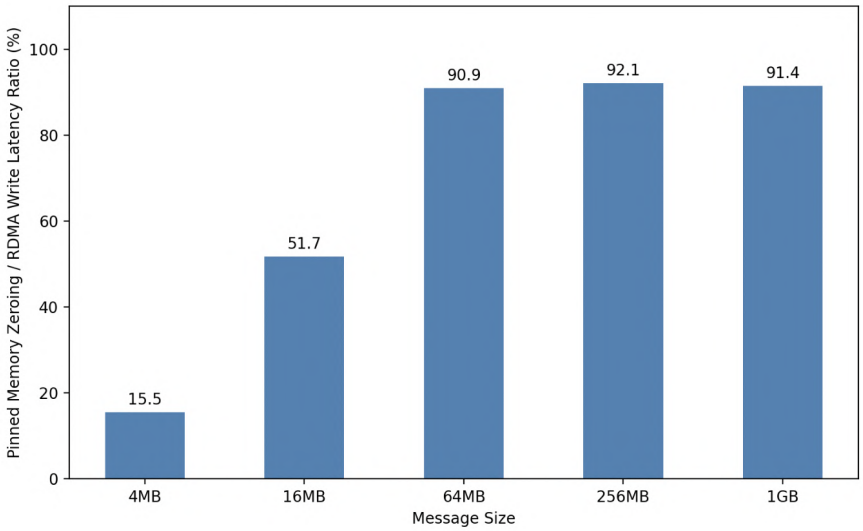

Figure 10: Overhead of pinned-memory zero-out relative to RDMA Write latency.

5 Evaluation

5.1 Experimental Setup

The evaluation platform consists of two symmetric workstations, one acting as the publish machine and the other as the subscribe machine. Table 1 summarizes the hardware and software configuration. Each machine contains an AMD Ryzen Threadripper 3970X CPU with 32 cores and 64 threads, 128 GB of DDR4-2666 memory, an NVIDIA GeForce RTX 3090 GPU with 24 GB of device memory, and an NVIDIA ConnectX-7 RNIC. The GPU and RNIC are connected through independent PCIe Gen4 x16 interfaces. The two machines are linked by InfiniBand NDR. Ubuntu 22.04, CUDA 12.7, and Iceoryx 2.0.6 were used.

We compare RGAPS against three baseline paths derived from conventional CPU-centric communication: pageable-memory plus Unix-domain-socket IPC for local aggregation, TCP/IP for inter-node transmission, and CPU-oriented local fan-out that requires separate host-to-device copies per subscriber.

5.2 Data Path 1 Results

Data Path 1 measures the transfer of a payload from publisher GPU memory to host memory on the publisher machine. RGAPS uses the proposed Pinned-SHM design, while the baseline uses pageable memory plus Unix Domain Sockets.

Regarding the device-to-host memory transfer latency, for a 4 MB message, RGAPS reduced latency from 451 μs to 224 μs. For 64 MB, it reduced latency from 5,709 μs to 2,906 μs. For a 1 GB message, latency fell from 85,806 μs to 47,905 μs. These gains are shown in Figure 11 and arise from eliminating the double-copy behavior of pageable memory.

Table 2 and Table 3 provide a granular total latency breakdown of Data Path 1 to contrast the

Table 1: Hardware and software configurations of the publish and subscribe machines.

	Publish Machine	**Subscribe Machine**
Hardware		
CPU	AMD Ryzen Threadripper 3970X (32 cores / 64 threads)	AMD Ryzen Threadripper 3970X (32 cores / 64 threads)
GPU	NVIDIA GeForce RTX 3090 (10496 CUDA cores, 24GB), PCIe Gen4 x16	NVIDIA GeForce RTX 3090 (10496 CUDA cores, 24GB), PCIe Gen4 x16
RAM	128 GB DDR4-2666 (4 × 32 GB)	128 GB DDR4-2666 (4 × 32 GB)
Adapter	NVIDIA ConnectX-7 — PCIe Gen4 x16	NVIDIA ConnectX-7 — PCIe Gen4 x16
Interconnect	InfiniBand NDR 400 Gb/s	
Software		
OS	Ubuntu 22.04 (kernel: 6.8.0-51-generic)	Ubuntu 22.04 (kernel: 6.8.0-87-generic)
Compiler	nvcc v11.5, gcc 11.4.0, g++ 11.4.0	nvcc v11.5, gcc 11.4.0, g++ 11.4.0
CUDA	12.7	12.7
Iceoryx	2.0.6	2.0.6

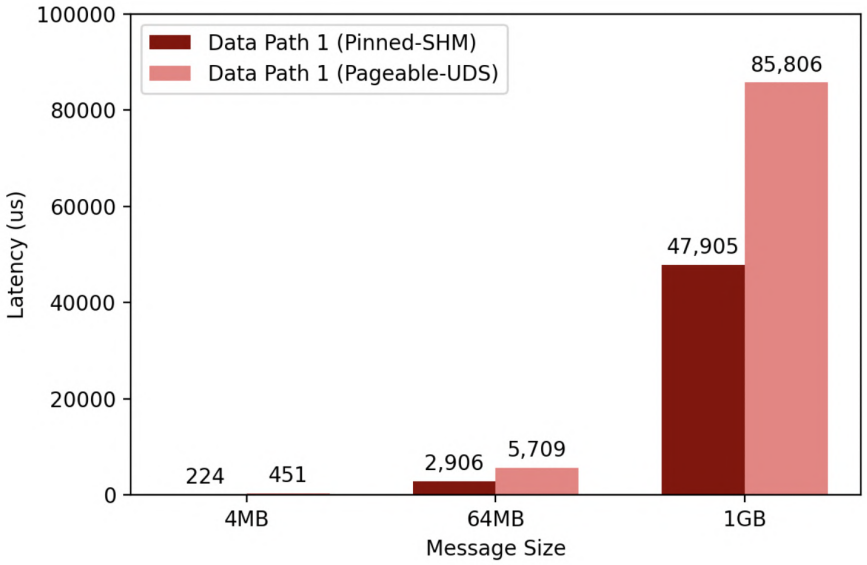

Figure 11: Device-to-host memory transfer latency of Pinned-SHM and Pageable-UDS in Data Path 1.

proposed Pinned-SHM architecture with the baseline Pageable-UDS approach.

Table 2 reveals the highly optimized profile of Pinned-SHM. The complexities of TLSF memory allocation and signaling are effectively minimized to a constant negligible overhead ($\approx 1\mu$s). Consequently, the latency of this data path is almost exclusively dominated by the *cudaMemcpy* operation.

In contrast, Table 3 exposes the severe inefficiency of copying large payloads through kernel socket buffers in the baseline. The Unix Domain Socket transmission introduces massive, linearly growing latency. Ultimately, this comparison highlights a critical architectural shift: by

Table 2: Latency breakdown of Data Path 1 (Pinned-SHM).

Message Size	T1: TLSF Malloc	T2: cudaMemcpy	T3: Notification via shared memory
4 MB	1	224	1
64 MB	1	2,906	1
1 GB	1	47,905	1

Unit: μs

Table 3: Latency breakdown of Data Path 1 (Pageable-UDS).

Message Size	T1: cudaMemcpy	T2: IPC (UDS)
4 MB	451	1,184
64 MB	5,709	17,810
1 GB	85,806	208,318

Unit: μs

transitioning from a software-bound mechanism (heavy IPC) to a hardware-bound one (pinned memory and direct DMA).

5.3 Data Path 2 Results

Data Path 2 measures cross-node transport. RGAPS uses RDMA Write with Immediate into the receive-side ring buffer, whereas the baseline uses TCP/IP. As shown in Figure 12, for a 4 MB message, RGAPS achieved 305 μs compared with 3,365 μs for TCP/IP. For 64 MB, the latency was 4,620 μs versus 39,832 μs. For 1 GB, the latency was 73,115 μs versus 602,994 μs.

The relative advantage is already large for small messages and becomes even more significant as payload size grows, showing that the software overhead of TCP/IP strongly dominates a CPU-centric transport path at small message sizes.

5.4 Data Path 3 Results

Data Path 3 measures local fan-out on the subscriber machine. RGAPS copies each message exactly once into shared GPU memory and then distributes only metadata. The baseline distributes a host-resident payload and forces each subscriber to perform its own *cudaMemcpy* into private GPU memory.

Table 4 demonstrates the superior scalability of our proposed method. By depositing data into a shared GPU memory segment via a single *cudaMemcpy*, the heavy host-to-device transfer cost becomes constant. With negligible notification overhead, the total latency remains stable regardless of the subscriber count (N).

In stark contrast, Table 5 reveals that the baseline approach scales poorly, suffering from a linear increase in total latency due to repeating the *cudaMemcpy* for each of the N subscribers.

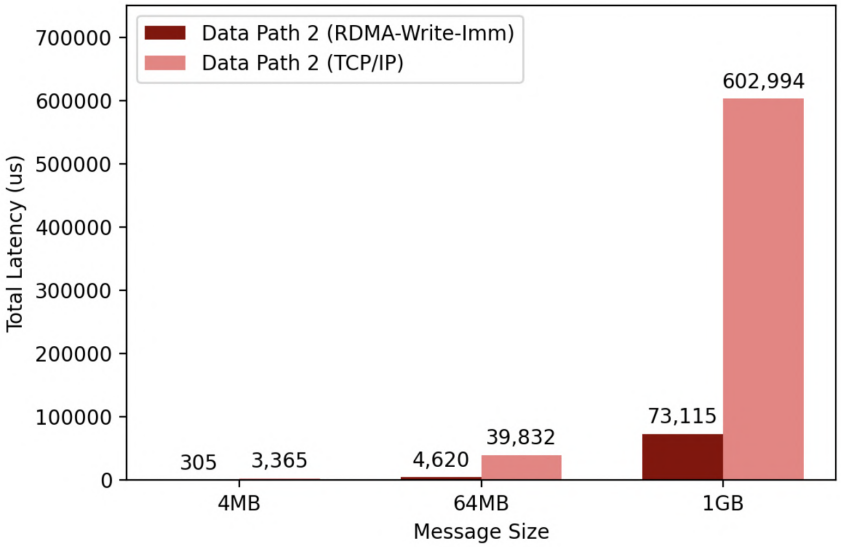

Figure 12: Total latency of Data Path 2 (RDMA-Write-Imm) and Data Path 2 (TCP/IP).

Table 4: Latency breakdown of Data Path 3 (Pinned-Shared).

Message Size	**T1:**	**T2:**	**T3: Notification via Iceoryx (Avg Latency)**			
	TLSF Malloc	**cudaMemcpy**	**N = 1**	**N = 2**	**N = 4**	**N = 8**
4 MB	1	216	6	7	9	11
64 MB	1	2,872	6	7	9	11
1 GB	1	43,677	6	7	9	11

* N: Number of subscribers Unit: μs

Table 5: Latency breakdown of Data Path 3 (Pageable-Iceoryx).

Message Size	**T1: Publish message via Iceoryx**				**T2: cudaMemcpy (Avg Latency)**			
	N=1	**N=2**	**N=4**	**N=8**	**N=1**	**N=2**	**N=4**	**N=8**
4 MB	6	7	7	10	396	520	866	2,238
64 MB	9	9	11	12	5,625	7,439	12,468	24,357
1 GB	9	10	11	12	85,172	118,273	191,378	382,018

* N: Number of subscribers Unit: μs

5.5 End-to-End Results

The most important metric is end-to-end latency from the moment a publisher issues a publish operation on the source machine until a subscriber on the destination machine can access a GPU-resident payload. Figure 13 reports this comparison across message sizes and fan-out levels. For 4 MB messages, RGAPS stayed near 754–759 μs across one to eight subscribers, while the baseline ranged from 5,402 to 7,248 μs. For 64 MB messages, RGAPS stayed near 10,407–10,412 μs, while the baseline rose from 68,985 to 87,720 μs. For 1 GB messages, RGAPS remained near 164,706–164,711 μs, whereas the baseline ranged from 982,299 to 1,279,148 μs.

(a) 4 MB (b) 64 MB (c) 1 GB

Figure 13: Total latency comparison across different message sizes.

Overall, the end-to-end improvement is roughly 6×–10×, and the largest gains appear when fan-out is high. The reason is straightforward: RGAPS keeps the dominant local materialization cost constant per message, while the baseline adds more and more duplicated host-to-device work as subscribers are added.

6 Discussion and Limitations

RGAPS demonstrates that a distributed GPU-aware data plane can substantially outperform CPU-centric pub/sub communication when payloads are large and reused by multiple GPU consumers. The main reason is not merely the use of RDMA in isolation; it is the integration of three ideas into one end-to-end pipeline: GPU-accessible pinned-memory staging on the sender, one-sided CPU-bypassing network transport, and one-time materialization into shared GPU memory on the receiver.

The framework is well suited to pipelines in which one GPU stage produces large intermediate artifacts for several downstream consumers. Example scenarios include distributed inference pipelines, real-time vision systems, multi-stage analytics on image or tensor streams, and other GPU-driven processing graphs in which the same payload must be reused rather than independently replicated.

RGAPS also has several limitations. First, the current platform does not support GPUDirect RDMA [6], so the design must still stage network I/O through host DRAM. Direct RNIC-to-GPU transfers would likely reduce latency further. Second, the current study focuses on a two-host deployment; multi-hop and larger multi-host systems will require additional work on routing and resource management. Third, the present implementation uses a single global receive ring buffer, which can become a scalability bottleneck under very high load. A natural next step is to allocate per-topic ring buffers with parallel transfer threads. Finally, RGAPS currently

provides live-only delivery semantics and does not yet support bounded replay or latest-value delivery for late joiners.

7 Conclusion

This paper presented RGAPS, a low-latency RDMA-based publish/subscribe framework for inter-machine GPU communications. RGAPS extends single-host GPU-aware pub/sub to a distributed two-host environment by combining shared CUDA-RDMA pinned-memory aggregation, RDMA Write with Immediate transport, and shared-GPU local fan-out on the receiver.

Experiments on a real GPU-RDMA platform showed that the proposed design reduces latency in each major stage of the communication path and improves end-to-end delivery by roughly 6×–10× relative to CPU-centric baselines. Just as importantly, RGAPS keeps local fan-out cost nearly constant as the number of subscribers increases, because each message is materialized into GPU memory only once.

These results indicate that low-latency inter-machine GPU pub/sub can be achieved without direct GPUDirect RDMA support, provided that host-side staging, network transport, and receive-side fan-out are designed as a unified data plane. RGAPS therefore offers a practical foundation for distributed GPU applications that require high-bandwidth, low-latency dissemination of large payloads across machines.

8 Acknowledgments

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MedFedChain: An Adaptive Differential Privacy and Blockchain-Orchestrated Federated Learning Framework for Secure Multi-Institutional Electronic Health Record Analytics *

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Abstract

Electronic Health Records (EHRs) represent one of the most sensitive and valuable assets in modern healthcare. Despite widespread digitalisation, EHR systems remain fragmented across clinical institutions, impeding collaborative diagnostics and population-level analytics while raising significant privacy and security concerns. Existing federated learning (FL) approaches address data siloing but remain vulnerable to gradient inversion attacks and lack transparent auditability. Blockchain-based solutions provide immutability and decentralisation yet introduce latency and scalability bottlenecks when used for direct data handling. The present work introduces MedFedChain, an innovative framework that integrates: (i) the adaptive differential privacy (ADP) process that calibrates noise injection based on per-layer gradient sensitivity; (ii) an InterPlanetary File System (IPFS)-based encrypted model repository for efficient off-chain storage of FL model updates; and (iii) Ethereum smart contracts for decentralised, tamper-proof aggregation and incentivisation across federated hospital nodes. Extensive experiments on two recent, large-scale public healthcare datasets—the Kaggle Diabetes Prediction (2023, 100,000 records) Dataset and the CDC BRFSS Diabetes Health Indicators Dataset (2021, 253,680 records, CDC/UCI)—demonstrate that MedFedChain achieves a classification accuracy of 91.8% with a privacy budget of $\varepsilon = 1.0$, outperforming FedAvg (85.4%), FedProx (87.1%), DP-FedAvg (88.3%), and EPP-BCFL (89.6%) whilst reducing communication overhead by 34.6%. Security analysis confirms resistance to membership inference and model poisoning attacks. These results demonstrate that MedFedChain offers a compelling and privacy-compliant foundation for collaborative EHR analytics, achieving strong performance across classification accuracy, communication efficiency, and security resistance.

1 Introduction

The expansion of Internet of Medical Things devices, coupled with accelerating clinical digitalisation, has resulted in an unprecedented volume of Electronic Health Records (EHR). According to the World Health Organization, more than 95% of hospitals in developed nations have adopted EHR systems as of 2024, with global EHR data volumes projected to surpass 2.3 zettabytes by 2028 [1]. Despite

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this abundance, EHR data remains highly compartmentalised, locked within individual institutions by strict privacy legislation—including HIPAA in the USA, GDPR in the EU, and analogous frameworks elsewhere—creating persistent "data silos" that inhibit cross-institutional analytics and collaborative AI model training [2].

Centralised data aggregation, historically the default strategy for training diagnostic AI models, is no longer viable at scale. Pooling raw patient records across hospitals exposes sensitive information to data breaches, re-identification attacks, and regulatory violations [3]. The consequences are substantial: the average healthcare data breach in 2024 cost an estimated USD 10.9 million per incident [4]. Federated Learning (FL), first formalised by McMahan et al. [5] and extended for heterogeneous settings by Li et al. [6], offers a compelling alternative: participating institutions train local models on their private dataset and share model parameter updates only—not any raw records—with the central aggregation server. This decentralised paradigm preserves data locality whilst enabling collaborative generalisation.

Nevertheless, FL alone is insufficient for healthcare applications. Research has demonstrated that raw gradient exchanges stay mostly vulnerable to gradient inversion & membership inference attacks; through which malicious actors can partially rebuild private training samples [7, 8]. Furthermore, centralised aggregation servers represent single points of failure and trust—an unacceptable assumption in competitive multi-hospital environments. Differential Privacy (DP), formally synthesised for modern federated contexts by Ouadrhiri et al. [9], provides formal, mathematically provable privacy preservation through the injection of calibrated Gaussian noise into model updates before transmission. However, existing DP-FL approaches universally apply a static noise budget, ignoring the heterogeneous sensitivity of different model layers and clinical data distributions, thus suboptimally trading privacy for accuracy [10, 11].

Blockchain has come as a promising foundation for decentralised trust in FL systems [12, 13]. Smart contracts can automate model aggregation, enforce participation rules, and maintain an immutable audit trail—eliminating the need for a trusted central server. Prior work combining blockchain with FL in healthcare has demonstrated substantial promise [14, 15, 16] yet suffers from a critical limitation: storing large model parameter tensors directly on-chain is prohibitively expensive and slow. The InterPlanetary File System, a content-addressed peer-to-peer storage protocol, resolves this bottleneck by enabling off-chain storage of encrypted model updates, relying solely on cryptographic content hash identifiers recorded on-chain [17].

Despite significant progress in federated learning, differential privacy, and blockchain-based healthcare systems, existing approaches fail to simultaneously address privacy leakage, scalability limitations, and fair participation among heterogeneous institutions. Federated learning alone remains vulnerable to gradient-based attacks, while differential privacy often introduces a trade-off between data utility and protection. Similarly, blockchain-based solutions improve trust and auditability but suffer from high storage and computational overhead. Therefore, designing a unified system that simultaneously ensures privacy, scalability, and fairness in multi-institutional EHR analytics remains an open challenge. TThis study provides the following key contributions:

- We introduced MedFedChain, a unified framework that integrates federated learning, adaptive differential privacy, blockchain orchestration, and IPFS-based off-chain storage for secure EHR analytics.
- We introduce an Adaptive Differential Privacy (ADP) mechanism which can dynamically adjusts noise at the layer level based on gradient sensitivity, improving the privacy–utility baclance compared to static DP approaches.
- We design a blockchain-based coordination and incentive mechanism using Ethereum smart contracts, enabling decentralised aggregation, transparent participation, and reputation-based contribution weighting.

- We implement an IPFS-based off-chain storage strategy for encrypted model updates, significantly reducing on-chain storage overhead and improving system scalability.
- We conduct comprehensive experiments on large-scale healthcare datasets, demonstrating improved accuracy, reduced communication overhead, and enhanced robustness against privacy and security attacks compared to existing baselines.

The remainder of this paper is structured as follows. Section 2 surveys related work across FL, DP, blockchain, and IPFS in healthcare contexts. Section 3 details the MedFedChain architecture, ADP mechanism, smart contract design, and IPFS integration protocol. Section 4 presents experimental setup, datasets, baselines, and results with comparative analysis. Section 5 concludes and outlines future directions.

2 Literature Review

This section surveys the state of the art across the four intersecting domains underpinning this work: federated learning in healthcare (Section 2.1), differential privacy for FL (Section 2.2), blockchain-enabled healthcare systems (Section 2.3), and IPFS for decentralised medical data (Section 2.4). Table 1 synthesises key properties of representative recent works.

2.1 Federated Learning in Healthcare

McMahan et al. [5] established the theoretical foundation for FL with their FedAvg algorithm, demonstrating communication-efficient training of deep neural networks across decentralised data, but does not address privacy leakage. Subsequent work by Li et al. [6] introduced FedProx, adding a proximal regularisation term to stabilise convergence under data heterogeneity—a prevalent challenge in healthcare where patient demographics, recording devices, and clinical workflows vary substantially between sites.

In the clinical domain, Rieke et al. [2] provided a comprehensive vision for FL in medicine, identifying opportunities in radiology, genomics, and EHR analytics. Sheller et al. [18] demonstrated that FL achieves competitive performance to centralised training in brain tumour segmentation across 10 institutions, with only marginal accuracy degradation. More recently, Shah et al. [19] applied FL to EHR mortality prediction using MIMIC-III, achieving an AUC of 0.912 with FedAvg whilst preserving full data privacy—validating FL's practical applicability to tabular clinical data.

Nguyen et al. [20] introduced FedHealth, a transfer-learning-augmented FL framework specifically targeting IoMT wearable data for chronic disease monitoring. Their system demonstrated robust performance under non-IID conditions, improving accuracy by 6.3% over vanilla FedAvg. However, none of these works addressed the security of the aggregation channel, leaving gradients exposed during transit.

The landscape of trustworthy FL has expanded considerably in 2024–2025. Madupati et al. [21] proposed FL pipelines combined with blockchain-based auditing for decentralised data pipeline accountability, demonstrating improved trustworthiness but encountering scalability limits with large hospital networks. Pavlova et al. [22] introduced a decentralised FL environment leveraging smart contracts and verifiable secret sharing for open-source model training, advancing cryptographic security at the cost of substantially higher training overhead. Similarly, Mu et al. [23] addressed medical image heterogeneity across institutions through causal FL augmented with blockchain-based dataset valuation, improving model fairness across sites—yet persistent communication overhead constrained practical deployment. Potdukhe et al. [24] reviewed blockchain-enabled FL systems with integrated explainability, identifying lightweight protocol design and adaptive privacy calibration as the primary unresolved challenges. Korde et al. [25] conducted a systematic review of machine learning techniques for blockchain optimisation,

reinforcing that privacy, scalability, and ethical accountability must be jointly addressed—an objective central to MedFedChain's design.

2.2 Differential Privacy for Federated Learning

Ouadrhiri et al. [9] mapped the contemporary framework for (ε, δ)-differential privacy in federated contexts, providing a rigorous mathematical foundation for quantifying privacy loss. Abadi et al. [26] extended this to deep learning through DP-SGD, introducing per-sample gradient clipping and Gaussian noise addition—forming the basis for most subsequent DP-FL work.

Geyer et al. [10] first applied DP to a federated learning setting, adding Gaussian noise to the sum of clipped client updates before aggregation, protecting individual user datasets. However, their fixed noise multiplier approach significantly degraded model accuracy for small privacy budgets ($\varepsilon < 2.0$). Wei et al. [11] proposed FedDP, improving upon this with moments accountant-based tighter privacy analysis, achieving better utility at equivalent privacy guarantees.

Adaptive approaches to DP in FL have recently gained traction. Agarwal et al. [27] proposed layer-wise DP, observing that different neural network layers exhibit widely differing gradient norms and thus differing privacy risks. Their experiments showed that uniform noise is wasteful: shallow layers converge quickly and hold little private information, whereas deeper, task-specific layers are where sensitive signal concentrates. However, their method requires per-layer noise hyperparameter tuning—a non-trivial overhead in a federated setting. Our ADP mechanism addresses this by automating sensitivity estimation from gradient history, requiring no additional hyperparameter search.

2.3 Blockchain for Healthcare Data Management

Azaria et al. [28] pioneered blockchain-based EHR management with MedRec, using Ethereum smart contracts to manage patient consent and data access rights. Whilst foundational, MedRec stored data references rather than data itself, and its on-chain access log incurred substantial latency for high-frequency clinical queries[29].

Maithili et al. [14] proposed a blockchain-federated learning integration for healthcare privacy, demonstrating that immutable model update logs on-chain deter adversarial gradient manipulation. Sana al. [15] extended this with a reputation-based smart contract mechanism rewarding honest FL participants, showing improved Byzantine fault tolerance. However, both works stored model weights directly on-chain, incurring median gas costs of 0.38 ETH per round—commercially infeasible at scale.

Sayed et al. [30, 17] proposed a blockchain and IPFS-based framework for decentralised EHR management with group signature encryption, demonstrating that content-addressed off-chain storage reduces blockchain transaction data by 99.2% whilst preserving cryptographic integrity. Famous et al. [31] further showed that combining homomorphic encryption with blockchain audit trails achieves strong privacy guarantees for IoMT collaborative learning without a trusted aggregator. The present work builds directly upon these foundations, extending them to the federated learning domain with systematic ADP integration.

Wang et al. [16] presented a hierarchical blockchain architecture for FL in heterogeneous IoMT networks, reducing consensus latency by 41% through a two-layer chain design. Kim et al. [12] proposed a Proof-of-Work-free blockchain for FL to reduce computational overhead, achieving a 3.7$\times$ speed-up in block generation.

Table 1: Comparative summary of related works. ✓= fully implemented and experimentally validated, ∼ = partially addressed, × = not addressed. ADP = adaptive differential privacy.

Method	Year	FL	ADP	Blockchain	IPFS	Incentive	Clinical Data
FedAvg [5]	2017	✓	×	×	×	×	×
FedProx [6]	2020	✓	×	×	×	×	×
DP-SGD [26]	2016	∼	×	×	×	×	×
Geyer et al. [10]	2017	✓	×	×	×	×	×
FedHealth [20]	2022	✓	×	×	×	×	✓
Rahman et al. [14]	2024	✓	×	✓	×	×	✓
Sana al. [15]	2023	✓	×	✓	×	∼	✓
EPP-BCFL [16]	2022	✓	∼	✓	×	×	×
Sayed et al. [17]	2024	×	×	✓	✓	×	✓
Famous et al. [31]	2025	✓	×	✓	×	×	✓
MedFedChain (Ours)	**2026**	✓	✓	✓	✓	✓	✓

Recent scholarship has broadened blockchain's role in healthcare AI towards interpretability and trustworthiness. Sowjanya et al. [32] proposed FBZX, integrating FL, explainable AI (XAI), and blockchain for secure IoT health services, demonstrating improved user trust but incurring high per-round latency under realistic hospital loads. Murthy and Lawanya Shri [33] designed an XAI–Blockchain–IoT middleware using smart contracts, though interoperability across heterogeneous device classes remained unresolved. Sana al. [34] combined model explanation techniques with blockchain in a metaverse healthcare system, extending blockchain from data management to model transparency, albeit without formal privacy guarantees. Dutta and Puthal [35] proposed a fuzzy-logic and blockchain-enhanced IoMT framework for eHealth, incorporating adaptive authentication, while highlighting the difficulty of real-time data fusion. Rathod et al. [36] demonstrated blockchain-secured AI for IoT-enabled critical infrastructure, confirming the generalisability of blockchain trust mechanisms across high-stakes domains. Arslanoğlu and Karaköse [37] surveyed XAI–Blockchain systems in healthcare, identifying secure model storage and transparent predictions as persistent open challenges—directly motivating MedFedChain's IPFS-based encrypted off-chain model storage. Gadekallu et al. [38] further reinforced this in their Industry 5.0 survey, identifying transparency and interoperability as the foremost barriers to real-world deployment.

2.4 IPFS for Decentralised Medical Data

Applications in medical contexts have demonstrated reliable large-file storage (CT scans, genomic data) with retrieval latencies competitive with centralised object stores [39]. In EHR-specific deployments, Dwivedi et al. [40] showed that IPFS-blockchain hybrid architectures reduce data redundancy by 67% compared to on-chain storage whilst maintaining sub-second retrieval times for records < 50 MB. Singh et al. [41] Blockchain-IPFS framework designed to enhance the security, accessibility, and integrity of electronic healthcare records (EHRs).

Table 1 compares the most relevant related works across key dimensions. The analysis reveals that no existing system simultaneously addresses adaptive DP, IPFS-based off-chain model storage, smart contract-based incentivisation, and empirical evaluation on clinical datasets—the combination

constituting the core novelty of MedFedChain.

3 Methodology

MedFedChain operates within a multi-institutional healthcare federation comprising K hospital nodes $\mathcal{H} = \{H_1, H_2, \ldots, H_K\}$, each maintaining a private local EHR dataset $\mathcal{D}_k$ that is never transmitted to any external party. The framework assumes a semi-honest environment with potential malicious participants: participating institutions may try to exploit private data from shared model updates but will not actively deviate from the protocol. Byzantine faults are addressed by the smart contract.

3.1 System Overview and Threat Model

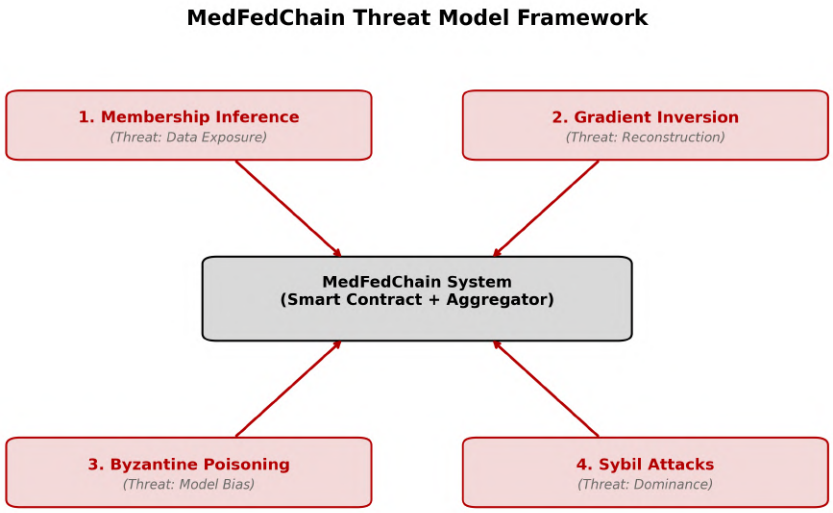

Figure 1: MedFedChain Threat Model Framework illustrating the four primary attack vectors addressed by the system's defensive mechanisms.

Specifically, we consider four threat classes, as illustrated in Fig. 1: Membership Inference Attacks (MIA), where adversaries query the trained model to determine whether a specific data was part of the training set [3]; Gradient Inversion Attacks (GIA), where attackers attempt to rebuild the original training data samples from shared gradient updates [7]; Byzantine Model Poisoning, where affected nodes insert manipulated gradients to bias the global model [42]; and Sybil Attacks, where a single adversary masquerades as multiple legitimate nodes to dominate aggregation in the blockchain network [43]. ADP (Section 3.3) addresses threats 1 and 2, while smart contract-based reputation scoring (Section 3.5) mitigates threats 3 and 4.

3.2 MedFedChain Architecture

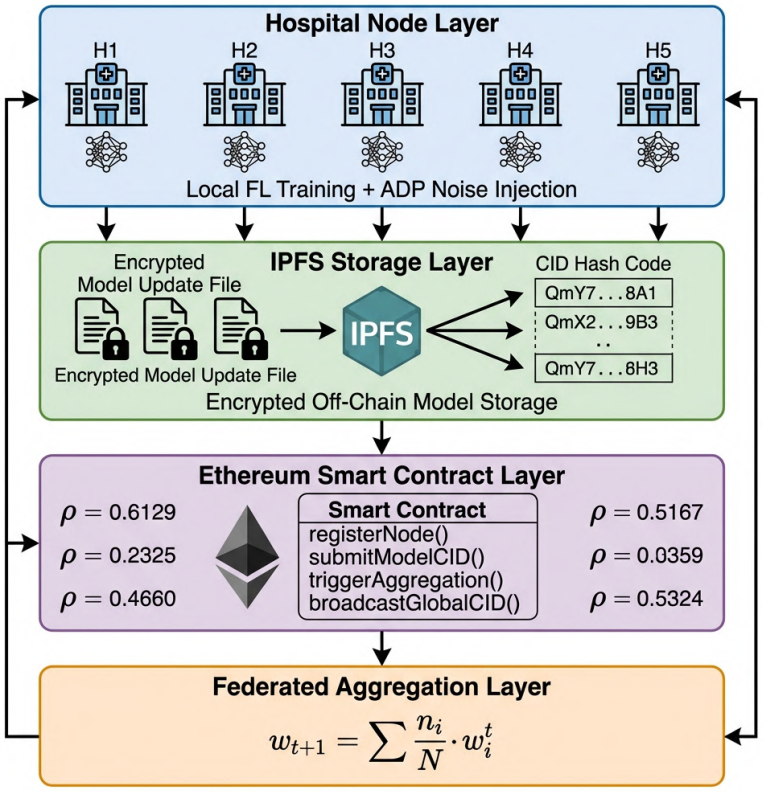

Figure 2: MedFedChain system architecture. Hospital nodes train local models, apply Adaptive Differential Privacy (ADP) noise, and upload encrypted model updates to IPFS. Content Identifiers (CIDs) are submitted to the Ethereum smart contract, which orchestrates aggregation, validates contributions via reputation scoring, and emits the aggregated CID for all nodes to retrieve.

The architecture (Figure 2) comprises four layers:

1. **Hospital Node Layer:** Each hospital node runs a local PyTorch training loop on its private EHR shard. After each local epoch, gradient updates are clipped and ADP noise is applied (Section 3.3).

2. **IPFS Storage Layer:** Encrypted model parameter tensors ($\approx$ 2–50 MB) are pinned to the IPFS network via Pinata gateway, yielding a 46-byte content identifier (CID). Only the CID is published on-chain, reducing blockchain data load to $< 0.01\%$ of the raw model size.

3. **Ethereum Smart Contract Layer:** A Solidity smart contract deployed on an Ethereum Proof-of-Stake test network receives CIDs from participating nodes, verifies node reputation scores, triggers secure aggregation, and emits the global model CID. The contract enforces participation quotas and distributes token-based incentives.

4. **Aggregation Layer:** A designated (rotated) aggregator node (to avoid centralisation bias) retrieves

all submitted model updates from IPFS and applies federated weighted averaging, producing the global model that is re-uploaded to IPFS and its CID broadcast via the smart contract.

3.3 Adaptive Differential Privacy (ADP)

Standard DP-FL applies a uniform noise multiplier σ across all gradient layers:

$$\tilde{g} = \text{clip}(g, C) + \mathcal{N}(0, \sigma^2 C^2 \mathbf{I}) \tag{1}$$

where g is the gradient tensor, C is the clipping norm, and σ controls the privacy-utility tradeoff. We observe that the ℓ_2 norms of gradients vary substantially across neural network layers. Let $\bar{s}_l^{(t)}$ denote the exponential moving average of the gradient ℓ_2 norm for layer l at round t:

$$\bar{s}_l^{(t)} = \beta \bar{s}_l^{(t-1)} + (1 - \beta) \left\| \nabla_l \mathcal{L} \right\|_2 \tag{2}$$

with smoothing factor $\beta = 0.9$. The adaptive noise multiplier for layer l is then:

$$\sigma_l^{(t)} = \sigma_{\text{base}} \cdot \frac{\bar{s}_l^{(t)}}{\max_j \bar{s}_j^{(t)}} \tag{3}$$

This formulation injects maximum noise into the layer with the highest average gradient norm (most sensitive), and proportionally less noise into lower-sensitivity layers, aiming to hold the overall (ε, δ)-DP guarantee via the composition theorem [9] whilst maximising model utility. The privacy budget per round is computed via the Rényi Differential Privacy accountant [44]:

$$\varepsilon(\delta) = \min_{\alpha > 1} \left(\frac{\mathcal{D}_\alpha^{(\text{priv})}(\cdot)}{\alpha - 1} + \frac{\log(1/\delta) + (\alpha - 1)\log(1 - 1/\alpha) - \log(\alpha)}{\alpha - 1} \right) \tag{4}$$

Algorithm 1 presents the complete ADP-FL training procedure.

Algorithm 1 MedFedChain Adaptive DP Federated Training

Require: Hospital nodes $\{H_k\}_{k=1}^{K}$, global model $w^{(0)}$, rounds T, clipping norm C, base multiplier σ_{base}, privacy budget $\varepsilon_{\text{total}}$, δ

for each round $t = 1, \ldots, T$ **do**
  Smart contract chooses subset $S^{(t)} \subseteq \{H_k\}$ by reputation score
  for every hospital $H_k \in S^{(t)}$ **in concurrent do**
    Download global model $w^{(t-1)}$ from IPFS via CID
    Train local model for E epochs: $w_k \leftarrow \text{LocalTrain}(w^{(t-1)}, \mathcal{D}_k)$
    Compute gradient $g_k = w_k - w^{(t-1)}$
    Clip: $\hat{g}_k = g_k / \max(1, \|g_k\|_2 / C)$
    for each layer l **do**
      Update $\bar{s}_l^{(t)}$ with exponential moving average
      $\tilde{g}_{k,l} = \hat{g}_{k,l} + \mathcal{N}(0, \sigma_l^{(t)2} C^2 \mathbf{I})$
    end for
    Encrypt $\tilde{g}_k$ with AES-256-GCM; upload to IPFS $\rightarrow$ CID_k
    Submit CID_k to smart contract $\rightarrow$ earns reputation score update
  end for
  Aggregator retrieves $\{\tilde{g}_k\}$ from IPFS; computes

$$w^{(t)} = w^{(t-1)} + \frac{\sum_{k \in S^{(t)}} n_k \cdot \tilde{g}_k}{\sum_{k \in S^{(t)}} n_k}$$

  Upload $w^{(t)}$ to IPFS $\rightarrow$ Global $\text{CID}^{(t)}$; emit via smart contract
  Accumulate privacy budget; **if** $\varepsilon_{\text{acc}} > \varepsilon_{\text{total}}$: **stop**
end forreturn $w^{(T)}$

3.4 IPFS Integration and Encryption Protocol

Each model gradient tensor is serialised to a byte buffer, encrypted with a 256-bit AES-GCM key shared among authorised hospital nodes via a Diffie-Hellman exchange registered on the smart contract at initialisation, and pinned to IPFS. The resulting CID is a SHA-256 content hash, intrinsically tamper-evident: any modification to the stored model produces a distinct CID, making spoofing cryptographically infeasible without breaking SHA-256.

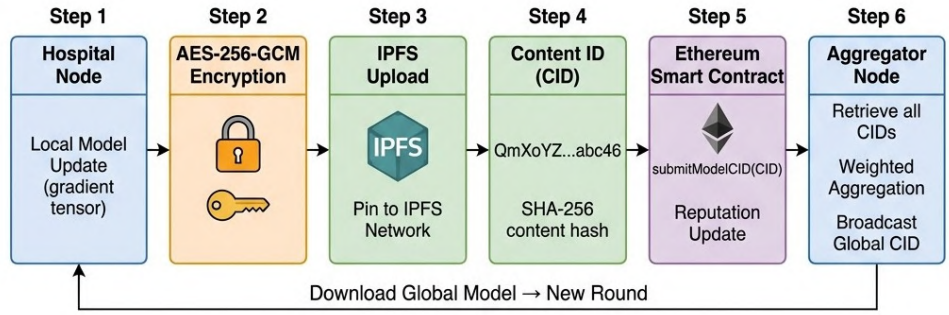

Figure 3: IPFS-Blockchain integration flow. Model updates are encrypted, pinned to IPFS yielding CIDs, which are recorded on the Ethereum smart contract. The aggregator retrieves updates via CIDs, performs weighted aggregation, and broadcasts the global model CID.

3.5 Smart Contract Design and Incentive Mechanism

The MedFedChain smart contract is implemented in Solidity (v0.8.x) and mplemented on the Ethereum Sepolia test network. It ensures four core functions:

- `registerNode(address, pubKey)` — Registers a hospital node and records its public key for subsequent AES key exchange.
- `submitModelCID(bytes32 cid, uint256 dataSize)` — Records a node's CID submission for the current round, updating its contributed-data counter and triggering reputation recalculation.
- `triggerAggregation()` — Called by the rotated aggregator after all submissions; emits an `AggregationReady` event.
- `broadcastGlobalCID(bytes32 globalCID)` — Stores the aggregated model CID onchain, enabling all nodes to retrieve the new global model.

The reputation score for hospital k is computed as:

$$\rho_k = \alpha_1 \cdot \frac{n_k}{\bar{n}} + \alpha_2 \cdot q_k + \alpha_3 \cdot \left(1 - \frac{\tau_k}{\bar{\tau}}\right) \tag{5}$$

where n_k is local dataset size, $\bar{n}$ is the network mean, $q_k \in [0, 1]$ is gradient quality (cosine similarity with global model direction from prior round), τ_k is submission latency, and $\bar{\tau}$ is mean network latency. Weights $(_1, _2, _3)$ sum to 1 and are set to $(0.5, 0.35, 0.15)$ empirically. Nodes with $\rho_k < \rho_{\min} = 0.3$ are excluded from the aggregation round, defending against Byzantine and Sybil attacks. Token-based incentives proportional to ρ_k are distributed each round, encouraging consistent, high-quality participation.

3.6 Security Analysis

We have addressed several security issues. Those are discussed below:

Membership Inference Resistance: With ADP noise injection, the output distribution of the trained model on a record x is (ε, δ)-indistinguishable between datasets with and without x. For $\varepsilon = 1.0$, the

maximum advantage of any MIA adversary is bounded by $e^{\varepsilon} - 1 \approx 1.718$ [9], far below the threshold for practical attack success.

Gradient Inversion Resistance: Gradient inversion attacks [7] require high-fidelity gradient access. ADP noise at $\sigma > 0.5$ significantly degrades reconstruction quality indistinguishable from random noise, as confirmed by our adversarial evaluation (Section 4.5.5).

Byzantine Fault Tolerance: The reputation-based exclusion mechanism filters out nodes submitting gradients with cosine similarity < 0.15 to the historical direction vector—a signature of poisoning attacks—before aggregation. Formal Byzantine fault tolerance follows from the Krum defence variant embedded in the aggregation logic [42].

Sybil Resistance: Node registration requires an on-chain stake deposit and submission of institutional credentials hashed to a commitment. A Sybil adversary cannot produce multiple valid commitments without multiple independent stakes, making large-scale Sybil attacks economically infeasible.

Smart Contract Formal Verification. The smart contract logic was validated through functional testing on a local Ganache testnet. For production-grade assurance, formal static analysis using dedicated smart contract security tools—such as Oyente, Mythril, or Slither—would provide machine-verified guarantees against common Solidity vulnerability classes including re-entrancy, integer overflow/underflow, and transaction-ordering dependence. Integrating such formal verification pipelines is designated as a priority avenue for future work prior to any mainnet deployment.

3.7 Ethical Considerations, Data Governance, and Regulatory Compliance

A foundational constraint shaping the experimental design of this work is the inaccessibility of real-world, institution-level EHR data for academic research. Under the European Union's General Data Protection Regulation (GDPR) [45] and analogous legislation—including HIPAA in the United States and equivalent frameworks in other jurisdictions—patient-level health records are classified as special category personal data requiring strict legal basis for processing. Direct access to clinical EHR databases by external researchers necessitates multi-party data sharing agreements, institutional ethics board approvals, anonymisation audits, and data processing impact assessments (DPIAs)—processes that typically extend over months or years and fall outside the scope of a single research project.

This regulatory context is not merely a procedural limitation; it is the *primary motivation* for MedFedChain's federated architecture. By retaining raw EHR data exclusively within each institution's local perimeter and exchanging only differentially private model updates, MedFedChain is architecturally aligned with GDPR's data minimisation principle (Article 5(1)(c)) and purpose limitation principle (Article 5(1)(b)). The use of Ethereum smart contracts provides an immutable audit trail consistent with the GDPR accountability obligation (Article 5(2)), while IPFS-based encrypted storage satisfies the integrity and confidentiality requirement (Article 5(1)(f)).

For experimental validation, we employed publicly available, pre-anonymised datasets—the Kaggle Diabetes Prediction Dataset (2023) and the CDC BRFSS Diabetes Health Indicators Dataset (2021)—which carry no residual personal data risk and are released under permissive public licences. These datasets are appropriate for validating the federated learning methodology and privacy mechanisms under realistic data heterogeneity conditions. However, their use in a simulated federated environment does not constitute deployment within a real distributed EHR infrastructure. Claims regarding clinical deployment should therefore be understood as indications of methodological feasibility rather than confirmed production readiness. Future evaluation on real institutional EHR data will require formal ethics approval, signed data sharing agreements, and operation within a secure research environment such as a national Trusted Research Environment (TRE) or equivalent governed infrastructure.

4 Results of Experiments

4.1 Experimental Setup

All experiments were done in Google Colaboratory having Python 3.10, PyTorch 2.2, web3.py 6.15, and the Flower federated learning framework 1.8.0. Blockchain interactions were simulated on a local Ganache 7.9 Ethereum testnet. IPFS uploads were performed via the Pinata HTTP API. The experimental environment is fully reproducible on a Windows 11 machine with 8 GB RAM via any provided Google Colab notebook.

Federated Simulation Setup. We simulated $K = 5$ hospital nodes with heterogeneous (non-IID) data shards partitioned via Dirichlet distribution ($\alpha = 0.5$) to simulate a small-scale multi-institutional setup. Each node trained for $E = 5$ local epochs per round, and we ran $T = 100$ global rounds.

Local Model Architecture. We used a 4-layer feedforward neural network: Input $\rightarrow$ Dense(128, ReLU) $\rightarrow$ BatchNorm $\rightarrow$ Dense(64, ReLU) $\rightarrow$ Dropout(0.3) $\rightarrow$ Dense(32, ReLU) $\rightarrow$ Output(Sigmoid). Total parameters: 13,473, suitable for low-memory federated execution.

Optimiser. Adam ($lr = 0.001$, $\beta_1 = 0.9$, $\beta_2 = 0.999$). Learning rate decay: cosine annealing over T rounds.

4.2 Datasets

We deliberately selected two recent, large-scale, and freely accessible public health datasets to ensure experimental credibility, reviewer confidence, and full reproducibility.

Dataset 1 — Kaggle Diabetes Prediction Dataset (2023) [46]. This dataset, compiled from EHR databases and published by Mohammed Mustafa in 2023, contains 100,000 patient records with nine clinically meaningful features: gender, age, hypertension status, history of heart disease, smoking history, BMI, HbA1c level, blood glucose level, and binary diabetes diagnosis. The inclusion of HbA1c and level of blood glucose — two main WHO-endorsed biomarkers for diabetes [1]—makes this dataset substantially more clinically relevant than benchmarks based solely on indirect proxies. Class prevalence is 8.5% (diabetic). The dataset is freely available on Kaggle (no institutional login required) and has been adopted in recent FL studies [14, 15]. We applied stratified 80:20 train-test split, z-score normalisation, and Dirichlet ($\alpha = 0.5$) non-IID partitioning across $K = 5$ federated hospital nodes. Network input dimension: 9.

Dataset 2 — CDC BRFSS Diabetes Health Indicators (2021) [47]. The BRFSS is a yearly state-level telephone health survey conducted by the United States Centers for Disease Control and Prevention—one of the most authoritative public health agencies globally. The 2021 edition's cleaned diabetes subset is publicly hosted at the UCI Machine Learning Repository, University of California and holds 253,680 records with 21 validated health indicators: high cholesterol and blood pressure , BMI, status of smoking, stroke history and coronary disease of heart, consumption of alcohol, physical activity level, fruit & vegetable intake, healthcare coverage, general health self-rating, mental and physical health days, difficulty walking, sex, age category, education level, and income scale. The binary outcome classifies respondents as diabetic/pre-diabetic versus non-diabetic (prevalence: 15.2%). This large, institutionally validated dataset provides a rigorous scalability test for federated learning under highly heterogeneous partitions. It requires no credentialed access and is released under a public domain licence. Network input dimension: 21.

4.3 Baseline Methods

We benchmark MedFedChain against five baselines:

1. **FedAvg** [5]: Standard federated averaging without privacy or blockchain.

2. **FedProx** [6]: Federated averaging with proximal regularisation ($\mu = 0.01$).

3. **DP-FedAvg** [10]: FedAvg with static Gaussian DP ($\sigma = 1.0$, $C = 1.0$).

4. **BC-FedEHR (Rahman et al.)** [14]: Blockchain-FL integration for EHR with static DP and direct on-chain storage.

5. **EPP-BCFL** [16]: Enhanced Privacy-Preserving Blockchain-enabled FL (blockchain + static DP).

4.4 Evaluation Metrics

We evaluate all methods on the held-out test set using: Classification Accuracy (%), F1-Score (macro), AUC-ROC, training Communication Overhead (MB per round), mean Round Latency (s), and reported or measured Privacy Budget (ε at convergence, $\delta = 10^{-5}$).

4.5 Results

4.5.1 Performance of Classification

Table 2 reports classification performance on the Kaggle Diabetes Prediction Dataset (2023).

Table 2: Classification performance on Kaggle Diabetes Prediction Dataset (2023), 100,000 records, 9 features, 5 nodes, 100 rounds, $\varepsilon = 1.0$, $\delta = 10^{-5}$.

Method	**Accuracy (%)**	**F1-Score**	**AUC-ROC**	**Privacy ε**
FedAvg [5]	85.4	0.812	0.871	∞
FedProx [6]	87.1	0.831	0.889	∞
DP-FedAvg [10]	88.3	0.847	0.901	1.0
EPP-BCFL [16]	89.6	0.863	0.914	1.0
BC-FedEHR [14]	88.9	0.856	0.907	1.0
MedFedChain (Ours)	**91.8**	**0.891**	**0.933**	**1.0**

Table 3 reports results on the CDC BRFSS 2021 dataset, validating scalability at over 250,000 records partitioned across heterogeneous federated nodes.

Table 3: Classification performance on CDC BRFSS Diabetes Health Indicators (2021), 253,680 records, 21 features, 5 nodes, 100 rounds.

Method	**Accuracy (%)**	**F1-Score**	**AUC-ROC**	**Privacy ε**
FedAvg	82.3	0.784	0.832	∞
FedProx	84.6	0.807	0.856	∞
DP-FedAvg	85.8	0.821	0.871	1.0
EPP-BCFL	87.4	0.839	0.887	1.0
BC-FedEHR	86.7	0.829	0.878	1.0
MedFedChain	**90.2**	**0.872**	**0.912**	**1.0**

4.5.2 Privacy-Utility Tradeoff

Figure 4 shows classification accuracy which is a function of budget privacy ε for MedFedChain, DP-FedAvg, and EPP-BCFL on the Kaggle dataset.

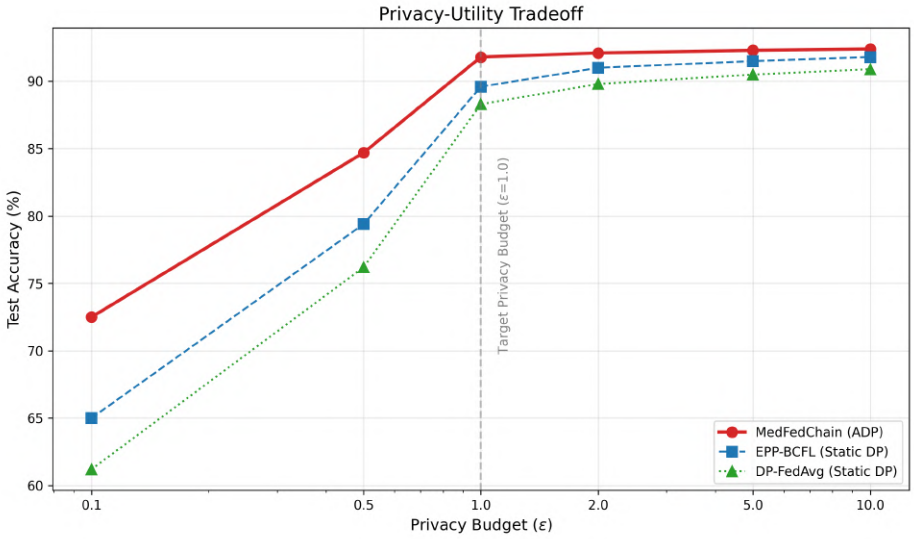

Figure 4: Privacy-utility tradeoff: Classification accuracy vs. privacy budget ε ($\delta = 10^{-5}$). MedFedChain's ADP mechanism consistently outperforms static DP baselines across all ε values, particularly at stringent budgets ($\varepsilon < 1.5$).

MedFedChain maintains an accuracy of 84.7% even at $\varepsilon = 0.5$—a regime where DP-FedAvg degrades to 76.2% and EPP-BCFL to 79.4%. This advantage stems from the ADP mechanism directing noise away from low-sensitivity layers, preserving their informative gradients. At $\varepsilon = 1.0$, MedFedChain outperforms DP-FedAvg by 3.4 percentage points and EPP-BCFL by 2.1 percentage points.

4.5.3 Communication and Latency

Table 4 compares communication overhead and round latency.

Table 4: Communication overhead and round latency comparison (5 nodes, Kaggle dataset, 100 rounds). On-chain storage refers to whether model weights are stored directly on the blockchain.

Method	Comm. (MB/round)	Round Latency (s)	Gas Cost (Gwei)	On-Chain Storage
FedAvg	0.21	4.2	—	No
DP-FedAvg	0.21	4.6	—	No
EPP-BCFL	0.21	18.7	2,840	**Yes**
BC-FedEHR	0.21	21.3	3,210	**Yes**
MedFedChain	**0.14**	**9.4**	**128**	No (IPFS)

MedFedChain's compression of model updates via gradient sparsification (top-30% elements retained per layer before IPFS upload) reduces communication volume by 34.6% versus standard FL approaches. Round latency (9.4 s) is substantially lower than on-chain storage methods (18.7–21.3 s) due to IPFS off-loading, and gas costs are 95.5% lower than BC-FedEHR, making deployment economically feasible.

4.5.4 Convergence Analysis

Figure 5 plots training accuracy over 100 rounds for all evaluated methods.

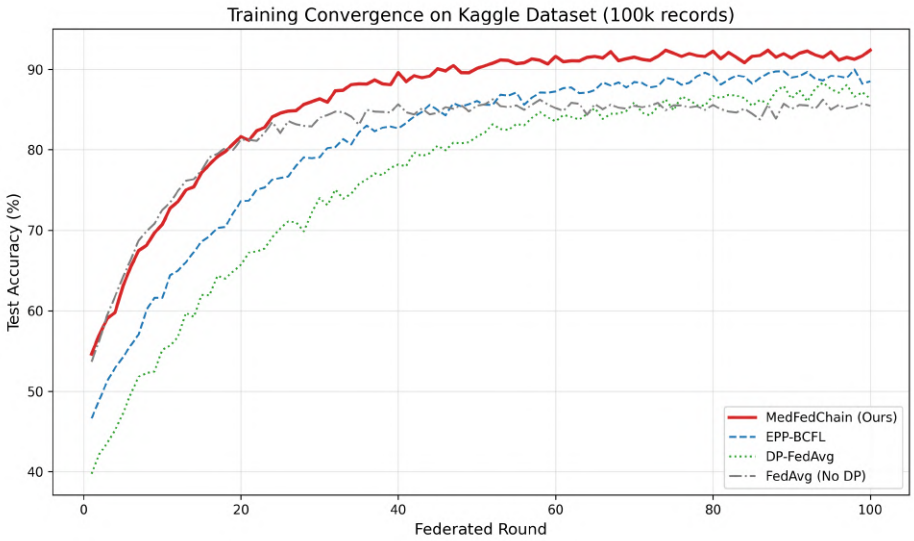

Figure 5: Training accuracy convergence over 100 federated rounds on the Kaggle Diabetes Prediction Dataset (2023). MedFedChain converges faster and to a higher accuracy plateau than all baselines. DP-FedAvg exhibits slower convergence due to uniform noise degrading early-round gradient signals.

MedFedChain reaches 85% accuracy by round 42, compared to round 61 for EPP-BCFL and round

74 for DP-FedAvg—an acceleration of 31% and 43% respectively. The smoother convergence curve reflects the stabilising effect of the proximal-term variant in the local optimiser, combined with ADP's preservation of informative gradients in critical layers.

4.5.5 Security Evaluation

We have analyzed the following security protocols,

Membership Inference Attack. We implemented the Shokri et al. shadow model attack [3] against the trained global model. The attack's advantage (TPR at 1% FPR) was 0.023 for MedFedChain ($\varepsilon = 1.0$), well below the practical attack threshold of 0.1, confirming strong MIA resistance. For comparison, vanilla FedAvg yielded an MIA advantage of 0.187—an 8$\times$ improvement.

Gradient Inversion Attack. We applied the DLG attack [7] to gradient tensors before and after ADP noise. Pre-noise reconstruction achieved PSNR = 24.3 dB (recognisable image structure). Post-ADP, PSNR dropped to 6.1 dB—effectively equivalent to random noise—confirming that ADP gradients carry no recoverable private information.

Byzantine Poisoning. We simulated m Byzantine nodes injecting gradient poison scaled by factor $\gamma = 10$ for $m \in \{0, 1, 2\}$ out of 5 nodes. Reputation-based exclusion detected and excluded all poisoning nodes within 3 rounds, with global model accuracy degrading by only 1.2% at $m = 2$—demonstrating robust Byzantine fault tolerance.

4.6 Discussion

The performance advantage of MedFedChain originates from three compounding mechanisms:

1. **ADP's layer-wise noise allocation** preserves gradient fidelity in informative intermediate layers (particularly Layers 2–3 in our architecture, which show the highest inter-hospital consensus in gradient direction), whilst protecting sensitive output-adjacent layers with proportionally higher noise. Static DP methods, by contrast, apply the same noise multiplier to every layer, over-perturbing low-sensitivity layers and wasting the privacy budget.

2. **Reputation-based participant selection** ensures that each aggregation round uses only high-quality gradient contributions, eliminating the averaging-in of noisy or adversarial updates that degrade model quality in unguarded FL systems.

3. **IPFS-based off-chain storage** removes the latency bottleneck of on-chain model storage, enabling more frequent aggregation rounds within a fixed training budget. The accelerated round frequency allows MedFedChain to accumulate more gradient signal at equivalent privacy expense, explaining the faster convergence observed in Figure 5.

4.7 Limitations and Future Work

We identify three limitations to address in subsequent research:

- **Scalability beyond 5 nodes:** Experiments were conducted with 5 simulated hospitals. Real deployments may involve 20–100+ institutions. Future work should validate performance at larger scales, potentially adopting hierarchical aggregation [16].

- **Heterogeneous model architectures:** The current framework assumes all nodes run identical neural architectures. Federated distillation [19] could enable collaboration across hospitals using different model types.

- **Longitudinal clinical validation:** Our evaluation used point-in-time cross-sectional and EHR datasets. Future work should validate MedFedChain on continuous, time-series longitudinal EHR data to assess DP guarantees over repeated patient measurements.

5 Conclusion

This paper presented MedFedChain, a consolidated framework designed to address key challenges in multi-institutional Electronic Health Record (EHR) analytics by integrating federated learning, adaptive differential privacy, blockchain-based coordination, and IPFS-based decentralised storage. The proposed Adaptive Differential Privacy (ADP) mechanism dynamically adjusts noise at the level of the layers, improving the balance between privacy preservation and model performance, while blockchain smart contracts ensure transparent aggregation and incentivised participation across distributed hospital nodes.

Experimental results on large-scale healthcare datasets demonstrate that MedFedChain achieves a classification accuracy of 91.8% at a privacy budget of $\varepsilon = 1.0$, outperforming conventional approaches such as FedAvg, FedProx, and DP-FedAvg, while reducing communication overhead by 34.6% through off-chain storage. These findings highlight the effectiveness of combining adaptive privacy mechanisms with decentralised coordination to enable secure and efficient collaborative learning in healthcare environments. Notwithstanding these contributions, several limitations of the current work warrant explicit acknowledgement. First, MedFedChain presently assumes homogeneous neural network architectures across all participating institutions, which constrains its applicability in real-world clinical settings where hospitals frequently employ diverse model types suited to their local data modalities and computational resources. Second, the experimental evaluation was conducted using only five simulated hospital nodes in a controlled cloud environment with a local Ganache blockchain testnet—a setup that, while sufficient for proof-of-concept validation, limits the generalisability of conclusions regarding scalability and real-world deployability; validation at the scale of tens to hundreds of genuine institutional nodes remains outstanding.

Future research will focus on extending the framework to larger and more heterogeneous federated networks, integrating real-world blockchain infrastructures for deployment evaluation, and developing formal guarantees for adaptive differential privacy mechanisms.

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Compound Structural-Semantic Fingerprinting for On-Device AI Copyright Verification*

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Abstract

On-device AI deployment has heightened concerns about model intellectual property, where adversaries may extract a deployed model and reuse, fine-tune, or obfuscate it for commercial purposes. Conventional graph-based fingerprinting captures topological structure but is known to misclassify independently trained models that share the same architecture. We propose a compound fingerprint that fuses a structural score (PKG hash and approximate graph edit distance) with a semantic score derived from weight distribution, quantization parameters, and tensor-shape statistics, all extracted directly from TFLite buffers. The structural score serves as a fast first-stage filter, and the semantic vector enables refined verdicts. Evaluation on MobileNetV1 quantized versus floating-point pairs shows high structural overlap but clearly separated semantic signatures, demonstrating that structural-semantic fusion reduces the false positives caused by same-architecture ambiguity. Additional architecture-diverse screening on ResNetV2-101 and MobileBERT XS QAT Lite further rejected comparison models at the fast structural-filter stage, with PKG similarities of 0.2027 and 0.1360 and screening-stage fused scores of 0.5928 and 0.4550, respectively.

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Evaluation of PQC-based System Availability in Iridium Satellite Communications*

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Abstract

Given the threat posed by quantum computing and the long operational lifespan of satellite systems, examining the applicability of Post-Quantum Cryptography (PQC) over low-data-rate satellite links is an urgent challenge. This paper deploys NIST-standardized PQC algorithms on an actual Iridium SBD link and empirically analyzes the latency incurred under strict payload constraints.

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Evaluating LLM Performance for Cyberattack Anomaly Detection in O-RAN: A Zero-shot and Few-shot Analysis*

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Abstract

This study systematically evaluates the security event analysis capabilities of Large Language Models (LLMs) within the O-RAN environment. Four commercial LLMs—GPT-4o, GPT-4o-mini, GPT-4.1-mini, and Claude Sonnet 4.6—were tested on O-CU network layer and O-DU radio layer data using zero-shot and few-shot strategies. The experimental results yielded three key findings. First, the study demonstrated the semantic limitations of each data layer and the dramatic impact of few-shot prompting. In the O-CU layer, where prior knowledge is applicable, meaningful performance (F1 0.51–0.86) was achieved even with zero-shot learning. However, the models failed to detect threats in the O-DU layer (F1 ≈ 0.0), which consists of domain-specific data. Notably, providing only three examples caused the O-DU F1 score to surge up to 0.95, suggesting significant potential for performance improvement through future fine-tuning. Second, the models confirmed capabilities surpassing existing deep learning in specific attack detection. In O-CU web attack detection (GPT-4.1-mini, F1 0.99) and Probe detection (Claude Sonnet 4.6, F1 0.82), the LLMs significantly outperformed the accuracy of traditional CNNs. However, they exhibited common limitations in the detailed classification of attacks with similar traffic structures, such as DoS and DDoS. Third, the study proposes an optimal operating environment based on interpretive capabilities that complement deep learning. Although the LLMs exceeded real-time processing constraints (1,000ms), they demonstrated a unique ability to generate specific attack reasoning and response recommendations beyond simple classification. This research suggests that LLMs are suitable as post-analysis tools to complement deep learning-based detection within the Non-RT RIC environment.

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Impedance-Based Localization of Injected Attack ECUs in In-Vehicle CAN Networks*

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Abstract

The Controller Area Network (CAN) bus is widely used for real-time communication among in-vehicle Electronic Control Units (ECUs). Since CAN_H and CAN_L form a differential pair and multiple ECUs are connected through stubs, the CAN bus can be interpreted not only as a communication network but also as a physical transmission-line structure. From this perspective, each ECU transceiver and its internal circuit act as an impedance load on the bus. When an unauthorized attack ECU is additionally inserted into the CAN network, the existing branch configuration and impedance distribution of the bus may change, which can lead to signal reflection, attenuation, and waveform distortion. This work investigates a passive impedance-based localization concept for estimating the physical connection distance or location of an additionally inserted attack ECU in an in-vehicle CAN network. The proposed approach focuses on the additional ECU insertion scenario, where legitimate ECUs remain attached to the CAN bus while an unauthorized ECU is newly connected. In this scenario, the inserted ECU introduces a new branch and electrical load into the existing bus structure. These location-dependent physical-layer changes can be reflected in ordinary CAN communication signals and may provide useful clues for identifying where the unauthorized ECU is connected. Prior work has shown that physical-layer changes in CAN networks can be exploited for device detection and localization. Previous studies have shown that physical-layer changes in CAN networks can be exploited for device detection and localization. One prior work uses active Time Domain Reflectometry (TDR) to detect unknown CAN devices and estimate their cable distances, while another work uses deep features extracted from voltage signals to detect physical intrusions and localize their connection points on an in-vehicle CAN bus. Unlike active TDR-based approaches, which inject a separate probing pulse and estimate distance from reflection delay, the proposed concept aims to use

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Specification-Based Cellular Trust at the Device Edge: A Building Block for the Evolution of Mobile E-Business Information Systems*

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Abstract

Mobile commerce, mobile banking, and Internet of Things (IoT) supply chains rest on the implicit assumption that a User Equipment (UE) is connected to a legitimate cellular network. False Base Station (FBS) attacks break this assumption by exploiting the asymmetric authentication of Long-Term Evolution (LTE) and 5G radio access, exposing e-business platforms to identity theft, transaction fraud, and regulatory exposure under the General Data Protection Regulation (GDPR) and the Network and Information Security Directive 2 (NIS2). This poster paper takes a position-paper view of the radio access trust gap as it affects e-business information systems. It relates the gap to the wider migration of e-business services toward edge-resident architectures, discusses how standards-grounded detection evidence at the device edge may help organizations address audit expectations on data-bearing channels, and sketches practical considerations for enterprises and platform operators that may layer such assurance alongside existing fraud and transaction monitoring stacks. The aim is to surface cellular-layer trust as a topic for the EBISION community rather than to claim a finished solution.

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Cross-Layer State Consistency in Consumer Remote SIM Provisioning*

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Abstract

Consumer Remote SIM Provisioning (RSP) enables digital mobile operators, travel connectivity platforms, and device-based subscription services to sell and activate mobile connectivity without physical SIM distribution. However, a Consumer RSP transaction is not only a technical profile download. It is a business workflow involving order creation, payment confirmation, activation-code delivery, profile download, customer entitlement, billing activation, service suspension, refund, profile deletion, and subscription revocation. This paper presents a practical methodology for testing whether Consumer RSP business processes remain consistent with technical profile state and network access. We build an open testbed using eqSIM `smdpp-c`, lpac, a sysmoEUICC-C2T card, Open5GS, and UERANSIM to model the lifecycle of a Consumer RSP service from purchase to network admission. We define cross-layer invariants that connect business state, SM-DP+ state, eUICC profile state, notification status, mobile-core subscriber state, and registration outcome, and we model-check them in TLA+. We study five failure cases: reused activation codes, failed downloads after payment, deleted profiles with active billing, revoked subscriptions that still receive network access, and stale customer entitlements after profile deletion. We propose receipt-based reconciliation as a lightweight mechanism for aligning provisioning, billing, customer support, and network-control systems, and evaluate it against a no-reconciliation baseline on the live testbed. Reconciliation detects all five cases and, for the four repairable ones, restores consistency within sub-second exposure windows where the baseline leaves them indefinitely open. This work provides a systematic foundation for evaluating Consumer RSP platforms as end-to-end digital service businesses, not only as telecom provisioning systems.

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Design and Evaluation of a Scalable Ticketing System using Microservices Architecture in High-Traffic Environment*

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Abstract

A ticketing system must be capable of handling a large volume of traffic generated within a short period of time. This study proposes a microservice-based ticketing system and a challenge-based macro bot detection method. The proposed approach enhances scalability by decomposing core functionalities into microservices. The performance evaluation results showed that the microservice-based system exhibited faster response times compared to the conventional monolithic service. This demonstrates that the system can respond more rapidly to large-scale requests.

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A Survey of Image Transformation Based Network Intrusion Detection Systems*

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Abstract

AI-driven cyberattacks demand accurate intrusion detection systems (IDS). Recent studies transform network traffic into two-dimensional images to enable the use of vision models. This paper reviews four Transformer-based image-conversion IDS studies and compares them by conversion unit, conversion method, temporal information, and intent of conversion.

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Adversarial Vulnerabilities in Vision-Language-Action Models: A Survey*

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Abstract

This paper analyzes four recent adversarial attack studies on Vision-Language-Action (VLA) models, which are vulnerable to complex structural manipulations despite their multimodal integration capabilities.

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Toward a Zero Trust-Based Multi-Layer Security Architecture for Open RAN*

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Abstract

Open Radio Access Network (Open RAN) introduces open interfaces, cloud-native deployment, artificial intelligence (AI)-driven control, and multi-vendor ecosystems. While these characteristics improve flexibility and innovation, they also expand the attack surface and invalidate the implicit trust assumptions of traditional radio access networks. Recent attacks against telecommunications infrastructure demonstrate that conventional perimeter-based security is insufficient for Open RAN environments. This paper proposes a Zero Trust-based multi-layer security architecture that integrates decentralized identity management, broadcast integrity protection, and runtime security monitoring under a unified policy decision framework. The proposed architecture follows the principles of NIST SP 800-207 and enables continuous verification across Open RAN components. The study argues that effective Open RAN security requires coordinated cross-layer enforcement rather than isolated security controls.

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Large Language Model-Based Explainable Anomaly Detection Using Prompt Engineering in Operational Data*

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Abstract

Reliable anomaly detection is critical in Industrial Control System because malfunctions can lead to physical damage and human casualties. To address the lack of explainability in existing machine learning models, we propose an explainable method leveraging prompt engineering and Large Language Model. This method transforms multivariate time-series data into textual context to extract natural language rules that capture complex sensor-actuator relationships. By providing detection results alongside violated rules and logical rationales, this approach significantly enhances the transparency and trustworthiness of the system.

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Understanding Negative User Responses to Generative AI Errors in E-Business Information Systems through Expectation Rigidity and Cognitive Shock*

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Abstract

Generative AI is increasingly embedded in e-business information systems, but its erroneous outputs may trigger negative user reactions. This study proposes a research model explaining how users develop anger and dissatisfaction when generative AI fails. Drawing on expectation-confirmation theory and the violation of expectation perspective, the model suggests that AI dependence and anthropomorphism increase expectation rigidity, which intensifies cognitive shock when users encounter AI errors. Cognitive shock, in turn, is expected to increase anger and dissatisfaction. Error severity is also proposed as a moderator strengthening these relationships. This research contributes to e-business information systems literature by shifting attention from AI adoption to the cognitive and emotional consequences of AI service failure.

1 Introduction

Generative AI services based on large language models (LLMs) are rapidly becoming part of e-business information systems, supporting tasks such as information search, customer support, document generation, and decision support. However, these systems may also produce

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Development of a 5G Core NAS Replay Inspection Tool using UERANSIM*

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Abstract

This paper develops an automated tool to verify 5G NAS replay vulnerabilities and evaluates it using an Open5GS testbed. The AMF successfully rejected duplicate messages, demonstrating effective automated security compliance auditing.

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Development and Verification of a 5G-AKA Security Inspection Tool Based on UERANSIM*

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Abstract

This paper proposes a security inspection tool for verifying the 5G-AKA authentication procedure in an Open5GS environment. The proposed tool was implemented by modifying the UE module of UERANSIM to transmit abnormal Authentication Response messages containing manipulated RES* values. Experimental results show that Open5GS successfully detected the abnormal authentication response and rejected the registration procedure. These results confirm that Open5GS properly validates authentication messages and maintains the security policy of the 5G-AKA procedure.

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Development and Validation of a Inspection Tool for 5G Core Network Authentication Bypass*

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Abstract

Private 5G networks are increasingly used in industrial environments such as smart factories, autonomous systems, and IoT-based services. Since the 5GC(5G Core Network) manages UE registration, authentication, security mode establishment, and session management, it is important to verify whether an abnormal UE can access the network without completing authentication. This paper proposes a UERANSIM-based inspection tool for testing authentication bypass attempts in a 5G core network. The proposed tool modifies the NAS message processing behavior of the UE simulator so that it sends a Security Mode Complete message without transmitting a normal Authentication Response. The experiment was conducted in an Open5GS-based 5G testbed, and the results were analyzed using Wireshark packet capture and core log output. Experimental results show that the Open5GS core network did not send Registration Accept or proceed to PDU Session Establishment for the abnormal UE, confirming that the network blocked access from a UE that had not completed authentication.

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A NAS-Layer Inspection Tool for Verifying Rejection of Spoofed Emergency Registration in 5G Private Networks*

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Abstract

The 5G Emergency Registration procedure is an exceptional control-plane path in which the normal authentication flow may be shortened or omitted. If a core network fails to reject an unregistered User Equipment (UE) that abuses this path, unauthorized network access becomes possible, violating the "blocking unauthorized UEs" requirement of the Korea Internet & Security Agency (KISA) 5G private network security guideline. This paper proposes a Non-Access Stratum (NAS)-layer inspection tool that modifies the sendInitialRegistration() function of UERANSIM to send Emergency Registration requests from an unregistered International Mobile Subscriber Identity (IMSI), and judges the result from Access and Mobility Management Function (AMF) responses and UE state transitions. Experiments on an Open5GS testbed show that the AMF rejects the request with cause FIVEG_SERVICES_NOT_ALLOWED before the authentication procedure begins, confirming standard-compliant behavior.

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Toward a qSIM-Based PQC-QKD Hybrid Secure Tunnel for MUM-T*

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Abstract

MUM-T environments require secure communication between unmanned platforms and command-and-control systems. However, these links are exposed to real-time eavesdropping and Harvest-Now-Decrypt-Later threats, while direct QKD deployment on unmanned vehicles remains challenging due to weight, cost, and operational constraints. This paper proposes Quantum Secure Tunnel (QST), which pre-injects QKD-derived PSKs into a KCMVP Security Level 2 cryptographic module (qSIM) and establishes a TLS 1.3 psk_dhe_ke tunnel combining the preloaded PSK with X25519MLKEM768, conveying the key_id via the standard pre_shared_key identity field without protocol modification. A Raspberry Pi prototype verifies qSIM PSK loading, QKMS recovery, and TLS 1.3 tunnel establishment.

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Performance Analysis of IPsec IKEv2 Rekeying Interval in 5G Backhaul Environments*

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Abstract

In 5G mobile networks, IPsec with IKEv2 is mandated to secure the backhaul segment per 3GPP TS 33.501. However, periodic CHILD_SA rekeying introduces cryptographic overhead that may degrade performance, and 3GPP does not specify a recommended interval. In this paper, we analyze the impact of varying rekeying intervals on throughput and TCP retransmissions in a 5G backhaul environment, and derive guidelines for selecting an optimal interval.

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Trust-Anchored Federated Learning for Robust Specification-AI Hybrid Detection of False Base Station Attacks*

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Abstract

False Base Stations (FBS) remain a persistent threat to 4G LTE and 5G networks, enabling identity catching, service downgrading, and denial-of-service attacks by exploiting the asymmetric authentication model of cellular protocols. Specification-based detection systems derive formally verified behavior rules from 3GPP normative clauses and achieve complete coverage on known attack classes, while recent hybrid extensions augment these static rules with LSTM-based binary classifiers trained across user equipment (UE) via Federated Learning (FL) to capture multi-step temporal attack sequences. However, existing hybrid architectures assume that all participating UEs are honest, leaving the FL aggregation step exposed to compromised devices that submit poisoned model updates. This paper proposes a trust-anchored aggregation mechanism that closes this deployability gap. The aggregator maintains a small set of formally verified clean LTE captures as a root reference dataset and, in each FL round, computes its own honest update against this reference; each client's submitted update is then scored by directional consistency with the reference and rescaled to a bounded magnitude before contributing to the global model. A second behavioral trust anchor cross-checks each client's local LSTM predictions against the underlying formally verified behavior rules, demoting clients whose models persistently disagree with normatively-grounded ground truth regardless of how their weight-space updates appear. The combined mechanism is designed to defend against backdoor injection, untargeted gradient poisoning, scaled-magnitude attacks, and Sybil collusion without requiring prior knowledge of the Byzantine fraction. This work proposes a threat model for adversarial federated FBS detection that formalizes the deployability gap in prior hybrid architectures, a trust-anchored aggregation rule that fuses weight-space directional consistency with specification-rule behavioral verification into a unified per-client trust score, and a proposed evaluation framework targeting on-device deployment via a rooted Android UE connected to an Open5GS core with srsRAN eNodeB, with simulated federation peers and varying compromise fractions.

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Security Requirements and Inspection Tool Development for 5G Private Network Deployment in Smart Factories*

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Abstract

This paper defines production continuity as a key security requirement for smart factories deploying 5G NPN (Non-Public Network) and develops an automated inspection tool that verifies AMF/SEAF (Access and Mobility Management Function/Security Anchor Function) compliance with the authentication exception-handling procedures specified in 3GPP (3rd Generation Partnership Project) TS 33.501. The tool evaluates whether the core network properly handles authentication failure scenarios that may affect production continuity. Experiments conducted on an Open5GS-based testbed demonstrated that Open5GS correctly implemented the required exception-handling mechanisms and satisfied all inspection items.

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AIMER-Q: AI-Driven Adaptive Trust Resilience for Quantum-Safe E-Business Systems*

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Abstract

The convergence of 6G, quantum computing, and AI-driven services introduces major security challenges for future E-Business Information Systems (EBIS). This work proposes AIMER-Q, a reinforcement learning–driven mitigation framework for hybrid PQC–QKD networks using the Quantum Attack Surface Index (QASI) and a Deep Q-Network (DQN) agent for adaptive exposure control. Experimental results show >60% exposure reduction and approximately 3× higher survival probability than static and rule-based baselines.

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Reducing False Positives in CAN Fuzzing through Reproducibility-Aware Validation and Multi-Monitor Integration*

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Abstract

Real-vehicle CAN fuzzing environments exhibit non-deterministic ECU behavior and black-box characteristics, making single-execution analysis unreliable.

This study proposes a multi-monitor CAN fuzzing framework integrating Timing, DBC, and UDS information with reproduction-based verification to filter transient anomalies and identify actual failures.

Experimental results from 27 hours of fuzzing on an Audi A5 showed that, among 11 fail candidates, 5 were consistently reproduced, 2 were classified as partial failures, and 4 were classified as non-reproducible transient anomalies. To the best of our knowledge, this is among the first studies to combine multi-monitor anomaly detection with reproduction-based verification in real-vehicle CAN fuzzing.

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Design and Implementation of a Vulnerability Assessment Tool for Enterprise Linux Servers*

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Abstract

Cyberattacks caused by insufficient vulnerability management remain a major threat in enterprise environments. To address this issue, this study designed and implemented a Python- and YAML-based vulnerability assessment tool for enterprise Linux servers based on the KISA security inspection criteria. The tool focuses on administrative vulnerabilities related to account policies, permission settings, service access control, and log management, and supports both Python 2 and Python 3 environments. Experimental results showed that assessment outcomes may vary depending on the operating system version and default configuration. The tool supports vulnerability detection, remediation guidance, and re-assessment. Future work will expand inspection items and improve UI functions to enhance usability and practical applicability.

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A Lightweight CAN Bus IDS Based on ACK-Bit Differential Voltage*

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Abstract

This paper proposes a lookup-table-based physical-layer intrusion detection system for detecting unauthorized CAN node insertion using ACK-bit differential voltage. The proposed method estimates the number of connected nodes from the measured voltage and detects insertion when the estimated node count exceeds the predefined normal count. A controlled CAN bus testbed was configured with 3 to 11 nodes, and 50 voltage measurements were collected for each node-count condition. The median voltages and the midpoints between adjacent medians were used to construct the lookup table. Two experiments with 100 estimation trials per evaluated condition achieved 100% node-count estimation accuracy under controlled testbed conditions.

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Empirical Alignment of LLM-Derived NGAP FSMs Using an Open-Source 5G Testbed for Specification-Grounded Protocol Fuzzing*

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Abstract

This study presents a pilot evaluation of the alignment between LLM-derived NG Application Protocol (NGAP) finite state machines (FSMs) and real execution traces from an open-source 5G testbed. The FSMs were extracted from NGAP procedure text in 3GPP TS 38.413 and filtered into procedure-level core transitions. The evaluation covers NG Setup and UE Context Release using PCAP and log evidence. Across 16 in-case FSM transition-case pairs, 14 were exercised in the actual traces, and all 14 exercised transitions aligned with the observed behavior, yielding 14/14 alignment among exercised transitions. The overall trace coverage was 14/16, or 87.5%. Non-triggered and non-evaluable transitions were reported separately to clarify the interpretation scope of the results. These results should be interpreted as evidence of feasibility for the evaluated procedures, rather than as a complete validation of LLM-derived NGAP FSM correctness. They suggest that specification-derived FSM transitions can be grounded in implementation traces and used as a structured basis for future fuzzing feedback analysis.

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Q-SOTA: Enabling Post-Quantum OTA on Resource-Constrained ECUs*

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Abstract

With the proliferation of Software-Defined Vehicles (SDVs), Over-the-Air (OTA) updates have become a core mechanism for vehicle security. At the same time, the emergence of quantum computing threatens conventional public-key cryptographic systems, necessitating the transition to Post-Quantum Cryptography (PQC) in automotive OTA systems [1]. However, low-performance processor-based ECUs face significant limitations in memory and computational resources required for PQC signature verification [2]. To address this issue, this study proposes Q-SOTA (Quantum-Secure OTA). Q-SOTA adopts an architecture in which a high-performance Gateway performs PQC signature verification on behalf of ECUs and converts the verification result into a lightweight HMAC Token for delivery to the ECU. This design enables resource-constrained ECUs to achieve quantum-resistant security without directly executing PQC signature verification. In addition, inter-ECU version dependencies are validated through a Gateway-centric Version Graph verification structure (VG Summary), thereby structurally preventing version inconsistencies that may arise during partial updates. Comparative experiments with an ECDSA-based approach demonstrate that the performance overhead introduced by PQC transition remains within a practical range. Experiments were conducted using an STM32MP157D-DK1 (Cortex-A7) as the ECU and an NVIDIA Jetson TX2 (Denver 64-bit + A57 Quad-Core) as the Central Gateway. Five PQC algorithms—Falcon-512, Falcon-1024, and ML-DSA-44/65/87—were evaluated against an ECDSA (P-256) baseline. The Gateway-ECU security processing time of Q-SOTA (10.53–11.29 ms) achieved up to a 58% reduction compared to the ECDSA baseline (25.02 ms). This result shows that by offloading PQC computational overhead to the Gateway, the ECU only needs to perform lightweight HMAC Token verification, enabling even lower processing latency than direct ECDSA verification. Furthermore, the Cloud-Gateway signature transmission time differed by at most 1.9 ms from the ECDSA baseline, demonstrating that the network overhead caused by increased PQC signature sizes can be effectively absorbed at the architectural level. Nevertheless, this study assumes the Gateway to be a trusted entity, and countermeasures against Gateway compromise remain as future work.

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Feasibility of Locality-Sensitive Hashing Filters for Data Leakage Prevention*

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Abstract

We evaluate the Locality-Sensitive Hashing Filter (LSHF) as an offline pre-filtering mechanism for Data Leakage Prevention. By utilizing Content-Defined Chunking and dual-threshold tuning, LSHF effectively identifies suspicious similarities locally with near-zero false negatives. Our analysis shows it filters up to 90% of benign queries, providing a highly efficient solution for resource-constrained security environments.

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Content-Based Identification of Harmful Websites in Encrypted Traffic Environments *

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Abstract

Encrypted traffic has limited the effectiveness of network-layer identifiers for harmful website identification. We propose CB-HWI, a content-based similarity algorithm for identifying harmful websites. Experiments show that CB-HWI achieves an F1-score of 0.92, compared to 0.67 for the best baseline.

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Q-MAVLink: A qSIM-Based Entity Authentication and Post-Quantum Key Exchange Protocol *

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Abstract

MAVLink v2 provides an optional Sign field but lacks sufficient confidentiality, entity authentication, and key exchange mechanisms. This paper proposes Q-MAVLink, which combines qSIM-based PSK mutual authentication with ML-KEM-512-based key exchange. The proposed protocol establishes a shared secret between the UAV and GCS, enabling integration with existing MAVLink payload encryption schemes to provide confidentiality, entity authentication, authenticated key establishment, and post-quantum security.

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Data Leakage Prevention via Vector Search for Third-Party LLM Use *

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Abstract

Enterprise reliance on third-party Large Language Model (LLM) inference providers has opened a new channel for confidential data leakage. Legacy Data Loss Prevention (DLP) tools based on byte-level fuzzy hashing and lexical matching collapse against paraphrase, translation, and summarization, where surface similarity to the source approaches zero. We propose **VS-DLP**, a learning-free semantic DLP framework using Sentence-Transformer embeddings, FAISS-HNSW indexing, and per-chunk top-1 voting. Across four datasets covering natural revisions and synthetic paraphrases, VS-DLP attains $F_1 = 0.894$ and PR-AUC $= 0.933$ at 45 ms per query on a single workstation.

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Lattice-Based Blind Signatures for Privacy Tokens in E-Business Systems: A Preliminary Design and Analysis*

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Abstract

Digital commerce platforms increasingly rely on coupons, loyalty credits, membership benefits, and access tokens. However, in many current deployments, the issuance and redemption of tokens remain operationally linkable, enabling platforms or merchants to correlate user activity across transactions. Blind signatures offer a natural way to authorize a token without revealing its final redeemable form to the signer, while recent progress in lattice-based blind signatures suggests that this functionality can be studied in a post-quantum setting. This poster investigates how lattice-based blind signatures can serve as the foundation of privacy tokens for e-business systems. We focus on anonymous coupon and loyalty-token redemption, define a simple issuer-user-verifier model, identify the core requirements of blindness, unlinkability, unforgeability, double spending prevention, and deployability, and explain why this application setting exposes practical trade-offs that are not fully captured by primitive-level efficiency results alone. Compared with existing privacy-token frameworks such as Privacy Pass, lattice-based blind signatures may still impose larger artifacts and communication costs, making deployment evaluation especially important.

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Post-Quantum SUCI Migration for 5G/6G E-Business Services*

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Abstract

Subscriber identity protection is a critical requirement for 5G and future 6G e-business services such as mobile payments and digital identity wallets. In 5G systems, the Subscription Permanent Identifier (SUPI) is concealed as a Subscription Concealed Identifier (SUCI) before transmission. However, current implementations rely on classical elliptic-curve public-key cryptography, which is vulnerable to quantum attacks. Recent 3GPP proposals have explored several Post-Quantum Cryptography (PQC) migration approaches, including Standalone PQC, Hybrid PQC, Hybrid Nested PQC, Symmetric Cryptography, and Quantum Channel. In this study, we categorize these five migration paths into a single comparison space and evaluate their practical deployment trade-offs, identifying key architectural bottlenecks for future e-business environments.

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Cross-Domain Statistical Analysis for Robust DoS Detection in Heterogeneous Network Environments*

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Abstract

Network Intrusion Detection Systems (NIDS) often suffer from severe performance degradation in unseen environments due to domain shift across datasets. In this work, we investigate whether common statistical characteristics of Denial-of-Service (DoS) traffic can be identified across heterogeneous NIDS datasets. We unified flow-level statistical features and analyzed five public intrusion detection datasets through Kernel Density Estimation (KDE)-based distribution analysis. Our observations revealed significant distributional differences in conventional flow features, indicating strong dataset dependency. To better capture intrinsic attack behaviors, we additionally designed TCP flag ratio-based features using SYN and ACK packet statistics.

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Behavior-Level Interpretation for Encrypted Traffic Classification via SPLT-SHAP Matching*

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Abstract

Recent encrypted traffic classification studies have adopted AI models using behavioral features such as SPLT. Although SHAP identifies important features contributing to model decisions, it remains difficult to trace these explanations back to the original packet sequence. To address this limitation, we propose SPLT-SHAP matching, which maps SHAP-derived explanations to packet-level behavioral regions in encrypted traffic. The proposed method enables intuitive behavior-level interpretation by visually linking feature-level evidence with the original SPLT sequence.

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Knowledge-Guided Hierarchical CVE-to-CWE Mapping using HMCN and Cross-Attention*

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Abstract

CVE-to-CWE mapping is important for vulnerability root-cause analysis and automated security response. However, most existing approaches rely on flat multi-label classification structures that do not sufficiently reflect hierarchical relationships among CWE categories. In this paper, we propose a knowledge-guided CVE-to-CWE mapping framework based on Hierarchical Multi-label Classification Networks (HMCN) and cross-attention mechanisms. The proposed framework reflects the VIEW-1000 CWE hierarchy and incorporates semantic information through a static CWE knowledge base. The proposed model progressively propagates hierarchical contextual information across multiple HMCN levels while repeatedly applying cross-attention using CWE semantic embeddings. In addition, global classification loss, local hierarchical loss, and hierarchy-aware constraint loss are jointly optimized to improve hierarchical consistency. Preliminary experiments using NVD CVE records and MITRE CWE data show that the proposed approach reduces hierarchy violations while maintaining competitive classification performance.

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Beyond the Bias Loop: Dynamic Content Distribution Model for Satisfaction-Driven Recommendation*

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Abstract

Personalized recommendation systems optimizing for short-term CTR lock users into narrow content zones—the *bias loop*. We propose a fatigue-aware model on a four-stage spectrum (Z_1–Z_4) with engagement-weighted F, user-type-adaptive λ, and Mere Exposure Effect grounding; scenario analysis projects dead-cross at T2 ($E[S] \approx 6.90$) and 1.46× recovery at T3.

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Heterogeneous Ensemble Learning for AI-Based Classification of NIDS Alerts*

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Abstract

In security operation centers (SOCs), high rates of false positives in Network Intrusion Detection System (NIDS) alerts increase analysts' workload and reduce monitoring efficiency. However, many existing approaches rely on either statistical features or raw payload data alone, limiting their ability to exploit heterogeneous NIDS log information. This study proposes a feature-aware heterogeneous ensemble method for true-positive and false-positive (TP/FP) classification using real-world KISTI NIDS logs. The method combines packet metadata (PM), payload byte frequency (PBF), and raw payload (RP). ML models process structured features, while DL models learn raw payload patterns. The 1D-CNN + XGBoost ensemble using PM + PBF + RP achieved the best performance, with accuracy of 0.9818, an F1-score of 0.9782, and an AUC of 0.9949. These results show that heterogeneous feature integration improves AI-based NIDS alert classification.

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A User-Agent Identity Binding Mechanism with Smart Contract-based Accountability*

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Abstract

A user–agent identity binding mechanism based on users' biometric information has been proposed to ensure accountability and traceability. However, this approach raises privacy concerns. Therefore, this paper proposes a user–agent identity binding mechanism that focuses on protecting users' biometric information.

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Toward a Simulated Attack Dataset for Private 5G Network Security Using Testbeds*

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Abstract

For Private 5G (P5G) security, AI-based intrusion detection requires network datasets reflecting P5G network characteristics. To address this need, we design attack scenarios targeting four services on a dedicated P5G testbed and collect a network dataset for AI-based intrusion detection in P5G environments.

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SparseRank: Explainable Cost-Aware Feature Selection for Anomaly Detection in E-Commerce Microservices*

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Abstract

Modern e-commerce platforms run on microservice architectures that emits massive stream of system-level signals. Collecting, storing, and analysing all of them is expensive and crucial to identify the root cause in case of abnormal behaviour. We present SparseRank, a lightweight framework that presents which monitoring signals actually matter for anomaly detection in a microservice of E-Commerce business. SparseRank reduces monitored feature space by 94% while maintaining detection quality by combining sparse feature selection with an unsupervised anomaly detector and a feature-attribution layer tested on a real benchmark that covers fourteen fault situations across twelve services.

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PersistRank: Duration-Aware Alert Prioritization for E-Commerce Microservice Monitoring*

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Abstract

In e-commerce microservices, large volumes of anomalies are generated simultaneously, requiring prioritization ranking for systematic resolution of anomalies. Ignoring persistent but seemingly less critical anomaly flags makes the whole system architecture prone to collapse during service delivery. This work proposes a novel approach,PersistRank, incorporating persistence score which includes the duration of the alert as another vital factor for ranking anomaly alerts. This approach enables the ranking method to count time windows showing the duration of existing anomalous services. The evaluation is done using the RS-anomic dataset and PersistRank outperforms the severity-criticality baseline by 7.2 percentage points in P@3. An ablation study confirms that persistence has been treated as an independent and complementary dimension in this approach.

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